1. On the Formation of the form System of the Ternary Biquadratic form

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(Extract from the author's dissertation.)

Work by Gordan, Maisano, and Pascal[1] deals with the form system of the ternary biquadratic form. Gordan sets up the complete form system, consisting of 54 formations, for the special automorphic form

$$f = x_1^3 x_2 + x_2^3 x_3 + x_3^3 x_1,$$

on the basis of principles similar to those which he had given for form systems in the binary domain.

In Maisano's work, for the general biquadratic form, the forms up to and including the fifth order[2] are set up, together with some invariants, covariants, and contravariants of higher order, according to the method used by Gordan in volume I of the *Mathematische Annalen* for the ternary cubic form. Pascal, using Maisano's results, is chiefly concerned with the question of the decomposition of the biquadratic form into factors[3].

The aim of my investigations is to set up the form system for the general ternary biquadratic form; to begin with, only the main foundations are given, and a so-called "relatively complete system"[4] is established.

The basic idea for forming systems of ternary forms is the same as in the binary domain. Starting from a first relatively complete system – the system of forms taken over from the binary case – one passes, according to a definite law, to systems with ever higher modulus, until either the system of a modulus – taking the modulus as ground form – becomes finite and known, or until a modulus can be reduced to forms having invariants as factors. By transvection of the relatively complete system with the system of the modulus, in the first case one obtains the absolutely complete system, while in the second case the relatively complete and absolutely complete systems coincide. By Hilbert's general proof the finiteness of form systems, this procedure must necessarily come to an end.

In our case, the modulus (abc) of the first relatively complete system can be led back to the moduli

$$\Delta = (abc)^2 a_x^2 b_x^2 c_x^2 \quad \text{and} \quad \nu = (abu)^4;$$

from this the relatively complete system modulo ν is readily obtained. As the sequence of moduli we now choose the forms

$$\nu; \qquad \nu^{(\nu)} = (\nu\nu_1 x)^4 = s_x^4,$$
$$\nu^{(s)} = (ss'u)^4 \ldots,$$

and adjoin, as further moduli, two quadratic forms occurring in the formation of the relatively complete system modulo s, namely u_σ^2 and t_x^2. One can then show that the modulus following s, namely $(ss'u)^4$, is reducible to the modulus (ϱ, t). Since, however, the simultaneous system of two quadratic forms is finite and known, the termination described above has thereby been reached. As the relatively complete system modulo (ϱ, t) one obtains 331 formations. —

[1] P. Gordan, "Über the volle Formensystem of the ternären biquadratischen form" $f = x_1^3x_2 + x_2^3x_3 + x_3^3x_1$ (Math. Annalen vol. XVII, pp. 217–233, 1880); G. Maisano, 1. "Sistemi completi dei primi cinque gradi della forma ternaria biquadratica e degl' invarianti, covarianti e contravarianti di sesto grado" (Giorn. di Battaglini XIX). 2. "Sui covarianti indipendenti di 6° grado nei coefficienti della forma biquadratica ternaria" (Rend. Circ. Mat. di Palermo I, 1887); E. Pascal, "Contributo alla teoria della forma ternaria biquadratica e delle sue varie decomposizioni in fattori" (Memoria premiata dalla R. Accademia delle scienze fisiche e matematiche di Napoli, 1905).

[2] By "order" the dimension in the coefficients is to be understood, and by "degree" the dimension in the variables.

[3] In his second short note Maisano attempts to prove a linear dependence among the three covariants of order 6 and degree 6. In contrast to this not quite complete proof, Pascal believes he has proved the linear independence of the three covariants of order 6, as well as that of the three invariants of order 9. That, however, a linear relation does in fact exist between the three covariants on the one hand and the three invariants on the other, emerged in the course of my computations; in Pascal's notation the explicit formulas are

$$0 = 20\Omega_1 + 6\Omega_2 - 3\Omega_3 + 4fC_1 - 12fC_2 - 4A \cdot \Delta,$$

$$0 = 4A^3 - 15AB + 30C - 90D + 9E.$$

[4] For the terminology, see Gordan–Kerschensteiner, *Vorlesungen über Invariantentheorie*, vol. II, p. 227.

I add a few details concerning the method used.

The fundamental process for producing forms, the folding process, can be defined in the ternary domain as follows:

If in a symbolic product

$$s_x^m t_x^n u_\sigma^\mu u_\tau^\nu$$

one replaces the pairs of factors

	$s_x t_x$	$u_\sigma u_\tau$	$s_x u_\sigma$ or $s_x u_\tau$	$t_x u_\tau$ or $t_x u_\sigma$
respectively by	(stu)	$(\sigma\tau x)$	s_σ or s_τ	t_τ or t_σ
folding	I	II	III	IV,

then the forms so arising have been obtained from the original form by folding[5].

Here one may still always put $s_\sigma = 0;\ t_\tau = 0$. For the relation among the individual foldings, the following theorem holds:

The foldings I and II are fundamental foldings, from which, independently of the order of composition, the foldings III and IV can be composed. In other words: to form all forms arising from a given expression by folding, one need only apply foldings I and II[6].

By a "form sequence" we mean an initial form together with all forms arising from it by folding into itself. By the theorem, a form sequence $s_x^m t_x^n u_\sigma^\mu u_\tau^\nu$ can be arranged in a rectangular scheme. In proceeding, one obtains:

1. by moving one column to the right, all forms arising by folding I from the adjacent forms;
2. by moving one row downward, all forms arising by folding II from the forms standing above, and hence all forms arising by folding.

Under these assumptions, the theorems on reducers can be stated in their most general form, under the definition

"A reducer is a reducible form sequence,"

as follows:

If the initial form of a form sequence is reducible because one of its terms has arisen by folding with a reducer, and if the final form of the form sequence has arisen from this same term by folding, then the whole form sequence is reducible.

Further reduction methods to be mentioned are:

1. the so-called "double reduction," that is, the reduction of a form in two ways in order to obtain a relation between the higher forms;
2. "folding with decomposed forms," which partly has the character of reduction by reducers and partly that of "double reduction."

[5] Gordan, Math. Annalen, vol. XVII, p. 219.

[6] The theorem has an exception in the case where n and μ, respectively m and ν, vanish simultaneously; the "form sequence" then reduces to the diagonal term.

2. On the Formation of the form System of the Ternary Biquadratic form

Journal for the reine and angewandte Mathematik 134 (1908), pp. 23–90 and two tables

Introduction.

Work by Gordan, Maisano, and Pascal[7][8] deals with the form system of the ternary biquadratic form. Gordan sets up the complete form system, consisting of 54 formations, of the special automorphic form

$$f = x_1^3 x_2 + x_2^3 x_3 + x_3^3 x_1$$

on the basis of principles similar to those that he had given for form systems in the binary domain.

In Maisano's work, for the general biquadratic form, the forms up to and including the fifth order[9] are set up, together with some invariants, covariants, and contravariants of higher order, according to the method used by Gordan in volume I of the *Mathematische Annalen* for the ternary cubic form. Pascal, using Maisano's results, is chiefly concerned with the question of the decomposition of the biquadratic form into factors[10].

The aim of the following investigations is to set up the form system for the general ternary biquadratic form; in this work, namely, only the principal foundations are given, and a so-called "relatively complete system"[11] *is established.*

The work is closely connected with Gordan's work; however, the principles that were stated there only quite generally first had to be worked out in detail. With the aid of a theorem on the connection among the foldings and by introducing the "form sequence" (§ 1), the reduction theorems are sharply formulated and completed (§ 3), while the recursive establishment of special series expansions (§ 2) gives the computational means for actually carrying out the reductions.

The basic idea for forming systems of ternary forms is the same as in the binary domain. Starting from a first relatively complete system – the system of forms taken over from the binary case – one passes, according to a definite law, to systems with ever higher modulus, until either the system of a modulus – the modulus taken as ground form – becomes finite and known, or until a modulus can be reduced to forms having invariants as factors. By transvection of the relatively complete system with the system of the modulus, in the first case the absolutely complete system arises, while in the second case the relatively complete and absolutely complete systems coincide. By Hilbert's general proof the finiteness of form systems, this procedure must necessarily come to an end.

In our case the modulus (abc) of the first relatively complete system (§ 4) is reduced to the moduli

$$\Delta = (abc)^2 a_x^2 b_x^2 c_x^2 \quad \text{and} \quad \nu = (abu)^4$$

(§ 5), and then the relatively complete system modulo ν is found (§ 6). These results are already given in Gordan's work for the general biquadratic form; our manner of deriving them, however, is more easily transferred to forms of higher degree.

As the sequence of moduli we now choose the forms (§ 7)

$$\nu, \qquad \nu(\nu) = (\nu\nu_1 x)^4 = s_x^4, \qquad \nu(s) = (ss'u)^4, \ldots.$$

[7]A short extract from the Introduction and Chapter I appeared in the *Sitzungsberichte of the physikalisch-medizinischen Sozietät Erlangen 1907*, pp. 176–179.

[8]P. Gordan, on the complete form system of the ternary biquadratic form

$$f = x_1^3 x_2 + x_2^3 x_3 + x_3^3 x_1.$$

(*Math. Annalen*, vol. XVII (1880), pp. 217–233.) G. Maisano, 1) *Sistemi completi dei primi cinque gradi della forma ternaria biquadratica e degl' invarianti, covarianti e contravarianti di sesto grado.* (Giorn. di Battaglini XIX.) 2) *Sui covarianti indipendenti di* 6° *grado nei coefficienti della forma biquadratica ternaria.* (Rend. Circ. Mat. di Palermo I, 1887.) E. Pascal, *Contributo alla teoria della forma ternaria biquadratica e delle sue varie decomposizioni in fattori.* (Memoria premiata dalla R. Accademia delle scienze fisiche e matematiche di Napoli. 1905.)

[9]By "order" the dimension in the coefficients is to be understood, and by "degree" the dimension in the variables.

[10]In his second short note, Maisano attempts to prove a linear dependence of the three covariants of order 6 and degree 6. In contrast to this not quite complete proof, Pascal believes he has proved the linear independence of the three covariants of order 6, and likewise that of the three invariants of order 9. That, however, a linear relation does in fact exist between the three covariants on the one hand and the three invariants on the other is shown by formulas (13), § 11, and (22), § 17 note, of the present work, where these relations are given explicitly. According to a communication from Pascal, the contradiction is explained by a numerical error in the expression for his contravariant p (called ϱ here), p. 46 of his paper.

[11]For the terminology see § 3a.

In forming the relatively complete system modulo s, two quadratic forms occurring in the system, u_ϱ^2 and t_x^2, are adjoined as moduli (§§ 9 and 10), and accordingly the relatively complete system modulo (s, ϱ, t) is formed. It is then shown (§ 17) that the modulus $(ss'u)^4$ is reducible to the moduli ϱ and t. The next higher system thereby passes over into a relatively complete system modulo (ϱ, t). Since, however, the simultaneous system of two quadratic forms is finite and known, and in the general case consists of 20 formations[12], the termination described above is reached with the establishment of the relatively complete system modulo (ϱ, t). The transvection of this system with the system of (ϱ, t) in order to form the absolutely complete system is reserved.

As the relatively complete system modulo (ϱ, t) one finds 331 formations, which in the appended table are ordered according to their degree in the variables x and u.

Finally we give an overview of the symbols introduced:

$$\begin{aligned}
f &= a_x^4,\\
\theta &= \theta_x^4 u_\vartheta^2 = (abu)^2 a_x^2 b_x^2,\\
K &= K_x^6 u_k^3 = (a\theta u) u_\vartheta^2 a_x^3 \theta_x^3,\\
j &= u_j^6 = (a\theta u)^4 u_\vartheta^2,\\
\Delta &= \Delta_x^6 = a_\vartheta^2 a_x^2 \theta_x^4 = (abc)^2 a_x^2 b_x^2 c_x^2,\\
N &= N_x^8 u_\eta = (a\Delta u) a_x^3 \Delta_x^5,\\
\nu &= u_\nu^4 = (abu)^4,\\
H &= H_x^2 u_\eta^4 = (\nu\nu_1 x)^2 u_\nu^2 u_{\nu_1}^2,\\
L &= L_x^3 u_l^6 = (\nu\eta x) H_x^2 u_\nu^3 u_\eta^3,\\
g &= g_x^6 = (\nu\eta x)^4 H_x^2,\\
\sigma &= u_\sigma^6 = H_\nu^2 u_\nu^2 u_\eta^4,\\
s &= s_x^4 = (\nu\nu_1 x)^4,\\
Z &= Z_x^4 u_\zeta^2 = (ss'u)^2 s_x^2 s_x'^2,\\
\varrho &= u_\varrho^2 = \theta_\nu^4 u_\vartheta^2,\\
t &= t_x^2 = a_\eta^4 H_x^2,\\
i &= a_\nu^4,\\
J &= s_\nu^4.
\end{aligned}$$

Chapter I. General Theorems on Ternary Forms.

§ 1. The folding process. form sequences.

The fundamental process for producing forms is the folding process, which in the ternary domain can be defined as follows. Let there be given a symbolic product

$$s_x^m t_x^n u_\sigma^\mu u_\tau^\nu.$$

If one replaces the pairs of factors

$$s_x t_x \qquad u_\sigma u_\tau \qquad s_x u_\sigma \text{ or } s_x u_\tau \qquad t_x u_\tau \text{ or } t_x u_\sigma$$

respectively by

$$(stu) \qquad (\sigma\tau x) \qquad s_\sigma \text{ or } s_\tau \qquad t_\tau \text{ or } t_\sigma,$$

folding

$$\text{I} \qquad \text{II} \qquad \text{III} \qquad \text{IV},$$

then the resulting forms have arisen from the original one by folding[13].

[12] Cf. Clebsch–Lindemann, *Vorlesungen über Geometrie.* (Vol. I, p. 288 ff.) System of two cogredient forms. Gordan, "Über Büschel of Kegelschnitten." (*Math. Ann.* vol. XIX, p. 530.) System of two contragredient forms.

[13] Gordan, *Math. Annalen* vol. XVII, p. 219.

Here the expression $s_x^m t_x^n u_\sigma^\mu u_\tau^\nu$ can be regarded either as an actual product of two forms $S \cdot T$, or as a single form with several rows of symbols:

$$s_x^m t_x^n u_\sigma^\mu u_\tau^\nu = A_x^{m+n} u_x^{\mu+\nu}.$$

In the first case we speak of the folding of the form S with T, in the second case of the "folding of the form A in itself." For brevity we always assume in what follows, as indeed is the case for all forms later to be considered, that

$$s_\sigma^\lambda = 0, \qquad t_\tau^\lambda = 0 \tag{a}$$

for any λ and x distinct from 0, and in conjunction with any other foldings. This says that the form $s_x^m t_y^n u_\sigma^\mu u_\tau^\nu$ is a so-called "normal form," that is, it vanishes under application of the Ω-process, both with respect to the variables x and u and with respect to the variables y and v.

Thus condition (a) represents no restriction, but can always be achieved[14].

For the connection among the individual foldings the following theorem holds:

Theorem I. Foldings I and II are fundamental foldings from which, independently of the order of composition, foldings III and IV can be composed. In other words: in order to form all forms arising by folding from a given expression, one has to apply only foldings I and II.

Proof: By the identity theorem, respectively the product theorem for matrices, taking (a) into account, one obtains

$$(\widehat{st}\sigma x) = -t_\sigma, \qquad (\widehat{\sigma\tau}su) = -s_\tau,$$
$$(\widehat{st}\tau x) = s_\tau, \qquad (\widehat{\sigma\tau}tu) = t_\sigma,$$
$$(stu)(\sigma\tau x) = s_\tau + t_\sigma - u_x\, s_\tau t_\sigma.$$

Thus by composing foldings I and II one obtains foldings III and IV, and this holds both when applying folding II to folding I, i.e. to the factors $(stu)u_\sigma u_\tau$, and when applying folding I to folding II, to the factors $(\sigma\tau x)s_x t_x$.

By a *form sequence*[15] we mean an initial form together with all forms that have arisen from it by folding in itself, and we denote the form sequence by the initial form. A "higher form" is to mean here any form with more folding in itself, while among the equally entitled foldings s_τ and t_σ one has to be normalized as higher. According to the law of formation of the form sequence it follows that:

1) Each form of the form sequence is linear in the symbols of the initial form.

2) The form sequence is uniquely determined for initial forms with two rows each of contragredient symbols; for initial forms with more rows of symbols it is to be uniquely normalized by distinguishing special foldings among those connected by the identity theorem.

By Theorem I, a form sequence $s_x^m t_x^n u_\sigma^\mu u_\tau^\nu$ $(m > n;\ \mu > \nu)$ can be arranged according to the following rectangular scheme:

$$\begin{array}{c|c|c|c}
s_x^m t_x^n u_\sigma^\mu u_\tau^\nu & (stu) & (stu)^2 & \cdots \\
(\sigma\tau x) & t_\sigma,\ s_\tau & t_\sigma(stu),\ s_\tau(stu) & \cdots \\
(\sigma\tau x)^2 & t_\sigma(\sigma\tau x),\ s_\tau(\sigma\tau x) & t_\sigma^2,\ t_\sigma s_\tau,\ s_\tau^2 & \cdots \\
\vdots & \vdots & \vdots & \ddots \\
(\sigma\tau x)^\nu & t_\sigma(\sigma\tau x)^{\nu-1},\ s_\tau(\sigma\tau x)^{\nu-1} & t_\sigma^2(\sigma\tau x)^{\nu-2},\ t_\sigma s_\tau(\sigma\tau x)^{\nu-2},\ s_\tau^2(\sigma\tau x)^{\nu-2} & \cdots \\
\vdots & t_\sigma(\sigma\tau x)^\nu & t_\sigma^2(\sigma\tau x)^{\nu-1},\ t_\sigma s_\tau(\sigma\tau x)^{\nu-1} & \cdots
\end{array}$$

and the correspondingly continued lower part[16]

$$\begin{array}{c|c|c}
(stu)^n & \cdots & \cdots \\
t_\sigma(stu)^{n-1},\ s_\tau(stu)^{n-1} & s_\tau(stu)^n & \cdots \\
t_\sigma^2(stu)^{n-2},\ t_\sigma s_\tau(stu)^{n-2},\ s_\tau^2(stu)^{n-2} & t_\sigma s_\tau(stu)^{n-1},\ s_\tau^2(stu)^{n-1} & s_\tau^2(stu)^n.
\end{array}$$

Here one obtains, when one moves

1) one column to the right, all forms arising by folding I from the adjacent forms,

[14] Cf. Gordan, *Math. Annalen* vol. V, p. 104.

[15] Cf. Clebsch, "Über eine Fundamentalaufgabe of the Invariantentheorie" (Abh. of the Gött. Ges. d. Wiss. vol. XVII): § 17 and end of § 18. The "form sequence" differs from the "properly reduced equivalent system" introduced there by the principle of ordering according to higher forms.

[16] Here, and similarly in what follows, the lower part marked with an asterisk is to be set at the correspondingly marked place at the upper right of the scheme.

2) one row downward, all forms arising by folding II from the forms above,

and hence all forms arising by folding.

An arbitrary form of the total form sequence, $t_\sigma^\kappa s_\tau^\lambda (stu)^\mu$ or $t_\sigma^\kappa s_\tau^\lambda (\sigma\tau x)^\nu$, forms the beginning of a new form sequence, consisting of all those forms of the rectangular scheme with $t_\sigma^\kappa s_\tau^\lambda (stu)^\mu$ or $t_\sigma^\kappa s_\tau^\lambda (\sigma\tau x)^\nu$ as left upper corner, which have the factor $t_\sigma^\kappa s_\tau^\lambda$.

Supplementary remark. Theorem I has an exception in the case where the numbers n and μ (respectively m and ν) become equal to zero, so that foldings I, II, and IV no longer exist, while folding III (respectively IV) still does. In this case we designate as the form sequence the diagonal member of the scheme:

$$\begin{array}{ccccc}
s_x^m u_\tau^\nu & & & & t_x^n u_\sigma^\mu \\
& s_\tau & & t_\sigma & \\
& s_\tau^2 & & t_\sigma^2 & \\
& & \ddots & & \\
& & s_\tau^\nu & & t_\sigma^\mu .
\end{array}$$

In accordance with the absence of foldings I and II, the series expansion according to polars of the form sequence in this case always reduces to the initial member; however, the theorems on reducers remain valid.

§ 2. Series expansions according to polars of the form sequence.

For forms with n variables two kinds of series expansions have been set up explicitly[17]:

1) for forms with one row each of contragredient variables, $s_x^m u_\tau^\nu$, a series progressing by powers of u_x, whose coefficients are "normal forms";

2) for forms with two rows of cogredient variables, $s_x^m t_x^n$, an expansion according to polars of the elementary covariants (polars of the form sequence $s_x^m t_x^n$), which in the ternary case reads as follows[18]:

$$s_x^m t_x^{n-\lambda} t_y^\lambda = \sum_\varrho \frac{\binom{m}{\varrho}\binom{\lambda}{\varrho}}{\binom{m+n-\varrho+1}{\varrho}} \left[(st\widehat{xy})^\varrho s_x^{m-\varrho} t_x^{n-\varrho}\right]_{y^{\lambda-\varrho}}. \tag{I}$$

We combine the two expansions by setting up, for forms with two rows each of contragredient variables,

$$s_x^m t_x^{n-\lambda} t_y^\lambda u_\sigma^\mu u_\tau^{\nu-\kappa} v_\tau^\kappa,$$

expansions according to powers of u_x, whose coefficients are composite polars of the form sequence $s_x^m t_x^n u_\sigma^\mu u_\tau^\nu$, taken with respect to the variables x and u.

It is enough to set up the series for two contragredient rows of symbols (respectively variables), since, by combining each two rows of symbols (variables) into a single new row, forms with more rows of symbols (variables) can be reduced to this case.

In order to ensure that all forms of the form sequence contain only two rows of symbols, respectively variables, we specialize:

$$\text{a) } \sigma = \widehat{st}, \quad \text{a') } y = \widehat{uv},$$
$$\text{b) } s = \widehat{\sigma\tau}, \quad \text{b') } v = \widehat{xy},$$

and carry out the expansion for the specialization a), a').

By direct transfer of expansion I one obtains composite polars for forms $(stu)^\mu s_x^m t_x^{n-\lambda} t_y^\lambda$, whereas for forms $(stu)^\mu u_\tau^{\nu-\kappa} v_\tau^\kappa s_x^m t_x^{n-\lambda} t_y^\lambda$, instead of composite polars, terms of such a kind arise whose evaluation makes necessary the introduction of ever new auxiliary variables that are to be set equal only at the end, thereby complicating the calculation unnecessarily.

We therefore take an indirect route, by representing the composite polars of all forms of the form sequence, that is, expressions

$$\left[s_\tau^\varrho (stu)^{\mu-\varrho+r}\right]_{y^\lambda}^{v^\kappa},$$

[17] Cf. Gordan, "Über Kombinanten," §§ 2 and 5 (*Math. Ann.* vol. V).

[18] Cf. Study, *Methoden to the Theorie of the ternären Formen* (Teubner 1889) II. § 7. In distinction to the derivation there, ours gives the method for the rapid computational determination of the numerical coefficients $C_{pr\varrho}$. The Clebsch–Study expansion, upon specialization a), a'), would read

$$(stu)^\mu u_\tau^{\nu-\kappa} v_\tau^\kappa s_x^m t_x^{n-\lambda} (tuv)^\lambda = \sum_{p,r,\varrho} C_{pr\varrho} \left[s_\tau^\varrho (stu)^{\mu+r-\varrho}\right]_{\widehat{uv}^{\lambda-r+\varrho}\ v^{\nu-\varrho},\ u^p,\ (vw=\widehat{xw})},$$

and therefore does not permit the simplification that occurs already in the course of our calculation, once one knows that all terms with the factor u_x are to be omitted.

by the initial members of these polars, and by inversion of the resulting recursive system of equations obtaining the desired expansion according to polars.

By the transfer principle, for the λ-th polar of a form $s_x^m t_x^n : [s_x^m t_x^n]_{y^\lambda}$ ($\lambda \leq n$) the following expansion holds (the case $\lambda > n$ is reduced to the first by the expansion of $[s_y^m t_y^n]_{x^{m+n-\lambda}}$)[19]:

$$[s_x^m t_x^n]_{y^\lambda} = \sum_r (-1)^r \frac{\binom{m}{r}\binom{\lambda}{r}}{\binom{m+n}{r}} (st\widehat{xy})^r s_x^{m-r} t_x^{n-\lambda} t_y^{\lambda-r},$$

and correspondingly for the κ-th polar of $u_\sigma^\mu u_\tau^\nu : [u_\sigma^\mu u_\tau^\nu]_{v^\kappa}$

$$[u_\sigma^\mu u_\tau^\nu]_{v^\kappa} = \sum_\varrho (-1)^\varrho \frac{\binom{\mu}{\varrho}\binom{\kappa}{\varrho}}{\binom{\mu+\nu}{\varrho}} (\sigma\tau\widehat{uv})^\varrho u_\sigma^{\mu-\varrho} u_\tau^{\nu-\kappa} v_\tau^{\kappa-\varrho}.$$

By multiplication it follows, with introduction of the specialization a) and a'),

$$[s_x^m t_x^n (stu)^\mu u_\tau^\nu]_{\widehat{uv}^\lambda}^{v^\kappa} = \sum_{r,\varrho} c_{r\varrho} (st\widehat{uv})^r (st\widehat{uv})^\varrho s_x^{m-r} t_x^{n-\lambda} (tuv)^{\lambda-r} (stu)^{\mu-\varrho} u_\tau^{\nu-\kappa} v_\tau^{\kappa-\varrho},$$

and from this, by expansion according to powers of u_x with the first member distinguished ($\sum'_{r,\varrho}$ means that the member $r=0$, $\varrho=0$ is to be omitted) and taking account of (a) on p. 26,

$$\begin{aligned}
&s_x^m t_x^{n-\lambda} (tuv)^\lambda (stu)^\mu u_\tau^{\nu-\kappa} v_\tau^\kappa \\
&\quad = [s_x^m t_x^n (stu)^\mu u_\tau^\nu]_{\widehat{uv}^\lambda}^{v^\kappa} \\
&\qquad - \sum_{r,\varrho}{}' c_{r\varrho}\, s_\tau^\varrho (stu)^{\mu-\varrho+r} u_\tau^{\nu-\kappa} v_\tau^{\kappa-\varrho} s_x^{m-r} t_x^{n-\lambda} (tuv)^{\lambda-r+\varrho} v_x^r \\
&\qquad + u_x \sum_\varrho \sum_{r=1}^{\lambda} c_{r\varrho}\, s_\tau^\varrho (stu)^{\mu-\varrho+r-1} (stv) u_\tau^{\nu-\kappa} v_\tau^{\kappa-\varrho} s_x^{m-r} t_x^{n-\lambda} (tuv)^{\lambda-r+\varrho} v_x^{r-1} \\
&\qquad \vdots \\
&\qquad - (-1)^\lambda u_x^\lambda \sum_\varrho c_{\lambda\varrho}\, s_\tau^\varrho (stu)^{\mu-\varrho} (stv)^\lambda u_\tau^{\nu-\kappa} v_\tau^{\kappa-\varrho} s_x^{m-\lambda} t_x^{n-\lambda} (tuv)^\varrho,
\end{aligned} \tag{II}$$

where

$$c_{r\varrho} = (-1)^{r+\varrho} \cdot \frac{\binom{m}{r}\binom{\lambda}{r}}{\binom{m+n}{r}} \cdot \frac{\binom{\mu}{\varrho}\binom{\kappa}{\varrho}}{\binom{\mu+\nu}{\varrho}} \qquad \begin{array}{l}\text{for } \lambda \leq n, \\ \kappa \leq \nu.\end{array}$$

and correspondingly for the remaining combinations of the numbers $\lambda, n; \kappa, \nu$. Thus the expression $s_x^m t_x^{n-\lambda} (tuv)^\lambda (stu)^\mu u_\tau^{\nu-\kappa} v_\tau^\kappa$ is reduced to a composite polar, and, by applying the identity

$$(stv) u_\tau = (stu) v_\tau - s_\tau (tuv),$$

to a sum of analogously formed expressions corresponding to higher forms of the form sequence.

By iteration one arrives at the expansion

$$s_x^m t_x^{n-\lambda} (tuv)^\lambda (stu)^\mu u_\tau^{\nu-\kappa} v_\tau^\kappa = \sum_{p,r,\varrho} u_x^p\, C_{pr\varrho} \cdot [s_\tau^\varrho (stu)^{\mu-\varrho+r}]_{\widehat{uv}^{\lambda-r+\varrho} v^{\kappa-\varrho+p},\, v_x^{r-p}}. \tag{III}$$

The numerical coefficients $C_{pr\varrho}$ are computed uniquely and recursively from the coefficients $c_{r\varrho}$, $c'_{r\varrho}$ of the system of equations arising by iteration of (II) (cf. the example on p. 42). Analogous expansions arise for the other specializations.

§ 3. Reduction theorems.

The known theorems on the reduction of forms and form systems shall now be brought into connection with §§ 1 and 2, for which some definitions taken over from the theory of binary forms are necessary.

a) We call a system of forms a *relatively complete system* modulo a prescribed sequence of forms if it has the property that all formations arising by folding an arbitrary product of these forms can be expressed as

[19] Gordan, *The Resultante binärer Formen* (Chap. I, § 5). Rend. Circ. Mat. di Palermo XXII. 1906.

integral rational functions of the forms of the system, up to an additive term consisting of expressions that have arisen by folding the system forms with the system of the modulus[20].

b) We call a form *reducible* when it can be expressed by forms having invariants as factors or by "higher forms," that is, by higher forms of the total form sequence to which the form belongs, or by forms containing the symbols of the modulus in higher order.

The theorems on reducers[21] can now be stated in their most general form as follows:

Definition: *A reducer is a reducible form sequence.*

Theorem II. If the initial form of a form sequence is reducible because one of its members has arisen by folding with a reducer (has a reducer as factor), and if the final form of the form sequence has arisen from exactly this member by folding, then the total form sequence is reducible[22].

Proof: The form sequence arises from the initial member by folding in itself, hence by higher folding with the reducer on the one hand, and by folding the reducer in itself on the other hand. In both cases, by § 2, all forms of the form sequence can be represented as polars (transvections) over the form sequence of the reducer.

The inference from the initial form to the total form sequence no longer holds for the remaining reduction methods. These are:

1) so-called "double reduction."

By this we mean the reduction, in two different ways, of an expression that has two mutually independent reducers as factors, whereby relations arise between the higher forms to which one has reduced.

By systematic application of "double reduction" one must, on the one hand, obtain all relations[23]; on the other hand, if "double reduction" applied to different expressions supplies no new relations, one has both a criterion for the independence of the higher forms and a check on the calculation.

2) *Folding with decomposable forms.*

By "decomposable forms" are to be understood – with a slight generalization of the usual concept – forms that can be expressed by products of forms of lower degree and by "higher" forms. Products of forms pass into products under one folding; likewise products of nonlinear forms under twofold folding in the specializations a') and b') (§ 2), according to the product theorem:

$$a_y b_y a_x b_x = \frac{1}{2} a_y^2 b_x^2 + \frac{1}{2} b_y^2 a_x^2 - \frac{1}{2} (ab\widehat{yx})^2.$$

First polars (transvections) and certain second polars of decomposable forms are therefore again decomposable forms. It follows that:

A member of a onefold (respectively twofold) transvection over a decomposable form, arising by onefold (respectively twofold) folding with a decomposable form, itself becomes a decomposable form:

a) if the next-higher forms in the form sequence of the decomposable form decompose or become reducible, or also if, under onefold folding, the specializations $y = \widehat{uv}$ or $v = \widehat{xy}$ occur;

b) if the remaining onefold foldings lead to higher forms (cf. § 12, B. H_k and $H_k(k\eta x)$).

Such a member of a transvection can also decompose:

c) if the remaining onefold foldings lead to lower forms that decompose independently of the folding with the decomposable form, that is, that can be expressed by products and by the form to be reduced, although in each case one must first verify by calculation that the numerical coefficient of the member concerned does not vanish identically. This is nothing other than "double reduction" of the lower form (cf. § 23, C. $H_q(\theta H u)(\theta s u)$).

Decomposable forms arise by all onefold foldings with functional determinants[24], and moreover by certain higher foldings. One has

$$(st\widehat{yx})s_y = \frac{1}{2}\, t\, s_y^2 - \frac{1}{2}\, s\, t_y^2 + \frac{1}{2} (st\widehat{yx})^2,$$

[20]Gordan–Kerschensteiner, *Vorlesungen über Invariantentheorie.* Vol. II. p. 227.

[21]Gordan, *Math. Annalen* vol. XVII, p. 222.

[22]The simplification that results from Theorem II can be seen in the following example. For the special form $f = x_1^3 x_2 + x_2^3 x_3 + x_3^3 x_1$, $a_y^2 a_x^2 u_y^2$ is a reducer. It follows by Theorem II that the form sequence

$$\theta_\nu (\vartheta \nu x) \theta_x^3 u_\vartheta u_\nu^2 = a_\nu^2 (abu) - a_\nu b_\nu (aba),$$

is reduced, since $\theta_\nu^4 = a_\nu^2 b_\nu^2 (abu)^2$ has arisen from the member $a_\nu^2 (abu)$ by folding. In Gordan, loc. cit., p. 231, by contrast, the reduction of the forms

$$\theta_\nu (\vartheta \nu x), \quad \theta_\nu (\vartheta \nu x)^2, \quad \theta_\nu^2, \quad \theta_\nu^2 (\vartheta \nu x), \quad \theta_\nu^2 (\vartheta \nu x)^2$$

is carried out individually.

[23]Thus, for example, the following were foand by "double reduction": the reduction of the form a_η^4 (§ 10), the only form included superfluously in Maisano's system; and also the relations mentioned in the note to the Introduction between the three covariants and the three invariants.

[24]cf. Gordan, loc. cit. § 3.

$$(st\widehat{yx})^2 s_y^2 = \frac{1}{3}\,t\,s_y^3 - \frac{1}{3}\,s\,t_y^3 + s_y(st\widehat{yx})^2 - \frac{1}{3}(st\widehat{yx})^3.$$

Analogous formulas hold for the dualistic forms

$$(\sigma\tau\widehat{\nu u})v_\sigma \quad \text{and} \quad (\sigma\tau\widehat{\nu u})v_\sigma^2.$$

From the theorems of this section the following rule follows for setting up all irreducible forms that arise from the folding of S with T:

Beginning with the lowest foldings, proceed along any previously fixed path (for example row by row, or symmetrically with respect to the diagonal member) in the formation of the form sequence ST until one reaches a form that has a reducer as factor. The form sequence defined by this form is to be omitted as soon as the final form satisfies the condition stated in Theorem II. After all reducible form sequences have been eliminated, all remaining reducible forms, and likewise all decomposable forms, are to be removed by applying double reduction. Places in the total form sequence that are covered twice by independently reducible form sequences give rise to the reduction of higher forms (cf. § 11, D).

Theorem III. *The forms taken over from the binary domain, that is, those forms which arise from the system forms of the corresponding binary form by bordering according to Clebsch's transfer principle, form a relatively complete system modulo* (abc).

Proof. To a binary relation asserting that the forms $Q_1', \ldots, Q_n'$ form a complete system of a binary ground form,

$$Q' - F(Q') = 0 = \sum_\lambda \{(ab)c_x + (bc)a_x + (ca)b_x\}\varphi_\lambda^*,$$

there corresponds, by bordering, the ternary relation

$$Q - F(Q_i) = \sum_\lambda u_x \cdot (abc)\varphi_\lambda.$$

This is precisely the defining equation for a relatively complete system modulo (abc). For the biquadratic form, the system of the forms taken over consists of the following forms:

$$f = a_x^4, \qquad \theta = \theta_x^4 u_\vartheta^2 = (abu)^2 a_x^2 b_x^2, \qquad K = K_x^6 u_k^3 = (a\theta u)u_\vartheta^2 a_x^3 \theta_x^3,$$
$$j = u_j^6 = (a\theta u)^4 u_\vartheta^2, \qquad \nu = u_\nu^4 = (abu)^4,$$

among which the first four form a relatively complete system modulo $((abc), \nu)$.

§ 5. Reduction of the modulus (abc) to the moduli Δ and ν.

The modulus (abc), given only by a bracket factor, is to be reduced, in agreement with the definition (§ 3, a), to forms of the system. In other words, the form sequence (abc) is to be normalized uniquely (cf. § 1).

$$a_s^2 a_x u_x = \frac{1}{3} a_s^3 u_x = \frac{1}{3} S \cdot u_x. \tag{2}$$

$$\left.\begin{aligned} (a\Delta u)^2 &= a_s - \frac{S}{6}u_x^2, \\ (a\Delta u)^3 &= 0. \end{aligned}\right\} \tag{3}$$

$$\left.\begin{aligned} \theta(s) &= (ss,x)^2 = 2a_t + \frac{1}{3}S \cdot \theta - \frac{1}{3}Tu_x^2, \\ \theta(t) &= (tt,x)^2 = -\frac{1}{3}S \cdot a_t + \frac{2}{3}T \cdot a_s + \frac{1}{18}S^2 \cdot \theta - \frac{1}{18}u_x^2 \cdot S \cdot T. \end{aligned}\right\} \tag{4}$$

Sequence of moduli: (abc); Δ; s; t (the system of the modulus t consists of the single form t).[25]

It will be shown that the forms

$$\Delta = (abc)^2 a_x^2 b_x^2 c_x^2, \qquad \nu = (abu)^4$$

suffice as moduli, that is, that the forms

$$\Delta = (abc)^2 a_x^2 b_x^2 c_x^2, \qquad a_\nu = (abc)(bcu)^3 a_x^3,$$
$$a_\nu^2 = (abc)^2 (bcu)^2 u_x^2, \qquad i = a_\nu^4 = (abc)^4$$

define the form sequence (abc). In the form sequence a_ν, the form a_ν^3 is missing according to the identity

$$a_\nu^3 = (abc)^3 a_x (bcu) = \frac{1}{3} u_x \cdot (abc)^4 = \frac{1}{3} i \cdot u_x.$$

For reducing a modulus to a higher one, according to the definition, we have to reduce the most general foldings, or all special foldings coming into consideration, with the modulus to foldings with the higher modulus.

In the present case the most general foldings arise from the expressions

$$(abc)a_x^3 b_y^3 c_z^3, \qquad (abc)a_x^2 b_y^2 c_z^2 a_z b_x c_x$$

(taken over from the cubic forms),

$$(abc)a_x b_y c_z a_x^2 b_x^2 c_x^2$$

[25] Gordan–Kerschensteiner, loc. cit., p. 134.

(taken over from the quadratic forms), and from the polarisation of these expressions.

For the calculation of these expressions by the polars of Δ and of the form sequence a_ν, respectively of their initial terms, we take the indirect route, as in § 2. We expand the polar terms

$$(abc)^2 a_x^2 b_y^2 c_z^2, \qquad a_\nu(\nu xy)(\nu yz)(\nu zx) a_x a_y a_z = (abc)(bc\widehat{x}u)(bc\widehat{y}z)(bc\widehat{z}x) a_x a_y a_z \quad \text{etc.}$$

as linear aggregates of expressions with the symbolic factor (abc) and obtain, by inverting the system of equations, the desired relations, as well as a relation between the polars taken according to different combinations of the variables. The identity theorem and the product theorem for determinants are used in the evaluation.

The system of equations is, for $(abc)a_x^3 b_y^3 c_z^3$:

$$1)\quad \sum_3 a_\nu(\nu yz)^3 a_x^3 = 6(abc)a_x^3 b_y^3 c_z^3 - 6\sum_3 (abc) a_x^3 b_y^2 b_z c_z^2 c_y^*,$$

$$\begin{aligned} 2)\quad 3(abc)^2 a_x^2 b_y^2 c_z^2 \cdot (xyz) - \frac{1}{2}\sum_3 a_\nu^2 (\nu yz)^2 \nu_x^2 (xyz) &= 2\sum_3 (abc) a_x^3 b_y^2 b_z c_z^2 c_y \\ &\quad - 4\sum_3 (abc) a_x a_y a_z b_y^2 b_z c_z^2 c_x, \end{aligned}$$

$$3)\quad a_\nu(\nu xy)(\nu yz)(\nu zx) a_x a_y a_z = 2\sum_3 (abc) a_x a_y a_z b_y^2 b_z c_z^2 c_x,$$

$$4)\quad \sum_3 a_\nu(\nu yz)^3 u_x^3 - \frac{3}{2}\sum_3 a_\nu^2(\nu yz)^2 \nu_x^2 (xyz) + \frac{1}{6} i \cdot (xyz)^3 = 6\sum_3 (abc) a_x a_y a_z b_y^2 b_z c_z^2 c_x.$$

It follows for

$$(abc)a_x^3 b_y^3 c_z^3,$$

and by analogous calculation for

$$(abc)a_x^2 b_y^2 c_z^2 a_z b_x c_x, \qquad (abc) a_x b_y c_z a_z^2 b_x^2 c_x^2 :$$

$$\begin{aligned} (abc)a_x^3 b_y^3 c_z^3 &= \frac{3}{2}(abc)^2 a_x^2 b_y^2 c_z^2 (xyz) + \frac{3}{2} a_\nu(\nu xy)(\nu yz)(\nu zx) a_x a_y a_z \\ &\quad - \frac{1}{12} i \cdot (xyz)^3, \\ \text{(IV.)}\quad (abc)a_x^2 b_y^2 c_z^2 a_z b_x c_x &= \frac{2}{3}(abc)^2 a_x^2 b_y^2 c_z^2 c_x \cdot (xyz) \\ &\quad + \frac{1}{3} a_\nu(\nu xy)(\nu yz)(\nu zx) a_x^3 - \frac{1}{6} a_\nu^2 (\nu xy)(\nu zx) a_x^2 \cdot (xyz), \\ (abc) a_x b_y c_z a_z^2 b_x^2 c_x^2 &= \frac{1}{6}(abc)^2 a_x^2 b_z^2 c_z^2 \cdot (xyz). \end{aligned}$$

We have therefore obtained a relatively complete system modulo (Δ, ν), consisting of the four forms f, θ, K, j. It follows from this that all transvections of f over θ not taken over from the binary domain, as well as the transvection $(a\theta u)^2$ reducible in the binary case to $(ab)^4$, can be expressed by the symbols Δ and ν.[26]

We obtain the reduction formulas fundamental for what follows by specializing expansion IV, or more shortly by direct calculation (interchanging the symbols), for $a_\vartheta(a\theta u)^2$, $a_\vartheta^2((a\theta u)^2$, $(a\theta u)^2$, according to the following ansatz:

$$\begin{aligned} a_\vartheta(a\theta u)^2 &= (abc)(bcu)(abu)^2 - \frac{1}{3}(abc)(bcu)\{(bcu)a_x - u_x(abc)\}^2, \\ a_\vartheta^2(a\theta u)^2 &= (abc)^2(abu)^2 - \frac{1}{3}(abc)^2\{(bcu)a_x - u_x(abc)\}^2, \end{aligned}$$

for $((a\theta u)^2)^*$:

$$\begin{aligned} (abu)a_y^3 b_x^3 &= \frac{3}{2} u_\vartheta(\vartheta yx)\theta_y^2 + \frac{1}{4}(\nu yx)^3, \\ ((abu)(acu)) b_x^3 &= \frac{3}{2} u_\vartheta(\vartheta\widehat{a}ux)(\theta au)^2 + \frac{1}{4}(\nu\widehat{a}ux)^3. \end{aligned}$$

[26] $\sum_3$ refers to the sum of three terms obtained from the initial term by cyclic interchange of the variables. The equations also hold individually for each term of the sum.

$$\left.\begin{aligned}
(a\theta u)^2 &= \frac{1}{6}\nu\cdot f - \frac{2}{3}u_x\cdot a_\nu + \frac{2}{3}u_x^2\cdot a_\nu^2 - \frac{i}{18}u_x^4,\\
a_\vartheta &= \frac{1}{3}u_x\cdot\Delta,\\
a_\vartheta(a\theta u) &= 0,\\
a_\vartheta^2 &= \Delta,\\
(a\theta u)^3 &= 0,\\
a_\vartheta(a\theta u)^2 &= -\frac{5}{6}a_\nu + \frac{7}{6}u_x\cdot a_\nu^2 - \frac{i}{9}u_x^3,\\
a_\vartheta^2(a\theta u) &= 0,\\
(a\theta u)^4 &= j,\\
a_\vartheta(a\theta u)^3 &= 0,\\
a_\vartheta^2(a\theta u)^2 &= \frac{2}{3}a_\nu^2 - \frac{i}{9}u_x^2.
\end{aligned}\right\} \tag{1}$$

We add the formula, likewise obtained from the reduction of the modulus (abc):

$$a_\nu^3 a_x u_\nu = \frac{1}{3}i\cdot u_x. \tag{2}$$

From formulas (1.) and (2.) and from the formulas formed analogously for the ground form ν, all later reduction formulas are derived by polarisation, except for the relations for the decomposed forms. From formulas (1.) we see:

The form sequence leads:

$$\begin{array}{rll}
a_\vartheta(a\theta u)^2 & \text{to} & \text{symbols } \nu \text{ alone},\\
a_\vartheta & \text{to} & \text{symbols } \Delta \text{ and } \nu,\\
(a\theta u)^2 & \text{to} & \text{symbols } j, \Delta \text{ and } \nu,\\
(a\theta u)^2 \text{ under folding I} & \text{to} & \text{symbols } j \text{ and } \nu,\\
(a\theta u)^2 \text{ under folding II} & \text{to} & \text{symbols } \Delta \text{ and } \nu.
\end{array}$$

We can therefore replace:

1. modulo (Δ, ν): foldings with j are replaced by corresponding foldings with $(a\theta u)^2$ or $(a\theta u)^3$;

2. modulo (ν): foldings with Δ are replaced by corresponding foldings with a_ϑ or $a_\vartheta(a\theta u)$, or also by special foldings with $(a\theta u)^2$ (which lead only to a single folding I).

In particular:

$$(a\theta u)^2\theta_y^2\theta_x^2 = \frac{1}{3}(jyx)^2 \pmod{\nu}, \qquad (a\theta u)^2 u_y\theta_y = -\frac{1}{6}(jyx)^2 \pmod{\nu},$$

$$(a\theta u)^2\nu_y^2 = \frac{1}{3}(\Delta u\nu)^2 \pmod{\nu}, \qquad (a\theta u)(a\theta\nu)u_\vartheta\nu_\vartheta = -\frac{1}{6}(\Delta u\nu)^2 \pmod{\nu},$$

$$(a\theta u)^2u_\nu^2\vartheta_\nu^2 = \frac{1}{3}(\Delta u\nu)^2\Delta_\nu \pmod{\nu}.$$

§ 6. Reduction of the modulus (Δ, ν) to the modulus (ν).

The reduction formulas of the preceding paragraph give us the means to set up the relatively complete system modulo ν directly; we shall see that to the system of the transferred forms one has only to add the forms Δ and $(a\Delta u) = N$ (analogously as for the cubic form, and indeed valid in general).

For the reduction of the modulus Δ we consider all special foldings coming into consideration, that is, the foldings of the relatively complete system modulo (Δ, ν) with the system of Δ, and begin with the forms lowest in the coefficients, the foldings of f with Δ.

For the reduction of the forms $(a\Delta u)^2$, $(a\Delta u)^3$, $(a\Delta u)^4$, we replace the symbols Δ by lower forms of the form sequence according to § 5; that is, we develop the expressions

$$a_\vartheta b_\vartheta(b\theta u)^2, \qquad a_\vartheta(a\theta u)b_\vartheta(b\theta u)^2, \qquad a_\vartheta(a\theta u)b_\vartheta(b\theta u)^3$$

1) by polars of the form sequence a_ϑ (symbols Δ and ν),

2) by polars of the form sequence $b_\vartheta(b\theta u)^2$ (symbols ν),

and obtain the reduction formulas (calculation below):

$$\begin{aligned}
&\text{a)}\quad (a\Delta u)^2 = \frac{1}{2}(\vartheta\nu x)^2 - \frac{2}{5}f\cdot a_\nu^2 - \frac{4}{5}u_x\cdot\theta_\nu(\vartheta\nu x)^2 \\
&\qquad\qquad\qquad + u_x^2\left\{\frac{1}{6}i\cdot f - \frac{1}{40}S\right\}, \\
&\text{b)}\quad (a\Delta u)^3 = -\frac{3}{10}\theta_\nu(\vartheta\nu x), \\
&\text{c)}\quad (a\Delta u)^4 = \frac{21}{10}\theta_\nu^2 - \frac{3}{10}\Pi - \frac{6}{5}u_x\cdot\theta_\nu^3 + \frac{1}{10}u_x^2\cdot\theta.
\end{aligned}\tag{3}$$

(The same results are also reached by the double reduction of the expressions $a_\vartheta^2(b\theta u)^2$, $a_\vartheta^2(b\theta u)^3$, $a_\vartheta^2((a\theta u)^2(b\theta u)^2$ and $a_\vartheta^2((a\theta u)(b\theta u))^3$ after elimination of a_ϑ^2.)

From formulas (3.) it follows that $(a\Delta u)^2$ is a reducer with respect to the modulus ν. Hence:

1) The reduction of the system of Δ: the form sequence $(\Delta\Delta' u)^2 = a_\vartheta^2((a\Delta u)^2$ has the reducer $(a\Delta u)^2$ as a factor.

2) The reduction of the system arising by folding with the modulus Δ:

The form sequence $(\theta\Delta u)^2$ has the reducer $(a\Delta u)^2$ as a factor.

For the forms $(\theta\Delta u)$ and Δ_ϑ one obtains:

$$(a\Delta u)^2(abu) = (\theta\Delta u) - u_x\cdot\Delta_\vartheta(\theta\Delta u),$$

$$(a\Delta u)(a\Delta b) = -\frac{1}{2}\Delta_\vartheta + \frac{1}{2}u_x\cdot\Delta_\vartheta^2.$$

From the reducer $(\theta\Delta u)$, however, follows the reduction of the system arising by folding with the modulus Δ.

The relatively complete system modulo ν therefore consists of the six forms:

$$f,\ \theta,\ K,\ j,\ \Delta,\ N.$$

As an example of the sequence expansion in § 2 we give the derivation of formula (3.)a in full; in later calculations the final formula III will be set down directly. (Polars of vanishing forms have already been omitted during the calculation; the first line gives the values inserted according to Expansion II, and the second line gives the final values found recursively, beginning with the last equation of the system.) It follows, according to Expansion II:

$$1)\quad m=3,\quad n=4,\quad \lambda=2,\quad \mu=0,\quad \nu=\chi=1,$$

$$\begin{aligned}
a_\vartheta b_\vartheta(b\theta u)^2 &= [a_\vartheta]_{b^2}b + \frac{6}{7}\{a_\vartheta b_\vartheta(a\theta u)(b\theta u) - u_x a_\vartheta b_\vartheta(a\theta b)(b\theta u)\} \\
&\quad - \frac{1}{7}\{a_\vartheta(a\theta u)^2 b_\vartheta - 2u_x a_\vartheta(a\theta u)(a\theta b)b_\vartheta + u_x^2 a_\vartheta(a\theta b)^2 b_\vartheta\} \\
&= \frac{2}{3}(a\Delta u)^2 + \frac{1}{5}[a_\vartheta(a\theta u)^2]b + \frac{1}{30}f[a_\vartheta^2(a\theta u)^2] - \frac{2}{5}u_x[a_\vartheta(a\theta u)^2]b^2 \\
&\quad - \frac{1}{15}u_x[a_\vartheta^2(a\theta u)^2]b + \frac{1}{5}u_x^2[a_\vartheta(a\theta u)^2]b^3 + \frac{1}{30}u_x^2[a_\vartheta^2(a\theta u)^2]b^2;
\end{aligned}$$

$$2)\quad m=2,\quad n=3,\quad \lambda=1,\quad \mu=1,\quad \nu=\chi=1,$$

$$\begin{aligned}
a_\vartheta(a\theta u)b_\vartheta(b\theta u) &= \frac{2}{5}\{a_\vartheta(a\theta u)^2 b_\vartheta - u_x a_\vartheta(a\theta u)(a\theta b)b_\vartheta\} + \frac{1}{2}a_\vartheta^2(b\theta u)^2 \\
&\quad - \frac{1}{5}\{a_\vartheta^2(b\theta u)(a\theta u) - u_x a_\vartheta^2(a\theta b)(b\theta u)\} \\
&= \frac{1}{2}(a\Delta u)^2 + \frac{2}{5}[a_\vartheta(a\theta u)^2]b + \frac{1}{15}[a_\vartheta^2(a\theta u)^2]f \\
&\quad - \frac{2}{5}u_x[a_\vartheta(a\theta u)^2]b^2 - \frac{1}{10}u_x[a_\vartheta^2(a\theta u)^2]b + \frac{1}{30}u_x^2[a_\vartheta^2(a\theta u)^2]b^2;
\end{aligned}$$

$$
\begin{aligned}
&\quad\;\; 3) \quad m=2, \quad n=3, \quad \lambda=1, \quad \mu=1, \quad \nu=1, \quad \chi=2,\\
a_\vartheta(a\theta b)b_\vartheta(b\theta u) &= \frac{2}{5}\{a_\vartheta(a\theta b)(a\theta u)b_\vartheta - u_x a_\vartheta(a\theta b)^2 b_\vartheta\}\\
&= \frac{2}{5}[a_\vartheta(a\theta u)^2]b^2 + \frac{1}{30}[a_\vartheta^2(a\theta u)^2]b - \frac{2}{5}u_x[a_\vartheta(a\theta u)^2]b^3 - \frac{1}{30}u_x[a_\vartheta^2(a\theta u)^2]b^2;
\end{aligned}
$$

$$
\begin{aligned}
&\quad\;\; 4) \quad m=1, \quad n=2, \quad \lambda=0, \quad \mu=2, \quad \nu=\chi=1,\\
a_\vartheta(a\theta u)^2 b_\vartheta &= [a_\vartheta(a\theta u)^2]b + \frac{2}{3}a_\vartheta^2(a\theta u)(b\theta u)\\
&= [a_\vartheta(a\theta u)^2]b + \frac{1}{6}[a_\vartheta^2(a\theta u)^2]f - \frac{1}{6}u_x[a_\vartheta^2(a\theta u)^2]b;
\end{aligned}
$$

$$
\begin{aligned}
&\quad\;\; 5) \quad m=1, \quad n=2, \quad \lambda=0, \quad \mu=2, \quad \nu=1, \quad \chi=2,\\
a_\vartheta(a\theta u)(a\theta b)b_\vartheta &= [a_\vartheta(a\theta u)^2]b^2 + \frac{1}{3}a_\vartheta^2(a\theta b)(b\theta u)\\
&= [a_\vartheta(a\theta u)^2]b^2 + \frac{1}{12}[a_\vartheta^2(a\theta u)^2]b - \frac{1}{12}u_x[a_\vartheta^2(a\theta u)^2]b^2;
\end{aligned}
$$

$$
\begin{aligned}
&\quad\;\; 6) \quad m=1, \quad n=2, \quad \lambda=0, \quad \mu=2, \quad \nu=1, \quad \chi=3,\\
a_\vartheta(a\theta b)^2 b_\vartheta &= [a_\vartheta(a\theta u)^2]b^3;
\end{aligned}
$$

$$
\begin{aligned}
&\quad\;\; 7) \quad m=2, \quad n=4, \quad \lambda=2, \quad \mu=\nu=\chi=0, \quad \text{(according to Expansion I)},\\
a_\vartheta^2(b\theta u)^2 &= [a_\vartheta^2]_{ub^2} + \frac{1}{10}\{[a_\vartheta^2(a\theta u)^2]f - 2u_x[a_\vartheta^2(a\theta u)^2]b + u_x^2[a_\vartheta^2(a\theta u)^2]b^2\};
\end{aligned}
$$

$$
\begin{aligned}
&\quad\;\; 8) \quad m=1, \quad n=3, \quad \lambda=1, \quad \mu=1, \quad \nu=\chi=0, \quad \text{(Expansion I)},\\
a_\vartheta^2(a\theta u)(b\theta u) &= \frac{1}{4}[a_\vartheta^2(a\theta u)^2]f - \frac{1}{4}u_x[a_\vartheta^2(a\theta u)^2]b;
\end{aligned}
$$

$$
\begin{aligned}
&\quad\;\; 9) \quad m=1, \quad n=3, \quad \lambda=1, \quad \mu=\chi=1, \quad \nu=0, \quad \text{(Expansion I)},\\
a_\vartheta^2(a\theta b)(b\theta u) &= \frac{1}{4}[a_\vartheta^2(a\theta u)^2]b - \frac{1}{4}u_x[a_\vartheta^2(a\theta u)^2]b^2.
\end{aligned}
$$

From 1) and 4) it follows:

$$
\begin{aligned}
(a\Delta u)^2 &= \frac{6}{5}[a_\vartheta(a\theta u)^2]b + \frac{1}{5}f[a_\vartheta^2(a\theta u)^2] + \frac{3}{5}u_x[a_\vartheta(a\theta u)^2]b^2 - \frac{3}{20}u_x[a_\vartheta^2(a\theta u)^2]b\\
&\quad - \frac{3}{10}u_x^2[a_\vartheta(a\theta u)^2]b^3 - \frac{1}{20}u_x^2[a_\vartheta^2(a\theta u)^2]b^2,\\
(a\Delta u)^2 &= -a_\nu b_\nu + \frac{3}{5}f a_\nu^2 + \frac{4}{5}u_x a_\nu^2 b_\nu - \frac{3}{20}u_x^2 a_\nu^2 b_\nu^2 - \frac{1}{12}u_x^2 if.
\end{aligned}
$$

(For conversion into the symbols θ cf. § 8 and § 9.) Formula (3.)a can be computed still more briefly by direct transfer of Expansion I, introducing the auxiliary variable $w = x\widehat{u}b$ whereas already for formulas (3.)b and (3.)c the path taken here becomes the simpler one; for (3.)b one knows in advance, by § 9, that all terms with factor u_x are to be omitted. For the calculation of (3.)b and (3.)c one obtains:

$$
\begin{aligned}
\text{(3.)b} \quad 1) \quad a_\vartheta(a\theta u)b_\vartheta(b\theta u)^2 &= \frac{1}{2}(a\Delta u)^3 + \frac{4}{5}[a_\vartheta(a\theta u)^2]_{ub}b + \frac{7}{360}[a_\vartheta^2(a\theta u)^2]_{ub^2},\\
2) \quad a_\vartheta(a\theta u)b_\vartheta(b\theta u)^2 &= [a_\vartheta(a\theta u)^2]_{ub}b + \frac{13}{36}[a_\vartheta^2(a\theta u)^2]_{ub^2};
\end{aligned}
$$

$$
\begin{aligned}
\text{(3.)c} \quad 1) \quad a_\vartheta(a\theta u)b_\vartheta(b\theta u)^3 &= \frac{1}{2}(a\Delta u)^4 + \frac{6}{5}[a_\vartheta(a\theta u)^2]_{ub^2}b + \frac{53}{120}[a_\vartheta^2(a\theta u)^2]_{ub^3}\\
&\quad - \frac{6}{5}u_x[a_\vartheta(a\theta u)^2]_{ub^3}b^2 - \frac{3}{5}u_x[a_\vartheta^2(a\theta u)^2]_{ub^2}b + \frac{19}{120}u_x^2[a_\vartheta^2(a\theta u)^2]_{ub^3}b^2,\\
2) \quad a_\vartheta(a\theta u)b_\vartheta(b\theta u)^3 &= \frac{3}{4}[a_\vartheta^2(a\theta u)^2]_{ub^2}.
\end{aligned}
$$

Supplementary remark: the transvectants of f over j that are reducible according to § 5 are computed analogously. One obtains

$$
\begin{aligned}
g_\nu &= -\theta_\nu + u_x\left(\frac{9}{2}\theta_\nu^2 - \frac{1}{2}H\right) - 3u_x^2\theta_\nu^3 + \frac{1}{2}u_x^3\theta,\\
g_\nu^2 &= \frac{21}{10}\theta_\nu^2 - \frac{3}{10}H - 2u_x\theta_\nu^3 + \frac{1}{2}u_x^2\theta,\\
g_\nu^3 &= -\frac{7}{10}\theta_\nu^3 + \frac{1}{2}u_x^2\theta, \qquad g_\nu^4 = \frac{3}{5}\theta.
\end{aligned} \tag{4}
$$

$$
g_\nu^3 = -\frac{7}{10}\theta_\nu^3 + \frac{1}{2}u_x^2\theta, \qquad g_\nu^4 = \frac{3}{5}\theta. \tag{4}
$$

From (3.) and (4.) it follows, by transfer from the binary domain, that

$$
(\theta\theta' u)^2 = \frac{1}{3}\, j\cdot f \quad (\mathrm{mod}\ \nu).
$$
[27]

The remaining forms of the form sequence $(\theta\theta' u)^2$ and the form θ'_ν are reducible to symbols ν. We can therefore replace:

mod ν : foldings with $f\cdot j$ by the corresponding foldings with $(\theta\theta' u)^2$.

Furthermore,

$$
(\vartheta\vartheta' x)^2 = \frac{4}{3} f\cdot\Delta, \qquad \theta_{g\nu}(\vartheta\vartheta' x) = -N.
$$

Chapter III. The relatively complete system modulo (s, ϱ, t).

§ 7. Overview of the formation of the system.

In order to pass from the relatively complete system modulo ν to the relatively complete system with the next higher modulus, one has, by Definition § 3, to form the system of ν with respect to this higher modulus and to fold it with the forms of the relatively complete system modulo ν. But $\nu = u_\nu^4$, as a contravariant of the fourth degree, is dual to the form f. If therefore we regard ν as the ground form, we obtain a relatively complete system of six forms modulo $\nu(\nu) = (\nu\nu_1 x)^4 = s_x^4$.

Thus, for the formation of the relatively complete system modulo s (System III), we have to transvect

System I: $f,\ \theta,\ K,\ j,\ \Delta,\ N$ over
System II: $\nu,\ H = (\nu\nu_1 x)^2 u_\nu^2 u_{\nu_1}^2,\ L = (\nu\eta x) H_x^2 u_\nu^3 u_\eta^3,$
$g = (\nu\eta x)^4 H_x^2,\ \sigma = H_\eta^2 u_\nu^2 u_\eta^4,\ (\nu\sigma x).$

It will be shown (formula (13.)) that the form g is reducible, and hence that System II consists of only five forms.

In the course of the calculation the quadratic forms

$$
\varrho = u_\varrho^2 = \theta_\nu^4 u_\vartheta^2, \qquad t = t_x^2 = a_\eta^4 H_x^2
$$

are adjoined as moduli, so that one obtains a relatively complete system modulo (s, ϱ, t); and the modulus (ϱ, t) is to be regarded as a higher modulus than the modulus s.

The arrangement of the individual forms is to follow the order in the coefficients. We normalize the folding

$$
\theta_\nu(K_\nu, \theta_\nu, \ldots)
$$

as higher than the folding

$$
H_y(H_y, L_y, \ldots).
$$

Consequently, by § 1 and § 3b, the order of the individual forms of the same order is uniquely determined.

The formation of the forms of the same order is carried out in tabular form:

I. Indication of which forms of Systems I and II are to be folded with each other in order to obtain the prescribed order in the coefficients.

[27] Gordan–Kerschensteiner, loc. cit., p. 181.

II. Listing of the irreducible forms according to the scheme of the form sequence.

III. Reduction of the reducible or decomposable form sequences or forms, that is, of those places where the total form sequence breaks off, thereby proving that all irreducible constructions have been exhausted by those indicated.

IV. Consequences: 1) indication of newly arising reducers; 2) indication of those forms which do not enter, in products with forms of the same system, into folding with forms of the other system.

It will be shown that only the following occur:

1) foldings of one form of System I with one form of System II;

2) foldings of powers of θ, respectively H, or of two forms of one system with one form of the second system.

We shall obtain the following scheme for folding:

System I		folded with	System II	
1. order	f		2. order	ν
2. ,,	θ		4. ,,	H
3. ,,	K, j, Δ		6. ,,	L, σ
4. ,,	$N, \theta^2, f \cdot j$		8. ,,	$(\nu\sigma x), H^2$
5. ,,	$\theta \cdot K$		10. ,,	$H \cdot L$
6. ,,	$\theta^3, \Delta \cdot j$		12. ,,	H^3.

For checking the calculation, besides the "double reduction", the two special forms serve:

$$\text{I.}\quad \begin{aligned} f &= \sum_3 x_1^3 x_2, & u_\nu^2 &= \tfrac{i}{6} u_x^2, & s &= \tfrac{i}{3} f,\\ a_\eta^2 &= -\tfrac{1}{9} i\theta, & H_\nu^2 &= \tfrac{i}{6}\theta, & \sigma &= -\tfrac{i}{6} j,\\ g &= -\tfrac{2}{3} i\Delta, & \varrho &= 0, & t &= 0,\end{aligned}$$

$$\text{II.}\quad \begin{aligned} f &= \sum_3 x_1^4, & s &= \tfrac{4}{3} i f, & a_\eta^2 &= \tfrac{2}{9} i\theta,\\ \Delta_\nu^2 &= \tfrac{1}{15} i\theta, & \sigma &= \tfrac{4}{3} i j, & g &= \tfrac{4}{3} i\Delta,\\ \varrho &= 0, & t &= 0.\end{aligned}$$

Supplementary remark: for the ground form ν, formulas (1.), (2.), (3.), (4.) correspond to the following:

$$\begin{array}{c|c|c} (\nu\eta x)^2 = A & H_\nu(\nu\eta x) = 0 & H_\nu^2 = \sigma \\ (\nu\eta x)^3 = 0 & H_\nu(\nu\eta x)^2 = B & H_\nu^2(\nu\eta x) = 0 \\ (\nu\eta x)^4 = g & H_\nu(\nu\eta x)^3 = 0 & H_\nu^2(\nu\eta x)^2 = C, \end{array} \tag{5}$$

$$A = \frac{1}{6} s\nu - \frac{2}{3} u_x s_\nu + \frac{2}{3} u_x^2 s_\nu^2 - \frac{J}{18} u_x^4, \quad B = -\frac{5}{6} s_\nu + \frac{7}{6} u_x s_\nu^2 - \frac{J}{9} u_x^3, \quad C = \frac{2}{3} s_\nu^2 - \frac{J}{9} u_x^2.$$

$$s_\nu^3 u_x s_\nu = \frac{1}{3} u_x s_\nu^4 = \frac{1}{3} J u_x. \tag{6}$$

$$\begin{aligned} &\text{a)}\quad (\nu\sigma x)^2 = \frac{1}{2}(Hsu)^2 - \frac{2}{5}\nu \cdot s_\nu^2 - \frac{4}{5} u_x\, s_\eta (Hsu)^2 + u_x^2 \left\{ \frac{1}{6} J \cdot \nu - \frac{1}{40}(ss'u)^4 \right\},\\ &\text{b)}\quad (\nu\sigma x)^3 = -\frac{3}{10} s_\eta (Hsu),\\ &\text{c)}\quad (\nu\sigma x)^4 = \frac{21}{10} s_\eta^2 - \frac{3}{10} Z - \frac{6}{5} u_x s_\eta^3 + \frac{1}{10} u_x^2 s_\eta^4. \end{aligned} \tag{7}$$

$$\begin{aligned} &\text{a)}\quad g_\nu = -s_\eta + u_x \left(\frac{9}{2} s_\eta^2 - \frac{1}{2} Z \right) - 3u_x^2 s_\eta^3 + \frac{1}{2} u_x^3 s_\eta^4,\\ &\text{b)}\quad g_\nu^2 = \frac{21}{10} s_\eta^2 - \frac{3}{10} Z - 2u_x s_\eta^3 + \frac{1}{2} u_x^2 s_\eta^4,\\ &\text{c)}\quad g_\nu^3 = -\frac{7}{10} s_\eta^3 + \frac{1}{2} u_x s_\eta^4,\\ &\text{d)}\quad g_\nu^4 = \frac{3}{5} s_\eta^4. \end{aligned} \tag{8}$$

§ 8. Forms of the third order (System III).

Folding of f with ν.
Irreducible forms:

$$\begin{array}{cccc} a_\nu & \cdot & \cdot & \cdot \\ \cdot & a_\nu^2 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & a_\nu^4 = i. \end{array}$$

Reduction:

$$a_\nu^3 = \frac{1}{3} i u_x \qquad \text{(formula (2.))}.$$

Consequences: 1) a_ν^3 is a reducer. 2) f enters into folding with ν only in products with contragredient forms (j). The same holds for ν with respect to f.

§ 9. Forms of the fourth order (System III).

Folding of θ with ν.
Irreducible forms:

$$\begin{array}{cccc} (\theta\nu x) & \theta_\nu & \cdot & \cdot \\ (\theta\nu x)^2 & \theta_\nu(\theta\nu x) & \theta_\nu^2 & \cdot \\ \cdot & \theta_\nu(\theta\nu x)^2 & \cdot & \theta_\nu^3. \end{array}$$

reductions: From the reducer a_ν^3, by double reduction of the expressions $a_\nu^3(abu)$, $a_\nu^3 b_\nu$, $a_\nu^3 b_\nu (abu)$ according to the ansatz:

$$\begin{aligned} \theta_\nu^2(\vartheta\nu x) &= a_\nu^2(\widehat{a}b\nu x)(abu) - \frac{1}{3}(\nu\nu_1 x)^3 = a_\nu b_\nu(\widehat{a}b\nu x)(abu) + \frac{1}{6}(\nu\nu_1 x)^3 \\ &= a_\nu^3(abu) - a_\nu^2 b_\nu(abu) = 2a_\nu^2 b_\nu(abu) = \frac{2}{3}a_\nu^3(abu), \\ \theta_\nu^2(\vartheta\nu x)^2 &= a_\nu^2(\widehat{a}b\nu x)^2 - \frac{1}{3}(\nu\nu_1 x)^4 = a_\nu b_\nu(\widehat{a}b\nu x)^2 + \frac{1}{6}(\nu\nu_1 x)^4 \\ &= if - 2a_\nu^3 b_\nu + a_\nu^2 b_\nu^2 - \frac{1}{3}s = 2a_\nu^3 b_\nu - 2a_\nu^2 b_\nu^2 + \frac{1}{6}s = \frac{2}{3}if - \frac{2}{3}a_\nu^3 b_\nu - \frac{1}{6}s, \\ \theta_\nu^3(\vartheta\nu x) &= a_\nu^2 b_\nu(\widehat{a}b\nu x)(abu) = a_\nu^3 b_\nu(abu). \end{aligned}$$

The formulas obtained are:

$$\left.\begin{array}{l} \theta_\nu^2(\theta\nu x) = 0 \\ \theta_\nu^2(\theta\nu x)^2 = \frac{4}{9}if - \frac{1}{6}s \end{array}\right| \begin{array}{l} \theta_\nu^3(\theta\nu x) = 0 \\ \end{array} \left| \begin{array}{l} \theta_\nu^4 u_x^2 = u_\varrho^2 = \varrho \quad \text{(adjoined modulus, § 7)}, \\ \end{array}\right. \tag{9}$$

Consequences: 1) $\theta_\nu^2(\theta\nu x)^2$ is a reducer. 2) ν enters into folding with θ only in products with contragredient forms (g).

§ 10. Forms of the fifth order (System III).

$$\text{Folding of } \begin{cases} K, j, \Delta \text{ with } \nu, \\ f \text{ with } H. \end{cases}$$

Irreducible forms:

$$\text{A)} \begin{array}{cccc} \cdot & K_\nu & \cdot & \cdot \\ (k\nu x)^2 & \cdot & K_\nu^2 & \cdot \\ (k\nu x)^3 & K_\nu(k\nu x)^2 & \cdot & \cdot \\ \cdot & K_\nu(k\nu x)^3 & \cdot & \cdot \end{array} \qquad \text{B)} \begin{array}{c} (j\nu x) \\ (j\nu x)^2 \\ (j\nu x)^3 \\ \cdot \end{array} \qquad \text{C)} \begin{array}{c} \Delta_\nu \\ \cdot \\ \cdot \\ \cdot \end{array}$$

$$\text{D)} \begin{array}{cccc} (aHu) & (aHu)^2 & \cdot & \cdot \\ a_\eta & a_\eta(aHu) & a_\eta(aHu)^2 & \cdot \\ \cdot & a_\eta^2 & \cdot & \cdot \end{array}$$

reductions:

A) form sequence K_ν^3: reducer a_ν^3.

form $K_\nu^2(k\nu x)^2$: reducer $\theta_\nu^2(\vartheta\nu x)^2$.

$$(k\nu x), \quad K_\nu(k\nu x), \quad K_\nu^2(k\nu x)$$

are decomposable forms according to § 3.

B) form

$$(j\nu x)^4 = (\widehat{a}\theta\nu x)^4 = a_\nu^4\theta + \theta_\nu^4 f - 4a_\nu^3\theta_\nu - 4a_\nu\{a_\nu\theta_x - (\widehat{a}\theta\nu x)\}^3$$
$$+ 6a_\nu^2\{a_\nu\theta_x - (\widehat{a}\theta\nu x)\}^2 = \varrho f + \frac{5}{3}i\theta - 6a_\nu^2(\widehat{a}\theta\nu x)^2 + 4a_\nu(\widehat{a}\theta\nu x)^3,$$

and therefore

$$(j\nu x)^4 = \varrho f + \frac{5}{3}i\theta \mod(\Delta, \nu), \qquad \text{(cf. § 5 end).}$$

C) form Δ_ν^4: reducer θ_ν^4.

forms Δ_ν^2 and Δ_ν^3: reducer a_ν^3 or θ_ν^4 by replacing the form Δ by lower forms of the form sequence (§ 5 end), i.e. by double reduction of the expressions

$$a_\vartheta a_\nu^3, \qquad a_\vartheta a_\nu^3\theta_\nu, \qquad a_\vartheta\theta_\nu^4.$$

The double calculation of Δ_ν^3 gives rise to the reduction of the higher form a_η^3.

Reduction formulas:

$$\begin{aligned}
&\text{a)}\quad \Delta_\nu^2 = \frac{3}{5}a_\eta^2 - \frac{3}{10}(asu)^2 + \frac{1}{3}i\theta + \frac{1}{5}u_x^2(2a_\varrho^2 - t),\\
&\text{b)}\quad \Delta_\nu^3 = -\frac{3}{10}a_\varrho + \frac{3}{5}u_x\left(\frac{13}{10}a_\varrho^2 - \frac{2}{5}t\right),\\
&\text{c)}\quad \Delta_\nu^4 = a_\varrho^2 - \frac{2}{5}t.
\end{aligned} \tag{10}$$

Derivation of the formulas:

$$\begin{aligned}
\text{a)}\quad a_\vartheta a_\nu^3 = \frac{1}{3}i\theta = [a_\vartheta]_{\nu^3} + \frac{6}{7}[a_\vartheta^2]_{\nu^2} + \frac{6}{5}[a_\vartheta(a\theta u)^2]_{\nu^2}\, ux^2 + \frac{11}{30}[a_\vartheta^2(a\theta u)^2]_\nu\, \nu x^2\\
- \frac{6}{7}u_x[a_\vartheta^2]_{\nu^3} - \frac{1}{6}u_x[a_\vartheta^2(a\theta u)^2]_{\nu^2}\, \nu x^2,\\
\frac{1}{3}i\theta = \Delta_\nu^2 - \frac{2}{3}u_x\Delta_\nu^3 - a_\nu a_{\nu_1}(\nu\nu_1 x)^2 + \frac{2}{5}a_\nu^2(\nu\nu_1 x)^2 + \frac{1}{5}u_x^2 a_\nu^2 a_{\nu_1}(\nu\nu_1 x)^2;
\end{aligned}$$

$$\begin{aligned}
\text{b)}\quad a_\vartheta a_\nu^3\theta_\nu = 0 = [a_\vartheta]_{\nu^4} + \frac{9}{14}[a_\vartheta^2]_{\nu^3} + \frac{3}{5}[a_\vartheta(a\theta u)^2]_{\nu^2}\, \nu x^2 + \frac{17}{120}[a_\vartheta^2(a\theta u)^2]_\nu\, \nu x^2\\
- \frac{9}{14}u_x[a_\vartheta^2]_{\nu^4} - \frac{1}{24}u_x[a_\vartheta^2(a\theta u)^2]_{\nu^2}\, \nu x^2,\\
0 = \frac{5}{6}\Delta_\nu^3 - \frac{1}{2}u_x\Delta_\nu^4 - \frac{1}{4}a_\eta^3 + \frac{1}{20}u_x a_\eta^4;
\end{aligned}$$

$$\begin{aligned}
\text{b')}\quad a_\vartheta\theta_\nu^4 = a_\varrho &= a_\vartheta\theta_\nu\{a_\nu - (a\theta\nu x)\}^3\\
&= -\frac{3}{2}[a_\vartheta]_{\nu^3} + \frac{3}{5}[a_\vartheta(a\theta u)^2]_{\nu^2}\, \nu x^2 - \frac{37}{120}[a_\vartheta^2(a\theta u)^2]_\nu\, \nu x^2\\
&\quad + \frac{3}{2}u_x[a_\vartheta^2]_{\nu^4} + \frac{49}{120}u_x[a_\vartheta^2(a\theta u)^2]_{\nu^2}\, \nu x^2,\\
a_\varrho &= -\frac{3}{2}\Delta_\nu^3 + \frac{3}{2}u_x\Delta_\nu^4 - \frac{11}{20}a_\eta^3 + \frac{7}{20}u_x a_\eta^4;
\end{aligned}$$

$$\begin{aligned}
\text{c)}\quad a_\vartheta^2\theta_\nu^4 = a_\varrho^2 = [a_\vartheta^2]_{\nu^4} + \frac{3}{5}[a_\vartheta^2(a\theta u)^2]_{\nu^2}\, \nu x^2,\\
a_\varrho^2 = \Delta_\nu^4 + \frac{2}{5}a_\eta^4.
\end{aligned}$$

D) reduction of a_η^3 by double reduction of Δ_ν^3 (C).

The remaining reductions are obtained by a computation dualistically opposite to that of § 9.

Reduction formulas:

$$\begin{array}{rl|rl}
a_\eta^2(aHu) & = -\dfrac{1}{6}(asu)^3, & a_\eta^2(aHu)^2 & = \dfrac{4}{9}i\nu - \dfrac{1}{6}(asu)^4, \\
a_\eta^3 & = -a_\varrho + \dfrac{3}{5}u_x a_\varrho^2 + \dfrac{1}{5}u_x t, & a_\eta^3(aHu) & = 0, \\
 & & a_\eta^4 & = t \quad \text{(adjoined as a modulus).}
\end{array} \tag{11}$$

Consequences:

1) $(j\nu x)^4$, Δ_ν^2, $a_\eta^2(aHu)$ are reducers.

2) ν enters only in products with contragredient forms in foldings with K, j, Δ; the same holds for f in relation to H, and for j in relation to ν, since $\theta_\nu(\theta \cdot j, \nu, x^3)$ becomes reducible according to § 11 B.

§ 11. Forms of the sixth order (System III).

$$\text{Folding of} \left\{ \begin{array}{l} \theta^2, f \cdot j, N \text{ with } \nu, \\ \theta \text{ with } H. \end{array} \right.$$

Irreducible forms:

$$\text{A) } (\vartheta^2\nu x)^3, \quad \theta_\nu(\vartheta^2\nu x)^3 \quad \text{B) } a_\nu(j\nu x), \quad a_\nu^2(j\nu x) \quad \text{C) } N_\nu$$

$$\text{D)} \quad \begin{array}{c|c|c|c}
 & (\theta H u) & (\theta H u)^2 & \cdot \\
(\vartheta\eta x) & \theta_\eta & H_\vartheta(\theta H u),\ \theta_\eta(\theta H u) & \theta_\eta(\theta H u)^2 \\
(\vartheta\eta x)^2 & H_\vartheta(\vartheta\eta x),\ \theta_\eta(\vartheta\eta x) & \theta_\eta^2 & \theta_\eta^2(\theta H u) \\
\cdot & \theta_\eta(\vartheta\eta x)^2 & \cdot & \cdot
\end{array}$$

reductions:

A) $(\vartheta^2\nu x)^4$ decomposes according to formula (3.)a:

$$(a\Delta\vartheta x)^4 = \frac{1}{2}(\vartheta^2\nu x)^4 - \frac{3}{5}f \cdot a_\nu^2(\nu\vartheta x)^2 = \frac{1}{3}\Delta^2 + f\Delta_\vartheta^2.$$

B) $a_\nu(j\nu x)^2$ decomposes according to formula (9.)a, indem wir according to § 6 end $f \cdot j$ by $(\theta\theta' u)^2$ ersetzen, and the reducibility of θ'_ϑ take account of:

$$\frac{1}{3}a_\nu(j\nu x)^2 = (\theta\theta'\nu x)^2\theta_\nu = \theta_\nu^3\theta - \theta_\nu^2\theta'_\nu = \theta_\nu^3\theta + \theta_\nu^2(\widehat{j\nu\theta' u}) = \theta_\nu^3\theta - \frac{2}{3}\theta_\nu^3\theta.$$

forms $a_\nu(j\nu x)^3$ and $a_\nu^2(j\nu x)^2$: reducer a_j and $(j\nu x)^4$:

$$a_\nu(j\nu x)^3 = a_j(j\nu x)^3 - (j\nu x)^3(j\nu\widehat{a}u),$$

$$a_\nu^2(j\nu x)^2 = a_j^2(j\nu x)^2 - 2a_j(j\nu x)^2(j\nu\widehat{a}u) + (j\nu x)^2(j\nu\widehat{a}u)^2.$$

C) form sequence N_ν^2; reducer Δ_ν^2. $(n\nu x)$ and $N_\nu(n\nu x)$ decompose as a transvection over functional determinants according to § 3.

D) *Overview of the reductions:*

a) form sequence $\theta_\eta^2 H_\vartheta$: reducer $a_\eta^2(aHu)$. (Leads to forms with higher symbols.)

b) form sequence $\theta_\eta H_\vartheta$: reducer $a_\eta^2(aHu)$ by double reduction. (Leads to forms with higher symbols and to higher forms of the total form sequence, i.e. to the form sequence θ_η^2.) Instead of $\theta_\eta H_\vartheta$ we consider the form sequence

$$\theta_\eta(\vartheta\eta x)(\theta H u) = \theta_\eta H_\vartheta + \theta_\eta^2,$$

ersetzen in it the symbols θ by the lower form (abu), gelangen so to the double reduction of the form sequence

$$a_\eta^2(aHu)(abu) = \theta_\eta H_\vartheta + \frac{3}{2}\theta_\eta^2 \mod s$$

up to the terminal form

$$a_\eta^3 b_\eta(bHu)(abu) = \theta_\eta^3 H_\vartheta + \frac{1}{2}\theta_\eta^4.$$

c) form sequence θ_η^3: By double reduction of those forms which simultaneously belong to the form sequences $\theta_\eta H_\vartheta$ and $\theta_\eta^2 H_\vartheta$ - i.e. the forms of the form sequence $\theta_\eta^2 H_\vartheta$ - to forms with higher symbols on the one hand, and to forms with higher symbols and to the higher form sequence θ_η^3 on the other hand.

d) forms $\theta_\eta^2(\vartheta\eta x)$, $\theta_\eta^2(\vartheta\eta x)^2$, $\theta_\eta^3(\vartheta\eta x)$: By double reduction by means of the reducers a analogous to the computation of § 9.

e) forms $\theta_\eta^2(\theta Hu)^2$ and $\theta_\eta^2(\vartheta\eta x)^2$: By double reduction of the expressions $\Delta_\nu^2(a\Delta u)^4$ and $(a\Delta\nu x)^4$ by means of the reducers Δ_ν^2 and $(a\Delta u)^4$.

f) forms H_ϑ and H_ϑ^2: Decomposition of forms. Is obtained for H_ϑ^2 by double reduction of the expression $\Delta_\nu^2(a\Delta u)^2$ or also by direct calculation of the products $(a_\nu^2)^2$, $a_\nu^2 b_{\nu_1}$ by forms of the system. (The analogous calculation of $(a_\nu)^2$ gives a relation between decomposable forms.)

g) Reduction of higher forms by the fact that forms of the form sequence $\theta \cdot H$ admit two reduction possibilities (belong to two reducible form sequences), namely:

g: by double reduction of $\theta_\eta^2(\vartheta\eta x)^2$; allows reduction d) and e) zu.

$s_\vartheta^2(\theta su)$: by double reduction of $\theta_\eta^3(\vartheta\eta x)$; allows reduction c) and d) zu.

$s_\vartheta^2(\theta su)^2$: by double reduction of $\theta_\eta^2 H_\vartheta^2$; allows reduction a) and has the reducers $\theta_\nu^2(\vartheta\nu x)^2$ as a factor.

s_ν^2: by double reduction of $\theta_\eta^3 H_\vartheta$; allows reduction a) and c) zu.

The proof of the independence of the remaining forms is obtained by double reduction of the form sequence $\Delta_\nu^2(a\Delta u)^2 = \theta_\eta^2$.

Execution of the reductions:

The existence of reductions a), b), c), d) is clear a priori, since according to the stated law of formation the relations arising from double reduction are independent of one another. For reductions e), f), g), it must be shown by computation that no identities arise.

Proceeding symmetrically with the diagonal term in the form sequence $\theta \cdot H$ up to the form θ_η, and partly indicating the calculation, we also give the formulas obtained by reductions a), b), c), d), since they are used partly in reductions e), f), g), and partly later.

1) H_ϑ and H_ϑ^2 according to f). According to the ansatz[28]

$$\begin{aligned} a_\nu b_{\nu_1}^2 &= \nu a_\nu b_\nu^2 - (aHu)(bHu)b_\eta \sim \nu a_\nu b_\nu^2 - f a_\eta (aHu)^2 - (abu)(aHu)b_\eta + (abu)^2 b_\eta, \\ a_\nu^2 b_{\nu_1}^2 &= b_\nu^2\{a_\nu u_\nu - (a\nu\nu_1 u)\}^2 \sim \nu\{a_\nu^2 b_\nu^2 - 2a_\nu b_\nu (aHu)(bHu) - \frac{1}{6}(asu)^2(bsu)^2\} \\ &\sim \nu a_\nu^2 b_\nu^2 - 2a^2(aHu)(abu) - \frac{1}{6}(asu)^2(bsu)^2 + (\vartheta\eta x)^2(\theta Hu)^2 \end{aligned}$$

after substitution of the value of $\theta_\eta H_\vartheta$, the formulas arise:

$$\begin{aligned} &\text{a)} \quad H_\vartheta \sim 2a_\nu a_\varrho^2 + 2\nu b_\nu(\vartheta\eta x)^2 + 2f a_\eta (aHu)^2 - 2\theta_\eta, \\ &\text{b)} \quad H_\vartheta^2 \sim (a^2)^2 + 2\theta_\eta^2 + \frac{2}{3}(\theta su)^2 + \frac{1}{12}s\nu - \frac{1}{9}if\nu - \frac{1}{6}f(asu)^4. \end{aligned} \tag{12}$$

2) $\theta_\eta H_\vartheta$ according to b):

$$\begin{aligned} a_\eta^2(aHu)(abu) &\sim [a_\eta^2(aHu)]_{bu} + \frac{1}{2}f[a_\eta^2(aHu)^2] = a_\eta b_\eta (aHu)(abu) \\ &+ a_\eta(\widehat{ab}\eta x)(abu)(aHu) \sim \theta_\eta(\vartheta\eta x)(\theta Hu) + \frac{1}{2}\theta_\eta^2 + \frac{1}{4}(\nu\eta x)^2. \end{aligned}$$

According to the formulas (5.) and (11.) follows:

$$\theta_\eta H_\vartheta \sim -\frac{3}{2}\theta_\eta^2 - \frac{1}{4}(\theta su)^2 - \frac{1}{12}s\nu + \frac{2}{9}if\nu.$$

3) $\theta_\eta H_\vartheta(\theta Hu)$ is computed according to b), analogously to $\theta_\eta H_\vartheta$:

$$\theta_\eta H_\vartheta(\theta Hu) \sim -\theta_\eta^2(\theta Hu) - \frac{1}{6}(\theta su)^3.$$

4) $\theta_\eta H_\vartheta(\vartheta\eta x)$ according to b); $\theta_\eta^2(\vartheta\eta x)$ according to d) (cf. § 9):

$$\theta_\eta H_\vartheta(\vartheta\eta x) \sim -\theta_\eta^2(\vartheta\eta x) - \frac{1}{6}s_\vartheta(\theta su) \sim -\frac{1}{3}(\vartheta\varrho x) - \frac{1}{6}s_\vartheta(\theta su),$$

[28]The sign $\sim$ in place of the equality sign means that products arising from u_x with higher forms, other than those arising by foldings III and IV, are omitted.

$$\theta_\eta^2(\vartheta\eta x) \sim \frac{2}{3}a_\eta^3(abu) \sim -\frac{1}{3}(\vartheta\varrho x).$$

5)

$\theta_\eta H_\vartheta^2$ according to b)	$\theta_\eta^2 H_\vartheta$ according to a)	$\theta_\eta^2 H_\vartheta$ according to b) and thereby θ_η^3 according to c)
from $a_\eta^2(aHb)(bHu)$	from $a_\eta^3(Hab)(abu)$	from $a_\eta^3(bHu)(abu)$

$$\theta_\eta H_\vartheta^2 \sim -\frac{3}{2}\theta_\eta^2 H_\vartheta - \frac{1}{4}s_\vartheta(\theta su)^2 + \frac{1}{6}s_\nu - \frac{4}{9}ia_\nu \sim -\theta_\varrho - \frac{1}{6}s_\nu - \frac{1}{9}ia_\nu,$$

$$\theta_\eta^2 H_\vartheta \sim \frac{2}{3}\theta_\varrho - \frac{1}{6}s_\vartheta(\theta su)^2 + \frac{2}{9}s_\nu - \frac{2}{9}ia_\nu,$$

$$\theta_\eta^3 H_\vartheta \sim \frac{2}{3}\theta_\varrho - \frac{2}{3}\theta_\eta^3 + \frac{5}{36}s_\nu, \qquad \theta_\eta^3 \sim \frac{1}{4}s_\vartheta(\theta su)^2 - \frac{1}{8}s_\nu + \frac{1}{3}ia_\nu.$$

6) $\theta_\eta^2(\theta Hu)^2$ according to e):

$$\Delta_\nu^2(a\Delta u)^2 = [(a\Delta u)^4]_{\nu^2} = -\frac{12}{5}\theta_\eta^2(\theta Hu)^2 - \frac{3}{10}\sigma - \frac{1}{20}(\theta su)^4 + \nu\varrho \quad \text{(according to formula (3.)c)},$$

$$\Delta_\nu^2(a\Delta u)^4 = [\Delta_\nu^2]_{aa}^4 = -\frac{3}{5}\theta_\eta^2(\theta Hu)^2 + \frac{1}{5}\sigma - \frac{3}{10}(\theta su)^4 + \frac{1}{3}ij \quad \text{(according to formula (10.)a)},$$

$$\theta_\eta^2(\theta Hu)^2 = -\frac{1}{6}\sigma + \frac{1}{12}(\theta su)^4 - \frac{1}{9}ij + \frac{1}{3}\nu\varrho.$$

7) $\theta_\eta^2(\vartheta\eta x)^2$ according to d) and e). From this reduction of the higher form g.

By a computation analogous to that in § 9, it follows:

$$\begin{aligned}\theta_\eta^2(\vartheta\eta x)^2 &= \frac{1}{2}if - \frac{1}{2}a_\eta^2 b_\eta - \frac{1}{12}g = \frac{2}{3}tf - \frac{2}{3}a_\eta^3 b_\eta - \frac{1}{6}g \\ &= -\frac{1}{6}g - \frac{1}{3}(\vartheta\varrho x)^2 + \frac{4}{15}fa_\varrho^2 + \frac{8}{15}ft \quad \text{(according to formula (11.))}.\end{aligned}$$

By the corresponding computation as above follows:

$$((a\Delta\nu x)^4) = [(a\Delta u)^4]_{\nu x^4} = \frac{21}{10}\theta_\eta^2(\vartheta\eta x)^2 + \frac{7}{10}s_\vartheta^2 - \frac{3}{10}g \quad \text{(according to formula (3.)c)},$$

$$\begin{aligned}(a\Delta\nu x)^4 &= -\frac{1}{3}i\Delta + 6[\Delta_\nu^2]a^2 - 4[\Delta_\nu^3]a + \Delta_\nu^4 f \\ &= -\frac{36}{5}\theta_\eta^2(\vartheta\eta x)^2 - \frac{9}{5}s_\vartheta^2 - \frac{3}{5}g - \frac{3}{5}(\vartheta\varrho x)^2 \\ &\quad + \frac{5}{3}i\Delta + \frac{37}{25}fa_\varrho^2 + \frac{74}{25}ft \quad \text{(according to formula (10.))}.\end{aligned}$$

Thus

$$\begin{aligned}\frac{93}{10}\theta_\eta^2(\vartheta\eta x)^2 &= -\frac{3}{10}g - \frac{5}{2}s_\vartheta^2 - \frac{3}{5}(\vartheta\varrho x)^2 \\ &\quad + \frac{5}{3}i\Delta + \frac{37}{25}fa_\varrho^2 + \frac{74}{25}ft,\end{aligned}$$

and by comparing the two values of $\theta_\eta^2(\vartheta\eta x)^2$ according to g):

$$0 = g - 2s_\vartheta^2 + 2(\vartheta\varrho x)^2 + \frac{4}{3}i\Delta - \frac{4}{5}fa_\varrho^2 - \frac{8}{5}ft. \tag{13}$$

8) $\theta_\eta^2 H_\vartheta(\theta Hu)$ according to a) and b), from this $\theta_\eta^3(\theta Hu)$ according to c) (forms with the Faktor u_x vanish):

$$\theta_\eta^2 H_\vartheta(\theta Hu) = -\frac{1}{6}s_\vartheta(\theta su)^3 - \frac{1}{3}(\nu\varrho x),$$

$$\theta_\eta^2 H_\vartheta(\theta Hu) = -\frac{1}{3}\theta_\eta^3(\theta Hu) - \frac{1}{3}(\nu\varrho x),$$

$$\theta_\eta^3(\theta Hu) = \frac{1}{2}s_\vartheta(\theta su)^3.$$

9) $\theta_\eta^2 H_\vartheta(\vartheta\eta x)$ according to a) and b), from this $\theta_\eta^3(\vartheta\eta x)$ according to c). $\theta_\eta^3(\vartheta\eta x)$ according to d), from this reduction of the higher form $s_\vartheta^2(\theta s u)$ according to g) (forms with the Faktor u_x vanish):

$$\begin{aligned}
\theta_\eta^2 H_\vartheta(\vartheta\eta x) &= \frac{1}{3}(atu),\\
\theta_\eta^2 H_\vartheta(\vartheta\eta x) &= \frac{1}{3}(atu) + \frac{1}{3}\theta_\eta^3(\vartheta\eta x) - \frac{1}{6}s_\vartheta^2(\theta s u),\\
\theta_\eta^3(\vartheta\eta x) &= \frac{1}{2}s_\vartheta^2(\theta s u),\\
\theta_\eta^3(\vartheta\eta x) &= \frac{2}{5}\theta_\varrho(\vartheta\varrho x) + \frac{1}{5}(atu),\\
s_\vartheta^2(\theta s u) &= \frac{4}{5}\theta_\varrho(\vartheta\varrho x) + \frac{2}{5}(atu).
\end{aligned} \tag{14}$$

10) $\theta_\eta^2 H_\vartheta^2$ according to a) and by reducer $\theta_\nu^2(\vartheta\nu x)^2$, $\theta_\eta^3 H_\vartheta$ according to a) and c), θ_η^4 according to c), from this reduction of the higher forms $s_\vartheta^2(\theta s u)^2$ and s_ν^2 according to g):

$$\begin{aligned}
\text{According to a)}\quad \theta_\eta^2 H_\vartheta^2 &= [a_\eta^2(aHu)^2]b^2 + \frac{1}{3}[a_\eta^4]_{bu^2} - \frac{1}{3}H_\nu^2(\nu\eta x)^2\\
&= \frac{4}{9}ia_\nu^2 - \frac{1}{6}s_\vartheta^2(\theta s u)^2 - \frac{5}{18}s_\nu^2 + \frac{1}{3}(atu)^2 + \frac{1}{54}Ju_x^2.
\end{aligned}$$

$$\begin{aligned}
\text{By } \theta_\nu^2(\vartheta\nu x)^2:\quad \theta_\eta^2 H_\vartheta^2 &= [\theta_\nu^2(\vartheta\nu x)^2]_{\nu_1} + \frac{1}{3}[\theta_\nu^4]_{\nu_1}\hat{\nu}_1 x^2 - \frac{1}{3}s_\vartheta^2(\theta s u)^2\\
&= \frac{4}{9}ia_\nu^2 - \frac{1}{6}s_\nu^2 - \frac{1}{3}s_\vartheta^2(\theta s u)^2 + \frac{1}{3}(\nu\varrho x)^2.
\end{aligned}$$

$$\begin{aligned}
\text{According to a)}\quad \theta_\eta^3 H_\vartheta &= -[a_\eta^2(aHu)]_{bu}b^2 - \frac{1}{2}[a_\eta^2(aHu)^2]b^2 - \frac{1}{3}[a_\eta^3]b + \frac{1}{12}[a_\eta^4]_{bu^2}\\
&\quad + \frac{1}{2}u_x[a_\eta^2(aHu)^2]b^3\\
&= -\frac{2}{9}ia_\nu^2 + \frac{1}{12}s_\nu^2 + \frac{4}{15}\theta_\varrho^2 - \frac{7}{90}(\nu\varrho x)^2 + \frac{1}{30}(atu)^2 + \frac{2}{27}i^2u_x^2 - \frac{1}{36}Ju_x^2.
\end{aligned}$$

$$\begin{aligned}
\text{According to c)}\quad \theta_\eta^3 H_\vartheta : a_\eta^3(Hab)(bHu) &= \frac{1}{2}(atu)^2 = \theta_\eta^2 H_\vartheta(\theta H u)(\vartheta\eta x) + \frac{1}{4}H_\nu^2(\nu\eta x)^2\\
&\quad + \frac{1}{2}\theta_\eta^2 H_\vartheta^2 = \frac{3}{2}\theta_\eta^2 H_\vartheta^2 + \theta_\eta^3 H_\vartheta - u_x[a_\eta^2(aHu)^2]b^3\\
&\quad + \frac{1}{4}H_\nu^2(\nu\eta x)^2,
\end{aligned}$$

and therefore

$$\begin{aligned}
\theta_\eta^3 H_\vartheta &= -\frac{2}{3}ia_\nu^2 + \frac{5}{12}s_\nu^2 + \frac{1}{2}(\nu\varrho x)^2 - \frac{1}{2}(atu)^2\\
&\quad + \frac{4}{27}i^2u_x^2 - \frac{1}{12}Ju_x^2.
\end{aligned}$$

According to c)

$$\begin{aligned}
\theta_\eta^4 &= \frac{4}{9}ia_\nu^2 - \frac{1}{6}s_\nu^2 + \frac{16}{15}\theta_\varrho^2\\
&\quad - \frac{14}{45}(\nu\varrho x)^2 + \frac{2}{15}(atu)^2.
\end{aligned}$$

According to g), from the two values for $\theta_\eta^2 H_\vartheta^2$:

$$\begin{aligned}
s_\vartheta^2(\theta s u)^2 &= \frac{2}{3}s_\nu^2 - 2(atu)^2 + 2(\nu\varrho x)^2 - \frac{1}{9}Ju_x^2\\
&= \frac{8}{9}ia_\nu^2 + \frac{8}{15}\theta_\varrho^2 - \frac{14}{15}(atu)^2 + \frac{38}{45}(\nu\varrho x)^2 - \frac{4}{27}i^2u_x^2.
\end{aligned} \tag{15}$$

According to g), from the two values for $\theta_\eta^3 H_\vartheta$:

$$s_\nu^2 = \frac{4}{3}ia_\nu^2 + \frac{4}{5}\theta_\varrho^2 + \frac{8}{5}(atu)^2 - \frac{26}{15}(\nu\varrho x)^2 + \frac{1}{6}Ju_x^2 - \frac{2}{9}i^2u_x^2. \tag{16}$$

Formula (16.) gives the basis for the reduction of the modulus $(ss'u)^4$, and formula (13.) gives the basis for the reduction of the system of s. (§ 17.)

Consequences:

1) $a_\nu(j\nu x)^3$, $\theta_\eta H_\vartheta$, $\theta_\eta^2(\vartheta\eta x)$, $\theta_\eta^2(\theta Hu)^2$ are reducers, $a_\nu(j\nu x)^2$, H_ϑ and H_ϑ^2 decomposable forms.

2) The form of the system II: $j(\nu) = g = (\nu\eta x)^4 H_x^2$ is reducible.

3) Since the only contragredient form g becomes reducible, ν does not enter at all in powers or in products with forms of the same system in folding with System I.

θ enters into foldings only as a contravariant in products or powers; correspondingly, H enters only as a covariant (cf. § 13 B).

§ 12. Forms of the seventh order.

$$\text{Folding of } \left\{ \begin{array}{l} \theta \cdot K \text{ with } \nu, \\ K, j, \Delta \text{ with } H, \\ f \text{ with } L, \sigma. \end{array} \right.$$

Irreducible forms:

$$\text{B)} \quad \begin{array}{c|c|c|c|c} \cdot & \cdot & (KHu)^2 & \cdot & \cdot \\ \cdot & K_\eta & \cdot & K_\eta(KHu)^2 & \cdot \\ (k\eta x)^2 & \cdot & K_\eta^2 & \cdot & \cdot \\ (k\eta x)^3 & H_k(k\eta x)^2,\ K_\eta(k\eta x)^2 & \cdot & \cdot & \cdot \\ \cdot & K_\eta(k\eta x)^3 & \cdot & \cdot & \cdot \end{array}$$

$$\text{C)} \quad \begin{array}{cc} (j\eta x) & H_j \\ \cdot & H_j(j\eta x) \\ \cdot & H_j^2 \end{array} \qquad \text{D)} \quad \begin{array}{c} (\Delta Hu) \\ \Delta_\eta \end{array}$$

$$\text{E)} \quad \begin{array}{cccc} a_i & (aLu)^2 & (aLu)^3 & \cdot \\ \cdot & \cdot & a_i(aLu)^2 & a_i(aLu)^3 \\ \cdot & a_i^2 & \cdot & \cdot \end{array}$$

reductions:

A) $(\vartheta \cdot k, \nu, x)^4$ decomposes as single Überschiebung over the decomposable form $(\vartheta^2\nu x)^4$.

B) form sequence $H_k K_\eta$: reducer $H_\vartheta\theta_\eta$.

form sequence $K_\eta^2(KHu)$: reducer $a_\eta^2(aHu)$.

form sequence $K_\eta^2(k\eta x)$: reducer $\theta_\eta^2(\vartheta\eta x)$.

The from this arising double reduction of the form sequence K_η^3 gives rise to the reduction of the higher form sequence $(j\eta x)^3$.

(KHu), $(k\eta x)$, $K_\eta(k\eta x)$ decomposable forms according to § 3.

$H_k(KHu)$, $K_\eta(KHu)$ decompose as arising by single folding with the decomposable form $K_\nu(k\nu x)$ and the functional determinant

$$K_\nu^2 = \theta_\nu^2(a\theta u) \equiv a_\nu^2(a\theta u) \mod \nu^2.$$

A relation between decomposable forms corresponds to the double representation of K_ν^2. One obtains:

$$H_k(KHu) = 2K_\nu(\nu\nu_1 k) = 2\theta_{\nu_1}(\nu\nu_1 a\theta) = 2\theta_\nu^2 a_\nu - a_\nu^2\theta_\nu + f\theta_\eta(\theta Hu)^2 - f\nu\theta_\vartheta^3 \mod \nu^3,$$

$$\begin{aligned} K_\eta(KHu) &= -K_\nu^2(\nu\nu_1 x) = -a_\nu^2(a\theta u)(\nu\nu_1 x) \\ &= a_\eta(aHu)^2\theta + a_\nu^2\theta_\nu = -\theta_\nu(\theta Hu)^2 f - \theta_\nu^2 a_\nu. \end{aligned}$$

Therefore

$$0 = a_\nu^2\theta_\nu + \theta_\nu^2 a_{\nu_1} + f\theta_\eta(\theta Hu)^2 + \theta a_\eta(aHu)^2 \mod \nu^3. \tag{17}$$

The forms H_k, $H_k(k\eta x)$, H_k^2, $H_k^2(k\eta x)$ decompose as arising by single folding with the decomposable forms H_ϑ and H_ϑ^2 (cf. § 3. 2), a) and b)).

Es is

$$H_k = H_\vartheta(a\theta u) \sim [H_\vartheta]_{ua} - fH_\vartheta(\theta Hu)$$

(specialization $y = uv$ of 2)a).

$$H_k(k\eta x) = H_\vartheta a_\eta - H_\vartheta\theta_\eta f, \qquad H_\vartheta a_\eta = \frac{5}{4}[H_\vartheta]a - \frac{1}{4}H_\vartheta a_\vartheta \quad (\S\ 3.2)b),$$

$$H_\vartheta^2(a\theta u) = [H_\vartheta^2]_{ua}, \qquad H_\vartheta^2 a_\eta = [H_\vartheta^2]a = H_k^2(k\eta x) + H_\vartheta^2\theta_\eta f.$$

C) form sequence $(j\eta x)^3$: Reducible by double reduction of the form sequence K_η^3 according to B); according to the ansatz:

$$0 = a_\eta^3(a\theta u) = \theta_\eta^3(a\theta u) = \theta_\eta + (a\theta\eta x)^3(a\theta u) - \theta_\eta^3(a\theta u) = \frac{1}{2}(j\eta x)^3 \mod(\nu^3, s, \varrho, t, i, J).$$

The analogous ansatz holds up to the terminal form of the form sequence:

$$0 = H_\vartheta^2(a\theta\eta x)a_\eta^3 = H_\vartheta^2(a\theta\eta x)\theta_\eta^3 - \frac{1}{2}H_\vartheta^2(j\eta x)^4 \mod(\nu^3, s, \varrho, t, i, J).$$

form $(j\eta x)^2$: decomposes 1) by twofold polarization of the relation for the decomposable form H_ϑ^2 ((12.) b);
2) by direct calculation of the products $a_\nu\theta_\nu^3$; thereby Decomposition of the higher form $(\Delta Hu)^2$. One obtains:

$$\left.\begin{aligned} &\text{a)}\quad \frac{1}{3}(\Delta Hu)^2 + \frac{1}{3}(j\eta x)^2 = \theta_\nu^2 a_{\nu_1}^2, \\ &\text{b)}\quad (j\eta x)^2 = \frac{4}{3}\theta_\nu^3 a_{\nu_1}, \end{aligned}\right\} \mod(\nu^3, s, \varrho, t, i, J). \tag{18}$$

according to the ansatz:

$$H_\vartheta^2(a\theta u)^2 - \frac{1}{3}(\Delta Hu)^2 = 2\theta_\eta^2(a\theta u)(aHu) + \theta_\nu^2 a_{\nu_1}^2 = (a\theta u)(aHu)\{a_\eta - (a\theta\eta x)^2\} + \theta_\nu^2 a_{\nu_1}^2,$$

$$\begin{aligned} a_{\nu_1}\theta_\nu^3 &= -(a\widehat{\nu}\nu_1 u)\theta_\nu^2(\theta\widehat{\nu}\nu_1 u) - \frac{1}{2}(u\nu\nu_1 u)\theta_\nu\theta_{\nu_1}(\vartheta\nu\nu_1 u) \\ &= -\frac{3}{2}\theta_\eta^2(\theta Hu)(aHu) = -\frac{3}{2}\theta_\eta^2(\theta Hu)(a\theta u) \\ &= -\frac{3}{2}\{a_\eta - (a\theta\eta x)^2\}\{(aHu) - (a\theta u)\}(a\theta u). \end{aligned}$$

D) form sequence $\Delta_\eta(\Delta Hu)$: reducer Δ_ν^2.
$(\Delta Hu)^2$ decomposable form according to C).
E) form sequence $a_i^2(aLu)$: reducer $a_\eta^2(aHu)$. forms (aLu) and $a_i(aLu)$: Decomposition as a transvectionen over functional determinants according to § 3.
F) form sequence a_σ^3: reducer a_ϱ^3.
forms a_σ^2 and a_σ^3: reducer a_ν^3 and a_η^3 by double reduction of the corresponding expressions

$$H_\nu a_\nu^3 \quad\text{and}\quad H_\nu a_\eta^3, \qquad H_\nu a_\nu^3 a_\nu \quad\text{and}\quad H_\nu a_\eta^4$$

(cf. § 10. C) reduction of Δ_ν^2 and Δ_ν^3).
From this reduction for the higher forms $a_\nu s_\nu(asu)^2$, $a_\nu^2 s_\nu(asu)^2$ and for forms in symbolsn $\varrho, t(a_\nu, a_\eta^2)$ taking account of (16.):

$$\left.\begin{aligned} a_\nu s_\nu(asu)^2 &= \frac{1}{3}i\theta_\nu^2 - \frac{1}{18}iH, \\ a_\nu^2 s_\nu(asu)^2 &= 0, \end{aligned}\right\} \mod(\varrho, t). \tag{19}$$

form a_σ: decomposes by polarization of (12.)b or, more briefly, by direct expansion of the products $a_\nu^2\theta_\nu^3$ according to the ansatz:

$$a_\nu^2\theta_{\nu_1}^3 = (a_\nu^2\theta_{\nu_1})a_{\nu_1} - (a\theta\nu_1 x)^2 = -2a_\eta(aHu)(a\theta H)(a\theta\eta x) + (a\theta\nu_1 x)^2 a_\nu^2\theta_{\nu_1}$$

and taking account of the reduction of $H_j(j\eta x)^2$ (C):

$$a_\sigma = 2a_\nu^2\theta_{\nu_1}^3 \mod(s, \varrho, t, i). \tag{20}$$

Consequences:
1) $(j\eta x)^3$, $\Delta_\eta(\Delta Hu)$, a_σ^2 are reducers, $(j\eta x)^2$, $(\Delta Hu)^2$, a_σ decomposable forms.
2) f enters into folding with L only in products with contragredient forms; therefore f, and likewise K and N, do not enter at all into folding with σ and consequently with $(\nu\sigma x)$. The form j enters only in products with contragredient forms; Δ does not enter at all in products in folding with H and L (this follows from $\Delta_\eta(\Delta Hu)$ and $(\Delta Hu)^2$). Likewise σ, and consequently $(\nu\sigma x)$, do not enter in products in folding with any form of System I. Thus, among products in System II, taking account of the consequences of § 11, there remain only powers of H and products of H with L.

§ 13. Forms of the eighth order (System III).

$$\text{Folding of }\begin{cases}\Delta\cdot j \text{ with } \nu,\\ \theta^2, f\cdot j, N \text{ with } H,\\ \theta \text{ with } L,\sigma.\end{cases}$$

Irreducible forms:

$$\text{A)}\quad \Delta_\nu(j\nu x) \qquad\qquad \text{B)}\quad \begin{matrix}(\vartheta^2\eta x)^3\\(\vartheta^2\eta x)^4\end{matrix}\quad H_\vartheta(\vartheta^2\eta x)^3,\ \theta_\eta(\vartheta^2\eta x)^3,$$

$$\text{C)}\quad (aHu)H_j,\quad a_\eta(aHu)H_j \qquad \text{D)}\quad H_n,\ N_\eta,$$

$$\text{E)}\quad \begin{array}{c|c|c|c} \cdot & (\theta Lu)^2 & (\theta Lu)^3 & \cdot\\ \theta_i & \cdot & L_\vartheta(\theta Lu)^2,\ \theta_i(\theta Lu)^2 & \theta_i(\theta Lu)^3\\ (\vartheta lx)^2 & \theta_i^2 & \cdot & \cdot\\ \cdot & \theta_i(\vartheta lx)^2 & \cdot & \cdot \end{array}$$

reductions:
A) $\Delta_\nu(j\nu x)^3$: reducer $a_\nu(j\nu x)^3$.

$\Delta_\nu(j\nu x)^2$ is reducible by the decomposable form $a_\nu(j\nu x)^2$ according to § 3. 2)b, taking account of the reducer $\theta_{\vartheta j}$. Indeed, according to § 11 B), one obtains:

$$\frac{3}{10}\Delta_\nu(j\nu x)^2+\frac{3}{5}a_\nu(j\nu x)a_\vartheta(j\nu x)+\frac{1}{10}a_\nu(j\nu x)^2=\frac{3}{5}\theta_\nu^3\theta_{\vartheta j}+\frac{2}{5}\theta_\nu^3\theta_{\vartheta j}\theta_{\vartheta j},$$

(reducer a_j and $\theta_{\vartheta j}$).
B) The forms $H_\vartheta(\vartheta'\eta x)^2$, $H_\vartheta^2(\vartheta'\eta x)$, $H_\vartheta^2(\vartheta'\eta x)^2$ decompose as arising by successive single folding with the decomposable forms H_ϑ and H_ϑ^2, respectively $H_\vartheta a_\eta$ and $H_\vartheta^2 a_\eta$ according to § 3. 2)b) and according to the relation:

$$H_\vartheta(\vartheta'\eta x)^2=2H_\vartheta a_\eta^2 f-2H_\vartheta a_\eta b_\eta.$$

C) form sequence $a_\eta(j\eta x)^2$: reducers a_j and $(j\eta x)^3$.

form $a_\eta(j\eta x)$ decomposes: reducer a_j and single folding with the decomposable form $(j\eta x)^2$ according to § 3, 2)a (specialization $y=uv$).

form $a_\eta H_j$ decomposes: 1) by single folding with the decomposable form $a_\nu(j\nu x)^2$;

2) by a special twofold folding with the decomposable form $(j\eta x)^2$; from this one obtains a relation between decomposable forms.

From $a_\nu(j\nu x)^2$:

$$0=\theta_\nu'\theta_\nu+2a_\eta H_j \mod(s,\varrho,t,i,J).$$

From $(j\eta x)^2$:

$$0=\theta_\nu'\theta_\nu-\frac{3}{2}a_\eta H_j \mod(s,\varrho,t,i,J)$$

(products with contravariants are omitted), according to the ansatz:

$$0=\frac{1}{2}a_\nu(abu)^2\theta_\nu^3+\frac{1}{2}u_x(ubu)\theta_{\nu_1}^3(\theta bu)-\frac{3}{4}(\widehat{j}\eta bu)^2+\frac{3}{4}H_j(\widehat{j}\eta bu)-\frac{1}{8}fH_j^2$$

and by interchanging the symbols in $a_\nu(ubu)(\theta bu)\theta_{\nu_1}^3$.
D) form sequence $N_\eta(NHu)$: reducer Δ_ν^2.

(NHu), $(n\eta x)$, $N_\eta(n\eta x)$, $H_\eta(NHu)$ decomposable forms (cf. § 12 B), $(NHu)^2$ decomposes by single folding with the form $(\Delta Hu)^2$.
E) form sequences $\theta_i L_\vartheta$, $\theta_i^2(\theta Lu)^2$, $\theta_i^2(\vartheta lx)$:

reducers: $\theta_\eta H_\vartheta$, $\theta_\eta^2(\theta Hu)^2$, $\theta_\eta^2(\vartheta\eta x)$.

forms (θLu), (ϑlx), L_ϑ, $L_\vartheta(\theta Lu)$, $\theta_i(\theta Lu)$, $L_\vartheta(\vartheta lx)$, $\theta_i(\vartheta lx)$, L_ϑ^2, $L_\vartheta^2(\theta Lu)$, $\theta_i^2(\theta Lu)$ decompose. (Dualistically opposite to the decomposable forms of § 12 B).
F) form sequence $\theta_\sigma(\vartheta\sigma x)$: reducer a_σ^2.

forms $(\vartheta\sigma x)$ and $(\vartheta\sigma x)^2$ decompose, as arising by single folding with the decomposable form a_σ; θ_σ, by twofold folding arising, decomposes, since the form sequence

$$a_{\nu_2}^2(a\theta u)\theta_{\nu_1}^3\equiv H_j^2(j\eta x)\equiv 0 \mod u_x(s,\varrho,t,i,J):$$

$$\theta_\sigma = a_\sigma (abu)^2 = a_\nu^2 (abu)^2 \theta_\nu^3.$$

Consequences:

θ^3 or $\theta^2 K$ does not enter into folding with H; θ enters into folding with L only as a contravariant in powers or products, and not at all into folding with σ and $(\nu\sigma x)$.

N does not enter in products in folding with irgend einer form of System II enters, ebensowenig wie with powers or products of forms of System II. There remain an products in System I, taking account of the consequences of the last paragraphs, only products $f \cdot j$, $\Delta \cdot j$, powers of θ and products of $\theta \cdot K$.

§ 14. Forms of the ninth order (System III).

$$\text{Folding of } \begin{cases} \theta \cdot K \text{ with } H, \\ K, j, \Delta \text{ with } L, \\ j, \Delta \text{ with } \sigma, \\ f \text{ with } H^2. \end{cases}$$

Irreducible forms:

A) $(\vartheta \cdot k, \eta, x)^4,\ H_k(\vartheta \cdot k, \eta, x)^4$ B) $(KLu)^3,\ K_i(KLu)^3,\ K_i^2,\ (klx)^3,\ K_i(Klx)^3,$

C) $L_j,\ L_j^2$ D) $\Delta_i,$

E) $(j\sigma x)$ G) $(aH^2u)^3,\ (aH^2u)^4,\ a_\eta(aH^2u)^3.$

reductions:

A) the forms $H_\vartheta(k\eta x)^3$, $H_\vartheta^2(k\eta x)^2$, $H_\vartheta^2(k\eta x)^3$ decompose as arising by successive single folding with decomposable forms.

B) form sequences K_iL_k, $K_i^2(KLu)$, $K_i^2(klx)$:

reducers: $\theta_\eta H_\vartheta$, $a_\eta^2(aHu)$, $\theta_\eta^2(\vartheta\eta x)$.

forms L_i, $K_i(KLu)$, $K_i(klx)$, $(KLu)^2$, $(klx)^2$ decompose as a transvectionen over functional determinants (§ 3).

forms $L_k,\ L_k(KLu),\ L_k(KLu)^2,\ L_k(klx),\ L_k(klx)^2,\ L_k^2,\ L_k^2(KLu),\ L_k^2(klx),\ L_k^3,\ K_i(KLu)^2,\ K_i(klx)^2$ decompose as arising by single folding with the decomposable forms:

$$L_\vartheta,\ H_k(KHu),\ L_\vartheta(\vartheta lx),\ H_k^2,\ L_\vartheta^2,\ L_\vartheta^2(\theta Lu),\ K_\eta(KHu),\ \theta_i(\vartheta lx)$$

according to § 3. 2), a) and b).

C) form sequence $(jlx)^3$: reducer $(j\eta x)^3$.

forms $(jlx)^2$ and $L_j(jlx)$: arising by single folding with the decomposable form $(j\eta x)^2$.

D) form sequence $\Delta_i(\Delta Lu)$: reducer Δ_ν^2.

forms $(\Delta Lu)^2$, $(\Delta Lu)^3$: Single folding with the decomposable form $(\Delta Hu)^2$.

E) form sequence $(j\sigma x)^2$: reducer a_σ^2 by replacing j by lower forms of the form sequence (§ 5):

$$a_\sigma^2(a\theta u)^2 = \frac{1}{3}(j\sigma x)^2, \qquad a_\sigma^2(a\theta\sigma x)^2 = \frac{1}{3}(j\sigma x)^4.$$

F) form sequence Δ_σ^2: reducer a_σ^2.

form Δ_σ: reducer a_σ^2 by replacing Δ by lower forms of the form sequence:

$$a_\sigma a_\varrho^2 = \frac{2}{3}\Delta_\sigma \mod \nu.$$

Consequences: Only the form j enters in folding with σ and $(\nu\sigma x)$ enters.

N does not enter in folding with L enters, since the Δ_i correspondinglye form N_i decomposes.

§ 15. Forms of the tenth order (System III).

$$\text{Folding of } \begin{cases} \theta^2,\ f \cdot j \text{ with } L, \\ \theta \text{ with } H^2. \end{cases}$$

Irreducible forms:

A) $(\vartheta^2lx)^4,\ \theta_i(\vartheta^2lx)^4$ B) $(aLu)^2L_j,\ a_i(aLu)^2L_j,$

C) $(\theta H^2u)^3,\ (\theta H^2u)^4,\ H_\vartheta(\theta H^2u),\ \theta_\eta(\theta H^2u)^3.$

reductions:

A) and B): Decomposition of the remaining forms by single folding with decomposable forms.

C) Decomposition of the remaining forms according to § 13 B) by the dualistically opposite consideration.

Consequences:

$\theta^2 K$ does not enter into folding with L; correspondingly, $H^2 L$ does not enter into folding with K. The forms H^3 and $H^2 L$ do not enter into folding with θ. The folding scheme of § 7 is therefore proved.

§ 16. Forms of the 11th, 12th, 13th, and 15th orders (System III).

11th order.

$$\text{Folding of } \begin{cases} \theta \cdot K \text{ with } L, \\ K, j \text{ with } H^2, \\ j \text{ with } (\nu\sigma x), \\ f \text{ with } H \cdot L. \end{cases}$$

Irreducible forms:

$$\text{A)} \quad (\vartheta \cdot k, l, x)^5, \ \theta_i(\vartheta \cdot k, l, x)^5 \qquad \text{B)} \quad (KH^2u)^4, \ K_\eta(KH^2u)^4,$$

$$\text{C)} \quad (\nu\sigma j), \qquad \text{D)} \quad (a_\eta H \cdot L, u)^4.$$

12th order.

$$\text{Folding of } \begin{cases} \theta^3 \text{ with } L, \\ \theta \text{ with } H \cdot L. \end{cases}$$

Irreducible forms:

$$\text{E)} \ (\vartheta^3 lx)^6, \qquad \text{F)} \ (\theta, H \cdot L, u)^4, \qquad L_\vartheta(\theta, H \cdot L, u^4).$$

13th order.

$$\text{Folding of } K \text{ with } H \cdot L.$$

Irreducible forms:

$$\text{G)} \ (K, H \cdot L, u)^5, \qquad K_\eta(K, H \cdot L, u)^5.$$

15th order.

$$\text{Folding of } K \text{ with } H^3.$$

Irreducible form:

$$\text{H)} \ (KH^3u)^6.$$

reductions:

The forms H_j^2, H_j^2 have reducer L_j^3, from the relation:

$$(\nu l x) L_x^3 = -\frac{1}{2} H^2 \ \operatorname{mod}(\sigma, s).$$

Decomposition or reducibility of the remaining forms follows from the considerations of the earlier paragraphs.

Consequences:

According to the folding scheme of § 7 and the consequences of §§ 8–15, the irreducible forms of System III (117 forms) are thereby exhausted.

Chapter IV. The relatively complete system mod (ϱ, t).

§ 17. Reduction of the modulus $(ss'u)^4$ and of the system of s.

For the formation of the next higher relatively complete system, one must, in analogy with § 7, form the system of s with respect to the next modulus

$$\nu(s) = (ss'u)^4$$

and fold it with the forms of the relatively complete system mod s. It will be shown that, after introduction of the modulus (ϱ, t) as the higher modulus,

A) the modulus $(ss'u)^4$ becomes reducible,

B) the system of s consists of the single form s.

A) The reduction of the modulus $(ss'u)^4$ follows at once from the reducer s_ν^2 found by formula (16.), § 11. From

$$s_\nu^2 = \frac{4}{3} i a_\nu^2 + \frac{4}{5}\theta_\varrho^2 + \frac{8}{5}(atu)^2 - \frac{26}{15}(\nu\varrho x)^2 + \frac{1}{6} J \cdot u_x^2 - \frac{2}{9} i^2 \cdot u_x^2$$

there follows, by polarization from x to ν_1,

$$\begin{aligned}(ss'u)^4 &= -\frac{2}{3} J \cdot \nu + 6 s_\nu^2 s_{\nu_1}^2 \\ &= \frac{24}{5}\theta_\nu^2\theta_\varrho^2 + \frac{48}{5} a_\nu^2 (atu)^2 - \frac{52}{5} H_\varrho^2 + \frac{4}{3} i \cdot (asu)^4 + \frac{1}{3} J \cdot \nu - \frac{4}{9} i^2 \cdot \nu,\end{aligned} \tag{21}$$

and with this the reduction of the modulus $(ss'u)^4$.

B) The reduction of

$$Z = (ss'u)^2 = \theta(s)$$

and hence the reduction of the system of s is obtained by replacing the form Z by the lower form g_ν^2, according to formula (8.)b, § 7, taking into account the relation

$$s_\eta^2 = s_\nu^2 (\nu\nu_1 x)^2 - \frac{1}{3} Z.$$

The form g_ν^2 has, by formula (13.), § 11, the reducer s_ν^2 as a factor. By formula (8.) and (16.) one obtains

$$g_\nu^2 = -Z + \frac{14}{5} i a_\eta^2 + \frac{14}{15} i (asu)^2 \mod(\varrho, t),$$

and by formulas (13.), (14.), (15.), (16.),

$$\begin{aligned} g_\nu^2 &= 2[s_\vartheta^2]_{\nu^2} - \frac{4}{3} i \Delta_\nu^2 = 2 s_\vartheta^2 s_\nu^2 - \frac{6}{5}[s_\vartheta^2(\theta s u)^2]\widehat{\nu x}^2 - \frac{4}{3} i \Delta_\nu^2 \\ &= \frac{4}{5} i \cdot a_\eta^2 - \frac{2}{5} i \cdot (asu)^2 + \frac{1}{3} J \cdot \theta \mod(\varrho, t).\end{aligned}$$

Thus

$$Z = 2i \cdot a_\eta^2 + \frac{4}{3} i \cdot (asu)^2 - \frac{1}{3} J \cdot \theta \mod(\varrho, t). \tag{23}$$

The relatively complete system mod $((ss'u)^4, \varrho, t)$ therefore passes into a relatively complete system mod (ϱ, t), and from this, by transvection over the known system of two quadratic forms, the absolutely complete system arises. In order to form the relatively complete system mod (ϱ, t), since by formula (23.) the system of s reduces to the form s, we must transvect the relatively complete system mod (s, ϱ, t) over s and over powers of s. It will be shown that powers of s enter into folding only with System I.

The announced relation is

$$\begin{aligned}(ass')^4 &= \frac{24}{5}\theta_\nu^2\theta_\varrho^2 a_\nu^2 a_\varrho^2 + \frac{48}{5} a_\nu^2 b_\nu^2 (tab)^2 - \frac{52}{5} H_\varrho^2 a_\nu^4 + \frac{5}{3} J \cdot i - \frac{4}{9} i^3, \\ (ass')^4 &= \frac{24}{5} a_\varrho^2 a_{\varrho'}^2 - \frac{12}{5} t_\varrho^2 + \frac{5}{3} J \cdot i - \frac{4}{9} i^3.\end{aligned} \tag{22}$$

[28] From formula (21.) there follows the relation, mentioned in the note to the introduction, between the three invariants of order 9, with use of formula (10.)c.

Among forms of the same order and the same degree, we define as "higher forms" those which have arisen from higher forms of the relatively complete system mod (s, ϱ, t) by folding with s, regarding the folding with s merely as a polarization of the original forms. Thus the order of the individual forms is uniquely determined (cf. § 1, form sequences 2). For instance,

$$(\theta_\eta(\theta H u), s, u)^2 \text{ is higher than } s_\eta((\theta H u)^2, s, u),$$

$$(\theta_\eta(\theta H u)^2, s, u) \text{ is higher than } (\theta_\eta(\theta H u), s, u)^2.$$

We divide the resulting system into the four subsystems

System s_I :	obtained by folding s and powers of s with System I;
System s_{II} :	obtained by folding s with System II;
System s_{III} :	obtained by folding s with the individual forms (without products) of System III and with products of one form each from Systems I and II;
System s_{IV} :	obtained by folding s with products of one form each from Systems I and III, II and III, III and III.

It should also be noted that from now on all formulas are to be understood mod (ϱ, t); from formula (27.) on, they are to be understood mod $(\varrho, t, i, J,$ higher forms), without this being expressly stated each time.

For the reduction we collect the formulas obtained earlier:

$$\begin{aligned}
s_\nu^2 &= \frac{4}{3} i a_\nu^2 + \frac{1}{6} J \cdot u_x^2 - \frac{2}{9} i^2 \cdot u_x^2, & s_\nu^3 &= \frac{1}{3} J \cdot u_x, \quad s_\nu^4 = J, \\
g &= 2 s_\vartheta^2 - \frac{4}{3} i \Delta, & s_\vartheta^2(\theta s u) &= 0, \\
s_\vartheta^2(\theta s u)^2 &= \frac{8}{9} i \cdot a_\nu^3 - \frac{4}{27} i^2 \cdot u_x^2, \\
Z &= 2i \cdot a_\eta^2 + \frac{4}{3} i (a s u)^2 - \frac{1}{3} J \cdot \theta, \\
g_\nu &= -s_\eta + u_x \left\{ 2 i a_\eta^2 - \frac{2}{3} i (a s u)^2 + \frac{2}{3} J \cdot \theta \right\}, \\
g_\nu^2 &= \frac{4}{5} i a_\eta^2 - \frac{2}{5} i (a s u)^2 + \frac{1}{3} J \cdot \theta, & g_\nu^3 &= 0, \qquad g_\nu^4 = 0.
\end{aligned} \tag{24}$$

Consequences:

$$s_\nu^3, \quad s_\vartheta^2(\theta s u), \quad s_\vartheta^2 s_\nu, \quad s_\vartheta^2 \theta_\nu$$

are reducers.

§ 18. System s_I.

$$\text{Folding of} \quad \begin{cases} s \text{ with } f;\ \theta;\ K, j, \Delta;\ f \cdot j, N;\ \Delta \cdot j, \\ s^3 \text{ with } \theta, K. \end{cases}$$

Irreducible forms.

Order 5:

$$(a s u), \qquad (a s u)^2, \qquad (a s u)^3, \qquad (a s u)^4.$$

Order 6:

$$\begin{array}{cccc}
(\theta s u) & (\theta s u)^2 & (\theta s u)^3 & (\theta s u)^4 \\
s_\vartheta & s_\vartheta(\theta s u) & s_\vartheta(\theta s u)^2 & s_\vartheta(\theta s u)^3 \\
\cdot & s_\vartheta^2 & \cdot & \cdot
\end{array}$$

Order 7:

$$\text{A)}\ \begin{array}{cccc}
\cdot & (K s u)^2 & (K s u)^3 & (K s u)^4 \\
s_k & s_k(K s u) & s_k(K s u)^2 & s_k(K s u)^3 \\
\cdot & s_k^2 & \cdot & \cdot
\end{array} \qquad \begin{array}{l} \text{B)}\ s_j, \\ \text{C)}\ (\Delta s u),\ (\Delta s u)^2. \end{array}$$

Order 8:

$$\text{A)}\quad s_j(asu),\ s_j(asu)^2,\ s_j(asu)^3,$$
$$\text{B)}\quad (Nsu)^2,\quad s_n,\quad s_n(Nsu).$$

Order 10:

$$\text{A)}\ s_j(\Delta su),\qquad \text{B)}\ s_\vartheta(\theta s'u)^4.$$

Order 11:

$$(Ks^2u)^6,\qquad s_k(Ks^2u)^5,\qquad s_k(Ks^2u)^6.$$

Reductions:

The form sequences

$$s_\vartheta^2(\theta su),\quad s_k^2(Ksu),\quad s_j^2,\quad (\Delta su)^3,\quad (Nsu)^3$$

have the reducer $s_\vartheta^2(\theta su)$ as a factor; s_j^2 and $(\Delta su)^3$ are obtained by going back from j and Δ to lower forms of the form sequence:

$$\begin{aligned}s_\vartheta^2(\theta su)(a\theta u)^3 &= -\frac{1}{2}s_j^2,\\ s_\vartheta^2(\theta su)(a\theta u)^2 &= \frac{1}{3}(\Delta su)^3,\\ s_\vartheta^2(\theta su)^2(a\theta u)^2 &= \frac{1}{3}(\Delta su)^4+\frac{1}{3}s_j^2.\end{aligned}$$

Forms $s_\vartheta^2 s_{\vartheta'}$ and $s_\vartheta^2 s_{\vartheta'}^2$: reducers $s_\vartheta^2(\theta su)$ and $\theta_{\vartheta'}$:

$$\begin{aligned}s_\vartheta^2 s_{\vartheta'} &= s_\vartheta^2\theta_{\vartheta'} - s_\vartheta^2(\theta s\widehat{\vartheta}'x),\\ s_\vartheta^2 s_{\vartheta'}^2 &= s_\vartheta^2 s_{\vartheta'}\theta_{\vartheta'} - s_\vartheta^2 s_{\vartheta'}(\theta s\widehat{\vartheta}'x).\end{aligned}$$

Form sequence $s_j(\Delta su)^2$: reducers Δ_j and $(\Delta su)^3$.

Consequences:

s does not enter into powers in folding with f, j, Δ, N. The forms f, j, Δ, N enter into folding only in products with contragredient forms; however, in products with f those forms may still be quadratic in x, and in products with Δ they may still be linear in x. Thus the following products of forms are to be considered:

$$f\cdot(0,\lambda),\quad f\cdot(1,\lambda),\quad f\cdot(2,\lambda),\quad j\cdot(\mu,0),\quad N,\quad \Delta\cdot(0,\lambda),\quad \Delta\cdot(1,\lambda)\quad (\lambda>0),$$

where (μ,λ) denotes a form of degree μ in x and of degree λ in u.

Neither θ nor K enters into powers or products with arbitrary forms in folding with s or s^2. For products with the functional determinant K, $K\cdot\varphi_x$ ($\varphi\neq f$ or θ), the relation between decomposing forms is

$$K\cdot\varphi_x = (a\theta u)\varphi_x = f\cdot(\varphi\theta u) - \theta\cdot(\varphi a u).$$

The above scheme for folding System s_I is thereby proved.

§ 19. System s_{II}.

Folding of s with ν; H; L; H^2.

Irreducible forms:

Order 6	Order 8	Order 10	Order 12
s_ν	(Hsu), $(Hsu)^2$	$(Lsu)^2$, $(Lsu)^3$	$(H^2su)^3$
.	s_η	s_l	.
.	$s_\nu^4 = J$	.	.

Reductions:

Form sequences $s_\eta(Hsu)$ and $s_l(Lsu)$: reducer s_ν^2.

Form sequence s_σ: reducer s_ν^2 from H, $s_\nu^2 = \dfrac{2}{3}s_\sigma$.

$(H^2su)^4$ decomposes by formula (7.)a (cf. § 11 A). Hence $(H\cdot L, s, u)^4$ also decomposes.

Consequences:

s^2 does not enter into folding with System II. The form ν enters only in products with contragredient forms. The form H, regarded as a covariant, enters into folding in products with forms $(1,\lambda)$, $(2,\lambda)$, $(3,\lambda)$, with arbitrary λ. The forms σ and $(\nu\sigma x)$ do not enter at all, and L, as a functional determinant, does not enter into products in folding; the full transvections over covariants of System III become reducible. The products of Systems I and II that remain are

$$f \cdot \nu, \qquad \Delta \cdot \nu.$$

§ 20. Forms of order 7 (System s_{III}).

Folding of s with $f \cdot \nu$, a_ν, a_ν^2.
Irreducible forms:

$$\begin{gathered} s_\nu(asu), \quad a_\nu(asu), \quad s_\nu(asu)^2, \quad a_\nu(asu)^2, \quad s_\nu(asu)^3, \quad a_\nu(asu)^3, \\ a_\nu s_\nu, \quad a_\nu s_\nu(asu), \quad a_\nu^2(asu), \quad a_\nu^2(asu)^2. \end{gathered}$$

Reductions:

The evident reductions that arise by direct folding with the reducers s_ν^2 and $s_\eta(Hsu)$ will not be mentioned, nor will the decomposition of transvections with functional determinants.

For the forms $a_\nu s_\nu(asu)^2$ and $a_\nu^2 s_\nu(asu)^2$, see formula (19.), § 12; moreover,

$$a_\nu s_\nu(asu)^3 = a_\nu(a\widehat{\nu}_1\nu_2 u)^3(\nu_1\nu_2\nu) = 0.$$

Form sequence $a_\nu^2 s_\nu$: reducer $s_\vartheta^2(\theta su)$, obtained by going back from a_ν^2 to lower forms, i.e. by double reduction of the expressions

$$s_\vartheta^2(a\theta s)(a\theta u), \qquad s_\vartheta^2(a\theta s)(a\theta u)^2$$

according to the Ansatz

$$\begin{aligned} s_\vartheta^2(a\theta s)(a\theta u) &= [(a\theta u)^2]s^3 + \frac{1}{5}[a_\vartheta(a\theta u)^2]s^2 + \frac{1}{60}[a_\vartheta^2(a\theta u)^2]s \\ &= \frac{1}{2}a_\nu^2 s_\nu - \frac{2}{3}a_\nu^2 s_\nu^2, \\ s_\vartheta^2(a\theta s)(a\theta u)^2 &= \frac{4}{5}[a_\vartheta(a\theta u)^2]_{us}s^2 + \frac{23}{180}[a_\vartheta^2(a\theta u)^2]_{us}s \\ &= -\frac{1}{2}a_\nu^2 s_\nu(asu). \end{aligned}$$

Here $a_\nu^2 b_\nu(abu) = 0$.

The formulas obtained are

$$\begin{aligned} a_\nu^2 s_\nu &= \frac{4}{9}i\, a_\nu^2 b_\nu, & a_\nu s_\nu(asu)^2 &= \frac{1}{3}i\theta_\nu^2 - \frac{1}{18}iH, \\ a_\nu s_\nu(asu)^3 &= 0, & a_\nu^2 s_\nu(asu) &= 0, \\ a_\nu^2 s_\nu(asu)^2 &= 0. \end{aligned} \tag{25}$$

Consequences:

1) $a_\nu s_\nu(asu)^2$ and $a_\nu^2 s_\nu$ are reducers.

2) The values of the forms $(\Delta su)^3$, $(\Delta su)^4$, already recognized as reducible in § 19, are thereby obtained; likewise for the corresponding form sequence $(agu)^2$, which, when folded with ν, gives rise to double reduction through the reducer g_ν:

$$\begin{aligned} &\text{a)} \quad (\Delta su)^3 = -\frac{6}{5}a_\nu^2(asu) + 3a_\nu s_\nu(asu) + 2i\theta_\nu(\theta\nu x), \\ &\text{b)} \quad (\Delta su)^4 = -\frac{2}{5}a_\nu^2(asu)^2 + \frac{4}{3}i\theta_\nu^2 + \frac{1}{9}iH. \end{aligned} \tag{26}$$

From formula (24.) and (26.), mod (i, J) (cf. p. 71),

$$\begin{aligned} &\text{a)} \quad (agu)^2 = \frac{2}{3}(\Delta su)^2 - \frac{4}{3}a_\nu s_\nu + \frac{3}{5}s \cdot a_\nu^2, \\ &\text{b)} \quad (agu)^3 = 2a_\nu s_\nu(asu) - a_\nu^2(asu)^2, \\ &\text{c)} \quad (agu)^4 = -a_\nu^2(asu)^2. \end{aligned} \tag{27}$$

§ 21. Forms of 8th order (System s_{III}).

Folding of s with the forms of 4th order (System III). (§ 9.)

Irreducible forms:

$$\begin{array}{ll} (\vartheta\nu x)s_\nu, & ((\vartheta\nu x), s, u)^2, \\ ((\vartheta\nu x)^2, s, u), & ((\vartheta\nu x)^2, s, u)^2,\ \theta s_\nu, \\ (\vartheta\nu x)^2 s_\nu, & ((\vartheta\nu x)^2, s, u)s_\nu, \end{array}$$

$$\begin{array}{ll} ((\vartheta\nu x), s, u)^3,\ (\theta_\nu su)^2, & ((\vartheta\nu x), s, u)^4,\ (\theta_\nu su)^3, \\ ((\vartheta\nu x)^2, s, u)^3,\ (\theta_\nu(\vartheta\nu x), s, u)^2,\ (\theta_\nu^2 su), & ((\vartheta\nu x), s, u)^3,\ (\theta_\nu^2 su)^2,\ (\theta_\nu(\vartheta\nu x), s, u)^4. \end{array}$$

Reductions:

$((\vartheta\nu x), s, u)s_\nu$ decomposes as a transvection over functional determinants, by § 3.

Form sequence $((\vartheta\nu x), s, u)^2 s_\nu$: reducers s_ν^2 and $s_\vartheta^2\theta_\nu$:

$$(\widehat{\vartheta}\nu su)s_\nu(\theta su) = s_\nu s_\vartheta(\theta su) = -\frac{1}{2}(\widehat{\vartheta}\nu su)^2(\theta su),$$

$$\theta_\nu(\widehat{\vartheta}\nu su)s_\nu = \theta_\nu s_\nu s_\vartheta = -\frac{1}{2}\theta_\nu(\widehat{\vartheta}\nu su)^2.$$

Form sequence $\theta_\nu s_\nu(\theta su)$: reducer $a_\nu s_\nu(asu)^2$ by double reduction of the expressions

$$a_\nu s_\nu(asu)^2(abu) \quad \text{and} \quad a_\nu s_\nu(asu)^2(abu)(sbu);$$

$\theta_\nu s_\nu(\theta su)^3$ by double reduction of $(agu)^3(abu)(bgu)^3$.

Form sequence $\theta_\nu(\vartheta\nu x)s_\nu$: reducer $a_\nu^2 s_\nu$ from $\theta_\nu(\vartheta\nu x)s_\nu = a_\nu^2(abu)s_\nu$.

Form sequence $(\theta_\nu(\vartheta\nu x)^2, s, u)$: double reduction of the forms of the sequence $\theta_\nu(\widehat{\vartheta}\nu su)s_\nu$, which at the same time belong to the form sequences $((\vartheta\nu x)su)^2 s_\nu$ and $\theta_\nu(\vartheta\nu x)s_\nu$.

The form $(\theta_\nu(\vartheta\nu x), s, u)$ decomposes as a single transvection over the functional determinant $\theta_\nu(\vartheta\nu x) = a_\nu^2(abu)$.

The form $((\vartheta\nu x)^2, s, u)^4$: double reduction of the expression $(a\Delta u)^2(\Delta su)^4$, or also of $(\theta gu)^4$, from formulas (27.) and analogously to the reduction of $(j\eta x)^4$ (§ 12 C)).

Double reduction gives the formulas:

$$\begin{aligned} 0 &= \theta_\nu s_\nu(\theta su) - \frac{1}{6}(\widehat{\vartheta}\nu su)^2(\theta su) - \frac{1}{12}(Hsu), \\ 0 &= \theta_\nu s(\theta su)^2 - \frac{1}{4}(\widehat{\vartheta}\nu su)^2(\theta su)^2 - \frac{1}{24}(Hsu)^2, \\ 0 &= (\widehat{\vartheta}\nu su)^2(\theta su)^2 + 2\theta_\nu(\widehat{\vartheta}\nu su)(\theta su)^2 + 3\theta_\nu^2(\theta su)^2 - \frac{1}{3}H(su)^2, \\ 0 &= \theta_\nu s_\nu(\theta su)^3 + \frac{1}{2}\theta_\nu(\widehat{\vartheta}\nu su)(\theta su)^3, \\ 0 &= \theta_\nu s_\nu s_\vartheta - \frac{1}{4}s_\eta, \quad 0 = \theta_\nu s_\nu s_\vartheta(\theta su), \quad 0 = \theta_\nu s_\nu s_\vartheta(\theta su)^2. \end{aligned} \tag{28}$$

The last group corresponds to $((\theta_\nu(\vartheta\nu x)^2, s, u)^2, \ldots)$.

Consequences:

1) $\theta(\vartheta\nu x)s_\nu$, $(\widehat{\vartheta}\nu su)(\theta su)s_\nu$, $\theta_\nu(\widehat{\vartheta}\nu su)^2$, $(\widehat{\vartheta}\nu su)^2(\theta su)^2$ are reducers.

2) The forms of 4th order $(5, \lambda)$, $(6, \lambda)$ etc. do not enter into folding with s^2.

3) For the form sequence $(\theta gu)^2$, formulas (27.), with the reductions of this paragraph taken into account, give the following formulas:

$$\begin{aligned} &\text{a)} \quad (\theta gu)^2 = \frac{4}{3}\theta_\nu s_\nu + \frac{2}{3}s_\nu s_\vartheta - \frac{1}{3}s\theta_\nu^2 - \frac{1}{9}sH - \frac{1}{3}f\,a_\nu^2(asu)^2 + \frac{1}{3}a_\nu^2(asu)^2, \\ &\text{b)} \quad (\theta gu)^3 = -\frac{1}{3}(\widehat{\vartheta}\nu su)^2(\theta su) - \theta_\nu(\widehat{\vartheta}\nu su)(\theta su) - \theta_\nu^2(\theta su), \\ &\text{c)} \quad (\theta gu)^4 = 2\theta_\nu^2(\theta su)^2 - \frac{1}{3}(Hsu)^2, \\ &g\vartheta(\theta gu) = (\widehat{\vartheta}\nu su)(\vartheta\nu x)s_\nu - \theta_\nu(\vartheta\nu x)(\widehat{\vartheta}\nu su), \\ &g\vartheta(\theta gu)^2 = \frac{2}{3}s\theta_\nu^3, \quad g\vartheta(\theta gu)^3 = \frac{1}{3}\theta_\nu^3(\theta su), \quad g\vartheta(\theta gu)^4 = 0. \end{aligned} \tag{29}$$

§ 22. Forms of 9th order (System s_{III}).

Folding of s with the forms of 5th order (System III) and the product $\Delta \cdot \nu$ (§ 10).
Irreducible forms:

A) $(k\nu x)^2 s_\nu, ((k\nu x)^3, s, u), (k\nu x)^3 s_\nu, ((k\nu x)^2, s, u)^2, K_\nu s_\nu,$
$((k\nu x)^3, s, u)^2, ((k\nu x)^2, s, u) s_\nu, (K_\nu (k\nu x)^3, s, u),$
$(K_\nu su)^2,\ (K_\nu su)^3,\ (K_\nu su)^4, ((k\nu x)^2, s, u)^3, (K_\nu^2 su)^4,$

B) $(j\nu x) s_\nu, ((j\nu x)^2, s, u), (j\nu x)^3, s, u, ((j\nu x)^2, s, u)^2,$

C) $s_\nu(\Delta su), \quad (\Delta_\nu su),$

D) $(aHu)s_\eta, (a_\eta su), ((aHu), s, u)^2, ((aHu)^2, s, u),$
$(aHu)^2 s_\eta, (a_\eta su)^2, (a_\eta^2 su), ((aHu), s, u)^3, ((aHu)^2, s, u)^2,$
$((aHu), s, u)^4, (a_\eta su)^3, (a_\eta (aHu), s, u)^2, (a_\eta (aHu)^2, s, u), (a_\eta (aHu), s, u)^3.$

Reductions:

A) The reductions follow directly from the reductions of § 21 by single folding. The form sequence $K_\nu s_\nu (Ksu)^2$ has the reducers $a_\nu s_\nu (asu)^2$ and $\theta_\nu s_\nu (\theta su)^2$ as factors. Hence the higher forms $K_\nu^2 (Ksu)^2$, $K_\nu^2 (Ksu)^3$ decompose according to the Ansatz:

$$0 = a_\nu s_\nu (asu)^2 (a\theta u) = K_\nu s_\nu (Ksu)^2 - \theta_\nu s_\nu (\theta su)^2 (a\theta u).$$

B) Form sequence $(j\nu x)^2 s_\nu$: reducer $a_\nu^2 s_\nu$ from

$$(j\nu x)^2 s_\nu = 3 a_\nu^2 s_\nu (a\theta u)^2.$$

$((j\nu x)^3, s, u)^2$: reducer $a_\nu^2 s_\nu$ from

$$((j\nu x)^3, s, u)^2 = -6 a_\nu^2 s_\nu (a\theta u)^2 (\theta su).$$

C) Forms $\Delta_\nu (\Delta su)^2$ and $s_\nu (\Delta su)^2$: reducer g_ν from formula (27.)a.

Form $\Delta_\nu s_\nu$: 1) reducer $a_\nu^2 s_\nu$ from $a_g a_\nu^2 s_\nu = \frac{2}{3} \Delta_\nu s_\nu$; 2) decomposes by formula (27.)a through double reduction of the expression $(\Delta s \widehat{\nu} x)^2$.

Hence the higher form $a_\eta s_\eta$ decomposes.

D) Form sequence $(aHu)^2 s_\eta (asu)$: reducer $a_\nu s_\nu (asu)^2$ by double reduction of the form sequence $[a_\nu s_\nu (asu)^2]_{\nu_1 x^2}$; reduction of $(aHu)^2 s_\eta (asu)^2$ from $a_\nu s_\nu (asu)^2$ and $a_\nu s_\nu (asu)^3$. Hence the reduction of $(a_\eta, s, u)^4$.

Form sequence $a_\eta (aHu) s_\eta$: reducers $a_\nu^2 s_\nu$, $a_\eta^2 s_\eta$ by double reduction of the expression $(\Delta s \widehat{\nu} x)^3$, since $a_\eta^2 s_\eta$ does not arise by folding from the reducible term $a_\nu^2 s_\nu (\nu \nu_1 x)$ of the form $a_\eta (aHu) s_\eta$.

Form sequence $(a_\eta^2 su)^2$: reducer $a_\nu^2 s_\nu$ from

$$a_\nu^2 s_\nu s_{\nu_1} = -\frac{1}{2} (a_\eta^2 (Hsu))^2.$$

The form $(a_\eta (aHu), s, u)$ decomposes as a transvection over a functional determinant.

The reduction formulas obtained are:

$$\begin{aligned}
&\text{a)} \quad 0 = (aHu)^2 (asu) s_\eta - a_\eta (asu)(Hsu)^2 - a_\eta (aHu)(asu)(Hsu), \\
&\text{b)} \quad 0 = (aHu)^2 (asu)^2 s_\eta - a_\eta (aHu)(asu)^2 (Hsu), \\
&\text{c)} \quad 0 = a_\eta (asu)^2 (Hsu)^2, \\
&\qquad 0 = a_\eta s_\eta + \frac{1}{4} a_\nu^2 g - \frac{1}{4} s \cdot a_\eta^2, \\
&\qquad 0 = a_\eta (aHu) s_\eta - \frac{1}{2} a_\eta^2 (Hsu), \quad 0 = a_\eta^2 (Hsu)^2, \ldots, \quad 0 = a_\eta^2 s_\eta.
\end{aligned} \tag{30}$$

Consequences:

1) $(j\nu x)^2 s_\nu$, $\Delta_\nu s_\nu$, $\Delta_\nu (\Delta su)^2$, and consequently $N_\nu (Nsu)^2$, $(aHu)^2 (asu) s_\eta$, $a_\eta (aHu) s_\eta$, $a_\eta^2 (Hsu)^2$, are reducers; $a_\eta s_\eta$ is a decomposable form that passes into decomposable forms under all single and double foldings.

2) The forms of 5th order $(5, \lambda)$, $(6, \lambda)$ etc. do not enter into folding with s^2.

§ 23. Forms of the tenth order (System s_{III}).

Folding of s with the forms of the sixth order (System III) (§ 11).
Irreducible forms:

A) $(\vartheta^2\nu x)^3 s_\nu, ((\vartheta^2\nu x)^3, s, u), ((\vartheta^2\nu x)^3, s, u)^2, (\vartheta^2\nu x)^3 s_\nu s_\vartheta, \theta_\nu(\vartheta^2\nu x)^3 s_\vartheta,$

B) $a_\nu(j\nu x)s_\nu, (a_\nu(j\nu x), s, u)^2, (a_\nu(j\nu x), s, u)^3, (a_\nu(j\nu x), s, u)^4, (a_\nu^2(j\nu x), s, u)^2, (a_\nu^2(j\nu x), s, u)^3,$

C) $((\vartheta\eta x)^2, s, u), ((\vartheta\eta x), s, u)^2, (\theta H u)s_\eta, (\theta_\eta s u), ((\theta H u), s, u)^2, ((\theta H u)^2, s, u),$
$((\theta\eta x), s, u)^3, (\theta_\eta s u)^2, (\theta_\eta^2 s u), ((\theta H u), s, u)^3, ((\theta H u)^2, s, u)^2, (H_\vartheta(\theta H u), s, u)^2,$
$(\theta_\eta(\theta H u), s, u)^2, (\theta_\eta(\theta H u)^2, s, u), ((\theta H u), s, u)^4, (\theta_\eta s u)^4, (\theta_\eta(\theta H u), s, u)^3, (\theta_\eta^2(\theta H u), s, u)^2.$

Reductions: A) The forms $((\vartheta^2\nu x)^3, s, u)^2$, $((\vartheta^2\nu x)^3, s, u)^3$ decompose:

$$(\vartheta\nu x)(\widehat{\vartheta\nu s u})(\widehat{\vartheta\nu s u}) = (\vartheta\nu x)s_\vartheta s_\nu - (\vartheta\nu x)s_\nu s_\vartheta = -2\theta(\vartheta\nu x)s_\nu s_\vartheta + (\vartheta\nu x)s_\vartheta^2.$$

The remaining forms either have reducers as factors or are single foldings with decomposed forms.

B) Reducer $a_\nu^2 s_\nu$ and $(\widehat{j\nu s u})s_\nu$.

C) 1. The form sequence $(\theta H u)^2 s_\eta$: a) reducer $(aHu)^2(asu)s_\eta$; b) double reduction of the form sequences $(\theta g u)^3 g_\nu$ and $g_\nu(agu)^3(bgu)^3(abu)$. Hence reduction of the higher forms $(\theta, s, u)^3$, $(H_\vartheta(\theta H u), s, u)^3$ and $(a_\nu b_{\nu_1}^2, s, u)^4$ [the latter form replacing $(H_\vartheta s u)^4$].

2. $(\vartheta\eta x, s, u)^4$: reducer $(a_\eta s u)^4$.

3. $((\vartheta\eta x)^2, s, u)^3$: reducer $s_\vartheta^2 s_\nu$ from $(\widehat{\vartheta\eta s u})^2(Hsu)$. The forms $(\vartheta\eta x)s_\eta$, $(\vartheta\eta x)^2 s_\eta$, $((\vartheta\eta x)^2, s, u)^2$, $\theta_\eta s_\eta$ decompose by folding with the decomposed form $a_\eta s_\eta$.

4. Form sequences $H_\vartheta(\theta H u)s_\eta$, $\theta_\eta(\theta H u)s_\eta$, $H_\vartheta(\vartheta\eta x)s_\eta$, $\theta_\eta(\vartheta\eta x)s_\eta$: reducers $a_\eta(aHu)s_\eta$ and $a_\eta^2 s_\eta$.

The form sequence $(\theta_\eta(\vartheta\eta x), s, u)^2$: reducer $a_\eta^2(Hsu)^2$ [except for $(\theta_\eta^2(\theta H u), s, u)^2$, which arises from the term $a_\eta^2(\theta s u)(Hsu)$ by folding].

5. $(H_\vartheta(\vartheta\eta x), s, u)$, $(H_\vartheta(\vartheta\eta x), s, u)^2$: double reduction of $a_\eta(aHu)s_\eta(abu)$ and $g_\vartheta(\theta g u)g_\nu$.

The form sequence $(H_\vartheta(\vartheta\eta x), s, u)^3$: reducer $((\vartheta\nu x)^2, s, u)^2 s_\nu$.

6. $(H_\vartheta(\theta H u), s, u)$ decomposes by single folding with the decomposed form $(\theta_\nu(\vartheta\nu x), s, u)$ according to § 3. 2), c), that is, by double reduction of the lower decomposed form $(H_\vartheta s u)^2$:[29]

$$\begin{aligned}
0 &= \theta_\nu(\vartheta\nu_1)(\theta s u) + \frac{2}{3}\theta_\nu^2(\vartheta\nu_1 x)(\theta s u) + \theta_\nu(\vartheta\nu x)(\theta s u)s_{\nu_1} \\
&\equiv -\frac{1}{2}H_\vartheta(\theta H u)(\theta s u) + \theta_\eta(\theta s u)(Hsu) + \frac{1}{2}H_\vartheta(\theta s u)(Hsu) \\
&\equiv -\frac{1}{2}H_\vartheta(\theta H u)(\theta s u) + \theta_\eta(\theta s u)(Hsu) + \frac{1}{2}[H_\vartheta]_{sx}^2 - \frac{1}{2}\frac{3}{5}H_\vartheta(\theta H u)(\theta s u) \\
&\equiv -\frac{3}{5}H_\vartheta(\theta H u)(\theta s u).
\end{aligned}$$

By double reduction, according to 1. and 5., the following formulae arise:

$$\begin{aligned}
0 &= \theta_\eta(\theta s u)(Hsu)^2 + \frac{1}{4}H_\vartheta(\theta H u)(\theta s u)^2 \\
&\quad - \frac{3}{4}\theta_\eta(\theta H u)(\theta s u)(Hsu) - \frac{5}{4}\theta_\eta(\theta H u)^2(\theta s u)^3 \\
&\quad - (asu)^3 b_\eta(\theta H u)^2 - a_\nu(asu)^3 b_\nu^2, \\
0 &= H_\vartheta(\theta H u)(\theta s u)^3 - 7\theta_\eta(\theta H u)(\theta s u)^2(Hsu), \\
0 &= a_\nu(asu)^2 b_{\nu_1}^2(bsu)^2 - \theta_\eta(\theta s u)^2(Hsu)^2 \\
&\quad + 6\theta_\eta(\theta H u)(\theta s u)^2(Hsu), \\
0 &\equiv (H_\vartheta(\vartheta\eta x), s, u), \qquad 0 \equiv H_\vartheta(\widehat{\vartheta\eta s u})(Hsu) - 4\theta_\eta^2(Hsu).
\end{aligned} \tag{31}$$

Consequences:

1. $(\theta H u)^2 s_\eta$, $H_\vartheta(\theta H u)s_\eta$, $\theta_\eta(\theta H u)s_\eta$, $H_\vartheta(\vartheta\eta x)s_\eta$, $\theta_\eta(\vartheta\eta x)s_\eta$, $(H_\vartheta(\vartheta\eta x), s, u)$, $(\theta_\eta(\vartheta\eta x), s, u)^2$ are reducers.

2. s^2 does not enter into folding with the forms $(5, \lambda)$, etc., of order 6.

[29]The sign $\equiv$ means that products of forms of lower degree have been omitted.

§ 24. Forms of the 11th order (System s_{III}).

Folding of s with the forms of the 7th order (System III) (§ 12).
Irreducible forms:

A) $((k\eta x)^3, s, u)$, $(K_\eta(k\eta x)^3, s, u)$, $(K_\eta su)^2$, $((KHu)^2 su)^2$,
$((KHu)^2, s, u)^3$, $((KHu)^2, s, u)^4$, $(K_\eta(KHu)^2, s, u)^3$,

B) $((j\eta x), s, u)^2$, $(H_j su)$, $((j\eta x), s, u)^3$,

C) $(\Delta_\eta su)$,

D) $(aLu)^2 s_\eta$, $(a_\eta su)^2$, $((aLu)^2, s, u)^2$, $((aLu)^3, s, u)^2$, $((aLu)^3, s, u)^3$, $(a_l(aLu)^2, s, u)^2$.

Reductions:

A) Reduction or decomposition by single folding with the corresponding forms in the symbols $\theta - H$.

$(K_\eta(KHu)^2, s, u)$ and $(K_\eta(KHu)^2, s, u)^2$: decomposition according to § 3. 2), a), by folding with $K_\nu^2(Ksu)$, $K_\nu^2(Ksu)^2$.

$(K_\eta(KHu), s, u)^4 \equiv (a_\nu^2\theta_{\nu_1}, s, u)^4$: decomposition according to § 3. 2), b), from the decomposed form $K_\nu^2(Ksu)^3$.

B) Form sequence $(j\eta x)s_\eta$: reducer $(j\nu x)^2 s_\nu$.

Form $H_j(\widehat{j\eta su})(Hsu)$: reducer $(j\nu x)(\widehat{j\nu su})^2$.

Form $H_j(j\eta x)(Hsu)$: decomposes by reducing the decomposed form $((j\eta x)^2, s, u)^2$ by the reducer $(j\nu x)^2 s_\nu$ (formula (18.)a):

$$0 = -2(j\nu x)^2 s_\nu s_{\nu_1} = [(j\eta x)^2]_{su^2} + H_j(j\eta x)(Hsu).$$

C) Reducers $\Delta_\nu s_\nu$ and $\Delta_\nu(\Delta su)^2$.

D) Reduction and decomposition by single folding with the corresponding forms in the symbols $f - H$.

E) From (30.)c one obtains the reduction of the decomposed form sequence $(a_\sigma su)^2$:

$$0 = a_\eta(Hsu)^2(as\widehat{\nu x})^2 = a_\eta(Hsu)^2(a\widehat{\nu\eta}u)^2 + 2a_\eta(a\widehat{\nu\eta}u)(\widehat{\eta\sigma}u)(Hsu)^2 = \frac{1}{3}a_\sigma(asu)^2,$$
$$0 = a_\eta a_\nu(Hsu)^2(as\widehat{\nu x})(asu) = a_\eta(a\widehat{\nu\eta}u)^2(Hsu)^2(asu) + a_\eta(a\widehat{\nu\eta}u)(\widehat{\eta\sigma}u)(Hsu)^2(asu) = \frac{1}{3}a_\sigma(asu)^3.$$

Consequence: s^2 does not enter into folding with the forms of order 7.

§ 25. Forms of the 12th, 13th, 14th, and 15th orders (System s_{III}).

Folding of s with the forms of orders 8, 9, 10, 11 (System III) (§§ 13–16).
Irreducible forms:
12th order.

A) $((\vartheta^2\eta x)^4, s, u)$, $\theta_\eta(\vartheta^2\eta x)^3 s_\vartheta$,

B) $((aHu)H_j, s, u)^2$, $((aH\cdot u)H_j, s, u)^3$,

C) $(\theta_\nu su)^2$, $((\theta Lu)^2, s, u)^2$, $((\theta Lu)^3, s, u)$, $((\theta Lu)^2, s, u)^3$, $(\theta_l(\theta Lu)^2, s, u)^2$.

Reductions: The reductions arising by direct folding with reducers or decomposed forms are no longer to be mentioned.

B) Decomposition of $((aHu)H_j, s, u)$ and $(a_\eta(aHu)H_j, s, u)$ as transvections over functional determinants.

Decomposition of $(a_\eta(aHu)H_j, s, u)^2$: double reduction of the decomposed form $[\theta_\nu\theta_{\nu_1}^3]_{su^3}$ (§ 13. C):

$$\begin{aligned}0 &\equiv [\theta_{\nu_1}^3\theta_\nu]_{su^3} \equiv -\frac{3}{4}\theta_\nu(\theta su)^2(\theta\theta' u)\theta_{\nu_1}^3\\ &\equiv -\frac{9}{8}(\theta\theta'\eta su)(\theta\theta' u)(\theta Hu)(\theta' Hu)\theta'_\nu(\theta su) \equiv -\frac{3}{8}a_\eta(aHu)H_j(asu)^2.\end{aligned}$$

C) From formulae (31.) follows the reduction of the decomposed forms

$$[L_\vartheta(\theta Lu)]_{su^3} \quad \text{and} \quad [\theta_l(\theta Lu)]_{s^2u^3},$$

that is, of the products $[\theta_\nu^2 H]_{su^3}$ and $[a_\nu^2(bHu)^2]_{su^3}$:

$$
\begin{aligned}
&0 \equiv \theta_\nu^2(\theta su)^2(Hsu), 0 \qquad\qquad \equiv [L_\vartheta(\theta Lu)]_{su^4} - 7[\theta_l(\theta Lu)]_{su^4},\\
&0 \equiv a_\nu^2(asu)^2(bHu)^2(bsu)\\
&\quad + 6\theta_l(\theta Lu)^2(\theta su)(Lsu),\\
&0 \equiv \theta_\nu^3(\theta su)(Hsu)^2\\
&\quad \text{from } 0 \equiv \theta_l^2(\theta Lu)(\theta su)(Lsu)^2\\
&\quad \text{(reducer } a_\eta^2(Hsu)^2).
\end{aligned}
\tag{32}
$$

13th order.

$$\text{A) } ((KLu)^3, s, u)^2,\ ((KLu)^3, s, u)^3, \qquad \text{B) } (L_j su)^2, \qquad \text{C) } ((aH^2u)^3, s, u)^2.$$

14th order.

$$\text{A) } ((aLu)^2 L_j, s, u)^2, \qquad \text{B) } ((\theta H^2 u)^3, s, u)^2.$$

15th order.

$$((KH^2u)^4, s, u)^2.$$

Reductions: Decomposition of $(K_l(KLu)^3, s, u)^2$, $(H_\vartheta(\theta H^2u)^3, s, u)$, $((K, H \cdot L, u)^5, s, u)$ according to § 3. 2), c); that is, with simultaneous consideration of the decomposed forms $(K_l(KLu)^2, s, u)^3$, $(H_\vartheta(\theta H'u)^2, s, u)^2$, $((K, H \cdot L, u)^4, s, u)^2$.
Consequence: s^2 does not enter into folding with individual forms of System III.

§ 26. System s_{IV}.

1) *Folding of s with System I–III.*
Reductions: According to the reductions of the last paragraphs ($a_\nu^2 s_\nu (aHu)^2(asu)s_\eta$, etc.), the forms $(0, \lambda)$, $(1, \lambda)$, $(2, \lambda)$ do not permit foldings I and III simultaneously; hence, by § 18, the forms f, Δ, N do not enter in products in folding to form System s_{IV}.

The form j does not enter into folding, since, by the same reductions, the foldings contragredient to j,

$$(K_\nu(k\nu x)^3, s, u), \qquad ((\vartheta^2\eta x)^4, s, u) \quad \text{etc.},$$

correspond to the foldings cogredient to j:

$$K_\nu(k\nu x)^2 s_k, \qquad (\vartheta^2\eta x)^3 s_\vartheta \quad \text{etc.}$$

2) *Folding of s with System II–III*, that is, of s with $H \cdot (1, \lambda)$, $H \cdot (2, \lambda)$, $H \cdot (3, \lambda)$.
Irreducible forms:

$$
\begin{aligned}
\text{A)}\quad & (a_\nu H, s, u)^4, ((aHu)^2H, s, u)^3, ((aHu)^2H, s, u)^4,\\
& ((aLu)^2H, s, u)^4, ((aLu)^3H, s, u)^3, ((aH^2u)^3H, s, u)^4,\\
\text{B)}\quad & (a_\nu^2 H, s, u)^3, (a_\eta(aHu)^2H, s, u)^3, (a_l(aLu)^2H, s, u)^4,\\
\text{C)}\quad & (\theta_\nu H, s, u)^4, ((\theta Hu)^2H, s, u)^3, ((\theta Hu)^2H, s, u)^4,\\
& ((\theta Lu)^2H, s, u)^4, ((\theta Lu)^3H, s, u)^3, ((\theta H^2u)^3H, s, u)^4,\\
\text{D)}\quad & (\theta_\eta(\theta Hu)^2H, s, u)^3, (\theta_l(\theta Lu)^2H, s, u)^4,\\
\text{E)}\quad & ((KLu)^3H, s, u)^4, ((KH^2u)^4H, s, u)^4,\\
\text{F)}\quad & ((j\nu x)^2H, s, u)^3, ((j\nu x)^2H, s, u)^4, ((j\eta x)H, s, u)^4,\\
& (H_jH, s, u)^3, (L_jH, s, u)^4.
\end{aligned}
$$

Reductions: ν does not enter into folding, analogously to j.

B) and D). $(a_\nu^2 H, s, u)^4$ and $(\theta_\nu^2 H, s, u)^4$ decompose by formula (27.)c) and (29.)c), taking into account $(gHu)^2 = -\frac{1}{3}s\sigma$:

$$(a_\nu^2 H, s, u)^4 = \frac{1}{3}(asu)^4\sigma, \qquad (\theta_\nu^2 H, s, u)^4 = -\frac{1}{6}(\theta su)^4\sigma;$$

hence decomposition of $(a_\eta(aHu)H, s, u)^4$, $(\theta_\eta(\theta Hu)H, s, u)^4$.

$(\theta_\nu^2 H, s, u)^3$, $(\theta_\nu^3 H, s, u)^3$, and hence $(\theta_\eta^2(\theta Hu)H, s, u)^4$, decompose according to § 25, forms of order 12, C).

3) *Folding of s with System III–III*, that is, of s with forms of System III:

$$(3,\lambda)(3,\lambda),\quad (3,\lambda)(2,\lambda),\quad (3,\lambda)(1,\lambda),\quad (2,\lambda)(2,\lambda),\quad (2,\lambda)(1,\lambda),\quad (1,\lambda)(1,\lambda).$$

Irreducible forms:

$$\begin{aligned}
&\text{A)}\quad (a_\nu^2 a_\nu^2, s, u)^3, (a_\nu^2 a_\nu^2, s, u)^4, (a^2 a_\eta (aHu)^2, s, u)^3,\\
&\text{B)}\quad (a_\nu H_j, s, u)^4, ((aHu)^2 H_j, s, u)^3, ((aLu)^2 H_j, s, u)^4, ((aLu)^3 H_j, s, u)^2, ((aH^2u)^3 H_j, s, u)^3.
\end{aligned}$$

Reductions:

Products II–III, for the same order in the symbols ν and f, are to be considered higher than products III–III.

The reductions arise partly from relations between products (syzygies), partly by replacing the products by the corresponding decomposed forms and reducing the folding of these forms with s. Products containing contravariants are always omitted, since they pass into products.

A) f quadratic, symbols ν of arbitrary order. By § 11, formulae (12.), and by analogous computation, one has:

$$\begin{aligned}
&1)\quad 0 = a_\nu b_{\nu_1} + \frac{1}{2} f (aHu)^2 - \frac{1}{4}\theta H + \frac{1}{2} u_x H_\vartheta - \frac{1}{4} u_x^2 H_\vartheta^2,\\
&2)\quad 0 = a_\nu b_{\nu_1}^2 + f a_\eta (aHu)^2 - \theta_\eta - \frac{1}{2} H_\vartheta,\\
&3)\quad 0 = a_\nu^2 b_{\nu_1}^2 - H_\vartheta^2 + 2\theta_\eta^2.
\end{aligned}$$

It follows that

$$\begin{aligned}
4)&\quad 0 = a_\nu^2 b_{\nu_1}^2 (j\nu_1 x),\\
5)&\quad L_\vartheta(\theta L u) = -a_\nu b_\eta (bHu)^2 + \frac{3}{4} a_\nu^2 (bHu)^2 + \frac{3}{4}\theta_\nu^2 H,\\
6)&\quad L_\vartheta(\theta L u) = -6 a_\nu b_\eta (bHu)^2 + 2 a_\nu^2 (bHu)^2 - \frac{1}{2}\theta_\nu^2 H,\\
7)&\quad 0 = a_\nu (bHu)^2 + f (aLu)^3 + \theta_\nu H,\\
8)&\quad 0 = a_\nu (bLu)^3 + \frac{3}{4} (\theta H u)^2 H,\\
9)&\quad 0 = (aHu)^2 (bHu)^2 + 2 (\theta H u)^2 H,\\
10)&\quad 0 = a_\eta (aHu)^2 b_\eta (bHu)^2,\\
11)&\quad 0 \equiv [a_\nu^2 b_\eta (bHu)]_{\widehat{su^4}}.
\end{aligned}$$

The reductions of

$$(H_\vartheta s u)^4,\qquad (L_\vartheta(\theta L u), s, u)^3,\qquad (L_\vartheta(\theta L u), s, u)^4$$

(formulae (31.) and (32.)) together with formulae 1) to 11) give the reduction of all foldings of s with products that are quadratic in the symbols f and arbitrary in ν.

B) Products of two of the symbols f, θ, j; ν in arbitrary order. By formulae (17.), (18.), (20.), and by direct computation, the relations are:

$$\begin{aligned}
1)&\quad 0 = a_\nu \theta_{\nu_1} + \frac{1}{4}\theta (aHu)^2 + \frac{1}{4} f (\theta H u)^2,\\
2)&\quad 0 = \theta_\nu \theta_{\nu_1} + \frac{1}{2}\theta (\theta H u)^2.
\end{aligned}$$

Therefore:

$$\begin{aligned}
&3)\quad 0 = 3 a_\nu (\theta H u)^2 + \theta (aLu)^3 + 2 f (\theta L u)^3,\\
&4)\quad 0 = 3\theta_\nu (aHu)^2 + f (\theta L u)^3 + 2\theta (aLu)^3,\\
&5)\quad 0 = a_\nu (\theta L u)^3, \qquad\qquad 0 = \theta_\nu (aLu)^3, \qquad 0 = (aHu)^2 (\theta H u)^2,\\
&6)\quad 0 = a_\nu \theta_\nu^2 + \theta_\nu a_\nu^2 + f\theta_\eta (\theta H u)^2 + \theta a_\eta (aHu)^2,\\
&7)\quad 0 = \theta_\nu \theta_\nu^2 + \theta\theta_\eta (\theta H u)^2 + \frac{1}{6} f H_j,
\end{aligned}$$

8) $0 = a_\nu (j\nu x)^2 + f H_j$,

9) $0 = (aHu)^2 (j\nu x)^2 - 2 a_\nu H_j$,

10) $0 = \theta_\nu (j\nu x)^2$, $\qquad 0 = \theta H_j$,

11) $0 = a_\nu \theta_\nu^3 - \frac{3}{4}(j\eta x)^2$,

12) $0 = a_\nu^2 \theta_\nu^2 - \frac{1}{3}(j\eta x)^2 - \frac{1}{3}(\Delta H u)^2$,

13) $0 = a_\nu^2 \theta_\nu^3 - \frac{1}{2} a_\sigma$,

14) $0 = \theta_\nu \theta_\nu^3$, $\qquad 0 = a_\eta H_j$.

The formulae have been carried far enough for the products to become polarizable: that is, by folding only one new product arises, because a) the folding of the product into itself becomes reducible, and b) the remaining products arising by folding become reducible or lead to contravariants (cf. § 3. 2), a) and b)).

Formulae 1) to 5) give the reduction of all foldings of s with products arising from $a_\nu \theta_\nu$ by folding. Formulae 6) to 10) give the reduction of those products arising from $a_\nu \theta_\nu^2$ by folding. By § 24 A), taking formulae 6) to 10) into account, the foldings of s with products arising from $a_\nu^2 \theta_\nu$ decompose.

One has:

$$0 \equiv a_\nu^2 (asu)^2 \theta_{\nu_1} (\theta su)^2, \qquad 0 \equiv a_\nu^2 (asu) \theta_{\nu_1} (\theta su)^3 - K_\eta (KHu)^2 (Ksu)^3,$$
$$0 \equiv a_\nu^2 (asu)^2 \theta_{\nu_1} (\theta su), \qquad 0 \equiv a_\nu^2 (asu) \theta_{\nu_1} (\theta su)^2,$$
$$0 \equiv a_\nu^2 (asu) \theta_{\nu_1} (\theta su).$$

The reducers

$$((j\eta x)^2, s, u)^3 \text{ [from } (\widehat{\vartheta \eta su})^2 (Hsu),\ \S\ 23.\ C3],$$
$$(H_j (j\eta x), s, u)^2 \ [\S\ 24,\ B], \quad ((\Delta H u)^2, s, u)^2 \ [\S\ 24,\ C], \quad (a_\sigma su)^2 \ [\S\ 24,\ E]$$

together with formulae 11) to 14) give the reduction of all products arising from $a_\nu \theta_{\nu_1}^3$, $a_\nu^2 \theta_{\nu_1}^2$, $a_\nu^2 \theta_{\nu_1}^3$.

Our preceding computations can be summarized in the following *conclusion*:

The relatively complete system modulo (ϱ, t) consists of the following 331 forms and subsystems, which are also reproduced below in tabular order.

Table I (cf. p. 90)

$u \backslash x$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
0	i				f		Δ		$K_\nu(k\nu x)^3$		$K_\eta(k\eta x)^3$				$(\vartheta^3\eta x)^4$	$H_k(k\vartheta, \eta x)^4$		$\theta_l(\vartheta k, lx)^5$				$(\vartheta^2 lx)^6$
1		u_x				$\theta_\nu(\vartheta\nu x)^2$		$\theta_\eta(\vartheta\eta x)^2$	N	$(k\nu x)^3$	$\theta_\nu(\vartheta^2\nu x)^3$	$(k\eta x)^3$	$H_\vartheta(\vartheta^2\eta x)^3$ $\theta_\eta(\vartheta^2\eta x)^3$		$\theta_l(\vartheta^2 lx)^4$		$(\vartheta k, \eta, x)^4$		$(\vartheta k, l, x)^5$			
2			a_ν^2		θ a_η^2		$(\vartheta\nu x)^2$	$K_\nu(k\nu x)^2$	$(\vartheta\eta x)^2$	$H_k(k\eta x)^2$ $K_\eta(k\eta x)^2$		$(\vartheta^2\nu x)^3$ $K_l(klx)^3$		$(\vartheta^2\eta x)^3$		$(\vartheta^2 lx)^4$						
3		θ_ν^3		a_ν	$\theta_\nu(\vartheta\nu x)$	Δ_ν a_η	K $H_\vartheta(\vartheta\eta x)$ $\theta_\eta(\vartheta\eta x)$	Δ_η	$(k\nu x)^2$ $\theta_l(\vartheta lx)^2$		$(k\eta x)^2$		$(klx)^3$									
4	ν		H θ_ν^2	$(j\nu x)^3$ $a_\eta(aHu)$	θ_η^2	$(\vartheta\nu x)$ a_l^2		N_ν $(\vartheta\eta x)$		H_η N_η $(\vartheta lx)^2$												
5		$a_\eta(aHu)^2$	$\theta_\eta^2(\theta Hu)$	θ_ν	$\frac{K_\nu^2}{(aHu)}$	θ_η	K_η^2 (ΔHu) a_l		Δ_l													
6	j σ		$(j\nu x)^2$ $(aHu)^2$	L $a_\nu^2(j\nu x)$ $H_\vartheta(\theta Hu)$ $\theta_\eta(\theta Hu)$	$\frac{K_\nu}{\theta_l^2}$			K_η														
7		$\theta_\eta(\theta Hu)^2$	$H_j(j\eta x)$ $a_l(aLu)^2$		$a_\nu(j\nu x)$ (θHu)		$\Delta_\nu(j\nu x)$ θ_l	K_l^2														
8	H_j^2 $a_l(aLu)^3$	$(\nu\sigma x)$ $(j\nu x)$	$(\theta Hu)^2$	$K_\eta(KHu)^2$ $(j\eta x)$ $(aLu)^2$																		
9		H_j $(aLu)^3$	$a_\eta(aHu)H_j$ $L_\vartheta(\theta Lu)^2$ $\theta_l(\theta Lu)^2$		(KHu)																	
10	$\theta_l(\theta Lu)^3$	L_j^2 $(j\sigma x)$ $a_\eta(aH^2u)^3$		$H_j(aHu)$ $(\theta Lu)^2$																		
11		$(\theta Lu)^3$	$K_l(KLu)^3$ L_j $(aH^3u)^3$																			
12	$(aH^2u)^4$	$a_\eta(aLu)^2L_j$ $H_\vartheta(\theta H^2u)^3$ $\theta_\eta(\theta H^2u)^3$		$(KLu)^3$																		
13	$(\nu\sigma j)$		$(aLu)^2L_j$ $(\theta H^3u)^3$																			
14	$(\theta H^2u)^4$	$K_\eta(KH^2u)^4$ $(a_\eta H\cdot L, u)^4$																				
15	$L_9(\theta, H\cdot L, u)^4$		$(KH^2u)^4$																			
16	$(\theta, H\cdot L, u)^5$																					
17	$K_\eta(K, H\cdot L, u)^5$																					
18		$(K, H\cdot L, u)^5$																				
19																						
20																						
21	$(KH^3u)^6$																					

(The numbers in the uppermost horizontal row denote the degree in the x; the numbers in the first vertical row denote the degree in the u.)

Table II (cf. p. 90)

$u \backslash x$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
0	J				s		s_j^3					s_ν	$(k\nu x)^3 s_\nu$	$(\vartheta^3\nu x)^2 s_\vartheta$ $(\vartheta^2\nu x)^2 s_\vartheta$		$\theta_\eta(\vartheta^2\eta x)^2 s_\eta$	
1							(asu)	s_η	s_l^2 (Δsu)	$s_\eta(Nsu)$ $(\vartheta\nu x)^2 s, u$	$((k\nu x)^3, s, u)_s$ $(K_\nu(k\nu x)^3, s, u)$		$K_\eta(k\eta x)^3, s, u$		$(\vartheta^2\nu x)^2 s_\nu$		$(\vartheta^2\eta x)^4, s, u$
2					$(asu)^2$	$s_\theta(\theta su)$	$(\Delta su)^2/a_\nu s_\nu$	$(\vartheta\nu x)^2, s, u$ $(\theta_l(\vartheta\nu x)^2, s, u)$		$\theta_\eta(\vartheta^2\eta x)^2, s, u$		$(k\eta x)^2 s_\nu$ $((k\nu x)^2, s, u)$		$(k\eta x)^3, s, u$			
3			$(asu)^3$	$s_\vartheta(\theta su)^2$ s_ν	$a_\nu s_\nu(asu)$ $a_l^2(asu)$	s_η	(θsu) $a_l^2 su$		$(Nsu)^2$ $(\vartheta\eta x)s_\nu$ $((\vartheta\nu x)^2, s, u)$	$((k\nu x)^2, s, u)^2$	$(\vartheta^2\eta x)^2, s, u$						
4	$(asu)^4$	$s_\vartheta(\theta su)^3$	$a_\nu^2(asu)^2$	$(\theta_\nu^2 su)$	$(\theta su)^2$	$s_k(Ksu)^3$ $a_\nu(asu)$ $s_\nu(asu)$	$((\vartheta\nu x)^2, s, u)^2$	$s_\nu(\Delta su)$ (Δsu) $(aHu)s_\eta$ $(a_\nu su)$		$(\Delta_\eta su)$							
5			$(\theta su)^3$	$s_k(Ksu)^3, s_j$ $s_\vartheta(\theta s'u)^4$ $s_\nu(asu)^2$ $a_\nu(asu)^2$	(Hsu) $((\vartheta\nu x)^2, s, u)$ $(\theta_\nu(\vartheta\nu x)^2, s, u)$ $(\theta_\nu^2 su)$	$((j\nu x)^2, s, u)$ $(aHu)^2 s_\nu$ $(a_\eta su)^2$	$(Ksu)^2$ s_η $(\theta_\eta^2 su)$		$((k\nu x)^2, s, u)^2$ $K_\nu s_\nu$								
6	$(\theta su)^4$	$s_\nu(asu)^3$ $a_\nu(asu)^3$	$(Hsu)^2$ $(\theta_\nu(\vartheta\nu x), s, u)^3$ $(\theta_\nu^2 su)^2$	$a_\eta s_\nu^2$ $(a_\eta(aHu), s, u)^2$ $(a_\eta(aH^2u), s, u)$	$(Ksu)^3$	$s_\eta(asu)$ $((\vartheta\nu x), s, u)^3$ $(\theta_\eta su)$	$((k\nu x)^2, s, u)^3$	$s_\nu(\Delta su)$ $a_\nu(j\nu x)s_\nu$ $(\theta Hu)s_\eta$ $(\theta_\eta su)$									
7	$\theta_\nu(\vartheta\nu x), s, u)^4$	$(a_\eta(aHu), s, u)^3$	$(Ksu)^4$ $(\theta_\eta^2(\theta Hu), s, u)^2$ $(a_\eta^2 - a_l^2, s, u)^3$	$s_j(asu)^3$ $s_k(K^2u)^3$ $((j\nu x), s, u)^2$ $(\theta_\eta su)^2$	$((j\nu x), s, u)$ $((j\eta x), s, u)$ $(aHu), s, u)$ $(aH^2u), s, u)$	$((\vartheta\eta x), s, u)^3$	$(aLu)^3 s_l/(asu)^3$										
8	$(a^2 - a_l^2, s, u)^4$	$s_j(asu)^3$ $s_k(K^2u)^3$ $((j\nu x), s, u)^4$ $(\theta_\eta su)^3$	$((j\nu x)^2, s, u)^2$ $(aHu), s, u)^3$ $((aHu)^2, s, u)^3$	$(Lsu)^2$ $(a_l^2(j\nu x), s, u)^3$ $H_\vartheta(\theta Hu), s, u)^3$ $\theta_l(\theta Hu), s, u)^3$	$(asu)^3$	$(K, su)^3$		$(K_\vartheta su)^2$									
9	$K_j^2 su^4$ $((aHu), s, u)^4$	$(Lsu)^4$ $(a_l^2(j\nu x), s, u)^4$ $(\theta_\nu(\theta Hu), s, u)^2$	$(K_\nu^3 u)^6$ $(a_\eta(aLu)^2, s, u)^3$ $(a_l^2 H, s, u)^3$	$(K, su)^3$	$(a_\nu(j\nu x), s, u)^3$ $(\theta Hu), s, u$ $(\theta Hu)^2, s, u$		$(\theta_l su)^3$										
10			$(K, su)^4$ $(a_l^2 a_\eta(aHu)^2, s, u)^3$	$((j\eta x), s, u)^3$ (H_j, su) $((aLu)^2, s, u)^2$ $((aLu)^3, s, u)$													
11	$(a_\nu(j\nu x), s, u)^4$ $((\theta Hu), s, u)^4$	$(K_l(KHu)^2, s, u)^3$ $((j\eta x), s, u)^3$ $((aLu)^2, s, u)^4$ $(a_\eta H, s, u)^3$	$(H^2 su)^3$ $\theta_l(\theta Lu)^2, s, u)^3$		$((KHu)^2, s, u)^3$												
12		$(a_\eta(aHu)^2 H, s, u)^3$	$((KHu)^3, s, u)^3$	$((aHu)H_j, s, u)^2$ $((\theta Lu)^2, s, u)^3$													
13	$((KHu)^3, s, u)^4$	$((aHu)H_j, s, u)^3$ $((\theta Lu)^2, s, u)^4$ $(\nu \cdot H, s, u)^3$	$(L_j su)^2$ $((aH^3u)^3, s, u)^3$ $((j\eta x)H, s, u)^3$ $((aHu)^2 H, s, u)^3$														
14	$((j\nu x)^2 H, s, u)^4$ $((aHu)^2 H, s, u)^4$	$(\theta_\eta(\theta Hu)^2 H, s, u)^3$		$((KLu)^3, s, u)^2$													
15	$(a_l(aLu)^3 H, s, u)^4$	$((KLu)^3, s, u)^3$	$((aLu)^2 L_j, s, u)^2$ $((\theta Hu)^3 H, s, u)^3$ $((\theta Hu)^2 H, s, u)^2$														
16	$((\theta Hu)^2 H, s, u)^4$ $(\sigma \cdot H_j, s, u)^4$	$((j\eta x)H, s, u)^4$ $(H_j H, s, u)^3$ $((aLu)^2 H, s, u)^4$ $((aLu)^3 H, s, u)^3$															
17	$(\theta(\theta Lu)^2 H, s, u)^4$		$((KH^3u)^3, s, u)^3$														
18		$((\theta Lu)^2 H, s, u)^4$ $((\theta Lu)^3 H, s, u)^3$ $(aHu)^2 H, s, u)^4$															
19	$((aH^3u)^3 H, s, u)^4$ $(L_j H, s, u)^5$																
20		$((KLu)^3 H, s, u)^5$	$((aLu)^3 H, s, u)^5$														
21	$((\theta H^3u)^3 H, s, u)^4$ $((aLu)^2 H_j, s, u)^4$																
22																	
23	$((KH^3u)^4 H, s, u)^4$	$((aH^3u)^4 H, s, u)^3$															

(The numbers in the uppermost horizontal row denote the degree in the x; the numbers in the first vertical row denote the degree in the u.)

3. On the Invariant Theory of Forms in n Variables[30]

Jahresbericht der DMV 19 (1910), pp. 101–104

In the projective invariant theory of forms in n variables the main problem has been solved: the finiteness of the system of forms has been proved. A series of further questions, however, have not been treated, namely those concerned with methods for investigating the connection among forms. I would like briefly to communicate the results that I have obtained with respect to such questions, but for the sake of clarity I shall first sketch the questions in the ternary domain, where they are known.

I have first in mind the two fundamental theorems of the symbolic method: the theorem that all invariant formations can be represented symbolically, and the second theorem that all relations existing among invariants are obtained by successive application of a small number of symbolic identities; thus the symbolic method gives all relations without leaving the domain of invariants.[31] Built upon this are the processes for generating the forms, the symbolic folding process and the nonsymbolic differentiation processes, the theory of series expansions, and the like. In this connection I should also mention a theorem proved in my dissertation,[32] namely that all foldings can be generated by successive application of the two possible "cogredient" foldings; by this, for a symbolic product $s_x t_x u_\sigma u_\tau$, I mean the two foldings (stu) and $(\sigma\tau x)$. On this theorem one can build the theory of reducers and of the reduction of systems of forms, in analogy with the binary domain, as I showed in my dissertation.

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[30] Lecture delivered at the Salzburg meeting of natural scientists, 1909.

[31] Study: *Methoden zur Theorie der ternären Formen.* II. § 6. (Leipzig, Teubner, 1889.)

[32] *Über die Bildung des Formensystems der ternären biquadratischen Form.* § 1. (Crelle's Journal Bd. 134. 1908.)

When we now pass to the n-ary domain, even the first theorem of the symbolic method holds only in a conditional measure. It is well known that already for quaternary forms there occur, besides the point and plane coordinates x and u, line coordinates p_{ik}, the subdeterminants from two rows of point coordinates. Analogously, in the n-ary domain we have rows of variables p_ρ, whose individual elements are the subdeterminants from ρ rows x, and rows of symbols S^ρ, whose elements behave like subdeterminants from ρ rows s cogredient to the u. The symbolic representation by determinants is known to hold only when the symbol rows S^ρ are resolved into the rows s and these are distributed among different determinants; only such aggregates of determinants as depend on the S^ρ then have real meaning. But which aggregates of determinants accomplish this cannot be surveyed, and thereby all the other theorems mentioned in the ternary domain are lost.

I would now like to state a simple principle by which one obtains a symbolic representation explicit in the rows S^ρ, and thereby also recovers the remaining theorems. This is an application of the familiar theorem on corresponding matrices: "To each row of variables p_ρ there corresponds one-to-one another row $q_{n-\rho}$ whose individual elements are subdeterminants from $n-\rho$ rows u connected with the ρ rows x by equations $(u \mid x) = 0$." Correspondingly, to each symbol row S^ρ there is assigned another $S_{n-\rho}$, behaving as if it were composed from $n-\rho$ rows cogredient to the x.

The theorem also permits a second formulation, suited for applications: "To every matrix product $(v^{(1)} \cdots v^{(\rho)} \mid y^{(1)} \cdots y^{(\rho)})$,[33] where the rows of the second matrix are contragredient to those of the first, there is assigned a determinant; and conversely each determinant can be transformed into a matrix product with a prescribed number ρ of rows." Thus determinant and matrix product appear as fully equivalent, and the first theorem of the symbolic method assumes the form: "All invariant formations can be represented symbolically by matrix products." From two symbol rows S^σ, T^τ one can now, with the aid of rows of variables, very easily form a matrix product explicitly containing these symbol rows and hence representing an invariant formation of the form $(S^\sigma \mid p_\sigma)(T^\tau \mid p_\tau)$, namely the following:

$$(q_{n-\sigma-\lambda} S^\sigma \,|\, T_{n-\tau} p_{\tau-\lambda}), \qquad (\lambda = 1, 2, \ldots, \tau \text{ or } n-\sigma). \tag{1}$$

More important is the converse: that all invariant formations can be represented explicitly by matrix products, so that the above process is the extension of the binary and ternary folding process. To prove this converse it is necessary to transfer the second theorem of the symbolic method to the n-ary domain, namely to set up the totality of the independent indecomposable identities in which all rows S^ρ, p_ρ are regarded as indecomposable. These identities arise from the known ones containing only rows x and u,[34] by replacing the product theorem for determinants by a system of product theorems for matrices, and by contracting different matrix products into a single one by means of precisely these product theorems. One obtains in this way two dualistic formulas, of which only one is given:

$$\begin{gathered}(S_1^{\rho_1} S_2^{\rho_2} \cdots S_k^{\rho_k} \,|\, x p_{\rho-1}) = \sum_{i=1}^{k} \varepsilon\, (S_1^{\rho_1} \cdots S_{i-1}^{\rho_{i-1}} S_{i+1}^{\rho_{i+1}} \cdots S_k^{\rho_k} q_{n-\rho_i+1} \,|\, S_{n-\rho_i} x), \\ \rho = \sum_{i=1}^{k} \rho_i < n, \qquad \varepsilon = \pm 1.\end{gathered} \tag{2}$$

The only specialization contained in these formulas, namely that a row x enters, cannot be avoided; for completely general symbol and variable rows we arrive at a "decomposition identity," an identity in which a row p_ρ is decomposed into the individual rows x. The simplest special case of the decomposition identity was already used by Clebsch in his "fundamental problem of invariant theory" to derive the elementary covariants in the theory of series expansions. With the aid of these identities, which mediate the transition from the non-explicit determinant expressions to the matrix product, one can in fact show that the desired explicit symbolic representation is obtained with the matrix representation.

For the folding process, however, a theorem entirely analogous to that in the ternary domain holds, and the reduction theorems can likewise be built upon it analogously. If in (1) one denotes the number λ as the defect of the folding, and calls foldings for which $\sigma = \tau$ cogredient, then all foldings can be generated by successive application of the cogredient foldings of defect 1. The proof of this theorem rests essentially on the fact that the most general forms can be replaced by "normal forms," that is, by forms that vanish identically under the application of all invariant differential operations. In the quaternary domain this latter fact was proved by Mertens;[35] our knowledge of the most general differential operations–which, as is well

[33] That is, the sum over the products of corresponding determinants, formed from the matrix of the ρ rows v and that of the ρ rows y.

[34] E. Pascal in *Memorie d. R. Acc. d. Lincei.* (4) 5 (1888), p. 375.

[35] Über invariante Gebilde quaternärer Formen. Wiener Berichte. Bd. 98. 1889.

known, arise from the folding processes by replacing the symbol rows by differential symbols–allows us, using the decomposition identity, to transfer Mertens's proof almost verbatim to the n-ary domain.

4. On the Invariant Theory of Forms in n Variables

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Introduction.

In the projective invariant theory of forms in n variables, the questions adjoining the main problem–the proof of finiteness of the system of forms–have scarcely been treated when one goes beyond the binary and ternary domains. These are the questions concerning the mutual dependence of the forms and the methods for controlling this connection among forms. These methods rest essentially–as has been completely proved in the binary and ternary domains–on two theorems designated by Study as the "fundamental theorems of the symbolic method," namely the theorem on the symbolic representability of the forms and the theorem on symbolic identities.[36] The latter theorem says that all relations existing among integral invariant formations are obtained by successive application of a small number of symbolic identities; thus the symbolic method, and only it, gives knowledge of the connection among forms without leaving the domain of invariants.

Gordan, in his Erlangen program,[37] already pointed to the reason for the difficulty in passing to n variables: it lies in the fact that the first theorem of the symbolic method is no longer valid in its general form. In the n-ary domain there occur, as is well known, variable rows of higher level p_ρ[38] and analogously symbol rows of higher level S^ρ; one arrives at these rows both when, as originally with Clebsch,[39] one starts from forms with arbitrarily many rows p_ρ, and when one starts, following F. Deruyts and A. Capelli,[40] from forms containing only arbitrarily many cogredient rows x. A symbolic representation that would contain the rows p_ρ, S^ρ explicitly[41] does not exist; it is necessary to resolve the p_ρ, S^ρ into the individual rows x and the s contragredient to them, and to distribute these among different determinants. It is indeed possible, by means of differential conditions (cf. formula (19.)), to verify that a given aggregate of determinants has the property of depending only on the rows p_ρ, S^ρ; but in this way one obtains no overview of the totality of invariant formations and of the processes for their generation.[42]

We shall now show in §§ 2 and 6 how, using the "theorem on corresponding matrices,"[43] the first fundamental theorem is recovered in its generality by replacing representation by determinants with representation by "matrix products." In § 2 it is shown that every matrix product explicitly containing the rows S^ρ, p_ρ is an invariant formation of the ground form; the converse given in § 6, that every invariant formation can be represented explicitly by matrix products, succeeds only through the transfer of the second fundamental theorem (§§ 3–5).

This theorem is known for forms that contain only variable rows x.[44] The general problem, to set up the totality of the indecomposable identities in which all rows S^ρ, p_ρ are regarded as indecomposable, was formulated by Study;[45] at the same time he sees in the number of identities that occur a measure of the difficulty of the methods in the n-ary domain. Since it is now possible to combine the totality of the identities in a single formula (36.), this difficulty is at least pressed down to a minimum. In applications, however, certain distinguished special cases of (36.) will often occur, such as (20.), (21.), characterized by the fact that they admit explicit representation without substitution of new rows; or formula (31.), designated as the "decomposition identity," since a row p_ρ appears to be decomposed into the rows y.

Upon the two fundamental theorems the further theory can now easily be built. The first theorem gives directly the processes for generating the forms, the symbolic folding process and the nonsymbolic differentiation processes,[46] while the decomposition identity eliminates, among these processes, all equivalent

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[36]E. Study, *Methoden zur Theorie der ternären Formen* (Leipzig, Teubner, 1889). II. § 6. For the occurrence of these fundamental theorems also in the invariant theory of special transformation groups, cf. Study, *Das Apollonische Problem* (Math. Ann. Bd. 49 [1897]) and *Zur Differentialgeometrie der analytischen Kurven* (Transactions of the American Math. Society, Bd. 1 [1909]).

[37]P. Gordan, *Über das Formensystem binärer Formen.* S. 46. (Leipzig, Teubner, 1875).

[38]For the notation see § 1.

[39]A. Clebsch, über eine Fundamentalaufgabe der Invariantentheorie. (Abh. der Ges. d. Wiss. zu Göttingen, Bd. XVII, 1872).

[40]Cf. A. Capelli, *Lezioni sulla teoria delle forme algebriche* (Napoli 1902), § 22–24, and the literature cited there.

[41]More precise definition of "explicit representation" in § 2.

[42]A computationally very valuable development of the symbolism due to Study (communicated by F. Engel, Ber. d. K. Sächs. Ges. d. Wiss., math.-phys. Kl., 1900, p. 126, and for the quaternary domain by E. Study, *Geometrie der Dynamen*, p. 135) is also not an explicit representation in our sense. It would, for example, hardly lead to the nonsymbolic differentiation processes for generating the forms.

[43]Apart from Grassmann's *Ausdehnungslehre* of 1862, no. 112, first in A. Brill, über zwei Berührungsprobleme, § 2 (Math. Ann. Bd. 4. 1871).

[44]E. Pascal, Giorn. di mat. 26 (1888), pp. 33, 102; Rendic. d. Accad. d. Linc. (4) 4 (1888), p. 119; ib. Mem. (4) 5 (1888), p. 375.

[45]"Methoden ... ", p. 204, note 17).

[46]In the paper "Invariante Gebilde von Nullsystemen" (Sitzungsber. d. Ak. d. Wiss. Wien, Bd. 97, Abt. IIa, 1888), Mertens arrived by another path, not directly generalizable, at the quaternary folding process. The alternating invariant of 6 rows S^2 occurring there differs from the one arising with us by a single application of substitution (11.) by an aggregate of even invariants. For a special case the quaternary folding process is also found

ones (§ 6). Knowledge of the differentiation processes further yields the transfer of the concept of "normal form" to the n-ary domain and, by means of a proof procedure given by Mertens[47] in the quaternary domain, the theorem that a system of invariant formations can be replaced by an equivalent system of normal forms (§ 7). Under this latter assumption I showed in my dissertation[48] that the ternary folding process can be generated by successive application of two fundamental foldings. On the basis of this theorem and the theory of series expansions, I then developed, by introducing the concept of the "form row," the principal reduction theorems for systems of forms. Entirely analogously, in the n-ary domain the folding process can be generated from $n-1$ fundamental foldings (§ 8), and the reduction theorems can be set up (§ 9).

Afterword. On the occasion of the Salzburg communication, Prof. E. Müller of Vienna drew my attention to the fact that the calculation with matrix products underlying this paper is, in principle, identical with the calculus of Grassmann's theory of extension.[49] Indeed, Grassmann's "combinatorial product" $[S^{\rho_1}S^{\rho_2}\cdots S^{\rho_i}]$ can be regarded as a matrix given by these rows (cf. § 2), specifically the "progressive product" as the matrix of the rows themselves, the "regressive" as the matrix of the assigned rows $S_{n-\rho_1}, S_{n-\rho_2}, \ldots, S_{n-\rho_i}$, while the matrix corresponding to the "mixed product" arises when several rows S are combined by substitution (9.) into new rows P, Q. Our "assigned row" corresponds to Grassmann's "complement," our "matrix product" to a "mixed product of level number 0."

Under this correspondence, the development given in the *Ausdehnungslehre* of 1862, no. 113, becomes identical with a formula dualistic to our formula (32.). In the special case $\rho = 1$, (32.) goes over into (17.), the dualistic formula into (16.). From these special cases of Grassmann's development, Müller[50] has recently derived the identities (20.), (21.) mentioned above.

In contrast to the Grassmann–Müller derivation, our derivation at the same time gives the proof of completeness of the identities, which appears to be the essential point for the questions considered here, and it proceeds from (20.), (21.) further to the most general indecomposable identities (36.), which do not seem to have been noticed before now.

§ 1. Notations and definitions. Summary of known results.

We denote, as usual, by the variable row x the row of elements $x_1, x_2, \ldots, x_n$, and analogously by the variable row p_ρ $(\rho = 1, 2, \ldots, n-1)$ the row of the $\binom{n}{\rho}$ elements $p_{i_1 i_2 \ldots i_\rho}$, where $p_{i_1 i_2 \ldots i_\rho}$ denotes the $\binom{n}{\rho}$ determinants formed from the matrix of the ρ rows $x^{(1)}, x^{(2)}, \ldots, x^{(\rho)}$,

$$\begin{vmatrix} x^{(1)}_{i_1} & x^{(1)}_{i_2} & \cdots & x^{(1)}_{i_\rho} \\ x^{(2)}_{i_1} & x^{(2)}_{i_2} & \cdots & x^{(2)}_{i_\rho} \\ \vdots & \vdots & & \vdots \\ x^{(\rho)}_{i_1} & x^{(\rho)}_{i_2} & \cdots & x^{(\rho)}_{i_\rho} \end{vmatrix}, \qquad (i_1 < i_2 < \cdots < i_\rho = 1, 2, \ldots, n).$$

According to the theorem on corresponding matrices,[51] to every row p_ρ there is assigned one-to-one a second row $q_{n-\rho}$ by the relation, valid for each element of the row,

$$p_{i_1 i_2 \ldots i_\rho} = (-1)^{\frac{\rho(\rho+1)}{2} + i_1 + i_2 + \cdots + i_\rho} q_{j_1 j_2 \ldots j_{n-\rho}}, \tag{1}$$

where the i and j are each in good order and together form a complete permutation of the numbers 1 through n. The row $q_{n-\rho}$ consists of the $\binom{n}{\rho}$ determinants of degree $n-\rho$, $q_{j_1 j_2 \ldots j_{n-\rho}}$, formed from the matrix of $n-\rho$ rows $u^{(1)}, u^{(2)}, \ldots, u^{(n-\rho)}$ contragredient to the x. These rows satisfy the $\rho(n-\rho)$ condition equations

$$(u^{(k)} \mid x^{(l)}) = \sum_{\alpha=1}^{n} u^{(k)}_\alpha x^{(l)}_\alpha = 0; \qquad (k = 1, 2, \ldots, n-\rho;\ l = 1, 2, \ldots, \rho). \tag{2}$$

in P. Gordan, *Das simultane System von zwei quadratischen quaternären Formen* (Math. Ann. Bd. 56. 1903).

[47] F. Mertens, Über invariante Gebilde quaternärer Formen. (Sitzber. d. Ak. d. Wiss. Wien. Bd. 98. Abt. IIa. 1889.)

[48] Über die Bildung des Formensystems der ternären biquadratischen Form. (Dieses Journal, Bd. 134. 1908.)

[49] Hermann Graßmanns gesammelte mathematische und physikalische Werke. Ersten Bandes zweiter Teil: Die Ausdehnungslehre von 1862. In Gemeinschaft mit H. Graßmann d. Jüngeren herausgegeben v. Fr. Engel (Leipzig, Teubner 1896).

[50] E. Müller, Beiträge zur Graßmannschen Ausdehnungslehre. 1. Mitteilung: Einige allgemeine Sätze. (Sitzungsber. d. Ak. d. Wiss. Wien; Bd. 118. Abt. IIa. Juli 1909), Satz 16 und 18.

[51] Cf. for example Gordan–Kerschensteiner, *Vorlesungen über Invariantentheorie*, Bd. I, p. 99.

We shall assume them normalized (with Clebsch[52]) so that the proportionality factor otherwise entering in (1.) is set equal to one.

According to Clebsch[53] (and likewise according to the later investigations of F. Deruyts and A. Capelli, cf. the Introduction), the most general forms can be reduced to those that contain at most one of each of the $n-1$ rows p_ρ, and hence can be represented symbolically as follows:

$$(S^1 \mid p_1)^{m_1}(S^2 \mid p_2)^{m_2} \cdots (S^{n-1} \mid p_{n-1})^{m_{n-1}}, \tag{3}$$

where, in analogy with (2.), we have set

$$(S^\rho \mid p_\rho) = \sum S^{i_1 i_2 \dots i_\rho} p_{i_1 i_2 \dots i_\rho}.$$

In consequence of the nonlinear identical relations existing between the elements of a row p_ρ, the symbol rows S^ρ can be normalized so that they satisfy the very same identities, hence so that they behave like the determinants of a ρ-rowed matrix whose rows $s^{(1)}, s^{(2)}, \dots, s^{(\rho)}$ are contragredient to the x.[54] Thus to each row S^ρ there can be assigned one-to-one another $S_{n-\rho}$ through the relation

$$S_{i_1 i_2 \dots i_{n-\rho}} = (-1)^{\frac{(n-\rho)(n-\rho+1)}{2} + i_1 + i_2 + \cdots + i_{n-\rho}} S^{j_1 j_2 \dots j_\rho}, \tag{4}$$

where again the i and j are each in good order and together exactly exhaust the numbers 1 through n. The elements $S_{i_1 i_2 \dots i_{n-\rho}}$ of the row $S_{n-\rho}$ behave like the determinants formed from $(n-\rho)$ rows $\sigma^{(1)}, \sigma^{(2)}, \dots, \sigma^{(n-\rho)}$ cogredient to the x, and the rows s and σ are linked by the $\rho(n-\rho)$ conditions

$$(s^{(k)} \mid \sigma^{(l)}) = 0, \qquad (k = 1, 2, \dots, \rho;\ l = 1, 2, \dots, n-\rho). \tag{5}$$

In consequence of the relations (1.) and (4.), each factor of (3.) is capable of the equivalent fourfold symbolic representation

$$(S^\rho \mid p_\rho) = (S^\rho q_{n-\rho}) = (S_{n-\rho} p_\rho) = (-1)^{\rho(n-\rho)}(q_{n-\rho} \mid S_{n-\rho}), \tag{6}$$

where, as always in what follows, $(S^\rho q_{n-\rho})$ and $(S_{n-\rho} p_\rho)$ are abbreviations for the determinants $(s^{(1)} \cdots s^{(\rho)} u^{(1)} \cdots u^{(n-\rho)})$ and $(\sigma^{(1)} \cdots \sigma^{(n-\rho)} x^{(1)} \cdots x^{(\rho)})$; by Laplace expansion these prove to depend only on the rows $S^\rho, q_{n-\rho}$ and respectively $S_{n-\rho}, p_\rho$, as the above notation is meant to indicate. The expressions $(S^\rho \mid p_\rho)$ and $(q_{n-\rho} \mid S_{n-\rho})$ mean, according to the preceding discussion, the products of the matrices of the rows s and x, respectively of the rows u and σ, where by matrix product is meant the sum over the products of corresponding determinants, that is, the value of the determinant of the product matrix.

The characteristic number $\rho(n-\rho)$ belonging to a symbol row (respectively variable row) will be called, following Clebsch, the weight of the row;[55] the number ρ, which gives the number of rows s, the upper weight index; the number $n-\rho$, which gives the number of rows σ, the lower weight index. It follows from the relations (4.) that a symbol row can occur in one and the same relation once with upper and once with lower weight index; the same holds for the occurrence of the corresponding rows p_ρ and $q_{n-\rho}$. It remains to note that we generally denote variable rows of higher level whose associated matrix consists of rows x by p, those whose matrix consists of rows u by q; symbol rows cogredient to the x by small Greek letters $\sigma, \tau, \alpha, \beta$; symbol rows cogredient to the u by small Latin letters s, t, a, b; all other symbol rows by capital Latin letters with weight index: S^ρ, T^τ, A^ρ or $S_{n-\sigma}$, $T_{n-\tau}$, $A_{n-\rho}$. Corresponding to an upper or lower weight index, we also write the individual elements of a row with upper or lower indices, as for instance in (4.).

§ 2. Representation by matrix products.

In what follows we always understand by "matrix product" the sum over the products of corresponding determinants of two matrices with the same number of rows, where the rows of the second matrix are contragredient to those of the first. The rows of each matrix may be: 1. individually given; 2. combined, as in (6.), into a row S^ρ; 3. united into different rows $S^\rho, S^\sigma \dots$. The matrix product is to be denoted by the symbol (|) used in (6.). The representation (6.) is a special application of a second formulation of the theorem on corresponding matrices: "To every ρ-rowed matrix product $(v^{(1)} v^{(2)} \cdots v^{(\rho)} \mid y^{(1)} \cdots y^{(\rho)})$ there is

[52] a. a. O. § 5.
[53] a. a. O. § 12.
[54] Clebsch, a. a. O. § 4.
[55] a. a. O. § 11.

assigned a determinant; and conversely to every determinant there can be assigned a matrix product with prescribed number ρ of rows ($\rho = 1, 2, \ldots, n-1$)."[56] Indeed, one has only to combine the rows y into a row p_ρ (respectively the rows v into a row q_ρ) and to carry out the substitutions given by (1.) in order to pass from a given matrix product to the determinant and conversely. The value of the determinant is, however, not given by its individual elements, but first results from Laplace expansion. Since determinant and matrix product therefore prove to be equivalent, the theorem on the symbolic representability of the forms (the first fundamental theorem) is expressed also as follows:

Theorem I. "All invariant formations can be represented symbolically by matrix products, and conversely all matrix products are invariant formations."[57]

Representation by matrix products allows us to overcome the essential difficulty of the non-explicit representation caused by the determinant representation.[58] By "explicit representation" we mean a representation into which only the rows $S^\rho, T^\tau, \ldots$ themselves enter, or rows R^ρ uniquely derivable from these, but in which the individual component rows $s, t, \ldots$ no longer occur. We shall obtain in § 6, for all invariant formations that depend only on the symbol rows $S^\rho, T^\tau, \ldots$, a representation explicit in these symbol rows, and thereby the transfer to the n-ary domain of the generating processes, the folding process and the differentiation processes. The possibility of explicit representation is evident from the fact that only the simplest matrix products can be assigned precisely to the determinants occurring in the determinant representation, so that all other matrix products are combinations of different determinants into a single expression. For if one resolves the symbol and variable rows into the individual rows s, u, x, then, according to the first fundamental theorem in its usual form, an aggregate of determinants must necessarily arise. The proof that, conversely, all determinant aggregates that represent invariant formations of prescribed ground forms can be combined into explicit matrix products follows by using the second fundamental theorem (§ 6).

In order to obtain from two symbol rows S^σ, T^τ, with the help of variable rows, an invariant formation of the ground form $(S^\sigma \mid p_\sigma)(T^\tau \mid p_\tau)$ that explicitly contains all rows (the folding process), we need only form a matrix product of the third kind according to the definition given at the beginning of the paragraph:

$$(q_{n-\sigma-\lambda} S^\sigma \,|\, T_{n-\tau} p_{\tau-\lambda}), \qquad (\lambda = 1, 2, \ldots, \tau, n-\sigma). \tag{7}$$

Expanded, (7.) gives:

$$\begin{aligned}(v^{(1)} \cdots v^{(n-\sigma-\lambda)} s^{(1)} \cdots s^{(\sigma)} \,|\, \alpha^{(1)} \cdots \alpha^{(n-\tau)} y^{(1)} \cdots y^{(\tau-\lambda)}) \\ = \sum_{i,k,l} \varepsilon\, q_{k_1 \ldots k_{n-\sigma-\lambda}} S^{k_{n-\sigma-\lambda+1} \ldots k_{n-\lambda}} T_{l_1 \ldots l_{n-\tau}} p^{l_{n-\tau+1} \ldots l_{n-\lambda}},\end{aligned} \tag{8}$$

that is, an expression depending only on the rows S, T, q, p. Here $\varepsilon = \pm 1$[59] and the sum is to be understood in such a way that the indices of each row $S \ldots$ are in good order, and that for each combination $i_1, i_2, \ldots, i_{n-\lambda}$ the k as well as the l individually run through all then possible permutations of the numbers i. The number λ entering into (7.) will be called the defect of the folding in the case $\sigma > \tau$; the reason for this latter restriction appears in § 6.[60]

The determinants of the two matrices entering in (7.) can be regarded as elements of new rows $Q^{n-\lambda}, P_{n-\lambda}$ whose associated rows are Q_λ, P^λ. According to (8.), these rows Q, P can be expressed through the original S and q, respectively T and p. We indicate this by the equations

$$\begin{aligned} q_{n-\sigma-\lambda} S^\sigma = Q^{n-\lambda} \sim Q_\lambda, &\qquad \text{or also} \quad Q_\lambda = [q_{n-\sigma-\lambda} S^\sigma]_\lambda, \\ T_{n-\tau} p_{\tau-\lambda} = P_{n-\lambda} \sim P^\lambda, &\qquad \text{or also} \quad P^\lambda = [T_{n-\tau} p_{\tau-\lambda}]^\lambda. \end{aligned} \tag{9}$$

Taking (4.) into account, one then has, analogously to (6.), the equivalent fourfold symbolic representation for the matrix product (7.):

$$(q_{n-\sigma-\lambda} S^\sigma \,|\, T_{n-\tau} p_{\tau-\lambda}) = (q_{n-\sigma-\lambda} S^\sigma P^\lambda) = (Q_\lambda T_{n-\tau} p_{\tau-\lambda}) = (-1)^{\lambda(n-\lambda)} (P^\lambda \,|\, Q_\lambda). \tag{10}$$

A further substitution of new rows can be performed on (7.) by setting

$$(q_{n-\sigma-\lambda} S^\sigma \,|\, T_{n-\tau} p_{\tau-\lambda}) = (R^{\sigma+\lambda} \,|\, p_{\sigma+\lambda})(R^{\tau-\lambda} p_{\tau-\lambda}). \tag{11}$$

[56] For $\rho > n$ the matrix product is known to vanish; for $\rho = n$ it goes into the product of two determinants.

[57] Invariant formations of prescribed ground forms are, of course, only those aggregates of matrix products that depend on the symbol rows $S^\rho \ldots$.

[58] Cf. the Introduction.

[59] This notation is to be retained throughout.

[60] The number λ always runs up to the smaller of the two last indicated numbers.

Here too (8.) shows that the products of the elements of the rows R can be expressed uniquely through the rows S and T. The substitutions (9.) and (11.) will be used especially when a further folding with other rows is performed with (7.).

The matrix representation enables us to state invariantly the quadratic relations between the elements of a row p_ρ (from which, as is known, all the others follow), hence, according to § 1, the relations linear in the coefficients of the ground form that the rows S^ρ satisfy. These are the identities

$$(q_{n-\rho-\lambda}S^\rho \,|\, S_{n-\rho}p_{\rho-\lambda}) = 0, \qquad (\lambda = 1, 2, \ldots, \rho, n-\rho), \tag{12}$$

where the p and q are to be regarded as arbitrary quantities. We shall see in § 5 that the identities with odd defect number λ are consequences of those with higher defect. The relations (12.) are a direct consequence of the relations (5.); for one has

$$(q_{n-\rho-\lambda}S^\rho \,|\, S_{n-\rho}p_{\rho-\lambda}) = (v^{(1)}\cdots v^{(n-\rho-\lambda)}s^{(1)}\cdots s^{(\rho)} \mid \sigma^{(1)}\cdots\sigma^{(n-\rho)}y^{(1)}\cdots y^{(\rho-\lambda)}),$$

and the application of the product theorem gives, in consequence of (5.), a vanishing $(n-\lambda)$-rowed determinant. The identities (12.) say that in order to find all types of foldings it suffices to fold the product of forms linear in each variable row; or also, by § 6, that the normalized form $(S^\rho \mid p_\rho)^m$ possesses no invariant formation linear in the coefficients.[61]

The conditions (12.) are also sufficient to characterize the rows p_ρ. For the property that the individual elements of p_ρ are determinants of a matrix is an invariant property, and therefore it must be expressible invariantly; but by Theorem VI, the conditions (12.) give the greatest possible invariant restriction for the p_ρ, namely that the linear form formed from the row p_ρ as coefficients possesses no invariant formations at all.

§ 3. The symbolic identities.

We now have to transfer the second fundamental theorem of the symbolic method, namely to set up the totality of the irreducible indecomposable identities in which all rows S^ρ, p_ρ are regarded as indecomposable. We also stipulate the following. Corresponding to the meaning of symbol and variable rows, those identities are to be regarded as composite that arise when one specializes the variable row p_ρ to $p_{\rho-1}y$ and applies one of the identities of the system to the resulting expressions; while, on the other hand, identities containing k symbol rows are to count as indecomposable even if they arise from identities with fewer symbol rows by the above process. The rows S^ρ, p_ρ need not necessarily enter the identities explicitly; for their interpretation as indecomposable quantities it suffices to show that the matrix products that occur depend only on these rows. We shall show, however, that in this case, partly by applying substitution (11.), an explicit representation can always be obtained. The general identities are obtained, by means of the principle of matrix representation, from the special identities given by Pascal, which contain only rows x and u and constitute a direct transfer of those given by Study for the ternary domain:[62]

$$\sum_{i=1}^{n}(-1)^{i+1}(a^{(i)} \,|\, x)(a^{(1)}\cdots a^{(i-1)}a^{(i+1)}\cdots a^{(n)}u) = (-1)^{n+1}(u \,|\, x)(a^{(1)}\cdots a^{(n)}), \tag{13}$$

$$\sum \pm (a^{(1)} \,|\, x^{(1)})(a^{(2)} \,|\, x^{(2)})\cdots(a^{(n)} \,|\, x^{(n)}) = (a^{(1)}\cdots a^{(n)})(x^{(1)}\cdots x^{(n)}), \tag{14}$$

$$\sum_{i=1}^{n}(-1)^{i+1}(u \,|\, a^{(i)})(a^{(1)}\cdots a^{(i-1)}a^{(i+1)}\cdots a^{(n)}x) = (-1)^{n+1}(u \,|\, x)(a^{(1)}\cdots a^{(n)}). \tag{15}$$

Two further identities arise from (13.) and (15.) through the substitutions $(a^{(i)} \mid x) = (a^{(i)}q_{n-1})$ and respectively $(u \mid a^{(i)}) = (p_{n-1}a^{(i)})$, and so by our interpretation they are not to be regarded as essentially different. (14.) represents the product theorem for determinants; we shall show how (13.) and (14.) (and, dualistically, (15.) and (14.)) can be replaced by the system of suitably formed product theorems for matrices. The product theorem, formed once for ρ-rowed matrices and once for $(\rho-1)$-rowed matrices, reads

$$(a^{(1)}a^{(2)}\cdots a^{(\rho)} \,|\, xp_{\rho-1}) = (a^{(1)}a^{(2)}\cdots a^{(\rho)} \mid xy^{(2)}\cdots y^{(\rho)}) = \sum \pm (a^{(1)} \,|\, x)(a^{(2)} \,|\, y^{(2)})\cdots(a^{(\rho)} \,|\, y^{(\rho)}),$$

$$\sum \pm (a^{(2)} \,|\, y^{(2)})\cdots(a^{(\rho)} \,|\, y^{(\rho)}) = (a^{(2)}\cdots a^{(\rho)} \mid y^{(2)}\cdots y^{(\rho)}) = (a^{(2)}\cdots a^{(\rho)}p_{\rho-1}) = (a^{(2)}\cdots a^{(\rho)}q_{n-\rho+1});$$

[61] Cf. an addendum by Study in Grassmann's works, first volume, second part, p. 510 (condition that a quantity S^ρ is simple).

[62] For formulation and literature of the problem, cf. the Introduction.

and from this follows the system of product theorems, where $\rho = 2, 3, \ldots, n$:

$$\begin{aligned}(a^{(1)}a^{(2)}\cdots a^{(\rho)} \,|\, xp_{\rho-1}) &= \sum_{i=1}^{\rho}(-1)^{i+1}(a^{(i)} \,|\, x)(a^{(1)}\cdots a^{(i-1)}a^{(i+1)}\cdots a^{(\rho)} \mid p_{\rho-1})\\ &= \sum_{i=1}^{\rho}(-1)^{i+1}(a^{(i)} \,|\, x)(a^{(1)}\cdots a^{(i-1)}a^{(i+1)}\cdots a^{(\rho)}q_{n-\rho+1}).\end{aligned} \tag{16}$$

For $\rho = n$, (16.) goes over into (13.); if we further specialize $p_{n-1} = y^{(2)}p_{n-2}$, $p_{n-2} = y^{(3)}p_{n-3}$, etc., and apply the identities (16.) to the expressions $(a^{(1)}\cdots a^{(i-1)}a^{(i+1)}\cdots a^{(n)} \mid y^{(2)}p_{n-2})$, $(a^{(1)}\cdots a^{(i-1)}a^{(i+1)}\cdots a^{(h-1)}a^{(h+1)}\cdots a^{(n)} \mid y^{(3)}p_{n-3})$, etc., then we pass from (13.) to (14.). Thus the identity (14.) is, by our definition, composite from the identities (16.); equivalently, the system (16.) is equivalent to the identities (13.) and (14.). The system (16.) satisfies one of the conditions required for the general identities: it contains the row $p_{\rho-1}$ as an indecomposable quantity and at the same time in explicit form. It is the only system satisfying this condition and only this condition. For we can form no further determinants or matrix products from the rows a, x, p; and between these no relations independent of (16.) can exist, since otherwise, by eliminating $(a^{(1)}\cdots a^{(\rho)} \mid xp_{\rho-1})$, there would arise a relation between ρ $(\rho < n)$ expressions $(a^{(i)} \mid x)(a^{(1)}\cdots a^{(i-1)}a^{(i+1)}a^{(\rho)}q_{n-\rho+1})$, whereas by (13.) at least $n+1$ such expressions must enter into a relation. From analogous considerations one obtains the system equivalent to (15.) and (14.):

$$\begin{aligned}(uq_{\rho-1} \,|\, a^{(1)}\cdots a^{(\rho)}) &= \sum_{i=1}^{\rho}(-1)^{i+1}(u \,|\, a^{(i)})(q_{\rho-1} \mid a^{(1)}\cdots a^{(i-1)}a^{(i+1)}\cdots a^{(\rho)})\\ &= \sum_{i=1}^{\rho}(-1)^{i+1}(u \,|\, a^{(i)})(p_{n-\rho+1}a^{(1)}\cdots a^{(i-1)}a^{(i+1)}\cdots a^{(\rho)}).\end{aligned} \tag{17}$$

In order to pass from (16.) to identities between arbitrary rows $A^{\rho_1}, B^{\rho_2}, \ldots, x, p$, we observe that these identities must become identical with (16.) as soon as we resolve $A^{\rho_1} = a^{(1)}a^{(2)}\cdots a^{(\rho_1)}$, $B^{\rho_2} = b^{(1)}b^{(2)}\cdots b^{(\rho_2)}$, etc., and that they can also be combined into final expressions only by the operations determined by (16.) as soon as the individual expressions are of the type entering into (16.). Thus we obtain, if we also set

$$\rho = \rho_1 + \rho_2 + \cdots + \rho_k \leq n, \tag{18}$$

$$\begin{aligned}&(a^{(1)}\cdots a^{(\rho_1)}b^{(1)}\cdots b^{(\rho_2)}\cdots k^{(1)}\cdots k^{(\rho_k)} \,|\, xp_{\rho-1}) = (A^{\rho_1}B^{\rho_2}\cdots K^{\rho_k} \,|\, xp_{\rho-1})\\ &\quad= \sum_{i=1}^{\rho_1}(-1)^{i+1}(a^{(i)} \,|\, x)(a^{(1)}\cdots a^{(i-1)}a^{(i+1)}\cdots a^{(\rho_1)}B^{\rho_2}\cdots K^{\rho_k}q_{n-\rho+1})\\ &\quad\quad+ (-1)^{\rho_1\rho_2}\sum_{i=1}^{\rho_2}(-1)^{i+1}(b^{(i)} \,|\, x)(b^{(1)}\cdots b^{(i-1)}b^{(i+1)}\cdots b^{(\rho_2)}A^{\rho_1}\cdots K^{\rho_k}q_{n-\rho+1})\\ &\quad\quad+ \cdots\\ &\quad\quad+ (-1)^{(\rho_1+\cdots+\rho_{k-1})\rho_k}\sum_{i=1}^{\rho_k}(-1)^{i+1}(k^{(i)} \,|\, x)(k^{(1)}\cdots k^{(i-1)}k^{(i+1)}\cdots k^{(\rho_k)}A^{\rho_1}\cdots K^{\rho_{k-1}}q_{n-\rho+1}).\end{aligned}$$

Each individual sum (but not already a part of the sum) does in fact contain the rows $A^{\rho_1}, B^{\rho_2}, \ldots$ as indecomposable quantities; for it satisfies the necessary and sufficient conditions[63]

$$\left(a^{(i+1)}\middle|\frac{\partial}{\partial a^{(i)}}\right) = 0;\ \ldots\ ;\left(k^{(j+1)}\middle|\frac{\partial}{\partial k^{(j)}}\right) = 0. \quad (i = 1, 2, \ldots, \rho_1 - 1;\ j = 1, 2, \ldots, \rho_k - 1) \tag{19}$$

We show that it also contains all rows explicitly. For if we set $B^{\rho_2}\cdots K^{\rho_k}q_{n-\rho+1} = q_{n-\rho_1+1}$, then by (16.) and (10.) we obtain

$$\begin{aligned}\sum(-1)^{i+1}(a^{(i)} \,|\, x)(a^{(1)}\cdots a^{(i-1)}a^{(i+1)}\cdots a^{(\rho_1)}q_{n-\rho+1}) &= (A^{\rho_1} \,|\, xp_{\rho_1-1})\\ &= (A_{n-\rho_1}xp_{\rho_1-1}) = (-1)^{(\rho_1+1)(n+1)}(q_{n-\rho_1+1} \,|\, A_{n-\rho_1}x)\end{aligned}$$

[63] Capelli, a. a. O. § 23, gives sufficient conditions: $\left(a^{(k)} \mid \frac{\partial}{\partial a^{(i)}}\right) = 0$, $(k > i,\ i = 1, 2, \ldots, \rho_1 - 1)$. These most simply imply the necessary and sufficient conditions by application of the Poisson bracket process after Wellstein, Von den Differentialgleichungen der projektiven Invarianten, Math. Ann. 67 (1909) § 3.

$$= (-1)^{(\rho_1+1)(n+1)}(B^{\rho_2}\cdots K^{\rho_k} q_{n-\rho+1} \,|\, A_{n-\rho_1} x).$$

If we compute the remaining sums analogously and subsequently set, since the rows are already characterized as distinct by the weight indices, $B^{\rho_2} = A^{\rho_2}$, $C^{\rho_3} = A^{\rho_3}, \ldots$, then we obtain the desired identities:

$$\begin{gathered}(A^{\rho_1} A^{\rho_2}\cdots A^{\rho_k} \,|\, x p_{\rho-1}) = \sum_{i=1}^{k} (-1)^{\delta_i} (A^{\rho_1}\cdots A^{\rho_{i-1}} A^{\rho_{i+1}}\cdots A^{\rho_k} q_{n-\rho+1} \,|\, A_{n-\rho_i} x),\\ \delta_i = (\rho_1+\rho_2+\cdots+\rho_{i-1})\rho_i + (\rho_i+1)(n+1),\end{gathered} \tag{20}$$

and from (17.) the dualistic identities:

$$\begin{gathered}(u q_{\sigma-1} \,|\, A_{\sigma_1} A_{\sigma_2}\cdots A_{\sigma_k}) = \sum_{i=1}^{k} (-1)^{\varepsilon_i} (u A^{n-\sigma_i} \,|\, p_{n-\sigma+1} A_{\sigma_1}\cdots A_{\sigma_{i-1}} A_{\sigma_{i+1}}\cdots A_{\sigma_k}),\\ \varepsilon_i = (\sigma_1+\sigma_2+\cdots+\sigma_{i-1})\sigma_i + (\sigma_i+1)(n+\sigma).\end{gathered} \tag{21}$$

Theorem II. With (20.) and (21.) the totality of the indecomposable identities between the rows A^{ρ_i}, x, p, respectively A_{σ_i}, u, q, is exhausted, and at the same time the totality of those indecomposable identities that admit explicit representation without carrying out substitution (11.).

The first part of the theorem follows from the considerations leading to (20.) and (21.); the second part from the impossibility of an identity between two symbol rows:

$$(q_\rho A^\sigma \,|\, B_{\rho+\sigma-\tau} p_\tau) = \varepsilon_1 (q_\rho B^{n-\rho-\sigma+\tau} \,|\, A_{n-\sigma} p_\tau) + \varepsilon_2 (q_\rho q_{n-\tau} \,|\, B_{\rho+\sigma-\tau} A_{n-\sigma}),$$

where $\rho \le \tau \le \sigma$ and none of the rows has weight $1\cdot(n-1)$. (Other assumptions about ρ, τ, σ can be reduced to this one by interchanging the row designations.) This impossibility follows, for example, by expanding (8.) under the specialization $q_\rho = l M^{\rho-1}$, $q_{n-\tau} = l N^{n-\tau-1}$, if one sets $l_1 = 1$, $M^{2,3,\ldots,\rho} = 1$, $A^{\rho+1,\ldots,\rho+\sigma} = 1$, all other elements of the rows l, M, A equal to zero, while N and B are arbitrary. Since identities between arbitrary rows, by resolving a row p_τ into $y^{(1)} y^{(2)}\cdots y^{(\tau)}$, pass into identities composite from (20.), the second part of the assertion is proved as soon as (20.) can be generated from identities with only two symbol rows. In fact, from

$$(A^{\rho_1} A^{\rho_2} \,|\, x p_{\rho-1}) = \varepsilon (A^{\rho_1} q_{n-\rho+1} \,|\, A_{n-\rho_2} x) + (A^{\rho_1} \,|\, x p_{\rho_1-1}), \quad \text{where } p_{\rho_1-1} \sim A^{\rho_2} q_{n-\rho+1}, \tag{20a}$$

by specializing $A^{\rho_1} = B^{\sigma_1} B^{\sigma_2}$, that is, by

$$(B^{\sigma_1} B^{\sigma_2} \,|\, x p_{\rho_1-1}) = \varepsilon (B^{\sigma_1} q_{n-\rho_1+1} \,|\, B_{n-\sigma_2} x) + (B^{\sigma_1} \,|\, x p_{\sigma_1-1}), \quad \text{where } p_{\sigma_1-1} \sim B^{\sigma_2} q_{n-\rho_1+1},$$

one obtains an identity (20.) in three symbol rows; and by further specializations of $B^{\sigma_1} = C^{\tau_1} C^{\tau_2}$, etc., the general identities (20.). The nonexistence of the explicit representation of an identity between two symbol rows and two arbitrary variable rows thus implies the nonexistence in the case of arbitrarily many symbol rows, provided that among the variable rows no row x or u is contained.

§ 4. The decomposition identities.

We now consider the most general case of identities among arbitrary symbol and variable rows in whose final expressions, without performing substitution (11.), a row p_τ occurs resolved into its individual rows $y^{(1)} \cdots y^{(\tau)}$; we call these identities "decomposition identities." Following the remark at the end of the preceding paragraph, we first determine them for the case of only two symbol rows, that is, we expand the expression $(q_\rho A^\sigma \,|\, B_{\rho+\sigma-\tau} p_\tau)$. We set:

$$q_{\rho-\alpha}^{(i_1 \dots i_\alpha)} \sim p_{n-\rho}\, y^{(i_1)} \cdots y^{(i_\alpha)}, \qquad p_{\tau-\alpha}^{(i_1 \dots i_\alpha)} = y^{(i_{\alpha+1})} \cdots y^{(i_\tau)}, \qquad p_\tau = y^{(1)} \cdots y^{(\tau)}. \tag{22}$$

With respect to the numbers ρ, σ, τ four cases must be distinguished:

$$\begin{aligned}
&1) \quad \rho + \sigma \le n, \quad \tau \le \rho, \quad \tau \le \sigma; \\
&2) \quad \rho + \sigma \le n, \quad \tau \ge \rho, \quad \tau \le \sigma; \\
&3) \quad \rho + \sigma \le n, \quad \tau \le \rho, \quad \tau > \sigma; \\
&4) \quad \rho + \sigma \le n, \quad \tau \ge \rho, \quad \tau \ge \sigma.
\end{aligned}$$

We shall single out the first case, and then briefly characterize the others from it. By (20.), (10.) and (9.) we obtain, when we decompose the row $y^{(1)} y^{(2)} \cdots y^{(\tau)} B_{\rho+\sigma-\tau}$ into $y^{(1)}$ and $y^{(2)} \cdots y^{(\tau)} B_{\rho+\sigma-\tau}$:

$$\begin{aligned}
(q_\rho A^\sigma \,|\, y^{(1)} y^{(2)} \cdots y^{(\tau)} B_{\rho+\sigma-\tau}) &= (q_\rho^{(1)} A^\sigma \,|\, y^{(2)} \cdots y^{(\tau)} B_{\rho+\sigma-\tau}) \\
&\quad + (-1)^\rho (q_\rho [A_{n-\sigma} y^{(1)}]^{\sigma-1} \,|\, y^{(2)} \cdots y^{(\tau)} B_{\rho+\sigma-\tau}).
\end{aligned} \tag{23}$$

The last term in (23.) now has exactly the form of the original expression–only σ is replaced by $\sigma - 1$, and τ by $\tau - 1$–and therefore again admits equation (23.). Thus, since $\tau \le \sigma$, we can apply operation (23.) τ times and, taking account of (10.), obtain the relation:

$$\begin{aligned}
(q_\rho A^\sigma \,|\, p_\tau B_{\rho+\sigma-\tau}) &= (-1)^{\rho\sigma + (\sigma+\tau)(n+1)} (q_\rho B^{n-\rho-\sigma+\tau} \,|\, A_{n-\sigma} p_\tau) \\
&\quad + \sum_{i_1=1}^{\tau} (-1)^{\rho(i_1-1)} (q_{\rho-1}^{(i_1)} [A_{n-\sigma} y^{(1)} \cdots y^{(i_1-1)}]^{\sigma-i_1+1} \,|\, y^{(i_1+1)} \cdots y^{(\tau)} B_{\rho+\sigma-\tau}).
\end{aligned} \tag{24}$$

Every term under the summation sign is again of the type of the original expression; ρ is replaced by $\rho - 1$, σ by $\sigma - i_1 + 1$, and τ by $\tau - i_1$, so that condition 1) is even more strongly satisfied. Since $\tau \le \rho$, we can therefore apply operation (24.) τ times and obtain the desired decomposition identity:

$$\begin{aligned}
(q_\rho A^\sigma \,|\, p_\tau B_{\rho+\sigma-\tau}) &= \varepsilon_{i_0} (q_\rho B^{n-\rho-\sigma+\tau} \,|\, A_{n-\sigma} p_\tau) \\
&\quad + \sum_{i_1=1}^{\tau} \varepsilon_{i_1} (q_{\rho-1}^{(i_1)} B^{n-\rho-\sigma+\tau} \,|\, A_{n-\sigma} p_{\tau-1}^{(i_1)}) \\
&\quad + \sum_{i_2=i_1+1}^{\tau} \varepsilon_{i_2} (q_{\rho-2}^{(i_1 i_2)} B^{n-\rho-\sigma+\tau} \,|\, A_{n-\sigma} p_{\tau-2}^{(i_1 i_2)}) + \cdots \\
&\quad + \varepsilon_{i_\tau} (q_{\rho-\tau}^{(i_1 \dots i_\tau)} B^{n-\rho-\sigma+\tau} \,|\, A_{n-\sigma}) \\
&= \sum_{\alpha=0}^{\tau} \sum_{i_\alpha = i_{\alpha-1}+1}^{\tau} \varepsilon_{i_\alpha} (q_{\rho-\alpha}^{(i_1 \dots i_\alpha)} B^{n-\rho-\sigma+\tau} \,|\, A_{n-\sigma} p_{\tau-\alpha}^{(i_1 \dots i_\alpha)}).
\end{aligned} \tag{25}$$

For the sign factor ε one obtains from (24.):

$$\begin{aligned}
\varepsilon_{i_\alpha} = (-1)^{\delta_{i_\alpha}}, \qquad \delta_{i_\alpha} &= \sum_{\beta=1}^{\alpha} (\rho - \beta + 1)(i_{\beta-1} + i_\beta + 1) \\
&\quad + (\rho - \alpha)(\sigma + i_\alpha + \alpha) + (\sigma + \tau + \alpha)(n+1),
\end{aligned} \tag{26}$$

where $i_0 = 0$ is to be put. The summation sign in (25.) is to be understood as follows: to each combination $i_1 i_2 \dots i_{\alpha-1}$, where $i_1 < i_2 < \cdots < i_{\alpha-1}$, all values i_α for which $i_{\alpha-1} < i_\alpha \le \tau$ are adjoined. Taking (26.) into account, one easily verifies that the individual sums in (25.) satisfy the differential equations analogous to (19.):

$$\left(\frac{\partial}{\partial y^{(i)}} \middle| y^{(i+1)} \right) = 0, \qquad (i = 1, 2, \dots, \tau - 1), \tag{27}$$

and hence depend only on the row p_τ. Thus the decomposition identity is an indecomposable identity among the rows A, B, q, p; we shall return to its explicit representation in the next paragraph.

In case 2), operation (23.) can likewise be applied τ times, but operation (24.) stops after the ρ-th application. Thus in the final expression of (25.) the first summation sign must be replaced by $\sum_{\alpha=0}^{\rho}$. In cases 3) and 4), operation (23.) can be carried out only σ times; in place of (24.) one then has the relation:

$$\begin{aligned}(q_\rho A^\sigma \,|\, p_\tau B_{\rho+\sigma-\tau}) = (-1)^{\rho\sigma}&(q_\rho^{(\sigma+1\ldots\tau)} B^{n-\rho+\tau-\sigma} \,|\, A_{n-\sigma} p_{\tau-\sigma}^{(\sigma+1\ldots\tau)}) \\ &+ \sum_{i_1=1}^{\sigma} (-1)^{\rho(i_1-1)} (q_{\rho-1}^{(i_1)} [A_{n-\sigma} y^{(1)} \cdots y^{(i_1-1)}]^{\sigma-i_1+1} \,|\, y^{(i_1+1)} \cdots y^{(\tau)} B_{\rho+\sigma-\tau});\end{aligned} \tag{28}$$

repeating (28.) $\tau-\sigma$ times, where the summation sign becomes $\sum_{i_\beta=i_{\beta-1}+1}^{\sigma+\beta-1}$–as the $(\sigma-i_1+1)$-fold application of (23.) to the terms under the summation sign in (28.) shows–gives all terms of the individual sum in (25.) corresponding to the index $\alpha=\tau-\sigma$, with the corresponding sign. Then cases 3) and 4) pass over into cases 1) and 2), for the numbers corresponding to σ, τ become $\sigma' = \sigma - i_{\tau-\sigma} + \tau - \sigma$, $\tau' = \tau - i_{\tau-\sigma} = \sigma'$; and as the procedure continues, $\tau' < \sigma'$. Thus in the final expression of (25.) the first summation sign must be replaced by $\sum_{\alpha=\tau-\sigma}^{\tau,\rho}$.

Analogously, from (21.) one obtains the dual decomposition identity, in which a row q_ρ is resolved into the individual rows $v^{(1)}, \ldots, v^{(\rho)}$. If we put

$$\dot{p}_{\tau-\beta}^{(i_1\ldots i_\beta)} \sim v^{(i_1)} \cdots v^{(i_\beta)} q_{n-\tau}; \qquad \dot{q}_{\rho-\beta}^{(i_1\ldots i_\beta)} = v^{(i_{\beta+1})} \cdots v^{(i_\rho)}; \qquad q_\rho = v^{(1)} \cdots v^{(\rho)}, \tag{29}$$

then

$$(q_\rho A^\sigma \,|\, p_\tau B_{\rho+\sigma-\tau}) = \sum_{\beta=\max(0,\tau-\sigma)}^{\min(\tau,\rho)} \sum_{i_\beta=i_{\beta-1}+1}^{\rho} \varepsilon(\dot{q}_{\rho-\beta}^{(i_1\ldots i_\beta)} B^{n-\rho-\sigma+\tau} \,|\, A_{n-\sigma} \dot{p}_{\tau-\beta}^{(i_1\ldots i_\beta)}), \tag{30}$$

where the summation index β runs from the larger of the lower values to the smaller of the upper values. All terms of an individual sum in (25.) and (30.) are polars of a single form, produced by the polar processes

$$\left(\frac{\partial}{\partial p_{\tau-\alpha}}\middle| p_{\tau-\alpha}^{(i_1\ldots i_\alpha)}\right), \qquad \left(q_{\rho-\alpha}^{(i_1\ldots i_\alpha)}\middle|\frac{\partial}{\partial q_{\rho-\alpha}}\right).$$

To express this, we denote by Π an aggregate of such polar operations, multiplied by constants, and obtain, in place of (25.) and (30.), the relation

$$(q_\rho A^\sigma \,|\, p_\tau B_{\rho+\sigma-\tau}) = \sum_{\alpha=\max(0,\tau-\sigma)}^{\min(\tau,\rho)} \Pi(q_{\rho-\alpha} B^{n-\rho-\sigma+\tau} \,|\, A_{n-\sigma} p_{\tau-\alpha}). \tag{31}$$

§ 5. The decomposition identities in explicit form.

If in (25), in case 4), we specialize τ to $\sigma+\rho$, we obtain

$$(q_\rho A^\sigma \,|\, y^{(1)} \cdots y^{(\rho+\sigma)}) = \sum_{1\le i_1<\cdots<i_\rho\le\rho+\sigma} \varepsilon_{i_\rho} (q_\rho \,|\, y^{(i_1)} \cdots y^{(i_\rho)})(A^\sigma \,|\, y^{(i_{\rho+1})} \cdots y^{(i_{\rho+\sigma})}), \tag{32}$$

where $\varepsilon_{i_\rho} = (-1)^{\delta_{i_\rho}}$ and $\delta_{i_\rho} = \rho(\rho+1)/2 + i_1 + \cdots + i_\rho$. This is a more general form of the product theorem for matrices than (16.) and (17.), and follows by Laplace expansion of the determinant of the product matrix

$$\sum \pm (v^{(1)} y^{(1)}) \cdots (v^{(\rho)} y^{(\rho)})(a^{(1)} y^{(\rho+1)}) \cdots (a^{(\sigma)} y^{(\rho+\sigma)}).$$

Thus the more general product theorem is composed out of the identities (20.) and (21.), respectively, and the selection of the special form (16.), (17.) is thereby explained.

Relation (32.) now gives the means for explicitly representing the decomposition identities. For this purpose we regard the forms occurring in (31.) as ground forms; that is, we perform substitution (11.), which changes nothing in the explicit representation in the rows A, B, since by (8.) the newly introduced rows R can be expressed through the rows A, B themselves and not through their component rows $a^{(1)} a^{(2)} \ldots$, $b^{(1)} b^{(2)} \ldots$. Thus let

$$(q_{\rho-\alpha} B^{n-\rho-\sigma+\tau} \,|\, A_{n-\sigma} p_{\tau-\alpha}) = (R_{\rho-\alpha} p_{n-\rho+\alpha} \,|\, R^{\tau-\alpha} p_{\tau-\alpha}). \tag{33}$$

Then, for the individual sum of (25.) corresponding to the index α, we get

$$\begin{aligned}&\sum \varepsilon_{i_\alpha}(R_{\rho-\alpha}p_{n-\rho}y^{(i_1)}\cdots y^{(i_\alpha)}\,|\,R^{\tau-\alpha}y^{(i_{\alpha+1})}\cdots y^{(i_\tau)})\\ &\quad=\sum \varepsilon_{i_\alpha}([R_{\rho-\alpha}p_{n-\rho}]^\alpha y^{(i_1)}\cdots y^{(i_\alpha)}\,|\,R^{\tau-\alpha}y^{(i_{\alpha+1})}\cdots y^{(i_\tau)})\\ &\quad=\varepsilon([R_{\rho-\alpha}p_{n-\rho}]^\alpha R^{\tau-\alpha}\,|\,p_\tau)=\varepsilon(R^{\tau-\alpha}q_{n-\tau}\,|\,R_{\rho-\alpha}p_{n-\rho}).\end{aligned} \tag{34}$$

Thus an explicit final expression is in fact obtained, and the symmetric form in which q_ρ and p_τ occur shows that (30.) leads to the same final expression, which is therefore independent of the original decomposition. There is a difference between (25.) and (30.) only as long as the rows y and v have independent meaning, so that the decomposition identity passes, by our definition, into a decomposable identity. To arrive at identities with more symbol rows, one need only, according to § 3 (end), resolve a row q_ρ into $q_{\rho_1}C^{\sigma_1}$, or p_τ into $p_{\tau_1}C_{\tau-\tau_1}$. The resulting expressions

$$(q_{\rho_1}C^{\sigma_1}\,|\,[R^{\tau-\alpha}q_{n-\tau}]_\lambda R_{\rho+\sigma_1-\alpha-\lambda}),\qquad ([R_{\rho-\alpha}p_{n-\rho}]^\lambda R^{\tau-\alpha}\,|\,p_{\tau_1}C_{\tau-\tau_1}) \tag{35}$$

are exactly of the type that occurs for two symbol rows, and therefore again admit explicit representation after a substitution corresponding to (33.). Repetition therefore gives explicit representation for arbitrarily many rows. It may also be noted that the first and last sums in (25.) and (30.), respectively, give the explicit representation without applying (33), solely by formula (32), or else reduce altogether to a single term containing all rows explicitly. The relation obtained from (32.) by resolving q_ρ into $q_{\rho_1}C^{\sigma_1}$ and so forth can be regarded as the product theorem in its most general form.

In summary we shall show:

Theorem III: With the identities

$$(q_\rho A^\sigma\,|\,p_\tau B_{\rho+\sigma-\tau})=\sum_{\alpha=\max(0,\tau-\sigma)}^{\min(\tau,\rho)}\varepsilon(R^{\tau-\alpha}q_{n-\tau}\,|\,R_{\rho-\alpha}p_{n-\rho}), \tag{36}$$

where the rows R are determined by (33), together with those obtained from them by resolving q_ρ, p_τ into $q_{\rho_1}C^{\sigma_1}$, etc., the totality of indecomposable identities is exhausted.

Indeed, the resulting identities are the most general of their kind, since the individual rows are no longer subject to any restriction as to weight or number, as was still the case in (20.) and (21.). Nor do any further identities among these rows exist; for when a row p_τ is resolved into $y^{(1)}\cdots y^{(\tau)}$, the most general identities are composed out of (20.), hence by § 3 (end) out of (20a.); and the composition discussed following (16.) is precisely the one carried out in § 4. The transition to arbitrarily many rows is supplied, as the discussion of (20a.) shows, by applying always the same operation to the expressions that arise from resolving q_ρ into $q_{\rho_1}C^{\sigma_1}$, and so on. It also follows that the direct transition from (20.) instead of from (20a.) gives nothing new. This is easily verified by calculation, by specializing once a row q_ρ in (25.), and another time carrying out the transition from (20.) by the method of § 4. In both cases the same sums arise, which pass by application of the general product theorem (see above) into the identities obtained from (36.). The identities (20.), (21.) correspond to the special cases $\tau=1$, $\rho=1$; only two individual sums then occur, and hence, in accordance with an earlier remark, explicit representation without the substitution (33.), agreeing with what was found before.

At the end let us briefly mention two special cases of the decomposition identity. Put in (31): $\tau=\sigma$, $\rho=n-\sigma$, and denote the row p_τ by $p'_\sigma\sim q'_{n-\sigma}$. Then

$$(A^\sigma\,|\,p_\sigma)(B^\sigma\,|\,p'_\sigma)-(A^\sigma\,|\,p'_\sigma)(B^\sigma\,|\,p_\sigma)=\sum_{\alpha=1}^{\min(\sigma,n-\sigma)}\Pi(q_{n-\sigma-\alpha}B^\sigma\,|\,A_{n-\sigma}p_{\sigma-\alpha}). \tag{37}$$

This is Clebsch's formula[64] for expanding a form with two cogredient rows p_σ, p'_σ into elementary covariants. The elementary covariants belonging to a form $(A^\sigma\,|\,p_\sigma)^m(B^\sigma\,|\,p'_\sigma)^n$ are then the following:

$$\left\{\sum_{\alpha=1}^{\min(\sigma,n-\sigma)}\varepsilon(q_{n-\sigma-\alpha}B^\sigma\,|\,A_{n-\sigma}p_{\sigma-\alpha})\right\}^\rho(A^\sigma\,|\,p_\sigma)^{m-\rho}(B^\sigma\,|\,p'_\sigma)^{n-\rho},\qquad(\rho=0,1,\dots,m,n). \tag{38}$$

[64] Clebsch, loc. cit., pp. 25–32. Clebsch proves that the individual terms on the right-hand sides depend only on the rows A, B; the explicit representation would be obtained by applying (32.) to his expressions Φ_h.

They arise, as we shall briefly say, from the ground form by "cogredient folding of defect ρ" (cf. § 2).

Second, put $\tau = \sigma - \lambda$, $\rho = n - \sigma - \lambda$, $B^\sigma = A^\sigma$. Then

$$\begin{aligned}(q_{n-\sigma-\lambda}A^\sigma \mid A_{n-\sigma}p_{\sigma-\lambda}) &= (-1)^\lambda (q_{n-\sigma-\lambda}A^\sigma \mid A_{n-\sigma}p_{\sigma-\lambda}) \\ &\quad + (-1)^{(n-\sigma)(\sigma-\lambda)} \sum_{\alpha=1}^{\min(\sigma-\lambda, n-\sigma-\lambda)} \Pi(q_{n-\sigma-\lambda-\alpha}A^\sigma \mid A_{n-\sigma}p_{\sigma-\lambda}).\end{aligned} \tag{39}$$

Thus, as noted in § 2, the conditions (12.) characterizing the symbol rows for odd defect λ are consequences of the conditions of higher defect. For a linear form $(A^\sigma \mid p_\sigma) = (B^\sigma \mid p_\sigma)$, (39.) implies that its irreducible invariant formations that are quadratic in the coefficients reduce to ones of even defect; this agrees with the known fact that a linear form in the binary and ternary domains has no invariant formations.

It should be noted that the general identities were originally derived under the assumption that the individual rows satisfy conditions (12.). Since, however, these individual rows enter the identities only linearly, the latter also hold for the more general rows that do not satisfy conditions (12.).

§ 6. Explicit representation of the invariant formations. Folding and differentiation processes.

The results of the preceding paragraphs allow us to complete the theorem stated in § 2 on symbolic representability (the first fundamental theorem) by adding explicit representability; namely, we show:

Theorem IV: "All invariant formations of arbitrary ground forms can be represented explicitly by matrix products; that is, aside from variable rows p_ρ, only the rows $S^\sigma, \ldots, T^\tau$ of the original forms, or the rows P, Q, R derived from them by substitution (9.) or (11.), enter the final expressions."

For the proof, suppose that an invariant formation depends, besides variable rows, on the rows $S^\sigma, \ldots, T^\tau$; and suppose that these rows have been resolved sufficiently far into individual rows $S^\sigma = S^{\sigma_1} \cdots S^{\sigma_i}, \ldots, T^\tau = T^{\tau_1} \cdots T^{\tau_k}$ to make representation by determinants possible. Let the rows be ordered in a sequence $S^\sigma, \ldots, T^\tau$ that is arbitrary in itself but fixed once and for all. We pick out an aggregate of determinants that depends on the first total row $S^\sigma = S^{\sigma_1} \cdots S^{\sigma_i}$, not merely on individual component rows. This dependence, however, is by § 3–§ 5 a consequence of the general identities. If we now resolve a variable row p_ρ into the rows y, v, or one of the later rows T into the individual rows t, respectively τ, then the most general identities are composed out of (20.) and (21.). But these latter identities always lead to an expression containing the row $S^\sigma = S^{\sigma_1} \cdots S^{\sigma_i}$ explicitly, namely the left-hand side of (20.) and (21.); for by (19.) one verifies that only the complete sum on the right–not a part of it corresponding to a term of (20a.)–has the property of depending only on S^σ. As the procedure continues, all rows that have once entered explicitly remain explicit; application of the identities can at most require the combining of several rows into a single one, i.e. substitution (9.). If finally only a single row T^τ remains to be brought to explicit form, two cases are possible. There may still occur a free variable row, that is, one not connected with other rows by substitution (9.); resolving it into component rows y or v completes the procedure and gives, as in (31.), polars of one or more forms containing all rows explicitly. If no free variable row remains, then from the manner in which they arose we recognize, in the totality of the expressions that contain the individual rows t or τ but depend on the row T^τ, the terms of a decomposition identity; by § 5, however, these always admit explicit representation in all rows after applying substitution (11.). This proves the theorem in full generality. It should also be noted that when only two symbol rows occur, substitution (11.) never appears. For, since $\sigma, n-\sigma, \tau, n-\tau < n$, no variable rows can enter only when the two symbol rows are united in a single determinant and thus occur explicitly; but as soon as variable rows enter, (34.) and (33.) show that substitution (11.) becomes unnecessary.

It follows from what has just been said that, in order to generate all invariant formations, it suffices to unite the symbol rows, after adjoining variable rows, into all possible matrix products. The most general matrix product is now of the form

$$(P^{\mu_1} \cdots P^{\mu_i} \mid Q_{\nu_1} \cdots Q_{\nu_k}), \qquad \sum \mu = \sum \nu, \tag{40}$$

where the P and Q may be rows of the original forms, or may have arisen from the original rows by a sequence of substitutions (9.) or (11.), or finally may be variable rows. But the expressions (40.) can be generated by successively uniting two symbol rows at a time into a matrix product, with the help of variable rows; for from

$$(q_{n-\mu-\lambda}P^\mu \mid Q_{\nu_1}p_{n-\mu-\nu_1-\lambda}) = (R_\lambda Q_{\nu_1} \mid p_{n-\mu-\nu_1-\lambda})$$

one obtains, by uniting with the row Q_{ν_2}, an expression

$$([R_\lambda Q_{\nu_1}]^{n-\lambda-\nu_1} \,|\, Q_{\nu_2} p_{n-\mu-\nu_1-\nu_2-\lambda}) = (q_{n-\mu-\lambda} P^\mu \,|\, Q_{\nu_1} Q_{\nu_2} p_{n-\mu-\nu_1-\nu_2-\lambda}),$$

and hence, by continuing the procedure, an expression (40.). If P and Q are not rows of the original forms, then (9.) and (11.), in conjunction with what was just shown, imply that they too arose through a sequence of unions of two symbol rows at a time. Thus:

Theorem V: The process for generating all invariant formations, the folding process, consists in successively uniting two symbol rows at a time into a matrix product, with the help of variable rows.

From two symbol rows S^σ, T^τ, where $\sigma \geq \tau$, the following matrix products can be formed:

$$\begin{aligned}
&(q_{n-\sigma-\lambda} S^\sigma \,|\, T_{n-\tau} p_{\tau-\lambda}), && (\lambda = 1, 2, \ldots, \min(\tau, n-\sigma)), && (41)\\
&(q_{n-\tau-\mu} T^\tau \,|\, S_{n-\sigma} p_{\sigma-\mu}), && (\mu = \sigma - \tau + \lambda), && (42)\\
&(q_{n-\tau-\nu} T^\tau \,|\, S_{n-\sigma} p_{\sigma-\nu}), && (\nu = 1, 2, \ldots, \sigma - \tau). && (43)
\end{aligned}$$

From (31.), however, the following values are obtained for (42.) and (43.):

$$(q_{n-\sigma-\lambda} S^\sigma \,|\, T_{n-\tau} p_{\tau-\lambda}) + \sum_\alpha \Pi(q_{n-\sigma-\lambda-\alpha} S^\sigma \,|\, T_{n-\tau} p_{\tau-\lambda-\alpha}), \tag{42a}$$

$$\Pi(q_{n-\sigma} S^\sigma)(T_{n-\tau} p_\tau) + \sum_\beta \Pi(q_{n-\sigma-\beta} S^\sigma \,|\, T_{n-\tau} p_{\tau-\beta}). \tag{43a}$$

Thus the forms (42.) can be expressed linearly by (41.) and polars of (41.), and the forms (43.) by polars of the ground form and of the forms (41.). Likewise (31.) gives the linear dependence of the forms (41.) on (42.) and polars of (42.). Hence, according to Clebsch's definition,[65] the forms (42.) are equivalent to the forms (41.), while (43.) leads back to the ground form. The generating process is therefore given with equal right by (41.) or (42.); we shall define the process (41.), which is simpler with respect to the number λ, as the folding process. The number λ, the defect of the folding (cf. § 2), gives the number of rows missing from the determinant product, namely in that matrix product which, for the given folding, contains the maximal number of rows. Since λ runs only up to the smaller of the two values $\tau, n - \sigma$, where $\sigma \geq \tau$, we have

$$\lambda \leq \frac{n}{2}, \ n \text{ even}; \qquad \lambda \leq \frac{n-1}{2}, \ n \text{ odd}. \tag{44}$$

The reduction of (42.) and (43.) to (41.) remains valid even when, through further folding, the rows p, q have passed into symbol rows; for by (36.) one has

$$(A^{n-\tau-\varkappa} T^\tau \,|\, S_{n-\sigma} B_{\sigma-\varkappa}) = \sum_{\alpha=\max(0,\varkappa-\sigma+\tau)}^{\min(\tau,n-\sigma)} \varepsilon(R^{\tau-\alpha} B^{n-\sigma+\varkappa} \,|\, A_{\tau+\varkappa} R_{n-\sigma-\alpha}),$$

where the rows R have arisen from S and T according to (33.). Thus these forms containing symbol rows only are obtained by folding $(q_\tau A^{n-\tau-\varkappa} \,|\, B_{\sigma-\varkappa} p_{n-\sigma})$ or $(q_{\tau-\alpha} B^{n-\sigma+\varkappa} \,|\, A_{\tau+\varkappa} p_{n-\sigma-\alpha})$–according as $\sigma - \tau \gtreqless 2\varkappa$–with the ground form and the forms (41.).

From the symbolic folding process one obtains, in the usual way, the invariant differentiation processes if one merely replaces the symbol rows $S^\sigma, T_{n-\tau}$ by the rows of differentiation symbols $\frac{\partial}{\partial p_\sigma}, \frac{\partial}{\partial q_{n-\tau}}$; as is well known, the product $\frac{\partial}{\partial p_{i_1\ldots i_\sigma}} \frac{\partial}{\partial q_{j_1\ldots j_{n-\tau}}}$ then has the real meaning $\frac{\partial^2}{\partial p_{i_1\ldots i_\sigma} \partial q_{j_1\ldots j_{n-\tau}}}$. Since the rows $\frac{\partial}{\partial p_\sigma}$ and $\frac{\partial}{\partial q_{n-\tau}}$ are cogredient with the rows $S^\sigma, T_{n-\tau}$, they satisfy the same identities as these, in particular also the ones existing between (42.) and (42a.), and between (43.) and (43a.). In summary we obtain:

Theorem VI: All invariant formations of arbitrary ground forms arise from these by repeated application of the symbolic folding process (41.), or also of the invariant differentiation processes

$$\left(q_{n-\sigma-\lambda} \frac{\partial}{\partial p_\sigma} \middle| \frac{\partial}{\partial q_{n-\tau}} p_{\tau-\lambda} \right), \qquad (\sigma \geq \tau, \ \lambda = 1, 2, \ldots, \min(\tau, n-\sigma)). \tag{45}$$

It remains to note that in every folding (41.) the total weight of the form, that is, the sum of the weights of the individual variable rows (cf. § 1), decreases by the positive number $2(\lambda^2 + \lambda(\sigma - \tau))$. This implies,

[65] Clebsch, loc. cit., § 7.

independently of the general finiteness proof, the finiteness of the form row, that is, of the totality of invariant formations of a given ground form that are linear in the coefficients.[66]

It should further be noted that Theorem VI still holds for forms that do not satisfy conditions (12.). For each factor in (3), $(S^\rho \,|\, p_\rho)^{m_\rho}$, can be replaced by a product of m_ρ linear factors $(A^\rho \,|\, p_\rho)(B^\rho \,|\, p_\rho) \cdots (M^\rho \,|\, p_\rho)$; the invariant formations thereby pass into those of a system of linear forms, which need only afterwards be set equal to one another. For every single linear form, however, conditions (12.)–which, when differential symbols are introduced, contain only second-order differential quotients–are always satisfied.

§ 7. Expansion of the general forms according to normal forms.

In what follows (§ 8) we shall derive a principal property of the folding process (41.), namely its composability from fundamental foldings. For this we need the transfer to the n-ary domain of an important theorem given by Mertens[67] in the quaternary domain. This theorem says that any finite system of forms can be replaced by an equivalent system of normal forms. By a "normal form" we mean–generalizing the binary and ternary definition[68]–a form all of whose invariant formations linear in the coefficients vanish identically, with the exception of the ground form itself. Thus by Theorem VI (§ 6) a normal form φ satisfies the system of differential equations

$$\left(q_{n-\sigma-\lambda}\frac{\partial}{\partial p_\sigma}\middle|\frac{\partial}{\partial q_{n-\tau}}p_{\tau-\lambda}\right)\varphi = 0, \qquad (\sigma \geq \tau,\ \lambda = 1, 2, \ldots, \min(\tau, n-\sigma)), \tag{46}$$

where the rows $q_{n-\sigma-\lambda}, p_{\tau-\lambda}$ are to be regarded as arbitrary quantities. In the quaternary domain–taking (39.) into account for $\sigma = \tau = 2$–these differential equations agree completely with the ten differential equations given by Mertens without introduction of the arbitrary rows p, q (loc. cit., formula 28). Their knowledge, together with Theorem VI, allows us to transfer his proof almost word for word to n variables. The only thing to add is the verification, obtained by the decomposition identity (30.), that the constructed forms are normal forms; in the quaternary domain this verification still follows without further transformation. It is therefore enough to give a short exposition of the proof, and for simplicity in the case relevant below, namely that of a single ground form f and its form row (cf. § 6, end); for invariant formations of arbitrary degree in the coefficients of arbitrarily many ground forms, the proof remains the same.

The basic idea of the proof is to replace, by introducing the transformed coefficients, a form with $n-1$ rows p_ρ by forms with n rows ξ cogredient to the x, namely the rows of the transformation quantities. From these forms the sought normal forms are obtained directly by invariant differentiation processes. The expansion of the general forms according to normal forms is mediated by a series expansion for forms with ρ ($\rho < n$) cogredient rows ξ; invariant differentiation processes lead back to the forms in the original rows p_ρ.

Let $\xi^{(1)}, \ldots, \xi^{(n)}$ be the rows of transformation quantities, so that

$$\begin{gathered} x_i = \xi_i^{(1)} x'_1 + \xi_i^{(2)} x'_2 + \cdots + \xi_i^{(n)} x'_n, \\ p_{i_1 \ldots i_\rho} = \sum_j (\xi_{i_1}^{(j_1)} \cdots \xi_{i_\rho}^{(j_\rho)}) p'_{j_1 \ldots j_\rho}. \end{gathered} \tag{47}$$

The transformed coefficients of the forms $f, \varphi, \ldots$ will be denoted by $(f), (\varphi), \ldots$, and let

$$f = (S^1 \,|\, p_1)^{m_1} (S^2 \,|\, p_2)^{m_2} \cdots (S^{n-1} \,|\, p_{n-1})^{m_{n-1}}.$$

Then

$$\begin{aligned} (f) = (S^1 \,|\, \xi^{(1)})^{m_{11}} \cdots (S^1 \,|\, \xi^{(n)})^{m_{1n}} (S^2 \,|\, \xi^{(1)}\xi^{(2)})^{m_{21}} \cdots (S^{n-1} \,|\, \xi^{(2)} \cdots \xi^{(n)})^{m_{n-1,n}} \\ \cdot (s^{(1)} \,|\, \xi^{(1)})^{k_1} (s^{(2)} \,|\, \xi^{(2)})^{k_2} \cdots (s^{(n)} \,|\, \xi^{(n)})^{k_n}, \end{aligned} \tag{48}$$

where $m_{11} + \cdots + m_{1n} = m_1$, and so on. From each coefficient (f) for which

$$k_1 \geq k_2 \geq k_3 \geq \cdots \geq k_n, \tag{49}$$

the differentiation process exhausting exactly the rows ξ,

$$\left(\frac{\partial}{\partial \xi^{(1)}}\middle| p_1\right)^{k_1-k_2} \left(\frac{\partial}{\partial \xi^{(1)}}\frac{\partial}{\partial \xi^{(2)}}\middle| p_2\right)^{k_2-k_3} \cdots \left(\frac{\partial}{\partial \xi^{(1)}} \cdots \frac{\partial}{\partial \xi^{(n-1)}}\middle| p_{n-1}\right)^{k_{n-1}-k_n} \left(\frac{\partial}{\partial \xi^{(1)}} \cdots \frac{\partial}{\partial \xi^{(n)}}\right)^{k_n} \tag{50}$$

[66] Cf. Clebsch, loc. cit., § 11 and 12: finiteness of the system of elementary covariants.

[67] F. Mertens, *Über invariante Gebilde quaternärer Formen* (see the Introduction), cited as "Mertens."

[68] Study, loc. cit., p. 54.

gives the form

$$\varphi = \left(s^{(1)} \mid p_1\right)^{k_1-k_2}\left(s^{(1)}s^{(2)} \mid p_2\right)^{k_2-k_3}\cdots\left(s^{(1)}\cdots s^{(n-1)} \mid p_{n-1}\right)^{k_{n-1}-k_n}\left(s^{(1)}s^{(2)}\cdots s^{(n)}\right)^{k_n}. \tag{51}$$

The forms φ are the sought normal forms. Indeed, they have the following properties defining normal forms.

1. They are invariant formations of the ground form f that are linear in the coefficients. This follows from the fact that they arise from f by the invariant differentiation processes $\left(\frac{\partial}{\partial p_\rho}\Big|\xi^{(i_1)}\xi^{(i_2)}\cdots\xi^{(i_\rho)}\right)$ and (50.). Thus (by § 6, end) they can be expressed linearly by polars of the finite form row of f. These polars are obtained by regarding the variables p_ρ entering φ as coefficients of a form decomposed into linear forms with the variable rows p_ρ:

$$H = \left(q_{n-1} \mid p_{n-1}\right)^{k_1-k_2}\left(q_{n-2} \mid p_{n-2}\right)^{k_2-k_3}\cdots\left(q_1 \mid p_1\right)^{k_{n-1}-k_n}. \tag{52}$$

The simultaneous invariants of the form rows f and H give all relevant polars, since the latter must be of the same dimension in the individual rows p_ρ as φ itself.

2. They satisfy the differential equations (46.). If the differential process (45.) is applied to φ, then

$$\varphi' = \left(q_{n-\sigma-\lambda}s^{(1)}\cdots s^{(\sigma)} \mid [s^{(1)}\cdots s^{(\tau)}]_{n-\tau}p_{\tau-\lambda}\right), \qquad (\sigma > \tau).$$

From the decomposition identity (30.) it follows, if the rows $q_\sigma = s^{(1)}\cdots s^{(\sigma)}$, $p_{n-\tau} = [s^{(1)}\cdots s^{(\tau)}]_{n-\tau}$ formed out of the rows s are identified with the q_ρ, p_τ occurring there, that

$$\varphi' = \sum_{\beta=\sigma-\tau+\lambda}^{n-\tau,\sigma}\sum_{i_\beta}\varepsilon\,(q_{\sigma-\beta}^{(i_1\ldots i_\beta)}q_{n-\tau+\lambda}\,|\,p_{\sigma+\lambda}p_{n-\tau-\beta}^{(i_1\ldots i_\beta)}).$$

Here

$$p_{n-\tau-\beta}^{(i_1\ldots i_\beta)} \sim s^{(1)}\cdots s^{(\tau)}s^{(i_1)}\cdots s^{(i_\beta)},$$

where $i_1,\ldots,i_\beta$ denote any β distinct numbers chosen from the numbers 1 to σ. Since $\beta > \sigma-\tau$, it follows that among these numbers $i_1,\ldots,i_\beta$ there must be some between 1 and τ; hence $p_{n-\tau-\beta}^{(i_1\ldots i_\beta)}$ vanishes identically, and therefore every individual matrix product, and with it φ', vanishes.

For the equivalence of the system of normal forms with the form row f, it remains only to show that all forms of the form row can be expanded in polars of the normal forms in the sense fixed under 1. Let Π be an aggregate of polar operations

$$\left(\frac{\partial}{\partial\xi^{(i)}}\bigg|\xi^{(k)}\right), \qquad (i \neq k;\ i = 1,2,\ldots,\rho;\ k = 1,2,\ldots,\rho), \tag{53}$$

and let Π^σ be an aggregate of the special operations (53.) for which $i = \sigma$ and $k < \sigma$. For a form F of the ρ rows $\xi^{(1)},\ldots,\xi^{(\rho)}$ that has degree k_ρ in the row $\xi^{(\rho)}$, the expansion[69] holds:

$$F = \sum_{\alpha=0}^{k_\rho}\Pi F_\alpha, \qquad (\rho \leq n), \tag{54}$$

where F_α has degree α in the last row $\xi^{(\rho)}$ and arises from F by the invariant differential operations

$$\left(\frac{\partial}{\partial\xi^{(1)}}\cdots\frac{\partial}{\partial\xi^{(\rho)}}\bigg|\xi^{(1)}\cdots\xi^{(\rho)}\right)^\alpha \tag{55}$$

and Π^ρ. If, in constructing F_α, one replaces process (55.) by

$$\left(\frac{\partial}{\partial\xi^{(1)}}\cdots\frac{\partial}{\partial\xi^{(\rho)}}\bigg|p_\rho\right)^\alpha, \tag{56}$$

then a form $F_\alpha(p_\rho)$ with only $\rho-1$ rows $\xi^{(1)},\ldots,\xi^{(\rho-1)}$ is obtained, to which (54.) can again be applied. Continuing the procedure leads to forms $G(p_\rho,p_{\rho-1},\ldots,p_1)$. In order to obtain from these the expansion of F according to (54.), one replaces the individual rows p_ρ by $\xi^{(1)},\ldots,\xi^{(\rho)}$ and applies to the resulting forms

[69] F. Mertens, *Über eine Formel der Determinantentheorie.* Sitzungsberichte der Akademie der Wissenschaften, Vienna, Math.-Naturw. Kl., vol. 91, 2nd section (1885), formula 20).

the polar process Π (53.); this gives (cf. (48.)) a sum of transformed coefficients (G). For $\rho = n$, the process (55.) remains at the first step. If c denotes numerical constants and we put, for brevity,

$$(\xi^{(1)}\xi^{(2)}\cdots\xi^{(n)}) \sim \Delta; \qquad \left(\frac{\partial}{\partial\xi^{(1)}}\frac{\partial}{\partial\xi^{(2)}}\cdots\frac{\partial}{\partial\xi^{(n)}}\right) = \nabla, \tag{57}$$

then, in the case $\rho = n$,

$$F = \sum c\Delta^{\alpha}(G), \tag{58}$$

whereas for $\rho < n$ only $\alpha = 0$, and hence $\sum c(G)$, appears on the right.

Applying (58.) to a transformed coefficient (f) (48.), and using the commutativity of the polar processes Π^{σ} with all operations (56.) for which $\rho \geq \sigma$, as well as the fact that polars of transformed coefficients again give transformed coefficients, one obtains

$$G = \left(\frac{\partial}{\partial\xi^{(1)}}\bigg|p_1\right)^{\beta_1}\left(\frac{\partial}{\partial\xi^{(1)}}\frac{\partial}{\partial\xi^{(2)}}\bigg|p_2\right)^{\beta_2}\cdots\left(\frac{\partial}{\partial\xi^{(1)}}\cdots\frac{\partial}{\partial\xi^{(n-1)}}\bigg|p_{n-1}\right)^{\beta_{n-1}}\nabla^{\beta_n}\Pi(f) = \sum c\varphi,$$

and from (58.) it follows that

$$(f) = \sum c\Delta^{\alpha}(\varphi) \qquad \text{(Mertens, formula 23)).} \tag{59}$$

To pass from (59.) to the expansion of f itself, we again regard the variables p_ρ as coefficients of a form H (52.) with the degrees $m_1, \ldots, m_{n-1}$ corresponding to f, and put $m = m_1 + \cdots + m_{n-1}$. Then, by the definition of invariants,

$$f \cdot \Delta^m = \sum(f)(H) = \sum c\Delta^{\alpha}(\varphi)(H),$$

and application of the process ∇^m that exactly exhausts the rows ξ gives

$$f = \sum c\Phi, \tag{60}$$

where the forms Φ are derived from φ and H by invariant differentiation processes with respect to the variables, and hence are simultaneous invariants of the form rows φ and H.[70] But since φ is a normal form, the form row φ reduces to the form φ itself; the form row H contains all possible invariant variable combinations. Thus the simultaneous invariants of φ and H are the polars of φ in the sense indicated under 1.; in other words, (60.) gives the expansion of f in polars of the normal forms. It remains to verify the same expansion for an arbitrary form θ of the form row f. Let Γ be the normal forms derived from (θ) by (50.), so that

$$(\theta) = \sum c\Delta^{\beta}(\Gamma). \tag{61}$$

The forms θ are characterized by the relation, valid for every transformed coefficient,

$$(\theta)\Delta^{\sigma} = \sum c(f) \qquad \text{(Mertens, formula 9)).}$$

But every form F of the n rows ξ which contains the factor Δ^{σ} also contains the second $\nabla^{\sigma}\Pi F$ (Mertens, formula 20)); hence

$$(\theta) = \sum c\nabla^{\sigma}(f), \qquad \text{therefore} \qquad \Gamma = \sum c\varphi,$$

and from (61.) we obtain the relation corresponding to (59.),

$$(\theta) = \sum c\Delta^{\beta}(\varphi),$$

which implies for θ the expansion (60.) in polars of the normal forms φ. Thus:

Theorem VII. Every form row f can be replaced by an equivalent system of normal forms.

[70] Instead of the direct conclusion, Mertens, § 7, introduces a series of further differential processes that essentially serve to form the form row H; this is done by our Theorem VI.

§ 8. Reduction of the folding process to fundamental foldings.

Following Theorem VII, we now carry out the reduction of the folding process to fundamental foldings mentioned at the beginning of § 7. We call (cf. § 5) every folding of two cogredient rows S^ρ, T^ρ a "cogredient folding", and specifically the cogredient folding of defect 1 of two rows S^ρ, T^ρ a "fundamental folding of type ρ". We then prove, completing Theorem VI:

Theorem VIII: The general folding process (41.) can be composed from the $(n-1)$ fundamental foldings of type ρ, where $\rho = 1, 2, \ldots, n-1$. In other words, to generate all invariant formations it suffices to apply successively the $n-1$ fundamental foldings.

In the binary domain the folding process (41.) coincides directly with the only possible fundamental folding; in the ternary domain I proved Theorem VIII in § 1 of my dissertation (cf. the Introduction). There, because of (44.), the general cogredient foldings are still identical with the fundamental foldings, and it is noteworthy that Theorem VIII for n variables gives the reduction to fundamental foldings, not merely the much weaker reduction to cogredient foldings. The proof is modeled in principle on the ternary proof: the most general foldings are constructed from the fundamental foldings, using the symbolic identities. We shall generate:

1. arbitrary foldings of defect 1 by cogredient foldings of defect 1, that is, by fundamental foldings (the analogue of the ternary case),
2. arbitrary foldings of defect λ by cogredient foldings of defect λ and arbitrary foldings of defect 1,
3. cogredient foldings of defect λ by cogredient foldings of defect $\lambda-1$ and arbitrary foldings of defect 1.

As is essential, we assume by Theorem VII that the two forms S and T to be folded are normal forms. Thus, by (46.), the relations hold:

$$\begin{aligned} (q_{n-\sigma_1-\lambda} S^{\sigma_1} \,|\, S_{n-\sigma_2} p_{\sigma_2-\lambda}) = 0, &\quad (\sigma_1 \geq \sigma_2,\ \lambda = 1, 2, \ldots, \sigma_2, n-\sigma_1), \\ (q_{n-\tau_1-\lambda} T^{\tau_1} \,|\, T_{n-\tau_2} p_{\tau_2-\lambda}) = 0, &\quad (\tau_1 \geq \tau_2,\ \lambda = 1, 2, \ldots, \tau_2, n-\tau_1). \end{aligned} \tag{62}$$

It is further sufficient, by § 2 (end), to suppose that the forms S and T are linear in all variable rows; that is, the constructed forms belong to the form row

$$f = (S^1 \,|\, p_1)(S^2 \,|\, p_2) \cdots (S^{n-1} \,|\, p_{n-1})(T^1 \,|\, p_1)(T^2 \,|\, p_2) \cdots (T^{n-1} \,|\, p_{n-1}).$$

1) Generation of $(q_{n-\sigma-1} S^\sigma \,|\, T_{n-\tau} p_{\tau-1})$, where $\sigma > \tau$, from fundamental foldings of type $\tau + \alpha$, $\alpha = 0, 1, \ldots, \sigma - \tau$. From the fundamental folding of type τ

$$(q_{n-\tau-1} S^\tau \,|\, T_{n-\tau} p_{\tau-1})$$

we obtain, by the substitution $w \sim T_{n-\tau} p_{\tau-1}$, a form of the form row f with three rows of upper weight index (cf. § 1) $\tau + 1$:

$$(w S^\tau \,|\, p_{\tau+1})(S^{\tau+1} \,|\, p_{\tau+1})(T^{\tau+1} \,|\, p_{\tau+1}). \tag{63}$$

A fundamental folding of type $\tau + 1$ applied to (63.) gives

$$(q_{n-\tau-2} w S^\tau \,|\, S_{n-\tau-1} p_\tau);$$

from (31.), by the substitution $q_{n-\tau-2} w = q'_{n-\tau-1}$, this has the value

$$\varepsilon (q_{n-\tau-2} w S^{\tau+1} \,|\, S^\tau p_\tau) + \sum_{\alpha=1}^{\tau, n-\tau-1} \Pi (q'_{n-\tau-1-\alpha} S^{\tau+1} \,|\, S_{n-\tau} p_{\tau-\alpha}).$$

The expression under the summation sign vanishes identically by (62.); thus we have reached a form

$$(w S^{\tau+1} \,|\, p_{\tau+2})(S^{\tau+2} \,|\, p_{\tau+2})(T^{\tau+2} \,|\, p_{\tau+2}),$$

which is obtained from (63.) by changing τ into $\tau + 1$. Repeating the process $\sigma - \tau$ times therefore leads to the desired folding:

$$(w S^\sigma \,|\, p_{\sigma+1}) = \varepsilon (q_{n-\sigma-1} S^\sigma \,|\, T_{n-\tau} p_{\tau-1}).$$

We indicate the process just carried out by the formula

$$T^\tau S^\tau, \quad S^{\tau+1}, \quad S^{\tau+2}, \ldots, S^\sigma; \tag{64}$$

and see that only the normal-form property of S, not that of T, is used. A process dual to (64.) can be performed, using only the normal-form property of T, in the reverse order, namely:

$$S^\sigma T^\sigma, \quad T^{\sigma-1}, \quad T^{\sigma-2}, \ldots, T^\tau, \tag{65}$$

according to the relations:

$$\begin{aligned}
(q_{n-\sigma-1}S^\sigma \,|\, T_{1-\sigma}p_{\sigma-1}) &= \varepsilon(T_{n-\sigma}y \,|\, p_{\sigma-1}), \quad y \sim q_{n-\sigma-1}S^\sigma,\\
(q_{n-\sigma}T^{\sigma-1} \,|\, T_{n-\sigma}yp_{\sigma-2}) &= \varepsilon(T_{n-\sigma+1}y \,|\, p_{\sigma-2}),\\
&\ \,\vdots\\
(q_{n-\tau-1}T^\tau \,|\, T_{n-\tau-1}yp_{\tau-1}) &= \varepsilon(q_{n-\sigma-1}S^\sigma \,|\, T_{n-\tau}p_{\tau-1}).
\end{aligned}$$

We should also point out the connection with the decrease of total weight given in § 6 (end): for foldings of defect 1 this decrease has the value $2(1+(\sigma-\tau))$, and for fundamental foldings the value $2\cdot 1$. Therefore, if composition out of fundamental foldings is possible at all, exactly $\sigma-\tau+1$ such foldings are necessarily required, as was carried out above.

2) Generation of general foldings from cogredient and fundamental foldings. The weight decrease for a general folding (41) is $2(\lambda^2+\lambda(\sigma-\tau)) = 2[\lambda^2+(\sigma-\tau)(1+(\lambda-1))]$. This gives the possibility of decomposing it into a cogredient folding of defect λ (weight decrease $2\lambda^2$) and $\sigma-\tau$ foldings of defect 1 with difference of weight indices $\lambda-1$ (weight decrease $2\{1+(\lambda-1)\}$). In the case $\lambda=1$ this decomposition agrees with the one given under 1); we show that it can indeed be realized.

From the cogredient folding of defect λ

$$(q_{n-\tau-\lambda}S^\tau \,|\, T_{n-\tau}p_{\tau-\lambda})$$

we obtain, by the substitution $R^\lambda \sim T_{n-\tau}p_{\tau-\lambda}$, a form with the two rows of upper weight indices $\tau+\lambda$ and $\tau+1$:

$$(R^\lambda S^\tau \,|\, p_{\tau+\lambda})(S^{\tau+1} \,|\, p_{\tau+1}). \tag{66}$$

The row with the lower weight index $\tau+1$ belongs to a normal form; hence (66.) admits a folding of defect 1 composed out of λ fundamental foldings, in the order (65):

$$(R^\lambda S^\tau)S^{\tau+\lambda}, \quad S^{\tau+\lambda-1}, \quad S^{\tau+\lambda-2}, \ldots, S^{\tau+1}.$$

Taking account of (31.) and (62.), this gives

$$(q_{n-\tau-\lambda-1}R^\lambda S^\tau \,|\, S_{n-\tau-1}p_\tau) = \varepsilon(R^\lambda S^{\tau+1} \,|\, p_{\tau+\lambda+1}),$$

that is, a form obtained from (66.) by replacing τ by $\tau+1$, just as in 1) for (63.). Repeating the process $\sigma-\tau$ times leads to the desired folding:

$$(R^\lambda S^\sigma \,|\, p_{\sigma+\lambda}) = \varepsilon(q_{n-\sigma-\lambda}S^\sigma \,|\, T_{n-\tau}p_{\tau-\lambda}).$$

In the whole process only the normal-form property of S was used. Viewing T as a normal form gives the dual process, with complete reversal of the order, according to the formulas

$$\begin{aligned}
(q_{n-\sigma-\lambda}S^\sigma \,|\, T_{n-\sigma}p_{\sigma-\lambda}) &= \varepsilon(T_{n-\sigma}R_\lambda \,|\, p_{\sigma-\lambda}), \quad R_\lambda \sim q_{n-\sigma-\lambda}S^\sigma,\\
(q_{n-\sigma}T^{\sigma-1} \,|\, T_{n-\sigma}R_\lambda p_{\sigma-\lambda-1}) &= \varepsilon(T_{n-\sigma+1}R_\lambda \,|\, p_{\sigma-\lambda-1}),\\
&\ \,\vdots\\
(q_{n-\tau-1}T^\tau \,|\, T_{n-\tau-1}R_\lambda p_{\tau-\lambda}) &= \varepsilon(q_{n-\sigma-\lambda}S^\sigma \,|\, T_{n-\tau}p_{\tau-\lambda}).
\end{aligned}$$

3) Generation of cogredient foldings of defect λ from cogredient foldings of defect $\lambda-1$ and fundamental foldings. The weight decrease for cogredient foldings of defect λ is $2\lambda^2 = 2((\lambda-1)^2+2\lambda-1)$, giving the possibility of a decomposition into a cogredient folding of defect $\lambda-1$ and $2\lambda-1$ fundamental foldings.

This is most easily found by first putting $n = 2\lambda$. The only possible cogredient folding of defect λ is $(S^\lambda \,|\, T_\lambda) = (S^\lambda \,|\, T_{n/2})$; it contains one row u and one row x fewer than the folding of defect $\lambda - 1$ of the same rows, $(uS^\lambda \,|\, T_\lambda x)$. Conversely, we compose the folding of defect λ out of fundamental foldings which effect the disappearance of a row u and a row x (weight decrease $2(n-1) = 2(2\lambda - 1)$) and at the same time generate a new symbol row R^λ, together with a cogredient folding of defect $\lambda - 1$. The simplest folding that does this is the following one of defect 1:

$$(sT^\lambda \,|\, \sigma p_\lambda) = \varepsilon(S^{2\lambda-1} \,|\, R_{\lambda-1} p_\lambda) = \varepsilon(R^\lambda \,|\, p_\lambda),$$

where

$$R^{\lambda+1} = sT^\lambda, \quad s = S^1, \quad \sigma = S_1, \quad R^\lambda \sim \sigma R_{\lambda-1}.$$

By (65.) and (64.) it is composed out of the $2\lambda - 1$ fundamental foldings

$$T^\lambda S^\lambda, S^{\lambda-1}, \ldots, S^1; \qquad R^{\lambda+1} S^{\lambda+1}, S^{\lambda+2}, \ldots, S^{2\lambda-1}. \tag{67}$$

A folding of defect $\lambda - 1$, taking account of (21.) and (62.), gives

$$(uS^\lambda \,|\, \sigma R_{\lambda-1} x) = \varepsilon(R^{\lambda+1} \,|\, S_\lambda x) = \varepsilon(sT^\lambda \,|\, S^\lambda x) = \varepsilon(S^\lambda \,|\, T_\lambda).$$

The general case, for two rows S^ρ, T^ρ and arbitrary n, is obtained directly from the special one. In place of (67.) one has the following $2\lambda - 1$ fundamental foldings:

$$T^\rho S^\rho, S^{\rho-1}, \ldots, S^{\rho-\lambda+1}; \qquad R^{\rho+1} S^{\rho+1}, S^{\rho+2}, \ldots, S^{\rho+\lambda-1}. \tag{68}$$

The first half of the foldings gives the form

$$(q_{n-\rho-1} T^\rho \,|\, S_{n-\rho+\lambda-1} p_{\rho-\lambda}),$$

from which, by the substitutions

$$S_{n-\rho+\lambda-1} p_{\rho-\lambda} \sim w, \qquad T^\rho w = R^{\rho+1},$$

and application of the second half of the foldings, one obtains

$$(q_{n-\rho-\lambda} S^{\rho+\lambda-1} \,|\, R_{n-\rho-1} p_\rho).$$

Substituting

$$q_{n-\rho-\lambda} S^{\rho+\lambda-1} \sim y,$$

one obtains, by a cogredient folding of defect $\lambda - 1$,

$$(q_{n-\rho-\lambda+1} S^\rho \,|\, y R_{n-\rho-1} p_{\rho-\lambda+1}). \tag{69}$$

We show that this is the desired folding of defect λ. Substitute

$$R_{n-\rho-1} p_{\rho-\lambda+1} = R_{n-\lambda}$$

and observe that resolving y gives

$$(q_{n-\rho-\lambda+1} R^\lambda \,|\, S_{n-\rho} y) = \varepsilon(q_{n-\rho-\lambda} S^{\rho+\lambda-1} \,|\, S_{n-\rho} p'_{\rho-1}) = 0.$$

Thus by (20.) the value of (69.) is

$$\varepsilon(S^\rho R^\lambda \,|\, p_{\rho+\lambda-1} y) = \varepsilon(q_{n-\rho-\lambda} S^{\rho+\lambda-1} \,|\, [S^\rho R^\lambda]_{n-\rho-\lambda} p_{\rho+\lambda-1}).$$

This folding is of type (43.), hence equivalent to

$$(q_{n-\rho-\lambda} S^\rho \,|\, R_{n-\rho-1} p_{\rho-\lambda+1}) = \varepsilon(wT^\rho \,|\, R_\lambda p_{\rho-\lambda+1}), \qquad R_\lambda \sim q_{n-\rho-\lambda} S^\rho.$$

Here R_λ is already of the desired final form; the expression corresponds to the form $(sT^\lambda \,|\, S_\lambda x)$ in the special case. Now, observing that resolving w and R_λ gives

$$(wR^{n-\lambda} \,|\, T_{n-\rho} p_{\rho-\lambda+1}) = \varepsilon(q'_{n-1} R^{n-\lambda} \,|\, S_{n-\rho+\lambda-1} p_{\rho-\lambda}) = \varepsilon(q''_{n-\sigma-1} S^\rho \,|\, S_{n-\rho+\lambda-1} p_{\rho-\lambda}) = 0,$$

we get by (21.) the value

$$(wq_{n-\rho+\lambda-1} \,|\, T_{n-\rho}R_\lambda) = \varepsilon(q_{n-\rho+\lambda-1}[T_{n-\rho}R_\lambda]^{n-\lambda} \,|\, S_{n-\rho+\lambda-1}p_{\rho-\lambda}) \quad \text{equivalent to} \quad (q_{n-\rho-\lambda}S^\rho \,|\, T_{n-\rho}p_{\rho-\lambda}).$$

In the whole process only the normal-form property of S, not that of T, was used. Thus the cogredient folding of defect $\lambda - 1$ used to generate (69.) can be replaced by $2\lambda - 3$ fundamental foldings and a cogredient folding of defect $\lambda - 2$, which in turn admits a decomposition into fundamental foldings, and so on. Analogously, the generation is obtained by using the normal-form property of T and applying the dualistic process, that is, the fundamental foldings

$$S^\rho T^\rho, \quad T^{\rho-1}, \ldots, T^{\rho-\lambda+1}; \qquad R^{\rho-1}T^{\rho-1}, \quad T^{\rho-2}, \ldots, T^{\rho-\lambda+1},$$

and a folding dual to (69.). The folding

$$(q_{n-\rho-\lambda}T^\rho \,|\, S_{n-\rho}p_{\rho-\lambda}),$$

equivalent to the preceding one, is then obtained.

Adding the decomposition obtained under 2), we have indeed obtained a generation of the most general foldings from fundamental foldings. It remains to show that this generation is unique, apart from permutation of the order, i.e. that to each folding there corresponds one and only one number α_ρ of fundamental foldings of type ρ, where $\rho = 1, 2, \ldots, n-1$.

Let m_ρ be the degrees in the variable rows p_ρ of the initial form f, and l_ρ the degrees of the form produced by folding. Each fundamental folding of type ρ transforms the degrees $m_{\rho-1}, m_\rho, m_{\rho+1}$ respectively into $m_{\rho-1}+1, m_\rho - 2, m_{\rho+1}+1$. Thus the α satisfy the system of equations

$$m_\rho - l_\rho = -\alpha_{\rho-1} + 2\alpha_\rho - \alpha_{\rho+1}, \qquad (\rho = 1, 2, \ldots, n-1), \tag{70}$$

where α_0 and $\alpha_n = 0$ are to be put, and whose determinant has value n.[71] The α are thereby uniquely determined, and indeed, by Theorem VIII, as positive integers; this gives conditions for the differences $m_\rho - l_\rho$.

The results of this paragraph hold by virtue of relations (62.); however, if the relations (62.) do not hold, the final expressions are by no means equivalent to the initial expressions. The definition of equivalence given in § 6 also admits the formulation:[72] "Two forms are equivalent if and only if they differ by forms of higher folding, i.e. by forms that exhibit a greater weight decrease." This greater weight decrease always occurs when the two rows $q_{n-\sigma-i}, p_{\tau-\lambda}$ entering the folding are really variable rows, as happens in (41.), (42.), in the foldings appearing in this paragraph as equivalent, and also in the equivalent forms of Theorem VII. But it need not occur when these rows are derived from symbol rows by substitution, as was the case for the forms of this paragraph which could be expressed in terms of one another by virtue of (62.). One easily sees that the vanishing forms even have, in part, a higher total weight.

§ 9. Form rows. Reduction theorems.

We now define the form row introduced in § 6 (end) somewhat more sharply as "the totality of invariant formations of a given initial form that are linear in the coefficients, i.e. the totality of forms that arise from the initial form by folding in itself, arranged according to higher forms."

To explain the higher forms, suppose by Theorem VII that the initial form is given as a product of two normal forms $S \cdot T$. By a higher form we mean forms with higher folding in itself, and among forms with the same folding in itself, specially normalized ones. More precisely: suppose the form A has arisen by α_ρ fundamental foldings of type ρ, and the form B by β_ρ such foldings. Then B is higher than A:

1. if, in the system of inequalities $\beta_\rho \geq \alpha_\rho$, $\rho = 1, 2, \ldots, n-1$, the inequality sign holds for at least one value of ρ;[73]

2. if the equality sign holds for all values ρ, but fewer symbol rows have been combined into a single matrix product. This normalization proves necessary because of the reductions in § 8. Accordingly, for example, $(S^{n-1} \,|\, T_{n-1})$ is higher than $(S^{n-1} \,|\, [S^1T^\rho]_{n-\rho-1}p_\rho)$;

[71] If Δ_n denotes the n-rowed determinant, then the recurrence formula holds: $\Delta_n = 2\Delta_{n-1} - \Delta_{n-2}$; that is, $\Delta_n - \Delta_{n-1} = \Delta_{n-1} - \Delta_{n-2} = \cdots = \Delta_2 - \Delta_1 = 1$, since $\Delta_2 = 3$, $\Delta_1 = 2$.

[72] Cf. the consideration at the end of the next paragraph.

[73] If in the system of inequalities the sign $>$ holds in part and the sign $<$ in part, then the forms are not comparable, since a form arising from another by folding always has a positive value system of fundamental foldings, hence in our case positive or vanishing values $\beta_\rho - \alpha_\rho$.

3. if the equality sign holds for all values ρ and the number of symbol rows combined into one matrix product is the same, then B is made higher than A by a normalization arbitrary in itself but fixed once and for all. This normalization is always possible because of the finite number of forms belonging to a value system α; it can be achieved, for instance, by privileging the one form S (larger number of symbol rows S, larger sum of the upper weight indices of the rows S, and so on).

It is useful to think of the form row as arranged as an $(n-1)$-dimensional manifold of lattice points, where the $n-1$ directions correspond to the $n-1$ fundamental foldings. To every form there then corresponds one and only one value system α, i.e. a positive integral lattice point, while the different forms belonging to a value system α are uniquely determined in their order by the above normalization. An arbitrary form of the form row gives the initial form of a new form row, contained as a part of the total form row; indeed it is contained as the part consisting of the forms obtained from the new initial form by proceeding in the $n-1$ positive directions.

By virtue of Theorem VIII and the general theory of series expansions, exactly the reduction theorems of the binary and ternary domains (cf. dissertation, § 3) hold for form rows. We shall briefly derive the principal theorem. We first state the definition:

"In a given system of forms, suppose a certain finite number of forms has been combined into a module. We call a form reducible if it can be expressed by forms that have invariants as factors, or by 'higher forms'; that is, by higher forms of the total form row to which the form belongs, or by forms that contain the symbols of the module in higher order. A reducible form row is called a reducer." Then:

Theorem IX: If the initial form of a form row is reducible by having arisen through folding with a reducer, then the entire form row arising from this initial form is reducible.

For the proof, observe that a form h arising by folding a form f with a second form g can, because of (45.), be regarded as a form f containing several cogredient variable rows p_ρ, p'_ρ. Such forms can, by the general theory of series expansion, be represented as polars of the elementary covariants of f, by applying the series expansion successively to each pair of cogredient variable rows p_ρ, p'_ρ. By (38.), the elementary covariants that occur are the forms generated by cogredient folding. Since, however, the order in which the series expansion is applied is still arbitrary, we may in particular choose an order corresponding to the generation of the general foldings from fundamental foldings. The resulting system of elementary covariants is then identical with the form row f; the forms that arise by cogredient folding of higher defect are arranged among those that arise by fundamental foldings. But one also obtains polars of the form row f for any order of the series expansion, to which a second system of elementary covariants may correspond. Since the form row exhausts the totality of invariant formations, every form of the second system can be expressed linearly by polars of the form row, and indeed by polars of those forms that show the same or a greater weight decrease. The latter follows because (cf. § 7) the polars can be regarded as simultaneous invariants of a form row H (52.), with the degree numbers corresponding to the form h, and of the form row f. Every folding of H in itself corresponds to a weight decrease which, after substitution (11.) is performed, is transferred to the symbol rows and hence to the forms of f.[74]

An arbitrary form of the form row h has arisen by folding h in itself; that is, on the one hand by a higher folding of g with f, and on the other by folding f or g in itself. In both cases the series expansion leads to polars of the form row f, just as was shown above for the initial form h. If in particular f is a reducer, then an expansion in polars of the reducer results; hence the form row h becomes reducible.

Applications of the theorem to the reduction of complete systems of forms follow analogously to the binary and ternary domains.

[74]This consideration also underlies the second formulation of the equivalence concept (§ 8, end). Further explanations of the different systems of elementary covariants in the ternary domain are found in Study, loc. cit., II, § 7, Theorems 7) and 8). By our Theorem VII the same results also hold for n variables.

5. Rational Function Fields

Jahresbericht der DMV 22 (1913), pp. 316–319

The following questions originally go back to conversations with E. Fischer. Some of the questions, moreover, were already raised and settled by E. Steinitz for the special case of one indeterminate — and indeed under more general assumptions on the coefficient field.[75]

By a "rational function field" I understand a field whose elements are rational functions of n indeterminates, with coefficients from a prescribed number field, which in particular may also comprise all complex numbers. Examples of such fields are the field of symmetric functions of n quantities, or more generally the totality of rational functions of n indeterminates that allow the permutations of a certain group (Lagrange's *Gattungsbereiche*); furthermore, the invariant field.

Between the first two fields just mentioned and the latter there is a characteristic difference: the former contain n algebraically independent functions, the elementary symmetric functions; whereas the number of algebraically independent invariants is always smaller than the number of indeterminates. It is therefore important that, in general, one can associate such fields of the second kind one-to-one with fields of the first kind, by replacing some of the indeterminates by numbers. In what follows I shall therefore restrict myself to fields of the first kind, i.e. to fields with n algebraically independent functions; by virtue of the association, the results then hold in general.

The questions group themselves around three basis concepts: rational basis, minimal basis, and integrality basis.

1. By a "rational basis" I understand a finite number of functions of the field such that every function of the field can be represented as a rational combination of this finite number, with coefficients from the prescribed coefficient field. By simple considerations — which in the case of one indeterminate are also found in E. Steinitz, loc. cit. — one proves the existence of a rational basis for every rational function field. For example, by Lagrange's theorem, the elementary symmetric functions and one function belonging to the group form a rational basis for the Lagrange *Gattungsbereiche* mentioned earlier.

2. Every rational basis must contain (for fields of the first kind) at least n functions; if it contains exactly n functions, which then must be algebraically independent, I call it a "minimal basis." The question of the existence of a minimal basis can be answered in part by theorems of Lüroth, Castelnuovo, and Enriques.[76] According to these, for fields of one indeterminate there always exists a minimal basis: the Lüroth function of the field, uniquely determined up to fractional linear transformation (cf. Steinitz § 24). For fields of two indeterminates, a minimal basis exists always, and in general only, if one permits an algebraic extension of the coefficient field, whereas for three and more indeterminates a minimal basis in general does not exist at all. No attempt has yet been made to characterize the special fields with minimal basis.

I have pursued further the question of the minimal basis for Lagrange's *Gattungsbereiche.* Here it acquires the following meaning: "If the Lagrange *Gattungsbereich* belonging to the group G possesses a minimal basis, then one can construct rationally, by a parametric representation, the totality of equations with prescribed group G; and indeed for every number field that contains the coefficient field of the minimal basis — hence in particular for any arbitrary number field if this coefficient field is the field of rational numbers."

The simplest example is provided by the symmetric group; here the elementary symmetric functions form a minimal basis with rational numerical coefficients. From this one obtains as parametric representation simply the equation with indeterminate coefficients. Hilbert's irreducibility theorem shows how one passes from this to arbitrarily many affectless equations over every number field.

Now the existence of the minimal basis for all groups occurring in equations of degrees 3 and 4 follows directly from the theorems of Lüroth and Castelnuovo; at the same time it further appears that this minimal basis has rational numerical coefficients. Thus, for every arbitrary number field, one can rationally construct the totality of equations of degrees 3 and 4 with prescribed group. For the cyclic and dihedral groups there exists a minimal basis with the field of the nth roots of unity as coefficient field; the same holds for a few further metacyclic groups. The question whether the minimal basis is at all characteristic for certain groups must remain undecided.

3. To arrive at the last basis concept, I consider the totality of polynomials contained in the field. By

[75] E. Steinitz: Algebraische Theorie der Körper, especially § 24. Crelle's Journal 137. 1910.

[76] J. Lüroth: Beweis eines Satzes über rationale Kurven. Math. Ann. 9. 1875. — G. Castelnuovo: Sulla razionalità delle involuzioni piane. Math. Ann. 44. 1893. — F. Enriques: Sopra una involuzione non razionale dello spazio. Rend. Acc. Linc. vol. 21. 21 Jan. 1912.

an "integrality basis" I understand a finite number of these polynomials such that every polynomial of the field can be represented as an integral rational combination of this finite number, with coefficients from the prescribed coefficient field. The question of the existence of an integrality basis — which, for example, contains as a special case the finiteness of the invariant system — was posed by Hilbert in his "Mathematical Problems"[77] in a somewhat different formulation, as the problem of relatively integral functions.

For the moment I can indicate only one class of fields for which an integrality basis exists. Namely, if the field contains a system of n polynomials such that the homogeneous resultant of the terms of highest dimension of these polynomials is different from zero, then an integrality basis exists; it is given by the coefficients of all u in the equation, irreducible in the field, for the linear form:

$$u_0 = u_1x_1 + u_2x_2 + \cdots + u_nx_n.$$[78]

If, in particular, the value of the resultant is equal to a unit of the coefficient field, then the integrality of the representation is also secured. An example of this class of fields is furnished by Lagrange's *Gattungsbereiche*, since the resultant of the elementary symmetric functions has the value ± 1. A further example (without integrality) is given by all fields that contain only one algebraically independent polynomial; here the integrality basis is identical with the Lüroth function of the smallest subfield containing all polynomials. Thus, as is well known, all integral rational projective invariants of a quadratic form in n variables are integral functions of the discriminant.

The condition just given is only sufficient, by no means necessary, for the existence of an integrality basis. One can easily construct fields in which the resultant of every n polynomials vanishes, and which nevertheless possess an integrality basis given by the coefficients of that irreducible equation. The question arises whether perhaps even in the most general case the coefficients of that equation furnish the integrality basis.

[77] D. Hilbert: Mathematische Probleme. Lecture, Paris 1900. Problem 14. Göttinger Nachr. 1900.

[78] This irreducible equation is found, in the case of one indeterminate, in E. Steinitz's proof of Lüroth's theorem.

6. Fields and Systems of Rational Functions

Math. Ann. 76 (1915), pp. 161–196

The present paper treats basis questions for arbitrary systems of rational and integral rational functions. The methods of field theory used here make the settlement of these questions for fields of rational functions — rational function fields[79] — appear as the essential matter, while the generalization of the results to arbitrary systems arises as a consequence.

Among basis questions for general systems, up to now only the existence of a module basis, guaranteed by Hilbert's theorem (Math. Ann. 36), has been known. In what follows, the question of rational representability is answered completely by the existence of a rational basis for every arbitrary system (§ 7); the rational basis of fields is already obtained in § 4. This existence of the rational basis permits us throughout to start from the abstractly defined field or system, and thereby to avoid difficulties that are caused only by the special choice of the rational basis and not by the system itself, such as the occurrence of special denominators or of fundamental points of the basis functions.

For fields the question of the minimal basis is also treated, i.e. of a rational basis consisting of algebraically independent functions (§ 6). The methods of field theory further lead to a distinguished rational basis, the involution basis[80] (§ 5), which becomes especially important for integrality domains consisting of polynomials (§ 8). In that setting it gives a representation with a fixed denominator (a power of a function determined by the integrality domain), as is known for example in the special case of the typical representation of invariants.

From rational representability we now draw, as far as possible, consequences for integral rational representability, i.e. for the question of finiteness in the narrower sense.

The finiteness theorems known up to now all rest on the fact that Hilbert's theorem on module bases guarantees finiteness for all systems in which a representation

$$F = A_1 f_1 + A_2 f_2 + \cdots + A_k f_k$$

entails a second representation

$$F = B_1 f_1 + B_2 f_2 + \cdots + B_k f_k,$$

where the B_i belong to the system and are of lower degree than F.

By contrast, the theorem on the rational basis and the methods of field theory permit one to infer finiteness under assumptions of an entirely different kind.

Thus, as a partial answer to a problem of Hilbert (Mathematical Problems 14), one obtains a class of relatively integral functions (relatively integral domains of the first kind) which are finite integrality domains and whose integrality basis is given by the involution basis mentioned above (§ 10). In analogy with Hilbert's proof of Kronecker's theorem on the fundamental system of algebraic integers (Complete invariant systems, § 2; Math. Ann. 42), it can further be shown that all regular systems of polynomials — i.e. all systems that can be mapped to ones without fundamental points — possess an integrality basis (§ 12), whereas nonregular systems without integrality basis can easily be specified. As a simple example of this fact one may mention the theorem: "In every arbitrary system of infinitely many polynomials in one indeterminate, there exists a finite number of these polynomials such that every polynomial of the system can be represented as an integral rational combination of this finite number." Further examples are found in § 13.

Finally, the last paragraphs (§§ 14 and 15) give a sharpening of the basis theorems with respect to integrality.

An essential tool of the investigation is the transfer principle of § 3, according to which it suffices to consider all basis questions raised here for those systems in which the number of indeterminates coincides with the number of algebraically independent functions of the system.

The impetus for the present work came from conversations with Mr. E. Fischer, in particular from a question of his concerning the minimal basis of Lagrange's *Gattungsbereiche*. The investigations relating to this, which led to the construction of equations with prescribed group (cf. the cited communication on "Rational Function Fields," part 2), are reserved for a further publication.

[79]Cf. a preliminary communication on the occasion of the Vienna meeting of natural scientists: Rationale Funktionenkörper, Jahresber. d. D. Math.-Ver. 22 (1913), where an overview is given of the questions and of the results concerning function fields.

[80]The name was chosen because, in its geometric interpretation, the rational function field represents the most general involution in a linear space.

The essential extension of the present paper over the original communication consists in the fact that the basis theorems stated there only for fields or special integrality domains are here transferred to arbitrary systems. This transfer succeeds in a simple way by means of the concept of the smallest containing field of an arbitrary system, as a generalization of the quotient field of an integrality domain (§ 7). I was led to form this concept by the fact that Mr. K. Hentzelt, after learning the rational basis of fields, was able to prove the existence of the rational basis for arbitrary systems by a detour through Hilbert's module basis. I was also led to the theorem on the integrality basis of regular systems by K. Hentzelt's observation that the arguments I had applied to integrality domains were valid for arbitrary systems subject to the same assumptions; likewise I owe to Mr. Hentzelt a few isolated smaller remarks.

Finally, let it be mentioned that in the case of one indeterminate the rational basis and minimal basis of fields are found in E. Steinitz's "Algebraische Theorie der Körper," § 24 (Crelle's Journal 137); there the coefficient field is taken to be a field in the most general sense. The question how far the theorems given here remain valid under these more general assumptions is not touched on in what follows; in any event the proofs require modification, since, for example, the tools from the theory of functional determinants fail for fields of characteristic p.

§ 1. Fields $\mathfrak{K}_{n\rho}$ and systems $\mathfrak{S}_{n\rho}$.

In what follows we are concerned with the investigation of systems of rational functions in n indeterminates, especially fields of rational functions.

By $f(x_1,\ldots,x_n)$, $g(x_1,\ldots,x_n)$, ..., or abbreviated $f(x)$, $g(x)$, ..., one should therefore always understand rational functions of $x_1,\ldots,x_n$; these are assumed to be in reduced representation — that is, numerator and denominator relatively prime. Integral rational functions (polynomials) will be denoted by capital letters, $F(x)$, $G(x)$, ... The coefficient field Ω is assumed to be some number field; in particular, it may also contain all complex numbers. The field Ω may also contain a finite number of parameters.

The quantities from Ω, hence all functions of degree zero, are always counted as belonging to the system.

With these stipulations, the field consisting of rational functions — the rational function field — can be defined abstractly.

Definition I: A system of rational functions is called a field if it satisfies the conditions:

1. *Together with $f(x)$, and for every quantity c from Ω, it also contains $c \cdot f(x)$.*

2. *Together with $f(x)$ and $g(x)$, it always also contains $f(x)+g(x)$, $f(x)\cdot g(x)$ and — for $g(x) \neq 0$ — the quotient $f(x) : g(x)$.*[81]

Special fields are those which arise by adjoining a finite number of rational functions

$$f_1(x),\ f_2(x),\ldots,f_k(x)$$

to Ω; they will be denoted by

$$\Omega(f_1(x),\ldots,f_k(x)) \quad \text{or} \quad \Omega(f_1,\ldots,f_k).$$

[82] Among these special fields we also mention the field arising by adjoining $x_1,\ldots,x_n$ to Ω,

$$\Omega(x_1,\ldots,x_n),$$

which is identical with the totality of rational functions of $x_1,\ldots,x_n$ with coefficients from Ω.

We also introduce (following E. Steinitz) the concept of an intermediate field:

> "A field $\mathfrak{M}$ lies between Ω_1 and Ω_2 if it contains Ω_1 and is contained in Ω_2;"

then Definition I also admits the following formulation:

The field of rational functions of n indeterminates is an intermediate field between Ω and $\Omega(x_1,\ldots,x_n)$ which effectively contains the n indeterminates in at least one function.

Beside the number of indeterminates, there arises for systems of rational functions a further characteristic number, the number of algebraically independent functions, or the algebraic rank, by the following definition.

[81] Here $f(x)$ and $g(x)$ may also be of degree zero, i.e. quantities from Ω.

[82] That these "special fields" already exhaust the totality of all fields follows in § 4.

Definition II: A system of rational functions has algebraic rank ρ if ρ functions of the system can be specified that are algebraically independent, but every $(\rho+1)$ functions of the system are algebraically dependent.[83]

Systems of (finitely or infinitely many) rational functions in n indeterminates and of algebraic rank ρ will be denoted by the double index

$$\mathfrak{S}_{n\rho}.$$

In particular, by

$$\mathfrak{K}_{n\rho}$$

one is to understand a field of n indeterminates and of algebraic rank ρ. The inequality holds:

$$1 \leq \rho \leq n.$$

We give a few more examples of fields $\mathfrak{K}_{nn}$ and $\mathfrak{K}_{n\rho}$:

1. Fields $\mathfrak{K}_{nn}$ are Lagrange's *Gattungsbereiche*, which can be defined as

> "the totality of rational (and integral rational) functions of $x_1, \ldots, x_n$ that allow the permutations of a permutation group G of $x_1, \ldots, x_n$."

They always contain the field of symmetric functions, hence also the n algebraically independent elementary symmetric functions.

2. Fields $\mathfrak{K}_{n\rho}$ are the projective invariant fields, which are formed from the totality of the rational (and integral rational) invariants of one or more ground forms.[84] Here $\rho < n$, since the number of algebraically independent invariants is always smaller than the number of coefficients.

The corresponding statement holds for the invariant fields with respect to subgroups of the projective group.

§ 2. Auxiliary theorem on algebraically dependent functions.

The auxiliary theorem of this paragraph will show in § 3 that the algebraic rank of a system is its properly characteristic number. Namely, the auxiliary theorem shows that by a successive numerical specialization of some indeterminates, such that ρ algebraically independent functions remain algebraically independent, an arbitrary function algebraically dependent on these ρ functions can neither vanish identically nor become indeterminate.

Let therefore the system

$$g_1(x),\ g_2(x), \ldots, g_\rho(x) \tag{1}$$

have algebraic rank ρ, where $\rho < n$, and let $h(x)$ be a rational function algebraically dependent on the functions (1). By known theorems, the rank of the functional matrix of (1) is equal to ρ. The indeterminates may therefore be named, for instance, so that

$$\frac{\partial(g_1, g_2, \ldots, g_\rho)}{\partial(x_1, x_2, \ldots, x_\rho)} = \frac{\Gamma(x)}{G(x)^{\rho+1}} \neq 0, \tag{2}$$

where $G(x)$, as usual, denotes the common denominator of the functions (1).

Now choose the number a so that the product $G(x)\cdot\Gamma(x)$ is not divisible by $(x_i - a)$, where i is understood to be one of the numbers $\rho+1, \rho+2, \ldots, n$. Thus, for $x_i = a$, the common denominator of the functions (1) cannot vanish, and the ρ functions

$$[g_1(x)]_{x_i=a}, \ldots, [g_\rho(x)]_{x_i=a} \tag{3}$$

are likewise algebraically independent. As we shall say, the algebraic rank of the system (1) is preserved under the substitution $x_i = a$.

[83] ρ functions $f_1(x), \ldots, f_\rho(x)$ are called algebraically independent if from every relation

$$F\Big(f_1(x), \ldots, f_\rho(x)\Big) = 0 \quad \text{identically in } x_1, \ldots, x_n$$

it follows that

$$F(\lambda_1, \ldots, \lambda_\rho) = 0 \quad \text{identically in } \lambda_1, \ldots, \lambda_\rho.$$

In the opposite case they are called algebraically dependent.

[84] It is expedient to consider only transformations of determinant 1; then every rational combination of arbitrarily many invariants again admits the transformation, and one does indeed obtain a field. The homogeneous, isobaric invariants taken by themselves do not form a field, but they do form a system $\mathfrak{S}_{n\rho}$.

Let the irreducible equation satisfied by $h(x)$ with respect to the functions (1) be given by

$$A_0(g_1,\ldots,g_\rho)\,h(x)^\tau + A_1(g_1,\ldots,g_\rho)\,h(x)^{\tau-1} + \cdots \\ + A_\tau(g_1,\ldots,g_\rho) = 0, \tag{4}$$

where $A_0(g_1,\ldots,g_\rho)$ and $A_\tau(g_1,\ldots,g_\rho)$ do not vanish identically. In order to pass from this to an identity between polynomials, we set

$$g_i(x) = G_i(x) : G(x); \qquad h(x) = H_1(x) : H_2(x), \tag{5}$$

and note that — because of the assumed reduced representation of $h(x)$ — $H_1(x)$ and $H_2(x)$ are relatively prime. By virtue of (5), (4) becomes

$$\begin{aligned} &B_0(G,G_1,\ldots,G_\rho)\,G(x)^{\sigma_0}H_1(x)^\tau \\ &\quad + B_1(G,G_1,\ldots,G_\rho)\,G(x)^{\sigma_1}H_1(x)^{\tau-1}H_2(x) + \cdots \\ &\quad + B_\tau(G,G_1,\ldots,G_\rho)\,G(x)^{\sigma_\tau}H_2(x)^\tau = 0, \end{aligned} \tag{6}$$

where the B_k are homogeneous forms in G, G_i, and at least one of the nonnegative exponents σ_k must be zero.

Suppose now that $H_1(x)$ were divisible by $(x_i - a)$. Then, by (6),

$$B_\tau(G,G_1,\ldots,G_\rho)\,G(x)^{\sigma_\tau}H_2(x)^\tau$$

would also be divisible by $(x_i - a)$. This is excluded by assumption for $G(x)$; hence from the divisibility of B_τ there would also follow the divisibility of

$$A_\tau = B_\tau : G^\lambda,$$

and between the functions (3) there would exist the relation $A_\tau = 0$, contrary to the assumed algebraic independence. Finally $H_2(x)$, being relatively prime to $H_1(x)$, also cannot be divisible by $(x_i - a)$;[85] thus the assumption that $H_1(x)$ is divisible by $(x_i - a)$ leads to a contradiction. In the same way one shows that $H_2(x)$ also cannot be divisible by $(x_i - a)$. Hence the function $[h(x)]_{x_i=a} = h'(x)$ can neither vanish identically nor become indeterminate.

The above conclusion can now be applied again to the system (3) and the function $h'(x)$, again assumed to be in reduced representation, and continuation of the procedure leads finally to the following auxiliary theorem.

Auxiliary theorem: The two systems

$$g_1(x),\ g_2(x),\ldots,g_\rho(x),$$

$$g_1(x),\ g_2(x),\ldots,g_\rho(x),\ h(x)$$

shall each have algebraic rank ρ, where $\rho < n$, and suppose, for instance, that

$$\frac{\partial(g_1,\ldots,g_\rho)}{\partial(x_1,\ldots,x_\rho)} = \frac{\Gamma(x)}{G(x)^{\rho+1}} \neq 0.$$

The numbers

$$a_n,\ a_{n-1},\ldots,a_{\rho+1}$$

are to be chosen so that, under the substitution

$$x_n = a_n, \quad x_{n-1} = a_{n-1},\ldots, \quad x_{\rho+1} = a_{\rho+1},$$

the product $G(x)\cdot\Gamma(x)$ does not vanish — that is, the algebraic rank of the system (1) is preserved. In the successive substitutions

$$[h(x)]_{x_n=a_n} = h^{(n-1)}(x), \qquad [h^{(n-1)}(x)]_{x_{n-1}=a_{n-1}} = h^{(n-2)}(x),\ldots, \\ [h^{(\rho+1)}(x)]_{x_{\rho+1}=a_{\rho+1}} = h^*(x_1,x_2,\ldots,x_\rho),$$

[85] This last conclusion, resting on the reduced representation, would no longer be possible for a substitution $x_i = a$, $x_k = b$ while the remaining assumptions are preserved. Then $H_1(x)$ and $H_2(x)$ could vanish simultaneously, and the quotient would become indeterminate under the substitution, equal to $\frac{0}{0}$; for example

$$h(x) = \frac{x_3(\alpha x_1 + \beta x_2) + x_4(\gamma x_1 + \delta x_2)}{x_3(\alpha' x_1 + \beta' x_2) + x_4(\gamma' x_1 + \delta' x_2)}$$

under the substitution $x_3 = 0$, $x_4 = 0$.

let the functions $h(x)$, $h^{(n-1)}(x)$, ..., $h^{(\rho+1)}(x)$ each be put into reduced representation. Under these assumptions, in the function $h^*(x_1, \ldots, x_\rho)$ neither numerator nor denominator can vanish identically, and the function also cannot become $\frac{0}{0}$.[86]

§ 3. One-to-one mapping of systems $\mathfrak{S}_{n\rho}$ onto systems $\mathfrak{S}^*_{\rho\rho}$.

From the auxiliary theorem we shall immediately obtain a mapping of the systems $\mathfrak{S}_{n\rho}$ onto systems $\mathfrak{S}^*_{\rho\rho}$, by replacing some of the indeterminates by numbers; the algebraic rank ρ thereby presents itself as the properly characteristic number.

Since $\mathfrak{S}_{n\rho}$ has algebraic rank ρ, there exist ρ functions from $\mathfrak{S}_{n\rho}$,

$$g_1(x), \ldots, g_\rho(x), \tag{1}$$

which are algebraically independent. Further, if $f(x)$ denotes an arbitrary function from $\mathfrak{S}_{n\rho}$, then the system

$$g_1(x), \ldots, g_\rho(x),\ f(x) \tag{2}$$

also has algebraic rank ρ, and the systems (1) and (2) satisfy the conditions of the auxiliary theorem.[87]

If, therefore, one determines the numbers

$$a_n,\ a_{n-1}, \ldots, a_{\rho+1}$$

according to the auxiliary theorem in such a way that the algebraic rank ρ of the system (1) is preserved by the substitution — which is possible in arbitrarily many ways — then in the function defined by

$$[f(x)]_{x_n=a_n} = f^{(n-1)}(x), \qquad [f^{(n-1)}(x)]_{x_{n-1}=a_{n-1}} = f^{(n-2)}(x); \ldots,$$
$$[f^{(\rho+1)}(x)]_{x_{\rho+1}=a_{\rho+1}} = f^*(x_1, \ldots, x_\rho),$$

namely

$$f^*(x_1, \ldots, x_\rho), \tag{3}$$

neither numerator nor denominator can vanish identically.

Now let $f(x)$ run through all (finitely or infinitely many) functions from $\mathfrak{S}_{n\rho}$. We consider the system consisting of the totality of all functions (3); it has the following properties:

1. It has algebraic rank ρ; for it contains the functions arising from (1),

$$g_1^*(x_1, \ldots, x_\rho), \ldots, g_\rho^*(x_1, \ldots, x_\rho),$$

which, because of the choice of $a_n, a_{n-1}, \ldots, a_{\rho+1}$, are algebraically independent. The system is therefore to be denoted by

$$\mathfrak{S}^*_{\rho\rho}.$$

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2. The correspondence of the functions $f(x)$ and $f^*(x)$ is without exception one-to-one.

For if $f_1(x)$ and $f_2(x)$ belong to $\mathfrak{S}_{n\rho}$, then their difference

$$d(x) = f_1(x) - f_2(x)$$

is also algebraically dependent on the system (1); moreover

$$d^*(x) = f_1^*(x) - f_2^*(x).$$

Since $d(x)$, together with the system (1), satisfies the conditions of the auxiliary theorem, it follows from

$$d(x) \neq 0$$

[86] For example, in the function $h(x)$ of the preceding note, the successive substitution gives

$$[h(x)]_{x_4=0} = h^{(3)}(x) = \frac{\alpha x_1 + \beta x_2}{\alpha' x_1 + \beta' x_2}, \qquad h^*(x_1, x_2) = \frac{\alpha x_1 + \beta x_2}{\alpha' x_1 + \beta' x_2}.$$

[87] If necessary, the indeterminates are to be renumbered.

[88] Correspondingly, by $f^*(x)$, $g^*(x)$, ... one should throughout understand mapping functions, i.e. functions of $x_1, \ldots, x_\rho$.

necessarily that

$$d^*(x) \neq 0;$$

in other words, different functions from $\mathfrak{S}_{n\rho}$ correspond to different functions from $\mathfrak{S}^*_{\rho\rho}$.

3. From every relation between functions from $\mathfrak{S}^*_{\rho\rho}$,

$$F\big(f_1^*(x), f_2^*(x), \ldots, f_k^*(x)\big) = 0 \quad \text{identically in } x_1, \ldots, x_\rho,$$

it follows, for the uniquely corresponding functions from $\mathfrak{S}_{n\rho}$, that

$$F\big(f_1(x), f_2(x), \ldots, f_k(x)\big) = 0 \quad \text{identically in } x_1, \ldots, x_n.$$

For with $f_1(x), f_2(x), \ldots, f_k(x)$, every rational combination of these functions is also algebraically dependent on the system (1).

It further follows from

$$\begin{aligned} h(x) &= F\big(f_1(x), \ldots, f_k(x)\big) : \\ h^*(x) &= F\big(f_1^*(x), \ldots, f_k^*(x)\big). \end{aligned} \tag{4}$$

By the auxiliary theorem, therefore,

$$h(x) \neq 0 \quad \text{implies} \quad h^*(x) \neq 0;$$

q.e.d.

In summary, we obtain

Theorem I: To every system $\mathfrak{S}_{n\rho}$ of (finitely or infinitely many) rational functions there can be assigned — in arbitrarily many ways — a one-to-one system $\mathfrak{S}^*_{\rho\rho}$, in such a way that all relations existing between functions from $\mathfrak{S}^*_{\rho\rho}$ are also preserved in $\mathfrak{S}_{n\rho}$ and conversely. In particular, by (4), a field $\mathfrak{K}_{n\rho}$ again corresponds to a field $\mathfrak{K}^*_{\rho\rho}$.

From Theorem I we draw the consequence that simplifies all further proofs: *properties of systems $\mathfrak{S}_{n\rho}$ which are characterized by finitely or infinitely many relations between functions of the system hold for the systems $\mathfrak{S}_{n\rho}$ as soon as they hold for a mapping system $\mathfrak{S}^*_{\rho\rho}$.*

We call this consequence briefly the “transfer principle.”

The simplest example of this mapping is furnished by the theory of algebraic invariants. In fact, by a linear transformation one can always numerically fix some coefficients of the ground form in such a way that all relations valid for the special form also remain valid in general.

§ 4. Rational basis of the fields $\mathfrak{K}_{n\rho}$.

In the following investigations, grouped around basis concepts, we shall consistently use the "transfer principle" of § 3. We first prove the existence of the basis for fields, but state the definitions generally:

Definition III: A finite number of functions from $\mathfrak{S}_{n\rho}$ is called a rational basis if every function from $\mathfrak{S}_{n\rho}$ can be represented as a rational combination of this finite number, with coefficients from Ω.

The rational basis must contain ρ algebraically independent functions, since the algebraic rank of a system derived from a rational basis is equal to the algebraic rank of the rational basis.

We first show that the existence proof for the rational basis admits the transfer principle. Namely, Definition III can be formulated as follows:

The functions from $\mathfrak{S}_{n\rho}$

$$g_1(x),\ g_2(x), \ldots, g_k(x)$$

form a rational basis if and only if the infinitely many relations

$$f(x) = \gamma(g_1(x), \ldots, g_k(x)) \tag{1}$$

are satisfied, where $f(x)$ runs through all functions from $\mathfrak{S}_{n\rho}$ and γ denotes a rational function of k arguments, varying with f.

Thus, if in the mapping system $\mathfrak{S}^*_{\rho\rho}$ a rational basis is given by

$$g^*_1(x),\ g^*_2(x), \ldots, g^*_k(x),$$

which entails the relations

$$f^*(x) = \gamma(g^*_1(x), \ldots, g^*_k(x)),$$

then by Theorem I the relations (1) are fulfilled for $\mathfrak{S}_{n\rho}$. Thus the *transfer principle for the rational basis* says:

From the existence of a rational basis for a mapping system $\mathfrak{S}^*_{\rho\rho}$ follows the rational basis for the original system $\mathfrak{S}_{n\rho}$.[89]

To prove the existence of the rational basis of all fields $\mathfrak{K}_{n\rho}$, we may therefore restrict ourselves to fields $\mathfrak{K}^*_{\rho\rho}$; here a direct generalization of an argument used by E. Steinitz leads to the goal.[90]

Let

$$g^*_1(x_1, \ldots, x_\rho), \ldots, g^*_\rho(x_1, \ldots, x_\rho) \tag{2}$$

be a system of ρ algebraically independent functions from $\mathfrak{K}^*_{\rho\rho}$. By eliminating $x_1, \ldots, x_\rho$ one obtains ρ relations:

$$\begin{aligned} &F_1(g^*_1(x), \ldots, g^*_\rho(x), x_1) = 0, \ldots, \\ &F_\rho(g^*_1(x), \ldots, g^*_\rho(x), x_\rho) = 0, \end{aligned} \tag{3}$$

where in each relation $F_i = 0$ the indeterminate x_i actually occurs; otherwise, contrary to the assumption, there would be an algebraic dependence among the functions (2). The relations (3) say that

$$x_1, x_2, \ldots, x_\rho$$

are algebraic elements with respect to

$$\Omega(g^*_1(x), \ldots, g^*_\rho(x)).$$

Hence the field arising by adjoining $x_1, \ldots, x_\rho$,

$$\Omega(g^*_1, \ldots, g^*_\rho, x_1, \ldots, x_\rho) = \Omega(x_1, \ldots, x_\rho),$$

is a finite algebraic extension of $\Omega(g^*_1, \ldots, g^*_\rho)$. By a known theorem,[91] $\mathfrak{K}^*_{\rho\rho}$, as an intermediate field between $\Omega(g^*_1, \ldots, g^*_\rho)$ and $\Omega(x_1, \ldots, x_\rho)$, is itself an algebraic extension of $\Omega(g^*_1, \ldots, g^*_\rho)$, while $\Omega(x_1, \ldots, x_\rho)$ is an

[89] The transfer principle is formulated correspondingly for every basis question treated here.

[90] E. Steinitz, Algebraische Theorie der Körper, § 24, Crelle's Journal 137 (1909).

[91] Compare, for example, Weber, Lehrbuch d. Algebra 1, § 144 (1st ed.) or § 55 (small edition).

algebraic extension of $\mathfrak{K}^*_{\rho\rho}$. Thus $\mathfrak{K}^*_{\rho\rho}$ arises from $\Omega(g_1^*, \ldots, g_\rho^*)$ by adjoining a finite number of elements from $\mathfrak{K}^*_{\rho\rho}$, or also by adjoining one suitably chosen function; i.e. one has

$$\mathfrak{K}^*_{\rho\rho} = \Omega(g_1^*, \ldots, g_\rho^*, g_{\rho+1}^*, \ldots, g_k^*) = \Omega(g_1^*, \ldots, g_\rho^*, g_0^*).$$

By the transfer principle we therefore have

Theorem II: Every field $\mathfrak{K}_{n\rho}$ possesses a rational basis of algebraic rank ρ; the rational basis can in particular be chosen so that it contains at most $\rho + 1$ functions.

Let an arbitrary system of ρ algebraically independent functions from $\mathfrak{K}_{n\rho}$ be given by $h_1(x), \ldots, h_\rho(x)$, and suppose it can be extended by the above method to a rational basis

$$h_1(x), \ldots, h_\rho(x); h_{\rho+1}(x), \ldots, h_\sigma(x).$$

By the defining property, relations exist:

$$\begin{aligned} h_i(x) &= \chi_i(g_1(x), \ldots, g_k(x)) \quad (i = 1, 2, \ldots, \sigma), \\ g_j(x) &= \psi_j(h_1(x), \ldots, h_\sigma(x)) \quad (j = 1, 2, \ldots, k), \end{aligned}$$

where χ_i and ψ_j denote rational functions of their arguments. Thus:

From any rational basis one obtains every further rational basis by a birational transformation; and:

If

$$h_1(x), \ldots, h_\rho(x)$$

is any system of ρ algebraically independent functions from $\mathfrak{K}_{n\rho}$, then $\mathfrak{K}_{n\rho}$ is a finite algebraic extension of

$$\Omega(h_1(x), \ldots, h_\rho(x)).$$

§ 5. Involution form and involution basis of the fields $\mathfrak{K}_{n\rho}$.

Whereas every rational basis constructed in § 4 had the disadvantage of depending on the arbitrarily chosen starting functions, we now show the existence of a *rational basis uniquely determined by the field.*

By § 4, $\Omega(x_1, \ldots, x_\rho)$ is an algebraic extension — say of degree i — of $\mathfrak{K}^*_{\rho\rho}$, and, because

$$\mathfrak{K}^*_{\rho\rho}(x_1, \ldots, x_\rho) = \Omega(x_1, \ldots, x_\rho),$$

arises by adjoining the elements $x_1, \ldots, x_\rho$, which are algebraic with respect to $\mathfrak{K}^*_{\rho\rho}$. We therefore consider the integral function $\Phi^*(z, u)$ irreducible with respect to $\mathfrak{K}^*_{\rho\rho}$, with the zero

$$z = u_1 x_1 + u_2 x_2 + \cdots + u_\rho x_\rho = u(x),$$

which is of ith degree in z. Further, as the product of the quantities conjugate to $[z - u(x)]$, $\Phi^*(z, u)$ is a homogeneous form of dimension i in $z, u_1, \ldots, u_\rho$; hence as coefficient of z^i it contains a function from $\mathfrak{K}^*_{\rho\rho}$ that is free of u. If we make this highest coefficient equal to 1 by division, then the irreducible function $\Phi^*(z, u)$ is uniquely determined. The homogeneous form so defined,

$$\Phi^*(z, u) = z^i + \sum{}' g^*_{\alpha\alpha_1\ldots\alpha_\rho}(x_1, \ldots, x_\rho) z^\alpha u_1^{\alpha_1} \cdots u_\rho^{\alpha_\rho}, \qquad (\alpha + \alpha_1 + \cdots + \alpha_\rho = i), \tag{1}$$

is to be called the *involution form*[92] of $\mathfrak{K}^*_{\rho\rho}$; from (1) one obtains

$$\Phi(z, u) = z^i + \sum{}' g_{\alpha\alpha_1\ldots\alpha_\rho}(x) z^\alpha u_1^{\alpha_1} \cdots u_\rho^{\alpha_\rho}, \qquad (\alpha + \alpha_1 + \cdots + \alpha_\rho = i), \tag{2}$$

as an *involution form* of $\mathfrak{K}_{n\rho}$.

The coefficients of the involution form $\Phi^*(z, u)$ will turn out to be a rational basis of $\mathfrak{K}^*_{\rho\rho}$; i.e. the relation

$$\Omega(g^*_{\alpha\alpha_1\ldots\alpha_\rho}(x_1, \ldots, x_\rho)) = \mathfrak{K}^*_{\rho\rho} \quad (\alpha + \alpha_1 + \cdots + \alpha_\rho = i) \tag{3}$$

[92]The reason for the name will be clear at the end of the paragraph; $\sum{}'$ means that the summation is to extend over all terms except z^i.

holds. For the proof we observe that $\Phi^*(z,u)$ has coefficients from $\Omega(g^*_{\alpha\alpha_1\dots\alpha_\rho}(x))$ and, because of its irreducibility with respect to $\mathfrak{K}^*_{\rho\rho}$, is all the more irreducible with respect to this subfield. Hence $\Phi^*(z,u)$ gives the irreducible equation of $u(x)$ with respect to $\Omega(g^*_{\alpha\alpha_1\dots\alpha_\rho}(x))$. It follows that $\Omega(x_1,\dots,x_\rho)$ is an algebraic extension — of degree i — of $\Omega(g^*_{\alpha\alpha_1\dots\alpha_\rho}(x))$. Since $\mathfrak{K}^*_{\rho\rho}$ lies between $\Omega(g^*_{\alpha\alpha_1\dots\alpha_\rho}(x))$ and $\Omega(x_1,\dots,x_\rho)$, the identity of degrees is known to imply the identity of the fields, hence relation (3). By the transfer principle, the coefficients of $\Phi(z,u)$ correspondingly yield a rational basis of $\mathfrak{K}_{n\rho}$; i.e. we have

Theorem III: The coefficients of the involution form constitute a rational basis determined by the field, the involution basis; for fields $\mathfrak{K}^*_{\rho\rho}$ this basis is uniquely determined.[93]

We add an irrational interpretation of this fact, which establishes the connection with the geometric theory of involutions.

The factorization of $\Phi^*(z,u)$ into linear factors is given in an extension field by

$$\begin{aligned}\Phi^*(z,u) = (z - u_1x_1 - \dots - u_\rho x_\rho)(z - u_1x_1^{(1)} - \dots - u_\rho x_\rho^{(1)})\cdots\\ \cdot (z - u_1x_1^{(i-1)} - \dots - u_\rho x_\rho^{(i-1)}).\end{aligned} \tag{4}$$

Comparison with (1) shows that the functions

$$g^*_{\alpha\alpha_1\dots\alpha_\rho}(x_1,\dots,x_\rho) \quad (\alpha+\alpha_1+\dots+\alpha_\rho = i)$$

are identical with the elementary symmetric functions of the rows of quantities

$$\begin{matrix} x_1 & x_2 & \cdots & x_\rho \\ x_1^{(1)} & x_2^{(1)} & \cdots & x_\rho^{(1)} \\ \vdots & \vdots & & \vdots \\ x_1^{(i-1)} & x_2^{(i-1)} & \cdots & x_\rho^{(i-1)}. \end{matrix} \tag{5}$$

Thus every function from $\mathfrak{K}^*_{\rho\rho}$ is a rational combination of these elementary functions, symmetric in the rows (5), and conversely every symmetric function of the rows (5) belongs to $\mathfrak{K}^*_{\rho\rho}$.

Thus, as

$$f^*(x) = \frac{F^*(x)}{H^*(x)}$$

runs through all functions from $\mathfrak{K}^*_{\rho\rho}$, there is a system of infinitely many relations:

$$\frac{F^*(x)}{H^*(x)} = \frac{F^*(x^{(1)})}{H^*(x^{(1)})} = \dots = \frac{F^*(x^{(i-1)})}{H^*(x^{(i-1)})}, \tag{6}$$

or, summarized,

$$F^*(x^{(k)})H^*(x^{(l)}) - F^*(x^{(l)})H^*(x^{(k)}) = 0 \quad (k,l = 0,1,\dots,i-1),$$

where neither $F^*(x^{(k)})$ nor $H^*(x^{(k)})$ vanishes identically, since they are conjugate to $F^*(x)$ and $H^*(x)$ and formally symmetric.

Conversely, for every row of quantities ξ satisfying the infinitely many relations

$$F^*(x)H^*(\xi) - H^*(x)F^*(\xi) = 0, \qquad H^*(\xi) \neq 0, \tag{7}$$

one obtains membership in the rows of quantities (5).

Indeed, if $G^*(x)$ denotes the common denominator of the coefficients of $\Phi^*(z,u)$, then by assumption (7) the function

$$G^*(\xi)\cdot\Phi^*(z,u;\xi),$$

which is also integral in ξ, does not vanish identically — that is, the coefficients of $\Phi^*(z,u;\xi)$ do not become indeterminate — and has the zero

$$z = u_1\xi_1 + u_2\xi_2 + \dots + u_\rho\xi_\rho.$$

Because of (7), $\Phi^*(z,u;x)$ therefore also has the zero $z = u(\xi)$; i.e. ξ belongs to the rows of quantities (5). The corresponding assertion holds by (6) if ξ satisfies a system of equations with respect to a conjugate row of quantities:

$$F^*(x^{(k)})H^*(\xi) - H^*(x^{(k)})F^*(\xi) = 0; \qquad H^*(\xi) \neq 0.$$

[93] For fields $\mathfrak{K}_{n\rho}$ the involution form may depend on the mapping; uniqueness can be obtained by a mapping with indeterminate coefficients.

Thus the rows of quantities (5) are defined as the solutions ξ of the system of equations

$$F^*(x)H^*(\xi) - H^*(x)F^*(\xi) = 0; \qquad H^*(\xi) \neq 0,$$

where $F^*(x) : H^*(x)$ runs through all functions from $\mathfrak{K}^*_{\rho\rho}$; here the row x may also be replaced by any conjugate row.

Geometrically this says the following, interpreting each row x as a point: the solutions of (7) represent, in a ρ-dimensional linear space, ∞^ρ groups of i points each, in such a way that any point of the group determines the individual group uniquely. Such a system of point-groups is called an "*involution in a linear space*."[94]

Correspondingly, a field $\mathfrak{K}_{n\rho}$ characterizes an involution in a linear space consisting of curves, surfaces, and so on.

By the preceding, the degree i of $\Omega(x_1, \ldots, x_\rho)$ with respect to $\mathfrak{K}^*_{\rho\rho}$ is equal to the number of solution systems of (7) satisfying the side condition $H^*(\xi) \neq 0$ — which we may call the solution systems of $\mathfrak{K}^*_{\rho\rho}$. Hence the argument used to prove the existence of the involution basis also permits the following irrational formulation:

"If $\mathfrak{L}^*$ is a divisor of $\mathfrak{K}^*_{\rho\rho}$ and every solution system of the field $\mathfrak{L}^*$ is also a solution system of $\mathfrak{K}^*_{\rho\rho}$, then $\mathfrak{L}^*$ and $\mathfrak{K}^*_{\rho\rho}$ are identical."

The corresponding statement holds, by the transfer principle, for divisors $\mathfrak{L}$ of $\mathfrak{K}_{n\rho}$.

§ 6. Minimal basis of the fields $\mathfrak{K}_{n\rho}$.

This paragraph gives an overview of known theorems, but in an expanded formulation made possible by the transfer principle and the existence of the rational basis.

Definition IV: A rational basis of $\mathfrak{K}_{n\rho}$ is called a minimal basis if it contains only the minimum number ρ of functions, which must then be algebraically independent.

From theorems of Lüroth, Castelnuovo, and Enriques[95] one obtains the facts:

I. Every field $\mathfrak{K}_{n1}$ possesses a minimal basis, the Lüroth function of the field, uniquely determined up to linear fractional transformation.

II. For fields $\mathfrak{K}_{n2}$, a minimal basis exists always, and in general only, if one permits an algebraic extension of the coefficient field Ω.

III. For fields $\mathfrak{K}_{n\rho}$ ($\rho \geq 3$), in general no minimal basis exists. A characterization of the special fields $\mathfrak{K}_{n\rho}$ with minimal basis has not yet been attempted.

We briefly indicate the proof of Lüroth's theorem for fields $\mathfrak{K}^*_{11}$ given by E. Steinitz:[96]

Let $\Phi^*(z, u)$ be the involution form of $\mathfrak{K}^*_{11}$, and let $\psi^*(x)$ be any coefficient of $\Phi^*(z, u)$ that actually contains the indeterminate x. It turns out that the degree of $\Omega(x)$ with respect to $\Omega(\psi^*(x))$ is equal to the degree of $\Omega(x)$ with respect to $\mathfrak{K}^*_{11}$, whence the identity of the fields follows. Further, if $\Omega(\chi^*(x))$ is equal to $\Omega(\psi^*(x))$, then the mutual rational dependence of $\chi^*(x)$ and $\psi^*(x)$ implies the linear fractional dependence of the two functions.

By the transfer principle, Lüroth's theorem therefore also admits the formulation:

The minimal basis of the fields $\mathfrak{K}_{n1}$ consists of any function of the involution basis; and any two functions of the involution basis of $\mathfrak{K}_{n1}$ depend on one another by a linear fractional transformation.

G. Castelnuovo proves Fact II, by means of the geometric theory of involutions, for fields $\mathfrak{K}^*_{22}$ derived from a rational basis of three functions. Since the transfer principle also remains valid under finite algebraic extension of the coefficient field Ω, the existence of the rational basis makes the proof complete for all fields $\mathfrak{K}_{n2}$.

Regarding III, it should be noted that Enriques constructs a nonrational involution in three-dimensional space; that is, a field $\mathfrak{K}_{33}$ which has no minimal basis.

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[94] The excluded solutions of (7), for which $F^*(\xi) = 0$, $H^*(\xi) = 0$, and likewise the excluded solutions of the system of equations corresponding to the involution basis, are called fundamental points of the involution; this also includes those arising by homogenization, the points "lying at infinity."

[95] J. Lüroth: Beweis eines Satzes über rationale Kurven, Math. Ann. 9 (1875). — G. Castelnuovo: Sulla razionalità delle involuzioni piane, Math. Ann. 44 (1893). — F. Enriques: Sopra una involuzione non razionale dello spazio, Rend. Acc. Linc., Vol. 21, 21 Jan. 1912.

[96] Loc. cit., § 24.

§ 7. Arbitrary system $\mathfrak{S}_{n\rho}$, linear family $\mathfrak{L}_{n\rho}$, and integrality domain $\mathfrak{J}_{n\rho}$ of rational functions. Existence of the rational basis.

From the existence of the rational basis of the fields $\mathfrak{K}_{n\rho}$ we shall now obtain the rational basis for arbitrary systems of rational and integral rational functions. We arrange these systems in the order in which they successively arise from one another through the elementary operations of addition, multiplication, and division.

I. As always, let $\mathfrak{S}_{n\rho}$ denote an arbitrary system of rational (or integral rational) functions of n indeterminates and of algebraic rank ρ. The quantities from Ω are counted as belonging to the system.

II. The system $\mathfrak{L}_{n\rho}$ is called a *linear family* if it satisfies the conditions:

1. Together with $f(x)$, $\mathfrak{L}_{n\rho}$ always also contains $c \cdot f(x)$, where c is an arbitrary quantity from Ω;
2. Together with $f(x)$ and $g(x)$, $\mathfrak{L}_{n\rho}$ always also contains $f(x) + g(x)$.

III. The linear family $\mathfrak{J}_{n\rho}$ is called an *integrality domain* under the further condition:

3. Together with $f(x)$ and $g(x)$, $\mathfrak{J}_{n\rho}$ always also contains $f(x) \cdot g(x)$.

IV. The integrality domain $\mathfrak{K}_{n\rho}$ is called a *field* under the further condition:

4. Together with $f(x)$ and $g(x)$ — where $g(x) \neq 0$ — $\mathfrak{K}_{n\rho}$ always also contains the quotient $f(x) : g(x)$.

Conditions 1 through 4 are those stated in § 1 for the field.[97]

These domains are now to be derived successively from one another.

a) An arbitrary system $\mathfrak{S}_{n\rho}$ is enlarged to a linear family $\mathfrak{L}_{n\rho}$ by the requirement:

$\mathfrak{L}_{n\rho}$ contains, in addition to $\mathfrak{S}_{n\rho}$, all functions $h(x)$ and only those that can be represented linearly with coefficients from Ω by a finite number of functions from $\mathfrak{S}_{n\rho}$:

$$h(x) = c_1 f_1(x) + c_2 f_2(x) + \cdots + c_k f_k(x),$$

where $f_1(x), \ldots, f_k(x)$ run through all functions from $\mathfrak{S}_{n\rho}$, and $c_1, \ldots, c_k$ through all quantities from Ω.

For adding rational combinations of functions of the system preserves the algebraic rank of a system. $\mathfrak{L}_{n\rho}$ satisfies conditions 1 and 2, and is therefore a linear family. Moreover, every linear family containing $\mathfrak{S}_{n\rho}$ also contains $\mathfrak{L}_{n\rho}$ by 1 and 2; hence $\mathfrak{L}_{n\rho}$ is the smallest linear family containing $\mathfrak{S}_{n\rho}$.

b) Correspondingly, from a linear family $\mathfrak{L}_{n\rho}$ arises the smallest containing integrality domain $\mathfrak{J}_{n\rho}$ by the requirement:

$\mathfrak{J}_{n\rho}$ contains, in addition to $\mathfrak{L}_{n\rho}$, all functions $h(x)$ and only those that can be represented as an *integral rational combination* — with coefficients from Ω — of a finite number of functions from $\mathfrak{L}_{n\rho}$, $f_1(x), \ldots, f_k(x)$, where $f_1(x), \ldots, f_k(x)$ run through all functions from $\mathfrak{L}_{n\rho}$.

c) Finally, from an integrality domain $\mathfrak{J}_{n\rho}$ arises the smallest containing field $\mathfrak{K}_{n\rho}$ by the requirement: $\mathfrak{K}_{n\rho}$ contains, in addition to $\mathfrak{J}_{n\rho}$, all functions $h(x)$ and only those that can be represented as quotients of two functions from $\mathfrak{J}_{n\rho}$.

Combining the three extensions into one step gives:

Every system $\mathfrak{S}_{n\rho}$ can be enlarged to a smallest containing field $\mathfrak{K}_{n\rho}$ by the requirement:

$\mathfrak{K}_{n\rho}$ contains, in addition to $\mathfrak{S}_{n\rho}$, all functions $h(x)$ and only those that can be represented as rational combinations — with coefficients from Ω — of a finite number of functions from $\mathfrak{S}_{n\rho}$, $f_1(x), \ldots, f_k(x)$, where $f_1(x), \ldots, f_k(x)$ run through all functions from $\mathfrak{S}_{n\rho}$.

From the latter fact the existence of the rational basis for arbitrary domains follows immediately:

Let $\mathfrak{K}_{n\rho}$ be the smallest field containing $\mathfrak{S}_{n\rho}$, and let a rational basis of $\mathfrak{K}_{n\rho}$ be given by

$$h_1(x),\ h_2(x), \ldots, h_\sigma(x). \tag{1}$$

By the definition of $\mathfrak{K}_{n\rho}$, relations exist:

$$\begin{aligned} h_1(x) &= r_1(g_1(x), \ldots, g_\lambda(x)); \ldots; \\ h_\sigma(x) &= r_\sigma(g_\mu(x), \ldots, g_\nu(x)), \end{aligned} \tag{2}$$

where $r_1, \ldots, r_\sigma$ are rational functions of their arguments, and

$$g_1(x), \ldots, g_\lambda(x), \ldots, g_\mu(x), \ldots, g_\nu(x) \tag{3}$$

all belong to $\mathfrak{S}_{n\rho}$. Because of the relations (2), (3), in addition to (1), also represents a rational basis of $\mathfrak{K}_{n\rho}$; because the functions (3) belong to $\mathfrak{S}_{n\rho}$, this is at the same time a rational basis of $\mathfrak{S}_{n\rho}$. We therefore have

[97] $f(x)$ and $g(x)$ may also be of degree zero, i.e. quantities from Ω.

Theorem IV: An arbitrary system $\mathfrak{S}_{n\rho}$ of rational (or integral rational) functions possesses a rational basis, consisting of a finite number of functions of the system, of algebraic rank ρ.

Now let $\mathfrak{L}_{n\rho}$ in particular be a linear family, and let

$$g_1(x), \ldots, g_\rho(x),\ g_{\rho+1}(x), \ldots, g_\nu(x) \tag{4}$$

be a rational basis of $\mathfrak{L}_{n\rho}$. Suppose, for instance, that

$$g_1(x), \ldots, g_\rho(x)$$

are algebraically independent. If one then sets

$$g(x) = c_1 g_{\rho+1}(x) + c_2 g_{\rho+2}(x) + \cdots + c_{\nu-\rho} g_\nu(x),$$

then the c_i can be chosen as numbers from Ω in such a way that

$$g_1(x), \ldots, g_\rho(x),\ g(x) \tag{5}$$

becomes a rational basis of the smallest containing field $\mathfrak{K}_{n\rho}$. But since $\mathfrak{L}_{n\rho}$ is a linear family, $g(x)$ belongs to $\mathfrak{L}_{n\rho}$, and (5) represents a rational basis of $\mathfrak{L}_{n\rho}$; that is,

Theorem V: The rational basis of every linear family $\mathfrak{L}_{n\rho}$ of rational (or integral rational) functions — hence in particular the rational basis of every integrality domain $\mathfrak{J}_{n\rho}$ and every field $\mathfrak{K}_{n\rho}$ — can be chosen in particular so that it contains at most $\rho + 1$ functions (and at least ρ).

The bound on the number of functions in the rational basis found in § 4 for fields therefore rests on the property of the field of being a linear family, whereas the restriction to ρ functions in the case of the existence of the minimal basis uses all assumptions of the field.

It remains to note that, by § 3 (4), all characteristic properties of the systems and smallest containing systems are preserved under the mapping.

§ 8. Involution basis of the integrality domains $\mathfrak{J}_{n\rho}$ consisting of polynomials.

In the following paragraphs we are concerned with systems $\mathfrak{S}_{n\rho}$ whose elements are integral rational functions (polynomials); first with integrality domains $\mathfrak{J}_{n\rho}$ consisting of polynomials. For these domains $\mathfrak{J}_{n\rho}$, the divisibility notions are first to be fixed.

Let $F(x), G(x)$ be two polynomials from $\mathfrak{J}_{n\rho}$, and let $L(x)$ be a common divisor of these polynomials, so that one has

$$F(x) = L(x) \cdot F_1(x), \qquad G(x) = L(x) \cdot G_1(x).$$

$L(x)$ is called a *common divisor in* $\mathfrak{J}_{n\rho}$ if and only if the three polynomials $L(x), F_1(x), G_1(x)$ belong to $\mathfrak{J}_{n\rho}$.

Two polynomials from $\mathfrak{J}_{n\rho}$ are called *relatively prime in* $\mathfrak{J}_{n\rho}$ if they possess no common divisor in $\mathfrak{J}_{n\rho}$.

Let $\mathfrak{K}_{n\rho}$ be the smallest field containing $\mathfrak{J}_{n\rho}$. Then, from the definition, every function from $\mathfrak{K}_{n\rho}$ has a representation

$$f(x) = F_1(x) : F_2(x),$$

where $F_1(x), F_2(x)$ belong to $\mathfrak{J}_{n\rho}$ and are relatively prime in $\mathfrak{J}_{n\rho}$. Correspondingly, several functions from $\mathfrak{K}_{n\rho}$ can be brought to a common denominator in $\mathfrak{J}_{n\rho}$:

$$f_1(x) = \frac{F_1(x)}{F(x)}, \ldots, f_k(x) = \frac{F_k(x)}{F(x)}.$$

$F(x)$ is called *a least common denominator in* $\mathfrak{J}_{n\rho}$ if and only if the polynomials from $\mathfrak{J}_{n\rho}$,

$$F(x), F_1(x), \ldots, F_k(x),$$

are relatively prime in $\mathfrak{J}_{n\rho}$.

Such a representation can be possible in different ways, since in $\mathfrak{J}_{n\rho}$ the laws of unique factorization into irreducible functions do not hold in general.

We now investigate the significance of the involution basis of $\mathfrak{K}_{n\rho}$ for $\mathfrak{J}_{n\rho}$, by transferring the notions of involution form and involution basis to integrality domains. Let

$$\Phi(z, u) = z^i + \sum{}' g_{\alpha\alpha_1\ldots\alpha_\rho}(x) z^\alpha u_1^{\alpha_1} \cdots u_\rho^{\alpha_\rho}, \qquad (\alpha + \alpha_1 + \cdots + \alpha_\rho = i)$$

be an involution form of $\mathfrak{K}_{n\rho}$, and let $G(x)$ be a least common denominator in $\mathfrak{J}_{n\rho}$ of the coefficients of $\Phi(z, u)$. By multiplication with $G(x)$, $\Phi(z, u)$ passes into a form $\Psi(z, u)$ whose coefficients are polynomials from $\mathfrak{J}_{n\rho}$ and are relatively prime in $\mathfrak{J}_{n\rho}$:

$$G(x) \cdot \Phi(z, u) = \Psi(z, u) = G(x) z^i + \sum{}' G_{\alpha\alpha_1\ldots\alpha_\rho}(x) z^\alpha u_1^{\alpha_1} \cdots u_\rho^{\alpha_\rho}, \qquad (\alpha + \alpha_1 + \cdots + \alpha_\rho = i).$$

Every form $\Psi(z, u)$ arising from an involution form $\Phi(z, u)$ of $\mathfrak{K}_{n\rho}$ by multiplication with a polynomial from $\mathfrak{J}_{n\rho}$ in such a way that the coefficients of $\Psi(z, u)$ are polynomials from $\mathfrak{J}_{n\rho}$ and are relatively prime in $\mathfrak{J}_{n\rho}$, shall be called an *involution form* of $\mathfrak{J}_{n\rho}$; and the coefficients of $\Psi(z, u)$ shall be called a corresponding *involution basis.*

It is first clear from the meaning of the involution basis of $\mathfrak{K}_{n\rho}$ that every involution basis of $\mathfrak{J}_{n\rho}$ is at the same time a rational basis. For the further investigation we consider a mapping domain $\mathfrak{J}^*_{\rho\rho}$ for whose smallest containing field $\mathfrak{K}^*_{\rho\rho}$ the irreducible equation with zero

$$z = u_1 x_1 + \cdots + u_\rho x_\rho$$

is given by $\Phi^*(z, u) = 0$.

According to § 5, the coefficients of

$$\Phi^*(z, u) = z^i + \sum{}' g^*_{\alpha\alpha_1\ldots\alpha_\rho}(x) z^\alpha u_1^{\alpha_1} \cdots u_\rho^{\alpha_\rho}, \qquad (\alpha + \alpha_1 + \cdots + \alpha_\rho = i)$$

represent the elementary symmetric functions of the rows of quantities conjugate with respect to $\mathfrak{K}^*_{\rho\rho}$,

$$x, \quad x^{(1)}, \ldots, x^{(i-1)}.$$

Let $F^*(x)$ be any polynomial from $\mathfrak{K}^*_{\rho\rho}$. Since

$$F^*(x) = \frac{1}{i}\Big(F^*(x) + F^*(x^{(1)}) + \cdots + F^*(x^{(i-1)})\Big),$$

it follows that $F^*(x)$ is an integral symmetric function of these rows of quantities; consequently it can be expressed as an integral rational combination of the elementary symmetric functions, that is, of the functions

$$g^*_{\alpha\alpha_1\ldots\alpha_\rho}(x).$$

If now

$$G^*(x)\cdot\Phi^*(z,u)=\Psi^*(z,u)=G^*(x)z^i+\sum{}'G^*_{\alpha\alpha_1\ldots\alpha_\rho}(x)z^\alpha u_1^{\alpha_1}\cdots u_\rho^{\alpha_\rho},\qquad(\alpha+\alpha_1+\cdots+\alpha_\rho=i)$$

is an involution form of $\mathfrak{J}^*_{\rho\rho}$, then every polynomial from $\mathfrak{K}^*_{\rho\rho}$, and in particular every polynomial from $\mathfrak{J}^*_{\rho\rho}$, is an integral rational combination of the quotients

$$G^*_{\alpha\alpha_1\ldots\alpha_\rho}(x):G^*(x).$$

Taking the transfer principle into account, this gives

Theorem VI: To every integrality domain of polynomials $\mathfrak{J}_{n\rho}$ there exists at least one distinguished rational basis, the involution basis, such that in the rational representation of all polynomials from $\mathfrak{J}_{n\rho}$ by the functions of the involution basis only a power of one fixed function (the highest coefficient of the corresponding involution form) occurs in the denominator.

§ 9. Relatively integral domains $\mathfrak{G}_{n\rho}$ consisting of polynomials.

In what follows we consider special integrality domains consisting of polynomials, which form an intermediate stage between integrality domains and fields.

An integrality domain of polynomials $\mathfrak{G}_{n\rho}$ is called a "relatively integral domain" under the condition:

4a) $\mathfrak{G}_{n\rho}$ contains, along with $F(x)$ and $G(x)$ — where $G(x)\neq 0$ — the quotient $F(x):G(x)$ always and only when this quotient is integral in the indeterminates, that is, is a polynomial.

First the identity of the relatively integral domains with the totality of the polynomials of a field $\mathfrak{K}_{n\sigma}$ $(\sigma\geq\rho)$ of rational functions is to be proved.

Let $\mathfrak{K}_{n\rho}$ be the smallest field containing $\mathfrak{G}_{n\rho}$. For every function $f(x)$ from $\mathfrak{K}_{n\rho}$ there is a representation as quotient of two polynomials from $\mathfrak{G}_{n\rho}$:

$$f(x)=F_1(x):F_2(x),$$

where polynomials of degree zero are also allowed. As soon as $f(x)$ becomes a polynomial, it belongs to $\mathfrak{G}_{n\rho}$ by condition 4a); and since $\mathfrak{G}_{n\rho}$ can contain no further polynomials, *every relatively integral domain $\mathfrak{G}_{n\rho}$ is identical with the totality of the polynomials of the smallest containing field.*

Conversely, let $\mathfrak{K}_{n\sigma}$ be any field of rational functions (therefore not necessarily the smallest containing field of a given system), and let $\mathfrak{J}$ be the totality of the polynomials contained in $\mathfrak{K}_{n\sigma}$. $\mathfrak{J}$ satisfies the conditions 1, 2, 3 of an integrality domain and condition 4a), and so is a relatively integral domain; that is, *the totality of the polynomials contained in a field $\mathfrak{K}_{n\sigma}$ forms a relatively integral domain $\mathfrak{G}_{n\rho}$ $(\rho\leq\sigma)$.*[98]

We further show the identity of the relatively integral domains with Hilbert's "integrality domains of relatively integral functions."[99]

Let

$$G_1(x),\ldots,G_k(x)\tag{1}$$

be a rational basis of $\mathfrak{G}_{n\rho}$. By conditions 1, 2, 3, 4a, every rational combination of the polynomials (1) which is integral in the indeterminates, that is, becomes a polynomial, belongs to $\mathfrak{G}_{n\rho}$. Conversely, every polynomial from $\mathfrak{G}_{n\rho}$ can, by the notion of rational basis, be rationally expressed through the polynomials (1); hence $\mathfrak{G}_{n\rho}$ is *identical with the totality of those rational combinations of the polynomials (1) which are integral in the indeterminates, that is, are polynomials.* Thus we have Hilbert's definition starting from the special rational basis, whereas our definition uses only abstract properties of the domain.

For relatively integral domains, the definition of a common divisor given in § 8 for general integrality domains simplifies:

Let $F(x),G(x)$ be two polynomials from $\mathfrak{G}_{n\rho}$ and let $L(x)$ be a common divisor of these polynomials. Then $L(x)$ is called a *common divisor in $\mathfrak{G}_{n\rho}$* if and only if $L(x)$ belongs to $\mathfrak{G}_{n\rho}$.

[98] It may also be $\rho=0$; that is, the totality of the polynomials from $\mathfrak{K}_{n\sigma}$ consists of elements of the coefficient domain Ω.

[99] D. Hilbert: Mathematische Probleme. Lecture Paris 1900. Problem 14 (Göttinger Nachrichten 1900).

For then the membership of the quotients $F(x) : L(x)$ and $G(x) : L(x)$ follows from condition 4a).

Another essential property of relatively integral domains is that the notion of being "algebraically integral with respect to $\mathfrak{G}_{n\rho}$" can be defined by means of any equation exactly as is known for algebraic number fields or for the field $\Omega(x_1, \dots, x_n)$.

Definition V: A quantity ξ is called algebraically integral with respect to $\mathfrak{G}_{n\rho}$ — or also relatively algebraically integral — if it satisfies some equation whose highest coefficient is the unit and whose remaining polynomials are from $\mathfrak{G}_{n\rho}$.

Suppose, namely, that the relatively algebraically integral quantity ξ satisfies, for example, the equation

$$L(z) = z^\lambda + G_1(x)z^{\lambda-1} + \cdots + G_\lambda(x) = 0.$$

$L(z)$ is divisible by the integral function irreducible with respect to the smallest containing field $\mathfrak{K}_{n\rho}$,

$$M(z) = z^\mu + H_1(x)z^{\mu-1} + \cdots + H_\mu(x).$$

But from this divisibility it follows by the known extension of Gauss's theorem[100] that the rational functions $H_1(x), \dots, H_\mu(x)$ are polynomials from $\mathfrak{K}_{n\rho}$, and therefore belong to $\mathfrak{G}_{n\rho}$. Thus the definition of the relatively algebraically integral quantity ξ is obtained from the irreducible equation. In particular, every polynomial $F(x)$ from $\mathfrak{G}_{n\rho}$ is also algebraically integral with respect to $\mathfrak{G}_{n\rho}$, since it satisfies the equation $z - F(x) = 0$.

It should further be mentioned that, for an integrality domain $\mathfrak{J}_{n\rho}$ of polynomials, analogously to the smallest containing field, the smallest containing relatively integral domain $\mathfrak{G}_{n\rho}$ is also defined by the requirement:

$\mathfrak{G}_{n\rho}$ contains, along with $\mathfrak{J}_{n\rho}$, all polynomials $H(x)$, and only those, which can be represented as quotients of two polynomials from $\mathfrak{J}_{n\rho}$.

From this definition it follows further: From every involution form and involution basis of $\mathfrak{J}_{n\rho}$ a corresponding one of $\mathfrak{G}_{n\rho}$ arises by splitting off, at most, a common divisor in $\mathfrak{G}_{n\rho}$ of all polynomials of the basis.

For $\mathfrak{J}_{n\rho}$ and $\mathfrak{G}_{n\rho}$ possess the same smallest containing field $\mathfrak{K}_{n\rho}$; and every involution form of $\mathfrak{J}_{n\rho}$ arises from an involution form of $\mathfrak{K}_{n\rho}$ by multiplication with a polynomial from $\mathfrak{J}_{n\rho}$, and so has coefficients from $\mathfrak{G}_{n\rho}$, which however may have a common divisor in $\mathfrak{G}_{n\rho}$.[101]

Finally it should be noted that the *mapping domain* of a relatively integral domain $\mathfrak{G}_{n\rho}$, which by § 7 is again an integrality domain $\mathfrak{J}^*_{\rho\rho}$ of polynomials, need not itself be a relatively integral domain. For if $F(x)$ and $G(x)$ are two polynomials from $\mathfrak{G}_{n\rho}$ whose quotient is not a polynomial and consequently does not belong to $\mathfrak{G}_{n\rho}$, then the quotient $F^*(x) : G^*(x)$ also does not belong to $\mathfrak{J}^*_{\rho\rho}$. But this quotient, arising by specialization of some indeterminates, may very well be a polynomial, and for $\mathfrak{J}^*_{\rho\rho}$ condition 4a) is then not fulfilled.[102]

§ 10. Relatively integral domains of the first kind and their integrality basis.

The notion introduced in § 9 (Definition V), "relatively algebraically integral," makes it possible to give a class of relatively integral domains which are finite integrality domains, and thereby to answer positively, at least for this class, a question raised by D. Hilbert (Math. Problems, Problem 14). We give the following

Definition VI: A finite number of polynomials from $\mathfrak{G}_{n\rho}$ is called an integrality basis if every polynomial from $\mathfrak{G}_{n\rho}$ can be represented as an integral rational combination of this finite number, with coefficients from Ω. An integrality domain of polynomials which possesses an integrality basis is called a finite integrality domain.

The class of relatively integral domains to be considered, for which the involution basis will prove to be an integrality basis, we designate as relatively integral domains of the first kind according to the following

[100] Compare, for instance, J. König: Einleitung in die allgemeine Theorie der algebraischen Größen (Leipzig, B. G. Teubner, 1903), Kap. IX, § 2, or Weber (small edition), § 20.

[101] For example, for the integrality domain $\mathfrak{J}_{22}$ consisting of all integral rational combinations of x_1^2 and x_1x_2, the following is obtained as an involution form:

$$\Psi(z,u) = x_1^2z^2 - x_1^4u_1^2 - 2x_1^3x_2u_1u_2 - x_1^2x_2^2u_2^2.$$

Since x_1^2 belongs to $\mathfrak{J}_{22}$, and therefore all the more to the smallest containing relatively integral domain $\mathfrak{G}_{22}$, and consequently is a common divisor in $\mathfrak{G}_{22}$, the following results as an involution form of $\mathfrak{G}_{22}$:

$$\Psi(z,u) = z^2 - x_1^2u_1^2 - 2x_1x_2u_1u_2 - x_2^2u_2^2.$$

[102] Let $\mathfrak{G}_{22}$ consist, for example, of all polynomials which are rational combinations of $F = x_1^2x_2$ and $G = x_1^2(1 + x_3)$. The admissible specialization $x_3 = 0$ gives $F^* : G^* = x_2$, whereas $F : G$ is not a polynomial. $\mathfrak{J}^*_{22}$ is therefore not a relatively integral domain.

Definition VII: A relatively integral domain $\mathfrak{G}_{n\rho}$ is called of the first kind if, as mapping domain, it possesses a relatively integral domain $\mathfrak{G}^*_{\rho\rho}$ such that every arbitrary polynomial in $x_1, \dots, x_\rho$ (with coefficients from Ω) is algebraically integral over $\mathfrak{G}^*_{\rho\rho}$.

According to this definition, the indeterminates $x_1, \dots, x_\rho$, and consequently also the linear form $u(x) = u_1x_1 + \cdots + u_\rho x_\rho$, are algebraically integral over $\mathfrak{G}^*_{\rho\rho}$. The irreducible integral function with zero $z = u(x)$ — the involution form of the smallest containing field $\mathfrak{K}^*_{\rho\rho}$ — therefore has the unit as highest coefficient and, as its further coefficients, polynomials from $\mathfrak{G}^*_{\rho\rho}$; it is therefore at the same time an involution form of $\mathfrak{G}^*_{\rho\rho}$, and in $\mathfrak{G}^*_{\rho\rho}$ no further involution form can exist. By the transfer principle, therefore, $\mathfrak{G}_{n\rho}$ also possesses an involution form whose highest coefficient is the unit; and since, by Theorem VI, in the rational representation of all polynomials from $\mathfrak{G}_{n\rho}$ by the corresponding involution basis only this highest coefficient enters into the denominator, the involution basis becomes an integrality basis. Thus:

Theorem VII. Every relatively integral domain of the first kind is a finite integrality domain; the integrality basis is given by the involution basis obtained from the characterizing mapping domain. In particular, the integrality basis of relatively integral domains of the first kind $\mathfrak{G}_{\rho\rho}$ is given by the uniquely defined involution basis of $\mathfrak{G}_{\rho\rho}$.

We add a criterion for relatively integral domains of the first kind. Let $\mathfrak{G}_{n\rho}$ be of the first kind and let an integrality basis of $\mathfrak{G}_{n\rho}$ be given by

$$F_1(x), F_2(x), \dots, F_k(x). \tag{1}$$

Then, by a Hilbert auxiliary theorem,[103] there exist ρ algebraically independent polynomials from $\mathfrak{G}_{n\rho}$,

$$G_1(x), G_2(x), \dots, G_\rho(x), \tag{2}$$

such that the polynomials (1), and hence every polynomial from $\mathfrak{G}_{n\rho}$, are algebraically integral over the polynomials (2). The same holds for every polynomial of the mapping domain $\mathfrak{G}^*_{\rho\rho}$ with respect to the corresponding polynomials

$$G^*_1(x), G^*_2(x), \dots, G^*_\rho(x). \tag{3}$$

By Definition VII, therefore, every arbitrary polynomial in $x_1, \dots, x_\rho$ is also algebraically integral over the polynomials (3).

Conversely, suppose a mapping domain $\mathfrak{J}^*_{\rho\rho}$ of an arbitrary relatively integral domain $\mathfrak{G}_{n\rho}$ contains ρ polynomials (3) over which every polynomial in $x_1, \dots, x_\rho$ is algebraically integral. Then we show that $\mathfrak{J}^*_{\rho\rho}$ thereby becomes a relatively integral domain $\mathfrak{G}^*_{\rho\rho}$, and further that $\mathfrak{G}^*_{\rho\rho}$, and consequently $\mathfrak{G}_{n\rho}$, are of the first kind.

Indeed, let $F^*(x)$ be a polynomial from the smallest field $\mathfrak{K}^*_{\rho\rho}$ containing $\mathfrak{J}^*_{\rho\rho}$. By assumption it is algebraically integral over the polynomials (3). Since the corresponding function from $\mathfrak{K}_{n\rho}$ is then algebraically integral over ρ polynomials from $\mathfrak{G}_{n\rho}$, it is a polynomial $F(x)$ and hence belongs to $\mathfrak{G}_{n\rho}$; therefore $F^*(x)$ belongs to $\mathfrak{J}^*_{\rho\rho}$. The mapping domain $\mathfrak{J}^*_{\rho\rho}$ therefore contains all polynomials of $\mathfrak{K}^*_{\rho\rho}$ and is a relatively integral domain $\mathfrak{G}^*_{\rho\rho}$. For $\mathfrak{G}^*_{\rho\rho}$, and hence also for $\mathfrak{G}_{n\rho}$, it follows immediately from Definitions V and VII and from the assumption that they are of the first kind; that is:

The relatively integral domains of the first kind $\mathfrak{G}_{n\rho}$ are characterized by the fact that in a mapping domain $\mathfrak{J}^*_{\rho\rho}$ there exist ρ polynomials over which every arbitrary polynomial in $x_1, \dots, x_\rho$ is algebraically integral. From this assumption the mapping domain itself results as a relatively integral domain.[104]

§ 11. Auxiliary theorem on algebraically integral dependence.

We now give a criterion for every arbitrary polynomial in $x_1, \dots, x_\rho$ to be algebraically integral over ρ polynomials in $x_1, \dots, x_\rho$,

$$G^*_1(x), \dots, G^*_\rho(x). \tag{1}$$

If the polynomials (1) are, in particular, homogeneous forms, then for every special system of values which makes (1) vanish, all algebraically integrally dependent forms vanish at the same time; hence in particular also $x_1, \dots, x_\rho$.

[103] D. Hilbert: Über die vollen Invariantensysteme, Math. Ann. 42 (1893), § 1. The auxiliary theorem stated there only for homogeneous polynomials holds likewise for inhomogeneous ones.

[104] For example, in the domain $\mathfrak{G}_{22}$ mentioned in the closing remark to § 9, the admissible specialization $x_1 = 1$ gives the two polynomials $F^* = x_2$, $G^* = 1 + x_3$, over which every arbitrary polynomial in x_2, x_3 is algebraically integral. $\mathfrak{G}_{22}$ is therefore of the first kind, indeed with $F(x)$ and $G(x)$ as integrality basis.

The forms (1), therefore, cannot vanish for any system of values other than

$$x_1 = 0, \ldots, x_\rho = 0;$$

their resultant must be different from zero.

Conversely, this condition is also sufficient and admits an extension to inhomogeneous polynomials according to the following

Auxiliary theorem: A sufficient condition that every arbitrary polynomial in $x_1, \ldots, x_\rho$ be algebraically integral over ρ polynomials in $x_1, \ldots, x_\rho$

$$G_1^*(x), \ldots, G_\rho^*(x)$$

is the non-vanishing of the resultant, taken with respect to $x_1, \ldots, x_\rho$, of the terms of highest dimension

$$H_1^*(x), \ldots, H_\rho^*(x)$$

of the polynomials (1):

$$R_0 = \begin{bmatrix} H_1^* & \cdots & H_\rho^* \\ x_1 & \cdots & x_\rho \end{bmatrix} \neq 0. \tag{2}$$

For homogeneous forms (1) the condition (2) is necessary.[105]

For the proof we form the polynomials

$$\begin{gathered} \bar{G}_1^*(x) = G_1^*(x) - \lambda_1, \ldots, \bar{G}_\rho^*(x) = G_\rho^*(x) - \lambda_\rho, \\ \bar{\Phi}^*(x) = \Phi^*(x) - \mu, \end{gathered} \tag{3}$$

where $\lambda_1, \ldots, \lambda_\rho, \mu$ denote indeterminates and $\Phi^*(x)$ denotes an arbitrary polynomial in $x_1, \ldots, x_\rho$. Let the resultant of the polynomials (3) with respect to $x_1, \ldots, x_\rho, 1$[106] be given by

$$R(\lambda, \mu) = \begin{bmatrix} \bar{G}_1^* & \cdots & \bar{G}_\rho^* & \bar{\Phi}^* \\ x_1 & \cdots & x_\rho & 1 \end{bmatrix}; \tag{4}$$

it is an integral rational function of $\lambda_1, \ldots, \lambda_\rho, \mu$. According to F. Mertens, in the expansion of (4) by powers of μ, the highest coefficient can be explicitly specified;[107] one obtains

$$(-1)^\tau R(\lambda, \mu) = R_0^\sigma \mu^\tau + A_1(\lambda)\mu^{\tau-1} + \cdots + A_\tau(\lambda), \tag{5}$$

where $A_1(\lambda), \ldots, A_\tau(\lambda)$ denote integral integer-valued functions of $\lambda_1, \ldots, \lambda_\rho$ and the coefficients of the polynomials (3). This expansion first shows that under the assumption (2), $R_0 \neq 0$, also $R(\lambda, \mu)$ cannot vanish identically. The identity in $x_1, \ldots, x_\rho, \lambda_1, \ldots, \lambda_\rho, \mu$,

$$R(\lambda, \mu) \equiv 0 \pmod{\bar{G}_1^*(x), \ldots, \bar{G}_\rho^*(x), \bar{\Phi}^*(x)},$$

passes, by the substitution

$$\lambda_i = G_i^*(x), \qquad \mu = \Phi^*(x),$$

into

$$\begin{gathered} R(G_i^*(x), \Phi^*(x)) = R_0^\sigma \Phi^{*\tau}(x) + A_1(G_i^*(x))\Phi^{*\tau-1}(x) + \cdots \\ + A_\tau(G_i(x)) = 0, \end{gathered} \tag{6}$$

and this proves the auxiliary theorem, since R_0 is, by assumption, a number from Ω.

[105]For resultant theory, compare F. Mertens, Zur Theorie der Elimination (I. and II. Teil), Sitzungsber. d. k. Ak. d. Wiss. Wien, Math.-naturw. Kl. Abt. IIa, Bd. 108 (1899). Partly reproduced in J. König, Theorie d. algebr. Größen, Kap. VI.

[106]That is, the homogeneous resultant with respect to $x_1, \ldots, x_\rho, x_0$, if the polynomials (3) are homogenized by means of an additional indeterminate x_0 so that the degree in the x is preserved.

[107]A. a. O. § 3 (proof of Theorem I), and explicitly § 6 (determination of the totality of all terms of the resultant R which contain a given power product $a_{\lambda\lambda}^{p_\lambda} a_{\mu\mu}^{p_\mu} \cdots a_{\sigma\sigma}^{p_\sigma}$ as factor). In our case $p_\lambda = 0$, $p_\mu = 0, \ldots, p_\rho = \tau$, $a_{\sigma\sigma} = (-\mu)$ is set. Reproduced in König, Kap. VI, § 9.

§ 12. The integrality basis of regular systems $\mathfrak{S}_{n\rho}$ consisting of polynomials.

The auxiliary theorem of § 11, together with the conclusion of § 10, immediately yields the fact:

If there exist, in a mapping domain $\mathfrak{J}^*_{\rho\rho}$ of $\mathfrak{G}_{n\rho}$, ρ polynomials such that the homogeneous resultant of their terms of highest dimension does not vanish identically — $R_0 \neq 0$ —, then $\mathfrak{G}_{n\rho}$ is of the first kind and consequently is a finite integrality domain, with the involution basis as integrality basis.

The condition $R_0 \neq 0$ now permits, by means of the auxiliary theorem, a second existence proof for the integrality basis, essentially due to D. Hilbert;[108] this proof, to be sure, does not make the involution basis recognizable as such, but it holds uniformly for all systems $\mathfrak{S}_{n\rho}$ with the indicated property.

Indeed, let the ρ polynomials from $\mathfrak{J}^*_{\rho\rho}$ for which $R_0 \neq 0$, and which consequently are algebraically independent, be given by

$$G_1^*(x), \ldots, G_\rho^*(x). \tag{1}$$

By the auxiliary theorem, every arbitrary polynomial in $x_1, \ldots, x_\rho$, hence in particular every polynomial of the smallest field $\mathfrak{K}^*_{\rho\rho}$ containing $\mathfrak{J}^*_{\rho\rho}$, is algebraically integral over the polynomials (1). If the corresponding polynomials in $\mathfrak{S}_{n\rho}$ are therefore

$$G_1(x), \ldots, G_\rho(x), \tag{2}$$

then, by the transfer principle, every polynomial of the smallest field $\mathfrak{K}_{n\rho}$ containing $\mathfrak{S}_{n\rho}$, and hence in particular every polynomial from $\mathfrak{S}_{n\rho}$, is algebraically integral over the polynomials (2). Conversely, every function from $\mathfrak{K}_{n\rho}$ which is algebraically integral over the polynomials (2) is also a polynomial. Thus, if one regards $\mathfrak{K}_{n\rho}$ (according to § 4) as an algebraic field over $\Omega(G_1, \ldots, G_\rho)$, the totality of the polynomials from $\mathfrak{K}_{n\rho}$ — the relatively integral domain $\mathfrak{G}_{n\rho}$ — is identical with the algebraically integral quantities of this field. Let $G(x)$ be a polynomial from $\mathfrak{K}_{n\rho}$ which completes (2) to a rational basis, and let $D(x)$ be the discriminant of the irreducible equation of degree k which $G(x)$ satisfies with respect to $\Omega(G_1, \ldots, G_\rho)$. Then every polynomial from $\mathfrak{K}_{n\rho}$, and in particular every polynomial $F(x)$ from $\mathfrak{S}_{n\rho}$, as an algebraically integral quantity, admits the representation

$$F(x) = \frac{\Gamma_1(G_i)G(x)^{k-1} + \Gamma_2(G_i)G(x)^{k-2} + \cdots + \Gamma_k(G_i)}{D(x)}. \tag{3}$$

Let $F(x)$ now run through all polynomials from $\mathfrak{S}_{n\rho}$. Then the application of Hilbert's Theorem I on the module basis (Math. Ann. 36) to the system formed from the corresponding functions

$$v_1\Gamma_1(G_i) + v_2\Gamma_2(G_i) + \cdots + v_k\Gamma_k(G_i)$$

shows the existence of a finite number of polynomials from $\mathfrak{S}_{n\rho}$,

$$F_1(x), F_2(x), \ldots, F_\lambda(x), \tag{4}$$

such that every polynomial from $\mathfrak{S}_{n\rho}$ admits a representation

$$F(x) = A_1(G_i)F_1(x) + A_2(G_i)F_2(x) + \cdots + A_\lambda(G_i)F_\lambda(x), \tag{5}$$

where $A_1, \ldots, A_\lambda$ denote polynomials in the $G_i(x)$. The polynomials (2) and (4) therefore form an integrality basis of $\mathfrak{S}_{n\rho}$; that is:

Theorem VIII: If there exist, in a mapping system $\mathfrak{J}^*_{\rho\rho}$ of the system of polynomials $\mathfrak{S}_{n\rho}$, ρ polynomials such that the resultant of the terms of highest dimension does not vanish identically — $R_0 \neq 0$ —, then $\mathfrak{S}_{n\rho}$ possesses an integrality basis.

This theorem allows a slight generalization. It is enough that the ρ polynomials (1) for which $R_0 \neq 0$ belong to the integrality domain $\mathfrak{J}^*_{\rho\rho}$ which is the smallest containing domain of $\mathfrak{S}^*_{\rho\rho}$. In fact, the arguments leading to Theorem VIII show that the representation (5) still holds then. But by the definition, every polynomial $L(x)$ of the smallest containing integrality domain $\mathfrak{J}_{n\rho}$ can be represented integrally and rationally by a finite number — depending on $L(x)$ — of polynomials from $\mathfrak{S}_{n\rho}$; hence there exist σ polynomials from $\mathfrak{S}_{n\rho}$,

$$K_1(x), K_2(x), \ldots, K_\sigma(x), \tag{6}$$

[108] Über die vollen Invariantensysteme, § 2. In Hilbert's case, under the assumption of the otherwise obtained existence of the integrality basis of the invariant field, the point is only an interpretation of this basis through the Kronecker fundamental system of the algebraically integral quantities of a field.

such that the polynomials $G_i(x)$ become integral rational combinations of (6). Thus (6) and (4) give an integrality basis. Introducing the following terminology:

Definition VIII: A system $\mathfrak{S}_{n\rho}$ of polynomials is called regular if in a mapping domain $\mathfrak{J}^*_{\rho\rho}$ of the smallest containing integrality domain there exist ρ polynomials such that the resultant of their terms of highest dimension does not vanish identically — $R_0 \neq 0$ —

we therefore have:

Theorem IX: Every regular system $\mathfrak{S}_{n\rho}$ of polynomials possesses an integrality basis.[109]

We further give, analogously to § 5 for the involution, a geometric interpretation of regular systems. For this purpose we consider, as there in (7), the system of equations

$$L^*(x) - L^*(\xi) = 0, \tag{7}$$

where $L^*(x)$ runs through all polynomials from $\mathfrak{J}^*_{\rho\rho}$. Homogenizing by means of x_0, ξ_0, let (7) pass into the system of equations between homogeneous forms

$$\xi_0^m M^*(x) - x_0^m M^*(\xi) = 0, \tag{8}$$

where, if $H^*(x)$ denotes the terms of highest dimension of $L^*(x)$, the relation

$$H^*(x) = M^*(x) \pmod{x_0} \tag{9}$$

holds.

As *fundamental points* of $\mathfrak{J}^*_{\rho\rho}$ (compare the note to § 5), we designate the solutions of (8) that are independent of x — thus different from the rows conjugate to x. Fundamental points are therefore the systems of values different from $\xi_1 = 0, \ldots, \xi_\rho = 0$ for which simultaneously

$$\xi_0^m = 0, \qquad M^*(\xi) = 0$$

holds, or, by (9),

$$\xi_0 = 0, \qquad H^*(\xi) = 0.$$

If now there exist ρ forms $H^*(\xi)$ for which $R_0 \neq 0$, hence with no common system of solutions, then $\mathfrak{J}^*_{\rho\rho}$ can possess no fundamental point. If, in particular, $\mathfrak{S}^*_{\rho\rho}$ consists of homogeneous forms, then for $R_0 \neq 0$ also $\mathfrak{S}^*_{\rho\rho}$ can possess no fundamental point, since such a point would at the same time be a fundamental point of $\mathfrak{J}^*_{\rho\rho}$. But the converse also holds. For this we consider the system $\mathfrak{J}(H^*)$ formed from the totality of the forms $H^*(x)$; this is again an integrality domain.

From Hilbert's Theorem I on the module basis and Hilbert's auxiliary theorem (compare the citation in § 10), there follows the existence of ρ forms from $\mathfrak{J}(H^*)$ whose vanishing has the vanishing of all forms from $\mathfrak{J}(H^*)$ as consequence. If now $\mathfrak{J}^*_{\rho\rho}$ possesses no fundamental point, these ρ forms cannot vanish simultaneously; their resultant $R_0 \neq 0$.

Thus:

The regular systems $\mathfrak{S}_{n\rho}$ of polynomials are characterized by the fact that a mapping domain $\mathfrak{J}^*_{\rho\rho}$ of the smallest containing integrality domain possesses no fundamental point. For regular systems $\mathfrak{S}^*_{\rho\rho}$ of homogeneous forms, the system itself already possesses no fundamental point.[110]

[109] That there are non-regular systems without integrality basis was shown by Hilbert with an example (Gött. Nachr. 1891, p. 233). The following is even simpler:

$$\mathfrak{S}_{22} = x_1, x_1x_2, x_1x_2^2, \ldots, x_1x_2^\nu, \ldots \quad \text{in inf.}$$

The existence of the integrality basis here would be equivalent to the existence of a positive integer ν such that the identities

$$x_1x_2^{\nu+\sigma} = \sum C \cdot x_1^{i_0}(x_1x_2)^{i_1}(x_1x_2^2)^{i_2}\cdots(x_1x_2^\nu)^{i_\nu} \quad (\sigma = 1, 2, \ldots)$$

hold. Comparing the dimensions in x_1, x_2 already for $\sigma = 1$ leads to the two conditional equations for the exponents

$$1 = i_0 + i_1 + \cdots + i_\nu, \qquad \nu + 1 = i_1 + 2i_2 + \cdots + \nu i_\nu,$$

which plainly have no solution in non-negative integers. Thus $\mathfrak{S}_{22}$ possesses no integrality basis.

[110] The non-regular system $\mathfrak{S}_{22}$ given in the preceding note possesses the fundamental point

$$\xi_1 : \xi_2 : \xi_0 = 0 : 1 : 0.$$

§ 13. Examples of relatively integral domains of the first kind and of regular systems.

1. As a first example of a relatively integral domain of the first kind $\mathfrak{G}_{nn}$, we consider the *totality of the polynomials in $x_1, \ldots, x_n$ that admit a given permutation group G* — thus the totality of the polynomials of a Lagrange genus domain. Since $x_1, \ldots, x_n$, by virtue of the identity

$$x_k^n - \sigma_1(x)x_k^{n-1} + \sigma_2(x)x_k^{n-2} + \cdots \pm \sigma_n(x) = 0 \qquad (k = 1, 2, \ldots, n), \tag{1}$$

are algebraically integral over the elementary symmetric functions belonging to $\mathfrak{G}_{nn}$, and hence by Definition V are algebraically integral over $\mathfrak{G}_{nn}$, $\mathfrak{G}_{nn}$ is of the first kind. Moreover $\mathfrak{G}_{nn}$, as a domain of the first kind, since its elements are homogeneous forms and $n = \rho$, forms a regular system (by the conclusion of § 10). Indeed, because of (1), the resultant R_0 of the elementary symmetric functions cannot vanish identically; by the rules of calculation of resultant theory, R_0 is computed to be ± 1. The *involution form* of $\mathfrak{G}_{nn}$ is uniquely determined — since $\mathfrak{G}_{nn}$ is of the first kind and $n = \rho$ — as the involution form of the corresponding Lagrange genus domain. Since the quantities conjugate to $x_1, \ldots, x_n$ with respect to this domain arise through the permutations of the group G, this involution form is therefore given by

$$\begin{aligned} \Psi(z, u) &= \prod_i (z - u_1 x_{i_1} - u_2 x_{i_2} - \cdots - u_n x_{i_n}) \\ &= z^i + \sum{}' G_{\alpha\alpha_1 \ldots \alpha_n}(x) z^\alpha u_1^{\alpha_1} \cdots u_n^{\alpha_n}, \end{aligned} \tag{2}$$

where the product is to be extended over all permutations of G. Formula (2) represents the "Galois form of the group G," and hence the following fact holds:

All polynomials in $x_1, \ldots, x_n$ which admit a permutation group G are integral rational combinations of the coefficients $G_{\alpha\alpha_1 \ldots \alpha_n}(x)$ of the Galois form of the group.

2. As an example of *regular systems*, consider the systems $\mathfrak{S}_{n1}$ of polynomials, that is, systems which contain only one algebraically independent polynomial.

Indeed, let $\mathfrak{S}_{11}^*$ be a mapping system of $\mathfrak{S}_{n1}$ and let

$$F^*(x) = a_0 x^k + a_1 x^{k-1} + \cdots + a_k \qquad (a_0 \neq 0)$$

be a polynomial from $\mathfrak{S}_{11}^*$. Then

$$R_0 = a_0 \neq 0. \tag{3}$$

Thus $\mathfrak{S}_{n1}$ is regular by Definition VIII, and consequently:

Every system $\mathfrak{S}_{n1}$ of polynomials, in particular every system of infinitely many polynomials in one indeterminate, possesses an integrality basis.[111]

3. We now specialize the systems $\mathfrak{S}_{n1}$ to relatively integral domains $\mathfrak{G}_{n1}$, which, as regular systems by § 12, are relatively integral domains of the first kind. Their integrality basis is therefore given by an involution basis, which is at the same time the involution basis of the smallest containing field $\mathfrak{K}_{n1}$. But by § 6, any function of the involution basis of $\mathfrak{K}_{n1}$ is at the same time a minimal basis — we called it the Lüroth function of the field — and every function of the involution basis depends on this Lüroth function by a fractional-linear transformation. In the present case the Lüroth function, since it belongs to $\mathfrak{G}_{n1}$, is a polynomial; every further polynomial of the involution basis therefore depends fractional-linearly, and by (3) algebraically integrally, hence integrally and linearly, on this Lüroth function $L(x)$, which thus becomes the integrality basis. The involution form of $\mathfrak{G}_{11}^*$, setting also $(u_1 = 1)$, is

$$\Psi^*(z) = L^*(z) - L^*(x),$$

from which $L(x)$ again appears as integrality basis. Thus:

The relatively integral domains $\mathfrak{G}_{n1}$ are of the first kind; their integrality basis consists of one polynomial, the Lüroth function of the smallest containing field.

4. As an example of a relatively integral domain $\mathfrak{G}_{n1}$, let us mention specifically the integral rational projective invariants of a quadratic form in s variables. In agreement with the preceding, it is well known that they can be represented as integral rational combinations of the determinant $|a_{ik}|$ of the form; and this determinant is at the same time the Lüroth function of the corresponding invariant field.

[111] The simplest possible non-regular systems without integrality basis are therefore systems $\mathfrak{S}_{22}$. That such systems actually exist is shown by Hilbert's example and by the example given in the note to § 12.

§ 14. Systems consisting of integer polynomials.

The basis theorems obtained are now to be sharpened with respect to integrality over the integers.

The coefficient domain Ω will therefore in what follows be assumed to be a (finite) algebraic number field; and let $[\Omega]$ denote the totality of the algebraic integers from Ω, which we shall briefly call whole numbers. By an integer polynomial is meant a polynomial with coefficients from $[\Omega]$; from now on $F(x), G(x), \ldots$ shall denote only integer polynomials.

In analogy with § 7, we consider the different systems made up of integer polynomials:

An arbitrary system of integer polynomials shall be denoted as an integer system $\mathfrak{S}_{n\rho}$.

The integer linear family $\mathfrak{L}_{n\rho}$ is an integer system which, along with $F_1(x)$ and $F_2(x)$, always contains $c_1F_1(x) + c_2F_2(x)$, where c_1, c_2 are arbitrary whole numbers from $[\Omega]$.

The integer integrality domain $\mathfrak{J}_{n\rho}$ is an integer linear family which, along with $F(x)$ and $G(x)$, always contains $F(x) \cdot G(x)$.

The integer relatively integral domain $\mathfrak{G}_{n\rho}$ is an integrality domain which, along with $F(x)$ and $G(x)$ — where $G(x) \neq 0$ — always and only contains the quotient $F(x) : G(x)$ when this quotient is an integer polynomial.

Correspondingly as in § 7, the smallest containing integer integrality domain $\mathfrak{J}_{n\rho}$ of an integer system $\mathfrak{S}_{n\rho}$ consists of all polynomials $H(x)$ which are integral rational, integer combinations of a finite number of polynomials from $\mathfrak{S}_{n\rho}$;

the smallest containing integer relatively integral domain $\mathfrak{G}_{n\rho}$ consists of all integer polynomials $H(x)$ which are rational combinations of a finite number of polynomials from $\mathfrak{S}_{n\rho}$.

We now pass to the transfer of the basis theorems. With respect to the rational basis, it is to be noted that the notions of field, smallest containing field, and rational basis, which rest only on "rationality," undergo no change. Furthermore, the mapping system $\mathfrak{S}^*_{\rho\rho}$ of an integer system $\mathfrak{S}_{n\rho}$ can in arbitrarily many ways again be chosen as an integer system. Since, also for the integer linear family, the arguments leading to the bound on the number of generators remain valid, one obtains:

Theorem V′: Every integer system $\mathfrak{S}_{n\rho}$ possesses a rational basis; the rational basis of an integer linear family $\mathfrak{L}_{n\rho}$ can in particular be chosen so that it contains at most $\rho + 1$ polynomials.

We now investigate the analogue of the involution basis of integrality domains of polynomials (§ 8). For this purpose the theorem on *symmetric functions of rows of quantities* must be extended with respect to *integrality over the integers* by the following

Auxiliary theorem. The integral integer symmetric functions of n rows of quantities $(yz \cdots t)$:

$$\begin{matrix} y_1 & z_1 & \cdots & t_1 \\ y_2 & z_2 & \cdots & t_2 \\ \vdots & \vdots & & \vdots \\ y_n & z_n & \cdots & t_n \end{matrix} \tag{1}$$

are integral integer functions of a finite number among them, namely of the elementary symmetric functions

$$\begin{gathered} \sigma_1(y), \sigma_2(y), \ldots, \sigma_n(y); \quad \sigma_1(z), \sigma_2(z), \ldots, \sigma_n(z); \quad \ldots; \\ \sigma_1(t), \ldots, \sigma_n(t); \end{gathered} \tag{2}$$

and of the functions

$$\chi_1(y, z, \ldots, t), \ldots, \chi_k(y, z, \ldots, t), \tag{3}$$

which are algebraically integral and integer over the functions (2).

Indeed, the n quantities $y_1, \ldots, y_n$ depend algebraically integrally and integrally over the integers on $\sigma_1(y), \ldots, \sigma_n(y)$; the corresponding assertion holds for $z_1, \ldots, z_n$ with respect to $\sigma_1(z), \ldots, \sigma_n(z)$, and so on. The field of symmetric functions of the rows of quantities (1) therefore forms an algebraic field over the field determined by the functions (2), in such a way that the algebraically integral and integer quantities of this field are identical with the integral integer symmetric functions of the rows of quantities (1). If one therefore takes (3) as Kronecker's fundamental system, the auxiliary theorem is proved.

Let now $\mathfrak{J}_{n\rho}$ be an integer integrality domain, $\mathfrak{J}^*_{\rho\rho}$ an integer mapping domain, and

$$\Psi^*(z, u) = G^*(x) z^i + \sum{}' G^*_{\alpha\alpha_1\ldots\alpha_\rho}(x) z^\alpha u_1^{\alpha_1} \cdots u_\rho^{\alpha_\rho}$$

an involution form of $\mathfrak{J}^*_{\rho\rho}$; here $G^*(x)$ may also have dimension zero, that is, be a number from $[\Omega]$. The elementary symmetric functions corresponding to the functions (2) are then given by certain ones among the quotients

$$G^*_{\alpha\alpha_1\ldots\alpha_\rho}(x) : G^*(x). \tag{4}$$

Since by the auxiliary theorem the functions (3) depend algebraically integrally and integrally over the integers on (2), the functions from $\mathfrak{K}^*_{\rho\rho}$ corresponding to (3) are given — by the extension of Gauss's theorem to polynomials with algebraically integral coefficients — by

$$K^*_\nu(x) : (G^*(x))^\sigma, \tag{5}$$

where $K^*_\nu(x)$ are integer polynomials from $\mathfrak{J}^*_{\rho\rho}$. Thus $K^*_\nu(x)$ belong to the domain if the case is that of relatively integral domains; in general they are quotients of polynomials from $\mathfrak{J}^*_{\rho\rho}$.

If now $F^*(x)$ is an arbitrary integer polynomial from $\mathfrak{J}^*_{\rho\rho}$, then one has

$$F^*(x) = \frac{1}{i}\Big(F^*(x) + F^*(x^{(1)}) + \cdots + F^*(x^{(i-1)})\Big),$$

and therefore $i \cdot F^*(x)$ becomes an integral integer symmetric function of the rows of quantities $x, x^{(1)}, \ldots, x^{(i-1)}$.

By the auxiliary theorem and the transfer principle, therefore:

Theorem VI′: For integer integrality domains $\mathfrak{J}_{n\rho}$, each involution form determines a distinguished rational basis in such a way that every polynomial $F(x)$ from $\mathfrak{J}_{n\rho}$ admits a representation

$$F(x) = \frac{\Gamma(H(x), H_1(x), \ldots, H_s(x))}{i \cdot H(x)^i},$$

where Γ denotes an integral integer function. For integer relatively integral domains $\mathfrak{G}_{n\rho}$, with integer relatively integral domain $\mathfrak{G}^*_{\rho\rho}$ as mapping domain, the denominator $H(x)$ is given by the highest coefficient of the corresponding involution form.

§ 15. Integer relatively integral domains of the first kind and regular systems.

For integer systems, the integer integrality basis takes the place of the integrality basis; that is, a finite number of integer polynomials of the system such that every integer polynomial of the system can be represented as an integral integer combination of this finite number.

The theorems on the integer integrality basis run essentially parallel to the earlier ones, and only the proofs require slight modifications.

First, Definition V (§ 9) transfers directly.

Definition V′: A quantity ξ is called algebraically integral and integer with respect to the integer relatively integral domain $\mathfrak{G}_{n\rho}$ if it satisfies some equation whose highest coefficient is a unit from $[\Omega]$ and whose remaining polynomials are from $\mathfrak{G}_{n\rho}$.

The extension of Gauss's theorem to polynomials with algebraically integral coefficients shows, as in § 9, that the definition at the irreducible equation can be obtained from this.

From Definition V′ one further obtains the transfer of Definition VII.

Definition VII′: An integer relatively integral domain $\mathfrak{G}_{n\rho}$ is called of the first kind if, as mapping domain, it possesses a relatively integral domain $\mathfrak{G}^*_{\rho\rho}$ such that every arbitrary integer polynomial in $x_1, \ldots, x_\rho$ is algebraically integral and integer over $\mathfrak{G}^*_{\rho\rho}$.

To show the finiteness of these domains of the first kind, we first note that, as in § 10, it follows from the definition that the highest coefficient of the involution form is a unit from $[\Omega]$.

By Theorem VI′ (§ 14), every polynomial $F(x)$ from $\mathfrak{G}_{n\rho}$ therefore admits a representation

$$F(x) = \frac{\Gamma(H_1(x), \ldots, H_\sigma(x))}{i}.$$

Applying here to the system formed from the integer polynomials $\Gamma(H_1, \ldots, H_\sigma)$ Hilbert's Theorem II on the integer module basis (Math. Ann. 36), originally stated for integral rational numerical coefficients, in its extension to algebraically integral polynomials,[112] the existence of the integer integrality basis follows; that is:

[112]If $w_1, \ldots, w_k$ denotes a fundamental system of the algebraic integers from Ω, and one sets

$$\Gamma(H_1, \ldots, H_\sigma) = \Gamma_1(H)w_1 + \cdots + \Gamma_k(H)w_k,$$

Theorem VII′: Every integer relatively integral domain of the first kind possesses an integer integrality basis.

We further transfer the definition of regular systems:

Definition VIII′: An integer system $\mathfrak{S}_{n\rho}$ is called integer-regular if, in a mapping domain $\mathfrak{J}^*_{\rho\rho}$ of the smallest containing integer integrality domain, there exist ρ polynomials such that the resultant of the terms of highest dimension is a unit from $[\Omega]$ — $R_0 = \varepsilon$.

In order to transfer the existence proof of the integrality basis, we note that the auxiliary theorem of § 11 admits the following sharper formulation, which follows from equations (5) and (6) in § 11 and from the fact that $A_1(\lambda), \ldots, A_\tau(\lambda)$ are integral integer functions of $\lambda_1, \ldots, \lambda_\rho$:

A sufficient condition that every arbitrary integer polynomial in $x_1, \ldots, x_\rho$ depend algebraically integrally and integrally over ρ integer polynomials is that the resultant of the terms of highest dimension of these polynomials be a unit from $[\Omega]$ — $R_0 = \varepsilon$.

From this auxiliary theorem, exactly as in § 12, for every polynomial $F(x)$ of an integer-regular system, representation (3) of § 12 follows, where now $\Gamma_1(G_i), \ldots, \Gamma_k(G_i)$ are integer polynomials in the G_i. Furthermore, as in § 12, the application of Hilbert's Theorem II (Math. Ann. 36) in its extension to algebraically integral polynomials, together with the definition of the smallest containing integer integrality domain, gives the existence of the integer integrality basis; that is:

Theorem IX′: Every integer-regular system possesses an integer integrality basis.

As an example of an integer relatively integral domain of the first kind, in analogy with example 1 in § 13, we refer to the totality $\mathfrak{G}_{nn}$ of integer polynomials of a Lagrange genus domain. That $\mathfrak{G}_{nn}$ is integer of the first kind follows from the identity

$$x_k^n - \sigma_1(x)x_k^{n-1} + \cdots \pm \sigma_n(x) = 0 \qquad (k = 1, 2, \ldots, n);$$

since the resultant of the elementary symmetric functions has the value ± 1, $\mathfrak{G}_{nn}$ is also an integer regular system.

A further example is given, by the auxiliary theorem of § 14, by the totality of the integral integer symmetric functions of n rows of quantities.

Erlangen, May 1914.

then $\Gamma_1(H), \ldots, \Gamma_k(H)$ have integral rational numerical coefficients. Applying Theorem II to the system formed from the forms $v_1\Gamma_1(H) + \cdots + v_k\Gamma_k(H)$ directly shows the validity of the extension.

7. The finiteness theorem for the invariants of finite groups

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In what follows an entirely elementary proof of finiteness for the invariants of finite groups is to be given — one based only on the theory of symmetric functions — and at the same time it gives an actual specification of the full system; whereas the usual proof, based on Hilbert's theorem on module bases (Ann. 36), is only an existence proof.[113]

Let the finite group $\mathfrak{H}$ consist of the h linear transformations (with non-vanishing determinant) $A_1, \dots, A_h$, where A_k denotes the linear transformation

$$x_1^{(k)} = \sum_{\nu=1}^{n} a_{1\nu}^{(k)} x_\nu, \dots, x_n^{(k)} = \sum_{\nu=1}^{n} a_{n\nu}^{(k)} x_\nu,$$

or, in abbreviated form, $(x^{(k)}) = A_k(x)$. The group $\mathfrak{H}$ therefore carries the row (x) with elements $x_1, \dots, x_n$ into the rows $(x^{(k)})$ with elements $x_1^{(k)}, \dots, x_n^{(k)}$. Since the identity must be contained among $A_1, \dots, A_h$, the row (x) is also contained among the rows $(x^{(k)})$. By an integral rational (absolute) invariant of the group is meant an integral rational function of $x_1, \dots, x_n$ which remains identically unchanged under the application of $A_1, \dots, A_h$; for such an invariant $f(x)$ one therefore has

$$f(x) = f(x^{(1)}) = \cdots = f(x^{(h)}) = \frac{1}{h} \sum_{k=1}^{h} f(x^{(k)}). \tag{1}$$

1.

Formula (1) says that $f(x)$ is an integral rational symmetric function of the rows of quantities $(x^{(1)}), \dots, (x^{(h)})$ — and indeed, since each summand $f(x^{(k)})$ contains only the one row $(x^{(k)})$, this is the simplest case, usually called the *one-form* case. By the known theorem on symmetric functions of rows of quantities,[114] $f(x)$ is therefore integrally and rationally representable by the elementary symmetric functions of these rows, that is, by the coefficients $G_{\alpha\alpha_1\dots\alpha_n}(x)$ of the "Galois resolvent"

$$\begin{aligned}\Phi(z,u) &= \prod_{k=1}^{h} (z + u_1 x_1^{(k)} + \cdots + u_n x_n^{(k)}) \\ &= z^h + \sum G_{\alpha\alpha_1\dots\alpha_n}(x) z^\alpha u_1^{\alpha_1} \cdots u_n^{\alpha_n},\end{aligned} \qquad \left(\begin{array}{c} \alpha + \alpha_1 + \cdots + \alpha_n = h \\ \alpha \neq h \end{array} \right),$$

where the $G_{\alpha\alpha_1\dots\alpha_n}(x)$ are invariants of degree $\alpha_1 + \cdots + \alpha_n$ in the x's. Thus it is proved:

The coefficients $G_{\alpha\alpha_1\dots\alpha_n}(x)$ of the Galois resolvent form a full system of invariants of the group, in such a way that every invariant can be represented integrally and rationally by these finitely many invariants.

2.

Starting from (1), one may also make the following still more elementary consideration,[115] which at the same time leads to a second full system. Put

$$f(x) = a + b x_1^{\mu_1} \cdots x_n^{\mu_n} + \cdots + c x_1^{\nu_1} \cdots x_n^{\nu_n},$$

where $a, b, \dots, c$ denote constants. Then by (1) one has

$$\begin{aligned} h \cdot f(x) &= h \cdot a + b \sum_{k=1}^{h} (x_1^{(k)})^{\mu_1} \cdots (x_n^{(k)})^{\mu_n} + \cdots + c \sum_{k=1}^{h} (x_1^{(k)})^{\nu_1} \cdots (x_n^{(k)})^{\nu_n} \\ &= h \cdot a + b \cdot J_{\mu_1\dots\mu_n} + \cdots + c \cdot J_{\nu_1\dots\nu_n}. \end{aligned}$$

Every invariant can therefore be composed integrally and linearly from the special invariants

$$J_{\mu_1\dots\mu_n} = \sum_{k=1}^{h} (x_1^{(k)})^{\mu_1} \cdots (x_n^{(k)})^{\mu_n},$$

[113] Cf., for example, Weber, *Lehrbuch der Algebra* (2nd ed.), vol. 2, § 57.

[114] Cf. the note to 2.

[115] This amounts to a proof of the theorem on symmetric functions of rows of quantities in the above-mentioned one-form case.

and it is enough to prove the assertion for these special ones. But, apart from a numerical factor, $J_{\mu_1 \dots \mu_n}$ is the coefficient of $u_1^{\mu_1} \cdots u_n^{\mu_n}$ in the expression

$$S_\mu = \sum_{k=1}^{h} (u_1 x_1^{(k)} + \cdots + u_n x_n^{(k)})^\mu,$$

where $\mu = \mu_1 + \cdots + \mu_n$, and this expression is the μ-th power sum of the h linear forms

$$\xi_1 = u_1 x_1^{(1)} + \cdots + u_n x_n^{(1)}, \dots, \xi_h = u_1 x_1^{(h)} + \cdots + u_n x_n^{(h)}.$$

It is known that the infinitely many power sums S_μ are integral rational combinations of

$$S_1, S_2, \dots, S_h,$$

whose coefficients are given by the invariants

$$J_{\mu_1 \dots \mu_n} \qquad (\mu_1 + \cdots + \mu_n \leq h).$$

Thus a second full system has been obtained:

A full system of invariants of the group is given by all invariants $J_{\mu_1 \dots \mu_n}$ for which $\mu_1 + \cdots + \mu_n \leq h$, where h denotes the order of the group.

The connection with 1. is established by observing that the power sums $S_1, \dots, S_h$ can be replaced by the elementary symmetric functions of $\xi_1, \dots, \xi_h$, which leads to the full system of the $G_{\alpha\alpha_1 \dots \alpha_n}(x)$ given there. Both results show that all invariants can be expressed integrally and rationally by those whose degree in the x's does not exceed the order h of the group.

3.

Consequences for rational representation can be drawn from what has just been proved. Every rational absolute invariant can be represented — as is seen immediately by multiplying numerator and denominator by the conjugates of the denominator with respect to the group — as the quotient of two integral rational absolute invariants, not necessarily relatively prime. It follows that every rational absolute invariant can be expressed rationally by the coefficients $G_{\alpha\alpha_1 \dots \alpha_n}(x)$ of the Galois resolvent $\Phi(z, u)$. This theorem can also be proved easily in various ways without passing through the finiteness theorem; it is already formulated in Weber II, § 58.[116]

4.

Finally, it should be mentioned that the results derived here are implicitly contained in my paper "Fields and systems of rational functions";[117] more precisely, the full system given in 1. is contained in Theorems VI and VII, and a proof of rational representation independent of this finiteness theorem is contained in Theorem III.[118] In the present special case, however, the course of the proof and the result simplify considerably in comparison with the general theory. I was led to apply the general investigations to the invariants of finite groups by conversations with E. Fischer, from whom, in substance, the proof given under 2 — in the general case the full system that appears there is not rationally definable — and the note to 3 also originate.

Erlangen, May 1915.

[116]The proof given there, however, is not sound. It is only shown that the function $\Psi(t)$ in formula (7) has invariants as coefficients, but not that these invariants are rationally expressible by the coefficients of $\Phi(t)$. This gap is avoided if, in representing the $x_i^{(k)}$ by ξ_k, one uses a known differentiation procedure instead of Lagrange's interpolation formula. This is the relation obtained by differentiating the identity $\Phi(-\xi_k, u) = 0$ with respect to all u:

$$\left[\frac{\partial \Phi}{\partial u_i} - x_i^{(k)} \frac{\partial \Phi}{\partial z}\right]_{z=-\xi_k} = 0.$$

Accordingly, for each rational function $\omega(x)$ the following formula must replace formulas (7) and (8) in Weber:

$$\omega(x_1^{(k)}, \dots, x_n^{(k)}) = \omega\left(\frac{\partial\Phi/\partial u_1}{\partial\Phi/\partial z}, \dots, \frac{\partial\Phi/\partial u_n}{\partial\Phi/\partial z}\right)\Big|_{z=-\xi_k},$$

which contains only coefficients of $\Phi(z, u)$, and whose summation over all k, for invariants $\omega(x)$, gives the desired representation.

[117]Math. Ann. 76, p. 161 (1915).

[118]For relative invariants, Theorems VIII and IX contain a proof of finiteness which is likewise based on a different foundation from the usual one, but is also only an existence proof; the reason is that the relative invariants do not form a field.

8. On integral rational representation of the invariants of a system of arbitrarily many ground forms

Math. Ann. 77 (1916), pp. 93–102

At the end of his contribution to the Schwarz Festschrift, “On the invariants of a system of arbitrarily many ground forms”, D. Hilbert states the conjecture that these invariants can be represented integrally and rationally by the finitely many invariants of the system (J, PJ), where J denotes the full system of invariants of the linearly independent ground forms, and the invariants PJ arise from J by polar processes.

I show below, in I, that this conjecture is in fact correct, and indeed as a simple consequence of the reduction theorem that every form in arbitrarily many rows of n variables each can be represented as a sum of polars of forms which contain only n fixed rows and which themselves are derived from the original form by polar processes. This reduction theorem is a special case, first formulated by F. Mertens, of the generalization, given by A. Capelli, F. Mertens, and J. Deruyts, of the Clebsch–Gordan series expansion to forms in arbitrarily many rows of n variables each; this generalization is proved by formal transformation of the differentiation processes.[119]

In II I also give a more conceptual proof of the reduction theorem, by treating the question as a problem concerning the equivalence of linear families of forms. This point of view, and also an essential part of the considerations used, I take from a paper “On algebraic module systems”[120] and from unpublished theorems of E. Fischer adjoining it, whose use he kindly permitted me.

Finally, in III, I show how the corresponding considerations also lead to the most general series expansions.

I.

Let θ be a form homogeneous in each of the $N > n$ rows

$$A_1, A_2, \ldots, A_N,$$

where in general the row A_k consists of the n elements

$$A_k^{(1)}, A_k^{(2)}, \ldots, A_k^{(n)}.$$

The coefficients of θ may be indeterminates or quantities of a given domain of rationality. Let P further denote an integral rational function — with rational integer coefficients — of the polar processes

$$P_{hk} = \left(A_h \frac{\partial}{\partial A_k}\right) = A_h^{(1)} \frac{\partial}{\partial A_k^{(1)}} + A_h^{(2)} \frac{\partial}{\partial A_k^{(2)}} + \cdots + A_h^{(n)} \frac{\partial}{\partial A_k^{(n)}},$$

where the product $P_{hk} P_{ij} \theta$ is to be understood as the application of P_{hk} to $P_{ij}\theta$.

Then the reduction theorem just mentioned is given by the identity

$$\theta = \sum PZ, \tag{1}$$

where the forms Z contain only the rows

$$A_1, A_2, \ldots, A_n$$

and are derived from θ by polar processes P.

We now consider the system of ground forms (F'), consisting of the N forms of the same order and the same number of variables

$$F_1, F_2, \ldots, F_N,$$

whose coefficients are taken respectively as the N rows

$$A_1, A_2, \ldots, A_N.$$

[119] F. Mertens: “Über eine Formel der Determinantentheorie.” (Formula 5), Sitzb. d. Ak. d. Wiss. Wien, vol. 91, II. Abt. (1885), and more fully (with applications) in: “Über ganze Funktionen von m Systemen von je n Unbestimmten”. Monatsh. f. Math. u. Phys. 4th year (1893). For some of Capelli’s proofs (since 1882) cf., for example, Math. Ann. 37, or the autographed “Lezioni sulla teoria delle forme algebriche” (1902). J. Deruyts: *Essai d’une théorie générale des formes algébriques*, Mém. d. 1. Soc. Roy. des Sciences de Liège (2) 17 (1892).

[120] E. Fischer: Über algebraische Modulsysteme und lineare homogene partielle Differentialgleichungen mit konstanten Koeffizienten, § 1–4, J. f. M. 140 (1911).

Let (J) denote the full system — consisting of finitely many invariants — of $F_1, \ldots, F_n$, in such a way that every invariant of $F_1, \ldots, F_n$ can be expressed integrally and rationally by the invariants (J). Hilbert's conjecture to be proved is then that, for the simultaneous invariants S of $F_1, \ldots, F_N$, a full system is given by the finitely many invariants (J, PJ), where PJ denotes all invariants derivable from J by polar processes P.

Indeed, every simultaneous invariant S with rational integer coefficients of (F') is a form homogeneous in each of the N rows $A_1, \ldots, A_N$ — with rational integer coefficients — and identity (1) can therefore be applied to it. Since application of the polar processes P to S is known to produce invariants again, the forms Z are also invariants, namely simultaneous invariants of $F_1, \ldots, F_n$, since they contain only the rows $A_1, \ldots, A_n$; hence, by assumption, they are integral rational functions of the invariants (J). Thus identity (1) becomes

$$S = \sum PY,$$

where the Y denote power products of the invariants (J). But, by the definition of P, the actual execution of the processes P on Y consists in the successive, finitely repeated execution of the simple polar operations P_{hk}, which, by the differentiation rule for products, leads to sums of products of invariants (J) and (PJ). The same holds for the application of P_{hk} to a product from (J) and (PJ); hence PY too yields only integral rational combinations of (J, PJ). *The integral rational representability of all invariants S by (J, PJ) is thereby proved.*

II.

Before proving the reduction theorem, we recall some facts about linear families of forms. By a *linear family of forms over K* we mean a system of forms of the same dimension, complete in the sense that together with θ_1 and θ_2 it always contains $c_1\theta_1 + c_2\theta_2$, where c_1, c_2 are quantities of a prescribed domain of rationality K, and where K contains the coefficient domain of the θ's. Among these forms there is always a finite number — say ρ — linearly independent over K, while any $\rho + 1$ forms satisfy a linear relation with coefficients from K; the family has *rank* ρ with respect to K.[121] We shall use the easily seen fact that if $\mathcal{L}$ and $\mathcal{T}$ are two linear families over K of the same rank ρ, and if $\mathcal{T}$ is a subfamily of $\mathcal{L}$, then $\mathcal{L}$ and $\mathcal{T}$ are identical (equivalent).

With this in hand, we pass to the reduction theorem. The identity from I,

$$\theta = \sum PZ,$$

passes into an analogous identity for $P\theta$ when one applies the same polar process P to both sides; thus the reduction theorem at the same time gives a representation of all polars of θ by sums of the special polars PZ. Since, conversely, these special polars are surely contained among all polars, the reduction theorem asserts the equivalence of these two linear families of forms; in particular, it also asserts the equivalence of those subfamilies whose forms have, in each individual row, the same dimension as θ. To prove this equivalence we insert an intermediate family determined by the forms Z. For simplicity we first give the proof for the case of three binary rows ($N = 3, n = 2$), and then briefly indicate the completely parallel general proof.

Thus let $\theta(x, y, z)$ have dimensions α, β, p respectively in the rows

$$x_1x_2; \quad y_1y_2; \quad z_1z_2;$$

and let the coefficients of θ first be indeterminates a (or rational numbers). We consider the following three linear families of forms.

1. The family $\mathcal{L}$, consisting of all forms $f(x, y, z)$ which arise from θ by polar processes P and which, in the individual rows x, y, z, have the same dimension as θ; $\mathcal{L}$ contains θ in particular. If $f(x, y, z)$ is regarded as a form in x, y, z and the indeterminates a, then the coefficient field is the field R of rational numbers; for K as well one must take R, since only for rational integral θ_1, θ_2 does $c_1P_1 + c_2P_2$ again become a process P; let $\mathcal{L}$ have rank ρ with respect to R.

2. The family $\mathcal{S}$, consisting of all forms $g(\lambda; x, y)$ defined by

$$g(\lambda; x, y) = f(x, y, \lambda_1 x + \lambda_2 y),$$

or, expressed by polar processes,

$$\begin{aligned} g(\lambda; x, y) &= \frac{1}{p!}\left[(\lambda_1 x + \lambda_2 y)\frac{\partial}{\partial z}\right]^p f(x, y, z) \\ &= g_0(xy)\lambda_1^p + g_1(x, y)\lambda_1^{p-1}\lambda_2 + \cdots + g_p(xy)\lambda_2^p, \end{aligned}$$

[121] More general linear families of forms — whose rank need not be finite — are obtained by dropping the restriction of equal dimension, or also by dropping the restriction that K contain the coefficient domain of the θ's.

where

$$i!(p-i)!\,g_i(xy) = \left(y\frac{\partial}{\partial z}\right)^i \left(x\frac{\partial}{\partial z}\right)^{p-i} f(xyz).$$

[122] Thus the $g_i(xy)$ give forms Z of identity (1). The g are rational-integer forms in x, y, λ and the indeterminates a; let $\mathcal{S}$ have rank σ with respect to R.

3. The family $\mathcal{T}$, obtained from $\mathcal{S}$ by the inverse polar process just as $\mathcal{S}$ is obtained from $\mathcal{L}$; the forms h of $\mathcal{T}$ are defined by

$$\begin{aligned} h(x,y,z) &= \frac{1}{p!}\left[z\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y}\right)\right]^p g(\lambda; xy) \\ &= h_0(xyz) + h_1(xyz) + \cdots + h_p(xyz), \end{aligned}$$

where, by 2.,

$$h_i(xyz) = \left(z\frac{\partial}{\partial x}\right)^{p-i} \left(z\frac{\partial}{\partial y}\right)^i g_i(xy).$$

The individual h_i, and consequently also h, are therefore forms PZ and sums of them; let $\mathcal{T}$ have rank τ with respect to R.

As the construction of the forms $h(x,y,z)$ of $\mathcal{T}$ shows, these forms arise from θ by polar processes P, and in the rows x, y, z have individually the same dimension as θ; hence the family $\mathcal{T}$ is a subfamily of $\mathcal{L}$. By the fact mentioned above, it remains only to prove equality of the ranks. In this proof no further use is made of the special structure of the $f(xyz)$ — namely, that they are polars of one and the same form θ.

a) For comparing the ranks of $\mathcal{L}$ and $\mathcal{S}$ we use Lemma a): from $f(x,y,\lambda_1 x + \lambda_2 y) = 0$ it follows necessarily that $f(x,y,z) = 0$. This follows immediately from the identical relation between three binary rows

$$(xy)z + (yz)x + (zx)y = 0, \qquad (xy) = (x_1y_2 - x_2y_1), \ldots,$$

which implies

$$f[x, y, (xy)x + (xz)y] = (xy)^p f(x,y,z).$$

Thus from $f(x,y,\lambda_1 x + \lambda_2 y) = 0$ (identically in λ) it follows, as is shown by the substitution $\lambda_1 = (zy)$, $\lambda_2 = (xz)$, and since $(xy) \neq 0$, that necessarily $f(x,y,z) = 0$.

By this lemma there is a one-to-one correspondence between the forms of $\mathcal{S}$ and those of $\mathcal{L}$. For from

$$f_1(x,y,\lambda_1 x + \lambda_2 y) = g(\lambda; xy), \qquad f_2(x,y,\lambda_1 x + \lambda_2 y) = g(\lambda; xy)$$

it follows necessarily that

$$f_1(xyz) = f_2(xyz).$$

Furthermore, every relation in $\mathcal{S}$ is preserved between the uniquely corresponding forms in $\mathcal{L}$ (and conversely). For from

$$c_1 g^{(1)}(\lambda; xy) + c_2 g^{(2)}(\lambda; xy) + \cdots + c_r g^{(r)}(\lambda; xy) = 0$$

it follows by the lemma that necessarily

$$c_1 f^{(1)}(x,y,z) + c_2 f^{(2)}(x,y,z) + \cdots + c_r f^{(r)}(x,y,z) = 0.$$

This proves equality of the ranks of $\mathcal{L}$ and $\mathcal{S}$; $\rho = \sigma$.

b) For comparing the ranks of $\mathcal{S}$ and $\mathcal{T}$ we use the corresponding Lemma b): from $h(x,y,z) = 0$ it follows necessarily that $g(\lambda; xy) = 0$. For the proof, observe that by 3. $h(x,y,z) = 0$ is equivalent to the vanishing of the individual power products

$$\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x_1} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y_1}\right)^{p_1} \left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x_2} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y_2}\right)^{p_2} g(\lambda; x, y), \qquad p_1 + p_2 = p.$$

[122] As in I,

$$\left(t\frac{\partial}{\partial x}\right) = t_1\frac{\partial}{\partial x_1} + t_2\frac{\partial}{\partial x_2},$$

and hence in particular

$$\left[(\lambda_1 x + \lambda_2 y)\frac{\partial}{\partial z}\right] = (\lambda_1 x_1 + \lambda_2 y_1)\frac{\partial}{\partial z_1} + (\lambda_1 x_2 + \lambda_2 y_2)\frac{\partial}{\partial z_2},$$

$$\left[z\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y}\right)\right] = z_1\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x_1} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y_1}\right) + z_2\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x_2} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y_2}\right).$$

Further differentiation then also yields the vanishing of the power products

$$\left(\frac{\partial}{\partial x_1}\right)^{\alpha_1}\left(\frac{\partial}{\partial x_2}\right)^{\alpha_2}\left(\frac{\partial}{\partial y_1}\right)^{\beta_1}\left(\frac{\partial}{\partial y_2}\right)^{\beta_2}$$
$$\cdot\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x_1}+\frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y_1}\right)^{p_1}\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x_2}+\frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y_2}\right)^{p_2} g(\lambda;xy).$$

Consequently every linear combination of these power products vanishes, and hence every form

$$f\left(\frac{\partial}{\partial x},\frac{\partial}{\partial y},\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x}+\frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y}\right)g(\lambda,xy).$$

Choosing f in particular so that

$$f(x,y,\lambda_1x+\lambda_2y)=g(\lambda;xy),$$

one also obtains

$$g\left(\frac{\partial}{\partial x};\frac{\partial}{\partial x},\frac{\partial}{\partial y}\right)g(\lambda,x,y)=0;$$

or, if

$$g(\lambda,xy)=\sum G_i\lambda_1^{\nu_1}\lambda_2^{\nu_2}x_1^{\alpha_1}x_2^{\alpha_2}y_1^{\beta_1}y_2^{\beta_2},$$

where the G_i are rational-integer linear forms in the indeterminates a, then

$$\sum\nu_1!\nu_2!\alpha_1!\alpha_2!\beta_1!\beta_2!\,G_i^2=0,$$

and hence $g(\lambda;xy)=0$.

It follows from Lemma b), exactly as in a), that *the ranks of $\mathcal{S}$ and $\mathcal{T}$ are equal; $\sigma=\tau$.*

Thus by a) and b) one has $\rho=\tau$, and consequently — since $\mathcal{T}$ is a subfamily of $\mathcal{L}$ — the equivalence of these two linear families is proved. Hence every form of $\mathcal{L}$, in particular also θ, can be represented as a linear rational-integer combination of ρ linearly independent forms $h(x,y,z)$:

$$\theta=\sum c_ih^{(i)}(x,y,z)=\sum PZ.$$

This identity, derived for indeterminate coefficients a of θ, remains valid when the indeterminates a are replaced by quantities of any domain of rationality; hence *the reduction theorem is generally proved for the case of three binary rows.*

The general proof, for N rows of n variables each, is completely parallel. Let $\theta(A)$ have dimension α_i in the row A_i. We again consider the three linear families of forms:

1. the family $\mathcal{L}$, consisting of all forms $f(A)$ which arise from $\theta(A)$ by polar processes P and, in each individual row A_i, have the same dimension as $\theta(A)$;
2. the family $\mathcal{S}$, consisting of all forms $g(\lambda;A_1\cdots A_n)$ which arise from $f(A)$ by the substitutions

$$A_{n+1}=\lambda_{n+1}^{(1)}A_1+\lambda_{n+1}^{(2)}A_2+\cdots+\lambda_{n+1}^{(n)}A_n,$$
$$\vdots$$
$$A_N=\lambda_N^{(1)}A_1+\lambda_N^{(2)}A_2+\cdots+\lambda_N^{(n)}A_n;$$

in this process, the coefficients of the individual power products of the λ's in $g(\lambda;A_1\cdots A_n)$ give forms Z of identity (1);

3. the family $\mathcal{T}$, consisting of all forms $h(A)$ defined by

$$h(A)=\left[A_{n+1}\left(\frac{\partial}{\partial\lambda_{n+1}^{(1)}}\frac{\partial}{\partial A_1}+\cdots+\frac{\partial}{\partial\lambda_{n+1}^{(n)}}\frac{\partial}{\partial A_n}\right)\right]^{\alpha_{n+1}}$$
$$\cdots\left[A_N\left(\frac{\partial}{\partial\lambda_N^{(1)}}\frac{\partial}{\partial A_1}+\cdots+\frac{\partial}{\partial\lambda_N^{(n)}}\frac{\partial}{\partial A_n}\right)\right]^{\alpha_N}g(\lambda;A_1\cdots A_n).$$

Here $h(A)$ is a sum of forms PZ.

Because of the identical relation between any $n+1$ rows, Lemma a) remains valid; Lemma b) likewise remains valid, since the number of variables did not enter into its proof. Thus $\mathcal{L}$ and $\mathcal{T}$ have the same rank and — since $\mathcal{T}$ is a subfamily of $\mathcal{L}$ — are identical. Consequently the identity

$$\theta=\sum PZ$$

holds for indeterminate coefficients, and hence for coefficients that are quantities of any domain of rationality; *the reduction theorem is thereby proved in general.*

III.

We still indicate briefly how analogous considerations also yield the *general series expansion* (for $N \leq n$). Let Z be a separately homogeneous form in the n rows $A_1, \ldots, A_n$ of n quantities each; and let Δ be the determinant of these rows. We first prove the identity corresponding to (1):

$$Z = \sum PH \pmod{\Delta}, \tag{2}$$

where the forms H contain only the rows $A_1, \ldots, A_{n-1}$ and are derived from Z by processes P.

The proof consists in converting the relations derived in II into congruences modulo Δ. For simplicity take three ternary rows x, y, z; the coefficients are assumed to be indeterminates.

The three linear families $\mathcal{L}, \mathcal{S}, \mathcal{T}$ are defined as in II for the case of binary rows; let their ranks be ρ, σ, τ, while ρ^* and τ^* denote the ranks of $\mathcal{L}$ and $\mathcal{T}$ modulo Δ (that is, there are ρ^*, but not ρ^*+1, forms of $\mathcal{L}$ that are linearly independent mod Δ). Then, as in II, one has

Lemma a): from $f(x, y, \lambda_1 x + \lambda_2 y) = 0$ it follows necessarily that $f(xyz) \equiv 0 \pmod{\Delta}$ (and conversely). For, by the relation between four ternary rows,

$$\Delta_1 x - \Delta_2 y + \Delta_3 z = \Delta t, \qquad \Delta_1 = (yzt), \ldots,$$

there is always a linear relation modulo Δ between three ternary rows:

$$\Delta_1 x - \Delta_2 y + \Delta_3 z \equiv 0 \pmod{\Delta},$$

and hence, as in II,

$$f(x, y, -\Delta_1 x + \Delta_2 y) = \Delta_3^p f(x, y, z) \pmod{\Delta}.$$

It follows from $f(x, y, \lambda_1 x + \lambda_2 y) = 0$ (identically in λ), since Δ is irreducible and Δ_3 is not divisible by Δ, that necessarily $f(x, y, z) \equiv 0 \pmod{\Delta}$. The converse is clear from $\Delta(x, y, \lambda_1 x + \lambda_2 y) = 0$. Hence, as in II, $\rho^* = \sigma$.

Lemma b) remains valid with its proof, and one has therefore $\sigma = \tau$.

For proving $\tau = \tau^*$ we use

Lemma c): from $h(x, y, z) \equiv 0 \pmod{\Delta}$ it follows necessarily that $h(x, y, z) = 0$. Namely, as a polar, $h(x, y, z)$ vanishes under the application of the known Ω-process; this is expressed by the differential equation

$$\Delta\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) h(x, y, z) = 0.$$

If now $h(x, y, z) = \Delta(x, y, z) \cdot k(x, y, z)$, then by further differentiation one also has the vanishing of

$$k\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) \Delta\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) = h\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) h(x, y, z),$$

and hence of $h(x, y, z)$. Thus $\rho^* = \sigma = \tau = \tau^*$; since $\mathcal{T}$ is a subfamily of $\mathcal{L}$, identity (2) is proved. The same argument gives the proof for n variables.[123]

Applying identity (2) to the multiplier of Δ, one obtains an expansion

$$Z = \varphi_0 + \Delta\varphi_1 + \Delta^2\varphi_2 + \cdots + \Delta^\mu\varphi_\mu,$$

where the φ_i each vanish under application of the Ω-process. The i-fold repetition of the process gives, since in general $\Omega(\Delta \cdot \psi) = c_1\psi + c_2\Delta \cdot \Omega(\psi)$,

$$c \cdot \Omega^i(Z) \equiv \varphi_i \pmod{\Delta};$$

on the other hand, by (2) one has

$$c \cdot \Omega^i(Z) \equiv \sum PH_i \pmod{\Delta},$$

[123] After 3. one has the corresponding relation, vanishing because of the determinant factor, for $\Omega(h)$; in operator notation it is the equation arising from products of

$$\left(\lambda_1 \frac{\partial}{\partial \xi} + \lambda_2 \frac{\partial}{\partial \eta}\right), \qquad \left[z\left(\frac{\partial}{\partial \lambda_1}\frac{\partial}{\partial \xi} + \frac{\partial}{\partial \lambda_2}\frac{\partial}{\partial \eta}\right)\right],$$

and it vanishes because of the vanishing of the determinant factor.

where the H_i are derived from the form $\Omega^i(Z)$ in the same way as H is derived from Z. By Lemma c) (extended to n variables) one therefore gets $\varphi_i = \sum PH_i$, and consequently

$$Z = \sum PH + \Delta \cdot \sum PH_1 + \Delta^2 \cdot \sum PH_2 + \cdots + \Delta^\mu \cdot \sum PH_\mu \tag{3}$$

as the most general series expansion of a form in n rows of n variables each, according to polars of forms in only $n-1$ rows.

The series expansion for forms in N rows of n variables ($N < n$) is then obtained, as is known, by a simple substitution. We put (for $N = 3$, $n > 3$)

$$u = \xi_1 x + \xi_2 y + \xi_3 z, \quad v = \eta_1 x + \eta_2 y + \eta_3 z, \quad w = \zeta_1 x + \zeta_2 y + \zeta_3 z,$$

and expand $H(u, v, w) = Z(\xi, \eta, \zeta)$ according to (3). Then, since ξ, η, ζ occur only in the combinations u, v, w, one has

$$\left(\eta \frac{\partial}{\partial \xi}\right) = \left(v \frac{\partial}{\partial u}\right) \cdots, \qquad \Omega = \sum_{ikl} \left(\frac{\partial}{\partial u} \frac{\partial}{\partial v} \frac{\partial}{\partial w}\right)_{ikl} (xyz)_{ikl} = \nabla_{uvw}(xyz).$$

Under the specialization $\xi_1 = \eta_2 = \zeta_3 = 1$; $\xi_2 = \xi_3 = \cdots = \zeta_2 = 0$, $H(u, v, w)$ goes over into $H(x, y, z)$, and $\nabla_{uvw}(xyz)$ into $\nabla_{xyz}(xyz) = \nabla_{xyz}$ up to a numerical factor, since the application of $(y\partial/\partial x)$ etc. to the determinants $(xyz)_{ikl}$ makes them vanish. Since the corresponding assertion holds for N rows, (3) gives the expansion

$$H = \sum P\Phi + \sum P\Phi_1 + \cdots + \sum P\Phi_\mu, \tag{4}$$

where the Φ_i arise from $\nabla^i_{A_1 \ldots A_N} H$ by processes P and contain only the rows $A_1, \ldots, A_{N-1}$, apart from the determinant combinations occurring in ∇^i, which, as mentioned, do not come into consideration under polarization.

If these determinants in ∇^i are then replaced by indeterminates p, the resulting forms can again be expanded according to (4); in this way one finally obtains an expansion

$$H = \left[\sum P\psi(p)\right]_{p = \text{determinants of } A_1 \cdots A_N}, \tag{5}$$

where the ψ arise from the original form by the processes P and $\nabla_{A_1 \ldots A_N}(p)$, $\nabla_{A_1 \ldots A_{N-1}}(p)$, and so on. With these expansions and their combinations — for example by applying (3), and then (4) and (5), to the forms Z occurring in (1) — all known expansions for forms in arbitrarily many rows of n cogredient variables each are exhausted.

Erlangen, 5 January 1915.

9. The Most General Domains from Integral Transcendental Numbers

Math. Ann. 77 (1916), pp. 103–128

By means of the well-ordering theorem, E. Zermelo constructed a domain of "integral transcendental numbers," that is, a numerical domain $\mathfrak{G}$ having the following abstract properties:

I. The sum, difference, and product of two numbers from $\mathfrak{G}$ is again a number from $\mathfrak{G}$.

II. Every real or complex number is the quotient of two numbers from $\mathfrak{G}$.

III. Every integral rational (respectively algebraic) number is an element of $\mathfrak{G}$.

IV. No non-integral rational (respectively algebraic) number belongs to $\mathfrak{G}$.[124]

The construction rests on the fact that the well-ordering theorem implies the existence of an *algebraic basis* of all numbers: namely, a *system H of numbers* η among which no algebraic relations exist, but by means of which all remaining numbers can be expressed algebraically. Because of their algebraic independence, these basis numbers η can be regarded as indeterminates, and the desired domain therefore becomes an integral domain of rational and algebraic functions of indeterminates η,[125] whose coefficients belong to the field K of all algebraic numbers. The special domain $\mathfrak{G}_\eta$ constructed by Zermelo is then characterized by the following basis properties:

$\mathfrak{G}_\eta$ contains:

1) all basis numbers η,

2) all polynomials in the η with integral[126] coefficients,

3) all integral algebraic functions of the η with integral coefficients.

$\mathfrak{G}_\eta$ excludes:

4) all rationally fractional functions of the η with integral coefficients,

5) all algebraically fractional functions of the η with integral coefficients.[127]

It is now easy to see (compare the examples in §§ 2 and 3) that there are domains $\mathfrak{G}$ essentially different from the Zermelo domain $\mathfrak{G}_\eta$: that is, domains $\mathfrak{G}$ for which, under no well-ordering, do all basis properties 1)–5) hold, and which consequently cannot be generated by Zermelo's construction under any well-ordering. In other words, there are domains $\mathfrak{G}$ that cannot be mapped one-to-one and isomorphically onto $\mathfrak{G}_\eta$. Thus the question arises of the most general domains from integral transcendental numbers; it will be answered completely in what follows.

For this purpose one must first determine which of the basis properties are consequences of the abstract defining properties and which are caused by the special construction. It is shown (§ 1) that for every prescribed domain $\mathfrak{G}$ there is a well-ordering such that basis properties 1) and 2) are satisfied; hence every domain $\mathfrak{G}$ can be constructed by means of an algebraic basis of all numbers. None of the further basis properties need be satisfied (§§ 2 and 3). In particular it follows that a further abstract property of the Zermelo domain does not belong to all domains $\mathfrak{G}$: algebraic integral closedness. It can be formulated as follows:

V. Every number that depends algebraically-integrally and integrally on finitely many numbers from $\mathfrak{G}$ belongs to $\mathfrak{G}$.

The special domains for which V holds in addition to I–IV will be called domains $\mathfrak{H}$ from algebraically-integral transcendental numbers. For the most general domains $\mathfrak{H}$ one obtains the simple results (§ 4):

[124] E. Zermelo, Über ganze transzendente Zahlen, Math. Ann. 75, p. 434 (1914).

[125] For this reason the considerations of the present paper partly run parallel to those of the author's earlier paper: Körper und Systeme rationaler Funktionen, Math. Ann. 76, p. 161 (1915).

[126] Throughout, "integral" is to mean that the coefficients belong to the integral domain $[K]$ of all algebraic integers. For the definition of $\mathfrak{G}_\eta$, it would also suffice in 1)–5) to consider only rational integer coefficients; for more general domains, however, the basis properties would then become less simple.

[127] Zermelo, loc. cit., § 3.

a) The intersection of all domains $\mathfrak{H}$ constructible by means of the same basis H is given by all integral algebraic functions of the basis, that is, by the Zermelo domain $\mathfrak{G}_\eta$, which is itself a domain $\mathfrak{H}$. (This also gives an abstract characterization of the Zermelo domain.)

b) Every domain $\mathfrak{H}$ arises by algebraically-integral adjunction[128] of an arbitrary system $\mathfrak{S}(\eta)$ to $\mathfrak{G}_\eta$, where $\mathfrak{S}(\eta)$ has to satisfy only the single condition that no algebraically fractional number enters the domain through the adjunction.

The results for the most general domains $\mathfrak{G}$, when an algebraic basis is taken as foundation, are less simple. Here the intersection of all domains $\mathfrak{G}$ is itself not a domain $\mathfrak{G}$, and the construction of the most general domains is consequently harder to survey (§ 5).

In order to obtain results analogous to those for the domains $\mathfrak{H}$, the algebraic basis is replaced by a *rational basis* Θ, that is, by a *system* Θ *of numbers* ϑ which permits all numbers to be expressed rationally, while under at least one well-ordering no basis number can be expressed rationally by finitely many preceding ones. This rational basis Θ must still be subject to the side condition that the integral domain $[\Theta]$ consisting of all integral polynomials in the ϑ contains no algebraically fractional number. (The construction of such a basis with the side condition is shown in § 7.) Then for the most general domains $\mathfrak{G}$ — which one could also call domains from rationally integral transcendental numbers — one again obtains the simple results (§ 6):

a) The intersection of all domains $\mathfrak{G}$ constructible by means of the same rational basis Θ is given by all integral rational functions of the basis, that is, by the domain $[\Theta]$, which is itself a domain $\mathfrak{G}$.

b) Every domain $\mathfrak{G}$ arises by rationally integral adjunction of an arbitrary system $\mathfrak{S}(\vartheta)$ to $[\Theta]$, where $\mathfrak{S}(\vartheta)$ has to satisfy only the single condition that no algebraically fractional number enters the domain through the adjunction.

To have complete analogy with the domains $\mathfrak{H}$, one would have to omit the algebraic numbers from the defining properties III and IV for the domains $\mathfrak{G}$. With this modification of III and IV, the construction of the integral elements by means of a rational basis can be transferred to an arbitrary field (§ 10), whereas Zermelo's construction presupposes the algebraic closedness of the field.

As H runs through every algebraic basis, respectively as Θ runs through every rational basis satisfying the side condition, the results a) and b) give the totality of all domains $\mathfrak{H}$ and $\mathfrak{G}$. If all isomorphic domains are collected into a class, then from this totality one can select a subset which represents the totality of all classes — at least countably infinitely many by § 9 (§ 8).

§ 1. Proof of basis properties 1) and 2) for all domains $\mathfrak{G}$.

Let $\mathfrak{G}$ be any prescribed domain from integral transcendental numbers, so that it has the abstract properties I–IV. Following the procedure applied by E. Zermelo to the field of all complex numbers, we select from $\mathfrak{G}$ an *algebraic basis* H *of all numbers from* $\mathfrak{G}$. Since by II every real or complex number can be represented as the quotient of two numbers from $\mathfrak{G}$, H is at the same time an algebraic basis of *all* numbers. Thus, for every prescribed domain $\mathfrak{G}$, the algebraic basis of all numbers can be chosen so that it belongs to $\mathfrak{G}$; hence basis property 1) is satisfied. Moreover, by III, $\mathfrak{G}$ contains the integral domain $[K]$ of all algebraic integers, and therefore, by I, also all polynomials in the basis elements η with coefficients from $[K]$, that is, all integral polynomials in the η, which form the integral domain $[H]$. Hence basis property 2) is also always satisfied; in other words, every domain $\mathfrak{G}$ can be constructed by means of an algebraic basis H of all numbers in such a way that $\mathfrak{G}$ contains the integral domain $[H]$ as a divisor.

Remark. Nevertheless, starting from a basis H, one can of course construct a domain $\mathfrak{G}$ for which basis properties 1) and 2) do not hold with respect to H itself, but do hold with respect to some other basis H'. For example, let $\mathfrak{G}'$ arise from the Zermelo domain $\mathfrak{G}_\eta$ by replacing, throughout the whole construction, some basis element η by a polynomial $f(\eta)$. Then $\mathfrak{G}'$ contains neither η nor $[H]$; but if in the basis H the basis element η is replaced by $\xi = f(\eta)$ — whereby H again goes over into an algebraic basis H' —, then basis properties 1) and 2) are satisfied for $\mathfrak{G}'$ with respect to H'.

§ 2. Exclusion of basis properties 4) and 5).

We now show that the domains $\mathfrak{G}$ need not satisfy any of the further basis properties. First 4) and 5) are excluded by replacing, in Zermelo's construction, the integral domain $[H]$ by a domain of integral rational

[128] Explanation of the expression in § 4.

"functionals."

According to Weber,[129] a *rational functional* means every rational function with rational number coefficients in the following uniquely determined representation:

$$A(\eta_1 \cdots \eta_t) = a \cdot \frac{E_1(\eta_1 \cdots \eta_t)}{E_2(\eta_1 \cdots \eta_t)},$$

where E_1, E_2 are relatively prime primitive polynomials with rational integer coefficients, and a is a positive rational number (the absolute value of A). If, in particular, a is a *rational integer*, then A is called an *integral rational functional.* As special cases, the integral rational functionals include the rational integers and the polynomials with rational integer coefficients, while rationally fractional numbers and polynomials with rationally fractional coefficients are to be counted among the fractional rational functionals.

Let the domain $\mathfrak{G}$ now consist of the totality $\mathfrak{F}$ of all integral rational functionals of finitely many basis elements η at a time, and of all functions which depend algebraically-integrally on $\mathfrak{F}$ — that is, which satisfy some equation whose leading coefficient is unity and whose remaining coefficients are integral rational functionals.

The proof of properties I–IV for $\mathfrak{G}$ runs exactly parallel to the corresponding proof for Zermelo and will only be indicated briefly. First, II and III are clear, since $\mathfrak{G}$ contains the Zermelo domain $\mathfrak{G}_\eta$ as a divisor. By Gauss's theorem on polynomials with rational integer coefficients, sums, differences, and products of integral rational functionals are again integral rational functionals; hence $\mathfrak{F}$ forms an integral domain, and by familiar arguments[130] $\mathfrak{G}$ also becomes an integral domain; thus I is fulfilled. Further, from the extended Gauss theorem[131] follow the two auxiliary propositions: "If z depends algebraically-integrally on $\mathfrak{F}$ by means of *some* equation, then it also does so by means of the *irreducible* equation which z satisfies with respect to the field of all rational functionals,"[132] and "the equation irreducible over the field of all rational numbers and satisfied by an algebraic number is also irreducible over the field of all rational functionals." Thus $\mathfrak{G}$ can contain no fractional algebraic number; hence IV, and therefore all abstract properties I–IV, have been verified.

On the other hand, under no well-ordering can a basis be selected from $\mathfrak{G}$ in such a way that $\mathfrak{G}$ excludes all rationally and algebraically fractional functions of the basis elements. For let $z(\eta)$ be any function of the domain to be chosen as a basis element, hence defined by an equation

$$z^k + A_1(\eta) z^{k-1} + \cdots + A_k(\eta) = 0,$$

where

$$A_i(\eta) = a_i \cdot \frac{E_1(\eta)}{E_2(\eta)}$$

are integral rational functionals. Then the function $\frac{a_k}{z}$ belongs to $\mathfrak{G}$, and likewise $\frac{a_k}{\sqrt{z}}$, $\frac{a_k}{\sqrt[3]{z}}$, etc.[133] Hence basis properties 4) and 5) are excluded for the totality of the domains $\mathfrak{G}$.

Remark. An example in § 3 will show that $\mathfrak{G}$ may, under every well-ordering, contain rationally fractional functionals without containing a rationally or algebraically fractional number.

§ 3. Exclusion of basis property 3).

For excluding basis property 3), we use a generalized *module domain* in which the arguments of the polynomials of the domain are no longer the indeterminates, but all integral algebraic functions of the indeterminates, [134] while the indeterminates themselves take over the role of a module basis.

Let $\mathfrak{G}$ consist of *all elements*

$$z = \alpha + g_1(\eta)\eta_1 + g_2(\eta)\eta_2 + \cdots + g_t(\eta)\eta_t, \tag{1}$$

where α runs through all algebraic integers, $\eta_1 \cdots \eta_t$ through all basis elements, and for each fixed combination $\eta_1 \cdots \eta_t$, the functions $g_1(\eta) \cdots g_t(\eta)$ individually run through all integral algebraic functions of the η.[135]

[129] H. Weber, Lehrbuch der Algebra. Kleine Ausgabe §§ 96 and 97.

[130] Compare, for example, Weber, § 97, 8.

[131] Weber, § 20, 5.

[132] Compare Weber, § 97, 4.

[133] Quite analogously, every algebraic function of the η can be transformed into an element of the functional domain $\mathfrak{G}$ by multiplication by a rational integer. Thus this domain has the property that every complex number can be represented as the quotient of an element from $\mathfrak{G}$ and a rational integer, in analogy with the algebraic numbers.

[134] By "integral algebraic functions" are always meant functions with *integral* coefficients; that is, functions satisfying some equation whose leading coefficient is unity and whose remaining coefficients are polynomials in the η with algebraic-integer coefficients — polynomials from $[H]$. In particular, all polynomials from $[H]$ belong to the integral algebraic functions.

[135] The module property belongs to the elements $z - \alpha$.

It is easy to verify that properties I–IV hold for this module domain. First I holds, since the sum, difference, and product of two elements z again takes the form (1). The domain $\mathfrak{G}$ also contains all basis elements η, and besides all elements $\eta \cdot g(\eta)$, where $g(\eta)$ runs through all integral algebraic functions of the η. Hence all integral algebraic functions of the η, and therefore all algebraic functions of the η altogether, can be represented as quotients of two elements of $\mathfrak{G}$, proving II. In the representation (1), all algebraic integers are also included in particular — for $g_1 = 0 \cdots g_t = 0$ —; but z cannot be equal to a fractional algebraic number. For if z represents an algebraic number β, then from

$$\beta = \alpha + g_1(\eta)\eta_1 + \cdots + g_t(\eta)\eta_t$$

identically in the η, necessarily $\beta = \alpha$. This is shown by the specialization

$$\eta_1 = 0 \cdots \eta_t = 0,$$

under which the multipliers $g_1(\eta) \cdots g_t(\eta)$ remain finite as integral algebraic functions, since they go over into integral algebraic functions or numbers.[136] Thus conditions I–IV are satisfied for $\mathfrak{G}$.

On the other hand, under no well-ordering can a basis be selected from $\mathfrak{G}$ in such a way that all integral algebraic functions of the basis elements belong to $\mathfrak{G}$. For let $z(\eta)$ be any function of the domain to be chosen as a basis element — which therefore must actually contain the indeterminates η —. We shall show that, from a certain finite value of ν onward, depending on z, the functions

$$\xi = \sqrt[\nu]{z - \alpha}$$

can no longer belong to $\mathfrak{G}$.

Indeed, suppose that ξ belongs to $\mathfrak{G}$. Then, by (1), it is of the form

$$\xi = \gamma + h_1(\eta)\eta_{i_1} + h_2(\eta)\eta_{i_2} + \cdots + h_\tau(\eta)\eta_{i_\tau},$$

where again $h_1(\eta) \cdots h_\tau(\eta)$ are integral algebraic functions of the η, and $\eta_{i_1} \cdots \eta_{i_\tau}$ are arbitrary basis elements, which in particular may also coincide with $\eta_1 \cdots \eta_t$. First it must be shown that the algebraic integer γ is zero. Since $h_1(\eta) \cdots h_\tau(\eta)$ are integral algebraic functions, ξ becomes equal to γ for $\eta_{i_1} = 0 \cdots \eta_{i_\tau} = 0$ with arbitrary values of the remaining η; in particular, then, ξ is equal to γ for

$$\eta_{i_1} = 0 \cdots \eta_{i_\tau} = 0; \qquad \eta_1 = 0 \cdots \eta_t = 0.$$

But under this specialization the function $(z - \alpha)$ vanishes by (1), and therefore so does ξ; hence $\gamma = 0$, and one has

$$\xi = h_1(\eta)\eta_{i_1} + h_2(\eta)\eta_{i_2} + \cdots + h_\tau(\eta)\eta_{i_\tau}.$$

Raising this to the ν-th power gives

$$\xi^\nu = z - \alpha = F_\nu(\eta),$$

where $F_\nu(\eta)$ denotes a homogeneous, not identically vanishing, form of ν-th dimension in $\eta_{i_1} \cdots \eta_{i_\tau}$, whose coefficients are integral algebraic functions of the η. Now $z - \alpha$, which by (1) is an integral algebraic function of the η, satisfies an equation irreducible with respect to $[H]$:

$$(z - \alpha)^\chi + B(\eta)(z - \alpha)^{\chi - 1} + \cdots + C(\eta) = 0,$$

where the last coefficient $C(\eta)$ — the product of $(z - \alpha)$ with its conjugates — is a polynomial; let its degree be λ. On the other hand, from $z - \alpha = F_\nu(\eta)$, by multiplying $F_\nu(\eta)$ by the corresponding conjugates, one obtains

$$C(\eta) = G_{\nu\chi}(\eta),$$

where $G_{\nu\chi}(\eta)$ denotes a homogeneous form of $\nu\chi$-th dimension in $\eta_{i_1} \cdots \eta_{i_\tau}$, whose coefficients are polynomials in the η. Thus $G_{\nu\chi}$ is at least of degree $\nu\chi$ in η, or $\lambda \geq \nu\chi$; that is, the functions $\sqrt[\nu]{z - \alpha}$ for which $\nu > \lambda/\chi$ do not belong to $\mathfrak{G}$. Basis property 3) is therefore excluded for the totality of the domains $\mathfrak{G}$.

Remark. A slight generalization of the domain just constructed leads to a domain $\mathfrak{G}$ which, under every well-ordering, also contains fractional functionals. Let $\mathfrak{G}$ consist of all elements

$$z = \alpha + \gamma_1(\eta)\eta_1 + \gamma_2(\eta)\eta_2 + \cdots + \gamma_t(\eta)\eta_t,$$

where now $\gamma(\eta)$ runs through all algebraically integral, but not necessarily integral, functions of the η. The proof of properties I–IV remains the same as above. But since $\mathfrak{G}$, besides every function z, also contains $a \cdot (z - \alpha)$, where a denotes any rational or algebraically fractional number, fractional functionals also occur under every well-ordering.

[136]This more detailed proof of IV is given because of the expanded example at the end of the paragraph. Here the remark would suffice that, by construction, z is an integral algebraic function.

§ 4. The most general domains from algebraically-integral transcendental numbers.

For a more convenient formulation of what follows, introduce the expression: "an element z depends algebraically-integrally on $\mathfrak{T}$," meaning that z satisfies some equation whose leading coefficient is unity and whose remaining coefficients are integral polynomials in the elements of $\mathfrak{T}$, where $\mathfrak{T}$ denotes any prescribed domain.[137]

A domain $\mathfrak{T}$ is called *algebraically-integrally closed* if it satisfies the condition (compare the introduction):

V. Every element that depends algebraically-integrally on $\mathfrak{T}$ belongs to $\mathfrak{T}$.

We first show that the abstract property V belongs to the Zermelo domain $\mathfrak{G}_\eta$. Let an element z depend algebraically-integrally on $\mathfrak{G}_\eta$ by means of an equation $f(z) = 0$. Then multiplication of $f(z)$ by its conjugates with respect to $[H]$ shows that z also depends algebraically-integrally on $[H]$, and hence belongs to $\mathfrak{G}_\eta$. The algebraically-integral closedness of the functional domain considered in § 2 is shown in the same way.

That property V does not belong to every domain $\mathfrak{G}$ is shown by the module domain considered in § 3, which indeed does not possess basis property 3). More precisely, § 3 showed that the module domain is algebraically-integrally closed with respect to no transcendental element of the domain, whereas, by III and IV, this is of course the case with respect to every algebraic element of the domain.

This leads us first to select, from the totality of domains $\mathfrak{G}$, those which also possess the abstract property V; these are to be called "domains $\mathfrak{H}$ from algebraically-integral transcendental numbers."

Let $\mathfrak{H}$ be such a domain, and let H be an algebraic basis of all numbers from $\mathfrak{H}$; by II, H is at the same time an algebraic basis of all numbers. By III, I, and V, $\mathfrak{H}$ contains, besides H, all integral algebraic functions of the η, that is, the Zermelo domain $\mathfrak{G}_\eta$. Thus, in analogy with § 1: every domain $\mathfrak{H}$ from algebraically-integral transcendental numbers can be constructed by means of an algebraic basis of all numbers in such a way that $\mathfrak{H}$ contains the Zermelo domain $\mathfrak{G}_\eta$ as a divisor.

Now let H be any algebraic basis of all numbers, and let $\mathfrak{M}(H)$ denote the totality of all domains $\mathfrak{H}$ containing H. Every domain $\mathfrak{H}$ in $\mathfrak{M}(H)$ contains, as just proved, the Zermelo domain $\mathfrak{G}_\eta$ as a divisor; and since $\mathfrak{G}_\eta$ itself is a domain $\mathfrak{H}$, $\mathfrak{G}_\eta$ is the greatest common divisor, or intersection, of all domains $\mathfrak{H}$ from $\mathfrak{M}(H)$. Thus $\mathfrak{G}_\eta$ has been defined abstractly. It also follows directly that $\mathfrak{M}(H)$ is closed under intersection: that is, the intersection of all domains $\mathfrak{H}$ of any subsystem of $\mathfrak{M}(H)$ again belongs to $\mathfrak{M}(H)$. For, as an intersection, I and V are satisfied; II and III follow because $\mathfrak{G}_\eta$ is a divisor; IV follows because the intersection is a divisor of $\mathfrak{H}$.

As the example of the functional domain in § 2 shows, the domains $\mathfrak{H}$ in general contain further functions besides $\mathfrak{G}_\eta$. Let $\psi(\eta)$ be such a rational or algebraic function of the η. Then, by I, $\mathfrak{H}$ also contains all integral rational combinations of ψ and finitely many functions from $\mathfrak{G}_\eta$ at a time. The integral domain so arising — consisting of all polynomials in ψ with coefficients from $\mathfrak{G}_\eta$ — will be denoted by $[\mathfrak{G}_\eta, \psi]$, and the process leading to it will be called "rationally integral adjunction of ψ to $\mathfrak{G}_\eta$." By V, $\mathfrak{H}$ further contains all functions which depend algebraically-integrally on $[\mathfrak{G}_\eta, \psi]$; the domain so arising will be denoted by $\{\mathfrak{G}_\eta, \psi\}$, and the process leading to it by "algebraically integral adjunction of ψ to $\mathfrak{G}_\eta$." The domain $\{\mathfrak{G}_\eta, \psi\}$ consists of all integral algebraic functions of ψ with coefficients from $\mathfrak{G}_\eta$, and is therefore an integral domain by familiar arguments.[138]

We show that $\{\mathfrak{G}_\eta, \psi\}$ is also algebraically-integrally closed, that is, satisfies condition V. Indeed, let z depend algebraically-integrally on $\{\mathfrak{G}_\eta, \psi\}$ by means of an equation

$$f(z; a \cdots c) = z^\nu + az^{\nu-1} + \cdots + c = 0,$$

where $a \cdots c$, as elements of $\{\mathfrak{G}_\eta, \psi\}$, satisfy equations with coefficients from $[\mathfrak{G}_\eta, \psi]$:

$$a^\lambda + A_1 a^{\lambda-1} + \cdots + A_\lambda = 0,$$
$$\vdots$$
$$c^\mu + C_1 c^{\mu-1} + \cdots + C_\mu = 0.$$

Let their roots be denoted by $a, a' \cdots a^{(\lambda-1)} \cdots c, c' \cdots c^{(\mu-1)}$. If one now forms the product over all combinations of the roots,

$$g(z) = f(z; a \cdots c) f(z; a' \cdots c') \cdots f(z; a^{(\lambda-1)} \cdots c^{(\mu-1)}),$$

[137] For an integral domain $\mathfrak{T}$, the leading coefficient is unity and the remaining coefficients are elements of $\mathfrak{T}$.

[138] Compare, for example, Weber, § 97, 6 and 7.

then $g(z)$ has unity as leading coefficient and elements of $[\mathfrak{G}_\eta, \psi]$ as its remaining coefficients. Consequently z belongs to $\{\mathfrak{G}_\eta, \psi\}$.[139]

Thus the domain $\{\mathfrak{G}_\eta, \psi\}$ has properties I and V; and, since $\mathfrak{G}_\eta$ is a divisor of the domain, it also has II and III. Whether IV is satisfied depends on the choice of ψ; for example, IV is not satisfied for $\psi = \frac{1}{n \cdot \eta}$, since then $\frac{1}{n}$ belongs to the domain. It is satisfied, however, by § 2 for $\psi = \frac{1}{\eta}$, or also for $\psi = \frac{\eta}{n}$. For in the latter case, if a function from $\{\mathfrak{G}_\eta, \psi\}$ represents an algebraic number, its value remains unchanged for $\eta = 0$; thereby it passes into a function from $\mathfrak{G}_\eta$, and consequently into an algebraic integer.

Thus $\{\mathfrak{G}_\eta, \psi\}$ becomes a domain $\mathfrak{H}$ as soon as one imposes on ψ the condition that no algebraically fractional number enters the domain through the algebraically integral adjunction.

Quite analogously one defines the rationally integral, respectively algebraically integral, adjunction of an arbitrary system $\mathfrak{S}$ of rational or algebraic functions of the η to $\mathfrak{G}_\eta$. The rationally integral adjunction leads to the integral domain $[\mathfrak{G}_\eta, \mathfrak{S}]$, consisting of all integral polynomials in finitely many elements from $\mathfrak{G}_\eta$ and $\mathfrak{S}$ at a time. The algebraically integral adjunction gives the domain $\{\mathfrak{G}_\eta, \mathfrak{S}\}$ consisting of all elements that depend algebraically-integrally on $[\mathfrak{G}_\eta, \mathfrak{S}]$. Exactly as above, one shows that the integral domain $\{\mathfrak{G}_\eta, \mathfrak{S}\}$ is algebraically-integrally closed and also satisfies II and III. Thus, as above:

$\{\mathfrak{G}_\eta, \mathfrak{S}\}$ becomes a domain $\mathfrak{H}$ as soon as one imposes on $\mathfrak{S}$ the condition that no algebraically fractional number enters the domain through the algebraically integral adjunction — that is, as soon as $\mathfrak{S}$ becomes an admissible system.[140]

It is now important that the converse also holds, so that the domains $\{\mathfrak{G}_\eta, \mathfrak{S}\}$ actually exhaust all domains $\mathfrak{H}$ from $\mathfrak{M}(H)$ as $\mathfrak{S}$ runs through all admissible systems. Indeed, let $\mathfrak{H}$ be any domain from $\mathfrak{M}(H)$ and denote by $\mathfrak{L} = \mathfrak{H} - \mathfrak{G}_\eta$ the system consisting of all functions from $\mathfrak{H}$ not belonging to $\mathfrak{G}_\eta$. By V, $\mathfrak{H}$ contains the domain $\{\mathfrak{G}_\eta, \mathfrak{L}\}$ as a divisor; but $\{\mathfrak{G}_\eta, \mathfrak{L}\}$ is also divisible by $\mathfrak{H}$, because it contains $\mathfrak{G}_\eta$ and $\mathfrak{L}$, and since these have no common element, it contains $\mathfrak{H}$ as well. Hence $\mathfrak{H} = \{\mathfrak{G}_\eta, \mathfrak{L}\}$; and since IV is satisfied for $\mathfrak{H}$, $\mathfrak{L}$ is of course an admissible system.[141]

As H runs through every possible algebraic basis, $\mathfrak{M}(H)$ runs, as proved at the beginning, through the totality of all domains $\mathfrak{H}$. Consequently the question of the most general domains $\mathfrak{H}$ from algebraically-integral transcendental numbers is completely answered by the results of this paragraph:

a) The intersection of all domains $\mathfrak{H}$ from $\mathfrak{M}(H)$ is given by the Zermelo domain $\mathfrak{G}_\eta$, which itself is a domain $\mathfrak{H}$; $\mathfrak{M}(H)$ is closed with respect to forming intersections.

b) Every domain $\mathfrak{H}$ from $\mathfrak{M}(H)$ arises by algebraically integral adjunction of an admissible system $\mathfrak{S}$ to $\mathfrak{G}_\eta$. As H runs through every possible algebraic basis, $\mathfrak{M}(H)$ runs through the totality of all domains $\mathfrak{H}$.[142]

§ 5. The most general domains made from integral transcendental numbers, on the basis of an algebraic basis.

Let H be any algebraic basis of all numbers, and let $\mathfrak{M}(\mathfrak{G})$ denote the totality of all domains $\mathfrak{G}$ which contain H and therefore $[H]$. As H runs through every possible basis, $\mathfrak{M}(\mathfrak{G})$ runs through the totality of all domains $\mathfrak{G}$; hence, as in § 4, we may restrict ourselves to considering the domains $\mathfrak{G}$ from $\mathfrak{M}(\mathfrak{G})$.

The domains $\mathfrak{G}$ from $\mathfrak{M}(\mathfrak{G})$ all contain $[H]$ as a divisor; but, since $[H]$ itself is not a domain $\mathfrak{G}$, one cannot conclude as in § 4 that $[H]$ is the greatest common divisor. That this is nevertheless the case is shown by a countable set of domains $\mathfrak{G}$ from $\mathfrak{M}(\mathfrak{G})$, obtained by a slight generalization from the module domain of § 3, and in fact having no element in common besides $[H]$.

For every fixed ν, let the domains $\mathfrak{G}_\nu$ ($\nu = 1, 2, \ldots$ in inf.) consist of all quantities

$$z = f(\eta) + g_1(\eta)\eta_1^\nu + g_2(\eta)\eta_2^\nu + \cdots + g_t(\eta)\eta_t^\nu, \tag{1}$$

139 Thus, by defining algebraic integrality by means of *some* equation, the concept of irreducibility with respect to a ground field becomes unnecessary for the proof of I and V. Compare also § 2, where, only for the proof of IV, the auxiliary proposition is needed which passes from the definition of algebraic integrality by an arbitrary equation to the irreducible equation. Since this auxiliary proposition no longer holds in general in the present case, IV has to be imposed on the function ψ as a special condition.

140 More generally, one shows analogously: if $\mathfrak{T}$ is any domain and $\{\mathfrak{T}\}$ denotes the totality of elements depending algebraically-integrally on $\mathfrak{T}$, then $\{\mathfrak{T}\}$ is algebraically-integrally closed.

141 The adjunction of a subsystem of $\mathfrak{L}$ can already suffice; for example, the functional domain of § 2 arises by algebraically integral adjunction of all integral rational functionals — with the exception of the polynomials — to $\mathfrak{G}_\eta$.

142 The admissible systems $\mathfrak{S}$ can be found as follows. Put all algebraically fractional functions of the η under some well-ordering Ω, and take into $\mathfrak{S}$ all functions except those which, when algebraically-integrally adjoined to $\mathfrak{G}_\eta$ and the previously admitted functions, generate an algebraically fractional number. This procedure can be broken off at any arbitrary place. As Ω runs through every well-ordering, and as the process is broken off at every arbitrary place for each fixed Ω, all admissible systems $\mathfrak{S}$ are thereby exhausted.

where $f(\eta)$ runs through all integral polynomials in the η; $\eta_1, \dots, \eta_t$ run through the basis quantities, and, for each fixed combination $\eta_1^\nu, \dots, \eta_t^\nu$, the functions $g_1(\eta), \dots, g_t(\eta)$ run independently through all algebraic integers functions of the η.

As in § 3, one sees that the domains $\mathfrak{G}_\nu$ have the abstract properties I–IV; $\mathfrak{G}_\nu$ contains only algebraic-integer functions and, for $\nu = 1$, is identical with the domain of § 3. We now show that the domains $\mathfrak{G}_\nu$ have no further common element besides the polynomials $f(\eta)$ from $[H]$. Indeed, let z be any algebraic-integer — but not rational — function of the η, satisfying the equation, irreducible with respect to $[H]$,

$$z^\chi + a_1(\eta)z^{\chi-1} + a_2(\eta)z^{\chi-2} + \cdots + a_\chi(\eta) = 0, \tag{2}$$

where $\chi > 1$; and let λ be the greatest of the degrees of the polynomials $a_1(\eta), \dots, a_\chi(\eta)$. If z belongs to $\mathfrak{G}_\nu$, then the quantity $\xi = z - f(\eta)$ satisfies the likewise irreducible equation, again with respect to $[H]$,

$$\xi^\chi + b_1(\eta)\xi^{\chi-1} + b_2(\eta)\xi^{\chi-2} + \cdots + b_\chi(\eta) = 0, \tag{3}$$

where, by (1), $b_1(\eta), b_2(\eta), \dots, b_\chi(\eta)$, as symmetric functions of the ξ, contain as terms of lowest dimension terms of at least the dimensions $\nu, 2\nu, \dots, \chi\nu$. Comparing (2) and (3) gives

$$b_1(\eta) = a_1(\eta) + \chi f(\eta).$$

If, therefore, the quantity $\xi = z + \frac{a_1(\eta)}{\chi}$ satisfies

$$\xi^\chi + c_2(\eta)\xi^{\chi-2} + \cdots + c_\chi(\eta) = 0, \tag{4}$$

then

$$\xi - \zeta = \frac{b_1(\eta)}{\chi}; \qquad c_i(\eta) \equiv b_i(\eta) \pmod{\frac{b_1(\eta)}{\chi}}, \tag{5}$$

where $b_1(\eta)$ begins with terms of at least dimension ν. Now the $c_i(\eta)$, as the comparison of (2) with (4) shows, are of degree χ in the coefficients $a_i(\eta)$ of (2), hence of degree at most $\chi\lambda$ in the η; on the other hand, all the $b_i(\eta)$ begin with terms of at least dimension ν. Thus, if one chooses $\nu > \chi\lambda$, (5) gives

$$c_i(\eta) = 0; \qquad \xi^\chi = 0 \quad \text{or} \quad \left(z + \frac{a_1(\eta)}{\chi}\right)^\chi = 0.$$

Thus z, contrary to the assumption, becomes rational in the η. Hence:

If z is an algebraic-integer, non-rational function of the η, then z can belong to no domain $\mathfrak{G}_\nu$ for which $\nu > \chi\lambda$, or:

The intersection of all domains $\mathfrak{G}$ from $\mathfrak{M}(\mathfrak{G})$ is given by the integral domain $[H]$, which itself is not a domain $\mathfrak{G}$; consequently $\mathfrak{M}(\mathfrak{G})$ is not closed with respect to forming intersections.

Consequently the construction of the most general domains $\mathfrak{G}$ by means of an algebraic basis is also harder to survey than that of the domains $\mathfrak{H}$. The domain $[H]$ has properties I, III and IV; if one adjoins any system $\mathfrak{S}$ rationally-integrally, then for the resulting integral domain $[H, \mathfrak{S}]$ the properties I and III remain valid, whereas IV forms a special condition on $\mathfrak{S}$. In order that a domain $\mathfrak{G}$ arise, II must also be imposed on $\mathfrak{S}$ as a condition; or, expressed differently: for every field $\mathfrak{K}$, finite over (H)[143], there must be a field $\mathfrak{L}$ over $\mathfrak{K}$, finite over $\mathfrak{K}$, such that $[H, \mathfrak{S}]$ contains a primitive element from $\mathfrak{L}$.[144] Conversely, every domain $\mathfrak{G}$ from $\mathfrak{M}(\mathfrak{G})$ can also be represented in the form $[H, \mathfrak{S}]$; hence one obtains all domains $\mathfrak{G}$ from $\mathfrak{M}(\mathfrak{G})$ when $\mathfrak{S}$ runs through all systems restricted in this way. The structure of the systems $\mathfrak{S}$ will appear more simply in the next paragraph by means of the "rational basis"; at the same time this gives a complete analogy with the theorems of § 4.

§ 6. The most general domains made from integral transcendental numbers, on the basis of a rational basis.

In analogy with the algebraic basis, the rational basis is to be defined as follows: "A system Θ of numbers ϑ is called a rational basis of all numbers if Θ permits every number to be expressed rationally — with

[143] (H) means the field consisting of all rational functions of the η with algebraic-number coefficients.

[144] Cf. § 7.

coefficients in K[145] — while, for at least one well-ordering, no basis number can be expressed rationally — with coefficients in K — by finitely many preceding basis numbers."[146]

Let a domain $\mathfrak{G}$ made from integral transcendental numbers be subjected to a well-ordering Ω. Then it can happen that a quantity z from $\mathfrak{G}$ is rationally expressible by finitely many preceding quantities from $\mathfrak{G}$; that is, it satisfies an equation $z = \varphi(\xi)$, where $\varphi(\xi)$ is a rational function, with coefficients in K, of $\xi_1, \ldots, \xi_t$. In particular, every algebraic number α of the domain satisfies the equation $z = \alpha$. The remaining quantities ϑ from $\mathfrak{G}$ form a system Θ which is to be shown to be a rational basis of all quantities from $\mathfrak{G}$. For by assumption each quantity ϑ is rationally independent of the preceding quantities from Θ. The induction argument[147] further shows that every relation $z = \varphi(\xi)$ entails a relation $z = \psi(\vartheta)$; for otherwise there would have to be a first element $z_0 = \varphi_0(\xi_1, \ldots, \xi_t)$ which cannot be expressed rationally by the ϑ, while the preceding $\xi_1, \ldots, \xi_t$ are rational functions of the ϑ; this gives a contradiction. By II, Θ is at the same time a rational basis of all numbers; by III and I, besides Θ, $\mathfrak{G}$ contains as a divisor the integral domain $[\Theta]$ consisting of all integral polynomials in the ϑ, whereas by IV $[\Theta]$ contains no algebraically fractional number. Thus the rational basis Θ of all numbers satisfies the side condition that the integral domain $[\Theta]$ derived from Θ contains no algebraically fractional number;[148] Θ is a *rational basis with side condition* for all numbers. Thus, in analogy with § 2: *Every domain $\mathfrak{G}$ can be constructed by means of a rational basis with side condition, in such a way that $\mathfrak{G}$ contains the integral domain $[\Theta]$ as a divisor.*

Let Θ be any rational basis with side condition for all numbers — whose existence is therefore guaranteed by the existence of the domains $\mathfrak{G}$. Let $\mathfrak{N}(\mathfrak{G})$ denote the totality of all domains $\mathfrak{G}$ which contain Θ and therefore $[\Theta]$. As Θ runs through every possible basis with side condition, $\mathfrak{N}(\mathfrak{G})$ runs, by the above, through the totality of all domains $\mathfrak{G}$; hence we may again restrict ourselves to the domains $\mathfrak{G}$ from $\mathfrak{N}(\mathfrak{G})$.

Now $[\Theta]$ itself is a domain $\mathfrak{G}$; for every number can be expressed rationally by Θ, and therefore as a quotient of two polynomials from $[\Theta]$; and by its construction $[\Theta]$ also satisfies the remaining conditions I, III, IV. *Thus* $[\Theta]$ *is the greatest common divisor, or intersection, of all domains* $\mathfrak{G}$ *from* $\mathfrak{N}(\mathfrak{G})$*; moreover* $\mathfrak{N}(\mathfrak{G})$ *is closed with respect to intersection; that is, the intersection of all domains* $\mathfrak{G}$ *of any subsystem of* $\mathfrak{N}(\mathfrak{G})$ *also belongs to* $\mathfrak{N}(\mathfrak{G})$. For I holds for the intersection; II and III follow from the fact that $[\Theta]$ is a divisor; and IV from the fact that the intersection domain is contained as a divisor in a domain $\mathfrak{G}$.

The rationally integral adjunction of any system $\mathfrak{S}$ of rational functions of the ϑ to $[\Theta]$ — every algebraic function of the ϑ can, by the property of the rational basis, be expressed rationally through certain other ϑ — gives an integral domain $[\Theta, \mathfrak{S}]$ having properties I, II, III, and hence becomes a domain $\mathfrak{G}$ as soon as $\mathfrak{S}$ is an "admissible system," i.e. as soon as it satisfies the condition that no algebraically fractional number enters the domain through the rationally integral adjunction. Conversely, every domain $\mathfrak{G}$ from $\mathfrak{N}(\mathfrak{G})$ permits the representation $[\Theta, \mathfrak{S}]$, provided only that $\mathfrak{S}$ denotes the residual system $\mathfrak{G} - [\Theta]$. Thus, for the most general domains $\mathfrak{G}$, in complete analogy with § 4, one has:

a) *The intersection of all domains* $\mathfrak{G}$ *from* $\mathfrak{N}(\mathfrak{G})$ *is given by the integral domain* $[\Theta]$*, which itself is a domain* $\mathfrak{G}$*;* $\mathfrak{N}(\mathfrak{G})$ *is closed with respect to forming intersections.*

b) *Every domain* $\mathfrak{G}$ *from* $\mathfrak{N}(\mathfrak{G})$ *arises by rationally integral adjunction of an admissible system* $\mathfrak{S}$ *to* $[\Theta]$*. As* Θ *runs through every possible rational basis with side condition,* $\mathfrak{N}(\mathfrak{G})$ *runs through the totality of all domains* $\mathfrak{G}$*.*[149]

Remark. If one extracts from Θ an algebraic basis H of all quantities from Θ, then H is at the same time an algebraic basis of all numbers; conversely, every algebraic basis can be extended to a rational one by putting H first in the well-ordering. Accordingly, the construction in § 5 of the domains $\mathfrak{G}$ by means of an algebraic basis can be interpreted as follows. Decompose the system $\mathfrak{S}$ to be adjoined to $[H]$ into two subsystems $\mathfrak{S}_1$ and $\mathfrak{S}_2$; the adjunction of $\mathfrak{S}_1$ amounts to extending H to a rational basis with side condition, to which an admissible system $\mathfrak{S}_2$ is then still rationally-integrally adjoined.

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[145] By rational functions one should always understand such functions with algebraic-number coefficients; these algebraic-number coefficients are included in the definition because of III and IV. It would be more appropriate to require rational-number coefficients both in III and IV and in the definition of the rational basis. The arguments of §§ 6 and 7 remain literally unchanged if one merely replaces the field K of all algebraic numbers by the field of all rational numbers.

[146] In contrast to the linear and algebraic bases, the ordering of the elements plays a role here. For example, in the ordering η, $\sqrt[2]{\eta}$ both elements have to be admitted, whereas in the reverse ordering only $\sqrt[2]{\eta}$ has to be admitted.

[147] Cf. Zermelo, loc. cit., § 1.

[148] That this is indeed a condition is shown, for instance, by the fact that the two quantities η, $\frac{1}{2\sqrt{\eta}}$ are not admissible as basis elements, whereas η, $\frac{1}{\sqrt{\eta}}$ are.

[149] For the most general admissible systems $\mathfrak{S}$, what was said in § 4 applies, with "rational" replacing "algebraic."

§ 7. Construction of the most general rational basis with side condition.

The results of the preceding paragraph still need a supplement, since there the *existence* of a rational basis with side condition for all numbers was proved, but not a method for constructing it from a prescribed well-ordering. On account of the side condition, this construction takes a more complicated form when, instead of a domain $\mathfrak{G}$, the field of all complex numbers is present.[150]

Let the field of all complex numbers be subjected to a well-ordering Ω. Then quantities of three kinds can occur:

1. Quantities y whose rationally integral adjunction to the preceding basis quantities — defined recursively in 3. — forces an algebraically fractional number to enter the resulting integral domain; this domain consists of all integral polynomials in y if no basis quantity precedes it. In particular, all algebraically fractional numbers are quantities y, since the resulting integral domain contains the quantities $y = \beta$.

2. Quantities z which do not have property 1, but which can be expressed rationally by finitely many preceding numbers different from the y; thus they satisfy a relation $z = \varphi(\xi_1, \ldots, \xi_t)$, where φ is a rational function with coefficients in K and none of the ξ are y. In particular, all algebraic integer numbers α belong here, since property 1 does not hold for them and they satisfy the relation $z = \alpha$.

3. Quantities ϑ for which neither 1 nor 2 holds and which are to be called basis quantities. In particular, the first transcendental number ϑ_0 in the well-ordering is such a basis quantity; for the integral polynomials in ϑ_0 cannot represent an algebraically fractional number, and ϑ_0 also cannot be rationally dependent on the preceding algebraic numbers.

The system Θ of all basis quantities ϑ is now to be shown to be a rational basis with side condition for all numbers.

Since no quantity y occurs among the ϑ, the integral domain $[\Theta]$ cannot contain any algebraically fractional number; hence the side condition is satisfied. Since, furthermore, no quantity z occurs among the ϑ, each ϑ is rationally independent of the preceding basis quantities. The induction argument shows, exactly as in § 6, that every relation $z = \varphi(\xi)$ — where no y occurs among the ξ — entails a relation $z = \psi(\vartheta)$, so that all quantities z are rationally expressible by the ϑ.

It remains only to prove the more difficult assertion that the y are also rationally expressible by the ϑ. First we show that the y depend algebraically on the ϑ. By assumption, for each y there exists at least one algebraically fractional number β admitting the representation

$$\beta = f_0(\vartheta) + f_1(\vartheta)y + \cdots + f_\chi(\vartheta)y^\chi, \tag{1}$$

where $f_0(\vartheta), \ldots, f_\chi(\vartheta)$ are polynomials from $[\Theta]$ and hence cannot represent algebraically fractional numbers. If y were algebraically independent of the ϑ, then (1) would hold identically in y and one would have $\beta = f_0(\vartheta)$, contradicting the properties of $[\Theta]$.

In order to infer rational dependence from algebraic dependence, we extract from Θ an algebraic basis H of all quantities from Θ. Then every y, as an algebraic function of the ϑ, is also algebraically dependent on the η; through $y = y(\eta)$ an algebraic field $(H, y(\eta))$ over (H) is therefore determined. Since all η are at the same time quantities ϑ, the proof of rational dependence amounts to showing that for every such field there are primitive elements which are rational functions of the ϑ; more precisely, it will be shown that $y(\eta)$ loses property 1 after multiplication by a polynomial in the η, and consequently passes into such an element.

It is known that between (H) and (H, y) there are only finitely many intermediate fields. Let $\Omega_0 = (H), \Omega_1, \ldots, \Omega_\sigma$ be those among them which possess a primitive element rationally expressible by the ϑ, so that every element of the field becomes a rational function of the ϑ; this condition is at least always satisfied for $\Omega_0 = (H)$. Let the primitive function belonging to Ω_i be given by $\frac{g_i(\vartheta)}{h_i(\vartheta)}$, where g_i, h_i are polynomials from $[\Theta]$. Then — with α_i denoting a suitably chosen integer — the polynomial from $[\Theta]$

$$\xi_i = g_i(\vartheta) + \alpha_i h_i(\vartheta)$$

[150] In § 6 b) it would suffice to let Θ run only through those special rational bases which consist of an algebraic basis and algebraically integral functions of the basis quantities; then the side condition is automatically satisfied. To see this, one need only, by § 6, extend an algebraic basis from $\mathfrak{G}$ to a rational one and, in the resulting basis, multiply each algebraically fractional function of the η by a suitably chosen polynomial in the η, whereby the basis property is preserved. This construction transfers unchanged to the field of all complex numbers. The question arising from § 6 concerning the most general rational basis with side condition, however, also has independent interest.

is a primitive function of Ω_i, or of an algebraic field over Ω_i,[151] and every function $\psi(\eta)$ from Ω_i admits the representation

$$\psi(\eta) = \frac{a_1(\eta)\xi_i^{\kappa-1} + a_2(\eta)\xi_i^{\kappa-2} + \cdots + a_\kappa(\eta)}{a(\eta)} = \frac{A(\eta,\xi_i)}{a(\eta)}, \tag{2}$$

where $a(\eta), a_1(\eta), \ldots, a_\kappa(\eta)$ are integral polynomials in the η, and hence $A(\eta,\xi_i)$ and $a(\eta)$ are quantities from $[\Theta]$.

The irreducible equation satisfied by y with respect to Ω_i is, by (2), of the form

$$f^{(i)}(\eta)y^{\lambda_i} + f_1^{(i)}(\eta,\xi_i)y^{\lambda_i-1} + \cdots + f_{\lambda_i}^{(i)}(\eta,\xi_i) = 0,$$

and all coefficients of this equation belong to $[\Theta]$. Then the function $y_i = f^{(i)}(\eta)\cdot y$ satisfies the likewise irreducible equation, with respect to Ω_i,

$$y_i^{\lambda_i} + f_1^{(i)}(\eta,\xi_i)y_i^{\lambda_i-1} + \cdots + \left(f^{(i)}(\eta)\right)^{\lambda_i-1} f_{\lambda_i}^{(i)}(\eta,\xi_i) = 0,$$

by virtue of which it depends algebraically-integrally on $[H,\xi_i]$; the same holds for the product of y_i with any polynomial in the η. In particular, the function

$$Y = f^{(0)}(\eta)f^{(1)}(\eta)\cdots f^{(\sigma)}(\eta)\cdot y \tag{3}$$

depends algebraically-integrally on all domains $[H,\xi_i]$ by means of the equation, irreducible in each case with respect to Ω_i,

$$Y^{\lambda_i} + F_1^{(i)}(\eta,\xi_i)Y^{\lambda_i-1} + \cdots + F_{\lambda_i}^{(i)}(\eta,\xi_i) = 0 \qquad (i = 0,1,\ldots,\sigma). \tag{4}$$

We now show that Y is the desired primitive element rationally expressible by the ϑ. Suppose, for example, that the algebraic number γ is represented by Y and $[\Theta]$:

$$\gamma = k_0(\vartheta) + k_1(\vartheta)Y + \cdots + k_\nu(\vartheta)Y^\nu, \tag{5}$$

where $k_0(\vartheta), \ldots, k_\nu(\vartheta)$ are polynomials from $[\Theta]$. We now determine the irreducible equation satisfied by Y with respect to the field $\mathfrak{K} = (H; k_0(\vartheta), \ldots, k_\nu(\vartheta))$ arising from the coefficient domain of (5). It is known to be given by the irreducible equation of Y with respect to the intersection field of $\mathfrak{K}$ and (H,Y), lying between (H) and (H,Y). Now (H,Y) is identical with (H,y) by (3); and since all elements of $\mathfrak{K}$, hence also all elements of the intersection field, can be expressed rationally by finitely many ϑ, this latter field is necessarily one of the fields $\Omega_0, \Omega_1, \ldots, \Omega_\sigma$, say Ω_τ. The desired irreducible equation is therefore, by (4), of degree λ_τ and is

$$Y^{\lambda_\tau} + F_1^{(\tau)}(\eta,\xi_\tau)Y^{\lambda_\tau-1} + \cdots + F_{\lambda_\tau}^{(\tau)}(\eta,\xi_\tau) = 0 \tag{6}$$

with polynomials from $[\Theta]$ as coefficients. By means of (6), (5) can be rewritten in the form

$$\gamma = K_0(\vartheta) + K_1(\vartheta)Y + \cdots + K_{\lambda_\tau-1}(\vartheta)Y^{\lambda_\tau-1}, \tag{7}$$

where $K_0(\vartheta), K_1(\vartheta), \ldots$ belong to $\mathfrak{K}$ and, on the other hand, are polynomials from $[\Theta]$. But since γ is an algebraic number, the relation (7) does not reach degree λ_τ in Y and is therefore satisfied *identically* in Y. Hence

$$\gamma = K_0(\vartheta),$$

i.e. γ belongs to $[\Theta]$ and is consequently an algebraic integer.

As γ runs through all algebraic numbers representable by Y and $[\Theta]$, the intersection field of (H,Y) and the corresponding field $\mathfrak{K}$ always runs only through the fields $\Omega_0, \Omega_1, \ldots, \Omega_\sigma$, and all the preceding arguments remain valid. That is, *by rationally integral adjunction of Y to $[\Theta]$ no algebraically fractional number can be represented.* Thus Y no longer has property 1; according to 2) or 3) it is a quantity z or ϑ, and therefore rationally representable by finitely many ϑ; by (3), the same holds for y: Θ *has been proved to be a rational basis of all numbers.* Conversely, since every rational basis with side condition can be constructed according to 1), 2), 3) by putting Θ first in the well-ordering, we have proved: *The construction given by 1), 2), 3) leads to the most general rational basis with side condition for all numbers.*

[151] Cf. the end of § 5.

§ 8. Division of the domains $\mathfrak{G}$ and $\mathfrak{H}$ into classes.

Alongside the question of the most general domains $\mathfrak{G}$ and $\mathfrak{H}$ there arises the further question of the *essentially different* domains, which — in a sense to be made more precise below — cannot be generated by the same construction principle. This leads to a division of these domains into classes.

Two domains will be called belonging to the same class, or isomorphic, if their elements can be put into one-to-one correspondence in such a way that the sum and product of any two elements of one domain are always assigned to the sum and product of the corresponding elements of the other. From this definition it follows that under every isomorphic relation zero and the unit correspond to themselves, and that all algebraic relations between finitely many elements are preserved for the corresponding elements, and conversely. In particular, besides the unit, all rational numbers correspond to themselves; whereas the totality of algebraic numbers — and likewise the totality of algebraic integers — goes over into itself, although an individual algebraic number can very well correspond to a conjugate.

First note that every domain from the totality $\mathfrak{M}(\mathfrak{G})$ of all domains $\mathfrak{G}$ containing a fixed algebraic basis H is isomorphic to a domain from the totality $\mathfrak{M}^*(\mathfrak{G})$ belonging to another algebraic basis H^*. For the field of all complex numbers arises from every algebraic basis by algebraic extension, and therefore has the same cardinality as the basis;[152] hence H and H^* have the same cardinality, and the desired isomorphic mapping is obtained by assigning the basis quantities η_i to η_i^*.[153] On the other hand, $\mathfrak{M}(\mathfrak{G})$ contains non-isomorphic, or essentially different, domains $\mathfrak{G}$, as follows from the examples in §§ 2, 3 and 5. For since all algebraic relations are preserved under the isomorphic mapping, an algebraic basis must again correspond to an algebraic basis; but §§ 2 and 3 showed that the functional and module domains possess no algebraic basis for which all the basis properties of the Zermelo domain would be fulfilled. Since the functional domain of § 2 is algebraically integrally closed, the subset $\mathfrak{M}(\mathfrak{H})$ also contains essentially different domains $\mathfrak{H}$.

We now show, analogously to the basis construction, how one can extract from $\mathfrak{M}(\mathfrak{G})$ a subset $\mathfrak{L}(\mathfrak{G})$ such that all domains $\mathfrak{G}$ from $\mathfrak{L}(\mathfrak{G})$ are essentially different, while every domain from $\mathfrak{M}(\mathfrak{G})$ — and hence every domain $\mathfrak{G}$ at all — is isomorphic to a domain from $\mathfrak{L}(\mathfrak{G})$. Subject $\mathfrak{M}(\mathfrak{G})$ to a well-ordering Ω; then a domain $\mathfrak{G}$ from $\mathfrak{M}(\mathfrak{G})$ may or may not be isomorphic to some preceding one. The domains $\mathfrak{G}$ isomorphic to no preceding one form a set $\mathfrak{L}(\mathfrak{G})$, which by construction contains only essentially different domains $\mathfrak{G}$. The induction argument also shows that every domain from $\mathfrak{M}(\mathfrak{G})$ is isomorphic to a domain from $\mathfrak{L}(\mathfrak{G})$. For if $\mathfrak{G}_1$ were the first domain for which this failed, then $\mathfrak{G}_1$ either belongs to $\mathfrak{L}(\mathfrak{G})$, or is isomorphic to a preceding domain $\mathfrak{G}_0$, which by assumption is isomorphic to a domain from $\mathfrak{L}(\mathfrak{G})$; hence the same would hold for $\mathfrak{G}_1$. Since every domain $\mathfrak{G}$ belongs to some set $\mathfrak{M}^*(\mathfrak{G})$ and is therefore isomorphic to a domain from $\mathfrak{M}(\mathfrak{G})$, *the totality of the classes of domains $\mathfrak{G}$ is represented by $\mathfrak{L}(\mathfrak{G})$.*

If, in a domain $\mathfrak{G}$ from $\mathfrak{L}(\mathfrak{G})$, one replaces the basis quantities η by those η^* of some other basis, then, as shown above, a domain isomorphic to $\mathfrak{G}$ arises. Conversely, let $\mathfrak{G}^*$ be any domain and suppose it is isomorphic to a domain $\mathfrak{G}$ from $\mathfrak{L}(\mathfrak{G})$. If the basis quantities η correspond to the quantities η^* from $\mathfrak{G}^*$, then these again form an algebraic basis, and $\mathfrak{G}^*$ contains the same algebraic functions of η^* as $\mathfrak{G}$ contains of η — or their conjugates. *Thus every domain isomorphic to $\mathfrak{G}$ arises by letting the algebraic basis H run through every possible algebraic basis in $\mathfrak{G}$ and in its conjugates with respect to (H)*[154] *— if these differ from $\mathfrak{G}$.* From each fixed domain $\mathfrak{G}_i$ in $\mathfrak{L}(\mathfrak{G})$ one obtains in this way the class $\mathfrak{R}_i$, which contains all and only the domains $\mathfrak{G}$ isomorphic to $\mathfrak{G}_i$, although possibly each domain occurs several times or even infinitely often. From $\mathfrak{R}_i$ one can again, by the procedure above, extract a subset $\mathfrak{R}_i^*$ which contains every domain isomorphic to $\mathfrak{G}_i$ once and only once. As $\mathfrak{R}_i^*$ runs through all classes, one obtains the totality of all domains $\mathfrak{G}$, every domain once and only once. To characterize the class $\mathfrak{R}_i$, one can of course use, instead of $\mathfrak{G}_i$, any other domain of the class from which all domains of $\mathfrak{R}_i$ can be derived as above. In summary:

From $\mathfrak{M}(\mathfrak{G})$ one can extract a subset $\mathfrak{L}(\mathfrak{G})$ such that the domains $\mathfrak{G}$ from $\mathfrak{L}(\mathfrak{G})$ are in one-to-one correspondence with the totality of classes (without repetition). From $\mathfrak{G}_i$ the corresponding class $\mathfrak{R}_i$ arises by letting H run through every possible algebraic basis in $\mathfrak{G}_i$ and its conjugates with respect to (H). If $\mathfrak{R}_i^*$ denotes all mutually distinct domains from $\mathfrak{R}_i$, and $\mathfrak{R}_i^*$ runs through all classes, then the totality of all domains $\mathfrak{G}$ arises, every domain once and only once.[155]

[152] Steinitz, §§ 23 and 24.

[153] This conclusion of course does not hold for the sets $\mathfrak{N}(\mathfrak{G})$ and $\mathfrak{N}^*(\mathfrak{G})$ belonging to two rational bases Θ and Θ^*; nor can the isomorphism of $\mathfrak{N}(\mathfrak{G})$ and $\mathfrak{N}^*(\mathfrak{G})$ be inferred from the mutual rational expressibility of Θ and Θ^*. What is obtained, however, is an isomorphic relation between the fields (Θ) and (Θ^*).

[154] The elements of these domains conjugate to $\mathfrak{G}$ can certainly be expressed rationally by the elements of $\mathfrak{G}$ by II, but not necessarily integrally and rationally; the conjugates can therefore differ from $\mathfrak{G}$.

[155] Here $\mathfrak{M}(\mathfrak{G})$ is to be constructed according to §§ 6 and 7: by § 7 one extends an algebraic basis H to every rational basis Θ with side condition arising from it, and for each Θ forms, by § 6 b), the totality $\mathfrak{N}(\mathfrak{G})$ of all domains $\mathfrak{G}$ containing Θ.

The analogous statement holds for the domains $\mathfrak{H}$ made from algebraically integral transcendental numbers.

§ 9. An example of countably infinitely many classes of domains $\mathfrak{G}$.

The cardinality of the constructions given in § 4 b) and § 6 b) can be read off from the remarks thereto as equal to the cardinality of all well-orderings, hence as 2^c, if c denotes the cardinality of the continuum. But from this no conclusion can be drawn about the cardinality of the totality of all domains $\mathfrak{G}$, counted once and only once, and still less about the cardinality of all classes. That the number of classes is certainly not finite will now be shown by an example of countably infinitely many domains $\mathfrak{G}$ — analogous to those of § 5 — which will all turn out to be essentially different and hence supply countably infinitely many classes.[156]

Analogously to § 5, (1), let the domain $\mathfrak{G}_\sigma$ consist of the totality of quantities

$$z = a_0 + a_1(\eta) + \cdots + a_{\sigma-1}(\eta) \pmod{\mathfrak{M}_\sigma(\eta)}, \tag{1}$$

where $a_i(\eta)$ denotes homogeneous forms of i-th dimension in the η; and the module $\mathfrak{M}_\sigma$ contains as basis all power products of σ-th dimension in the η[157] and as multipliers all algebraic-integer functions of the η. We show that under an isomorphic relation between two domains $\mathfrak{G}_\sigma$ and $\mathfrak{G}_\tau$ ($\sigma \neq \tau$) the modules $\mathfrak{M}_\sigma$ and $\mathfrak{M}_\tau$ must necessarily correspond isomorphically; the impossibility then follows easily.

Let the indeterminates in $\mathfrak{G}_\tau$ be denoted by ξ; under the isomorphic relation, let the quantities $f(\eta)$ from $\mathfrak{G}_\sigma$ correspond to these ξ. Since the isomorphic relation is preserved under forming quotients, the domain $\mathfrak{G}_\xi$ of all algebraic-integer functions of the ξ corresponds to an algebraically integrally closed domain $\mathfrak{T}$ consisting of all functions algebraically integral over the $f(\eta)$, and hence also algebraically integral in the η.

Suppose (1) corresponds to a quantity from $\mathfrak{M}_\tau$. Then, by the module property of $\mathfrak{M}_\tau$, the product of (1) with an arbitrary quantity $\psi(\eta)$ from $\mathfrak{T}$ must also belong to $\mathfrak{G}_\sigma$; in formulas:

$$(a_0 + a_1(\eta) + \cdots + a_{\sigma-1}(\eta))\psi(\eta) \equiv b_0 + b_1(\eta) + \cdots + b_{\sigma-1}(\eta) \pmod{\mathfrak{M}_\sigma}. \tag{2}$$

Setting all η equal to zero gives $a_0\psi(0) = b_0$, so (2) becomes

$$(a_0 + a_1(\eta) + \cdots + a_{\sigma-1}(\eta))(\psi(\eta) - \psi(0)) \equiv c_1(\eta) + \cdots + c_{\sigma-1}(\eta) \pmod{\mathfrak{M}_\sigma}. \tag{3}$$

Now, besides $\psi(\eta)$, the functions

$$\chi_\nu(\eta) = \sqrt[\nu]{\psi(\eta) - \psi(0)}$$

also belong to $\mathfrak{T}$ for every ν; and since $\chi_\nu(0) = 0$, the formulas analogous to (3) are:

$$(a_0 + a_1(\eta) + \cdots + a_{\sigma-1}(\eta))\chi_\nu(\eta) \equiv c_1^{(\nu)}(\eta) + \cdots + c_{\sigma-1}^{(\nu)}(\eta) \pmod{\mathfrak{M}_\sigma} \quad (\nu = 1, 2, \ldots). \tag{4}$$

Let $A(\eta)$, of degree λ, denote the product of $\chi_1 = \psi(\eta) - \psi(0)$ with its $\kappa - 1$ conjugates. Since $\chi_1(0) = 0$, $A(\eta)$ must begin with terms of at least first dimension. From (4) one obtains

$$a_0\chi_\nu(\eta) \equiv 0 \pmod{\mathfrak{M}_1}; \qquad a_0^\nu\chi_1(\eta) \equiv 0 \pmod{\mathfrak{M}_\nu}; \qquad a_0^{\nu\kappa}A(\eta) \equiv 0 \pmod{\mathfrak{M}_{\nu\kappa}}, \tag{5}$$

and therefore, for $a_0 \neq 0$, as in § 3, $\lambda > \nu\kappa$. But since $\nu < \frac{\lambda}{\kappa}$ contradicts the algebraically integral closedness of $\mathfrak{T}$, necessarily $a_0 = 0$. From (4) one now obtains

$$a_1(\eta)\chi_\nu(\eta) \equiv c_1^{(\nu)}(\eta) \pmod{\mathfrak{M}_2}; \qquad [a_1(\eta)]^{\nu\kappa}A(\eta) \equiv [c_1^{(\nu)}(\eta)]^{\nu\kappa} \pmod{\mathfrak{M}_{\nu\kappa+1}}, \tag{6}$$

and from this, since $A(\eta)$ begins with terms of at least first dimension, $c_1^{(\nu)}(\eta) = 0$. Thus, in complete analogy with (5), for every ν:

$$a_1(\eta)\chi_\nu(\eta) \equiv 0 \pmod{\mathfrak{M}_2}; \qquad [a_1(\eta)]^{\nu\kappa}A(\eta) \equiv 0 \pmod{\mathfrak{M}_{2\nu\kappa}},$$

and hence $a_1(\eta) = 0$. Continuing in this way, the alternating application of (5) and (6) gives

$$a_0 = 0, \qquad a_1(\eta) = 0, \quad \ldots, \quad a_{\sigma-1}(\eta) = 0.$$

[156] The domains $\mathfrak{G}$ of § 6 likewise give countably infinitely many classes, but the proof is somewhat more complicated.

[157] Naturally, in each individual z only finitely many power products of the η occur in the module.

Since the analogous consideration also holds with respect to $\mathfrak{G}_\tau$: *Under every isomorphic relation between* $\mathfrak{G}_\sigma$ *and* $\mathfrak{G}_\tau$ $(\sigma \neq \tau)$*, the modules* $\mathfrak{M}_\sigma$ *and* $\mathfrak{M}_\tau$ *correspond isomorphically.*

Suppose, for instance, that $\sigma > \tau$. To the indeterminates ξ there corresponded the quantities $f(\eta)$ from $\mathfrak{G}_\sigma$; and, since all quantities from $\mathfrak{G}_\tau$ can be represented by polynomials in the ξ mod. $\mathfrak{M}_\tau$, the correspondence of $\mathfrak{M}_\tau$ and $\mathfrak{M}_\sigma$ implies that all quantities from $\mathfrak{G}_\sigma$ are obtained as polynomials in the $f(\eta)$ mod. $\mathfrak{M}_\sigma$. Since $\sigma > \tau \geq 1$, the indeterminates η certainly do not belong to $\mathfrak{M}_\sigma$, and hence, being integrally and rationally expressible by the $f(\eta)$ mod. $\mathfrak{M}_\sigma$, there must be an $f(\eta)$ with non-vanishing linear terms. Thus let

$$\xi : f(\eta) \equiv a_0 + a_1(\eta) \pmod{\mathfrak{M}_2} \quad [a_1(\eta) \neq 0],$$

$$\begin{aligned} \xi^\tau : [f(\eta)]^\tau &\equiv a_0^\tau \pmod{\mathfrak{M}_1} \quad \text{for } a_0 \neq 0, \\ &\equiv [a_1(\eta)]^\tau \pmod{\mathfrak{M}_{\tau+1}} \quad \text{for } a_0 = 0. \end{aligned}$$

Now ξ^τ belongs to $\mathfrak{M}_\tau$, while $[f(\eta)]^\tau$ begins with non-vanishing terms of zeroth or τ-th dimension and, since $\tau < \sigma$, cannot belong to $\mathfrak{M}_\sigma$, contrary to the correspondence of $\mathfrak{M}_\sigma$ and $\mathfrak{M}_\tau$ derived above. The case $\tau > \sigma$ is dealt with at the same time by interchanging η and ξ. It is therefore shown that the domains $\mathfrak{G}_\sigma$ and $\mathfrak{G}_\tau$ are *not isomorphic, or essentially different.* Thus *the domains* $\mathfrak{G}_i$ $(i = 1, 2, \ldots$ *in inf.*$)$ *give an example of countably infinitely many classes of domains* $\mathfrak{G}$*.*

Remark. Although any two domains $\mathfrak{G}_i$ are non-isomorphic, it can very well happen, for two domains, that each is isomorphic to a part of the other.[158] Let, for example, $\tau = 1$ and σ be arbitrary:

$$\mathfrak{G}_1 : a_0 + \mathfrak{M}_1(\xi); \qquad \mathfrak{G}_\sigma : a_0 + a_1(\eta) + \cdots + a_{\sigma-1}(\eta) + \mathfrak{M}_\sigma(\eta).$$

The assignment $\xi = \eta$ makes $\mathfrak{G}_\sigma$ a proper divisor of $\mathfrak{G}_1$; conversely, the mutually one-to-one assignment in $\mathfrak{G}_1$ and $\mathfrak{G}_\sigma$ given by $\xi = \eta^\sigma$, $\eta = \sqrt[\sigma]{\xi}$ makes $\mathfrak{G}_1$ a proper divisor of $\mathfrak{G}_\sigma$. Thus every $\mathfrak{G}_i$ is also isomorphic to a proper divisor of itself. In contrast with the notion of intersection, the notion of an intersection class — that is, a class such that each of its domains would be isomorphic to a proper divisor of every domain of every other class — does not exist, at least not as a uniquely determined notion.

§ 10. The integral quantities of an arbitrary field.

The developments of the preceding paragraphs have, for the domains $\mathfrak{G}$ — provided that the algebraic numbers are not included in the defining properties III and IV for the domains $\mathfrak{G}$ — relied only on the fact that the totality of all complex numbers forms a field;[159] for the domains $\mathfrak{H}$, they relied on the further fact that this field is algebraically closed. The results obtained therefore directly admit the *generalization to the most general field*, which, following Weber and E. Steinitz,[160] is abstractly defined as "a system of elements with two operations, addition and multiplication, which are subject to the associative and commutative laws, are connected by the distributive law, and admit unrestricted and unique inverse operations."[161]

Every such field contains a unit element, and consequently the "prime field" derived from it by the operations of addition and multiplication and their inverses. This prime field is either of the type of the rational numbers (characteristic 0), or of the type of the residue-class system of a prime number p (characteristic p).[162] The *integral quantities of the prime field* are then well-defined as the quantities arising from the unit element by addition and subtraction; for prime fields of characteristic p, the integral quantities already exhaust the prime field, and there are no fractional quantities of the prime field. Every algebraically closed field contains an "absolutely algebraic field," arising from the prime field by adjoining all elements algebraic over the prime field; for fields of characteristic 0 this is therefore of the type of the field of all algebraic numbers. The *algebraically integral quantities of the absolutely algebraic field* are then well-defined as all quantities algebraically integral over the integral quantities of the prime field (or, what is the same thing, over the unit); for fields of characteristic p there are therefore also no algebraically fractional quantities of the absolutely algebraic field. After these preliminary remarks, the definition of the "integral" and "algebraically integral" quantities can be transferred to arbitrary fields, respectively to algebraically closed fields: *The*

[158] This behavior can also occur for fields; cf. Steinitz, loc. cit., conclusion.

[159] Only in § 7 was the further assumption used that the "prime field" of the rational numbers contains infinitely many elements; in the case of a prime field with only finitely many elements, however, this paragraph simply disappears, as we shall see.

[160] loc. cit., introduction.

[161] Only division by zero is to be excluded.

[162] Steinitz, § 4.

integral quantities of a field $\mathfrak{K}$ are defined as an integral domain $\mathfrak{G}$ of quantities from $\mathfrak{K}$ whose quotient field[163] *exhausts $\mathfrak{K}$, and which contains the unit element but no fractional quantity of the prime field.*

The algebraically integral quantities of an algebraically closed field $\mathfrak{K}$ are defined as an algebraically integrally closed integral domain $\mathfrak{H}$ of quantities from $\mathfrak{K}$ whose quotient field exhausts $\mathfrak{K}$, and which contains the unit element but no algebraically fractional quantity of the absolutely algebraic field.

For fields of characteristic zero, the results obtained in the preceding paragraphs apply literally, once one merely replaces the "algebraic numbers" by the quantities of the prime field, respectively of the absolutely algebraic field. For fields of characteristic p, the results simplify still further because the prime field contains no fractional quantities, so that the side condition for the rational basis and for the systems to be adjoined simply drops away.[164] Thus the result is: *The abstract definition permits as integral quantities of a field of characteristic p: every integral domain derived from an arbitrary rational basis, the field itself, and every integral domain lying between these domains and the field. Correspondingly, as algebraically integral quantities of an algebraically closed field of prime characteristic one has: every algebraically integrally closed integral domain lying between the domain derived from an algebraic basis and the field itself — including the two boundary domains.* For the most general fields of prime characteristic, therefore, as with the corresponding prime fields, the distinction between integral and fractional is effaced.

Erlangen, 30 March 1915.

[163]That is, the field consisting of all quotients of pairs of quantities from $\mathfrak{G}$.
[164]Cf. the first note to this paragraph.

10. The Functional Equations of the Isomorphic Mapping

Math. Ann. 77 (1916), pp. 536–545

Following Dedekind,[165] by an *isomorphic mapping of the field* $\mathfrak{A}$ *onto a system* $\mathfrak{B}$ – which subsequently also proves to be a field – one understands *such a single-valued correspondence that to every element of* $\mathfrak{A}$ *there corresponds one and only one element of* $\mathfrak{B}$; *and that to the sum, the difference, the product, and the quotient of any two elements of* $\mathfrak{A}$ *there are again assigned the sum, difference, product, and quotient of the uniquely corresponding elements of* $\mathfrak{B}$.

Let such a mapping be mediated by a – by the preceding, single-valued – function $f(z)$. If z then runs through all elements $x, y, \ldots$ of the field $\mathfrak{A}$, then $f(z)$ is characterized by the functional equations

$$(1)\quad f(x+y)=f(x)+f(y);\quad (2)\quad f(x-y)=f(x)-f(y);$$
$$(3)\quad f(x\cdot y)=f(x)\cdot f(y);\qquad (4)\quad f\left(\frac{x}{y}\right)=\frac{f(x)}{f(y)}.$$

In what follows the most general single-valued solutions $f(z)$ *of these functional equations are to be given when* $x, y, \ldots$ *run through all real and complex numbers; thus, the most general function* $f(z)$ *which effects an isomorphic mapping of the field of all complex numbers.*[166] The solution for arbitrary fields follows quite analogously.

G. Hamel[167] constructed the most general solutions of the functional equation (1) by means of a *linear* basis of all numbers. The construction given here for the solutions of (1) to (4) – hence of the mapping function relative to which the rational and algebraic operations are invariant – uses the *rational* and *algebraic* basis of all numbers. After the isomorphic mapping is discussed more closely in 1, necessary conditions for $f(z)$ are given in 2; in 3 these are recognized as sufficient and lead to the actual construction.[168] – The solutions of (1) to (4), with the known exceptions $f(z)=z$ or $\overline{z}$, are discontinuous; in 4 it is further shown that they are "extremely discontinuous" – a fact which G. Hamel proved for the real solutions of (1), but which need not be fulfilled there in the complex domain.[169]

1. We first draw some consequences from the functional equations (1) to (4), directly for general fields $\mathfrak{A}$. By (1), single-valuedness implies $f(0)=0$. If $f(y)$ does not vanish, then the original y cannot vanish either, and the functional equations (1) to (4) therefore show that *the mapping system* $\mathfrak{B}$ *of all values* $f(z)$ *again forms a field.* Conversely, if the initial condition $f(0)=0$ is prescribed, then (2) implies the single-valuedness of the function $f(z)$: *the requirement of single-valuedness and the initial condition* $f(0)=0$ *are therefore equivalent.* From (4) it follows that $f(z)$ cannot vanish identically; from (3) further, that $f(z)$ vanishes only for $z=0$. For from $f(y)=0$ and $y\neq 0$ it would follow, since z/y also belongs to $\mathfrak{A}$, that

$$f(z)=f\left(\frac{z}{y}\cdot y\right)=f\left(\frac{z}{y}\right)\cdot f(y)=0,$$

which contradicts (4) by making $f(z)$ vanish identically. From $f(x)=f(y)$ it follows therefore that $x=y$; that is, to each value of $f(z)$ there also corresponds one and only one value of z: $f(z)$ *is a reversibly single-valued function of* z. As the functional equations show, the mapping mediated by the inverse function is again isomorphic. Since every rational operation is composed of a finite number of additions, multiplications, and their inverses, one can also say: *By the isomorphic mapping a one-to-one relation is established between the elements of* $\mathfrak{A}$ *and of* $\mathfrak{B}$, *in such a way that all rational relations holding among finitely many elements of* $\mathfrak{A}$,

$$F(x, y, \ldots, t)=0,$$

[165] Cf. for example Dirichlet-Dedekind, *Lectures on Number Theory* (4th ed.), Supplement XI, § 161. The designation "isomorphic mapping" or "isomorphism", modeled on group theory, occurs only in the later literature; Dedekind speaks of the "permutations of a field".

[166] This question was occasionally put to me by Mr. Landau. As I noticed afterward, part of the solutions already occurs in H. Lebesgue, Sur les transformations ponctuelles, transformant les plans en plans, Atti Acad. Torino, 1906/07; and likewise implicitly in A. Ostrowski, Über einige Fragen der allgemeinen Körpertheorie, Crelle's Journal 143 (1913), § 2.

[167] G. Hamel, Eine Basis aller Zahlen und die unstetigen Lösungen der Funktionalgleichung: $f(x+y)=f(x)+f(y)$, Math. Ann. 60, p. 459 (1905).

[168] Cf. the parallel considerations in my paper: Die allgemeinsten Bereiche aus ganzen transzendenten Zahlen. Math. Ann. 77, p. 103 (1915), especially §§ 8 and 10.

[169] Hamel's term "totally discontinuous" is used in a different sense in the rest of the literature.

are preserved in $\mathfrak{B}$, *and conversely.*

It should also be noted that for a field $f(z)$ is already characterized by the functional equations (1) and (3), by the requirement that $f(z)$ not vanish identically, and by the initial condition $f(0) = 0$. Indeed, (2) follows from (1) by replacing x by the element $(x - y)$ belonging to $\mathfrak{A}$; then (2) and $f(0) = 0$ give the single-valuedness of $f(z)$. As was shown above, it follows further from (3), by replacing x by the element x/y belonging to $\mathfrak{A}$, that $f(y)$ cannot vanish for $y \neq 0$, and hence the validity of (4) and at the same time one-to-oneness. If instead of a field one takes an integral domain $\mathfrak{S}$ as basis, x/y need not belong to $\mathfrak{S}$, and from (3) one cannot infer one-to-oneness. Here, however, the following most familiar formulation suffices: an isomorphic mapping is a *one-to-one* correspondence between the elements of $\mathfrak{S}$ and $\mathfrak{S}'$ such that to the sum and the product there always correspond sum and product.[170]

2. From now on, for simplicity, instead of $\mathfrak{A}$ let the field $\mathfrak{C}$ of all real and complex numbers be taken as the basis. The first matter is to discuss the systems of values which $f(z)$ assumes. From (3) or (4) it follows that $f(1) = 1$, and thence, by the functional equations, for every *rational number* α, $f(\alpha) = \alpha$; *therefore every rational number corresponds to itself under the mapping.* Since an *algebraic number* β satisfies an irreducible equation with rational numerical coefficients $F(\beta, \alpha) = 0$, no. 1 and the preceding observation give, for $f(\beta)$, the equation $F(f(\beta), \alpha) = 0$; *every algebraic number therefore goes into itself or into one of its conjugates*, and by one-to-oneness the totality of algebraic numbers again corresponds to the field of all algebraic numbers. In order to recognize the behavior of the *transcendental* numbers, and at the same time to separate the conjugate values, we take as basis a *rational basis of all numbers*,[171] i.e. a system Θ of numbers ϑ which permits all numbers to be expressed rationally – with rational numerical coefficients[172] – while in at least one well-ordering no basis number is rationally expressible through finitely many preceding ones. The existence of this basis follows exactly analogously to linear and algebraic bases from the well-ordering theorem.

Let $\mathfrak{C}$ be subjected to a well-ordering Ω. Then every number is either expressible rationally through finitely many preceding numbers, or is rationally independent of the preceding ones. These latter numbers form the basis Θ, and induction shows that indeed every number is rationally expressible through finitely many ϑ's.

For each basis number ϑ, the "segment field $\mathfrak{K}(\vartheta)$" is defined: it consists of all rational combinations of those basis numbers which precede ϑ in the well-ordering. As the segment field of the first basis number one is to regard the field $\mathfrak{R}$ of all rational numbers. The number ϑ can display two sorts of behavior with respect to $\mathfrak{K}(\vartheta)$: it can be algebraically dependent on $\mathfrak{K}(\vartheta)$, or algebraically independent of it. Since, by no. 1, all relations valid in $\mathfrak{C}$ are also valid in the mapping field, and conversely, the totality of values $f(\vartheta)$ corresponding to the ϑ's forms a basis of the mapping range, well-ordered by means of the well-ordering of Θ itself. In particular, the segment field $\mathfrak{K}(\vartheta)$ corresponds to the segment field $\mathfrak{K}(f(\vartheta))$; and according as ϑ is algebraically dependent on, or independent of, $\mathfrak{K}(\vartheta)$, the same holds for $f(\vartheta)$ with respect to $\mathfrak{K}(f(\vartheta))$. In the dependent case the irreducible equations for ϑ and $f(\vartheta)$ also correspond. The totality of those values ϑ which are in each case algebraically independent of $\mathfrak{K}(\vartheta)$ form – as induction again shows – an *algebraic basis of all numbers*,[173] i.e. a system H of numbers η which permits all numbers to be expressed algebraically, while no algebraic relations hold among the η's themselves. To this algebraic basis H there corresponds in the mapping range again an algebraic basis Z, which by one-to-oneness has the same cardinality as H, namely the cardinality of the continuum, since algebraic extension is known not to increase cardinality.[174] From the knowledge of the values $f(\vartheta)$, however, $f(z)$ is immediately obtained for all systems of values, since from $z = \psi(\vartheta)$ there always follows $f(z) = \psi(f(\vartheta))$.

3. We now show that the necessary conditions found above actually also provide the construction of $f(z)$. The basis Z corresponding one-to-one to the algebraic basis H is constructed and assigned to H as follows. Put a second well-ordering Ω' in place, which supplies an algebraic basis Z' of all numbers. Let Z, with elements ξ, be a subset of Z' of the same cardinality as Z', and hence as H. To each element η_i of H assign one and only one element ξ_i, with the same numbering, by the rule: ξ_i is the *first* of all elements of Z in the well-ordering Ω' which has not been assigned to any element of H preceding η_i.

Now define $f(z)$ as follows:

(a) *for all rational numbers* α, *by* $f(0) = 0$, $f(\alpha) = \alpha$;

(b) *for all quantities of the algebraic basis* H, *by* $f(\eta_i) = \xi_i$;

[170] Cf. on no. 1: Dedekind, loc. cit., § 161.

[171] Cf. my cited paper on "whole transcendental numbers", § 6.

[172] By a "rational function" I shall always mean one whose coefficients are also rational numbers.

[173] Cf. E. Zermelo, Über ganze transzendente Zahlen, § 1, Math. Ann. 75 (1914).

[174] Cf. E. Steinitz, Algebraische Theorie der Körper, Crelle's Journal 137 (1910), § 24.

(c) *for all remaining quantities ϑ of the rational basis Θ – those algebraically dependent on the segment field $\mathfrak{K}(\vartheta)$, and hence satisfying an equation $F(\vartheta; \vartheta_{i_1}, \ldots, \vartheta_{i_\nu}) = 0$ irreducible in $\mathfrak{K}(\vartheta)$ – as a root of the equation $F(f(\vartheta); f(\vartheta_{i_1}), \ldots, f(\vartheta_{i_\nu})) = 0$;*

(d) *for all remaining values z, which by the definition of Θ are representable as $z = \psi(\vartheta_{k_1}, \ldots, \vartheta_{k_\nu})$, by $f(z) = \psi(f(\vartheta_{k_1}), \ldots, f(\vartheta_{k_\nu}))$.*[175]

To show that the so-defined $f(z)$ actually satisfies the functional equations (1) to (4), by no. 1 it suffices to prove this for (1) and (3), since $\mathfrak{C}$ is a field and $f(z)$, because of (a) and (b), cannot vanish identically and also satisfies the initial condition $f(0) = 0$. By means of the rational basis Θ, express $x, y, x + y$ as

$$x = \psi_1(\vartheta_1 \cdots \vartheta_t), \quad y = \psi_2(\vartheta_1 \cdots \vartheta_t), \quad (x + y) = \psi_3(\vartheta_1 \cdots \vartheta_t).$$

Thus the proof that $f(z)$ satisfies the functional equation (1),

$$f(x) + f(y) - f(x + y) = 0,$$

is, by (d), identical with showing that from

$$\psi_1(\vartheta) + \psi_2(\vartheta) - \psi_3(\vartheta) = \frac{H_1(\vartheta)}{H_2(\vartheta)} = 0$$

there always follows

$$\psi_1(f(\vartheta)) + \psi_2(f(\vartheta)) - \psi_3(f(\vartheta)) = \frac{H_1(f(\vartheta))}{H_2(f(\vartheta))} = 0,$$

where H_1, H_2 denote integral rational functions of their arguments. The functional equation (3) leads to precisely the same form of condition; hence the satisfaction of all the functional equations is identical with the following condition – necessary also by nos. 1 and 2: for *every* integral rational function $H(\vartheta)$, from $H(\vartheta) = 0$ there always follows $H(f(\vartheta)) = 0$, and from $H(\vartheta) \neq 0$ there follows $H(f(\vartheta)) \neq 0$, the latter because of the occurrence of the denominator.

That this is indeed the case is shown by induction. Suppose that in $H(\vartheta_1 \cdots \vartheta_t)$ the basis quantity ϑ_t is the last, in the well-ordering, among $\vartheta_1 \cdots \vartheta_t$.[176] Then $H(\vartheta_1 \cdots \vartheta_t) = 0$ may be regarded as a relation satisfied by ϑ_t with respect to the segment field $\mathfrak{K}(\vartheta_t)$, and likewise $H(f(\vartheta_1) \cdots f(\vartheta_t)) = 0$ as a relation satisfied by $f(\vartheta_t)$ with respect to $\mathfrak{K}(f(\vartheta_t))$. If not all relations $H(\vartheta) = 0$ were also fulfilled by $f(\vartheta)$, and conversely, then there would have to be a first basis quantity, say ϑ_0, for which at least one relation valid with respect to $\mathfrak{K}(\vartheta_0)$ failed to be preserved in the mapping field, or conversely, while the preceding segment fields $\mathfrak{K}(\vartheta_0)$ and $\mathfrak{K}(f(\vartheta_0))$ were isomorphic. In particular, by (a), the segment field of the first basis quantity of Θ, the field $\mathfrak{R}$ of all rational numbers, is isomorphic to its mapping field. Suppose now that $H(\vartheta_0)$ has the form

$$H(\vartheta_0) = a_0(\vartheta) \cdot \vartheta_0^{\chi} + a_1(\vartheta) \cdot \vartheta_0^{\chi - 1} + \cdots + a_{\chi}(\vartheta),$$

where the $a_i(\vartheta)$ are quantities from $\mathfrak{K}(\vartheta_0)$. Then there are two possibilities.

1) ϑ_0 is algebraically independent of $\mathfrak{K}(\vartheta_0)$. Then $H(\vartheta_0) = 0$ is fulfilled identically in ϑ_0,[177] so that $a_i(\vartheta) = 0$ for all coefficients, and hence by the isomorphism of $\mathfrak{K}(\vartheta_0)$ and $\mathfrak{K}(f(\vartheta_0))$ also $a_i(f(\vartheta)) = 0$. From $H(\vartheta_0) = 0$ there follows, therefore, always $H(f(\vartheta_0)) = 0$. Conversely, suppose $H(f(\vartheta_0)) = 0$. Since ϑ_0, being algebraically independent of $\mathfrak{K}(\vartheta_0)$, belongs to the algebraic basis H, $f(\vartheta_0)$ belongs by (b) to the algebraic basis Z and is algebraically independent of $\mathfrak{K}(f(\vartheta_0))$. Thus all $a_i(f(\vartheta))$ vanish, and by the isomorphism all $a_i(\vartheta)$ vanish. From $H(f(\vartheta_0)) = 0$ follows $H(\vartheta_0) = 0$, or from $H(\vartheta_0) \neq 0$ also $H(f(\vartheta_0)) \neq 0$.

2) ϑ_0 is algebraically dependent on $\mathfrak{K}(\vartheta_0)$. Then $H(\vartheta_0)$ is divisible by the equation $F(\vartheta_0)$ irreducible with respect to $\mathfrak{K}(\vartheta_0)$; that is, identically in t,

$$H(t; a_i(\vartheta)) = F(t; \vartheta) \cdot G(t; \vartheta_i).$$

Since all the coefficients which occur belong to $\mathfrak{K}(\vartheta_0)$, the corresponding relation is fulfilled in the isomorphic field $\mathfrak{K}(f(\vartheta_0))$:

$$H(t; a_i(f(\vartheta))) = F(t; f(\vartheta)) \cdot G(t; f(\vartheta)).$$

[175] In (d), (a) is contained as a special case.

[176] The expressions $H(\vartheta)$ which are of degree zero in the ϑ's go over into themselves under the mapping and therefore need not be investigated further.

[177] Because $x, y, \ldots$ can be expressed by the ϑ's, this form of relation can very well occur.

But by (c), $f(\vartheta_0)$ was chosen to be a root of $F(t; f(\vartheta))$. Hence from $H(\vartheta_0) = 0$ there follows $H(f(\vartheta_0)) = 0$.

Conversely, if $H(f(\vartheta_0)) = 0$, then the divisibility of $H(t; a_i(f(\vartheta)))$ by the function $F(t; f(\vartheta))$, irreducible with respect to $\mathfrak{K}(f(\vartheta_0))$, implies for the isomorphic field $\mathfrak{K}(\vartheta_0)$ the divisibility of $H(t; a_i(\vartheta))$ by $F(t; \vartheta)$. And since $F(t; f(\vartheta))$ arose from the irreducible equation for ϑ_0 by the isomorphic relation, and this relation is one-to-one, $F(t; \vartheta)$ necessarily represents the irreducible function whose root is ϑ_0. Thus from $H(f(\vartheta_0)) = 0$ follows $H(\vartheta_0) = 0$, or from $H(\vartheta_0) \neq 0$ also $H(f(\vartheta_0)) \neq 0$.

From isomorphic segment fields $\mathfrak{K}(\vartheta_0)$ and $\mathfrak{K}(f(\vartheta_0))$, therefore, isomorphic fields also arise by adjunction of ϑ_0 and $f(\vartheta_0)$, respectively; the assumption that this fails for some ϑ_0 leads to a contradiction. It is thus proved that *the function $f(z)$ defined by (a) to (d) actually represents a solution of the functional equations (1) to (4), and, by 2, the most general one.*

All considerations used in constructing $f(z)$ have used only the fact that the totality $\mathfrak{C}$ of all numbers forms a field; hence they hold for every abstractly defined field $\mathfrak{A}$. Here one should note that the field $\mathfrak{R}$ of rational numbers is replaced by the "prime field" of $\mathfrak{A}$, i.e. by the field derived from the unit element of $\mathfrak{A}$ by the operations of addition and multiplication and their inverses. If, therefore, in (a) to (d) one replaces the rational numbers by the elements of the prime field, then *the function $f(z)$ so obtained effects the most general isomorphic mapping of an arbitrary abstractly defined field.*

4. It remains to show that the functions $f(z)$ defined for the field $\mathfrak{C}$ of all real and complex numbers – which, except for $f(z) = z$ or $\overline{z}$, are known to be discontinuous – are *extremely discontinuous*; that is, in every neighborhood of any real or complex number z_0 there are values z for which $f(z)$ comes arbitrarily close to any prescribed value Z_0. This extreme discontinuity belongs, as G. Hamel proved loc. cit., to the *real* solutions $\varphi(x)$, defined for all *real* numbers, of the functional equation (1). But, as the examples below show, it need not belong to the solutions defined on the complex numbers. Hamel argues as follows. If $\varphi(x)$ differs from $C \cdot x$, then there are at least two values of the linear basis, say x_1 and x_2, such that $\varphi(x_1) : x_1 \neq \varphi(x_2) : x_2$. Then the two equations

$$\alpha x_1 + \beta x_2 = a; \qquad \alpha\varphi(x_1) + \beta\varphi(x_2) = b$$

have a solution in real numbers α, β; consequently a and b can be approximated arbitrarily closely by rational numbers α, β. Because α, β are rational, $\alpha\varphi(x_1) + \beta\varphi(x_2)$ is then actually $\varphi(\alpha x_1 + \beta x_2)$. This argument breaks down for complex systems of values. In fact one can give, also in the complex domain, solutions of (1) which are discontinuous but not extremely discontinuous; for example,

$$\varphi(z) = \varphi(x + iy) = \varphi(x) + i(y + \varphi(x)),$$

where $\varphi(x)$ denotes a real solution which is extremely discontinuous in the real domain; or, still more simply, $\varphi(z) = \varphi(x)$. In both cases the real part of $\varphi(z)$ is extremely discontinuous, while the imaginary part is determined by the real part.

This behavior, and at the same time the behavior of the function $f(z)$, becomes completely clear if one introduces the concept of rank. We put $z = x + iy$, $\varphi(z) = X + iY$, and call $\varphi(z)$ of rank four, three, two, or one, respectively, according as there exist no, one, two, or three linearly independent relations

$$c_1x + c_2y + c_3X + c_4Y = 0,$$

where of course the c's are real numbers.

1) Let $\varphi(z)$ have *rank four.* Then there exist at least four values of the linear basis, say z_1, z_2, z_3, z_4, such that the determinant

$$\begin{vmatrix} x_1 & x_2 & x_3 & x_4 \\ y_1 & y_2 & y_3 & y_4 \\ X_1 & X_2 & X_3 & X_4 \\ Y_1 & Y_2 & Y_3 & Y_4 \end{vmatrix}$$

does not vanish. The equations

$$\begin{aligned} \alpha x_1 + \beta x_2 + \gamma x_3 + \delta x_4 &= a, \\ \alpha y_1 + \cdots + \delta y_4 &= b, \\ \alpha X_1 + \cdots + \delta X_4 &= A, \\ \alpha Y_1 + \cdots + \delta Y_4 &= B \end{aligned}$$

therefore have a solution in real numbers $\alpha, \ldots, \delta$, and a, b, A, B can be approximated arbitrarily closely by rational numbers $\alpha, \beta, \gamma, \delta$. Moreover one has

$$\alpha(X_1 + iY_1) + \beta(X_2 + iY_2) + \gamma(X_3 + iY_3) + \delta(X_4 + iY_4) = \varphi(\alpha z_1 + \beta z_2 + \gamma z_3 + \delta z_4).$$

"In general", therefore, the solutions $\varphi(z)$ are extremely discontinuous also in the complex domain; the *discontinuity values* of $\varphi(z)$ in the neighborhood of $z_0 = a + ib$ *form a two-dimensional linear manifold.*

2) Let $\varphi(z)$ have *rank three.* Then there are at least three systems of values of the linear basis, say z_1, z_2, z_3, such that the above determinant has rank three. The values a, b, A, B satisfying the relation

$$c_1 a + c_2 b + c_3 A + c_4 B = 0,$$

and only these, can be approximated arbitrarily closely; the *discontinuity values* form a *one-dimensional linear manifold* determined by $z_0 = a + ib$. As the examples given show, this case can actually occur.

3) Let $\varphi(z)$ have *rank two.* Since one can always choose two complex numbers z_1, z_2 such that

$$\begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix} \neq 0,$$

the relations have the form

$$X = \lambda_1 x + \lambda_2 y; \qquad Y = \mu_1 x + \mu_2 y.$$

The resulting expression

$$\varphi(z) = (\lambda_1 x + \lambda_2 y) + i(\mu_1 x + \mu_2 y)$$

is precisely the most general *continuous* solution of the functional equation (1) in the domain of complex numbers.

4) Let $\varphi(z)$ have *rank one.* This case cannot occur in the domain of complex numbers; in the domain of real numbers it means the *continuous* solution $\varphi(x) = C \cdot x$.

We now show that the *solutions $f(z)$ of the functional equations (1) to (4) defined for all real and complex numbers can have only rank two or rank four.* Here rank two gives only the two *continuous* functions $f(z) = z$ or $\overline{z}$, as the specialization $f(1) = 1$ and $f(i) = \pm i$, combined with 3), shows. Since rank one has already been excluded, it remains to exclude rank three. We therefore assume that at least one relation exists,

$$c_1 x + c_2 y + c_3 X + c_4 Y = 0,$$

so that $f(z)$ has rank three, or possibly lower rank, and show that under this assumption the latter case always occurs. As again the specialization $f(1) = 1$, $f(i) = \pm i$ shows, the relation becomes

$$c_3(X - x) + c_4(Y \mp y) = 0,$$

where c_3 and c_4 do not both vanish. First suppose, say, $c_4 = 0$. The relation $(X - x) = 0$ then means that to a purely imaginary z there also corresponds a purely imaginary $f(z)$. Thus for every real y one has $f(iy) = i \cdot \mathfrak{Y}$, where $\mathfrak{Y}$ is again real. From this, because

$$f(iy) = \pm i f(y),$$

one obtains also $f(y) = \pm\mathfrak{Y}$; hence to every real z there also corresponds a real $f(z)$, and because of $(X - x)$ all real values go into themselves. Thus one has only $f(z) = z$ or $\overline{z}$, so actually rank two. The conclusion is entirely analogous when $c_3 = 0$, $c_4 \neq 0$.

Now let c_3 and c_4 be different from zero, so that the relation can be written in the form

$$(Y \mp y) = c \cdot (X - x)$$

with real, nonzero, finite c. But this means that for $y = \pm cx$ one also has $Y = c \cdot X$. Hence, taking account of the functional equations, for every real x one has

$$f\{x \cdot (1 \pm ic)\} = \mathfrak{X}(1 + ic) = f(x) \cdot (1 + i f(c)),$$

where $\mathfrak{X}$ is again real. But here, by no. 1, the factor of $f(x)$ cannot vanish identically, and division gives

$$f(x) = \mathfrak{X}(\sigma + i\tau)$$

with real σ and τ, independent of x and $\mathfrak{X}$. In particular, to $x = 1$ there also belongs a real X_0, and the condition $1 = X_0(\sigma + i\tau)$ shows that necessarily $\tau = 0$, so that $f(x) = \sigma \cdot \mathfrak{X}$. Thus to every real z there also corresponds a real $f(z)$; equivalently, from $y = 0$ there necessarily follows $Y = 0$. Hence the linear relation for real z becomes $X - x = 0$;[178] every real value corresponds to itself. One again has only $f(z) = z$ or $\overline{z}$, so actually rank two. Rank three is thereby excluded in all cases. Since, however – as the investigations of the first sections show – discontinuous solutions actually exist, and these must therefore necessarily have rank four, the following theorem holds by 1): *"All solutions $f(z)$ of the functional equations (1) to (4) – with the exception of the continuous solutions $f(z) = z$ or $\overline{z}$ – are extremely discontinuous in the real and in the complex domain."*[179]

Erlangen, 30 October 1915.

[178] Here one uses the fact that $c \neq 0$, so that neither c_3 nor c_4 vanishes, whereas above only $c_4 \neq 0$ was used; therefore the case where one of these two vanishes had to be dealt with in advance.

[179] Lebesgue shows loc. cit. that the real and imaginary parts of $f(z)$ are both "non-measurable" functions of x and y; as the example at the beginning of no. 4, $\varphi(z) = \varphi(x) + i(y + \varphi(x))$, shows, this does not yet imply extreme discontinuity in the complex domain.

11. Equations with Prescribed Group

Math. Ann. 78 (1918), pp. 221–229

The problem of constructing equations with prescribed group can be attacked in two directions, which one may briefly call the "irrational" direction, characterizing the roots, and the "rational" direction, characterizing the coefficients.

In the "irrational" direction, which works function-theoretically and arithmetically, lies Kronecker's theorem that all Abelian fields in the domain of the rational numbers are cyclotomic fields, and the corresponding theorems for relatively Abelian fields with respect to quadratic number fields. These theorems give, on the one hand, the realizability of all Abelian groups with respect to these special domains of rationality; on the other hand, they supply the totality of the equations and at the same time give a deeper insight into the structure of the number fields thereby defined. Each individual theorem, however, is restricted to a special domain of rationality; and every new domain of rationality requires an entirely new treatment.

The "rational", algebraic direction rests on Hilbert's irreducibility theorem, according to which in the coefficient representation one may restrict oneself to a parameter representation. To this direction belongs, for example, the existence of arbitrarily many equations with alternating group with respect to any prescribed domain of rationality. Here the insight, described above, into the structure of the fields defined by the equations is naturally absent; but the advantage of this "rational" direction lies in the fact that the realizability of the groups is proved for *all domains of rationality simultaneously*; nevertheless, the parameter representations known up to now do not in general yield the totality of the equations.[180]

What follows is a contribution to the "rational" direction; *sufficient conditions* are given for the *existence of parameter representations which, for any domain of rationality, supply the totality of the equations with prescribed group.* These conditions consist in this: the invariant field belonging to the group – Lagrange's genus domain – can be represented rationally by independent functions of the domain, that is, it possesses a "minimal basis"; or, expressed otherwise, it is isomorphic to the field of all rational functions of n indeterminates. As will still be shown in § 2, this is satisfied, for example, for all groups occurring in equations of the third and fourth degrees. The actual construction and more exact discussion of these equations of the third and fourth degrees is found in the dissertation of F. Seidelmann,[181] an extract from which immediately follows this communication.

The question of the minimal basis of Lagrange's genus domains was – as already mentioned in the paper "Körper und Systeme rationaler Funktionen" (Math. Ann. 76) – raised to me by E. Fischer independently of the connection just described, and gave the impetus for the present investigations.[182]

§ 1.

Sufficient Conditions for the Parameter Representation.

1. The possibility of reducing the construction of equations with prescribed group to parameter representation follows from Hilbert's irreducibility theorem, which says the following. Suppose that the equation

$$f(x) = x^n + F_1(\lambda_1 \cdots \lambda_r)x^{n-1} + \cdots + F_n(\lambda_1 \cdots \lambda_r) = 0 \tag{1}$$

– where $F_i(\lambda_1 \cdots \lambda_r)$ denotes rational functions of the independent parameters $\lambda_1 \cdots \lambda_r$, with coefficients from a number field[183] Ω – has, with respect to the field $\Omega(\lambda_1 \cdots \lambda_r)$ of all rational functions of $\lambda_1 \cdots \lambda_r$ with coefficients from Ω, the group Γ. Then one can replace the parameters λ by arbitrarily many rational numbers in such a way that the resulting numerical equation

$$g(x) = x^n + a_1x^{n-1} + \cdots + a_n = 0 \tag{2}$$

[180] For solvable equations, the parameter representation of the coefficients can be replaced by that of the roots. Recently F. Mertens has given, for certain groups of degree 8, most general root representations: "Gleichungen 8ten Grades mit Quaternionengruppen", Sitzb. d. Ak. d. Wiss., Wien, Abt. IIa, Bd. 125, p. 735. As I gather from that note, G. Bucht had already earlier given the most general root expressions for always constructible normal forms of equations of degree 3 and 4 and of equations of degree 8 with the groups mentioned: "Über einige algebraische Körper achten Grades", Arkiv for Math., Astron. och Physik, Bd. 6, Nr. 30. [Addition at correction.]

[181] Die Gesamtheit der kubischen und biquadratischen Gleichungen mit Affekt bei beliebigem Rationalitätsbereich. Erlangen 1916.

[182] Cf. Part 2 of the preliminary communication on "Rationale Funktionenkörper", Jahresb. d. d. Math. Vereinig., Bd. 22, 1913.

[183] Instead of the number field, a general domain of rationality could also occur.

has, with respect to Ω, precisely the group Γ. Such a $g(x)$ shall be denoted more precisely by g_Γ.

The question now is whether *such a parameter representation* (1) *can be given that the resulting numerical equation* (2) *runs through the totality of all* g_Γ *when* $\lambda_1 \cdots \lambda_r$ *run through all quantities from* Ω – with the exclusion of those quantities from Ω which cause a reduction of the group.[184] In other words: the nonlinear system of equations

$$F_1(\lambda_1 \cdots \lambda_r) = a_1; \quad \cdots ; \quad F_n(\lambda_1 \cdots \lambda_r) = a_n \tag{3}$$

should possess, for every system of values $a_1 \cdots a_n$ corresponding to a g_Γ, a solution, and indeed in quantities $\lambda_1 \cdots \lambda_r$ from Ω. Since the system (3) is nonlinear, one must from the outset introduce a restriction when demanding that the totality of the g_Γ be obtained through a parameter representation. Namely, in such nonlinear systems there can always occur singular systems of values $a_1 \cdots a_n$, satisfying an algebraic relation $H(a_1 \cdots a_n) = 0$, for which *no* solution exists;[185] for them the parameter representation (1) fails and a supplementary representation is necessary. But in the present case these singular values of $a_1 \cdots a_n$, as will be shown, can be characterized simply.

2. To obtain such a parameter representation, we impose the stronger requirement that the λ_i be *natural irrationalities belonging to the group, identically in the roots regarded as indeterminates*. By this we mean the following. If in (1) one replaces the parameters $\lambda_1 \cdots \lambda_r$ by suitably chosen *rational* functions $\varphi_1(x) \cdots \varphi_r(x)$ of the indeterminates $x_1 \cdots x_n$ – with coefficients from Ω – then (1) is to pass into

$$h(x) = x^n - \sigma_1(x)x^{n-1} + \sigma_2(x)x^{n-2} \cdots \pm \sigma_n(x), \tag{4}$$

where $\sigma_1(x) \cdots \sigma_n(x)$ denote the elementary symmetric functions of $x_1 \cdots x_n$; and $h(x)$, with respect to the domain of rationality $\Omega(\varphi_1(x) \cdots \varphi_r(x))$ thereby arising from $\Omega(\lambda_1 \cdots \lambda_r)$, is to have the group Γ. Because of the independence of the parameters, $\varphi_1(x) \cdots \varphi_r(x)$ are also to be algebraically independent functions.

To make this requirement precise, one must clarify the connection of $\Omega(\varphi_1(x) \cdots \varphi_r(x))$ with the group Γ. For this we first determine the most general domain of rationality K with respect to which $h(x)$ becomes h_Γ. Let $\mathfrak{L}_\Gamma$ denote the field of all rational functions with rational integral coefficients of $x_1 \cdots x_n$ which admit Γ – also called the "Lagrangian genus domain" belonging to Γ, or the invariant field of Γ – and let $\Omega(\mathfrak{L}_\Gamma)$ denote the corresponding field arising by adjoining $\mathfrak{L}_\Gamma$ to Ω. Thus $\mathfrak{L}_\Gamma$ is a subfield of the field $\mathfrak{R}(x_1 \cdots x_n)$ of all rational functions with rational integral coefficients of $x_1 \cdots x_n$. According to the definition of the group, the most general domain K is then given by *any field which contains* $\mathfrak{L}_\Gamma$ *as a subfield and whose intersection with* $\mathfrak{R}(x_1 \cdots x_n)$ *is exactly* $\mathfrak{L}_\Gamma$. Correspondingly, for every domain K containing Ω, the intersection with $\Omega(x_1 \cdots x_n)$ is given by $\Omega(\mathfrak{L}_\Gamma)$. Therefore, in particular, since by our requirement $\Omega(\varphi_1(x) \cdots \varphi_r(x))$ is to be a domain K, the intersection of $\Omega(\varphi_1(x) \cdots \varphi_r(x))$ with $\Omega(x_1 \cdots x_n)$ is given by $\Omega(\mathfrak{L}_\Gamma)$; and since on the other hand $\Omega(\varphi_1(x) \cdots \varphi_r(x))$ is a subfield of $\Omega(x_1 \cdots x_n)$, $\Omega(\varphi_1(x) \cdots \varphi_r(x))$ is exactly equal to $\Omega(\mathfrak{L}_\Gamma)$. The requirement therefore says that *Lagrange's genus domain* $\Omega(\mathfrak{L}_\Gamma)$ *is to arise by adjoining the algebraically independent functions* $\varphi_1(x) \cdots \varphi_r(x)$ *to* Ω; here one must have $r = n$, since $\mathfrak{L}_\Gamma$ contains the n algebraically independent elementary functions, while on the other hand no more than n functions can be algebraically independent. The functions $\varphi_1(x) \cdots \varphi_n(x)$ shall be said, as we shall also put it, to form a *minimal basis* of $\Omega(\mathfrak{L}_\Gamma)$; or also, $\mathfrak{L}_\Gamma$ possesses a minimal basis with coefficients from Ω.

3. It is now essential that this whole consideration can be reversed, and thus actually leads to sufficient conditions[186] for the desired parameter representation. We therefore assume that $\varphi_1(x) \cdots \varphi_n(x)$ is a minimal basis of $\Omega(\mathfrak{L}_\Gamma)$; and let Ω be the smallest field containing the coefficients of $\varphi_1(x) \cdots \varphi_n(x)$. Then Ω is necessarily an algebraic number field, and can in particular be the field $\mathfrak{R}$ of all rational numbers. Since $\sigma_1(x), \ldots, \sigma_n(x)$ belong to $\Omega(\mathfrak{L}_\Gamma)$, there exist identities in $x_1 \cdots x_n$ – where the F_i denote rational functions of their arguments:

$$\sigma_1(x) = F_1(\varphi_1(x) \cdots \varphi_n(x)); \quad \cdots ; \quad \sigma_n(x) = F_n(\varphi_1(x) \cdots \varphi_n(x)); \tag{5}$$

whereas conversely $\varphi_1(x) \cdots \varphi_n(x)$, and hence also $\varphi(x) = \mu_1\varphi_1(x) + \cdots + \mu_n\varphi_n(x)$, depend algebraically on $\sigma_1(x) \cdots \sigma_n(x)$ by means of the equation irreducible with respect to $\Omega(\sigma_1(x) \cdots \sigma_n(x))$:

$$G(\varphi(x); \sigma_1(x) \cdots \sigma_n(x)) = 0, \tag{6}$$

which is again an identity in the x.

[184] The occurrence of a double root of $g(x) = 0$ is also to be included here.

[185] This is to be carried out shortly in general in a paper on the "alternative in nonlinear systems of equations."

[186] The requirement in 2. was enlarged relative to 1., so that the conditions need not be necessary.

By virtue of (5), this identity (6) passes into

$$G(\varphi(x); F_1(\varphi_1 \cdots \varphi_n); \cdots ; F_n(\varphi_1 \cdots \varphi_n)) = H(\varphi_1(x) \cdots \varphi_n(x)) = 0; \tag{7}$$

and because of the algebraic independence of $\varphi_1(x) \cdots \varphi_n(x)$, (7) is fulfilled not only identically in x, but also in the φ_i; that is, identically in independent parameters λ one has

$$H(\lambda_1 \cdots \lambda_n) = G(\mu_1\lambda_1 + \cdots + \mu_n\lambda_n; F_1(\lambda_1 \cdots \lambda_n); \cdots ; F_n(\lambda_1 \cdots \lambda_n)) = 0. \tag{8}$$

From the identity (8) it follows that the equation

$$f(x) = x^n - F_1(\lambda_1 \cdots \lambda_n)x^{n-1} + F_2(\lambda_1 \cdots \lambda_n)x^{n-2} \cdots \pm F_n(\lambda_1 \cdots \lambda_n) = 0 \tag{9}$$

has the group Γ *with respect to* $\Omega(\lambda_1 \cdots \lambda_n)$. For this identity shows that the rationally known $\lambda = \mu_1\lambda_1 + \cdots + \mu_n\lambda_n$ is a function of the roots belonging to Γ; but no further function belonging to a subgroup of Γ can be rationally known, as is shown by the substitution $\lambda_i = \varphi_i(x)$; and under the same substitution λ is also different from all its conjugates. If further Ω^* is any number field containing Ω, then $f(x)$ *has the group* Γ *with respect to* $\Omega^*(\lambda_1 \cdots \lambda_n)$; for by 2. the group is not reduced under the substitution $\lambda_i = \varphi_i(x)$ even after adjoining Ω^*.

It remains to prove that, in the sense modified in 1., $f(x)$ actually runs through the totality of all g_Γ. For this, observe that by (5) the $F_i(\lambda)$ are algebraically independent functions of their arguments λ. The system of equations

$$F_1(\lambda_1 \cdots \lambda_n) = -a_1; \quad F_2(\lambda_1 \cdots \lambda_n) = a_2; \quad \cdots ; \quad F_n(\lambda_1 \cdots \lambda_n) = \pm a_n \tag{10}$$

therefore possesses solutions for every arbitrary system of values $a_1 \cdots a_n$ – with the exception of the singular ones still to be specified. For every system of values $a_1 \cdots a_n$ corresponding to a g_Γ with respect to Ω^*, the corresponding λ_i, as natural irrationalities belonging to the group,[187] become quantities from Ω^*. Thus as $\lambda_1 \cdots \lambda_n$ run through all systems of values, $g(x)$ runs through the totality of all equations (in the modified sense); and this totality contains the totality of the g_Γ to which values from Ω^* correspond. Since, conversely, by Hilbert's irreducibility theorem the values λ from Ω^* "in general" also correspond to equations g_Γ, one actually obtains, in the sense modified in 1., through the parameter representation (9) the totality of all g_Γ with respect to Ω^*.

It remains now to determine the singular systems of values for which the parameter representation fails. Split into numerator and denominator, let the functions of the minimal basis be given by

$$\varphi_i(x) = \frac{\psi_i(x)}{\chi_i(x)}.$$

Substituting this into the identities (5), suppose that these identities become, without cancellation,

$$\sigma_k(x) = \frac{G_k(x)}{H_k(x)} \quad \text{or} \quad H_k(x) \cdot \sigma_k(x) = G_k(x), \tag{11}$$

so that the product over all $H_k(x)$ is divisible by the product of all denominators $\chi_i(x)$. For all root systems $\alpha_1 \cdots \alpha_n$ for which the product

$$H_1(\alpha) \cdot H_2(\alpha) \cdots H_n(\alpha) \neq 0$$

and consequently no denominator $\chi_i(\alpha)$ vanishes either, one can divide (11) by $H_k(\alpha)$ and obtains

$$\sigma_k(\alpha) = \frac{G_k(\alpha)}{H_k(\alpha)} = F_k(\varphi_1(\alpha); \cdots ; \varphi_n(\alpha)), \tag{12}$$

where the $\varphi_i(\alpha)$, because of the nonvanishing of the denominators, assume finite values – and for equations g_Γ become quantities from Ω^*. Equation (12) represents the solution system of (10), as soon as $\alpha_1 \cdots \alpha_n$ are determined so that $a_k = (-1)^k\sigma_k(\alpha)$.[188]

[187] Cf. also below!

[188] The solution of the equation is thereby reduced to the solution of the system, consisting of independent functions,

$$\varphi_1(\alpha_1 \cdots \alpha_n) = \lambda_1; \quad \cdots ; \quad \varphi_n(\alpha_1 \cdots \alpha_n) = \lambda_n.$$

Thus only such coefficient systems $a_k = (-1)^k \sigma_k(\alpha)$ can become *singular* for which

$$H_1(\alpha) \cdot H_2(\alpha) \cdots H_n(\alpha) = 0;$$

for these, (11), instead of becoming a representation, passes into $0 = 0$. A special investigation is then needed as to whether among the singular $a_1 \cdots a_n$ so defined there are also such which correspond to equations g_Γ, and for which number fields.[189] In summary, we obtain the following.

Theorem. If the Lagrangian genus domain belonging to the group Γ possesses a minimal basis[190] *with coefficients from Ω, then for every number field Ω^* containing Ω, the totality of equations with prescribed group Γ with respect to Ω^* can be constructed rationally by the parameter representation (2) – possibly with the exception of those which are singular with respect to the parameter representation and can be given separately. The parameters are to run through all values from Ω^* which do not cause a reduction of the group. The construction is possible for every number field if the coefficient field Ω of the minimal basis is equal to the field $\mathfrak{R}$ of all rational numbers.*

Since, by Hilbert's irreducibility theorem, the parameter manifold is not lowered by excluding those values from Ω^* which cause a reduction of the group, it follows further: *from the existence of a minimal basis it follows that the manifold of equations with the group Γ with respect to Ω^* is equal to the manifold of the affect-free equations; the manifold is not lowered by the affect.*

§ 2.

Application to Special Equations.

We now prove the existence of the minimal basis for all groups occurring in equations of the third and fourth degrees; and for this we first show quite generally that, for equations of degree n, the problem can be reduced to one of $n-2$ indeterminates. The first reduction corresponds to the linear Tschirnhaus transformation for removing the second-highest term. We take as first basis function $\varphi_1(x) = \sigma_1(x)$; and we observe further that $\mathfrak{L}_\Gamma$ can be obtained rationally from all *homogeneous* forms in $x_1 \cdots x_n$ admitting Γ. Let $p(x_1 \cdots x_n)$ be such a form; then also

$$q(x) = p\left(x_1 - \frac{\sigma_1(x)}{n}; \cdots; x_n - \frac{\sigma_1(x)}{n}\right) = p\left(x_i - \frac{\sigma_1(x)}{n}\right) = p(y_i)$$

belongs to $\mathfrak{L}_\Gamma$, since it admits Γ, and moreover one has

$$p(x) \equiv p\left(x_i - \frac{\sigma_1(x)}{n}\right) \bmod \sigma_1(x)$$

or

$$p(x) = q(x) + \sigma_1(x) \cdot p_1(x)$$

with $p_1(x)$ belonging to $\mathfrak{L}_\Gamma$, and of smaller dimension than $p(x)$. Continuing the procedure thus shows that all homogeneous forms from $\mathfrak{L}_\Gamma$ – and consequently all rational functions from $\mathfrak{L}_\Gamma$ at all – can be rationally expressed by $\varphi_1(x) = \sigma_1(x)$ and by homogeneous forms

$$q(x) = p\left(x_i - \frac{\sigma_1(x)}{n}\right) = p(y_i).$$

Between these quantities y_i there is, however, the linear relation $\sum y_i = 0$; hence the $q(x)$ *depend only on* $(n-1)$ *arguments.*

The second reduction rests on the fact that the $q(x)$ are homogeneous forms of their arguments. We take as second basis function

$$\varphi_2(x) = \frac{\tau_3(x)}{\tau_2(x)}, \quad \text{where} \quad \tau_3(x) = \sigma_3\left(x_i - \frac{\sigma_1(x)}{n}\right) = \sigma_3(y_i), \quad \tau_2(x) = \sigma_2\left(x_i - \frac{\sigma_1(x)}{n}\right) = \sigma_2(y_i);$$

[189] For example, as F. Seidelmann has shown at the cited place, only for the alternating group of equations of the third and fourth degrees do such equations – corresponding to singular systems of values – occur, and indeed only for those number fields Ω^* which contain the field of third roots of unity. In each case a simple supplementary representation can be given.

[190] From the existence of one minimal basis there follows at once the existence of arbitrarily many, derivable from the original one by reversible rational transformations.

thus $\varphi_2(x)$ is a homogeneous fractional function of first degree of $(n-1)$ arguments, namely of the y_i with $\sum y_i = 0$. But by means of $\varphi_2(x)$ all $q(x)$ can be expressed rationally by the functions, likewise belonging to $\mathfrak{L}_\Gamma$,

$$r(x) = \frac{q(x)}{\varphi_2(x)^\lambda} = \frac{q(x) \cdot \tau_2(x)^\lambda}{\tau_3(x)^\lambda} = \frac{p(y_i) \cdot \sigma_2(y_i)^\lambda}{\sigma_3(y_i)^\lambda},$$

where λ denotes the degree of $q(x)$. Thus $r(x)$ is homogeneous of degree zero and therefore depends only on $(n-2)$ arguments, the ratios $y_1 : y_2 : \cdots : y_n$ with $\sum y_i = 0$. Hence $\mathfrak{L}_\Gamma$ can be rationally expressed by

$$\varphi_1(x) = \sigma_1(x); \quad \varphi_2(x) = \frac{\tau_3(x)}{\tau_2(x)}$$

and by the *field of all functions* $r(x)$ *from* $\mathfrak{L}_\Gamma$ *depending on these* $(n-2)$ *arguments*,[191] whereby the desired reduction is accomplished.

For equations of the third and fourth degrees, this gives in particular a reduction to function fields of one and two arguments respectively. But for these a minimal basis always exists by the theorems of Lüroth and Castelnuovo.[192] In fact, in the case of one indeterminate the minimal basis can be derived by rational operations, and therefore, specifically for $\mathfrak{L}_\Gamma$, contains only rational numerical coefficients. In the case of two indeterminates, according to Castelnuovo, one might still be dealing with a basis with coefficients from an algebraic number field Ω; the actual investigation (cf. F. Seidelmann, loc. cit.) shows that here too, for every $\mathfrak{L}_\Gamma$, one obtains a basis with rational integral coefficients. One sees this almost without computation already from the fact that for the four group such a rational-integral basis can be chosen, namely

$$\varphi_1(x) = \sigma_1(x); \quad \varphi_2(x) = \frac{vw}{u}; \quad \varphi_3(x) = \frac{u \cdot w}{v}; \quad \varphi_4(x) = \frac{u \cdot v}{w},$$

where

$$u = x_1 + x_2 - x_3 - x_4; \quad v = x_1 - x_2 + x_3 - x_4; \quad w = x_1 - x_2 - x_3 + x_4,$$

such that the alternating group passes into the cyclic group of the three basis functions $\varphi_2, \varphi_3, \varphi_4$, and each group of order eight into the symmetric group of two of these functions, whereby a reduction to the groups of equations of the third and second degrees is effected. For the cyclic group, however, a rational-integral basis is already known from Abel, although not formulated as such.[193] Thus it is proved: *The totality of equations of the third and fourth degrees with prescribed group – possibly with the exception of those singular with respect to the parameter representation – can be constructed rationally by parameter representation for any prescribed domain of rationality; their manifold is equal to the manifold of affect-free equations.*

The actual investigation shows (cf. F. Seidelmann, loc. cit.) that a supplementary representation is needed only for the alternating group and only with respect to domains of rationality containing the field of third roots of unity, whereas in all other cases the parameter representation supplies the totality without qualification.

[191] For an equation of degree n without second term,

$$f(x) = x^n + a_2 x^{n-2} + a_3 x^{n-3} + \cdots + a_n = 0,$$

this second reduction corresponds to the substitution, always possible for $a_2 \neq 0$, $a_3 \neq 0$,

$$y = \frac{x \cdot a_2}{a_3},$$

by which $f(x)$, up to a factor, passes into

$$g(y) = y^n + \frac{a_2^3}{a_3^2} y^{n-2} + \frac{a_2^3}{a_3^2} y^{n-3} + \frac{a_4 \cdot a_2^4}{a_3^4} y^{n-4} + \cdots + a_n \frac{a_2^n}{a_3^n} = 0;$$

that is, into an equation containing only $n-2$ parameters:

$$g(y) = y^n + c \cdot y^{n-2} + c \cdot y^{n-3} + c_4 y^{n-4} + \cdots + c_n = 0.$$

[192] J. Lüroth: Beweis eines Satzes über rationale Kurven, Math. Ann. 9 (1875); G. Castelnuovo: Sulla razionalità delle involuzioni piane, Math. Ann. 44 (1893). That for fields of more than two indeterminates a minimal basis need not exist was shown by F. Enriques by a counterexample, Rend. Acc. Linc., Vol. 21, 21 Jan. 1912.

[193] Cf. Weber Algebra (small edition), § 93, Formula (12).

It should also be noted that the minimal basis is known for all *Abelian groups*.[194] Here, however, the basis has coefficients from the cyclotomic field derived from the group characters. Thus, in this case, no new results can be obtained for the construction of equations with prescribed group.

Göttingen, July 1916.

[194] E. Fischer: Die Isomorphie der Invariantenkörper der endlichen Abelschen Gruppen linearer Transformationen. Gött. Nachr. 1915, and Zur Theorie der endlichen Abelschen Gruppen. Math. Ann. 77 (1915).

12. Invariants of Arbitrary Differential Expressions

Nachr. v. d. Ges. d. Wiss. zu Göttingen 1918, pp. 37–44

Presented in the session of 25 January 1918 by F. Klein.

By a differential expression I mean a function

$$f(x, dx) = f(x_1, \ldots, x_n;\, dx_1, \ldots, dx_n)$$

of the n variables $x_1, \ldots, x_n$ and of their differentials $dx_1, \ldots, dx_n$, which is assumed to be analytic in the two systems of n arguments, but not necessarily real. I now take as underlying group the group of all analytic transformations of the variables, to which corresponds the group of all linear transformations of the differentials:

$$x_i = x_i(y_1, \ldots, y_n); \qquad dx_i = \sum_k \frac{\partial x_i}{\partial y_k}\, dy_k; \qquad \delta x_i = \sum_k \frac{\partial x_i}{\partial y_k}\, \delta y_k, \tag{1}$$

under which $f(x, dx)$ is to pass over into $g(y, dy)$. By an invariant of $f(x, dx)$ I understand an invariant with respect to this group, and with respect to the group thereby induced for the higher differentials $d^2x, ddx, \ldots$ and for the derivatives of $f(x, dx)$; thus an (analytic) function J such that, for arbitrary f, and by means of (1), the following identity holds in the variables and in all occurring differentials:

$$\begin{aligned} &J\left(f, \frac{\partial f}{\partial dx} \cdots \frac{\partial^{\rho+\sigma} f}{\partial x^\rho \partial dx^\sigma} \cdots dx, \delta x, d^2x, \ldots\right) \\ &\qquad = J\left(g, \frac{\partial g}{\partial dy} \cdots \frac{\partial^{\rho+\sigma} g}{\partial y^\rho \partial dy^\sigma} \cdots dy, \delta y, d^2y, \ldots\right).\end{aligned}$$[195]

In precisely the same way one defines the invariant of a simultaneous system of differential expressions. If, in particular, such an invariant contains only first differentials $dx, \delta x$, and only derivatives with respect to these differentials, not with respect to the variables, then the occurrence of the variables in the functions and the linear transformation of the differentials do not enter into consideration; these special invariants are invariants with respect to the group of all linear transformations of the differentials $dx, \delta x$ with undetermined coefficients, and shall be called projective invariants of the simultaneous system.

For quadratic homogeneous differential forms Christoffel (Crelle 70) and Ricci (Math. Annalen 54) have now set up such a system of invariant formations that all invariants become projective invariants of this simultaneous system, whereby the questions concerning the totality of invariants and concerning equivalence are reduced to questions of linear invariant theory. This is what I should like to call the "reduction theorem." The complete system consists of countably infinitely many forms; but for invariants of any finite order ρ (that is, those which contain no higher than ρ-th derivatives with respect to the variables), it consists only of finitely many forms.

In what follows I show the validity of the "reduction theorem" for arbitrary differential expressions; and here again the issue is a complete system of countably infinitely many invariant formations, but only finitely many of them for each finite order. Whereas, however, Christoffel and Ricci base the generation of the invariants and the proof of the reduction theorem on eliminations that are difficult to survey, I generate the invariants directly by invariant differentiation and variation processes, the latter of which can simply be regarded as differentiation processes with respect to other parameters. The algorithm of these invariant differentiation processes was developed, following Lagrange (*Mécanique analytique*, part 2, section IV), by Riemann (Works XXII) and Lipschitz (Crelle 70, 72) for quadratic differential forms, without, however, advancing as far as

[195] $\frac{\partial^{\rho+\sigma} f}{\partial x^\rho \partial dx^\sigma}$ will throughout be used as an abbreviation for

$$\frac{\partial^{\rho+\sigma} f}{\partial x_{i_1} \cdots \partial x_{i_\rho} \partial dx_{k_1} \cdots \partial dx_{k_\sigma}},$$

where the indices denote any of the numbers 1 to n.

the reduction theorem. The auxiliary means for proving the reduction theorem are provided by the "normal coordinates" introduced by Riemann for quadratic forms (Works XIII); these transform the extremals of the associated variational problem emanating from a point into straight lines, but they exist only for homogeneous differential expressions $f(x, \varkappa \cdot dx) = \Phi(\varkappa) \cdot f(x, dx)$[196]. The inhomogeneous case, however, can be reduced to this one.

I. Generation of invariants.

A sequence of projective invariants of $f(x, dx)$, in general not terminating, is furnished by the polars. Because of the linearity of the transformation of the differentials, from $f(x, dx) = g(y, dy)$ it also follows that $f(x, dx + \lambda\delta x) = g(y, dy + \lambda\delta y)$; and from this, by expansion in λ, there arise the relations characteristic of invariance:

$$\begin{gathered} f(dx) = g(dy), \\ f_\delta = \sum \frac{\partial f(dx)}{\partial dx}\,\delta x = \sum \frac{\partial g(dy)}{\partial dy}\,\delta y, \\ \vdots \\ f_{\delta^\sigma} = \sum \frac{\partial^\sigma f(dx)}{\partial dx^\sigma}\,\delta x^\sigma = \sum \frac{\partial^\sigma g(dy)}{\partial dy^\sigma}\,\delta y^\sigma, \\ \vdots \end{gathered} \tag{2}$$

Each polar, by application of a variation process $\bar{\delta}$, gives rise to a non-terminating sequence of general invariants:

$$\begin{gathered} \bar{\delta}^\rho f(x, dx) = \bar{\delta}^\rho g(y, dy), \\ \vdots \\ \bar{\delta}^\rho f_{\delta^\sigma} = \bar{\delta}^\rho \sum \frac{\partial^\sigma f(dx)}{\partial dx^\sigma}\,\delta x^\sigma = \bar{\delta}^\rho \sum \frac{\partial^\sigma g(dy)}{\partial dy^\sigma}\,\delta y^\sigma, \\ \vdots \\ (\rho = 0, 1, 2, \ldots). \end{gathered} \tag{3}$$

By means of (1), the relations (2) and (3) allow the transformation laws for the derivatives of $f(x, dx)$ to be read off; this, however, will not be needed in what follows. From the invariants (3) one can now obtain by linear combination the "normal form of the ρ-th variation," which is found for $\rho = 1$ in Lagrange and for $\rho = 2$ in Riemann; it is characterized by the fact that the mixed differentials of order $\rho + 1$, namely $d^\rho\delta, d^{\rho-1}\delta^2, \ldots$, are eliminated. One obtains for it:

$$\begin{gathered} \Omega_1 = \delta f(dx) - df_\delta(dx), \\ \Omega_2 = \delta^2 f - \delta df_\delta + \frac{1}{2} d^2 f_{\delta^2}, \\ \vdots \\ \Omega_\rho = \delta^\rho f - \delta^{\rho-1} df_\delta + \frac{1}{2}\delta^{\rho-2} d^2 f_{\delta^2} + \cdots + \frac{(-1)^\rho}{\rho!}\, d^\rho f_{\delta^\rho}, \\ \vdots \end{gathered} \tag{4}$$

From the invariants (4), by generalizing an ansatz of Riemann, one can now arrive at invariants that contain only first differentials; such invariants shall be called "fundamental functions." Namely, if in Ω_1 the differentials are replaced by the differential quotients, then $\Omega_1 = 0$ (identically in δx) gives exactly the Lagrange equations of the variational problem belonging to

$$f\left(\frac{dx}{dt}\right).$$

I now impose the invariant condition that the direction $(d + \lambda\delta)$ satisfy these Lagrange equations:

$$\Omega_1(d + \lambda\delta) = \bar{\delta} f(dx + \lambda\delta x) - (d + \lambda\delta) f_{\bar{\delta}}(dx + \lambda\delta x) = 0 \quad \text{(identically in } \delta x\text{)}, \tag{5}$$

[196]It is easy to see that $\Phi(\varkappa) = \varkappa^c$, where c denotes an arbitrary real or complex quantity.

from which, by the substitution $\bar{\delta} = d + \lambda\delta$, there still follows for homogeneous $f(dx)$:

$$(d+\lambda\delta)f(dx+\lambda\delta x) = 0, \tag{6}$$

except in the case of homogeneity of first order. In the latter case (6) is to be added as a new condition, since then the system of equations (5) represents only $n-1$ independent conditions. If in (5), respectively in (5) and (6), one sets the coefficients of the zeroth, first, and second powers of λ equal to zero, one thus obtains three invariant systems of equations $\varphi = 0$, $\psi = 0$, $\chi = 0$ (identically in δx), by means of which $d^2x, ddx, \delta^2 x$ can be expressed through first differentials. The further invariant systems of equations

$$\begin{gathered} d^\sigma\varphi = 0, \qquad d^\sigma\psi = 0, \qquad d^\sigma\chi = 0, \qquad d^{\sigma-1}\delta\chi = 0, \\ d^{\sigma-2}\delta^2\chi = 0, \ldots, \delta^\sigma\chi = 0 \qquad (\sigma = 0, 1, 2, \ldots) \end{gathered} \tag{7}$$

then suffice precisely to express all higher differentials by first ones.[197] The functions (4), combined with the invariant system of equations (7), therefore lead to fundamental functions $[\Omega_\rho]$ with only first differentials; moreover $[\Omega_1]$ vanishes identically.

Similarly, from the differential $\delta h(dx, \delta x)$ of a fundamental function h, one can by means of (7) again obtain a fundamental function, which is to be called the "covariant derivative" $[h^{(1)}]$ of h; in general let $[h^{(\nu)}]$ denote the covariant derivative of $[h^{(\nu-1)}]$. These two processes thus lead to a doubly infinite sequence of fundamental functions:

$$\begin{array}{lllll} [\Omega_2], & [\Omega_2^{(1)}], & [\Omega_2^{(2)}], & \ldots, & [\Omega_2^{(\nu)}], \quad \ldots \\ \vdots & & & & \vdots \\ [\Omega_\rho], & [\Omega_\rho^{(1)}], & \ldots, & \ldots, & [\Omega_\rho^{(\nu)}], \quad \ldots \\ \vdots & \vdots & & & \vdots \end{array} \tag{8}$$

It further turns out that the left-hand sides of the equations which express the higher differentials through first ones are cogredient to the first differentials, that is, they undergo the same linear transformations. If, therefore, instead of the higher differentials, one introduces these left-hand sides $p, q, r, \ldots$ as arguments of the invariant, then, apart from the derivatives, it depends only on quantities cogredient to one another. The formation of the functions (8) indicated above amounts simply to the introduction of these new quantities; every invariant thereby passes over into a sum of invariants, and from this sum one selects the one which contains precisely $dx, \delta x$, but not $p, q, r, \ldots$.

In what follows it is shown that under the homogeneity assumption

$$f(\varkappa \cdot dx) = \Phi(\varkappa) \cdot f(dx)$$

the fundamental functions (8) exhaust the complete system sought.

II. Normal coordinates, equivalence, and the reduction theorem.

I arrive at the normal coordinates by integrating the variational problem belonging to $f\left(\frac{dx}{dt}\right)$:

$$\begin{gathered} \Omega_1 = \delta f\left(\frac{dx}{dt}\right) - \frac{d}{dt} f_\delta\left(\frac{dx}{dt}\right) = 0 \quad \text{(identically in } \delta x\text{)}, \\ \frac{d}{dt} f\left(\frac{dx}{dt}\right) = 0 \quad \text{(as a consequence or additional condition).} \end{gathered} \tag{9}$$

Because of the assumed homogeneity of $f(dx)$, this system of equations goes over into itself under the parameter substitution $t = \varkappa \cdot \tau$. Hence, if by means of (9) one expresses $\frac{d^\rho x_i}{dt^\rho}$ as a function $\varphi_\rho^{(i)}\left(\frac{dx}{dt}\right)$ of the first derivatives, then $\varphi_\rho^{(i)}$ is homogeneous of order ρ:

$$\varphi_\rho^{(i)}\left(\varkappa \cdot \frac{dx}{dt}\right) = \varkappa^\rho \varphi_\rho^{(i)}\left(\frac{dx}{dt}\right). \tag{10}$$

If, in integrating (9), one now prescribes for $t = t_0$ the initial values

$$x = (x)_0, \qquad \frac{dx}{dt} = \left(\frac{dx}{dt}\right)_0$$

[197] Only the linear form $\sum A_i\, dx_i$ requires special treatment, since here the system of equations (5) contains no second differentials.

and sets

$$u_i = (t - t_0)\left(\frac{dx_i}{dt}\right)_0, \tag{11}$$

where, by virtue of $f\left(\frac{dx}{dt}\right) = \text{const.}$, the parameter $(t - t_0)$ becomes proportional to the "length of the extremal," then the expansion in powers of $(t - t_0)$, by (10), gives

$$x_i - (x_i)_0 = u_i + \frac{1}{2}\varphi_2^{(i)}(u) + \cdots + \frac{1}{\rho!}\varphi_\rho^{(i)}(u) + \cdots. \tag{12}$$

This expansion can be regarded as a transformation of variables $x_i = x_i(u)$, where the quantities u are precisely the Riemann normal coordinates. Thus, by (11) and (12), these coordinates transform the extremals passing through $(x)_0$ into straight lines. Moreover, to an arbitrary transformation of variables $x = x(y)$ there corresponds, by (11), a linear transformation $u = u(v)$ of the associated normal coordinates, where the coefficients depend only on the arbitrary system of values $(x)_0$; hence the differentials $du, \delta u$ are subject to the same transformation as the u themselves.

By means of (12), let $f(x, dx)$ pass over into $F(u, du)$, or, expanded in powers of the u,

$$F(u, du) = F_0(u, du) + F_1(u, du) + \cdots + F_\rho(u, du) + \cdots, \tag{13}$$

where $F_\rho(u, du)$ denotes a homogeneous form of dimension ρ in u. Because of the linearity of the transformation $u = u(v)$, the identity $F(u, du) = G(v, dv)$ also carries the individual homogeneous components into the corresponding components:

$$F_\rho(u, du) = G_\rho(v, dv). \tag{14}$$

If one therefore takes any fixed constant system of values for $(x)_0$, then (13) and (14) show that the equivalence of $f(dx)$ with $g(dy)$ is equivalent to the equivalence of the associated systems of functions $F_\rho(u, du)$ with respect to linear transformation. Furthermore, the $F_\rho(u, du)$, or the corresponding $F_\rho(\delta u, du)$, represent invariants of $f(dx)$, taken at the point $x = (x)_0$; indeed, they form a complete system of fundamental functions for this point $x = (x)_0$. Namely,

$$F_\rho(\delta u, du) = \frac{1}{\rho!}\sum \frac{\partial F_\rho}{\partial \delta u^\rho}\,\delta u^\rho; \qquad \frac{\partial F_\rho}{\partial \delta u^\rho} = \left(\frac{\partial F(du)}{\partial u^\rho}\right)_0,$$

and this latter derivative can, by means of (12), be expressed in just the same way through derivatives of $f(dx)$, taken for $x = (x)_0$, as the correspondingly formed derivative of $G(dv)$ can be expressed through derivatives of $g(dy)$. Because

$$\left(\frac{\partial^{\rho+\sigma} F(du)}{\partial u^\rho \partial du^\sigma}\right)_0 = \frac{\partial^{\rho+\sigma} F_\rho(\delta u, du)}{\partial \delta u^\rho \partial du^\sigma},$$

every invariant of $f(dx) = F(du)$, taken at the point $x = (x)_0$, also becomes a projective invariant of the $F_\rho(\delta u, du)$, once the higher differentials have merely been replaced by the $p, q, r, \ldots$ cogredient to $dx, \delta x$. But now $(x)_0$ can be regarded as a parameter; to every invariant at the point $x = (x)_0$ there corresponds an invariant of $f(dx)$, if only $(x)_0$ is replaced again by x. Thus, if the $F_\rho(\delta u, du)$ become invariants $\Psi_\rho(\delta x, dx)$, taken for $x = (x)_0$ — for $x = (x)_0$, $du, \delta u$ pass over into $dx, \delta x$ —, then the Ψ_ρ themselves form the fundamental functions of a complete system. It remains only to express this system of fundamental functions by explicit invariants of $f(dx)$, namely by the functions (8). That this is possible is shown by their generation through differentiation processes. For the terms of highest order these processes pass into simple polarization processes, and a transformation analogous to the Clebsch–Gordan series expansion, combined with induction on account of the neglected terms, proves the assertion. If one is dealing with invariants of a simultaneous system, then the covariant derivatives of the remaining differential expressions must still be added to the fundamental functions (8), as is shown by introducing into these expressions the normal coordinates belonging to $f(dx)$.

The equivalence of $f(dx)$ with $g(dy)$, which had been reduced to (14), is therefore also equivalent to the equivalence of the Ψ_ρ, or equally of the functions (8), with respect to linear transformation of the differentials, without any special integrability conditions being required; this follows from the possibility of reduction to (14). Thus the reduction theorem is proved in all its parts. In particular, the identical vanishing of the sequence of functions $[\Omega_2], [\Omega_3], \ldots, [\Omega_\rho], \ldots$ is necessary and sufficient for $f(dx)$ to be transformable into an expression with constant coefficients.

Finally, if instead of the group of all analytic transformations one takes a subgroup as the basis, then the group of the corresponding linear transformations of the u also passes over into a subgroup of the projective group; the invariants of $f(dx)$ again become invariants of the functions (8) with respect to linear transformation, but now with respect to this subgroup. Thus, by homogenizing, the case of nonhomogeneous functions $f(dx)$ can be reduced to the affine group; the complete system can here be derived from the functions (8) formed for one additional variable.

From the reduction theorem one obtains the special case of quadratic forms, and more generally of forms of p-th dimension, from the fact that here the system of functions terminates with $[\Omega_2]$, more generally with $[\Omega_p]$ and its covariant derivatives, since all later fundamental functions become projective invariants of these. This explains the distinguished position of the Riemannian curvature form $[\Omega_2]$ for quadratic forms. The question of the equivalence of quadratic forms, respectively of forms of p-th dimension, comes down precisely to the equivalence of $[\Omega_2]$, respectively $[\Omega_2] \dots [\Omega_p]$, and of their covariant derivatives under linear transformation; and the identical vanishing of $[\Omega_2]$, respectively of $[\Omega_2] \dots [\Omega_p]$, is necessary and sufficient for transformation into forms with constant coefficients, which, for quadratic forms, states one of the best-known results.

A more detailed presentation is to appear shortly in the *Mathematische Annalen*.

13. Invariant Variational Problems

Nachr. v. d. Ges. d. Wiss. zu Göttingen 1918, pp. 235–257

Presented by F. Klein in the session of 26 July 1918.[198]

The subject is variational problems that admit a continuous group (in Lie's sense); the consequences which arise from this for the corresponding differential equations find their most general expression in the theorems formulated in §1 and proved in the following sections. For these differential equations arising from variational problems, much more precise assertions can be made than for arbitrary differential equations admitting a group, which form the subject of Lie's investigations. What follows therefore rests on a combination of the methods of the formal calculus of variations with those of Lie group theory. For special groups and variational problems this combination of methods is not new; I mention Hamel and Herglotz for special finite groups, Lorentz and his students (for example Fokker), Weyl and Klein for special infinite groups.[199] In particular, the second note of Klein and the present developments have mutually influenced one another; for this I may refer to the closing remarks of Klein's note.

§1. Preliminaries and formulation of the theorems

All functions occurring below are to be assumed analytic, or at least continuous and continuously differentiable a finite number of times, and single-valued in the domain under consideration.

By a "transformation group" one understands, as usual, a system of transformations such that for each transformation there exists an inverse transformation contained in the system, and such that the composition of any two transformations of the system again belongs to the system. The group is called a finite continuous $\mathfrak{G}_\rho$ when its transformations are contained in a most general transformation depending analytically on ρ essential parameters ε (that is, the ρ parameters are not to be representable as ρ functions of fewer parameters). Correspondingly, by an infinite continuous $\mathfrak{G}_{\infty\rho}$ one understands a group whose most general transformations depend on ρ essential arbitrary functions $p(x)$ and their derivatives, analytically or at least continuously and with a finite number of continuous derivatives. An intermediate case between the two is the group depending on infinitely many parameters but not on arbitrary functions. Finally, a mixed group denotes one that depends both on arbitrary functions and on parameters.[200]

[198] The final version of the manuscript was submitted only at the end of September.

[199] Hamel: *Math. Ann.* vol. 59 and *Zeitschrift f. Math. u. Phys.* vol. 50. Herglotz: *Ann. d. Phys.* (4) vol. 36, especially §9, p. 511. Fokker, Verslag d. Amsterdamer Akad., 27 Jan. 1917. For the further literature compare the second note of Klein: Göttinger Nachrichten, 19 July 1918. In a paper by Kneser that has just appeared (*Math. Zeitschrift* vol. 2), the issue is the construction of invariants by a similar method.

[200] Lie defines, in *Grundlagen für die Theorie der unendlichen kontinuierlichen Transformationsgruppen* (Reports of the Royal Saxon Society of Sciences, 1891), the infinite continuous group as a transformation group whose transformations are given by the most general solutions of a system of partial differential equations, once these solutions do not depend only on a finite number of parameters. This gives one of the types indicated above, different from the finite group; conversely, however, the limiting case of infinitely many parameters need not necessarily satisfy a system of differential equations.

Let $x_1, \dots, x_n$ be independent variables, and let $u_1(x), \dots, u_\mu(x)$ be functions depending on them. If one subjects the x's and u's to the transformations of a group, then, because the transformations are assumed invertible, among the transformed quantities there must again be exactly n independent ones, denoted $y_1, \dots, y_n$; the remaining quantities depending on these are denoted $v_1(y), \dots, v_\mu(y)$. The transformations may also contain the derivatives of the u's with respect to the x's, thus $\partial u/\partial x$, $\partial^2 u/\partial x^2, \dots$.[201] A function is called an invariant of the group if a relation holds:

$$P\left(x, u, \frac{\partial u}{\partial x}, \frac{\partial^2 u}{\partial x^2}, \dots\right) = P\left(y, v, \frac{\partial v}{\partial y}, \frac{\partial^2 v}{\partial y^2}, \dots\right).$$

In particular, an integral I is therefore an invariant of the group if a relation holds:

$$\begin{aligned} I &= \int \cdots \int f\left(x, u, \frac{\partial u}{\partial x}, \frac{\partial^2 u}{\partial x^2}, \dots\right) \mathrm{d}x \\ &= \int \cdots \int f\left(y, v, \frac{\partial v}{\partial y}, \frac{\partial^2 v}{\partial y^2}, \dots\right) \mathrm{d}y, \end{aligned} \tag{1}$$

where the integration is over an arbitrary real x-domain and the corresponding y-domain.[202]

On the other hand, for an arbitrary integral I, not necessarily invariant, I form the first variation δI, and transform it by partial integration according to the rules of the calculus of variations. As is well known, once δu, together with all its derivatives occurring at the boundary, is taken to vanish at the boundary, but is otherwise arbitrary, one obtains:

$$\delta I = \int \cdots \int \delta f \, \mathrm{d}x = \int \cdots \int \left(\sum_i \psi_i\left(x, u, \frac{\partial u}{\partial x}, \dots\right) \delta u_i \right) \mathrm{d}x, \tag{2}$$

where the ψ's denote the Lagrange expressions, that is, the left-hand sides of the Lagrange equations of the corresponding variational problem $\delta I = 0$. This integral relation corresponds to an identity free of integrals in δu and its derivatives, which arises when one writes the boundary terms as well. As partial integration shows, these boundary terms are integrals over divergences, that is, over expressions

$$\operatorname{Div} A = \frac{\partial A_1}{\partial x_1} + \cdots + \frac{\partial A_n}{\partial x_n},$$

where A is linear in δu and its derivatives. Thus one obtains:

$$\sum_i \psi_i \, \delta u_i = \delta f + \operatorname{Div} A. \tag{3}$$

In particular, if f contains only first derivatives of the u's, then, in the case of the simple integral, the identity (3) is identical with Herr's so-called "Lagrangian central equation":

$$\sum_i \psi_i \, \delta u_i = \delta f - \frac{\mathrm{d}}{\mathrm{d}x}\left(\sum_i \frac{\partial f}{\partial u_i'} \delta u_i \right), \qquad \left(u_i' = \frac{\mathrm{d}u_i}{\mathrm{d}x} \right), \tag{4}$$

whereas for the n-fold integral (3) becomes:

$$\sum_i \psi_i \, \delta u_i = \delta f - \frac{\partial}{\partial x_1}\left(\sum_i \frac{\partial f}{\partial\left(\frac{\partial u_i}{\partial x_1}\right)} \delta u_i \right) - \cdots - \frac{\partial}{\partial x_n}\left(\sum_i \frac{\partial f}{\partial\left(\frac{\partial u_i}{\partial x_n}\right)} \delta u_i \right). \tag{5}$$

For the simple integral and κ derivatives of the u's, (3) is given by:

$$\begin{aligned} \sum_i \psi_i \, \delta u_i = \delta f &- \frac{\mathrm{d}}{\mathrm{d}x}\left\{ \sum_i \left[\binom{1}{1} \frac{\partial f}{\partial u_i^{(1)}} \delta u_i + \binom{2}{1} \frac{\partial f}{\partial u_i^{(2)}} \delta u_i^{(1)} + \cdots + \binom{\kappa}{1} \frac{\partial f}{\partial u_i^{(\kappa)}} \delta u_i^{(\kappa-1)} \right] \right\} \\ &+ \frac{\mathrm{d}^2}{\mathrm{d}x^2}\left\{ \sum_i \left[\binom{2}{2} \frac{\partial f}{\partial u_i^{(2)}} \delta u_i + \binom{3}{2} \frac{\partial f}{\partial u_i^{(3)}} \delta u_i^{(1)} + \cdots + \binom{\kappa}{2} \frac{\partial f}{\partial u_i^{(\kappa)}} \delta u_i^{(\kappa-2)} \right] \right\} \\ &+ \cdots + (-1)^\kappa \frac{\mathrm{d}^\kappa}{\mathrm{d}x^\kappa}\left\{ \sum_i \binom{\kappa}{\kappa} \frac{\partial f}{\partial u_i^{(\kappa)}} \delta u_i \right\}. \end{aligned} \tag{6}$$

[201] I suppress the indices, as far as possible also in summations; thus, for example, $\partial^2 u/\partial x^2$ denotes $\partial^2 u_\nu/\partial x_i \partial x_k$, and so on.

[202] I write $\mathrm{d}x, \mathrm{d}y$, for short, for $\mathrm{d}x_1 \cdots \mathrm{d}x_n, \mathrm{d}y_1 \cdots \mathrm{d}y_n$. All arguments $x, u, \varepsilon, p(x)$ occurring in the transformations are to be taken as real, whereas the coefficients may be complex. But since the final results are identities in the x, u, the parameters, and the arbitrary functions, they also hold for complex values as soon as all occurring functions are assumed analytic. Moreover, a large part of the results can be founded without integrals, so that the restriction to the real is not necessary even for the justification. By contrast, the considerations at the end of §2 and the beginning of §5 do not seem to be feasible without integrals.

A corresponding identity holds for the n-fold integral; in particular A contains δu up to the $(\kappa - 1)$-st derivative. That (4), (5), and (6) do in fact define the Lagrange expressions ψ_i follows from the fact that, in the combinations on the right-hand sides, all higher derivatives of δu are eliminated, whereas the relation (2) is fulfilled, to which partial integration leads uniquely.

In what follows the issue is the two theorems:

I. If the integral I is invariant with respect to a $\mathfrak{G}_\rho$, then ρ linearly independent combinations of the Lagrange expressions become divergences; conversely, from this follows the invariance of I with respect to a $\mathfrak{G}_\rho$. The theorem also holds in the limiting case of infinitely many parameters.

II. If the integral I is invariant with respect to a $\mathfrak{G}_{\infty\rho}$ in which the arbitrary functions occur up to the σ-th derivative, then there exist ρ identical relations between the Lagrange expressions and their derivatives up to order σ; here too the converse holds.[203]

For mixed groups, the assertions of both theorems apply; hence there occur both dependencies and divergence relations independent of them.

Passing from these identities to the corresponding variational problem, and thus setting $\psi = 0$,[204] Theorem I says in the one-dimensional case — where the divergence becomes a total differential — that ρ first integrals exist, although nonlinear dependencies may occur among them;[205] in the multidimensional case one obtains the divergence equations that recently have often been called "conservation laws". Theorem II says that ρ of the Lagrange equations are consequences of the remaining ones.

The simplest example of Theorem II — without the converse — is the Weierstrass parameter representation; here, under homogeneity of first order, the integral is known to be invariant if one replaces the independent variable x by an arbitrary function of x, while leaving u unchanged ($y = p(x);\ v_i(y) = u_i(x)$). Thus one arbitrary function occurs, but without derivatives; and to this corresponds the well-known linear relation between the Lagrange expressions themselves:

$$\sum_i \psi_i \frac{\mathrm{d}u_i}{\mathrm{d}x} = 0.$$

A further example is offered by the physicists' "general theory of relativity". Here one is dealing with the group of all transformations of the x's, $y_i = p_i(x)$, while the u's (denoted $g_{\mu\nu}$ and q) are subjected to the transformations thereby induced for the coefficients of a quadratic and a linear differential form, transformations that contain the first derivatives of the arbitrary functions $p(x)$. To this correspond the known n dependencies between the Lagrange expressions and their first derivatives.[206]

If, in particular, one specializes the group by admitting no derivatives of $u(x)$ in the transformations, and in addition by letting the transformed independent quantities depend only on the x's and not on the u's, then, as will be shown in §5, the invariance of I implies the relative invariance of $\sum_i \psi_i \delta u_i$,[207] and likewise of the divergences occurring in Theorem I, as soon as the parameters are subjected to suitable transformations. It follows further that the first integrals mentioned above also admit the group. For Theorem II one similarly obtains the relative invariance of the left-hand sides of the dependencies, combined by means of the arbitrary functions; and, as a consequence, also a function whose divergence vanishes identically and which admits the group — the function that in the physicists' theory of relativity mediates the connection between dependencies and energy theorem.[208] Finally, in group-theoretic form, Theorem II gives the proof of a related assertion of Hilbert concerning the failure of proper energy theorems in "general relativity". With these supplementary remarks Theorem I contains all the theorems on first integrals known in mechanics and so forth, whereas Theorem II can be described as the greatest possible group-theoretic generalization of the "general theory of relativity".

§2. Divergence relations and dependencies

Let $\mathfrak{G}$ be a continuous group, finite or infinite. It can always be arranged that the identical transformation corresponds to the value zero of the parameters ε, respectively of the arbitrary functions $p(x)$.[209] Thus the

[203] For certain trivial exceptional cases compare §2, second note.

[204] Somewhat more generally one may also set $\psi_i = T_i$; compare §3, first note.

[205] Compare the end of §3.

[206] Compare, for example, Klein's presentation.

[207] That is, $\sum_i \psi_i \delta u_i$ acquires a factor under transformations.

[208] Compare the second note of Klein.

[209] Compare, for example, Lie, *Grundlagen*, p. 331. If arbitrary functions are involved, then the special values a^σ of the parameters are to be replaced by fixed functions $p^\sigma,\ \partial p^\sigma/\partial x, \ldots$; and correspondingly the values $a^\sigma + \varepsilon$ by $p^\sigma + p(x),\ \partial p^\sigma/\partial x + \partial p/\partial x$, and so on.

most general transformation will be of the form

$$y_i = A_i\Big(x, u, \frac{\partial u}{\partial x}, \dots\Big) = x_i + \Delta x_i + \cdots, \qquad v_i(y) = B_i\Big(x, u, \frac{\partial u}{\partial x}, \dots\Big) = u_i + \Delta u_i + \cdots,$$

where $\Delta x_i, \Delta u_i$ denote the terms of lowest dimension in ε, respectively in $p(x)$ and its derivatives; they are to be assumed linear in these quantities. As will be shown later, this is no restriction of generality.

Now let the integral I be an invariant with respect to $\mathfrak{G}$, so that relation (1) is fulfilled. In particular, I is then also invariant with respect to the infinitesimal transformation contained in $\mathfrak{G}$:

$$y_i = x_i + \Delta x_i, \qquad v_i(y) = u_i + \Delta u_i.$$

For this, relation (1) becomes:

$$\begin{aligned} 0 = \Delta I = \int \cdots \int f\Big(y, v(y), \frac{\partial v}{\partial y}, \dots\Big)\, \mathrm{d}y \\ - \int \cdots \int f\Big(x, u(x), \frac{\partial u}{\partial x}, \dots\Big)\, \mathrm{d}x. \end{aligned} \tag{7}$$

The first integral is to be extended over the $x + \Delta x$-domain corresponding to the x-domain. But this integration can also be transformed into an integration over the x-domain by means of the transformation valid for infinitesimal Δx:

$$\begin{aligned} \int \cdots \int f\Big(y, v(y), \frac{\partial v}{\partial y}, \dots\Big)\, \mathrm{d}y = \int \cdots \int f\Big(x, v(x), \frac{\partial v}{\partial x}, \dots\Big)\, \mathrm{d}x \\ + \int \cdots \int \mathrm{Div}(f \cdot \Delta x)\, \mathrm{d}x. \end{aligned} \tag{8}$$

If one therefore introduces, in place of the infinitesimal transformation Δu, the variation

$$\bar{\delta} u_i = v_i(x) - u_i(x) = \Delta u_i - \sum_\lambda \frac{\partial u_i}{\partial x_\lambda} \Delta x_\lambda, \tag{9}$$

then (7) and (8) become:

$$0 = \int \cdots \int \{\bar{\delta} f + \mathrm{Div}(f \cdot \Delta x)\} \mathrm{d}x. \tag{10}$$

Since the relation (10) is fulfilled when integrating over every arbitrary domain, the integrand must vanish identically. Thus the Lie differential equations for the invariance of I pass over into the relation

$$\bar{\delta} f + \mathrm{Div}(f \cdot \Delta x) = 0. \tag{11}$$

If one expresses $\bar{\delta} f$ here, by (3), in terms of the Lagrange expressions, one obtains

$$\sum_i \psi_i \bar{\delta} u_i = \mathrm{Div}\, B, \qquad (B = A - f \cdot \Delta x), \tag{12}$$

and this relation therefore represents, for every invariant integral I, an identity in all occurring arguments; it is the desired form of the Lie differential equations for I.[210]

First assume $\mathfrak{G}$ to be a finite continuous group $\mathfrak{G}_\rho$. Since, by assumption, Δu and Δx are linear in the parameters $\varepsilon_1, \dots, \varepsilon_\rho$, by (9) the same holds for $\bar{\delta} u$ and its derivatives; hence A and B are linear in the ε. I therefore put

$$B = B^{(1)} \varepsilon_1 + \cdots + B^{(\rho)} \varepsilon_\rho, \qquad \bar{\delta} u = \bar{\delta} u^{(1)} \varepsilon_1 + \cdots + \bar{\delta} u^{(\rho)} \varepsilon_\rho,$$

where $\bar{\delta} u^{(1)}, \dots$ are functions of $x, u, \partial u/\partial x, \dots$. From (12) follow the desired divergence relations:

$$\sum_i \psi_i \bar{\delta} u_i^{(1)} = \mathrm{Div}\, B^{(1)}, \dots, \qquad \sum_i \psi_i \bar{\delta} u_i^{(\rho)} = \mathrm{Div}\, B^{(\rho)}. \tag{13}$$

Thus ρ linearly independent combinations of the Lagrange expressions become divergences. The linear independence follows from the fact that, by (9), $\bar{\delta} u = 0, \Delta x = 0$ would imply $\Delta u = 0, \Delta x = 0$, and hence

[210](12) becomes $0 = 0$ for the trivial case — which can occur only if $\Delta x, \Delta u$ also depend on derivatives of u — when $\mathrm{Div}(f \cdot \Delta x) = 0,\ \bar{\delta} u = 0$. These infinitesimal transformations must therefore always be split off from the groups, and in the formulation of the theorems only the number of the remaining parameters or arbitrary functions is to be counted. Whether the remaining infinitesimal transformations still always form a group must be left open.

a dependency between the infinitesimal transformations. But by hypothesis no such dependency holds for any parameter value; otherwise the $\mathfrak{G}_\rho$ arising again by integration from the infinitesimal transformations would depend on fewer than ρ essential parameters. The further possibility $\bar{\delta} u = 0,\ \mathrm{Div}(f\Delta x) = 0$ has been excluded. These conclusions remain valid in the limiting case of infinitely many parameters.

Now let $\mathfrak{G}$ be an infinite continuous group $\mathfrak{G}_{\infty\rho}$. Then again $\bar{\delta} u$ and its derivatives, hence also B, are linear in the arbitrary functions $p(x)$ and their derivatives.[211] Substitution of the values of $\bar{\delta} u$, still independently of (12), gives:

$$\sum_i \psi_i \bar{\delta} u_i = \sum_{\lambda,i} \psi_i \left\{ a_i^{(\lambda)}(x,u,\ldots) p^{(\lambda)}(x) + b_i^{(\lambda)}(x,u,\ldots) \frac{\partial p^{(\lambda)}}{\partial x} + \cdots + c_i^{(\lambda)}(x,u,\ldots) \frac{\partial^\sigma p^{(\lambda)}}{\partial x^\sigma} \right\}.$$

Now, analogously to the formula for partial integration, by means of the identity

$$\varphi(x,u,\ldots) \frac{\partial^\tau p(x)}{\partial x^\tau} = (-1)^\tau \frac{\partial^\tau \varphi}{\partial x^\tau}\, p(x) \quad \text{mod divergences},$$

the derivatives of p can be replaced by p itself and by divergences that are linear in p and its derivatives. Thus:

$$\sum_i \psi_i \bar{\delta} u_i = \sum_\lambda \left\{ a_i^{(\lambda)} \psi_i - \frac{\partial}{\partial x}(b_i^{(\lambda)} \psi_i) + \cdots + (-1)^\sigma \frac{\partial^\sigma}{\partial x^\sigma}(c_i^{(\lambda)} \psi_i) \right\} p^{(\lambda)} + \mathrm{Div}\, \Gamma, \tag{14}$$

with summation over i in the brace. In combination with (12) this gives:

$$\sum_\lambda \left\{ a_i^{(\lambda)} \psi_i - \frac{\partial}{\partial x}(b_i^{(\lambda)} \psi_i) + \cdots + (-1)^\sigma \frac{\partial^\sigma}{\partial x^\sigma}(c_i^{(\lambda)} \psi_i) \right\} p^{(\lambda)} = \mathrm{Div}(B - \Gamma). \tag{15}$$

I now form the n-fold integral over (15), extended over an arbitrary domain, and choose the $p(x)$'s so that they, together with all derivatives occurring in $B - \Gamma$, vanish at the boundary. Since the integral over a divergence reduces to a boundary integral, the integral over the left-hand side of (15) also vanishes for arbitrary $p(x)$'s that vanish at the boundary with sufficiently many derivatives. From the usual arguments it follows that the integrand vanishes for every $p(x)$, giving the ρ relations

$$\sum_i \left\{ a_i^{(\lambda)} \psi_i - \frac{\partial}{\partial x}(b_i^{(\lambda)} \psi_i) + \cdots + (-1)^\sigma \frac{\partial^\sigma}{\partial x^\sigma}(c_i^{(\lambda)} \psi_i) \right\} = 0, \qquad (\lambda = 1, 2, \ldots, \rho). \tag{16}$$

These are the desired dependencies between the Lagrange expressions and their derivatives when I is invariant with respect to $\mathfrak{G}_{\infty\rho}$. The linear independence follows as above, since the converse leads back to (12); and since one can again conclude from the infinitesimal transformations to the finite ones, as is carried out more fully in §4. Accordingly, for a $\mathfrak{G}_{\infty\rho}$, ρ arbitrary transformations already occur in the infinitesimal transformations. From (15) and (16) it follows further that $\mathrm{Div}(B - \Gamma) = 0$.

If, corresponding to a "mixed group", one takes Δx and Δu to be linear in the ε's and the $p(x)$'s, then by setting once the $p(x)$'s and once the ε's equal to zero, one sees that both the divergence relations (13) and the dependencies (16) exist.

§3. The converse in the case of the finite group

To prove the converse, the preceding considerations must first be run through, in essence, in reverse order. From the validity of (13), multiplying by the ε's and adding gives the validity of (12); and by means of the identity (3) one obtains a relation

$$\bar{\delta} f + \mathrm{Div}(A - B) = 0.$$

Thus, if one sets $\Delta x = \frac{1}{f}(A - B)$, one thereby arrives at (11); by integration this finally gives (7), $\Delta I = 0$, hence the invariance of I with respect to the infinitesimal transformation determined by $\Delta x, \Delta u$, where the Δu's are determined by (9) from Δx and $\bar{\delta} u$, and $\Delta x, \Delta u$ become linear in the parameters. But $\Delta I = 0$ implies, in the known way, the invariance of I with respect to the finite transformations arising by integration of the simultaneous system:

$$\frac{\mathrm{d}x_i}{\mathrm{d}t} = \Delta x_i, \qquad \frac{\mathrm{d}u_i}{\mathrm{d}t} = \Delta u_i, \qquad \left\{ \begin{array}{c} x_i = y_i, \\ u_i = v_i \end{array} \right\} \quad \text{for } t = 0. \tag{17}$$

[211] That it is no restriction to assume the p's independent of $u, \partial u/\partial x, \ldots$ is shown by the converse.

These finite transformations contain ρ parameters $a_1, \ldots, a_\rho$, namely the combinations $t\varepsilon_1, \ldots, t\varepsilon_\rho$. From the assumption that there are to be ρ, and only ρ, linearly independent divergence relations (13), it follows further that the finite transformations, as soon as they do not contain the derivatives $\partial u/\partial x$, always form a group. In the contrary case, at least one infinitesimal transformation arising by Lie's bracket process would not be a linear combination of the remaining ρ; and since I also admits this transformation, there would be more than ρ linearly independent divergence relations. Or else this infinitesimal transformation would be of the special form $\bar{\delta}u = 0,\ \mathrm{Div}(f\Delta x) = 0$. But then Δx or Δu would depend on derivatives, contrary to the assumption. Whether this case can occur when derivatives appear in Δx or Δu must be left open; then all functions Δx for which $\mathrm{Div}(f\Delta x) = 0$ must still be added to the Δx determined above in order to obtain the group property again, but by agreement the parameters thereby added are not to be counted. This proves the converse.

It also follows from this converse that Δx and Δu may indeed be assumed linear in the parameters. If Δu and Δx were forms of higher degree in ε, then, because of the linear independence of the power products of the ε's, entirely corresponding relations (13), only in larger number, would follow; by the converse, these imply invariance of I with respect to a group whose infinitesimal transformations contain the parameters linearly. If this group is to contain exactly ρ parameters, then there must be linear dependencies among the divergence relations originally obtained from the terms of higher degree in ε.

It should also be noted that, in the case where Δx and Δu also contain derivatives of the u's, the finite transformations may depend on infinitely many derivatives of the u's; for in this case the integration of (17), when determining

$$\frac{\mathrm{d}^2 x_i}{\mathrm{d}t^2}, \quad \frac{\mathrm{d}^2 u_i}{\mathrm{d}t^2},$$

leads to

$$\Delta\left(\frac{\partial u}{\partial x_\lambda}\right) = \frac{\partial\, \Delta u}{\partial x_\lambda} - \sum_\nu \frac{\partial u}{\partial x_\nu}\frac{\partial\, \Delta x_\nu}{\partial x_\lambda},$$

so that in general the number of derivatives of u grows at every step. An example is:

$$f = \frac{1}{2}u'^2, \qquad \psi = -u'', \qquad \psi \cdot x = \frac{\mathrm{d}}{\mathrm{d}x}\{(u - u'x)\}, \qquad \bar{\delta}u = x\varepsilon,$$

$$\Delta x = -\frac{2u}{u'}\,\varepsilon, \qquad \Delta u = \left(x - \frac{2u}{u'}\right)\varepsilon.$$

Since the Lagrange expressions of a divergence vanish identically, the converse finally shows the following: if I admits a $\mathfrak{G}_\rho$, then every integral that differs from I only by a boundary integral, that is, by an integral over a divergence, also admits a $\mathfrak{G}_\rho$ with the same $\bar{\delta}u$'s, while its infinitesimal transformations will in general contain derivatives of the u's. Thus, corresponding to the preceding example,

$$f^* = \frac{1}{2}\left\{u'^2 - \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{u^2}{x}\right)\right\}$$

admits the infinitesimal transformation $\Delta u = x\varepsilon,\ \Delta x = 0$; whereas derivatives of u occur in the infinitesimal transformations corresponding to f.

If one passes to the variational problem, that is, if one sets $\psi_i = 0$,[212] then (13) becomes the equations

$$\mathrm{Div}\, B^{(1)} = 0, \ldots, \qquad \mathrm{Div}\, B^{(\rho)} = 0,$$

which are often called "conservation laws". In the one-dimensional case this gives:

$$B^{(1)} = \text{const.}, \ldots, \qquad B^{(\rho)} = \text{const.}.$$

Here the B's contain at most $(2\kappa-1)$-st derivatives of u (by (6)), as soon as Δu and Δx contain no derivatives higher than the κ-th derivatives occurring in f. Since ψ contains, in general, 2κ-th derivatives,[213] one thus has the existence of ρ first integrals. That nonlinear dependencies may exist among them is shown again by the above f. To the linearly independent $\Delta u = \varepsilon_1,\ \Delta x = \varepsilon_2$ correspond the linearly independent relations

$$u'' = \frac{\mathrm{d}}{\mathrm{d}x}u', \qquad u''u' = \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}x}(u')^2,$$

[212] $\psi_i = 0$, or somewhat more generally $\psi_i = T_i$, where the T_i's are newly adjoined functions, are called "field equations" in physics. In the case $\psi_i = T_i$, the identities (13) pass into the equations $\mathrm{Div}\, B^{(\lambda)} = \sum_i T_i \bar{\delta}u_i^{(\lambda)}$, which in physics are also still called conservation laws.

[213] As soon as f is nonlinear in the κ-th derivatives.

whereas among the first integrals $u' = \text{const.}$, $u'^2 = \text{const.}$ there is a nonlinear dependency. This is the elementary case in which $\Delta u, \Delta x$ contain no derivatives of u.[214]

§4. The converse in the case of the infinite group

First it is to be shown that the assumption of linearity of Δx and Δu is no restriction. Here this follows, even without the converse, from the fact that $\mathfrak{G}_{\infty\rho}$ depends formally on ρ, and only ρ, arbitrary functions. Namely, in the nonlinear case, when transformations are composed, the terms of lowest order add and the number of arbitrary functions would increase. Indeed, suppose for instance that

$$\begin{aligned} y &= A\Big(x, u, \frac{\partial u}{\partial x}, \ldots; p\Big) \\ &= x + \sum a(x,u,\ldots)p^\nu + b(x,u,\ldots)p^{\nu-1}\frac{\partial p}{\partial x} + c\,p^{\nu-2}\left(\frac{\partial p}{\partial x}\right)^2 + \cdots + d\left(\frac{\partial p}{\partial x}\right)^\nu + \cdots, \end{aligned}$$

where

$$p^\nu = (p^{(1)})^{\nu_1}\cdots(p^{(\rho)})^{\nu_\rho},$$

and correspondingly

$$v = B\Big(x, u, \frac{\partial u}{\partial x}, \ldots; p\Big).$$

Then, by composition with

$$z = A\Big(y, v, \frac{\partial v}{\partial y}, \ldots; q\Big),$$

the terms of lowest order give

$$z = x + \sum a(p^\nu + q^\nu) + b\left\{p^{\nu-1}\frac{\partial p}{\partial x} + q^{\nu-1}\frac{\partial q}{\partial x}\right\} + c\left\{p^{\nu-2}\left(\frac{\partial p}{\partial x}\right)^2 + q^{\nu-2}\left(\frac{\partial q}{\partial x}\right)^2\right\} + \cdots.$$

If here some coefficient other than a and b is different from zero, so that a term

$$p^{\nu-\sigma}\left(\frac{\partial p}{\partial x}\right)^\sigma + q^{\nu-\sigma}\left(\frac{\partial q}{\partial x}\right)^\sigma$$

actually occurs for $\sigma > 1$, then it cannot be written as the differential quotient of a single function, nor as a product of powers of such a quotient. Thus the number of arbitrary functions has increased against the hypothesis. If all coefficients other than a and b vanish, then, depending on the values of the exponents $\nu_1, \ldots, \nu_\rho$, the second term is the differential quotient of the first (as, for example, always for a $\mathfrak{G}_{\infty 1}$), so that linearity actually occurs; or else the number of arbitrary functions increases here as well. Hence, because of linearity in $p(x)$, the infinitesimal transformations satisfy a system of linear partial differential equations; and, since the group property is fulfilled, they form an "infinite group of infinitesimal transformations" in Lie's sense (Grundlagen, §10).

The converse now results similarly to the case of the finite group. The existence of the dependencies (16), after multiplication by $p^{(\lambda)}(x)$ and addition, leads by means of the identical transformation (14) to

$$\sum_i \psi_i \bar{\delta} u_i = \operatorname{Div}\Gamma.$$

From this, as in §3, one obtains the determination of Δx and Δu, and the invariance of I with respect to these infinitesimal transformations, which in fact depend linearly on ρ arbitrary functions and their derivatives up to order σ. That these infinitesimal transformations, when they contain no derivatives $\partial u/\partial x, \ldots$, certainly form a group follows as in §3, since otherwise more arbitrary functions would occur by composition, whereas by assumption there are to be only ρ dependencies (16). Thus they form an "infinite group of infinitesimal transformations". But such a group consists (Grundlagen, Theorem VII, p. 391) of the most general

[214] Otherwise one has additionally $u''u'^{\lambda-1} = \text{const.}$ for every λ, corresponding to

$$u''(u')^{\lambda-1} = \frac{1}{\lambda}\frac{\mathrm{d}}{\mathrm{d}x}(u')^\lambda.$$

infinitesimal transformations of a certain thereby defined "infinite group $\mathfrak{G}$ of finite transformations" in Lie's sense. Every finite transformation is thereby generated from infinitesimal ones (Grundlagen, §7),[215] and therefore arises by integration of the simultaneous system:

$$\frac{\mathrm{d}x_i}{\mathrm{d}t} = \Delta x_i, \qquad \frac{\mathrm{d}u_i}{\mathrm{d}t} = \Delta u_i, \qquad \left\{ \begin{array}{l} x_i = y_i, \\ u_i = v \end{array} \right\} \quad \text{for } t = 0.$$

It may be necessary, however, to let the arbitrary $p(x)$'s still depend on t. Thus $\mathfrak{G}$ in fact depends on ρ arbitrary functions. In particular, if it is sufficient to take $p(x)$ free of t, then this dependency is analytic in the arbitrary functions $q(x) = t\,p(x)$.[216] If derivatives $\partial u/\partial x, \ldots$ occur, it may be necessary to adjoin infinitesimal transformations $\bar{\delta} u = 0$, $\operatorname{Div}(f\Delta x) = 0$ before drawing the same conclusions.

Following an example of Lie (Grundlagen, §7), I now indicate a rather general case in which one can get as far as explicit formulae, which also show that the derivatives of the arbitrary functions occur up to order σ; in this case the converse is therefore complete. These are groups of infinitesimal transformations to which there corresponds the group of all transformations of the x's and the transformations of the u's thereby "induced"; that is, transformations of the u's in which Δu, and hence u, depends only on the arbitrary functions occurring in Δx. It is still assumed that the derivatives $\partial u/\partial x, \ldots$ do not occur in Δu. Thus:

$$\Delta x_i = p^{(i)}(x), \qquad \Delta u_i = \sum_{\lambda=1}^{n} \left\{ a_i^{(\lambda)}(x,u) p^{(\lambda)} + b_i^{(\lambda)} \frac{\partial p^{(\lambda)}}{\partial x} + \cdots + c_i^{(\lambda)} \frac{\partial^\sigma p^{(\lambda)}}{\partial x^\sigma} \right\}.$$

Since the infinitesimal transformation $\Delta x = p(x)$ generates every transformation $x = y + g(y)$ with arbitrary $g(y)$, one can in particular determine $p(x)$ as a function of t so that the one-parameter group is generated:

$$x_i = y_i + t\,g_i(y), \tag{18}$$

which for $t = 0$ becomes the identity and for $t = 1$ becomes the desired $x = y + g(y)$. Differentiation of (18) gives

$$\frac{\mathrm{d}x_i}{\mathrm{d}t} = g_i(y) = p^{(i)}(x,t), \tag{19}$$

where $p(x,t)$ is determined from $g(y)$ by inversion of (18); conversely, (18) arises from (19) by means of the side condition $x_i = y_i$ for $t = 0$, by which the integral is uniquely fixed. By means of (18), the x's in Δu can be replaced by the integration constants y and by t; the $g(y)$'s then occur exactly up to their σ-th derivatives, since in

$$\frac{\partial p}{\partial x} = \sum_k \frac{\partial g}{\partial y_k} \frac{\partial y_k}{\partial x}$$

the $\partial y/\partial x$ are expressed by $\partial x/\partial y$, and in general $\partial^\sigma p/\partial x^\sigma$ is replaced by its value in

$$\frac{\partial g}{\partial y}, \ldots, \frac{\partial^\sigma g}{\partial y^\sigma}, \ldots, \frac{\partial^\sigma x}{\partial y^\sigma}.$$

For the determination of the u's one therefore obtains the system of equations

$$\frac{\mathrm{d}u_i}{\mathrm{d}t} = F_i\left(g(y), \frac{\partial g}{\partial y}, \ldots, \frac{\partial^\sigma g}{\partial y^\sigma}, u, t \right), \qquad (u_i = v_i \text{ for } t = 0),$$

in which only t and u are variables, while $g(y), \ldots$ belong to the coefficient domain; hence integration gives

$$u_i = v_i + B_i\left(v, g(y), \frac{\partial g}{\partial y}, \ldots, \frac{\partial^\sigma g}{\partial y^\sigma}, t \right)_{t=1}.$$

Thus one obtains transformations that depend exactly on σ derivatives of the arbitrary functions. The identity is contained in this for $g(y) = 0$, by (18); and the group property follows from the fact that the procedure just described yields every transformation $x = y + g(y)$, whereby the induced transformation of the u's is uniquely determined, so that the group $\mathfrak{G}$ is exhausted.

[215] From this it follows in particular that the group $\mathfrak{G}$ generated from the infinitesimal transformations $\Delta x, \Delta u$ of a $\mathfrak{G}_{\infty\rho}$ leads back again to $\mathfrak{G}_{\infty\rho}$. For $\mathfrak{G}_{\infty\rho}$ contains no infinitesimal transformations depending on arbitrary functions and different from $\Delta x, \Delta u$, and cannot contain independent transformations depending on parameters either, since otherwise it would be a mixed group. But by the preceding argument the finite transformations are determined by the infinitesimal ones.

[216] The question whether this latter case perhaps always occurs was raised by Lie in another formulation (Grundlagen, §7 and §13, end).

It also follows incidentally from the converse that it is no restriction to assume that the arbitrary functions depend only on the x's, and not on $u, \partial u/\partial x, \ldots$. In the latter case, in the identical transformation (14), and hence also in (15), besides the $p^{(\lambda)}$'s there would occur the derivatives $\partial p^{(\lambda)}/\partial u,\ \partial p^{(\lambda)}/\partial(\partial u/\partial x), \ldots$. If one now successively takes the $p^{(\lambda)}$'s to be of zeroth, first, and higher degrees in $u, \partial u/\partial x, \ldots$, with arbitrary functions of x as coefficients, then dependencies (16) again arise, only in greater number. But by the preceding converse they can be brought back to the earlier case by being combined with arbitrary functions depending only on x. Similarly one shows that mixed groups correspond to the simultaneous occurrence of dependencies and independent divergence relations.[217]

§5. Invariance of the individual components of the relations

If the group $\mathfrak{G}$ is specialized to the simplest case, the one ordinarily considered, by admitting no derivatives of the u's in the transformations and by letting the transformed independent variables depend only on the x's and not on the u's, then one can infer the invariance of the individual components in the formulae. First, by familiar arguments, one obtains the invariance of

$$\int \cdots \int \Big(\sum_i \psi_i \delta u_i \Big) \, \mathrm{d}x,$$

hence the relative invariance of $\sum_i \psi_i \delta u_i$, where δ denotes any variation. Namely, on the one hand,

$$\delta I = \int \cdots \int \delta f\Big(x, u, \frac{\partial u}{\partial x}, \ldots\Big) \, \mathrm{d}x = \int \cdots \int \delta f\Big(y, v, \frac{\partial v}{\partial y}, \ldots\Big) \, \mathrm{d}y.$$

On the other hand, for $\delta u, \delta(\partial u/\partial x), \ldots$ vanishing at the boundary, to which, because of the linear homogeneous transformation of $\delta u, \delta(\partial u/\partial x), \ldots$, there corresponds $\delta v, \delta(\partial v/\partial y), \ldots$ also vanishing at the boundary, one has

$$\int \cdots \int \delta f\Big(x, u, \frac{\partial u}{\partial x}, \ldots\Big) \, \mathrm{d}x = \int \cdots \int \Big(\sum_i \psi_i(u, \ldots) \delta u_i \Big) \, \mathrm{d}x,$$

$$\int \cdots \int \delta f\Big(y, v, \frac{\partial v}{\partial y}, \ldots\Big) \, \mathrm{d}y = \int \cdots \int \Big(\sum_i \psi_i(v, \ldots) \delta v_i \Big) \, \mathrm{d}y.$$

Thus, for $\delta u, \delta(\partial u/\partial x), \ldots$ vanishing at the boundary,

$$\int \cdots \int \Big(\sum_i \psi_i(u, \ldots) \delta u_i \Big) \, \mathrm{d}x = \int \cdots \int \Big(\sum_i \psi_i(v, \ldots) \delta v_i \Big) \, \mathrm{d}y$$

$$= \int \cdots \int \Big(\sum_i \psi_i(v, \ldots) \delta v_i \Big) \Big| \frac{\partial y_i}{\partial x_k} \Big| \, \mathrm{d}x.$$

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[217] As in §3, it follows here too from the converse that besides I, every integral I^* differing from it by an integral over a divergence also admits an infinite group with the same $\bar{\delta}u$'s; here, however, Δx and Δu will in general contain derivatives of the u's. Einstein introduced such an integral I^* into general relativity in order to obtain a simpler formulation of the energy theorems. I give the infinitesimal transformations admitted by this I^*, using exactly the notation of Klein's second note. The integral $I = \int \cdots \int K \, \mathrm{d}\omega = \int \cdots \int \mathfrak{R} \, \mathrm{d}S$ admits the group of all transformations of the w's and the group thereby induced for the $g_{\mu\nu}$'s. To this correspond the dependencies ((30) in Klein):

$$\sum \mathfrak{R}_{\mu\nu} g^{\alpha\beta}_{\mu\nu} + 2 \sum \frac{\partial g^{\alpha\beta}_{\mu\nu} \mathfrak{R}_{\mu\nu}}{\partial w^\sigma} = 0.$$

Now $I^* = \int \cdots \int \mathfrak{R}^* \, \mathrm{d}S$, where $\mathfrak{R}^* = \mathfrak{R} + \mathrm{Div}$, and consequently $\mathfrak{R}^*_{\mu\nu} = \mathfrak{R}_{\mu\nu}$, where $\mathfrak{R}^*_{\mu\nu}, \mathfrak{R}_{\mu\nu}$ denote the Lagrange expressions respectively. The given dependencies are therefore also dependencies for $\mathfrak{R}^*_{\mu\nu}$; and after multiplication by p^τ and addition, by reversing the transformations of product differentiation, one obtains

$$\sum \mathfrak{R}_{\mu\nu} p^{\mu\nu} + 2 \, \mathrm{Div}\Big(\sum g^{\mu\sigma} \mathfrak{R}_{\mu\tau} p^\tau \Big) = 0,$$

$$\delta \mathfrak{R}^* + \mathrm{Div}\Big(\sum 2 g^{\mu\sigma} \mathfrak{R}_{\mu\tau} p^\tau - \frac{\partial \mathfrak{R}^*}{\partial g^\sigma_{\mu\nu}} p^{\mu\nu} \Big) = 0.$$

Comparison with the Lie differential equation $\delta \mathfrak{R}^* + \mathrm{Div}(\mathfrak{R}^* \Delta w) = 0$ gives

$$\Delta w^\sigma = \frac{1}{\mathfrak{R}^*} \Big(\sum 2 g^{\mu\sigma} \mathfrak{R}_{\mu\tau} p^\tau - \frac{\partial \mathfrak{R}^*}{\partial g^\sigma_{\mu\nu}} p^{\mu\nu} \Big), \qquad \Delta g^{\mu\nu} = p^{\mu\nu} + \sum g^{\mu\nu}_\sigma \Delta w^\sigma,$$

as infinitesimal transformations admitted by I^*. These infinitesimal transformations therefore depend on the first and second derivatives of the $g^{\mu\nu}$'s, and contain the arbitrary p's up to the first derivative.

If one expresses $y, v, \delta v$ in the third integral by $x, u, \delta u$, and sets it equal to the first, then one obtains a relation

$$\int \cdots \int \left(\sum_i \chi_i(u, \ldots)\delta u_i \right) \mathrm{d}x = 0$$

for δu vanishing at the boundary but otherwise arbitrary. From this it follows, as is known, that the integrand vanishes for arbitrary δu; hence the relation

$$\sum_i \psi_i(u, \ldots)\delta u_i = \left| \frac{\partial y_i}{\partial x_k} \right| \left(\sum_i \psi_i(v, \ldots)\delta v_i \right)$$

holds identically in δu. This expresses the relative invariance of $\sum_i \psi_i \delta u_i$, and consequently the invariance of

$$\int \cdots \int \left(\sum_i \psi_i \delta u_i \right) \mathrm{d}x.$$

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To apply this to the derived divergence relations and dependencies, one must first show that the $\bar{\delta} u$ derived from $\Delta u, \Delta x$ actually satisfies the transformation laws for the variation δu, provided only that the parameters, respectively arbitrary functions, in $\bar{\delta} v$ are determined as they correspond to the similar group of infinitesimal transformations in y, v. Let T_q denote the transformation that carries x, u into y, v; and let T_p be an infinitesimal transformation in x, u. Then the similar one in y, v is given by $T_r = T_q T_p T_q^{-1}$, where the parameters, respectively arbitrary functions, r are determined by p and q. In formulae this is expressed as follows:

$$T_p: \ \xi = x + \Delta x(x, p), \qquad u^* = u + \Delta u(x, u, p);$$
$$T_q: \ y = A(x, q), \qquad v = B(x, u, q);$$
$$T_q T_p: \ \eta = A(x + \Delta x(x, p), q), \quad v^* = B(x + \Delta x(p),\, u + \Delta u(p), q).$$

But from this arises $T_r = T_q T_p T_q^{-1}$, hence

$$\eta = y + \Delta y(r), \qquad v^* = v + \Delta v(r),$$

where, by means of the inverse of T_q, one regards the x's as functions of the y's and retains only the infinitesimal terms. Thus one has the identity

$$\eta = y + \Delta y(r) = y + \sum \frac{\partial A(x, q)}{\partial x} \Delta x(p), \tag{20a}$$

$$v^* = v + \Delta v(r) = v + \sum \frac{\partial B(x, u, q)}{\partial x} \Delta x(p) + \sum \frac{\partial B(x, u, q)}{\partial u} \Delta u(p). \tag{20b}$$

If, in this, one replaces $\xi = x + \Delta x$ by $\xi - \Delta\xi$, so that ξ goes back into x and hence Δx vanishes, then by the first formula (20) η also goes back into $y = \eta - \Delta\eta$. If under this substitution $\Delta u(p)$ goes over into $\bar{\delta} u(p)$, then $\Delta v(r)$ also goes over into $\bar{\delta} v(r)$, and the second formula (20) gives

$$v + \bar{\delta} v(y, v, \ldots, r) = v + \sum \frac{\partial B(x, u, q)}{\partial u} \, \bar{\delta} u(p),$$

$$\bar{\delta} v(y, v, \ldots, r) = \sum \frac{\partial B}{\partial u_\kappa} \, \bar{\delta} u_\kappa(x, u, p).$$

Thus the transformation formulae for variations are indeed satisfied, provided only that $\bar{\delta} v$ is taken to depend on the parameters, respectively arbitrary functions, r.[219]

[218]These arguments fail if y also depends on the u's, because then $\delta f(y, v, \partial v/\partial y, \ldots)$ also contains terms $\sum(\partial f/\partial y)\delta y$, so that the divergence transformation does not lead to the Lagrange expressions. They likewise fail if derivatives of the u's are admitted; then the δv's become linear combinations of $\delta u, \delta(\partial u/\partial x), \ldots$, and hence only after a further divergence transformation lead to an identity $\int \cdots \int (\sum \chi_i(u, \ldots)\delta u_i)\mathrm{d}x = 0$, so that on the right the Lagrange expressions again do not occur. The question whether the invariance of $\int \cdots \int (\sum \psi_i \delta u_i)\mathrm{d}x$ already allows one to conclude the existence of divergence relations is, by the converse, equivalent to the question whether it implies invariance of I with respect to a group that need not lead to the same $\Delta u, \Delta x$, but does lead to the same $\bar{\delta} u$. In the special case of the simple integral and only first derivatives in f, for the finite group one can infer the existence of first integrals from the invariance of the Lagrange expressions (compare, for example, Engel, Gött. Nachr. 1916, p. 270).

[219]It is again seen that y must be assumed independent of u, and so on, if the conclusions are to hold. As an example, one may cite the $\delta g^{\mu\nu}$ and δq_σ indicated by Klein, which satisfy the transformations for variations as soon as the p's are subjected to a vector transformation.

It follows in particular that $\sum_i \psi_i \bar{\delta} u_i$ is relatively invariant; hence also, by (12), since the divergence relations are also fulfilled in y, v, the relative invariance of $\operatorname{Div} B$; and further, by (14) and (13), the relative invariance of $\operatorname{Div}\Gamma$ and of the left-hand sides of the dependencies combined with the $p^{(\lambda)}$'s, where in the transformed formulae the arbitrary $p(x)$'s (respectively the parameters) are everywhere to be replaced by the r's. From this follows also the relative invariance of $\operatorname{Div}(B-\Gamma)$, that is, of the divergence of a system of functions $B-\Gamma$ that does not vanish identically, but whose divergence vanishes identically.

From the relative invariance of $\operatorname{Div} B$, in the one-dimensional case and for a finite group, one can draw a further conclusion about the invariance of the first integrals. The transformation of the parameters corresponding to the infinitesimal transformation becomes, by (20), linear and homogeneous; and because all transformations are invertible, the ε's are also linear and homogeneous in the transformed parameters ε^*. This invertibility is certainly preserved when one sets $\psi = 0$, since no derivatives of u occur in (20). By equating the coefficients of the ε^* in

$$\operatorname{Div} B(x, u, \ldots, \varepsilon) = \frac{\mathrm{d}y}{\mathrm{d}x} \operatorname{Div} B(y, v, \ldots, \varepsilon^*)$$

the quantities $\frac{\mathrm{d}}{\mathrm{d}y} B^{(\lambda)}(y, v, \ldots)$ also become linear homogeneous functions of the $\frac{\mathrm{d}}{\mathrm{d}x} B^{(\lambda)}(x, u, \ldots)$. Thus from

$$\frac{\mathrm{d}}{\mathrm{d}x} B^{(\lambda)}(x, u, \ldots) = 0 \quad \text{or} \quad B^{(\lambda)}(x, u) = \text{const.}$$

it follows that

$$\frac{\mathrm{d}}{\mathrm{d}y} B^{(\lambda)}(y, v, \ldots) = 0 \quad \text{or} \quad B^{(\lambda)}(y, v) = \text{const.}$$

The ρ first integrals corresponding to a $\mathfrak{G}_\rho$ therefore each admit the group, so that the further integration is also simplified. The simplest example of this is that f is free of x or of a u, corresponding to the infinitesimal transformations $\Delta x = \varepsilon, \Delta u = 0$, respectively $\Delta x = 0, \Delta u = \varepsilon$. Then $\bar{\delta} u = -\varepsilon\, \mathrm{d}u/\mathrm{d}x$, respectively ε; and since B is derived from f and $\bar{\delta} u$ by differentiation and rational operations, it is also free of x, respectively u, and admits the corresponding groups.[220]

§6. A theorem of Hilbert

From the foregoing there finally results the proof of an assertion of Hilbert concerning the connection between the failure of proper energy theorems and "general relativity" (Klein's first note, Göttinger Nachr. 1917, Reply, first paragraph), in a generalized group-theoretic formulation.

Suppose the integral I admits a $\mathfrak{G}_{\infty\rho}$, and let $\mathfrak{G}_\sigma$ be any finite group arising by specialization of the arbitrary functions, thus a subgroup of $\mathfrak{G}_{\infty\rho}$. To the infinite group $\mathfrak{G}_{\infty\rho}$ there correspond dependencies (16), to the finite group $\mathfrak{G}_\sigma$ divergence relations (13); conversely, from the existence of any divergence relations follows the invariance of I with respect to a finite group, which is identical with $\mathfrak{G}_\sigma$ if and only if the $\bar{\delta} u$'s are linear combinations of those arising from $\mathfrak{G}_\sigma$. Hence the invariance with respect to $\mathfrak{G}_\sigma$ can lead to no divergence relations different from (13). But since from the existence of (16) follows the invariance of I with respect to the infinitesimal transformations $\Delta u, \Delta x$ of $\mathfrak{G}_{\infty\rho}$ for arbitrary $p(x)$, it follows in particular that I is already invariant with respect to the infinitesimal transformations of $\mathfrak{G}_\sigma$ arising from these by specialization, and hence with respect to $\mathfrak{G}_\sigma$. The divergence relations

$$\sum_i \psi_i \bar{\delta} u_i^{(\lambda)} = \operatorname{Div} B^{(\lambda)}$$

must therefore be consequences of the dependencies (16), which can also be written

$$\sum_i \psi_i a_i^{(\lambda)} = \operatorname{Div} \chi^{(\lambda)},$$

[220] In those cases where the existence of first integrals already follows from the invariance of $\int (\sum \psi_i \delta u_i) \mathrm{d}x$, these do not admit the complete group $\mathfrak{G}_\rho$. For example, $\int (u'' \delta u) \mathrm{d}x$ admits the infinitesimal transformation $\Delta x = \varepsilon_2, \Delta u = \varepsilon_1 + x\varepsilon_3$; whereas the first integral $u - u'x = \text{const.}$, corresponding to $\Delta x = 0, \Delta u = x\varepsilon_3$, does not admit the two other infinitesimal transformations, since it contains both u and x explicitly. To this first integral there correspond infinitesimal transformations for f that contain derivatives. Thus one sees that the invariance of $\int \cdots \int (\sum \psi_i \delta u_i) \mathrm{d}x$ in any case accomplishes less than the invariance of I, a point relevant to the question raised in a preceding note.

where the $\chi^{(\lambda)}$'s are linear combinations of the Lagrange expressions and their derivatives. Since the ψ's enter linearly both in (13) and in (16), the divergence relations must in particular be linear combinations of the dependencies (16). Thus

$$\mathrm{Div}\, B^{(\lambda)} = \mathrm{Div}\left(\sum \alpha_\mu \chi^{(\mu)}\right),$$

and the $B^{(\lambda)}$'s themselves are thus composed linearly from the χ's, that is, from the Lagrange expressions and their derivatives, and from functions whose divergence vanishes identically, as for instance the $B-\Gamma$ occurring at the end of §2, for which $\mathrm{Div}(B-\Gamma)=0$, and where the divergence has at the same time the character of an invariant. I shall call divergence relations in which the $B^{(\lambda)}$'s can be composed in the indicated way from the Lagrange expressions and their derivatives "improper", and all the remaining ones "proper".

Conversely, if the divergence relations are linear combinations of the dependencies (16), hence "improper", then the invariance with respect to $\mathfrak{G}_\sigma$ follows from that with respect to $\mathfrak{G}_{\infty\rho}$; $\mathfrak{G}_\sigma$ becomes a subgroup of $\mathfrak{G}_{\infty\rho}$. Thus the divergence relations corresponding to a finite group $\mathfrak{G}_\sigma$ become improper if and only if $\mathfrak{G}_\sigma$ is a subgroup of an infinite group with respect to which I is invariant.

By specialization of the groups, Hilbert's original assertion follows from this. By the "translation group" one means the finite group

$$y_i = x_i + \varepsilon_i, \qquad v_i(y) = u_i(x);$$

hence

$$\Delta x_i = \varepsilon_i, \qquad \Delta u_i = 0, \qquad \bar{\delta} u_i = -\sum_\lambda \frac{\partial u_i}{\partial x_\lambda}\varepsilon_\lambda.$$

Invariance with respect to the translation group says, as is known, that in

$$I = \int \cdots \int f\left(x, u, \frac{\partial u}{\partial x}, \ldots\right) \mathrm{d}x$$

the x's do not occur explicitly in f. The associated n divergence relations

$$\sum_i \psi_i \frac{\partial u_i}{\partial x_\lambda} = \mathrm{Div}\, B^{(\lambda)} \qquad (\lambda = 1, 2, \ldots, n)$$

are to be called "energy relations", since the "conservation laws" $\mathrm{Div}\, B^{(\lambda)} = 0$ corresponding to the variational problem correspond to the "energy theorems", and the $B^{(\lambda)}$'s to the "energy components". Thus one obtains: if I admits the translation group, then the energy relations become improper if and only if I is invariant with respect to an infinite group containing the translation group as a subgroup.[221]

An example of such infinite groups is given by the group of all transformations of the x's and of the induced transformations of the $u(x)$'s in which only derivatives of the arbitrary functions $p(x)$ occur. The translation group arises by the specialization $p^{(i)}(x) = \varepsilon_i$. It must remain undecided, however, whether by this means — and by the groups arising through changing I by a boundary integral — the most general such groups have already been given. Induced transformations of the indicated kind arise, for example, by subjecting the u's to the coefficient transformations of a "total differential form", that is, of a form

$$\sum a\, \mathrm{d}^2 x_i + \sum b\, \mathrm{d}x_i \mathrm{d}x_k + \cdots,$$

which contains, besides the $\mathrm{d}x$'s, also higher differentials. More special induced transformations, in which the $p(x)$'s occur only in first derivatives, are given by the coefficient transformations of ordinary differential forms $\sum c\, \mathrm{d}x_{i_1} \cdots \mathrm{d}x_{i_\lambda}$; these are the ones that have usually been considered.

A further group of the indicated kind — which, because of the occurrence of the logarithmic term, cannot be a coefficient transformation — is, for example, the following:

$$y = x + p(x), \qquad v_i = u_i + \lg(1 + p'(x)) = u_i + \lg \frac{\mathrm{d}y}{\mathrm{d}x};$$

$$\Delta x = p(x), \qquad \Delta u_i = p'(x), \qquad \bar{\delta} u_i = p'(x) - u_i' p(x).$$

Here the dependencies (16) become

$$\sum_i \left(\psi_i u_i' + \frac{\mathrm{d}\psi_i}{\mathrm{d}x}\right) = 0,$$

[221]The energy theorems of classical mechanics, and likewise those of the old "theory of relativity" (where $\sum \mathrm{d}x^2$ goes over into itself), are "proper", since no infinite groups occur here.

and the improper energy relations become

$$\sum_i \left(\psi_i u_i' + \frac{\mathrm{d}(\psi_i + \text{const.})}{\mathrm{d}x} \right) = 0.$$

A simplest invariant integral of the group is

$$I = \int \frac{e^{-2u_1}}{u_1' - u_2'} \, \mathrm{d}x.$$

The most general I is determined by integration of the Lie differential equation (11)

$$\bar{\delta} f + \frac{\mathrm{d}}{\mathrm{d}x}(f \cdot \Delta x) = 0,$$

which, after substituting the values for Δx and $\bar{\delta} u$, becomes, as soon as one assumes f to depend on only first derivatives of the u's,

$$\frac{\partial f}{\partial x} p(x) + \left\{ \sum_i \frac{\partial f}{\partial u_i} - \sum_i \frac{\partial f}{\partial u_i'} u_i' + f \right\} p'(x) + \left\{ \sum_i \frac{\partial f}{\partial u_i'} \right\} p''(x) = 0$$

(identically in $p(x), p'(x), p''(x)$). Already for two functions $u(x)$, this system of equations has solutions that actually contain the derivatives, namely

$$f = (u_1' - u_2') \, \Phi \left(u_1 - u_2, \frac{e^{-u_1}}{u_1' - u_2'} \right),$$

where Φ denotes an arbitrary function of the indicated arguments.

Hilbert states his assertion by saying that the failure of proper energy theorems is a characteristic feature of the "general theory of relativity". For this assertion to hold literally, the designation "general relativity" must therefore be understood more broadly than usual, and extended also to the preceding groups depending on n arbitrary functions.[222]

14. The Arithmetic Theory of Algebraic Functions of One Variable in Its Relation to the Other Theories and to Algebraic Number Theory

Jahresbericht der Deutschen Mathematiker-Vereinigung 38 (1919), pp. 182–203

The following report arose from the wish to have a connecting link between the individual theories of algebraic functions; at the same time it may be regarded as a supplement to the report by Brill–Noether,[223] in which the discussion of the arithmetic theory is essentially absent.[224]

For discussions of the individual theories, besides Brill–Noether, the following encyclopedia articles, with the literature indicated there, come into consideration:

For the theory of Riemann: Wirtinger (II B 2), nos. 1–19;

for the theory of Weierstrass: Wirtinger (II B 2), nos. 23, 32, 33, 34;

for the algebraic-geometric theory of Brill–Noether: Wirtinger (II B 2), nos. 21–31; Berzolari (III C 4), nos. 23–38;

for the arithmetic theory of Dedekind–Weber and Hensel–Landsberg: Hensel (II C 5), [cited as Hensel];[225]

more generally, for the arithmetic theory of algebraic quantities: Landsberg (I C 5), and for the arithmetic theory of algebraic functions of several variables: Jung (II C 6).

For algebraic number theory compare Hilbert's "Zahlbericht"[226] and the lectures on number theory by Dirichlet–Dedekind [cited as Dedekind–number theory].[227]

[222] This again confirms the correctness of a remark of Klein, that the term "relativity" customary in physics should be replaced by "invariance relative to a group". (Über die geometrischen Grundlagen der Lorentzgruppe. *Jhrber. d. d. Math. Vereinig.* vol. 19, p. 287, 1910; reprinted in the *Physikalische Zeitschrift.*)

[223] The development of the theory of algebraic functions in earlier and more recent times. *Jahresbericht der Deutschen Mathematiker-Vereinigung* vol. 3 (1894) [cited as Brill–Noether].

[224] With the exception of a discussion of Kronecker's investigation of the discriminant. (*J. f. M.* vol. 91.)

[225] The parts concerning Dedekind–Weber are due to A. Ostrowski.

[226] The theory of algebraic number fields. *Jahresbericht der Deutschen Mathematiker-Vereinigung* vol. 4 (1894/95).

[227] Braunschweig, Vieweg, 4th ed. 1894.

Contents

1. Foundations of the various theories

Riemann's theory is the only purely transcendental one, resting on existence theorems (Dirichlet's principle, Schwarz–Neumann methods). The integrals of algebraic functions are here primary, the algebraic functions themselves secondary. The functions on the Riemann surface are characterized by their zeros and poles, corresponding to the polygons (respectively divisors) of the arithmetic theory; whereas Weierstrass also speaks of the functions themselves together with their zeros and poles. The main problem, apart from the existence theorems, is the setting up of all functions with given poles, which is accomplished by integrals of the first and second kind; and, further, the question of the number of arbitrary constants that still enter when the points at infinity are prescribed. This latter question is answered by the Riemann–Roch theorem; corresponding to it in the arithmetic theory is the question of the dimension of a polygon class (respectively divisor class), and in the algebraic-geometric theory the question of the dimension of a complete linear system, whose answer is also provided here by the Riemann–Roch theorem. The actual construction of the functions is carried out in the arithmetic theory by means of the basis theorems; in the algebraic-geometric theory, by the residue theorem. In particular, the zeros of the integrands of the first kind correspond to the differential class in the arithmetic theory, and to the doctrine of special groups (that is, the pencil of point groups cut out by the adjoint curves of order $(n-3)$, the φ-curves) in the algebraic-geometric theory.

Just as Riemann's theory historically preceded the other theories, even though the rigorous proof of the existence theorems was obtained only later, so also later, with its transcendental aids, algebraic questions were settled which to this day are not yet accessible to a purely algebraic or arithmetic treatment; here above all Hurwitz's theory of singular correspondences is to be mentioned.[228]

Analogies with the transcendental questions in the theory of algebraic number fields will be discussed in the last section.

For the remaining theories the algebraic functions are primary, the integrals secondary. In this respect the Weierstrass theory rests on the transcendental foundations of power-series expansion and the residue theorem; but it then proceeds algebraically, by explicit specification of all functions under consideration.

The arithmetic theory in the form of Hensel–Landsberg also uses the transcendental foundations of series expansion. But by assigning divisors to the individual points on this foundation, it then proceeds purely arithmetically and can thus be viewed as a fusion of Weierstrass and Dedekind–Weber. In contrast with the latter, it simultaneously supplies methods for the actual construction of all basis functions in question. In detail it exhibits a number of simplifications compared with Dedekind–Weber, essentially by the introduction of "fractional" divisors. Recently Hensel[229] has given an arithmetic foundation for the theory of series expansions (apart from a majorant method for the convergence proof), whereby the theory has become almost purely arithmetic.

The algebraic-geometric theory of Brill–Noether presupposes only the concept of "successive points" of a branch, which is reduced to the series expansion at an ordinary point[230] and corresponds to the Weierstrass

[228] A. Hurwitz, Über algebraische Korrespondenzen und das erweiterte Korrespondenzprinzip. *Math. Ann.* vol. 28. (Compare Brill–Noether, section 10.)

[229] *Mathematische Zeitschrift* vol. 4; the foundation is also used as the basis of his report.

[230] For successive "singular" points compare Brill–Noether, section 6, especially nos. 10 and 11. Ordinary successive points are, for example, the points of intersection of the tangent with the curve or higher points of contact (branch points).

element concept; it then continues with algebraic-rational methods, essentially those of ternary elimination theory; it too reaches explicit representations.

The original, purely arithmetic theory of Dedekind–Weber[231] is akin to Riemann's in that its theorems again have the character of existence proofs; it preserves complete purity of method.[232] The arithmetic foundation of the theory, where the variables play only the role of indeterminates, runs parallel to the theory of algebraic numbers, as will be explained more closely in 2.[233] This foundation gives the means of introducing the point concept arithmetically (compare Hensel no. 10a) and thus of setting up the concept of the absolute Riemann surface without any consideration of continuity. The theory thereby acquires a formal character; for example, the concept of the differential is also introduced purely formally. On the basis obtained by the point concept, the close kinship with the methods and concepts of the algebraic-geometric theory can be demonstrated, since the latter too, apart from the element concept, is free of transcendental aids; this is to be done in the following sections.

2. Parallelism between algebraic numbers and algebraic functions. (Conjugate elements, basis theorems, module concept)

An algebraic number field is defined as a field of degree n over the field R of rational numbers; a field of algebraic functions of one variable as a field of degree n over the field $K(z)$ of all rational functions of one indeterminate z with arbitrary complex coefficients. Thus for both fields all theorems hold in common which abstract completely from the special nature of the ground field; these are the theorems on bases, norms, traces, and discriminants.[234] All these concepts are introduced by Dedekind–Weber without the use of conjugate elements. It should be noted, however, that the concept of the conjugate field can be defined in an entirely formal way, as Steinitz showed following Kronecker.[235] Namely, if K is an arbitrary field and $\varphi(x)$ a polynomial in x irreducible over K, then the system of residue classes modulo $\varphi(x)$ forms a field. If one assigns to the residue class of x an element ξ, then under the isomorphic assignment the element $f(\xi)$ corresponds to the residue class represented by the polynomial $f(x)$. Since all polynomials divisible by $\varphi(x)$ represent the zero class, $\varphi(\xi)$ is assigned to the zero class; the element ξ can thus symbolically be regarded as a zero of $\varphi(x) = 0$ and generates the algebraic field $K(\xi)$. Repetition of the procedure leads to the n conjugate fields $K(\xi), K(\xi'), \ldots, K(\xi^{(n-1)})$. Thus one may in particular speak of conjugate elements also for algebraic functions, without leaving the purely arithmetic theory.

The two ground fields R and $K(z)$ have the property in common that in them the system of all integral quantities is defined, as rational integers and as integral rational functions of one indeterminate, respectively. These integral quantities have the characteristic property that any two a and b possess a greatest common divisor d that is representable in the form $ma + nb$, where d, m, n are also integral quantities from R or $K(z)$; and indeed d is the absolutely smallest number, respectively the polynomial of lowest degree, representable in this form. On this property alone, which entails the unique decomposition into prime factors for rational integers and for functions of one indeterminate, rest the theorems on the integral quantities of the algebraic or upper field. The argument proving the existence of the greatest common divisor also gives the existence of a fundamental system or a basis of the integral quantities of the algebraic field. If one considers only such bases of the field as consist of integral quantities, then their discriminant becomes a rational integer, respectively a rational function of one indeterminate. Every basis that corresponds to the absolutely smallest number, respectively to the function of lowest degree, forms a fundamental system.

A more precise insight into the basis questions, also for more general fields, is given by the introduction of the module concept, which shall be formulated so as to include the definitions that occur differently in the literature according to the nature of the elements. A module — with respect to a basic domain of integrity $\mathfrak{S}$ — is a system of elements such that the difference of two elements again belongs to the system, and likewise the product of an element with an arbitrary quantity from $\mathfrak{S}$. (If $\mathfrak{S}$ consists of the rational integers, then the second requirement is a consequence of the first.)[236]

[231] *J. f. M.* 92 (cited as D.–W.).

[232] For this reason it is also more suitable for comparison with the other theories than, for instance, the theory of Hensel–Landsberg, which will occur only occasionally in the following comparisons. — The theory of Abelian integrals and their inversion, insofar as it presupposes continuity, is however not treated in D.–W. nor in the further development of the algebraic theory (M. Noether, *Ann.* 37).

[233] A comparison of the theory of Hensel–Landsberg with number theory, especially the theory of p-adic numbers, is found in Hensel.

[234] D.–W., § 1 and 2; Hensel no. 4 (with note 5).

[235] Steinitz, Algebraische Theorie der Körper. (*J. f. M.* 137, § 6 and 8.)

[236] The module concept is due to Dedekind, who first introduced the number module ($\mathfrak{S}$ = rational integers) in the second edition of the number theory; the module of integral linear forms is also implicitly contained there. In Dedekind–Weber the "function modules" occur, whose elements consist of algebraic functions, whereas $\mathfrak{S}$ consists of all integral rational functions of one indeterminate. — The module of homogeneous forms in n indeterminates, more generally of polynomials, occurs first in Hilbert (*Annalen* 36); $\mathfrak{S}$ consists of all polynomials, and since the elements are also

Every module that has a basis of finitely many elements through which every element can be represented linearly with coefficients from $\mathfrak{S}$ is called finite (finitely generated); in particular, every one-term module is therefore of the form $c \cdot \alpha$, where α denotes the basis element and c runs through all quantities from $\mathfrak{S}$. *A module whose elements themselves belong to $\mathfrak{S}$ is called an ideal in $\mathfrak{S}$;* whence follows the concept of a finite ideal, and in particular of a one-term ideal (principal ideal).

Now all basis theorems for integral quantities and ideals rest on the following theorem:

If M is a module with respect to $\mathfrak{S}$, whose elements consist of linear forms in n indeterminates with coefficients from $\mathfrak{S}$, and if every ideal in $\mathfrak{S}$ is finite, then M is also a finite module. If in particular every ideal in $\mathfrak{S}$ is a principal ideal, then M possesses a basis of linearly independent forms.[237]

Indeed, if the elements of M are given by $l(x) = a_1x_1 + \cdots + a_nx_n$, then the totality of the coefficients a_1 runs through an ideal in $\mathfrak{S}$, which by hypothesis is of the form $b_1a^{(1)} + \cdots + b_\rho a^{(\rho)}$. The linear form belonging to M,

$$m(x) = l(x) - b_1l^{(1)}(x) - \cdots - b_\rho l^{(\rho)}(x),$$

therefore depends only on $(n-1)$ indeterminates, from which the assertion follows by finitely many repetitions. If in each case $\rho = 1$, then the basis consists of n forms in respectively $n, n-1, \ldots, 1$ indeterminates, which gives the linear independence of the non-vanishing forms.

The theorem on the fundamental system follows from this in the various cases as follows:

If the algebraic field is obtained by adjoining the integral algebraic element δ to the ground field, and if $\mathfrak{S}$ denotes the totality of all integral quantities of the ground field, then every integral quantity of the upper field admits the representation (compare, for instance, Zahlbericht § 3)

$$\omega = \frac{a_1\delta^{n-1} + a_2\delta^{n-2} + \cdots + a_n}{\Delta(\delta)},$$

where the a_i are quantities from $\mathfrak{S}$ and $\Delta(\delta)$ denotes the discriminant of δ. By the substitution

$$x_i = \frac{\delta^{n-i}}{\Delta(\delta)}$$

the module of all integral quantities is transformed into a module of linear forms; hence the existence of the fundamental system follows in all cases where every ideal in $\mathfrak{S}$ is finite. The same argument shows more generally that every ideal in the domain of integral quantities of the upper field is finite. Now in particular for rational integers and for functions of one indeterminate every ideal is a principal ideal; hence it follows that the fundamental systems in algebraic number fields and in fields of algebraic functions of one indeterminate consist of exactly n quantities, when n denotes the degree. And since, in these fields, every ideal is finite by the preceding argument, this at the same time yields the fundamental system for relative fields, which in general will consist of more quantities than the relative degree indicates.[238] Further, if $\mathfrak{S}$ denotes the domain of all polynomials in several indeterminates, every ideal in $\mathfrak{S}$ is also finite;[239] hence the existence of the fundamental system is obtained also for algebraic function fields of several indeterminates, and it will again consist of more quantities than the degree indicates.

3. Parallelism and differences in the ideal theory of algebraic numbers and functions

The proof of the fundamental theorem of ideal theory, that every ideal can be represented uniquely as a product of powers of prime ideals, can be carried out jointly for algebraic numbers and functions; solely on the basis of the property that, for rational integers and for functions of one indeterminate, every ideal is a principal ideal.[240] The main point of the proof lies in showing that, from the divisibility of an ideal $\mathfrak{c}$ by an

polynomials, this module falls under our ideal definition. Kronecker's "module systems" of polynomials start from the basis, not from the abstract definition.

[237] The theorem appears in the literature separately for the different coefficient domains $\mathfrak{S}$. For rational integers compare, for instance, Hilbert, Zahlbericht § 3 and 4; for polynomials in several indeterminates: Hilbert, Über die vollen Invariantensysteme: end of § 2 (*Math. Ann.* 42).

[238] According to Steinitz's investigations on modules of linear forms, where $\mathfrak{S}$ consists of the integers of a (finite) algebraic number field, $\nu + 1$ basis elements suffice when ν denotes the relative degree (*Math. Ann.* 71, § 4).

[239] By Hilbert's theorem on the module basis (*Math. Ann.* 36). The modules of polynomials here called ideals for the sake of consistency are usually called "form modules" or simply "modules". Hilbert's theorem that such a module N is finite can also be reduced to the theorem on linear forms. Namely, let f be a polynomial of degree n contained in N in the $r+1$ indeterminates $x_1, \ldots, x_r, x_{r+1}$, containing the term x_{r+1}^n (which can always be achieved by a linear transformation); then all polynomials from N are congruent mod. f to those representable in the form $a_1(x)x_{r+1}^{n-1} + \cdots + a_n(x)$; hence, by $x_{r+1}^{n-i} = X_i$, they form a module of linear forms for which every ideal in the domain of r indeterminates may be assumed finite, since this is the case for one indeterminate.

[240] Compare a remark in the introduction to Hurwitz: Über die Theorie der Ideale (*Gött. Nachr.* 1894). In Weber's textbook of algebra the proof, following Kronecker, is carried out in parallel for both fields by means of the introduction of functionals. [The connection between Kronecker's theory and ideal theory is mediated by the "extended Gauss theorem"; Hurwitz, loc. cit.]

ideal $\mathfrak{a}$ (that is, every element of $\mathfrak{c}$ is contained in $\mathfrak{a}$), the product representation $\mathfrak{c} = \mathfrak{ab}$ follows.[241] In Hurwitz this proof rests on the "extended Gauss theorem", which says that an integral quantity ω which divides every coefficient of the product of two polynomials φ and ψ also divides the product of all coefficients of φ and ψ; in Dedekind it rests on a module theorem equivalent to this theorem.[242]

Going further, the concept of the complementary module (Hensel no. 12), and from it the ramification ideal (ground ideal, different) $\mathfrak{D}$, can be defined jointly (its norm becomes the field discriminant D); and the fundamental theorem holds that the ramification ideal $\mathfrak{D}$ is the *greatest common divisor of all principal ideals* $F'(\delta)$; here δ runs through all integral quantities of the field and $F(\delta) = 0$ denotes in each case the corresponding irreducible equation. From this it follows that the *field discriminant contains all and only those rational primes, respectively linear factors in z, which are divisible by the square of a prime ideal.* For $F'(\delta)$ itself one obtains the representation

$$F'(\delta) = \mathfrak{f} \cdot \mathfrak{D},$$

and by taking norms

$$\Delta(\delta) = R^2 \cdot D;$$

here $\mathfrak{f}$ denotes the conductor of the ring derived from δ (order, domain of integrity), that is, $\mathfrak{f}$ consists of the totality of the (integral) quantities which, after multiplication by any integral quantity, are divisible by the module $[1, \delta, \ldots, \delta^{n-1}]$. $R^2 = \operatorname{Norm} \mathfrak{f}$ represents the square of the determinant of substitution that carries the basis $1, \delta, \ldots, \delta^{n-1}$ into a basis of the integral quantities.[243]

Differences in the further development are conditioned by special properties of the ground fields. Thus, in the case of algebraic numbers, the fact that the *elements of the ground field are at the same time exponents* leads to the possibility that the ramification ideal may be divisible by a higher power than the $(e-1)$-st power of a prime ideal $\mathfrak{p}$ which occurs in p to the e-th power (namely when e is divisible by p); and this leads to the further concepts of the inertia field and ramification field,[244] which have no analogue in the theory of algebraic functions. Above all, however, because of the *existence of infinitely many constants* in the ground field, the finiteness of the number of ideal classes disappears for algebraic functions, and with it all questions connected with determining the class number. Nevertheless, in a certain sense the domain of all (transcendental) prime forms[245] may be regarded as a transcendental analogue of the class field; for here every point is generated by a single form, a principal ideal.

On the other hand, the existence of infinitely many constants in the ground field $K(z)$ leads, for algebraic functions, to theorems and questions which have no analogue for algebraic numbers. First, a certain simplification in the arrangement of the proof follows from this; for example in the proof given by D.–W. of the unique factorization of ideals into prime ideals, where the "extended Gauss theorem" or a theorem equivalent to it can be avoided by using the fact that every linear form $(z - c)$ has infinitely many residue classes, namely all constants (whereas the number of residue classes modulo a prime number is finite).[246]

The further fact that every polynomial with integral rational functions of z as coefficients decomposes into linear factors mod. $(z - \alpha)$ (which is not the case for an integral polynomial mod. p) leads to genuinely new results; this fact also still holds if, instead of all constants, one takes only all algebraic numbers.[247] From this it follows, namely, that every prime ideal is an ideal of first degree;[248] and further, that the field discriminant D becomes the intersection of all discriminants $\Delta(\delta)$, where δ runs through all integral quantities;[249] both are in contrast to algebraic numbers. Subsequently it again follows from this that the ramification ideal is the intersection of all principal ideals $F'(\delta)$.

Further differences are conditioned by the fact that, for algebraic numbers, the ground field R of rational numbers is something absolutely distinguished, namely the prime field contained in the field (that is, the field derived from the unit); whereas the ground field $K(z)$ is something relative. Every nonconstant function ξ

[241] Emphasized especially by Dedekind (number theory); Supplement XI, theorem VII, § 177; compare the note on p. 554. The theorem uses the fact that algebraic fields are involved; it no longer holds for ideals (modules) of polynomials, where the least common multiple takes the place of the product.

[242] Number theory, Suppl. XI; theorem VI, § 173. Dedekind points out the equivalence of the two theorems (Über die Begründung der Idealtheorie, *Göttinger Nachr.* 1895).

[243] Dedekind, Über die Diskriminante endlicher Körper (*Abhandl. der Gött. Ges. d. Wissensch.* vol. 29, 1882). The definitions given there for algebraic numbers can be transferred directly to algebraic functions; the proofs too can be transferred almost exactly. The order of proof in D.–W. is different: see below. For the cited theorems compare also Zahlbericht, § 31 and 32. ($F'(\delta)$ is there called the "different" of the integer δ.)

[244] Compare Zahlbericht, § 39–45.

[245] Klein, Zur Theorie der Abelschen Funktionen (*Ann.* 36, 1890).

[246] D.–W. § 9. Compare the notes on pp. 212 and 213.

[247] D.–W. note in the introduction that in this case the whole theory remains intact.

[248] D.–W., § 9, no. 7.

[249] D.–W., p. 224.

may be chosen as independent variable; the field then becomes an algebraic field over $K(\xi)$. The theorem that every prime ideal $\mathfrak{p}$ is an ideal of first degree, hence every function is congruent modulo $\mathfrak{p}$ to a constant, combined with the possibility of taking an arbitrary function as independent variable, leads to the most important concept going beyond algebraic numbers, the point concept in purely arithmetic form. (Compare Hensel 10a.) "Every point is represented here by a field of constants isomorphic to the function field."[250] The totality of points so defined forms the "absolute Riemann surface". This definition of point depends only on the field, not on a special variable. The existence proof, however, is carried out by distinguishing some function as independent variable z; a point P is generated by a prime ideal $\mathfrak{p}$ by assigning to every function its constant residue modulo $\mathfrak{p}$; thus in particular the functions divisible by $\mathfrak{p}$ vanish at P, and conversely. As $\mathfrak{p}$ runs through all prime ideals in z, all points in which z is finite are thereby obtained; the remaining points arise by introducing the independent variable $\xi = 1/z$ and correspond to the prime ideals which divide the principal ideal ξ. Naturally, the concepts attached to the point concept also can have no analogue for algebraic numbers; such as the concept of a polygon, of a polygon class, of the dimension of a polygon class, and so on.

It should also be noted that the free choice of the independent variable leads to a further interpretation of the decomposition $F'(\vartheta) = \mathfrak{f} \cdot \mathfrak{D}$. Namely, if one considers ϑ as independent variable, then correspondingly one obtains

$$\left(\frac{\partial F}{\partial z}\right) = \mathfrak{f} \cdot \mathfrak{a},$$

since the ring derived from ϑ was defined symmetrically with respect to z and ϑ; and $\mathfrak{f}$ becomes the greatest common divisor of the two principal ideals $\frac{\partial F}{\partial z}$ and $\frac{\partial F}{\partial \vartheta}$; the conductor $\mathfrak{f}$ becomes at the same time the "double-point ideal" with respect to ϑ, z.[251]

4. Connection of the arithmetic foundation of series expansion with ideal theory

In Hensel–Landsberg the transcendental aid of series expansion and the point concept borrowed from function theory take the place of ideal theory. Hensel's recently given arithmetic foundation of series expansion for algebraic numbers and functions[252] can therefore be regarded as an equivalent of ideal theory. In fact this arithmetic foundation amounts to passing, for each prime ideal, to such a (transcendental) extension field in which this prime ideal becomes a principal ideal, so that the usual decomposition theorems hold; and it suffices to carry out this consideration for all prime ideals occurring in one prime number p, respectively one linear function $z - a$.

First one introduces a transcendental extension field formed by all series expansions in integral powers of $z - a$, respectively p (p-adic numbers), with in each case at most finitely many negative powers. In this field the irreducible equation defining the algebraic number field or function field decomposes into the product of ρ irreducible factors, corresponding to the decomposition of p, respectively $z - a$, into the product of the ρ "primary" ideals $\mathfrak{q}_1, \ldots, \mathfrak{q}_\rho$. The further representation of each primary ideal as a power of a prime ideal,

$$\mathfrak{q}_i = \mathfrak{p}_i^{\ell_i},$$

corresponds in Hensel to the introduction of a further field, algebraic with respect to the transcendental extension field. And in the case of algebraic functions this field can be defined by a pure equation; in the case of algebraic numbers, more generally, by an algebraically solvable equation. The simplification for algebraic functions is due to the fact that here no inertia and ramification fields (compare no. 3) occur.

The "terminating series" in a prime element π[253] that occurs in these investigations (and which does not yet require a transcendental convergence proof)

$$v = c_0 + c_1\pi + \cdots + c_{\sigma-1}\pi^{\sigma-1} - v_\sigma\pi^\sigma,$$

where $c_0, \ldots, c_{\sigma-1}$ are constants and v_σ denotes an integral element of the field, is found in exactly corresponding form in D.–W.[254] There it is derived from ideal theory and gives the basis for defining the order of vanishing (respectively becoming infinite) of a function at a point.

[250] Two fields are, as is well known, called "isomorphic" if they can be put in one-to-one correspondence in such a way that sum, difference, product, and quotient also correspond.

[251] D.–W. p. 263.

[252] Eine neue Theorie der algebraischen Zahlen (*Math. Zeitschr.* vol. 2, 1918) and Neue Begründung der arithmetischen Theorie der algebraischen Funktionen einer Variablen (*Math. Zeitschr.* vol. 4, 1919). For a more detailed exposition compare Hensel no. 44ff.

[253] Neue Begründung ..., formula (11).

[254] D.–W., p. 231 and § 15.

5. Comparison of the arithmetic theory of algebraic functions with the geometric theory. Different concept of invariance

"Invariant" means any concept characterized by the field alone. In the arithmetic theory this invariance is in general already given in the definition, which is stated with respect to all elements of the field, not with respect to some basis or distinguished variable; the point concept given above is an example. By contrast, the remaining theories start from some special variable or basis and show afterwards the independence of the concept from this special choice; invariance here becomes invariance under birational transformation.

Namely, suppose that the field arises on the one hand by adjoining the algebraic element s to $K(z)$, and on the other by adjoining σ to $K(\xi)$, where $f(z,s)=0$, respectively $F(\xi,\sigma)=0$, denote the irreducible equations for s, respectively σ; or, still more generally, by adjoining a finite number of elements to $K(\eta)$, with the corresponding irreducible equations

$$g_1(\eta,\tau_1)=0, \qquad g_2(\eta,\tau_1,\tau_2)=0,\ldots.$$

Then, by definition, all elements can be expressed rationally by z and s, just as by ξ and σ, and also by $\eta,\tau_1,\tau_2,\ldots$. In particular z and s are therefore rationally expressible by ξ and σ, and conversely; likewise z and s rationally by $\eta,\tau_1,\tau_2,\ldots$, and conversely. The plane curve $f(z,s)=0$ thus goes by "birational transformation" into the plane curve $F(\xi,\sigma)=0$, or also into the curve in the space of variables $\eta,\tau_1,\tau_2,\ldots$ defined by $g_1(\eta,\tau_1)=0; g_2(\eta,\tau_1,\tau_2)=0,\ldots$. Invariance therefore appears in the fact that a definition stated for one special curve remains valid for all curves obtained from it by birational transformation (in a space of arbitrarily many dimensions); for all curves of the "class" in Riemann's sense. A "point" is here defined as a coherent system of values $z=a, s=b$, so that $f(a,b)=0$; under birational transformation this point generally goes again into a coherent system of values α,β of $F(\xi,\sigma)=0$, but in special cases, when a,b was a "singular" point, it can correspond to several "points" on $F(\xi,\sigma)=0$; and one can always specify curves on which a "singular" point of $f(z,s)=0$ corresponds to a finite number of simple points lying separately or successively (resolution of singularities).[255] Likewise a point group again goes into a point group; and the number of its points is invariant as soon as one counts a singular point as as many points as the simple points corresponding to it under a suitable transformation. Since every curve can be transformed birationally into one with "ordinary multiple" points,[256] the geometric theory develops its concepts only for such special curves and then proves the invariance under every birational transformation, including one which leads to the most general singular points.

In particular, as a distinguished curve of the "class" — excluding the hyperelliptic case — one can introduce the so-called "normal curve of the φ's" in the space of $(p-1)$ dimensions, which has no singular points; here the φ's are the p "adjoint curves of order $(n-3)$" for the homogenized $f(z,s)=0$. A birational transformation of $f(z,s)=0$ into $F(\xi,\sigma)=0$ corresponds to a linear transformation of the φ's, so that this is invariance in the projective sense. The representation of all functions of the field as rational functions of the φ's is called the "invariant representation" in the geometric theory.[257]

The linear system of the φ's, or also of the "special groups" cut out by them, therefore goes into itself under every birational transformation, and is consequently characterized by the field alone. It corresponds to the differential class in the arithmetic theory, where the invariance is again shown directly in the definition, which is given by properties with respect to every function of the field as independent variable.

6. Continuation of the comparison. Singular points

The different conception of the point concept is connected with the fact that the "singular points" of curves, which cause special difficulties in the algebraic-geometric theory, do not occur at all in the foundations of the arithmetic theory; and consequently the "adjoint functions" are not required for the construction of the arithmetic theory, as they are in the geometric theory, but belong to special higher parts of the theory. On the other hand, the branch points do not occur in the foundations of the geometric theory; in fact, they also drop out in the representation of the integrals in Aronhold's normal form (Hensel no. 27, formula 80).

The definition of singular point given in 5 can be formulated as follows: $f(z,s)=0$ (respectively a curve in a higher-dimensional space) has a singular point at $z=a, s=b$ (respectively $z=a; s_i=b_i$) if this system of values is assumed by more than one point of the absolute Riemann surface (point in the arithmetic sense), or

[255] Compare, for instance, Brill–Noether, section 6. More precise information on "singular" points appears in the next section.

[256] Ibid.

[257] Compare, for instance, Brill–Noether, section 8. For algebraic functions of several variables the question whether invariance can be reduced to such invariance under linear transformation is not treated.

if it is assumed at one or several points with higher order. In the latter case one has a point of the ramification polygon of z as well as of s; this corresponds to the "successive points" of the geometric theory; the point becomes a (higher) cusp point of $f(z,s)=0$ (respectively of the curve in the higher-dimensional space).

Since by hypothesis every arbitrary function η of the field can be represented rationally by z and s, because of the isomorphism every point of the absolute Riemann surface at which $z=a, s=b$ is obtained by assigning to every function η, represented by z and s, its value for $z=a, s=b$. Thus, for the point to be singular, there must be at least one function η which assumes several values or one system of values several times for $z=a, s=b$. And because of rational representability by z and s, this can be the case only when in this representation numerator and denominator vanish simultaneously for $z=a, s=b$. It follows that the above definition is identical with the usual one, namely that

$$f, \quad \frac{\partial f}{\partial z}, \quad \frac{\partial f}{\partial s}$$

vanish for $z=a, s=b$. For in this case

$$\eta = \frac{\frac{\partial f}{\partial z}}{\frac{\partial f}{\partial s}}$$

is a function of the indicated kind which is multivalued for $z=a, s=b$, whereas in the opposite case every function, even if it becomes $0/0$, assumes only one value.

With the arithmetic point concept the singular point cannot occur, since two points are regarded as different as soon as only one function is assigned different systems of values. But even in the ideal theory serving to found the arithmetic point concept, the concept of a singular point does not occur, precisely because the totality of all integral quantities of the field is always considered at the same time. All singular points in the finite part of $F(z,\delta)=0$, where δ is algebraically integral with respect to z, are generated (according to the second definition and the end of no. 3) by the "double-point ideal" $\mathfrak{f}$. Now, since by the fundamental theorem the ramification ideal $\mathfrak{D}$ is the greatest common divisor of all principal ideals $F'(\delta)$, for every prime ideal $\mathfrak{p}$ one can specify a δ such that the corresponding $\mathfrak{f}$ is not divisible by $\mathfrak{p}$, and consequently — since $\mathfrak{f}$ is the greatest common divisor of $F'(\delta)$ and $F'(z)$ — at least one of the two principal ideals $F'(\delta)$ and $F'(z)$ is not divisible by $\mathfrak{p}$. Thus there is *no system of values for which all $F'(\delta)$ and $F'(z)$ vanish simultaneously*, and consequently the system of values $z=a, \delta_i=b_i$ is assumed at only one point of the absolute Riemann surface; the *totality of integral algebraic functions of the field* (which may be interpreted as a curve in a space of infinitely many dimensions) *has no singular point in the finite part*; only finite values enter into consideration in ideal theory.

But the basis functions $\omega_1,\dots,\omega_n$ of all integral quantities also have no singular points in the finite part. By the preceding, each point in the arithmetic sense is already uniquely determined by a system of constants isomorphically assigned to all integral functions. If, therefore, a "space curve" defined by any integral functions s_i has a singular point in the finite part for $s_i=\alpha_i$, then there must be at least one integral function η which assumes several systems of values for $s_i=\alpha_i$. If now $\omega_1,\dots,\omega_n$ denotes a basis of all integral quantities, then, because $\delta=a_1\omega_1+\cdots+a_n\omega_n$ with integral rational a_i, each system of values of the ω's corresponds to only one system of values of δ; hence the space curve defined by $\omega_1,\dots,\omega_n$ can possess no singular point in the finite part; and *every point of the absolute Riemann surface at which z is finite is already uniquely defined by a system of constants isomorphically assigned to the ω's. The basis functions are precisely the functions η which become multivalued for $s_i=\alpha_i$.* Thus by introducing the basis of integral quantities, for example in place of a basis $1,\delta,\dots,\delta^{n-1}$, the difficulty of singular points is avoided.

In the geometric theory the task of generating all points of the absolute Riemann surface uniquely — more generally, of forming all functions which become infinite at given points — is accomplished by means of adjoint forms and their quotients. A form ψ is called adjoint if $\psi=0$ has at least an $(i-1)$-fold point at each i-fold point of the base curve $f(z,s)=0$; a higher singular point must first be resolved into ordinary multiple points. It was shown above that, for the most general function $\eta=\frac{\psi}{\chi}$, numerator and denominator must vanish at all singular points of $f(z,s)=0$; that they must be adjoint is shown by considering the everywhere finite differentials. That, conversely, the conditions of adjointness are also sufficient for forming the most general function is shown by the "residue theorem", which will be discussed more closely below. Thus the adjoint functions take the place of the basis functions of the arithmetic theory.

Arithmetically, an adjoint form corresponds to the numerator of a function of the field divisible by the "double-point ideal" $\mathfrak{f}$, as soon as one assumes the singular points — which can always be achieved by transformation — all to lie in the finite part. From this the representation of the basis quantities $\omega_1,\dots,\omega_n$

as quotients of adjoint forms, and hence the connection among the various ways of representing them, is shown formally as follows:

Let $R(z)$ denote the determinant of substitution of a basis $1, \delta, \ldots, \delta^{n-1}$ into a basis $\omega_1, \ldots, \omega_n$. Conversely, every ω can then be expressed as a quotient of an integral function of z and δ by $R(z)$:

$$\omega_i = \frac{G_i(z, \delta)}{R(z)}.$$

Since $R(z)\omega_i$ belongs to the ring derived from δ, by the definition of the conductor $R(z)$ is divisible by $\mathfrak{f}$ [from $(R(z))^2 = \operatorname{Norm} \mathfrak{f}$ this divisibility follows only for $(R(z))^2$]. Since now ω_i, as an integral function, remains finite for all finite z, the numerator function $G_i(z, \delta)$ must also be divisible by $\mathfrak{f}$; numerator and denominator of all ω_i are therefore adjoint. From this representation of the ω's follows, however, the representation of all functions of the field by quotients of adjoints.

According to the preceding, the geometric theory can be characterized as the *theory of the ring derived from a quantity* δ (order), in contrast with the arithmetic theory, which considers *all integral quantities* simultaneously. Thus it is clear that the conductor of the ring — the singular points — plays a decisive role. Further analogies between the arithmetic theory of rings (orders) and the geometric theory can also be given. Thus the geometric theory does not count the intersection points that fall into the singular points in the intersection with adjoint curves, but considers only the point groups different from these. Corresponding to this is that Dedekind[258] considers only such ring ideals as are prime to the conductor ideal $\mathfrak{f}$, and for this domain of ideals sets up all divisibility laws, including unique decomposition into prime ideals.

Also the "fundamental theorem of algebraic functions"[259] underlying the geometric theory, or at least an immediate consequence of it, has in a certain sense an analogue in the arithmetic theory of rings. This theorem gives the conditions for a representability

$$\psi(z, s) = A(z, s)f(z, s) + B(z, s)\varphi(z, s),$$

where all quantities occurring are integral rational in s, z (more generally integral and homogeneous in x_1, x_2, x_3). From these conditions it follows,[260] by virtue of $f(z, s) = 0$, that the quotient $\frac{\psi}{\varphi}$ is always equal to an integral rational function $B(z, s)$ when it is adjoint to f. Corresponding to this is the theorem from no. 3, which underlies all these comparisons, that the conductor $\mathfrak{f}$ of a ring derived from δ is equal to the double-point ideal of $f(z, \delta) = 0$. For, by the definition of the conductor, every algebraic integral quantity divisible by $\mathfrak{f}$ belongs to the ring, and is therefore integral and rationally representable by z and δ; whereas the cited theorem says that such a quantity is an adjoint function.[261]

7. Comparison between the residue theorem and theorems of ideal theory

The just mentioned consequence of the "fundamental theorem of algebraic functions" leads in the geometric theory directly to the "residue theorem", and thus to the foundation on which the whole geometric theory is built. In a certain respect, however, the residue theorem again has a more elementary character than the remaining foundations, insofar as it can be stated arithmetically as a direct consequence of the unique factorization of ideals into prime ideals, or rather of the facts underlying this decomposition theorem, and therefore does not rest on the higher theory of rings. One is led to this arithmetic formulation as follows:

Assume the base curve $f(z, \delta) = 0$ — as can always be achieved by a linear fractional transformation of the variables — to be such that the singular points and the finitely many point groups under consideration lie in the finite part; and further that δ depends algebraically integrally on z, which can always be achieved by a subsequent linear transformation integral in the variables. Then to a prime ideal $\mathfrak{p}$ generating a point P of the absolute Riemann surface at which $z = a, \delta = b$, there corresponds the point $z = a, \delta = b$ of $f(z, \delta) = 0$; more generally, an ideal $\mathfrak{a} = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_\rho^{e_\rho}$ generates the point group consisting of the points $z = a_i, \delta = b_i$ with the corresponding multiplicity; and all points of $f(z, \delta) = 0$ lying in the finite part are so generated. Since at a point P all functions $g_1, g_2, \ldots$ divisible by $\mathfrak{p}$ vanish, and since conversely this totality of functions g generates the ideal $\mathfrak{p}$, the corresponding point on f is the intersection of

$$g_1(z, \delta) = 0, \quad g_2(z, \delta) = 0, \ldots$$

[258] Über die Anzahl der Idealklassen in den verschiedenen Ordnungen eines endlichen Körpers. (Festschrift Braunschweig 1877); § 3–5. (The theorems stated in these paragraphs for algebraic numbers hold literally for algebraic functions of one variable.)

[259] M. Noether, Über einen Satz aus der Theorie der algebraischen Funktionen. *Math. Ann.* vol. 6 (1873).

[260] Brill–Noether, no. 55.

[261] Following Dedekind, the assumption is made here throughout that δ is an algebraic integer (which can always be achieved by transformation). The case more exactly corresponding to geometry, $f(z, s) = 0$, where s may also depend algebraically fractionally on z, is treated by Hensel–Landsberg; compare Hensel no. 26 and 29.

with $f = 0$; the same holds for the point group corresponding to an ideal $\mathfrak{a}$. Indeed, by the theorem that every ideal is the greatest common divisor of two principal ideals, two such curves always suffice to cut out a point group. In particular, the point group corresponding to a principal ideal is cut out by a curve $g(z, \delta) = 0$ and conversely; the principal ideal corresponds to the complete intersection.

Two point groups A and B are called *corresidual* in the geometric theory if they can be completed with the same point group R to a full intersection. If these point groups are generated respectively by the ideals $\mathfrak{a}, \mathfrak{b}, \mathfrak{r}$, then between these ideals one has the relation

$$\mathfrak{a} \cdot \mathfrak{r} = (\alpha), \qquad \mathfrak{b} \cdot \mathfrak{r} = (\beta),$$

which shows that the ideals $\mathfrak{a}$ and $\mathfrak{b}$ are equivalent. Conversely, from the equivalence of two ideals, $\mathfrak{a} = \eta\mathfrak{b}$, this relation always follows on the basis of the theorem that every ideal can be transformed into a principal ideal by multiplication by another ideal;[262] *equivalent ideals and corresidual point groups are therefore identical concepts.*

The residue theorem now says: *If A and B are corresidual, then they can be cut out by adjoint curves with the same residue (outside the singular points).* Since the zeros and poles of a function are always corresidual, this proves at the same time that the most general functions of the field are representable as quotients of adjoint forms. In the language of ideal theory the residue theorem simply amounts to saying that, besides $\mathfrak{a}$ and $\mathfrak{b}$, $\mathfrak{a}\mathfrak{f}$ and $\mathfrak{b}\mathfrak{f}$ are also equivalent, where $\mathfrak{f}$ denotes the double-point ideal. For then, by the above, there always exists an ideal $\mathfrak{m}$ such that

$$\mathfrak{a}\mathfrak{f}\mathfrak{m} = (\alpha), \qquad \mathfrak{b}\mathfrak{f}\mathfrak{m} = (\beta)$$

become principal ideals; $\alpha = 0$ and $\beta = 0$ are the adjoint curves which cut out the point groups A and B, respectively their subgroups A' and B' different from the singular points.[263] The proof of the residue theorem is therefore arithmetically contained in the possibility of assigning points to prime ideals, and in the proof that equivalent ideals can be transformed into principal ideals by multiplication by one and the same ideal.

Corresidual point groups need not consist of the same number of points; just as equivalent ideals need not be of the same degree. Thus, for example, all principal ideals are equivalent, and all point groups generated by complete intersections are corresidual. This difference as against equivalent polygons (divisors)[264] is explained by the fact that those points at which z becomes infinite are not generated by prime ideals in z. In the representation of a function as an ideal quotient $\eta = \frac{\mathfrak{a}}{\mathfrak{b}}$, which states the equivalence of $\mathfrak{a}$ and $\mathfrak{b}$, the zeros or poles of η at which z becomes infinite therefore do not appear. For instance, the principal ideal (α) becomes the product of the prime ideals corresponding to the zeros of the integral function α; the poles of α do not occur, since α becomes infinite only at points where z becomes infinite. In this respect ideal theory is the exact analogue of form theory in the geometric theory, where there are likewise only zeros and no poles.

The passage from forms to functions is made geometrically by forming quotients of forms of the same order; arithmetically, this corresponds to the passage from ideal classes to the invariantly defined polygon (divisor) classes. For, since no variable is distinguished here any longer, the representation of a function by polygon quotients gives all zeros and poles. The relation of ideal class (class of corresidual point groups) to polygon class is the following: if $\mathcal{A}$ and $\mathcal{B}$ are the polygons generated by the ideals $\mathfrak{a}$ and $\mathfrak{b}$, and $\mathcal{N}$ is the polygon of the points at which z becomes infinite (the denominator polygon of z and of the integral functions of z), then from the equivalence of $\mathfrak{a}$ and $\mathfrak{b}$ follows the equivalence of $\mathcal{A}$ and $\mathcal{B}$ if and only if $\mathfrak{a}$ and $\mathfrak{b}$ have the same degree; in general only the equivalence of $\mathcal{A}\mathcal{N}^{\nu}$ with $\mathcal{B}\mathcal{N}^{\nu'}$ $(\nu \neq \nu')$ follows. The division of ideals (respectively of corresidual point groups) into classes is thus a gathering together of infinitely many polygon classes into one class; it may also be viewed as a division of the polygons into classes modulo a power of $\mathcal{N}$.[265]

The residue theorem for the case of corresidual point groups consisting of different numbers of points occurs only in purely geometric questions. In the application to algebraic functions, in fact only the case corresponding to polygon classes occurs, that of corresidual point groups of the same number. The "complete linear system" generated by a point group, that is, the totality of point groups corresidual to this point group and of the same number of points, corresponds exactly to the polygon class generated by the corresponding

[262] Compare Dedekind, number theory § 181; namely, if $\mathfrak{r}$ is chosen so that $\mathfrak{a}\mathfrak{r}$ becomes a principal ideal (α), then $\mathfrak{b}\mathfrak{r} = \eta^{-1} \cdot (\alpha)$; and since $\mathfrak{b}\mathfrak{r}$ contains only integral quantities, the same holds for $\eta^{-1} \cdot (\alpha)$, which is therefore also a principal ideal (β).

[263] If $\mathfrak{a} = \mathfrak{f}\mathfrak{a}_1$, $\mathfrak{b} = \mathfrak{f}\mathfrak{b}_1$, $\mathfrak{r} = \mathfrak{f}\mathfrak{r}_1$, where $\mathfrak{a}_1, \mathfrak{b}_1, \mathfrak{r}_1$ are relatively prime, then already $\mathfrak{a}_1\mathfrak{f}\mathfrak{n}$, $\mathfrak{b}_1\mathfrak{f}\mathfrak{n}$ become principal ideals, where $\mathfrak{n}$ is prime to $\mathfrak{f}$. For every ideal can be transformed into a principal ideal by multiplication with an ideal that is prime to a given one (D.–W. p. 214; or number theory, § 178, theorem XI).

[264] A polygon A consists of finitely many points of the absolute Riemann surface; two polygons A and B are called equivalent if they are the zeros and poles of a function η; η then admits the representation by polygon quotients $\eta = \frac{A}{B}$. (D.–W. § 17 and 18.)

[265] Hensel–Landsberg, 25th lecture, p. 438. — If one represents the two variables z and s by polygon quotients, then this can be understood as binary homogenizing of each of these variables; if one chooses in particular z and s so that they have the same denominator polygon, then the ternary homogenizing customary in geometry arises.

polygon; the concepts "sum of two complete systems" and "product of two polygon classes" also coincide. That such a complete system contains only finitely many linearly independent point groups is a direct consequence of the residue theorem, which reduces the dimension of the system (the number of linearly independent point groups) to the dimension of all adjoint curves of arbitrary but fixed order passing through the residual group. In order to arrive at the actual determination of this dimension, the Riemann–Roch theorem, the geometric theory first derives from the residue theorem the "reduction theorem" (theorem on the fixed points);[266] and from this the Riemann–Roch theorem.

Quite correspondingly, the arithmetic theory shows that every class of polygons is of finite dimension, and determines its dimension by a special basis of the integral quantities, a normal basis defined by means of the complementary module; thus it obtains the Riemann–Roch theorem as a consequence of the connection of the complementary module with the ramification polygon. In the presentation of D.–W. the reduction theorem also occurs,[267] though not as a foundation as in the geometric theory. In consequence of the correspondence mentioned in no. 6 between basis and adjoint functions, a certain correspondence of the auxiliary means in the two theories can also be established here. In summary, one may say that the formations of concepts in the two theories essentially coincide, even though, apart from formulation, they differ in their arrangement. The two theories arose independently of one another, the geometric one a decade earlier. It should also be noted that the Riemann–Roch theorem originally, in Riemann and Roch and also in Weierstrass, occurs as a rank theorem; what is at issue is the rank of the matrix $(\varphi_i(x_j))$, where the φ's run through the p adjoint curves of order $n-3$, and the x's through the points of the residual group.

8. Transcendental questions for algebraic functions and algebraic numbers

Hilbert in particular pointed out analogies between transcendental questions for algebraic functions and for algebraic numbers;[268] these are essentially existence questions. Thus Riemann's question concerning the existence of an algebraic function for a given Riemann surface (hence also for given branch points) corresponds to the question of the existence of algebraic number fields with prescribed group and discriminant.[269] Whereas in the case of algebraic functions the existence proof can be carried out in general, in the domain of algebraic number fields it has been achieved only for special ground fields (rational numbers, quadratic number fields) and only for Abelian groups.[270] This concerns Kronecker's theorem that every Abelian number field over the rational numbers can be assembled from fields of roots of unity, and the corresponding theorems of Fueter[271] and Hecke[272] for relatively Abelian number fields. Thus the desired number fields are actually supplied by a transcendental means, exponential functions, modular functions; in contrast with the pure existence proof for algebraic functions.

The existence proof of algebraic functions is preceded by the existence of everywhere finite integrals. An analogue of this is the existence of the "singular primary numbers" of an algebraic number field defined with respect to a prime number ℓ, whose ℓ-th roots are relatively unramified and supply the first building blocks for the class field.[273] This existence proof is carried out with essential help from the theorem that in the number field there always exist prime ideals with prescribed residue characters; this theorem may therefore be regarded as an analogue of the boundary-value problem on which the existence of the everywhere finite integrals rests.

The everywhere finite integrals provide a transcendental criterion for determining whether two polygons belong to the same class (two point groups of the same number are corresidual), namely by Abel's theorem, which says that the integrals of the first kind, extended over "corresidual paths", vanish; or by the corresponding differential theorem together with its converse.[274]

As the analogue of Abel's theorem for algebraic number fields one must take the "reciprocity law for ℓ-th

[266] Brill–Noether, no. 58.

[267] D.–W., § 30, no. 2.

[268] Mathematische Probleme, problem 12. (*Göttinger Nachrichten* 1900).

[269] The treatment of algebraic functions suggested thereby, starting from the group, was carried out by R. König as a special case of more general questions; compare especially: Riemannsche Funktionen- und Differentialsysteme in der Ebene (*J. f. M.* vol. 148) and *Math. Ann.* 78, 79. In place of the algebraic field, König has a special finite module with respect to $K(z)$: there is only "class property".

[270] Only the simple case of the groups occurring for equations of third and fourth degree, and some quite special groups for equations of degree 8, can be handled for arbitrary ground fields: G. Bucht, Über einige algebraische Körper 8. Grades (*Arkiv f. Math., Astron. och Physik*, vol. 6, no. 30.) Compare also E. Noether, Gleichungen mit vorgeschriebener Gruppe and the Seidelmann dissertation cited there. (*Math. Ann.* 78).

[271] *Math. Ann.* 75.

[272] *Math. Ann.* 76.

[273] Furtwängler, Reziprozitätsgesetze für Potenzreste mit Primzahlexponenten in algebraischen Zahlkörpern. *Math. Ann.* 67 and 72.

[274] Compare, for example, the formulation in M. Noether (*Ann.* 37). Since the residue theorem historically grew out of Abel's theorem, the geometric theory speaks of the "sum" of point groups and linear systems, following the integral sums; in contrast to the "product" of the classes in the arithmetic theory.

power residues", which gives, for the field itself or at least for a suitable upper field, a criterion that two ideals belong to the same class. Its content can also be formulated by saying that in this upper field the singular primary numbers have the same residue character with respect to all ideals of the same class. This latter fact also holds in the field itself, but there it does not coincide with the reciprocity law.[275]

Finally, Hensel's theory of p-adic numbers supplies the analogue of the power-series expansion of algebraic functions; for this, see Hensel's report.

[275] Furtwängler, loc. cit.

15. The Finiteness of the System of Integral Invariants of Binary Forms

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Presented by F. Klein in the session of 27 March 1919.

In what follows, as an extension of the known finiteness theorem, the following theorem is proved: *All polynomial integral invariants of a binary system of ground forms are polynomial rational functions, with integral coefficients, of a finite number among them.* Here, as usual, by an "invariant" one always means a polynomial rational invariant of a ground form or of a system of ground forms, namely an invariant with respect to the coefficient transformation induced by the group of all linear transformations.

The proof rests on an integrality sharpening of the Mertens-Hilbert proof of finiteness.[276] This proof of finiteness may be characterized as follows. One decomposes the ground forms into their linear factors - more precisely, one associates to each ground form a product of linear forms - whereby each invariant I is changed into an invariant of these linear forms, hence into a function of the homogenized roots. This function has to satisfy three conditions: 1) the condition of invariance, 2) weight conditions, and 3) symmetry conditions. Conditions 2) and 3) are necessary for I to become polynomial and rational in the coefficients of the ground forms. Condition 1) is satisfied by representing I as a polynomial rational function of the determinants of the linear forms; condition 2) by taking, instead of the individual determinants, certain products of powers of them as new arguments. Condition 3) then simply says that I becomes a polynomial rational symmetric function of rows of quantities, and that conversely every such symmetric function of rows of quantities becomes an invariant of the ground forms. The elementary symmetric functions of these rows of quantities therefore give the complete system.

If one considers only integral invariants, condition 1) can be satisfied just as above as soon as, for invariants of linear forms, the sharper theorem is proved that *all integral invariants of binary linear forms are polynomial integral functions of the determinants of these linear forms.* I show that this is in fact the case in § 3, in the form that the totality of all relations modulo any prime between these determinants coincides with the totality of all identical relations; that is, the module corresponding to these relations remains a prime module modulo every prime number. This verification is carried out by normalizing the residue classes with respect to the module of quadratic forms corresponding to the quadratic relations between the determinants; these quadratic forms thereby turn out to form an integral basis of the module, modulo every prime, whereas previously they were known as a module basis only when integrality was disregarded.

To satisfy condition 2) no alteration is needed, since the introduction of products of powers does not affect the integer coefficients. Condition 3) now says that $n_1! \cdots n_\mu! I$ (where $n_1, \dots, n_\mu$ are the degrees of the individual ground forms) becomes an integral symmetric function of rows of quantities, and conversely every integral symmetric function of these rows becomes an integral invariant. But, as is easily shown, these integral symmetric functions of rows possess an integral basis, for which the elementary functions alone no longer suffice. The application of Hilbert's theorem on the integral module basis then also yields an integral basis for the I themselves.

Section 1 collects the special finiteness theorems corresponding to conditions 1), 2), and 3), from which the proof of finiteness follows in § 2. The special finiteness theorems corresponding to condition 3) also yield the finiteness of the integral invariants of finite groups of integral or algebraic-integral substitutions, as is briefly carried out in § 4.

§ 1. Special Finiteness Theorems

I first state the following special finiteness theorems, which will serve to satisfy the conditions of invariance, weight, and symmetry mentioned in the introduction.

I. Every integral invariant of a system of binary linear forms is a polynomial rational function, with integral coefficients, of the determinants of these linear forms. The proof will be given in § 3.

II. Every system $\mathfrak{S}$ of products of powers of n variables with nonnegative exponents, which together with any two products of powers f and g also contains their quotient whenever this quotient is polynomial in the

[276] F. Mertens, Wiener Sitzungsberichte Jan. 1889; D. Hilbert, *Math. Ann.* vol. 33 (1889) [dated March 1888]. The two proofs, which arose independently of one another following Mertens, Crelle vol. 100, agree in content.

variables, has a finite multiplicative basis $f_1, \ldots, f_k$ consisting of functions from $\mathfrak{S}$, such that for every f in $\mathfrak{S}$ one has a representation

$$f = f_1^{\lambda_1} \cdots f_k^{\lambda_k}$$

with nonnegative exponents $\lambda_1, \ldots, \lambda_k$.

Indeed, by Hilbert's theorem on the module basis there exists a finite number of functions from $\mathfrak{S}$, say $f_1, \ldots, f_k$, each of which really contains the x's, such that every nonconstant f from $\mathfrak{S}$ belongs to the module generated by them:

$$f \equiv 0 \pmod{(f_1, \ldots, f_k)}.$$

It follows that for every $f = x_1^{i_1} \cdots x_n^{i_n}$ there is at least one $f_\nu = x_1^{\nu_1} \cdots x_n^{\nu_n}$ such that $\nu_1 \leq i_1, \ldots, \nu_n \leq i_n$. In the representation

$$f = x_1^{i_1 - \nu_1} \cdots x_n^{i_n - \nu_n} f_\nu = A f_\nu$$

the factor A belongs to $\mathfrak{S}$ by hypothesis, but it has lower degree in the variables than f. Hence by repeating the argument finitely many times the assertion follows. In order that the representation also hold for $f = 1$, one need only put all exponents λ equal to zero.[277]

III. All polynomial integral symmetric functions of n rows of quantities are polynomial integral functions of a finite number among them.

Let $\tau_1(x), \ldots, \tau_n(x)$ denote the elementary symmetric functions of the indeterminates $x_1, \ldots, x_n$. Then for every exponent k there is an identity

$$x_i^k = G_0^{(k)}(\tau) + x_i G_1^{(k)}(\tau) + \cdots + x_i^{n-1} G_{n-1}^{(k)}(\tau), \qquad (i = 1, 2, \ldots, n),$$

where $G_0^{(k)}, \ldots, G_{n-1}^{(k)}$ are polynomial integral functions of the $\tau(x)$. If $x_i, y_i, \ldots, z_i$ are the elements of the i-th row, then by these identities and the corresponding ones for $y, \ldots, z$, all special, one-rowed symmetric functions

$$\sum_{i=1}^{n} x_i^a y_i^b \cdots z_i^c$$

become polynomial integral functions of the finitely many

$$\sum_{i=1}^{n} x_i^a y_i^b \cdots z_i^c \qquad (0 \leq a < n,\ 0 \leq b < n,\ \ldots,\ 0 \leq c < n)$$

and of the elementary symmetric functions $\tau(x), \tau(y), \ldots, \tau(z)$.[278] If one has the most general symmetric functions of the n rows - a case not occurring below - it is shown in the same way that the functions

$$\sum x_{i_1}^{a_1} y_{i_1}^{b_1} \cdots z_{i_1}^{c_1} x_{i_2}^{a_2} y_{i_2}^{b_2} \cdots z_{i_2}^{c_2} \cdots x_{i_n}^{a_n} y_{i_n}^{b_n} \cdots z_{i_n}^{c_n}$$
$$(0 \leq a_\nu < n,\ 0 \leq b_\nu < n,\ \ldots,\ 0 \leq c_\nu < n),$$

where $i_1, i_2, \ldots, i_n$ runs through all permutations, together with $\tau(x), \tau(y), \ldots, \tau(z)$, form an integral basis.

IV. Let $F_1(x), \ldots, F_k(x)$ be polynomial integral functions of $x_1, \ldots, x_n$, and let a be a fixed integer. Then all functions

$$\frac{G(F)}{a},$$

where G denotes a polynomial integral function, which become integral in the x's, are polynomial integral functions of a finite number among them.

Indeed, by Hilbert's theorem on the integral module basis, for each such $G(F)/a$ one has a representation

$$\frac{G(F)}{a} = A_1(F_1 \ldots F_k) \frac{G_1(F)}{a} + \cdots + A_\rho(F_1 \ldots F_k) \frac{G_\rho(F)}{a},$$

[277] This representation can also be proved directly, and conversely Hilbert's theorem follows from it: compare P. Gordan, Neuer Beweis des Hilbertschen Satzes über homogene Funktionen, Gött. Nachr. 1899. Another direct proof of this representation, from which Theorem II is also derived, is in A. Ostrowski, Über die Existenz einer endlichen Basis bei gewissen Funktionensystemen, *Math. Ann.* 78 (1916), § 2,1 and § 3,2. The special case of Theorem II used in § 2, where the exponents run through all nonnegative solutions of a Diophantine system of equations, is already found in Gordan-Kerschensteiner, *Vorlesungen über Invariantentheorie*, vol. I, p. 199.

[278] Hilbert, loc. cit., takes the power sums instead of the $\tau(x), \ldots, \tau(z)$ into the basis; this makes the basis consist only of one-rowed functions, but the integrality of the representation is then lost.

where $A_1, \ldots, A_\rho$ are polynomial integral functions of the F's, and

$$\frac{G_1(F)}{a}, \ldots, \frac{G_\rho(F)}{a}$$

belong to the system. The functions

$$F_1 = \frac{aF_1}{a}, \ldots, F_k = \frac{aF_k}{a}; \quad \frac{G_1(F)}{a}, \ldots, \frac{G_\rho(F)}{a}$$

therefore form the required basis.

§ 2. Finiteness of the Integral Binary Invariants

Let x, y be binary variables subjected to a transformation with indeterminate coefficients

$$x = s_{11}x' + s_{12}y', \qquad y = s_{21}x' + s_{22}y', \tag{1}$$

and let f be a binary ground form whose coefficients are indeterminates. Then by (1),

$$f = a_0x^n + a_1x^{n-1}y + \cdots + a_ny^n = a_0'x'^n + a_1'x'^{n-1}y' + \cdots + a_n'y'^n. \tag{2}$$

By an integral invariant of f one understands, as is well known, a polynomial integral function $I(a)$ such that

$$I(a') = |s_{ik}|^\rho I(a), \qquad \text{identically in } a, s_{ik}. \tag{3}$$

The integral invariants of a system of ground forms are defined correspondingly; one may include homogeneity in the coefficients of the individual ground forms in the definition, since from such individually homogeneous invariants the remaining ones are composed in an integral-linear way.

In addition to (2), consider a special binary form g, namely the product of n linear forms whose coefficients are again indeterminates:[279]

$$\begin{aligned} g &= (\alpha_1x + \beta_1y)(\alpha_2x + \beta_2y)\cdots(\alpha_nx + \beta_ny) \\ &= \sigma_0(\alpha,\beta)x^n + \sigma_1(\alpha,\beta)x^{n-1}y + \cdots + \sigma_n(\alpha,\beta)y^n \\ &= (\alpha_1'x' + \beta_1'y')(\alpha_2'x' + \beta_2'y')\cdots(\alpha_n'x' + \beta_n'y') \\ &= \sigma_0'x'^n + \sigma_1'x'^{n-1}y' + \cdots + \sigma_n'y'^n. \end{aligned} \tag{4}$$

Thus $\sigma_0(\alpha,\beta), \sigma_1(\alpha,\beta), \ldots, \sigma_n(\alpha,\beta)$ denote the homogenized elementary symmetric functions, and the transformed coefficients σ_i' become equal to $\sigma_i(\alpha',\beta')$.

Because of the algebraic independence of these elementary functions, to every relation

$$G(\sigma_0, \sigma_1, \ldots, \sigma_n) = 0 \quad \text{identically in } \alpha, \beta \tag{5}$$

there corresponds a further relation

$$G(\sigma_0, \sigma_1, \ldots, \sigma_n) = 0 \quad \text{identically in } \sigma_0, \ldots, \sigma_n, \tag{6}$$

and hence, for indeterminates $a_0, \ldots, a_n$,

$$G(a_0, a_1, \ldots, a_n) = 0 \quad \text{identically in } a_0, \ldots, a_n. \tag{7}$$

Now every integral invariant $I(a)$ of f, by the substitution $a_i = \sigma_i(\alpha,\beta)$, gives an integral invariant $I(\sigma) = H(\alpha,\beta)$ of g, where H becomes an integral function of α, β. For this invariant the relation (3) becomes

$$I(\sigma') = |s_{ik}|^\rho I(\sigma) \quad \text{identically in } \sigma, s_{ik}, \tag{8}$$

and since $\sigma' = \sigma(\alpha',\beta')$, (8) gives

$$H(\alpha',\beta') = |s_{ik}|^\rho H(\alpha,\beta) \quad \text{identically in } \alpha, \beta; s_{ik}. \tag{9}$$

[279] If, as is often done in invariant theory, one writes f with binomial coefficients, $f = b_0x^n + \binom{n}{1}b_1x^{n-1}y + \cdots + b_ny^n$, then the domain of all integral invariants $I(b)$ is enlarged relative to that of the $I(a)$, and the finiteness arguments used here fail; $I(b)$ is no longer an integral function of the roots.

Thus $I(\sigma) = H(\alpha, \beta)$ becomes an integral invariant of the n linear forms $(\alpha_i x + \beta_i y)$. Conversely, suppose $H(\alpha, \beta)$ is an integral invariant of these n linear forms, hence satisfies (9), and is at the same time equal to an integral function $I(\sigma)$ of the σ's. Since then

$$H(\alpha', \beta') = I(\sigma(\alpha', \beta')) = I(\sigma'),$$

equation (9) can also be written

$$I(\sigma') = |s_{ik}|^{\rho} I(\sigma) \quad \text{identically in } \alpha, \beta; s_{ik}. \tag{9a}$$

It follows from (5) and (6) that (8) holds, and from (7) that (3) holds. Therefore to every integral invariant $I(a)$ there corresponds an integral invariant $H(\alpha, \beta) = I(\sigma)$ of the n linear forms, and conversely. Moreover, if $I_1(a), \dots, I_k(a)$ is an integral basis of all $I(a)$, then $H_1(\alpha, \beta), \dots, H_k(\alpha, \beta)$ is an integral basis of all $H(\alpha, \beta)$ which are at the same time integral functions $I(\sigma)$; here too the converse holds by (5), (6), and (7). Thus, to prove the existence of an integral basis for all $I(a)$, it suffices to prove it for all integral invariants $H(\alpha, \beta)$ which are at the same time integral functions $I(\sigma)$; and the basis

$$H_1(\alpha, \beta) = I_1(\sigma), \dots, H_k(\alpha, \beta) = I_k(\sigma)$$

then supplies the basis $I_1(a), \dots, I_k(a)$ for the $I(a)$.

Let $H(\alpha, \beta) = I(\sigma)$ run through the totality of all integral invariants of (4). Since by (3) every invariant is homogeneous in the σ's, and by (4) each σ is homogeneous and linear in the coefficients of each linear form, $H(\alpha, \beta)$ is homogeneous in each row α_i, β_i and of the same dimension in each row; moreover it is of course a symmetric function of the n rows. Conversely, by the fundamental theorem on symmetric functions, every integral symmetric function $H(\alpha, \beta)$ of the n rows, homogeneous separately and of the same dimension in each row, is a polynomial integral function of the σ's. Thus, by the preceding argument, it is an invariant $I(\sigma)$ as soon as $H(\alpha, \beta)$ is invariant. The totality of the integral invariants of (4) is therefore characterized as the totality of integral functions $H(\alpha, \beta)$ with the following properties:

1) integral invariant of the n linear forms $\alpha_i x + \beta_i y$;
2) separately homogeneous and of the same dimension in each of the n rows α_i, β_i;
3) symmetric in the n rows α_i, β_i.

By the special finiteness theorem I, $H(\alpha, \beta)$, because of 1), becomes a polynomial integral function of the determinants

$$(ik) = \alpha_i \beta_k - \beta_i \alpha_k;$$

hence

$$H(\alpha, \beta) = G((ik)).$$

In this representation the symmetry can also be expressed formally by applying all permutations of the n rows to $G((ik))$:

$$n!\, H(\alpha, \beta) = G^{(1)}((ik)) + G^{(2)}((ik)) + \dots + G^{(n!)}((ik)) = G^*((ik)). \tag{10}$$

Along with each product of powers P of the (ik), G^* therefore also contains the sum $Q = P^{(1)} + \dots + P^{(n!)}$ over all permutations of P. In fact G^* becomes an integral linear combination of finitely many Q's, $G^* = \sum cQ$. Since the Q's satisfy conditions 1), 2), and 3), and hence themselves become invariants, it suffices to consider the Q's in place of $G^*((ik))$.

Now let P run through the system $\mathfrak{S}$ of all products of powers of the (ik) occurring in the Q's, i.e. satisfying condition 2). If the quotient of two P's is polynomial in the (ik), then it also satisfies condition 2), and hence belongs to $\mathfrak{S}$. Therefore, by Theorem II, there are finitely many such P's, say $P_1, \dots, P_\rho$, such that every P is representable in the form

$$P = P_1^{\lambda_1} \cdots P_\rho^{\lambda_\rho}$$

with nonnegative exponents λ. Applying all permutations gives

$$Q = P_1^{(1)\lambda_1} \cdots P_\rho^{(1)\lambda_\rho} + \dots + P_1^{(n!)\lambda_1} \cdots P_\rho^{(n!)\lambda_\rho}.$$

Thus Q becomes an integral symmetric function of the $n!$ rows of quantities, each with ρ elements:

$$P_1^{(1)} \dots P_\rho^{(1)}; \quad P_1^{(2)} \dots P_\rho^{(2)}; \quad \dots; \quad P_1^{(n!)} \dots P_\rho^{(n!)};$$

hence, by Theorem III, it is a polynomial integral function of finitely many symmetric functions of these rows. These basis functions are themselves quantities Q or elementary symmetric functions of the $n!$ elements $P_i^{(1)}, P_i^{(2)}, \ldots, P_i^{(n!)}$, and therefore satisfy conditions 1), 2), and 3) and become integral invariants

$$H_1(\alpha, \beta) = I_1(\sigma); \ldots; \quad H_l(\alpha, \beta) = I_l(\sigma). \tag{11}$$

By (10), since $G^* = \sum cQ$, every $H(\alpha, \beta) = I(\sigma)$ is therefore of the form

$$H(\alpha, \beta) = I(\sigma) = \frac{1}{n!}\Phi(I_1, \ldots, I_l),$$

where Φ denotes a polynomial integral function; conversely, $\frac{1}{n!}\Phi$ is an integral invariant as soon as it is polynomial in α, β. Hence by Theorem IV finitely many further invariants $I_{l+1}(\sigma), \ldots, I_k(\sigma)$ can be adjoined to (11) in such a way that

$$H_1(\alpha, \beta) = I_1(\sigma); \ldots; H_l(\alpha, \beta) = I_l(\sigma);$$
$$H_{l+1}(\alpha, \beta) = I_{l+1}(\sigma); \ldots; H_k(\alpha, \beta) = I_k(\sigma)$$

form an integral basis of all invariants $H(\alpha, \beta) = I(\sigma)$. This proves the theorem for the case of one ground form.

If one is dealing with a system of ground forms, then, as noted above, one may assume homogeneity of the invariants in the coefficients of each ground form. One again replaces the general ground forms by ones which are respectively products of linear forms. Their invariants then become: 1) invariants of these linear forms; 2) separately homogeneous in the rows formed from the coefficients of the linear forms, and of equal dimension in the rows belonging to each individual ground form; 3) symmetric in the rows belonging to each individual ground form. Conversely, each integral invariant of all the linear forms satisfying these conditions again corresponds to an integral invariant of the system of ground forms. The proof is absolutely parallel to that just given. To each product of powers $P = P_1^{\lambda_1} \cdots P_\rho^{\lambda_\rho}$ of the individual determinants satisfying the weight conditions (Theorems I and II) one must apply all

$$N = n_1! n_2! \cdots n_\mu!$$

permutations, where $n_1, \ldots, n_\mu$ denote the degrees of the ground forms. In this way $N \cdot I$ is converted into an integral symmetric function of N rows, each with ρ elements. From this one obtains the integral basis for all $N \cdot I$, and consequently, by Theorem IV, the integral basis for all I.

§ 3. The Basis of the Integral Invariants of Binary Linear Forms

It remains to supply the proof of the special finiteness theorem I. By a known theorem, first proved by Clebsch, all polynomial rational invariants of the n linear forms $\alpha_i x + \beta_i y$ are polynomial rational functions of the determinants (ik). In particular, therefore, all integral invariants of these linear forms become polynomial rational, though not necessarily integral, functions of these determinants.

It will be shown below that, in view of the identical relations existing between the (ik), this representation is always possible integrally.

In formulas, and in the language of module theory, this is expressed as follows. Let ξ_{ik}, $i < k$, be indeterminates. Then the totality of all integral polynomials $F(\xi_{ik})$ with the property that $F((ik))$ vanishes identically in α, β forms a module $\mathfrak{M}$; the sum of two such F's, and also the product of such an F with an arbitrary integral polynomial in the ξ_{ik}, again belongs to this totality. In formulas:

$$F((ik)) = 0 \quad \text{identically in } \alpha, \beta$$

implies

$$F(\xi_{ik}) \equiv 0 \pmod{\mathfrak{M}} \quad \text{identically in } \xi_{ik},$$

and conversely.

I now show that the following completely analogous statement holds. From

$$F((ik)) \equiv 0 \pmod{p} \quad \text{identically in } \alpha, \beta$$

$$\text{it follows that} \qquad F(\xi_{ik}) \equiv 0 \pmod{(\mathfrak{M}, p)} \quad \text{identically in } \xi_{ik}, \tag{1}$$

and conversely, for every prime number p. Hence $\mathfrak{M}$ is also identical with the module $\mathfrak{M}_p$ corresponding to the totality of all relations modulo p among the (ik); this module consists of all integral polynomials $F(\xi_{ik})$ for which $F((ik))$ vanishes identically modulo p in α, β.

That (1) asserts the integral representability is seen as follows. It can also be written in the form: from $F((ik)) = p \cdot H(\alpha, \beta)$, identically in α, β, it follows that

$$F(\xi_{ik}) - p \cdot G(\xi_{ik}) \equiv 0 \pmod{\mathfrak{M}} \quad \text{identically in } \xi_{ik},$$

and hence, by the definition of $\mathfrak{M}$,

$$F((ik)) = p \cdot G((ik)) \quad \text{identically in } \alpha, \beta.^{280}$$

Thus $p \cdot H(\alpha, \beta) = p \cdot G((ik))$, or

$$H(\alpha, \beta) = G((ik)).$$

If, therefore, a representation is given

$$H(\alpha, \beta) = \frac{1}{p_1^{\lambda_1} \cdots p_\rho^{\lambda_\rho}} F((ik)),$$

where $p_1, \ldots, p_\rho$ are prime numbers, then it follows also that

$$H(\alpha, \beta) = \frac{1}{p_1^{\lambda_1 - 1} \cdots p_\rho^{\lambda_\rho}} G((ik)),$$

and by repeating this finitely many times one obtains the integrality of the representation.

For the proof of (1), denote by

$$f_{ik,rs} = \xi_{ik}\xi_{rs} - \xi_{ir}\xi_{ks} + \xi_{is}\xi_{kr}, \qquad i < k < r < s,$$

the forms in the indeterminates ξ_{ik} corresponding to the quadratic relations between the (ik),

$$(ik)(rs) - (ir)(ks) + (is)(kr) = 0,$$

and denote by $\mathfrak{N} = (f_{ik,rs})$ the module of integral polynomials generated from them. From

$$F(\xi_{ik}) \equiv 0 \pmod{\mathfrak{N}}$$

it follows that

$$F((ik)) = 0 \quad \text{identically in } \alpha, \beta,$$

and

$$\begin{array}{rcl} \text{from } F(\xi_{ik}) & \equiv & 0 \pmod{(\mathfrak{N}, p)} \\ \text{it follows that } F((ik)) & \equiv & 0 \pmod{p} \quad \text{identically in } \alpha, \beta. \end{array} \tag{2}$$

I shall now show, by normalizing the residue classes with respect to $\mathfrak{N}$, that the converse of (2) also holds. Then $\mathfrak{N} = \mathfrak{M} = \mathfrak{M}_p$, and (1) will be proved. The equality $\mathfrak{N} = \mathfrak{M}$ is obtained as the special case $\mathfrak{N} = \mathfrak{M}_p$ for $p = 0$; the case $p = 0$ is always included in the following considerations. In other words, the $f_{ik,rs}$ form an integral module basis of $\mathfrak{M}_p$, and in particular also of $\mathfrak{M}_0 = \mathfrak{M}$.

a) The case of two or three rows α_i, β_i

Since no four different indices occur, $\mathfrak{N}$ is the zero module. It must therefore be shown that from

$$F((ik)) \equiv 0 \pmod{p} \quad \text{identically in } \alpha, \beta \tag{3}$$

it follows that

$$F(\xi_{ik}) \equiv 0 \pmod{p} \quad \text{identically in } \xi_{ik}.$$

This proves (1) with $\mathfrak{M} = \mathfrak{M}_p = 0$.

Since the condition $F((ik)) = p \cdot H(\alpha, \beta)$ must be satisfied separately by the components homogeneous in the individual rows, it is enough throughout to assume $F((ik))$ homogeneous in each individual row, and

[280] This latter conclusion simultaneously proves the converse of (1).

hence $F(\xi_{ik})$ homogeneous in each index. In the case of two rows there is only the one determinant (12) and the corresponding indeterminate ξ_{12}. By the assumed homogeneity,

$$F = c\,\xi_{12}^{\nu}.$$

From

$$c\,(12)^{\nu} \equiv 0 \pmod{p} \quad \text{identically in } \alpha, \beta$$

it follows, since $(12) \not\equiv 0 \pmod{p}$ identically in α, β, that $c \equiv 0 \pmod{p}$; hence

$$F(\xi_{ik}) \equiv 0 \pmod{p} \quad \text{identically in } \xi_{ik}.$$

This proves (3), and hence (1).

In the case of three rows there are the determinants $(12), (13), (23)$, corresponding to the indeterminates $\xi_{12}, \xi_{13}, \xi_{23}$. Thus

$$F = \sum c\,\xi_{12}^{\tau_{12}}\xi_{13}^{\tau_{13}}\xi_{23}^{\tau_{23}}.$$

Let F have respectively the degrees $\sigma_1, \sigma_2, \sigma_3$ in the indices $1, 2, 3$. Then

$$\sigma_1 = \tau_{12} + \tau_{13}, \qquad \sigma_2 = \tau_{12} + \tau_{23}, \qquad \sigma_3 = \tau_{13} + \tau_{23},$$

with the unique inverse

$$\tau_{12} = \frac{1}{2}(\sigma_1 + \sigma_2 - \sigma_3), \quad \tau_{13} = \frac{1}{2}(\sigma_1 - \sigma_2 + \sigma_3), \quad \tau_{23} = \frac{1}{2}(-\sigma_1 + \sigma_2 + \sigma_3).$$

Thus each separately homogeneous F contains at most one term,

$$F = c\,\xi_{12}^{\tau_{12}}\xi_{13}^{\tau_{13}}\xi_{23}^{\tau_{23}},$$

from which, as above, (3), and consequently (1), follows.

b) The case of four rows α_i, β_i

The six indeterminates $\xi_{12}, \xi_{13}, \xi_{14}, \xi_{23}, \xi_{24}, \xi_{34}$, and the corresponding determinants, occur; and $\mathfrak{N} = (f_{12,34})$. Since

$$\xi_{14}\xi_{23} \equiv \xi_{13}\xi_{24} - \xi_{12}\xi_{34} \pmod{\mathfrak{N}}, \tag{4}$$

every product of powers of the ξ_{ik} is congruent modulo $\mathfrak{N}$ to an integral linear combination of such products of powers in which the product $\xi_{14}\xi_{23}$ does not occur. Thus for every polynomial $F(\xi_{ik})$,

$$F(\xi_{ik}) \equiv F_0(\xi_{ik}) \pmod{\mathfrak{N}}, \tag{5}$$

where F_0 contains no product $\xi_{14}\xi_{23}$. The polynomial F_0 will be called the normalized polynomial of the residue class of F. Since (4) is separately homogeneous, a separately homogeneous F again corresponds to a separately homogeneous F_0.

Now put

$$F_0(\xi_{ik}) = G(\xi_{ik}) + \xi_{34}F_1(\xi_{ik}), \tag{6}$$

where G contains no product of powers divisible by ξ_{34}, and further put

$$[G]_{\xi_{14}=\xi_{13}} = K(\xi_{12}, \xi_{13}, \xi_{23}). \tag{7}$$

From

$$F((ik)) \equiv 0 \pmod{p} \quad \text{identically in } \alpha, \beta$$

it follows by (5), taking account of (2), that

$$F_0((ik)) \equiv 0 \pmod{p} \quad \text{identically in } \alpha, \beta. \tag{8}$$

By specializing $\alpha_4 = \alpha_3,\ \beta_4 = \beta_3$, (6) and (7) give

$$K((ik)) \equiv 0 \pmod{p} \quad \text{identically in } \alpha, \beta.$$

By the result proved under a), therefore,

$$K(\xi_{ik}) \equiv 0 \pmod p \quad \text{identically in } \xi_{ik}.$$

From this follows what remains to be shown below:

$$G(\xi_{ik}) \equiv 0 \pmod p \quad \text{identically in } \xi_{ik}.$$

Thus (6) gives

$$F_0(\xi_{ik}) \equiv \xi_{34} F_1(\xi_{ik}) \pmod p \quad \text{identically in } \xi_{ik}.$$

Since $(34) \not\equiv 0 \pmod p$ identically in α, β, it follows from (8) that

$$F_1((ik)) \equiv 0 \pmod p \quad \text{identically in } \alpha, \beta,$$

where $F_1(\xi_{ik})$ is again normalized and of lower degree in ξ_{34}. Repeating the procedure finitely many times yields

$$F_0(\xi_{ik}) \equiv \xi_{34}^{\lambda} F_\lambda(\xi_{ik}) \pmod p,$$

where F_λ is of degree zero in ξ_{34}, and hence vanishes modulo p by what has just been proved. Therefore

$$\begin{array}{rcll} F_0(\xi_{ik}) & \equiv & 0 \pmod p \quad \text{identically in } \xi_{ik}, & \text{or} \\ F(\xi_{ik}) & \equiv & 0 \pmod{(\mathfrak{N}, p)} \quad \text{identically in } \xi_{ik}. & \end{array} \tag{9}$$

This proves the converse of (2), and hence (1), at the same time in the sharper assertion for $F_0(\xi_{ik})$.[281]

It remains only to add that $G(\xi_{ik}) \equiv 0 \pmod p$ follows from $K(\xi_{ik}) \equiv 0 \pmod p$; equivalently, from $G(\xi_{ik}) \not\equiv 0 \pmod p$ it follows that $K(\xi_{ik}) \not\equiv 0 \pmod p$. The normalization is needed only for this part of the induction. Under the substitution $\xi_{14} = \xi_{13}$, the individual products of powers of G do not vanish. A vanishing of K could therefore occur only because distinct products of powers in G correspond to equal products in K; and that in turn can happen only when these products of powers differ merely by an interchange of the indices 3 and 4. Suppose that such a normalized product of powers is, for example,

$$\xi_{12}^{\tau_{12}} \xi_{13}^{\tau_{13}} \xi_{14}^{\tau_{14}} \xi_{24}^{\tau_{24}}.$$

Because of the assumed separate homogeneity in each index, a replacement of 3 by 4 in one ξ_{ik} must be accompanied by a replacement of 4 by 3 in another; a relevant term would therefore contain, instead of $\xi_{13}\xi_{24}$, the product $\xi_{14}\xi_{23}$, which is excluded by the normalization. In exactly the same way, against

$$\xi_{12}^{\tau_{12}} \xi_{13}^{\tau_{13}} \xi_{14}^{\tau_{14}} \xi_{24}^{\tau_{24}}$$

only a term containing the factor $\xi_{14}\xi_{23}$ could cancel, which is again excluded by the normalization. Hence $K(\xi_{ik}) \equiv 0 \pmod p$ implies $G(\xi_{ik}) \equiv 0 \pmod p$.

c) The case of n rows α_i, β_i

The case of n rows is now settled by complete induction, as an exact generalization of the just treated case of four rows.[282] First I again show that for every polynomial F there is a congruence

$$F(\xi_{ik}) \equiv F_0(\xi_{ik}) \pmod{\mathfrak{N}} \quad \text{identically in } \xi_{ik}, \tag{10}$$

where $\mathfrak{N} = (f_{ik,rs})$ and where $F_0(\xi_{ik})$ is normalized. The normalization is to consist in the following: if i, k, r, s are four different indices and $i < k < r < s$, then F_0 is not to contain the product $\xi_{is}\xi_{kr}$; that is, the indices of one ξ are not both to lie between those of the other. This normalization imposes no restriction in the case of two and three rows, where every polynomial is already normalized; in the case of four rows, as shown, there also exists a normalized polynomial for every residue class. The existence of a normalized polynomial for every residue class may therefore be assumed proved for $(n-1)$ rows.

Let an arbitrary product of powers in F be written in the form

$$\xi_{i_1 n} \cdots \xi_{i_\nu n}\, P,$$

[281] This sharper assertion means that each residue class contains only one normalized polynomial.

[282] The proof still applies for $n = 3$.

where P no longer contains the index n. By the induction hypothesis, P is congruent modulo $\mathfrak{N}$ to an integral normalized polynomial P_0:

$$\xi_{i_1 n} \cdots \xi_{i_\nu n} P \equiv \xi_{i_1 n} \cdots \xi_{i_\nu n} P_0 \pmod{\mathfrak{N}}.$$

If P_0 contains no ξ_{rs} such that for at least one index i_λ the two indices r, s lie between i_λ and n, i.e. $i_\lambda < r < s < n$, then the normalization has already been reached, since the hypotheses are fulfilled for the product of any two ξ's from P_0. Otherwise let i_λ be such an index, $i_\lambda < r < s < n$, and let P_0^* be the product of powers from P_0 containing ξ_{rs}. Since

$$\xi_{i_\lambda n} \xi_{rs} \equiv \xi_{i_\lambda s} \xi_{rn} - \xi_{i_\lambda r} \xi_{sn} \pmod{\mathfrak{N}},$$

one obtains

$$\begin{aligned} &\xi_{i_1 n} \cdots \xi_{i_\lambda n} \cdots \xi_{i_\nu n} P_0^* \\ &\equiv \xi_{i_1 n} \cdots \xi_{rn} \cdots \xi_{i_\nu n} Q + \xi_{i_1 n} \cdots \xi_{sn} \cdots \xi_{i_\nu n} R \pmod{\mathfrak{N}} \\ &\equiv \xi_{i_1 n} \cdots \xi_{rn} \cdots \xi_{i_\nu n} Q_0 + \xi_{i_1 n} \cdots \xi_{sn} \cdots \xi_{i_\nu n} R_0 \pmod{\mathfrak{N}}, \end{aligned}$$

where Q_0 and R_0 again denote the normalized polynomials of the residue classes of Q and R. These exist by the induction hypothesis, since Q and R do not contain the index n, and they are certainly integral polynomials, since Q and R are simply products of powers. Since $i_\lambda < r < s < n$, one also has

$$((n-i_1) + \cdots + (n-r) + \cdots + (n-i_\nu)) < ((n-i_1) + \cdots + (n-i_\lambda) + \cdots + (n-i_\nu)),$$

$$((n-i_1) + \cdots + (n-s) + \cdots + (n-i_\nu)) < ((n-i_1) + \cdots + (n-i_\lambda) + \cdots + (n-i_\nu)).$$

Thus, in passing from P_0^* to Q_0, R_0, this sum of differences has decreased; since the sum is necessarily $\geq \sigma$, the procedure must terminate after finitely many steps for each product of powers P_0^*. Hence

$$F(\xi_{ik}) = \sum \xi_{i_1 n} \cdots \xi_{i_\nu n} P^{(1)} \equiv \sum \xi_{j_1 n} \cdots \xi_{j_\sigma n} S_0^{(j)} = F_0(\xi_{ik}) \pmod{\mathfrak{N}},$$

where the integral polynomial $F_0(\xi_{ik})$ is necessarily normalized, since otherwise the difference sum could be lowered further. This proves (10). Since, by the induction hypothesis for $(n-1)$ indices, every separately homogeneous F has a separately homogeneous normalized F_0, the procedure just given shows that this is also the case for n indices. Hence in what follows only separately homogeneous polynomials need be considered.

As in b), set

$$F_0(\xi_{ik}) = G(\xi_{ik}) + \xi_{n-1,n} F_1(\xi_{ik}),\text{[283]} \tag{11}$$

where G contains no product of powers divisible by $\xi_{n-1,n}$, and put further

$$[G]_{\xi_{in} = \xi_{i,n-1}} = K(\xi_{ik}). \tag{12}$$

As the definition of normalization shows, K, as well as G, is normalized.

Again there is a one-to-one correspondence between G and K. Distinct products of powers from G can correspond to the same product in K only if they differ merely by an interchange of the indices n and $(n-1)$. Suppose, for instance, that G is of degree ρ in the index $(n-1)$ and of degree σ in the index n, and that

$$\chi_1 \leq \chi_2, \ldots, \leq \chi_\rho \leq \chi_{\rho+1}, \ldots, \leq \chi_{\rho+\sigma}$$

are the indices connected with $(n-1)$ and n in a product of powers of G. Their number must be $\rho + \sigma$, since by (11) the indeterminate $\xi_{n-1,n}$ does not occur in G. By the normalization, the product of powers is therefore of the form

$$\xi_{\chi_1, n-1} \cdots \xi_{\chi_\rho, n-1} \xi_{\chi_{\rho+1}, n} \cdots \xi_{\chi_{\rho+\sigma}, n} P(\xi_{ik}),$$

[283]The identity corresponding to (10) and (11),

$$F((ik)) = F_0((ik)) = G((ik)) + (n-1,n) F_1((ik)),$$

is an analogue of the Clebsch-Gordan series expansion, as soon as the two rows $(n-1)$ and (n) are replaced by rows of variables x, y, so that $(n-1,n)$ is replaced by (xy). But whereas Clebsch-Gordan involves an expansion by polars, the normalization of F_0 corresponds to "initial terms" of polars, and thereby forces the integrality which does not hold for Clebsch-Gordan. The known proof that, when integrality is disregarded, the $f_{ik,rs}$ form a basis of $\mathfrak{M}$ is carried out precisely by means of the Clebsch-Gordan series expansion (Gordan-Kerschensteiner, *Vorlesungen über Invariantentheorie*, p. 132).

where $P(\xi_{ik})$ no longer contains the indices n and $(n-1)$. Each interchange of n with $(n-1)$ would, for $i \neq j$, give the factor $\xi_{in}\xi_{j,n-1}$ with $i < j$; hence $j, n-1$ lies between i and n, and the term would not be normalized. Thus, for fixed $\chi_1, \chi_2, \ldots, \chi_{\rho+\sigma}$ and fixed P, G contains only one product of powers. This proves that different products of powers from G correspond to different products in K. Therefore

$$K(\xi_{ik}) \equiv 0 \pmod{p} \quad \Longrightarrow \quad G(\xi_{ik}) \equiv 0 \pmod{p}. \tag{13}$$

From

$$F((ik)) \equiv 0 \pmod{p} \quad \text{identically in } \alpha, \beta$$

it follows now from (10), taking account of (2), that

$$F_0((ik)) \equiv 0 \pmod{p} \quad \text{identically in } \alpha, \beta. \tag{14}$$

Specializing $\alpha_n = \alpha_{n-1},\ \beta_n = \beta_{n-1}$, (11) and (12) give

$$K((ik)) \equiv 0 \pmod{p} \quad \text{identically in } \alpha, \beta. \tag{15}$$

Since $K(\xi_{ik})$ is normalized and contains only $(n-1)$ indices, it follows by what was proved under a) and under b), formula (9), that

$$K(\xi_{ik}) \equiv 0 \pmod{p} \quad \text{identically in } \xi_{ik}.$$

Then by (13)

$$G(\xi_{ik}) \equiv 0 \pmod{p} \quad \text{identically in } \xi_{ik}.$$

Hence (11) gives

$$F_0(\xi_{ik}) \equiv \xi_{n-1,n} F_1(\xi_{ik}) \pmod{p} \quad \text{identically in } \xi_{ik}.$$

Since $(n, n-1) \not\equiv 0 \pmod{p}$ identically in α, β, it follows from (14) that

$$F_1((ik)) \equiv 0 \pmod{p} \quad \text{identically in } \alpha, \beta,$$

where $F_1(\xi_{ik})$ is again normalized and has lower degree in $\xi_{n-1,n}$ than F_0. Repeating the process finitely many times yields

$$F_0(\xi_{ik}) \equiv \xi_{n-1,n}^{\lambda} F_\lambda(\xi_{ik}) \pmod{p},$$

where F_λ is of degree zero in $\xi_{n-1,n}$, and hence vanishes modulo p by what has just been proved. Thus one obtains

$$\begin{array}{rcl} F_0(\xi_{ik}) & \equiv & 0 \pmod{p} \quad \text{identically in } \xi_{ik}, \quad \text{or} \\ F(\xi_{ik}) & \equiv & 0 \pmod{(\mathfrak{N}, p)} \quad \text{identically in } \xi_{ik}. \end{array} \tag{16}$$

This proves the converse of (2), and hence (1). At the same time, it proves in the case of n indices, and therefore in general, the sharper assertion used in the induction step for $F_0(\xi_{ik})$.

§ 4. Finiteness of the Integral Invariants of Finite Groups

The elementary proof of finiteness given in my note "The finiteness theorem for invariants of finite groups"[284] admits, by means of the special finiteness theorems III and IV of § 1, the sharpening to integrality, whereas the usual proof based on Hilbert's theorem on the module basis does not supply this integrality. Even the use of the integral module basis only gives a representation in which an arbitrarily high power of the integer h, the order of the group, appears in the denominator.

I use throughout the notation of the note just mentioned. Let the finite group $\mathfrak{H}$ consist of the h linear transformations $A_1, \ldots, A_h$ of nonzero determinant,[285] where A_k denotes the linear transformation

$$x_1^{(k)} = \sum_{\nu=1}^{n} a_{1\nu}^{(k)} x_\nu, \quad \ldots, \quad x_n^{(k)} = \sum_{\nu=1}^{n} a_{n\nu}^{(k)} x_\nu,$$

[284] *Math. Ann.* 77, p. 89 (1915).

[285] It would suffice to suppose that $\mathfrak{H}$ is simply isomorphic to an abstract finite group; that is, if a transformation determinant of the A's vanishes, then $\mathfrak{H}$ should be similar to a group $\begin{pmatrix} \mathfrak{S} & 0 \\ 0 & 0 \end{pmatrix}$, where the determinants of the substitutions of $\mathfrak{S}$ do not vanish. In this case the invariants of $\mathfrak{H}$ coincide with the invariants of $\mathfrak{S}$, so the condition of the nonvanishing of the determinants is in fact no restriction of generality.

or, briefly, $(x^{(k)}) = A_k(x)$. Thus the group $\mathfrak{H}$ sends the row (x) with elements $x_1, \dots, x_n$ into the rows $(x^{(k)})$ with elements $x_1^{(k)}, \dots, x_n^{(k)}$. The coefficients $a_{i\nu}^{(k)}$ are assumed to be algebraic integers; this is no restriction of generality, since by a theorem of J. Schur[286] every finite group of linear transformations is similar to one for which the $a_{i\nu}^{(k)}$ become algebraic integers.

By an integral absolute invariant of the group is meant a polynomial rational function of $x_1, \dots, x_n$, with algebraic integers of the number field $\mathfrak{K}$ generated by the $a_{i\nu}^{(k)}$ as coefficients, which remains identically unchanged under application of $A_1, \dots, A_h$. For such an invariant $f(x)$ one has

$$f(x) = f(x^{(1)}) = \cdots = f(x^{(h)}) = \frac{1}{h}\sum_{k=1}^{h} f(x^{(k)}). \tag{1}$$

Conversely, every integral symmetric function of the h rows of quantities $(x^{(k)})$ also becomes an integral invariant of the group.

Let $\omega_1, \omega_2, \dots, \omega_\rho$ denote a basis of all integers of $\mathfrak{K}$. Then every invariant can be represented as

$$f(x) = \omega_1 f_1(x) + \cdots + \omega_\rho f_\rho(x),$$

where $f_1(x), \dots, f_\rho(x)$ have rational integer coefficients. Thus by (1)

$$h\, f(x) = \omega_1 \sum_{k=1}^{h} f_1(x^{(k)}) + \cdots + \omega_\rho \sum_{k=1}^{h} f_\rho(x^{(k)}).$$

Here the individual sums are symmetric functions of the h rows of quantities $(x^{(k)})$, and indeed with rational integer coefficients. By Theorem III they can therefore be expressed polynomially and rationally, with rational integer coefficients, in terms of finitely many integral-rational symmetric functions of these rows. By the preceding argument, these basis functions $I_1(x), \dots, I_\sigma(x)$ become integral invariants of the group, in general with algebraic-integral coefficients, since $\mathfrak{H}$ need not contain together with A all of its algebraically conjugate transformations.

There is therefore a representation

$$h \cdot f(x) = \omega_1 G_1(I) + \cdots + \omega_\rho G_\rho(I), \tag{2}$$

where $G_1, \dots, G_\rho$ have rational integer coefficients.

Now let $w_1, \dots, w_\rho$ be indeterminates, and let

$$L = w_1 G_1(I) + \cdots + w_\rho G_\rho(I)$$

run through all linear forms in the w's which, by the substitution $w_i = \omega_i/h$, correspond to invariants $f(x)$ in the representation (2). Hilbert's theorem on the integral module basis gives

$$L = A_1(I)L_1 + \cdots + A_\tau(I)L_\tau, \tag{3}$$

where the A_i have rational integer coefficients, and where, since all L's are linear in the w's, the A_i no longer contain the w's. If the substitution $\omega_i = w_i/h$ sends $L_1, \dots, L_\tau$ to the invariants $\mathfrak{I}_1(x), \dots, \mathfrak{I}_\tau(x)$, then (3) becomes

$$f(x) = A_1(I)\mathfrak{I}_1(x) + \cdots + A_\tau(I)\mathfrak{I}_\tau(x). \tag{4}$$

This representation shows that

$$I_1(x), \dots, I_\sigma(x); \mathfrak{I}_1(x), \dots, \mathfrak{I}_\tau(x)$$

form a basis of all integral invariants of the group $\mathfrak{H}$, such that every invariant is representable as a polynomial rational function, with rational integer coefficients, of these finitely many invariants.

If, in particular, the group $\mathfrak{H}$ has the property that with each transformation A it also contains all $A', A'', \dots$ with algebraically conjugate number coefficients, then the integral-rational symmetric functions of the h rows $(x^{(k)})$ become functions of x with rational integer coefficients. This holds in particular also for the basis functions of the symmetric functions. Thus, if one restricts oneself to invariants $f(x)$ with rational integer coefficients, then the basis functions

$$I_1(x), \dots, I_\sigma(x); \mathfrak{I}_1(x), \dots, \mathfrak{I}_\tau(x)$$

also have rational integer coefficients. Under this special assumption, which is in general not satisfied by the groups $\mathfrak{H}$, integral finiteness therefore also holds for the subdomain of invariants with rational integer coefficients.

[286] Über Gruppen linearer Substitutionen mit Koeffizienten aus einem algebraischen Zahlkörper. Satz VII. *Math. Ann.* 71, p. 555 (1912).

16. On Series Expansion in Form Theory

Math. Ann. 81 (1920), pp. 25–30

By the "series expansion of form theory" one understands, as is well known, on the one hand the generalization of the Clebsch–Gordan series expansion – that is, the expansion of a form containing two binary rows according to powers of the determinant of these rows, where the coefficients are polars of forms in only one row of variables, or, what is identical with this, vanish under application of the Ω-process – to forms in arbitrarily many series of n variables each; and, on the other hand, as a generalization of the normal forms occurring for the "complex coordinates" t_{ik}, an expansion according to normal forms for forms which contain simultaneously the quantities of different levels $t_i, t_{ik}, t_{ikl}, \ldots$. This second kind of expansion can, however, be obtained from the first by invariant differentiation processes, as was carried out above all by Mertens and Deruyts.[287]

The various expansions of the first kind all arise – as is briefly sketched, for instance, in Math. Ann. 77, loc. cit. III – as consequences of the expansion of a form in n rows of n variables each according to powers of the determinant of these rows, where the coefficients are polars of forms in only $(n-1)$ rows, which themselves are derived from the initial form by polar formation and the Ω-process.[288] The original Clebsch–Gordan expansion $(n = 2)$ had gone one step further: the special polar processes which occur there are given explicitly, even including the numerical coefficients.

What I now wish to add to the indicated note – where the series expansion is conceived more conceptually, as a problem of linear pencils of forms – is an explicit representation up to numerical coefficients, obtained by specializing E. Fischer's investigations concerning general module systems.[289]

This classification is led to by interpreting the normal form in the t_{ik}. Namely, if one replaces the "complex coordinates" t_{ik} by the "line coordinates"

$$p_{ik} = (x_i y_k - x_k y_i),$$

then, because of the identical relations existing among the p_{ik}, the representation of a form $F(p_{ik})$ is no longer unique. Uniqueness can be obtained by requiring that among the coefficients of F there hold the same linear relations as among the power products of the p_{ik} occurring in F.[290] Thus for this normal form Φ one has:

$$\begin{aligned} &F(p_{ik}) = \Phi(p_{ik}) \quad \text{[identically in } x, y], \quad \text{or} \\ &F(t_{ik}) \equiv \Phi(t_{ik}) \qquad \operatorname{mod} M, \end{aligned} \tag{1}$$

where M denotes the module of all identical relations among the p_{ik}, that is, consists of the totality of all forms $G(t_{ik})$ for which $G(p_{ik})$ vanishes identically in x, y. $\Phi(t_{ik})$ is the "normal form" in complex coordinates, while the basis functions of M become quadratic forms in the t_{ik}. By iterating the procedure with respect to the factors of these basis functions, the complete series expansion arises. Corresponding considerations hold when $t_i, t_{ik}, t_{ikl}, \ldots$ occur simultaneously.

By an analogous line of thought one can reduce the expansion according to polars. If, for example, in a form $F(x, y, z)$ one replaces the three ternary rows $x = (x_1, x_2, x_3), y, z$ by rows ξ, η, ζ which are linearly composed of only two of them (for instance $\xi = x, \eta = y; \zeta = \lambda_1 x + \lambda_2 y$), then the representation of $F(\xi, \eta, \zeta)$ is again not unique. One can again prescribe for the coefficients the same linear relations as for the power products of ξ, η, ζ occurring in F. Since here the module M of all identical relations has as basis function the determinant $\Delta = (xyz)$ of the three rows (loc. cit. III, auxiliary lemma a)), [1] is replaced by:

$$\begin{aligned} &F(\xi, \eta, \zeta) = \Phi(\xi, \eta, \zeta) \quad \text{[identically in } x, y], \quad \text{or} \\ &F(x, y, z) \equiv \Phi(x, y, z) \qquad \operatorname{mod} \Delta. \end{aligned} \tag{2}$$

It follows that Φ arises by polar formation (compare formula (2) loc. cit.); the complete series expansion is again obtained by iteration.

[287] F. Mertens, Über invariante Gebilde quaternärer Formen (Sitzber. d. Ak. d. Wiss. Wien 98, Abt. IIa (1889)) for the quaternary case. The general normal forms are found in J. Deruyts: Sur la réduction des fonctions invariantes dans le système des variables géométriques, Bulletins de l'Ac. R. de Belgique 24 (1892). Without knowing this work, and in close dependence on Mertens, I gave the general expansion according to normal forms in § 7 of the paper: Zur Invariantentheorie der Formen von n Variabeln, J. f. M. 139 (1910).

[288] Compare the bibliographical references loc. cit. Since then there has appeared, also based on the formal method: Schouten, Over reeksontwikkelingen van algebraische vormen met verschillende rijen van variabelen van verschillenden graad, Verslag Ak. Amsterdam 27 (1919), p. 1481.

[289] E. Fischer, Über die Differentiationsprozesse der Algebra, J. f. M. 148 (1917).

[290] Compare Clebsch, Über eine Fundamentalaufgabe der Invariantentheorie, § 4, Abh. d. K. G. d. Wissensch. zu Göttingen 17 (1872).

The formation of the normal forms Φ is therefore subordinated in both cases to the following problem: if in $F(z)$ the indeterminates z_i are replaced by polynomials $f_i(\xi)$ in parameters ξ, so that the representation $F(f_i(\xi))$ is therefore no longer unique, then the uniqueness is to be fixed by requiring that among the coefficients there hold the same linear relations as among the power products of the $f_i(\xi)$ occurring in F. Thus one has

$$\begin{aligned} F(f_i(\xi)) &= \Phi(f_i(\xi)) \quad [\text{identically in } \xi], \quad \text{or} \\ F(z) &\equiv \Phi(z) \qquad \bmod M, \end{aligned} \tag{3}$$

where M denotes the module of all identical relations among the $f_i(\xi)$.

For this normal form $\Phi(z)$, E. Fischer, in § 11, I of the cited paper, gave an explicit expansion up to numerical coefficients, namely by means of linear operations. At the same time, however, in Introduction III and in § 6 II, he also gave, for the difference $F - \Phi$ belonging to M – assuming knowledge of a module basis – an explicit expression in the same sense; and, by combining these developments in § 11 II, he thereby replaced formula [3] by an explicit representation. The question of the numerical coefficients, still open here, is according to § 12 of the same paper identical with the determination of the eigenvalues and eigenfunctions of the linear operations (formula 82).

I now specialize Fischer's formulas, in 1., to the case of the series expansion according to polars, and obtain from them, by iteration, the series expansion. The operation for determining Φ thereby passes into definite polar processes, which is considerably more precise than the previously known formulation given above. In the operation for determining $F - \Phi$, multiplication by the determinant Δ and the Ω-process occur in combination. To get to the usual, non-explicit form, one must also note that $\Omega\Delta$ can be expressed by polar processes.

At this point a correction should also be added. In the work cited above (p. 101, line 6 from the top) I inadvertently assumed as generally valid the identical transformation

$$\Omega(\Delta\psi) = c_1\psi + c_2\Delta \cdot \Omega(\psi),$$

which holds only in the binary domain, and thereby made the passage from the fundamental formula (2), which gives the expression for the normal form Φ in [2], to the series expansion (3). The considerations there therefore yield only formula (2), and as a special case of it the reduction theorem (1); but, since an explicit expression for the difference $F - \Phi$ is missing, they do not suffice to justify the series expansion.

In 2. I similarly obtain, by specialization, the normal form $\Phi(t_{ik})$ of [1]; more generally, the normal form $\Phi(t_i, t_{ik}, t_{ikl}, \ldots)$, which leads to known explicit formulas. But since setting up the complete series expansion requires knowing a module basis of M, the invariant-theoretic route (compare the cited literature) is preferable here; it does not require this knowledge, but on the contrary supplies a module basis by means of the series expansion.

1. According to Fischer, § 11 (formula (19) and the following lines), the normal form $\Phi(z)$ of [3] – there denoted $N(\zeta)$ – is given by:

$$\Phi(z) = (a_1 S + a_2 S^2 + \cdots + a_r S^r) F(z), \tag{4}$$

where the constants $a_1 \ldots a_r$ belong to the coefficient domain of the $f(\xi)$ and depend only on the degree of $F(z)$; the operator S is defined by

$$\begin{aligned} S = AB; \qquad BF(z) &= F(f(\xi)); \\ AF(f(\xi)) &= \left\{\sum z_i f_i\left(\frac{\partial}{\partial\xi}\right)\right\}^p F(f(\xi))^{291}. \end{aligned} \tag{5}$$

If in [2] one regards F as a form of p-th dimension in z alone, then $f_i(\xi)$ specializes to $\lambda_1 x_i + \lambda_2 y_i$; consequently one obtains:

$$\begin{aligned} BF &= F(x, y, \lambda_1 x + \lambda_2 y), \\ SF &= \left\{\sum z_i \left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x_i} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y_i}\right)\right\}^p F(x, y, \lambda_1 x + \lambda_2 y), \end{aligned}$$

or, written in polar processes (compare loc. cit. II, 2 and 3; also for the abbreviated notation):

$$SF(x, y, z) = \frac{1}{p!}\left[z\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y}\right)\right]^p \left[(\lambda_1 x + \lambda_2 y)\frac{\partial}{\partial z}\right]^p F(x, y, z). \tag{6}$$

[291] Compare formula (75). In the case of [3], the sum α runs over all power products of p-th dimension, and (75) therefore specializes to [5].

Equations [4] *and* [6] *give the desired explicit expression for* $\Phi(x,y,z)$; at the same time $\Phi(x,y,z)$ falls under the general form mentioned at the start ($\sum PH$ in (2), loc. cit.) – polars of expressions which depend on only 2 rows and are obtained from F by polar processes. The still undetermined coefficients a_i are rational numbers, since here the $f_i(\xi)$ have integral-rational numerical coefficients. If [2] is written in the form

$$F = \Phi + \Delta F_1, \tag{2a}$$

then specialization of the formula given by Fischer in Introduction III and in § 6 II (p. 41) gives

$$F_1 = (c_1\Omega + c_2\Omega\Delta\Omega + \cdots + c_i\Omega\Delta\Omega\cdots\Delta\Omega)F^{292}, \qquad \left[\Omega = \left(\frac{\partial}{\partial x}\frac{\partial}{\partial y}\frac{\partial}{\partial z}\right)\right], \tag{7}$$

where the c_i are again rational numbers depending only on the degree of F.

The iteration leads to:

$$F_1 = \Phi_1 + \Delta F_2; \quad F_2 = \Phi_2 + \Delta F_3; \quad \ldots; \quad F_i = \Phi_i + \Delta F_{i+1}; \quad \ldots,$$

where in each case the Φ_i are the normal forms corresponding to the F_i. Since the F_i have degree $(p-i)$ in z, [4] and [6] give:

$$\begin{gathered}\Phi_i = (a_{i1}S_i + a_{i2}S_i^2 + \cdots)F_i,\\ S_iF_i = \frac{1}{(p-i)!}\left[z\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y}\right)\right]^{p-i}\left[(\lambda_1 x + \lambda_2 y)\frac{\partial}{\partial z}\right]^{p-i} F_i.\end{gathered} \tag{8}$$

Formula [7] corresponds to

$$F_i = (\alpha_{i1}\Omega + \alpha_{i2}\Omega\Delta\Omega + \cdots)F_{i-1}, \quad \text{and by iteration}$$

$$F_i = (\beta_1\Omega^i + \beta_2\Omega^i\Delta\Omega + \cdots + \beta_\rho\Omega^{k_1}\Delta\Omega^{k_2}\cdots\Delta\Omega^{k_\lambda} + \cdots)F, \quad (k_1 + k_2 + \cdots + k_\lambda - \lambda + 1 = i). \tag{9}$$

By means of [8] *and* [9], *the expansion*

$$F = \Phi + \Delta\Phi_1 + \Delta^2\Phi_2 + \cdots + \Delta^i\Phi_i + \cdots + \Delta^p\Phi_p$$

has reached an explicit representation in the stated sense.

If one applies to [7] the identical transformation $\Omega\Delta F = PF$, where P denotes polar processes,[293] then, since these processes commute with the Ω-process, one obtains the known formula

$$F_1 = P\Omega F; \quad \ldots; \quad F_i = P\Omega^i F;$$

and from this, in conjunction with [8] or with the considerations loc. cit., follows the known formulation of the series expansion, as for instance in (3) loc. cit. If one is dealing with n rows of n variables, $x^{(1)}, x^{(2)}, \ldots, x^{(n-1)}, z$, then correspondingly

$$f_i(\xi) = \lambda_1 x_i^{(1)} + \lambda_2 x_i^{(2)} + \cdots + \lambda_{n-1}x_i^{(n-1)};$$

and likewise Δ and Ω are to be defined for n rows.

2. For the normal form of $F(t_i, t_{ik}, t_{ikl}, \ldots)$, the functions $f_i(\xi), f_{ik}(\xi), \ldots$ are given by the one-row, two-row, up to $(n-1)$-row determinants

$$\xi_i^{(1)}, \quad (\xi^{(1)}\xi^{(2)})_{ik}, \quad \ldots, \quad (\xi^{(1)}\xi^{(2)}\cdots\xi^{(n-1)})_{i_1i_2\cdots i_{n-1}}.$$

Thus, as a specialization of [5], one obtains:

$$\begin{gathered}S = AB; \qquad BF = F(\xi_i^{(1)}, (\xi^{(1)}\xi^{(2)})_{ik}, \ldots),\\ ABF = \left(\sum t_i\frac{\partial}{\partial\xi_i^{(1)}}\right)^{\lambda_1}\left(\sum t_{ik}\left(\frac{\partial}{\partial\xi^{(1)}}\frac{\partial}{\partial\xi^{(2)}}\right)_{ik}\right)^{\lambda_2}\cdots F(\xi_i^{(1)}, (\xi^{(1)}\xi^{(2)})_{ik}, \ldots),\end{gathered}$$

where $\lambda_1, \lambda_2, \ldots$ denote the degrees in $t_i, t_{ik}, \ldots$.

[292] Kerim uses this specialized formula in his Erlangen dissertation, soon to appear, in the application to "inertia forms".

[293] Compare, for example, F. Mertens, Über ganze Funktionen von m Systemen von je n Unbestimmten, Monatsh. f. Math. u. Phys. 4, formula (5).

From invariant-theoretic considerations (namely, all invariants of F which are linear in the coefficients can be linearly composed from Φ and from forms out of M, thus in particular also the SF not belonging to M)[294] the congruence follows here: $\Phi \equiv a_1 SF \mod M$; and because of the normal-form property one obtains from this, instead of [4], the simple relation

$$\Phi = a_1 SF = a_1 S\Phi. \tag{10}$$

The second equality holds because applying S makes every form from M vanish. Formula [10] shows that the normal form Φ is itself at the same time an eigenfunction; and indeed, by the invariant-theoretic consideration above, there are no further eigenfunctions which are at the same time invariants of F (compare, concerning [10], Deruyts, formulas (10) and (10')).

Corrections.

1. In the paper "Die allgemeinsten Bereiche aus ganzen transzendenten Zahlen" (Math. Ann. 77), on p. 121, line 8 from the top, line 9 from the top, and line 11 from the bottom, the field (H, Y) is to be replaced by the "field $(H, Y, Y', \ldots, Y^{(\lambda_0-1)})$, where $Y', \ldots, Y^{(\lambda_0-1)}$ denote the conjugates of Y with respect to H." Correspondingly, on p. 120, line 5 from the top, and p. 121, line 9 from the top, (H, y) is to be replaced by $(H, y, y', \ldots, y^{(\lambda_0-1)})$.

2. In the paper "Gleichungen mit vorgeschriebener Gruppe" (Math. Ann. 78), in the formulation of Hilbert's irreducibility theorem – p. 222, line 3 from the bottom – instead of "number field Ω" it must more precisely say "finite algebraic number field Ω", as Mr. Ostrowski has pointed out to me. In fact, with respect to the field of all algebraic numbers, for instance, the group of every equation with algebraic numerical coefficients is the identity group. – Correspondingly, in the introduction, "arbitrary domain of rationality" is to be understood as a finite algebraic number field to which finitely many parameters can still be adjoined, or as one of the intermediate fields which arise in this way.

(Accepted September 1919.)

[294] This follows, for instance, from § 6 and § 7, 2. of my cited paper "Zur Invariantentheorie der Formen von n Variabeln".

17. Together with W. Schmeidler: Modules in Noncommutative Domains, Especially from Difference Expressions

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Contents.

Introduction.

The formal theory of differential expressions in one variable leads to a decomposition into irreducible factors which is unique in a certain sense.[1] But whereas the corresponding theorem in algebra can be carried over to polynomials in several variables, this is no longer possible for partial differential expressions.[2] For differential expressions in one variable, however, alongside the product representation there is also a representation as least common multiple, with similar laws holding; for, in two different decompositions, the individual constituents can be assigned to one another pairwise: they are "of the same kind".[3]

This concept of the least common multiple can now be interpreted as a module concept and so transferred to n variables; in the present paper this gives the natural generalization of the theorems just mentioned. Beyond this, however, we also note that for differential expressions only the following fact is used: they can be interpreted as polynomials with noncommutative multiplication, for which the degree of a product is equal to the sum of the degrees of the factors. These conditions are fulfilled, for example, by difference expressions as well, and hence the theorems also hold for these, where until now they were likewise known only for one variable.[4]

We therefore develop, in general, a theory of modules whose elements are polynomials with noncommutative multiplication; in doing so we reduce the study of the modules to that of residue classes.[5] In contrast with the commutative special case, one can indeed speak here of the sum of residue classes, but not of their product; one can speak only of the product of a residue class with a polynomial. The concept of a "residue group" is therefore to be modified relative to the commutative case. In the special case of polynomials in one variable, the residue group is moreover "finite", that is, it possesses only a finite number of linearly independent residue classes. In general there is no such finite "order"; the groups are "infinite". For our methods, however, this difference is immaterial.

Just as in the commutative case, to the representation of a module as the least common multiple of relatively prime modules there corresponds the simpler process of additive decomposition of the residue group. In this connection the isomorphism of residue groups proves essential; upon specialization to differential expressions in one variable this isomorphism becomes the concept of the same kind for the associated modules, while upon specialization to commutative modules it becomes identity. Incidentally, the concept of the same

[1] Cf. E. Landau, Ein Satz über die Zerlegung homogener linearer Differentialausdrücke in irreduzible Faktoren, Journal für die reine und angewandte Mathematik 124 (1902), pp. 115–120, and subsequently A. Loewy, Über reduzible lineare homogene Differentialgleichungen, Mathematische Annalen 56 (1903), pp. 549–584.

[2] Cf. a counterexample of Landau published in H. Blumberg, Über algebraische Eigenschaften von linearen homogenen Differentialausdrücken, dissertation, Göttingen 1912, 52 pp., pp. 51–52.

[3] Cf., besides the papers just cited, A. Loewy, Über vollständig reduzible lineare homogene Differentialgleichungen, Mathematische Annalen 62 (1906), pp. 89–117; cited as Loewy.

[4] Cf. G. Wallenberg–A. Guldberg, Theorie der linearen Differenzengleichungen, Leipzig and Berlin 1911.

[5] For the case of commutative domains, cf. W. Schmeidler, Über Moduln und Gruppen hyperkomplexer Größen, Mathematische Zeitschrift 3 (1919), pp. 29–42.

kind can also be transferred to n variables, and its agreement with the isomorphism concept can be proved. The fact that in the additive decomposition of the residue group only finitely many summands ever occur follows by transferring Hilbert's theorem on the module basis, using Gordan's method of proof; Hilbert's original method of proof would require restrictions on the multiplication laws (in formula (6), a on the right instead of a_i, which is fulfilled for differential expressions but not for difference expressions).

In order to arrive at a uniqueness theorem, we must specialize the modules – as Loewy also does – to "completely reducible" ones, that is, to those representable as least common multiples of prime modules. Given two different decompositions, the constituents are then either pairwise identical, or at least to every constituent of one decomposition there corresponds an isomorphic constituent of the other decomposition, and conversely; at the same time, each decomposition then contains at least two isomorphic constituents. Furthermore, from the occurrence of two isomorphic constituents in one and the same decomposition there follows the existence of infinitely many decompositions; this even holds without the restriction to prime modules, which until now was not known even in the special case of differential expressions in one variable.

Finally, for modules from differential expressions, we discuss the connection with integrals and show that, for finite residue groups, the order is equal to the largest number of linearly independent integrals. Thus arbitrary functions occur in the general integral precisely when the residue group is not finite. The concept that two modules are of the same kind means, here as in the one-variable case, a "birational" relation between the integrals. Some examples conclude the paper.

The first impulse for the present paper was given several years ago by a question of E. Landau to E. Noether concerning the generalization of the product theorem. At that time E. Noether only observed that the question was one of module theory in the noncommutative polynomial domain, but could not yet be attacked. The Lasker module theorems then known were not directly transferable to this domain, because Lasker's method of proof rests on the theorem that a polynomial is uniquely decomposable into irreducible factors.[6] Only Schmeidler's methods offered the possibility of a transfer, which we then carried out jointly. In detail, let it be mentioned that the generalization of the residue group, the transfer and application of Hilbert's finiteness theorem, and the existence of infinitely many decompositions are due to E. Noether; the generalization of Loewy's concept of complete reducibility, the concept of isomorphism, and its connection with the concept of the same kind are due to W. Schmeidler.

§ 1. Formal Matters Concerning Differential and Difference Expressions.

1. For differential expressions in one variable, a symbolic product formation has been introduced. Let

$$a_0(x)\frac{\mathrm{d}^n y}{\mathrm{d}x^n} + a_1(x)\frac{\mathrm{d}^{n-1} y}{\mathrm{d}x^{n-1}} + \cdots + a_n(x)y = A(y),$$

$$b_0(x)\frac{\mathrm{d}^m y}{\mathrm{d}x^m} + b_1(x)\frac{\mathrm{d}^{m-1} y}{\mathrm{d}x^{m-1}} + \cdots + b_m(x)y = B(y).$$

Then one defines the symbolic product $AB(y)$ as the differential expression

$$a_0(x)\frac{\mathrm{d}^n B(y)}{\mathrm{d}x^n} + a_1(x)\frac{\mathrm{d}^{n-1} B(y)}{\mathrm{d}x^{n-1}} + \cdots + a_n(x)B(y).$$

This product formation is in reality a multiplication of two operators; it is most simply written by replacing, as usual, $\frac{\mathrm{d}^n}{\mathrm{d}x^n}$ by $\left(\frac{\mathrm{d}}{\mathrm{d}x}\right)^n$. Then AB becomes

$$\left(a_0\left(\frac{\mathrm{d}}{\mathrm{d}x}\right)^n + \cdots + a_n\right)\left(b_0\left(\frac{\mathrm{d}}{\mathrm{d}x}\right)^m + \cdots + b_m\right)y,$$

where the ordinary rules for differentiating sums and products apply in carrying out the product. Every such operator can now be regarded as a polynomial in the single variable $\frac{\mathrm{d}}{\mathrm{d}x} = \xi$, by not writing explicitly the function to which the operator is applied and by fixing for the variable ξ the multiplication rule corresponding to product differentiation:

$$\frac{\mathrm{d}}{\mathrm{d}x}(a(x)y) = a(x)\frac{\mathrm{d}}{\mathrm{d}x}(y) + a'(x)y,$$

or, in our notation ($n = 1, m = 0$),

$$\xi \cdot a = a \cdot \xi + a'. \tag{1}$$

[6] Recently E. Noether observed that the Lasker theorems too can be established without using this theorem, by methods related to those used here.

One proceeds in exactly the same way for linear partial differential expressions

$$\left(\sum a_{\alpha_1\dots\alpha_n}(x_1,\dots,x_n)\frac{\partial^{\alpha_1+\dots+\alpha_n}}{\partial x_1^{\alpha_1}\cdots\partial x_n^{\alpha_n}}\right)y$$

of a function y. For the n operators $\frac{\partial}{\partial x_1},\dots,\frac{\partial}{\partial x_n}$ we introduce variables $\xi_1,\dots,\xi_n$ and thereby obtain the polynomial

$$\sum a_{\alpha_1\dots\alpha_n}(x_1,\dots,x_n)\xi_1^{\alpha_1}\cdots\xi_n^{\alpha_n}.$$

The product of two polynomials is then computed according to the laws

$$\xi_i a = a\xi_i + a_i \qquad (i=1,\dots,n), \tag{2}$$

where a_i denotes the partial derivative of a with respect to x_i; the products of the ξ's among themselves are commutative. Apart from the commutative law of multiplication, all laws of addition and multiplication hold for the polynomials so defined just as in ordinary algebra. In particular, for the product of two such polynomials the total degree, and the degree in each individual variable, is equal to the sum of the corresponding degrees of the two factors.

2. For difference expressions (cf. Wallenberg, loc. cit.) of the form

$$a_x^{(0)}y_{x+n} + a_x^{(1)}y_{x+n-1} + \dots + a_x^{(n)}y_x = A_x,$$

$$b_x^{(0)}y_{x+m} + b_x^{(1)}y_{x+m-1} + \dots + b_x^{(m)}y_x = B_x$$

one has a corresponding product definition

$$(AB)_x = a_x^{(0)}B_{x+n} + a_x^{(1)}B_{x+n-1} + \dots + a_x^{(n)}B_x,$$

which can be carried out with the aid of the multiplication rule

$$(ay)_{x+1} = a_{x+1}y_{x+1}.$$

If we introduce the operator ξ for the transition from x to $x+1$, then we obtain $\xi\{ay\} = a^*\xi\{y\}$, where $a^* = a_{x+1}$. Again one can omit the function y to which the operator is applied and obtains the law

$$\xi a = a^*\xi, \tag{3}$$

where, along with a, also a^* does not vanish identically. In exactly the same way, for n variables,

$$\xi_i a = a_i\xi_i \qquad (i=1,\dots,n), \tag{4}$$

where ξ_i stands for the transition from x_i to x_i+1, and a_i denotes the function obtained from a by replacing x_i by x_i+1; hence it does not vanish identically if a does not. With these conventions, all rules of addition and multiplication, including the rule concerning the degree of a product, again hold, except for the commutative law.

Both cases will be subordinated, in the following section, to a quite generally defined polynomial domain with arbitrary coefficients.

§ 2. The General Polynomial Domain.

We start from an arbitrary abstractly defined field P (whose elements we denote by small Latin letters), for which all rules of algebra hold, in particular also the commutative law of multiplication with unique inversion. To this field we adjoin n indeterminates $\xi_1,\dots,\xi_n$; that is, we consider the integral domain J consisting of the products $a\xi_1^{\nu_1}\xi_2^{\nu_2}\cdots\xi_n^{\nu_n}$ and their finite sums, where all laws of addition and multiplication are to hold with the exception of the commutative law of multiplication. In particular, we call finite sums of the form

$$F(\xi_1,\dots,\xi_n) = \sum a_{\nu_1\dots\nu_n}\xi_1^{\nu_1}\cdots\xi_n^{\nu_n}$$

"polynomials" (which throughout are to be denoted by capital Latin letters). In place of the commutative law, we now subject the elements of J to product laws such that *the product of two polynomials is again a* *polynomial*; thus the domain of all polynomials from J already exhausts J. For this it is plainly necessary and sufficient that, for the elements $\xi_1, \ldots, \xi_n$ and an arbitrary element a of P, product rules of the form

$$\xi_i a = F_i(\xi_1, \ldots, \xi_n) \qquad (i = 1, \ldots, n) \tag{5}$$

hold, where $F_i(\xi_1, \ldots, \xi_n)$ denotes some polynomial depending naturally on ξ_i and a. (In particular, the rules (2) and (4) for differential and difference expressions, respectively, are of this form.) For since $\xi_i a$ is not in itself a polynomial, while by hypothesis it belongs to the domain, the equation is necessary. Conversely, by finitely repeated application of formula (5), every element of the domain J can be transformed into a polynomial.

For the unit of the field, which we denote by 1, we shall further require $\xi_i \cdot 1 = 1 \cdot \xi_i = \xi_i$. We shall also stipulate that the multiplication of $\xi_1, \ldots, \xi_n$ among themselves is commutative.[7] Every polynomial in J then writes itself just like an ordinary polynomial in $\xi_1, \ldots, \xi_n$ with coefficients from P, and two polynomials agree only when all their coefficients agree.

More specifically, in what follows[8] we require that, in place of (5), the rule

$$\xi_i a = a_i \xi_i + b_i \qquad (a_i \neq 0) \quad (i = 1, \ldots, n) \tag{6}$$

shall hold. Here too, in particular, the laws (2) and (4) are included. Then the degree of a product is equal to the sum of the degrees of the factors; this holds both for total degree and for the degree in each individual variable, so that the product of two polynomials which do not vanish identically is likewise different from 0.

§ 3. One-Sided Modules and Their Residue Groups.

For our domain J, because multiplication is noncommutative, the following kinds of modules[9] can be distinguished:[10]

1. The *right-sided module*, which, along with M, contains also FM for an arbitrary polynomial F, and along with M_1 and M_2 contains also $M_1 + M_2$.

2. The *left-sided module*, which, along with M, contains also MF, and along with M_1 and M_2 contains also $M_1 + M_2$.

We refer to these two kinds by the common name *one-sided modules*; modules which are both right-sided and left-sided will be called *two-sided*. In what follows we restrict ourselves to right-sided modules and note that the theory of the left-sided ones can be built up in exactly the same way.

We call two polynomials F and G congruent modulo $\mathfrak{M}$, written $F \equiv G(\mathfrak{M})$, if their difference is divisible by $\mathfrak{M}$, that is, if it belongs to $\mathfrak{M}$. The totality of all polynomials congruent to a given polynomial forms a *residue class* modulo $\mathfrak{M}$. (Residue classes are to be denoted by small German letters.) Since from

$$F_1 \equiv F_2(\mathfrak{M}) \qquad \text{and} \qquad G_1 \equiv G_2(\mathfrak{M})$$

it follows that $F_1 + G_1 \equiv F_2 + G_2(\mathfrak{M})$, the sum of two residue classes $\mathfrak{x}$ and $\mathfrak{y}$ is defined and is again a residue class $\mathfrak{x} + \mathfrak{y}$. By contrast, because the module is one-sided, one cannot speak of the product of two residue classes. Indeed, from

$$F_1 = F_2 + M_1, \qquad G_1 = G_2 + M_2$$

there follows only

$$F_1 G_1 = F_2 G_2 + F_2 M_2 + M_1 G_2 + M_1 M_2;$$

but since $M_1 G_2$ need not belong to $\mathfrak{M}$ when M_1 belongs to $\mathfrak{M}$, $F_1 G_1$ and $F_2 G_2$ need not be congruent in general. On the other hand, if $M_1 = 0$, hence $F_1 = F_2 = F$, one sees that $G_1 \equiv G_2$ implies $FG_1 \equiv FG_2$. Thus an arbitrary residue class $\mathfrak{y}$ can be multiplied from the front by a polynomial F, and this gives a well-defined residue class $F\mathfrak{y} = \mathfrak{z}$.

For computation with residue classes, the commutative and associative laws of addition hold, as does the law of unique reversal of addition; moreover, for multiplication of residue classes by polynomials and addition, the distributive law $F(\mathfrak{y}_1 + \mathfrak{y}_2) = F\mathfrak{y}_1 + F\mathfrak{y}_2$ holds, as follows directly from computing with polynomials modulo $\mathfrak{M}$.

[7] This is needed only from § 6 onward for the finiteness theorem and its consequences.

[8] This too is used only from § 6 onward.

[9] We denote modules by capital German letters.

[10] One-sided ideals were already considered by A. Hurwitz; cf. Vorlesungen über die Zahlentheorie der Quaternionen (Berlin, Springer 1919), Lecture 6. There, however, what is essentially involved are principal ideals; the same holds for the works of Du Pasquier cited there in note 1.

The residue class to which the unit belongs will be called the unit class, or unit, and denoted $\mathfrak{e}$. Then $F\mathfrak{e} = \mathfrak{x}$ is the residue class to which the polynomial F belongs. In contrast to the general residue classes, however, here the product $\mathfrak{x}\mathfrak{e}$ is also well-defined and equal to $\mathfrak{x}$; for from

$$F_1 = F_2 + M_1, \qquad G_1 = 1 + M_2$$

and $M_1 \cdot 1 = M_1$ it follows that

$$F_1 G_1 = F_2 + M.$$

The totality of the residue classes modulo $\mathfrak{M}$ is called the *residue group of* $\mathfrak{M}$. A residue group therefore has the following properties: 1. Along with every residue class $\mathfrak{x}$ and an arbitrary polynomial F, it contains the residue class $F\mathfrak{x}$; and along with the residue classes $\mathfrak{x}_1$ and $\mathfrak{x}_2$ it contains the residue class $\mathfrak{x}_1 + \mathfrak{x}_2$.[11] 2. It possesses a unit $\mathfrak{e}$, so that $F\mathfrak{e}$ represents, for every polynomial F, the residue class $\mathfrak{x}$ of F. *Thus, as F runs through all polynomials, $F\mathfrak{e}$ runs through the whole residue group.*

§ 4. Reducibility of the Residue Group into Two Subgroups and Its Relation to the Module.

For one-sided modules, the concepts *divisibility*, *greatest common divisor*, and *least common multiple* remain as in the two-sided case, as the following definitions show.

A module $\mathfrak{M}$ is said to be divisible by another module $\mathfrak{N}$ if from $M \equiv 0(\mathfrak{M})$ it follows also that $M \equiv 0(\mathfrak{N})$.

For two modules $\mathfrak{M}$ and $\mathfrak{N}$, the *greatest common divisor* $(\mathfrak{M}, \mathfrak{N})$ is defined as the totality of all polynomials representable in the form $M + N$, where $M \equiv 0(\mathfrak{M})$ and $N \equiv 0(\mathfrak{N})$; it is again a module. The same holds for several modules.

The totality of all polynomials divisible both by $\mathfrak{M}$ and by $\mathfrak{N}$ forms a module $[\mathfrak{M}, \mathfrak{N}]$, the *least common multiple* of $\mathfrak{M}$ and $\mathfrak{N}$. This definition too applies correspondingly to several modules, indeed even to a set of modules of arbitrary cardinality.

Two modules are called relatively prime when their greatest common divisor is the unit module, that is, the totality of all polynomials.

We now start from a module $\mathfrak{M}$ which is representable as the least common multiple of two relatively prime modules $\mathfrak{N}$ and $\mathfrak{L}$, both assumed to be proper divisors:

$$\mathfrak{M} = [\mathfrak{N}, \mathfrak{L}]. \tag{7}$$

We investigate the residue group $\mathfrak{G}$ of $\mathfrak{M}$. By hypothesis,

$$1 \equiv N + L(\mathfrak{M}) \tag{8}$$

for two polynomials $N \equiv 0(\mathfrak{N})$ and $L \equiv 0(\mathfrak{L})$. From this we show

$$\mathfrak{L} = (\mathfrak{M}, L), \qquad \mathfrak{N} = (\mathfrak{M}, N), \tag{9}$$

where, for example, $(\mathfrak{M}, L)$ means the module consisting of all polynomials of the form $M + FL$. In fact, first $(\mathfrak{M}, L) \equiv 0(\mathfrak{L})$. Conversely, from (8) it follows, for an arbitrary polynomial F, that

$$F \equiv FN(\mathfrak{L}). \tag{10}$$

Now if $F \equiv 0(\mathfrak{L})$, then $FN \equiv 0(\mathfrak{L})$, hence $\equiv 0(\mathfrak{M})$, and consequently $F \equiv FL(\mathfrak{M})$. In the same way one proves $\mathfrak{N} = (\mathfrak{M}, N)$. It follows incidentally, since $\mathfrak{N}$ and $\mathfrak{L}$ are proper divisors of $\mathfrak{M}$, that neither $L \equiv 0$ nor $N \equiv 0(\mathfrak{M})$ can hold.

Let $\mathfrak{a}$ be the residue class of N modulo $\mathfrak{M}$, and $\mathfrak{b}$ the residue class of L modulo $\mathfrak{M}$. Then by (8)

$$\mathfrak{e} = \mathfrak{a} + \mathfrak{b} \qquad (\mathfrak{a} \neq 0,\ \mathfrak{b} \neq 0).$$

Since every residue class is of the form $F\mathfrak{e}$, the distributive law gives, for every residue class,

$$F\mathfrak{e} = F\mathfrak{a} + F\mathfrak{b} \qquad (\mathfrak{a} \neq 0,\ \mathfrak{b} \neq 0). \tag{11}$$

This representation of an arbitrary residue class as a sum of two residue classes of the form $G\mathfrak{a}$ and $H\mathfrak{b}$ is, however, *unique*. For from a relation $0 = G\mathfrak{a} + H\mathfrak{b}$, if K is a polynomial in the residue class $G\mathfrak{a} = -H\mathfrak{b}$, then

[11] This property could, in a more general sense, be called the "module property" of the residue group.

plainly $K \equiv GN(\mathfrak{M})$, $K \equiv -HL(\mathfrak{M})$, hence $K \equiv 0(\mathfrak{N})$, $K \equiv 0(\mathfrak{L})$, and therefore $K \equiv 0(\mathfrak{M})$; that is, $G\mathfrak{a} = 0$, $H\mathfrak{b} = 0$.

Conversely, suppose now that relation (11) is fulfilled for every polynomial F, and indeed uniquely in the above sense. Let N then be a polynomial from $\mathfrak{a}$ and L a polynomial from $\mathfrak{b}$. We form the two modules (9) and assert (7) and (8). The latter follows from (11) for $F = 1$. Furthermore, by the definition of $\mathfrak{N}$ and $\mathfrak{L}$, $\mathfrak{M}$ is a common multiple of these two modules, and indeed the least one. For from $F \equiv 0(\mathfrak{L})$ and $F \equiv 0(\mathfrak{N})$ there follows $F \equiv HL \equiv GN(\mathfrak{M})$; hence $G\mathfrak{a} = H\mathfrak{b}$, $G\mathfrak{a} - H\mathfrak{b} = 0$, and therefore, by uniqueness of the representation, $G\mathfrak{a} = 0$, $H\mathfrak{b} = 0$, $F \equiv 0(\mathfrak{M})$. It has therefore been shown that the *representation of a module $\mathfrak{M}$ as the least common multiple of two relatively prime modules is equivalent to the unique additive decomposition of the residue group $\mathfrak{G}$ of $\mathfrak{M}$, as given by formula* (11).

We now investigate the system $\mathfrak{A}$ of residue classes of the form $F\mathfrak{a}$, which we wish to relate to the residue group $\overline{\mathfrak{A}}$ of $\mathfrak{L}$. By (11), every polynomial F in $\mathfrak{L}$ belongs to the residue class $F\mathfrak{a}$ when $\overline{\mathfrak{a}}$ is the residue class of N modulo $\mathfrak{L}$, and hence the unit class of $\overline{\mathfrak{A}}$. If we now associate the residue classes $F\mathfrak{a}$ in $\mathfrak{A}$ with the residue classes $F\overline{\mathfrak{a}}$ in $\overline{\mathfrak{A}}$, then to every residue class of $\mathfrak{A}$ there is assigned a residue class of $\overline{\mathfrak{A}}$, in such a way that the sum of two residue classes of $\mathfrak{A}$ corresponds to the sum of the assigned residue classes of $\overline{\mathfrak{A}}$, and the product of a residue class of $\mathfrak{A}$ with a polynomial corresponds to the product of the assigned residue class of $\overline{\mathfrak{A}}$ with the same polynomial. This assignment is moreover one-to-one: from $F\mathfrak{a} = 0$ follows $F \equiv 0(\mathfrak{L})$ by (11), hence $F\overline{\mathfrak{a}} = 0$; conversely, $F\overline{\mathfrak{a}} = 0$, hence $F \equiv 0(\mathfrak{L})$, also says by (10) that $FN \equiv 0(\mathfrak{L})$, hence $FN \equiv 0(\mathfrak{M})$, that is, $F\mathfrak{a} = 0$.

This leads us to a fundamental concept, which we define as follows:

A one-to-one correspondence between two systems $\mathfrak{A}$ and $\overline{\mathfrak{A}}$ of residue classes is called isomorphic if the sum of two residue classes of $\mathfrak{A}$ is assigned the sum of the corresponding residue classes of $\overline{\mathfrak{A}}$, and the product of an arbitrary residue class of $\mathfrak{A}$ with a polynomial is assigned the product of the corresponding residue class of $\overline{\mathfrak{A}}$ with the same polynomial.

A subsystem $\mathfrak{A}$ of residue classes in $\mathfrak{G}$ which is isomorphic to the residue group of a module is called a *subgroup* of $\mathfrak{G}$. As the proof above shows, the assignment here is specifically such that every residue class of $\mathfrak{A}$ arises from the corresponding residue class of $\overline{\mathfrak{A}}$ by adjoining certain polynomials, namely all polynomials belonging to $\mathfrak{L}$ but not to $\mathfrak{M}$.

Similarly one shows that the totality of all residue classes of the form $F\mathfrak{b}$ forms a subgroup $\mathfrak{B}$ of $\mathfrak{G}$ which is isomorphic to the residue group $\overline{\mathfrak{B}}$ of $\mathfrak{N}$.

A residue group $\mathfrak{G}$ whose residue classes admit a unique additive decomposition (11) will be called *reducible* into the two subgroups $\mathfrak{A}$ and $\mathfrak{B}$, written $\mathfrak{G} = \mathfrak{A} + \mathfrak{B}$; a group which is not reducible is called *irreducible.* We also occasionally call the associated modules reducible or irreducible. Thus the following holds.

Theorem 1.[12] *A module $\mathfrak{M}$ is representable as the least common multiple of two relatively prime proper divisors $\mathfrak{L}$ and $\mathfrak{N}$ if and only if its residue group $\mathfrak{G}$ is reducible into two subgroups $\mathfrak{A}$ and $\mathfrak{B}$. In this case $\mathfrak{A}$ and $\mathfrak{B}$ are respectively isomorphic to the residue groups of $\mathfrak{L}$ and $\mathfrak{N}$.*

[12] Cf. Schmeidler, loc. cit., p. 39.

17. Together with W. Schmeidler: Modules in Noncommutative Domains, Especially from Differential and Difference Expressions

Math. Zs. 8 (1920), pp. 1–35

§ 5. Computation with Units and Reducibility into k Subgroups.

The residue classes $\mathfrak{a}$ and $\mathfrak{b}$ of $\mathfrak{G}$ share with the unit $\mathfrak{e}$ the property that the product of $\mathfrak{a}$ and $\mathfrak{b}$ with an arbitrary residue class $\mathfrak{x}$ is defined; we therefore likewise call $\mathfrak{a}$ and $\mathfrak{b}$ units of $\mathfrak{G}$. Indeed, from (11), for $F = M$, there follows

$$0 = M\mathfrak{e} = M\mathfrak{a} + M\mathfrak{b},$$

and hence, by uniqueness, $M\mathfrak{a} = 0$, $M\mathfrak{b} = 0$, and therefore

$$(F + M)\mathfrak{a} = F\mathfrak{a} = \mathfrak{x}\mathfrak{a},$$

where $\mathfrak{x}$ denotes the residue class of F. In place of relation (11), we may therefore write the equivalent class relation

$$\mathfrak{x}\mathfrak{e} = \mathfrak{x}\mathfrak{a} + \mathfrak{x}\mathfrak{b}.$$

In particular, putting $\mathfrak{x} = \mathfrak{a}$ gives

$$\mathfrak{a} = \mathfrak{a}^2 + \mathfrak{a}\mathfrak{b},$$

and hence, by uniqueness, $\mathfrak{a}^2 = \mathfrak{a}$, $\mathfrak{a}\mathfrak{b} = 0$. Similarly one obtains $\mathfrak{b}^2 = \mathfrak{b}$, $\mathfrak{b}\mathfrak{a} = 0$.

Corresponding to the representation of a residue group as a sum of two constituents, we may now also define such a representation as a sum of k constituents. Namely, suppose a unique representation is given

$$F\mathfrak{e} = F\mathfrak{a}_1 + F\mathfrak{a}_2 + \cdots + F\mathfrak{a}_k \qquad (\mathfrak{a}_1 \neq 0, \ldots, \mathfrak{a}_k \neq 0),$$

where from $0 = G_1\mathfrak{a}_1 + \cdots + G_k\mathfrak{a}_k$ it also follows that $G_i\mathfrak{a}_i = 0$. Then, in particular for $F = M$, one obtains the relation $M\mathfrak{a}_i = 0$, so that the representation may also be written in the form[1]

$$\mathfrak{x}\mathfrak{e} = \mathfrak{x}\mathfrak{a}_1 + \cdots + \mathfrak{x}\mathfrak{a}_k \qquad (\mathfrak{a}_1 \neq 0, \ldots, \mathfrak{a}_k \neq 0). \tag{12}$$

This representation may also be regarded as having arisen from a successive decomposition into two constituents. If this can be continued without limit, with all $\mathfrak{a}_i \neq 0$, then we speak of a sum of infinitely many constituents.

Putting $\mathfrak{x} = \mathfrak{a}_i$ in (12) gives

$$\mathfrak{a}_i^2 = \mathfrak{a}_i, \qquad \mathfrak{a}_i\mathfrak{a}_j = 0 \quad (i \neq j), \qquad i, j = 1, \ldots, k, \tag{13}$$

with $\mathfrak{a}_i \neq 0$. Conversely, suppose a system of residue classes $\mathfrak{a}_1, \ldots, \mathfrak{a}_k$ in $\mathfrak{G}$ is present for which the orthogonality conditions (13) are fulfilled and for which

$$\mathfrak{e} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_k \tag{14}$$

holds. We shall show that this gives a representation of the group as a sum of k constituents. First, from the hypothesis, for every polynomial A_i from $\mathfrak{a}_i$, since $A_i\mathfrak{a}_i = (A_i + M)\mathfrak{a}_i$, it follows that $M\mathfrak{a}_i = 0$; hence $\mathfrak{a}_i$ can be multiplied by every class. Thus (14) implies representation (12), and this representation is unique. For if $0 = \mathfrak{x}_1\mathfrak{a}_1 + \cdots + \mathfrak{x}_k\mathfrak{a}_k$, then right multiplication by $\mathfrak{a}_i$ gives, by (13), $0 = \mathfrak{x}_i\mathfrak{a}_i$, proving uniqueness and hence the assertion. The residue classes $\mathfrak{a}_1, \ldots, \mathfrak{a}_k$ occurring in the decomposition of the unit class will be called the units belonging to the decomposition; equivalently they are characterized by the orthogonality relations (13) and condition (14). The later developments will essentially amount to computation with units.

It remains to establish the connection between the additive decomposition of the group into k constituents and the corresponding decomposition of the module. As in the case $k = 2$, a representation of the module as the least common multiple of k relatively prime modules will result, by induction.

Let the decomposition (12) be given, which we write in the form

$$\mathfrak{x}\mathfrak{e} = \mathfrak{x}\mathfrak{b} + \mathfrak{x}\mathfrak{a}_k, \tag{15}$$

[1] For the additive decomposition of residue groups, the commutative and associative laws hold; in contrast to the commutative case, however, $\mathfrak{U}_1 + \mathfrak{U}_2 = \mathfrak{U}_1 + \mathfrak{U}_3$ does not imply $\mathfrak{U}_2 = \mathfrak{U}_3$; cf. example 1 in § 12.

$$\mathfrak{x}\mathfrak{b} = \mathfrak{x}\mathfrak{a}_1 + \cdots + \mathfrak{x}\mathfrak{a}_{k-1}. \tag{16}$$

From (15) it follows by Theorem I that the residue classes $\mathfrak{x}\bar{\mathfrak{b}}$ are isomorphic to the residue group of the module

$$\mathfrak{N} = (\mathfrak{M}, A_k). \tag{17}$$

For their residue classes $\mathfrak{x}\bar{\mathfrak{b}}$, the representation corresponding to (16) is therefore also valid:

$$\mathfrak{x}\bar{\mathfrak{b}} = \mathfrak{x}\bar{\mathfrak{a}}_1 + \cdots + \mathfrak{x}\bar{\mathfrak{a}}_{k-1}. \tag{18}$$

Suppose now it has already been proved that

$$\mathfrak{N} = [\mathfrak{M}_1, \ldots, \mathfrak{M}_{k-1}], \tag{19}$$

where

$$\begin{cases} \mathfrak{M}_1 = (\mathfrak{N}, A_2 + \cdots + A_{k-1}), \\ \mathfrak{M}_2 = (\mathfrak{N}, A_1 + A_3 + \cdots + A_{k-1}), \\ \qquad\vdots \\ \mathfrak{M}_{k-1} = (\mathfrak{N}, A_1 + \cdots + A_{k-2}). \end{cases} \tag{20}$$

For two summands this assertion agrees with our earlier result. Moreover, the polynomials A_i may here be chosen specifically as polynomials from $\bar{\mathfrak{a}}_i$ which at the same time are contained in $\mathfrak{a}_i$, since by § 4 the polynomials of the latter residue class are contained in the former. Then, by (15),

$$\mathfrak{M} = [\mathfrak{N}, \mathfrak{M}_k], \tag{21}$$

where

$$\mathfrak{M}_k = (\mathfrak{M}, A_1 + \cdots + A_{k-1}).$$

From (19) and (21) one obtains

$$\mathfrak{M} = [[\mathfrak{M}_1, \ldots, \mathfrak{M}_{k-1}], \mathfrak{M}_k] = [\mathfrak{M}_1, \ldots, \mathfrak{M}_{k-1}, \mathfrak{M}_k]$$

by the definition of least common multiple; and from (17) and (20) one gets

$$\mathfrak{M}_1 = (\mathfrak{M}, A_k, A_2 + \cdots + A_{k-1}).$$

It follows that

$$\mathfrak{M}_1 = (\mathfrak{M}, A_2 + \cdots + A_k),$$

for the latter module contains, by (13), also the polynomial

$$A_k = A_k(A_2 + \cdots + A_k) \pmod{\mathfrak{M}}.$$

Corresponding expressions are obtained for $\mathfrak{M}_2, \ldots, \mathfrak{M}_{k-1}$. Furthermore, the modules $\mathfrak{M}_1, \ldots, \mathfrak{M}_k$ are relatively prime, since by (14)

$$\frac{1}{k-1}\Big((A_2 + \cdots + A_k) + (A_1 + A_3 + \cdots + A_k) + \cdots + (A_1 + \cdots + A_{k-1})\Big) \equiv 1 \pmod{\mathfrak{M}}.$$

They moreover form a totally relatively prime system, that is, the least common multiple of any of them is relatively prime to the least common multiple of the remaining ones.[2] This follows directly by grouping the summands in formula (12) into two suitable groups.

It has thus been shown that to a decomposition of the residue group into k summands there corresponds a representation of the module as the least common multiple of k modules forming a totally relatively prime system. We now prove the converse theorem, likewise by induction.

Suppose therefore that

$$\mathfrak{M} = [\mathfrak{M}_1, \ldots, \mathfrak{M}_k] = [[\mathfrak{M}_1, \ldots, \mathfrak{M}_{k-1}], \mathfrak{M}_k] = [\mathfrak{N}, \mathfrak{M}_k],$$

[2] This is a generalization of a concept introduced by Loewy.

where $\mathfrak{M}_1, \ldots, \mathfrak{M}_k$ form a totally relatively prime system. In combination with Theorem I this gives representation (15) for the residue group. The modules $\mathfrak{M}_1, \ldots, \mathfrak{M}_{k-1}$ also form a totally relatively prime system. For if, under some division of the modules of this system into two groups, the least common multiple of one group and the least common multiple of the other had a divisor in common, this divisor would remain if the module $\mathfrak{M}_k$ were added to one of the two groups and the least common multiple were then formed.[3] For $k = 2$, "totally relatively prime" means the same as "relatively prime"; hence our assertion may be assumed proved up to $k-1$. Thus for the residue class $\mathfrak{r}\bar{\mathfrak{b}}$ the unique representation (18) holds, hence, by the isomorphism, also (16), and so the assertion follows.

Theorem II. *A module $\mathfrak{M}$ is representable as the least common multiple of k modules $\mathfrak{M}_1, \ldots, \mathfrak{M}_k$ which form a totally relatively prime system if and only if its residue group $\mathfrak{G}$ is reducible into k subgroups $\mathfrak{U}_1, \ldots, \mathfrak{U}_k$. With suitable notation, $\mathfrak{U}_i$ is then isomorphic to the residue group of $\mathfrak{M}_i$.*

§ 6. Finiteness Theorems.

In this paragraph it will be shown that every residue group can be represented as a sum of finitely many irreducible constituents. To do this we first transfer Hilbert's theorem on the module basis.

From now on we use the more special hypotheses formulated in § 2: the multiplication of $\xi_1, \ldots, \xi_n$ among themselves is commutative, and the multiplication law of the polynomials has the form

$$\xi_i a = a_i \xi_i + b_i, \qquad (a_i \neq 0) \quad (i = 1, \ldots, n). \tag{6}$$

It suffices to prove the assertion for modules only. For if $\mathfrak{S}$ is any system of polynomials, we form the module $\mathfrak{M}$ derived from it by adjoining to $\mathfrak{S}$ all polynomials which can be composed from finitely many polynomials $G_1, \ldots, G_k$ of $\mathfrak{S}$ in the form $A_1 G_1 + \cdots + A_k G_k$. If $F_1, \ldots, F_l$ is then a module basis of $\mathfrak{M}$, and if

$$F_i = \sum_j A_{ij} G_j \qquad (i = 1, \ldots, l),$$

then the finitely many polynomials G_j occurring in these representations form a basis of $\mathfrak{S}$.

We prove the module-basis theorem following Gordan (Göttingen Nachrichten 1899). The proof splits into two parts. The first concerns a system $\mathfrak{P}$ of infinitely many power products and remains essentially unchanged because of the commutativity of $\xi_1, \ldots, \xi_n$. The power products

$$P = \xi_1^{a_1} \cdots \xi_n^{a_n},$$

each of which is entered into the system only once, are to be ordered by increasing degree and, within the same degree, in some fixed way. We now consider the subsystem $\mathfrak{Q}$ of $\mathfrak{P}$ consisting of all power products $P = Q$ not divisible by any preceding P. Then no Q is divisible by a later Q either, since later ones are either of higher degree or, in the same degree, different from the earlier ones. On the other hand, every P of $\mathfrak{P}$ is divisible by at least one Q. For suppose, to the contrary, that P_m is the first P divisible by no Q. Then P_m cannot itself be a Q, and so it is divisible by at least one preceding P; since by hypothesis this preceding P is divisible by some Q, P_m itself is divisible by a Q, a contradiction.

We now assert that the number of the Q is finite. Indeed, let

$$Q_0 = \xi_1^{h_1} \cdots \xi_n^{h_n}$$

be the first Q. Since an arbitrary $Q_\lambda = \xi_1^{h_{1\lambda}} \cdots \xi_n^{h_{n\lambda}}$ is not divisible by Q_0, for every Q at least one $h_{\rho\lambda} < h_\rho$ must hold; let λ denote the smallest index for which this is the case. We group all Q for which $h_{\rho\lambda}$ has the same fixed value c_λ into one class $K(c_\lambda)$. Since $c_\lambda < h_\lambda$, the number of these classes is finite. But each class contains only finitely many elements. For if we put a Q in the class $K(c_\lambda)$ equal to $\xi_\lambda^{c_\lambda} Q'$, then the Q' also form a system of power products none of which is divisible by another, and they contain only $n-1$ variables. For $n = 2$ the Q' contain only one variable; their number is then 1, hence finite, so that the number may also be assumed finite for $n-1$ variables. This proves the assertion.

Second, let $\mathfrak{M}$ be a module of polynomials whose terms are lexicographically ordered. Let $\mathfrak{P}$ be the system of the power products which occur in the highest term, each written only once. Then to the system $\mathfrak{Q} = (Q_1, \ldots, Q_\rho)$ there corresponds a system of finitely many polynomials $F_1, \ldots, F_\rho$, obtained by assigning

[3] Continuing this argument shows that among $\mathfrak{M}_1, \ldots, \mathfrak{M}_k$ every two are relatively prime; the converse, that this would imply $\mathfrak{M}_1, \ldots, \mathfrak{M}_k$ to form a totally relatively prime system, is generally valid only in the commutative case; cf. the counterexample 1 in § 12.

to each Q_i some polynomial from $\mathfrak{M}$ whose highest term is equal to b_iQ_i, so that $F_i = b_iQ_i + \cdots$ with lower terms.

Now let $F = bP + \cdots$ be an arbitrary polynomial from $\mathfrak{M}$, where $P = \xi_1^{\beta_1} \cdots \xi_n^{\beta_n} Q_i$. Form the product

$$\xi_1^{\beta_1} \cdots \xi_n^{\beta_n} F_i = \xi_1^{\beta_1} \cdots \xi_n^{\beta_n}(b_iQ_i + \cdots) = c\,\xi_1^{\beta_1} \cdots \xi_n^{\beta_n} Q_i + \cdots = cP + \cdots .$$

Here $c \neq 0$ by (6), and all following terms are lower terms in the lexicographic order, since multiplication is commutative as regards the exponents, up to lower terms.[4] The difference

$$F - \frac{b}{c}\,\xi_1^{\beta_1} \cdots \xi_n^{\beta_n} F_i$$

then contains only lower terms and again belongs to $\mathfrak{M}$. The procedure therefore terminates after finitely many steps, so that $F_1, \ldots, F_\rho$ are recognized as a module basis.

Theorem III. *Every system $\mathfrak{S}$ of infinitely many polynomials has a finite module basis $G_1, \ldots, G_k$, so that every polynomial G from $\mathfrak{S}$ admits a representation $G = A_1G_1 + \cdots + A_kG_k$.*

The following is immediate.

Corollary. *Every system $\mathfrak{S}$ of infinitely many residue classes mod $\mathfrak{M}$ has a basis $\mathfrak{x}_1, \ldots, \mathfrak{x}_k$, so that every $\mathfrak{x}$ from $\mathfrak{S}$ is representable in the form $\mathfrak{x} = A_1\mathfrak{x}_1 + \cdots + A_k\mathfrak{x}_k$.*

Namely, assign to each residue class some representative and consider the system of these representatives together with all polynomials from $\mathfrak{M}$.

We now turn to the proof of the following theorem.

Theorem IV. *Every residue group can be represented as a sum of finitely many irreducible constituents; equivalently, by Theorem II, every module can be represented as the least common multiple of finitely many irreducible modules forming a totally relatively prime system.*

Suppose, to the contrary, that $\mathfrak{G}$ is a sum of infinitely many constituents $\mathfrak{U}_1, \mathfrak{U}_2, \ldots$ (cf. § 5), and let $\mathfrak{a}_1, \mathfrak{a}_2, \ldots$ be the corresponding units, all of which are nonzero by definition. We form the system $\mathfrak{S}$ of all these units; by the corollary to Theorem III it has a finite basis $\mathfrak{a}_{i_1}, \ldots, \mathfrak{a}_{i_v}$ with $i_1 < \cdots < i_v$. Let $\mu > i_v$. Then

$$\mathfrak{a}_\mu = \mathfrak{x}_1\mathfrak{a}_{i_1} + \cdots + \mathfrak{x}_v\mathfrak{a}_{i_v}.$$

This is a linear relation between units, which is excluded by the definition. Hence in every additive decomposition of a residue group an irreducible constituent occurs after finitely many steps; because of commutativity of the decomposition, we can place it at the beginning. Thus, without loss of generality, we may assume that the $\mathfrak{U}_i$ themselves are irreducible. But by the preceding argument only finitely many constituents $\mathfrak{U}_i$ can occur. The assertion is proved.

§ 7. A Case of Unique Decomposition.

Let

$$\mathfrak{G} = \mathfrak{U}_1 + \cdots + \mathfrak{U}_k = \mathfrak{B}_1 + \cdots + \mathfrak{B}_l$$

be a residue group whose constituents $\mathfrak{U}_1, \ldots, \mathfrak{U}_k$, $\mathfrak{B}_1, \ldots, \mathfrak{B}_l$ are irreducible. Then

$$\mathfrak{x}\mathfrak{e} = \mathfrak{x}\mathfrak{a}_1 + \cdots + \mathfrak{x}\mathfrak{a}_k = \mathfrak{x}\mathfrak{b}_1 + \cdots + \mathfrak{x}\mathfrak{b}_l, \tag{22}$$

where $\mathfrak{a}_\chi$ is the residue class of $\mathfrak{G}$ isomorphically assigned to the unit class of $\mathfrak{U}_\chi$, and $\mathfrak{b}_\lambda$ is the one assigned to the unit class of $\mathfrak{B}_\lambda$. Replacing $\mathfrak{x}$ by $\mathfrak{x}\mathfrak{b}_\lambda$, with λ a fixed index in the range $1, \ldots, l$, gives

$$\mathfrak{x}\mathfrak{b}_\lambda = \mathfrak{x}\mathfrak{b}_\lambda\mathfrak{a}_1 + \cdots + \mathfrak{x}\mathfrak{b}_\lambda\mathfrak{a}_k. \tag{23}$$

This representation of the group $\mathfrak{B}_\lambda$ is unique, but it is not an additive decomposition of $\mathfrak{B}_\lambda$, since the individual residue classes on the right need not belong to $\mathfrak{B}_\lambda$. Since $\mathfrak{b}_\lambda$ does not vanish, not every product $\mathfrak{b}_\lambda\mathfrak{a}_\chi$ can be zero. Without loss of generality, let

$$\mathfrak{b}_\lambda\mathfrak{a}_1 \neq 0, \ldots, \mathfrak{b}_\lambda\mathfrak{a}_\rho \neq 0, \qquad \mathfrak{b}_\lambda\mathfrak{a}_{\rho+1} = 0, \ldots, \mathfrak{b}_\lambda\mathfrak{a}_k = 0.$$

[4] This shows that in place of law (6) one could also use the somewhat less restrictive law

$$\xi_i a = a_i\xi_i + a_{i,i+1}\xi_{i+1} + \cdots + a_{in}\xi_n + b_i \qquad (a_i \neq 0)$$

for some fixed numbering of the ξ.

Then

$$\mathfrak{x}\mathfrak{b}_\lambda = \mathfrak{x}\mathfrak{b}_\lambda\mathfrak{a}_1 + \cdots + \mathfrak{x}\mathfrak{b}_\lambda\mathfrak{a}_\rho.$$

For a fixed χ in the range $1, \ldots, \rho$, consider the totality of those residue classes $\mathfrak{x}\mathfrak{b}_\lambda$ which are nonzero and for which also $\mathfrak{x}\mathfrak{b}_\lambda\mathfrak{a}_\chi \neq 0$. The two systems $\mathfrak{x}\mathfrak{b}_\lambda$ and $\mathfrak{x}\mathfrak{b}_\lambda\mathfrak{a}_\chi$ are isomorphic. First, the assignment is one-to-one: from $\mathfrak{x}\mathfrak{b}_\lambda\mathfrak{a}_\chi = 0$ follows, by hypothesis, $\mathfrak{x}\mathfrak{b}_\lambda = 0$, and from the latter, by uniqueness of representation (23), $\mathfrak{x}\mathfrak{b}_\lambda\mathfrak{a}_\chi = 0$. Furthermore, sums correspond to sums, and products with arbitrary polynomials to the corresponding products. Keeping χ fixed, the same argument can be made for every λ with $\mathfrak{b}_\lambda\mathfrak{a}_\chi \neq 0$. For each $\mathfrak{a}_\chi$ there must be at least one such $\mathfrak{b}_\lambda$, since $\mathfrak{a}_\chi = (\mathfrak{b}_1 + \cdots + \mathfrak{b}_l)\mathfrak{a}_\chi \neq 0$.

We first assume that for each χ there is only one λ, and for each λ only one χ, such that $\mathfrak{b}_\lambda\mathfrak{a}_\chi \neq 0$.

This correspondence implies $k = l$; moreover the notation may be chosen so that $\mathfrak{b}_\chi\mathfrak{a}_\chi \neq 0$ and $\mathfrak{b}_\lambda\mathfrak{a}_\chi = 0$ for $\lambda \neq \chi$. We then say that the products $\mathfrak{b}_\lambda\mathfrak{a}_\chi$ form a diagonal scheme. In this case, by (23), the unit is $\mathfrak{b}_\chi = \mathfrak{b}_\chi\mathfrak{a}_\chi$, and since

$$\mathfrak{a}_\chi = \mathfrak{b}_1\mathfrak{a}_\chi + \cdots + \mathfrak{b}_l\mathfrak{a}_\chi = \mathfrak{b}_\chi\mathfrak{a}_\chi = \mathfrak{b}_\chi, \tag{24}$$

one has, in general, $\mathfrak{U}_\chi = \mathfrak{B}_\chi$. Thus there is uniqueness of decomposition, and the products $\mathfrak{a}_\chi\mathfrak{b}_\lambda$ form a diagonal scheme.

If, for example, the commutative law holds specifically for the residue classes $\mathfrak{a}_\chi\mathfrak{b}_\lambda$, then the preceding hypothesis is always fulfilled. For in (23) each constituent $\mathfrak{x}\mathfrak{b}_\lambda\mathfrak{a}_\chi = \mathfrak{x}\mathfrak{a}_\chi\mathfrak{b}_\lambda$ is then contained in $\mathfrak{B}_\lambda$, so that (23) gives an additive decomposition of the group $\mathfrak{B}_\lambda$. Since $\mathfrak{B}_\lambda$ is irreducible, all but one of the $\mathfrak{b}_\lambda\mathfrak{a}_\chi$ are zero. By (24), correspondingly, for fixed χ only one $\mathfrak{b}_\lambda\mathfrak{a}_\chi$ is different from zero, since otherwise (24), by interchangeability of $\mathfrak{a}_\chi$ and $\mathfrak{b}_\lambda$, would yield an additive decomposition of $\mathfrak{U}_\chi$, which is supposed to be irreducible.[5]

The same conclusions hold in the case where the conditions $\mathfrak{b}_\lambda\mathfrak{a}_\lambda \neq 0$, $\mathfrak{b}_\lambda\mathfrak{a}_\chi = 0$ for $\lambda \neq \chi$, $\chi = 1, \ldots, k$, and likewise, for each $\mu = 1, \ldots, l$, $\mathfrak{b}_\mu\mathfrak{a}_\chi = 0$, $\mathfrak{b}_\mu\mathfrak{a}_\mu \neq 0$, $\chi = 1, \ldots, \sigma$, are fulfilled not for all, but only for part of the $\mathfrak{b}_\lambda$ ($\lambda = 1, \ldots, \sigma$). One finds $\mathfrak{U}_\chi = \mathfrak{B}_\chi$ for $\chi = 1, \ldots, \sigma$; and if one puts

$$\mathfrak{U}_{\sigma+1} + \cdots + \mathfrak{U}_k = \mathfrak{U}, \qquad \mathfrak{B}_{\sigma+1} + \cdots + \mathfrak{B}_l = \mathfrak{B},$$

then, also for the units $\mathfrak{a}$ and $\mathfrak{b}$ of $\mathfrak{U}$ and $\mathfrak{B}$, the conditions $\mathfrak{b}\mathfrak{a}_\chi = 0$, $\mathfrak{b}_\chi\mathfrak{a} = 0$ ($\chi = 1, \ldots, \sigma$), and $\mathfrak{b}\mathfrak{a} \neq 0$ are fulfilled, whence $\mathfrak{U} = \mathfrak{B}$.

§ 8. Additive Decomposition of Completely Reducible Groups into Prime Groups. The Theorem on Isomorphism.

We now consider another case, in which uniqueness does not hold, but isomorphism of the different decompositions does. This is the generalization of the case considered by Loewy.

Definition. *A module is called a prime module if it has no divisor other than itself and the unit module containing all polynomials. The residue group of a prime module is called a prime group.*

Every prime group is, a fortiori, irreducible. There are also infinite prime groups (cf. the definition in § 11 and example § 12, 2).

Let the given residue group $\mathfrak{G}$ now be representable in two ways as a sum of finitely many prime groups $\mathfrak{U}_1, \ldots, \mathfrak{U}_k$ and $\mathfrak{B}_1, \ldots, \mathfrak{B}_l$. A residue group which admits at least one such representation as a sum of prime groups, and the corresponding module $\mathfrak{M}$, will be called completely reducible, following Loewy's terminology.

We consider the products $\mathfrak{a}_\chi\mathfrak{b}_\lambda$ ($\chi = 1, \ldots, k$, $\lambda = 1, \ldots, l$), and show that either $\mathfrak{a}_\chi\mathfrak{b}_\lambda = 0$, or else $\mathfrak{x}\mathfrak{a}_\chi\mathfrak{b}_\lambda$ runs through the whole group $\mathfrak{B}_\lambda$ as $\mathfrak{x}$ runs through all residue classes of $\mathfrak{G}$. Indeed, the module belonging to $\mathfrak{B}_\lambda$ is

$$(\mathfrak{M}, \mathfrak{b}_1 + \cdots + \mathfrak{b}_{\lambda-1} + \mathfrak{b}_{\lambda+1} + \cdots + \mathfrak{b}_l) = \mathfrak{M}_{\mathfrak{b}_\lambda};$$

it arises from $\mathfrak{M}$ by adjoining all polynomials of the residue class $\mathfrak{b}_1 + \cdots + \mathfrak{b}_{\lambda-1} + \mathfrak{b}_{\lambda+1} + \cdots + \mathfrak{b}_l$. This module has the divisor

$$(\mathfrak{M}, \mathfrak{b}_1 + \cdots + \mathfrak{b}_{\lambda-1} + \mathfrak{b}_{\lambda+1} + \cdots + \mathfrak{b}_l, \mathfrak{a}_\chi\mathfrak{b}_\lambda).$$

But since $\mathfrak{M}_{\mathfrak{b}_\lambda}$ is, by hypothesis, a prime module, the divisor is either equal to $\mathfrak{M}_{\mathfrak{b}_\lambda}$ or to the unit module. In the first case $\mathfrak{a}_\chi\mathfrak{b}_\lambda$ belongs to $\mathfrak{M}_{\mathfrak{b}_\lambda}$; hence there is a relation

$$\mathfrak{a}_\chi\mathfrak{b}_\lambda = \mathfrak{x}(\mathfrak{b}_1 + \cdots + \mathfrak{b}_{\lambda-1} + \mathfrak{b}_{\lambda+1} + \cdots + \mathfrak{b}_l),$$

[5] The same applies in the noncommutative two-sided case treated by Schmeidler, loc. cit., where the product of every residue class of one subgroup with every residue class of the other subgroup vanishes. Then, since $\mathfrak{b}_\lambda\mathfrak{a}_\chi = \mathfrak{b}_\lambda\mathfrak{a}_\chi\mathfrak{b}_1 + \cdots + \mathfrak{b}_\lambda\mathfrak{a}_\chi\mathfrak{b}_l$ and $\mathfrak{b}_\lambda(\mathfrak{a}_\chi\mathfrak{b}_\mu) = 0$ for $\lambda \neq \mu$, one has $\mathfrak{b}_\lambda\mathfrak{a}_\chi = \mathfrak{b}_\lambda\mathfrak{a}_\chi\mathfrak{b}_\lambda$, hence this class is contained in $\mathfrak{B}_\lambda$; by irreducibility of $\mathfrak{B}_\lambda$, it follows that for only one, and hence for exactly one, χ the product $\mathfrak{b}_\lambda\mathfrak{a}_\chi \neq 0$. In the same way one shows that for every $\mathfrak{a}_\chi$ there is one and only one $\mathfrak{b}_\lambda$ with $\mathfrak{a}_\chi\mathfrak{b}_\lambda \neq 0$, whence $k = l$. Thus the $\mathfrak{b}_\lambda\mathfrak{a}_\chi$ form a diagonal scheme, proving the uniqueness established there.

which gives $\mathfrak{a}_\chi \mathfrak{b}_\lambda = 0$. In the second case there are two classes $\mathfrak{x}_0$ and $\mathfrak{x}_1$ such that

$$\mathfrak{x}_0 \mathfrak{a}_\chi \mathfrak{b}_\lambda + \mathfrak{x}_1(\mathfrak{b}_1 + \cdots + \mathfrak{b}_{\lambda-1} + \mathfrak{b}_{\lambda+1} + \cdots + \mathfrak{b}_l) = \mathfrak{e} = \mathfrak{b}_1 + \cdots + \mathfrak{b}_{\lambda-1} + \mathfrak{b}_{\lambda+1} + \cdots + \mathfrak{b}_l + \mathfrak{b}_\lambda,$$

and therefore, by uniqueness, $\mathfrak{x}_0 \mathfrak{a}_\chi \mathfrak{b}_\lambda = \mathfrak{b}_\lambda$. Thus $F\mathfrak{x}_0 \mathfrak{a}_\chi \mathfrak{b}_\lambda$, and hence also $\mathfrak{x}\mathfrak{a}_\chi \mathfrak{b}_\lambda$, runs through the whole group $\mathfrak{B}_\lambda$ as F or $\mathfrak{x}$ runs through all polynomials or all residue classes of $\mathfrak{G}$. This proves the assertion.

If $\mathfrak{a}_\chi \mathfrak{b}_\lambda \neq 0$, then $\mathfrak{U}_\chi$ is also isomorphic to $\mathfrak{B}_\lambda$. For by § 7 the totality of all classes $\mathfrak{x}\mathfrak{a}_\chi \neq 0$ for which $\mathfrak{x}\mathfrak{a}_\chi \mathfrak{b}_\lambda \neq 0$ is isomorphic to the system of classes $\mathfrak{x}\mathfrak{a}_\chi \mathfrak{b}_\lambda$; and these exhaust the subgroup $\mathfrak{B}_\lambda$. It remains only to show that the indicated classes $\mathfrak{x}\mathfrak{a}_\chi$ exhaust $\mathfrak{U}_\chi$. To this end, consider all residue classes $\mathfrak{x}\mathfrak{a}_\chi$ for which $\mathfrak{x}\mathfrak{a}_\chi \mathfrak{b}_\lambda = 0$. This system has the property that the sum of two of its classes and the product with an arbitrary polynomial again belong to the system. If all these residue classes are adjoined to the module $\mathfrak{M}_{\mathfrak{a}_\chi}$, one again obtains a module which is a divisor of $\mathfrak{M}_{\mathfrak{a}_\chi}$; thus it is either $\mathfrak{M}_{\mathfrak{a}_\chi}$ or the unit module. In the latter case, however, $\mathfrak{a}_\chi$ itself would be contained in it, so that $\mathfrak{a}_\chi \mathfrak{b}_\lambda = 0$, contrary to the hypothesis. Hence there are no classes $\mathfrak{x}\mathfrak{a}_\chi \neq 0$ for which $\mathfrak{x}\mathfrak{a}_\chi \mathfrak{b}_\lambda = 0$, and the isomorphism of $\mathfrak{U}_\chi$ with $\mathfrak{B}_\lambda$ is proved.

Suppose now, for a fixed χ, that $\mathfrak{a}_\chi \mathfrak{b}_{\lambda_1} \neq 0, \ldots, \mathfrak{a}_\chi \mathfrak{b}_{\lambda_i} \neq 0$ with $i > 1$. Then $\mathfrak{U}_\chi$ is isomorphic to the groups $\mathfrak{B}_{\lambda_1}, \ldots, \mathfrak{B}_{\lambda_i}$, so these groups are mutually isomorphic.

We now show that at least two of the $\mathfrak{U}$ must then also be isomorphic. For if in the products $\mathfrak{b}_\lambda \mathfrak{a}_\chi$ there belonged to each $\mathfrak{b}_\lambda$ only a single $\mathfrak{a}_\chi$ with $\mathfrak{b}_\lambda \mathfrak{a}_\chi \neq 0$, so that $\mathfrak{b}_\lambda = \mathfrak{b}_\lambda \mathfrak{a}_\chi$, then $\mathfrak{B}_\lambda = \mathfrak{U}_\chi$, since $\mathfrak{U}_\chi$, as a prime group, cannot contain a proper subgroup. Thus $k = l$ and, with a correct numbering, $\mathfrak{a}_\chi = \mathfrak{b}_\chi$; but then the scheme $\mathfrak{a}_\chi \mathfrak{b}_\lambda$ would be a diagonal scheme, which is not the case. The following theorem is therefore proved.

Theorem V. *If a completely reducible group $\mathfrak{G}$ is represented in two ways as a sum of prime groups,*

$$\mathfrak{G} = \mathfrak{U}_1 + \cdots + \mathfrak{U}_k = \mathfrak{B}_1 + \cdots + \mathfrak{B}_l,$$

then every group $\mathfrak{U}$ is isomorphic to at least one group $\mathfrak{B}$, and conversely. If every $\mathfrak{U}$ is isomorphic to exactly one $\mathfrak{B}$, then the decompositions are identical. In the other case at least two groups $\mathfrak{U}$ are mutually isomorphic, and likewise at least two groups $\mathfrak{B}$ are mutually isomorphic.

§ 9. Connection Between the Concepts “Isomorphic” and “of the Same Kind”.

In this paragraph we show that the concept “of the same kind”, known for differential expressions in one variable, when generalized in the natural way, leads to the isomorphism of the associated residue groups.

Definition.[6] *Two modules $\mathfrak{M}$ and $\mathfrak{N}$ are called of the same kind if there are two polynomials P and Q, with P relatively prime to $\mathfrak{M}$ and Q relatively prime to $\mathfrak{N}$, such that for every $M \equiv 0(\mathfrak{M})$ and $N \equiv 0(\mathfrak{N})$*

$$MQ \equiv 0(\mathfrak{N}), \tag{25}$$
$$NP \equiv 0(\mathfrak{M}) \tag{26}$$

hold. Relative primeness here means the existence of two polynomials X and Y such that

$$YQ \equiv 1(\mathfrak{N}), \tag{27}$$
$$XP \equiv 1(\mathfrak{M}) \tag{28}$$

hold.

We now have the following theorem.

Theorem VI. *Two modules are of the same kind if and only if their residue groups are isomorphic.*

1. Let $\mathfrak{A}$ be the residue group of $\mathfrak{M}$, $\mathfrak{B}$ the residue group of $\mathfrak{N}$, and suppose $\mathfrak{A}$ is isomorphic to $\mathfrak{B}$. Let $\mathfrak{a}$ and $\mathfrak{b}$ be the unit classes of $\mathfrak{A}$ and $\mathfrak{B}$. Further, let $Q\mathfrak{b}$ be the class in $\mathfrak{B}$ corresponding to the unit $\mathfrak{a}$ in $\mathfrak{A}$, and let $P\mathfrak{a}$ be the class in $\mathfrak{A}$ corresponding to the unit $\mathfrak{b}$ of $\mathfrak{B}$. We write this assignment as

$$\mathfrak{a} \sim Q\mathfrak{b}, \tag{29}$$
$$P\mathfrak{a} \sim \mathfrak{b}. \tag{30}$$

It follows that

$$0 = M\mathfrak{a} \sim MQ\mathfrak{b}, \qquad \text{that is,} \qquad MQ \equiv 0(\mathfrak{N}),$$

which proves (25). Likewise,

$$NP\mathfrak{a} \sim N\mathfrak{b} = 0, \qquad \text{that is,} \qquad NP \equiv 0(\mathfrak{M}),$$

[6] For differential expressions in one variable, cf. Blumberg’s formal definition, loc. cit., p. 25.

which proves (26). Moreover, by (29), $P\mathfrak{a} \sim PQ\mathfrak{b}$; since by (30) $P\mathfrak{a} \sim \mathfrak{b}$ was assumed, uniqueness of the assignment gives $PQ\mathfrak{b} = \mathfrak{b}$, hence $PQ \equiv 1(\mathfrak{N})$, and similarly $QP \equiv 1(\mathfrak{M})$. Thus (27) and (28) are proved, with $X = Q$, $Y = P$.

2. Conversely, suppose $\mathfrak{M}$ and $\mathfrak{N}$ are of the same kind, so that relations (25) through (28) hold. We assign to an arbitrary class $F\mathfrak{a}$ of $\mathfrak{A}$ the class $FQ\mathfrak{b}$ of $\mathfrak{B}$. This assigns to each class in $\mathfrak{A}$ only one class in $\mathfrak{B}$: for from $F\mathfrak{a} = 0$, hence $F = M$, it follows by (25) that $FQ\mathfrak{b} = 0$. Moreover, the whole group $\mathfrak{B}$ is exhausted by this assignment, since the special class $Y\mathfrak{a}$ corresponds to $YQ\mathfrak{b}$, which equals $\mathfrak{b}$ by (27). Thus we have an unambiguous map Φ from $\mathfrak{A}$ onto $\mathfrak{B}$; it has not yet been shown to be one-to-one. Similarly, equations (26) and (28) give an unambiguous map Ψ from $\mathfrak{B}$ onto $\mathfrak{A}$, which again has not yet been shown to be one-to-one.

If either of these two maps is one-to-one, then the isomorphism is proved, since the requirements concerning sums and products are fulfilled. But if both maps Φ and Ψ were not one-to-one, then there would be classes $G\mathfrak{a}$ in $\mathfrak{A}$ corresponding to the class $GQ\mathfrak{b} = 0$ in $\mathfrak{B}$. The classes $FQ\mathfrak{b}$ corresponding to all the other residue classes $F\mathfrak{a}$ would then already exhaust the group $\mathfrak{B}$, and consequently the map Ψ would return the whole group $\mathfrak{A}$ in the classes $FQP\mathfrak{a}$. As we know, there are also nonzero classes in $\mathfrak{B}$ which, under Ψ, correspond to the zero class in $\mathfrak{A}$; we denote by $G_1\mathfrak{a}$ all those classes for which $G_1\mathfrak{a} \neq 0$ but $G_1QP\mathfrak{a} = 0$. All remaining polynomials F have the property that $FQP\mathfrak{a}$ exhausts the group $\mathfrak{A}$. Those polynomials F for which $FQP\mathfrak{a} = G_1\mathfrak{a}$ will be denoted by G_2; such polynomials exist whenever polynomials G_1 exist. For each G_2 one has $G_2QP\mathfrak{a} \neq 0$, but $G_2(QP)^2\mathfrak{a} = 0$. Correspondingly, for every integer ν there are classes $G_\nu\mathfrak{a}$ such that $G_\nu(QP)^{\nu-1}\mathfrak{a} \neq 0$ but $G_\nu(QP)^\nu\mathfrak{a} = 0$.

Now consider the system $\mathfrak{S}$ consisting of all polynomials $G_1, G_2, \ldots$. It has a finite module basis $G_{\lambda_1}, \ldots, G_{\lambda_e}$. Choose ν greater than the largest of the indices $\lambda_1, \ldots, \lambda_e$, which we denote by λ. Then

$$G_\nu = A_1G_{\lambda_1} + \cdots + A_eG_{\lambda_e}.$$

Multiplying by $(QP)^\lambda$ gives

$$G_\nu(QP)^\lambda \equiv 0(\mathfrak{M}),$$

a contradiction to our hypothesis. Hence the isomorphism follows, and therefore, by part 1, the sharper relations (27), (28) hold with $X = Q$, $Y = P$.

By Theorem VI, Theorem V may also be expressed as follows.

Theorem VII.[7] *If a completely reducible module $\mathfrak{M}$ is representable in two ways as the least common multiple of prime modules $\mathfrak{P}_1, \ldots, \mathfrak{P}_k$ and $\mathfrak{D}_1, \ldots, \mathfrak{D}_l$, each forming a totally relatively prime system, then every $\mathfrak{P}$ is of the same kind as at least one $\mathfrak{D}$, and conversely. If every $\mathfrak{P}$ is of the same kind as exactly one $\mathfrak{D}$, then the decompositions are identical. In the other case at least two of the modules $\mathfrak{P}$ are of the same kind as one another, and likewise at least two of the modules $\mathfrak{D}$ are of the same kind as one another.*

[7]Cf. Loewy, pp. 100–101.

§ 10. Existence of Infinitely Many Decompositions of a Group.

If the representation of a completely reducible group as a sum is not unique, then by Theorem V every representation contains at least two mutually isomorphic subgroups. We now consider the subgroup formed from such a system of isomorphic subgroups,

$$\mathfrak{H} = \mathfrak{A}_1 + \cdots + \mathfrak{A}_\alpha,$$

where $\mathfrak{A}_1, \mathfrak{A}_2, \ldots, \mathfrak{A}_\alpha$ are all isomorphic, and assert that $\mathfrak{H}$ then admits infinitely many distinct decompositions of this kind, provided only that the rationality domain P contains infinitely many "constants", that is, quantities a for which $\xi_i a = a\xi_i$. This theorem is valid even when no further hypothesis is imposed on the groups $\mathfrak{A}_i$, not even irreducibility. Thus we have:

Theorem VIII. *If $\mathfrak{G}$ can be represented as a sum of finitely many mutually isomorphic subgroups, then this representation is possible in infinitely many distinct ways, provided only that the rationality domain P contains infinitely many constants. Here constants mean those quantities a of the domain for which $\xi_i a = a\xi_i$.*

Proof. Let

$$\mathfrak{G} = \mathfrak{A}_1 + \cdots + \mathfrak{A}_\alpha,$$

and let $\mathfrak{A}_i$ be isomorphic to $\mathfrak{A}_\varkappa$ for $i, \varkappa = 1, \ldots, \alpha$ ($\alpha \geq 2$). In order to specify a definite isomorphic assignment, we first single out one subgroup, say $\mathfrak{A}_1$. Let the isomorphism between $\mathfrak{A}_1$ and $\mathfrak{A}_i$ be fixed by

$$a_1 \sim t_{1i} a_i, \qquad a_i \sim t_{i1} a_1, \qquad t_{11} a_1 = a_1 \quad (i, \varkappa = 1, \ldots, \alpha), \tag{31}$$

so that, in the group $\mathfrak{A}_1$, each residue class is assigned to itself. Since $Ma_1 = 0$ and $Ma_i = 0$, it follows that $Mt_{1i}a_i = 0$ and $Mt_{i1}a_1 = 0$; hence the classes $t_{1i}a_i$ and $t_{i1}a_1$, analogously to units, admit multiplication by classes, not merely by polynomials. It follows from the definition of isomorphism that, for two assigned classes $\mathfrak{y}$ and $\bar{\mathfrak{y}}$ admitting this class multiplication, the classes $\mathfrak{r}\mathfrak{y}$ and $\mathfrak{r}\bar{\mathfrak{y}}$ are again assigned to each other. Therefore the relations $a_r a_s = 0$ ($r \neq s$) ($r = 1, \ldots, \alpha$; $s = 1, \ldots, \alpha$) give, for $s = 1$,

$$\begin{cases} & a_r t_{1\varkappa} a_\varkappa = 0 & (\varkappa = 1, \ldots, \alpha,\ r \neq 1), \\ \text{and for } s > 1 & a_r t_{s1} a_1 = 0 & (r \neq s), \\ \quad\text{hence also} & t_{1i}a_i = a_1 t_{1i} a_i, \quad t_{i1}a_1 = a_i t_{i1} a_1. & \end{cases} \tag{32}$$

Furthermore, from (31), generalizing (27) and (28), the uniqueness of the assignment gives

$$t_{1i} t_{i1} a_1 = a_1, \qquad t_{\varkappa 1} t_{1\varkappa} a_\varkappa = a_\varkappa. \tag{33}$$

Let the isomorphism between $\mathfrak{A}_i$ and $\mathfrak{A}_\varkappa$ now be the relation obtained by composing (31); such a stipulation is necessary, since the individual groups may admit isomorphisms into themselves. From (31), (32), and (33) we therefore get

$$a_i \sim t_{i1} t_{1\varkappa} a_\varkappa = t_{i\varkappa} a_\varkappa, \qquad t_{i\varkappa} a_i = a_i, \qquad a_r t_{s\varkappa} a_\varkappa = 0 \quad (r \neq s), \tag{34}$$

where $t_{i1} t_{1\varkappa} = t_{i\varkappa}$ has been put. From (33) and (34) one obtains further

$$t_{i\varkappa} t_{\varkappa l} a_l = t_{i1} t_{1\varkappa} t_{\varkappa 1} t_{1l} a_l = t_{i1} t_{1l} a_l = t_{il} a_l;$$

thus the singling out of the group $\mathfrak{A}_1$ is again removed. The isomorphic relation between $\mathfrak{A}_i$ and $\mathfrak{A}_\varkappa$ remains the same, whichever group $\mathfrak{A}_\varkappa$ is used in place of $\mathfrak{A}_1$ for the definition. Hence the $t_{i\varkappa}$ are classes which, in summary, satisfy the relations

$$Mt_{i\varkappa} a_\varkappa = 0; \quad a_r t_{i\varkappa} a_\varkappa = 0 \ (r \neq i); \quad t_{i\varkappa} t_{\varkappa l} a_l = t_{il} a_l; \quad t_{\varkappa i} a_i = t_{\varkappa i} t_{i\varkappa} a_\varkappa = a_\varkappa. \tag{35}$$

These relations say precisely that the modules $\mathfrak{M}_i$ and $\mathfrak{M}_\varkappa$ belonging to $\mathfrak{A}_i$ and $\mathfrak{A}_\varkappa$ are of the same kind (cf. the definition in § 9); in place of the P and Q there we have here the classes $t_{i\varkappa}$ and $t_{\varkappa i}$, and the first two relations (35), taken together, correspond to formula (25) or (26). Conversely, these relations imply the isomorphism by Theorem VI.[20]

From the classes $t_{i\varkappa} a_\varkappa$ we now compose classes b_ρ which will again turn out to be units of groups $\mathfrak{B}_\rho$, whence a representation

$$\mathfrak{G} = \mathfrak{B}_1 + \cdots + \mathfrak{B}_\alpha$$

[20]And in fact without applying the finiteness theorem, since the formulas already display respectively a map $\varphi_{i\varkappa}$ and its inverse $\varphi_{\varkappa i}$. For from $ra_i = ra_i t_{i\varkappa} a_\varkappa t_{\varkappa i} a_i$ it follows that $ra_i \neq 0$ also implies $ra_i t_{i\varkappa} a_\varkappa \neq 0$.

will follow. Let $\lambda_{i\varkappa}$ be constants whose determinant $|\lambda_{i\varkappa}|$ is nonzero, and let $\lambda^{\varkappa r}$ be the corresponding cofactors divided by $|\lambda_{i\varkappa}|$, so that the relations

$$\begin{cases} \sum_{r=1}^{\alpha} \lambda_{ri}\lambda^{r\varkappa} = \delta_{i\varkappa} = \begin{cases} 0, & i \neq \varkappa, \\ 1, & i = \varkappa, \end{cases} \\ \sum_{r=1}^{\alpha} \lambda_{ir}\lambda^{\varkappa r} = \delta_{i\varkappa} \end{cases} \tag{36}$$

hold. We then define

$$b_\rho = \sum_{i,\varkappa} \lambda_{\rho i}\lambda^{\rho\varkappa} t_{i\varkappa} a_\varkappa \qquad (\rho = 1, \dots, \alpha). \tag{37}$$

From (35), (36), and (37) one obtains $Mb_\rho = 0$ and

$$\sum_{\rho=1}^{\alpha} b_\rho = \sum_{i,\varkappa,\rho} \lambda_{\rho i}\lambda^{\rho\varkappa} t_{i\varkappa} a_\varkappa = \sum_{i,\varkappa} \delta_{i\varkappa} t_{i\varkappa} a_\varkappa = \sum_i t_{ii} a_i = \sum_i a_i = e, \tag{38}$$

and also

$$\begin{aligned} b_\rho b_\sigma &= \sum_{\substack{i,\varkappa \\ \mu,\nu}} \lambda_{\rho i}\lambda^{\rho\varkappa}\lambda_{\sigma\mu}\lambda^{\sigma\nu} t_{i\varkappa} a_\varkappa t_{\mu\nu} a_\nu \\ &= \sum_{i,\mu,\nu} \lambda_{\rho i}\lambda^{\rho\mu}\lambda_{\sigma\mu}\lambda^{\sigma\nu} t_{i\nu} a_\nu = \delta_{\rho\sigma} \sum_{i,\nu} \lambda_{\rho i}\lambda^{\sigma\nu} t_{i\nu} a_\nu = \begin{cases} 0 & (\rho \neq \sigma), \\ b_\rho & (\rho = \sigma), \end{cases} \end{aligned}$$

$(\rho, \sigma = 1, \dots, \alpha)$. Thus the representation $\mathfrak{r}e = \mathfrak{r}b_1 + \cdots + \mathfrak{r}b_\alpha$ is an additive decomposition of the group $\mathfrak{G}$. The groups $\mathfrak{B}_\rho$ are distinct from the groups $\mathfrak{A}_\sigma$ as soon as the $\lambda_{i\varkappa}$ do not merely form a diagonal matrix. For if we form $a_\sigma b_\rho$, then by (35) we get

$$a_\sigma b_\rho = \sum_{i,\varkappa} \lambda_{\rho i}\lambda^{\rho\varkappa} a_\sigma t_{i\varkappa} a_\varkappa = \lambda_{\rho\sigma} \sum_\varkappa \lambda^{\rho\varkappa} t_{\sigma\varkappa} a_\varkappa \neq 0 \quad \text{for } \lambda_{\rho\sigma} \neq 0,$$

since otherwise each individual summand in the last sum would have to vanish, which is not the case. Thus if, for fixed σ, there are at least two ρ with $\lambda_{\rho\sigma} \neq 0$, then $\mathfrak{A}_\sigma$ cannot be identical with any $\mathfrak{B}_\rho$.

But the groups $\mathfrak{B}_\rho$ are also mutually isomorphic and isomorphic to the groups $\mathfrak{A}_\sigma$. We first show the existence of classes $r_{\rho\sigma}a_\sigma$ and $r^{\sigma\rho}b_\rho$ which mediate the isomorphism of $\mathfrak{A}_\sigma$ and $\mathfrak{B}_\rho$ $(\rho, \sigma = 1, \dots, \alpha)$. Indeed, we show that for the classes

$$r_{\rho\sigma} a_\sigma = \sum_i \lambda_{\rho i} t_{i\sigma} a_\sigma, \qquad r^{\sigma\rho} = \sum_\varkappa \lambda^{\rho\varkappa} t_{\sigma\varkappa} a_\varkappa = r^{\sigma\rho} b_\rho$$

the relations "of the same kind" are fulfilled between the modules

$$(\mathfrak{M}, b_1 + \cdots + b_{\rho-1} + b_{\rho+1} + \cdots + b_\alpha)$$

and

$$(\mathfrak{M}, a_1 + \cdots + a_{\sigma-1} + a_{\sigma+1} + \cdots + a_\alpha):$$

$$\begin{cases} b_\varkappa r_{\rho\sigma} a_\sigma = 0 \quad (\varkappa \neq \rho), & r^{\sigma\rho} r_{\rho\sigma} a_\sigma = a_\sigma, \\ a_\varkappa r^{\sigma\rho} b_\rho = 0 \quad (\varkappa \neq \sigma), & r_{\rho\sigma} r^{\sigma\rho} b_\rho = b_\rho, \end{cases} \tag{39}$$

from which, as in (35), the isomorphism follows through the assignment

$$b_\rho \sim r_{\rho\sigma} a_\sigma, \qquad a_\sigma \sim r^{\sigma\rho} b_\rho. \tag{40}$$

Indeed,

$$\begin{aligned} b_\varkappa r_{\rho\sigma} a_\sigma &= \sum_{\mu,\nu,i} \lambda_{\varkappa\mu}\lambda^{\varkappa\nu} t_{\mu\nu} a_\nu \lambda_{\rho i} t_{i\sigma} a_\sigma \\ &= \sum_{\mu,i} \lambda_{\varkappa\mu}\lambda^{\varkappa i}\lambda_{\rho i} t_{\mu\sigma} a_\sigma = \delta_{\varkappa\rho} \sum_\mu \lambda_{\varkappa\mu} t_{\mu\sigma} a_\sigma = \delta_{\varkappa\rho} r_{\rho\sigma} a_\sigma = 0 \quad (\varkappa \neq \rho), \end{aligned}$$

and further

$$r^{\sigma\rho} r_{\rho\sigma} a_\sigma = \sum_{\nu,i} \lambda^{\rho\nu} t_{\sigma\nu} a_\nu \lambda_{\rho i} t_{i\sigma} a_\sigma = \sum_i \lambda^{\rho i}\lambda_{\rho i} a_\sigma = a_\sigma.$$

Conversely, in the same way one finds

$$a_\varkappa r^{\sigma\rho} b_\rho = a_\varkappa \sum_\mu \lambda^{\rho\mu} t_{\sigma\mu} a_\mu = 0 \quad (\varkappa \neq \sigma),$$

and

$$r_{\rho\sigma} r^{\sigma\rho} b_\rho = \sum_{i,\mu} \lambda_{\rho i} t_{i\sigma} a_\sigma \lambda^{\rho\mu} t_{\sigma\mu} a_\mu = \sum_{i,\mu} \lambda_{\rho i} \lambda^{\rho\mu} t_{i\mu} a_\mu = b_\rho.$$

From the relations of the same kind just proved, the isomorphism of $\mathfrak{A}_\sigma$ and $\mathfrak{B}_\rho$ follows (again without the finiteness theorem). The isomorphism between $\mathfrak{B}_\rho$ and $\mathfrak{B}_\tau$ is obtained from this by composition; we record the formulas. Put

$$\ell_{\rho\tau} b_\tau = r_{\rho\sigma} r^{\sigma\tau} b_\tau = \sum_{i,j} \lambda_{\rho i} t_{i\sigma} a_\sigma \lambda^{\tau j} t_{\sigma j} a_j = \sum_{i,j} \lambda_{\rho i} \lambda^{\tau j} t_{ij} a_j;$$

then the assignment is $b_\rho \sim \ell_{\rho\tau} b_\tau$. Here too the relations of the same kind can again be verified formally; if the $\ell_{\rho\tau}$ are put in place of the $t_{i\varkappa}$, they satisfy (35). Indeed,

$$b_\sigma \ell_{\rho\tau} b_\tau = b_\sigma r_{\rho\sigma} r^{\sigma\tau} b_\tau = 0 \quad (\sigma \neq \rho),$$

$$\ell_{\rho\sigma} \ell_{\sigma\tau} b_\tau = r_{\rho\varkappa} r^{\varkappa\sigma} r_{\sigma\lambda} r^{\lambda\tau} b_\tau = r_{\rho\varkappa} a_\varkappa r^{\varkappa\tau} b_\tau = \ell_{\rho\tau} b_\tau.$$

This proves Theorem VIII, since one need only choose the $\lambda_{i\varkappa}$ in infinitely many ways so that they form no diagonal matrix and also are not transformed into one another by diagonal matrices.

A consequence of Theorem VIII, in connection with Theorem V or VII, is:

Theorem IX.[21] *Let the module $\mathfrak{M}$ be completely reducible and the least common multiple of prime modules $\mathfrak{P}_1, \dots, \mathfrak{P}_k$ which form a totally relatively prime system, and suppose the rationality domain P contains infinitely many constants. Then the necessary and sufficient condition that $\mathfrak{M}$ be representable in infinitely many distinct ways as the least common multiple of prime modules is that at least two of the modules $\mathfrak{P}_1, \dots, \mathfrak{P}_k$ be of the same kind.*

We shall now derive general criteria for a module $\mathfrak{M}$ to be divisible by infinitely many prime modules, and first prove the following:

Auxiliary theorem.[22] *If a completely reducible module $\mathfrak{M} = [\mathfrak{P}_1, \dots, \mathfrak{P}_k]$ is divisible by a prime module $\mathfrak{P}'$ different from all the $\mathfrak{P}_i$, then at least two of the $\mathfrak{P}_i$ are of the same kind.*

We first select from $\mathfrak{P}_1, \dots, \mathfrak{P}_k$ a totally relatively prime system by successively deleting every one of these prime modules that is contained in the least common multiple of those not yet deleted. Hence assume now that $\mathfrak{P}_1, \dots, \mathfrak{P}_k$ themselves form a totally relatively prime system. Let a' be a class of $\mathfrak{M}$ corresponding to the unit class modulo $\mathfrak{P}'$. Then

$$Fa' = Fa'a_1 + \cdots + Fa'a_k$$

for every polynomial F. At least two of the products $a'a_i$ are nonzero, say $a'a_1$ and $a'a_2$; for if, for instance, only $a'a_1 \neq 0$, then $Fa' = Fa'a_1$, and the residue classes Fa' would form a subgroup of $\mathfrak{A}_1$, hence, since $\mathfrak{A}_1$ is a prime group, would be identical with $\mathfrak{A}_1$, so that the modules $\mathfrak{P}'$ and $\mathfrak{P}_1$ would also coincide. By the reasoning of § 8, it follows that $\mathfrak{A}'$ is isomorphic to $\mathfrak{A}_1$ and $\mathfrak{A}_2$.

Let $\mathfrak{M}$ now be an arbitrary module. We consider the least common multiple $\mathfrak{N}$ of all prime modules by which $\mathfrak{M}$ is divisible.[23] Two cases are possible. Either $\mathfrak{N}$ is not representable as the least common multiple of finitely many prime modules; in this case $\mathfrak{N}$, and hence also $\mathfrak{M}$, has infinitely many prime modules as divisors. Or $\mathfrak{N}$ is representable as the least common multiple of finitely many prime modules; then $\mathfrak{N}$ is called the greatest completely reducible module of $\mathfrak{M}$ (following a corresponding construction of Loewy). If among these finitely many prime modules two are of the same kind, then by Theorem IX the module $\mathfrak{N}$, and therefore $\mathfrak{M}$, is divisible by infinitely many prime modules. Conversely, if the latter is the case, then there is at least one prime divisor of $\mathfrak{N}$ different from all the finitely many modules whose least common multiple represents $\mathfrak{N}$. By the auxiliary theorem, $\mathfrak{N}$, hence $\mathfrak{M}$, has two prime divisors of the same kind. Thus we have proved:

Theorem X. *If the rationality domain P contains infinitely many constants, then the necessary and sufficient condition that a module be divisible by infinitely many distinct prime modules is that either no greatest completely reducible module exists, or, if such a module exists, that it be divisible by at least two distinct prime modules of the same kind.*

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[21] Cf. Loewy, pp. 106–107.

[22] Cf. Loewy, p. 96.

[23] If there are infinitely many of these, then by Theorem IV no additive decomposition of the residue group can correspond to this representation. Cf. also the example § 12, 4.

§ 11. Relations to Systems of Differential Equations and Their Integrals.

By virtue of the finiteness theorem, to every module of differential expressions there can be assigned as a basis a system of finitely many differential expressions, so that the totality of simultaneous integrals of all differential equations obtained by setting a differential expression of the module equal to zero coincides with the totality of integrals of the finite system of differential equations obtained by setting the basis differential expressions equal to zero. This gives the connection with the theory of systems of linear partial differential equations; we pursue this connection a little further in the case where the residue group of the module is "finite".

Definition. *A residue group is called finite if there is a number μ, the order of the group, such that between any $\mu + 1$ residue classes there is a linear relation with coefficients from P, while there is at least one system of μ residue classes between which no relation exists, and which we shall call a linear basis of the residue group. In the other case the residue group is called infinite.*

The sum of two finite residue groups has finite order, namely the sum of their orders. The sum of two infinite groups, or of a finite and an infinite group, is an infinite group. Examples of finite groups are furnished by all residue groups of modules in one variable; but for several variables the order can also be finite, for instance for the module with basis ξ_1^2, ξ_2^3, where all residue classes can be linearly composed from those of $1, \xi_1, \xi_2, \xi_1\xi_2$. For examples of infinite groups see § 12.

We now have the following:

Theorem XI. *If a module has a finite residue group, then every system of differential equations obtained by setting a basis of the module equal to zero admits exactly as many linearly independent integrals as the order of its residue group indicates. Conversely, for a given system of finitely many linear partial differential equations that possess only a finite number of linearly independent integrals, this number is equal to the order of the residue group of the module generated by the differential expressions.*

Here the rationality domain P consists of analytic functions of n variables $x_1, \ldots, x_n$, with the occurring singularities, inside a certain n-dimensional domain, forming at most manifolds of lower dimensions, so that there are arbitrarily many points with regular n-dimensional neighborhoods.

Let $\mathfrak{M}$ be a module with finite residue group. We show that there is a linear basis of power products with the property that *when an arbitrary residue class is expressed through this basis, only powers of finitely many different quantities from P can occur in the denominator.*

For this purpose we order all power products $\xi_1^{a_1} \cdots \xi_n^{a_n}$ lexicographically and take as representatives $Q_1, \ldots, Q_m$ of the residue classes all those that are linearly independent modulo $\mathfrak{M}$ of the preceding ones. By hypothesis there are only finitely many such products. All the others can be expressed linearly through these.

Let $\mathfrak{S}$ be the system of all power products not admitted into the linear basis. Then, by the first part of the proof of Theorem III, $\mathfrak{S}$ has a finite subsystem $\mathfrak{T}$ of power products such that every power product from $\mathfrak{S}$ is divisible by at least one from $\mathfrak{T}$. Let $P_1, \ldots, P_\rho$ be the power products of $\mathfrak{T}$, and let $M_1 = 0, \ldots, M_\rho = 0$ $(\mathfrak{M})$ be the relations by means of which $P_1, \ldots, P_\rho$ are each expressed modulo $\mathfrak{M}$ by a linear combination of lower power products; for example,

$$M_i = b_i P_i + \text{lower terms.}$$

If P is an arbitrary power product from $\mathfrak{S}$, hence of the form

$$\xi_1^{h_1} \cdots \xi_n^{h_n} P_i,$$

then

$$\begin{aligned} \xi_1^{h_1} \cdots \xi_n^{h_n} M_i &= b_i \xi_1^{h_1} \cdots \xi_n^{h_n} P_i + \text{lower terms} \\ &= b_i P + \text{lower terms.} \end{aligned}$$

Assume now that it has already been proved for all power products lower than P that only powers of $b_1, \ldots, b_\rho$ occur in denominators, while the numerators are formed from the coefficients of the M_i by addition, multiplication, and differentiation. Then the same holds for P itself. This assumption is permitted, since it is fulfilled for P_1, the first element from $\mathfrak{S}$ and $\mathfrak{T}$.

By Theorem III, moreover, $M_1, \ldots, M_\rho$ is a module basis of $\mathfrak{M}$. Now consider a system of values $\beta_1, \ldots, \beta_n$ of $x_1, \ldots, x_n$ for which $b_1 \neq 0, \ldots, b_\rho \neq 0$ and the remaining coefficients of the M_i are regular, that is, a regular point of the system of differential equations $M_1 = 0, \ldots, M_\rho = 0$. The power products $Q_i = \xi_1^{h_{i1}} \cdots \xi_n^{h_{in}}$ $(i = 1, \ldots, m)$ correspond to the partial derivatives

$$\frac{\partial^{h_{i1} + \cdots + h_{in}}}{\partial x_1^{h_{i1}} \cdots \partial x_n^{h_{in}}} y,$$

for which we prescribe arbitrary constant values $(\gamma_1, \ldots, \gamma_m)$ at the point $(\beta_1, \ldots, \beta_n)$. Then the differential equations $M_i = 0$ determine uniquely, as finite values at this point, the values of all higher derivatives of y, since only the b_i occur in denominators; and it is known that this implies the existence of m linearly independent analytic integrals.

On the other hand, under our hypotheses there cannot exist more than m linearly independent integrals, since otherwise at a regular point one could prescribe arbitrarily the values of $m+1$ suitably chosen derivatives, contrary to the fact that linear dependencies modulo $\mathfrak{M}$ must exist among any $m+1$ power products.

Conversely, if a module has an infinite residue group, the same method shows that there are infinitely many linearly independent integrals; therefore, in the case of only m linearly independent integrals, the residue group is finite, and its order is, by the above, equal to m.

To close this paragraph we show that, for two modules of differential expressions, the concept of isomorphism, or equivalently of being of the same kind, has for the integrals of the associated systems of differential equations the same meaning as Poincaré's concept of kind for one variable (cf. Blumberg, loc. cit., Appendix I).

Theorem XII. *If two modules $\mathfrak{M}$ and $\mathfrak{N}$ of differential expressions are of the same kind, then there are two differential expressions P and Q such that $P(y)$ and $Q(z)$ run through all integrals of $\mathfrak{N}$ and $\mathfrak{M}$, respectively, when y runs through all integrals of $\mathfrak{M}$ and z through all integrals of $\mathfrak{N}$.*

By hypothesis, namely (by § 9), for two suitably chosen polynomials P and Q the relations

$$MQ = 0(\mathfrak{N}), \qquad PQ = 1(\mathfrak{N}),$$
$$NP = 0(\mathfrak{M}), \qquad QP = 1(\mathfrak{M})$$

hold. Thus, since $NP(y) = 0$, the function $P(y)$ is an integral of $\mathfrak{N}$. Likewise $Q(z)$ is an integral of $\mathfrak{M}$. Since $PQ(z) = z$ and $QP(y) = y$, $P(y)$ runs through all integrals of $\mathfrak{N}$, and $Q(z)$ all integrals of $\mathfrak{M}$. In this way, to each nonzero y there corresponds a nonzero z, and conversely.

§ 12. Examples.

1. As a first example we consider an ordinary linear differential equation of second order whose coefficients are represented as symmetric functions of the integrals. The rationality domain P consists here of all rational functions of the integrals and their derivatives; we restrict ourselves to a sufficiently small neighborhood of a regular point of the differential equation. We define the module $\mathfrak{M}$ as the totality of all differential expressions that have y_1 and y_2 as integrals, and for which we again write polynomials in one indeterminate ξ. The module basis is then one-membered, and is given by the polynomial

$$M = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} \xi^2 - \begin{vmatrix} y_1 & y_2 \\ y_1'' & y_2'' \end{vmatrix} \xi + \begin{vmatrix} y_1' & y_2' \\ y_1'' & y_2'' \end{vmatrix},$$

which is uniquely determined up to a factor from P. Now

$$\mathfrak{M} = [\mathfrak{M}_1, \mathfrak{M}_2], \tag{41}$$

where $\mathfrak{M}_i$ consists of all differential expressions having the integral y_i, and whose basis is determined up to a factor from P by

$$M_i = y_i\xi - y_i' \qquad (i = 1, 2).$$

Indeed, $\mathfrak{M}$ is divisible by $\mathfrak{M}_1$ and $\mathfrak{M}_2$, since it vanishes for the integrals y_1 and y_2, and hence is divisible by $[\mathfrak{M}_1, \mathfrak{M}_2]$; and since the order of the basis polynomial of $[\mathfrak{M}_1, \mathfrak{M}_2]$ must be at least 2, one has $\mathfrak{M} = [\mathfrak{M}_1, \mathfrak{M}_2]$. Further, $\mathfrak{M}_1$ and $\mathfrak{M}_2$ are relatively prime, because

$$\frac{y_2}{D}(y_1\xi - y_1') - \frac{y_1}{D}(y_2\xi - y_2') = 1, \qquad D = D(y) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}. \tag{42}$$

They are also prime modules, since their residue group has order 1.

But besides y_1 and y_2, $\mathfrak{M}$ also has all integrals

$$z_1 = \alpha_{11}y_1 + \alpha_{12}y_2, \qquad z_2 = \alpha_{21}y_1 + \alpha_{22}y_2, \qquad A = |\alpha_{ik}| \neq 0,$$

and hence is identical with the least common multiple of the two relatively prime modules $\mathfrak{N}_1$ and $\mathfrak{N}_2$, which have respectively the integrals z_1 and z_2, with basis polynomials

$$N_i = z_i\xi - z_i' = (\alpha_{i1}y_1 + \alpha_{i2}y_2)\xi - (\alpha_{i1}y_1' + \alpha_{i2}y_2').$$

Thus $\mathfrak{M}$ admits infinitely many distinct representations as the least common multiple of relatively prime prime modules; by Theorem VII the two modules $\mathfrak{N}_1$ and $\mathfrak{N}_2$ are therefore of the same kind as one another and as all $\mathfrak{M}_1$ and $\mathfrak{M}_2$. Indeed, the residue groups of all these modules have order 1, and so are isomorphic.

We shall now write the additive decomposition of the residue group $\mathfrak{G} = \mathfrak{A}_1 + \mathfrak{A}_2$ corresponding to the representation (41). To do this we start, as in § 4, from relation (42), which expresses the relative primeness of $\mathfrak{M}_1$ and $\mathfrak{M}_2$. For the units a_1 and a_2 we obtain the residue classes generated by the polynomials

$$-\frac{y_1}{D}(y_2\xi - y_2') \quad \text{and} \quad \frac{y_2}{D}(y_1\xi - y_1').$$

Thus

$$\mathfrak{M}_1 = \left(\mathfrak{M}, \frac{y_2}{D}(y_1\xi - y_1')\right), \qquad \mathfrak{M}_2 = \left(\mathfrak{M}, -\frac{y_1}{D}(y_2\xi - y_2')\right).$$

If we now form in the same way the units b_1 and b_2 corresponding to the decomposition $\mathfrak{M} = [\mathfrak{N}_1, \mathfrak{N}_2]$, replacing y_i by z_i, we get

$$-\frac{z_1}{D(z)}(z_2\xi - z_2') = -\frac{\alpha_{11}y_1 + \alpha_{12}y_2}{AD(y)}((\alpha_{21}y_1 + \alpha_{22}y_2)\xi - (\alpha_{21}y_1' + \alpha_{22}y_2')),$$

$$\frac{z_2}{D(z)}(z_1\xi - z_1') = \frac{\alpha_{21}y_1 + \alpha_{22}y_2}{AD(y)}((\alpha_{11}y_1 + \alpha_{12}y_2)\xi - (\alpha_{11}y_1' + \alpha_{12}y_2')),$$

and therefore

$$\begin{cases} b_1 = \dfrac{\alpha_{11}y_1 + \alpha_{12}y_2}{y_1}\left(\dfrac{\alpha_{22}}{A}\right)a_1 + \dfrac{\alpha_{11}y_1 + \alpha_{12}y_2}{y_2}\left(-\dfrac{\alpha_{21}}{A}\right)a_2 = \dfrac{z_1}{y_1}\alpha^{11}a_1 + \dfrac{z_1}{y_2}\alpha^{12}a_2, \\ b_2 = \dfrac{\alpha_{21}y_1 + \alpha_{22}y_2}{y_1}\left(-\dfrac{\alpha_{12}}{A}\right)a_1 + \dfrac{\alpha_{21}y_1 + \alpha_{22}y_2}{y_2}\left(\dfrac{\alpha_{11}}{A}\right)a_2 = \dfrac{z_2}{y_1}\alpha^{21}a_1 + \dfrac{z_2}{y_2}\alpha^{22}a_2. \end{cases} \tag{43}$$

Here (α^{ik}) is the inverse matrix of (α_{ik}).

Conversely, exchanging z_i with y_i gives

$$\begin{cases} a_1 = \dfrac{y_1}{z_1}\alpha_{11}b_1 + \dfrac{y_1}{z_2}\alpha_{12}b_2, \\ a_2 = \dfrac{y_2}{z_1}\alpha_{21}b_1 + \dfrac{y_2}{z_2}\alpha_{22}b_2. \end{cases} \tag{44}$$

These formulas display the isomorphism of the groups $\mathfrak{A}$ and $\mathfrak{B}$. Namely, by our general considerations, if $a_1 b_\sigma \neq 0$ the corresponding assignment is $a_1 \sim a_1 b_\sigma$, and hence by (44)

$$a_1 \sim \frac{y_1}{z_\sigma}\alpha_{1\sigma}b_\sigma, \qquad \text{provided } \alpha_{1\sigma} \neq 0,$$

and by (43) likewise

$$b_\sigma \sim \frac{z_\sigma}{y_2}\alpha^{\sigma 2}a_2, \qquad \text{provided } \alpha^{\sigma 2} \neq 0,$$

so that

$$\frac{y_1}{z_\sigma}\alpha_{1\sigma}b_\sigma \sim \frac{y_1}{z_\sigma}\alpha_{1\sigma}\frac{z_\sigma}{y_2}\alpha^{\sigma 2}a_2 = \frac{y_1}{y_2}\alpha_{1\sigma}\alpha^{\sigma 2}a_2.$$

Hence

$$a_1 \sim \alpha_{1\sigma}\alpha^{\sigma 2}\frac{y_1}{y_2}a_2 \quad \text{for } \alpha_{1\sigma}\alpha^{\sigma 2} \neq 0, \qquad \frac{1}{\alpha_{1\sigma}\alpha^{\sigma 2}}\frac{y_2}{y_1}a_1 \sim a_2,$$

which gives the inverse. If (α_{ik}) is not a diagonal matrix, this condition $\alpha_{1\sigma}\alpha^{\sigma 2} \neq 0$ is always fulfilled for some σ. The case of a diagonal matrix must be excluded, since then the modules are identical. If, for example, only $\alpha_{12} = 0$, then $\mathfrak{A}_1 = \mathfrak{B}_1$, but not $\mathfrak{A}_2 = \mathfrak{B}_2$. Thus from $\mathfrak{A}_1 + \mathfrak{A}_2 = \mathfrak{B}_1 + \mathfrak{B}_2$ and $\mathfrak{A}_1 = \mathfrak{B}_1$ it does not follow that $\mathfrak{A}_2 = \mathfrak{B}_2$ (cf. note 12). The three modules $\mathfrak{M}_1$, $\mathfrak{M}_2$, and $\mathfrak{N}_2$ are then pairwise relatively prime, but do not form a totally relatively prime system, since $[\mathfrak{M}_1, \mathfrak{M}_2]$ is divisible by $\mathfrak{N}_2$ (cf. note 15).

Putting further

$$t_{12} = \alpha_{1\sigma}\alpha^{\sigma 2}\frac{y_1}{y_2}a_2, \qquad t_{21} = \frac{1}{\alpha_{1\sigma}\alpha^{\sigma 2}}\frac{y_2}{y_1}a_1,$$

one verifies by direct calculation that the relations (35) are fulfilled.

2. Every module in more than one variable whose basis consists of a single polynomial gives an example of an infinite group. The following somewhat more complicated example is meant to show that infinite groups can also occur for multi-membered modules, that is, for modules whose basis contains more than one polynomial.[24]

$$\mathfrak{M} = [(\xi + x\eta), (\xi + 1)] = [(P), (Q)] \qquad \left(\xi = \frac{\partial}{\partial x},\ \eta = \frac{\partial}{\partial y}\right).$$

The multiplication rule is that for differential expressions. The residue group is here infinite because the residue groups of $(\xi + x\eta)$ and $(\xi + 1)$ are both infinite, whereas the modules $(\xi + x\eta)$ and $(\xi + 1)$ are relatively prime, since

$$1 = (-x\xi - x^2\eta + 1)Q + (x\xi + x - 1)P.$$

It is now very easy to show that $\mathfrak{M}$ possesses no polynomial of the second degree: one sets

$$(u_1\xi + u_2\eta + u_3)(\xi + x\eta) = (v_1\xi + v_2\eta + v_3)(\xi + 1),$$

which gives $u_1 = u_2 = u_3 = v_1 = v_2 = v_3 = 0$. If one next looks for polynomials of the third degree in $\mathfrak{M}$, one finds two linearly independent ones, for example

$$(\xi^2 + x\xi\eta + \xi + (2 + x)\eta)Q = (\xi^2 + 2\xi + 1)P$$

and

$$(\xi^2 + 2x\xi\eta + x^2\eta^2 + \eta)Q = (\xi^2 + x\xi\eta + \xi + (x - 1)\eta)P.$$

If $\mathfrak{M}$ were a one-membered module, then both would have to be divisible by the basis polynomial, and, since $\mathfrak{M}$ contains no polynomial of the second degree, by a polynomial of the third degree; they would therefore have to be proportional, which they are not. The multi-memberedness of $\mathfrak{M}$ is the internal reason for the failure of Landau's product theorems, which are valid in the case of one variable.

3. A further example shows that infinite groups can also occur for prime modules.[25] Let P be the domain of all rational functions of two variables x, y. The module $\mathfrak{M}$ with basis

$$\xi - \frac{y}{x} = \xi - a$$

has an infinite group, since this group contains the residue class of every power of y; on the other hand, $\mathfrak{M}$ is a prime module. Indeed, we show that every divisor

$$\mathfrak{M}_1 = (\xi - a, F(\xi, \eta))$$

is equal to the unit module. Every polynomial $F(\xi, \eta)$ can be represented modulo $(\xi - a)$ by a polynomial $G(\eta)$:

$$F(\xi, \eta) = c_0\eta^\nu + c_1\eta^{\nu-1} + \cdots + c_\nu \quad (\mathfrak{M}).$$

Without loss of generality let $c_0 = 1$. By the multiplication law, $\xi\eta^\nu - \eta^\nu\xi = 0$; hence, since

$$\xi \equiv a(\mathfrak{M}_1), \qquad \eta^\nu \equiv F_1(\mathfrak{M}_1),$$

where $F_1 = -(c_1\eta^{\nu-1} + \cdots + c_\nu)$, one has

$$\xi F_1 - \eta^\nu a \equiv 0(\mathfrak{M}_1).$$

Expanding

$$\eta^\nu a = a\eta^\nu + \nu\frac{\partial a}{\partial y}\eta^{\nu-1} + \cdots,$$

and again replacing η^ν by F_1, one obtains

$$\xi(c_1\eta^{\nu-1} + \cdots + c_\nu) - a(c_1\eta^{\nu-1} + \cdots + c_\nu) + \nu\frac{\partial a}{\partial y}\eta^{\nu-1} + \cdots \equiv 0(\mathfrak{M}_1),$$

[24] Cf. the example of Landau communicated by Blumberg (loc. cit., pp. 51–52). Here and below we denote the independent variables by x and y.

[25] Our definition of "prime module" differs from what is called a prime module in the commutative case, since there divisors of lower dimension always exist, except when the dimension is 0, that is, when the residue group is finite.

or

$$c_1\eta^{\nu-1}a + \frac{\partial c_1}{\partial x}\eta^{\nu-1} + \cdots - ac_1\eta^{\nu-1} + \cdots + u\frac{\partial a}{\partial y}\eta^{\nu-1} + \cdots \equiv 0(\mathfrak{M}_1),$$

or finally

$$\left(\frac{\partial c_1}{\partial x} + \nu\frac{\partial a}{\partial y}\right)\eta^{\nu-1} + \cdots \equiv 0(\mathfrak{M}_1).$$

On the left now stands a polynomial in η of exact degree $\nu - 1$ which does not vanish identically, for the highest coefficient

$$\frac{\partial c_1}{\partial x} + \nu\frac{\partial a}{\partial y} = \frac{\partial c_1}{\partial x} + \nu\frac{1}{x} \quad \text{is } \neq 0,$$

since otherwise $c_1 = -\nu\log x + \varphi(y)$ would not be rational. Thus the procedure can be continued and leads to a non-identically-vanishing polynomial of degree zero, proving that the unit is also contained in $\mathfrak{M}_1$.[26]

4. On the other hand, if one adjoins the function $\log x$ to the rationality domain, then $\mathfrak{M}$ has the divisors

$$\left(\xi - \frac{y}{x},\ \eta - \log x + \varphi(y)\right),$$

where $\varphi(y)$ runs through all rational functions of y. These divisors are all distinct, the integrability condition is fulfilled for each, and the corresponding integral is

$$y\log x - \int\varphi(y)\,dy + c.$$

Therefore the unit cannot be contained in the module, since otherwise the zero function would be the only possible integral. On the other hand, ξ and η are congruent, with respect to the module, to a quantity of the domain; hence the residue group is finite and of order 1. Further, any two of the infinitely many divisors are relatively prime, and the given module $\mathfrak{M}$ is equal to the least common multiple $\mathfrak{N}$ of all these divisors. Indeed, it is divisible by all divisors, hence also by $\mathfrak{N}$. If $\mathfrak{N}$ were a proper divisor of $\mathfrak{M}$, then $\mathfrak{N}$ would still have to contain at least one polynomial F, say of degree ρ, depending only on η alone. Letting $\varphi(y)$ run through the functions $y, \ldots, y^{\rho+1}$, there is no linear relation with coefficients independent of y between the corresponding integrals

$$y\log x - \frac{y^{i+1}}{i+1} + c_i \qquad (i = 1, \ldots, \rho+1).$$

But the differential equation of order ρ, $F = 0$, cannot possess $\rho + 1$ linearly independent integrals.

The module $\mathfrak{M} = \mathfrak{N}$ is, however, not completely reducible.[27] For otherwise it would be the least common multiple of finitely many of these prime modules, and its residue group would therefore be of finite order; yet the residue group of $\xi - y/x$ contains the residue classes of all powers of η, and so is infinite.

Göttingen, August 1, 1919.

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[26]This procedure amounts to proving that the integrability conditions necessary for a system of differential equations are not fulfilled.

[27]Thus we have an example of the first case of Theorem X.

18. On a Work of K. Hentzelt, Fallen in the War, on Elimination Theory

Jahresbericht der DMV 30 (1921), p. 101

11th meeting, Friday, September 23, 1921, 11 a.m.
(Chair: Loewy.)

1. E. Noether: On a work of K. Hentzelt, fallen in the war, on elimination theory.

Kurt Hentzelt has been missing before Dixmuiden since October 1914 and must be counted among the dead. I intend soon to publish, in a free redaction in the *Mathematische Annalen*, the essential parts of his Erlangen dissertation, which he left behind built up without gaps but almost incomprehensibly presented.

The matter concerns the general problem of **elimination theory**: to determine the common zeros of all polynomials of an ideal $\mathfrak{M}$ consisting of polynomials. If the variables are first subjected to a linear transformation with indeterminate coefficients, then the main result can be stated as follows:

To the ideal $\mathfrak{M}$ there can be assigned uniquely a resultant form

$$R_{\mathfrak{M}} = R^{(1)}(x_1, \dots, x_n) \cdots R^{(i)}(x_i, \dots, x_n) \cdots R^{(m)}(x_n) = 0(\mathfrak{M}),$$

such that $R_{\mathfrak{M}}$ vanishes for all zeros of $\mathfrak{M}$ and only for these. If $\mathfrak{N}$ is a divisor of $\mathfrak{M}$ and $R_{\mathfrak{N}}$ and $R_{\mathfrak{M}}$ agree, then $\mathfrak{N}$ and $\mathfrak{M}$ themselves also agree.

The second part of the theorem shows that the resultant form supplies the zeros in characteristic multiplicity. It rests on the fact that $\mathfrak{M}$ can be regarded as a module of linear forms in the power products of $x_1, \dots, x_{i-1}$; the individual factors $R^{(i)}(x_i, \dots, x_n)$ of $R_{\mathfrak{M}}$ then become the norms of the respective "basic module" with respect to this module, when $x_{i+1}, \dots, x_n$ is adjoined to the coefficient domain. If one invokes abstract ideal theory, this yields the following interpretation: if

$$\mathfrak{M} = [\mathfrak{Q}, \mathfrak{Q}_1, \dots, \mathfrak{Q}_r] = [\mathfrak{Q}, \mathfrak{L}]$$

is a decomposition of $\mathfrak{M}$ into primary ideals $\mathfrak{Q}_i$, and $\mathfrak{L}$ is the complement of $\mathfrak{Q}$, then to the ideal $\mathfrak{Q}$ there corresponds a primary factor $Q^{(i)}(x_i, \dots, x_n)$ of $R^{(i)}(x_i, \dots, x_n)$, which, in the sense specified above, represents the norm of $\mathfrak{L}$ with respect to $\mathfrak{Q}$.

19. Ideal Theory in Ring Domains

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Table of Contents.

Introduction.

The content of the present paper is the *transfer of the decomposition theorems for the rational integers, respectively of the ideals in algebraic number fields, to ideals in arbitrary integral domains, and more generally ring domains.* To understand this transfer, let us first state the decomposition theorems for the rational integers in a form somewhat different from the usual formulation.

If in

$$a = p_1^{\varrho_1} p_2^{\varrho_2} \cdots p_\sigma^{\varrho_\sigma} = q_1 q_2 \cdots q_\sigma$$

one regards the prime-power factors q_i as the components of the decomposition, then these components have the following characteristic properties:

1. They are *pairwise coprime*; but no q_i can be represented as a product of pairwise coprime numbers, so that in this sense irreducibility holds. From the pairwise coprimeness it further follows that the product $q_1 \cdots q_\sigma$ is equal to the least common multiple $[q_1 \cdots q_\sigma]$.

2. Any two of the components, q_i and q_k, are *relatively prime*; that is, if bq_i is divisible by q_k, then b is divisible by q_k. In this sense too irreducibility holds.

3. Every q is *primary*; that is, if a product $b \cdot c$ is divisible by q, but b is not divisible, then a power[1] of c is divisible. The representation is moreover one by *greatest primary components*, since the product of two distinct q is no longer primary. Also with respect to decomposition into greatest primary components the q are irreducible.

4. Every q is *irreducible* in the sense that it cannot be represented as the least common multiple of two proper divisors.

The connection of these primary numbers q with the prime numbers p consists in this: to each q there exists one, and apart from sign only one, p which is a divisor of q and a power of which is divisible by q: the associated prime number. If p^ϱ is the lowest power of this kind – ϱ the exponent of q – then here, in particular, p^ϱ is equal to q. The uniqueness theorem may now be stated as follows:

For two different decompositions of a rational integer into the irreducible, greatest primary components q, the number of components, the associated prime numbers (up to sign), and the exponents coincide. Since $p^\varrho = q$, it follows from this also that the q themselves coincide (up to sign).

The indeterminacy arising from the sign is, as is well known, removed when one considers, instead of the numbers, the ideals derived from them (all numbers divisible by a); then the formulation holds in exactly the same way for the unique decomposition of the ideals of finite algebraic number fields into powers of prime ideals.

[1] If this power is always the first, then one is, as is well known, dealing with prime numbers.

In what follows (§ 1), the basis will be a general ring domain satisfying only the finiteness condition that each ideal of the domain possess a finite ideal basis. Without such a finiteness condition, irreducible and prime ideals need not exist at all, as is shown by the domain of all algebraic integers, in which there is no decomposition into prime ideals.

It turns out that – corresponding to the four characteristic properties of the components q – in general four separate decompositions exist, each arising from the preceding one by subdivision. The decomposition into coprime-irreducible ideals is a product representation; the other three decompositions are reduced (§ 2) representations as least common multiples. The connection between a primary ideal – the irreducible ideals too are primary – and the associated prime ideal also persists: to every primary ideal $\mathfrak{Q}$ there is uniquely determined an associated prime ideal $\mathfrak{P}$ which is a divisor of $\mathfrak{Q}$ and a power of which is divisible by $\mathfrak{Q}$. If $\mathfrak{P}^\varrho$ is the lowest such power – ϱ the exponent of $\mathfrak{Q}$ – then here $\mathfrak{P}^\varrho$ need not coincide with $\mathfrak{Q}$. The uniqueness theorem is expressed as follows:

Decompositions 1 and 2 are unique; in two different decompositions 3 or 4 the number of components and the associated prime ideals coincide;[2] the isolated ideals occurring among the components (§ 7) are uniquely determined.

For the proof of the decomposition theorems, the finiteness condition first yields the "theorem on the finite chain", first stated by Dedekind for finite number modules; from it the representation 4 of each ideal as the least common multiple of finitely many irreducible ideals is derived. By transforming the concept of reducibility of a component, one obtains from this the fundamental uniqueness theorem for decomposition 4 into irreducible ideals. By combining finitely many components at a time one rises to the remaining decompositions, whose uniqueness theorems then follow from the uniqueness theorem 4.

Finally it is shown (§ 9) that the representation by finitely many irreducible components still holds under weaker hypotheses; commutativity of the ring domain is not required, and it is enough to consider, in place of an ideal, a module with respect to the domain. In this more general case the equality of the number of components in two different decompositions still holds, whereas the notions prime and primary are tied to commutativity and to the ideal concept; by contrast, the notion coprime remains valid for ideals in noncommutative domains.

The simplest ring domain for which the four separate decompositions actually occur is the domain of all polynomials in n variables with arbitrary complex coefficients. Here the individual decompositions can be interpreted, irrationally, by the behavior of algebraic configurations, and the uniqueness theorem for the associated prime ideals corresponds to the fundamental theorem of elimination theory on the unique decomposability of algebraic configurations into irreducible ones. Further examples are given by all finite integral domains made of polynomials (§ 10). But even the simple domain of all even numbers, more generally of all numbers divisible by a fixed number, already gives an example of partially separated decompositions (§ 11). An example of ideal theory in noncommutative domains is provided by elementary-divisor theory (§ 12), where unique decomposition into irreducible ideals, respectively classes, holds. These irreducible classes completely characterize the irreducible constituents of the elementary divisors, and may perhaps be viewed as their equivalent for domains in which the usual elementary-divisor theory fails.

On the existing literature the following is to be said. The decomposition into greatest primary ideals for the polynomial domain with arbitrary complex, respectively integral, coefficients was given by Lasker, and in individual points further developed by Macaulay.[3] Both rely on elimination theory, and hence use the fact that a polynomial can be represented uniquely as a product of irreducible polynomials. In fact the decomposition theorems for ideals are independent of this prerequisite, as ideal theory in algebraic number fields suggests and as the present paper shows. The primary ideal is also defined by Lasker and Macaulay on the basis of concepts from elimination theory.

The decomposition into irreducible ideals and the decomposition into relatively-prime irreducible ideals do not seem to have been noticed in the literature even for the polynomial domain; only in Macaulay is there a remark on the uniqueness of isolated primary ideals.

The decomposition into coprime-irreducible ideals was given for the polynomial domain by Schmeidler,[4] using elimination theory for the proof of finiteness. There, however, the uniqueness theorem is stated only for classes of ideals, not for the ideals themselves. This latter uniqueness theorem appears in a joint paper,[5]

[2] Presumably, beyond this, the exponents also coincide, and still more generally corresponding components are isomorphic.

[3] E. Lasker, Zur Theorie der Moduln und Ideale. Math. Ann. 60 (1905), p. 20, Theorems VII and XIII. – F. S. Macaulay, On the Resolution of a given Modular System into Primary Systems including some Properties of Hilbert Numbers. Math. Ann. 74 (1913), p. 66.

[4] W. Schmeidler, Über Moduln und Gruppen hyperkomplexer Größen. Math. Zeitschr. 3 (1919), p. 29.

[5] E. Noether–W. Schmeidler, Moduln in nichtkommutativen Bereichen, insbesondere aus Differential- und Differenzenausdrücken. Math. Zeitschr. 8 (1920), p. 1.

where ideals in noncommutative polynomial domains are involved. Only the finite ideal basis is used there; hence theorems and methods remain valid for general ring domains, and are sharpened in the present paper with respect to equality of number (§ 11). The present investigations are a strong generalization and further development of the conceptual formations underlying those two papers. The essential feature of the two papers is the passage from representation as a least common multiple to an additive decomposition of the system of residue classes. Here, for simplicity of presentation, we remain with the least common multiple; the additive decomposition is then matched by the transformation of the notion of reducibility into a property of the complement (§ 3). Yet by considerations essentially corresponding to those of the joint paper, all the theorems stated here can also be understood as additive decomposition theorems for the system of residue classes and certain partial systems. This system of residue classes forms a ring of the same generality as the ring originally taken as basis; indeed every ring may be regarded as the system of residue classes of the ideal corresponding to the totality of identical relations among the ring elements; or also of a partial system of these relations, if the remaining relations are assumed to be satisfied already in the domain.

This remark also gives the place of Fraenkel's papers.[6] Fraenkel considers additive decompositions of rings that are subjected to such restrictive conditions (existence of regular elements, division by these, decomposability condition) that, for the corresponding ideal, the four decompositions coincide. Because of this coincidence, his finiteness condition, that the ideal should have only finitely many proper divisors – from which he in part departs – also means no stronger restriction than ours. Fraenkel's starting point is conditioned by the different, essentially algebraic, aims of his papers; by algebraic extension he then arrives at more general rings with less restrictive conditions.

§ 1. Ring domain, ideal, finiteness condition.

1. The domain Σ taken as basis is to be a (commutative) ring in the abstract definition;[7] that is, Σ consists of a system of elements $a, b, c, \dots, f, g, h, \dots$ in which a relation satisfying the usual conditions is defined as equality; and in which by two operations (modes of composition), addition and multiplication, from any two ring elements a and b a third is always uniquely obtained, respectively as sum $a + b$ and as product $a \cdot b$. The ring and the otherwise wholly arbitrary operations must satisfy the following laws:

1. The associative law of addition: $(a + b) + c = a + (b + c)$.
2. The commutative law of addition: $a + b = b + a$.
3. The associative law of multiplication: $(a \cdot b)c = a(b \cdot c)$.
4. The commutative law of multiplication: $a \cdot b = b \cdot a$.
5. The distributive law: $a(b + c) = ab + ac$.
6. The law of unrestricted and unique subtraction. There is in Σ a single element x satisfying the equation $a + x = b$. (One denotes $x = b - a$.)

From these properties follows the existence of zero; but a ring need not possess a unit; and the product of two elements can vanish without one of the factors vanishing. Rings for which, from the vanishing of a product, the vanishing of a factor always follows, and which in addition possess a unit, will be called proper integral domains. For the finite sum $a + a + \cdots + a$ we introduce the usual abbreviated notation na, where the integers n are to be regarded merely as abbreviating signs, not as ring elements, and are defined recursively by $a = 1a$, $na + a = (n + 1)a$.

2. By an ideal $\mathfrak{M}$[8] in Σ is meant a system of elements from Σ satisfying the two conditions:

1. $\mathfrak{M}$ contains, together with f, also $a \cdot f$, where a is an arbitrary element of Σ.
2. $\mathfrak{M}$ contains, together with f and g, also the difference $f - g$; hence with f also nf for every integer n.

[6] A. Fraenkel, Über die Teiler der Null und die Zerlegung von Ringen. J. f. M. 145 (1914), p. 139. Über gewisse Teilbereiche und Erweiterungen von Ringen. Habilitationsschrift, Leipzig, Teubner, 1916. Über einfache Erweiterungen zerlegbarer Ringe. J. f. M. 151 (1920), p. 121.

[7] The definition is taken from Fraenkel's habilitation thesis, omitting his restrictive conditions 6, I and II; for this the commutative law of addition had to be included. Thus these are the laws defining a field with the invertibility of multiplication omitted.

[8] Ideals are denoted by capital German letters. $\mathfrak{M}$ is meant to recall the example of the ideal in polynomials usually called a "module" or form module. Incidentally, §§ 1–3 use only the module property and not the ideal property; compare § 9.

If f is an element of $\mathfrak{M}$, we express this as usual by $f \equiv 0(\mathfrak{M})$ and say that f is divisible by $\mathfrak{M}$. If every element of $\mathfrak{N}$ is at the same time an element of $\mathfrak{M}$, and hence divisible by $\mathfrak{M}$, we say: $\mathfrak{N}$ is divisible by $\mathfrak{M}$; in symbols, $\mathfrak{N} \equiv 0(\mathfrak{M})$. $\mathfrak{M}$ is called a proper divisor of $\mathfrak{N}$ when it contains elements different from those of $\mathfrak{N}$, and hence is not conversely divisible by $\mathfrak{N}$. From $\mathfrak{N} \equiv 0(\mathfrak{M})$ and $\mathfrak{M} \equiv 0(\mathfrak{N})$ follows $\mathfrak{N} = \mathfrak{M}$.

The other familiar notions are also retained word for word. By the greatest common divisor of two ideals $\mathfrak{A}$ and $\mathfrak{B}$ – $\mathfrak{D} = (\mathfrak{A}, \mathfrak{B})$ – we mean the totality of elements that can be represented in the form $a + b$, where a runs through all elements of $\mathfrak{A}$ and b through all elements of $\mathfrak{B}$; $\mathfrak{D}$ is again an ideal. Likewise, the greatest common divisor of infinitely many ideals – $\mathfrak{D} = (\mathfrak{A}_1, \mathfrak{A}_2, \ldots, \mathfrak{A}_\nu, \ldots)$ – is defined as the totality of elements d representable as sums of elements from finitely many of the ideals at a time: $d = a_{i_1} + a_{i_2} + \cdots + a_{i_n}$; here too $\mathfrak{D}$ is again an ideal.

If, in particular, the ideal $\mathfrak{M}$ contains a finite number of elements $f_1, f_2, \ldots, f_\varrho$ such that

$$\mathfrak{M} = (f_1 \ldots f_\varrho), \qquad f = a_1 f_1 + \cdots + a_\varrho f_\varrho + n_1 f_1 + \cdots + n_\varrho f_\varrho$$

for every $f \equiv 0(\mathfrak{M})$, where the a_i are quantities of the ring domain and the n_i are integers, then $\mathfrak{M}$ is called a finite ideal; $f_1, \ldots, f_\varrho$ an ideal basis.

In what follows we take as basis only rings Σ satisfying the finiteness condition: *Every ideal in Σ is finite, hence possesses an ideal basis.*

3. From the finiteness condition there follows directly the result on which all subsequent considerations rest.

Theorem I (theorem on the finite chain).[9] *Let $\mathfrak{M}, \mathfrak{M}_1, \mathfrak{M}_2, \ldots, \mathfrak{M}_\nu, \ldots$ be a countably infinite system of ideals in Σ, each of which is divisible by the following one. Then from some finite index n onward all ideals are identical, $\mathfrak{M}_n = \mathfrak{M}_{n+1} = \cdots$. In other words: if $\mathfrak{M}, \mathfrak{M}_1, \mathfrak{M}_2, \ldots, \mathfrak{M}_s, \ldots$ form a simply ordered chain of ideals such that every ideal is a proper divisor of the one immediately preceding it, then the chain breaks off after finitely many steps.*

Indeed, let $\mathfrak{D} = (\mathfrak{M}, \mathfrak{M}_1, \mathfrak{M}_2, \ldots, \mathfrak{M}_\nu, \ldots)$ be the greatest common divisor of the system, and let $f_1, \ldots, f_\varrho$ be a basis of $\mathfrak{D}$, which always exists by the finiteness condition. From the divisibility assumption it follows that every element of $\mathfrak{D}$ is at the same time an element of one ideal of the chain; for from

$$f = g + h, \qquad g \equiv 0(\mathfrak{M}_r), \qquad h \equiv 0(\mathfrak{M}_s), \qquad (r < s)$$

one gets $g \equiv 0(\mathfrak{M}_s)$ and hence $f \equiv 0(\mathfrak{M}_s)$. The same holds when f is a sum of several constituents. Hence there is a finite index n such that

$$f_1 \equiv 0(\mathfrak{M}_n), \quad f_2 \equiv 0(\mathfrak{M}_n), \quad \ldots, \quad f_\varrho \equiv 0(\mathfrak{M}_n).$$

Since conversely $\mathfrak{M}_n \equiv 0(\mathfrak{D})$, we have $\mathfrak{M}_n = \mathfrak{D}$; and since further $\mathfrak{M}_n \equiv 0(\mathfrak{D})$ and $\mathfrak{D} = \mathfrak{M}_n \equiv 0(\mathfrak{M}_r)$, we also have $\mathfrak{M}_r = \mathfrak{D} = \mathfrak{M}_n$ for every $r > n$, proving the theorem.

We note that conversely the existence of an ideal basis again follows from this theorem, so that the finiteness condition could also have been stated in this basis-free form.

§ 2. Representation of an ideal as the least common multiple of finitely many irreducible ideals.

The least common multiple $[\mathfrak{B}_1, \mathfrak{B}_2, \ldots, \mathfrak{B}_k]$ of the ideals $\mathfrak{B}_1, \mathfrak{B}_2, \ldots, \mathfrak{B}_k$ is defined as usual as the totality of the elements divisible by $\mathfrak{B}_1$, by $\mathfrak{B}_2$, …, and by $\mathfrak{B}_k$; in symbols,

$$\text{from } f \equiv 0(\mathfrak{B}_i), \quad (i = 1, 2, \ldots, k), \quad \text{follows} \quad f \equiv 0([\mathfrak{B}_1, \mathfrak{B}_2, \ldots, \mathfrak{B}_k]),$$

and conversely. The least common multiple is again an ideal; the ideals $\mathfrak{B}_i$ are also called the components of the decomposition.

Definition I. A representation $\mathfrak{M} = [\mathfrak{B}_1 \cdots \mathfrak{B}_k]$ is called a reduced representation if no $\mathfrak{B}_i$ is contained in the least common multiple $\mathfrak{U}_i$ of the remaining ideals, and if no $\mathfrak{B}_i$ can be replaced by a proper divisor.[10] If the conditions are satisfied only for the ideal $\mathfrak{B}_i$, the representation is called reduced with respect to $\mathfrak{B}_i$. The

[9] First stated for number modules by Dedekind: Zahlentheorie, Suppl. XI, § 172, Theorem VIII (4th ed.); our proof and the designation “chain” are taken from there. For ideals in polynomials, see Lasker, loc. cit., p. 56 (lemma). But in both cases the theorem finds only isolated application. Our applications rest throughout on the principle of choice.

[10] An example of a non-reduced representation is $(x, xy) = [(x), (x^2, xy, y^\lambda)]$ for every exponent $\lambda gtr 2$; the representation $[(x), (x^2, y)]$ corresponding to $\lambda = 1$ is an associated reduced one.

least common multiple $\mathfrak{U}_i = [\mathfrak{B}_1, \dots, \mathfrak{B}_{i-1}, \mathfrak{B}_{i+1}, \dots, \mathfrak{B}_k]$ is called the complement of $\mathfrak{B}_i$. Representations in which only the first condition is satisfied are called shortest representations.

For representing an ideal as a least common multiple it therefore suffices to restrict to reduced representations, by the following result.

Lemma I. *Every representation of an ideal as the least common multiple of finitely many ideals can be replaced, in at least one way, by a reduced representation; such a representation can in particular be attained by successive decomposition.*

Indeed, let $\mathfrak{M} = [\mathfrak{B}_1 \cdots \mathfrak{B}_k]$ be an arbitrary representation of $\mathfrak{M}$. We omit, in order, those $\mathfrak{B}_i$ which are contained in the least common multiple of the ideals retained. Since the remaining ideals still yield $\mathfrak{M}$, in the resulting representation

$$\mathfrak{M} = [\mathfrak{B}_1 \cdots \mathfrak{B}_k] = [\mathfrak{U}_i, \mathfrak{B}_i]$$

the first condition is satisfied, hence it is a shortest representation; and this condition remains satisfied if any $\mathfrak{B}_i$ is replaced by a proper divisor. The second condition, however, is always attainable by the theorem on the finite chain (Theorem I). For if

$$\mathfrak{B}_i \supset \mathfrak{B}_i' \supset \mathfrak{B}_i'' \supset \cdots \supset \mathfrak{B}_i^{(\nu)} \supset \cdots, \qquad \mathfrak{M} = [\mathfrak{U}_i, \mathfrak{B}_i] = [\mathfrak{U}_i, \mathfrak{B}_i'] = \cdots = [\mathfrak{U}_i, \mathfrak{B}_i^{(\nu)}] = \cdots,$$

where every $\mathfrak{B}_i^{(\nu)}$ is a proper divisor of the one immediately preceding, then by that theorem the chain $\mathfrak{B}_i, \mathfrak{B}_i', \dots, \mathfrak{B}_i^{(\nu)}, \dots$ must terminate after finitely many steps; hence in the representation $\mathfrak{M} = [\mathfrak{U}_i, \mathfrak{B}_i^{(\nu)}]$ the ideal $\mathfrak{B}_i^{(\nu)}$ can no longer be replaced by a proper divisor; and this holds a fortiori if $\mathfrak{U}_i$ is replaced by a proper divisor. Applying the procedure successively to each $\mathfrak{B}_i$, always forming the complement with the already reduced $\mathfrak{B}$, gives a reduced representation.[11]

In order to obtain such a representation successively, it remains to show that from the individual reduced representations

$$\mathfrak{M} = [\mathfrak{B}_1, \mathfrak{C}_1], \qquad \mathfrak{C}_1 = [\mathfrak{B}_2, \mathfrak{C}_2], \quad \dots, \quad \mathfrak{C}_{k-1} = [\mathfrak{B}_k, \mathfrak{C}_k]$$

it follows that the resulting representation $\mathfrak{M} = [\mathfrak{B}_1 \cdots \mathfrak{B}_k, \mathfrak{C}_k]$ is reduced. For this it suffices to show that from reduced representations

$$\mathfrak{M} = [\mathfrak{B}, \mathfrak{C}], \qquad \mathfrak{C} = [\mathfrak{C}_1, \mathfrak{C}_2]$$

one obtains a reduced representation $\mathfrak{M} = [\mathfrak{B}, \mathfrak{C}_1, \mathfrak{C}_2]$. In fact, by hypothesis no $\mathfrak{B}$ lies in its complement; if this happened for a $\mathfrak{C}_i$, then, contrary to the hypothesis in the first representation, $\mathfrak{C}$ would be replaceable by a proper divisor, since by the second reduced representation $\mathfrak{C}_1$ and $\mathfrak{C}_2$ are proper divisors of $\mathfrak{C}$; so the representation is shortest. Moreover, by hypothesis no $\mathfrak{B}$ can be replaced by a proper divisor; if this were possible for a $\mathfrak{C}_i$, it would, contrary to the hypothesis, correspond to replacing $\mathfrak{C}$ by a proper divisor, since the representation for $\mathfrak{C}$ is reduced. Thus the lemma is proved.

Definition II. An ideal $\mathfrak{M}$ is called reducible if it can be represented as the least common multiple of two proper divisors; in the contrary case $\mathfrak{M}$ is called irreducible.

We now prove, by means of Theorem I on the finite chain and using reduced representations:

Theorem II. *Every ideal is representable as the least common multiple of finitely many irreducible ideals.*[12]

For an arbitrary ideal $\mathfrak{M}$ is either irreducible; then $\mathfrak{M} = [\mathfrak{M}]$ is a representation of the kind required by Theorem II; or else $\mathfrak{M} = [\mathfrak{B}_1, \mathfrak{C}_1]$, where $\mathfrak{B}_1$ and $\mathfrak{C}_1$ are proper divisors of $\mathfrak{M}$, and the representation may, by Lemma I, be assumed reduced. For $\mathfrak{C}_1$ the same alternative holds: either it is irreducible, or there is a reduced representation $\mathfrak{C}_1 = [\mathfrak{B}_2, \mathfrak{C}_2]$.

Continuing in this way, one obtains the series of reduced representations

$$\mathfrak{M} = [\mathfrak{B}_1, \mathfrak{C}_1] = [\mathfrak{B}_1, \mathfrak{B}_2, \mathfrak{C}_2] = \cdots = [\mathfrak{B}_1, \dots, \mathfrak{B}_n, \mathfrak{C}_n] = \cdots. \tag{1}$$

In the chain $\mathfrak{C}_1, \mathfrak{C}_2, \dots, \mathfrak{C}_n, \dots$ each $\mathfrak{C}_n$ is a proper divisor of the one immediately preceding, and therefore the chain terminates after finitely many steps; there is an index n such that $\mathfrak{C}_n$ is irreducible. By Lemma I, moreover, the representation $\mathfrak{M} = [\mathfrak{U}_n, \mathfrak{C}_n]$ is reduced; $\mathfrak{C}_n$ therefore cannot lie in its complement $\mathfrak{U}_n$, and in the representation $\mathfrak{M} = [\mathfrak{U}_n, \mathfrak{C}_n]$ the ideal $\mathfrak{U}_n$ cannot be replaced by a proper divisor. Replacing, if necessary,

[11] That such a representation is not uniquely defined by the given representation is shown by the previous example. For $(x^2, xy) = [(x), (x^2, xy, y^\lambda)]$, $\lambda \geq 2$, besides $[(x), (x^2, y)]$ also $[(x), (x^2, ux + y)]$, for arbitrary u, is a reduced representation.

[12] That such a representation is not unique in general is shown by the previous example: $(x^2, xy) = [(x), (x^2, ux + y)]$ for arbitrary u. The two components are irreducible for arbitrary u. Indeed, all divisors of (x) are of the form $(x, g(y))$, where $g(y)$ denotes a polynomial in y; the least common multiple of any two therefore also has this form, hence is a proper divisor of (x). The ideal $(x^2, ux + y)$ possesses only the single divisor (x, y), and therefore is necessarily irreducible as well.

$\mathfrak{U}_n$ by a proper divisor,[13] so that the representation becomes reduced, shows that every reducible ideal admits a reduced representation as the least common multiple of an irreducible ideal and a complementary ideal. In the series (1), therefore, all $\mathfrak{B}_i$ may be assumed irreducible without loss of generality; repeating the preceding argument gives the existence of an irreducible $\mathfrak{C}_n$, proving Theorem II.

§ 3. Equality of the number of components in two different decompositions into irreducible ideals.

For the proof of equality of number, reducibility or irreducibility of an ideal must first be expressed by properties of its complement.

Theorem III.[14] *Let the shortest representation $\mathfrak{M} = [\mathfrak{U}, \mathfrak{C}]$ be reduced with respect to $\mathfrak{C}$. Then the necessary and sufficient condition that $\mathfrak{C}$ be reducible is the existence of two ideals $\mathfrak{N}_1$ and $\mathfrak{N}_2$, proper divisors of $\mathfrak{M}$, such that*

$$\mathfrak{N}_1\mathfrak{N}_2 \equiv 0(\mathfrak{M}), \qquad \mathfrak{N}_i \equiv 0(\mathfrak{U}), \qquad [\mathfrak{N}_1, \mathfrak{N}_2] = \mathfrak{M}. \tag{2}$$

It also follows: if conditions (2) are satisfied and $\mathfrak{C}$ is irreducible, then at least one $\mathfrak{N}_i$ is not a proper divisor of $\mathfrak{M}$; $\mathfrak{N}_i = \mathfrak{M}$.

Let $\mathfrak{C} = [\mathfrak{C}_1, \mathfrak{C}_2]$, where $\mathfrak{C}_1, \mathfrak{C}_2$ are proper divisors of $\mathfrak{C}$. Then

$$\mathfrak{M} = [\mathfrak{U}, \mathfrak{C}] = [\mathfrak{U}, \mathfrak{C}_1, \mathfrak{C}_2] = [[\mathfrak{U}, \mathfrak{C}_1], [\mathfrak{U}, \mathfrak{C}_2]].$$

Here the ideals $[\mathfrak{U}, \mathfrak{C}_i]$ are proper divisors of $\mathfrak{M}$, since otherwise $[\mathfrak{U}, \mathfrak{C}]$ would not be reduced with respect to $\mathfrak{C}$. Since divisibility by $\mathfrak{U}$ is also fulfilled, condition (2) is proved necessary. (The representation (2) is not reduced, since one $[\mathfrak{U}, \mathfrak{C}_i]$ can be replaced by $\mathfrak{C}_i$.)

Conversely, suppose (2) is satisfied. We form the ideals

$$\mathfrak{C}_1 = (\mathfrak{C}, \mathfrak{N}_1), \qquad \mathfrak{C}_2 = (\mathfrak{C}, \mathfrak{N}_2), \qquad \mathfrak{C}^* = [\mathfrak{C}_1, \mathfrak{C}_2].$$

Then $\mathfrak{C}$ is divisible by both $\mathfrak{C}_1$ and $\mathfrak{C}_2$, hence also by their least common multiple $\mathfrak{C}^*$. To show divisibility of $\mathfrak{C}^*$ by $\mathfrak{C}$, let

$$f \equiv 0(\mathfrak{C}^*); \quad f \equiv 0(\mathfrak{C}_1); \quad f \equiv 0(\mathfrak{C}_2), \quad \text{or also} \quad f = c + n_1 = t + n_2,$$

where c, t, n_1, n_2 are quantities in $\mathfrak{C}, \mathfrak{C}, \mathfrak{N}_1, \mathfrak{N}_2$, and in particular n_1 is divisible by $\mathfrak{N}_1$. Hence the difference

$$g = c - t = n_2 - n_1$$

is divisible both by $\mathfrak{C}$ and by $\mathfrak{U}$, and hence by $\mathfrak{M}$. Since $n_1 = n_2 + m$, furthermore, n_1 (and likewise n_2) is divisible by both $\mathfrak{N}_1$ and $\mathfrak{N}_2$, hence by $\mathfrak{M}$. Thus

$$f = c + m, \qquad f \equiv 0(\mathfrak{C}), \qquad \mathfrak{C}^* = \mathfrak{C}.$$

The ideals $\mathfrak{C}_1$ and $\mathfrak{C}_2$ are proper divisors of $\mathfrak{C}$; for if $\mathfrak{C}_1 = (\mathfrak{C}, \mathfrak{N}_1) = \mathfrak{C}$, then $\mathfrak{N}_1$ would be divisible by $\mathfrak{C}$, and hence, because of the divisibility by $\mathfrak{U}$, equal to $\mathfrak{M}$, contrary to the hypothesis. Thus $\mathfrak{C} = [\mathfrak{C}_1, \mathfrak{C}_2]$ has been recognized as reducible; Theorem III is proved.

Almost the same argument also shows the following.

Lemma II. *If in a shortest representation $\mathfrak{M} = [\mathfrak{U}, \mathfrak{C}]$ the ideal $\mathfrak{C}$ can be replaced by a proper divisor, then $\mathfrak{C}$ is reducible.*

Thus suppose

$$\mathfrak{M} = [\mathfrak{U}, \mathfrak{C}] = [\mathfrak{U}, \mathfrak{C}_1],$$

and set

$$\mathfrak{C}^* = [\mathfrak{C}_1, (\mathfrak{U}, \mathfrak{C})].$$

Again $\mathfrak{C}$ is divisible by $\mathfrak{C}^*$. From $f \equiv 0(\mathfrak{C}^*)$ follows

$$f = c_1 = a + c.$$

[13] In fact the representation is also reduced with respect to $\mathfrak{U}_n$, as will be shown in § 3 (Lemma IV) as the converse of Lemma I.

[14] Theorem III corresponds to the passage from modules to residue groups in the papers of Schmeidler and Noether–Schmeidler (compare the Introduction). The residue group corresponds to $\mathfrak{C}$, and $\mathfrak{N}_1$ and $\mathfrak{N}_2$ to the subgroups into which the residue group is decomposed.

The difference $a = c_1 - c$ is therefore divisible both by $\mathfrak{U}$ and by $\mathfrak{C}_1$, hence by $\mathfrak{M}$; thus $c_1 = c + m$, $f \equiv 0(\mathfrak{C})$, $\mathfrak{C}^* = \mathfrak{C}$. Since both $\mathfrak{C}_1$ and $(\mathfrak{U}, \mathfrak{C})$ are, by hypothesis, proper divisors of $\mathfrak{C}$, the ideal $\mathfrak{C} = \mathfrak{C}^*$ is recognized as reducible.

An irreducible $\mathfrak{C}$ therefore cannot be replaced by a proper divisor.

Now let two different shortest representations of $\mathfrak{M}$ as the least common multiple of finitely many irreducible ideals be given:

$$\mathfrak{M} = [\mathfrak{B}_1 \cdots \mathfrak{B}_k] = [\mathfrak{D}_1 \cdots \mathfrak{D}_l].$$

By the remark after Lemma II, these representations are at the same time reduced. We first prove:

Lemma III. *For every complement $\mathfrak{U}_i = [\mathfrak{B}_1 \cdots \mathfrak{B}_{i-1}\mathfrak{B}_{i+1} \cdots \mathfrak{B}_k]$ there exists an ideal $\mathfrak{D}_j$ such that $\mathfrak{M} = [\mathfrak{U}_i, \mathfrak{D}_j]$.*

Indeed, putting $\mathfrak{M} = [\mathfrak{B}_i, \mathfrak{C}_1]$, $\mathfrak{C}_1 = [\mathfrak{D}_1, \mathfrak{C}_2]$, and so on, we get

$$\mathfrak{M} = [\mathfrak{U}_i, \mathfrak{B}_i] = [\mathfrak{U}_i, [\mathfrak{D}_1, \mathfrak{C}_2]] = [\mathfrak{B}_i, [\mathfrak{U}_i, \mathfrak{D}_1], \mathfrak{C}_2].$$

Here, for $\mathfrak{B}_i = [\mathfrak{U}_i, \mathfrak{D}_1]$ and $\mathfrak{C}_i = [\mathfrak{U}_i, \mathfrak{C}_2]$, the conditions (2) of Theorem III are satisfied, since $\mathfrak{M} = [\mathfrak{U}_i, \mathfrak{B}_i]$ is reduced with respect to $\mathfrak{B}_i$ and the representation is shortest. Since $\mathfrak{B}_i$ was assumed irreducible, one $\mathfrak{N}_i$ must necessarily become equal to $\mathfrak{M}$.

If $[\mathfrak{U}_i, \mathfrak{D}_1] = \mathfrak{M}$, the lemma is proved; if $[\mathfrak{U}_i, \mathfrak{C}_2] = \mathfrak{M}$, then correspondingly $\mathfrak{M} = [[\mathfrak{U}_i, \mathfrak{D}_1], [\mathfrak{U}_i, \mathfrak{C}_2]]$, where by the same argument again one component must become equal to $\mathfrak{M}$. Continuing in this way, either $\mathfrak{M} = [\mathfrak{U}_i, \mathfrak{D}_j]$ with $j < l$, or $\mathfrak{M} = [\mathfrak{U}_i, \mathfrak{D}_1, \ldots, \mathfrak{D}_{l-1}]$; since $\mathfrak{D}_1 \cdots \mathfrak{D}_{l-1} = \mathfrak{D}_l$, the lemma is proved.

It follows that:

Theorem IV. *In two different shortest representations of an ideal as the least common multiple of irreducible ideals, the number of components is the same.*

From the lemma, for $i = 1$,

$$\mathfrak{M} = [\mathfrak{B}_1 \cdots \mathfrak{B}_k] = [\mathfrak{U}_1, \mathfrak{D}_{j_1}] = [\mathfrak{D}_{j_1}, \mathfrak{B}_2, \ldots, \mathfrak{B}_k].$$

Considering now the two decompositions

$$\mathfrak{M} = [\mathfrak{D}_{j_1}, \mathfrak{B}_2, \ldots, \mathfrak{B}_k] = [\mathfrak{D}_1, \mathfrak{D}_2, \ldots, \mathfrak{D}_l]$$

and repeating the previous argument with respect to the complement $\mathfrak{U}_2 = [\mathfrak{D}_{j_1}, \mathfrak{B}_3, \ldots, \mathfrak{B}_k]$ of $\mathfrak{B}_2$, one obtains

$$\mathfrak{M} = [\mathfrak{U}_2, \mathfrak{B}_2] = [\mathfrak{U}_2, \mathfrak{D}_{j_2}] = [\mathfrak{D}_{j_1}, \mathfrak{D}_{j_2}, \mathfrak{B}_3, \ldots, \mathfrak{B}_k],$$

and, continuing the procedure,

$$\mathfrak{M} = [\mathfrak{D}_{j_1}, \mathfrak{D}_{j_2}, \ldots, \mathfrak{D}_{j_k}].$$

Since by hypothesis the representation $\mathfrak{M} = [\mathfrak{D}_1 \cdots \mathfrak{D}_l]$ is shortest, so that no $\mathfrak{D}$ can be omitted, the distinct ideals among the $\mathfrak{D}_{j_i}$ must exhaust all $\mathfrak{D}$; hence $k \geq l$. If in the lemma and the following arguments the $\mathfrak{B}$ and $\mathfrak{D}$ are interchanged throughout, one obtains correspondingly $l \geq k$, and therefore $k = l$, proving equality of number. It also follows that the ideals $\mathfrak{D}_{j_i}$ are all mutually distinct, since otherwise in a shortest representation by the $\mathfrak{D}_{j_i}$ fewer than k components would occur; thus the notation can be chosen so that $\mathfrak{D}_{j_i} = \mathfrak{D}_i$. For the same reason, all intermediate representations $\mathfrak{M} = [\mathfrak{D}_{j_1} \cdots \mathfrak{D}_{j_r}, \mathfrak{B}_{r+1}, \ldots, \mathfrak{B}_k]$ are shortest and hence, by the remark after Lemma II, reduced.

Equality of number leads to a converse of Lemma I:

Lemma IV. *If in a reduced representation one combines the components into groups and forms their least common multiples, then the resulting representation is reduced. In other words, from a reduced representation*

$$\mathfrak{M} = [\mathfrak{C}_1 \cdots \mathfrak{C}_k]$$

it follows that $\mathfrak{M} = [\mathfrak{N}_1, \mathfrak{N}_2]$ is also reduced, if $\mathfrak{N}_1 = [\mathfrak{C}_1, \ldots, \mathfrak{C}_h]$ and $\mathfrak{N}_2 = [\mathfrak{C}_{h+1}, \ldots, \mathfrak{C}_k]$.

First, $\mathfrak{N}_1$ cannot lie in its complement $\mathfrak{N}_2$, since this is true of none of its divisors $\mathfrak{C}_i$; the representation is therefore shortest. To show that $\mathfrak{N}_1$ cannot be replaced by a proper divisor, decompose the $\mathfrak{C}$ into irreducible ideals $\mathfrak{B}$,[15] giving the reduced representations, by Lemma I,

$$\mathfrak{N}_1 = [\mathfrak{B}_1 \cdots \mathfrak{B}_\lambda], \qquad \mathfrak{N}_2 = [\mathfrak{B}_{\lambda+1} \cdots \mathfrak{B}_\kappa].$$

[15] This always means a shortest, hence reduced, representation by the $\mathfrak{B}$.

Let $\mathfrak{M} = [\mathfrak{N}_1, \mathfrak{N}_2]$ be reduced with respect to $\mathfrak{N}_2$, and let $\mathfrak{N}_1^*$ be a proper divisor of $\mathfrak{N}_1$. By Lemma II,

$$\mathfrak{N}_1 = [\mathfrak{N}_1^*, (\mathfrak{N}_1, \mathfrak{N}_2)],$$

and this representation is necessarily reduced with respect to $\mathfrak{N}_1^*$, since otherwise $\mathfrak{N}_1$ could also be replaced in $\mathfrak{M}$ by a proper divisor. If necessary, replace $(\mathfrak{N}_1, \mathfrak{N}_2)$ by a proper divisor as well so that a reduced representation for $\mathfrak{N}_1$ is obtained. Decomposing the two components of $\mathfrak{N}_1$ into irreducible ideals, the number λ of irreducible ideals of $\mathfrak{N}_1$ is then additively composed from the numbers of the components; the number of irreducible ideals corresponding to the proper divisor $\mathfrak{N}_1^*$ is therefore necessarily less than λ. But then decomposing $\mathfrak{M} = [\mathfrak{N}_1^*, \mathfrak{N}_2]$ into irreducible ideals would lead to fewer than κ ideals, contradicting equality of number. As the special case $\varrho = 2$, it follows also that the representation $\mathfrak{M} = [\mathfrak{N}_1, \mathfrak{N}_2]$ is reduced with respect to the complement $\mathfrak{N}_2$ as well.

§ 4. Primary ideals. Uniqueness of the associated prime ideals in two different decompositions into irreducible ideals.

In what follows the matter at issue is the connection between primary and irreducible ideals.

Definition III. An ideal $\mathfrak{Q}$ is called primary if from $a \cdot b \equiv 0(\mathfrak{Q})$ and $a \not\equiv 0(\mathfrak{Q})$ it necessarily follows that $b^{\varkappa} \equiv 0(\mathfrak{Q})$, where the exponent $\varkappa$ is a finite number.

The definition may also be stated as follows: if a product $a \cdot b$ is divisible by $\mathfrak{Q}$, then either one factor is divisible or a power of each factor is divisible. If in particular $\varkappa$ is always equal to 1, the ideal is called a prime ideal.

From the definition of the primary (respectively prime) ideal, by virtue of the existence of bases, there follows the formulation involving only products of ideals:[16]

Definition IIIa. An ideal $\mathfrak{Q}$ is called primary if from $\mathfrak{A}\mathfrak{B} \equiv 0(\mathfrak{Q})$ and $\mathfrak{A} \not\equiv 0(\mathfrak{Q})$ it necessarily follows that $\mathfrak{B}^{\varkappa} \equiv 0(\mathfrak{Q})$. If $\varkappa$ is always equal to 1, the ideal is called a prime ideal. For a prime ideal $\mathfrak{P}$, therefore, from $\mathfrak{A}\mathfrak{B} \equiv 0(\mathfrak{P})$ and $\mathfrak{A} \not\equiv 0(\mathfrak{P})$ it always follows that $\mathfrak{B} \equiv 0(\mathfrak{P})$.

Since IIIa contains Definition III as the special case $\mathfrak{A} = (a)$, $\mathfrak{B} = (b)$, every ideal primary according to IIIa is also primary according to III. Conversely, let $\mathfrak{Q}$ be primary according to III, and suppose the hypothesis of IIIa is fulfilled: $\mathfrak{A}\mathfrak{B} \equiv 0(\mathfrak{Q})$. Then either $\mathfrak{A} \equiv 0(\mathfrak{Q})$ follows, or else there exists at least one quantity $a \not\equiv 0(\mathfrak{Q})$ such that $a\mathfrak{B} \equiv 0(\mathfrak{Q})$ and $a \not\equiv 0(\mathfrak{Q})$. If $b_1, \dots, b_s$ is an ideal basis of $\mathfrak{B}$, then by Definition III, since $ab_i \equiv 0(\mathfrak{Q})$,

$$b_i^{\varkappa_i} \equiv 0(\mathfrak{Q}) \qquad (i = 1, \dots, s).$$

Consequently, because

$$b = n_1 b_1 + \cdots + n_s b_s + a_1 b_1 + \cdots + a_s b_s,$$

for $\varkappa = \varkappa_1 + \cdots + \varkappa_s$ every product of $\varkappa$ quantities b is divisible by $\mathfrak{Q}$, proving that ideals primary according to III satisfy Definition IIIa. In the special case of prime ideals $\mathfrak{P}$, from $a\mathfrak{B} \equiv 0(\mathfrak{P})$, hence also $ab \equiv 0(\mathfrak{P})$ for every $b \equiv 0(\mathfrak{B})$, and $a \not\equiv 0(\mathfrak{P})$, it follows that $b \equiv 0(\mathfrak{P})$ and therefore $\mathfrak{B} \equiv 0(\mathfrak{P})$. Thus the equivalence of the two definitions is proved.

The connection between primary and prime ideals is made by the observation that the totality $\mathfrak{P}$ of all elements p with the property that some power of p is divisible by $\mathfrak{Q}$ forms a prime ideal. First it is clear that $\mathfrak{P}$ is an ideal, since together with p_1 and p_2 the elements ap_1 and $p_1 - p_2$ also have the stated property. By the same basis argument as was applied in the definition of IIIa, there further follows the existence of a number λ such that $\mathfrak{P}^{\lambda} \equiv 0(\mathfrak{Q})$.

Now let

$$ab \equiv 0(\mathfrak{P}), \qquad a \not\equiv 0(\mathfrak{P}).$$

Then, by the definition of $\mathfrak{P}$,

$$a^{\lambda} b^{\lambda} \equiv 0(\mathfrak{Q}), \qquad a^{\lambda} \not\equiv 0(\mathfrak{Q}),$$

and hence, by the definition of $\mathfrak{Q}$,

$$b^{\lambda\varkappa} \equiv 0(\mathfrak{Q}), \qquad \text{and consequently} \qquad b \equiv 0(\mathfrak{P}),$$

so that $\mathfrak{P}$ is proved to be a prime ideal. $\mathfrak{P}$ is also defined as the greatest common divisor of all ideals $\mathfrak{B}$ having the property that some power of $\mathfrak{B}$ is divisible by $\mathfrak{Q}$. For each of these $\mathfrak{B}$ is divisible by $\mathfrak{P}$, hence so

[16] If $\mathfrak{A} = (a_1, \dots, a_r)$ and $\mathfrak{B} = (b_1, \dots, b_s)$, then $\mathfrak{A}\mathfrak{B}$ is understood as the ideal having the products $a_i b_k$ as basis; that these and only these elements form the product ideal follows from the basis representation, by commutativity.

is their greatest common divisor $\mathfrak{D}$. Conversely $\mathfrak{P}$ itself is an ideal $\mathfrak{B}$, and hence is divisible by $\mathfrak{D}$, which proves $\mathfrak{P} = \mathfrak{D}$. Thus $\mathfrak{P}$ is a prime ideal which is a divisor of $\mathfrak{Q}$ and a power of which is divisible by $\mathfrak{Q}$; this property defines it uniquely. For from

$$\mathfrak{Q} \equiv 0(\mathfrak{P}), \qquad \mathfrak{P}^{\varrho} \equiv 0(\mathfrak{Q}), \qquad \mathfrak{Q} \equiv 0(\mathfrak{B}), \qquad \mathfrak{B}^{\sigma} \equiv 0(\mathfrak{Q})$$

follows

$$\mathfrak{P}^{\varrho} \equiv 0(\mathfrak{B}), \qquad \mathfrak{B}^{\sigma} \equiv 0(\mathfrak{P}),$$

hence, by the property of prime ideals,

$$\mathfrak{P} \equiv 0(\mathfrak{B}), \qquad \mathfrak{B} \equiv 0(\mathfrak{P}), \qquad \mathfrak{B} = \mathfrak{P}.$$

In summary:

Theorem V. *To every primary ideal $\mathfrak{Q}$ there exists one and only one prime ideal $\mathfrak{P}$ which is a divisor of $\mathfrak{Q}$ and a power of which is divisible by $\mathfrak{Q}$; $\mathfrak{P}$ will be called the "associated prime ideal".*[17] *$\mathfrak{P}$ is defined as the greatest common divisor of all ideals $\mathfrak{B}$ with the property that a power of $\mathfrak{B}$ is divisible by $\mathfrak{Q}$. If ϱ is the smallest number such that $\mathfrak{P}^{\varrho} \equiv 0(\mathfrak{Q})$, then ϱ is called the exponent of $\mathfrak{Q}$.*[18]

We now prove the connection between primary and irreducible:

Theorem VI. *Every non-primary ideal is reducible; in other words, every irreducible ideal is primary.*[19]

Let $\mathfrak{K}$ be a non-primary ideal, so that by Definition III there exists at least one pair of quantities a, b such that

$$ab \equiv 0(\mathfrak{K}), \qquad a \not\equiv 0(\mathfrak{K}), \qquad b^{\varkappa} \not\equiv 0(\mathfrak{K}) \quad \text{for every } \varkappa. \tag{3}$$

We now form the two ideals

$$\mathfrak{K}_1 = (\mathfrak{K}, a), \qquad \mathfrak{N}_1 = (\mathfrak{K}, b),$$

which by (3) are proper divisors of $\mathfrak{K}$ and for which, by (3),

$$\mathfrak{K}_1 \mathfrak{N}_1 \equiv 0(\mathfrak{K}). \tag{4}$$

For the elements f of the least common multiple $\mathfrak{K}_2 = [\mathfrak{K}_1, \mathfrak{N}_1]$, the following alternative holds.

Either from

$$f \equiv 0(\mathfrak{K}_1), \qquad f \equiv 0(\mathfrak{N}_1), \qquad \text{that is} \qquad f \equiv a_1 b \ (\mathfrak{K}),$$

there always follows a representation

$$f \equiv l_1 b \ (\mathfrak{K}), \qquad l_1 \equiv 0(\mathfrak{K}_1),$$

and hence, by (4), $f \equiv 0(\mathfrak{K})$, so that $\mathfrak{K}_2 \equiv 0(\mathfrak{K})$; since also $\mathfrak{K} \equiv 0(\mathfrak{K}_2)$, one has $\mathfrak{K} = \mathfrak{K}_2$, and $\mathfrak{K}$ is recognized as reducible.

Or there is at least one $f \equiv 0(\mathfrak{K}_2)$ for which no such l_1 exists. With the corresponding a_1 we then form

$$\mathfrak{K}_2 = (\mathfrak{K}, a, a_1), \qquad \mathfrak{N}_2 = (\mathfrak{K}, b^2).$$

Then because $a_1 b \equiv 0(\mathfrak{K}_1)$, (4) gives again

$$\mathfrak{K}_2 \mathfrak{N}_2 \equiv 0(\mathfrak{K}), \tag{4'}$$

and $\mathfrak{K}_2$ is a proper divisor of $\mathfrak{K}_1$.

The same alternative holds for the elements f of $\mathfrak{K}_3 = [\mathfrak{K}_2, \mathfrak{N}_2]$. Either from

$$f \equiv 0(\mathfrak{K}_2), \qquad f \equiv 0(\mathfrak{N}_2), \qquad \text{that is} \qquad f \equiv a_2 b^2 \ (\mathfrak{K}),$$

there always follows

$$f \equiv l_2 b^2 \ (\mathfrak{K}), \qquad l_2 \equiv 0(\mathfrak{K}_2),$$

and hence, by (4'), $f \equiv 0(\mathfrak{K})$.

[17]That the converse does not hold is shown by the example $\mathfrak{M} = (x^2, xy)$. The prime ideal (x) satisfies all conditions, but $\mathfrak{M}$ is not primary.

[18]That in general $\mathfrak{P}^{\varrho} = \mathfrak{Q}$ need not hold, as it does in the domain of rational, respectively algebraic, integers, is shown by the example $\mathfrak{Q} = (x^2, y)$; $\mathfrak{P} = (x, y)$; $\mathfrak{P}^2 = (x^2, xy, y^2) \equiv 0(\mathfrak{Q})$, but $\mathfrak{Q} \not\equiv 0(\mathfrak{P}^2)$; thus $\mathfrak{Q}$ is different from $\mathfrak{P}^2$.

[19]That the converse does not hold here is shown, for example, by $\mathfrak{Q} = (x^2, xy, y^{\lambda}) = [(x^2, y), (x, y^{\lambda})]$, where $\lambda \geq 2$. Here $\mathfrak{Q}$ is primary but reducible. That $\mathfrak{Q}$ is primary follows from the fact that it contains all power products of dimension λ in x, y; thus for every polynomial without constant term some power is divisible by $\mathfrak{Q}$. But if, in $ab \equiv 0(\mathfrak{Q})$, the polynomial b contains a constant term – hence $b^{\varkappa} \not\equiv 0(\mathfrak{Q})$ for every $\varkappa$ – then, because the basis polynomials of $\mathfrak{Q}$ are homogeneous and every homogeneous component of ab is divisible by $\mathfrak{Q}$, a must be divisible by $\mathfrak{Q}$.

Or for at least one f there is no such l_2, which leads to the formation of $\mathfrak{K}_3 = (\mathfrak{K}, a, a_1, a_2)$ and $\mathfrak{N}_3 = (\mathfrak{K}, b^3)$, with $\mathfrak{K}_3\mathfrak{N}_3 \equiv 0(\mathfrak{K})$, where $\mathfrak{K}_3$ is a proper divisor of $\mathfrak{K}_2$. Continuing in this way, we define in general

$$\mathfrak{K}_n = (\mathfrak{K}, a, a_1, \ldots, a_{n-1}), \qquad \mathfrak{N}_n = (\mathfrak{K}, b^n),$$

$$\mathfrak{K}_n\mathfrak{N}_n \equiv 0(\mathfrak{K}),$$

where the a_n are defined by the fact that there is an f such that

$$f \equiv 0(\mathfrak{K}_n), \quad f \equiv 0(\mathfrak{N}_n), \quad f \equiv a_n b^n\ (\mathfrak{K}), \quad \text{but} \quad f \not\equiv l_n b^n\ (\mathfrak{K}), \quad l_n \equiv 0(\mathfrak{K}_n).$$

Thus, in general, $\mathfrak{K}_n\mathfrak{N}_n \equiv 0(\mathfrak{K})$, $\mathfrak{N}_n$ is a proper divisor of $\mathfrak{K}$ by (3), and $\mathfrak{K}_n$ is a proper divisor of $\mathfrak{K}_{n-1}$. By Theorem I on the finite chain, the chain of the $\mathfrak{K}_n$ must therefore terminate after finitely many steps, say with $\mathfrak{K}_n$. For every $f \equiv 0(\mathfrak{K}_n)$ and $f \equiv 0(\mathfrak{N}_n)$, then, one has $f \equiv l_n b^n\ (\mathfrak{K})$ with $l_n \equiv 0(\mathfrak{K}_n)$, and consequently, by the preceding argument, $\mathfrak{K} = [\mathfrak{K}_n, \mathfrak{N}_n]$, so that $\mathfrak{K}$ is recognized as reducible.

The *uniqueness of the associated prime ideals* follows from what has just been proved.

Let

$$\mathfrak{M} = [\mathfrak{B}_1 \cdots \mathfrak{B}_k] = [\mathfrak{D}_1 \cdots \mathfrak{D}_k]$$

be two shortest, hence reduced, representations of $\mathfrak{M}$ as the least common multiple of irreducible ideals, whose number agrees by Theorem IV. Then by that theorem also the intermediate representations occurring there (where, as noted there, the index $j_i = i$ may be put)

$$\mathfrak{M} = [\mathfrak{D}_1 \cdots \mathfrak{D}_{i-1}\mathfrak{B}_i\mathfrak{B}_{i+1} \cdots \mathfrak{B}_k] = [\mathfrak{D}_1 \cdots \mathfrak{D}_{i-1}\mathfrak{D}_i\mathfrak{B}_{i+1} \cdots \mathfrak{B}_k] = [\mathfrak{U}_i, \mathfrak{B}_i] = [\mathfrak{U}_i, \mathfrak{D}_i]$$

are shortest representations. Thus

$$\mathfrak{U}_i\mathfrak{B}_i \equiv 0(\mathfrak{D}_i), \qquad \mathfrak{U}_i \not\equiv 0(\mathfrak{D}_i), \qquad \mathfrak{U}_i\mathfrak{D}_i \equiv 0(\mathfrak{B}_i), \qquad \mathfrak{U}_i \not\equiv 0(\mathfrak{B}_i).$$

Since by Theorem VI the irreducible ideals $\mathfrak{B}_i$ and $\mathfrak{D}_i$ are primary, there follows the existence of two numbers λ_i and μ_i such that

$$\mathfrak{B}_i^{\lambda_i} \equiv 0(\mathfrak{D}_i), \qquad \mathfrak{D}_i^{\mu_i} \equiv 0(\mathfrak{B}_i). \tag{5}$$

Let $\mathfrak{P}_i$ and $\bar{\mathfrak{P}}_i$ denote the associated prime ideals of $\mathfrak{B}_i$ and $\mathfrak{D}_i$, respectively; thus $\mathfrak{P}_i^{\varrho_i} \equiv 0(\mathfrak{B}_i)$ and $\bar{\mathfrak{P}}_i^{\sigma_i} \equiv 0(\mathfrak{D}_i)$. From (5) then follows

$$\mathfrak{P}_i^{\lambda_i\varrho_i} \equiv 0(\bar{\mathfrak{P}}_i), \qquad \bar{\mathfrak{P}}_i^{\mu_i\sigma_i} \equiv 0(\mathfrak{P}_i),$$

and by the property of prime ideals

$$\mathfrak{P}_i \equiv 0(\bar{\mathfrak{P}}_i), \qquad \bar{\mathfrak{P}}_i \equiv 0(\mathfrak{P}_i), \qquad \mathfrak{P}_i = \bar{\mathfrak{P}}_i.$$

Thus:

Theorem VII. *In two different shortest representations of an ideal as the least common multiple of irreducible ideals, the associated prime ideals coincide; among them equal ones can also occur,*[20] *and indeed occur equally often in every decomposition. Consequently the ideals themselves can be paired in at least one way such that in each case a power of one ideal $\mathfrak{B}_i$ is divisible by the assigned ideal $\mathfrak{D}_i$, and conversely. Their number coincides by Theorem IV.*[21]

§ 5. Representation of an ideal as the least common multiple of greatest primary ideals. Uniqueness of the associated prime ideals.

Definition IV. *A shortest representation $\mathfrak{M} = [\mathfrak{Q}_1 \cdots \mathfrak{Q}_\alpha]$ will be called a least common multiple of greatest primary ideals if all $\mathfrak{Q}$ are primary, but the least common multiple of two $\mathfrak{Q}$ is no longer primary.*

That at least one such representation always exists follows from the representation of $\mathfrak{M}$ as a least common multiple of irreducible ideals. These ideals are primary; either there is already a representation by greatest primary ideals, or the least common multiple of some two ideals is again primary. Since the number of

[20]This is shown, for example, by the example in footnote 19: $(x^2, xy, y^\lambda) = [(x^2, y), (x, y^\lambda)]$, where $\lambda \geq 2$. The two associated prime ideals here are (x, y).

[21]For the uniqueness of the "isolated" ideals contained among the irreducible ideals, compare § 7.

ideals has thereby decreased by one, repetition of the procedure leads after finitely many steps to the desired representation.

This representation is reduced by Lemma IV. Conversely, every reduced representation by greatest primary ideals arises in this way, as the decomposition of the $\mathfrak{Q}$ into irreducible ideals shows.

In order to derive here from Theorem VII a corresponding uniqueness theorem, the connection with the associated prime ideals has to be investigated.

Theorem VIII. *If the primary ideals $\mathfrak{N}_1, \mathfrak{N}_2, \ldots, \mathfrak{N}_\lambda$ all possess the same associated prime ideal $\mathfrak{P}$, then their least common multiple $\mathfrak{Q} = [\mathfrak{N}_1, \mathfrak{N}_2, \ldots, \mathfrak{N}_\lambda]$ is also primary and has $\mathfrak{P}$ as associated prime ideal. Conversely, if $\mathfrak{Q} = [\mathfrak{N}_1 \cdots \mathfrak{N}_\lambda]$ is a reduced representation of the primary ideal $\mathfrak{Q}$, then all $\mathfrak{N}_i$ are primary and have, as associated prime ideal, the associated prime ideal $\mathfrak{P}$ of $\mathfrak{Q}$.*

For the proof of the first part, first observe that from $\mathfrak{P}^{\varrho_i} \equiv 0(\mathfrak{N}_i)$ for every i it also follows that $\mathfrak{P}^\tau \equiv 0(\mathfrak{Q})$, where τ denotes the largest of the indices ϱ_i. Since $\mathfrak{P}$ is also a divisor of $\mathfrak{Q}$, $\mathfrak{P}$ is necessarily the associated prime ideal if $\mathfrak{Q}$ is primary. From

$$\mathfrak{A}\mathfrak{B} \equiv 0(\mathfrak{Q}), \qquad \mathfrak{B}^k \not\equiv 0(\mathfrak{Q}) \quad \text{(for every } k)$$

it follows that

$$\mathfrak{B} \not\equiv 0(\mathfrak{P}), \quad \text{hence} \quad \mathfrak{B}^k \not\equiv 0(\mathfrak{N}_i) \quad \text{(for every } k),$$

and consequently $\mathfrak{A} \equiv 0(\mathfrak{N}_i)$; hence $\mathfrak{A} \equiv 0(\mathfrak{Q})$. Thus $\mathfrak{Q}$ is proved primary, and $\mathfrak{P}$ the associated prime ideal.

Conversely, let first

$$\mathfrak{Q} = [\mathfrak{N}_1 \cdots \mathfrak{N}_\lambda] = [\mathfrak{N}_i, \mathfrak{N}_i']$$

be a shortest representation of $\mathfrak{Q}$ by primary ideals $\mathfrak{N}_i$, and let $\mathfrak{P}_i$ be the associated prime ideals. From

$$\mathfrak{N}_i\mathfrak{N}_i' \equiv 0(\mathfrak{Q}), \qquad \mathfrak{N}_i' \not\equiv 0(\mathfrak{N}_i)$$

(because the representation is shortest) follows

$$\mathfrak{N}_i' \not\equiv 0(\mathfrak{P}_i), \qquad \text{but} \qquad \mathfrak{N}_i' \equiv 0(\mathfrak{P}).$$

Since $\mathfrak{P}_i$ is at the same time a divisor of $\mathfrak{Q}$, $\mathfrak{P}_i$ thus agrees, for every i, with the associated prime ideal $\mathfrak{P}$ of $\mathfrak{Q}$.

It remains to show that in every reduced representation $\mathfrak{Q} = [\mathfrak{N}_1 \cdots \mathfrak{N}_\lambda]$ the $\mathfrak{N}_i$ are primary.[22] To this end, decompose the $\mathfrak{N}_i$ into their irreducible ideals $\mathfrak{B}$; in the resulting shortest representation $\mathfrak{Q} = [\mathfrak{B}_1 \cdots \mathfrak{B}_s]$, every ideal is primary and, by what has just been proved, has $\mathfrak{P}$ as associated prime ideal. Then by the first part of the theorem proved above the same holds for each $\mathfrak{N}_i$, completing the proof of Theorem VIII.

Addendum. It also follows that a prime ideal is necessarily irreducible. For from the reduced representation $\mathfrak{P} = [\mathfrak{N}, \mathfrak{K}]$ one obtains by Theorem VIII $\mathfrak{N} \equiv 0(\mathfrak{P})$; hence, since $\mathfrak{P} \equiv 0(\mathfrak{N})$, also $\mathfrak{P} = \mathfrak{N}$, and similarly $\mathfrak{P} = \mathfrak{K}$. The irreducibility of $\mathfrak{P}$ also follows directly: for from $\mathfrak{P} = [\mathfrak{N}, \mathfrak{K}]$ follows $\mathfrak{N}\mathfrak{K} \equiv 0(\mathfrak{P})$, $\mathfrak{N} \not\equiv 0(\mathfrak{P})$, $\mathfrak{K} \not\equiv 0(\mathfrak{P})$, contrary to the defining property of a prime ideal.

Now let

$$\mathfrak{M} = [\mathfrak{Q}_1 \cdots \mathfrak{Q}_\alpha] = [\mathfrak{Q}_1' \cdots \mathfrak{Q}_\beta']$$

be two reduced representations of $\mathfrak{M}$ as the least common multiple of greatest primary ideals. Decomposing the $\mathfrak{Q}$ into irreducible ideals $\mathfrak{B}$, and the $\mathfrak{Q}'$ correspondingly into irreducible ideals $\mathfrak{D}$, gives two reduced representations of $\mathfrak{M}$ as the least common multiple of irreducible ideals; by Theorem VII, both the number of components and the associated prime ideals coincide. By Theorem VIII, all irreducible ideals $\mathfrak{B}$ occurring for a fixed $\mathfrak{Q}_i$ have the same associated prime ideal $\mathfrak{P}_i$; whereas the $\mathfrak{P}_j$ associated to $\mathfrak{Q}_j$ is necessarily different from this, since otherwise, by Theorem VIII, there would be no representation by greatest primary ideals. Hence the number α of the $\mathfrak{Q}$ is equal to the number of different associated prime ideals $\mathfrak{P}$ of the $\mathfrak{B}$; these different $\mathfrak{P}$ form the associated prime ideals of the $\mathfrak{Q}$. The same holds for the $\mathfrak{Q}'$ with respect to their decomposition into the $\mathfrak{D}$. Theorem VII therefore gives equality of number for the $\mathfrak{Q}$ and $\mathfrak{Q}'$, and agreement of their associated prime ideals. At the same time it is seen that the grouping of the irreducible ideals into greatest primary ideals described at the beginning of the paragraph consists in combining all, and only those, with the same associated prime ideal. Theorem VIII further shows the irreducibility property of the greatest primary ideals: they admit no reduced representation as the least common multiple of greatest primary ideals.

[22]That the reduced representation is essential here is shown by the example of the non-reduced representation $\mathfrak{Q} = [x^2, xy, y^\lambda] = [(x^2, xy, y^\lambda, yz), (x, y^\lambda)]$, where $\lambda \geq 2$. Here $(x^2, xy, y^\lambda, yz) = [(x^2, y), (x, y^\lambda, z)]$ is not primary by what has just been proved; for the latter representation is a shortest one by primary ideals, but the associated prime ideals (x, y) and (x, y, z) are different. ($\mathfrak{Q}$ is primary by footnote 19.)

In summary:

Theorem IX. *In two reduced representations of an ideal as the least common multiple of greatest primary ideals, the number of components and the associated prime ideals, all of which are distinct from each other, coincide. In other words, to each* $\mathfrak{Q}$ *there can be assigned one and only one* $\mathfrak{Q}'$ *such that a power of* $\mathfrak{Q}$ *is divisible by* $\mathfrak{Q}'$*, and conversely.*[23] *The* $\mathfrak{Q}$ *and* $\mathfrak{Q}'$ *have the irreducibility property with respect to decomposition into greatest primary ideals.*

Addendum. It should be noted that Theorem IX remains essentially valid if one assumes only a shortest representation instead of a reduced one. If, say,

$$\mathfrak{M} = [\mathfrak{Q}_1 \cdots \mathfrak{Q}_i^* \cdots \mathfrak{Q}_\alpha]$$

is reduced with respect to $\mathfrak{Q}_i^*$, and $\mathfrak{Q}_i^*$ is a proper divisor of $\mathfrak{Q}_i$, while $\mathfrak{U}_i$ is the complement of $\mathfrak{Q}_i$, then by Lemma IV

$$\mathfrak{Q}_i = [\mathfrak{Q}_i^*, (\mathfrak{U}_i, \mathfrak{Q}_i)],$$

and this representation is reduced with respect to $\mathfrak{Q}_i^*$. By Theorem VIII – replacing $(\mathfrak{U}_i, \mathfrak{Q}_i)$ by a proper divisor if necessary when applying the theorem – $\mathfrak{Q}_i^*$ is therefore primary and has the same associated prime ideal $\mathfrak{P}_i$ as $\mathfrak{Q}_i$. Continuing the procedure shows that to every such representation there can be assigned a reduced representation by greatest primary ideals in such a way that the number of components and the associated prime ideals coincide. Thus even for shortest representations it holds that in two different representations the number of components and the associated prime ideals coincide.

The associated, mutually distinct prime ideals thereby uniquely defined will be called simply “the associated prime ideals of $\mathfrak{M}$.”

§ 6. Unique representation of an ideal as the least common multiple of relatively-prime irreducible ideals.

Definition V. *An ideal* $\mathfrak{R}$ *is called relatively prime to* $\mathfrak{S}$ *if from*

$$\mathfrak{T}\mathfrak{R} \equiv 0(\mathfrak{S})$$

it necessarily follows that $\mathfrak{T} \equiv 0(\mathfrak{S})$*. If both* $\mathfrak{R}$ *is relatively prime to* $\mathfrak{S}$ *and* $\mathfrak{S}$ *is relatively prime to* $\mathfrak{R}$*, then* $\mathfrak{R}$ *and* $\mathfrak{S}$ *are called mutually relatively prime.*[24]

An ideal is called relatively-prime irreducible if it cannot be represented as the least common multiple of mutually relatively prime proper divisors.

In particular, if instead of $\mathfrak{T}$ one takes the greatest common divisor $\mathfrak{T}_0$ of all $\mathfrak{T}$ for which $\mathfrak{T}\mathfrak{R} \equiv 0(\mathfrak{S})$, then also $\mathfrak{T}_0\mathfrak{R} \equiv 0(\mathfrak{S})$ and $\mathfrak{S} \equiv 0(\mathfrak{T}_0)$. Thus $\mathfrak{T}_0 = \mathfrak{S}$ when $\mathfrak{R}$ is relatively prime to $\mathfrak{S}$; and $\mathfrak{T}_0$ is a proper divisor of $\mathfrak{S}$ when $\mathfrak{R}$ is not relatively prime to $\mathfrak{S}$.[25]

The uniqueness proof rests on:

Theorem X. *1. If* $\mathfrak{R}$ *is relatively prime to the ideals* $\mathfrak{S}_1, \ldots, \mathfrak{S}_\lambda$*, then* $\mathfrak{R}$ *is also relatively prime to their least common multiple* $\mathfrak{S}$*.*

2. If the ideals $\mathfrak{S}_1, \ldots, \mathfrak{S}_\lambda$ *are relatively prime to* $\mathfrak{R}$*, then their least common multiple* $\mathfrak{S}$ *is also relatively prime to* $\mathfrak{R}$*.*

3. If $\mathfrak{R}$ *is relatively prime to* $\mathfrak{S}$ *and* $\mathfrak{S} = [\mathfrak{S}_1 \cdots \mathfrak{S}_\lambda]$ *is a reduced representation for* $\mathfrak{S}$*, then* $\mathfrak{R}$ *is also relatively prime to each* $\mathfrak{S}_i$*.*

4. If $\mathfrak{S}$ *is relatively prime to* $\mathfrak{R}$*, then every divisor* $\mathfrak{S}_i$ *of* $\mathfrak{S}$ *is also relatively prime to* $\mathfrak{R}$*.*

1. Since $\mathfrak{T}\mathfrak{R} \equiv 0(\mathfrak{S})$ necessarily gives $\mathfrak{T}\mathfrak{R} \equiv 0(\mathfrak{S}_i)$, the hypothesis gives $\mathfrak{T} \equiv 0(\mathfrak{S}_i)$, and hence $\mathfrak{T} \equiv 0(\mathfrak{S})$.

2. Put

$$\widehat{\mathfrak{S}}_1 = [\mathfrak{S}_2 \cdots \mathfrak{S}_\lambda], \quad \widehat{\mathfrak{S}}_{12} = [\mathfrak{S}_3 \cdots \mathfrak{S}_\lambda], \quad \ldots, \quad \widehat{\mathfrak{S}}_{12\ldots\lambda-1} = \mathfrak{S}_\lambda.$$

From $\mathfrak{T}\mathfrak{S} \equiv 0(\mathfrak{R})$ it follows that $\mathfrak{T}\mathfrak{S}_1\widehat{\mathfrak{S}}_1 \equiv 0(\mathfrak{R})$; hence, by the hypothesis, $\mathfrak{T}\widehat{\mathfrak{S}}_1 \equiv 0(\mathfrak{R})$. Applying the same argument successively gives $\mathfrak{T}\widehat{\mathfrak{S}}_{12} \equiv 0(\mathfrak{R})$ and finally $\mathfrak{T}\widehat{\mathfrak{S}}_{12\ldots\lambda-1} = \mathfrak{T}\mathfrak{S}_\lambda \equiv 0(\mathfrak{R})$; hence $\mathfrak{T} \equiv 0(\mathfrak{R})$.

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[23]An example of different representations is the one given in footnote 12 for Theorem II: $(x^2, xy) = [(x), (x^2, \mu x + y)]$ for arbitrary μ. Since the associated prime ideals $\mathfrak{P}_1 = (x)$ and $\mathfrak{P}_2 = (x, y)$ are distinct, these are greatest primary ideals. – For the uniqueness of the “isolated” greatest primary ideals, compare § 7.

[24]The condition of being relatively prime is not symmetric. For example, $\mathfrak{R} = (x^2, y)$ is relatively prime to $\mathfrak{S} = (x)$; but $\mathfrak{S}$ is not relatively prime to $\mathfrak{R}$, since $\mathfrak{S}^2 \equiv 0(\mathfrak{R})$, whereas $\mathfrak{S} \not\equiv 0(\mathfrak{R})$.

[25]The $\mathfrak{T}_0$ so defined agrees with Lasker’s “residual module”; and, in its extension to number modules in place of ideals, with Dedekind’s “quotient” of two modules. Lasker, loc. cit., p. 49; Dedekind (Zahlentheorie), p. 504.

3. Let $\widehat{\mathfrak{S}}_i$ denote the complement of $\mathfrak{S}_i$. From $\mathfrak{T}_0\mathfrak{R} \equiv 0(\mathfrak{S}_i)$, together with $\widehat{\mathfrak{S}}_i\mathfrak{R} \equiv 0(\mathfrak{S}_i)$, it follows that $[\mathfrak{T}_0, \widehat{\mathfrak{S}}_i]\mathfrak{R} \equiv 0(\mathfrak{S})$. If $\mathfrak{T}_0$ were a proper divisor of $\mathfrak{S}_i$, then, since $\mathfrak{S} = [\mathfrak{S}_i, \widehat{\mathfrak{S}}_i]$ is reduced, $[\mathfrak{T}_0, \widehat{\mathfrak{S}}_i]$ would be a proper divisor of $\mathfrak{S}$, contrary to the hypothesis.

4. From $\mathfrak{T}\mathfrak{S}_i \equiv 0(\mathfrak{R})$, where $\mathfrak{T}$ is a proper divisor of $\mathfrak{R}$, it follows also that $\mathfrak{T}\mathfrak{S} \equiv 0(\mathfrak{R})$; hence $\mathfrak{T}$ would be a proper divisor of $\mathfrak{R}$, contrary to the hypothesis.

Definition V shows in particular that every ideal is relatively prime to the unit ideal $\mathfrak{O}$ consisting of all elements of Σ;[26] the following theorems, however, apply only to ideals distinct from $\mathfrak{O}$.

From Theorem X one first obtains:

Lemma V. *Every representation of an ideal as the least common multiple of mutually relatively prime ideals distinct from $\mathfrak{O}$ is reduced.*

Let

$$\mathfrak{M} = [\mathfrak{R}_1\mathfrak{R}_2 \cdots \mathfrak{R}_\sigma] = [\mathfrak{R}_i, \mathfrak{L}_i]$$

be such a representation. By Theorem X, 2, $\mathfrak{L}_i$ is also relatively prime to $\mathfrak{R}_i$; hence $\mathfrak{R}_i$ cannot be contained in $\mathfrak{L}_i$, and the representation is shortest. For from $\mathfrak{L}_i \equiv 0(\mathfrak{R}_i)$, with $\mathfrak{L}_i$ relatively prime to $\mathfrak{R}_i$, it would follow from $\mathfrak{O}\mathfrak{L}_i \equiv 0(\mathfrak{R}_i)$ that $\mathfrak{O} \equiv 0(\mathfrak{R}_i)$, hence $\mathfrak{R}_i = \mathfrak{O}$, contrary to the hypothesis.

If now $\mathfrak{R}_i$ could be replaced by a proper divisor $\mathfrak{R}_i^*$, then by Lemma II $\mathfrak{R}_i$ would be reducible and

$$\mathfrak{R}_i = [\mathfrak{R}_i^*, (\mathfrak{R}_i, \mathfrak{L}_i)].$$

If necessary, replace $(\mathfrak{R}_i, \mathfrak{L}_i)$ by a proper divisor $(\mathfrak{R}_i, \mathfrak{L}_i)^*$ and likewise replace $\mathfrak{R}_i^*$ by a proper divisor so that a reduced representation of $\mathfrak{R}_i$ is obtained. Then Theorem X, 3 shows that $\mathfrak{L}_i$ is also relatively prime to $(\mathfrak{R}_i, \mathfrak{L}_i)^*$, and this ideal is distinct from $\mathfrak{O}$ because the original representation was shortest. But $(\mathfrak{R}_i, \mathfrak{L}_i)^*$ is contained in $(\mathfrak{R}_i, \mathfrak{L}_i)$ and $(\mathfrak{R}_i, \mathfrak{L}_i)$ is contained in $\mathfrak{L}_i$, giving the contradiction.

Theorem X further gives the following theorem, which mediates the connection with the associated prime ideals.

Theorem XI. *If $\mathfrak{R}$ is relatively prime to $\mathfrak{S}$ and $\mathfrak{S}$ is distinct from $\mathfrak{O}$, then no associated prime ideal of $\mathfrak{R}$[27] is divisible by an associated prime ideal of $\mathfrak{S}$. Conversely, if no such divisibility holds, then $\mathfrak{R}$ is relatively prime to $\mathfrak{S}$, and of course $\mathfrak{S}$ is distinct from $\mathfrak{O}$.*

Let

$$\mathfrak{R} = [\mathfrak{Q}_1 \cdots \mathfrak{Q}_\alpha], \qquad \mathfrak{S} = [\mathfrak{Q}_1^* \cdots \mathfrak{Q}_\beta^*]$$

be reduced representations of $\mathfrak{R}$ and $\mathfrak{S}$ by greatest primary ideals, and let $\mathfrak{P}_1, \ldots, \mathfrak{P}_\alpha$ and $\mathfrak{P}_1^*, \ldots, \mathfrak{P}_\beta^*$ be their associated prime ideals. We prove the assertion in the form: if some $\mathfrak{P}$ is divisible by some $\mathfrak{P}^*$, then $\mathfrak{R}$ cannot be relatively prime to $\mathfrak{S}$, and conversely.

Suppose then that $\mathfrak{P}_\mu \equiv 0(\mathfrak{P}_\nu^*)$. Then also $\mathfrak{Q}_\mu^\varrho \equiv 0(\mathfrak{P}_\nu^*)$, and by the definition of $\mathfrak{P}_\nu^*$ this gives $\mathfrak{Q}_\mu^\varrho \equiv 0(\mathfrak{Q}_\nu^*)$, hence $\mathfrak{R}^\varrho \equiv 0(\mathfrak{Q}_\nu^*)$. Let $\mathfrak{R}^\tau$ be the lowest power of $\mathfrak{R}$ divisible by $\mathfrak{Q}_\nu^*$. If $\tau = 1$, then $\mathfrak{R}$ is divisible by $\mathfrak{Q}_\nu^*$; but since $\mathfrak{S} \neq \mathfrak{O}$, $\mathfrak{R}$ is not relatively prime to $\mathfrak{Q}_\nu^*$. If $\tau \geq 2$, then $\mathfrak{R}^{\tau-1}\mathfrak{R} \equiv 0(\mathfrak{Q}_\nu^*)$ while $\mathfrak{R}^{\tau-1} \not\equiv 0(\mathfrak{Q}_\nu^*)$, so again $\mathfrak{R}$ is not relatively prime to $\mathfrak{Q}_\nu^*$;[28] in both cases Theorem X, 3 then implies that $\mathfrak{R}$ is not relatively prime to $\mathfrak{S}$.

Conversely, if $\mathfrak{R}$ is not relatively prime to $\mathfrak{S}$, then by Theorem X, 1 it is not relatively prime to at least one $\mathfrak{Q}_\nu^*$. Thus

$$\mathfrak{T}_0\mathfrak{R} \equiv 0(\mathfrak{Q}_\nu^*), \qquad \mathfrak{T}_0 \not\equiv 0(\mathfrak{Q}_\nu^*),$$

and since $\mathfrak{Q}_\nu^*$ is primary,

$$\mathfrak{R}^\varrho \equiv 0(\mathfrak{Q}_\nu^*), \qquad \mathfrak{Q}_1^\varrho \cdots \mathfrak{Q}_\alpha^\varrho \equiv 0(\mathfrak{Q}_\nu^*).$$

For the associated prime ideals this implies

$$\mathfrak{P}_1^{\varrho_1} \cdots \mathfrak{P}_\alpha^{\varrho_\alpha} \equiv 0(\mathfrak{P}_\nu^*),$$

and therefore, by the property of prime ideals, $\mathfrak{P}_\nu^*$ is contained in at least one of the $\mathfrak{P}$. This proves Theorem XI.

Theorems X and XI now give the existence[29] and uniqueness of decomposition into relatively-prime irreducible ideals as follows.

[26] $\mathfrak{O}$ plays the role of the unit only with respect to divisibility and least common multiple, not with respect to product formation. For instance, $\mathfrak{O} = (x)$ for the domain of all integral polynomials in x without constant term; and $\mathfrak{O} = (2)$ for the domain of all even numbers.

[27] The associated prime ideals of $\mathfrak{R}$ were defined at the end of § 5.

[28] $\mathfrak{R}^0$ is not defined, because Σ need not contain a unit; hence the case $\tau = 1$ had to be dealt with first. The case $\tau = 0$ is also excluded, if Σ has a unit, by the hypothesis on $\mathfrak{S}$.

[29] The existence of the decomposition can also be proved directly, in exact analogy with the proof in § 2 of the existence of a decomposition into finitely many irreducible ideals.

Let $\mathfrak{M} = [\mathfrak{Q}_1 \cdots \mathfrak{Q}_\alpha]$ be a reduced (or at least shortest) representation of $\mathfrak{M}$ by greatest primary ideals, and let $\mathfrak{P}_1, \dots, \mathfrak{P}_\alpha$ be the associated prime ideals. We collect the $\mathfrak{P}$ into groups so that no ideal of one group is divisible by an ideal of a different group, while a single group cannot be split into two subgroups both having this property.

To construct such a grouping, note that by definition a single group G must contain, together with each ideal $\mathfrak{P}$, all divisors and multiples of $\mathfrak{P}$ occurring among $\mathfrak{P}_1, \dots, \mathfrak{P}_\alpha$. Let, for example, $\mathfrak{P}_j^{(i)}$ be all multiples of $\mathfrak{P}$, $\mathfrak{P}_{j_1}^{(i_1)}$ all divisors of $\mathfrak{P}_j^{(i)}$, $\mathfrak{P}_{j_1}^{(i_1 i_2)}$ all multiples of $\mathfrak{P}_{j_1}^{(i_1)}$, and so on. Since altogether only finitely many ideals $\mathfrak{P}$ occur, the procedure must stop after finitely many steps. The set of ideals so obtained is then a group G with the required properties: it contains all divisors and all multiples prescribed by construction, and no ideal outside it can be a divisor or a multiple of an ideal inside it.

The group G also satisfies the irreducibility condition. If it were divided into two subgroups $G^{(1)}$ and $G^{(2)}$, and if $G^{(1)}$ contained one of the ideals occurring at some stage of the alternating construction, then it would contain all preceding ones, since these are alternately multiples and divisors (or divisors and multiples), hence also $\mathfrak{P}$ and therefore the whole group G. Repeating the construction with the ideals not contained in G yields a decomposition into groups $G_1, \dots, G_\sigma$ of all the $\mathfrak{P}$, and the same argument shows that this grouping is unique.

Let $\mathfrak{Q}_{i\mu}$, where μ runs from 1 to λ_i, denote the ideals $\mathfrak{Q}$ whose associated prime ideals are collected in a group G_i. Since the $\mathfrak{P}$ are all distinct, the primary ideals $\mathfrak{Q}_{i\mu}$ belonging to a shortest representation are uniquely determined. Put

$$\mathfrak{R}_i = [\mathfrak{Q}_{i1} \cdots \mathfrak{Q}_{i\lambda_i}], \qquad \mathfrak{M} = [\mathfrak{R}_1 \cdots \mathfrak{R}_\sigma].$$

Theorem XI remains applicable even if $\mathfrak{R}_i$ is only a shortest representation, since by the addendum to Theorem IX its associated prime ideals are already well-defined. Because no associated prime ideal of $\mathfrak{R}_i$ is divisible by one of $\mathfrak{R}_j$, and conversely, Theorem XI shows that $\mathfrak{R}_i$ and $\mathfrak{R}_j$ are mutually relatively prime; and each $\mathfrak{R}_i$ is relatively-prime irreducible by the irreducibility property of the group G_i. By Lemma V the representation is reduced. Conversely, every decomposition into relatively-prime irreducible ideals leads, by Theorem XI, to the grouping just described.

Now let

$$\mathfrak{M} = [\bar{\mathfrak{R}}_1 \cdots \bar{\mathfrak{R}}_\tau]$$

be a second representation of $\mathfrak{M}$ by relatively-prime irreducible ideals. It is reduced by Lemma V. Decomposing the $\mathfrak{R}$ into greatest primary ideals shows that the associated prime ideals agree, and hence the groupings of these prime ideals agree. Thus $\tau = \sigma$, and the notation may be chosen so that $\mathfrak{R}_i$ and $\bar{\mathfrak{R}}_i$ belong to the same group. If

$$\mathfrak{M} = [\mathfrak{R}_i, \mathfrak{L}_i] = [\bar{\mathfrak{R}}_i, \bar{\mathfrak{L}}_i]$$

is the representation by ideal and complement, then, since $\bar{\mathfrak{R}}_i$ is assigned to the same group as $\mathfrak{R}_i$, Theorem XI gives that $\mathfrak{L}_i$ is relatively prime to $\bar{\mathfrak{R}}_i$, and $\bar{\mathfrak{L}}_i$ is relatively prime to $\mathfrak{R}_i$. From

$$\mathfrak{R}_i \mathfrak{L}_i \equiv 0(\bar{\mathfrak{R}}_i), \qquad \bar{\mathfrak{R}}_i \bar{\mathfrak{L}}_i \equiv 0(\mathfrak{R}_i)$$

therefore follows

$$\mathfrak{R}_i \equiv 0(\bar{\mathfrak{R}}_i), \qquad \bar{\mathfrak{R}}_i \equiv 0(\mathfrak{R}_i), \qquad \mathfrak{R}_i = \bar{\mathfrak{R}}_i.$$

Thus:

Theorem XII. *Every ideal can be represented uniquely as the least common multiple of finitely many mutually relatively prime and relatively-prime irreducible ideals.*

§ 7. Uniqueness of the isolated ideals.

Definition VI. *If the shortest representation $\mathfrak{M} = [\mathfrak{R}, \mathfrak{L}]$ is reduced with respect to $\mathfrak{L}$, then $\mathfrak{R}$ is called an isolated ideal if no associated prime ideal of $\mathfrak{R}$ is contained in an associated prime ideal of $\mathfrak{L}$; equivalently, if $\mathfrak{L}$ is relatively prime to $\mathfrak{R}$.*

Then the representation $\mathfrak{M} = [\mathfrak{R}, \mathfrak{L}]$ satisfies the conditions of the representation $\mathfrak{M} = [\mathfrak{R}_i, \mathfrak{L}_i]$ in Lemma V, and Lemma V proves:

Lemma VI. *If $\mathfrak{R}$ is an isolated ideal of the shortest representation $\mathfrak{M} = [\mathfrak{R}, \mathfrak{L}]$, reduced with respect to $\mathfrak{L}$, then the representation is also reduced with respect to $\mathfrak{R}$.*

Thus isolated ideals occur only in reduced representations. The associated prime ideals arising in the decompositions of $\mathfrak{R}$ and $\mathfrak{L}$ into irreducible ideals complement one another to the uniquely determined associated prime ideals arising in the corresponding decomposition of $\mathfrak{M}$. Hence no associated prime ideal belonging to the decomposition of $\mathfrak{R}$ into irreducible ideals is contained in the remaining associated prime ideals of $\mathfrak{M}$. Conversely, if this condition is satisfied and $\mathfrak{R}$ occurs in at least one representation $\mathfrak{M} = [\mathfrak{R}, \mathfrak{L}]$ reduced with respect to $\mathfrak{R}$, and hence also in a reduced representation $\mathfrak{M} = [\mathfrak{R}, \mathfrak{L}^*]$, then $\mathfrak{R}$ is isolated by Definition VI. This gives the following definition, independent of the particular complement $\mathfrak{L}$.

Definition VIa. *$\mathfrak{R}$ is called an isolated ideal if the prime ideals belonging to its decomposition into irreducible ideals are not contained in the other associated prime ideals belonging to the corresponding decomposition of $\mathfrak{M}$, and if $\mathfrak{R}$ occurs in at least one representation $\mathfrak{M} = [\mathfrak{R}, \mathfrak{L}]$ reduced with respect to $\mathfrak{R}$.*[30]

Suppose

$$\mathfrak{M} = [\mathfrak{R}, \mathfrak{L}] = [\bar{\mathfrak{R}}, \bar{\mathfrak{L}}]$$

are two representations of $\mathfrak{M}$ by isolated ideals $\mathfrak{R}$ and $\bar{\mathfrak{R}}$, with complements $\mathfrak{L}$ and $\bar{\mathfrak{L}}$, such that the associated prime ideals of $\mathfrak{R}$ and $\bar{\mathfrak{R}}$ agree. Replace $\mathfrak{L}$ and $\bar{\mathfrak{L}}$ by divisors $\mathfrak{L}^*$ and $\bar{\mathfrak{L}}^*$ so that the representations become reduced. Then the associated prime ideals of $\mathfrak{L}^*$ and $\bar{\mathfrak{L}}^*$ also agree; by Theorem XI, $\mathfrak{L}^*$ is relatively prime to $\bar{\mathfrak{R}}$ and $\bar{\mathfrak{L}}^*$ is relatively prime to $\mathfrak{R}$. Since

$$\mathfrak{R}\mathfrak{L}^* \equiv 0(\bar{\mathfrak{R}}), \qquad \bar{\mathfrak{R}}\bar{\mathfrak{L}}^* \equiv 0(\mathfrak{R}),$$

we obtain

$$\mathfrak{R} \equiv 0(\bar{\mathfrak{R}}), \qquad \bar{\mathfrak{R}} \equiv 0(\mathfrak{R}), \qquad \mathfrak{R} = \bar{\mathfrak{R}}.$$

Thus isolated ideals are uniquely determined by their associated prime ideals. This sharpens Theorems VII and IX on decompositions into irreducible, respectively greatest primary, ideals; by the addendum to Theorem IX it is enough to assume a shortest representation. In summary:

Theorem XIII. *In every shortest representation of an ideal as the least common multiple of irreducible, respectively greatest primary, ideals, the isolated irreducible, respectively greatest primary, ideals are uniquely determined; the ambiguity concerns only the non-isolated irreducible, respectively greatest primary, ideals.*[31] *In general, isolated ideals are uniquely determined by their associated prime ideals.*

If in such a shortest representation by irreducible, respectively greatest primary, ideals the ideals $\mathfrak{B}_i$, respectively $\mathfrak{Q}_j$, are non-isolated, then by definition their complements $\mathfrak{U}_i$, respectively $\mathfrak{L}_j$, are divisible by $\mathfrak{P}_i$, respectively $\mathfrak{P}_j$. Hence

$$\mathfrak{U}_i^{\varrho_i} \equiv 0(\mathfrak{B}_i) \qquad \text{respectively} \qquad \mathfrak{L}_j^{\sigma_j} \equiv 0(\mathfrak{Q}_j).$$

Conversely, if these relations hold, then $\mathfrak{P}_i$, respectively $\mathfrak{P}_j$, is contained in at least one associated prime ideal of the complement, and hence, by the addendum to Theorem IX, in an associated prime ideal of the divisor $\mathfrak{L}_j^*$ of $\mathfrak{L}_j$ that supplies a representation reduced with respect to $\mathfrak{L}_j^*$. Thus $\mathfrak{B}_i$, respectively $\mathfrak{Q}_j$, is non-isolated. In particular, irreducible ideals $\mathfrak{B}_i$ whose associated prime ideal occurs more than once in the decomposition of $\mathfrak{M}$ are always non-isolated. *Non-isolated primary ideals are therefore also characterized by the fact that a power of every complement is divisible by them; isolated primary ideals by the fact that this cannot occur.*

§ 8. Unique representation of an ideal as a product of coprime-irreducible ideals.

If the underlying ring domain Σ has a unit, that is, an element ε such that $\varepsilon a = a$ for every element of Σ,[32] then coprime ideals can be defined as follows.

Definition VIII. *Two ideals $\mathfrak{R}$ and $\mathfrak{S}$ are called coprime if their greatest common divisor is the unit ideal $\mathfrak{O} = (\varepsilon)$ consisting of all elements of Σ. An ideal is called coprime-irreducible if it cannot be represented as the least common multiple of pairwise coprime ideals.*

Note that two coprime ideals are always mutually relatively prime. By definition there are elements $r \equiv 0(\mathfrak{R})$ and $s \equiv 0(\mathfrak{S})$ with $\varepsilon = r + s$. From $\mathfrak{T}\mathfrak{R} \equiv 0(\mathfrak{S})$ it follows that $\mathfrak{T}r \equiv 0(\mathfrak{S})$, hence $\mathfrak{T}\varepsilon = \mathfrak{T} \equiv 0(\mathfrak{S})$; correspondingly, from $\mathfrak{T}\mathfrak{S} \equiv 0(\mathfrak{R})$ it follows that $\mathfrak{T} \equiv 0(\mathfrak{R})$.[33] Thus Lemma V implies that every representation by pairwise coprime ideals is also reduced.

The uniqueness proof rests on the analogue of Theorem X:

[30] If one introduced ordinary associated prime ideals (end of § 5), then one would have to add the special condition that those of $\mathfrak{L}$ are all distinct from those of $\mathfrak{R}$. With Definition VIa, the representation no longer has to be assumed reduced with respect to the complement.

[31] For ideals in polynomial domains, this theorem was already stated without proof by Macaulay in the case of decomposition into greatest primary ideals; his definition of isolated and non-isolated (imbedded) primary ideals can be regarded as an irrational form of the one given below.

[32] Because multiplication is commutative, Σ can of course have at most one unit: for any two units ε_1 and ε_2, one has $\varepsilon_1\varepsilon_2 = \varepsilon_2 = \varepsilon_1$.

[33] The converse is false: for example, $\mathfrak{R} = (x)$ and $\mathfrak{S} = (y)$ are mutually relatively prime, but not coprime.

Theorem XIV. *If $\mathfrak{R}$ is coprime to each of the ideals $\mathfrak{S}_1, \ldots, \mathfrak{S}_\lambda$, then $\mathfrak{R}$ is also coprime to $\mathfrak{S} = [\mathfrak{S}_1 \cdots \mathfrak{S}_\lambda]$. Conversely, if $\mathfrak{R}$ and $\mathfrak{S}$ are coprime, then $\mathfrak{R}$ is coprime to each $\mathfrak{S}_j$. If $\mathfrak{R} = [\mathfrak{R}_1 \cdots \mathfrak{R}_\mu]$ and every $\mathfrak{R}_i$ is coprime to every $\mathfrak{S}_j$, then $\mathfrak{R}$ and $\mathfrak{S}$ are coprime; here too the converse holds.*

Indeed, if $\mathfrak{R}$ is coprime to every $\mathfrak{S}_j$, then there are elements s_j such that

$$s_j \equiv 0(\mathfrak{S}_j), \qquad s_j \equiv \varepsilon(\mathfrak{R}).$$

Consequently

$$s_1 s_2 \cdots s_\lambda \equiv 0(\mathfrak{S}), \qquad s_1 s_2 \cdots s_\lambda \equiv \varepsilon(\mathfrak{R}), \qquad (\mathfrak{R}, \mathfrak{S}) = (\varepsilon).$$

Since $(\mathfrak{R}, \mathfrak{S})$ is divisible by each $(\mathfrak{R}, \mathfrak{S}_j)$, the converse follows. Repeated application of the same argument gives the second part. For if $\mathfrak{R}_i$ is coprime to $\mathfrak{S}_1, \ldots, \mathfrak{S}_\lambda$ for a fixed i, then $\mathfrak{R}_i$ is coprime to $\mathfrak{S}$; if this holds for every i, then, since coprimeness is mutual, $\mathfrak{S}$ is coprime to $\mathfrak{R}$. Conversely, coprimeness of $\mathfrak{R}$ and $\mathfrak{S}$ gives coprimeness of $\mathfrak{S}$ with $\mathfrak{R}_i$, and hence of $\mathfrak{R}_i$ with $\mathfrak{S}$.

The proof of existence and uniqueness[34] of the decomposition into coprime-irreducible ideals is, as in the corresponding proof for relatively-prime ideals, a matter of a unique grouping. Since, however, the relatively-prime irreducible ideals $\mathfrak{R}_1, \ldots, \mathfrak{R}_\sigma$ of $\mathfrak{M}$ are uniquely defined by Theorem XII, it is unnecessary here to go back to the associated prime ideals.

We collect the uniquely defined relatively-prime irreducible ideals $\mathfrak{R}_1, \ldots, \mathfrak{R}_\sigma$ of $\mathfrak{M}$ into groups so that *every ideal of one group is coprime to every ideal of any different group, while a single group cannot be split into two subgroups such that every ideal of one subgroup is coprime to every ideal of the other subgroup.* Such a grouping is obtained as follows. By definition a group G must contain, together with every ideal $\mathfrak{R}$, all ideals not coprime to $\mathfrak{R}$. Let these be $\mathfrak{R}_{i_1}$; let $\mathfrak{R}_{i_1 i_2}$ be the ideals not coprime to these; in general let $\mathfrak{R}_{i_1 \ldots i_\lambda}$ be not coprime to $\mathfrak{R}_{i_1 \ldots i_{\lambda-1}}$. Since only finitely many ideals occur, this process terminates after finitely many steps. The set of ideals so obtained forms a group G with the desired properties: all ideals outside G are, by construction, coprime to all ideals in G. If G were split into two subgroups $G^{(1)}$ and $G^{(2)}$, and if $\mathfrak{R}_{i_1 \ldots i_\lambda}$ belonged to $G^{(1)}$, then, because coprimeness is mutual, $G^{(1)}$ would also have to contain $\mathfrak{R}_{i_1 \ldots i_{\lambda-1}}, \ldots, \mathfrak{R}_{i_1}$, hence also $\mathfrak{R}$ and therefore the whole group G. This proves irreducibility. Repeating the construction with the remaining ideals gives a grouping $G_1, \ldots, G_\tau$ of all $\mathfrak{R}$.

The grouping is unique: if $G'_1, \ldots, G'_\tau$ is a second grouping and $\mathfrak{R}_{i_1 \ldots i_\lambda}$ is an element of G'_i, then G'_i contains $\mathfrak{R}$ and hence G_1, and, by the irreducibility condition, cannot contain elements different from G_1; therefore $G'_i = G_1$.

Let $\mathfrak{T}_i$ be the least common multiple of the ideals $\mathfrak{R}$ united in a group G_i. We show that

$$\mathfrak{M} = [\mathfrak{T}_1 \cdots \mathfrak{T}_\tau]$$

is a representation of $\mathfrak{M}$ by coprime-irreducible ideals. Decomposing the $\mathfrak{T}$ into the $\mathfrak{R}$ first shows that $[\mathfrak{T}_1 \cdots \mathfrak{T}_\tau]$ really represents $\mathfrak{M}$. By Theorem XIV the $\mathfrak{T}$ are pairwise coprime, and each $\mathfrak{T}$ is coprime-irreducible. This proves the existence of such a representation.

For uniqueness, let $\mathfrak{M} = [\bar{\mathfrak{T}}_1 \cdots \bar{\mathfrak{T}}_\tau]$ be a second representation of this type. Decomposing the $\bar{\mathfrak{T}}$ into relatively-prime irreducible ideals $\mathfrak{R}$, Theorem XIV shows that ideals $\mathfrak{R}$ occurring in different $\bar{\mathfrak{T}}$ are coprime to one another, hence also mutually relatively prime; therefore they agree with the uniquely defined relatively-prime irreducible ideals $\mathfrak{R}$ of $\mathfrak{M}$. The ideals $\bar{\mathfrak{T}}_i$ also induce, by Theorem XIV, a grouping G'_i of the $\mathfrak{R}$ with the stated properties. Since this grouping is unique and each $\bar{\mathfrak{T}}_i$ is uniquely determined by the group $G'_i = G_i$, one has $\bar{\mathfrak{T}}_i = \mathfrak{T}_i$; uniqueness is proved.

For pairwise coprime ideals, the least common multiple is moreover equal to the product. For by Theorem XIV, if $\mathfrak{M} = [\mathfrak{T}_1 \cdots \mathfrak{T}_\tau]$, the complement $\mathfrak{L}_i$ is also coprime to $\mathfrak{T}_i$; hence there are elements

$$t_i \equiv 0(\mathfrak{T}_i), \qquad l_i \equiv 0(\mathfrak{L}_i), \qquad \varepsilon = t_i + l_i.$$

From $f \equiv 0(\mathfrak{T}_i)$ and $f \equiv 0(\mathfrak{L}_i)$ it follows that

$$f = f\varepsilon = ft_i + fl_i, \qquad f \equiv 0(\mathfrak{L}_i \mathfrak{T}_i).$$

Conversely $\mathfrak{L}_i \mathfrak{T}_i$ is divisible by $[\mathfrak{L}_i, \mathfrak{T}_i]$, whence $[\mathfrak{L}_i, \mathfrak{T}_i] = \mathfrak{L}_i \mathfrak{T}_i$; continuing the procedure on $\mathfrak{L}_i$ gives

$$\mathfrak{M} = [\mathfrak{T}_1 \cdots \mathfrak{T}_\tau] = \mathfrak{T}_1 \mathfrak{T}_2 \cdots \mathfrak{T}_\tau.$$

[34]The existence of the decomposition can again be proved directly in analogy with § 2; the uniqueness proof too can be carried out directly (compare what was said in the introduction about Schmeidler and Noether–Schmeidler). The proof given here also gives insight into the structure of coprime-irreducible ideals.

Therefore:

Theorem XV. *Every ideal can be represented uniquely as a product of finitely many pairwise coprime and coprime-irreducible ideals.*

§ 9. Extension of the investigation to modules. Equality of the number of components in decompositions into irreducible modules.

We now show that the content of the first three paragraphs, which concerns irreducible ideals and not primary and prime ideals, remains valid under weaker hypotheses. Those paragraphs do not use the commutative law of multiplication and refer only to the property of ideals of being modules. They therefore remain valid for modules with respect to non-commutative domains, which are now to be defined. The definition of modules is to be based on a double domain (Σ, T) with the following properties.

Σ is an abstractly defined non-commutative ring, that is, Σ is a system of elements $a, b, c, \ldots$ for which two modes of combination are defined, ring addition $(+)$ and ring multiplication $(\times)$, satisfying the laws stated in § 1, with the exception of the commutative law 4 for ring multiplication.

T is a system of elements $\alpha, \beta, \gamma, \ldots$, for which, in connection with Σ, two combinations are likewise defined: addition, which from any two elements α, β produces a unique third element $\alpha + \beta$; and multiplication of an element α of T by an element c of Σ, which uniquely produces an element $c\alpha$ of T.[35]

For these combinations the following laws hold:

1. The associative law of addition: $(\alpha + \beta) + \gamma = \alpha + (\beta + \gamma)$.
2. The commutative law of addition: $\alpha + \beta = \beta + \alpha$.
3. The law of unrestricted and unique subtraction: there is in T one and only one element ξ satisfying the equation $\alpha + \xi = \beta$ (one writes $\xi = \beta - \alpha$).
4. The associative law of multiplication: $a(b\gamma) = (a \times b)\gamma$.
5. The distributive law:
$$(a + b)\gamma = a\gamma + b\gamma, \qquad c(\alpha + \beta) = c\alpha + c\beta.$$

From these conditions there follow, as is well known, the existence of the zero element and the validity of the distributive law also for the combination of subtraction and multiplication:

$$(a - b)\gamma = a\gamma - b\gamma, \qquad c(\alpha - \beta) = c\alpha - c\beta,$$

where $(-)$ denotes subtraction in Σ. If Σ contains a unit ε, then $\varepsilon\alpha = \alpha$ is to hold for every element α of T.

By a module M in (Σ, T) we mean a system of elements of T satisfying the two conditions:

1. along with α, M contains $c\alpha$, where c is any element of Σ;
2. along with α and β, M contains the difference $\alpha - \beta$, hence also $n\alpha$ for every integer n.[36]

By this definition, T itself forms a module in (Σ, T). If in particular the domain T and the combinations established there coincide with the domain Σ and the combinations valid there, then the module M becomes a right ideal M in Σ. If Σ is also assumed commutative, the ordinary concept of ideal results, and is thus a special case of the concept of module.[37]

For modules all definitions of § 1 remain valid. Thus $\alpha \equiv 0\,(M)$, respectively $N \equiv 0\,(M)$, means that α, respectively every element of N, is an element of M; in other words, α, respectively N, is divisible by M. M is a proper divisor of N if M contains elements different from those of N; from $N \equiv 0\,(M)$ and $M \equiv 0\,(N)$ follows $M = N$. The definitions of greatest common divisor and least common multiple likewise remain literally the same. If the module M contains, in particular, a finite number of elements $\alpha_1, \ldots, \alpha_\rho$ such that

$$M = (\alpha_1, \ldots, \alpha_\rho),$$

[35] Thus here one is dealing with a "right-sided" multiplication, a "right-sided" domain T, and consequently with "right-sided" modules and ideals. If one were to base T instead on a left-sided multiplication αc, a corresponding theory of left-sided modules and ideals would result; M would contain αc along with α. The associative law would then have the form $(\gamma b)a = \gamma(b \times a)$.

[36] The integers are again to be understood as abbreviating symbols, not as ring elements.

[37] The simplest example of a module is the module of integral linear forms: here Σ consists of all rational integers and T of all integral linear forms. A somewhat more general module arises when, in Σ and T, one takes algebraic integers instead of rational integers, or for instance all even integers. If, instead of the linear forms, one considers each complex of coefficients as an element, then the combinations in Σ and T are in fact different. Ideals in non-commutative polynomial ring domains are the subject of the joint paper of Noether–Schmeidler. Ideals in other special non-commutative domains are treated in Hurwitz's lectures on the number theory of quaternions (Berlin, Springer 1919) and in the works of Du Pasquier cited there.

that is,

$$\alpha = c_1\alpha_1 + \cdots + c_\rho\alpha_\rho + n_1\alpha_1 + \cdots + n_\rho\alpha_\rho$$

for every $\alpha \equiv 0\,(M)$, where the c_i are elements of Σ and the n_i are integers, then M is called a finite module and $\alpha_1, \dots, \alpha_\rho$ a module basis.

In what follows we assume, just as in § 1, only those domains (Σ, T) that satisfy the finiteness condition: every module in (Σ, T) is finite, and hence has a module basis.

For such a domain (Σ, T), Theorem I on finite chains holds also for modules, as the proof given there shows, and therefore all hypotheses for §§ 2 and 3 are satisfied. Definition I and Lemma I on shortest and reduced representations carry over directly; likewise Theorem II on the representability of every module as the least common multiple of finitely many irreducible modules, where Lemma II shows that every shortest representation of this kind is at the same time reduced. Theorem III also remains valid, expressing reducibility of a module through properties of its complement; from it follows Lemma III, and finally Theorem IV, which asserts equality of the number of components in two different shortest representations of a module as the least common multiple of irreducible modules. Theorem IV also gives Lemma IV as the converse of Lemma I on reduced representations.

Addendum. The same argument shows that all these theorems and definitions remain valid if, for two-sided domains T, that is, domains which are both right- and left-sided, one everywhere understands by a module a two-sided module: one which, along with α, also contains $c\alpha$ and αc, and along with α and β also contains $\alpha - \beta$.

While all these theorems rest only on the concept of divisibility and of least common multiple, the further uniqueness theorems rest essentially on the product concept and therefore do not admit a direct transfer. Thus the definition of a primary ideal and of a prime ideal cannot be transferred to modules, since the product of two elements of T is not defined. The formally possible transfer to non-commutative rings loses its meaning, since here the existence of the associated prime ideal cannot be proved[38] and the proof that an irreducible ideal is primary also fails. These two facts form the basis of the following uniqueness theorems. By contrast, if the non-commutative ring has a unit, coprime and coprime-irreducible ideals can be defined, and the arguments used in the proof of Theorem II show that every ideal can be represented as the least common multiple of finitely many pairwise coprime and coprime-irreducible ideals.[39]

Finally, we mention a sufficient criterion for the finiteness condition to hold in (Σ, T): if Σ contains a unit and satisfies the finiteness condition, and if T itself is a finite module in (Σ, T), then every module in (Σ, T) is finite.

Indeed, from the existence of the unit in Σ, for

$$\mathfrak{M} = (f_1, \dots, f_e), \qquad f = \bar{b}_1 f_1 + \cdots + \bar{b}_e f_e + n_1 f_1 + \cdots + n_e f_e,$$

where the $\bar{b}_i$ are elements of Σ and the n_i are integers, there follows also a representation

$$f = b_1 f_1 + \cdots + b_e f_e$$

with the b_i in Σ. For since $f_i = \varepsilon f_i$, one has $n_i f_i = n_i \varepsilon f_i$, and since $n_i\varepsilon = (\varepsilon + \cdots + \varepsilon)$ belongs to Σ, $b_i = \bar{b}_i + n_i\varepsilon$ is an element of Σ. The hypothesis on T says that every element α of T admits a representation

$$\alpha = \bar{a}_1\tau_1 + \cdots + \bar{a}_k\tau_k + n_1\tau_1 + \cdots + n_k\tau_k,$$

and hence

$$\alpha = a_1\tau_1 + \cdots + a_k\tau_k,$$

the second representation following from the first just as above because $\varepsilon\tau_i = \tau_i$.

As α runs through the elements of a module M in (Σ, T), the coefficients a_k of τ_k run through an ideal $\mathfrak{M}_k$ of Σ; by the preceding, for every $a_k \equiv 0\,(\mathfrak{M}_k)$ one has

$$a_k = b_1 a_k^{(1)} + \cdots + b_e a_k^{(e)}.$$

Let $\alpha^{(i)}$ be an element of M for which the coefficient of τ_k is $a_k^{(i)}$. Then $\alpha - b_1\alpha^{(1)} - \cdots - b_e\alpha^{(e)}$ is an element belonging to M which depends only on $\tau_1, \dots, \tau_{k-1}$. The procedure can be repeated for the totality of these elements, which no longer contain τ_k and which form a module M', and the assertion is proved by finitely many repetitions.

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[38] For from $p_1^\rho \equiv 0\,(\mathfrak{Q})$ and $p_2^\sigma \equiv 0\,(\mathfrak{Q})$ it does not follow here that $(p_1 - p_2)^{\rho+\sigma} \equiv 0\,(\mathfrak{Q})$; likewise, from $(ab)^\rho \equiv 0\,(\mathfrak{Q})$ it does not follow that $a^\rho b^\rho \equiv 0\,(\mathfrak{Q})$. Thus $\mathfrak{P}$ can be proved neither to be an ideal nor to have the property of a prime ideal. For two-sided ideals $\mathfrak{P}$ can indeed be proved to be an ideal, but not a prime ideal.

[39] For special, "completely reducible" ideals, uniqueness theorems can be set up here too; compare the joint paper Noether–Schmeidler.

§ 10. Special case of the polynomial domain.

1. Let the underlying ring domain Σ consist of all polynomials in $x_1, \ldots, x_n$ with arbitrary complex coefficients. By Hilbert's basis theorem (Ann. 36) the finiteness condition holds for it. The issue is the relation of our theorems to the known theorems of elimination theory and module theory.

This relation is established by the following special case of a known theorem of Hilbert:[40]

If f vanishes for every finite system of values of $x_1, \ldots, x_n$ that is a zero of all polynomials of a prime ideal $\mathfrak{P}$ (a zero of $\mathfrak{P}$), then f is divisible by $\mathfrak{P}$. In other words, a prime ideal $\mathfrak{P}$ consists of the totality of polynomials which vanish at its zeros.[41]

Thus, if a product fg vanishes for all zeros of $\mathfrak{P}$, at least one factor vanishes; the zeros form an irreducible algebraic variety. Conversely, if this definition of irreducible variety is adopted, then the totality of polynomials vanishing on an irreducible variety forms a prime ideal. Prime ideal and irreducible variety therefore correspond one-to-one. Since $\mathfrak{P}^\rho \equiv 0\,(\mathfrak{Q})$ and $\mathfrak{Q} \equiv 0\,(\mathfrak{P})$, the zeros of a primary ideal coincide with those of its associated prime ideal.[42] Therefore the representation of an ideal as the least common multiple of greatest primary ideals gives a resolution of all zeros of the ideal into irreducible varieties; and, as Lasker showed, the converse also holds. The proof of uniqueness of the associated prime ideals thus corresponds here to the fundamental theorem of elimination theory on the unique decomposability of an algebraic variety into irreducible varieties, and for special polynomial domains, where no unique product representation of the polynomials by irreducible polynomials of the domain and consequently no elimination theory exists, it can serve as an equivalent of that theorem of elimination theory.

The irreducible varieties corresponding to the isolated primary ideals are exactly those which occur in the "minimal resolvent";[43] for the zeros of every non-isolated greatest primary ideal are at the same time zeros of at least one isolated one, namely one whose associated prime ideal is divisible by that of the non-isolated ideal. The uniqueness of isolated primary ideals therefore gives, in the exponents, new invariant multiplicity numbers. The uniqueness theorems for the resolution of primary ideals into irreducible ones can likewise be regarded as supplements to elimination theory in the sense of multiplicity.

After these remarks the different decompositions can be interpreted in terms of their behavior with respect to algebraic varieties. Pairwise coprime ideals correspond to varieties having no common zero; for mutually relatively prime ideals, no irreducible variety of one ideal is at the same time a common zero; the greatest primary ideals vanish only on irreducible varieties which are all distinct from one another; in the decomposition into irreducible ideals the same irreducible varieties may occur with multiplicity.

It should also be noted that, instead of the general polynomial domain, the domain of all homogeneous forms may also be taken as the basis; for one readily verifies that the general theorems remain valid for the combinations there in force, where addition is defined only for forms of the same dimension.[44]

A simplest example of the four different decompositions - for which the formulas follow below - is given, according to the preceding, by a line and two intersecting lines skew to it, one of the latter containing, with higher multiplicity, a point different from the point of intersection. The decomposition into coprime-irreducible ideals corresponds to the decomposition into the line and the variety skew to it; in the decomposition into relatively-prime irreducible ideals this variety splits into the two lines; the decomposition into greatest primary ideals corresponds to separating off the point of higher multiplicity, while the decomposition into irreducible ideals forces a resolution of that point.

Taking this point as the origin, the running line as the y-axis, the line intersecting it parallel to the x-axis, and the line skew to it parallel to the z-axis, such a configuration is represented, for example, by the following irreducible ideals:[45]

$$\mathfrak{B}_1 = (x-1, y), \quad \mathfrak{B}_2 = (y-1, z), \quad \mathfrak{B}_3 = (x, z), \quad \mathfrak{B}_4 = (x^3, y, z), \quad \mathfrak{B}_5 = (x^2, y^2, z).$$

[40] Über die vollen Invariantensysteme. Math. Ann. 42 (1893), § 3, p. 313.

[41] As Lasker showed [Math. Ann. 60 (1905), p. 607], this special case can also be proved directly for homogeneous forms; conversely, Hilbert's theorem follows again from it in the homogeneous and inhomogeneous cases. We can express that theorem as follows: if an ideal $\mathfrak{R}$ vanishes at all zeros of M, then a power of $\mathfrak{R}$ is divisible by M. This theorem, and also the special case, holds only if the range of values of the x is algebraically closed; hence it cannot follow from our theorems alone, but must use the existence of roots, for example at the point where one uses that an ideal whose zeros consist only of $x_1 = 0, \ldots, x_n = 0$ contains all products of powers of the x from some dimension on. The remaining proof can be somewhat simplified in comparison with Lasker by using our theorems. For Lasker must also appeal to Hilbert's theorem to prove the decomposition of an ideal into greatest primary ideals.

[42] Macaulay (compare the introduction) takes this property of a primary ideal, namely to possess an irreducible variety, as the definition; Lasker includes only the concept of the manifold of a variety in the definition and otherwise defines abstractly. The primary ideals vanishing only for $x_1 = 0, \ldots, x_n = 0$ occupy a special position for Lasker.

[43] Compare, for example, J. König, Einleitung in die allgemeine Theorie der algebraischen Größen (Leipzig, Teubner, 1903), p. 235.

[44] That in the ambiguous case inhomogeneous decompositions can exist beside homogeneous ones is shown by the example $(x^3, xy, y^3) = [(x^3, y), (y^3, x)] = [(xy, x^3, y^3, x + y^3), (xy, x^3, y^3, y + x^2)]$.

[45] Of these, the first three are irreducible as prime ideals; $\mathfrak{B}_4$ because it has only the divisors (x^2, y, z) and (x, y, z); $\mathfrak{B}_5$ because every divisor contains the polynomial xy.

The associated prime ideals are

$$\mathfrak{P}_1 = \mathfrak{B}_1, \quad \mathfrak{P}_2 = \mathfrak{B}_2, \quad \mathfrak{P}_3 = \mathfrak{B}_3, \quad \mathfrak{P}_4 = \mathfrak{P}_5 = (x, y, z).$$

The greatest primary ideals are

$$\mathfrak{Q}_1 = \mathfrak{B}_1, \quad \mathfrak{Q}_2 = \mathfrak{B}_2, \quad \mathfrak{Q}_3 = \mathfrak{B}_3, \quad \mathfrak{Q}_4 = [\mathfrak{B}_4, \mathfrak{B}_5] = (x^3, y^2, x^2y, z).$$

The relatively-prime irreducible ideals are

$$\mathfrak{R}_1 = \mathfrak{Q}_1, \quad \mathfrak{R}_2 = \mathfrak{Q}_2, \quad \mathfrak{R}_3 = [\mathfrak{Q}_3, \mathfrak{Q}_4] = (x^3, x^2y, xy^2, z).$$

The coprime-irreducible ideals are

$$\mathfrak{S}_1 = \mathfrak{R}_1, \quad \mathfrak{S}_2 = [\mathfrak{R}_2, \mathfrak{R}_3] = ((y-1)x^3, (y-1)x^2y, (y-1)xy^2, z).$$

Thus the total ideal is

$$\begin{aligned} \mathfrak{M} &= [\mathfrak{S}_1, \mathfrak{S}_2] = \mathfrak{S}_1\mathfrak{S}_2 \\ &= ((x-1)(y-1)x^3, (y-1)x^2y, (y-1)xy^2, (x-1)z, y(y-1)x^3, yz), \end{aligned}$$

where

$$1 = -(y-1)x^3 + (y-1)(x^3-1) + y,$$
$$(y-1)(x^3-1) + y \equiv 0\,(\mathfrak{S}_1), \qquad -(y-1)x^3 \equiv 0\,(\mathfrak{S}_2).$$

Here the ideals $\mathfrak{B}_1$, $\mathfrak{B}_2$, $\mathfrak{B}_3$ are isolated, and hence are uniquely determined also in the decompositions into irreducible and greatest primary ideals. The ideals $\mathfrak{B}_4$ and $\mathfrak{B}_5$, respectively $\mathfrak{Q}_4$, are non-isolated; they are not uniquely determined, but can, for instance, be replaced by

$$\mathfrak{D}_4 = (x^3, y + \lambda x^2, z), \qquad \mathfrak{D}_5 = (x^2 + \mu xy, y^2, z),$$

and $\mathfrak{Q}_4$ can likewise be replaced by

$$\overline{\mathfrak{Q}}_4 = (x^3, x^2y, y^2 + \lambda xy, z).$$

2. Just as the general, and the integral, polynomial domains satisfy the finiteness condition, so does every finite integral domain of polynomials, by Hilbert's theorem on module bases;[46] the coefficients may be assigned to an arbitrary field. We give one more example, for the domain of all even polynomials, the simplest domain in which, because $x^2y^2 = (xy)^2$, no unique product representation of the polynomials by irreducible polynomials of the domain exists. It is the same configuration as in the preceding example, now given by the irreducible ideals

$$\mathfrak{B}_1 = (x^2-1, xy, y^2, yz), \quad \mathfrak{B}_2 = (y^2-1, xz, yz, z^2),$$
$$\mathfrak{B}_3 = (x^2, xy, xz, yz, z^2), \quad \mathfrak{B}_4 = (x^4, x^3y, y^2, xz, yz, z^2),$$
$$\mathfrak{B}_5 = (x^2, y^2, xz, yz, z^2).$$

The associated prime ideals are

$$\mathfrak{P}_1 = \mathfrak{B}_1, \quad \mathfrak{P}_2 = \mathfrak{B}_2, \quad \mathfrak{P}_3 = \mathfrak{B}_3, \quad \mathfrak{P}_4 = \mathfrak{P}_5 = (x^2, xy, y^2, xz, yz, z^2).$$

That $\mathfrak{P}_1$ is a prime ideal follows from the fact that every polynomial of the domain has the form

$$f = \varphi(z^2) + xz\,\psi(z^2) \pmod{\mathfrak{P}_1}.$$

Thus if

$$f_1 = \varphi_1(z^2) + xz\,\psi_1(z^2), \qquad f_2 = \varphi_2(z^2) + xz\,\psi_2(z^2),$$

then from $f_1f_2 \equiv 0\,(\mathfrak{P}_1)$ one obtains the equations

$$\varphi_1\varphi_2 + z^2\psi_1\psi_2 = 0, \qquad \varphi_2\psi_1 + \varphi_1\psi_2 = 0,$$

[46] Conversely, if a polynomial domain satisfies the finiteness condition and every polynomial has at least one representation in which the multipliers are of lower degree in x, then the domain is a finite integral domain.

and hence $f_1 \equiv 0\,(\mathfrak{P}_1)$ or $f_2 \equiv 0\,(\mathfrak{P}_1)$. In exactly the same way one sees that $\mathfrak{P}_2$ is a prime ideal; $\mathfrak{P}_3$ is a prime ideal because every polynomial of the domain modulo $\mathfrak{P}_3$ is congruent to a polynomial in y^2; $\mathfrak{P}_4$ consists of all polynomials of the domain. Thus $\mathfrak{B}_1$, $\mathfrak{B}_2$, and $\mathfrak{B}_3$ are irreducible as prime ideals, while $\mathfrak{B}_4$ and $\mathfrak{B}_5$ each have only the single proper divisor $\mathfrak{P}_4$, and hence are necessarily irreducible.

From the irreducible ideals one obtains the greatest primary ideals

$$\mathfrak{Q}_1 = \mathfrak{B}_1, \quad \mathfrak{Q}_2 = \mathfrak{B}_2, \quad \mathfrak{Q}_3 = \mathfrak{B}_3, \quad \mathfrak{Q}_4 = [\mathfrak{B}_4, \mathfrak{B}_5] = (x^4, x^3y, y^2, xz, yz, z^2),$$

the relatively-prime irreducible ideals

$$\mathfrak{R}_1 = \mathfrak{Q}_1, \quad \mathfrak{R}_2 = \mathfrak{Q}_2, \quad \mathfrak{R}_3 = [\mathfrak{Q}_3, \mathfrak{Q}_4] = (x^4, x^3y, x^2y^2, xy^3, xz, yz, z^2),$$

and the coprime-irreducible ideals

$$\mathfrak{S}_1 = \mathfrak{R}_1, \quad \mathfrak{S}_2 = [\mathfrak{R}_2, \mathfrak{R}_3] = ((y^2-1)x^4, (y^2-1)x^3y, (y^2-1)x^2y^2, (y^2-1)xy^3, xz, yz, z^2).$$

As in the first example,

$$[\mathfrak{B}_4, \mathfrak{B}_5] = [\mathfrak{D}_4, \mathfrak{D}_5], \qquad [\mathfrak{Q}_3, \mathfrak{Q}_4] = [\mathfrak{Q}_3, \overline{\mathfrak{Q}}_4],$$

where

$$\mathfrak{D}_4 = (x^4, xy + \lambda x^2, \ldots), \qquad \mathfrak{D}_5 = (x^2 + \mu xy, \ldots), \qquad \lambda\mu \neq 1,$$

and

$$\overline{\mathfrak{Q}}_4 = (x^4, x^3y, y^2 + \lambda xy, \ldots).$$

The remaining irreducible, respectively greatest primary, ideals $\mathfrak{B}_1$, $\mathfrak{B}_2$, $\mathfrak{B}_3$ are uniquely determined as isolated ideals.

§ 11. Examples from number theory and from the theory of differential expressions.

1. Let the underlying domain Σ consist of all even rational integers. Thus Σ can be put in one-to-one correspondence with the domain of all rational integers, by assigning to each number $2a$ in Σ the number a. It follows at once that every ideal in Σ is a principal ideal $(2a)$, where in the basis representation $2c = n \cdot 2a$ for every element $2c$ of the ideal, the odd integers n are only abbreviations for finite sums.

The prime ideals of the domain are $\mathfrak{P}_0 = \mathfrak{D} = (2)$ and $\mathfrak{P} = (2p)$, where p denotes an odd prime number; every prime ideal is therefore divisible by $\mathfrak{P}_0$, but by no further prime ideal. The primary ideals are

$$\mathfrak{Q}_0 = (2 \cdot 2^{e_0}) \qquad \text{and} \qquad \mathfrak{Q}_p = (2p^e).$$

They are at the same time irreducible ideals, and by what was said about the prime ideals, two ideals belonging to different odd prime numbers are mutually relatively prime, but no $\mathfrak{Q}$ is relatively prime to any $\mathfrak{Q}_0$.[47] Thus the $\mathfrak{Q}_0$ are the only non-isolated primary ideals. The unique decomposition of a into prime powers corresponds to the unique representation of the ideal $(2a)$ by greatest primary, at the same time irreducible, ideals:

$$(2a) = [(2 \cdot 2^{e_0}), (2p_1^{e_1}), \ldots, (2p_\mu^{e_\mu})].$$

Thus here, in contrast to the examples from the polynomial domain, even the non-isolated greatest primary ideals are uniquely determined. If $e_0 = 0$ this is at the same time a representation by mutually relatively prime ideals, while if $e_0 > 0$ the ideal is relatively-prime-irreducible.

Thus, while in the domain of all integers the four different decompositions coincide, here this happens only for the two decompositions into greatest primary and into irreducible ideals on the one hand, and, for $e_0 > 0$, for coprime-irreducible (every ideal is coprime-irreducible, since the domain has no unit) and relatively-prime-irreducible ideals on the other; for $e_0 = 0$, however, the coprime-irreducible and relatively-prime-irreducible decompositions become different. At the same time one already obtains here an example in which one prime ideal can be divisible by another without being identical with it; more generally, one sees that product representation does not follow from divisibility. The latter fact - a consequence of the absence of a unit in the domain - is also why no unique product representation of the numbers of the domain by irreducible numbers of the domain exists, even though every ideal is a principal ideal; the introduction of least common multiples

[47] Indeed, from $2b \cdot 2p_1^{e_1} \equiv 0\,(2p_2^{e_2})$ for odd $p_1 \neq p_2$ one always obtains $2b \equiv 0\,(2p_2^{e_2})$; but from $2b \cdot 2p^e \equiv 0\,(2 \cdot 2^{e_0})$ one obtains only $2b = 2 \cdot 2^{e_0 - 1}$ $(\mathrm{mod}\ 2 \cdot 2^{e_0})$.

therefore proves necessary. It should also be noted that the situation remains exactly the same if, instead of all even integers, one takes all integers divisible by a fixed prime number or power of a prime.

Irreducible and primary ideals also become different, however, if Σ consists of all numbers divisible by a composite number $g = p_1^{\nu_1} \cdots p_s^{\nu_s}$. Again every ideal is a principal ideal $(g \cdot a)$, and the prime ideals are again $\mathfrak{P}_0 = \mathfrak{D} = (g)$ and $\mathfrak{P} = (g \cdot p)$, where p is a prime different from the primes occurring in g. But the irreducible ideals are $\mathfrak{B}_{\lambda_i} = (g \cdot p_i^{\lambda_i})$ and $\mathfrak{B}_e = (g \cdot p^e)$; the primary ideals are $\mathfrak{Q}_e = \mathfrak{B}_e$ and the ideals, different from the irreducible ones,

$$\mathfrak{Q}_{\lambda_1 \ldots \lambda_s} = (g \cdot p_1^{\lambda_1} \cdots p_s^{\lambda_s}),$$

where the $\mathfrak{B}_{\lambda_i}$ and $\mathfrak{Q}_{\lambda_1 \ldots \lambda_s}$ all have the same associated prime ideal $\mathfrak{P}_0 = (g)$. Here too the decomposition into irreducible ideals is unique, and consequently so is the decomposition into greatest primary ideals; thus here again the non-isolated ideals are uniquely determined.

2. An example of a non-commutative ring domain is supplied by the ideal theory in non-commutative polynomial domains treated in the paper of Noether–Schmeidler. In particular one has “completely reducible” ideals, that is, ideals for which the components of the decomposition are pairwise coprime and have no proper divisors; the components are therefore a fortiori irreducible. Thus, in addition to the isomorphism proved there according to § 9, one obtains equality of the number of components in two different decompositions. For the decomposition of systems of partial or ordinary linear differential expressions, which arises as a special case of that paper, this gives a result which does not seem to have been noticed even in the known case of a single ordinary linear differential expression.

At the same time the system T of all residue classes of a fixed ideal M, together with the non-commutative polynomial domain Σ, supplies a double domain (Σ, T), where T has the module property with respect to Σ. For the difference of two residue classes is again a residue class, and likewise the product of a residue class by an arbitrary polynomial; whereas the product of two residue classes does not exist (loc. cit., § 3). Thus the systems of residue classes called “subgroups” there furnish examples of modules in double domains (Σ, T), where the underlying ring domain Σ is non-commutative.

§ 12. Example from elementary divisor theory.

This is a conception of elementary divisor theory dictated by the general developments, but the theory itself is assumed as known.

Let Σ be the domain of all integral matrices with n^2 elements, with addition and multiplication defined in the usual sense for matrices. Then Σ is a non-commutative ring domain; the ideals are therefore in general one-sided, with two-sided ideals only as a special case.[48]

We first show that every ideal is principal. For this purpose, in the case of right ideals, to each matrix $A = (a_{ik})$ we assign the module

$$A = (a_{11}\xi_1 + \cdots + a_{1n}\xi_n, \ldots, a_{n1}\xi_1 + \cdots + a_{nn}\xi_n)$$

of integral linear forms. Conversely, to this module there corresponds every matrix that supplies a basis of A, hence, beside A, also UA, where U is unimodular. More generally, to the product PA there corresponds a module B which is a multiple of A. A single linear form from A is given by such matrices P as contain only one non-zero row.

Let now $A_1, A_2, \ldots, A_\nu, \ldots$ be all elements of an ideal $\mathfrak{M}$; let $A_1, A_2, \ldots, A_\nu, \ldots$ be the modules assigned to them; let A be their greatest common divisor and UA the most general matrix assigned to this module A. To each single linear form in A there then corresponds, by the definition of greatest common divisor, a matrix $P_1 A_{i_1} + \cdots + P_\sigma A_{i_\sigma}$, where, as above, the P have only one non-zero row. It follows that the matrix A corresponding to a basis of A also admits such a representation, now with general P, and hence becomes an element of $\mathfrak{M}$. Since, furthermore, every module A_i is divisible by A, every matrix A_i is divisible by A; A, and in general UA, forms a basis of $\mathfrak{M}$. If one is dealing with left ideals, then correspondingly the columns of each matrix are to be regarded as the basis of a module; every ideal is a principal ideal, for which, beside A, also AV is a basis, with V an arbitrary unimodular matrix.

For the connection with elementary divisor theory, we now base the following on two-sided ideals; hence in particular PAQ belongs to the ideal whenever A does. By the preceding, the most general basis of such an ideal is UAV, where U and V are unimodular. The basis elements therefore exhaust a class of equivalent

[48]The ideal theory of these domains is the subject of the works of Du Pasquier: Zahlentheorie der Tettarionen, dissertation, Zürich, Vierteljahrsschr. d. Naturf. Ges. Zürich, 51 (1906); Zur Theorie der Tettarionenideale, ibid., 52 (1907). The content of the second paper is the proof that every ideal is a principal ideal.

matrices,[49] and there is a one-to-one relation between ideal and class, hence also between ideal and the elementary divisor system $(a_1 \mid a_2 \mid \cdots \mid a_n)$ of the class, where the a_i are, as is well known, non-negative integers, each dividing the following one. Thus the matrix of the class that occurs in the normal form determined by the elementary divisors may be regarded as a special basis of the ideal; divisibility of ideals, respectively of classes, implies divisibility of the elementary divisors, and conversely.

But Du Pasquier has shown, loc. cit. § 11, that for every two-sided ideal the rank is n and all elementary divisors agree. To be able to consider the case of general elementary divisors, we must therefore start not from the ideals but directly from the two-sided classes (which are likewise to be denoted by capital German letters). A class $\mathfrak{A} = UAV$ is divisible by another class $\mathfrak{B}$ if $A = PBQ$.

In general: *the least common multiple (the greatest common divisor) of two classes is obtained by forming the least common multiple (the greatest common divisor) of the corresponding elementary divisor systems.*[50] For let

$$(a_1 \mid a_2 \mid \cdots \mid a_n), \quad (b_1 \mid b_2 \mid \cdots \mid b_n), \quad (c_1 \mid c_2 \mid \cdots \mid c_n),$$

where $c_i = [a_i, b_i]$, be respectively the elementary divisor systems of $\mathfrak{A}$, $\mathfrak{B}$, and $\mathfrak{C}^*$, and let $\mathfrak{C} = [\mathfrak{A}, \mathfrak{B}]$. Then $\mathfrak{C}^*$ is divisible by $\mathfrak{A}$ and $\mathfrak{B}$, hence by $\mathfrak{C}$; conversely, the elementary divisors of $\mathfrak{C}$ are divisible by those of $\mathfrak{C}^*$, so $\mathfrak{C}$ is divisible by $\mathfrak{C}^*$ and therefore $\mathfrak{C} = \mathfrak{C}^*$. The proof for the greatest common divisor is obtained correspondingly.

The unique representation of the elementary divisors a_i as least common multiples of prime powers therefore corresponds to a representation of $\mathfrak{A}$ as the least common multiple of classes $\mathfrak{Q}$ whose elementary divisors are given by powers of one prime number, in symbols

$$\mathfrak{Q} \sim (p^{r_1} \mid p^{r_2} \mid \cdots \mid p^{r_\rho} \mid 0 \mid \cdots \mid 0), \qquad r_1 \leq r_2 \leq \cdots \leq r_\rho.$$

If in particular the rank ρ is equal to n, then one has here a decomposition into coprime and coprime-irreducible classes which is unique despite the non-commutative domain.

The classes $\mathfrak{Q}$ can be further decomposed into classes corresponding to the elementary divisor systems

$$\mathfrak{B}_1 \sim (p^{r_1} \mid \cdots \mid p^{r_1}), \quad \mathfrak{B}_2 \sim (1 \mid p^{r_2} \mid \cdots \mid p^{r_2}), \quad \ldots,$$
$$\mathfrak{B}_\nu \sim (1 \mid \cdots \mid 1 \mid p^{r_\nu} \mid \cdots \mid p^{r_\nu}), \quad \mathfrak{B}_{\rho+1} \sim (1 \mid \cdots \mid 1 \mid 0 \mid \cdots \mid 0),$$

where in $\mathfrak{B}_\nu$ the number 1 occurs $(\nu - 1)$ times. If, for example, $r_1 = r_2 = \cdots = r_\mu = 0$, then $\mathfrak{B}_1, \ldots, \mathfrak{B}_\mu$ are equal to the unit class and are to be omitted from the decomposition; the same holds for $\mathfrak{B}_{\rho+1}$ if $\rho = n$. If, furthermore, say $r_\nu = r_{\nu+1} = \cdots = r_{\nu+\lambda}$, then $\mathfrak{B}_{\nu+1}, \ldots, \mathfrak{B}_{\nu+\lambda}$ are proper divisors of $\mathfrak{B}_\nu$, and are likewise to be discarded. Denote by $\mathfrak{B}_{i_1}, \ldots, \mathfrak{B}_{i_k}$ those that remain and now give a shortest representation. Then

$$\mathfrak{Q} = [\mathfrak{B}_{i_1}, \ldots, \mathfrak{B}_{i_k}]$$

is the unique decomposition of $\mathfrak{Q}$ into irreducible classes. Indeed, suppose

$$\mathfrak{B}_\nu \sim (1 \mid \cdots \mid 1 \mid p^{r_\nu} \mid \cdots \mid p^{r_\nu})$$

can be represented as the least common multiple of

$$\mathfrak{C} \sim (1 \mid \cdots \mid 1 \mid p^{s_\nu} \mid \cdots \mid p^{s_i}) \quad \text{and} \quad \mathfrak{D} \sim (1 \mid \cdots \mid 1 \mid p^{t_\nu} \mid \cdots \mid p^{t_i}).$$

Then in the elementary divisor systems of $\mathfrak{C}$ and $\mathfrak{D}$ the number 1 must stand in the first $(\nu - 1)$ places; in the ν-th place one of the exponents s_ν or t_ν, say s_ν, must become equal to r_ν. But since

$$r_\nu = s_\nu \leq s_{\nu+1} \leq \cdots \leq s_i \leq r_\nu,$$

one obtains $\mathfrak{C} = \mathfrak{B}_\nu$, and $\mathfrak{B}_\nu$ is therefore irreducible.[51] The same holds for $\mathfrak{B}_{\rho+1}$, where p is to be replaced by 0. But, as the formation of the least common multiple shows, each of these irreducible classes gives

[49] The basis elements of one-sided ideals correspond to right or left classes.

[50] In the paper Zur Theorie der Moduln, Math. Ann. 52 (1899), p. 1, E. Steinitz defines the least common multiple (the greatest common divisor) of classes by the least common multiples and greatest common divisors of the elementary systems. Independently, the least common multiple of classes occurs as "congruence composition" in H. Brandt, Komposition der binären quadratischen Formen relativ einer Grundform, J. f. M. 150 (1919), p. 1.

[51] The $\mathfrak{B}$ are still reducible into one-sided classes; here there is no longer a unique relation between elementary divisors and class, and hence no unique decomposition into irreducible one-sided classes. The following example, supplied to me by H. Brandt, shows this for decomposition into right-sided classes (where the classes are represented by a basis matrix, or by the corresponding module):

$$\mathfrak{B} = [\mathfrak{C}_1, \mathfrak{C}_2] = [\mathfrak{D}_1, \mathfrak{D}_2], \quad \mathfrak{B} \sim \begin{pmatrix} p & 0 \\ 0 & p \end{pmatrix}, \quad \mathfrak{C}_1 \sim \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}, \quad \mathfrak{C}_2 \sim \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix},$$
$$\mathfrak{D}_1 \sim \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}, \quad \mathfrak{D}_2 \sim \begin{pmatrix} p & 0 \\ (p-1) & 1 \end{pmatrix}.$$

Indeed, the modules $(\xi, p\eta)$, $(p\xi, \eta)$, and $(p\xi, (p-1)\xi + \eta)$ each have as their only proper divisor the module (ξ, η), and are therefore irreducible and mutually distinct.

a definite exponent and the place where this exponent first occurs in the elementary divisor system of $\mathfrak{O}$, respectively $\mathfrak{A}$, while $\mathfrak{B}_{\rho+1}$ gives the rank. Since these numbers are uniquely determined by the elementary divisor system of $\mathfrak{O}$, respectively $\mathfrak{A}$, and since the relation between elementary divisors and class is one-to-one, the decomposition of $\mathfrak{O}$, and likewise of any class, into irreducible classes is unique.

In summary: *Every two-sided class $\mathfrak{A}$ of integral matrices with bounded number of entries can be represented uniquely as the least common multiple of finitely many irreducible two-sided classes. Each irreducible class represents a fixed prime divisor of the elementary divisor system of $\mathfrak{A}$, an associated exponent, and the place where that exponent first occurs. The irreducible class corresponding to the divisor 0 gives the rank of $\mathfrak{A}$.*

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20. An Algebraic Criterion for Absolute Irreducibility

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A polynomial – with coefficients from an arbitrary, abstractly defined field – is called *absolutely irreducible* if it remains irreducible in the algebraically closed field to which the coefficient domain can be enlarged.[1] I show in what follows that absolute irreducibility, in contrast with irreducibility relative to a prescribed field, can be grasped algebraically. More precisely, the theorem is:

To every polynomial in $n \geq 2$ variables with indeterminate coefficients there can be assigned an integral rational function, with integral coefficients, of these coefficients and of further indeterminates – the reducibility form – in such a way that, for every special polynomial of the same degree, the necessary and sufficient condition for absolute irreducibility is the non-vanishing of the reducibility form. In other words: for every special system of values of the coefficients for which the reducibility form vanishes identically in the indeterminates and which does not cause a lowering of degree, the special polynomial is reducible, and conversely.

If, instead of polynomials, one considers homogeneous forms in one more variable, the degree condition is replaced by the simpler condition that the coefficient system not vanish identically.

As a direct consequence of the theorem one also obtains the following, first stated by A. Ostrowski:[2]

Every absolutely irreducible polynomial with algebraic numbers as coefficients remains irreducible modulo every prime ideal of a finite algebraic field Ω containing the coefficient domain, with the exception of at most finitely many prime ideals of Ω. In particular, Ω may also be the field of rational numbers.

I intend to discuss further applications to the theory of relatively integral functions[3] elsewhere.

1. We first show that *every polynomial in $n \geq 2$ variables with indeterminate coefficients is absolutely irreducible.* Set

$$\begin{aligned} F(x) &= \sum A_{i_1\dots i_n} x_1^{i_1}\cdots x_n^{i_n} && (i_1+\cdots+i_n \leq l), \\ G(x) &= \sum B_{j_1\dots j_n} x_1^{j_1}\cdots x_n^{j_n} && (j_1+\cdots+j_n \leq m), \\ H(x) = F(x)G(x) &= \sum C_{k_1\dots k_n}(A,B) x_1^{k_1}\cdots x_n^{k_n} && (k_1+\cdots+k_n \leq l+m), \end{aligned} \tag{1}$$

where the A and B denote indeterminates, and the $C(A,B)$ become integral bilinear combinations of the A and B. The quotients

$$D = C(A,B)/C_{0\dots0}(A,B)$$

are therefore rational functions of the quotients $A/A_{0\dots0}$ and $B/B_{0\dots0}$; but the number of these functions is greater than the number of their arguments, since these numbers are respectively

$$\binom{l+m+n}{n}-1, \qquad \binom{l+n}{n}-1, \qquad \binom{m+n}{n}-1,$$

[1] That every field can be enlarged, essentially uniquely, to an algebraically closed one, i.e. to one in which every polynomial in one variable splits into linear factors, was shown by E. Steinitz in his *Algebraische Theorie der Körper* (J. f. M. 137 (1910), p. 167); he thereby gave the rational equivalent of the fundamental theorem of algebra.

[2] Zur arithmetischen Theorie der algebraischen Größen. Gött. Nachr. 1919, p. 279 (lemma p. 296). For the case of fields consisting of ordinary numbers, Ostrowski, as is known to me, also arrived at the main theorem above; he will publish his investigations on this soon. That, in my arrangement of the proof, the main theorem holds for arbitrary, abstractly defined fields rests on the fact that the reducibility form constructed for indeterminates has integral coefficients independent of the particular field taken as ground field.

[3] D. Hilbert, Mathematische Probleme. Gött. Nachr. 1900, p. 253, Problem 14.

and

$$\binom{l+m+n}{n}+1>\binom{l+n}{n}+\binom{m+n}{n} \quad \text{for } n\geq 2,\ l>0,\ m>0.$$

Thus there is at least *one* algebraic relation among the $D(A,B)$ and consequently also among the $C(A,B)$:

$$\Phi(Z)\neq 0 \quad [\text{identically in } Z_{k_1\dots k_n}], \qquad \Phi(C(A,B))=0 \quad [\text{identically in } A,B], \tag{2}$$

where Φ denotes an integral polynomial.

A polynomial with indeterminate Z as coefficients,

$$E(x)=\sum Z_{k_1\dots k_n}x_1^{k_1}\cdots x_n^{k_n} \qquad (k_1+\cdots+k_n\leq l+m), \tag{3}$$

is therefore necessarily absolutely irreducible for $n\geq 2$.[4]

2. The totality of these polynomials $\Phi(Z)$ which vanish identically in A,B under the substitution $Z=C(A,B)$ forms an *ideal of polynomials*,[5] more precisely a prime ideal $\mathfrak{P}$; for if, under the substitution, a product $\Phi_1\Phi_2$ vanishes, then at least one factor vanishes. Since (2) is satisfied identically in A,B, it remains valid for every special system of values A,B, where A,B and consequently also $\Gamma=C(A,B)$ denote quantities of some field. *The coefficients* Γ *of every polynomial reducible in an extension field therefore satisfy the conditions* $\Phi(\Gamma)=0$*; they are zeros of* $\mathfrak{P}$*,* or also of the finitely many basis polynomials of $\mathfrak{P}$. It remains to prove the converse.

For this it should be noted that all relations among A,B,C are preserved when, by introducing a new variable y, the polynomials (1) are transformed into homogeneous forms $F(x,y)$, $G(x,y)$, $H(x,y)$ of dimensions $l,m,l+m$. Thus coefficients Γ also become zeros of $\mathfrak{P}$ when, for instance, $F(x,y)$[6] becomes a power of y; for them the passage to inhomogeneous polynomials causes a lowering of degree of $\bar{H}(x)$ and not a factorization. It will be shown that, with the exception of this lowering of degree, the converse always holds; in particular, that for *homogeneous forms the converse holds without exception, provided not all* Γ *vanish; in other words, every zero of* $\mathfrak{P}$ *is supplied by the parameter representation* $\Gamma=C(A,B)$ *with quantities* A,B *from an extension field.*[7]

I prove this by the direct construction of finitely many functions $\Phi(Z)$: by means of Kronecker's substitution

$$x_i=\xi^{d^{n-i}}$$

I assign to the polynomials (1) polynomials in *one* variable; I show that gaps occur here in the exponents, and by forming the norm of the corresponding coefficients I arrive at the functions $\Phi(Z)$ and at the desired criterion.

3. Put:[8]

$$d=l+m+1, \qquad i=i_1d^{n-1}+i_2d^{n-2}+\cdots+i_n, \quad j=j_1d^{n-1}+\cdots+j_n, \quad k=k_1d^{n-1}+\cdots+k_n. \tag{4}$$

Then, by Kronecker's substitution, the polynomials F,G,H in (1) correspond respectively to the polynomials

$$\begin{aligned} f(\xi)&=\sum A_{i_1\dots i_n}\xi^i && (i_1+\cdots+i_n\leq l),\\ g(\xi)&=\sum B_{j_1\dots j_n}\xi^j && (j_1+\cdots+j_n\leq m),\\ h(\xi)&=\sum C_{k_1\dots k_n}(A,B)\xi^k && (k_1+\cdots+k_n\leq l+m). \end{aligned} \tag{5}$$

This correspondence is known to be one-to-one, since for all induced exponents occurring in (5), $k=0$ implies $k_1=0,\dots,k_n=0$; and therefore $i+j=i'+j'$ also implies $i_1+j_1=i'_1+j'_1;\dots;i_n+j_n=i'_n+j'_n$. Hence distinct products $\xi^i\xi^j$ can produce the same exponent ξ^k only when the corresponding products $x_1^{i_1}\cdots x_n^{i_n}x_1^{j_1}\cdots x_n^{j_n}$ also lead to the same exponent. Thus one obtains

$$h(\xi)=f(\xi)g(\xi), \tag{6}$$

[4] For, in the terminology of 2, the indeterminates Z are not zeros of $\mathfrak{P}$.

[5] Such totalities are usually called modules; but in fact their characteristic property – that with Φ_1 and Φ_2 they contain also $\Phi_1-\Phi_2$, and with Φ also $\Psi\Phi$, where Ψ is an arbitrary integral polynomial in Z – is the ideal property.

[6] Polynomials and forms which arise by replacing the indeterminates in $F,G,\dots,f,g,\dots$ by special systems of values will throughout be denoted by overlining: $\bar{F},\bar{G},\dots,\bar{f},\bar{g},\dots$.

[7] That "in general" the zeros of $\mathfrak{P}$ are supplied by the parameter representation follows from the property of the prime ideal $\mathfrak{P}$ of defining an irreducible algebraic variety. But, as is well known, this does not imply – something I had originally overlooked and to which A. Ostrowski called my attention some time ago – that the parameter representation cannot fail for any special system of values. The proof given here rests on more elementary notions.

[8] Festschrift, p. 11.

and from the validity of a relation (6), in such a way that in $f(\xi)$ and $g(\xi)$ no exponents other than the induced ones occur, one obtains back again

$$H(x) = F(x)G(x). \tag{7}$$

Here *gaps actually occur in the induced exponents in* $f(\xi)$ *and* $g(\xi)$. The highest exponent of $f(\xi)$ is ld^{n-1}, while the second highest is $(l-1)d^{n-1} + d^{n-2}$; their difference is $d^{n-2}(d-1) > 1$, since $d = l + m + 1 \geq 3$ and $n \geq 2$; and the same holds for $g(\xi)$.

The relations (6) and (7), valid for the indeterminates A, B, remain valid for every special system of values A, B. To preserve the degrees, I homogenize (6) and (7) by means of the variables η and y. *A homogeneous form* $H(x, y)$ *therefore splits in an algebraic extension field if and only if, in that field, the corresponding binary form* $h(\xi, \eta)$ *can be split into two factors in such a way that no exponents other than the induced ones occur in the factors.*

4. In order to carry out the formation of the norm mentioned at the end of 2, let

$$a_1, b_1; \ldots; a_p, b_p; \quad a_{p+1}, b_{p+1}; \ldots; a_{p+q}, b_{p+q}$$

be new indeterminates. Set

$$\begin{aligned}
\varphi(\xi, \eta) &= (a_1\xi + b_1\eta) \cdots (a_p\xi + b_p\eta) = r_p\xi^p + r_{p-1}\xi^{p-1}\eta + \cdots + r_0\eta^p, \\
\psi(\xi, \eta) &= (a_{p+1}\xi + b_{p+1}\eta) \cdots (a_{p+q}\xi + b_{p+q}\eta) = s_q\xi^q + s_{q-1}\xi^{q-1}\eta + \cdots + s_0\eta^q, \\
\chi(\xi, \eta) &= \varphi(\xi, \eta)\psi(\xi, \eta) = t_{p+q}\xi^{p+q} + t_{p+q-1}\xi^{p+q-1}\eta + \cdots + t_0\eta^{p+q},
\end{aligned} \tag{8}$$

so that r, s, t denote the homogenized elementary symmetric functions; that is, the symmetric functions of the rows a, b that are linear in each individual row and from which all integral symmetric functions that are homogeneous and of the same dimension in each individual row are built up integrally and with integral coefficients – and in only one way.

Now form, with further indeterminates $u_{\mu\nu}$, the linear form

$$\zeta(u, a, b) = \sum u_{\mu\nu} r_\mu s_\nu, \tag{9}$$

where the summation is extended over prescribed indices μ and ν. The product of $\zeta(u)$ with all its conjugates arising from permutation of the row a, b which are different from $\zeta(u)$ and from one another – the norm $N(\zeta(u))$ of $\zeta(u)$ with respect to the field of the $t(a, b)$, when $\zeta(u)$ is regarded as a function of the field of the $r_\mu(a, b)$ and $s_\nu(a, b)$ – is then an integral symmetric function of all rows a, b, homogeneous and of the same dimension in each row. Hence, by what was said above, it is a uniquely determined integral function, with integral coefficients, of the $t(a, b)$ and $u_{\mu\nu}$:

$$N(\zeta(u, a, b)) = T(t(a, b), u) \qquad \text{[identically in } a, b, u\text{]}. \tag{10}$$

The same reasoning shows that also

$$N(z - \zeta(u)) = z^\delta + T_{\delta-1}(t, u)z^{\delta-1} + \cdots + T(t, u) \qquad \text{[identically in } a, b, u, z\text{]}$$

where all T are integral functions, with integral coefficients, of t and u. Both sides of this identity vanish for $z = \zeta(u)$; and indeed, because of the algebraic independence of the elementary symmetric functions $r(a, b)$ and $s(a, b)$, they vanish identically in r, s when $t(a, b)$ is set, according to (8), equal to an integral bilinear combination $w(r, s)$ of the r and s, and $\zeta(u)$ is regarded as a function of r and s. Thus

$$\zeta(u)^\delta + T_{\delta-1}(w(r, s), u)\zeta(u)^{\delta-1} + \cdots + T(w(r, s), u) = 0 \quad \text{[identically in } r, s, u\text{]}. \tag{11}$$

[9]

The identities (10) and (11) remain valid for all special systems of values α, β of a, b, respectively ϱ, σ of r, s. If ϱ, σ are chosen in particular so that all products $\varrho_\mu\sigma_\nu$ vanish – where μ, ν denote the indices occurring in (9) – so that $\zeta(u)$ vanishes under the substitution, then, putting $w(\varrho, \sigma) = \tau$, (11) gives

$$T(\tau, u) = 0 \qquad \text{[identically in } u\text{]}. \tag{12}$$

[9] If the summation in (9) is extended over all indices, (11) represents Kronecker's extension of Gauss's theorem, essentially in Kronecker's arrangement of the proof (Zur Theorie der Formen höherer Stufen, Berliner Ber. 1883, Werke, 2, p. 417).

Conversely, let $\tau_0, \ldots, \tau_{p+q}$ be systems of values, not all zero, for which (12) is satisfied. Since in an algebraic extension field there is a decomposition

$$\bar{\chi}(\xi,\eta) = \tau_{p+q}\xi^{p+q} + \cdots + \tau_0\eta^{p+q} = (\alpha_1\xi + \beta_1\eta)\cdots(\alpha_{p+q}\xi + \beta_{p+q}\eta), \tag{12'}$$

in which α and β do not vanish simultaneously in any factor,[10] although some α or β may vanish, we have $\tau = t(\alpha,\beta)$. Substituting these values into (10), using (12), shows that $\zeta(u,\alpha,\beta)$ or one of its conjugate factors vanishes identically in u. Thus, after a suitable numbering of α, β, setting $r(\alpha,\beta) = \varrho$ and $s(\alpha,\beta) = \sigma$ says precisely that $\bar{\chi}(\xi,\eta)$ admits at least one decomposition

$$\bar{\chi}(\xi,\eta) = \bar{\varphi}(\xi,\eta)\bar{\psi}(\xi,\eta) = \sum \varrho_\mu \xi^\mu \eta^{p-\mu} \cdot \sum \sigma_\nu \xi^\nu \eta^{q-\nu}, \tag{13}$$

in such a way that all products $\varrho_\mu\sigma_\nu$ occurring in $\zeta(u)$ vanish, but not all ϱ or all σ vanish.

5. The degrees p, q in (8) are now taken to be ld^{n-1} and md^{n-1}, i.e. the degrees of $f(\xi)$ and $g(\xi)$ in (5). The indices μ, ν are split into $\mu^{(1)}, \nu^{(1)}, \mu^{(2)}, \nu^{(2)}$: here $\mu^{(1)}$ runs through all exponents missing from $f(\xi)$, $\nu^{(1)}$ through all exponents of ξ in $\psi(\xi,\eta)$; correspondingly, $\nu^{(2)}$ runs through all exponents missing from $g(\xi)$, and $\mu^{(2)}$ through all exponents of ξ in $\varphi(\xi,\eta)$. Further, let

$$e(\xi) = \sum Z_{k_1\ldots k_n}\xi^k \qquad (k_1 + \cdots + k_n \le l+m)$$

be the polynomial in one variable corresponding to the polynomial (3), and let

$$S(Z,u)$$

be the integral polynomial in Z and u obtained from $T(t,u)$ – the uniquely determined right hand side of (10), where the summation in (9) is extended over the indicated indices $\mu^{(1)}, \nu^{(1)}, \mu^{(2)}, \nu^{(2)}$ – by replacing the t by the coefficients Z and the corresponding zeros of $e(\xi)$.

If now r, s are replaced by the coefficients A, B and the corresponding zeros of $f(\xi)$ and $g(\xi)$, then by this substitution $\zeta(u)$ vanishes for the indicated summation. Moreover $t = w(r,s)$ passes over into the coefficients $C(A,B)$ and the corresponding zeros of $h(\xi)$; hence $T(t,u)$ passes over into $S(C(A,B),u)$. By the identity (11) we therefore get

$$S(C(A,B),u) = 0 \qquad \text{[identically in } A, B, u\text{]}. \tag{14}$$

Since (14) is preserved for every special system of values, the coefficients $\Gamma = C(A,B)$ of such special $H(x)$, or more generally $H(x,y)$, that split in an extension field into two factors satisfy the condition

$$S(\Gamma,u) = 0 \qquad \text{[identically in } u\text{]}. \tag{15}$$

Conversely, suppose that for the coefficients Γ, not all zero, of a polynomial $\bar{H}(x)$, respectively form $\bar{H}(x,y)$, the condition (15) is satisfied. By the above this is equivalent to $T(\tau,u) = 0$, where τ denotes all coefficients of $\bar{h}(\xi,\eta)$. Hence by (13) there exists in an extension field of the Γ a decomposition

$$\bar{h}(\xi,\eta) = \bar{f}(\xi,\eta)\cdot\bar{g}(\xi,\eta),$$

where, because of the prescribed summation, no non-induced exponents occur in $\bar{f}$ and $\bar{g}$; consequently there is a relation

$$\bar{H}(x,y) = \bar{F}(x,y)\cdot\bar{G}(x,y).$$

If, for $y = 1$, no factor becomes of degree zero – in particular if the Γ do not cause a lowering of the degree of $H(x)$ – it follows further that

$$\bar{H}(x) = \bar{F}(x)\cdot\bar{G}(x).$$

Condition (15) is therefore necessary and sufficient for $\bar{H}(x,y)$, and, when degree lowering is excluded, also for $\bar{H}(x)$, to split into two factors in an extension field.

It follows moreover that $S(Z,u)$ cannot vanish identically in Z and u, since in 1 it was shown that for $n \ge 2$ polynomials with indeterminates as coefficients, hence also the corresponding homogeneous forms in one more variable, are absolutely irreducible. Thus, with U denoting power products of the u,

$$S(Z,u) = \sum \Phi_\lambda(Z)U_\lambda,$$

[10] This rational “fundamental theorem of algebra” is the existence theorem on which the exceptionless validity of the converse stated at the end of 2 rests.

where the $\Phi_\lambda(Z)$ are, by (14), polynomials belonging to $\mathfrak{P}$. It is thereby shown that $\mathfrak{P}$ has no further zeros beyond those given by the parameter representation $\Gamma = C(A, B)$, where A and B belong to an algebraic extension field of the Γ. The converse stated at the end of 2 is therefore proved.

6. *The condition of absolute irreducibility* can now be formulated directly. One need only decompose the degree of $E(x)$ in (3), in all possible ways, into two positive summands $l + m$, and form the form $S(Z, u)$ for each such decomposition. The product over these forms,

$$R(Z, u) = \prod S(Z, u), \tag{16}$$

then forms the reducibility form of $E(x)$; for by the above, every factorization of $E(x)$, respectively $E(x, y)$, corresponds to the vanishing of one factor of (16), and conversely. Thus the main theorem stated in the introduction is proved, and at the same time it is shown how the reducibility form can actually be constructed in finitely many steps.

7. From the existence of the reducibility form (16) the theorem of Ostrowski mentioned in the introduction follows immediately. Let

$$\bar{E}(x) = \sum \Gamma_{k_1 \ldots k_n} x_1^{k_1} \cdots x_n^{k_n} \qquad (k_1 + \cdots + k_n \leq v)$$

be an absolutely irreducible polynomial with algebraic integers Γ as coefficients; let Ω denote a finite algebraic number field containing all Γ, and let $\mathfrak{p}$ be a prime ideal of Ω.

Then the reducibility form $R(Z, u)$ of exact degree for $\bar{E}(x)$ remains different from zero under the substitution $Z = \Gamma$; that is,

$$R(\Gamma, u) = \sum P_\lambda U_\lambda \neq 0 \qquad \text{[identically in } u\text{]}.$$

The ideal $\mathfrak{r} = (P_1, P_2, \ldots)$ is therefore divisible by only finitely many prime ideals of Ω. If $\mathfrak{p}$ is different from these finitely many, then $R(\Gamma, u)$ does not vanish identically modulo $\mathfrak{p}$; and since the residue classes modulo $\mathfrak{p}$ again form a field, it follows that $\bar{E}(x)$ modulo $\mathfrak{p}$ is irreducible, more precisely absolutely irreducible. The same holds for $\bar{E}(x, y)$, so that no lowering of degree modulo $\mathfrak{p}$ occurs.

If the coefficients of $\bar{E}(x)$ are algebraically fractional numbers, then $\mathfrak{p}$ must also be taken different from the finitely many prime ideals dividing the common denominator Λ; modulo all remaining prime ideals, $E(x)$ may be replaced by $\Lambda \bar{E}(x)$, so that the preceding hypotheses are satisfied. This proves the theorem.

Erlangen, August 1921.

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21. Formal Variational Calculus and Differential Invariants

Encyclopedia of the Mathematical Sciences II, 3 (1922), pp. 68–71 (in: R. Weitzenböck, Differential Invariants)

28. Formal Variational Calculus and Differential Invariants.[149]

The generation of all differential invariants and the derivation of the main theorems can be carried out most uniformly by the formal methods of the calculus of variations. This principle goes back to Riemann[72)] and received its chief form-theoretic development from Lipschitz[73)].

Start from the variational problem belonging to a homogeneous form $f(dx)$ – where

$$\left|\frac{\partial^2 f}{\partial dx_i\,\partial dx_k}\right| \neq 0$$

is assumed – namely

$$\delta\int_{t_0}^{t_1} f(x')\,dt = 0, \qquad \left(x_i' = \frac{dx_i}{dt}\right). \tag{140}$$

By partial integration one is led to the invariant system of equations

$$2\psi_i = \frac{d}{dt}\frac{\partial f}{\partial x_i'} - \frac{\partial f}{\partial x_i} = 0,$$

where the Lagrange expressions ψ_i are contragredient to the dx. Formally this is seen most simply if the ψ_i are defined by the integral-free formal identity corresponding to partial integration (Lagrange central equation[150])

$$\delta f - df_\delta = -2\sum\psi_i(d,d)\,\delta x_i, \qquad \left(f_\delta = \sum\frac{\partial f}{\partial dx_i}\,\delta x_i\right),$$

where both f and df_δ are invariants, and hence the right hand side is one as well. In particular, for a quadratic form one obtains

$$\delta\sum g_{ik}dx_idx_k - 2d\sum g_{ik}dx_i\delta x_k = -2\sum\psi_\mu(d,d)\,\delta x_\mu.$$

Replacing d here by the cogredient $(d+\lambda\delta)$ and comparing powers of λ, one obtains the corresponding system of identities:

$$\begin{cases} D\sum g_{ik}dx_idx_k - 2d\sum g_{ik}dx_iDx_k = -2\sum\psi_\mu(d,d)\,Dx_\mu, \\ D\sum g_{ik}dx_i\delta x_k - \delta\sum g_{ik}dx_iDx_k - d\sum g_{ik}\delta x_iDx_k = -2\sum\psi_\mu(d,\delta)\,Dx_\mu, \\ D\sum g_{ik}\delta x_i\delta x_k - 2\delta\sum g_{ik}\delta x_iDx_k = -2\sum\psi_\mu(\delta,\delta)\,Dx_\mu. \end{cases} \tag{141}$$

It follows that $\psi_\mu(d,\delta)$, and in particular also $\psi_\mu(d,d)$ and $\psi_\mu(\delta,\delta)$, are contragredient vectors. From (141) one computes $\psi_\mu(d,\delta)$ as

$$\psi_\mu(d,\delta) = \sum g_{i\mu}\,d\delta x_i + \sum\left[{ik \atop \mu}\right]dx_i\delta x_k,$$

and from this one obtains the cogredient vector

$$p^\sigma(d,\delta) = \sum g^{\sigma\mu}\psi_\mu = d\delta x_\sigma + \sum\left\{{ik \atop \sigma}\right\}dx_i\delta x_k. \tag{142}$$

Putting $p^\nu(d,d)=0$ evidently gives the equations of the geodesic lines.

Knowledge of the cogredient vector $p^\sigma(d,\delta)$ leads directly to the covariant derivative (compare No. 19). If, for instance, $h(d,\delta)$ is a form of the two rows dx and δx, then the expression

$$h^{(1)}(dx,\delta x) = \delta h - \sum\frac{\partial h}{\partial dx_\sigma}p^\sigma(d,\delta) - \sum\frac{\partial h}{\partial\delta x_\sigma}p^\sigma(\delta,\delta) \tag{143}$$

is itself an invariant as a difference of invariants, and is formed so that the second differentials cancel. Thus $h^{(1)}(dx,dx)$ is a form involving only first differentials: the covariant derivative of h. The corresponding statement holds when h contains more rows $d^{(1)}x,\dots,d^{(\lambda)}x$.[151]

[149] This number is due to E. Noether.

[150] For special f the designation is due to Heun; see his article IV, 1 11, p. 447.

[151] Lipschitz, J. f. Math. 72 (1870), p. 17.

The formation of the curvature form by Riemann and Lipschitz rests on the same principle. One passes from $\delta^2 f$ to the "normal form of the second variation"

$$\Omega = \delta^2 \sum g_{ik} dx_i dx_k - 2d\delta \sum g_{ik} dx_i \delta x_k + d^2 \sum g_{ik} \delta x_i \delta x_k, \tag{144}$$

which, as an aggregate of invariants, is itself again an invariant, and in which, now generalizing the Lagrange central equation, the third differentials are eliminated. The elimination of the second differentials is carried out, in analogy with covariant differentiation, by forming the difference

$$K = \Omega - 2\sum\{p^\sigma(d,\delta)\psi_\sigma(d,\delta) - p^\sigma(\delta,\delta)\psi_\sigma(d,d)\}, \tag{145}$$

which may also be written as[152]

$$\begin{aligned}\frac{1}{2}K = d\sum \psi_\sigma(d,\delta)\delta x_\sigma - \delta \sum \psi_\sigma(d,d)\delta x_\sigma \\ - \sum\{p^\sigma(d,\delta)\psi_\sigma(d,\delta) - p^\sigma(\delta,\delta)\psi_\sigma(d,d)\}.\end{aligned} \tag{146}$$

The law of formation of K may also be stated by saying that in Ω the second differentials are eliminated by setting $p(d,\delta)$ and $p(\delta,\delta)$ – respectively the right hand sides of (141) – equal to zero. This is Riemann's definition of the curvature form (compare No. 21), but it must be explicitly noted – as (146) makes especially clear – that this setting to zero may take place only after the differentiation has been performed; Lipschitz's formal mode of formation avoids this difficulty. Correspondingly, the covariant derivative $h^{(1)}$ (143) could also be defined by eliminating in δh the second differentials by setting $p(d,\delta),\ldots$ equal to zero.

From the curvature form one obtains, by successive formation of covariant derivatives, a sequence of forms [$\Phi_4, \Phi_5, \ldots$ in Christoffel], whose projective invariants, after adjoining f, exhaust according to Christoffel and Ricci the totality of the differential invariants of the quadratic form, and whose equivalence under linear transformations, without any further integrability condition, expresses the equivalence of two quadratic differential forms (compare No. 22). The inner reason for this reduction theorem rests on the existence of the Riemannian normal coordinates arising from the variational problem (140) (compare No. 20): these transform the geodesic lines through a point into straight lines, and consequently transform linearly under arbitrary transformations of the x. The formation of all invariants and the equivalence problem are thereby reduced to the corresponding question for the individual homogeneous components in the expansion of f in normal coordinates, and one shows that these components are linearly composed from $f, \Phi_4, \Phi_5, \ldots$. For quadratic forms this is carried out in detail by Vermeil[92)], following a note by E. Noether[93)]. According to that note, the theorems and methods remain valid when an arbitrary differential expression is taken as the basis (compare Nos. 22 and 23); the scope of the methods of formal variational calculus as against those of pure elimination theory is thereby made apparent.[153]

Levi-Civita[134)] interpreted the setting $p(d,\delta) = 0$ geometrically as "parallel displacement of a vector" (compare also Hessenberg[133)]). This concept, cast axiomatically by Weyl, is essentially restricted to quadratic forms, but it admits a generalization of another kind. It remains valid when the group is enlarged (multiplication of the g_{ik} by an arbitrary function), as soon as, besides the quadratic form, a linear form is also made basic. Then $p(d,\delta)$ is uniquely determined, invariant constructions corresponding to those above can be carried out, and geodesic coordinates can be defined; but $p(d,d) = 0$ now no longer arises from a variational problem.[154] Questions about the reduction theorem and equivalence have not yet been attacked here in general; only for invariants of lowest order has R. Weitzenböck given a reduction theorem.[155]

Finally, attention should be drawn to the general "invariant variational problems", i.e. to those in which the simple or multiple integral J is invariant with respect to some group in Lie's sense. Special physically interesting cases are treated in works by Hamel[156], Herglotz[157], Lorentz[158], Fokker[159], Weyl[160] and Klein[161]. The principled formulation by E. Noether[162] shows that to the invariance of J under a G_ρ (a finite group

[152] Lipschitz, J. f. Math. 82 (1876), § 1.

[153] Compare these remarks, and for further geometric interpretations the seminar lectures of F. Klein listed under "Literature."

[154] H. Weyl, Reine Infinitesimalgeometrie, Math. Ztschr. 2 (1918), p. 384–411, and Raum, Zeit, Materie (3rd and 4th ed.).

[155] Wien. Ber. 129 (1920), p. 683 and 697.

[156] Math. Ann. 59 (1904), p. 416–434; Ztschr. Math. Phys. 50 (1904), p. 1–54.

[157] Ann. d. Phys. (4) 36 (1911), p. 511, especially § 9.

[158] Verslag der Amsterd. Akad., April, Sept. 1916, Oct. 1916, May 1917.

[159] Ibid., Jan. 1917.

[160] Ann. d. Phys. (4) 54 (1917), p. 117, § 2.

[161] Gött. Nachr. 1917, p. 469–482; ibid. 1918, p. 171–189.

[162] Gött. Nachr. 1918, p. 235–257.

with ρ essential parameters) there correspond ρ linearly independent divergences; to the invariance under an infinite group containing ρ arbitrary functions up to the σ-th derivative there correspond ρ dependencies between the Lagrange expressions and their derivatives up to order σ. In both cases the converse holds. Since the Lagrange expressions become relative invariants of the group, one has at the same time a process generating invariants.

(Completed in March 1921.)

22. Edited version of K. Hentzelt: on the Theory of Polynomial Ideals and Resultants

Math. Ann. 88 (1923), pp. 53–79

In what follows the question is the *general problem of elimination theory, namely to determine the common zeros of all polynomials of an ideal of polynomials*; or, what is identical with this, the zeros of any finite number of polynomials that form a basis of the ideal. At the same time a conceptual interpretation of the multiplicities that occur will emerge. The coefficients of the polynomials may be assigned to any field; the zeros then belong to the algebraically closed field derived from it. Moreover, the ideal may be assumed, without restriction of generality, to be transformed (§ 3). Then the essential result is the following:

To every ideal $\mathfrak{m}$ *of polynomials one can assign a resultant form:*

$$R_{\mathfrak{m}} = R^{(1)}(x_1, \dots, x_n) \cdots R^{(i)}(x_i, \dots, x_n) \cdots R^{(n)}(x_n) \equiv 0(\mathfrak{m}),$$

in such a way that $R_{\mathfrak{m}}$ *vanishes for all zeros of* $\mathfrak{m}$ *and only for these. If* $\mathfrak{n}$ *is a divisor of* $\mathfrak{m}$*, and if the resultant forms* $R_{\mathfrak{n}}$ *and* $R_{\mathfrak{m}}$ *agree, then the ideals* $\mathfrak{n}$ *and* $\mathfrak{m}$ *also agree.*[1]

The second part of the main theorem shows that the resultant form supplies the zeros in characteristic multiplicity. This second part is analogous to the fact that two ideals of an algebraic number field, one of which is a divisor of the other, agree if and only if their *norms* agree; the inner reason for the two theorems is also the same.

Namely, $\mathfrak{m}$ may be conceived as a module of linear forms $\mathfrak{M}_{i-1}$, by regarding each polynomial as a linear form in the power products of $x_1, \dots, x_{i-1}$ and adjoining $x_{i+1}, \dots, x_n$ to the coefficient domain. The resultant factor $R^{(i)}$ then becomes equal to the *norm* of the corresponding fundamental module (§ 1) $\mathfrak{G}_{i-1}$ with respect to $\mathfrak{M}_{i-1}$; this also immediately gives an *interpretation of the multiplicity – product of degree and exponent of a factor of* $R^{(i)}$ *– by the number of linearly independent residue classes*[2], a fact that until now was known only in the special case in which the resultant can be defined for indeterminate coefficients[3].

For the derivation of these results, §§ 1 and 2 give theorems on modules of linear forms whose coefficients are rational integers or polynomials in one indeterminate. The connection with the ideals of polynomials introduced in § 3 is supplied by the isomorphism theorem of § 4. Section 5 gives the above-mentioned second part of the main theorem; § 7 gives the theorem on the zeros, preceded in § 6 by a general elimination theory. There, at the same time, when new indeterminates are introduced, the decomposition of the resolvent into linear forms in these indeterminates is proved; and thus, for the elimination given here, an unobjectionable proof is supplied for an assertion of Kronecker[4].

§ 1. Modules of Linear Forms.

By integral quantities $a, b, c, \dots$ in the first two sections we shall mean rational integers or polynomials in one indeterminate with coefficients from an arbitrary, abstractly defined field P; by linear forms $a(\xi), b(\xi), \dots$ we shall mean such forms in finitely many indeterminates $\xi_1, \dots, \xi_k$ with integral quantities as coefficients.

A *module* $\mathfrak{A}$ *of linear forms* is defined by the conditions: together with $a(\xi)$ and $b(\xi)$, $\mathfrak{A}$ contains the difference $a(\xi) - b(\xi)$; together with $a(\xi)$ it contains $c \cdot a(\xi)$, where c denotes an arbitrary integral quantity.

By the *rank of* $\mathfrak{A}$ we mean, as usual, the maximal number of linearly independent linear forms from $\mathfrak{A}$; by the λ-th *determinantal divisor* d_λ of $\mathfrak{A}$, the greatest common divisor of all determinants of order λ from $\mathfrak{A}$

[1] Kurt Hentzelt has been missing before Dixmuiden since October 1914 and must be counted among the dead. The present article is a completely free reworking of the most essential part of his dissertation "Zur Theorie der Formenmoduln und Resultanten," with which he received his doctorate under E. Fischer in Erlangen in the summer of 1914. This dissertation, composed entirely on the basis of his own ideas, is built up without gaps; but lemma follows lemma, all concepts are paraphrased by formulas with four and five indices, and the text is almost wholly absent, so that the greatest difficulties are presented to understanding. He did not himself get to the planned reworking. I give the work again in a purely conceptual form, by which a great simplification is obtained of proofs whose fundamental ideas throughout go back to Hentzelt, and by which, as I hope, the beauty of the work becomes evident. – The parts of the dissertation that – for a given basis – concern the question of forming the functions that occur by finitely many steps are to remain reserved for a separate publication. (E. N.)

[2] These latter results show their true significance when the decomposition of ideals into primary ideals is brought in; the multiplicity corresponding to a primary ideal is then defined as the number of linearly independent residue classes of the complement modulo this ideal. This is to be taken up elsewhere. (E. N.)

[3] Cf. Macaulay, *The algebraic theory of modular systems*, Cambridge Tracts 19 (Cambridge University Press, 1916), No. 67; also Lasker, Zur Theorie der Moduln und Ideale, Math. Ann. 60 (1905), pp. 20–116; p. 98. That in this case the resultant and the resultant form coincide is shown in the still unpublished theorems of Hentzelt mentioned under 1). (E. N.)

[4] The attempted proof in König, *Algebraische Größen* (Leipzig 1903, Teubner), V, § 4 – for Kronecker's elimination theory – is known not to have succeeded. That even for the multiplicities introduced by König into Kronecker's elimination theory the second part of Hentzelt's main theorem does not hold is shown by Macaulay, loc. cit., p. 23; there it is further shown that Kronecker's elimination theory does not correspond to decomposition into primary ideals. (E. N.)

(that is, all determinants of order λ formed from the coefficient matrix of any λ linear forms from $\mathfrak{A}$). If ρ is the rank of $\mathfrak{A}$, so that $d_1, \ldots, d_\rho$ are the non-zero determinantal divisors, then the *elementary divisors* of $\mathfrak{A}$ are defined by $e_1 = d_1$, $e_2 = d_2/d_1, \ldots, e_\rho = d_\rho/d_{\rho-1}$, $e_{\rho+i} = 0$. It is known that $\mathfrak{A}$ is a finite module having a module basis of exactly ρ linear forms $a_1(\xi), \ldots, a_\rho(\xi)$, by which every linear form from $\mathfrak{A}$ can be represented linearly, with integral quantities as coefficients: $\mathfrak{A} = (a_1(\xi), \ldots, a_\rho(\xi))$. If A denotes the coefficient matrix of these linear forms, then the determinantal and elementary divisors of $\mathfrak{A}$ agree with those of A; hence first it follows that *each elementary divisor e_i divides the following one e_{i+1}*. The matrix equation furnished by the elementary-divisor theory,

$$PAQ = \begin{pmatrix} e_1 & & & 0 \\ & e_2 & & \\ & & \ddots & \\ 0 & & & e_\rho \end{pmatrix},$$

further shows – if new indeterminates $\eta_1, \ldots, \eta_k$ are introduced by the unimodular transformation $\xi = Q(\eta)$ – that the module equation holds:

$$\mathfrak{A} = (e_1\eta_1, e_2\eta_2, \ldots, e_\rho\eta_\rho). \tag{1}$$

The concept most essential for what follows is that of the fundamental module[5], according to

Definition I. *A module $\mathfrak{G}$ is called a fundamental module if it has no proper divisor of the same rank*[6].

Thus $\mathfrak{G}$ is a fundamental module if and only if from

$$c \cdot g(\xi) \equiv 0(\mathfrak{G}); \quad c \neq 0 \quad \text{there always follows:} \quad g(\xi) \equiv 0(\mathfrak{G}). \tag{2}$$

The connection between an arbitrary module and a fundamental module is given by

Theorem I. *Every module $\mathfrak{A}$ is divisible by one and only one fundamental module $\mathfrak{G}$ of the same rank; $\mathfrak{G}$ is defined as the smallest divisor of $\mathfrak{A}$ having the same rank, and is represented by*

$$\mathfrak{G} = (\eta_1, \ldots, \eta_\rho); \tag{3}$$

$\mathfrak{G}$ shall be called the fundamental module of $\mathfrak{A}$.

The condition that $\mathfrak{G}$ is to be the smallest divisor of the same rank is expressed as follows: $\mathfrak{G}$ contains all and only the linear forms $g(\xi)$ that satisfy a relation

$$b \cdot g(\xi) \equiv 0(\mathfrak{A}); \quad b \neq 0. \tag{4}$$

It follows first that $\mathfrak{G}$ is a module, since together with

$$b_1 g_1(\xi) \equiv 0(\mathfrak{A}); \quad b_1 \neq 0; \qquad b_2 g_2(\xi) \equiv 0(\mathfrak{A}); \quad b_2 \neq 0$$

also

$$b_1 b_2 (g_1(\xi) - g_2(\xi)) \equiv 0(\mathfrak{A}); \quad b_1 b_2 \neq 0$$

is fulfilled; and of course also $b_1 \cdot c g_1(\xi) \equiv 0(\mathfrak{A})$. But $\mathfrak{G}$ is also a fundamental module; for from

$$c \cdot g(\xi) \equiv 0(\mathfrak{G}); \quad c \neq 0$$

there follows, by (4),

$$bcg(\xi) \equiv 0(\mathfrak{A}); \quad bc \neq 0,$$

and hence again by (4) also $g(\xi) \equiv 0(\mathfrak{G})$.

There is moreover no fundamental module $\mathfrak{G}_1$ different from the smallest divisor $\mathfrak{G}$ of the same rank that satisfies the conditions. For $\mathfrak{G}_1$ would have to be divisible by $\mathfrak{G}$, but as a fundamental module it has no proper divisor of the same rank, and hence is identical with $\mathfrak{G}$. Finally, the module represented by the right hand side of (3) is a fundamental module, since every proper divisor must contain a further indeterminate η, and so has higher rank. Thus (3) gives the uniquely defined fundamental module of $\mathfrak{A}$, and *Theorem I is proved in all its parts.*

[5] Cf. Steinitz, Rechteckige Systeme und Moduln in algebraischen Zahlkörpern II. Math. Ann. 72 (1912), pp. 297–345, No. 36. Hentzelt seems in any case, independently of Steinitz – with whom points of contact are also found elsewhere – to have reached, for his simpler case, the concept in the form of formula (4). (E. N.)

[6] Divisor understood in the module sense: $\mathfrak{A}$ is divisible by $\mathfrak{B}$, $\mathfrak{A} \equiv 0(\mathfrak{B})$, if every element of $\mathfrak{A}$ is contained in $\mathfrak{B}$; $\mathfrak{B}$ is called a proper divisor if it contains elements different from those of $\mathfrak{A}$.

A further concept essential for what follows is Dedekind's concept of the quotient of two modules[7]:

Definition II. *The quotient $\mathfrak{c} = \mathfrak{A}/\mathfrak{B}$ of two modules of linear forms is defined as the totality of the integral quantities c satisfying the condition*

$$c \cdot \mathfrak{B} \equiv 0(\mathfrak{A}).$$

$\mathfrak{c}$ is an ideal of integral quantities, hence a principal ideal. Correspondingly, the quotient $\mathfrak{C} = \mathfrak{A}/\mathfrak{b}$ of a module $\mathfrak{A}$ of linear forms by an ideal $\mathfrak{b}$ of integral quantities is defined as the totality of the linear forms $c(\xi)$ satisfying the condition

$$c(\xi) \cdot \mathfrak{b} \equiv 0(\mathfrak{A}).$$

$\mathfrak{C}$ is again a module of linear forms, and indeed, as soon as $\mathfrak{b}$ is different from the zero ideal, a module of the same rank as $\mathfrak{A}$.

One need only remark that by definition it is clear that the module property belongs to the quotient in each case; systems of integral quantities with the module property, however, are ideals.

The quotient concept leads to a connection between the fundamental module and the highest elementary divisor, namely:

Theorem II. *Between the fundamental module and the highest elementary divisor of $\mathfrak{A}$ there is the reciprocal relation*

$$(e_\rho) = \mathfrak{A}/\mathfrak{G}; \qquad \mathfrak{G} = \mathfrak{A}/(e_\rho), \tag{5}$$

where (e_ρ) denotes the principal ideal derived from e_ρ.

For since every elementary divisor divides e_ρ, from (1) and (3) we have

$$e_\rho\mathfrak{G} \equiv 0(\mathfrak{A}) \quad \text{and consequently} \quad (e_\rho) \cdot \mathfrak{G} \equiv 0(\mathfrak{A}); \tag{6}$$

hence (e_ρ) is divisible by the quotient $\mathfrak{A}/\mathfrak{G}$. But the converse also holds; for from

$$b\mathfrak{G} \equiv 0(\mathfrak{A}) \quad \text{there follows} \quad b\eta_\rho \equiv 0(\mathfrak{A}) \quad \text{and hence} \quad b \equiv 0((e_\rho)),$$

which proves the first formula (5)[8]. Formula (6) further shows that $\mathfrak{G}$ is divisible by the quotient $\mathfrak{A}/(e_\rho)$; this quotient is a divisor of $\mathfrak{A}$ of the same rank, hence, by the definition of the fundamental module of $\mathfrak{A}$, conversely divisible by $\mathfrak{G}$. Thus the second formula (5) is also proved.

If d denotes any determinant of order ρ from $\mathfrak{A}$ which does not vanish identically, then d is divisible by the ρ-th determinantal divisor d_ρ, and hence by e_ρ; consequently $d\mathfrak{G} \equiv 0(\mathfrak{A})$, and by the preceding conclusion $\mathfrak{G} = \mathfrak{A}/(d)$. For what follows, however, it is essential that this latter quotient representation for $\mathfrak{G}$ can be proved without going back to the elementary divisors, and hence has a more general validity, according to

Theorem III. *If by integral quantities one understands polynomials in several indeterminates*[9]*, then the quotient representation still holds*

$$\mathfrak{G} = \mathfrak{A}/(d), \tag{7}$$

where d denotes any determinant of order ρ from $\mathfrak{A}$ which does not vanish identically.

First it is clear that the notions of rank, fundamental module and module quotient – which now is just no longer a principal ideal – are preserved under this extension, and so is Theorem I, with the exception of the basis representation.

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[7]In Dedekind the matter concerns modules of numbers, for which multiplication is also defined, so that Dedekind's quotient does not agree with the quotient defined here. (E. N.)

[8]Set $\mathfrak{A}_0 = \mathfrak{A}, \dots, \mathfrak{A}_\lambda = (\mathfrak{A}, \eta_\rho, \dots, \eta_{\rho-\lambda+1})$, where in each case $\mathfrak{A}_i$ has highest elementary divisor $e_{\rho-i}$. Then by the same conclusions one obtains

$$(e_\rho) = \mathfrak{A}_0/\mathfrak{A}_1, \dots, (e_{\rho-\lambda}) = \mathfrak{A}_\lambda/\mathfrak{A}_{\lambda+1}, \dots, (e_1) = \mathfrak{A}_{\rho-1}/\mathfrak{A}_\rho.$$

More generally, set $\zeta_\lambda = b_{\lambda 1}\eta_1 + \dots + b_{\lambda\lambda}\eta_\lambda$, assume $(b_{\lambda\lambda}, e_\lambda) = 1$, and let

$$\mathfrak{B}_0 = \mathfrak{A}, \dots, \mathfrak{B}_\lambda = (\mathfrak{A}, \zeta_\rho, \dots, \zeta_{\rho-\lambda+1}).$$

Then $\mathfrak{B}_i$ also has highest elementary divisor $e_{\rho-\lambda}$, and hence

$$(e_{\rho-\lambda}) = \mathfrak{B}_\lambda/\mathfrak{B}_{\lambda+1}.$$

[9]In fact only this is used: that the integral quantities reproduce themselves under addition, subtraction and multiplication, with the usual rules of calculation holding, so that they form a ring; and further, that in this ring a product vanishes only when one factor vanishes, so that the ring may be extended to a field by quotient formation (adjunction of pairs of elements). On the other hand no basis representation of the module is used; thus (7) holds for example also for the domain of all algebraic integers as coefficients. (E. N.)

Let now

$$a_1(\xi) = a_{11}\xi_1 + \cdots + a_{1k}\xi_k, \ldots, a_\rho(\xi) = a_{\rho 1}\xi_1 + \cdots + a_{\rho k}\xi_k$$

be ρ linearly independent linear forms from $\mathfrak{A}$, which in general will not form a module basis. Let the indeterminates ξ be so designated that

$$d = |a_{ij}| \neq 0; \qquad i, j = 1, 2, \ldots, \rho.$$

It follows that ρ linear forms of special shape belong to $\mathfrak{A}$:

$$d\xi_\lambda + b_{\lambda,\rho+1}\xi_{\rho+1} + \cdots + b_{\lambda,k}\xi_k \equiv 0(\mathfrak{A}) \quad (\lambda = 1, 2, \ldots, \rho). \tag{8}$$

But $\mathfrak{A}$ can contain no linear form depending only on $\xi_{\rho+1}, \ldots, \xi_k$, say $c_{\rho+1}\xi_{\rho+1} + \cdots + c_k\xi_k$, since otherwise at least one determinant of order $\rho + 1$, namely $d \cdot c_\alpha$, would be non-zero.

Now the quotient $\mathfrak{A}/(d)$, as a divisor of $\mathfrak{A}$ of the same rank, is divisible by $\mathfrak{G}$; but the converse also holds. For every $g(\xi) \equiv 0(\mathfrak{G})$ satisfies, by definition,

$$c \cdot g(\xi) \equiv 0(\mathfrak{A}); \quad c \neq 0 \quad \text{and consequently} \quad d \cdot c \cdot g(\xi) \equiv 0(\mathfrak{A}).$$

By (8), however,

$$d \cdot g(\xi) = g_{\rho+1}\xi_{\rho+1} + \cdots + g_k\xi_k \equiv 0(\mathfrak{A}),$$

and consequently

$$cg_{\rho+1}\xi_{\rho+1} + \cdots + cg_k\xi_k \equiv 0(\mathfrak{A}).$$

By what has just been observed this implies $c \cdot g_{\rho+1} = 0, \ldots, c \cdot g_k = 0$, and because $c \neq 0$, also $g_{\rho+1} = 0, \ldots, g_k = 0$; hence $d \cdot g(\xi) \equiv 0(\mathfrak{A})$, proving (7).

It should be noted that $d \cdot g(\xi) \equiv 0(\mathfrak{A})$ also follows directly from the fact that the two modules $(a_1(\xi), \ldots, a_\rho(\xi))$ and $(g(\xi), a_1(\xi), \ldots, a_\rho(\xi))$ have the same rank ρ, so that – putting $g(\xi) = c_1\xi_1 + \cdots + c_k\xi_k$ – the determinant of order $\rho + 1$

$$\begin{vmatrix} a_1(\xi) & a_{11} & \cdots & a_{1\rho} \\ \vdots & \vdots & & \vdots \\ a_\rho(\xi) & a_{\rho 1} & \cdots & a_{\rho\rho} \\ g(\xi) & c_1 & \cdots & c_\rho \end{vmatrix}$$

vanishes. The proof given above, however, recurs in § 4 following formula (30).

§ 2. Decomposition Theorems for Norms and Modules.

We now return to modules of linear forms with respect to rational integers or polynomials in one indeterminate with coefficients from P.

Definition III. *If, as usual, all linear forms in ξ that are congruent to one another modulo $\mathfrak{A}$ are collected into one residue class, then the symbol $\mathfrak{G} \mid \mathfrak{A}$ shall denote the system of residue classes of $\mathfrak{G}$ modulo $\mathfrak{A}$, that is, the system of all residue classes modulo $\mathfrak{A}$ that consist of elements of $\mathfrak{G}$. The norm of $\mathfrak{G}$ with respect to $\mathfrak{A}$ – in symbols $N(\mathfrak{G} \mid \mathfrak{A})$ – is defined as the determinant of the transition substitution from $\mathfrak{G}$ to $\mathfrak{A}$, that is, as the determinant of the substitution which expresses any linearly independent module basis of $\mathfrak{A}$ by such a basis of the fundamental module $\mathfrak{G}$ of $\mathfrak{A}$.*

From (1) and (3), respectively from the remark to Theorem II, one obtains:

$$N(\mathfrak{G} \mid \mathfrak{A}) = e_1 e_2 \cdots e_\rho = d_\rho; \qquad \mathfrak{G} = \mathfrak{A}/(N(\mathfrak{G} \mid \mathfrak{A})). \tag{9}$$

From (1), (3) and (9) it follows in the familiar way that, in the case of rational integers, $N(\mathfrak{G} \mid \mathfrak{A})$ represents the number of residue classes of $\mathfrak{G}$ modulo $\mathfrak{A}$, whereas in the case of polynomials in one indeterminate – where the number of residue classes is in general infinite – the degree of $N(\mathfrak{G} \mid \mathfrak{A})$ becomes equal to the *number of residue classes linearly independent with respect to P*. If $\mathfrak{B}$ is a divisor of $\mathfrak{A}$ of the same rank, then, together with a representative of any such residue class, $\mathfrak{B}$ also contains all elements of the residue class; hence $\mathfrak{B}$ arises from $\mathfrak{A}$ by adjoining finitely many residue classes, or their linear combinations. It follows at once that:

Theorem IV. *If $\mathfrak{B}$ is a divisor of $\mathfrak{A}$ of the same rank, so that the fundamental modules agree, and if moreover $N(\mathfrak{G} \mid \mathfrak{B}) = N(\mathfrak{G} \mid \mathfrak{A})$, then also $\mathfrak{B} = \mathfrak{A}$.*

On the connection between the decomposition of norms and modules one has

Theorem V. *Let*

$$N(\mathfrak{G} \mid \mathfrak{A}) = r \cdot s, \qquad (r, s) = 1; \tag{10}$$

then there exists one and only one module $\mathfrak{R}$, *and likewise one and only one module* $\mathfrak{S}$, *both divisors of* $\mathfrak{A}$ *of the same rank, such that*

$$r = N(\mathfrak{G} \mid \mathfrak{R}), \qquad s = N(\mathfrak{G} \mid \mathfrak{S})$$

holds. $\mathfrak{R}$ *and* $\mathfrak{S}$ *are defined by*

$$\mathfrak{R} = \mathfrak{A}/(s); \qquad \mathfrak{S} = \mathfrak{A}/(r),$$

and $\mathfrak{A}$ *is equal to the least common multiple of* $\mathfrak{R}$ *and* $\mathfrak{S}$.[10] *If one sets* $r_\rho = (r, e_\rho)$, $s_\rho = (s, e_\rho)$, *then the reciprocities hold*

$$(s_\rho) = \mathfrak{A}/\mathfrak{R}; \quad \mathfrak{R} = \mathfrak{A}/(s_\rho); \qquad (r_\rho) = \mathfrak{A}/\mathfrak{S}; \quad \mathfrak{S} = \mathfrak{A}/(r_\rho). \tag{11}$$

Indeed, setting generally $r_i = (r, e_i)$, $s_i = (s, e_i)$, one obtains from (9) and (10):

$$(r_i, s_i) = 1; \qquad e_i = r_i s_i; \qquad r = r_1 \cdots r_\rho; \qquad s = s_1 \cdots s_\rho,$$

and each r_i respectively s_i divides the following r_{i+1} respectively s_{i+1}. Thus, if one sets

$$\mathfrak{R} = (r_1\eta_1, \ldots, r_\rho\eta_\rho); \qquad \mathfrak{S} = (s_1\eta_1, \ldots, s_\rho\eta_\rho),$$

then $\mathfrak{R}$ and $\mathfrak{S}$ are divisors of $\mathfrak{A}$ of the same rank; owing to the relative primeness of r_i and s_i, $\mathfrak{A}$ becomes the least common multiple of $\mathfrak{R}$ and $\mathfrak{S}$, and moreover

$$N(\mathfrak{G} \mid \mathfrak{R}) = r_1 \cdots r_\rho = r; \qquad N(\mathfrak{G} \mid \mathfrak{S}) = s_1 \cdots s_\rho = s.$$

Furthermore

$$s_\rho \mathfrak{R} \equiv 0(\mathfrak{A}); \qquad r_\rho \mathfrak{S} \equiv 0(\mathfrak{A});$$

and from $c\mathfrak{R} \equiv 0(\mathfrak{A})$ follows $cr_\rho\eta_\rho \equiv 0(\mathfrak{A})$, hence $c \equiv 0((s_\rho))$, proving $(s_\rho) = \mathfrak{A}/\mathfrak{R}$, and similarly $(r_\rho) = \mathfrak{A}/\mathfrak{S}$. From

$$b(\eta) = b_1\eta_1 + \cdots + b_\rho\eta_\rho \equiv 0(\mathfrak{A}/(s_\rho))$$

it further follows, because all r_i are relatively prime to s_ρ, that $b_i \equiv 0((r_i))$; hence $\mathfrak{R} = \mathfrak{A}/(s_\rho)$ and similarly $\mathfrak{S} = \mathfrak{A}/(r_\rho)$. Thus the reciprocities (11) are proved; in the special case $r = 1$ they pass into (5).

It remains to show the *unique determination* of $\mathfrak{R}$ and $\mathfrak{S}$. Let $\overline{\mathfrak{R}}$ and $\overline{\mathfrak{S}}$ be modules satisfying the conditions of Theorem V, and let $\bar{r}_i$ and $\bar{s}_i$ be their elementary divisors. Since $\bar{r}_i$ and $\bar{s}_i$ are divisors of e_i, it follows, because

$$r = \bar{r}_1 \cdots \bar{r}_\rho, \qquad s = \bar{s}_1 \cdots \bar{s}_\rho, \qquad (\bar{r}_i, \bar{s}_i) = 1,$$

that also

$$e_i = \bar{r}_i\bar{s}_i, \qquad \bar{r}_i = (r, e_i) = r_i, \qquad \bar{s}_i = (s, e_i) = s_i.$$

Thus $\overline{\mathfrak{R}}$ and $\overline{\mathfrak{S}}$ agree with $\mathfrak{R}$ and $\mathfrak{S}$ in elementary divisors and in fundamental module. If therefore, by a unimodular substitution $\eta = Q(\xi)$, new indeterminates ξ are introduced, then

$$\overline{\mathfrak{R}} = (r_1\xi_1, \ldots, r_\rho\xi_\rho);$$

$$\mathfrak{A} = \big(r_1 s_1(q_{11}\xi_1 + \cdots + q_{1\rho}\xi_\rho), \ldots, r_\rho s_\rho(q_{\rho 1}\xi_1 + \cdots + q_{\rho\rho}\xi_\rho)\big).$$

From $\mathfrak{A} \equiv 0(\overline{\mathfrak{R}})$ one thus obtains

$$r_i s_i(q_{i1}\xi_1 + \cdots + q_{i\rho}\xi_\rho) \equiv 0(\mathfrak{R});$$

therefore $r_i s_i q_{ij} \equiv 0((r_i))$. Since $(r, s) = 1$, this gives $r_i q_{ij} \equiv 0((r_i))$, or $\overline{\mathfrak{R}} \equiv 0(\mathfrak{R})$. But since $N(\mathfrak{G} \mid \mathfrak{R}) = N(\mathfrak{G} \mid \overline{\mathfrak{R}})$, Theorem IV gives $\mathfrak{R} = \overline{\mathfrak{R}}$, and correspondingly $\mathfrak{S} = \overline{\mathfrak{S}}$. Theorem V is thereby proved in all its parts.

[10] Least common multiple $[\mathfrak{R}, \mathfrak{S}]$ understood in the module sense: the totality of linear forms that are divisible both by $\mathfrak{R}$ and by $\mathfrak{S}$. Correspondingly, the greatest common divisor $(\mathfrak{R}, \mathfrak{S})$ in the module sense is to be understood as the totality of linear forms that can be represented as the sum of an element from $\mathfrak{R}$ and an element from $\mathfrak{S}$.

§ 3. Ideals of Polynomials.

Let $\overline{\mathfrak{m}}$ be an *ideal of polynomials* in the n indeterminates $y_1, \dots, y_n$, with coefficients from an arbitrary, abstractly defined field P; that is, together with $\Phi_1(y)$ and $\Phi_2(y)$, $\overline{\mathfrak{m}}$ also contains the difference $\Phi_1(y) - \Phi_2(y)$, and together with $\Phi(y)$ it also contains $A(y) \cdot \Phi(y)$, where $A(y)$ denotes an arbitrary polynomial with coefficients from P[11].

Let the y be subjected to a transformation with indeterminates as coefficients,

$$y = U(x); \quad \text{for instance} \quad \begin{cases} y_1 = x_1, \\ y_2 = u_{21}x_1 + x_2, \\ \cdots \\ y_n = u_{n1}x_1 + \cdots + u_{n,n-1}x_{n-1} + x_n, \end{cases} \tag{12}$$

and adjoin the indeterminates $u_{\mu\nu}$ to the coefficient domain of the polynomials, which thereby passes into a field $P(u)$. By this adjunction let $\overline{\mathfrak{m}}$ pass into $\overline{\mathfrak{m}}_{(u)}$; hence $\overline{\mathfrak{m}}_{(u)}$ contains, besides the polynomials $\Phi_\lambda(y)$ from $\overline{\mathfrak{m}}$, all linear combinations $\sum \alpha_\lambda(u)\Phi_\lambda(y)$ of finitely many Φ_λ, where $\alpha_\lambda(u)$ denotes rational functions of the $u_{\mu\nu}$ with coefficients from P. By multiplication with a suitable polynomial in u, the $\alpha_\lambda(u)$ may, without restriction of generality, be assumed to be power products in u. Thus one has

$$\sum \alpha_\lambda(u)\Phi_\lambda(y) \equiv 0(\overline{\mathfrak{m}}_{(u)}); \qquad \Phi_\lambda(y) \equiv 0(\overline{\mathfrak{m}}) \quad \text{and conversely.} \tag{13}$$

By means of (12), $\overline{\mathfrak{m}}_{(u)}$ passes into an ideal $\mathfrak{m}$ of polynomials in $x_1, \dots, x_n$ with coefficients from $P(u)$:

$$\overline{\mathfrak{m}}_{(u)} = \mathfrak{m} \quad \text{by means of} \quad y = U(x). \tag{14}$$

The combination of the relations (14) and (13) says the following:

$$\text{From } F(x) \equiv 0(\mathfrak{m}); \quad F(x) = \Phi(y) = \sum \alpha_\lambda(u)\Phi_\lambda(y) = \sum \alpha_\lambda(u)F_\lambda(x) \text{ follows } F_\lambda(x) \equiv 0(\mathfrak{m}); \tag{15}$$

whereas (14) and (15) conversely yield (13) again.

Definition IV. *Ideals $\mathfrak{m}$ of polynomials in $x_1, \dots, x_n$ with coefficients from $P(u)$ which satisfy the relations (15), and hence arise from $\overline{\mathfrak{m}}_{(u)}$ by means of $y = U(x)$, shall be called transformed ideals.*

Transformed ideals are distinguished by the existence of regular polynomials; *a polynomial of degree r is called regular with respect to x_i if it contains the term x_i^r with non-zero coefficient.* One sees that the polynomials $F_i(x)$ are regular with respect to x_1. In what follows the main point will be to regard these transformed ideals as modules of linear forms in certain power products of the x. For this the concept of the fundamental ideal must be fixed; later it will pass into the fundamental module. We first give a notation that will be maintained throughout from now on:

Notation. *By $F^{(i)}, f^{(i)}, a^{(i)}, \dots$ we shall always mean polynomials in the x that are free of $x_1, \dots, x_{i-1}$.*

Definition V. *To every ideal $\mathfrak{m}$ there are defined n fundamental ideals $\mathfrak{g}_0, \mathfrak{g}_1, \dots, \mathfrak{g}_{n-1}$ by the following stipulation: the fundamental ideal of stage $(i-1)$, $\mathfrak{g}_{i-1}$, contains all and only the polynomials $G(x)$ for which there is a polynomial $b^{(i)}$ – in general varying with $G(x)$ – such that*

$$b^{(i)}G(x) \equiv 0(\mathfrak{m}); \qquad b^{(i)} \neq 0. \tag{16}$$

That the $\mathfrak{g}$ are in fact ideals corresponds exactly to the proof of the module property for $\mathfrak{G}$ following formula (4), with (16) playing the analogous role. The ideal $\mathfrak{g}_i$ is divisible by $\mathfrak{g}_{i-1}$, since every polynomial $b^{(i+1)}$ is at the same time a $b^{(i)}$.

For the fundamental ideals one has:

Theorem VI. *The fundamental ideals of transformed ideals are transformed ideals*[12].

The proof rests on a known theorem of Dedekind–Mertens[13]: let a and b be indeterminates, and set

$$\sum a_{i_1,\dots,i_s} z_1^{i_1} \cdots z_s^{i_s} \cdot \sum b_{j_1,\dots,j_s} z_1^{j_1} \cdots z_s^{j_s} = \sum c_{k_1,\dots,k_s} z_1^{k_1} \cdots z_s^{k_s},$$

[11]The conceptually resulting designation "ideal" instead of the formerly generally customary "module" or "module of forms" already proves necessary here in order to distinguish these from the modules of linear forms. (E. N.)

[12]Theorem VI is used only in § 7.

[13]Dedekind: Über einen arithmetischen Satz von Gauß, Mitt. d. deutsch. math. Ges. zu Prag 1892. F. Mertens: Über einen algebraischen Satz, Ber. d. Ak. d. Wissensch. Wien 101 (1892), pp. 1560–1566. – Kronecker's extension of Gauss's theorem to polynomials with indeterminate coefficients is an immediate consequence of this theorem.

$$\sum i \leq \nu, \qquad \sum j \leq \mu, \qquad \sum k \leq \nu + \mu.$$

If $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}$ denote the modules derived respectively from the a, b, c by means of rational integers, then there exists an exponent q such that the identity holds:

$$\mathfrak{A}^q \mathfrak{B} = \mathfrak{A}^{q-1} \mathfrak{C}. \tag{17}$$

Here $\mathfrak{A}^q$ consists of the power products of q-th dimension in the a and their integral linear combinations.

For the proof of Theorem VI, now put

$$\begin{cases} G(x) = \Gamma(y) = \sum \alpha_\lambda(u) \Gamma_\lambda(y) = \sum \alpha_\lambda(u) G_\lambda(x), \\ b^{(i)}(x) = \varphi(y) = \sum \beta_\mu(u) \varphi_\mu(y), \end{cases} \tag{18}$$

where $\alpha_\lambda(u)$ and $\beta_\mu(u)$ are power products in the u. Then, by (16), (14) and (18),

$$\sum \beta_\mu(u) \varphi_\mu(y) \cdot \sum \alpha_\lambda(u) \Gamma_\lambda(y) \equiv 0(\overline{\mathfrak{m}}_{(u)}); \qquad \sum \beta_\mu(u) \varphi_\mu(y) \neq 0. \tag{19}$$

On the left stands the product of two polynomials in u whose coefficients are the polynomials $\varphi_\mu(y)$ and $\Gamma_\lambda(y)$; the coefficients of the product polynomial in u are, by the right hand side of (19), polynomials from $\overline{\mathfrak{m}}$. If, therefore, in (17) one replaces the a, b, c by these polynomials in y, one obtains

$$\left\{ \sum \beta_\mu(u) \varphi_\mu(y) \right\}^q \cdot \Gamma_\lambda(y) \equiv 0(\overline{\mathfrak{m}}_{(u)}),$$

which by (18) is equivalent to:

$$b^{(i)}(x)^q \cdot G_\lambda(x) \equiv 0(\mathfrak{m}); \qquad G_\lambda(x) \equiv 0(\mathfrak{g}_{i-1}). \tag{20}$$

The relations (16), (18) and (20) show that the *fundamental ideals* $\mathfrak{g}_{i-1}$ *are indeed transformed ideals.*

22. Edited version of K. Hentzelt: On the Theory of Polynomial Ideals and Resultants (continued)

Math. Ann. 88 (1923), pp. 53–79

§ 4. Connection between ideals of polynomials and modules of linear forms.

To regard an ideal $\mathfrak{m}$ of polynomials, which is always assumed transformed, as a module $\mathfrak{M}_{i-1}$ ($i = 1, \ldots, n$) of linear forms, the individual polynomials $F(x)$ are to be considered as linear forms in the power products ξ_λ of $x_1 \ldots x_{i-1}$:

$$F(x) = \sum a_\lambda^{(i)} \xi_\lambda,$$

where, according to the notation of § 3, the $a_\lambda^{(i)}$ are polynomials in $x_i \ldots x_n$. From the definition of the ideal it follows at once that $\mathfrak{M}_{i-1}$ has the module property with respect to these integral quantities $a^{(i)}, b^{(i)}, \ldots$; and since the infinitely many power products ξ of $x_1 \ldots x_{i-1}$ occur only linearly, by reason of their linear independence they play for $\mathfrak{M}_{i-1}$ the role of indeterminates. Each module $\mathfrak{M}_{i-1}$ contains infinitely many indeterminates ξ, but in each individual linear form only finitely many indeterminates occur. Likewise each fundamental ideal $\mathfrak{g}_{i-1}$ is to be regarded as a module $\mathfrak{G}_{i-1}$ of linear forms, where the indeterminates are again the power products of $x_1 \ldots x_{i-1}$. For $i = 1$ there is only one indeterminate ξ_0, corresponding to the unit.

One can now, and this is the basis of everything that follows, restrict oneself, despite these infinitely many indeterminates ξ, to modules of linear forms in finitely many indeterminates, according to the following theorem.

Theorem VII. *The residue-class system* $\mathfrak{G}_{i-1} \mid \mathfrak{M}_{i-1}$ *is isomorphic to a residue-class system* $\mathfrak{G}^*_{i-1} \mid \mathfrak{M}^*_{i-1}$, *where* $\mathfrak{M}^*_{i-1}$ *is a module in finitely many indeterminates and* $\mathfrak{G}^*_{i-1}$ *is its fundamental module. The module* $\mathfrak{M}_{i-1}$ *has – after adjoining* $x_{i+1} \ldots x_n$ *to* $P(u)$ *– only finitely many elementary divisors different from* 1, *namely the elementary divisors of* $\mathfrak{M}^*_{i-1}$ *that are different from* 1.

The elementary divisors of $\mathfrak{M}_{i-1}$ are to be defined as in note 8). The isomorphism occurring in Theorem VII rests on the following definition.

Definition V. *Two residue-class systems are called isomorphic if they can be put in one-to-one correspondence in such a way that the difference of two classes is assigned to the difference of the corresponding classes, and the product of a class by an integral quantity to the product of the corresponding class by the same integral quantity.*

The proof of Theorem VII is obtained by complete induction. For $i = 1$ the theorem is evident: $\mathfrak{M}_0$ is a module of linear forms in one indeterminate ξ_0, corresponding to the power product of dimension zero of the x, that is, to the unit; the integral quantities are all polynomials in $x_1 \dots x_n$. The fundamental ideal $\mathfrak{g}_0$ is the unit ideal consisting of all polynomials in $x_1 \dots x_n$, and hence $\mathfrak{G}_0$ is the module (ξ_0), the fundamental module of $\mathfrak{M}_0$; thus here $\mathfrak{G}_0^* \mid \mathfrak{M}_0^*$ coincides with $\mathfrak{G}_0 \mid \mathfrak{M}_0$. Finally, since $\mathfrak{M}_0$ has rank 1, it can have at most one elementary divisor different from 1.

We first pass from $i = 1$ to $i = 2$, in order to use the insight thereby obtained into the structure of $\mathfrak{G}_1 \mid \mathfrak{M}_1$ for the general induction step. For this purpose we first show that for $\mathfrak{G}_0 \mid \mathfrak{M}_0$ an x_1-bounded system of representatives may be chosen; because of the divisibility of $\mathfrak{g}_1$ by $\mathfrak{g}_0$, this will then lead to the finitely many indeterminates of $\mathfrak{G}_1^*$.

As a transformed ideal, $\mathfrak{m}$ possesses by § 3 at least one polynomial $C^{(1)}(x)$ regular in x_1, say of dimension k; hence $\mathfrak{M}_0$ contains the linear form $C^{(1)}\xi_0$. By the regularity of $C^{(1)}(x)$, every polynomial $H(x)$ admits a representation

$$H(x) = b_0^{(2)} + b_1^{(2)} x_1 + \dots + b_{k-1}^{(2)} x_1^{k-1} \quad (C^{(1)}(x)); \tag{21}$$

therefore, since $C^{(1)}\xi_0$ belongs to $\mathfrak{M}_0$, one can indeed choose for $\mathfrak{G}_0 \mid \mathfrak{M}_0$ a system of representatives that does not reach the k-th dimension in x_1.

If now, in particular, for the polynomials $G(x)$ from $\mathfrak{g}_1$ and $F(x)$ from $\mathfrak{m}$ one writes

$$\begin{cases} G(x) = g_0^{(2)} + g_1^{(2)} x_1 + \dots + g_{k-1}^{(2)} x_1^{k-1} \quad (C^{(1)}(x)), \\ F(x) = a_0^{(2)} + a_1^{(2)} x_1 + \dots + a_{k-1}^{(2)} x_1^{k-1} \quad (C^{(1)}(x)), \end{cases} \tag{22}$$

and if $\mathfrak{G}_1^*$ denotes the system of all linear forms $g(\xi) = g_0^{(2)}\xi_0 + \dots + g_{k-1}^{(2)}\xi_{k-1}$, and correspondingly $\mathfrak{M}_1^*$ the system of the linear forms $a(\xi) = a_0^{(2)}\xi_0 + \dots + a_{k-1}^{(2)}\xi_{k-1}$, then the ideal property of $\mathfrak{g}_1$ and $\mathfrak{m}$ shows that $\mathfrak{G}_1^*$ and $\mathfrak{M}_1^*$ are modules of linear forms in $\xi_0 \dots \xi_{k-1}$ with respect to the integral quantities $b^{(2)}, c^{(2)}, \dots$. Likewise the principal ideal derived from $C^{(1)}(x)$ gives a module with respect to these integral quantities,

$$\mathfrak{G}_1 = (\xi_0, \xi_1, \dots, \xi_r, \dots),$$

where $\xi_r = x_1^r C^{(1)}(x)$. The ξ_r are linearly independent with respect to these integral quantities $b^{(2)}, c^{(2)}, \dots$, both among themselves and from the finitely many ξ. From the divisibility of $\mathfrak{G}_1$ by $\mathfrak{M}_1$, and hence by $\mathfrak{G}_1$, it follows that $\mathfrak{G}_1^*$ is divisible by $\mathfrak{G}_1$ and $\mathfrak{M}_1^*$ by $\mathfrak{M}_1$. Taking (22) into account, one obtains

$$\mathfrak{G}_1 = (\mathfrak{G}_1^*, \mathfrak{G}_1), \qquad \mathfrak{M}_1 = (\mathfrak{M}_1^*, \mathfrak{G}_1). \tag{23}$$

From (23) and from the linear independence of the ξ and ξ, Theorem VII for $i = 2$ follows as follows. Because $\mathfrak{G}_1$ is divisible by $\mathfrak{M}_1$, one first has $\mathfrak{G}_1 \mid \mathfrak{M}_1 = \mathfrak{G}_1^* \mid \mathfrak{M}_1$. To prove the isomorphism with $\mathfrak{G}_1^* \mid \mathfrak{M}_1^*$, let certain linear forms $g^*(\xi)$ from $\mathfrak{G}_1^*$ give a system of representatives of $\mathfrak{G}_1^* \mid \mathfrak{M}_1^*$. Because of the linear independence of the ξ and ξ, from $g^*(\xi) \equiv 0(\mathfrak{M}_1)$ it follows also that $g^*(\xi) \equiv 0(\mathfrak{M}_1^*)$; the $g^*(\xi)$ are therefore also incongruent modulo $\mathfrak{M}_1$, form a system of representatives of $\mathfrak{G}_1 \mid \mathfrak{M}_1$, and the correspondence from $\mathfrak{G}_1 \mid \mathfrak{M}_1$ to $\mathfrak{G}_1^* \mid \mathfrak{M}_1^*$ mediated by this system of representatives is isomorphic in the sense of Definition V.

In exactly the same way one obtains: from $b^{(2)}g(\xi) \equiv 0(\mathfrak{M}_1)$, $b^{(2)} \neq 0$, follows $b^{(2)}g(\xi) \equiv 0(\mathfrak{M}_1^*)$, $b^{(2)} \neq 0$. Since this is satisfied for all $g(\xi)$ from $\mathfrak{G}_1^*$, and only for these – for by (23) $\mathfrak{G}_1^*$ consists of all polynomials of the fundamental ideal $\mathfrak{g}_1$ that are at the same time linear forms in ξ – $\mathfrak{G}_1^*$ is thereby characterized as the fundamental module of $\mathfrak{M}_1^*$.

Adjoining finally $x_3 \dots x_n$ to $P(u)$, one gets

$$\begin{cases} \mathfrak{G}_1^* = (\eta_1, \dots, \eta_\rho), & \mathfrak{M}_1^* = (e_1\eta_1, \dots, e_\rho\eta_\rho), \\ \mathfrak{G}_1 = (\eta_\rho, \dots, \eta_1, \xi_0, \dots, \xi_r, \dots), & \mathfrak{M}_1 = (e_\rho\eta_\rho, \dots, e_1\eta_1, \xi_0, \dots, \xi_r, \dots), \end{cases} \tag{24}$$

and hence

$$(e_\rho) = \mathfrak{M}_1/\mathfrak{G}_1 = \mathfrak{M}_1/(\mathfrak{M}_1, \eta_\rho), \qquad (e_{\rho-1}) = (\mathfrak{M}_1, \eta_\rho)/\mathfrak{G}_1 \quad \text{etc.}$$

Thus, taking note 8) into account, $e_\rho, e_{\rho-1}, \ldots, e_1, 1, \ldots, 1, \ldots$ are recognized as the elementary divisors of $\mathfrak{M}_1$. This proves all parts of Theorem VII for $i = 2$.

It should be noted that (22) and (23) use only that $C^{(1)}(x)$ is regular in x_1. These formulae, and the arguments attached to them leading to Theorem VII, therefore remain valid if, in place of (12), the special transformation

$$y_1 = x_1; \quad y_2 = u_{21}x_1 + z_2; \quad \ldots; \quad y_n = u_{n1}x_1 + z_n \tag{25}$$

is taken as the basis; by composition with

$$\begin{cases} z = U_1(x) \quad \text{or} \\ z_2 = x_2; \quad z_3 = u_{32}x_2 + x_3; \quad \ldots; \quad z_n = u_{n2}x_2 + \cdots + u_{n,n-1}x_{n-1} + x_n, \end{cases} \tag{26}$$

this again gives (12). If $\widetilde{\mathfrak{m}}_{(u)}$ passes by means of (25) into $\widetilde{\mathfrak{m}}$, and if $\widetilde{\mathfrak{g}}_1, \widetilde{\mathfrak{G}}_1, \ldots$ have for $\widetilde{\mathfrak{m}}$ the same meaning as $\mathfrak{g}_1, \mathfrak{G}_1, \ldots$ have for $\mathfrak{m}$, then

$$\widetilde{\mathfrak{G}}_1 = (\widetilde{\mathfrak{G}}_1^*, \widetilde{\mathfrak{G}}_1), \qquad \widetilde{\mathfrak{M}}_1 = (\widetilde{\mathfrak{M}}_1^*, \widetilde{\mathfrak{G}}_1). \tag{27}$$

Here (27) arises from (23) simply by inversion of (26). This is evident for $\mathfrak{m}$ and $C^{(1)}(x)$, and hence also for $\mathfrak{G}_1$ and $\mathfrak{M}_1$. It is also true for $\mathfrak{g}_1$, since

$$b^{(2)}(x)G(x) \equiv 0(\mathfrak{m}), \quad b^{(2)} \neq 0, \qquad \widetilde{b}^{(2)}(z)\widetilde{G}(x,z) \equiv 0(\widetilde{\mathfrak{m}}), \quad \widetilde{b}^{(2)} \neq 0$$

mutually condition one another through $z = U_1(x)$. Thus $\widetilde{\mathfrak{g}}_1 = \mathfrak{g}_1$ by means of (26); hence also $\widetilde{\mathfrak{G}}_1 = \mathfrak{G}_1$. Consequently, by the definition given through (22), the same holds for $\mathfrak{G}_1^*$ and $\mathfrak{M}_1^*$ as for $\widetilde{\mathfrak{G}}_1^*$ and $\widetilde{\mathfrak{M}}_1^*$, and so for the whole representation (27).

For the general induction step, suppose the following hypotheses hold:

$$\begin{cases} \mathfrak{G}_{i-1} = (\mathfrak{G}_{i-1}^*, \mathfrak{G}_{i-1}), \qquad \mathfrak{M}_{i-1} = (\mathfrak{M}_{i-1}^*, \mathfrak{G}_{i-1}), \\ \mathfrak{G}_{i-1} = (\xi_0, \xi_1, \ldots, \xi_r, \ldots), \end{cases} \tag{28}$$

where $\mathfrak{G}_{i-1}^*$ and $\mathfrak{M}_{i-1}^*$ are modules, with respect to the integral quantities $b^{(i)}, c^{(i)}, \ldots$, of linear forms in finitely many indeterminates ξ, certain power products of $x_1, \ldots, x_{i-1}$; and where the ξ are linearly independent, both among themselves and from the ξ, with respect to these integral quantities. A representation corresponding to (28), and the linear independence of the ξ and ξ, is also to hold when, instead of (12), one uses the special transformation

$$\begin{aligned} &y_1 = x_1; \quad \ldots; \quad y_{i-1} = u_{i-1,1}x_1 + \cdots + x_{i-1}; \\ &y_i = u_{i,1}x_1 + \cdots + u_{i,i-1}x_{i-1} + z_i; \quad \ldots; \quad y_n = u_{n,1}x_1 + \cdots + u_{n,i-1}x_{i-1} + z_n, \end{aligned} \tag{29}$$

which can be composed to give (12) by means of

$$z = U_{i-1}(x) \quad \text{or} \quad z_i = x_i; \quad z_{i+1} = u_{i+1,i}x_i + x_{i+1}; \quad \ldots; \quad z_n = u_{n,i}x_i + \cdots + u_{n,n-1}x_{n-1} + x_n.$$

Moreover this special representation is to arise from (28) simply by inversion of $z = U_{i-1}(x)$.

From the hypotheses (28) and the linear independence of the ξ and ξ, the isomorphism of $\mathfrak{G}_{i-1} \mid \mathfrak{M}_{i-1}$ with $\mathfrak{G}_{i-1}^* \mid \mathfrak{M}_{i-1}^*$ follows by assignment to the same system of representatives, by exactly the same arguments as those attached above to (23). Likewise it follows that $\mathfrak{G}_{i-1}^*$ becomes the fundamental module of $\mathfrak{M}_{i-1}^*$, and that $\mathfrak{M}_{i-1}$ has only finitely many elementary divisors different from 1, namely those elementary divisors of $\mathfrak{M}_{i-1}^*$ that are different from 1.

To carry out the induction step it remains first to show again that for $\mathfrak{G}_{i-1}^* \mid \mathfrak{M}_{i-1}^*$, and hence also for $\mathfrak{G}_{i-1} \mid \mathfrak{M}_{i-1}$, an x_i-bounded system of representatives may be chosen. This follows from the quotient representation proved in (7), Theorem III:

$$\mathfrak{G}_{i-1}^* = \mathfrak{M}_{i-1}^*/(C^{(i)}), \tag{30}$$

where $C^{(i)}$ means any non-identically vanishing determinant of degree ρ from $\mathfrak{M}_{i-1}^*$, ρ being the rank of $\mathfrak{M}_{i-1}^*$. From the part of the hypotheses referring to the special transformation (29), it follows that $C^{(i)}$ may always be assumed regular with respect to x_i. For the module $\widetilde{\mathfrak{M}}_{i-1}^*$ corresponding to $\mathfrak{M}_{i-1}^*$ has the same rank; if $\widetilde{C}^{(i)}$ denotes any non-identically vanishing determinant of degree ρ from $\widetilde{\mathfrak{M}}_{i-1}^*$ which passes by $z = U_{i-1}(x)$ into a regular $C^{(i)}$, then by hypothesis $C^{(i)}$ is a determinant of degree ρ from $\mathfrak{M}_{i-1}^*$.

From the existence of $C^{(i)}$ it follows, as in (8), with a suitable numbering of the ξ, that ρ linear forms of the special shape

$$L_\lambda(\xi) = C^{(i)}\xi_\lambda + c^{(i)}_{\lambda,1}\xi_{\rho+1} + \cdots + c^{(i)}_{\lambda,s-\rho}\xi_s \qquad (\lambda = 1 \ldots \rho)$$

belong to $\mathfrak{M}^*_{i-1}$. Every linear form in ξ can now be brought into the form

$$H(\xi) = a^{(i)}_1\xi_1 + \cdots + a^{(i)}_\rho\xi_\rho + b^{(i)}_1\xi_{\rho+1} + \cdots + b^{(i)}_{s-\rho}\xi_s \quad (L_1(\xi), \ldots, L_\rho(\xi)), \tag{31}$$

where $a^{(i)}_1 \ldots a^{(i)}_\rho$ are bounded in x_i and do not reach the degree k of $C^{(i)}$. This representation is unique; for from

$$a^{(i)}_1\xi_1 + \cdots + a^{(i)}_\rho\xi_\rho + b^{(i)}_1\xi_{\rho+1} + \cdots + b^{(i)}_{s-\rho}\xi_s \equiv 0 \quad (L_1(\xi), \ldots, L_\rho(\xi))$$

it follows, by comparing coefficients in $\xi_1 \ldots \xi_\rho$ and considering the degrees in x_i, that all $a^{(i)}$ and $b^{(i)}$ vanish identically.

Let now in particular

$$H(\xi) \equiv 0(\mathfrak{G}^*_{i-1}); \qquad \text{hence} \qquad C^{(i)}H(\xi) \equiv 0(\mathfrak{M}^*_{i-1}) \quad \text{by (30).}$$

As in the proof of Theorem III one obtains

$$C^{(i)}H(\xi) - \sum_{\tau=1}^{s-\rho}\{C^{(i)}b^{(i)}_\tau - \sum_\lambda a^{(i)}_\lambda c^{(i)}_{\lambda,\tau}\}\xi_{\rho+\tau} \equiv 0(\mathfrak{M}^*_{i-1}),$$

and therefore, since as in Theorem III the coefficients of $\xi_{\rho+1} \ldots \xi_s$ must vanish,

$$C^{(i)}b^{(i)}_\tau = \sum_\lambda a^{(i)}_\lambda c^{(i)}_{\lambda,\tau} \qquad (\tau = 1 \ldots s - \rho).$$

Thus the $b^{(i)}_\tau$ are also bounded and do not reach the maximal degree of the $c^{(i)}_{\lambda,\tau}$. The existence of an x_i-bounded system of representatives for $\mathfrak{G}^*_{i-1} \mid \mathfrak{M}^*_{i-1}$, that is for $\mathfrak{G}_{i-1} \mid \mathfrak{M}_{i-1}$, not reaching a certain fixed degree r, is thereby proved.

Let $\vartheta_1 \ldots \vartheta_t$ denote the finitely many power products

$$x_i^\alpha\xi_\lambda \quad (\alpha = 0, 1, \ldots, k-1;\ \lambda = 1, 2, \ldots, \rho), \qquad x_i^\beta\xi_{\rho+\tau} \quad (\beta = 0, 1, \ldots, r-1;\ \tau = 1, 2, \ldots, s-\rho)$$

and arrange the countably infinite expressions

$$x_i^\gamma L_\lambda \quad (\gamma = 0, 1, \ldots \text{ in inf.};\ \lambda = 1, 2, \ldots, \rho), \qquad x_i^\gamma\xi_\nu \quad (\gamma, \nu = 0, 1, \ldots \text{ in inf.})$$

in a simply infinite sequence $\omega_0, \omega_1, \ldots, \omega_\mu, \ldots$. Then the ϑ and ω are linearly independent, both each among themselves and from one another, with respect to the integral quantities $b^{(i+1)}, c^{(i+1)}, \ldots$. For from

$$\sum c^{(i+1)}_\alpha x_i^\alpha\xi_\lambda + \sum c^{(i+1)}_\beta x_i^\beta\xi_{\rho+\tau} + \sum d^{(i+1)}_\gamma x_i^\gamma L_\lambda(\xi) + \sum e^{(i+1)}_{\gamma\nu}x_i^\gamma\xi_\nu = 0,$$

each sum extending over finitely many terms, the assumed linear independence of the ξ and ξ, and of the ξ among themselves with respect to the integral quantities $b^{(i)}$, gives the vanishing of all $e^{(i+1)}_{\gamma\nu}$. The uniqueness of the representation (31) further gives the vanishing of the $c^{(i+1)}_\alpha$ and $c^{(i+1)}_\beta$, and finally, because of the linear independence of the $L_\lambda(\xi)$, the vanishing of the $d^{(i+1)}_\gamma$.

If the module $(\mathfrak{G}_{i-1}, L_1(\xi), \ldots, L_\rho(\xi))$ is now regarded as a module $\mathfrak{G}_i$ with respect to the integral quantities $b^{(i+1)}$, then $\mathfrak{G}_i$ is divisible by $\mathfrak{M}_i$, because $\mathfrak{G}_{i-1}, L_1(\xi), \ldots, L_\rho(\xi)$ are divisible by $\mathfrak{M}_{i-1}$, and

$$\mathfrak{G}_i = (\omega_0, \omega_1, \ldots, \omega_\mu, \ldots).$$

For each $H(x) \equiv 0(\mathfrak{g}_{i-1})$, (28), (31), and what has just been proved give

$$H(x) = b^{(i+1)}_1\vartheta_1 + \cdots + b^{(i+1)}_t\vartheta_t \quad (\mathfrak{G}_i)$$

as a module equation with respect to the integral quantities $b^{(i+1)}, c^{(i+1)}$. Since $\mathfrak{g}_i$ is divisible by $\mathfrak{g}_{i-1}$ and $\mathfrak{m}$ is divisible by $\mathfrak{g}_i$, if one now sets, in particular, for every $G(x)$ from $\mathfrak{G}_i$ and every $F(x)$ from $\mathfrak{M}_i$,

$$G(x) = g^{(i+1)}_1\vartheta_1 + \cdots + g^{(i+1)}_t\vartheta_t \quad (\mathfrak{G}_i), \qquad F(x) = a^{(i+1)}_1\vartheta_1 + \cdots + a^{(i+1)}_t\vartheta_t \quad (\mathfrak{G}_i),$$

and denotes by $\mathfrak{G}_i^*$ the module consisting of all $g(\vartheta)$, and by $\mathfrak{M}_i^*$ the module consisting of all $a(\vartheta)$, then exactly the arguments leading to (23) give

$$\mathfrak{G}_i = (\mathfrak{G}_i^*, \mathfrak{S}_i), \qquad \mathfrak{M}_i = (\mathfrak{M}_i^*, \mathfrak{S}_i), \qquad \mathfrak{S}_i = (\omega_0, \omega_1, \ldots, \omega_\mu, \ldots). \tag{32}$$

In deriving (32), again only the regularity of $C^{(i)}$ in x_i was used, so the whole argument remains valid if the special transformation (29) is taken with the index increased by one. The same considerations as those attached to (27) then show that this special representation arises from (32) simply by the inversion of $z = U_i(x)$.

Thus all hypotheses (28), etc., of the general induction step have been proved for the next index; and since, as shown there, Theorem VII follows directly from these hypotheses, Theorem VII is proved in general. At the same time it has been shown that the arbitrariness in $(\mathfrak{G}_{i-1}^*, \mathfrak{M}_{i-1}^*)$ caused by the arbitrary choice of $C^{(i)}$ is again removed by the isomorphism of the residue-class system $\mathfrak{G}_{i-1}^* \mid \mathfrak{M}_{i-1}^*$.

In particular, if the field P used in § 3 as coefficient domain has characteristic zero – that is, if the prime field contained in P and derived from the unit is of the type of the rational numbers – then the indeterminates $u_{\mu\nu}$ can be specialized to quantities $\bar{u}_{\mu\nu}$ from P in such a way that for $i = 1 \ldots n$ each $C^{(i)}$ remains regular in x_i. The above arguments show that Theorem VII remains valid under this specialization as well.[14] The fundamental module and elementary divisors need not, however, arise necessarily from the general case by the specialization $u_{\mu\nu} = \bar{u}_{\mu\nu}$.[15]

§ 5. Resultant form and elementary-divisor form of an ideal of polynomials.

Let $x_{i+1} \ldots x_n$ be adjoined to $P(u)$, so that $\mathfrak{G}_{i-1}$ and $\mathfrak{M}_{i-1}$, and therefore also $\mathfrak{G}_{i-1}^*$ and $\mathfrak{M}_{i-1}^*$, pass into modules of linear forms in which the integral quantities can be regarded as polynomials in one indeterminate x_i. By Theorem VII, in this case the elementary divisors different from 1 of $\mathfrak{M}_{i-1}$, together with at most finitely many elementary divisors equal to 1, agree with those of $\mathfrak{M}_{i-1}^*$. The product of these elementary divisors, the ρ-th determinantal divisor of $\mathfrak{M}_{i-1}^*$, was equal to the determinant of the transition substitution from $\mathfrak{G}_{i-1}^*$ to $\mathfrak{M}_{i-1}^*$, hence to $N(\mathfrak{G}_{i-1}^* \mid \mathfrak{M}_{i-1}^*)$ by Definition III. Formula (24), or the analogous formula following from (28) for general i, shows that the product of these elementary divisors is also equal to the determinant of the transition substitution from $\mathfrak{G}_{i-1}$ to $\mathfrak{M}_{i-1}$, and is therefore to be denoted by $N(\mathfrak{G}_{i-1} \mid \mathfrak{M}_{i-1})$. Thus

$$N(\mathfrak{G}_{i-1} \mid \mathfrak{M}_{i-1}) = N(\mathfrak{G}_{i-1}^* \mid \mathfrak{M}_{i-1}^*) = R^{(i)}(x),$$

where the polynomial $R^{(i)}(x)$, that is, the greatest common divisor, in the polynomial sense, of all ρ-rowed determinants from $\mathfrak{M}_{i-1}^*$, may be assumed integral and primitive in $x_{i+1} \ldots x_n$, and consequently, as a divisor of $C^{(i)}$, becomes regular in x_i. Likewise the highest elementary divisor $E^{(i)}(x)$ of $\mathfrak{M}_{i-1}$, respectively of $\mathfrak{M}_{i-1}^*$, may be assumed integral and primitive in $x_{i+1} \ldots x_n$ and hence regular in x_i. This leads to the following definition.

Definition VI. *The polynomial $R^{(i)}(x)$ is called the resultant of the i-th stage of $\mathfrak{m}$, and $E^{(i)}(x)$ the elementary divisor of the i-th stage. Thus the resultant of the i-th stage is the norm of the fundamental ideal $\mathfrak{g}_{i-1}$ with respect to $\mathfrak{m}$, interpreted as the norm of the module $\mathfrak{G}_{i-1}$ with respect to $\mathfrak{M}_{i-1}$, where $x_{i+1} \ldots x_n$ are adjoined to $P(u)$; likewise $E^{(i)}(x)$, under the same adjunction, becomes equal to the quotient $\mathfrak{M}_{i-1}/\mathfrak{G}_{i-1}$. The resultant form and elementary-divisor form of $\mathfrak{m}$ are the products*

$$R_{\mathfrak{m}} = R^{(1)} R^{(2)} \cdots R^{(n)}, \qquad E_{\mathfrak{m}} = E^{(1)} E^{(2)} \cdots E^{(n)}.$$

For resultant form and elementary-divisor form one has:

Theorem VIII. *The resultant form is divisible by the elementary form; conversely, a power of $E_{\mathfrak{m}}$ is divisible by $R_{\mathfrak{m}}$. The forms $E_{\mathfrak{m}}$ and $R_{\mathfrak{m}}$ are divisible by $\mathfrak{m}$; more generally, $E^{(i)} \cdots E^{(n)} \mathfrak{g}_{i-1} \equiv 0(\mathfrak{m})$ and consequently $R^{(i)} \cdots R^{(n)} \mathfrak{g}_{i-1} \equiv 0(\mathfrak{m})$.*

[14] Hentzelt takes, instead of an arbitrary P, only the field of all complex numbers and treats chiefly this latter case, whereas he uses indeterminates only in the actual elimination theory. Hentzelt then further shows, essentially by the arguments given here, that in order to form the resultant form it suffices to go, in each $\mathfrak{M}_{i-1}$, only up to some finite degree in the x that exceeds a fixed bound. What is missing is the isomorphism of $\mathfrak{G}_{i-1} \mid \mathfrak{M}_{i-1}$ with $\mathfrak{G}_{i-1}^* \mid \mathfrak{M}_{i-1}^*$ given in Theorem VII, which explains the independence from the degree numbers. (E. N.)

[15] For $\mathfrak{m} = (x^2, ux + y)$ one has $\mathfrak{g}_0 = (1)$, $\mathfrak{g}_1 = (1)$. If $C^{(1)} = x^2$ is chosen, then $\mathfrak{G}_1^* = (1, x)$, $\mathfrak{M}_1^* = (ux + y, xy)$; this $\mathfrak{M}_1^*$ has elementary divisors y^2 and 1, and $C^{(2)} = y^2$ can be chosen. For $u = 0$, $C^{(1)}$ and $C^{(2)}$ remain regular in x and y respectively; but $\mathfrak{M}_1^* = (y, xy)$ has elementary divisors y, y. Thus $\mathfrak{G}_1 \mid \mathfrak{M}_1$ is still isomorphic to the residue-class system of a finite module, but is not isomorphic to $\mathfrak{G}_1 \mid \mathfrak{M}_1$. If instead one had chosen $C^{(1)} = ux + y$ and specialized so that $C^{(1)}$ remained regular, the isomorphism would also have been preserved. (E. N.)

Indeed, since the norm is divisible by the highest elementary divisor and conversely a power of this highest elementary divisor is divisible by the norm, this divisibility follows for $R^{(i)}(x)$ and $E^{(i)}(x)$ after adjoining $x_{i+1}\dots x_n$. But $R^{(i)}(x)$ and $E^{(i)}(x)$ are assumed primitive polynomials with respect to $x_{i+1}\dots x_n$; hence the divisibility holds with respect to all indeterminates $x_i, x_{i+1}, \dots, x_n$. Thus $R_{\mathfrak{m}}$ is divisible by $E_{\mathfrak{m}}$, and some power of $E_{\mathfrak{m}}$ by $R_{\mathfrak{m}}$, according to the definition of these expressions in all indeterminates x.

Furthermore, by Theorems VII and II, after adjoining $x_{i+1}\dots x_n$, the highest elementary divisor $E^{(i)}(x)$ is defined as the quotient $\mathfrak{M}_{i-1}/\mathfrak{G}_{i-1}$. Thus, returning to polynomials in all indeterminates x, for every

$$G(x) \equiv 0(\mathfrak{g}_{i-1}) \quad \text{there is a} \quad b^{(i+1)} \neq 0 \quad \text{such that} \quad b^{(i+1)}E^{(i)}(x)G(x) \equiv 0(\mathfrak{m}). \tag{33}$$

In particular $\mathfrak{g}_0$ is the unit ideal, whence

$$b^{(2)}E^{(1)}(x) \equiv 0(\mathfrak{m}), \quad b^{(2)} \neq 0, \qquad \text{and thus} \qquad E^{(1)}(x) \equiv 0(\mathfrak{g}_1).$$

In general, from (33), applied to

$$E^{(1)}(x)\cdots E^{(i-1)}(x) \equiv 0(\mathfrak{g}_{i-1}),$$

there exists $b^{(i+1)} \neq 0$ such that $b^{(i+1)}E^{(1)}\cdots E^{(i)} \equiv 0(\mathfrak{m})$; hence $E^{(1)}\cdots E^{(i)} \equiv 0(\mathfrak{g}_i)$. Since $\mathfrak{g}_n = \mathfrak{m}$, one obtains

$$E_{\mathfrak{m}} = E^{(1)}\cdots E^{(n)} \equiv 0(\mathfrak{m}) \quad \text{and consequently} \quad R_{\mathfrak{m}} = R^{(1)}\cdots R^{(n)} \equiv 0(\mathfrak{m}). \tag{34}$$

If in (33) $G(x)$ runs through the finitely many polynomials of an ideal basis of $\mathfrak{g}_{i-1}$, and if $c^{(i+1)}(x)$ denotes the product of the corresponding $b^{(i+1)}(x)$, then

$$c^{(i+1)}E^{(i)}(x)\mathfrak{g}_{i-1} \equiv 0(\mathfrak{m}); \qquad c^{(i+1)} \neq 0, \quad \text{and hence} \quad E^{(i)}(x)\mathfrak{g}_{i-1} \equiv 0(\mathfrak{g}_i),$$

and from this, as above, by finite repetition,

$$E^{(n)}(x)\cdots E^{(i)}(x)\mathfrak{g}_{i-1} \equiv 0(\mathfrak{m})$$

and consequently also

$$R^{(n)}(x)\cdots R^{(i)}(x)\mathfrak{g}_{i-1} \equiv 0(\mathfrak{m}).$$

This proves all parts of Theorem VIII.

Theorem VIII depends essentially on properties of the elementary-divisor form; the characteristic significance of the resultant form for $\mathfrak{m}$ is shown by

Theorem IX. *If $\mathfrak{n}$ is a divisor of $\mathfrak{m}$ – divisor understood in the ideal sense – and if the resultant forms $R_{\mathfrak{n}}$ and $R_{\mathfrak{m}}$ agree, then the ideals $\mathfrak{n}$ and $\mathfrak{m}$ agree.*

Let $\mathfrak{g}_0\dots\mathfrak{g}_{n-1}, \mathfrak{g}_n = \mathfrak{m}$ and $\mathfrak{h}_0\dots\mathfrak{h}_{n-1}, \mathfrak{h}_n = \mathfrak{n}$ denote the fundamental ideals of $\mathfrak{m}$ and $\mathfrak{n}$. First $\mathfrak{g}_0$ and $\mathfrak{h}_0$ are both the unit ideal, hence $\mathfrak{g}_0 = \mathfrak{h}_0$. Assume now that $\mathfrak{g}_{i-1} = \mathfrak{h}_{i-1}$, so that $\mathfrak{M}_{i-1}$ and $\mathfrak{N}_{i-1}$ have the same fundamental module $\mathfrak{G}_{i-1}$. The hypothesis $R_{\mathfrak{n}} = R_{\mathfrak{m}}$, after adjoining $x_{i+1}\dots x_n$, may be written as

$$N(\mathfrak{G}_{i-1} \mid \mathfrak{N}_{i-1}) = N(\mathfrak{G}_{i-1} \mid \mathfrak{M}_{i-1}) \quad \text{or} \quad N(\mathfrak{G}^*_{i-1} \mid \mathfrak{N}^*_{i-1}) = N(\mathfrak{G}^*_{i-1} \mid \mathfrak{M}^*_{i-1}).$$

Here, in accordance with (32), $\mathfrak{N}_{i-1} = (\mathfrak{N}^*_{i-1}, \mathfrak{G}_{i-1})$, so that $\mathfrak{N}^*_{i-1}$ is likewise a module of linear forms in ξ, with $\mathfrak{G}^*_{i-1}$ as its fundamental module. Since $\mathfrak{M}_{i-1}$ is divisible by $\mathfrak{N}_{i-1}$, which entails the divisibility of $\mathfrak{M}^*_{i-1}$ by $\mathfrak{N}^*_{i-1}$, Theorem IV shows that, under this adjunction, the two modules $\mathfrak{M}^*_{i-1}$ and $\mathfrak{N}^*_{i-1}$ agree, and consequently so do $\mathfrak{N}_{i-1}$ and $\mathfrak{M}_{i-1}$. But adjoining $x_{i+1}\dots x_n$ means that one adds to $\mathfrak{M}_{i-1}$, respectively to $\mathfrak{m}$, all polynomials $G(x)$ for which

$$b^{(i+1)}(x)G(x) \equiv 0(\mathfrak{m}), \qquad b^{(i+1)} \neq 0.$$

Thus, by the adjunction, $\mathfrak{M}_{i-1}$, respectively $\mathfrak{m}$, passes into $\mathfrak{g}_i$, and correspondingly $\mathfrak{n}$ into $\mathfrak{h}_i$; therefore $\mathfrak{g}_i = \mathfrak{h}_i$. Hence finally $\mathfrak{g}_n = \mathfrak{h}_n$, i.e. $\mathfrak{m} = \mathfrak{n}$.

Finally the same transfer principle, applied to Theorem V, gives the following.

Theorem X. *Let $R^{(i)} = S^{(i)}T^{(i)}$ be a decomposition of $R^{(i)}$ into relatively prime polynomials $S^{(i)}$ and $T^{(i)}$ in the polynomial sense. Then there exists one and only one ideal $\mathfrak{j}$, and likewise one and only one ideal $\mathfrak{t}$, both divisors of $\mathfrak{m}$ with the same fundamental ideal of the $(i-1)$-st stage, $\mathfrak{g}_{i-1}$, such that $S^{(i)} = N(\mathfrak{G}_{i-1} \mid \mathfrak{S}_{i-1})$ and $T^{(i)} = N(\mathfrak{G}_{i-1} \mid \mathfrak{T}_{i-1})$. The ideals $\mathfrak{j}$ and $\mathfrak{t}$ are defined as the totality of polynomials $S(x)$ and $T(x)$, respectively, such that*

$$s^{(i+1)}(x)T^{(i)}(x)S(x) \equiv 0(\mathfrak{m}), \quad s^{(i+1)} \neq 0,$$

$$t^{(i+1)}(x)S^{(i)}(x)T(x) \equiv 0(\mathfrak{m}), \quad t^{(i+1)} \neq 0,$$

and $\mathfrak{g}_i$ *becomes the least common multiple of* $\mathfrak{j}$ *and* $\mathfrak{t}$,

$$\mathfrak{g}_i = [\mathfrak{j}, \mathfrak{t}].$$

It remains only to note that $s^{(i+1)}$ and $t^{(i+1)}$ can be chosen fixed for $\mathfrak{j}$ and $\mathfrak{t}$, as the product of the $s^{(i+1)}$ and $t^{(i+1)}$ corresponding to the basis elements of $\mathfrak{j}$ and $\mathfrak{t}$, and that, as remarked above, by adjoining $x_{i+1} \dots x_n$ the ideal $\mathfrak{m}$ is transformed into $\mathfrak{g}_i$. In particular, the decomposition of $R^{(i)}$ can be continued up to powers of irreducible polynomials – primary factors – which entails a corresponding representation of $\mathfrak{g}_i$ as a least common multiple.

§ 6. Proper elimination theory.

By Theorem VIII, respectively formula (34), the resultant form and the elementary-divisor form are divisible by $\mathfrak{m}$, and therefore vanish at all zeros of $\mathfrak{m}$. Here by "zeros of $\mathfrak{m}$" are meant value systems of $x_1 \dots x_i$ belonging to the algebraically closed field derived from $P(u; x_{i+1} \dots x_n)$[16], such that every polynomial from $\mathfrak{m}$ vanishes at these value systems, while $x_{i+1} \dots x_n$ are thought of as adjoined to the coefficient field ($i = 1, \dots, n$). These adjoined $x_{i+1} \dots x_n$ can, as will be shown, also be replaced by quantities from $P(u)$. The content of elimination theory, conversely, is the derivation of the zeros of $\mathfrak{m}$ from those of the resultant form. This converse rests on the successive elimination to be developed in this section; since in the next section the divisibility of the form $D^{(i)}(x)$ occurring in it by $E^{(i)}(x)$ will be proved, the desired converse follows from the divisibility of a power of $E^{(i)}(x)$ by $R^{(i)}(x)$.

The theory of successive elimination to be given here rests on the following result.

Theorem XI. *Let* $\mathfrak{b}$ *be an ideal of polynomials in one indeterminate* t *with coefficients in* P*, hence a principal ideal; let* $h(t)$ *be a polynomial from* $\mathfrak{b}$ *of degree* k*, and let* $\mathfrak{B}$ *be the module of linear forms in* $1, t, \dots, t^{k-1}$ *into which* $\mathfrak{b}$ *passes modulo* $h(t)$*. Then* $\mathfrak{B}$ *has rank* $k - p$*, where* p *is the degree of the basis polynomial* $f(t)$ *of* $\mathfrak{b}$.

By hypothesis $h(t) = h_1(t)f(t)$, where $h_1(t)$ has degree $k - p$. The residue classes of $\mathfrak{b}$ modulo $h(t)$ represented by $f(t), tf(t), \dots, t^{k-p-1}f(t)$ are therefore linearly independent; and every residue class of $\mathfrak{b}$ modulo $h(t)$ can be represented linearly by these $k - p$ special classes, with coefficients in P. But modulo $h(t)$ the ideal $\mathfrak{b}$ passes into a module of linear forms in $1, t, \dots, t^{k-1}$, whose rank is therefore exactly $k - p$.

Now let $\mathfrak{a} = \mathfrak{a}_1$ be a transformed ideal of polynomials, and let $A^{(1)}(x)$ be a polynomial from $\mathfrak{a}_1$, regular in x_1, of degree k_1. Let $\mathfrak{A}_1$ be the module of linear forms in $1, x_1, \dots, x_1^{k_1-1}$, with polynomials $a^{(2)}(x)$ as coefficients, into which $\mathfrak{a}_1$ passes modulo $A^{(1)}(x)$. Let $\mathfrak{a}_2$ denote the ideal in $x_2 \dots x_n$ derived from all k_1-rowed determinants from $\mathfrak{A}_1$. By Theorem XI, $\mathfrak{a}_2$ is the zero ideal if and only if the polynomials from $\mathfrak{a}_1$ have a common divisor, in the polynomial sense, of degree $p_1 > 0$ in x_1. If $\mathfrak{a}_2$ is different from the zero ideal, then, since $\mathfrak{a}$ is assumed transformed, it contains a polynomial $A^{(2)}(x)$ regular in x_2, of degree k_2, as is seen directly by composing the transformation (12) from the special transformations (25) and (26). Let $\mathfrak{A}_2$ again denote the module of linear forms in $1, x_2, \dots, x_2^{k_2-1}$ into which $\mathfrak{a}_2$ passes modulo $A^{(2)}(x)$, and $\mathfrak{a}_3$ the ideal in $x_3 \dots x_n$ derived from all k_2-rowed determinants from $\mathfrak{A}_2$; this ideal must contain a polynomial $A^{(3)}(x)$ regular in x_3.

In this way one continues the process until one reaches a zero ideal, or until $\mathfrak{a}_{n+1}$ becomes the unit ideal; for since $\mathfrak{a}_{n+1}$ is free of the x, it can only be the zero ideal or the unit ideal. It is seen that the ideals $\mathfrak{a}_1, \mathfrak{a}_2, \dots$ are uniquely determined by $\mathfrak{a}$, independently of the chosen polynomials $A^{(\lambda)}(x)$. This is clear for $\mathfrak{a}_1 = \mathfrak{a}$, and can therefore be assumed for $\mathfrak{a}_1, \dots, \mathfrak{a}_i$. If $\mathfrak{a}_i$ passes modulo $A^{(i)}(x)$ into the module of linear forms $\mathfrak{A}_i$, then $\mathfrak{A}_i$ can also be characterized as the totality of the polynomials from $\mathfrak{a}_i$ that do not reach degree k_i in x_i; thus $\mathfrak{A}_i$ depends only on the degree k_i of $A^{(i)}(x)$. Let $\bar{A}^{(i)}(x)$ be a polynomial from $\mathfrak{a}_i$, regular in x_i, of degree $\bar{k}_i > k_i$, let $\overline{\mathfrak{A}}_i$ be the corresponding module of linear forms, and let $\bar{\mathfrak{a}}_{i+1}$ be the ideal derived from the $\bar{k}_i$-rowed determinants from $\overline{\mathfrak{A}}_i$. Then the module equation holds:

$$\overline{\mathfrak{A}}_i = (\mathfrak{A}_i, A^{(i)}, x_i A^{(i)}, \dots, x_i^{\bar{k}_i - k_i - 1} A^{(i)}).$$

Since, by the regularity of $A^{(i)}(x)$ in x_i, the polynomials $A^{(i)}, x_i A^{(i)}, \dots$ can be introduced as new indeterminates of the linear forms, it follows that the k_i-rowed determinants from $\mathfrak{A}_i$ agree with the $\bar{k}_i$-rowed determinants from $\overline{\mathfrak{A}}_i$, and therefore $\bar{\mathfrak{a}}_{i+1} = \mathfrak{a}_{i+1}$.[17]

[16]Steinitz showed, in his Algebraische Theorie der Körper (J. f. M. 137 (1910), pp. 167–309), that every field can be extended, essentially uniquely, to an algebraically closed one, thus giving the rational equivalent of the fundamental theorem of algebra. In fact, for each $i = 1, \dots, n$ one is dealing with a finite extension field of $P(u, x_{i+1}, \dots, x_n)$. (E. N.)

[17]This is in principle the same inference on which the isomorphism of $\mathfrak{G}^*_{i-1} \mid \mathfrak{M}^*_{i-1}$ with $\mathfrak{G}_{i-1} \mid \mathfrak{M}_{i-1}$ rested.

Furthermore $\mathfrak{a}_{i+1}$ is always divisible by $\mathfrak{a}_i$. This is clear if $\mathfrak{a}_{i+1}$ is the zero ideal; in the opposite case $\mathfrak{A}_i$ has rank k_i, so that $\mathfrak{a}_{i+1}$ is the ideal which by definition consists of the k_i-rowed determinants from $\mathfrak{A}_i$, and is divisible by $\mathfrak{A}_i$ and hence by $\mathfrak{a}_i$. Thus every $\mathfrak{a}_{i+1}$ is divisible by $\mathfrak{a}$; if therefore $\mathfrak{a}_{n+1}$ becomes the unit ideal, then $\mathfrak{a}$ itself is the unit ideal.

Let now $\mathfrak{a}$ be different from the unit ideal, with $\mathfrak{a}_1 \neq 0, \dots, \mathfrak{a}_i \neq 0$, and $\mathfrak{a}_{i+1} = 0$. Let $D^{(i)}(x)$ be the greatest common divisor, existing by Theorem XI, of all polynomials from $\mathfrak{a}_i$, where divisor is understood in the polynomial sense; as a divisor of $A^{(i)}(x)$, $D^{(i)}(x)$ is itself regular in x_i of degree $p_i > 0$. Adjoin $x_{i+1} \dots x_n$ to $P(u)$; then $D^{(i)}(x)$ has, in an algebraic extension field of $P(u; x_{i+1} \dots x_n)$, a decomposition into p_i linear factors. Let $x_i - \bar{x}_i$ be one such linear factor, so that all polynomials from $\mathfrak{a}_i$ vanish for $x_i = \bar{x}_i$; then, by Theorem XI, the polynomials from $\mathfrak{a}_{i-1}$ acquire for this specialization a greatest common divisor $D^{(i-1)}(x)$. As a divisor of $A^{(i-1)}(x)$, $D^{(i-1)}(x)$ is again regular in x_{i-1} of degree $p_{i-1} > 0$; in particular it cannot vanish identically, and in an extension field of $P(u, \bar{x}_i, x_{i+1}, \dots, x_n)$ it decomposes into p_{i-1} linear factors. If $x_{i-1} - \bar{x}_{i-1}$ is such a linear factor, there corresponds to it a greatest common divisor from $\mathfrak{a}_{i-2}$. Continuing in this way, one reaches finitely many common zeros of $\mathfrak{a}$ lying in an algebraic extension field of $P(u; x_{i+1} \dots x_n)$. Conversely, because $\mathfrak{a}_i$ is divisible by $\mathfrak{a}$, every zero of $\mathfrak{a}$ is also a zero of $\mathfrak{a}_i$. If, in particular, $x_1 = \bar{x}_1, \dots, x_i = \bar{x}_i, x_{i+1} = x_{i+1}, \dots, x_n = x_n$ is a zero of $\mathfrak{a}$, then $\mathfrak{a}_{i+1} = 0$ follows, and the polynomials from $\mathfrak{a}_i$ acquire a greatest common divisor with the linear factor $x_i - \bar{x}_i$. Summarizing:

Theorem XII. *Let $\mathfrak{a}$ be different from the unit ideal, with $\mathfrak{a}_1 \neq 0, \dots, \mathfrak{a}_i \neq 0$ and $\mathfrak{a}_{i+1} = 0$. Then, after adjoining $x_{i+1} \dots x_n$ to $P(u)$, there exist finitely many associated value systems $\bar{x}_1 \dots \bar{x}_i$, lying in an algebraic extension field of $P(u, x_{i+1}, \dots, x_n)$, such that all polynomials from $\mathfrak{a}$ vanish for $x_1 = \bar{x}_1, \dots, x_i = \bar{x}_i, x_{i+1} = x_{i+1}, \dots, x_n = x_n$. Conversely, if such a zero of $\mathfrak{a}$ is present, then $\mathfrak{a}_{i+1} = 0$ follows, as does the occurrence of a common divisor $D^{(i)}(x)$ of all polynomials from $\mathfrak{a}_i$ that has the linear factor $x_i - \bar{x}_i$.*

Theorem XII shows in particular that every ideal $\mathfrak{a}$ having no zero is the unit ideal.

To carry out a decomposition that also explicitly exhibits the association of the value systems, introduce new indeterminates v, which may be adjoined to $P(u, x_{i+1}, \dots, x_n)$. Put

$$z_i = -v_1 x_1 - \dots - v_{i-1} x_{i-1} + x_i. \tag{35}$$

Then the composition of (12) with (35) is again a substitution of type (12), in which now only the u_{ik} are replaced by $w_{ik} = u_{ik} - v_k$, and more generally $u_{i+h,k}$ by $w_{i+h,k} = u_{i+h,k} - u_{i+h,i} v_k$. If the substitution (35) carries the, by hypothesis transformed, ideal $\mathfrak{a}$ into $\mathfrak{b}$, then $\mathfrak{b}$ is obtained from $\mathfrak{a}$ simply by replacing the $u_{\mu\nu}$ by $w_{\mu\nu}$ and adjoining the v; and by the same process the ideals $\mathfrak{b}_1, \mathfrak{b}_2, \dots$ defined by $\mathfrak{b}$ arise from $\mathfrak{a}_1, \mathfrak{a}_2, \dots$. Conversely $\mathfrak{a}_i$ is obtained from $\mathfrak{b}_i$ by $v = 0$; therefore the ideals $\mathfrak{a}_i$ and $\mathfrak{b}_i$ are simultaneously zero ideals or simultaneously different from the zero ideal.

Now again let $\mathfrak{a}_1 \neq 0; \dots; \mathfrak{a}_i \neq 0; \mathfrak{a}_{i+1} = 0$, and hence also $\mathfrak{b}_1 \neq 0; \dots; \mathfrak{b}_i \neq 0; \mathfrak{b}_{i+1} = 0$. Let $\bar{x}_1 \dots \bar{x}_i, x_{i+1} \dots x_n$ be an associated zero-system of $\mathfrak{a}$. Define $\bar{z}_i = \bar{x}_i + v_1 \bar{x}_1 + \dots + v_{i-1} \bar{x}_{i-1}$. Then $\bar{x}_1 \dots \bar{x}_{i-1}, \bar{z}_i, x_{i+1} \dots x_n$ is, by (35), a zero of $\mathfrak{b}$. Hence if $H^{(i)}(z, x)$ denotes the greatest common divisor of all polynomials from $\mathfrak{b}_i$, which may be assumed integral and primitive in the v, then by the last part of Theorem XII, $H^{(i)}(z, x)$ has the factor

$$z_i - \bar{z}_i = z_i - (\bar{x}_i + v_1 \bar{x}_1 + \dots + v_{i-1} \bar{x}_{i-1}),$$

and to every zero of $\mathfrak{a}$ occurring after adjoining $x_{i+1} \dots x_n$ there corresponds such a factor. It must be shown that this exhausts the factorization of $H^{(i)}(z, x)$, i.e. that no factor $H_1^{(i)}$ can be separated from $H^{(i)}$ which does not decompose into linear factors in z and v. If this were the case, then because of the assumed primitivity in v, $H^{(i)}$ would contain at least one linear factor $z_i - \bar{z}_i$, where $\bar{z}_i$ belongs to an algebraic extension field of $P(u, v, x_{i+1}, \dots, x_n)$. If $\bar{x}_1 \dots \bar{x}_{i-1}, \bar{z}_i, x_{i+1} \dots x_n$ is the corresponding zero of $\mathfrak{b}$, then it must coincide with one of the finitely many zeros of $\mathfrak{a}$ that occur after adjoining $x_{i+1} \dots x_n$, and so it must be independent of v. Hence $\bar{z}_i$ necessarily contains the v linearly, not algebraically or rationally non-linearly; contrary to the assumption, another linear factor $z_i - (\bar{x}_i + v_1 \bar{x}_1 + \dots + v_{i-1} \bar{x}_{i-1})$ in z and v can be split from $H_1^{(i)}$. Thus the explicit decomposition is

$$H^{(i)}(z, x) = \prod \{z_i - (\bar{x}_i + v_1 \bar{x}_1 + \dots + v_{i-1} \bar{x}_{i-1})\}^{\alpha_\lambda}.$$

Since the degree of $H^{(i)}$ and the individual exponents α_λ must agree with the corresponding ones for $D^{(i)}(x)$ – for $H^{(i)}$ arises from $D^{(i)}$ by the specialization $u_{\mu\nu} = w_{\mu\nu}$, and $D^{(i)}$ from $H^{(i)}$ by the specialization $v = 0$ – this decomposition also shows that, because the indeterminates $u_{\mu\nu}$ are adjoined to P, every zero $\bar{x}_i$ of $D^{(i)}$

occurring in the elimination starting from $\mathfrak{a}$ corresponds to a greatest common divisor from $\mathfrak{a}_{i-1} \dots \mathfrak{a}_1$ that is respectively linear, or to a power of such; hence in each case only one associated zero-system $\bar{x}_1 \dots \bar{x}_i, x_{i+1} \dots x_n$ of $\mathfrak{a}$ is obtained.[18]

§ 7. Resultant form and elimination theory.

To establish the connection between the resultant form and elimination theory, apply the successive elimination of § 6 specifically to the quotient $\mathfrak{a} = \mathfrak{m}/\mathfrak{g}_{i-1}$ of ideal by fundamental ideal of the $(i-1)$-st stage. It must first be shown that this quotient is a transformed ideal. Now $\mathfrak{m}$ is transformed and hence, by Theorem VI of § 3, so is $\mathfrak{g}_{i-1}$. From

$$H(x) \equiv 0(\mathfrak{m}/\mathfrak{g}_{i-1}); \qquad H(x) = \overline{H}(y) = \sum \alpha_i(u)\overline{H}_i(y) = \sum \alpha_i(u) H_i(x)$$

one obtains

$$\overline{H}(y)\overline{\mathfrak{g}}_{i-1,(u)} \equiv 0(\overline{\mathfrak{m}}_{(u)}), \quad \text{hence also} \quad \overline{H}(y)\overline{\mathfrak{g}}_{i-1} \equiv 0(\overline{\mathfrak{m}}_{(u)}),$$

or

$$\overline{H}_i(y)\overline{\mathfrak{g}}_{i-1} \equiv 0(\overline{\mathfrak{m}}), \quad \text{hence} \quad H_i(x)\mathfrak{g}_{i-1} \equiv 0(\mathfrak{m}).$$

This proves the divisibility of $H_i(x)$ by $\mathfrak{m}/\mathfrak{g}_{i-1}$ and hence that this ideal is transformed, so that the elimination method of § 6 applies.

But (30), taking account of (28) – § 4 – shows that $C^{(i)}(x)$ is divisible by $\mathfrak{m}/\mathfrak{g}_{i-1}$, where $C^{(i)}(x)$ denotes any non-identically vanishing determinant of degree ρ from $\mathfrak{M}^*_{i-1}$. Since for $i > 1$ this $C^{(i)}$ has degree zero in x_1, the module $\mathfrak{A}_1$ corresponding to the ideal $\mathfrak{a} = \mathfrak{a}_1$ contains the polynomials $C^{(i)}, x_1 C^{(i)}, \dots, x_1^{k_1-1}C^{(i)}$, and therefore $(C^{(i)})^{k_1}$ is divisible by $\mathfrak{a}_2$; similarly $(C^{(i)})^{k_1 k_2}$ is divisible by $\mathfrak{a}_3$, and finally $(C^{(i)})^{k_1 k_2 \cdots k_{i-1}}$ by $\mathfrak{a}_i$. Thus $\mathfrak{a}_1, \dots, \mathfrak{a}_i$ are different from the zero ideal, as also holds for $i = 1$. Let $D^{(i)}(x)$ now be the greatest common divisor of all polynomials from $\mathfrak{a}_i$, which has degree $p_i > 0$ only when $\mathfrak{a}_{i+1}$ is the zero ideal. By the definition of $D^{(i)}(x)$, and taking into account the divisibility of $\mathfrak{a}_i$ by $\mathfrak{a}$, one obtains

$$d^{(i+1)}D^{(i)}(x) \equiv 0(\mathfrak{m}/\mathfrak{g}_{i-1}); \qquad d^{(i+1)} \neq 0.$$

Thus $d^{(i+1)}D^{(i)}$ is a polynomial, free of $x_1 \dots x_{i-1}$, belonging to $\mathfrak{m}/\mathfrak{g}_{i-1}$. But $E^{(i)}(x)$ was defined (Theorem II and § 4), as the highest elementary divisor of $\mathfrak{M}_{i-1}$ or $\mathfrak{M}^*_{i-1}$, to be the greatest common divisor, in the polynomial sense, of all polynomials from $\mathfrak{m}/\mathfrak{g}_{i-1}$ that are free of $x_1 \dots x_{i-1}$. Therefore $d^{(i+1)}D^{(i)}$, and because of the regularity of $E^{(i)}(x)$ in x_i also $D^{(i)}(x)$, is divisible by $E^{(i)}(x)$.

The same argument can be carried out if the transformation (12) had from the beginning been composed with (35), that is, if the indeterminates $u_{\mu\nu}$ had been replaced by $w_{\mu\nu}$; this gives the divisibility of $H^{(i)}(z,x)$ by the corresponding $\bar{E}^{(i)}(z,x)$. Since, just as $D^{(i)}(x)$ and $H^{(i)}(z,x)$ pass into one another by specialization, so too do $E^{(i)}(x)$ and $\bar{E}^{(i)}(z,x)$, and also $R^{(i)}(x)$ and $\bar{R}^{(i)}(z,x)$, the multiplicity numbers in the decomposition into linear factors must also agree mutually. Taking into account, furthermore, that a power of $E^{(i)}(x)$ is divisible by $R^{(i)}(x)$, one obtains the following summary.

Theorem XIII. *If $R^{(i)}(x)$ is decomposed in an algebraic extension field of $P(u, x_{i+1}, \dots, x_n)$ into linear factors $x_i - \bar{x}_i$, then $\bar{x}_i$ can be completed in one and only one way to an associated value system $\bar{x}_1 \dots \bar{x}_i, x_{i+1} \dots x_n$ that is a zero of $\mathfrak{m}/\mathfrak{g}_{i-1}$ and consequently of $\mathfrak{m}$. The finitely many zeros of $\mathfrak{m}$ arising in this way from the linear factors of $R^{(i)}$ – finitely many after adjoining $x_{i+1} \dots x_n$ – are collected by the explicit decomposition*

$$\bar{R}^{(i)}(z,x) = \prod \{z_i - (\bar{x}_i + v_1\bar{x}_1 + \dots + v_{i-1}\bar{x}_{i-1})\}^{\alpha_\lambda}. \tag{36}$$

Because of the regularity of $\bar{R}^{(i)}(z,x)$ in z_i, this decomposition is preserved if the $x_{i+1} \dots x_n$ adjoined to $P(u)$ are replaced by arbitrary special value systems $\bar{x}_{i+1} \dots \bar{x}_n$ from $P(u)$ or from the algebraically closed field derived from it.

This also accomplishes a separation of the zeros of $\mathfrak{m}$ according to the individual factors of the resultant form; the number of indeterminates x adjoined to $P(u)$ is also called the dimension of the corresponding algebraic object.

[18] It should be pointed out that the successive elimination given here, in contrast to Kronecker's method, depends only on the given ideal $\mathfrak{a}$ and is independent of every ideal basis; in particular this also holds for the exponents α_λ. A complete execution of the elimination by this method would, according to Hentzelt, require the continuation of the procedure on the quotient $\mathfrak{a}/D^{(i)}$, which may be omitted in view of the next section. (E. N.)

If, in the decomposition of $\bar{R}^{(i)}(z,x)$ or of the corresponding $R^{(i)}(x)$, the factors conjugate over $P(u, x_{i+1} \dots x_n)$ are collected together – factors that for general P may be wholly or partially identical – one obtains a decomposition of $R^{(i)}(x)$ into powers of polynomials irreducible over $P(u, x_{i+1} \dots x_n)$:

$$R^{(i)}(x) = S_1^{(i)}(x)^{\beta_1} \cdots S_\nu^{(i)}(x)^{\beta_\nu}.$$

By Theorem X, this decomposition corresponds to a representation $\mathfrak{g}_i = [\mathfrak{j}_1 \dots \mathfrak{j}_\nu]$ such that $S_\mu^{(i)}(x)^{\beta_\mu}$ is equal to the norm of $\mathfrak{g}_{i-1}$ with respect to the ideal $\mathfrak{j}_\mu$, respectively to the norm of the module $\mathfrak{G}_{i-1}$ with respect to the module corresponding to $\mathfrak{j}_\mu$. This clarifies the meaning of the exponents β_μ: in particular, if γ_μ is the degree of $S_\mu^{(i)}$, then the multiplicity $\beta_\mu \gamma_\mu$ is equal to the number of residue classes of $\mathfrak{g}_{i-1}$ modulo $\mathfrak{j}_\mu$ that are linearly independent over $P(u, x_{i+1} \dots x_n)$.

Finally consider briefly the special case where P has characteristic zero. If the $u_{\mu\nu}$ are specialized to quantities $\bar{u}_{\mu\nu}$ from P in such a way that the n determinants $C^{(i)}(x)$ $(i = 1, 2, \dots, n)$ remain regular in x_i,[19] then the regularity of $R^{(i)}$ is also preserved. Under this specialization, $R^{(i)}$ and likewise $\bar{R}^{(i)}(z,x)$ become a common divisor of all ρ-rowed determinants from $\mathfrak{M}_{i-1}^*$, though not necessarily the greatest common divisor. But the specialization can always be chosen so that this latter property also remains true. For, as the existence of the module basis shows, $R^{(i)}(z,x)$ can also be characterized as the greatest common divisor of finitely many ρ-rowed determinants from $\mathfrak{M}_{i-1}^*$. The decomposition of the so specialized $\bar{R}^{(i)}(z,x)$ is then obtained simply by replacing the $u_{\mu\nu}$ by $\bar{u}_{\mu\nu}$ in (36), so that the whole elimination theory is preserved.

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23. Algebraic and Differential Invariants

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If I am to report on the development of algebraic and differential invariants, then in both areas I should like to restrict myself to the critical period which, according to a remark of Hilbert, follows the naive and formal period. This critical period is characterized, for the algebraic invariants, by the name of Hilbert himself; for the differential invariants, by the name of Riemann — or, in substantive terms: for the algebraic invariants by the arithmetical methods of algebra, which essentially developed around these questions and were able here to show their full sharpness; for the differential invariants by the methods of the formal calculus of variations. Thus I should like to report on the arithmetical methods with their significance beyond the special subject, while in the second part I can restrict myself to the subordination of the differential invariants to the algebraic ones, which is accomplished precisely in connection with the methods of the calculus of variations.

1. The *algebraic invariants* are, as is well known, based on the *general linear group* in $x_1, \dots, x_n$:

$$x_i = \sum_k s_{ik} x_k' \qquad \text{or briefly:} \qquad x = S(x'), \tag{1}$$

where the s_{ik} denote indeterminates, so that $\Delta = |s_{ik}|$ is non-zero. If now $f(a,x)$, $g(b,x), \dots$ are forms in x, of dimensions $\alpha, \beta, \dots$, with indeterminates $a, b, \dots$ as coefficients, then by means of (1) the *induced transformation* of the $a, b, \dots$ arises; for from

$$f(a,x) = \sum a_{i_1 \dots i_n} x_1^{i_1} \cdots x_n^{i_n} = \sum a'_{j_1 \dots j_n} {x'_1}^{j_1} \cdots {x'_n}^{j_n} = f(a', x'),$$
$$g(b,x) = \sum b_{r_1 \dots r_n} x_1^{r_1} \cdots x_n^{r_n} = \sum b'_{s_1 \dots s_n} {x'_1}^{s_1} \cdots {x'_n}^{s_n} = g(b', x')$$

it follows that the $a', b', \dots$ become linear homogeneous functions of the $a, b, \dots$, whose coefficients are of degrees $\alpha, \beta, \dots$ in the s_{ik}:

$$a' = A_\alpha(a); \qquad b' = B_\beta(b) \dots. \tag{2}$$

By an *invariant* one now understands an invariant with respect to the transformations (1) and (2); that is, an integral rational function of the indeterminates $a, b, \dots, x, y, \dots$ — where also $y = S(y')$ — which, under application of the transformations, reproduces itself up to a factor, a power of the substitution determinant:

$$I(a', b', \dots, x', y', \dots) = \Delta^\rho I(a, b, \dots, x, y, \dots). \tag{3}$$

[19] Hentzelt assumes this specialization of the $u_{\mu\nu}$ from the start and therefore has to call upon much more complicated considerations to prove the decomposability of $H^{(i)}(z,x)$ and $\bar{R}^{(i)}(z,x)$ into linear factors in z and v. The exponents that occur there need not necessarily agree with the ones above. (E. N.)

The coefficients of I may be assumed to be rational, or even rational integers, since every invariant with coefficients in an arbitrary number field P can be expressed linearly, with coefficients from P, by finitely many such special ones. Likewise it is no restriction to assume I separately homogeneous in each series, since here again every invariant can be assembled as a sum of finitely many such special ones.

It should be noted that conversely (1) and (2) are again determined by the requirement (3). As Ostrowski and Schur have recently shown,[20] the induced transformations can also be characterized as the totality of the linear transformations, separated in each individual series, which — apart from certain specifiable exceptions — allow an arbitrary invariant I free of the $x, y, \ldots$.

The product of two invariants is again an invariant, but the sum is one if and only if the weight — the exponent ρ in (3) — agrees. If, however, one restricts oneself to transformations of determinant one, then any sum is again an invariant and also exhausts all invariants with respect to the so restricted group. One therefore obtains an integral domain, indeed a *homogeneous integral domain*, that is, a domain for which membership of a polynomial implies membership of its homogeneous components separately — which here again amounts to the separately homogeneous invariants according to (3).

Here arises the fundamental problem around which invariant theory, and arithmetical algebra in general, has developed: *Is this integral domain finite*, that is, does it possess a *finite integral basis* $I_1, \ldots, I_k$ such that every invariant can be represented integrally and rationally through $I_1, \ldots, I_k$, $I = G(I_1, \ldots, I_k)$? Hilbert's solution of the problem — Gordan's proof for the binary case does not admit an extension because of its unmanageable symbolic computations, and the Mertens–Hilbert proof does not do so because of its use of the decomposition of a binary form into linear forms — rests on reducing the question of the integral basis to that of the ideal basis. I should like to sketch the train of thought here, emphasizing the arithmetical fundamental ideas, and then to go on to three connected circles of questions: actual construction of the basis by finitely many steps; finiteness questions for subgroups; finiteness questions taking integrality into account.

2. I recall the *definition of an ideal* for finite algebraic number fields. A system $\mathfrak{a}$ of numbers of the field is called an ideal if

1. $\mathfrak{a}$ belongs to the domain $\mathfrak{o}$ of all integers of the field,
2. together with α and β, the difference $\alpha - \beta$ also belongs to $\mathfrak{a}$,
3. together with α, $\lambda\alpha$ also belongs to $\mathfrak{a}$, where λ is any element of $\mathfrak{o}$.

And, as is well known, the theorem holds that every ideal has an *ideal basis*: $\mathfrak{a} = (\alpha_1, \ldots, \alpha_k)$; that is, there are finitely many elements $\alpha_1, \ldots, \alpha_k$ from $\mathfrak{a}$ such that, by means of 2. and 3., $\mathfrak{a}$ is derivable from $\alpha_1, \ldots, \alpha_k$; thus $\alpha = \lambda_1\alpha_1 + \cdots + \lambda_k\alpha_k$ exhausts all elements from $\mathfrak{a}$.[21]

This definition of an ideal rests only on the fact that $\mathfrak{o}$ forms an integral domain; the concept of ideal, and likewise that of ideal basis, therefore remains literally the same when $\mathfrak{o}$ is replaced by an arbitrary integral domain.

Hilbert's finiteness proof now decomposes into the following three theorems:

1. In the domain of all polynomials in n indeterminates with coefficients in a field of rationality, the ideal-basis theorem holds for every ideal — and consequently also for every arbitrary system $\mathfrak{S}$.
2. A homogeneous integral domain $\mathfrak{J}$ of polynomials is finite if and only if the ideal-basis theorem holds in $\mathfrak{J}$.
3. For the domain of invariants $\mathfrak{J}$, the ideal-basis theorem holds in the sharpened form that every ideal basis formed with respect to the domain of all polynomials is at the same time an ideal basis in $\mathfrak{J}$, so that beside

$$I = A_1I_1 + \cdots + A_kI_k \qquad \text{one always has} \qquad I = i_1I_1 + \cdots + i_kI_k,$$

where the A denote polynomials in the $a, b, \ldots, x, y, \ldots$, and the i denote invariants.

The proof of 1. rests, in principle, on the same arguments as in the algebraic number field; in both cases the existence of the ideal basis follows directly from a general theorem on modules of linear forms.[22] From 1. the existence of the ideal basis follows directly for every finite integral domain of polynomials; for homogeneous domains, the converse is also easy to see. Only in 3. are special properties of invariants used; namely, in

[20] A. Ostrowski and I. Schur, Über eine fundamentale Eigenschaft der Invarianten einer binären Form, Math. Zeitschr. 15 (1922), and related, not yet published work of Ostrowski.

[21] That k can be reduced to two is irrelevant for what follows.

[22] Compare my comparative report on the arithmetical theory of algebraic functions. Jahresber. 28 (1920), p. 187 and note 2), p. 188.

Hilbert, 3. follows from a known theorem on the Ω-process, whereas recently E. Fischer has shown that 3. can — on the basis of other general theorems — already be obtained from the property of the general linear group that, along with every transformation, it also contains the contragredient one, respectively its conjugate-complex one.[23]

3. The *actual construction of the basis* rests on a further arithmetical transformation of the notion of finiteness by drawing on the theory of *function fields.* The following holds:

> 4. An integral domain $\mathfrak{J}$ of polynomials is finite if and only if there is contained in $\mathfrak{J}$ a finite subdomain $\mathfrak{J}'$ such that $\mathfrak{J}$ depends algebraically integrally on $\mathfrak{J}'$; every integral basis of $\mathfrak{J}'$ is then supplemented by a fundamental system of these algebraically integral quantities to a basis of $\mathfrak{J}$.

[24] If, specifically, as in the case of the invariants, one deals with homogeneous domains $\mathfrak{J}$ for which the ideal-basis theorem holds in the sharper form 3., then such a domain $\mathfrak{J}'$ is derivable from some always-existing finitely many polynomials $I_1, \ldots, I_s$ as an integral basis, from whose vanishing the vanishing of all polynomials from $\mathfrak{J}$ follows. This fact is again based on a general theorem of ideal theory:

> 5. To every ideal $\mathfrak{a}$ of polynomials there belongs an ordinary rational integer r such that for every ideal $\mathfrak{b}$ which vanishes at all zeros of $\mathfrak{a}$, $\mathfrak{b}^r$ is divisible by $\mathfrak{a}$.

In the case of invariants, one can determine an *upper bound for the degrees* of $I_1, \ldots, I_s$, and thus also for their number s, depending only on the degrees of the basic forms under consideration. At this point, again, special properties of invariants are used; the relevant theorem says that a numerically specialized system of basic forms has a non-zero invariant if and only if $\Delta = |s_{ik}|$ depends algebraically integrally on the transformed coefficients.

From this upper bound K. Hentzelt[25] has determined an *upper bound for the degrees of the invariants of the integral basis*, again depending only on the degrees of the basic forms. He succeeds in doing this by giving, for the exponent r in 5., an upper bound depending only on the number and the highest degree of the polynomials of an ideal basis of $\mathfrak{a}$, independent of the coefficients of these polynomials, and computable from the given numbers in finitely many steps. In principle the problem posed is thereby completely settled; admittedly the bound given is quite high. Thus, for a ternary quadratic form, where the discriminant — that is, an invariant of degree 3 — already forms the basis of all invariants free of the x, according to Hilbert and Hentzelt one reaches high into the millions.

4. In the finiteness question for invariants of *subgroups* of the general linear group, only partial results have so far been reached. The method rests almost exclusively on reducing the invariants of subgroups to simultaneous invariants of the general linear group, according to the procedure used by Study for the orthogonal group. This has been carried out by Weitzenböck for motion invariants and affine invariants, and by Deruyts for semi-invariants and shear invariants;[26] the question of the reach of this method is still undecided. Fischer's remark mentioned at the end of 2. likewise solves the finiteness problem for a series of subgroups; in particular, it thereby gives the simplest proof for the orthogonal group. But the example of semi-invariants shows — as Fischer has recently pointed out[27] — that for subgroups, despite finiteness, the sharper form 3. for the ideal basis need not be fulfilled.

Interpreted in the sense of field theory, the finiteness question present here leads to Hilbert's problem of *relatively integral functions*, that is, to the question whether such integral domains as consist of the totality of the polynomials of a field of rational functions are always finite. In this direction lies the elementary provable fact that for the invariants of finite groups a distinguished integral basis can be given directly — the system of coefficients of the Galois resolvent.[28]

5. The question whether the finiteness theorem for invariants can be sharpened with respect to *integrality* is also not yet settled in general. To be sure, as Hilbert showed, the ideal-basis theorem also holds for the domain

[23] E. Fischer, Über die Endlichkeit der Invarianten. Göttinger Nachrichten 1915, and Leipzig lecture.

[24] Hilbert shows (Über die vollen Invariantensysteme, Math. Annalen 42 (1893), §§ 1 and 2) that these facts follow from the assumed finiteness; conversely, that finiteness follows from algebraically integral dependence follows from the existence of the rational basis (E. Noether, Körper und Systeme rationaler Funktionen, Math. Annalen 76 (1915), § 12).

[25] I shall shortly publish the work from the estate.

[26] For literature references, compare Weitzenböck's encyclopedia article on invariant theory; III E 1.

[27] Leipzig lecture and Math. Zeitschrift.

[28] E. Noether, Der Endlichkeitssatz der Invarianten endlicher Gruppen. Math. Ann. 77 (1916).

of all integral polynomials; and likewise the connection 2. between ideal basis and integral basis remains valid;[29] but the proof for the — unnecessary — sharpened version 3. fails. For the case of binary invariants, integral finiteness can be proved;[30] the proof, which represents an integrality sharpening of the binary Mertens–Hilbert finiteness proof, uses, besides the theorem on the integral ideal basis, also the decomposition of binary forms into linear forms, and therefore does not permit transfer to more indeterminates. On similar foundations, by contrast, integral finiteness for the invariants of finite groups follows, as a sharpening of the proof mentioned above, although now again only an existence proof and no actual specification of the basis results.[31] Here too, only a further development of the theory of function fields and of general ideal theory will presumably lead to the goal.

6. I now pass to the *differential invariants* and to the question raised in the introduction concerning their reduction to the theory of algebraic invariants. The underlying group is, as is well known, defined by means of $x_i = x_i(y)$ as:

$$\mathrm{d}x_i = \sum_k \frac{\partial x_i}{\partial y_k}\mathrm{d}y_k; \qquad \mathrm{d}\delta x_i = \sum_k \frac{\partial x_i}{\partial y_k}\mathrm{d}\delta y_k + \sum_{k,l} \frac{\partial^2 x_i}{\partial y_k \partial y_l}\mathrm{d}y_k\,\delta y_l;\ \ldots. \tag{4}$$

To pass to the transformations induced by (4), one may begin with an arbitrary function $f(x, \mathrm{d}x) = \varphi(y, \mathrm{d}y)$; by requiring the invariance of $\delta f, \delta^2 f, \ldots$ with respect to (4), one obtains the transformations for the derivatives of f with respect to x and $\mathrm{d}x$. If one restricts oneself, for instance, to forms homogeneous in the $\mathrm{d}x$:

$$f(x, \mathrm{d}x) = \sum a_{i_1 \ldots i_n}(x)\mathrm{d}x_1^{i_1} \cdots \mathrm{d}x_n^{i_n} = \sum a'_{i_1 \ldots i_n}(y)\mathrm{d}y_1^{i_1} \cdots \mathrm{d}y_n^{i_n} = \varphi(y, \mathrm{d}y),$$

then the a' are linear in the a, the $\frac{\partial a'}{\partial y}$ are linear in the a and $\frac{\partial a}{\partial x}$, and so on:

$$a' = A_0(a), \qquad \frac{\partial a'}{\partial y} = A_1\left(a, \frac{\partial a}{\partial x}\right), \qquad \frac{\partial^2 a'}{\partial y^2} = A_2\left(a, \frac{\partial a}{\partial x}, \frac{\partial^2 a}{\partial x^2}\right), \ldots \tag{5}$$

and the question is one of invariance with respect to the transformations (4) and (5). Here all the quantities occurring,

$$a,\ \frac{\partial a}{\partial x}, \ldots,\ \mathrm{d}x,\ \delta x,\ \mathrm{d}\delta x, \ldots,\ \frac{\partial x}{\partial y},\ \frac{\partial^2 x}{\partial y^2}, \ldots,$$

are to be regarded as indeterminates, so the determinant of each individual transformation is certainly non-zero. Only when the formally completed theory is applied to special functions does the question arise of restriction to such functions that the transformations remain uniquely reversible in a certain domain and that a sufficient number of differential quotients exist, exactly as in the algebraic case, under numerical specialization, the determinant $|s_{ik}|$ must be assumed non-zero.

Under this treatment of all arguments occurring as indeterminates, one is thus dealing with a special linear group, not directly accessible to the ordinary algebraic methods, in which the transformations of $\mathrm{d}x$ and $a' = A_0(a)$ agree with (1) and the induced (2). The question arises whether, perhaps quite generally, by the *introduction of new arguments*, the transformations (4) and (5) can be reduced to the general linear group (1) and the transformations (2) thereby *induced.*

This is in fact the case. The higher differentials $\mathrm{d}^2x, \delta\mathrm{d}x, \ldots$ are to be replaced by the Lagrange derivatives arising from the variational problem belonging to $f(x, \mathrm{d}x)$, and by further combinations formed from them by polar formation (transmissions in the sense of Weyl and Schouten), all of which are cogredient to the $\mathrm{d}x$, so that their transformation is given by the general linear group (1). The quantities $\frac{\partial a}{\partial x}, \frac{\partial^2 a}{\partial x^2}, \ldots$ (or, more generally, the derivatives of the homogeneously assumed f with respect to x and $\mathrm{d}x$), however, are to be replaced by the coefficients of certain forms arising from the "normal forms of the higher variations" by the above elimination of the $\mathrm{d}^2x, \delta\mathrm{d}x, \ldots$, and of their "covariant derivatives"

$$[\Omega_i], \quad [\Omega_i^{(1)}], \quad [\Omega_i^{(2)}], \ldots \qquad (i = 1, 2, \ldots);$$

and since the $[\Omega_i^{(k)}]$ are invariants with respect to (4) and (5) which depend only on the $\mathrm{d}x$ and δx, these coefficients satisfy the algebraically induced transformations (2).[32] Through this *reduction theorem*, the theory of differential invariants is subordinated to algebraic invariant theory.

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[29] From this connection it follows in particular that the decomposition theorems of general ideal theory hold for all finite integral domains, as I developed them in Math. Ann. 83 (1921).

[30] E. Noether, Die Endlichkeit des Systems der ganzzahligen Invarianten binärer Formen. Göttinger Nachrichten 1919.

[31] § 4 of the work cited under 11).

[32] E. Noether, Invarianten beliebiger Differentialausdrücke. Göttinger Nachrichten 1918.

The series of the $[\Omega_i]$ breaks off with $[\Omega_\alpha]$ if $f(x, \mathrm{d}x)$ is a homogeneous form of α-th dimension in the $\mathrm{d}x$. For quadratic forms, $[\Omega_2]$ becomes identical with Riemann's curvature form, and the formation process indicated here is exactly the one given by Riemann in the Latin prize essay, which Lipschitz, independently of this and in connection with Riemann's habilitation lecture, also developed.[33] The proof of the reduction theorem, which Christoffel and Ricci had carried out computationally for quadratic differential forms, rests on the introduction of "Riemann normal coordinates," which transform the extremals emanating from a point into straight lines; since thereby to an arbitrary transformation of the x there corresponds a linear transformation of the normal coordinates, these determine the passage to the general linear group.

The question arises whether such a reduction theorem also exists when, following Weyl and Schouten, the group is defined not by a variational problem but only by a transmission; that is, when, corresponding to the Lagrange derivatives, expressions $\psi_i(\mathrm{d}\delta x, \mathrm{d}x, \delta x)$ are set up through which the elimination of the higher differentials is achieved, with the restriction, unnecessary in itself, that the ψ_i should be linear in the $\mathrm{d}x$ being added in analogy with quadratic differential forms (for arbitrary homogeneous $f(x, \mathrm{d}x)$, the polarized Lagrange derivatives become homogeneous of first order only in $\mathrm{d}x$). By the requirement of cogredience of ψ to the $\mathrm{d}x$, the transformation of their coefficients is induced, and thus, in analogy with (5), the group is fixed. For invariants of the lowest orders, Weitzenböck has confirmed the reduction theorem computationally in the Weyl case; a general Ansatz is still lacking.[34]

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24. Elimination Theory and General Ideal Theory

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In what follows the question is the placement of elimination theory – in the arithmetical form that I gave to Hentzelt's presentation[35] and that may be described as an ideal theory in the polynomial domain – within general ideal theory,[36] and, at the same time, a new foundation of the parts referring to zeroes. The basic concepts of both theories are briefly assembled in § 1, as are those of field theory,[37] so that I can refer to them here.

The classification and the new foundation group themselves around four questions: the arithmetical formulation of the concept of dimension; the theory of zeroes; the connection between the decomposition theorems for norms and elementary divisors and those of general ideal theory; the characterization of prime ideals and primary ideals by norm and elementary-divisor form.

The arithmetical formulation of the *concept of dimension* (§ 4) is given by chains of prime ideals, the arithmetical equivalent of the fact that on surfaces there are curves, and on curves there are points. This arithmetical formulation, which is proved identical with the parameter definition of elimination theory, can be transferred with a slight modification to arbitrary ring domains and, together with the transfer of certain parts of the other questions, gives there an exact insight into the structure of ideals, as I shall show elsewhere.

In H.-N. the question of *zeroes* is, as usual, attacked directly for the given ideal, namely by going back to successive elimination, whereby the character of verification is not entirely avoided. In contrast stands the route always taken for polynomials in one variable: one represents the given polynomial as a product of powers of prime functions (primary functions) and constructs, for each individual prime function, the field of zeroes as a field isomorphic to the residue-class field. The degree of the prime function gives the degree of this field; the degree of the primary function, however, whose zeroes coincide with those of the prime function, gives the degree of the ring of residue classes modulo this primary function. If, for each prime function, one passes to the Galois field, one obtains the decomposition into linear factors; and with this, for the originally given polynomial, the question of zeroes, decomposition into linear factors, and multiplicity is settled. Quite correspondingly, here (§ 3) the system of residue classes modulo a prime ideal is extended, by forming quotients, to the residue-class field, and a field of zeroes is constructed that is isomorphic to it. This field of zeroes contains a subfield generated by adjoining indeterminates – say $x_{i+1}, \ldots, x_n$; the degree of the norm gives the degree with respect to this subfield; at the same time, by passage to the Galois field, the decomposition of the elementary-divisor form and hence of the norm into linear factors is obtained. For the

[33] Compare also my account of Riemann and Lipschitz in no. 28 of the second part of Weitzenböck's encyclopedia article mentioned under 7).

[34] Compare: Weyl, Raum, Zeit, Materie; Schouten, Math. Zeitschr. 13 (1922); Weitzenböck, Sitzungsber. d. Wiener Akademie, 1920 and 1921.

[35] Kurt Hentzelt, Zur Theorie der Polynomideale und Resultanten. Edited by E. Noether. Math. Ann. 88 (1922), pp. 53–79; cited H.-N.

[36] E. Noether, Idealtheorie in Ringbereichen, Math. Ann. 83 (1921), pp. 24–66; cited Ideal Theory.

[37] E. Steinitz, Algebraische Theorie der Körper, J. f. M. 137 (1910), pp. 167–309; cited Steinitz.

corresponding primary ideal, however, the zeroes are the same; the decomposition is also given along with it, since the norm becomes a power of the elementary-divisor form of the prime ideal (§ 2); the degree of the norm gives the degree of the ring of residue classes modulo the primary ideal with respect to the subfield derived from the indeterminates.

Thus the question of zeroes of an arbitrary ideal is reduced to the connection between the decomposition theorems for norms and elementary divisors and those of general ideal theory (§ 5). The fact that the zeroes of the ideal are composed from those of its individual associated prime ideals corresponds to the decomposition of the norm into greatest primary factors, which become equal to powers of the elementary-divisor form of the individual associated prime ideals;[38] thereby the decomposition into linear factors is accomplished for the norm in general. Multiplicity, however, receives its interpretation in H.-N. by means of the fundamental ideals and their components; the degree of a greatest primary factor of the norm is made equal to the number of linearly independent residue classes – after adjoining $x_{i+1}, \dots, x_n$ – of the fundamental ideal of $(i-1)$-st stage modulo a component of the fundamental ideal of i-th stage. Now the fundamental ideal of $(i-1)$-st stage is proved identical with the isolated component of the decomposition that is uniquely defined by all prime ideals of dimension higher than the $(n-i)$-th; and a component of the fundamental ideal of i-th stage is proved identical with the isolated component of the decomposition defined by the prime ideal corresponding to the factor of the norm and by all prime ideals of higher dimension. The degree of the corresponding factor of the norm becomes – after adjoining $x_{i+1}, \dots, x_n$ – equal to the degree of that subring of the residue classes of this component that consists only of classes of the fundamental ideal of $(i-1)$-st stage.

The characterization of the prime ideals and proper primary ideals (§ 6) results as a direct consequence of the consideration of the residue-class fields. If the original coefficient domain is a perfect field, then prime functions correspond to the prime ideals as norm and elementary-divisor form, proper primary functions to the proper primary ideals, and conversely. For an imperfect field as coefficient domain only conditions can be given that are either necessary or sufficient; as examples show, the norm of a prime ideal can become properly primary, and the elementary-divisor form of a proper primary ideal can become a prime function. More precisely, in characteristic p the norm of a prime ideal becomes the p^f-th power ($f > 0$) of its elementary-divisor form; whereas, for proper primary ideals whose elementary-divisor form is a prime function, the norm can become an arbitrary power, as examples again show. Finally, absolute prime ideals are considered (§ 7), that is, those that remain prime ideals when the coefficient domain is extended to an algebraically closed field. For absolute prime ideals with coefficients from a (finite) algebraic number field, the characterization leads to the transfer of a theorem first stated by A. Ostrowski for absolute prime functions: the property of being a prime ideal is preserved modulo every prime ideal of the number field, with at most finitely many exceptions.

Let the analogy of the last question with ideal theory in algebraic number fields also be pointed out. As the elementary-divisor form of such an ideal one has to regard the least integral rational number divisible by the ideal (note 10), which for a prime ideal therefore always becomes a prime number, whereas conversely an ideal becomes a prime ideal as soon as its norm is a prime number; the proofs also run quite in parallel. The examples mentioned above thus show that, over an imperfect field, the analogue of prime ideals of higher degree and ramification ideals occurs, whereas over a perfect field there are only prime ideals of first degree and no ramification ideals.

§ 1. Basic Concepts of Elimination Theory, General Ideal Theory, and Field Theory

1. The domain. Let $\mathfrak{S}$ denote the integral domain of all polynomials in $y_1, \dots, y_n$ with coefficients from an abstractly defined field P. Let the y be subjected to a transformation with indeterminates $u_{\mu\nu}$ as coefficients:[39]

$$y_1 = u_{11}x_1 + \dots + u_{1n}x_n, \quad \dots, \quad y_n = u_{n1}x_1 + \dots + u_{nn}x_n, \tag{1}$$

with inverse

$$x_1 = t_{11}y_1 + \dots + t_{n1}y_n, \quad \dots, \quad x_n = t_{1n}y_1 + \dots + t_{nn}y_n. \tag{1'}$$

The $u_{\mu\nu}$, or – what is equivalent because of mutual rational expressibility – the $t_{\mu\nu}$, are adjoined to P, so that the coefficient domain $P(u) = P(t)$ arises; let $\mathfrak{S}'$ denote the integral domain of all polynomials in $x_1, \dots, x_n$ with coefficients from $P(u)$. The domains $\mathfrak{S}$ and $\mathfrak{S}'$ underlie elimination theory; ideals $\mathfrak{a}, \mathfrak{b}, \dots$ in $\mathfrak{S}$ and $\mathfrak{a}', \mathfrak{b}', \dots$ in $\mathfrak{S}'$ are considered. An ideal in an arbitrary ring is here defined in the usual way by the requirement

[38]This parallelism between elimination theory and general ideal theory is not fulfilled in Kronecker's elimination theory; compare H.-N., note *).

[39]The transformation is, with a view to § 3 and so on, somewhat more general than in H.-N.; all results there are thereby retained all the more.

that, together with α and β, it also contain $\alpha - \beta$, and, together with α, also $\lambda\alpha$, where λ is an arbitrary ring element; $\mathfrak{o}$ will always denote the unit ideal consisting of all elements.

2. Transformed ideals. An ideal $\mathfrak{m}$ in $\mathfrak{S}'$ is called transformed if it arises from an ideal $\overline{\mathfrak{m}}$ in $\mathfrak{S}$ by means of (1) after adjoining the $u_{\mu\nu}$, or the $t_{\mu\nu}$. Thus $\mathfrak{m}$ consists, when $\bar{f}(y)$ or $\bar{g}(y)$ runs through all polynomials in $\overline{\mathfrak{m}}$, of all linear combinations

$$f(x) = \sum U_i(u)\bar{f}_i(y) = \sum U_i(u)f_i(x)$$

or also

$$g(x) = \sum T_i(t)\bar{g}_i(y) = \sum T_i(t)g_i(x).$$

In particular all $f_i(x) = \bar{f}_i(y)$ are contained, or, what amounts to the same thing, all $g_i(x) = \bar{g}_i(y)$. Since $f(x)$ respectively $g(x)$ are fixed only up to quantities from $P(u) = P(t)$, the U_i, T_i may, without loss of generality, be assumed to be power products of the u, respectively the t. The polynomials $f(x), g(x)$ again pass into polynomials of $\mathfrak{m}$ if U, T are replaced by arbitrary other power products, in particular by interchanging the columns of (1), respectively the rows of (1'); hence $\mathfrak{m}$ contains, together with any polynomial, all those arising from it by interchanging the x, provided only that the corresponding interchanges are carried out in the coefficients depending on u, respectively t. All ideals occurring in what follows are to be assumed transformed unless the contrary is expressly stated. Non-transformed ideals pass, by composing (1) with

$$x_i = v_{i1}z_1 + \cdots + v_{in}z_n,$$

into transformed ideals in the polynomial domain of the z with coefficients from $P(u, v)$, whereas for transformed ideals this composition amounts merely to replacing the $u_{\mu\nu}$ by bilinear combinations of the u, v.

3. Notation. By $f^{(i)}, g^{(i)}, \ldots, a^{(i)}, b^{(i)}, \ldots$ we shall throughout understand polynomials in $\mathfrak{S}'$ that are free of $x_i, \ldots, x_{i-1}$.

4. Fundamental ideals. To every ideal $\mathfrak{m}$ there are assigned n fundamental ideals $\mathfrak{g}_0, \mathfrak{g}_1, \ldots, \mathfrak{g}_{n-1}$ by the following stipulation: the fundamental ideal of $(i-1)$-st stage, $\mathfrak{g}_{i-1}$, contains all and only those polynomials $G(x)$ for which there exists a polynomial $b^{(i)} \neq 0$ – in general varying with $G(x)$ – such that $b^{(i)}G(x) \equiv 0 \pmod{\mathfrak{m}}$. From the existence of the ideal basis follows also the existence of at least one $B^{(i)}$, fixed for all $G(x)$, so that

$$B^{(i)}\mathfrak{g}_{i-1} \equiv 0 \pmod{\mathfrak{m}}, \qquad B^{(i)} \neq 0. \tag{2}$$

The fundamental ideals of transformed ideals themselves become transformed ideals (H.-N., Theorem VI). The fundamental ideal of i-th stage $\mathfrak{g}_i$ is also defined as the ideal into which $\mathfrak{m}$ passes when $x_{i+1}, \ldots, x_n$ are adjoined to $P(u)$, provided one restricts oneself – which then means no loss of generality – to polynomials integral in $x_{i+1}, \ldots, x_n$; $\mathfrak{g}_0$ is always equal to the unit ideal $\mathfrak{o}$.

5. Module representation of ideals, elementary divisors and individual norms. If every polynomial $f(x)$ is regarded as a linear form in the power products ξ of $x_1, \ldots, x_{i-1}$, with polynomials $a^{(i)}$ as coefficients, then the ideal $\mathfrak{m}$ passes into a module $\mathfrak{M}_{i-1}$ of linear forms with respect to the domain of the $a^{(i)}$; that is, $\mathfrak{M}_{i-1}$ contains, together with any two linear forms, also their difference, and, together with a linear form $l(\xi)$, also $a^{(i)}l(\xi)$ for arbitrary $a^{(i)}$. As fundamental module of such a module $\mathfrak{M}$ one denotes the totality of linear forms $g(\xi)$ for which $b^{(i)}g(\xi) \equiv 0 \pmod{\mathfrak{M}}$, $b^{(i)} \neq 0$; hence the fundamental ideal $\mathfrak{g}_{i-1}$ passes, under the module representation, into the fundamental module $\mathfrak{G}_{i-1}$ of $\mathfrak{M}_{i-1}$. $\mathfrak{M}_{i-1}$ and $\mathfrak{G}_{i-1}$ depend on infinitely many indeterminates ξ, in such a way that in each individual linear form only finitely many of these indeterminates occur. $\mathfrak{M}_{i-1}$ has the characteristic property that, after adjoining $x_{i+1}, \ldots, x_n$ to $P(u)$, it has only finitely many elementary divisors different from unity; that is, after this adjunction $\mathfrak{G}_{i-1}$ and $\mathfrak{M}_{i-1}$ admit a basis representation – with ζ denoting new indeterminates, connected with the ξ by invertible linear transformations integral in x_i, in such a way that in $\zeta_i = r_i(\xi)$ only finitely many of these indeterminates occur at a time –

$$\begin{aligned}\mathfrak{G}_{i-1} &= (\zeta_0, \zeta_1, \ldots, \zeta_\sigma, \zeta_{\sigma+1}, \ldots, \zeta_{\sigma+\nu}, \ldots),\\ \mathfrak{M}_{i-1} &= (E_0^{(i)}\zeta_0, E_1^{(i)}\zeta_1, \ldots, E_\sigma^{(i)}\zeta_\sigma, \zeta_{\sigma+1}, \ldots, \zeta_{\sigma+\nu}, \ldots),\end{aligned} \tag{3}$$

where each $E_\mu^{(i)}$ divides the preceding one (H.-N. (24) and (28)).

The highest elementary divisor $E^{(i)}$ is also defined as the greatest common divisor – in the polynomial sense – of all $B^{(i)}$ occurring in (2); it can be assumed integral and primitive in $x_{i+1}, \ldots, x_n$, and is then regular in x_i, that is, $E^{(i)}$ contains the term x_i^λ with non-zero coefficient, if it is of degree λ in the x.

The product $R^{(i)}$ of all elementary divisors occurring in (3) is called the norm of $\mathfrak{G}_{i-1}$ with respect to $\mathfrak{M}_{i-1}$, or also the individual norm of the ideal $\mathfrak{m}$; in symbols, taking account of the fact that $\mathfrak{m}$ passes into $\mathfrak{g}_i$ when $x_{i+1}, \ldots, x_n$ are adjoined:

$$R^{(i)} = E_0^{(i)} E_1^{(i)} \cdots E_\sigma^{(i)} = N(\mathfrak{G}_{i-1} \mid \mathfrak{M}_{i-1}) = N(\mathfrak{g}_{i-1} \mid \mathfrak{g}_i). \tag{5}$$

As (3) shows, after adjoining $x_{i+1}, \ldots, x_n$, $R^{(i)}$ becomes equal to the determinant of the transition substitution from $\mathfrak{G}_{i-1}$ to $\mathfrak{M}_{i-1}$, and the degree of $R^{(i)}$ gives the number of residue classes, linearly independent over $P(u, x_{i+1}, \ldots, x_n)$, of $\mathfrak{G}_{i-1}$ modulo $\mathfrak{M}_{i-1}$, hence of $\mathfrak{g}_{i-1}$ modulo $\mathfrak{g}_i$; $R^{(i)}$ can be assumed integral and primitive in $x_{i+1}, \ldots, x_n$, and is then regular in x_i. $R^{(i)}$ can be computed as the greatest common divisor – in the polynomial sense – of all ϱ-rowed determinants of a module $\mathfrak{M}^*_{i-1}$, always existing, of rank ϱ, having the property that the residue-class system $\mathfrak{G}^*_{i-1}/\mathfrak{M}^*_{i-1}$ is isomorphic to $\mathfrak{G}_{i-1}/\mathfrak{M}_{i-1}$, where $\mathfrak{G}^*_{i-1}$ is understood as the fundamental module of $\mathfrak{M}^*_{i-1}$ (H.-N., Theorem VII and § 5).

6. Elementary-divisor form and norm (resultant form) of ideals. The product $E_\mathfrak{m} = E^{(1)} E^{(2)} \cdots E^{(n)}$ of all highest elementary divisors is called the elementary-divisor form, and the product $R_\mathfrak{m} = R^{(1)} R^{(2)} \cdots R^{(n)}$ of all individual norms the norm (resultant form)[40] of $\mathfrak{m}$. The elementary-divisor form and the norm are divisible by the ideal; the norm is divisible by the elementary-divisor form, and a power of the latter by the norm. More generally one still has:

$$\begin{aligned} &E^{(i)} \mathfrak{g}_{i-1} \equiv 0 \pmod{\mathfrak{g}_i}, && E^{(n)} \cdots E^{(i)} \mathfrak{g}_{i-1} \equiv 0 \pmod{\mathfrak{m}}, \\ &R^{(i)} \mathfrak{g}_{i-1} \equiv 0 \pmod{\mathfrak{g}_i}, && R^{(n)} \cdots R^{(i)} \mathfrak{g}_{i-1} \equiv 0 \pmod{\mathfrak{m}}, \end{aligned} \tag{4}$$

(H.-N., Theorem VIII), and further: if $\mathfrak{m}$ is divisible by $\mathfrak{n}$, and the norms $R_\mathfrak{m}$ and $R_\mathfrak{n}$ agree, then the ideals $\mathfrak{m}$ and $\mathfrak{n}$ also agree (H.-N., Theorem IX). Taking into account that, for a divisor of $\mathfrak{m}$, agreement of $R^{(1)}, R^{(2)}, \ldots$ entails agreement of the fundamental ideals $\mathfrak{g}_1, \mathfrak{g}_2, \ldots$, and that always $\mathfrak{g}_0 = \mathfrak{o}$, one obtains correspondingly: if $\mathfrak{n}$ is a proper divisor of $\mathfrak{m}$, then the first individual norm of $\mathfrak{n}$ that differs from the corresponding one of $\mathfrak{m}$ becomes a proper divisor.[41] If $\mathfrak{m}$ contains a polynomial $G^{(i)} \neq 0$, then by definition $\mathfrak{g}_0, \mathfrak{g}_1, \ldots, \mathfrak{g}_{i-1}$ are equal to the unit ideal; consequently, by the property of individual norms stated in 5, namely to determine the number of linearly independent residue classes, $R^{(1)}, \ldots, R^{(i-1)}$ become equal to unity; hence also $E^{(1)}, \ldots, E^{(i-1)}$ become equal to unity.

7. Decomposition of the individual norms and elementary divisors; components of the fundamental ideals. By a prime function one understands a polynomial irreducible with respect to $P(u)$; by a primary function, the power of a prime function. A proper primary function is one that is not at the same time a prime function, where therefore powers higher than the first are involved. A decomposition of a polynomial into greatest primary factors is one in which every factor is a primary function, but the product of any two factors is no longer primary.[42] To the decomposition of the individual norm $R^{(i)} = N(\mathfrak{g}_{i-1} \mid \mathfrak{g}_i)$ into greatest primary factors $Q_\nu^{(i)}$ there corresponds a representation of $\mathfrak{g}_i$ as a least common multiple

$$\mathfrak{g}_i = [\mathfrak{r}_1, \mathfrak{r}_2, \ldots, \mathfrak{r}_s],$$

such that $Q_\nu^{(i)} \mathfrak{g}_{i-1} \equiv 0 \pmod{\mathfrak{r}_\nu}$; here $\mathfrak{r}_\nu$ possesses $\mathfrak{g}_{i-1}$ as its fundamental ideal of $(i-1)$-st stage, and agrees with its fundamental ideal of i-th stage; the $\mathfrak{r}_\nu$ are called the components of the fundamental ideals. The highest elementary divisor of $\mathfrak{r}_\nu$ becomes equal to the greatest common divisor – in the polynomial sense – of $Q_\nu^{(i)}$ and $E^{(i)}$, hence equal to a greatest primary factor of $E^{(i)}$. The $\mathfrak{r}_\nu$ are uniquely determined by their behavior, stated above, with respect to the fundamental ideals and by the requirement that either the individual norm or the highest elementary divisor should be a greatest primary factor of $R^{(i)}$ respectively $E^{(i)}$ (H.-N., Theorem X).

8. Concept of dimension. To the ideal $\mathfrak{m}$ the highest dimension $n-i$ is assigned if $R^{(i)}$ is the first individual norm different from unity – or, what by 6 is equivalent, if $\mathfrak{g}_i$ is the first fundamental ideal different from the unit ideal; the dimension -1 is assigned to the unit ideal $\mathfrak{o}$. If there is only one individual norm $R^{(i)}$ different from unity, so that $\mathfrak{g}_i = \mathfrak{g}_{i+1} = \cdots = \mathfrak{m}$, then the highest dimension is also called simply the dimension of $\mathfrak{m}$. If $\mathfrak{m}$ contains a polynomial $G^{(i)} \neq 0$, then it is of highest dimension at most $n-i$, as the

[40] In H.-N. only the term resultant form is used; the term norm corresponds to the arithmetical properties.

[41] On the other hand, later factors of $R_\mathfrak{n}$ can become multiples of the corresponding factors of $R_\mathfrak{m}$; for instance always when $\mathfrak{m}$ is a prime ideal and $\mathfrak{n}$ is not the unit ideal.

[42] The definition of prime function and primary function could also be formulated in exact analogy with 11 by replacing only the word ideal by polynomial. The greatest primary factors are the analogue of the greatest primary components of the decomposition in 12.

remark at the end of 6 shows. If $\mathfrak{m}$ has highest dimension $n-i$ and $\mathfrak{n}$ is a divisor of $\mathfrak{m}$, then $\mathfrak{n}$ also has highest dimension at most $n-i$.

9. Norm (resultant form) and zeroes. By zeroes of an ideal $\mathfrak{m}$ one understands all systems of values of $x_1,\dots,x_i$ belonging to a suitable algebraic extension field of $P(u;x_{i+1},\dots,x_n)$ $(i=1,2,\dots,n)$ for which – after adjoining $x_{i+1},\dots,x_n$ to the coefficient domain – all polynomials from $\mathfrak{m}$ vanish. To each factor $R^{(i)}$ of the norm (resultant form), under this adjunction, there correspond as many zeroes of $\mathfrak{m}$ as the degree of $R^{(i)}$ indicates; and $R^{(i)}$, written in bilinear combinations $w_{\mu\nu}$ of the $u_{\mu\nu}$ and further indeterminates $v_{\mu\nu}$, admits the explicit decomposition

$$R^{(i)}=\prod_{\nu}(z_i-z_i^{(\nu)})=\prod_{\nu}(v_{i1}x_1+\dots+v_{in}x_n-z_i^{(\nu)}),\tag{5}$$

where $x_1^{(\nu)},\dots,x_i^{(\nu)}$ denotes a system of associated zeroes (H.-N., Theorem XIII; a new foundation will be given in § 3). Thus among the zeroes of an ideal of highest dimension $n-i$ there are zeroes depending on $(n-i)$ parameters, but none depending on more parameters; this proves the agreement of the concept of dimension given in 8 with the usual formulation.

10. The ring domain of general ideal theory. The underlying domain is to be a commutative ring in which the theorem of the finite chain holds: every chain of ideals $\mathfrak{a}_1,\mathfrak{a}_2,\dots,\mathfrak{a}_k,\dots$, where each $\mathfrak{a}_i$ is a proper divisor – divisor in the ideal sense – of the immediately preceding one, breaks off after finitely many terms. This requirement is equivalent to the other, that every ideal possesses an ideal basis (Ideal Theory, Theorem I), and is therefore fulfilled in particular for the polynomial domain. In the following numbers 11, 12, 13, this general ring domain is assumed.

11. Prime ideals and primary ideals. An ideal $\mathfrak{p}$ is called a prime ideal if from the divisibility of a product by $\mathfrak{p}$ follows the divisibility of at least one factor; $\mathfrak{q}$ is called a primary ideal – or primary – if from the divisibility of a product by $\mathfrak{q}$ follows the divisibility of one factor or of a power of every factor. In symbols:

$$a\not\equiv 0\pmod{\mathfrak{p}},\quad b\not\equiv 0\pmod{\mathfrak{p}}\quad\text{implies}\quad ab\not\equiv 0\pmod{\mathfrak{p}};$$

$$a\not\equiv 0\pmod{\mathfrak{q}},\quad b^x\not\equiv 0\pmod{\mathfrak{q}}\text{ for every power }x\quad\text{implies}\quad ab\not\equiv 0\pmod{\mathfrak{q}}.$$

To every primary ideal $\mathfrak{q}$ there is one and only one associated prime ideal $\mathfrak{p}$, which is a divisor of $\mathfrak{q}$ and a power of which is divisible by $\mathfrak{q}$, $\mathfrak{p}\neq\mathfrak{o}$, $\mathfrak{p}^\rho\equiv 0\pmod{\mathfrak{q}}$; here $\mathfrak{p}$ consists of all ring elements of which a power is divisible by $\mathfrak{q}$. As the definition shows, every prime ideal is at the same time a primary ideal; proper primary ideals mean those that are not at the same time prime ideals, so that $\rho>1$.

12. The decomposition theorem. A representation $\mathfrak{a}=[\mathfrak{q}_1,\dots,\mathfrak{q}_s]$ of an ideal as a least common multiple is called shortest if no $\mathfrak{q}$ is contained in the least common multiple of the others – in its complement; it is called a representation by greatest primary components if every $\mathfrak{q}$ is primary, but the least common multiple of any two $\mathfrak{q}$'s is not primary. Every ideal admits a shortest representation by finitely many greatest primary components; for two different such representations, the number of components and the associated prime ideals, all of which are mutually distinct, agree. The prime ideals thus uniquely determined are to be called the prime ideals, or associated prime ideals, of the ideal (Ideal Theory, Theorem IX).

13. Isolated components of the decomposition. An ideal $\mathfrak{r}=[\mathfrak{q}_1,\dots,\mathfrak{q}_i]$ is called an isolated component of the decomposition of $\mathfrak{a}$ if the $\mathfrak{q}_i$ occur in at least one shortest representation of $\mathfrak{a}$ by greatest primary components and satisfy the condition that none of their associated prime ideals is contained in one of the other prime ideals of $\mathfrak{a}$. The isolated components of the decomposition are uniquely determined by their prime ideals; in particular the isolated greatest primary components are uniquely determined (Ideal Theory, Theorem XIII).

14. Characteristic, perfect and imperfect fields, extension of the first kind. A field Ω is said to have characteristic zero if the prime field contained in Ω and derived from the unit is of the type of the field of rational numbers; it has characteristic p if this prime field is of the type of the residue-class system modulo a prime number p. A field Ω is called perfect if every prime function with coefficients from Ω decomposes in a suitable extension field into distinct linear factors; otherwise it is called imperfect. Every field of characteristic zero is perfect; a field of characteristic p is perfect if and only if, together with every element, it also contains its p-th root. A prime function in an imperfect field is an integral function of degree r in x^{p^f}, and decomposes in a suitable extension field into p^f-th powers of r distinct linear factors; f is called the exponent. A prime function is said to be of the first kind if it decomposes in a suitable extension field into distinct linear factors, so that the exponent f is zero; an extension is called of the first kind if every element is of the first kind, that is, is a zero of a prime function of the first kind. Every extension that arises by adjoining an element of the first kind is of the first kind; over perfect fields there are only extensions of the first kind (Steinitz, § 11 and § 13).

15. Reduced degree and exponent of a finite extension. If $\mathcal{K}$ is a finite extension of an imperfect field Ω, then $\mathcal{K}$ contains a subfield $\mathcal{K}_0$ of degree r that is of the first kind with respect to Ω and consists of all and only the elements of the first kind in $\mathcal{K}$. The p^f-th power of every element of $\mathcal{K}$ belongs to $\mathcal{K}_0$, and there are elements in $\mathcal{K}$ that are exactly of degree rp^f. r is called the reduced degree, f the exponent of the finite extension (Steinitz, § 14, 1 and § 14, 5).

§ 2. Elementary-Divisor Form and Norm of the Prime Ideals and Primary Ideals. Divisors of the Prime Ideals

From now on we shall throughout be concerned with the polynomial domain defined in § 1, 1. First it is to be shown that, also for prime ideals and primary ideals, one may restrict oneself to transformed ideals, according to

Lemma I. The transformed ideal $\mathfrak{p}$ of a prime ideal $\bar{\mathfrak{p}}$ in $\mathfrak{S}$ becomes a prime ideal; the transformed $\mathfrak{q}$ of a primary ideal $\bar{\mathfrak{q}}$ in $\mathfrak{S}$ becomes a primary ideal. In other words, the property of being a prime ideal or a primary ideal is preserved when the $u_{\mu\nu}$ are adjoined to P.

Indeed suppose that $\mathfrak{a} \not\equiv 0 \pmod{\mathfrak{p}}$, $\mathfrak{b} \not\equiv 0 \pmod{\mathfrak{p}}$, where $\mathfrak{a}$ and $\mathfrak{b}$ need not be transformed ideals; say $a(x) \not\equiv 0 \pmod{\mathfrak{p}}$ and $b(x) \not\equiv 0 \pmod{\mathfrak{p}}$, with $a(x)$ and $b(x)$ polynomials from $\mathfrak{a}$ and $\mathfrak{b}$. By inversion of (1) according to (1') one obtains – with T understood, without loss of generality, as power products of the $t_{\mu\nu}$ –

$$a(x) = \sum T_i a_i(y), \qquad b(x) = \sum T'_i b_i(y),$$

and at least one $a_i(y)$ and one $b_i(y)$ must fail to be divisible by $\bar{\mathfrak{p}}$. Let, for example, $a_h(y) \not\equiv 0 \pmod{\bar{\mathfrak{p}}}$ and $b_k(y) \not\equiv 0 \pmod{\bar{\mathfrak{p}}}$ be the respective highest terms under some lexicographic ordering of the T's; then also $a_h(y)b_k(y) \not\equiv 0 \pmod{\bar{\mathfrak{p}}}$, and consequently $a(x)b(x) \not\equiv 0 \pmod{\mathfrak{p}}$ and hence $\mathfrak{ab} \not\equiv 0 \pmod{\mathfrak{p}}$, proving that $\mathfrak{p}$ is prime. The proof evidently rests on the fact that the residue-class system modulo a prime ideal forms a ring without zero divisors, a property preserved by adjoining indeterminates.

Correspondingly suppose that $\mathfrak{a} \not\equiv 0 \pmod{\mathfrak{q}}$ and $\mathfrak{b}^x \not\equiv 0 \pmod{\mathfrak{q}}$ for every power x; then from the existence of the ideal basis it follows that there must exist an $a(x)$ from $\mathfrak{a}$ and a $b(x)$ from $\mathfrak{b}$ such that $a(x) \not\equiv 0 \pmod{\mathfrak{q}}$ and $b(x)^x \not\equiv 0 \pmod{\mathfrak{q}}$ for every power x; for under the contrary assumption, a power of $\mathfrak{b}$ would be divisible by $\mathfrak{q}$.

Putting $a(x)b(x) = c(x)$, let $\mathfrak{a}^*, \mathfrak{b}^*, \mathfrak{c}^*$ denote the ideals respectively derived from the coefficients $a_i(y), b_j(y), c_k(y)$ of $a(x), b(x), c(x)$. Then $\mathfrak{b}^{*x} \not\equiv 0 \pmod{\bar{\mathfrak{q}}}$ for every power x, since otherwise $b(x)^x$ would be divisible by $\mathfrak{q}$; because $\mathfrak{a}^* \not\equiv 0 \pmod{\bar{\mathfrak{q}}}$, one must also have $\mathfrak{a}^*\mathfrak{b}^{*\lambda} \not\equiv 0 \pmod{\bar{\mathfrak{q}}}$ for every power λ. But by the Dedekind-Mertens theorem[43] there exists an exponent λ such that $\mathfrak{a}^*\mathfrak{b}^{*\lambda} \equiv \mathfrak{c}^*\mathfrak{b}^{*\lambda-1}$; hence $\mathfrak{c}^* \not\equiv 0 \pmod{\bar{\mathfrak{q}}}$. This gives $c(x) \not\equiv 0 \pmod{\mathfrak{q}}$ and consequently $\mathfrak{ab} \not\equiv 0 \pmod{\mathfrak{q}}$, proving that $\mathfrak{q}$ is primary.

Also in representing an ideal as a least common multiple one may restrict oneself to transformed ideals, according to

Lemma II. From a representation $\overline{\mathfrak{m}} = [\mathfrak{c}_1, \ldots, \mathfrak{c}_s]$ in $\mathfrak{S}$ follows a representation $\mathfrak{m} = [\mathfrak{c}'_1, \ldots, \mathfrak{c}'_s]$ in $\mathfrak{S}'$, where $\mathfrak{c}'_i$ denotes the transformed ideal of $\mathfrak{c}_i$. In particular, therefore, in every shortest representation by greatest primary components these may be assumed transformed.

Indeed $\mathfrak{m}$ is divisible by $[\mathfrak{c}'_1, \ldots, \mathfrak{c}'_s]$, since every polynomial $f(x)$ from $\mathfrak{m}$ – after multiplication, if necessary, by a suitable power of $U(u)$ – is of the form $\sum U_i(u) f_i(y)$, where each $f_i(y)$, as a polynomial from $\overline{\mathfrak{m}}$, is by assumption divisible by all the $\mathfrak{c}_i$. Conversely, if $f(x)$ is divisible by every $\mathfrak{c}'_i$, then it has the above form and is therefore divisible by $\mathfrak{m}$. Thus if one starts from a decomposition into greatest primary components in $\mathfrak{S}$, one obtains a decomposition $\mathfrak{m} = [\mathfrak{q}'_1, \ldots, \mathfrak{q}'_s]$ in $\mathfrak{S}'$; here the $\mathfrak{q}'_i$, as transformed ideals of primary ideals, are primary by Lemma I, and they are greatest primary components, since distinct associated prime ideals $\bar{\mathfrak{p}}$ in $\mathfrak{S}$ correspond to distinct $\mathfrak{p}$ in $\mathfrak{S}'$.

We now need a first characterization of prime ideals and primary ideals by elementary-divisor form and norm, still without complete separation of prime and primary. In exact analogy with ideal theory in algebraic number fields,[44] one has

[43] Compare H.-N., p. 63 and note 10).

[44] As the highest elementary divisor of ideals in algebraic number fields one has to regard the least integral rational number divisible by the ideal, hence in particular the prime number divisible by a prime ideal or the prime-power divisible by a primary ideal (a power of a prime ideal). In fact every ideal $\mathfrak{a}^*$ is also a module A^* of linear forms in the elements $\omega_1, \ldots, \omega_n$ of a field basis, with $(\omega_1, \ldots, \omega_n)$ the fundamental module of A^*; the least integral rational number divisible by $\mathfrak{a}^*$ therefore becomes the highest elementary divisor of this module A^*, whereas the norm of $\mathfrak{a}^*$ becomes the product of all elementary divisors of A^*.

Theorem I. The elementary-divisor form of a prime ideal becomes equal to a prime function, and the norm to a power of this prime function. For primary ideals, elementary-divisor form and norm become primary functions, namely powers of the same prime function, which itself is the elementary-divisor form of the associated prime ideal. For prime ideals and primary ideals, highest dimension coincides with dimension simpliciter; this dimension agrees for a primary ideal and its associated prime ideal.

Theorem I is evidently fulfilled for the unit ideal, whose elementary-divisor form and norm become unity. Now let the prime ideal $\mathfrak{p}$ have highest dimension $(n-i)$; by (4), § 1, 6, one obtains

$$E^{(i)}\mathfrak{g}_{i-1} \not\equiv 0 \pmod{\mathfrak{p}}, \quad \text{but} \quad E^{(i)}\mathfrak{g}_i \equiv 0 \pmod{\mathfrak{p}},$$

since otherwise $\mathfrak{p}$ would, by § 1, 8, have highest dimension at most $n-i-1$. Hence $\mathfrak{g}_i \equiv 0 \pmod{\mathfrak{p}}$, so that $\mathfrak{p}$ is identical with its fundamental ideal of i-th stage and consequently also with $\mathfrak{g}_{i+1}, \mathfrak{g}_{i+2}, \ldots$. Thus $R^{(i+1)}, \ldots, R^{(n)}$ become equal to unity, and hence also $E^{(i+1)}, E^{(i+2)}, \ldots$, as divisors of $R^{(i+1)}, R^{(i+2)}, \ldots$; therefore the elementary-divisor form becomes equal to $E^{(i)}$, which must be equal to a prime function $P^{(i)}$. For by § 1, 5 and taking into account that $\mathfrak{g}_{i-1}$ becomes the unit ideal, $E^{(i)}$ is defined as the greatest common divisor – in the polynomial sense – of all $B^{(i)}$ divisible by $\mathfrak{p}$; from $E^{(i)} = E_1^{(i)} E_2^{(i)} \equiv 0 \pmod{\mathfrak{p}}$ it would follow that $E^{(i)}$ must be contained in one of its factors $E_1^{(i)}$ or $E_2^{(i)}$. Since, however, a power of $E^{(i)}$ is divisible by $R^{(i)}$, $R^{(i)}$ becomes a power – possibly the first power – of this prime function $P^{(i)}$; at the same time $R^{(i)}$ becomes the norm of $\mathfrak{p}$. Since only one individual norm different from unity occurs, the highest dimension of $\mathfrak{p}$ finally agrees with its dimension simpliciter.

Correspondingly, let the primary ideal $\mathfrak{q}$ have highest dimension $n-i$; then, as above,

$$(E^{(i)}\mathfrak{g}_{i-1})^x \not\equiv 0 \pmod{\mathfrak{q}}, \quad \text{but} \quad (E^{(i)}\mathfrak{g}_i)^x \equiv 0 \pmod{\mathfrak{q}}$$

for every power x; and consequently, as above, $\mathfrak{q} = \mathfrak{g}_i$, its elementary-divisor form is $E^{(i)}$, its norm is $R^{(i)}$, and its highest dimension is its dimension simpliciter. This dimension agrees with that of the associated prime ideal; for from $\mathfrak{p}^\rho \equiv 0 \pmod{\mathfrak{q}}$, $\mathfrak{q} \equiv 0 \pmod{\mathfrak{p}}$ it follows that the dimension $(n-i)$ of $\mathfrak{p}$ is at most equal to that of $\mathfrak{q}$, and that of $\mathfrak{q}$ at most equal to that of $\mathfrak{p}^\rho$; from $(P^{(i)})^\rho \equiv 0 \pmod{\mathfrak{p}^\rho}$, where $P^{(i)}$ denotes the elementary-divisor form of $\mathfrak{p}$, the latter highest dimension is found to be at most $n-i$, and thus $n-i$ is the dimension of $\mathfrak{p}$ and $\mathfrak{q}$. From $(P^{(i)})^\rho \equiv 0 \pmod{\mathfrak{q}}$ it follows further that $E^{(i)}$ – as the greatest common divisor of all $B^{(i)}$ from $\mathfrak{q}$ – becomes a power of $P^{(i)}$, which may also be the first power, and that the same holds for $R^{(i)}$. This proves Theorem I in all its parts.

A characterization of prime ideals by means of the concept of dimension is given by

Theorem II. A prime ideal has no proper divisor of the same highest dimension; conversely, every ideal with this property is prime.

The proof rests on

Lemma III. If a prime ideal $\mathfrak{p}$ of polynomials with coefficients from a field Ω has only finitely many residue classes linearly independent over Ω, then $\mathfrak{p}$ has no proper divisor different from the unit ideal; in other words, the residue-class system modulo $\mathfrak{p}$ forms a field.

The hypothesis says that there is a finite number k such that among any $(k+1)$ residue classes there is a linear dependence with coefficients from Ω – more precisely, with coefficients from the residue-class field (Ω) represented by all elements of Ω – but that at least one system of k residue classes linearly independent over Ω exists. It follows immediately that all residue classes can be expressed linearly by any chosen system of k linearly independent ones. Now let $\mathfrak{a}$ be a proper divisor of $\mathfrak{p}$, and let $a(z) \not\equiv 0 \pmod{\mathfrak{p}}$ be a polynomial from $\mathfrak{a}$; from $c_1 f_1(z) + \cdots + c_k f_k(z) \equiv 0 \pmod{\mathfrak{p}}$ it follows that

$$c_1 a(z) f_1(z) + \cdots + c_k a(z) f_k(z) \equiv 0 \pmod{\mathfrak{p}}.$$

That means that, together with the residue classes of $f_1, \ldots, f_k$, the residue classes of $af_1, \ldots, af_k$ also form a system of k linearly independent ones, through which in particular the unit class can be linearly represented. Hence $\mathfrak{a}$ is the unit ideal; expressed differently, the residue-class system forms a field, since for $a(z) \not\equiv 0 \pmod{\mathfrak{p}}$ there is always an $f(z)$ such that $1 \equiv a(z)f(z) \pmod{\mathfrak{p}}$.

For the proof of the first part of Theorem II it remains only to observe that every prime ideal before dimension $(n-i)$ – more generally every ideal of highest dimension $(n-i)$ – has only finitely many residue classes linearly independent over $P(u; x_{i+1}, \ldots, x_n)$, namely, by § 1, 5, as many as the degree of $R^{(i)}$ indicates, since here $\mathfrak{g}_{i-1}$ becomes the unit ideal. Every proper divisor $\mathfrak{a}$ of $\mathfrak{p}$ therefore becomes the unit ideal when $x_{i+1}, \ldots, x_n$ are adjoined to $P(u)$; expressed differently, the fundamental ideal of i-th stage of $\mathfrak{a}$ becomes the unit ideal, which means that the highest dimension of $\mathfrak{a}$ is at most $n-i-1$. In order that the assertion also hold for a non-transformed ideal $\mathfrak{a}$ as divisor, one has only to pass, according to § 1, 2, to a new transformed domain.

Conversely suppose now that $\mathfrak{p}$ has no proper divisor of the same highest dimension $n-i$, and let $\mathfrak{a} \not\equiv 0 \pmod{\mathfrak{p}}$ and $\mathfrak{b} \not\equiv 0 \pmod{\mathfrak{p}}$, with $\mathfrak{a}$ and $\mathfrak{b}$ again assumed to be transformed ideals. Then, by hypothesis, since $(\mathfrak{a}, \mathfrak{p})$ and $(\mathfrak{b}, \mathfrak{p})$, as greatest common divisors – in the ideal sense – become proper divisors of $\mathfrak{p}$, one has

$$E^{(n)} \cdots E^{(i+1)} \mathfrak{g}_i \equiv 0 \pmod{(\mathfrak{a}, \mathfrak{p})}, \qquad E^{(n)} \cdots E^{(i+1)} \mathfrak{g}_i \equiv 0 \pmod{(\mathfrak{b}, \mathfrak{p})}.$$

This gives

$$E^{(n)} \cdots E^{(i+1)} \mathfrak{g}_i \equiv 0 \pmod{(\mathfrak{ab}, \mathfrak{p})},$$

and consequently necessarily $\mathfrak{ab} \not\equiv 0 \pmod{\mathfrak{p}}$, since otherwise $\mathfrak{p}$ would have smaller highest dimension than $n-i$. Since $\mathfrak{a}, \mathfrak{b}$ are transformed ideals, $\mathfrak{p}$, and by Lemma I also $\bar{\mathfrak{p}}$, are prime ideals. This proves Theorem II.

§ 3. Zeroes of the Prime Ideals and Primary Ideals

The passage to a direct foundation of the theory of zeroes (§ 1, 9), at first for prime ideals, is formed by the following lemma, which extends Lemma III and is valid for prime ideals in arbitrary rings.

Lemma IV. The residue-class system modulo every prime ideal different from the unit ideal can be extended by quotient formation to a field, the residue-class field of the prime ideal.

The residue-class system modulo an arbitrary ideal forms a ring, since sum, difference and product again belong to the system, and the associative, commutative and distributive laws are preserved on passage to residue classes. If the ideal is specifically prime, this ring has no zero divisors; for the definition of prime ideals (§ 1, 11) says that the product of non-vanishing residue classes likewise does not vanish. If the prime ideal is different from the unit ideal, then this ring contains at least one element different from the zero element, and can therefore be extended to a field by quotient formation – adjunction of pairs of elements;[45] thus the residue-class field is defined.

We first fix the notation that will be kept throughout what follows.

Notation. Residue classes shall generally be denoted by putting any representative of the class in parentheses, $(x), \ldots, (a(x)), \ldots$; similarly, fields of residue classes shall be denoted by parentheses. In particular, (P), $(P(u))$, $(P(u, x_{i+1}, \ldots, x_n))$ denote the residue-class fields consisting of all and only those residue classes that can be represented by elements from P, $P(u)$, $P(u, x_{i+1}, \ldots, x_n)$.

We also recall the following. A field $\mathcal{R}$ is said to have algebraic rank (transcendence degree) k with respect to a base field Ω if $\mathcal{R}$ contains at least one system of k quantities algebraically independent over Ω – transcendental quantities – but every $k+1$ quantities from $\mathcal{R}$ are algebraically dependent over Ω. If $\mathcal{R}$ can be represented as an algebraic extension field of a subfield $\Omega(z_1, \ldots, z_s)$, where the z are algebraically independent – transcendental – over Ω, then necessarily $s = k$;[46] likewise the algebraic rank of $\mathcal{R}(u)$ over $\Omega(u)$ is equal to k when u denotes indeterminates not contained in $\mathcal{R}$.

The construction of the field of zeroes now proceeds in complete analogy with the usual route for prime functions of one indeterminate,[47] according to

Theorem III. To the residue-class field $(\mathcal{R})$ of a prime ideal $\mathfrak{p}$ of dimension $n-i$ there can be assigned, isomorphically, a field of zeroes R of $\mathfrak{p}$, which has algebraic rank $n-i$ – depends on $n-i$ parameters – and in particular can be regarded as a finite extension field of $P(u, x_{i+1}, \ldots, x_n)$. The isomorphism between $(\mathcal{R})$ and R includes that between (P) and P, $(P(u))$ and $P(u)$, and, when $x_{i+1}, \ldots, x_n$ are taken as parameters, also that between $(P(u, x_{i+1}, \ldots, x_n))$ and $P(u, x_{i+1}, \ldots, x_n)$. Conversely, every field of zeroes of algebraic rank $n-i$ – depending on $n-i$ parameters – is isomorphic to the residue-class field $(\mathcal{R})$.

Theorem III yields as a corollary the familiar theorem: if an ideal $\mathfrak{a}$ vanishes in all zeroes of a prime ideal $\mathfrak{p}$, then $\mathfrak{a}$ is divisible by $\mathfrak{p}$.

For the proof, first note that the residue-class field $(\mathcal{R})$ has algebraic rank $n-i$ with respect to $(P(u))$. Indeed, the classes $(x_{i+1}), \ldots, (x_n)$ are algebraically independent with respect to $(P(u))$, since $\mathfrak{p}$, being of dimension $n-i$, contains no polynomial free of $x_1, \ldots, x_i$; every further class, however, is dependent on these. For from the membership of the elementary-divisor form $P^{(i)}$ in $\mathfrak{p}$ it follows, by § 1, 2, also for $\lambda = 1, 2, \ldots, i$

[45] Steinitz, § 3. The assumption of a non-zero element in the original integral domain suffices, since then the existence of the unit is secured by quotient formation, and hence the original domain becomes a subdomain of the field; Steinitz assumes the existence of the unit in the integral domain.

[46] For a proof of this familiar fact valid in arbitrary characteristic, compare Steinitz, § 22, 9. – On adjoining the u, the algebraic rank cannot increase, since only rational combinations of the original elements with coefficients from $\Omega(u)$ are involved; nor can it decrease, since every relation between elements of $\mathcal{R}$ with coefficients from $\Omega(u)$ decomposes into finitely many relations with coefficients from Ω.

[47] Steinitz, § 6.

that polynomials belong to $\mathfrak{p}$ which depend respectively on $x_\lambda, x_{i+1}, \dots, x_n$. Thus $(\mathcal{R})$ becomes a finite extension field of $(P(u, x_{i+1}, \dots, x_n))$; by § 1, 5 – taking into account that $\mathfrak{g}_{i-1}$ is equal to the unit ideal and $\mathfrak{g}_i$, by Theorem I, to $\mathfrak{p}$ – the degree over this subfield is equal to the degree of the norm.

To pass to an isomorphic field of zeroes, one further observes that $P(u, x_{i+1}, \dots, x_n)$ and $(P(u, x_{i+1}, \dots, x_n))$ are related isomorphically by assigning to each element of $P(u, x_{i+1}, \dots, x_n)$ the residue class represented by that element. First, under this assignment, the integral domains consisting of polynomials in $u, x_{i+1}, \dots, x_n$ and respectively in $(u), (x_{i+1}), \dots, (x_n)$ correspond isomorphically, since $\mathfrak{p}$ contains no polynomial free of $x_1, \dots, x_i$, and therefore distinct polynomials from $P(u, x_{i+1}, \dots, x_n)$ correspond to distinct classes; from isomorphic integral domains, however, isomorphic fields arise by quotient formation. The isomorphic relation between $P(u, x_{i+1}, \dots, x_n)$ and $(P(u, x_{i+1}, \dots, x_n))$ includes that between P and (P), $P(u)$ and $(P(u))$. To extend this isomorphism to one between $(\mathcal{R})$ and a field of zeroes R, let certain new elements $\xi_1, \dots, \xi_i$ be assigned isomorphically to the classes $(x_1), \dots, (x_i)$; that is, to each polynomial $(a(x))$ there corresponds a polynomial $a(\xi_1, \dots, \xi_i, x_{i+1}, \dots, x_n)$, and sums and products correspond to sums and products. By quotient formation – where one may restrict oneself to denominators from the subfield $(P(u, x_{i+1}, \dots, x_n))$ respectively $P(u, x_{i+1}, \dots, x_n)$ – to the residue-class field $(\mathcal{R})$ there then corresponds a field R, a finite extension of $P(u, x_{i+1}, \dots, x_n)$ of the degree of the norm; it can be called the field of zeroes. For the assignment says that $f(\xi_1, \dots, \xi_i, x_{i+1}, \dots, x_n)$ vanishes as soon as $f(x) \equiv 0 \pmod{\mathfrak{p}}$. In place of $(P(u, x_{i+1}, \dots, x_n))$, any other subfield of $(\mathcal{R})$ arising by adjoining $n - i$ algebraically independent quantities could occur throughout the argument.

Conversely it is easy to show that every field of zeroes R of algebraic rank $n - i$ is isomorphic to the residue-class field. Since R can be derived rationally, with coefficients from $P(u)$, from the quantities $\xi_1, \dots, \xi_i$, among these quantities there must be $n - i$ algebraically independent, hence transcendental, elements with respect to $P(u)$. In particular this must hold for $\xi_{i+1}, \dots, \xi_n$, since, because $P^{(i)} \equiv 0 \pmod{\mathfrak{p}}$, it follows as above that R is an algebraic extension field of $P(u, \xi_{i+1}, \dots, \xi_n)$. Since by hypothesis $f(\xi)$ vanishes for every polynomial $f(x)$ in $\mathfrak{p}$, to every class of the polynomial domain of the (x) contained in $(\mathcal{R})$ there corresponds one and only one element of R which is a polynomial in the ξ; and sums and products correspond to sums and products. But different classes correspond to different elements of R; otherwise a non-zero class, and hence after suitable multiplication also a class represented by a polynomial from $(P(u, x_{i+1}, \dots, x_n))$, would correspond to the element zero in R, and contrary to the hypothesis the quantities $\xi_{i+1}, \dots, \xi_n$ would satisfy an algebraic dependence. This assignment exhausts all polynomials in the ξ contained in R; by quotient formation – where one may again restrict oneself to denominators from $(P(u, x_{i+1}, \dots, x_n))$ respectively $P(u, x_{i+1}, \dots, x_n)$ – isomorphic fields arise from these isomorphic integral domains.

Now let the ideal $\mathfrak{a}$ vanish in all zeroes of $\mathfrak{p}$, in particular also in all zeroes depending on $n - i$ parameters. If $a(x)$ is any polynomial from $\mathfrak{a}$, then $a(\xi)$ vanishes; because of the isomorphism between R and $(\mathcal{R})$, the class $(a(x))$ is thus the zero class, and $a(x)$, hence $\mathfrak{a}$, is divisible by $\mathfrak{p}$. This proves Theorem III and its corollary.

To pass from Theorem III to the decomposition given in § 1, 9, first for prime ideals, and in order to be able to make more precise statements about the exponents, a further investigation of the elementary-divisor form is needed, according to

Lemma V. If in characteristic p the elementary-divisor form $E^{(i)}$ of a prime ideal or primary ideal – more generally, of an ideal $\mathfrak{a}$ for which highest dimension and dimension simpliciter coincide – has exponent f with respect to x_i, hence is an integral function of $x_i^{p^f}$, then it also has exponent f with respect to $x_{i+1}, \dots, x_n$ and with respect to the indeterminates $u_{\mu\nu}$ respectively $t_{\mu\nu}$ occurring in $E^{(i)}$. In particular, if the coefficient domain is a perfect field, the elementary-divisor form $P^{(i)}$ of a prime ideal always has exponent zero with respect to x_i, hence is a prime function of the first kind.

For this it is to be observed that $P(u, x_{i+1}, \dots, x_n)$ becomes imperfect as soon as P is a perfect field of characteristic p, so that it does not follow directly from § 1, 14 that $P^{(i)}$ is of the first kind. Let $E^{(i)}$ have exponent f with respect to x_i, but let it contain another indeterminate x_{i+1} with exponent f'. Then – by interchanging the $u_{\mu\nu}$ respectively $t_{\mu\nu}$ – there also exists, by § 1, 2, a non-zero polynomial $G^{(i)}$, divisible by the ideal, that is an integral function of $x_{i+1}^{p^{f'}}$, and which, by definition of the elementary-divisor form, is divisible by $E^{(i)}$. But since $G^{(i)}$ has the same degree as $E^{(i)}$, it must agree with $E^{(i)}$ up to a factor from $P(u)$; hence necessarily $f' = f$. Thus $E^{(i)}$ – which may be assumed integral and primitive in the $t_{\mu\nu}$ – is of the form

$$E^{(i)} = \sum c_\rho(t)\, x_i^{\alpha_\rho p^f} x_{i+1}^{\beta_\rho p^f} \cdots x_n^{\gamma_\rho p^f} = \sum c_\rho(t)\, X_\rho^{p^f},$$

where the X_ρ are power products of the x, and the X_ρ are divisible by $\mathfrak{a}$. Therefore one obtains a polynomial again belonging to $\mathfrak{a}$ if in the X_ρ every exponent of the individual t is replaced by the greatest multiple of p^f

contained in it; as the displayed representation shows, this amounts to replacing, in $c_\rho(t)$, every exponent of the t by the greatest multiple of p^f contained in it. By this substitution $E^{(i)}$ passes into a polynomial in $x_i^{p^f}, \ldots, x_n^{p^f}$ which, by definition, is divisible by $E^{(i)}$, and, because of the assumed primitivity, also with respect to the t; hence it must be identical with $E^{(i)}$, since the degree has increased in no argument. In particular, if P is perfect, then $E^{(i)}$, as an integral function of $x_i^{p^f}$, is a p^f-th power; hence, if a prime ideal is involved, necessarily $f = 0$, and the elementary-divisor form $P^{(i)}$ becomes a prime function of the first kind.

The decomposition is now given by

Theorem IV. The elementary-divisor form $P^{(i)}$ of a prime ideal admits, in the Galois field derived from the field of zeroes, the explicit decomposition

$$P^{(i)} = \prod_\nu (x_i - t_{i+1}x_{i+1}^{(\nu)} - \cdots - t_n x_n^{(\nu)})^e = \prod_\nu (t_i(y_i - y_i^{(\nu)}) + \cdots + t_n(y_n - y_n^{(\nu)}))^e,$$

where $y_1^{(\nu)}, \ldots, y_n^{(\nu)}$ denotes an associated system of zeroes of the original indeterminates y, independent of $t_i, \ldots, t_n$; distinct ν correspond to distinct linear factors; and the exponent e has the value one when P is perfect and the value p^f $(f > 0)$ when it is imperfect. Thereby, after adjoining new indeterminates v, the decomposition of the elementary-divisor form written in bilinear combinations of the u and v in place of the $u_{\mu\nu}$ is also determined:

$$P^{(i)}(w) = \prod_\nu (z_i - z_i^{(\nu)})^e = \prod_\nu (v_{i1}(y_1 - y_1^{(\nu)}) + \cdots + v_{in}(y_n - y_n^{(\nu)}))^e,$$

where e has the same meaning as above. Likewise the decomposition of the norm of a prime ideal, or of a primary ideal with $\mathfrak{p}$ as associated prime ideal – as a power of $P^{(i)}$ – is also given.

As a corollary of Theorem IV one also obtains: if two prime ideals agree in their elementary-divisor form $P^{(i)}$, then they are identical.

The proof of Theorem IV will amount to proving that the elementary-divisor form $P^{(i)}$ is identical with the fundamental equation of the extension field $(P(u, x_{i+1}, \ldots, x_n), (x_i))$ over $(P(u, x_{i+1}, \ldots, x_n))$. First, to gain insight into the dependence on the indeterminates $u_{\mu\nu}$ respectively $t_{\mu\nu}$, pass to the residue-class field $(\mathcal{S})$ of $\mathfrak{p}$, which, by what was said in the definition of algebraic rank, must have algebraic rank $n - i$ with respect to (P), where (P) denotes the subfield represented by elements of P. For $(\mathcal{S})$ arises by adjoining the indeterminates u respectively t to a subfield isomorphic to $(\mathcal{R})$, obtained by quotient formation from all and only those classes representable by polynomials from $\mathfrak{S}$, with (P) and (P) corresponding to one another. The algebraic rank of $(\mathcal{S})$ with respect to $(P(u))$ is therefore equal to that of this subfield with respect to (P), and hence, by the isomorphism, to that of $(\mathcal{R})$ with respect to (P). Since $(\mathcal{S})$ can be derived rationally, with coefficients from (P), from the classes $(y_1), \ldots, (y_n)$, there must be among these classes $n - i$ algebraically independent ones, and $(\mathcal{S})$ arises as a finite algebraic extension field of the subfield generated by adjoining these $n - i$ classes to (P).[48]

If one adjoins only the indeterminates $t_{\mu\nu}$ occurring in $x_{i+1}, \ldots, x_n$, then again a residue-class field arises which contains the classes $(y_1), \ldots, (y_n)$ and $(x_{i+1}), \ldots, (x_n)$ and becomes a finite algebraic extension field of $(P(t_{\mu\nu}, x_{i+1}, \ldots, x_n))$; these are fields isomorphic to subfields of $(\mathcal{S})$, not identical with them, since for different indeterminates the residue classes are represented by the same elements but are not identical, because they do not contain the same elements. Thus in the dependence of the classes (y) on $(P(t_{\mu\nu}, x_{i+1}, \ldots, x_n))$ only the indeterminates $t_{\mu\nu}$ enter. Now form, by adjoining $t_{i1}, \ldots, t_{in}$, the class $(x_i) = (t_{i1}y_1 + \cdots + t_{in}y_n)$; let $(x_{i\nu}) = (t_{i1}y_{1\nu} + \cdots + t_{in}y_{n\nu})$ be the r distinct conjugate values lying in the Galois field with respect to $(P(t_{\mu\nu}, x_{i+1}, \ldots, x_n))$. Then – according as one is dealing with extensions of the first kind or not – the product over the factors $x_i - (x_{i\nu})$, or over their p^f-th powers, is a prime function (G) with respect to $(P(t_{\mu\nu}, x_{i+1}, \ldots, x_n))$; for, since there are elements of degree $r \cdot p^f$ (§ 1, 15), (x_i) must necessarily be one such element, since every element arises by specializing the $t_{\mu\nu}$; but (x_i) is a zero of (G). Passing to the isomorphic field of zeroes of $\mathfrak{p}$ and to the Galois field derived from it, G therefore admits a decomposition into linear factors $x_i - t_{i1}y_{1\nu} - \cdots - t_{in}y_{n\nu}$ respectively their p^f-th powers. Now, since (G) vanishes for $x_i = (x_i)$, $G(x_i)$ is divisible by $\mathfrak{p}$ and hence by $P^{(i)}$; and, as a prime function with respect to $P(t_{\mu\nu}, x_{i+1}, \ldots, x_n)$ and as primitive in x_i, G must become identical with $P^{(i)}$. Thus the first decomposition of Theorem IV is proved.

To pass to the second decomposition, note that the property of being, or not being, an extension of the first kind is preserved when indeterminates are adjoined. Therefore, after adjoining $v_1, \ldots, v_n$ and all the

[48]This remark shows that Theorem III with its corollary could have been stated and proved directly for the field of zeroes R of $\mathfrak{p}$. Only by passing to the transformed ideal is it achieved that in the greatest primary components of equal dimension of an arbitrary ideal the same parameters occur among the zeroes; and only thereby does the connection with norm and elementary-divisor form of an arbitrary ideal result.

$t_{\mu\nu}$, form the class $(z) = (v_1x_1 + \cdots + v_ix_i)$ and the conjugate classes $(z_\nu) = (v_1x_{1\nu} + \cdots + v_ix_{i\nu})$. Then the product over all $z - (z_\nu)$, or over their p^f-th powers, is again a prime function (H) that vanishes for $z = (z)$. Hence H is representable as a product of linear factors, as above, and is identical with the elementary-divisor form of the prime ideal derived from $\mathfrak{p}$ by composing the transformation (1) with $z = v_1x_1 + \cdots + v_nx_n$; this elementary-divisor form, however – because of the mutual specialization – is obtained from $P^{(i)}$ by replacing the $u_{\mu\nu}$ by certain bilinear combinations of the u and v.[49] With these decompositions, the decomposition of the norm as a power of $P^{(i)}$ is also given,[50] and likewise that of the norm and elementary-divisor form of a primary ideal with $\mathfrak{p}$ as associated prime ideal.

Finally, if two prime ideals agree in their elementary-divisor form $P^{(i)}$, then, as a consequence of the above decomposition, they agree in their zeroes; hence they are mutually divisible by one another and are therefore identical.

§ 4. Arithmetical Formulation of the Concept of Dimension

A first formulation of the concept of dimension that is independent of the introduction of transformed ideals is given by

Theorem V. *The dimension of a prime ideal is given by the algebraic rank of its residue-class field. The highest dimension of an ideal is equal to the highest of the dimension numbers of its associated prime ideals.*

The first part of Theorem V was proved in the first remark to Theorem IV, where it was shown that the algebraic rank of the residue-class field $(\mathcal{R})$ of $\mathfrak{p}$ is equal to that of the residue-class field $(\mathcal{R})$ of $\mathfrak{p}$, hence equal to the dimension of $\mathfrak{p}$, in agreement with § 1, 8. For the proof of the second part, let $\mathfrak{m} = [\mathfrak{q}_1, \ldots, \mathfrak{q}_a]$ be a shortest representation by greatest primary components. Then, first of all, no $\mathfrak{q}$ as a divisor of $\mathfrak{m}$ can possess higher dimension than $\mathfrak{m}$. But the product of all $\mathfrak{q}$ is divisible by $\mathfrak{m}$, and therefore so is the product of all elementary-divisor forms of the individual $\mathfrak{q}$'s. If, then, $n - i$ is the highest of the dimension numbers of the $\mathfrak{q}$, then $\mathfrak{m}$ contains a polynomial $G^{(i)} \neq 0$ and can therefore not have dimension higher than $n - i$. The dimension numbers of the $\mathfrak{q}$ agree, by Theorem I, with those of their associated prime ideals. Thus if, without introducing transformed ideals, one defines the dimension of a prime ideal by the algebraic rank of its residue-class field, then the second part of Theorem V gives the definition of the highest dimension of an arbitrary ideal.

In order, on the other hand, to grasp the concept of dimension by chains of prime ideals – again independently of the introduction of transformed ideals – one has, supplementing Theorem II, to show:

Theorem VI. *Every prime ideal different from the unit ideal possesses – after possible adjunction of indeterminates – at least one prime ideal as divisor whose dimension has decreased by exactly one unit.*

Indeed, if $\mathfrak{p}$ has dimension $(n - i) > 0$ and v denotes a new indeterminate, then

$$\mathfrak{c} = (\mathfrak{p}, x_i - v)$$

is an ideal with the required property. This follows from an isomorphic correspondence. One has $\mathfrak{c} = (\mathfrak{p}^*, x_i - v)$, where $\mathfrak{p}^*$ arises from $\mathfrak{p}$ by replacing x_i by the indeterminate v and adjoining v to the coefficient domain; the residue classes are likewise obtained by replacing the class (x_i) by (v). But $\mathfrak{p}$ passes into itself when x_i is adjoined to $P(u)$; for from

$$h(x_i)f(x) \equiv 0 \pmod{\mathfrak{p}}, \qquad h(x_i) \neq 0$$

one necessarily obtains $f(x) \equiv 0 \pmod{\mathfrak{p}}$. Otherwise $\mathfrak{p}$ would contain a polynomial depending only on x_i, and therefore also one depending only on x_n, contrary to the supposition it would have dimension at most zero. Under the assignment $v \sim x_i$, therefore, the zero class modulo $\mathfrak{c}$ necessarily corresponds to the zero class modulo $\mathfrak{p}$, and of course conversely as well. Hence there is an isomorphism between the residue-class system modulo $\mathfrak{p}$ and that modulo $\mathfrak{c}$; the latter has no zero divisors, so $\mathfrak{c}$ is a prime ideal. Under this isomorphism the algebraic dependence between (x_i) and $(P(u, x_{i+1}, \ldots, x_n))$ corresponds to a relation between $(x_{i+1}), \ldots, (x_n)$ with respect to $(P(u, v))$, whereas for $\lambda = 1, \ldots, i-1$ the dependences between (x_λ) and $(P(u, x_{i+1}, \ldots, x_n))$ remain as such, and hence do not further lower the algebraic rank. The algebraic rank has decreased by exactly one unit. If $\mathfrak{p}$ was of dimension zero, $\mathfrak{c}$ becomes the unit ideal; the above result therefore remains valid. In the original domain $\bar{\mathfrak{S}}$, consequently,

$$\bar{\mathfrak{a}} = (\bar{\mathfrak{p}}, v + v_1y_1 + \cdots + v_ny_n),$$

[49] H.-N., § 7.

[50] For extensions of the first kind, hence in particular over perfect fields, the first power is involved; over imperfect fields the p^f-th power is involved, as will be shown in § 6, Theorems XIII and XIV.

where v_λ is put for $-t_{i\lambda}$, is an ideal of the required kind. If, in particular, P contains infinitely many elements, then $v, v_1, \ldots, v_n$ can always be specialized to elements of P in such a way that norm and elementary-divisor form, and hence the dimension of $\bar{\mathfrak{a}}$, become equal to those of the specialized ideal $\mathfrak{b}$;[51] the elementary-divisor form need not, however, remain a prime function. By Theorem V, $\mathfrak{b}$ nevertheless possesses at least one associated prime ideal of the desired dimension; passing back again to $\bar{\mathfrak{S}}$, Theorem VI therefore holds there without adjoining any indeterminates.

Theorem II and Theorem VI immediately give the desired chain formulation:

Theorem VII. *A prime ideal $\mathfrak{p}$ has dimension $n-i$ if – after possible adjunction of indeterminates – there exists at least one chain of $1+(n-i)+1$ prime ideals*

$$\mathfrak{p}_0 = \mathfrak{p}, \quad \mathfrak{p}_1, \ldots, \mathfrak{p}_{n-i}, \quad \mathfrak{p}_{n-i+1},$$

each of which is a proper divisor of the immediately preceding one, whereas no such chain with a larger number of members exists. This definition also holds in the original domain, and without introducing any indeterminates, when P contains infinitely many elements.[52]

First it is clear that the last member of the chain must be equal to the unit ideal – which satisfies the definition of a prime ideal – since otherwise the chain could be lengthened by adjoining $\mathfrak{o}$; for $\mathfrak{o}$ itself Theorem VII gives dimension -1, in agreement with § 1, 8. Now let $\mathfrak{p}$ be different from $\mathfrak{o}$ and of dimension $n-i$. Then the number of members of the chain can be at most $1+(n-i)+1$, since by Theorem II two prime ideals of the same dimension cannot occur. Conversely, by Theorem VI, after possible adjunction of new indeterminates, there exists at least one chain of this length, since the dimension has decreased by exactly one unit at every member of the chain. If one takes Theorem VII as the definition of dimension – a definition which plainly can be stated in exactly the same way in the original domain, and for which, by Theorem VI, the introduction of indeterminates becomes unnecessary when P contains infinitely many elements – then the definition of the highest dimension of an arbitrary ideal is again given by the second part of Theorem V.

§ 5. Classification of the Fundamental Ideals and of Their Components in General Ideal Theory. Zeroes of Ideals

The question is the characterization of the fundamental ideals and their components (§ 1, 7) by isolated components of the decomposition (§ 1, 13) in the sense of general ideal theory. This characterization will show the parallelism between the decomposition theorems for norms and those of general ideal theory, and will make it possible to state, for arbitrary ideals, the theory of zeroes given in § 3 for prime ideals and primary ideals. For the fundamental ideals one has

Theorem VIII. *The fundamental ideals $\mathfrak{g}_i$ of i-th stage are the isolated components of the decomposition that are uniquely determined by the totality of the associated prime ideals of dimensions $n-1, n-2, \ldots, n-i$. If all associated prime ideals have the same dimension, then the ideal also has this dimension; that is, only one factor $R^{(i)}$ of the norm occurs. In general the norm has as many factors $R^{(i)}, R^{(i+\sigma)}, R^{(i+\tau)}, \ldots$ different from unity as different dimension numbers occur among the associated prime ideals.*

Precede the proof by

Lemma VI. *If an ideal is represented as a least common multiple, $\mathfrak{m} = [\mathfrak{a}_1, \ldots, \mathfrak{a}_a]$, then its fundamental ideal $\mathfrak{g}_i$ is the least common multiple of the fundamental ideals $\mathfrak{h}_i$ of i-th stage of the $\mathfrak{a}$.*

For from

$$b^{(i+1)} f \equiv 0 \pmod{\mathfrak{m}}, \qquad b^{(i+1)} \neq 0$$

one obtains

$$b^{(i+1)} f \equiv 0 \pmod{\mathfrak{a}_\lambda}, \qquad b^{(i+1)} \neq 0;$$

therefore $\mathfrak{g}_i$ is divisible by $[\mathfrak{h}_{i1}, \ldots, \mathfrak{h}_{ia}]$. Conversely, from

$$c_\lambda^{(i+1)} f \equiv 0 \pmod{\mathfrak{a}_\lambda}, \qquad c_\lambda^{(i+1)} \neq 0$$

one obtains

$$c_1^{(i+1)} \cdots c_a^{(i+1)} f \equiv 0 \pmod{\mathfrak{m}}, \qquad c_1^{(i+1)} \cdots c_a^{(i+1)} \neq 0,$$

and hence the converse divisibility, proving the lemma.

[51] Compare H.-N., the last paragraph of § 7.

[52] This chain definition of dimension gives, in the case of an algebraic number field, dimension zero for the ordinary prime ideals, dimension -1 for the unit ideal, and dimension one for the zero ideal. Indeed the latter also satisfies the definition of prime ideals: in a field a product of non-vanishing quantities is always different from zero. Theorem II remains valid, with this definition of dimension, in algebraic number fields.

Now let the associated prime ideals of $\mathfrak{m}$ be ordered by dimension (some dimensions may of course be absent, $j_\lambda = 0$):

$$\mathfrak{p}_{1,1}, \ldots, \mathfrak{p}_{1,j_1}; \quad \mathfrak{p}_{2,1}, \ldots, \mathfrak{p}_{2,j_2}; \quad \cdots \quad \mathfrak{p}_{n,1}, \ldots, \mathfrak{p}_{n,j_n},$$

where in general $\mathfrak{p}_{\lambda,\kappa}$ denotes a prime ideal of dimension $n-\lambda$; let $\mathfrak{q}_{\lambda,\kappa}$ be the corresponding primary ideal occurring in a fixed given decomposition of $\mathfrak{m}$. Then by definition the fundamental ideal of i-th stage is equal to the unit ideal for all $\mathfrak{q}_{\lambda,\kappa}$ for which $\lambda > i$, whereas for $\lambda \leq i$ it is, by Theorem I, equal to $\mathfrak{q}_{\lambda,\kappa}$. Therefore, by Lemma VI,

$$\mathfrak{g}_i = [\mathfrak{q}_{1,1}, \ldots, \mathfrak{q}_{1,j_1}; \ldots; \mathfrak{q}_{i,1}, \ldots, \mathfrak{q}_{i,j_i}], \tag{5}$$

and this representation recognizes $\mathfrak{g}_i$ as an isolated component of the decomposition, since none of the prime ideals belonging to $\mathfrak{g}_i$ can be contained in one of the remaining prime ideals, all of which have lower dimension.

If, in particular, only associated prime ideals of dimension $n-i$ occur, then by (5) one has $\mathfrak{g}_0, \ldots, \mathfrak{g}_{i-1}$ equal to the unit ideal, while $\mathfrak{g}_i = \mathfrak{g}_{i+1} = \cdots = \mathfrak{m}$; hence $R_\mathfrak{m} = R^{(i)}$. If, however, prime ideals of different dimensions occur, say $j_i, j_{i+\sigma}, j_{i+\tau}, \ldots$ are non-zero, then by (5)

$$\mathfrak{g}_0 \neq \mathfrak{g}_i \neq \mathfrak{g}_{i+\sigma} \neq \mathfrak{g}_{i+\tau} \neq \cdots,$$

whereas

$$\mathfrak{g}_0 = \mathfrak{g}_1 = \cdots = \mathfrak{g}_{i-1} = \mathfrak{o}, \qquad \mathfrak{g}_i = \mathfrak{g}_{i+1} = \cdots = \mathfrak{g}_{i+\sigma-1}, \ldots;$$

therefore the factors $R^{(i)}, R^{(i+\sigma)}, R^{(i+\tau)}$, and only these, are different from unity.[53]

By means of Theorem VIII, Theorem I sharpens to

Theorem IX. *An ideal is primary if and only if its elementary-divisor form – and therefore its norm – is a primary function.*

That the condition is fulfilled for primary ideals was shown in Theorem I. Conversely, let the elementary-divisor form of $\mathfrak{m}$ be given by $Q^{(i)} = (P^{(i)})^\rho$, where $P^{(i)}$ denotes a prime function. By Theorem VIII all prime ideals of $\mathfrak{m}$ then have the same dimension $n-i$. From $\mathfrak{m} = [\mathfrak{q}_1, \ldots, \mathfrak{q}_a]$ one obtains

$$Q^{(i)} \equiv 0 \pmod{\mathfrak{q}_\lambda}; \quad \text{hence} \quad (P^{(i)})^\rho \equiv 0 \pmod{\mathfrak{p}_\lambda}, \quad \text{and therefore} \quad P^{(i)} \equiv 0 \pmod{\mathfrak{p}_\lambda}.$$

Since all $\mathfrak{p}_\lambda$ have dimension $n-i$ and $P^{(i)}$ is a prime function, $P^{(i)}$ becomes the elementary-divisor form of each $\mathfrak{p}_\lambda$. Thus, by the corollary to Theorem IV, all $\mathfrak{p}_\lambda$ coincide. Hence, since greatest primary components are involved, only one $\mathfrak{p}$ can occur; $\mathfrak{m}$ is primary, the special prime case for $\rho = 1$ not being excluded. Theorems VIII and IX lead to the classification of the components of the fundamental ideals as follows.

Theorem X. *The component $\mathfrak{r}_\lambda$ of the fundamental ideal $\mathfrak{g}_i$ corresponding to the greatest primary factor $Q_\lambda^{(i)} = (P_\lambda^{(i)})^{\rho_\lambda}$ of $R^{(i)}$ is given by the isolated component of the decomposition which is uniquely determined by the associated prime ideals of dimensions $n-1, \ldots, n-i+1$ and by that prime ideal of dimension $n-i$ whose elementary-divisor form becomes equal to $P_\lambda^{(i)}$. Associated prime ideals and greatest primary factors of the norm correspond one-to-one.*

By § 1, 7 the $\mathfrak{r}_\lambda$ agree with their fundamental ideal of i-th stage and possess $\mathfrak{g}_{i-1}$ as fundamental ideal of $(i-1)$-st stage. Hence, by Theorem VIII respectively (5), they admit a shortest representation

$$\mathfrak{r}_\lambda = [\mathfrak{g}_{i-1}, \mathfrak{q}_{\lambda,1}, \ldots, \mathfrak{q}_{\lambda,t_\lambda}],$$

where all $\mathfrak{q}$ have dimension $n-i$ and belong to distinct prime ideals. By the definition of $Q_\lambda^{(i)}$ – taking into account that $\mathfrak{r}_\lambda$ agrees with its fundamental ideal of i-th stage – one has

$$Q_\lambda^{(i)} \mathfrak{g}_{i-1} \equiv 0 \pmod{\mathfrak{q}_{\lambda,\kappa}}, \quad (\kappa = 1, \ldots, t_\lambda),$$

and consequently

$$(Q_\lambda^{(i)})^{\rho_\lambda} \mathfrak{r}_\lambda \equiv 0 \pmod{\mathfrak{q}_{\lambda,\kappa}}; \quad \text{hence} \quad (P_\lambda^{(i)})^{\rho_\lambda} \mathfrak{r}_\lambda \equiv 0 \pmod{\mathfrak{p}_{\lambda,\kappa}},$$

and so $P_\lambda^{(i)} \equiv 0 \pmod{\mathfrak{p}_{\lambda,\kappa}}$. As in the proof of Theorem IX it follows that in the representation of $\mathfrak{r}_\lambda$ only one $\mathfrak{q}_\lambda$ and associated $\mathfrak{p}_\lambda$ of dimension $n-i$ can occur, and that the elementary-divisor form of $\mathfrak{p}_\lambda$ is given by $P_\lambda^{(i)}$. It remains to prove that $\mathfrak{p}_\lambda$ is an associated prime ideal of $\mathfrak{m}$ and that $\mathfrak{q}_\lambda$ actually occurs in at least one

[53] Taking Lemma II into account, Theorem VIII again implies that the fundamental ideals – as least common multiples of transformed ideals – become transformed ideals. Since Theorem VIII rests only on theorems from H.-N. where this fact is not used, this gives at the same time a new proof which reveals the internal reason. In H.-N. the theorem is used only in the theory of zeroes, which is newly developed here.

representation of $\mathfrak{m}$ by greatest primary components. Then $\mathfrak{r}_\lambda$ is also recognized as the isolated component of the decomposition, since no prime ideal can be absorbed in one of equal or lower dimension, namely as the component determined by $\mathfrak{p}_\lambda$ and by all associated prime ideals of higher dimension.

For this proof it suffices to show that $\mathfrak{q}_\lambda$ occurs in a shortest representation of $\mathfrak{g}_i$, since such a representation can always, by Theorem VIII, be completed to a shortest representation of $\mathfrak{m}$. Thus it is enough to show that

$$\mathfrak{g}_i = [\mathfrak{r}_1, \ldots, \mathfrak{r}_a] = [\mathfrak{g}_{i-1}, \mathfrak{q}_1, \ldots, \mathfrak{q}_a]$$

becomes a shortest representation. If, for instance, $\mathfrak{q}_\lambda$ were absorbed by its complement, so that $\mathfrak{g}_{i-1}\mathfrak{q}_1 \cdots \mathfrak{q}_{\lambda-1}\mathfrak{q}_{\lambda+1} \cdots \mathfrak{q}_a$ were divisible by $\mathfrak{q}_\lambda$, then, because $\mathfrak{g}_{i-1} \not\equiv 0 \pmod{\mathfrak{q}_\lambda}$, it would follow that a power of some $\mathfrak{q}_\mu$ is divisible by $\mathfrak{q}_\lambda$ for $\mu \neq \lambda$. Thus the associated prime ideals $\mathfrak{p}_\mu$ and $\mathfrak{p}_\lambda$, both of the same dimension, would have to coincide by Theorem II; but this is impossible, since their elementary-divisor forms $P_\mu^{(i)}$ and $P_\lambda^{(i)}$ are distinct by definition. The representation of $\mathfrak{g}_i$, to which by Theorem VIII all associated prime ideals of dimension $n - i$ correspond, further shows that to every such prime ideal there really corresponds a greatest primary factor of the norm. The one-to-one correspondence is therefore proved.

Theorem X immediately gives the transfer of Theorem IV:

Theorem XI. *The individual greatest primary factors $Q_\lambda^{(i)} = (P_\lambda^{(i)})^{\rho_\lambda}$ of the norm admit a product representation*

$$Q_\lambda^{(i)} = \prod \left(x_i - t_{1i}\bar{y}_{1\nu} - \cdots - t_{ni}\bar{y}_{n\nu}\right)^{\delta_\lambda \rho_\lambda} = \prod \left(t_{1i}(y_1 - \bar{y}_{1\nu}) + \cdots + t_{ni}(y_n - \bar{y}_{n\nu})\right)^{\delta_\lambda \rho_\lambda},$$

where $\bar{y}_{1\nu}, \ldots, \bar{y}_{n\nu}$ represents an associated system of zeroes of the associated prime ideal $\mathfrak{p}_\lambda$, in the original indeterminates y, independent of $t_{1i}, \ldots, t_{ni}$, and δ_λ has the value one or p^{f_λ}. Thus the decomposition of the norm into linear factors is completely given. The degree of $Q_\lambda^{(i)}$ gives the degree, over the residue-class subfield $(P(u, x_{i+1}, \ldots, x_n))$ obtained by quotient formation, of the subring of residue classes, represented by residue classes from $\mathfrak{g}_{i-1}$, modulo the isolated component $\mathfrak{r}_\lambda$; for primary ideals this is therefore the ring of all residue classes. When new indeterminates v are adjoined, one further obtains the decomposition

$$Q_\lambda^{(i)}(u, v) = \prod (z - v_1\bar{x}_{1\nu} - \cdots - v_i\bar{x}_{i\nu})^{\delta_\lambda \rho_\lambda}.$$

Since $P_\lambda^{(i)}$ has been recognized by Theorem X as the elementary-divisor form of $\mathfrak{p}_\lambda$, this is in fact only the substitution of the decomposition of Theorem IV. The remark on the degree is only another formulation of what was said in § 1, 5 and 7.[54]

§ 6. Characterization of Prime Ideals and Proper Primary Ideals by Elementary-Divisor Form and Norm

In Theorem IX primary ideals were characterized by elementary-divisor form and norm, but only in such a way that the prime case was to be regarded as a special case of primary. We now need to separate prime ideals from proper primary ideals; the considerations will run parallel to those of § 3. Conditions that are either necessary or sufficient are easily stated:

Theorem XII. *A necessary condition for a prime ideal is that its elementary-divisor form be a prime function; a sufficient condition is that its norm be a prime function. In other words, the elementary-divisor form of a prime ideal is always a prime function, and the norm of a proper primary ideal is always a proper primary function.*

That the elementary-divisor form of a prime ideal is a prime function was already shown in Theorem I. Suppose now that the norm $R_\mathfrak{q}$ is equal to a prime function. It is to be shown that, under this hypothesis, every proper divisor $\mathfrak{a}$ of $\mathfrak{q}$ has lower highest dimension; then $\mathfrak{q}$ is recognized as a prime ideal by Theorem II. Indeed, by § 1, 6 the first factor of the norm $R_\mathfrak{a}$ that differs from the corresponding one of $R_\mathfrak{q}$ is a proper divisor. Since $R_\mathfrak{q} = P^{(i)}$, the factor $R^{(i)}$ of $R_\mathfrak{a}$ must therefore be equal to a proper divisor of $P^{(i)}$, hence to unity; $\mathfrak{a}$ has lower highest dimension.

A second simple proof rests on the following lemma, on which the next theorem also rests.

Lemma VII. *The system of residue classes represented by polynomials $H^{(i)}$ modulo the elementary-divisor form $E^{(i)}$ of a primary ideal – more generally, of an ideal that has only associated prime ideals of*

[54] Likewise the formulation of multiplicity mentioned in note 2 of H.-N. is an immediate consequence of Theorem X. For, after adjoining $x_{i+1}, \ldots, x_n$, the residue-class system $\mathfrak{g}_{i-1} \mid \mathfrak{r}_\lambda$ is isomorphic to $\mathfrak{t}_\lambda \mid \mathfrak{r}_\lambda$, where $\mathfrak{t}_\lambda = [\mathfrak{g}_{i-1}, \mathfrak{r}_1, \ldots, \mathfrak{r}_{\lambda-1}, \mathfrak{r}_{\lambda+1}, \ldots, \mathfrak{r}_a]$ is put, as follows from passing back to modules of linear forms (H.-N., Theorem V) in view of the relative primality of the factors $Q_\mu^{(i)}$. Directly it is shown that $\mathfrak{t}_\lambda \mid \mathfrak{r}_\lambda$ is isomorphic to $\mathfrak{t}_\lambda \mid \mathfrak{q}_\lambda$; here $\mathfrak{t}_\lambda$ arises from the complement of $\mathfrak{q}_\lambda$ after adjoining $x_{i+1}, \ldots, x_n$.

dimension $n-i$ – is isomorphic to a subsystem of the residue classes modulo this ideal. The degree of the elementary-divisor form gives the degree of this residue-class system with respect to the quotient field $(P(u, x_{i+1}, \ldots, x_n))$.

Indeed, one defines a correspondence by assigning to one another the classes represented by the same polynomials $H^{(i)}$; sums and products correspond to sums and products. The assignment is also one-to-one. For all polynomials $H^{(i)}$ belonging to the zero class modulo the ideal are, by definition, divisible by $E^{(i)}$; hence the zero class corresponds to the zero class modulo $E^{(i)}$, and conversely as well.

Now let the norm of an ideal be a prime function $P^{(i)}$, hence identical with the elementary-divisor form. The degrees of the residue-class systems modulo the elementary-divisor form $P^{(i)}$ and modulo $\mathfrak{q}$ therefore agree with respect to $(P(u, x_{i+1}, \ldots, x_n))$; the subsystem isomorphic to the residue-class system modulo $P^{(i)}$ therefore exhausts the residue classes modulo $\mathfrak{q}$, and since this system can have no zero divisors, $\mathfrak{q}$ is a prime ideal.

In general Theorem XII cannot be sharpened to necessary and sufficient conditions, as will be shown below by examples. This is possible, however, when the coefficient domain P is a perfect field (§ 1, 14):

Theorem XIII. *If the coefficient domain P is a perfect field, then norm and elementary-divisor form of a prime ideal become prime functions and consequently identical; norm and elementary-divisor form of a proper primary ideal become proper primary functions. Thus, over a perfect field, the condition that the elementary-divisor form be a prime function is necessary and sufficient for a prime ideal; in the condition the norm may also stand in place of the elementary-divisor form.*

Taking Theorem XII into account, Theorem XIII will be proved once it is shown that over a perfect field the ideal becomes prime as soon as its elementary-divisor form is a prime function, and that then the norm is equal to this prime function. In Lemma V, § 3, it was shown that, over a perfect field, the elementary-divisor form becomes a prime function of the first kind as soon as it is a prime function at all. Therefore let $(\mathcal{R})$ denote the residue-class field obtained by adjoining (x_i) to $(P(u, x_{i+1}, \ldots, x_n))$, that is, $(P(u, x_i, x_{i+1}, \ldots, x_n))$ modulo $P^{(i)}$. Then (x_i) is a zero of the prime function of the first kind $(P^{(i)})$, and hence, as for example Lagrange's formula shows, $(y_1), \ldots, (y_n)$ are contained in $(\mathcal{R})$. The subsystem of the residue classes modulo the ideal $\mathfrak{q}$ under consideration that is isomorphic, by Lemma VII, to $(\mathcal{R})$ – where only denominators from $(P(u, x_{i+1}, \ldots, x_n))$ occur – therefore contains the residue classes $(y_1), \ldots, (y_n)$ and hence exhausts the residue-class system modulo $\mathfrak{q}$. Since, being isomorphic to $(\mathcal{R})$, it can have no zero divisors, $\mathfrak{q}$ is a prime ideal. The degree of this residue-class field $(\mathcal{R})$ of $\mathfrak{q}$ with respect to $(P(u, x_{i+1}, \ldots, x_n))$ is equal to the degree of the norm of $\mathfrak{q}$; on the other hand, by the isomorphism with $(\mathcal{R})$, it is equal to the degree of the elementary-divisor form $P^{(i)}$. Thus norm and elementary-divisor form must agree.

For imperfect coefficient domains Theorem XII still allows the following sharpening:

Theorem XIV. *If the coefficient domain P is an imperfect field of characteristic p, then the norm of a prime ideal is the p^g-th power of the elementary-divisor form; one has $0 \leq g \leq (i-1)f$, where f denotes the exponent of the elementary-divisor form and $n-i$ the dimension of the prime ideal.*

Theorem XIV amounts to the assertion that the residue-class field $(\mathcal{R})$ of $\mathfrak{p}$ has degree p^g $(g \geq 0)$ with respect to the subfield $(\mathcal{R}') = (P(u, x_i, x_{i+1}, \ldots, x_n))$ isomorphic to $(\mathcal{R})$. This follows directly from the Steinitz theorems stated in § 1, 15, more precisely from

Lemma VIII. *Let Λ be a finite extension of the imperfect field Ω of reduced degree r and exponent f; let Λ_0 be the field of the first kind contained in Λ, and let M be an intermediate field between Λ_0 and Λ. Then the degree of every element of Λ with respect to M is a power of p, where p denotes the characteristic of Ω.*

For the $p^{f'}$-th power $(f' \leq f)$ of every element z of Λ belongs to Λ_0; hence z is a zero of a prime function $G(t) = (t-z)^{p^{f'}}$ with coefficients in Λ_0. Let $H(t) = (t-z)^{\delta}$ be the prime function in M with zero z. Then $H(t)$ also has coefficients in $\Lambda_0(z)$, and therefore the coefficients of $H(t)$ belong to the intersection field of M and $\Lambda_0(z)$, hence to an intermediate field between Λ_0 and $\Lambda_0(z)$. The degree of z with respect to M is therefore equal to the degree with respect to this intermediate field, and hence, as a divisor of $p^{f'}$, is a power of p.

For the proof of Theorem XIV it remains only to observe that the field $(\mathcal{R}) = (P(u, x_i, x_{i+1}, \ldots, x_n))$ must have degree $r \cdot p^f$ with respect to $(P(u, x_{i+1}, \ldots, x_n))$, if r denotes the reduced degree and f the exponent of $(\mathcal{R})$; and consequently the field of the first kind $(\mathcal{R}_0)$ contained in $(\mathcal{R})$ must be a subfield of $(\mathcal{R}')$. Since there are elements of degree $r \cdot p^f$ in $(\mathcal{R})$ (§ 1, 15), (x_i) must necessarily be of this degree, since all elements of $(\mathcal{R})$ arise by specializing the $t_{\mu\nu}$ in (x_i). The exponent of $(\mathcal{R})$ agrees with the exponent of the elementary-divisor form, and $(x_i)^{p^f}$ has degree r and is an element of $(\mathcal{R}_0)$, hence a primitive element of $(\mathcal{R}_0)$. Since $(\mathcal{R})$ arises by adjoining finitely many elements – say $(x_1), \ldots, (x_{i-1})$ – to $(\mathcal{R}')$, repeated finite application of Lemma

VIII gives the desired proof and at the same time the estimate for g.[55]

We now add the examples which show that Theorem XII cannot in general be sharpened beyond Theorem XIV. They concern a prime ideal whose norm is equal to the square of the elementary-divisor form ($g > 0$), and proper primary ideals whose elementary-divisor forms are prime functions. The latter example also shows that no analogue of Theorem XIV exists here, but that the norm can become an arbitrary power of the elementary-divisor form.

Let the coefficient domain P arise by adjoining two indeterminates λ, μ to the field of residue classes modulo 2, and therefore let P be an imperfect field of characteristic two. Consider the ideal

$$\bar{\mathfrak{p}} = (y_1^2 + \lambda,\ y_2^2 + \mu),$$

which must be a prime ideal, since the residue-class system $(\mathcal{R})$ is isomorphic to the field $P(\sqrt{\lambda}, \sqrt{\mu})$. The field $(\mathcal{R})$ has degree four with respect to (P), reduced degree $r = 1$, and exponent $f = 2$. Therefore the elementary-divisor form $P^{(2)}$ is of the second degree, and the norm is of the fourth degree and consequently the square of $P^{(2)}$, as can also be checked directly from the module representation (3), § 1, 5. One obtains

$$P^{(2)} = (u_{11}u_{22} + u_{12}u_{21})^2 x_2^2 + (u_{11}^2\mu + u_{21}^2\lambda),$$

or, after a suitable multiplication,

$$x_2^2 + (t_{12}^2\lambda + t_{22}^2\mu).$$

Let P now arise by adjoining the one indeterminate λ to the field of residue classes modulo 2, and take as basis the ideal

$$\bar{\mathfrak{q}} = (y_1^2 + \lambda,\ y_2^2 + \lambda) = (y_1^2 + \lambda,\ (y_1 + y_2)^2).$$

As elementary-divisor form, by specializing $\mu = \lambda$ in the one just given, one obtains

$$P^{(2)} = x_2^2 + \lambda(t_{12}^2 + t_{22}^2),$$

hence a prime function, since $t_{12} = (0),\ t_{22} = (1)$ leads to the prime function $x_2^2 + \lambda$. But $\bar{\mathfrak{q}}$ is proper primary, since it contains $(y_1 + y_2)^2$ but not $(y_1 + y_2)$. The norm must be equal to the square, hence to the p-th power, of $P^{(2)}$, since the number of residue classes of $\bar{\mathfrak{q}}$ linearly independent over P is four.

That this last analogue of Theorem XIV is not always fulfilled is shown by the following example. Let P arise by adjoining the indeterminate λ to the residue-class field modulo 3, and put

$$\mathfrak{q} = (y_1^3 + \lambda,\ (y_1 - y_2)^2).$$

Then, as above, $\mathfrak{q}$ is a proper primary ideal. The elementary-divisor form becomes – taking into account that $y_2^3 + \lambda$ is also divisible by $\mathfrak{q}$ –

$$P^{(2)} = x_2^3 + \lambda(t_{12} + t_{22})^3,$$

a prime function; but the norm is equal to the square of $P^{(2)}$, since there are six linearly independent residue classes modulo $\mathfrak{q}$. Thus the p^g-th power of the elementary-divisor form does not occur here.

The first example shows that, over an imperfect field as coefficient domain, just as in algebraic number fields, prime ideals of degree higher than the first may occur, where p^g is to be called the degree of the prime ideal. The further examples are to be understood as analogues of ramification ideals: a prime number may be divisible by a higher power of a prime ideal, by a primary ideal. For, as shown in note 10, the elementary-divisor form corresponds to the least integral rational number divisible by an ideal in an algebraic number field.

§ 7. Absolute Prime Ideals

A prime ideal with coefficients in P is called an *absolute prime ideal* if it remains a prime ideal in the algebraically closed field A to which P can be extended.[56] Correspondingly, a prime function P with coefficients in P is called an *absolute prime function* – absolutely irreducible polynomial – if P remains a prime function in A. The characterization of absolute prime ideals is given by

[55] Compare the parallel considerations in A. Ostrowski, Zur arithmetischen Theorie der algebraischen Größen, Gött. Nachr. 1919, pp. 279–298, especially p. 288, where the corresponding theorem is stated without proof.

[56] An algebraically closed field is, as is well known, a field in which every polynomial in one indeterminate decomposes into linear factors. For the existence and essential uniqueness of the algebraically closed field belonging to an arbitrary field, see Steinitz.

Theorem XV. *A necessary and sufficient condition for an absolute prime ideal is that its elementary-divisor form – which may be assumed integral and primitive in the u – become an absolute prime function as a polynomial in the x and u. The elementary-divisor form becomes identical with the norm, so that the condition may also be stated for the norm.*

For the proof, observe first that in general norm and elementary-divisor form are preserved under algebraic extension of the coefficient domain. Indeed the individual factors $R^{(i)}$ respectively $E^{(i)}$ are defined as greatest common divisors in the polynomial sense, a property preserved by algebraic extension of the coefficient domain. Thus $P^{(i)}$ remains the elementary-divisor form of $\mathfrak{p}$ when passing from P to A, hence when passing to a perfect field as coefficient domain, since every algebraically closed field is perfect, as the definition immediately shows. By Theorem XIII, a necessary and sufficient condition for $\mathfrak{p}$ to be an absolute prime ideal is that $P^{(i)}$ become a prime function with respect to $A(u)$; that is, an absolute prime function as a polynomial in the x and u, if $P^{(i)}$ is assumed integral and primitive in the u. At the same time, by Theorem XIII, $P^{(i)}$ becomes identical with the norm of $\mathfrak{p}$ as soon as A is taken as coefficient domain. Hence, by what was just noted, the same holds over P. This proves Theorem XV.

For absolute prime functions P with coefficients in a (finite) algebraic number field $\mathfrak{K}$, one now has the theorem[57] that P remains a prime function, more precisely an absolute prime function, modulo every prime ideal of $\mathfrak{K}$, with at most finitely many prime ideals of $\mathfrak{K}$ excepted. The transfer of this theorem to absolute prime ideals rests on

Theorem XVI. *If, as coefficient domain, one takes, in place of a field, the ring $\mathfrak{o}^*$ of all algebraic integers of a (finite) algebraic number field $\mathfrak{K}$, then norm and elementary-divisor form of a polynomial ideal are preserved as norm and elementary-divisor form modulo every prime ideal of $\mathfrak{K}$, with at most finitely many prime ideals of $\mathfrak{K}$ excepted.*

For the proof the polynomial domain on which everything is based is to be modified relative to § 1, 1. Let $\bar{\mathfrak{S}}$ consist of all polynomials in $y_1, \ldots, y_n$ with coefficients in $\mathfrak{o}^*$, hence algebraic integers of $\mathfrak{K}$; let the coefficient domain of $\mathfrak{S}$ be $\mathfrak{o}^*[u]$, that is, all polynomials in u with coefficients in $\mathfrak{o}^*$. If $\bar{\mathfrak{m}}$ is an ideal in $\bar{\mathfrak{S}}$, it passes by (1) into a transformed ideal $\mathfrak{m}$ in $\mathfrak{S}$. It is to be shown that only finitely many polynomials from $\mathfrak{o}^*[u]$ occur in the denominators in forming the norm; then Theorem XVI follows directly.

First note that the module representation (3) in § 1, 5, for the formation of the individual norm, amounts to reducing powers of x_{i-1} modulo a polynomial regular in x_{i-1},

$$C^{(i-1)} = U_{i-1}(u)x_{i-1}^{e_{i-1}} + \text{lower terms},$$

[58] where all coefficients of $C^{(i-1)}$, in particular U_{i-1}, may be assumed to be elements of $\mathfrak{o}^*[u]$. Since this is a matter of successive reduction of $x_1, \ldots, x_{i-1}$, the module representation integral in the x is also integral in $\mathfrak{o}^*[u]$ except for power products

$$U_1^{\lambda_1} \cdots U_{i-1}^{\lambda_{i-1}}$$

in the denominator. The product $U_1 U_2 \cdots U_n$, considered as a polynomial in the u, defines by its coefficients an ideal $\mathfrak{r}^*$ of $\mathfrak{K}$, which is therefore divisible by only finitely many prime ideals of $\mathfrak{K}$. If $\mathfrak{p}^*$ is chosen different from these finitely many, and if the residue-class field modulo $\mathfrak{p}^*$ is used as coefficient domain, the original module representation is preserved by replacing each number of $\mathfrak{o}^*$ by its residue class modulo $\mathfrak{p}^*$.

Furthermore, since norm and elementary-divisor form are determined only up to factors from $\mathfrak{o}^*[u]$, they may also be assumed divisible by $\mathfrak{m}$ with respect to $\mathfrak{o}^*[u]$. Suppose, under this convention, that for instance

$$R^{(i)} = V_i(u)x_i^{e_i} + \text{lower terms},$$

so that, in the divisibility of all ρ-rowed determinants of the module $\mathfrak{M}_{i-1}^*$ of rank ρ determined by $C^{(i-1)}$, only powers of V_i occur in the denominator besides the $U_1^{\lambda_1} \cdots U_{i-1}^{\lambda_{i-1}}$. Choose $\mathfrak{p}^*$ further distinct from the finitely many prime ideals of $\mathfrak{K}$ that occur in the ideal determined by $V_1 V_2 \cdots V_n$. Then $R^{(i)}$ remains a common divisor of these determinants when the residue-class field modulo $\mathfrak{p}^*$ is taken as coefficient domain, and the rank of $\mathfrak{M}_{i-1}^*$ is preserved, since $C^{(i)}$ was a ρ-rowed determinant from $\mathfrak{M}_{i-1}^*$.

Finally $\mathfrak{p}^*$ is to be chosen so that $R^{(i)}$ remains the greatest common divisor in each case. Let $D^{(i)}$ run through all ρ-rowed determinants from $\mathfrak{M}_{i-1}^*$; by hypothesis

$$U_1^{\lambda_1} \cdots U_{i-1}^{\lambda_{i-1}} V_i^{\rho} D^{(i)} = R^{(i)} T^{(i)}$$

[57] A. Ostrowski, loc. cit. (note 21), lemma p. 296. A simpler proof is in E. Noether, Ein algebraisches Kriterium für absolute Irreduzibilität, Math. Ann. 85 (1922), pp. 26–33, no. 7.

[58] Compare H.-N., § 4.

with the $T^{(i)}$ having no common divisor containing x in the polynomial sense. Consequently the ideal derived from all $T^{(i)}$ contains a polynomial

$$G^{(i+1)} = W_i(u)x_{i+1}^{m_i} + \text{lower terms}, \qquad W_i \neq 0.$$

That $G^{(i+1)}$ may be assumed regular in x_{i+1} follows from composing the transformation (1) with one of $x_{i+1}, \ldots, x_n$ having new indeterminates as coefficients. If, then, $\mathfrak{p}^*$ is also chosen distinct from the finitely many prime ideals contained in the ideal determined by $W_1 W_2 \cdots W_n$, then when the residue-class field modulo $\mathfrak{p}^*$ is used as coefficient domain, $R_{\mathfrak{m}}$ is in fact preserved as norm. Quite correspondingly $\mathfrak{p}^*$ can be chosen still further so that $E_{\mathfrak{m}}$ also remains the elementary-divisor form. This proves Theorem XVI.

From Theorems XV and XVI, as a generalization of the theorem on absolute prime functions and using that theorem, one obtains

Theorem XVII. *If $\mathfrak{p}$ is an absolute prime ideal with algebraic integers from a (finite) algebraic number field $\mathfrak{K}$ as coefficients, then $\mathfrak{p}$ remains a prime ideal, more precisely an absolute prime ideal, modulo every prime ideal of $\mathfrak{K}$, with at most finitely many prime ideals of $\mathfrak{K}$ excepted.*

For the proof, let the norm $R_{\mathfrak{p}}$ of $\mathfrak{p}$ be chosen, as in Theorem XVI, so that it is divisible by $\mathfrak{p}$ also with respect to $\mathfrak{o}^*[u]$. Then

$$R_{\mathfrak{p}} = T(u)P^{(i)}(x),$$

where $P^{(i)}(x)$ may be assumed integral and primitive in the u. By Theorem XV, $P^{(i)}$, regarded as a polynomial in u and x with coefficients in $\mathfrak{K}$, is an absolute prime function. By the theorem on absolute prime functions it retains this property modulo every prime ideal of $\mathfrak{K}$, with at most finitely many prime ideals of $\mathfrak{K}$ excepted. Thus choose $\mathfrak{p}^*$ different from these finitely many prime ideals and, by Theorem XVI, also different from finitely many further prime ideals of $\mathfrak{K}$. Then $P^{(i)}$ – its coefficients from $\mathfrak{o}^*$ being replaced by the corresponding residue classes modulo $\mathfrak{p}^*$ – becomes the norm of $\mathfrak{p}$ when the residue-class field modulo $\mathfrak{p}^*$ is taken as coefficient domain P, and this norm is an absolute prime function. Taking P as coefficient domain, $\mathfrak{p}$ therefore remains an absolute prime ideal by Theorem XV. This proves Theorem XVII.

Göttingen, May 8, 1923.

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25. Elimination Theory and Ideal Theory

J. Ber. d. DMV 33 (1924), pp. 116–120

In what follows I would like to show how elimination theory can be built up in exact parallel with the theory of zeros for polynomials in one indeterminate. This construction rests on a connection between Hentzelt's elimination theory and the methods of field theory, as developed by Steinitz, and ideal theory, as I myself have developed it.[59]

I first want to sketch the question in the case of polynomials in one indeterminate. As coefficient domain let a fixed field P be taken once for all, to which the concepts prime function, etc., refer. Here P may be assumed to be abstractly defined in the sense of Steinitz, but in the special case it may naturally coincide with the field of rational numbers or the field of all complex numbers – according to the assumption formerly usual in elimination theory. The theory of zeros for polynomials in one indeterminate now splits into four steps:

1. *The decomposition theorem*, which permits the reduction of the question to one for prime functions – polynomials irreducible in P. For from

$$a(x) = p_1(x)^{\varrho_1} \cdots p_r(x)^{\varrho_r} = q_1(x) \cdots q_r(x)$$

– where the $p(x)$ denote distinct prime functions and the $q(x)$ therefore denote "largest primary functions" – it follows that $a(x)$ vanishes at all and only the zeros of the prime functions $p(x)$.

2. *The construction of the zero field for a prime function $p(x)$.* Such a zero field $\mathfrak{N} = P(\alpha)$ – where α denotes a zero of $p(x)$ – can always, by Steinitz, be constructed as an extension field of P isomorphic to the residue-class field $(\mathfrak{N})$ of $p(x)$. Conversely, every zero field lying in an extension field of P is isomorphic to the residue-class field $(\mathfrak{N})$; zero field and residue-class field are thus abstractly identical.

[59] Detailed presentation and literature references are in Math. Ann. 90 and in a forthcoming paper.

3. *The decomposition of $p(x)$ into linear factors in the Galois field determined by $p(x)$*. By repeating the construction given under 2. one arrives at the essentially uniquely determined Galois field $P(\alpha_1, \ldots, \alpha_m)$, which is precisely sufficient for $p(x)$ to split into linear factors.

4. *The meaning of multiplicity for the original polynomial $a(x)$*. By 1., 2., 3., the question of the zeros, and the corresponding decomposition into linear factors, is settled for an arbitrary polynomial $a(x)$. The multiplicity may be regarded as the degree of the individual primary function $q(x)$ – that is, the number of residue classes modulo $q(x)$ linearly independent with respect to P; for in the case of the algebraically closed coefficient domain P, where the $p(x)$ become linear, this number is precisely the multiplicity of the zeros.

I now pass to general elimination theory, and therefore take as basis the domain $\mathfrak{S}$ of all polynomials in $x_1, \ldots, x_n$ with coefficients from P; elimination theory can then be defined as the *theory of the zeros of an ideal* from this polynomial domain. By an ideal $\mathfrak{a}$ I understand, as usual, a system of polynomials such that together with f and g also $f - g$, and together with f also Af, belongs to $\mathfrak{a}$, where A is understood to be an arbitrary polynomial from $\mathfrak{S}$. By a zero of $\mathfrak{a}$ I understand every value system $\alpha_1, \ldots, \alpha_n$, belonging to an extension field of P, such that $f(\alpha)$ vanishes for every polynomial f belonging to the ideal. If $f_1, \ldots, f_t$ denotes an always existing ideal basis of $\mathfrak{a}$ – so that $f = A_1 f_1 + \cdots + A_t f_t$ holds for every f from $\mathfrak{a}$ – then the zeros of $\mathfrak{a}$ are plainly identical with the zeros of $f_1, \ldots, f_t$. Since, conversely, every system of finitely many polynomials can be regarded as a basis of the ideal derived from it, the definition of zeros just given for elimination theory is identical with the customary one, but avoids the difficulty lying in the arbitrary singling out of a basis. For polynomials in one indeterminate, where only principal ideals are involved, this difficulty does not arise, since the basis polynomial is uniquely determined up to a unit – a quantity from P as factor – but even there the ideal-theoretic conception will give a more exact insight.

Elimination theory now splits, correspondingly to the case of one indeterminate, into four steps:

I. *The decomposition theorem* – the representation of an ideal by largest primary components – which permits the reduction of the question to one for prime ideals. Here an ideal is called a prime ideal if it divides a product only when it divides at least one factor; it is called a primary ideal if it divides at least one factor or a power of each factor. To every primary ideal $\mathfrak{q}$ there belongs one and only one associated prime ideal $\mathfrak{p}$, which is a divisor of $\mathfrak{q}$ and a power of which is divisible by $\mathfrak{q}$. One has the representation as least common multiple:

$$\mathfrak{a} = [\mathfrak{q}_1, \ldots, \mathfrak{q}_r]; \qquad \mathfrak{p}_1^{\sigma_1} \cdots \mathfrak{p}_r^{\sigma_r} \equiv 0 \pmod{\mathfrak{a}}.$$

Here the $\mathfrak{q}$ are largest primary components; that is, each $\mathfrak{q}$ is primary, but the least common multiple of any two $\mathfrak{q}$'s is not primary. Moreover, without loss of generality it may be assumed that no $\mathfrak{q}$ divides the least common multiple of the remaining ones, so that no $\mathfrak{q}$ can simply be omitted; the $\mathfrak{p}$ are the associated prime ideals of the $\mathfrak{q}$. Now it is not the $\mathfrak{q}$, but – and herein lies the analogue to the case of one indeterminate – their associated prime ideals $\mathfrak{p}$ that are uniquely determined by $\mathfrak{a}$; and since each $\mathfrak{p}$ is a divisor of $\mathfrak{a}$, and conversely a product of powers of the $\mathfrak{p}$ is divisible by $\mathfrak{a}$, the zeros of the ideal therefore coincide with those of its associated prime ideals.

II. *The construction of the zero field for a prime ideal* $\mathfrak{p}$. By definition the residue classes modulo a prime ideal form a ring without zero divisors, containing at least one element distinct from the zero element as soon as $\mathfrak{p}$ is different from the unit ideal; by forming quotients this ring can therefore be extended to the residue-class field $(\mathfrak{R})$ of $\mathfrak{p}$. One can always construct an extension field $\mathfrak{R}$ of P isomorphic to $(\mathfrak{R})$ which becomes the zero field of $\mathfrak{p}$; that is, $\mathfrak{R} = P(\alpha_1, \ldots, \alpha_n)$, where $\alpha_1, \ldots, \alpha_n$ denotes a zero of $\mathfrak{p}$. Here $\mathfrak{R}$ has transcendence degree k $(0 \leq k < n)$ over P, hence contains k elements algebraically independent over P, whereas any $(k+1)$ elements become algebraically dependent over P. Conversely, every zero field of transcendence degree k lying in an extension field of P is isomorphic to the residue-class field $(\mathfrak{R})$; zero field of transcendence degree k and residue-class field are therefore abstractly identical.

III. *Decomposition of the norm of $\mathfrak{p}$ into linear factors in the Galois field determined by $\mathfrak{p}$*. If the indeterminates $x_1, \ldots, x_n$ are regarded as generated from original indeterminates $y_1, \ldots, y_n$ by a linear transformation with indeterminates as coefficients, then the zero field $\mathfrak{R}$ of transcendence degree k can be regarded as a finite algebraic extension field of $P(x_{i+1}, \ldots, x_n)$ $(k = n - i)$. One can therefore pass to the Galois field $\mathfrak{R}$ over $P(x_{i+1}, \ldots, x_n)$, in which the prime function $P^{(i)}(x_i, x_{i+1}, \ldots, x_n)$ with the zero α_i splits into linear factors. This prime function $P^{(i)}$, when one returns to the original indeterminates y, plays the role of the fundamental equation; for α_i becomes equal to the fundamental form $u_1\beta_1 + \cdots + u_n\beta_n$, where β denotes a zero system in the y algebraically dependent on the parameters $x_{i+1}, \ldots, x_n$. The decomposition into linear factors thus gives the product over all zeros. But the prime function $P^{(i)}$, or a power of it, may also be regarded as the norm $N(\mathfrak{p})$ of $\mathfrak{p}$, by considering $\mathfrak{p}$ as a module of linear forms in the power products of $x_1, \ldots, x_{i-1}$, taking $P(x_{i+1}, \ldots, x_n)$ instead of P as coefficient domain, and defining the norm as the product over all elementary divisors – of which only finitely many are distinct from the unit. For a perfect field P as coefficient domain one always obtains $N(\mathfrak{p}) = P^{(i)}$; for an imperfect field a power of $P^{(i)}$ may occur as the norm.

IV. *The norm of the original ideal and the meaning of multiplicity.* If correspondingly one regards the ideal $\mathfrak{a}$ successively as a module of linear forms in the power products of $x_1, \ldots, x_{i-1}$ $(i = 1, 2, \ldots, n)$, then it can be shown that, on taking $P(x_{i+1}, \ldots, x_n)$ as basis, this module has in each case only finitely many elementary divisors distinct from the unit; this is a finiteness condition following from the existence of an ideal basis. The product over these finitely many elementary divisors gives the individual norms; the norm $N(\mathfrak{a})$ is defined as the product of all individual norms. The largest primary factors $Q^{(i)}(x_i, x_{i+1}, \ldots, x_n)$ of $N(\mathfrak{a})$ correspond one-to-one to the associated prime ideals $\mathfrak{p}$ of $\mathfrak{a}$, in such a way that $Q^{(i)}$ becomes a power of $P^{(i)}$, where $P^{(i)}$ (or, if necessary, a power of $P^{(i)}$) denotes the norm $N(\mathfrak{p})$ of $\mathfrak{p}$. Thus there arises a decomposition

$$N(\mathfrak{a}) = N(\mathfrak{p}_1)^{\varrho_1} \cdots N(\mathfrak{p}_r)^{\varrho_r} = Q_1 \cdots Q_r,$$

– where if necessary $N(\mathfrak{p})$ is to be replaced by the highest elementary divisor $E(\mathfrak{p}) = P^{(i)}$ – and thus, by I, II, III, the decomposition of the norm into linear factors has been effected for all zeros of $\mathfrak{a}$. The multiplicity of the zeros corresponding to the individual prime ideal $\mathfrak{p}$ of $\mathfrak{a}$ can be regarded as the degree of $Q^{(i)}$, which can be interpreted as the number of residue classes linearly independent over $P(x_{i+1}, \ldots, x_n)$. More precisely, these are the residue classes of certain isolated components of the decomposition uniquely determined by $\mathfrak{p}$ and by the prime ideals of higher dimension, namely the residue classes of the ground ideal of the $(i-1)$-st stage modulo the component, determined by $\mathfrak{p}$, of the ground ideal of the i-th stage; the latter component has $\mathfrak{p}$ and all prime ideals of higher dimension as associated prime ideals, while the former has only the prime ideals of higher dimension.

A complete *insertion of the case of one indeterminate* is obtained when one passes here too to the principal ideal. The decomposition given under 1. then splits – according to whether $a(x)$ is regarded as basis polynomial or norm – into two separate decompositions: into the representation of the principal ideal as the least common multiple of largest primary components according to I, which here amounts to a representation as a product of powers of prime ideals; or into the decomposition of the norm as a product of powers of the norms of the individual prime ideals according to IV. In the construction of the zero field according to 2., what is in question in reality is the zero field of the prime ideal defined by $p(x)$, while in 3. the norm is decomposed into linear factors. The definition of multiplicity according to 4. is subordinated to that given in IV, since the ground ideal of the zeroth stage is equal to the unit ideal, while for one indeterminate the ground ideal of the first stage already equals the ideal itself.

The *insertion into the usual elimination theory* is given by the fact that the norm of the ideal has the meaning of a resultant. In this interpretation, the old problem of elimination theory thus becomes a part of the ideal theory of the polynomial domain, namely that part which, beside the decomposition theorems for ideals, also treats the decomposition of their norms and the connection between the two decompositions.

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26. Abstract Construction of Ideal Theory in the Algebraic Number Field

J. Ber. d. DMV 33 (1924), p. 102

4. E. Noether, Göttingen: Abstract construction of ideal theory in the algebraic number field.

The abstract properties were stated that are completely equivalent to ideal theory in the algebraic number field, and that therefore characterize all rings which possess such an ideal theory. They are the rings with identity element and without zero divisors in which the double chain condition holds – every divisor chain of ideals breaks off in the finite, and likewise every multiple chain modulo every ideal different from the zero ideal – and which are integrally closed with respect to their quotient field. If the latter assumption is dropped, then theorems result concerning ideal theory in a finite order, some of which are already found without proof in Dedekind (3rd edition).

A detailed presentation will appear in the Mathematische Annalen.

27. Hilbert Numbers in Ideal Theory

J. Ber. d. DMV 34 (1925), p. 101

E. Noether: Hilbert numbers in ideal theory. By Hilbert numbers one generally understands numbers that refer to the enumeration of residue classes of ideals. Hilbert himself, in his "characteristic function," gave a function which, for the ideals of the polynomial domain, gives from a certain bound onward the exact number of linearly independent residue classes. For the lower degree numbers Macaulay supplemented this function by a generating function, which Ostrowski was able to evaluate explicitly. But Macaulay also gave numbers of another kind, which have proved to be the essential ones for general ideal theory, since they can be formulated abstractly. In accordance with the general decomposition theorems, it is enough to give the definition for primary ideals $\mathfrak{q}$. What is involved is the number of residue classes, linearly independent with respect to the residue-class field of the associated prime ideal $\mathfrak{p}$. These numbers split into partial systems, each given by the number of irreducible components of $\mathfrak{q}, \mathfrak{q}/\mathfrak{p}, \mathfrak{q}/\mathfrak{p}^2, \ldots$. The number of the i-th partial system is also characterized by the number of terms of a composition series which runs from $\mathfrak{q}/\mathfrak{p}^i$ to $\mathfrak{q}/\mathfrak{p}^{i-1}$. The total composition series and its Hilbert numbers form the equivalent of the representation of primary ideals as powers of prime ideals, which is no longer valid in the general case – for instance in finite orders of an algebraic number field.

28. Group Characters and Ideal Theory

J. Ber. d. DMV 34 (1925), p. 144

3. E. Noether, Göttingen: Group characters and ideal theory.

The Frobenius theory of group characters – that is, of representations of finite groups – is conceived as the ideal theory of a completely reducible ring, the group ring. Therefore the isomorphism theorems for the additive decomposition of a completely reducible ring are developed in general; the representation as a direct sum of directly indecomposable, simple ideals is unique up to isomorphism, and the indecomposable one-sided ideals belonging to the same two-sided ideal are mutually isomorphic. If therefore isomorphic ideals are combined into one class, then there are as many distinct indecomposable ideal classes as there are two-sided indecomposable ideals.

In the specialization to completely reducible systems of hypercomplex quantities – in particular to the group ring – indecomposable ideal classes and irreducible representation classes correspond mutually and uniquely. Furthermore, two-sided indecomposable ideals and characters correspond; and the number of additively indecomposable components of a two-sided ideal is equal to the rank of the indecomposable ideals. Thus the Frobenius theory has been classified within this framework.

A detailed presentation is to appear in the Math. Ann.

29. The Finiteness Theorem for the Invariants of Finite Linear Groups of Characteristic p

Nachr. v. d. Ges. d. Wiss. zu Göttingen 1926, pp. 28–35

Presented by R. Courant in the session of 30 July 1926.

I show in what follows that the well-known finiteness theorem for the invariants of finite groups of linear substitutions[60] remains valid when, as coefficient domain, instead of an ordinary number field one takes an arbitrary field – hence in particular one of characteristic p.[61] However, the known proofs of finiteness fail as soon as the order of the group is divisible by p; deeper arithmetical facts must be brought in, namely the following criterion, previously known only for characteristic zero:

Finiteness criterion[62]: An integral domain $\mathfrak{S}$ of polynomials – more generally of rational functions – with coefficients from a field is finite if and only if $\mathfrak{S}$ contains a finite subring $\mathfrak{R}$ such that $\mathfrak{S}$ depends algebraically and integrally on $\mathfrak{R}$. Each integral basis of $\mathfrak{R}$ is then completed, by a module basis of $\mathfrak{S}$ with respect to a certain subring of $\mathfrak{R}$, to an integral basis of $\mathfrak{S}$.

This criterion – which is of independent interest in itself – rests on the existence of the rational basis and on the fact that the divisor-chain theorem is preserved in passing to the integral quantities of a finite

[60] For an elementary proof of finiteness see E. Noether, Der Endlichkeitssatz der Invarianten endlicher Gruppen. Math. Ann. 77 (1916), pp. 89–92.

[61] For the concepts of field theory see E. Steinitz, Algebraische Theorie der Körper. J. f. M. 137 (1910), pp. 167–309. Characteristic in § 4, root field in § 12, quotient field in § 3.

[62] Compare E. Noether, Algebraische und Differentialinvarianten. Jahresber. d. d. Math.-Ver. 32 (1923), pp. 177–184, no. 3, Theorem 4, and the literature cited there. In the last sentence of the criterion there it incorrectly says "module basis with respect to $\mathfrak{R}$" (fundamental system of integral quantities) instead of "with respect to a certain subring of $\mathfrak{R}$".

extension. This second theorem was proved in the preceding note of Artin and van der Waerden for arbitrary characteristic, only under a certain finiteness assumption on the original domain, an assumption fulfilled in the case of finite groups (the root ring represents a finite extension). For the existence of the rational basis I give a proof valid for arbitrary fields as coefficient domain; hence I can prove the finiteness criterion under the above restriction.

That, for invariants of finite groups, the finiteness criterion is fulfilled rests on the principle of symmetric functions. Whereas in characteristic zero the coefficients of the Galois resolvent already furnish the integral basis, in the general case they generate only the finite subring required in the finiteness criterion. From the existence of this finite subring the finiteness of the "relative" invariants and of the "modular" invariants follows by the same arguments.

The invariants considered here include, as special cases, the "projective invariants mod p" treated by Dickson and his school.[63] Here the finiteness theorem was known for modular invariants, but not for general "formal" invariants.

§ 1. The Finiteness Criterion

1. **Theorem on the existence of the rational basis.** First formulation: Every system S of rational functions of n indeterminates, with coefficients from a field P, possesses a finite rational basis; that is, there are finitely many functions of the system such that every function of the system can be expressed rationally, with coefficients from P, through these finitely many.

Second formulation: Let $\mathfrak{K}$ be an intermediate field between P and $P(x_1, \ldots, x_n)$ – where $x_1, \ldots, x_n$ denote indeterminates – of transcendence degree $t \leq n$; let $y_1, \ldots, y_t$ be an irreducible system[64] from $\mathfrak{K}$. Then $\mathfrak{K}$ becomes a finite algebraic extension of $P(y_1, \ldots, y_t)$.

The two formulations are in fact equivalent. It is enough to prove the existence of the rational basis for all intermediate fields; for the field derived from a system S becomes such an intermediate field $\mathfrak{K}$, whose rational basis immediately gives a rational basis of S, since every element of $\mathfrak{K}$ becomes a rational combination of finitely many functions from S. From the second formulation it follows, however, that an arbitrary irreducible system $y_1, \ldots, y_t$ from $\mathfrak{K}$, enlarged by the finitely many functions that characterize $\mathfrak{K}$ as a finite extension of $P(y_1, \ldots, y_t)$, gives such a rational basis. Conversely, if according to the first formulation a rational basis of $\mathfrak{K}$ is given, then by the definition of transcendence degree each basis element must be algebraically dependent on $P(y_1, \ldots, y_t)$; hence $\mathfrak{K}$ is thereby characterized as a finite algebraic extension of $P(y_1, \ldots, y_t)$.

The theorem will be proved in the second formulation. First let $\mathfrak{K}$ have the same transcendence degree n as $P(x_1, \ldots, x_n)$; let $y_1, \ldots, y_n$ be an irreducible system from $\mathfrak{K}$ and consequently from $P(x_1, \ldots, x_n)$. Because the transcendence degree of $y_1, \ldots, y_n$ is n, each x_i becomes algebraic over $P(y_1, \ldots, y_n)$; thus $P(x_1, \ldots, x_n)$, and hence $\mathfrak{K}$ as an intermediate field, becomes a finite algebraic extension of $P(y_1, \ldots, y_n)$.

Now let $t < n$; let the numbering of the x be chosen so that $y_1, \ldots, y_t, x_{t+1}, \ldots, x_n$ form an irreducible system from $P(x_1, \ldots, x_n)$. Then $x_{t+1}, \ldots, x_n$ are irreducible with respect to $P(y_1, \ldots, y_t)$ and hence, by the transitive law of algebraic dependence, also irreducible with respect to $\mathfrak{K}$ and to every intermediate field $\mathfrak{M}$ between $P(y_1, \ldots, y_t)$ and $\mathfrak{K}$. This means that $\overline{\mathfrak{K}} = \mathfrak{K}(x_{t+1}, \ldots, x_n)$, respectively $\overline{\mathfrak{M}} = \mathfrak{M}(x_{t+1}, \ldots, x_n)$, becomes a purely transcendental extension of $\mathfrak{K}$, respectively $\mathfrak{M}$; hence every element of $\overline{\mathfrak{K}}$ not belonging to $\mathfrak{K}$ is transcendental with respect to $\mathfrak{K}$, and similarly every element of $\overline{\mathfrak{M}}$ not belonging to $\mathfrak{M}$ is transcendental with respect to $\mathfrak{M}$. Likewise let $\overline{P}$ be set equal to the purely transcendental extension $P(x_{t+1}, \ldots, x_n)$; then $y_1, \ldots, y_t$ are irreducible with respect to $\overline{P}$.

The proof can now be reduced to the case already settled by taking $\overline{P}$ as coefficient domain. From $P < \mathfrak{K} < P(x_1, \ldots, x_t, x_{t+1}, \ldots, x_n)$[65] it follows that $\overline{P} < \overline{\mathfrak{K}} < \overline{P}(x_1, \ldots, x_t)$, where $\overline{\mathfrak{K}}$ has the same transcendence degree t over $\overline{P}$ as $\overline{P}(x_1, \ldots, x_t)$. Hence $\overline{\mathfrak{K}}$ has a finite rational basis with respect to $\overline{P}$ as coefficient domain, say $y_1, \ldots, y_t; z_1, \ldots, z_s$. Here the z can be chosen from $\mathfrak{K}$, since $\overline{\mathfrak{K}}$ is the union field of $\mathfrak{K}$ and $\overline{P}$. It remains to show that $\mathfrak{K}$ equals $\mathfrak{M} = P(y_1, \ldots, y_t; z_1, \ldots, z_s)$. Now $\mathfrak{M}$ is an intermediate field between $P(y_1, \ldots, y_t)$ and $\mathfrak{K}$; hence every element of $\mathfrak{K}$ is algebraic with respect to $\mathfrak{M}$. On the other hand, $\overline{\mathfrak{M}}$ equals $\overline{\mathfrak{K}}$, and hence $\mathfrak{K}$ is contained in $\overline{\mathfrak{M}}$. Thus $\mathfrak{K}$ is an intermediate field between $\mathfrak{M}$ and $\overline{\mathfrak{M}}$; since every element of $\overline{\mathfrak{M}}$ not belonging to $\mathfrak{M}$ is transcendental with respect to $\mathfrak{M}$, it necessarily follows that $\mathfrak{K}$ is identical

[63] Compare Dickson, On Invariants and the theory of Numbers. The Madison Colloquium 1913. More recent literature is in the volumes of the Transactions of the American Mathematical Society that have appeared since then.

[64] A system is called irreducible, according to Steinitz, if it consists of algebraically independent functions. A system is said to have transcendence degree t if it contains at least one irreducible system of t elements, but none of more elements. For the simple theorems on irreducible systems used below, see Steinitz, § 22.

[65] $P < \mathfrak{K}$ means: P is contained in $\mathfrak{K}$, hence is a subfield of $\mathfrak{K}$.

with $\mathfrak{M}$.

Corollary: If $P < \mathfrak{K} < \mathfrak{L} < P(x_1, \ldots, x_n)$ and $\mathfrak{L}$ is algebraic with respect to $\mathfrak{K}$, then $\mathfrak{L}$ is a finite algebraic extension of $\mathfrak{K}$. For every element of a rational basis of $\mathfrak{L}$ is then algebraic with respect to $\mathfrak{K}$; hence $\mathfrak{L}$ is characterized as a finite extension of $\mathfrak{K}$.

2. **Proof of the finiteness criterion.** The condition is plainly necessary; for if $\mathfrak{S}$ is a finite integral domain, then $\mathfrak{S}$ itself can be regarded as the finite subring $\mathfrak{R}$ occurring in the criterion.

The condition is sufficient under the assumption that the root field $P^{1/p}$ is a finite extension of the coefficient field[66] whenever the latter has characteristic p; for characteristic zero no restrictive assumption is needed. To prove this one has to show that $\mathfrak{S}$ has a finite module basis with respect to a subring of $\mathfrak{R}$.

Let $\mathfrak{K}$ denote the quotient field[67] of $\mathfrak{R}$, and let $\mathfrak{L}$ denote the quotient field of $\mathfrak{S}$. These quotient fields exist since the rings involved have no zero divisors; and one has $P < \mathfrak{K} < \mathfrak{L} < P(x_1, \ldots, x_n)$. Hence by the corollary under 1., $\mathfrak{L}$ becomes a finite algebraic extension field of $\mathfrak{K}$. Moreover, by hypothesis every element of $\mathfrak{S}$ is integral with respect to $\mathfrak{R}$, and thus $\mathfrak{S}$ becomes a subring of the ring $\mathfrak{S}'$ consisting of all elements of $\mathfrak{L}$ integral over $\mathfrak{R}$. But since nothing is assumed about the integral closedness of $\mathfrak{R}$ in $\mathfrak{K}$, the fact that the divisor-chain theorem is preserved under finite extension cannot be applied directly. Rather, one must first pass to a likewise finite subring $\mathfrak{T}$ of $\mathfrak{R}$ for which this integral closedness is fulfilled.

First suppose that the coefficient domain P contains infinitely many elements. Then one can select from $\mathfrak{R}$ an irreducible system $g_1(x), \ldots, g_t(x)$ in such a way that $\mathfrak{R}$, and hence $\mathfrak{S}$, is integral over $\mathfrak{T} = P[g_1(x), \ldots, g_t(x)]$. Namely, let $f_1(x), \ldots, f_k(x)$ be the integral basis of $\mathfrak{R}$ existing by hypothesis; if this does not form an irreducible system, then – after a possible renumbering of the indices – let $F(f_k; f_1, \ldots, f_{k-1}) = 0$ be a relation satisfied by f_k. If one replaces $f_1, \ldots, f_{k-1}$ by

$$f_1 + t_1 f_k, \ldots, f_{k-1} + t_{k-1} f_k,$$

then f_k, and hence also $f_1, \ldots, f_{k-1}$, is integral over these new functions when the t_i are chosen as indeterminates. Since P has infinitely many elements, the t_i can be specialized to elements of P in such a way that this integral dependence is preserved. After finitely many steps, using the transitive law of integral dependence, one therefore arrives at the required ring $\mathfrak{T} = P[g_1(x), \ldots, g_t(x)]$. Then $\mathfrak{L}$ is a finite extension field of the quotient field $P(g_1(x), \ldots, g_t(x))$ of $\mathfrak{T}$, and the ring $\mathfrak{S}'$ of all elements of $\mathfrak{L}$ integral over $\mathfrak{R}$ is identical with the ring of all elements of $\mathfrak{L}$ integral over $\mathfrak{T}$.

In $\mathfrak{T}$ the divisor-chain theorem holds for the system of all ideals, and $\mathfrak{T}$ is integrally closed in its quotient field, both because $\mathfrak{T}$ is isomorphic to the polynomial domain in t indeterminates. From the assumption that $P^{1/p}$ is finite over P it follows further that $\mathfrak{T}^{1/p}$ is finite over $\mathfrak{T}$.[68][69] Hence in $\mathfrak{S}'$ the divisor-chain theorem holds for the system of all $\mathfrak{T}$-modules; in particular, $\mathfrak{S}$, as a subring of $\mathfrak{S}'$ containing $\mathfrak{T}$, has a finite $\mathfrak{T}$-module basis. The existence of the finite $\mathfrak{T}$-module basis for $\mathfrak{S}$ continues to hold without any restrictive assumption on the coefficient domain P if $\mathfrak{L}$ is an extension of the first kind of $P(g_1(x), \ldots, g_t(x))$, hence in particular always when P has characteristic zero.[70]

Let $h_1(x), \ldots, h_s(x)$ be a $\mathfrak{T}$-module basis of $\mathfrak{S}$. Then the representation

$$f(x) = A_1(g)h_1(x) + \cdots + A_s(g)h_s(x)$$

for an arbitrary $f(x)$ from $\mathfrak{S}$ – where the A, as elements of $\mathfrak{T}$, denote polynomials in the g – shows that $g_1(x), \ldots, g_t(x), h_1(x), \ldots, h_s(x)$ form the required integral basis.

If the field P contains only finitely many elements, then the field $P(u)$ obtained by adjoining an indeterminate u – that is, an element transcendental over $\mathfrak{S}$ – contains infinitely many elements. Thus the ring $\mathfrak{S}(u)$ has a finite integral basis with respect to $P(u)$ as coefficient domain; and since $\mathfrak{S}(u)$ is the union ring of $\mathfrak{S}$ and $P(u)$, this basis can be chosen from $\mathfrak{S}$ by splitting according to powers of u, although the basis elements g_i of $\mathfrak{S}$ will now no longer form an irreducible system. Accordingly every polynomial $f(x)$ from $\mathfrak{S}$ admits a representation $C(u)f(x) = G(u, g_i, h_i)$, where $C(u)$ denotes a nonzero polynomial from $P[u]$ and G becomes integral in u, g_i, h_i. Comparing the coefficients of a suitable power of u shows that $g_i(x)$ and $h_i(x)$ give an integral basis of $\mathfrak{S}$. Hence the **finiteness criterion is proved in full generality**.

[66] See p. 28, note 2.

[67] See p. 28, note 2.

[68] Compare Hilbert, Über die vollen Invariantensysteme, § 1. Math. Ann. 42 (1893), pp. 313–373.

[69] Compare the note of Artin and van der Waerden cited in the introduction.

[70] Compare, for example, E. Noether, Abstrakter Aufbau der Idealtheorie in algebraischen Zahl- und Funktionenkörpern, § 3. Math. Ann. 96 (1926), pp. 26–61.

§ 2. Finiteness of the Invariants

1. By a **finite linear group of characteristic** p is meant a group $\mathfrak{H}$ of h linear substitutions $A_1, \ldots, A_h$ (of nonzero determinant) with coefficients from a field P of characteristic p. Here A_k is the linear substitution

$$x_1^{(k)} = a_{11}^{(k)} x_1 + \cdots + a_{1n}^{(k)} x_n, \ldots, x_n^{(k)} = a_{n1}^{(k)} x_1 + \cdots + a_{nn}^{(k)} x_n,$$

or, in abbreviated form, $(x^{(k)}) = A_k(x)$. Thus the group $\mathfrak{H}$ carries the row (x) with elements $x_1, \ldots, x_n$ into the rows $(x^{(k)})$ with elements $x_1^{(k)}, \ldots, x_n^{(k)}$. Since the identity must be contained among $A_1, \ldots, A_h$, the row (x) is also contained among the rows $(x^{(k)})$.

By an **integral rational (absolute) invariant of the group** is meant such an integral rational function of $x_1, \ldots, x_n$ – with coefficients from P[71] – that remains identically unchanged under application of $A_1, \ldots, A_h$. Thus all symmetric functions, with coefficients from P, of the rows $(x^{(k)})$ belong to these invariants. Conversely, for every such invariant one has

$$f(x) = f(x^{(1)}) = \cdots = f(x^{(h)}) \quad \text{or} \quad h \cdot f(x) = f(x^{(1)}) + \cdots + f(x^{(h)}).$$

It follows that in the case where the order h of the group is not divisible by the characteristic – in particular in characteristic zero – the symmetric functions of the rows $(x^{(k)})$, with coefficients from P, exhaust the totality of invariants. Since these symmetric functions possess a finite integral basis, the proof of finiteness is accomplished in this case; at the same time the integral basis is obtained as the system $G_{\alpha\alpha_1\ldots\alpha_n}(x)$ of coefficients of the Galois resolvent:[72]

$$\Phi(z, u) = \prod_{k=1}^{h} (z - u_1 x_1^{(k)} - \cdots - u_n x_n^{(k)})$$
$$= z^h + \sum G_{\alpha\alpha_1\ldots\alpha_n}(x) z^\alpha u_1^{\alpha_1} \cdots u_n^{\alpha_n} \quad (\alpha \neq h;\ \alpha + \alpha_1 + \cdots + \alpha_n = h).$$

2. In the case where h is divisible by p, it is not necessary that every invariant of the group be a symmetric function of the rows $(x^{(k)})$, since one can no longer conclude from $h \cdot f(x)$ to $f(x)$. But the ring $\mathfrak{R}$ derived from the finitely many invariants $G_{\alpha\alpha_1\ldots\alpha_n}(x)$ – the coefficients of the Galois resolvent – forms a finite subring of the system $\mathfrak{S}$ of all invariants, and it remains to show that $\mathfrak{S}$, with respect to $\mathfrak{R}$, satisfies the hypotheses of the finiteness criterion.

First, without loss of generality,[73] the subfield $\overline{P}$ of P derived from all substitution coefficients $a_{\mu\nu}^{(k)}$ may be taken as coefficient domain. This field, derived from finitely many elements, is thus obtained by adjoining finitely many algebraic or transcendental elements to the prime field; and since the prime field is perfect, it follows that $\overline{P}^{1/p}$ is finite with respect to $\overline{P}$.[74]

It remains to show that $\mathfrak{S}$ depends algebraically and integrally on $\mathfrak{R}$. By 1., among the h rows $x^{(k)}$ there also occurs the original $x_1, \ldots, x_n$; hence the Galois resolvent $\Phi(z, u)$ vanishes identically in u for $z = u_1 x_1 + \cdots + u_n x_n$. If one specializes in each case all u to zero except u_i, which is set equal to one, one obtains a relation

$$x_i^h + \sum x_i^\alpha G_{\alpha 0 \ldots \alpha_i \ldots 0}(x) = 0 \quad (\alpha \neq h;\ \alpha + \alpha_i = h),$$

which shows that each x_i depends algebraically and integrally on $\mathfrak{R}$. But then the same holds for every polynomial in x; in particular $\mathfrak{S}$ is integral over $\mathfrak{R}$. Thus the hypotheses of the finiteness criterion are fulfilled, and the **finiteness of the invariants is proved**.

3. It should be noted that the same arguments give the **finiteness of the relative and modular invariants**. A polynomial $f(x)$ is called a **relative invariant** of the group if all $f(x^{(k)})$ differ from $f(x)$ only by a nonzero factor from P. The system of absolute invariants – in particular the ring $\mathfrak{R}$ derived from the $G_{\alpha\alpha_1\ldots\alpha_n}(x)$ – therefore forms a finite subring of the ring derived from all relative invariants; and, as above, the hypotheses of the finiteness criterion with respect to $\mathfrak{R}$ are fulfilled.

A polynomial $f(x)$ is called a **modular invariant** of the group if $f(x)$ becomes equal to $f(x^{(k)})$ for every special value system $x_1 = \alpha_1, \ldots, x_n = \alpha_n$ from P. Every absolute invariant is at the same time a modular invariant; if, however, P contains only finitely many elements, the converse need not hold. Here too the hypotheses of the finiteness criterion are fulfilled with respect to the subring $\mathfrak{R}$ derived from the $G_{\alpha\alpha_1\ldots\alpha_n}(x)$.

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[71] This is not a restriction of generality inasmuch as every invariant with coefficients from a field of characteristic p – which must possess a common overfield with P in order for the invariants to be defined – can be linearly decomposed into finitely many invariants with coefficients from P. In the general case one need only express the coefficients through ones that are linearly independent over P; then the property of invariance is satisfied identically in these independent coefficients.

[72] See p. 28, note 1.

[73] See p. 33, note.

[74] See p. 32, note 2.

30. Abstract Construction of Ideal Theory in Algebraic Number and Function Fields

Math. Ann. 96 (1927), pp. 26–61.

In what follows an abstract characterization is given of all those rings whose ideal theory agrees with the ideal theory of all integers of an algebraic number field: that is, whose ideals can be represented uniquely as products of powers of prime ideals. Among the best-known examples of such rings are also the ring of all integers of an algebraic function field in one indeterminate, and, more generally, in several indeterminates as soon as, following Kronecker, one restricts to ideals of highest dimension, i.e. passes to a certain quotient ring, the functional domain.

For the underlying commutative ring, the axioms completely equivalent to this ideal theory are the following:[75]

I. Divisor-chain condition. Every chain of ideals in which each ideal is a proper divisor of the preceding one terminates after finitely many steps; in other words, in every divisor chain of ideals there is an index from which on all ideals are equal.

II. Multiple-chain condition modulo every nonzero ideal. Every chain of ideals — all of which are divisors of a fixed ideal different from the zero ideal — in which each ideal is a proper multiple of the preceding one terminates after finitely many steps.

III. Existence of the identity element for multiplication.

IV. Ring without zero divisors.

V. Integral closedness in the quotient field. Every element of the quotient field that is integral with respect to the ring belongs to the ring.

From these axioms I develop, step by step, an increasingly restricted ideal theory up to the desired one; conversely, I show that all these axioms are in fact satisfied there.

From the divisor-chain condition I it follows, as I showed in *Ideal Theory* §4, that an ideal is representable as the least common multiple of finitely many primary ideals belonging to distinct prime ideals. I recall this proof briefly here (§6), both because it can be simplified by using the concept of the ideal quotient, and because I want to correct an oversight: the original proof uses, at one point, the not-assumed existence of the identity element of multiplication.[76]

The assumption of the multiple-chain condition has the effect (§7) that, in the residue class ring modulo every ideal different from the zero ideal, a prime ideal cannot have a proper divisor, except for the unit ideal containing all elements. Consequently the primary components of these ideals are determined uniquely, except for those belonging to the unit ideal. In a somewhat less sharp form this result is already found in Masazo Sono;[77] his assumption of the existence of a composition series in the residue class ring is equivalent to the "double chain condition" in this ring, as I show briefly at the end (§10). By double chain condition I mean the validity of both the divisor-chain and multiple-chain conditions without restriction to ideals different from the zero ideal.

If axiom III — existence of the identity element — is also satisfied, the unit ideal does not occur as an associated prime ideal. The primary components are therefore uniquely determined; they are pairwise coprime, and their least common multiple equals their product. Axioms I, II, III hold for all finite orders of an algebraic number field; already Dedekind (Dirichlet–Dedekind, *Zahlentheorie*, 3rd ed., 11th supplement, §172), without giving the proof in full detail, stated here the unique representation of an ideal as a product of pairwise coprime one-kind ideals (primary components).

From assumption IV — absence of zero divisors — it follows that the different powers of an ideal distinct from both the zero ideal and the unit ideal are all distinct, and that the quotient field exists. If finally axiom V of integral closedness in the quotient field is also satisfied, the primary ideals — uniquely determined

[75] For the definition of the basic concepts of ideal theory, compare for instance E. Noether, *Idealtheorie in Ringbereichen*, Math. Ann. 83 (1921), pp. 24–66 (cited below as *Ideal Theory*). Incidentally, the most essential concepts are also formulated briefly in the present paper, especially in §§1, 4, and 5. For completeness, some material is included that is not needed for the immediate purposes of this paper.

[76] P. Urysohn called my attention to this oversight; my own alteration of the proof was somewhat more cumbersome than the one given here, which is due to E. Artin. He had already filled the gap for himself earlier, and, independently of me, had also observed the simplification with the ideal quotient. — I take a further simplification, which avoids the introduction of the "reduced representation," from a related question in W. Krull, *Algebraische Theorie der Ringe III*, Math. Ann. 92 (1924), pp. 181–213, Theorem I.

[77] Masazo Sono, *On the Reduction of Ideals*, Memoirs of the College of Science, Kyoto University, Series A, 7 (1924), pp. 191–204.

because of IV — become powers of prime ideals, and the usual factorization theorem is obtained (§8). This latter proof rests on a consequence of integral closedness already drawn by Dedekind (3rd ed., §172) in the case of number fields: a genuinely fractional element can be represented as a quotient of integers in such a way that even the square of the numerator is not divisible by the denominator. In later presentations this consequence is replaced — likewise as a consequence of integral closedness — by the much less transparent generalized Gaussian theorem or by corresponding module theorems that say somewhat more; all of these are theorems whose application presupposes basis elements, whereas here one works directly with the ideals themselves.

In the converse direction §9, the axioms distinct from axiom V follow almost directly; V follows from the proof that a genuinely fractional element can always be represented so that no power of the numerator is divisible by the denominator, which is therefore a strengthening of the consequence originally derived from V. This proof succeeds only after one draws from the ideal representation the usual conclusions: representation by principal ideals modulo a fixed ideal, and the theory of fractional ideals, i.e. of ideal quotients in the field. The fact that integral closedness follows from the representation by powers of prime ideals was also already known to Dedekind, as is shown by a remark in the 3rd edition, §172, p. 522, bottom.

Axioms I through V imply that the four decompositions which are generally distinct — into least pairwise coprime ideals, into mutually prime ideals, into largest primary components, and into irreducible ideals — coincide; conversely, from this coincidence and axioms I–IV, integral closedness follows. For rings with zero divisors, fulfillment of the remaining axioms — with V defined as integral closedness in the quotient ring — does not necessarily imply that the decompositions coincide, as the example of the residue class ring modulo certain ideals of a finite order shows §9.

In §1 I sketch the theory of quantities integral with respect to a ring, up to Dedekind's consequence from integral closedness. In §2 I show how the chain conditions are transferred from the ring to modules consisting of linear forms, with this ring as coefficient and multiplier domain.[78] From this I draw (§3) the conclusion that, on passing to the integral elements of an algebraic extension field of the first kind, axioms I to IV remain valid for every order, while for the order consisting of all integral elements, V is also valid. This fact — of course familiar for the algebraic number field — permits the subordination of the function fields mentioned at the beginning; in particular, by the simple proof that the functional domain derived from integral polynomials in several indeterminates satisfies the axioms, Kronecker's ideal theory is fully subsumed.[79]

The ideal theory subsequently developed does not presuppose these §§2 and 3, which serve only for classification. In §4 and §5 the foundations independent of axioms I through V are given; this brings out more sharply than in the original foundation which parts of the theory are independent of finiteness assumptions.

§1. Theory of Integral Quantities

Let there be given a commutative ring[80] T without zero divisors, with identity element e for multiplication (axioms III and IV); in T a fixed subring R containing the identity element is distinguished. The concepts module, order, integral are throughout to refer to this fixed ring R, which later will be omitted from the notation for brevity.[81] Elements of R will be denoted by Latin letters, elements of T in general by Greek letters.

1. A system M of elements of T is called, as usual, an R-module — briefly, a module — if, together with two elements α and β, it also contains their difference, and with α also $r\alpha$, where r is any element of R. M is said to be divisible by N,

$$M \equiv 0 \pmod{N},$$

[78]This form of the module theorems is due to Prüfer (Neue Begründung der algebraischen Zahlentheorie, §2, Math. Ann. 94 (1924), pp. 198–243) and is more transparent than the customary transfer of the basis theorems.

[79]The difference appears only in the discriminant theorems, because the residue class ring of the functional domain modulo a prime number becomes an imperfect field, and hence extensions of the second kind can occur.

[80]As is known, a ring is defined once an equality relation is given in a set, and in addition two operations, addition and multiplication, are given satisfying the usual laws: addition is associative, commutative, and uniquely reversible; multiplication is associative — commutative when the ring is commutative — and distributive with respect to addition. If the underlying definition of equality is not set-theoretic identity, then, so that the same definition of equality is retained in every subsystem, such a subsystem (subring, module, ideal) must contain with any element all elements equal to it. Likewise, in every extension it must be assumed that the new definition of equality includes the old one. All concepts such as "finitely many elements," "unique assignment," etc. are meant in the sense of the equality definition; for example, "there is one and only one element" means that there is only one element up to equality. Incidentally, one can always pass from equality to set-theoretic identity by collecting equal elements into a class and introducing these classes as the new elements. The remarks given here on the definition of equality are due to R. Hölzer.

[81]The theory of integral quantities remains valid if zero divisors are admitted, provided that in their place one assumes the divisor-chain condition in R, which by §2 also makes the proof of the lemma possible. Since this form of the theory of integral quantities is not used later, I only point to it in footnotes. All definitions under discussion remain valid in the presence of zero divisors; most of them require — as is known and easy to see — even weaker assumptions. What is not retained is Dedekind's formulation: "A number is integral when it has a hull." For with zero divisors present, the quotient of two finite modules need not remain finite; hence here the order is defined directly and not as a module quotient.

and M is called a multiple, N a divisor, if every element of M is also contained in N. R-modules all of whose elements belong to R are called ideals in R.

Clearly the intersection — the least common multiple $[\ldots, M_i, \ldots]$ — of any number of modules from T is again a module; likewise T itself is a module. Thus the module derived from any system Σ of elements of T exists uniquely as the intersection of all modules containing Σ. The sum — the greatest common divisor $(\ldots, M_i, \ldots)$ — of any system of modules is the module derived from their union. The product MN of two modules is defined as the module derived from the elements $\mu\nu$, where μ runs through all elements of M and ν through all elements of N. The product therefore satisfies the associative and commutative laws, since these hold for the ring elements. The distributive law in T further gives, for modules, the distributive law

$$(\ldots, M_i, \ldots)N = (\ldots, M_i N, \ldots),$$

since by exactly this law in T both sides are the module derived from all elements $\mu_i\nu$.

Since R itself is a module, the condition that R is the multiplier domain can be expressed by $RM \equiv 0 \pmod{M}$. Since by hypothesis R contains the identity element, one also has $M \equiv 0 \pmod{RM}$, and hence $M = RM$.

A module M is called finite if there is at least one system Σ consisting of finitely many elements from which M is derived; the elements $\mu_1, \ldots, \mu_n$ of Σ are called a module basis of

$$M = (\mu_1, \ldots, \mu_n).$$

The sum and product of two — and hence of finitely many — finite modules are again finite, since they are derived from the union, respectively from the products, of the basis elements. The latter assertion follows from the representation $r_1\mu_1 + \cdots + r_n\mu_n$ of all elements μ of M by the basis and from the commutative law of multiplication in T.

2. An extension ring of R lying in T — hence one that contains all elements of R — is called an R-order, briefly an order. The intersection of arbitrarily many orders in T is again an order, and likewise T is an order; hence the order derived from any system Σ of elements of T exists uniquely. Sums and products of orders are defined correspondingly as for modules; in particular $O^2 = O$ for every order.

Every order is at the same time an R-module; an order is called finite when it is a finite module. Since sums and products arise by forming module sums and products, sums and products of finite orders are again finite orders by 1.; further, the following transitive law holds: if A is a finite R-order and B is a finite A-order, then B is also a finite R-order. For B is at the same time an R-order, hence an R-module. If $\alpha_1, \ldots, \alpha_m$ form a module basis of A with respect to R, and $\beta_1, \ldots, \beta_n$ one of B with respect to A, then the products $\alpha_i\beta_j$ plainly form a module basis of B with respect to R.

3. An element α of T is called integral with respect to R, briefly integral, if the order R_α derived from α is finite — equivalently, if the divisor chain of modules

$$U_n = (\alpha^0, \alpha, \ldots, \alpha^{n-1})$$

terminates after finitely many steps, $U_n = U_{n+1} = \cdots$.

The definition also admits a second formulation, to be used in 4.: an element α of T is called integral if there is at least one principal module C different from the zero module — i.e. C is derived from an element $\gamma \neq 0$ — such that the product of the order R_α with C is a finite module; in other words, such that the chain of modules CU_n terminates after finitely many steps.[82]

Finally the definition is also identical with the usual formulation: an element α of T is called integral if it satisfies at least one equation

$$\alpha^n + r_1\alpha^{n-1} + \cdots + r_n = 0,$$

where the r's are elements of R.

These definitions are in fact identical. Since R_α agrees with the module derived from all powers of α, when R_α is finite one can always choose a module basis from these powers. If such a basis is given, for instance, by $e = \alpha^0, \alpha, \ldots, \alpha^{n-1}$, this means that the chain of the U_n terminates at U_n; conversely, $U_n = U_{n+1} = \cdots$ gives this basis. But the termination of the chain of the U_n is also identical with the existence of the equation

$$\alpha^n + r_1\alpha^{n-1} + \cdots + r_n = 0.$$

[82] If zero divisors are admitted in R, then the existence of a regular principal module must be required; that is, γ must be assumed to be a non-zero-divisor, a regular element, which in rings without zero divisors is simply $\gamma \neq 0$.

If R_α is finite, then CR_α is finite, so the chain CU_n terminates; conversely, from the finiteness of CR_α, hence from the termination of the CU_n, it follows that the U_n also terminate and hence that R_α is finite. Indeed, since C is assumed to be a principal module distinct from the zero module, $CU_n = CU_{n+1}$ also gives $U_n = U_{n+1}$.

The ring property of the integral quantities and the transitive law of integral dependence rest on the following

Lemma. Let O be a finite order in T, and let α be any element of O. Then the order R_α derived from α is finite; in other words, all elements of a finite order are integral.

The proof follows by the usual arguments. Let $\omega_1, \dots, \omega_n$ be a module basis of O. Since O is a ring, $\alpha\omega_i$ belongs to O. Thus

$$\alpha\omega_i = r_{i1}\omega_1 + \cdots + r_{in}\omega_n,$$

and hence

$$\det(r_{ij} - e_{ij}\alpha) = 0,$$

where $e_{ij} = 0$ for $i \neq j$ and $e_{ii} = e$; this proves that α is integral. For the vanishing of the determinant one uses the hypothesis that T, and hence O, has no zero divisors.[83]

The system G of all integral quantities in T forms an order. It contains R, for the elements of R are integral, since R is a finite R-module with the identity element as basis. But G also has the ring property: for the order $R_{\alpha,\beta}$ derived from two arbitrary integral quantities α, β is equal to the product $R_\alpha R_\beta$, and is therefore finite. Hence, by the lemma, $\alpha + \beta$ and $\alpha\beta$ are integral as elements of a finite order.

Transitive law of integral dependence. If O is an order consisting of integral quantities, then every element of T integral with respect to O is also integral with respect to R. By definition α satisfies an equation

$$\alpha^m + \omega_1\alpha^{m-1} + \cdots + \omega_m = 0,$$

and is therefore also integral with respect to the order $O' = R_{\omega_1,\dots,\omega_m}$ derived from $\omega_1, \dots, \omega_m$, which is finite as the product $R_{\omega_1} \cdots R_{\omega_m}$. By the transitive law given in 2., the finite O'-order O'_α derived from α is therefore also a finite R-order, and α is integral by the lemma as an element of a finite order.

4. The ring R is called integrally closed in T if every element of T integral with respect to R belongs to R. According to the second formulation of the definition of integral quantities in 3., an element α of T belongs to R if and only if there is at least one principal module C different from the zero module such that CR_α is a finite module.[84]

From the concept of an integrally closed ring Dedekind (3rd ed., §172) drew two important consequences, which make it possible to use this concept for ideal theory:

Dedekind consequence I. Let R be integrally closed in T, and assume further that the divisor-chain condition for ideals (axiom I) holds in R. If β is a nonintegral element of T, and $c \neq 0$ an element of R, then there is an exponent $\rho > 0$ such that in the sequence of elements

$$c,\ c\beta, \dots, c\beta^\nu, \dots$$

the elements $c, \dots, c\beta^\rho$ are integral, and all the remaining ones are nonintegral.

If all elements $c\beta^\nu$ were integral, then by the integral closedness of R they would belong to R. Then the chain of modules $C, CB_1, \dots, CB_\nu, \dots$ — where C denotes the module derived from c, and B_ν the one derived from $1, \beta, \dots, \beta^\nu$ — would become a chain of ideals $C_\nu = (c, c\beta, \dots, c\beta^\nu)$, which terminates by hypothesis. But then, contrary to the assumption, R_β would be finite and β integral; hence some of the elements $c\beta^\nu$ are nonintegral. Furthermore, with any one nonintegral element all subsequent ones are nonintegral: for if $c\beta^r$ is integral, hence belongs to R, then for $\rho < r$ the element $c\beta^\rho$ satisfies

$$(c\beta^\rho)^r - c^{r-\rho}(c\beta^r)^\rho = 0$$

and is therefore integral. Thus if $c\beta^{\rho+1}$ is the first nonintegral element, then — since c is integral — $\rho > 0$, and this is the required exponent.

[83] If the divisor-chain condition is assumed in R but zero divisors are admitted, then the lemma is an immediate consequence of the module theorem in §2, according to which the chain condition transfers to the modules from O. This proof too is due, for the number field, to Dedekind (4th ed., §173, II) and is the first application of chain conditions in the literature. In the consequences to be given under 4., Dedekind instead uses the more special fact that the number of residue classes modulo an ideal in an algebraic number field is finite.

[84] If zero divisors are admitted in R, C must again be assumed to be a regular principal module; likewise the element c in Consequence I must be assumed regular.

Specializing the extension ring T to the quotient field of R — that is, to the field obtained by adjoining all pairs of elements[85] — gives the following:

Dedekind consequence II. Let R be integrally closed in its quotient field K, and let the divisor-chain condition for ideals hold in R. If β is a nonintegral element of K, then β admits a representation as a quotient of elements of R,

$$\beta = m/n,$$

such that m^2/n is also nonintegral.

Put $\beta = b/c$ with $c \neq 0$. Then in the sequence $c, c\beta = b, c\beta^2, \dots$ the first two elements are certainly integral, so $\rho > 1$, where ρ is the exponent of Consequence I. If one sets

$$m = c\beta^\rho, \qquad n = c\beta^{\rho-1},$$

then $m/n = \beta$, and $m^2/n = c\beta^{\rho+1}$ is nonintegral.

The transitive law of integral closedness has the following form: if R is any ring in T and G denotes the system of all quantities in T integral with respect to R, then G also consists of all quantities of T integral with respect to G. For every quantity integral with respect to G is also integral with respect to R, by the transitive law in 3., and hence belongs to G.

§2. Chain Conditions in Finite Module Domains

The module theorem by which the chain conditions are transferred occurs, in the application given here, only for modules of an extension ring. Since the proof is the same in the case of a general module domain, such a domain will be taken as the starting point.

A system M of elements $\alpha, \beta, \dots$ is called a module domain with respect to a ring R if two operations are given in M, each leading uniquely to elements of M: an additive combination of the elements and a multiplication of the elements by elements of R; if M is an Abelian group with respect to addition, while multiplication satisfies the associative law; and if the distributive law holds in both forms.[86]

For a module domain, all module definitions given in §1, 1. that do not refer to multiplication plainly remain valid. In particular, M is called a finite module domain if there are finitely many elements $\xi_1, \dots, \xi_k$ of M such that M equals the module derived from $\xi_1, \dots, \xi_k$, hence equals the system of all linear forms $r_1\xi_1 + \cdots + r_k\xi_k$ when R has an identity element; otherwise additional terms $n_i\xi_i$ occur.

Module theorem. Let M be a finite module domain with respect to a commutative ring R with identity element, and suppose that in R the divisor- respectively multiple-chain condition for ideals holds. Then in M the divisor- respectively multiple-chain condition for modules holds.

The proof rests on assigning uniquely to every module in M a system of finitely many ideals from R in such a way that to a proper divisor respectively proper multiple of the module there corresponds each time at least one proper divisor respectively multiple ideal.

An element of M is said to have length i if it can be written in at least one way as a linear form in $\xi_1, \dots, \xi_i$, while it has no representation as a linear form in $\xi_1, \dots, \xi_{i-1}$. To an arbitrary module W in M assign now k ideals $\mathfrak{a}_1, \dots, \mathfrak{a}_k$ from R: let W_i mean the module of all elements of W of length $\leq i$, and let $\mathfrak{a}_i$ denote the system of all coefficients of ξ_i in W_i. First one obtains:

If B is a proper divisor of W, and if $\mathfrak{b}_1, \dots, \mathfrak{b}_k$ are the ideals assigned to B, then among the $\mathfrak{b}_i$ there is at least one proper divisor of the corresponding $\mathfrak{a}_i$. For from $B \equiv 0 \pmod{W}$ it follows that $B_i \equiv 0 \pmod{W_i}$, and hence $\mathfrak{b}_i \equiv 0 \pmod{\mathfrak{a}_i}$ for $i = 1, 2, \dots, k$. By hypothesis there must be elements in B not contained in W, and hence such elements of smallest length ℓ. Then $\mathfrak{b}_\ell$ is a proper divisor of $\mathfrak{a}_\ell$; indeed $\mathfrak{b}_\ell \equiv 0 \pmod{\mathfrak{a}_\ell}$ when

$$\beta = b_1\xi_1 + \cdots + b_\ell\xi_\ell$$

is such an element of smallest length in B. If $\mathfrak{b}_\ell = \mathfrak{a}_\ell$, there exists an element

$$\alpha = a_\ell\xi_\ell + \cdots + a_1\xi_1 \equiv 0 \pmod{W},$$

and therefore an element $\beta - \alpha \in B$, $\beta - \alpha \notin W$, of smaller length than ℓ.

[85] In Steinitz's familiar formulation, J. f. M. 137. If zero divisors are admitted in R, then the quotient ring must replace the quotient field, by adjoining all quotient pairs whose denominator is a regular element of R. Here Dedekind consequence II is no longer valid, for n will not in general be a regular element.

[86] Compare *Ideal Theory*, §9.

Now let a divisor or multiple chain

$$W^{(1)}, W^{(2)}, \ldots, W^{(\nu)}, \ldots$$

be given; suppose the corresponding chain condition holds in R. If $\mathfrak{a}_{1\nu}, \ldots, \mathfrak{a}_{k\nu}$ are the ideals assigned to $W^{(\nu)}$, then the sequences

$$\mathfrak{a}_{i1}, \mathfrak{a}_{i2}, \ldots$$

are divisor respectively multiple chains for $i = 1, \ldots, k$. Hence by hypothesis there is an index ν_0 such that the ideal $\mathfrak{a}_{i\nu_0}$ is equal to all subsequent ideals for every i. But then, by what was just proved, the module $W^{(\nu_0)}$ is equal to all subsequent modules; thus the chain conditions are transferred.

This immediately gives the following:

Consequence of the module theorem. If in R the multiple-chain condition holds modulo every ideal different from the zero ideal, then in M the multiple-chain condition holds modulo every module C whose assigned ideals $\mathfrak{c}_1, \ldots, \mathfrak{c}_k$ are all different from the zero ideal. Under these hypotheses the multiple chains $\mathfrak{a}_{i1}, \ldots, \mathfrak{a}_{i\nu}, \ldots$ terminate after finitely many steps.

Additional remark. If R is a ring without identity element, then the divisor-chain condition and the multiple-chain condition modulo ideals different from the zero ideal are still transferred. Namely, instead of M one considers the system M' of all elements of the form

$$a_1\xi_1 + \cdots + a_k\xi_k + n_1\xi_1 + \cdots + n_k\xi_k,$$

which again passes into M when the additional integral multiples $n_i\xi_i$ are replaced by elements of M. To this M' one can assign, correspondingly as above, $2k$ ideals, where the $\mathfrak{a}_i$ are ideals from R and the $\mathfrak{n}_i$ are ideals of the rational integers. Since for the $\mathfrak{n}_i$ both the divisor-chain condition and the multiple-chain condition modulo every nonzero ideal are satisfied, the proof for the divisor-chain condition remains valid for M' and hence for M; for the multiple-chain condition, however, one must require that the $2k$ ideals assigned to C respectively C' are all different from the zero ideal.

§3. Passage to finite extension fields

For the classification of number fields and function fields it remains to show how the axioms I through V formulated in the introduction are transferred on passing to finite extension fields.

1. *Suppose that in $\mathfrak{R}$ all axioms I through V are satisfied except axiom II – the multiple-chain condition. Let $\mathfrak{K}$ denote the quotient field of $\mathfrak{R}$, and let $\mathfrak{L}$ be a finite extension field of the first kind over $\mathfrak{K}$.*[87] *Then in the system $\mathfrak{S}$ of all elements of $\mathfrak{L}$ integral with respect to $\mathfrak{R}$ the same axioms are satisfied, and in every order from $\mathfrak{S}$ all these axioms are still satisfied with the exception of integral closedness.*

By §1, 3, $\mathfrak{S}$ forms a ring for which axioms III and IV – existence of the identity element and absence of zero divisors – are satisfied. By the conclusion of §1, 4, $\mathfrak{S}$ is integrally closed in $\mathfrak{L}$; at the same time $\mathfrak{L}$ becomes the quotient field of $\mathfrak{S}$, since every element of $\mathfrak{L}$ is carried into an element of $\mathfrak{S}$ by multiplication with a suitable element of $\mathfrak{R}$. Thus axiom V is satisfied.

The proof of I follows by familiar arguments. As an extension of the first kind, $\mathfrak{L}$ is obtained by adjoining to $\mathfrak{K}$ a single element α, which by the preceding may be assumed integral with respect to $\mathfrak{R}$. Besides α, the conjugate quantities $\alpha', \alpha'', \ldots$ contained in the Galois field of $\mathfrak{L}$ over $\mathfrak{K}$ are also integral; hence so is their product of differences, and so is the square D of this product of differences. Since $\alpha, \alpha', \ldots$ are all distinct, D is a nonzero quantity of $\mathfrak{K}$, hence, by the integral closedness of $\mathfrak{R}$ in $\mathfrak{K}$, an element of $\mathfrak{R}$. By the usual argument, because $\mathfrak{R}$ is integrally closed, $\mathfrak{S}$ is contained in the finite $\mathfrak{R}$-module domain M generated by the elements $\xi_i = \alpha^i/D$; for this one need only pass, in the representation of an element of $\mathfrak{S}$ by the ξ_i, to the conjugate elements and solve the resulting system of linear equations for the coefficients of the representation. By the module theorem of §2, the divisor-chain condition holds for all $\mathfrak{R}$-modules from M, hence in particular for all ideals of $\mathfrak{S}$. The divisor-chain condition likewise holds for the ideals of any order in $\mathfrak{S}$, since here too one is dealing with $\mathfrak{R}$-modules in M; for every order, moreover, axioms III and IV are satisfied.

2. *If all axioms I through V are satisfied in $\mathfrak{R}$, the same is true in $\mathfrak{S}$. In every order from $\mathfrak{S}$ all axioms still hold except integral closedness.*

[87] Following Steinitz, an algebraic extension field is called "of the first kind" if every element is a zero of a prime function which, in a suitable extension field, splits into pairwise distinct linear factors.

In view of 1, it remains only to prove axiom II. By the corollary to the module theorem in §2 it is enough to prove the following. If a nonzero ideal of $\mathfrak{S}$ is regarded as an $\mathfrak{R}$-module $\mathfrak{C}$ in M, then the ideals $\mathfrak{c}_i$ of $\mathfrak{R}$ associated with $\mathfrak{C}$ are all different from the zero ideal. Now $\mathfrak{C}$ has the same finite linear rank over $\mathfrak{K}$ as M; for along with any element γ, $\mathfrak{C}$ contains also $\gamma\alpha^i$. Thus for each ξ_i there is an element $c_i \neq 0$ of $\mathfrak{R}$ such that $c_i\xi_i$ belongs to $\mathfrak{C}$; by the uniqueness of the representation by the ξ_i – where the index is to run only through the values less than the degree of the field – c_i belongs to $\mathfrak{c}_i$, and therefore $\mathfrak{c}_i$ is not the zero ideal. This argument remains valid for all orders of the same rank as M; for an order of smaller rank one passes to its quotient field, which as an intermediate field between $\mathfrak{K}$ and $\mathfrak{L}$ is also of the first kind, and whose degree over $\mathfrak{K}$ agrees with the linear rank of the order. The same argument is therefore preserved. Finally, it should be noted that an order in $\mathfrak{S}$ different from $\mathfrak{S}$ is never integrally closed: since every order also contains $\mathfrak{R}$, integral closedness would force it to contain $\mathfrak{S}$.

For the subordination of number fields and function fields, it is therefore enough to verify axioms I through V for the basic domains: the rational integers, polynomials in one indeterminate, and the functional domain of polynomials in several indeterminates.

3. *For the ring $\mathfrak{R}$ of rational integers, respectively of polynomials in one indeterminate with coefficients in a field, axioms I through V are satisfied.* Here I and II follow either from the fact that, modulo any nonzero number or any such polynomial in one indeterminate, there are only finitely many, respectively finitely many linearly independent, residue classes; or from the fact that every ideal is principal. For from this follows the unique decomposition of ideals as a product of powers of prime ideals, and hence a nonzero ideal is divisible by only finitely many ideals. Axioms III and IV are plainly satisfied, the latter as a consequence of the definition of equality. Integral closedness V follows from the fact that, by unique factorization of elements of $\mathfrak{R}$ into irreducibles up to units – divisors of one – every element of the quotient field not contained in $\mathfrak{R}$ can be written as a quotient of two relatively prime elements of $\mathfrak{R}$; hence no power of the numerator is divisible by the denominator. Such an element can therefore satisfy no equation characterizing it as integral with respect to $\mathfrak{R}$.

4. *Let $\mathfrak{R}$ now denote the ring of all polynomials in several indeterminates with coefficients in a field, or with rational integer coefficients.* Then axioms III, IV, V are satisfied as in 3, for here too the factorization of elements into irreducibles is unique up to units. The divisor-chain condition I is an immediate consequence of Hilbert's theorem on the existence of an ideal basis, whereas the multiple-chain condition II does not hold here.

However, by passing to the *functional domain*, which amounts to restricting to ideals of highest dimension, one can obtain the validity of axiom II as well; the validity of I can then be proved as in 3, without using Hilbert's theorem. Namely, adjoin to $\mathfrak{R}$ an indeterminate u and consider the ring $\mathfrak{R}^*$ of all polynomials in u with coefficients in $\mathfrak{R}$, equality being defined coefficientwise. As usual, call a polynomial in $\mathfrak{R}^*$ primitive (with respect to $\mathfrak{R}$) if the greatest common divisor of its coefficients – divisor in the sense of a polynomial in $\mathfrak{R}$, not an ideal divisor – is a unit of $\mathfrak{R}$. Then every polynomial in $\mathfrak{R}^*$ is the product of an element of $\mathfrak{R}$ and a primitive polynomial in $\mathfrak{R}^*$, and the product of primitive polynomials in $\mathfrak{R}^*$ is again primitive.

The totality of elements of the quotient field of $\mathfrak{R}^*$ whose denominators are primitive polynomials in $\mathfrak{R}^*$ therefore forms a ring, the functional domain $\mathfrak{F}$ of $\mathfrak{R}$. In this functional domain $\mathfrak{F}$, the units are exactly those elements whose numerator and denominator are primitive polynomials in $\mathfrak{R}^*$; every element of $\mathfrak{F}$ is therefore uniquely representable as the product of an element of $\mathfrak{R}$ and a unit of $\mathfrak{F}$. Hence the factorization theorem for the elements of $\mathfrak{R}$ transfers to the elements of $\mathfrak{F}$; for $\mathfrak{F}$, therefore, in addition to axioms III and IV, axiom V is also satisfied as in 3.

Moreover, in $\mathfrak{F}$ every ideal is principal. For together with any element of $\mathfrak{F}$ an ideal contains also the element of $\mathfrak{R}$ obtained from it by multiplying by a unit; and together with any two elements of $\mathfrak{R}$ it contains their greatest common divisor. Indeed, with $f(x) = t(x)\overline{f}(x)$ and $g(x) = t(x)\overline{g}(x)$ it also contains $t(x)(\overline{f}(x)+u\overline{g}(x))$, and hence $t(x)$, if $\overline{f}$ and $\overline{g}$ are relatively prime and their linear combination is consequently a unit. Thus, starting from an arbitrary element $f(x)$ of the ideal, if there is in the ideal an element $g(x)$ not divisible by it, then the greatest common divisor $t(x)$ also belongs to the ideal and is a proper divisor of $f(x)$. Repeating this finitely many times leads to a basis element of the ideal, and so the ideal is recognized as principal. Thus in $\mathfrak{F}$ the unique decomposition of ideals as products of powers of prime ideals holds, corresponding to the factorization of the elements of $\mathfrak{R}$; as in 3, it follows that axioms I and II are satisfied. The extension fields of the quotient field of $\mathfrak{F}$ that come into consideration here are only those that can be generated by adjoining an element of $\mathfrak{S}$.[88] Taking 1 and 2 into account, this establishes the validity of axioms I through V for number

[88]The system of quantities integral with respect to $\mathfrak{F}$ in these extension fields is also obtained by passing from $\mathfrak{S}$ to a corresponding functional domain, just as one passes from $\mathfrak{R}$ to $\mathfrak{F}$. Compare J. König, *Algebraische Größen*, Leipzig 1903, pp. 468/69.

fields and function fields.

§4. Isomorphism theorems. Direct sums

In what follows only the ring property, respectively the module property, is assumed; no further axiom is presupposed.

1. If M and $\overline{M}$ are module domains with respect to $\mathfrak{R}$ (§2), M is called homomorphic to $\overline{M}$ (more precisely, module-homomorphic), $M \sim \overline{M}$, if to every element of M there corresponds one and only one element of $\overline{M}$, in such a way that $\overline{M}$ is exhausted;[89] and if, under this correspondence, difference and multiplication by the same element of $\mathfrak{R}$ correspond. Thus from $\beta \sim \overline{\beta}$ and $\gamma \sim \overline{\gamma}$ it always follows that $(\beta - \gamma) \sim (\overline{\beta} - \overline{\gamma})$ and $r\beta \sim r\overline{\beta}$.

An $\mathfrak{R}$-module $\mathfrak{B}$ in M therefore corresponds homomorphically to an $\mathfrak{R}$-module $\overline{\mathfrak{B}}$ in $\overline{M}$; and the totality $\mathfrak{C}^*$ of elements in M corresponding to elements of an $\mathfrak{R}$-module $\overline{\mathfrak{C}}$ in $\overline{M}$ forms an $\mathfrak{R}$-module in M uniquely determined by $\overline{\mathfrak{C}}$, the module $\mathfrak{C}^*$ associated with $\overline{\mathfrak{C}}$, with $\mathfrak{C}^* \sim \overline{\mathfrak{C}}$. If in particular $\mathfrak{A}$ is the module in M associated with the zero element of $\overline{M}$, then $\mathfrak{C}^*$ is a divisor of $\mathfrak{A}$ and splits into classes of elements of M congruent modulo $\mathfrak{A}$, in such a way that these classes correspond one-to-one to the elements of $\overline{\mathfrak{C}}$. If one passes from $\mathfrak{B}$ in M to $\overline{\mathfrak{B}}$ in $\overline{M}$ and then back to $\mathfrak{B}^*$, one obtains $\mathfrak{B}^* = (\mathfrak{B}, \mathfrak{A})$, the greatest common divisor of $\mathfrak{B}$ and $\mathfrak{A}$; thus $\mathfrak{B}^* = \mathfrak{B}$ if $\mathfrak{B}$ is a divisor of $\mathfrak{A}$.

If the correspondence between the elements of M and $\overline{M}$ is reversible one-to-one, the domains are called isomorphic (more precisely, module-isomorphic), $M \cong \overline{M}$.

If $\mathfrak{A}$ is an arbitrary $\mathfrak{R}$-module in M, then a module domain $\overline{M}$ homomorphic to M arises – the residue-class module $M \mid \mathfrak{A}$ – by taking congruence modulo $\mathfrak{A}$ as the new equality relation. To every element of M there are associated all and only the elements equal to it in $\overline{M}$. Passing in $\overline{M}$ from the equality definition to identity means collecting all equal elements in $\overline{M}$ into one class – a residue class – and taking these residue classes as the new elements of $\overline{M}$.

Every homomorphism is produced by passage to a residue-class module; for if $M \sim \overline{M}$ and if $\mathfrak{A}$ is the module associated with the zero element of $\overline{M}$, then, as shown above, $\overline{M}$ is isomorphic to the residue-class module $M \mid \mathfrak{A}$.

2. **First isomorphism theorem.** *Let $\overline{M}$ be the residue-class module $M \mid \mathfrak{A}$, and let $\mathfrak{C}$ be a divisor of $\mathfrak{A}$. Then there is an isomorphism $\overline{M} \mid \overline{\mathfrak{C}} \cong M \mid \mathfrak{C}$.* Indeed, congruence modulo $\mathfrak{A}$ is at the same time congruence modulo $\mathfrak{C}$; elements equal modulo $\mathfrak{A}$ remain equal modulo $\mathfrak{C}$. Thus one may form the residue-class module $M \mid \mathfrak{C}$ – that is, equality modulo $\mathfrak{C}$ – by first identifying elements modulo $\mathfrak{A}$, passing to $\overline{M}$, and then collecting in $\overline{M}$ those elements that are equal modulo $\mathfrak{C}$; this is the same as identifying modulo $\overline{\mathfrak{C}}$ in $\overline{M}$, i.e. forming $\overline{M} \mid \overline{\mathfrak{C}}$.

Second isomorphism theorem. *If $\mathfrak{B}$ and $\mathfrak{A}$ are modules in M, then* $(\mathfrak{B}, \mathfrak{A}) \mid \mathfrak{A} \cong \mathfrak{B} \mid [\mathfrak{B}, \mathfrak{A}]$. For by 1, $\mathfrak{B}$ becomes homomorphic to $\overline{\mathfrak{B}}$ when $(\mathfrak{B}, \mathfrak{A}) \mid \mathfrak{A}$ is put equal to $\overline{\mathfrak{B}}$; and since precisely the elements of $[\mathfrak{B}, \mathfrak{A}]$ correspond to the zero element in $\overline{\mathfrak{B}}$, the asserted isomorphism again follows from 1.

3. If $\mathfrak{R}$ and $\overline{\mathfrak{R}}$ are (commutative) rings, $\mathfrak{R}$ is called homomorphic to $\overline{\mathfrak{R}}$ (more precisely, ring-homomorphic), $\mathfrak{R} \sim \overline{\mathfrak{R}}$, if to each element of $\mathfrak{R}$ there corresponds one and only one element of $\overline{\mathfrak{R}}$ in such a way that $\overline{\mathfrak{R}}$ is exhausted, and if differences and products correspond under this assignment. If the correspondence of elements is reversible one-to-one, the rings are called isomorphic (more precisely, ring-isomorphic), $\mathfrak{R} \cong \overline{\mathfrak{R}}$.

All the arguments in 1 and 2 remain valid when the modules are replaced by ideals in $\mathfrak{R}$,[90] and when module-homomorphism and module-isomorphism are replaced by ring-homomorphism and ring-isomorphism. In particular, if $\mathfrak{c}$ is a divisor of $\mathfrak{a}$, the residue-class ring $\mathfrak{c} \mid \mathfrak{a}$ is obtained by introducing congruence modulo $\mathfrak{a}$ as the new equality relation; here one must note that products of residue classes are now also defined.

Thus $\mathfrak{c}$ is homomorphic to $\mathfrak{c} \mid \mathfrak{a}$; every homomorphism is produced in this way, and the isomorphism theorems hold as in 2:

First isomorphism theorem. If $\overline{\mathfrak{R}}$ denotes the residue-class ring $\mathfrak{R} \mid \mathfrak{a}$, and $\mathfrak{c}$ is a divisor of $\mathfrak{a}$, then $\overline{\mathfrak{R}} \mid \overline{\mathfrak{c}} \cong \mathfrak{R} \mid \mathfrak{c}$ in the sense of ring isomorphism.

Second isomorphism theorem. If $\mathfrak{b}$ and $\mathfrak{a}$ are ideals of $\mathfrak{R}$, then the ring-isomorphism $(\mathfrak{b}, \mathfrak{a}) \mid \mathfrak{a} \cong \mathfrak{b} \mid [\mathfrak{b}, \mathfrak{a}]$ holds.[91]

[89] In the sense of the definition of equality prevailing in $\overline{M}$; compare note 6.

[90] For noncommutative rings, two-sided ideals must be used.

[91] These isomorphism theorems, familiar from group theory, occur for modules in a somewhat more special form first in Dedekind (compare 3rd ed., p. 484). The above form for ideals occurs in Masazo Sono, *On Congruences. I, II, III, IV*, Memoirs of the College of Science, Kyoto Imperial University, 2 (1917), 3 (1918, 1919), compare 2, p. 215. In each case only the second isomorphism theorem is stated explicitly.

4. I now collect the calculation rules for coprime ideals[92] from which the theorems on direct sums in 5 follow. In the commutative ring $\mathfrak{R}$, assume the existence of a multiplicative identity element e. Thus $\mathfrak{a} = \mathfrak{o}\mathfrak{a}$ for every ideal of $\mathfrak{R}$, where $\mathfrak{o}$ denotes the unit ideal. Two ideals $\mathfrak{a}, \mathfrak{b}$ are called coprime if their greatest common divisor $(\mathfrak{a}, \mathfrak{b})$ is $\mathfrak{o}$.

4α. If $(\mathfrak{a}, \mathfrak{b}) = \mathfrak{o}$*, then* $(\mathfrak{a}, \mathfrak{b}\mathfrak{c}) = (\mathfrak{a}, \mathfrak{c})$. For $(\mathfrak{a}, \mathfrak{c}) = (\mathfrak{a}, \mathfrak{b})(\mathfrak{a}, \mathfrak{c}) = (\mathfrak{a}^2, \mathfrak{a}\mathfrak{b}, \mathfrak{a}\mathfrak{c}, \mathfrak{b}\mathfrak{c})$ is divisible by $(\mathfrak{a}, \mathfrak{b}\mathfrak{c})$ and conversely. Hence:

4β. From $\mathfrak{c}\mathfrak{b} \equiv 0 \pmod{\mathfrak{a}}$ *and* $(\mathfrak{a}, \mathfrak{b}) = \mathfrak{o}$ *it follows that* $\mathfrak{c} \equiv 0 \pmod{\mathfrak{a}}$. Indeed, $\mathfrak{c}\mathfrak{b} \equiv 0 \pmod{\mathfrak{a}}$ is equivalent to $(\mathfrak{a}, \mathfrak{b}\mathfrak{c}) = \mathfrak{a}$; because $(\mathfrak{a}, \mathfrak{b}) = \mathfrak{o}$, this also gives $(\mathfrak{a}, \mathfrak{c}) = \mathfrak{a}$, or $\mathfrak{c} \equiv 0 \pmod{\mathfrak{a}}$. Furthermore:

4γ. If each of the ideals $\mathfrak{a}_1, \ldots, \mathfrak{a}_r$ *is coprime to each of the ideals* $\mathfrak{b}_1, \ldots, \mathfrak{b}_s$*, then the product of the* $\mathfrak{a}_i$ *is coprime to the product of the* $\mathfrak{b}_i$. For from $(\mathfrak{a}, \mathfrak{b}) = \mathfrak{o}$ and $(\mathfrak{a}, \mathfrak{c}) = \mathfrak{o}$ one obtains $(\mathfrak{a}, \mathfrak{b}\mathfrak{c}) = \mathfrak{o}$, and finite repetition gives the assertion.

From 4α and 4γ follows:

4δ. If the ideals $\mathfrak{b}_1, \ldots, \mathfrak{b}_r$ *are pairwise coprime, i.e.* $(\mathfrak{b}_i, \mathfrak{b}_k) = \mathfrak{o}$ *for* $i \neq k$*, then their least common multiple equals their product.* Let $(\mathfrak{a}, \mathfrak{b}) = \mathfrak{o}$ and $\mathfrak{v} = [\mathfrak{a}, \mathfrak{b}]$. Then $\mathfrak{v} = \mathfrak{o}\mathfrak{v} = (\mathfrak{a}\mathfrak{v}, \mathfrak{b}\mathfrak{v}) = 0(\mathfrak{a}, \mathfrak{b})$; since $\mathfrak{a}\mathfrak{b} \equiv 0 \pmod{\mathfrak{v}}$, it follows that $\mathfrak{v} = \mathfrak{a}\mathfrak{b}$. Finite repetition, using 4γ, gives the claim.

4ε. If the ideals $\mathfrak{b}_1, \ldots, \mathfrak{b}_r$ *are pairwise coprime and* $\mathfrak{a}_i = [\mathfrak{b}_1, \ldots, \mathfrak{b}_{i-1}, \mathfrak{b}_{i+1}, \ldots, \mathfrak{b}_r] = \mathfrak{b}_1 \cdots \mathfrak{b}_{i-1}\mathfrak{b}_{i+1} \cdots \mathfrak{b}_r$*, then* $(\mathfrak{a}_1, \ldots, \mathfrak{a}_{i-1}, \mathfrak{a}_{i+1}, \ldots, \mathfrak{a}_r) = \mathfrak{b}_i$*, and hence* $(\mathfrak{a}_1, \ldots, \mathfrak{a}_r) = \mathfrak{o}$. For by the distributive law $(\mathfrak{a}_1, \mathfrak{a}_2) = \mathfrak{b}_3 \cdots \mathfrak{b}_r(\mathfrak{b}_2, \mathfrak{b}_1) = \mathfrak{b}_3 \cdots \mathfrak{b}_r$; hence $(\mathfrak{a}_1, \mathfrak{a}_2, \mathfrak{a}_3) = \mathfrak{b}_4 \cdots \mathfrak{b}_r(\mathfrak{b}_3, \mathfrak{b}_1\mathfrak{b}_2) = \mathfrak{b}_4 \cdots \mathfrak{b}_r$, and finite repetition, after a suitable numbering, proves the assertion.

The ideal $\mathfrak{a}_i$ is called the complement of $\mathfrak{b}_i$ in the representation $\mathfrak{m} = [\mathfrak{b}_1, \ldots, \mathfrak{b}_r]$.

5. Again let the commutative ring $\mathfrak{R}$ have a multiplicative identity element. The ring $\mathfrak{R}$ is called the direct sum of the ideals $\mathfrak{a}_1, \ldots, \mathfrak{a}_r$, in symbols $\mathfrak{R} = \mathfrak{a}_1 + \mathfrak{a}_2 + \cdots + \mathfrak{a}_r$, if each element c of $\mathfrak{R}$ can be represented in one and only one way in the form $c = a_1 + a_2 + \cdots + a_r$, with a_i an element of $\mathfrak{a}_i$.

If the zero ideal is the least common multiple of the pairwise coprime ideals $\mathfrak{b}_1, \ldots, \mathfrak{b}_r$*, and if* $\mathfrak{a}_i$ *is the complement of* $\mathfrak{b}_i$*, then* $\mathfrak{R}$ *is the direct sum of the ideals* $\mathfrak{a}_1, \ldots, \mathfrak{a}_r$. Since $\mathfrak{o} = (\mathfrak{a}_1, \ldots, \mathfrak{a}_r)$, every element of $\mathfrak{R}$ admits at least one additive representation by the $\mathfrak{a}_i$. Since $[\mathfrak{a}_i, \mathfrak{b}_i] = (0)$, this representation is unique: from $0 = a_1 + a_2 + \cdots + a_r = a_i + b_i$ one obtains $a_i = 0$. By the second isomorphism theorem the $\mathfrak{a}_i$ are isomorphic to the residue-class rings $\mathfrak{R} \mid \mathfrak{b}_i$. The orthogonality relations $\mathfrak{a}_i\mathfrak{a}_k = (0)$ for $i \neq k$, and $\mathfrak{a}_i^2 = \mathfrak{a}_i$, hold; the latter because $\mathfrak{a}_i = \mathfrak{o}\mathfrak{a}_i$.

Conversely, if there is a representation of $\mathfrak{R}$ *as a direct sum,* $\mathfrak{R} = \mathfrak{a}_1 + \mathfrak{a}_2 + \cdots + \mathfrak{a}_r$*, and if one sets* $\mathfrak{b}_i = (\mathfrak{a}_1, \ldots, \mathfrak{a}_{i-1}, \mathfrak{a}_{i+1}, \ldots, \mathfrak{a}_r) = \mathfrak{a}_1 + \cdots + \mathfrak{a}_{i-1} + \mathfrak{a}_{i+1} + \cdots + \mathfrak{a}_r$*, then the* $\mathfrak{b}_i$ *are pairwise coprime and their least common multiple is the zero ideal.* For from $\mathfrak{o} = (\mathfrak{a}_i, \mathfrak{b}_i)$ follows $\mathfrak{o} = (\mathfrak{b}_k, \mathfrak{b}_i)$ for $k \neq i$, since $\mathfrak{b}_k$ is a divisor of $\mathfrak{a}_i$. If, moreover, c is divisible by all $\mathfrak{b}_i$, then

$$c = a_2^{(1)} + \cdots + a_r^{(1)} = a_1^{(2)} + a_3^{(2)} + \cdots + a_r^{(2)} = \cdots = a_1^{(r)} + \cdots + a_{r-1}^{(r)},$$

where in each case $a_i^{(\lambda)} \equiv 0 \pmod{\mathfrak{a}_i}$. By the uniqueness of the representation, since in each representation one component is zero, it follows that $0 = a_1^{(\lambda)}, \ldots, 0 = a_r^{(\lambda)}$ for every λ, and hence $c = 0$.[93]

§5. Prime ideals and primary ideals

Let $\mathfrak{R}$ again be a commutative ring for which no further axiom, not even the existence of an identity element, is assumed. We collect the basic facts about prime ideals and primary ideals.[94]

1. An ideal $\mathfrak{p}$ of $\mathfrak{R}$ is called a *weak prime ideal* if the residue-class ring $\mathfrak{R} \mid \mathfrak{p}$ is a ring without zero divisors; that is, if from $a \not\equiv 0 \pmod{\mathfrak{p}}$ and $b \not\equiv 0 \pmod{\mathfrak{p}}$ it always follows that $ab \not\equiv 0 \pmod{\mathfrak{p}}$. An ideal $\mathfrak{p}^*$ of $\mathfrak{R}$ is called a *strong prime ideal* if the residue-class ring $\mathfrak{R} \mid \mathfrak{p}^*$ is a ring without zero-divisor ideals; that is, if from $\mathfrak{a} \not\equiv 0 \pmod{\mathfrak{p}^*}$ and $\mathfrak{b} \not\equiv 0 \pmod{\mathfrak{p}^*}$ it always follows that $\mathfrak{a}\mathfrak{b} \not\equiv 0 \pmod{\mathfrak{p}^*}$.

Every strong prime ideal is at the same time a weak prime ideal, as the specialization of $\mathfrak{a}, \mathfrak{b}$ to principal ideals shows. Conversely, *every weak prime ideal is also a strong prime ideal*; one can therefore simply speak of prime ideals. For let $\mathfrak{p}$ be a weak prime ideal; let $\mathfrak{a} \not\equiv 0 \pmod{\mathfrak{p}}$ and $\mathfrak{b} \not\equiv 0 \pmod{\mathfrak{p}}$; choose elements a

[92] In the form going back to Dedekind (4th ed., §178, III to VIII). The calculation with the identity element that appears throughout more recent accounts is much more cumbersome.

[93] The theorems given under 5 are special cases of a general theorem on the connection between direct sum and direct intersection. Compare H. Prüfer, Theorie der Abelschen Gruppen I, Math. Zeitschr. 20 (1924), pp. 165–187, §6. The arguments given there remain valid also for such a general formulation of the concept of group that modules and ideals fall under it as special cases.

[94] In *Idealtheorie*, §4, axiom I, the divisor-chain condition, is assumed; therefore it is not made sharp there which properties of prime and primary ideals are independent of this axiom.

and b in $\mathfrak{a}$ and $\mathfrak{b}$ such that $a \not\equiv 0 \pmod{\mathfrak{p}}$ and $b \not\equiv 0 \pmod{\mathfrak{p}}$. Then $ab \not\equiv 0 \pmod{\mathfrak{p}}$, hence $\mathfrak{ab} \not\equiv 0 \pmod{\mathfrak{p}}$, so $\mathfrak{p}$ is also a strong prime ideal.

2. An ideal $\mathfrak{q}$ of $\mathfrak{R}$ is called a *weak primary ideal* if in the residue-class ring $\mathfrak{R} \mid \mathfrak{q}$ some power of every zero divisor vanishes; equivalently, if from $a \not\equiv 0 \pmod{\mathfrak{q}}$ and $b^x \not\equiv 0 \pmod{\mathfrak{q}}$ for every x it always follows that $ab \not\equiv 0 \pmod{\mathfrak{q}}$. The system $\mathfrak{p}$ of all elements of $\mathfrak{R}$ that become zero divisors in $\mathfrak{R} \mid \mathfrak{q}$ forms an ideal, indeed a prime ideal; it is a divisor of $\mathfrak{q}$ and is called the associated prime ideal. For along with a, also ra is a zero divisor; along with a and b, so is $a-b$, since with a^x and b^λ the element $(a-b)^{x+\lambda}$ is always divisible by $\mathfrak{q}$. If furthermore a is not a zero divisor, then by definition no power of a is a zero divisor; from $a \not\equiv 0 \pmod{\mathfrak{p}}$ and $b \not\equiv 0 \pmod{\mathfrak{p}}$ follows $a^x b^\lambda = (ab)^x \not\equiv 0 \pmod{\mathfrak{q}}$, hence $ab \not\equiv 0 \pmod{\mathfrak{p}}$.

An ideal $\mathfrak{q}^*$ of $\mathfrak{R}$ is called a *strong primary ideal* if in the residue-class ring $\mathfrak{R} \mid \mathfrak{q}^*$ some power of every zero-divisor ideal vanishes; that is, if from $\mathfrak{a} \not\equiv 0 \pmod{\mathfrak{q}^*}$ and $\mathfrak{b}^x \not\equiv 0 \pmod{\mathfrak{q}^*}$ for every x it always follows that $\mathfrak{ab} \not\equiv 0 \pmod{\mathfrak{q}^*}$. As for weak primary ideals one shows: the greatest common divisor of all ideals of $\mathfrak{R}$ that become zero-divisor ideals in $\mathfrak{R} \mid \mathfrak{q}^*$ forms a prime ideal, a divisor of $\mathfrak{q}^*$, the associated prime ideal. Conversely, all primary ideals with the same associated prime ideal $\mathfrak{p}$ are said to belong to $\mathfrak{p}$.

Every strong primary ideal is at the same time a weak primary ideal, as the specialization of $\mathfrak{a}, \mathfrak{b}$ to principal ideals shows. In general, however, the converse is false.[95] *If, however, axiom I, the divisor-chain condition, is assumed in $\mathfrak{R}$, then every weak primary ideal is also a strong primary ideal;* one can therefore simply speak of primary ideals. Let $\mathfrak{q}$ be a weak primary ideal; suppose $\mathfrak{a} \not\equiv 0 \pmod{\mathfrak{q}}$ and $\mathfrak{b}^x \not\equiv 0 \pmod{\mathfrak{q}}$ for every x. Then there exist elements a of $\mathfrak{a}$ and b of $\mathfrak{b}$ such that $a \not\equiv 0 \pmod{\mathfrak{q}}$ and $b^x \not\equiv 0 \pmod{\mathfrak{q}}$ for every x; hence $ab \not\equiv 0 \pmod{\mathfrak{q}}$ and therefore $\mathfrak{ab} \not\equiv 0 \pmod{\mathfrak{q}}$. For if to every b in $\mathfrak{b}$ there were an exponent λ such that $b^\lambda \equiv 0 \pmod{\mathfrak{q}}$, then $\mathfrak{b}^{\lambda_1+\cdots+\lambda_r} \equiv 0 \pmod{\mathfrak{q}}$, with $\lambda_1, \ldots, \lambda_r$ the exponents of the finitely many basis elements.[96] The same argument also shows that there is a smallest exponent ϱ such that $\mathfrak{p}^\varrho \equiv 0 \pmod{\mathfrak{q}}$, where $\mathfrak{p}$ is the associated prime ideal; ϱ is called the exponent of $\mathfrak{q}$.

3. In what follows the divisor-chain condition is again *not* assumed. The least common multiple $[\mathfrak{a}_1, \ldots, \mathfrak{a}_r]$ of finitely many ideals is called a *shortest representation* if no $\mathfrak{a}_i$ is contained in the least common multiple of the remaining ideals, i.e. if no $\mathfrak{a}_i$ can be omitted.

The least common multiple of finitely many weak, respectively strong, primary ideals belonging to the same prime ideal $\mathfrak{p}$ is again a weak primary ideal with $\mathfrak{p}$ as associated prime ideal. A shortest representation by finitely many weak, respectively strong, primary ideals belonging to distinct prime ideals is not a weak, respectively strong, primary ideal. Let $\mathfrak{f} = [\mathfrak{q}_1, \ldots, \mathfrak{q}_r]$, and let $\mathfrak{p}$ be the associated prime ideal of the weak primary ideals $\mathfrak{q}_i$. Then $\mathfrak{p}$ also consists exactly of the elements of which some power is divisible by $\mathfrak{f}$. From $a \not\equiv 0 \pmod{\mathfrak{f}}$ and $b^x \not\equiv 0 \pmod{\mathfrak{f}}$ for every x it follows that $b \not\equiv 0 \pmod{\mathfrak{p}}$ and $a \not\equiv 0 \pmod{\mathfrak{q}_i}$ for at least one index i; hence $ab \not\equiv 0 \pmod{\mathfrak{q}_i}$, and therefore $ab \not\equiv 0 \pmod{\mathfrak{f}}$. If one replaces elements throughout by ideals, one can no longer infer $\mathfrak{b} \not\equiv 0 \pmod{\mathfrak{p}}$.

Now let $\mathfrak{m} = [\mathfrak{q}_1, \ldots, \mathfrak{q}_r]$ be a shortest representation by weak primary ideals with $\mathfrak{p}_i \neq \mathfrak{p}_k$ for $i \neq k$, and let $r \geq 2$. There is then at least one associated prime ideal, say $\mathfrak{p}_1$, which is not contained in any of the others; for every proper multiple-chain of the $\mathfrak{p}$'s must terminate after at most r steps. Hence there are elements a_i of $\mathfrak{p}_i$ such that $a_i^{\varrho_i} \equiv 0 \pmod{\mathfrak{q}_i}$, while $a_2 \not\equiv 0 \pmod{\mathfrak{p}_1}, \ldots, a_r \not\equiv 0 \pmod{\mathfrak{p}_1}$. If further $q_1 \not\equiv 0 \pmod{\mathfrak{m}}$ is an element of $\mathfrak{q}_1$, then $q_1 a_2^{\varrho_2} \cdots a_r^{\varrho_r} \equiv 0 \pmod{\mathfrak{m}}$, but no power of $a_2^{\varrho_2} \cdots a_r^{\varrho_r}$ is divisible by $\mathfrak{m}$, since otherwise divisibility by $\mathfrak{p}_1$ would follow. Thus $\mathfrak{m}$ is not a weak primary ideal. If elements are replaced throughout by ideals, that is q_1 by $\mathfrak{q}_1$ and a_i by $\mathfrak{a}_i[\not\equiv 0 \pmod{\mathfrak{p}_1}$, but $\mathfrak{a}_i^{\varrho_i} \equiv 0 \pmod{\mathfrak{q}_i}]$, the proof for strong primary ideals is obtained.[97]

4. *Module quotient and ideal quotient.* Let $\mathfrak{T}$ be an extension ring of $\mathfrak{R}$, and let $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \ldots$ be $\mathfrak{R}$-modules in $\mathfrak{T}$. The quotient $\mathfrak{C} = \mathfrak{A} : \mathfrak{B}$ is defined as a module satisfying the conditions $\mathfrak{CB} \equiv 0 \pmod{\mathfrak{A}}$ and, if $\mathfrak{DB} \equiv 0 \pmod{\mathfrak{A}}$, then $\mathfrak{D} \equiv 0 \pmod{\mathfrak{C}}$; thus $\mathfrak{C}$ is the smallest module for which $\mathfrak{CB} \equiv 0 \pmod{\mathfrak{A}}$ holds. This determines the quotient uniquely as the greatest common divisor of all modules $\mathfrak{D}$ for which $\mathfrak{DB} \equiv 0 \pmod{\mathfrak{A}}$; such $\mathfrak{D}$ exist, since the zero module is certainly one of them.

From the definition it follows: if $\mathfrak{B}$ is a multiple of $\overline{\mathfrak{B}}$, then $\mathfrak{A} : \overline{\mathfrak{B}}$ is a multiple of $\mathfrak{A} : \mathfrak{B}$. If $\mathfrak{A}$ is a multiple of $\overline{\mathfrak{A}}$, then $\mathfrak{A} : \mathfrak{B}$ is a multiple of $\overline{\mathfrak{A}} : \mathfrak{B}$. Also

$$\mathfrak{A} : \mathfrak{BC} = (\mathfrak{A} : \mathfrak{B}) : \mathfrak{C} = (\mathfrak{A} : \mathfrak{C}) : \mathfrak{B}.$$

[95] This is shown by the following example communicated to me by R. Hölzer. Let $\mathfrak{R}$ be the polynomial domain in countably many indeterminates x_i with coefficients in a field, and let $\mathfrak{q} = (x_1^2, x_2^3, \ldots, x_\nu^{\nu+1}, \ldots, x_i x_k\ [i \neq k] \ldots)$. The zero divisors in the residue-class ring are exactly the polynomials divisible by $\mathfrak{p} = (x_1, x_2, \ldots, x_\nu, \ldots)$, and in each case a power of such a zero divisor is divisible by $\mathfrak{q}$; therefore $\mathfrak{q}$ is a weak primary ideal with associated prime ideal $\mathfrak{p}$. But $\mathfrak{q}$ is not a strong primary ideal: putting $\mathfrak{a} = (x_1, x_3, x_5, \ldots, x_{2\nu+1}, \ldots)$ and $\mathfrak{b} = (x_2, x_4, \ldots, x_{2\nu}, \ldots)$, one has $\mathfrak{ab} \equiv 0 \pmod{\mathfrak{q}}$, but no power of $\mathfrak{a}$ or $\mathfrak{b}$ is divisible by $\mathfrak{q}$.

[96] In order to infer the finite ideal basis from axiom I, a fixed well-ordering in $\mathfrak{R}$ must be used; compare §6.

[97] I owe this modification of my original proof, which makes the divisor-chain condition unnecessary, to B. L. van der Waerden.

If the extension ring $\mathfrak{T}$ coincides with $\mathfrak{R}$, the module quotient becomes the ideal quotient $\mathfrak{a}:\mathfrak{b}$; hence the preceding rules also hold here. Since $\mathfrak{ab}\equiv 0 \pmod{\mathfrak{a}}$, one also has $\mathfrak{a}\equiv 0 \pmod{\mathfrak{a}:\mathfrak{b}}$.

5. *If $\mathfrak{a}:\mathfrak{b}=\mathfrak{a}$, then $\mathfrak{b}$ is called prime to $\mathfrak{a}$.* Thus an ideal $\mathfrak{b}$ is prime to $\mathfrak{a}$ if and only if $\mathfrak{db}\equiv 0 \pmod{\mathfrak{a}}$ always implies $\mathfrak{d}\equiv 0 \pmod{\mathfrak{a}}$. From the calculation rules under 4 it follows: if $\mathfrak{b}$ and $\mathfrak{c}$ are prime to $\mathfrak{a}$, then so are their product and their least common multiple. If $\mathfrak{b}$ is prime to $\mathfrak{a}$ and $\mathfrak{a}$ is prime to $\mathfrak{b}$, then $\mathfrak{a}$ and $\mathfrak{b}$ are called mutually prime. From §4, 4β, it follows that coprime ideals are also mutually prime.

§6. Ideal theory under the assumption of the divisor-chain condition

Let the basis be a commutative ring $\mathfrak{R}$ for which axiom I, the divisor-chain condition, is satisfied. In what follows assume also a fixed well-ordering of all elements of $\mathfrak{R}$. This also determines a well-ordering of all ideals of $\mathfrak{R}$. Indeed, the two assumptions imply at once that every ideal of $\mathfrak{R}$ possesses an ideal basis consisting of finitely many elements. Passing from the well-ordering of the elements of $\mathfrak{R}$ to a quasi-lexicographic well-ordering of all finite subsets of $\mathfrak{R}$, [26] and assigning to each ideal, as distinguished basis, the first among its possible bases in this well-ordering of finite subsets, one obtains a well-ordering of the ideals together with that of the finite subsets.

Theorem I. *Under the assumption of the divisor-chain condition, every ideal of $\mathfrak{R}$ admits a representation as the least common multiple of finitely many irreducible ideals, i.e. of ideals which cannot be represented as the least common multiple of two proper divisors.*

For the proof it is enough to show: if Theorem I is false for an ideal $\mathfrak{m}$, then $\mathfrak{m}$ possesses a proper divisor for which Theorem I is likewise false; from this one can construct, contrary to the divisor-chain condition, a divisor chain not terminating after finitely many steps. Indeed $\mathfrak{m}$ must be reducible, since otherwise $\mathfrak{m}=[\mathfrak{m}]$ would already give the desired representation. If $\mathfrak{m}=[\mathfrak{a},\mathfrak{b}]$, Theorem I cannot hold for both $\mathfrak{a}$ and $\mathfrak{b}$ at the same time; otherwise the corresponding representation would follow for $\mathfrak{m}$. Hence there are proper divisors of $\mathfrak{m}$ for which Theorem I is false; let $\mathfrak{a}_1$ be the first of these in the well-ordering of ideals. Constructing in the same way, from $\mathfrak{a}_1$, a proper divisor $\mathfrak{a}_2$, and in general from $\mathfrak{a}_i$ a proper divisor $\mathfrak{a}_{i+1}$, one obtains a well-defined divisor chain which does not terminate, contradicting the assumption. [27]

Because the divisor-chain condition is assumed, §5, 2 permits us simply to speak of primary ideals. The connection between primary and irreducible ideals is the following.

Theorem II. *Under the assumption of the divisor-chain condition, every irreducible ideal is primary; equivalently, every non-primary ideal is reducible.*

Passing to a residue-class ring $\mathfrak{R}/\mathfrak{m}$ preserves the divisor-chain condition by homomorphism. The representation of $\mathfrak{m}$ as a least common multiple corresponds to the representation of the zero ideal; and by the first isomorphism theorem (§4, 3), primary divisors of $\mathfrak{m}$ correspond again to primary ideals in $\mathfrak{R}/\mathfrak{m}$ and conversely. Thus it is enough to consider the decomposition of the zero ideal in the residue-class ring. Since that ring is again a ring of the same generality, one may from the outset suppose the ideal to be decomposed to be the zero ideal of $\mathfrak{R}$.

Suppose then that the zero ideal of $\mathfrak{R}$ is not primary. There is at least one pair of ideals $\mathfrak{a},\mathfrak{b}$ – and $\mathfrak{b}$ may be assumed principal – such that

$$\mathfrak{a}\neq(0),\qquad \mathfrak{b}^x\neq(0)\quad\text{for every } x,\qquad\text{but}\qquad \mathfrak{ab}=(0).$$

Form the chain of ideal quotients

$$\mathfrak{a},\quad \mathfrak{a}:\mathfrak{b},\quad \dots,\quad \mathfrak{a}:\mathfrak{b}^\nu,\quad \dots.$$

This chain terminates; let, for instance,

$$\mathfrak{t}=\mathfrak{a}:\mathfrak{b}^{m-1}=\mathfrak{a}:\mathfrak{b}^m=\cdots.$$

[26] This is to be understood as follows. If the elements $a_1,\dots,a_n$ of a finite subset $\mathfrak{W}$ are denoted so that the ordering of the indices agrees with the prescribed well-ordering in $\mathfrak{R}$, then every subset with n elements precedes one with $m>n$ elements. If two subsets have the same number of elements, the first precedes the second when the first element at which their ordered lists differ precedes the corresponding element of the other list. Thus every system of finite subsets has a first element. Given a well-ordering of $\mathfrak{R}$, no further choice postulate is needed; in particular, if $\mathfrak{R}$ is countable, as in the case of an algebraic number field, no choice postulate is involved. The role of the choice postulate in ideal theory was mentioned, without further detail, in *Idealtheorie*, note 5.

[27] Theorem I plainly holds in the same way for module domains, if the divisor-chain condition is assumed for the system of all modules. The theorem has a purely set-theoretic character; it is the only place in the proof where the well-ordering of ideals is used. Abstractly: in a set M let a family Σ of subsets be distinguished and well-ordered. Assume the chain condition for Σ-sets: every chain $X_1,X_2,\dots$ in which X_i is a proper superset of X_{i-1} terminates. A Σ-set is reducible if it is the intersection of two Σ-sets which are both proper supersets, otherwise irreducible. Then every Σ-set is an intersection of finitely many irreducible Σ-sets.

Thus $\mathfrak{t} = \mathfrak{t} : \mathfrak{b}$, or $\mathfrak{b}$ is prime to $\mathfrak{t}$. Further, $\mathfrak{t}$ is a divisor of $\mathfrak{a}$ different from the zero ideal, and $\mathfrak{b}^x$ is different from zero for every exponent x. It therefore suffices, for the proof of Theorem II, to establish

$$(0) = [\mathfrak{t}, \mathfrak{b}^{m+1}].$$

By definition,

$$\mathfrak{b}^{m-1}\mathfrak{t} \equiv 0 \pmod{\mathfrak{a}}, \qquad \text{hence} \qquad \mathfrak{b}^m\mathfrak{t} = (0),$$

so it remains only to show

$$\mathfrak{c} = [\mathfrak{t}, \mathfrak{b}^{m+1}] \equiv 0 \pmod{\mathfrak{b}^m\mathfrak{t}}.$$

This follows from the assumptions that $\mathfrak{b}$ is principal and prime to $\mathfrak{t}$. Every element c of $\mathfrak{c}$, because of its divisibility by $\mathfrak{b}^{m+1}$, has a representation – where n denotes an integer symbol –

$$c = kb^{m+1} + nb^{m+1} = rb^m,$$

where r is again an element of $\mathfrak{R}$. From $c \equiv 0 \pmod{\mathfrak{t}}$ follows $rb^m \equiv 0 \pmod{\mathfrak{t}}$, and hence $r \equiv 0 \pmod{\mathfrak{t}}$, since $\mathfrak{t} : \mathfrak{b} = \mathfrak{t}$. Therefore $c \equiv 0 \pmod{\mathfrak{t}\mathfrak{b}^m}$, and Theorem II is proved.

Theorem III. *Under the assumption of the divisor-chain condition, every ideal admits a shortest representation as the least common multiple of finitely many greatest primary components belonging to distinct prime ideals.*

To obtain such a representation, replace, whenever possible, the representation by irreducible ideals furnished by Theorem I – hence, by Theorem II, by primary ideals – by a shortest representation. If the primary ideals belonging to the same prime ideal are collected together, §5, 3 gives the asserted representation. The primary components may be called greatest because the least common multiple of any of them is no longer primary. [28]

§7. Ideal theory under the assumption of the double-chain condition

The simplifications, compared with the theory developed so far, rest on the auxiliary results given under 1 and 2.

1. *If, in a commutative ring without zero divisors, the multiple-chain condition is satisfied, then the ring is already a field.* It is to be shown that, for $a \neq 0$, the equation $ax = b$ always has a solution in the ring. That it can have at most one solution follows from the absence of zero divisors.

Let $\mathfrak{a}$ be the principal ideal generated by a. By assumption the sequence of powers terminates; let $\mathfrak{a}^m$ be equal to all following powers. If $\mathfrak{b}$ denotes the principal ideal generated by b, then

$$\mathfrak{a}^m\mathfrak{b} = \mathfrak{a}^{m+1}\mathfrak{b}, \qquad \mathfrak{a}^{m+1}\mathfrak{b} = (a^{m+1}b).$$

In particular the element $a^m b$ has a representation – again with n an integer symbol –

$$a^m b = ka^{m+1}b + na^{m+1}b = a^{m+1}c,$$

where c is an element of $\mathfrak{R}$. Since no zero divisors exist, this gives $b = ac$, proving that the ring is a field.

2. From 1 follows at once the auxiliary result: *If the multiple-chain condition is satisfied in a commutative ring, then a prime ideal has no proper divisor different from $\mathfrak{o}$.*

Likewise one obtains the supplement: *If only axiom II is satisfied, that is, the multiple-chain condition modulo every ideal different from the zero ideal, then every prime ideal different from the zero ideal has no proper divisor different from $\mathfrak{o}$.*

For the residue-class ring modulo a prime ideal $\mathfrak{p}$ is, by definition, a ring without zero divisors; since the multiple-chain condition is also preserved there by homomorphism, it is a field by 1. From the one-to-one correspondence between the divisors of $\mathfrak{p}$ and the ideals in the residue-class ring, $\mathfrak{p}$ has no proper divisor other than $\mathfrak{o}$. [29]

Theorem IV. *If the double-chain condition is satisfied in a commutative ring, then in every shortest representation of an ideal the primary components not belonging to the unit ideal $\mathfrak{o}$ are uniquely determined.*

[28]The corresponding prime ideals and the isolated components are uniquely determined. Compare *Idealtheorie*, or W. Krull, "Ein neuer Beweis für die Hauptsätze der allgemeinen Idealtheorie," Math. Ann. 90 (1923), pp. 55–64. In §7 a direct proof of uniqueness is given for the simple special case occurring there. The primary ideals there recognized as unique are isolated components.

[29]No identity element need exist in $\mathfrak{R}$, nor therefore in $\mathfrak{o}$, the ideal consisting of all elements of $\mathfrak{R}$, although an identity class exists in the residue-class ring by 1. The simplest example of this kind, in the weaker case of the supplement, is furnished by the system of all even integers.

Supplement. Under the assumptions of axioms I and II, Theorem IV holds for every ideal different from the zero ideal.

Let

$$\mathfrak{m} = [\mathfrak{q}, \mathfrak{q}_1, \dots, \mathfrak{q}_r] = [\overline{\mathfrak{q}}, \overline{\mathfrak{q}}_1, \dots, \overline{\mathfrak{q}}_s]$$

be shortest representations of $\mathfrak{m}$ by greatest primary components, with associated prime ideals

$$\mathfrak{o}, \mathfrak{p}_1, \dots, \mathfrak{p}_r, \qquad \mathfrak{o}, \overline{\mathfrak{p}}_1, \dots, \overline{\mathfrak{p}}_s.$$

The component $\mathfrak{q}$, respectively $\overline{\mathfrak{q}}$, is to be omitted when no primary ideal belonging to $\mathfrak{o}$ occurs. Since

$$\mathfrak{o}\,\mathfrak{p}_1 \cdots \mathfrak{p}_r \equiv 0 \pmod{\mathfrak{m}},$$

each $\mathfrak{p}_i$ must be contained in some $\overline{\mathfrak{p}}_j$, hence by the auxiliary result is identical with it. Conversely each $\overline{\mathfrak{p}}_j$ must coincide with some $\mathfrak{p}_i$; the prime ideals different from $\mathfrak{o}$ therefore agree. Further,

$$\mathfrak{q}\,\mathfrak{q}_1 \cdots \mathfrak{q}_r \equiv 0 \pmod{\mathfrak{q}_i}$$

implies $\overline{\mathfrak{q}}_i \equiv 0 \pmod{\mathfrak{q}_i}$, because the remaining components are not divisible by $\mathfrak{p}_i$ and hence are prime to $\mathfrak{q}_i$. The converse congruence follows in the same way, so the non-$\mathfrak{o}$ primary components are unique. If $\mathfrak{q}$ actually occurs, then $\overline{\mathfrak{q}}$ must actually occur as well, because the representations are shortest; this also gives the uniqueness of the associated prime ideals. The proof also shows that every representation without a primary component belonging to $\mathfrak{o}$ is already shortest.

3. If to the double-chain condition one adds the existence of an identity element, Theorem IV sharpens to the following.

Theorem V. *In a commutative ring with identity in which the double-chain condition is satisfied, every ideal can be represented uniquely as a product of finitely many pairwise coprime primary ideals.*

Supplement. Under axioms I, II, III, Theorem V holds for every ideal different from the zero ideal.

Because an identity element exists, $\mathfrak{o}^2 = \mathfrak{o}$; hence every primary ideal belonging to $\mathfrak{o}$ is equal to $\mathfrak{o}$ and cannot occur in a shortest representation of an ideal different from $\mathfrak{o}$. Thus, by Theorem IV, the primary components are uniquely determined, and at the same time every representation becomes shortest after omitting $\mathfrak{o}$. By the auxiliary result under 2, two distinct prime ideals are always coprime; by the calculation rules of §4, 4, the same holds for their primary components. The least common multiple therefore becomes their product.

From the same assumptions follows the auxiliary result: *In a commutative ring with identity and the double-chain condition, the prime ideals prime to an ideal* $\mathfrak{m}$ *are exactly those, apart from the unit ideal, which are not contained in* $\mathfrak{m}$. *The notions prime and coprime coincide.*

Supplement. Under axioms I, II, III, the ideal $\mathfrak{m}$ must be assumed different from the zero ideal.

If $\mathfrak{p}$ is not contained in $\mathfrak{m}$, then it is different from all prime ideals belonging to $\mathfrak{m}$, and by the auxiliary result under 2 it is not divisible by any of them. Hence it is prime to each primary component and therefore to $\mathfrak{m}$; at the same time it is coprime to each component and hence to $\mathfrak{m}$.

If, however, $\mathfrak{p} \neq \mathfrak{o}$ is contained in $\mathfrak{m}$, then it must coincide with one of the associated prime ideals. Let it belong to the component $\mathfrak{q} = \mathfrak{q}_1$. Then $\mathfrak{q} : \mathfrak{p}$ is a proper divisor of $\mathfrak{q}$, since

$$\mathfrak{p}^{\varrho-1} \equiv 0 \pmod{\mathfrak{q} : \mathfrak{p}}, \qquad \mathfrak{p}^{\varrho-1} \not\equiv 0 \pmod{\mathfrak{q}}.$$

At the same time $\mathfrak{q} : \mathfrak{p}$, as a divisor of $\mathfrak{p}^{\varrho-1}$, is divisible only by the prime ideal $\mathfrak{p} \neq \mathfrak{o}$ and is therefore primary. By the uniqueness of the decomposition, the product

$$(\mathfrak{q} : \mathfrak{p})\mathfrak{q}_2 \cdots \mathfrak{q}_r,$$

which is divisible by $\mathfrak{m} : \mathfrak{p}$, is a proper divisor of $\mathfrak{m}$; hence $\mathfrak{p}$ is not prime to $\mathfrak{m}$.

Thus an ideal $\mathfrak{t}$ is prime to $\mathfrak{m}$ if and only if none of the prime ideals belonging to $\mathfrak{t}$ is contained in $\mathfrak{m}$; but in that case $\mathfrak{t}$ is also coprime to $\mathfrak{m}$. Conversely, coprime ideals are always mutually prime (§5, 5), so the two concepts coincide under these assumptions.

§8. Ideal theory under integral closedness in the quotient field

The ideal theory developed so far is now to be sharpened to the usual one by adjoining the last two axioms.

1. *Auxiliary result.* Let $\mathfrak{R}$ be a commutative ring without zero divisors and with an identity element. If the divisor-chain condition is assumed in $\mathfrak{R}$ (axioms I, III, IV), then from $\mathfrak{a} = \mathfrak{a}\mathfrak{b}$ and $\mathfrak{a} \neq (0)$ it follows that $\mathfrak{b} = \mathfrak{o}$. Hence

$$\mathfrak{c} \neq \mathfrak{c}\mathfrak{t}$$

for all ideals different from the zero and unit ideals. [30] The proof is the same as for the auxiliary result §1, 3, since $\mathfrak{a}\mathfrak{b}$ may be regarded as a finite module over $\mathfrak{b}$. Let $\alpha_1, \ldots, \alpha_n$ be an ideal basis of $\mathfrak{a}$ existing by I. From $\mathfrak{a} = \mathfrak{a}\mathfrak{b}$ one obtains the system

$$\alpha_i = b_{i1}\alpha_1 + \cdots + b_{in}\alpha_n, \qquad b_{ij} \equiv 0 \pmod{\mathfrak{b}}, \quad i = 1, \ldots, n.$$

Since $\mathfrak{R}$ has no zero divisors, the determinant $|b_{ij} - e_{ij}|$ vanishes; with $e_{ij} = 0$ for $i \neq j$ and $e_{ii} = e$. From

$$e - b_{11} - \cdots \equiv 0 \pmod{\mathfrak{b}}$$

it follows that $e \equiv 0 \pmod{\mathfrak{b}}$ and therefore $\mathfrak{b} = \mathfrak{o}$.

2. It remains to show only that, after adjoining integral closedness in the quotient field (axiom V) to axioms I through IV, every primary ideal different from the zero ideal is a power of its associated prime ideal. The uniqueness of the resulting representation

$$\mathfrak{m} = \mathfrak{p}_1^{\varrho_1} \cdots \mathfrak{p}_r^{\varrho_r}$$

for every ideal different from the zero ideal has already been proved by Theorem V and the auxiliary result under 1. At the same time it has been shown that these prime ideals have no proper divisor distinct from the unit ideal. The proof is first given for exponent two, using Dedekind's corollary II (§1, 4), and then in general by complete induction.

Auxiliary result. Under axioms I through V there are no primary ideals of exponent two other than $\mathfrak{q} = \mathfrak{p}^2$.

It is to be shown that from

$$\mathfrak{p}^2 \equiv 0 \pmod{\mathfrak{q}}, \qquad \mathfrak{q} \equiv 0 \pmod{\mathfrak{p}}, \qquad \mathfrak{q} \not\equiv 0 \pmod{\mathfrak{p}^2}$$

there necessarily follows $\mathfrak{q} = \mathfrak{p}^2$. Choose

$$\mathfrak{c} \equiv 0 \pmod{\mathfrak{q}}, \qquad \mathfrak{c} \not\equiv 0 \pmod{\mathfrak{p}}, \qquad \mathfrak{c} \equiv 0 \pmod{\mathfrak{p}^2}.$$

From the auxiliary result §7, 3, it follows that $\mathfrak{c} : \mathfrak{p}$ is a proper divisor of $\mathfrak{c}$; hence there is an element b such that

$$b\mathfrak{p} \equiv 0 \pmod{\mathfrak{c}} \equiv 0 \pmod{\mathfrak{q}}, \qquad b \not\equiv 0 \pmod{\mathfrak{c}}.$$

Thus $y = b/c$ belongs to the quotient field but not to $\mathfrak{R}$; it is not integral. We must prove $b \not\equiv 0 \pmod{\mathfrak{p}}$; then, since $\mathfrak{q}$ is primary and belongs to $\mathfrak{p}$, it follows that $\mathfrak{p} \equiv 0 \pmod{\mathfrak{q}}$.

By Dedekind's corollary II there exist elements m, n of $\mathfrak{R}$ such that

$$y = b/c = m/n, \qquad m^2/n \text{ is not integral;}$$

that is,

$$bn = mc, \qquad m \not\equiv 0 \pmod{\mathfrak{n}}, \qquad m^2 \not\equiv 0 \pmod{\mathfrak{n}}.$$

Multiplying $b\mathfrak{p}$ by m gives

$$mb\mathfrak{p} \equiv 0 \pmod{\mathfrak{n}c}, \qquad \text{hence} \qquad m\mathfrak{p} \equiv 0 \pmod{\mathfrak{n}}.$$

Therefore $m \not\equiv 0 \pmod{\mathfrak{p}}$, because $m^2 \not\equiv 0 \pmod{\mathfrak{n}}$ and $\mathfrak{n} \not\equiv 0 \pmod{\mathfrak{p}}$; the latter again follows from §7, 3, since $\mathfrak{n} : \mathfrak{p}$ is a proper divisor of $\mathfrak{n}$. The four relations

$$m \not\equiv 0 \pmod{\mathfrak{p}}, \qquad n \not\equiv 0 \pmod{\mathfrak{p}}, \qquad c \equiv 0 \pmod{\mathfrak{p}}, \qquad c \not\equiv 0 \pmod{\mathfrak{p}^2}, \qquad bn = mc$$

[30] On the other hand, under the same assumptions one cannot infer $\mathfrak{b} = \mathfrak{c}$ from $\mathfrak{a}\mathfrak{b} = \mathfrak{a}\mathfrak{c}$. For example, let $\mathfrak{R}$ be the polynomial domain in x, y over a field, with $\mathfrak{a} = (xy)$, $\mathfrak{b} = (x^3, xy, y^2)$, $\mathfrak{c} = (x^2, y^2)$. Then $\mathfrak{a}\mathfrak{b} = \mathfrak{a}\mathfrak{c} = \mathfrak{a}^2$, but $\mathfrak{b} \neq \mathfrak{c}$.

now imply $b \not\equiv 0 \pmod{\mathfrak{p}}$. For, since $\mathfrak{p}^2$ is primary, one first gets $mc \not\equiv 0 \pmod{\mathfrak{p}^2}$, and then $b \not\equiv 0 \pmod{\mathfrak{p}}$ from $bn = mc$. This proves the auxiliary result.

3. *Auxiliary result. Under axioms I through V there are no primary ideals of exponent ϱ other than $\mathfrak{q} = \mathfrak{p}^\varrho$.*

This follows from the auxiliary result under 2 without further use of the axioms. [31] First one proves:

$$\text{if } \mathfrak{c} \equiv 0 \pmod{\mathfrak{p}},\ \mathfrak{c} \not\equiv 0 \pmod{\mathfrak{p}^2}, \quad \text{then} \quad \mathfrak{p}^\varrho = (\mathfrak{c}^\varrho, \mathfrak{p}^{\varrho+1})$$

for every ϱ. By the result under 2, $\mathfrak{p} = (\mathfrak{c}, \mathfrak{p}^2)$. Assuming $\mathfrak{p}^{\varrho-1} = (\mathfrak{c}^{\varrho-1}, \mathfrak{p}^\varrho)$, multiplication by $\mathfrak{p}$ and the distributive law give

$$\mathfrak{p}^\varrho = (\mathfrak{c}^\varrho, \mathfrak{p}^{\varrho+1}).$$

The desired result may be put in the following second form: if

$$\mathfrak{q} \equiv 0 \pmod{\mathfrak{p}^\varrho}, \qquad \mathfrak{q} \not\equiv 0 \pmod{\mathfrak{p}^{\varrho+1}}, \qquad \mathfrak{p}^{\varrho+1} \equiv 0 \pmod{\mathfrak{q}}, \quad \varrho \geq 1,$$

then already $\mathfrak{p}^\varrho \equiv 0 \pmod{\mathfrak{q}}$. Indeed, for every primary ideal there is such an exponent $\varrho > 1$; the condition $\mathfrak{q} \not\equiv 0 \pmod{\mathfrak{p}^{\varrho+1}}$ follows from $\mathfrak{p}^\varrho \equiv 0 \pmod{\mathfrak{q}}$ and $\mathfrak{p}^\varrho \neq \mathfrak{p}^{\varrho+1}$ by the result under 1. Repeated application of the second form gives $\mathfrak{p}^\varrho \equiv 0 \pmod{\mathfrak{q}}$ and hence $\mathfrak{q} = \mathfrak{p}^\varrho$.

From the assumption of the second form and from the representation of $\mathfrak{p}^\varrho$, each element q of $\mathfrak{q}$ can be written

$$q = bc^\varrho + p^{\varrho+1}, \qquad b \equiv 0 \pmod{\mathfrak{p}}, \quad \text{or} \quad bc^\varrho \equiv 0 \pmod{\mathfrak{q}, \mathfrak{p}^{\varrho+1}}.$$

Since $(\mathfrak{q}, \mathfrak{p}^{\varrho+1})$ is primary, it follows that

$$c^\varrho \equiv 0 \pmod{\mathfrak{q}, \mathfrak{p}^{\varrho+1}}, \qquad \text{or} \qquad c^{\varrho+1} \equiv 0 \pmod{\mathfrak{q}, \mathfrak{p}^{\varrho+2}},$$

and hence $\mathfrak{p}^\varrho \equiv 0 \pmod{\mathfrak{q}}$.

4. Summarizing, one obtains:

Theorem VI. *If axioms I through V hold in a commutative ring, then every ideal different from the zero and unit ideals can be represented uniquely as a product of powers of finitely many prime ideals different from the zero and unit ideals. These prime ideals have no proper divisor different from the unit ideal.*

§9. The axioms as consequences of the assumed decomposition

Conversely, in order to infer the axioms from the existence of the usual ideal decomposition – which by Theorem VI follows from axioms I through V – the decomposition must be assumed in the following form.

Assumption. In the commutative ring $\mathfrak{R}$, every ideal not generated by the zero or the unit element is representable uniquely as a product of powers of prime ideals. These prime ideals are simple ideals not generated by the unit element, i.e. they have no proper divisor different from $\mathfrak{o}$; conversely all simple ideals are prime ideals. The uniqueness is to hold in the sharp form that

$$\mathfrak{a}^\varrho \neq \mathfrak{a}^{\varrho+1}$$

for every ideal not generated by the zero or by the unit element.

The existence of an identity element is not assumed here. If no identity element exists, the condition "not generated by the unit element" is no restriction.

1. *Proof of axiom III, the existence of the identity element.* Suppose axiom III is not satisfied. It is to be shown that $\mathfrak{o}^2$ becomes a simple ideal without being prime, contrary to the assumption. By the sharp uniqueness assumption, $\mathfrak{o} \neq \mathfrak{o}^2$; hence $\mathfrak{o} \equiv 0 \pmod{\mathfrak{o}^2}$, but $\mathfrak{o}^2 \not\equiv 0 \pmod{\mathfrak{o}^2}$, and so $\mathfrak{o}^2$ is not prime. It is, however, simple: it cannot be divisible by any prime ideal different from $\mathfrak{o}$, and by the assumed product representation it has no divisors except $\mathfrak{o}$ and $\mathfrak{o}^2$.

2. *Proof of axiom IV, absence of zero divisors.* If a is a zero divisor, say $ab = 0$ with $a \neq 0$ and $b \neq 0$, then multiplication of the product representations of the principal ideals generated by a and b gives

$$(0) = \mathfrak{p}_1^{\varrho_1} \cdots \mathfrak{p}_r^{\varrho_r},$$

where the $\mathfrak{p}_i$ are different from the zero and unit ideals, the latter because the identity element has already been proved to exist. Take the representation to be shortest, in the sense that deleting any factor yields a

[31] The proof of auxiliary result 3 under the assumption of auxiliary result 2 is found in Masazo Sono, *On Congruences II*, §§9 and 10. Compare note 17.

proper divisor of the zero ideal; this need not be automatic, since no uniqueness assumption has been made about the zero ideal. If only one prime ideal occurs, then $(0) = \mathfrak{p}^{\varrho} = \mathfrak{p}^{\varrho+1}$, contrary to the sharp uniqueness demand. If more than one prime ideal occurs, then

$$\begin{aligned}\mathfrak{p}_1^{\varrho_1} &= (\mathfrak{p}_1^{\varrho_1}, \mathfrak{p}_2^{\varrho_2} \cdots \mathfrak{p}_r^{\varrho_r})\mathfrak{p}_1^{\varrho_1} \\ &= \mathfrak{p}_1^{\varrho_1}\mathfrak{p}_2^{\varrho_2} \cdots \mathfrak{p}_r^{\varrho_r},\end{aligned}$$

because, by the existence of the identity element, $\mathfrak{p}_1$ is coprime to all the other prime ideals. This again contradicts the sharp uniqueness condition.

3. *Proof of axioms I and II, the chain conditions.* The assumptions give, for every ideal different from the zero ideal: divisibility is reflected in the product representation. Thus from $\mathfrak{m} \equiv 0 \pmod{\mathfrak{a}}$ and $\mathfrak{a} \neq \mathfrak{o}$, with

$$\mathfrak{m} = \mathfrak{p}_1^{\varrho_1} \cdots \mathfrak{p}_r^{\varrho_r}, \qquad \mathfrak{a} = \overline{\mathfrak{p}}_1^{\bar{\varrho}_1} \cdots \overline{\mathfrak{p}}_s^{\bar{\varrho}_s},$$

it follows that every $\overline{\mathfrak{p}}_j$ is identical with some $\mathfrak{p}_i$ and that $0 \leq \bar{\varrho}_i \leq \varrho_i$. Hence there are only finitely many divisors of $\mathfrak{m}$, corresponding to the finitely many exponent combinations. Therefore the double-chain condition holds in the residue-class ring modulo $\mathfrak{m}$. Axioms I and II are thus proved: the divisor-chain condition also holds in $\mathfrak{R}$ itself, since every proper divisor of the zero ideal is different from the zero ideal.

The proof of axiom V, integral closedness in the quotient field, uses the standard consequences of ideal decomposition, which must therefore first be derived.

4. *The residue-class ring modulo every ideal different from the zero ideal is a principal-ideal ring.* The ideal may be assumed different from the unit ideal, since for the residue-class ring consisting only of the zero element the assertion is immediate.

If the ideal is first primary, $\mathfrak{q} = \mathfrak{p}^{\varrho}$, and if $\mathfrak{c} \equiv 0 \pmod{\mathfrak{p}}$ but $\mathfrak{c} \not\equiv 0 \pmod{\mathfrak{p}^2}$, then by 3 one has $\mathfrak{c} = \mathfrak{p}\mathfrak{a}$ with $(\mathfrak{a}, \mathfrak{p}) = \mathfrak{o}$. Consequently

$$\mathfrak{p}^{\sigma} = (\mathfrak{c}^{\sigma}, \mathfrak{p}^{\sigma+1}) \qquad (\sigma < \varrho),$$

so every ideal in the residue-class ring modulo a primary ideal is principal. In general, if

$$\mathfrak{m} = \mathfrak{q}_1 \cdots \mathfrak{q}_r$$

is the representation and

$$\mathfrak{R}/\mathfrak{m} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_r$$

the corresponding representation of the residue-class ring as a direct sum (§4, 5), then $\mathfrak{a}_i$ is isomorphic to $\mathfrak{R}/\mathfrak{q}_i$; every ideal of $\mathfrak{a}_i$ is therefore principal. If $\mathfrak{c}$ is any ideal of the residue-class ring, then $\mathfrak{a}_i\mathfrak{c}$ is a principal ideal $\mathfrak{a}_i c_i$, and

$$\mathfrak{c} = \mathfrak{a}_1 c_1 + \cdots + \mathfrak{a}_r c_r = \mathfrak{o}c, \qquad c = c_1 + \cdots + c_r.$$

Indeed $\mathfrak{a}_i\mathfrak{o}c_i$ and $\mathfrak{a}_i\mathfrak{c}$ agree in $\mathfrak{a}_i$, since $\mathfrak{a}_i\mathfrak{a}_k = (0)$ for $i \neq k$ and $c_i \equiv 0 \pmod{\mathfrak{a}_i}$.

The theorem on principal-ideal residue rings also has the following form. If $\mathfrak{c}$ is any ideal, it can be transformed into a principal ideal by multiplication with an ideal prime to a prescribed ideal $\mathfrak{b}$. For, putting $\mathfrak{m} = \mathfrak{b}\mathfrak{c}$, the ideal $\mathfrak{c}$ becomes principal modulo $\mathfrak{m}$; that is,

$$\mathfrak{c} = (\mathfrak{o}c, \mathfrak{m}) = (\mathfrak{c}\mathfrak{a}, \mathfrak{c}\mathfrak{b}) = \mathfrak{c}(\mathfrak{a}, \mathfrak{b}),$$

so that $\mathfrak{o}c = \mathfrak{c}\mathfrak{a}$ and $(\mathfrak{a}, \mathfrak{b}) = \mathfrak{o}$.

5. *Theory of fractional ideals.* A fractional ideal means a finite $\mathfrak{R}$-module in the quotient field $\mathfrak{K}$.

The nonzero finite $\mathfrak{R}$-modules in $\mathfrak{K}$ form an abelian group under multiplication.

The product of two finite $\mathfrak{R}$-modules is again finite; multiplication is associative and commutative; since $\mathfrak{K} = \mathfrak{o}\mathfrak{K}$, the unit ideal is the identity element. It remains only to show that the equation

$$\mathfrak{U}X = \mathfrak{o}$$

always has a solution in the system of finite $\mathfrak{R}$-modules.

Preliminary remark: if the equation $\mathfrak{A}T = \mathfrak{B}$ has one and only one solution, then T is the module quotient $\mathfrak{B} : \mathfrak{A}$ (§5, 4). For

$$T \equiv 0 \pmod{\mathfrak{B} : \mathfrak{A}}, \qquad \mathfrak{B} = \mathfrak{A}T \equiv 0 \pmod{\mathfrak{A}(\mathfrak{B} : \mathfrak{A})} \equiv 0 \pmod{\mathfrak{B}}.$$

If first $\mathfrak{A}$ is principal, $\mathfrak{A} = (\alpha)$, then $X = (e/\alpha)$ is a solution of $\mathfrak{A}X = \mathfrak{o}$, and X is again a finite $\mathfrak{R}$-module. It follows that every finite $\mathfrak{R}$-module is the module quotient of two ideals of $\mathfrak{R}$, which justifies the term fractional ideal. Indeed, if $t_1, \dots, t_n$ is a module basis of T, put $t_i = \tau_i/a$ and let $\mathfrak{t}$ be the ideal generated by the τ_i in $\mathfrak{R}$. Then $\mathfrak{o}a\,T = \mathfrak{t}$, whence by the preliminary remark $T = (\mathfrak{t} : \mathfrak{o}a)$, the quotient being taken in $\mathfrak{K}$.

The solution of $\mathfrak{A}X = \mathfrak{o}$ can now be given generally. Put $\mathfrak{A} = \mathfrak{a} : \mathfrak{o}c$, and choose $\mathfrak{b}$ so that $\mathfrak{ab}$ is the principal ideal $\mathfrak{o}a$. Then

$$\mathfrak{A}\,\mathfrak{o}c = \mathfrak{a}, \qquad \mathfrak{A}\,\mathfrak{b}\mathfrak{o}c = \mathfrak{o}a, \qquad \mathfrak{A}(\mathfrak{b}\mathfrak{o}c : \mathfrak{o}a) = \mathfrak{o}.$$

Here X, as the quotient of an ideal by a principal ideal, is again a finite $\mathfrak{R}$-module. The group property is proved. It also follows that the module quotient of any two ideals, $\mathfrak{a} : \mathfrak{b}$, is a finite $\mathfrak{R}$-module, as the solution of $\mathfrak{b}X = \mathfrak{a}$.

6. From the group property follow further consequences.

6 α. *Cancellation is possible: from $\mathfrak{T} = \mathfrak{AM} : \mathfrak{BM}$ follows $\mathfrak{T} = \mathfrak{A} : \mathfrak{B}$, and conversely.* For $\mathfrak{TBM} = \mathfrak{AM}$ gives $\mathfrak{TB} = \mathfrak{A}$, and conversely.

6 β. *Every fractional ideal admits a representation $\mathfrak{C} = \mathfrak{a} : \mathfrak{b}$, where $\mathfrak{a}$ and $\mathfrak{b}$ are coprime; this condition determines $\mathfrak{a}$ and $\mathfrak{b}$ uniquely.* Existence follows from 5 and 6 α. From $\mathfrak{C} = \mathfrak{a} : \mathfrak{b} = \bar{\mathfrak{a}} : \bar{\mathfrak{b}}$ follows $\mathfrak{a}\bar{\mathfrak{b}} = \mathfrak{b}\bar{\mathfrak{a}}$, and hence uniqueness when both pairs $\mathfrak{a}, \mathfrak{b}$ and $\bar{\mathfrak{a}}, \bar{\mathfrak{b}}$ are assumed coprime.

6 γ. *Every principal module generated by a nonintegral element, i.e. by an element of $\mathfrak{K}$ not in $\mathfrak{R}$, admits a representation as a quotient of principal ideals such that no power of the numerator is divisible by the denominator.* Let the reduced representation be $\mathfrak{o}n = \mathfrak{b} : \mathfrak{c}$, so $(\mathfrak{b}, \mathfrak{c}) = \mathfrak{o}$; choose $\mathfrak{a}$ according to 4 so that $\mathfrak{o}b = \mathfrak{ba}$ and $(\mathfrak{a}, \mathfrak{c}) = \mathfrak{o}$. Then

$$\mathfrak{o}n = \mathfrak{ab} : \mathfrak{ac} = \mathfrak{o}b : \mathfrak{ac},$$

and since $\mathfrak{ac} = \mathfrak{o}b : \mathfrak{o}n$, the ideal $\mathfrak{ac}$ is principal as well; write

$$\mathfrak{o}n = \mathfrak{o}b : \mathfrak{o}c.$$

No power of b is divisible by c, because $(\mathfrak{a}, \mathfrak{c}) = \mathfrak{o}$ and $(\mathfrak{b}, \mathfrak{c}) = \mathfrak{o}$.

7. *Proof of axiom V, integral closedness in the quotient field.* By 6 γ, every nonintegral element of $\mathfrak{K}$ has a quotient representation

$$\eta = a/c$$

such that, in $\mathfrak{R}$, no power of the numerator is divisible by the denominator. Such an element cannot satisfy an equation characterizing it as integral over $\mathfrak{R}$; for from

$$\eta^n + r_1\eta^{n-1} + \cdots + r_n = 0$$

one would obtain $a^n \equiv 0 \pmod{\mathfrak{o}c}$ in $\mathfrak{R}$.

8. *Consequence.* If axioms I through IV hold in a commutative ring $\mathfrak{R}$, and every primary ideal is irreducible, then axiom V, integral closedness in the quotient field, also holds. Because of axioms I through III, every prime-ideal power $\mathfrak{p}^\varrho$ is primary; in particular $\mathfrak{p}^2$ is primary and therefore irreducible by assumption. From this one must prove

$$\mathfrak{p} = (\mathfrak{o}c, \mathfrak{p}^2).$$

Then, by the auxiliary result §8, 3, every primary ideal is a prime-ideal power; together with the other assumptions, 7 gives integral closedness.

By the divisor-chain condition,

$$\mathfrak{p} = (\mathfrak{o}c_1, \dots, \mathfrak{o}c_n),$$

where only residue classes modulo $\mathfrak{p}$ are relevant as multipliers of the c_i, and the c_i may be assumed linearly independent over the residue-class field modulo $\mathfrak{p}$. If $n > 1$, this linear independence gives a representation

$$\mathfrak{p}^2 = [\mathfrak{q}_1, \mathfrak{q}_2], \qquad \mathfrak{q}_1 = (\mathfrak{o}c_1, \mathfrak{p}^2), \qquad \mathfrak{q}_2 = (\mathfrak{o}c_2, \dots, \mathfrak{o}c_n, \mathfrak{p}^2),$$

with both $\mathfrak{q}_1$ and $\mathfrak{q}_2$ proper divisors of $\mathfrak{p}^2$; hence $\mathfrak{p}^2$ would be reducible, contrary to the assumption. Thus $\mathfrak{p} = (\mathfrak{o}c, \mathfrak{p}^2)$, and the asserted consequence is proved.

Conversely, as a consequence of axioms I through V, every primary ideal is immediately irreducible: a prime-ideal power $\mathfrak{p}^\varrho$ cannot be the least common multiple of proper divisors of the form $\mathfrak{p}^\sigma$ and $\mathfrak{p}^\tau$.

9. If zero divisors are allowed in the ring, then axioms I through III and integral closedness in the quotient ring do not imply that every primary ideal is irreducible. [32] Let $\mathfrak{T}$ be a ring in which all axioms I through IV

[32] Compare note 18.

except integral closedness hold; such rings exist, for instance the finite orders distinct from the full system of integral quantities in a finite extension field of the quotient field of $\mathfrak{R}$, when axioms I through V hold in $\mathfrak{R}$ (§3, 2). By the consequence under 8 there is in $\mathfrak{T}$ a reducible primary ideal $\mathfrak{q}$. Let $\mathfrak{p}^{\varrho}$ be a proper multiple of $\mathfrak{q}$, and let $\mathfrak{R}$ be the residue-class ring $\mathfrak{T}/\mathfrak{p}^{\varrho}$, in which the ideal corresponding to $\mathfrak{q}$ is a nonzero reducible primary ideal. In $\mathfrak{R}$ axioms I through III hold, and integral closedness in the quotient ring also holds. For every element of $\mathfrak{T}$ not divisible by $\mathfrak{p}$ becomes coprime to $\mathfrak{p}^{\varrho}$; hence all regular elements of $\mathfrak{R}$ are units, and $\mathfrak{R}$ coincides with its quotient ring.

§10. The double-chain condition and composition series

We now show that, for arbitrary module domains (§2) assumed to be well-ordered, the validity of the double-chain condition – every divisor chain and every multiple chain of modules terminates after finitely many steps – is equivalent to the existence of a composition series. [33] Specializing the module domain to a commutative ring gives the corresponding facts for the system of all ideals of the ring; in 2, throughout, module isomorphism is then to be replaced by ring isomorphism.

A module domain $\mathfrak{G}$ is called *simple* if in $\mathfrak{G}$ there are no $\mathfrak{R}$-modules besides $\mathfrak{G}$ itself and the zero module $\mathfrak{E}$. A module $\mathfrak{A}$ is called *simple in* $\mathfrak{G}$ if $\mathfrak{G}/\mathfrak{A}$ is simple.

A multiple chain

$$\mathfrak{G}, \mathfrak{U}_1, \ldots, \mathfrak{U}_r, \mathfrak{E}$$

is called a composition series of $\mathfrak{G}$ of length r if all modules in the chain are distinct and if each module is simple in the preceding one; equivalently, if the residue-class modules, the quotient groups, $\mathfrak{U}_{i-1}/\mathfrak{U}_i$ are simple module domains, as are $\mathfrak{G}/\mathfrak{U}_1$ and $\mathfrak{U}_r/\mathfrak{E}$. By forming the residue-class module $\mathfrak{G}/\mathfrak{F}$, the case in which a composition series exists from $\mathfrak{G}$ down to $\mathfrak{F} \neq \mathfrak{G}$ is reduced to the absolute case.

1. *If the double-chain condition is assumed in $\mathfrak{G}$, then a composition series exists in $\mathfrak{G}$*, provided $\mathfrak{G} \neq \mathfrak{E}$. From the assumed well-ordering of $\mathfrak{G}$ and the divisor-chain condition, one obtains as in §6, by quasi-lexicographic ordering, a well-ordering of the system of all modules. This well-ordering, together with the divisor-chain condition, gives the existence of at least one module simple in $\mathfrak{G}$. Namely, let $\mathfrak{G}_1$ be the first proper divisor of $\mathfrak{G}$ in the well-ordering; generally let $\mathfrak{G}_i$ be the first proper divisor of $\mathfrak{G}_{i-1}$. The well-determined chain

$$\mathfrak{G}, \mathfrak{G}_1, \ldots, \mathfrak{G}_i, \ldots$$

terminates after finitely many steps and therefore leads necessarily to a module $\mathfrak{A}$ simple in $\mathfrak{G}$. If $\mathfrak{U}_1 \neq \mathfrak{E}$, the same procedure gives a module $\mathfrak{U}_2$ simple in $\mathfrak{U}_1$; in general, if $\mathfrak{U}_{i-1} \neq \mathfrak{E}$, there is a module $\mathfrak{U}_i$ simple in $\mathfrak{U}_{i-1}$. This gives a well-determined multiple chain

$$\mathfrak{G}, \mathfrak{U}_1, \ldots, \mathfrak{U}_i,$$

which terminates by the multiple-chain condition and is therefore a composition series of $\mathfrak{G}$.

2. *If a composition series exists in $\mathfrak{G}$, then the double-chain condition holds in $\mathfrak{G}$.* The proof is by complete induction and does not require a well-ordering of $\mathfrak{G}$. In the course of the induction one obtains at the same time a proof of the Jordan–Hölder theorem under the sole assumption that a composition series exists. [34]

If $\mathfrak{G}$ is simple, so that $r = 0$ and the only composition series is $\mathfrak{G}, \mathfrak{E}$, then the following assertions hold.

α) Through every module simple in $\mathfrak{G}$ one can draw a composition series.

β) Jordan–Hölder theorem: every composition series has the same length and the system of quotient groups (residue-class modules) agrees up to order; corresponding quotient groups are isomorphic.

γ) Through every module different from $\mathfrak{G}$ one can draw a composition series.

δ) The multiple-chain condition holds.

ε) The divisor-chain condition holds.

[33] The modules of the domain are abelian groups with respect to addition, in fact generalized abelian groups, since the multiplier domain consists of the elements of $\mathfrak{R}$. Since they are abelian groups, the concept of composition series coincides here with that of principal series. The theorems of this paragraph remain valid if, under "modules," one understands the totality of normal divisors of an arbitrary noncommutative, even generalized, group. The notation, slightly different from the earlier one, is meant to recall group theory.

[34] Compare Masazo Sono (note 21), *On Congruences I*, §§11–13, for the case of rings. There the analogue of a composition series in noncommutative groups is also treated: series $\mathfrak{R}, \mathfrak{U}_1, \ldots, \mathfrak{U}_r, \mathfrak{E}$, where $\mathfrak{U}_i$ is an ideal and simple in $\mathfrak{U}_{i-1}$, without necessarily being an ideal in $\mathfrak{R}$. The present arguments remain valid for noncommutative groups if, by a multiple chain, one means a chain in which each $\mathfrak{U}_i$ is normal in $\mathfrak{U}_{i-1}$, and divisor chains are defined correspondingly. Compare also Dedekind, "Über die von drei Moduln erzeugte Dualgruppe," Math. Ann. 53 (1900), pp. 371–403. There, for much more general domains, the Jordan–Hölder theorem is likewise proved under the existence assumption alone; in place of the second isomorphism theorem there occurs a somewhat weaker correspondence relation, and consequently the statement of the theorem is also somewhat weaker. The principle of the proof is identical with the one given here.

Assume therefore that $\alpha)$–$\varepsilon)$ have been proved for every module domain possessing a composition series of length $r < r+1$. We prove them under the assumption that in $\mathfrak{G}$ there is a composition series

$$\mathfrak{G}, \mathfrak{A}, \mathfrak{A}_1, \ldots, \mathfrak{A}_r, \mathfrak{E}$$

of length $r+1$.

2α. If there is no module simple in $\mathfrak{G}$ other than $\mathfrak{A}$, there is nothing to prove. Let $\mathfrak{B}$ be simple in $\mathfrak{G}$ and $\mathfrak{B} \neq \mathfrak{A}$. Since $\mathfrak{A}$ and $\mathfrak{B}$ are both simple in $\mathfrak{G}$ and are distinct, necessarily $(\mathfrak{A}, \mathfrak{B}) = \mathfrak{G}$. Put $\mathfrak{M} = [\mathfrak{A}, \mathfrak{B}]$. By the second isomorphism theorem (§4, 2),

$$\mathfrak{G}/\mathfrak{B} \cong \mathfrak{A}/\mathfrak{M}, \qquad \mathfrak{G}/\mathfrak{A} \cong \mathfrak{B}/\mathfrak{M}.$$

Thus $\mathfrak{M}$ is simple in $\mathfrak{A}$ and in $\mathfrak{B}$. Since $\mathfrak{A}$ has a composition series of length $r < r+1$, the induction hypotheses α and δ give in $\mathfrak{A}$ a composition series through $\mathfrak{M}$, say

$$\mathfrak{A}, \mathfrak{M}, \mathfrak{M}_1, \ldots, \mathfrak{M}_{r-1}, \mathfrak{E}.$$

Consequently

$$\mathfrak{G}, \mathfrak{B}, \mathfrak{M}, \mathfrak{M}_1, \ldots, \mathfrak{M}_{r-1}, \mathfrak{E}$$

is a composition series of $\mathfrak{G}$ passing through $\mathfrak{B}$.

2β. To prove the Jordan–Hölder theorem, let

$$(1) \quad \mathfrak{G}, \mathfrak{A}, \mathfrak{A}_1, \ldots, \mathfrak{A}_r, \mathfrak{E}, \qquad \text{and} \qquad (2) \quad \mathfrak{G}, \mathfrak{B}, \mathfrak{B}_1, \ldots, \mathfrak{B}_s, \mathfrak{E}$$

be two composition series of $\mathfrak{G}$. If $\mathfrak{A} = \mathfrak{B}$, the result follows from the induction assumption for length $r < r+1$. Otherwise compare with the series constructed under 2α,

$$(3) \quad \mathfrak{G}, \mathfrak{B}, \mathfrak{M}, \mathfrak{M}_1, \ldots, \mathfrak{M}_{r-1}, \mathfrak{E}, \qquad (4) \quad \mathfrak{G}, \mathfrak{A}, \mathfrak{M}, \mathfrak{M}_1, \ldots, \mathfrak{M}_{r-1}, \mathfrak{E}.$$

By the induction hypothesis, the system of quotient groups of (1) and (3) agrees; likewise that of (2) and (4), since (4) is a composition series of length r through $\mathfrak{B}$ after passing to the appropriate quotient. Thus $s = r$. The isomorphism under 2α gives agreement between (3) and (4), and hence between (1) and (2). This proves the Jordan–Hölder theorem.

2γ. Preliminary remark. Let $\mathfrak{A}, \mathfrak{B}, \mathfrak{H}$ be modules of $\mathfrak{G}$, and suppose $\mathfrak{B}$ is simple in $\mathfrak{A}$. Then the modules $(\mathfrak{B}, \mathfrak{H})$ and $(\mathfrak{A}, \mathfrak{H})$ either coincide, or $(\mathfrak{B}, \mathfrak{H})$ is simple in $(\mathfrak{A}, \mathfrak{H})$. By the second isomorphism theorem,

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{H})/(\mathfrak{B}, \mathfrak{H}) \cong \mathfrak{A}/[\mathfrak{A}, (\mathfrak{B}, \mathfrak{H})].$$

Since $\mathfrak{B} \equiv 0 \pmod{\mathfrak{A}}$, this is

$$(\mathfrak{A}, \mathfrak{H})/(\mathfrak{B}, \mathfrak{H}) \cong \mathfrak{A}/\mathfrak{C}, \qquad \mathfrak{C} = [\mathfrak{A}, (\mathfrak{B}, \mathfrak{H})].$$

Here $\mathfrak{C} \equiv 0 \pmod{\mathfrak{A}}$; on the other hand $\mathfrak{B} \equiv 0 \pmod{\mathfrak{C}}$, because $\mathfrak{B} \equiv 0 \pmod{\mathfrak{A}}$ and $\mathfrak{B} \equiv 0 \pmod{(\mathfrak{B}, \mathfrak{H})}$. Hence $\mathfrak{C}$ lies between $\mathfrak{A}$ and $\mathfrak{B}$ and is therefore, by assumption, identical with $\mathfrak{A}$ or $\mathfrak{B}$. This proves the preliminary remark.

Now let

$$\mathfrak{G}, \mathfrak{U}, \mathfrak{U}_1, \ldots, \mathfrak{U}_r, \mathfrak{E}$$

be a composition series of $\mathfrak{G}$, and let $\mathfrak{H}$ be an arbitrary module of $\mathfrak{G}$ different from $\mathfrak{G}$. To show that a composition series passes through $\mathfrak{H}$, form

$$\mathfrak{G}, (\mathfrak{U}, \mathfrak{H}), (\mathfrak{U}_1, \mathfrak{H}), \ldots, (\mathfrak{U}_r, \mathfrak{H}), \mathfrak{H}.$$

By the preliminary remark, the distinct modules among these form a composition series from $\mathfrak{G}$ to $\mathfrak{H}$. Thus the first proper term is simple in $\mathfrak{G}$ (this is the special case 2α when it coincides with $\mathfrak{B}$). By 2α there is in $\mathfrak{G}$ a composition series of length $r < r+1$ through that term, and consequently, by the induction hypothesis, a composition series through $\mathfrak{H}$; adjoining $\mathfrak{G}$ gives a composition series of $\mathfrak{G}$ through $\mathfrak{H}$. Repeating the argument shows that such a series may in particular be chosen as an extension of the series just constructed from $\mathfrak{G}$ to $\mathfrak{H}$.

2δ. Proof of the multiple-chain condition. Suppose again that a composition series of length $r+1$ exists in $\mathfrak{G}$, and let

$$\mathfrak{G}, \mathfrak{B}, \mathfrak{B}_1, \dots, \mathfrak{B}_i, \dots$$

be a multiple chain, each term being a proper multiple of the preceding one. That the chain begins with $\mathfrak{G}$ is no restriction of generality, since $\mathfrak{G}$ may always be adjoined as first term. By 2γ there is a composition series through $\mathfrak{B}$, and hence $\mathfrak{B}$ has a composition series of shorter length. By the induction hypothesis, the chain

$$\mathfrak{B}, \mathfrak{B}_1, \dots, \mathfrak{B}_i,$$

terminates after finitely many steps. Hence the same is true after adjoining $\mathfrak{G}$.

2ε. Proof of the divisor-chain condition. Again assume a composition series of length $r+1$ in $\mathfrak{G}$, and let

$$\mathfrak{G}, \mathfrak{C}, \mathfrak{C}_1, \dots, \mathfrak{C}_i,$$

be a divisor chain, each term being a proper divisor of the preceding one. Again there is no restriction in assuming that the chain begins with $\mathfrak{G}$. By 2γ there is a composition series through $\mathfrak{C}$, and hence in $\mathfrak{G}/\mathfrak{C}$ there is a composition series of shorter length. By the induction hypothesis the divisor chain

$$\mathfrak{G}/\mathfrak{C},\ \mathfrak{C}_1/\mathfrak{C}, \dots, \mathfrak{C}_i/\mathfrak{C},$$

terminates after finitely many steps. By the one-to-one correspondence furnished by the first isomorphism theorem (§4, 2), the same holds for the original chain beginning with $\mathfrak{G}$, and hence also for the chain with $\mathfrak{E}$ adjoined. The equivalence of the two assumptions – existence of a composition series and the double-chain condition – is thereby proved.

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31. The Discriminant Theorem for Orders of an Algebraic Number Field or Function Field

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Dedekind's discriminant theorem says, as is well known: a prime number enters the field discriminant if and only if it is divisible at least by the square of a prime ideal.

I show in what follows that this theorem remains valid for all finite orders of a finite algebraic number field, that is, for all such rings of algebraic integers which contain the ring of all rational integers, in the following form: a prime number p enters the discriminant of an order if and only if, in the decomposition valid in the order of the principal ideal (p) into greatest primary components, at least one proper primary ideal occurs, that is, one which is not a prime ideal.[35]

Since for the principal order, the system of all integers of the field, there are no further primary ideals except powers of prime ideals, Dedekind's theorem follows by specialization.

The proof of the theorem is reduced, by passing to the residue-class ring of the order modulo (p), to the decomposition theory of those rings which are of finite rank with respect to a field contained in the ring, that is, which form a commutative system of hypercomplex quantities with principal identity and coefficients from that same field. Here, in particular, the coefficient field is the residue-class field of the rational integers modulo p, and the decomposition of (p) corresponds in the residue-class ring to the decomposition of the zero ideal.

Thus if such a ring of finite rank is called completely reducible when its zero ideal can be represented as the least common multiple of finitely many prime ideals, the question is to find criteria for complete reducibility. As such a criterion one first obtains that, for a perfect field as coefficient domain, the ring is completely reducible if and only if the extension ring with algebraically closed coefficient domain is completely reducible. For this extension ring, however, the nonvanishing of the discriminant is very easily recognized as a necessary and sufficient criterion for complete reducibility;[36] and since the discriminant of the order modulo each prime number passes into the discriminant of the residue-class ring, the discriminant theorem is thereby proved.[37]

If one considers function fields instead of number fields, the order must now contain the ground domain of all rational integral functions, respectively, for algebraic functions of several indeterminates or for integral algebraic functions, the ground domain of all integral functionals with rational integral coefficients. In that case, the case of an imperfect coefficient domain can also occur in the residue-class ring, for example when, in the case of an integral functional domain, one takes the residue-class ring modulo a prime number. For an imperfect field as coefficient domain one must distinguish between prime ideals of the first and of the second kind according as the residue field modulo the prime ideal is an extension field of the first or of the second kind,[38] and correspondingly between complete reducibility of the first and of the second kind. The criteria stated above refer throughout to complete reducibility of the first kind, which alone occurs in the perfect case. Hence, in the most general case, the discriminant theorem says that a prime element of the ground domain enters the discriminant of an order if and only if, in the ideal decomposition of (p), at least one properly primary component or one prime ideal of the second kind occurs.

For the principal order this fact was first noticed in examples by Kronecker, after the Festschrift had still given, erroneously, the formulation valid for the principal order of a number field, though without a complete proof. For the principal order Ostrowski[39] gave a proof of the above discriminant theorem and further ramification theorems attached to it; at the same time a detailed survey of the literature is found there. Finally, the discriminant theorem for all orders of relative fields follows as a direct consequence, by passing to the quotient ring with respect to each prime ideal of the ground domain. The idea of this passage, which is known in the special case, was emphasized in principle by W. Krull;[40] a systematic theory of the quotient ring in the framework of more general investigations is given by H. Grell.[41]

[35] Cf. E. Noether, *Abstrakter Aufbau der Idealtheorie in algebraischen Zahl- und Funktionenkörpern*, Math. Ann. 96 (1926), pp. 26–61. For all definitions reference is made to this paper, cited below as "Ideal Theory". In what follows, "order" always means "finite order", and "ring" a commutative ring.

[36] In a special case Hilbert argues similarly: Zur Theorie der aus n Haupteinheiten gebildeten komplexen Größen, Gött. Nachr. 1896.

[37] Related investigations, also in the noncommutative case, but restricted to the principal order (maximal domain) and assuming rational integral number coefficients, are found in the recently published paper of A. Speiser, *Allgemeine Zahlentheorie*, Naturf. Gesellschaft Zürich 71 (1926), pp. 8–48. Cf. also the American literature cited there.

[38] In Steinitz's familiar formulation, J. f. M. 137. Cf. the note to §2, no. 4, of this paper.

[39] A. Ostrowski, Zur arithmetischen Theorie der algebraischen Größen, Nachr. Gött. Ges. d. Wissensch. 1919.

[40] Cf. his Danzig lecture, Jhrbr. d. D. Math. Ver. 34, p. 121.

[41] *Beziehungen zwischen den Idealen verschiedener Ringe* (to appear in Math. Annalen).

The uniform treatment of all orders given here, and of perfect and imperfect coefficient domains, rests, besides the use of ideal theory, on taking from the outset the theory of rings of finite rank, and hence the general concept of a finite extension, as the basis, and not the concept of an extension generated by a primitive element. The module basis of an order, together with the relations among its elements, therefore takes the place of the powers of a primitive element of the field and of the defining equation,[42] respectively of the powers of the fundamental form and of the fundamental equation.

Such a module basis passes, modulo every prime element of the ground domain, into a basis of the residue-class ring, whereas this need not hold even for the principal order when the basis consists of powers of a primitive element; nor for the basis consisting of powers of the fundamental form as soon as integral algebraic functions of several indeterminates are involved.[43] Expressed differently: the residue-class ring of the polynomial domain in one indeterminate with coefficients in the ground domain, modulo an arbitrary defining equation, or even modulo the fundamental equation, need not be isomorphic modulo p to the residue-class ring of the principal order modulo (p). The main difficulties in the customary presentations lie in this disappearance of the isomorphism. Since this auxiliary polynomial domain does not occur here, these difficulties are thereby avoided automatically. At the same time the additive decomposition theory of the residue-class ring can be understood as the equivalent of the multiplicative decomposition of the equation; here too, decomposition into relatively prime factors corresponds to an additive decomposition in the polynomial residue-class ring.

It should be noted that the discriminant theorem can represent only the first theorem of a general ramification theory of orders, just as in the principal order the discriminant theorem is accompanied by the more refined theory of the ramification ideal, which there permits a more precise determination of exponents. This determination of exponents, which rests on the power representation of the primary factors of the equations, can be interpreted as the determination of the length of a composition series from p to $q = p^t$, and will presumably occur in this form in the general ramification theory.

§1. Extension rings of fields

1. We first place in front a few remarks on arbitrary rings. If T is a ring and S a system of elements of T, then by the "ring derived from S in T" is meant the intersection of all rings in T which contain S. Correspondingly one defines the "ideal derived from S in T", or the "P-module derived from S in T", when P is a subring of T.[44]

If R is a subring of T and S a system of elements of T, the ring derived from R and S in T is denoted as the ring $R[S]$ arising by ring-adjunction of S to R. This ring consists of all polynomials in the elements of S with coefficients from R, where the equality and operation relations are fixed by the equality and operation relations in T. Because of these already fixed equality and operation relations it may suffice, in a special case, to form only a subsystem of all polynomials, for instance only all linear forms in S with coefficients from R. Such a special case occurs, for example (cf. no. 2), when R is a ring with identity element and S is a field with the same identity element: all linear combinations $\sum c_i\gamma_i$, where c_i belongs to S and γ_i to R, form the ring $R[S]$.

But when only the ring R is given, and in addition a system S of elements not contained in R with equality relations, or also mutual operation relations, one can also construct the extension ring $R[S]$ arising by adjunction of S to R. One shows that there is always a ring containing R and S, namely the ring formally consisting of all polynomials in S with coefficients from R, with the usual operation rules fixed. For the definition of equality, aside from the operation rules, full freedom is still possible, except for the restriction that the equality definitions valid in R and S must be included. Thus it is necessary only that those polynomials be set equal which, by virtue of equality in R and S and by virtue of the operations, can be formed from equal elements of R and S in the same way. The ring T so constructed, containing R and S, is then identical with the ring derived in T from R and S.[45]

2. Extension rings of fields. From now on assume rings $\mathfrak{R}$ with identity element, containing a field P as a subring, and therefore over-rings of fields. The identity element e of $\mathfrak{R}$ is then also the identity element of P, and the ring is a P-module.

If $\overline{P}$ is an overfield of P, the extension ring $\overline{\mathfrak{R}} = \mathfrak{R}[\overline{P}]$ arising by ring-adjunction of $\overline{P}$ to $\mathfrak{R}$ is to be

[42] This point of view is closely connected with Dedekind's view of higher complex numbers: Zur Theorie der aus n Haupteinheiten gebildeten komplexen Größen, Gött. Nachr. 1885.

[43] Cf. Ostrowski, loc. cit., "irregularity of the first kind".

[44] Cf. Ideal Theory, §1.

[45] Cf. Steinitz's construction of the quotient field.

explained as follows.

It is assumed that the set-theoretic intersection $[\mathfrak{R}, \overline{P}]$ of $\mathfrak{R}$ and $\overline{P}$ is P, and further that between the elements of $\mathfrak{R}$ and $\overline{P}$ not belonging to P no operation relations exist. This assumption can always be achieved by replacing $\overline{P}$ by an equivalent extension field of P, or also by replacing $\mathfrak{R}$ by an equivalent extension ring.[46]

As elements of $\overline{\mathfrak{R}}$ one declares all formally formed linear combinations

$$\bar{c}_{i_1}\gamma_{i_1} + \cdots + \bar{c}_{i_s}\gamma_{i_s},$$

where the $\bar{c}_i$ run through all elements of $\overline{P}$ and the γ_i through all elements of $\mathfrak{R}$.[47] The definition is to be unique; that is, if the γ are replaced by elements equal to them in $\mathfrak{R}$, and the $\bar{c}$ by elements equal to them in $\overline{P}$, then the element in $\overline{\mathfrak{R}}$ is also to be declared equal.

As sum one defines, uniquely in the sense that replacing a summand by an equal one in the equality definition to be fixed gives the same result,

$$\sum \bar{c}_i\gamma_i + \sum \bar{d}_i\gamma_i = \sum (\bar{c}_i + \bar{d}_i)\gamma_i.$$

Here zeros may occur among the $\bar{c}$ and $\bar{d}$, whereby formally the same γ_i occur.

As product one defines, in the same sense uniquely,

$$\sum \bar{c}_i\gamma_i \cdot \sum \bar{d}_k\gamma_k = \sum \bar{c}_i\bar{d}_k\gamma_i\gamma_k.$$

For the equality definition it is declared that, if $\gamma_{i_1}, \dots, \gamma_{i_s}$ are linearly independent with respect to P, then two elements

$$\bar{c}_{i_1}\gamma_{i_1} + \cdots + \bar{c}_{i_s}\gamma_{i_s} \quad \text{and} \quad \bar{d}_{i_1}\gamma_{i_1} + \cdots + \bar{d}_{i_s}\gamma_{i_s}$$

are equal if and only if $\bar{c}_{i_k}$ is equal to $\bar{d}_{i_k}$ in $\overline{P}$. Expressed differently: elements of $\mathfrak{R}$ linearly independent with respect to P are declared to be linearly independent with respect to $\overline{P}$.

By the definitions of sum and product the equality definition for all elements of $\overline{\mathfrak{R}}$ is thereby given. Indeed their expressibility by linearly independent elements from $\mathfrak{R}$ follows. For from a relation

$$\mu = c_1\gamma_1 + \cdots + c_s\gamma_s \quad \text{in } \mathfrak{R}$$

one obtains, by multiplication with $\bar{b} = be$ according to the product definition,

$$\bar{b}\mu = \bar{b}e \cdot (c_1\gamma_1 + cdots + c_s\gamma_s) = \bar{b}c_1\gamma_1 + \cdots + \bar{b}c_s\gamma_s,$$

and hence, by the sum definition, the expressibility of all elements by linearly independent elements from $\mathfrak{R}$.

Consequently, if all elements have been expressed by linearly independent elements from $\mathfrak{R}$, the equality definition is equality of coefficients in $\overline{P}$. It therefore agrees, for elements belonging simultaneously to $\mathfrak{R}$ and $\overline{\mathfrak{R}}$ or to $\overline{P}$ and $\overline{\mathfrak{R}}$, with the equality valid there; whereas for the remaining elements of $\overline{\mathfrak{R}}$ the equality valid in $\mathfrak{R}$ and $\overline{P}$ says nothing beyond the uniqueness required in the definition, because of the intersection assumption and the further condition.[48] Similarly, the requirement that sum and product be unique entails no further relations, since equal elements may be assumed coefficientwise equal, and hence formally give the same sum and product expressions.

One now verifies at once that all axioms of the ring operation are fulfilled; thus the extension ring $\mathfrak{R}[\overline{P}]$ generated by ring-adjunction always exists.

3. Ideals in $\mathfrak{R}$ and $\overline{\mathfrak{R}}$. If $\mathfrak{a}$ is an ideal in $\mathfrak{R}$, then the module product $\overline{\mathfrak{R}}\mathfrak{a}$ is an ideal $\bar{\mathfrak{a}}$ in $\overline{\mathfrak{R}}$, the extension ideal of $\mathfrak{a}$. At the same time $\bar{\mathfrak{a}}$ is the ideal derived from $\mathfrak{a}$ in $\overline{\mathfrak{R}}$; it consists of all linear combinations $\sum \bar{c}_i a_i$, the sum in each case being over finitely many elements, where the a_i run through all elements of $\mathfrak{a}$ and the $\bar{c}_i$ through all elements of $\overline{P}$.

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[46]Two extension fields of P are called "equivalent", following Steinitz, if they can be placed in isomorphism with one another in such a way that P corresponds elementwise to itself. Equivalent extension rings of P are to be defined correspondingly. By introducing new symbols one can always produce equivalent extensions; this freedom in the use of equivalent extensions is essential for what follows.

[47]In general, elements from $\mathfrak{R}$ or $\overline{\mathfrak{R}}$ are denoted by Greek letters, and elements from P or $\overline{P}$ by Latin letters.

[48]Without this assumption there would be at least one element $\bar{c}$ belonging to $\overline{P}$ but not to P, for which $\bar{c} = \bar{c}e = \gamma$ would hold. Then, by the equality definition in $\mathfrak{R}$, e and γ would indeed be linearly independent with respect to P, but dependent with respect to $\overline{P}$. Because of the relations between elements of $\overline{P}$ and $\mathfrak{R}$, the above equality definition would then not be possible; think, for instance, of the lowering of degree of a finite extension field when the ground field is replaced by an intermediate field. The above extension is, by contrast, characterized by the possibility of the appearance of new zero divisors. Thus, for example, the residue-class ring of the polynomial domain with rational coefficients P modulo x^2+1 is a ring without zero divisors, isomorphic to the field $P(\sqrt{-1})$, whereas on passing to the polynomial domain with coefficients from $P(\sqrt{-1})$ zero divisors occur, corresponding to $(x+i)(x-i) = x^2+1$, although no factor is divisible by x^2+1. The corresponding statement holds, for example, for every finite number field; this is the conception first expressed by Dedekind for the higher complex numbers. Cf. on this point §6, no. 4, at the end.

If $\overline{\mathfrak{b}}$ is an ideal in $\overline{\mathfrak{R}}$, then the intersection $[\overline{\mathfrak{b}}, \mathfrak{R}]$ is an ideal $\mathfrak{b}$ in $\mathfrak{R}$, the contraction ideal of $\overline{\mathfrak{b}}$.[49] Every ideal $\mathfrak{a}$ of $\mathfrak{R}$ is a contraction ideal, namely of its extension ideal $\overline{\mathfrak{a}}$. Indeed, every element a of $[\mathfrak{R}, \overline{\mathfrak{a}}]$ is of the form $\sum \bar{c}_i a_i$, where the a_i may be assumed linearly independent with respect to P or $\overline{P}$. Thus the elements a, a_i become linearly dependent in $\overline{\mathfrak{R}}$, and hence, by the equality definition, also in $\mathfrak{R}$; therefore a belongs to $\mathfrak{a}$. Since the converse also holds, $\mathfrak{a} = [\mathfrak{R}, \overline{\mathfrak{a}}]$.

If $\overline{\mathfrak{p}}$ is a prime ideal in $\overline{\mathfrak{R}}$, then $\mathfrak{p} = [\mathfrak{R}, \overline{\mathfrak{p}}]$ is a prime ideal in $\mathfrak{R}$. For, by the second isomorphism theorem,[50] the residue-class ring $\mathfrak{R}/\mathfrak{p}$ is isomorphic to a subring of $\overline{\mathfrak{R}}/\overline{\mathfrak{p}}$, and hence is a ring without zero divisors. Namely, there is the ring isomorphism

$$(\mathfrak{R}, \overline{\mathfrak{p}})/\overline{\mathfrak{p}} \simeq \mathfrak{R}/[\mathfrak{R}, \overline{\mathfrak{p}}],$$

where $(\mathfrak{R}, \overline{\mathfrak{p}})$ denotes the sum, the greatest common divisor, of the modules $\mathfrak{R}$ and $\overline{\mathfrak{p}}$.

§2. Rings of finite rank. Representation as a direct sum

1. Rings of finite rank. If $\mathfrak{R}$ is a finite P-module, then $\mathfrak{R}$ is of finite rank, say k, with respect to P: that is, there are k, and no more, elements in $\mathfrak{R}$ linearly independent with respect to P, and every system of k elements of $\mathfrak{R}$ linearly independent with respect to P consequently forms a module basis of $\mathfrak{R}$. If a P-module $\mathfrak{B}$ of finite rank is divisible by another $\mathfrak{A}$ of the same finite rank, $\mathfrak{B} = O(\mathfrak{A})$, then they are therefore identical.

From now on assume that $\mathfrak{R}$ has finite rank n with respect to P. Then every P-module in $\mathfrak{R}$ has finite rank. Thus in every proper chain of divisors or multiples of P-modules in $\mathfrak{R}$, the rank of each module must be larger, respectively smaller, than the rank of the immediately preceding one; the chains break off after finitely many steps. In particular, the "double-chain theorem" holds for the system of all ideals in $\mathfrak{R}$.

It follows from this[51] that a prime ideal $\mathfrak{p}$ of $\mathfrak{R}$ has no proper divisor different from the identity ideal; hence the residue-class ring $\mathfrak{R}/\mathfrak{p}$ modulo a prime ideal different from the identity ideal is a field. If $\mathfrak{q}$ is a primary ideal belonging to $\mathfrak{p}$, that is, $\mathfrak{q} = O(\mathfrak{p})$ and $\mathfrak{p}^\rho = O(\mathfrak{q})$, then the residue-class ring $\mathfrak{R}/\mathfrak{q}$ is a primary ring, that is, a ring in which a power, here in any case already the ρ-th, of every zero divisor vanishes; and $\mathfrak{R}/\mathfrak{q}$ contains only zero divisors and units. The latter follows immediately from the fact that every ideal not divisible by $\mathfrak{p}$ is relatively prime to $\mathfrak{p}$, and hence also to $\mathfrak{q}$.

From the double-chain theorem and from the existence of the identity element it follows further[52] that every ideal $\mathfrak{a}$ of $\mathfrak{R}$ can be represented uniquely as a product of finitely many pairwise relatively prime primary ideals. Because of relative primeness this product is equal to the least common multiple and corresponds to a representation of the residue-class ring $\mathfrak{R}/\mathfrak{a}$ as a direct sum of primary rings.[53] That is, $\mathfrak{R}/\mathfrak{a}$ is the greatest common divisor of these rings, and the sum representation of every element of $\mathfrak{R}/\mathfrak{a}$ is unique.

2. Representation of rings of finite rank as direct sums of primary rings. Suppose that the representation of the zero ideal of $\mathfrak{R}$ is given by

$$(0) = \mathfrak{q}_1\mathfrak{q}_2\cdots\mathfrak{q}_r = [\mathfrak{q}_1, \mathfrak{q}_2, \dots, \mathfrak{q}_r],$$

and let the corresponding representation as a direct sum be

$$\mathfrak{R} = \mathfrak{R}_1 + \mathfrak{R}_2 + \cdots + \mathfrak{R}_r, \qquad \mathfrak{R}_i\mathfrak{R}_k = (0)\ (i \neq k), \qquad \mathfrak{R}_i^2 = \mathfrak{R}_i, \qquad \mathfrak{R}\mathfrak{R}_i = \mathfrak{R}_i.$$

Then $\mathfrak{R}_i$ is an ideal of $\mathfrak{R}$, namely

$$\mathfrak{R}_i = \mathfrak{q}_1\cdots\mathfrak{q}_{i-1}\mathfrak{q}_{i+1}\cdots\mathfrak{q}_r = [\mathfrak{q}_1, \dots, \mathfrak{q}_{i-1}, \mathfrak{q}_{i+1}, \dots, \mathfrak{q}_r],$$

$$\mathfrak{q}_i = \mathfrak{R}_1 + \cdots + \mathfrak{R}_{i-1} + \mathfrak{R}_{i+1} + \cdots + \mathfrak{R}_r,$$

and $\mathfrak{R}_i$ is isomorphic to the residue-class ring $\mathfrak{R}/\mathfrak{q}_i$: to each element γ_i of $\mathfrak{R}_i$ corresponds the class represented by γ_i in $\mathfrak{R}/\mathfrak{q}_i$.[54] If $\mathfrak{a}$ is any ideal of $\mathfrak{R}$, then the distributive law gives

$$\mathfrak{a} = \mathfrak{R}\mathfrak{a} = \mathfrak{R}_1\mathfrak{a} + \cdots + \mathfrak{R}_r\mathfrak{a} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_r;$$

[49] In H. Grell's notation, in the work cited in note 6 on p. 83 of the original.

[50] Ideal Theory, §4, no. 3. The theorem is there stated only for ideals of the same ring; the same consideration shows its validity in the above form for arbitrary ideals $\overline{\mathfrak{a}}$ of $\overline{\mathfrak{R}}$. The ring homomorphism $\overline{\mathfrak{R}} \to \overline{\mathfrak{R}}/\overline{\mathfrak{a}}$ induces, for the subring $\mathfrak{R}$ of $\overline{\mathfrak{R}}$, the ring homomorphism $\mathfrak{R} \sim (\mathfrak{R}, \overline{\mathfrak{a}})/\overline{\mathfrak{a}}$. Under this homomorphism all and only the elements of $[\mathfrak{R}, \overline{\mathfrak{a}}]$ correspond to the zero element, and hence the ring isomorphism $(\mathfrak{R}, \overline{\mathfrak{a}})/\overline{\mathfrak{a}} \simeq \mathfrak{R}/[\mathfrak{R}, \overline{\mathfrak{a}}]$ follows. The classes can be represented on the right and on the left by the same elements of $\mathfrak{R}$, for at each step of the proof the elements are assigned to the classes represented by them.

[51] Ideal Theory, §7, no. 2.

[52] Ideal Theory, §7, no. 3, Theorem V.

[53] Ideal Theory, §4, no. 5. What is added in the present compilation is found scattered in many places in the literature.

[54] Ideal Theory, §4, no. 5.

here, since $\mathfrak{a}_i = \mathfrak{R}_i\mathfrak{a} = \mathfrak{R}\mathfrak{a}_i$, the components $\mathfrak{a}_i$ are ideals both in $\mathfrak{R}_i$ and in $\mathfrak{R}$. As ideals, the $\mathfrak{a}_i$, like the $\mathfrak{R}_i$ themselves, are finite P-modules; because the sum is direct, their ranks add to the rank of $\mathfrak{a}$, respectively of $\mathfrak{R}$.

If the sum representation of the identity element is given by

$$e = \varepsilon_1 + \cdots + \varepsilon_r, \qquad \varepsilon_i\varepsilon_k = 0\ (i \neq k), \qquad \varepsilon_i^2 = \varepsilon_i, \qquad \varepsilon_i \equiv e \pmod{\mathfrak{q}_i},$$

then the representation of every element γ of $\mathfrak{R}$ is obtained uniquely as

$$\gamma = \gamma e = \gamma\varepsilon_1 + \cdots + \gamma\varepsilon_r = \gamma_1 + \cdots + \gamma_r,$$

where the component $\gamma_i = \gamma\varepsilon_i$ is an element of $\mathfrak{R}_i$; at the same time $\mathfrak{R}_i = \mathfrak{R}\varepsilon_i$. From the "orthogonality relations" one obtains

$$\gamma\delta = (\gamma_1\varepsilon_1 + \cdots + \gamma_r\varepsilon_r)(\delta_1\varepsilon_1 + \cdots + \delta_r\varepsilon_r) = \gamma_1\delta_1\varepsilon_1 + \cdots + \gamma_r\delta_r\varepsilon_r.$$

Thus $\gamma_i\delta_i$ is the i-th component of the product, and similarly $\gamma_i + \delta_i$ is the i-th component of the sum, facts which also follow from the homomorphism from $\mathfrak{R}$ to $\mathfrak{R}_i$.

The ring $\mathfrak{R}_i$ contains a subfield $P_i = P\varepsilon_i$ isomorphic to P and is of finite rank with respect to P_i; this rank is equal to the rank possessed by $\mathfrak{R}_i$ as a P-module. For for every element c of P, one has $c\varepsilon_i \equiv ce \equiv c \pmod{\mathfrak{q}_i}$, and hence $c\varepsilon_i \neq 0$ as soon as $c \neq 0$; the homomorphism between P and P_i becomes an isomorphism. Further, every linear dependence with respect to P of elements of $\mathfrak{R}_i$ passes, by multiplication with ε_i, into such a dependence with respect to P_i; conversely, because $P_i = P\varepsilon_i$, every relation with respect to P_i is at the same time a relation with respect to P. Thus the ranks with respect to P and P_i agree.

3. Direct sum representation of extension rings. If

$$\mathfrak{R} = \mathfrak{R}_1 + \cdots + \mathfrak{R}_r,$$

then for $\overline{\mathfrak{R}}$ a representation

$$\overline{\mathfrak{R}} = \overline{\mathfrak{R}}_1 + \cdots + \overline{\mathfrak{R}}_r$$

holds, where $\overline{\mathfrak{R}}_i$ denotes the extension ideal of $\mathfrak{R}_i$ in $\overline{\mathfrak{R}}$, at the same time the ring derived from $\mathfrak{R}_i$ in $\overline{\mathfrak{R}}$. Indeed, from $(0) = \mathfrak{q}_1 \cdots \mathfrak{q}_r$, multiplication by $\overline{\mathfrak{R}}$ gives the product representation $(0) = \overline{\mathfrak{q}}_1 \cdots \overline{\mathfrak{q}}_r$, where the $\overline{\mathfrak{q}}_i$ are again pairwise relatively prime. Hence one obtains a direct-sum representation for $\overline{\mathfrak{R}}$, and since the i-th component of this representation is given by

$$\overline{\mathfrak{q}}_1 \cdots \overline{\mathfrak{q}}_{i-1}\, \overline{\mathfrak{q}}_{i+1} \cdots \overline{\mathfrak{q}}_r,$$

it is equal to the extension ideal of $\mathfrak{R}_i$. But by §1, no. 3, this extension ideal is identical with the extension ring $\mathfrak{R}_i[\overline{P}_i]$ formed from $\mathfrak{R}_i$ with respect to $\overline{P}_i = \overline{P}\varepsilon_i$; therefore in extensions it suffices to extend the individual components. In general $\overline{\mathfrak{R}}_i$ will then no longer be a primary ring.

4. Prime ideals of the first and second kind. By no. 1, the residue-class ring $\mathfrak{R}/\mathfrak{p}$ modulo every prime ideal different from the identity ideal is a field. This field possesses a subfield $(P) = (P, \mathfrak{p})/\mathfrak{p} = P/[P, \mathfrak{p}] \simeq P$ isomorphic to P, since $\mathfrak{p}$ can contain no nonzero element from P; hence (P) consists of all and only the classes which can be represented by elements of P.[55] Thus $\mathfrak{R}/\mathfrak{p}$ is of finite rank with respect to (P), hence is a finite algebraic extension field. According as this extension is of the first or second kind,[56] $\mathfrak{p}$ is to be called a prime ideal of the first or of the second kind.

§3. Completely reducible rings

1. The ring $\mathfrak{R}$ is called *completely reducible* if, in the product representation of its zero ideal (§2, no. 2), all primary components become prime ideals; or – what is identical with this by §2, nos. 1 and 2 – if $\mathfrak{R}$ can be represented as a direct sum of finitely many fields. The completely reducible ring is called completely reducible of the first kind if all prime ideals are of the first kind (§2, no. 4); in the opposite case it is completely reducible of the second kind.

[55] Cf. the note to §1, no. 3, on the second isomorphism theorem.

[56] In the familiar terminology of Steinitz: an algebraic extension is called of the first kind if every element is a zero of a prime function which, in a suitable extension field, decomposes into distinct linear factors; otherwise it is of the second kind.

If P is a perfect field, then only complete reducibility of the first kind occurs; in particular this is always the case for an algebraically closed field. In what follows $\overline{P}$ denotes specifically the algebraic closure derived from P, and $\overline{\mathfrak{R}}$ is equal to $\mathfrak{R}[\overline{P}]$.

2. **Theorem.** *The ring $\mathfrak{R}$ is completely reducible of the first kind if and only if the extension ring $\overline{\mathfrak{R}} = \mathfrak{R}[\overline{P}]$ with algebraically closed $\overline{P}$ is completely reducible.*

The proof proceeds in three steps.

2a. From complete reducibility of $\overline{\mathfrak{R}}$ – more generally, of any extension ring – there follows complete reducibility (of the first or second kind) of $\mathfrak{R}$. For by hypothesis the zero ideal of $\overline{\mathfrak{R}}$ has a representation as the least common multiple of prime ideals,

$$(0) = [\overline{\mathfrak{p}}_1, \ldots, \overline{\mathfrak{p}}_s].$$

By contraction one obtains in $\mathfrak{R}$ the representation of the zero ideal

$$(0) = [[\mathfrak{R}, \overline{\mathfrak{p}}_1], \ldots, [\mathfrak{R}, \overline{\mathfrak{p}}_s]],$$

and the ideals $[\mathfrak{R}, \overline{\mathfrak{p}}_i]$ are prime ideals by §1, no. 3.

2b. From complete reducibility of the first kind of $\mathfrak{R}$ there follows complete reducibility of the first kind of $\overline{\mathfrak{R}}$, and more generally complete reducibility of the first kind of every intermediate ring between $\mathfrak{R}$ and $\overline{\mathfrak{R}}$.

Since, by §2, no. 3, a representation of $\mathfrak{R}$ as a direct sum gives a corresponding representation of $\overline{\mathfrak{R}}$ such that each component $\overline{\mathfrak{R}}_i$ becomes the extension ring $\mathfrak{R}_i[\overline{P}_i]$ – where $\overline{P}_i$ is an extension field isomorphic to $\overline{P}$ of the subfield P_i of the component $\mathfrak{R}_i$ isomorphic to P – it suffices to consider the individual component $\mathfrak{R}_i$. Thus, without loss of generality, $\mathfrak{R}$ may be assumed to be a field, indeed a finite extension field of the first kind over its subfield P. Then $\mathfrak{R}$ is generated by adjoining a primitive element of the first kind to P; equivalently, $\mathfrak{R}$ is isomorphic to the residue-class ring $P[x]/f(x)$, where $f(x)$ is a prime function of the first kind in the polynomial domain $P[x]$ in one indeterminate x. If Q is an extension field of P, then $Q[x]/f(x)$ is isomorphic to $\mathfrak{R}[Q]$; for, in passing from $P[x]$ to $Q[x]$, the rank of the residue-class ring is preserved, equal to the degree of $f(x)$, so that precisely the construction of the extension ring given in §1, no. 2 is involved. In $Q[x]$ the polynomial $f(x)$ decomposes into pairwise coprime prime functions of the first kind, each of which generates a prime ideal in $Q[x]$; hence $Q[x]/f(x)$, and therefore by the isomorphism also $\mathfrak{R}[Q]$, is completely reducible of the first kind. Since this holds for every extension field Q, in particular also for the algebraically closed field $\overline{P}$, 2b is proved.

2c. If $\mathfrak{R}$ is completely reducible of the second kind, then $\overline{\mathfrak{R}}$ is not completely reducible.

As in 2b, it suffices to suppose that $\mathfrak{R}$ is a field, and indeed a finite extension field of the second kind over P. It must be shown that in the representation of $\overline{\mathfrak{R}}$ as a direct sum of primary rings at least one proper primary component occurs. For this it is enough to exhibit a non-zero element of $\overline{\mathfrak{R}}$ some power of which vanishes. For if $\bar{\beta} \neq 0$, at least one component $\bar{\beta}_i$ must be non-zero; and if $\bar{\beta}^\rho = 0$, then $\bar{\beta}_i^\rho = 0$ (§2, no. 2, the homomorphism from $\overline{\mathfrak{R}}$ to $\overline{\mathfrak{R}}_i$), and hence $\overline{\mathfrak{R}}_i$ is not a field.

Let γ be an element of the second kind in $\mathfrak{R}$, of exponent $f \geq 1$, and let

$$F(\gamma) = \gamma^{kp^f} + c_1\gamma^{(k-1)p^f} + \cdots + c_k = 0$$

be the equation of lowest degree satisfied by γ over P, where p denotes the characteristic of P. The elements

$$e, \gamma, \gamma^2, \ldots, \gamma^{kp^f-1}$$

of $\mathfrak{R}$ are therefore linearly independent over P and consequently remain, by the definition of equality, linearly independent over $\overline{P}$ in $\overline{\mathfrak{R}}$. If one now sets

$$G(\gamma) = \gamma^{kp^{f-1}} + \sqrt[p]{c_1}\,\gamma^{(k-1)p^{f-1}} + \cdots + \sqrt[p]{c_k},$$

then $G(\gamma)$ is a non-zero element of $\overline{\mathfrak{R}}$; for, since $p > 1$ and $f \geq 1$, the exponent kp^{f-1} is always smaller than kp^f, and no dependence among the powers of γ occurring in $G(\gamma)$ can exist. But $G(\gamma)^p = F(\gamma)$ and hence vanishes. Thus 2c is proved.

From 2a and 2c it follows that complete reducibility of the first kind of $\overline{\mathfrak{R}}$ implies complete reducibility of the first kind of $\mathfrak{R}$; 2b gives the converse. This proves the theorem.

3. **Corollary.** *If $\mathfrak{R}$ is completely reducible of the first kind, then $\overline{\mathfrak{R}}$ is a direct sum of rings of rank one, hence of fields isomorphic to $\overline{P}$. Such a decomposition into rings of rank one already exists in a ring $\mathfrak{R}[Q]$, where Q is a finite extension field of P.*

For by 2b the ring $\overline{\mathfrak{R}}$ becomes a direct sum of fields which, as finite extensions of subfields isomorphic to $\overline{P}$, coincide with those subfields because $\overline{P}$ is algebraically closed. Thus one obtains

$$\overline{\mathfrak{R}} = \overline{P}\bar{e}_1 + \cdots + \overline{P}\bar{e}_n, \qquad e = \bar{e}_1 + \cdots + \bar{e}_n;$$

the components $\bar{e}_i$ of e at the same time form a linearly independent module basis of $\overline{\mathfrak{R}}$.

If the $\bar{e}_i$ are represented, according to §1, no. 2, as linear combinations of elements γ of $\mathfrak{R}$ with coefficients in $\overline{P}$, then the system of all these coefficients determines a finite extension field Q of P. Hence the $\bar{e}_i$ already belong to $\mathfrak{R}[Q]$, and from the module-basis property one obtains

$$\mathfrak{R}[Q] = Q\bar{e}_1 + \cdots + Q\bar{e}_n.$$

Thus already in $\mathfrak{R}[Q]$ there is a decomposition into rings of rank one, and in every intermediate ring between $\mathfrak{R}[Q]$ and $\overline{\mathfrak{R}}$ the indecomposable components arise as extension rings of the $Q\bar{e}_i$.[57]

§4. Matrix representation, traces, and discriminants for rings of finite rank

In this and the next paragraph the assumptions on the underlying ring are slightly more general than before, so as also to include orders in number fields and function fields. What is involved is a general and uniform formulation of essentially known facts.

Assumption. $\mathfrak{R}$ is an extension ring of a domain $\mathfrak{Z}$; the identity element of $\mathfrak{R}$ already lies in $\mathfrak{Z}$. For every $\mathfrak{Z}$-module in $\mathfrak{R}$ – in particular for $\mathfrak{R}$ itself – there exists at least one finite module basis consisting of elements linearly independent over $\mathfrak{Z}$. Besides $\alpha_1, \ldots, \alpha_s$, the elements $\beta_1, \ldots, \beta_s$ form a linearly independent module basis of the same $\mathfrak{Z}$-module if and only if the transformation determinant is a unit of $\mathfrak{Z}$.

The assumption is plainly fulfilled for the rings of finite rank considered up to now, where $\mathfrak{Z}$ is the field P. It is also fulfilled for the orders in number fields and function fields, where $\mathfrak{Z}$ is the ring of rational integers, respectively the functional domain.[58]

1. *Matrix rings homomorphic to* $\mathfrak{R}$. Let $\alpha_1, \ldots, \alpha_s$ be a linearly independent $\mathfrak{Z}$-module basis of the ideal $\mathfrak{a}$, and let γ be any element of $\mathfrak{R}$. Then $\gamma\alpha_i$ also belongs to $\mathfrak{a}$, and there are equations with coefficients in $\mathfrak{Z}$:

$$\gamma\alpha_i = c_{i1}\alpha_1 + \cdots + c_{is}\alpha_s, \qquad i = 1, \ldots, s.$$

In matrix form, if $(\tau_1, \ldots, \tau_s)$ denotes the one-row matrix formed from the elements in question,

$$(\gamma\alpha_1, \ldots, \gamma\alpha_s) = (\alpha_1, \ldots, \alpha_s)\begin{pmatrix} c_{11} & \cdots & c_{1s} \\ \vdots & & \vdots \\ c_{s1} & \cdots & c_{ss} \end{pmatrix} = (\alpha_1, \ldots, \alpha_s)C.$$

As γ runs through all elements of $\mathfrak{R}$, the totality of matrices C assigned by means of the basis α_i forms a ring $R_{\mathfrak{a}}$ homomorphic to $\mathfrak{R}$; $R_{\mathfrak{a}}$ is isomorphic to the residue-class ring $\mathfrak{R}/((0) : \mathfrak{a})$, where $(0) : \mathfrak{a}$ denotes the ideal quotient of the zero ideal by $\mathfrak{a}$. For to the difference $\beta - \gamma$ there is assigned the difference $B - C$, and the same holds for products, since

$$(\beta\gamma\alpha_1, \ldots, \beta\gamma\alpha_s) = (\gamma\alpha_1, \ldots, \gamma\alpha_s)B = (\alpha_1, \ldots, \alpha_s)CB.$$

To the elements c of $\mathfrak{Z}$ there correspond in particular the diagonal matrices cE. Exactly those elements γ for which $\gamma\mathfrak{a}$ is the zero ideal are assigned the zero matrix.

A different basis $\bar{\alpha}_1, \ldots, \bar{\alpha}_s$ of the same ideal generates a matrix ring isomorphic to $R_{\mathfrak{a}}$, indeed equivalent to it:

$$R_{\bar{\mathfrak{a}}} = P^{-1}R_{\mathfrak{a}}P.$$

For if $(\bar{\alpha}_1, \ldots, \bar{\alpha}_s) = (\alpha_1, \ldots, \alpha_s)P$, then the corresponding matrices C' and C satisfy $C' = P^{-1}CP$.

2. *Ideal classes and representation classes.* Two ideals $\mathfrak{a}$ and $\mathfrak{b}$ of $\mathfrak{R}$ belong to the same ideal class if they are isomorphic as $\mathfrak{R}$-modules, i.e. if their elements can be put into a one-to-one correspondence such that

[57] The smallest such extension field Q may be characterized as the compositum of the Galois fields determined by the components $\mathfrak{R}_i = \mathfrak{R}e_i$ of $\mathfrak{R}$. Namely, choose $f_i(x)$ as in 2b so that $P[x]/f_i(x)$ is isomorphic to the component $\mathfrak{R}_i$, and let $Q^{(i)}$ be the Galois extension field lying in $\overline{P}$ which just suffices to split $f_i(x)$ into linear factors. Then Q is the compositum of all $Q^{(i)}$. The splitting of $f_i(x)$ into linear factors corresponds to a representation of $Q^{(i)}[x]/f_i(x)$ as a direct sum of fields of rank one; $Q^{(i)}$ is the smallest extension field in which this happens. By isomorphism the same holds for $\mathfrak{R}_i[Q^{(i)}\bar{e}_i]$ and hence for $\mathfrak{R}[Q]$.

[58] Cf. *Ideal Theory*, §3. Elements of $\mathfrak{Z}$ are denoted by Latin letters.

difference and multiplication by the same element of $\mathfrak{R}$ correspond. Two rings $R_{\mathfrak{a}}$ and $R_{\mathfrak{b}}$ produced by matrix representation as in no. 1 belong to the same representation class if they are equivalent, i.e. if there is a unimodular matrix P such that

$$R_{\mathfrak{b}} = P^{-1} R_{\mathfrak{a}} P.$$

Ideal classes and representation classes correspond one-to-one. Different module bases of the same ideal give equivalent matrix rings; and different ideals in the same ideal class likewise give equivalent matrix rings, because corresponding basis elements yield the same linear coefficients when multiplied by any $\gamma \in \mathfrak{R}$. Conversely, if $R_{\mathfrak{a}}$ and $R_{\mathfrak{b}}$ belong to the same representation class, the transformation matrix supplies a basis of $\mathfrak{b}$ in which to the element $a = c_1\alpha_1 + \cdots + c_s\alpha_s$ of $\mathfrak{a}$ there corresponds $b = c_1\beta_1 + \cdots + c_s\beta_s$ of $\mathfrak{b}$; equality of the matrix rings then gives $\gamma a \leftrightarrow \gamma b$. The class of the unit ideal is called the principal class.

3. *Trace of an element with respect to a class.* Let C be the matrix assigned to the element γ of $\mathfrak{R}$ in $R_{\mathfrak{a}}$. Passing from $R_{\mathfrak{a}}$ to an equivalent ring shows that $|C|$, and more generally $|tE - C|$, where t is an indeterminate, is an invariant of the representation class. Hence it is also an invariant of the ideal class. The coefficient of $(-t^{s-1})$ in $|tE - C|$ is to be called the trace of γ with respect to the class:

$$S_{(\mathfrak{a})}(\gamma) = c_{11} + c_{22} + \cdots + c_{ss}.$$

The constant coefficient $\pm|C|$ is called the norm of γ with respect to the class.

For a fixed class the traces $S_{(\mathfrak{a})}(\gamma)$ form a $\mathfrak{Z}$-module homomorphic to $\mathfrak{R}$ viewed as a $\mathfrak{Z}$-module. Indeed, by the ring homomorphism proved in no. 1,

$$S_{(\mathfrak{a})}(\beta - \gamma) = S_{(\mathfrak{a})}(\beta) - S_{(\mathfrak{a})}(\gamma), \qquad S_{(\mathfrak{a})}(c\beta) = cS_{(\mathfrak{a})}(\beta).$$

If $\mathfrak{R}$ is a ring without zero divisors, then the trace of an element is defined independently of the class; one may simply speak of the trace. The same holds for the norm. For every non-zero ideal then has the rank of the ring, and every basis of an ideal is simultaneously a basis of the unit ideal in the quotient field.

4. *Discriminant of an ideal with respect to a class.* It is not the discriminant itself, but only the principal ideal derived from it in $\mathfrak{Z}$, that becomes an invariant of ideal and class. Let $\omega_1, \ldots, \omega_s$ and $\bar{\omega}_1, \ldots, \bar{\omega}_s$ be two linearly independent $\mathfrak{Z}$-module bases of an ideal $\mathfrak{t}$ of $\mathfrak{R}$. The determinants

$$|S_{(\mathfrak{a})}(\omega_i\omega_j)|, \qquad |S_{(\mathfrak{a})}(\bar{\omega}_i\bar{\omega}_j)|$$

differ only by the square of a unit of $\mathfrak{Z}$. The same is true for the two determinants of the basis products; and because of the module homomorphism, the same linear relations hold among the traces as among the products, so that under the cogredient transformation the determinant acquires the same quadratic factor.

Every determinant $|S_{(\mathfrak{a})}(\omega_i\omega_j)|$ is called a discriminant of $\mathfrak{t}$ with respect to the class $(\mathfrak{a})$, determined up to the square of a unit; the principal ideal derived from it in $\mathfrak{Z}$ is called the discriminant ideal. If $\mathfrak{R}$ is a ring without zero divisors, then the discriminant of an ideal is independent of the class used.

§5. Direct sums of classes

1. *Direct sum of the ideal class or representation class.* If the ideal $\mathfrak{a}$ is a direct sum, $\mathfrak{a} = \mathfrak{b} + \mathfrak{c}$, then from the isomorphism it follows that every ideal in the class is also a direct sum, $\bar{\mathfrak{a}} = \bar{\mathfrak{b}} + \bar{\mathfrak{c}}$. Here $\bar{\mathfrak{b}}$ is respectively isomorphic to $\mathfrak{b}$ and $\bar{\mathfrak{c}}$ to $\mathfrak{c}$, the ideals being regarded as $\mathfrak{R}$-modules; one may therefore speak of the direct sum of the ideal class and of its component classes.

If a ring $R_{\mathfrak{a}}$ in a representation class is a direct sum of subrings which are at the same time ideals in $R_{\mathfrak{a}}$, then by isomorphism the same is true for every ring $R_{\mathfrak{b}}$ in the representation class, and the components are equivalent rings. Thus one may speak of the direct sum of the representation class and of the component classes. The direct sum of the ideal class corresponds to the direct sum of the representation class; in this sense one speaks of a direct sum of the class.[59]

Let $\beta_1, \ldots, \beta_m$ be a basis of $\mathfrak{b}$ and $\gamma_1, \ldots, \gamma_n$ a basis of $\mathfrak{c}$; because the sum is direct, the β's and γ's together form a linearly independent basis of $\mathfrak{a}$. If $R_{\mathfrak{a}}$ is the matrix ring generated by this basis, then the matrix in $R_{\mathfrak{a}}$ assigned to any element δ of $\mathfrak{R}$ has the form

$$\begin{pmatrix} D^{\mathfrak{b}} & 0 \\ 0 & D^{\mathfrak{c}} \end{pmatrix}.$$

[59]The question to what extent all direct sums of representation classes can be produced by direct sums of ideal classes is not taken up here.

The systems of matrices containing only the first, respectively only the second, block form ideals $R^{\mathfrak{b}}$ and $R^{\mathfrak{c}}$ in $R_{\mathfrak{a}}$; their product vanishes, and $R_{\mathfrak{a}} = R^{\mathfrak{b}} + R^{\mathfrak{c}}$ is a direct sum. The assignment of $D^{\mathfrak{b}}$ to the corresponding block matrix in $R^{\mathfrak{b}}$ gives an isomorphism with the ring $R_{\mathfrak{b}}$ generated from $\mathfrak{b}$, and similarly for $\mathfrak{c}$.

2. *Behavior of trace and discriminant under direct sums of classes.* If $\mathfrak{a} = \mathfrak{b} + \mathfrak{c}$ and one uses the block matrix to determine the trace $S_{(\mathfrak{a})}(\delta)$, one reads off: the trace of an element taken with respect to a class is the sum of the traces taken with respect to the component classes. Because the product $\mathfrak{b}\mathfrak{c}$ vanishes, one further reads off: if β is an element of $\mathfrak{b}$, then $S_{(\mathfrak{a})}(\beta) = S_{(\mathfrak{b})}(\beta)$; the trace of β with respect to the class $(\mathfrak{a})$ equals its trace with respect to the class $(\mathfrak{b})$.

It follows that the discriminant ideal of the component ideal $\mathfrak{b}$, taken with respect to the class $(\mathfrak{a})$, equals the discriminant ideal of $\mathfrak{b}$ taken with respect to the component class $(\mathfrak{b})$. Finally, the discriminant ideal of $\mathfrak{a}$, taken with respect to $(\mathfrak{a})$, equals the product of the discriminant ideals of the components, taken with respect to the component classes. For with the basis adapted to the direct sum the discriminant matrix is block diagonal:

$$\begin{vmatrix} S_{(\mathfrak{b})}(\beta_i\beta_j) & 0 \\ 0 & S_{(\mathfrak{c})}(\gamma_i\gamma_j) \end{vmatrix} = |S_{(\mathfrak{b})}(\beta_i\beta_j)|\,|S_{(\mathfrak{c})}(\gamma_i\gamma_j)|.$$

3. *Passage to extension rings.* Let $\mathfrak{X}$ be an extension ring without zero divisors of $\mathfrak{Z}$ such that the intersection of $\mathfrak{R}$ and $\mathfrak{X}$ is $\mathfrak{Z}$; let the extension ring $\overline{\mathfrak{R}} = \mathfrak{R}[\mathfrak{X}]$ obtained by adjoining $\mathfrak{X}$ be defined as a ring of the same rank. If $\mathfrak{a}$ and $\mathfrak{t}$ are ideals in $\mathfrak{R}$ and $\overline{\mathfrak{a}}$ and $\overline{\mathfrak{t}}$ their extension ideals in $\overline{\mathfrak{R}}$, then the discriminant ideal of $\overline{\mathfrak{t}}$, taken with respect to $(\overline{\mathfrak{a}})$, is also the extension ideal in $\mathfrak{X}$ of the discriminant ideal of $\mathfrak{t}$ taken with respect to $(\mathfrak{a})$. For every basis of $\mathfrak{a}$ or $\mathfrak{t}$ is also an $\mathfrak{X}$-module basis of $\overline{\mathfrak{a}}$ or $\overline{\mathfrak{t}}$. In particular, the discriminant ideal of $\overline{\mathfrak{R}}$ with respect to the principal class is the extension ideal of that of $\mathfrak{R}$ with respect to the principal class.

§6. Discriminant criterion for complete reducibility of the first kind

From now on $\mathfrak{R}$ is again assumed to be an extension ring of a field P, and $\overline{P}$ is an extension field of P. The non-vanishing of the discriminant will turn out to be the necessary and sufficient condition for complete reducibility of the first kind. By §5, no. 3, the discriminant ideal of $\mathfrak{R}$ can be determined by means of the extension ring $\overline{\mathfrak{R}} = \mathfrak{R}[\overline{P}]$; and by §5, no. 2, it suffices to restrict attention to the primary rings of which $\overline{\mathfrak{R}}$ is additively composed.

1. *Special P-module bases.* If $\mathfrak{a}$ and $\mathfrak{t}$ are ideals of $\mathfrak{R}$ and $\mathfrak{a}$ is divisible by $\mathfrak{t}$, then every basis of $\mathfrak{a}$ can be completed, by adjoining elements, to a linearly independent module basis of $\mathfrak{t}$. If $\mathfrak{R}$ is a primary ring, $\mathfrak{p}$ the prime ideal belonging to the zero ideal, and ρ the exponent, so that $\mathfrak{p}^{\rho} = (0)$ and $\mathfrak{p}^{\rho-1} \neq (0)$, then a special P-module basis of $\mathfrak{R}$ is obtained by successively completing a basis of $\mathfrak{p}^{\rho-1}$ to one of $\mathfrak{p}^{\rho-2}$, and in general a basis of $\mathfrak{p}^{i}$ to one of $\mathfrak{p}^{i-1}$, with $\mathfrak{R}$ counted as $\mathfrak{p}^{0}$.

With such a basis one can show: if $\mathfrak{R}$ is primary, then the trace, taken with respect to the principal class, of every element divisible by the corresponding prime ideal $\mathfrak{p}$ vanishes. Namely, let

$$\eta_1, \ldots, \eta_s$$

be such a special basis, ordered by the powers of $\mathfrak{p}$. From $\pi \equiv 0 \pmod{\mathfrak{p}}$ it follows that multiplication by π moves each basis block into the next higher power of $\mathfrak{p}$; hence the assigned matrix has only zeros in the diagonal, and the trace of π vanishes.[60]

2. *Discriminant ideals of primary rings with algebraically closed subring P.* If the zero ideal of $\mathfrak{R}$ is a prime ideal and P is algebraically closed, then $\mathfrak{R}$ is of rank one and hence identical with P (§3, no. 3); the identity element e may be taken as module basis. Thus

$$S_{(\mathfrak{R})}(e^2) = S_{(\mathfrak{R})}(e) = e,$$

and the discriminant ideal is the unit ideal.

The discriminant ideal of a proper primary ring with algebraically closed subring P is the zero ideal. If one uses the special module basis of no. 1 – or merely completes any basis of $\mathfrak{p}$ to a basis of $\mathfrak{R}$ – the first basis element may be taken to be the identity element e, and all basis products different from e^2 are divisible by $\mathfrak{p}$; their trace vanishes. Thus the discriminant vanishes as a determinant of size at least two in which all entries except $S(e^2)$ vanish.

[60]The matrix representation furnished by such a basis has the block-triangular form belonging to the ideal factors. Since in complete reducibility the indecomposable components have no ideal different from the zero and unit ideals, such a non-trivial representation cannot occur in the representation class of these component ideals.

3. **Theorem.** *The ring $\mathfrak{R}$ of finite rank over a subfield P is completely reducible of the first kind if and only if its discriminant ideal is the unit ideal in P; in other words, if and only if the discriminant, determined up to the square of a unit, is non-zero.*

Complete reducibility of the first kind is, by §3, no. 2, identical with the assertion that the extension ring $\overline{\mathfrak{R}} = \mathfrak{R}[\overline{P}]$ is completely reducible, that is, that its indecomposable components are fields isomorphic to $\overline{P}$. By §5, no. 3, the discriminant ideal of $\mathfrak{R}$ and at the same time that of $\overline{\mathfrak{R}}$ is either the zero ideal or the unit ideal; and by §5, no. 2, the discriminant ideal of $\overline{\mathfrak{R}}$ is the product of the discriminant ideals of its components, taken with respect to these components. The individual discriminant ideals arise from those considered in no. 2 by replacing, in each case, the field $P_i = \overline{P}e_i$ isomorphic to $\overline{P}$ by $\overline{P}$ itself. Hence, if $\overline{\mathfrak{R}}$ is completely reducible, its discriminant ideal, being a product of unit ideals, is the unit ideal; whereas if $\overline{\mathfrak{R}}$ is not completely reducible, at least one component has discriminant ideal equal to the zero ideal, and consequently the discriminant ideal of $\overline{\mathfrak{R}}$ is zero. This proves the theorem.

4. *Representation of trace, norm, and discriminant by conjugate elements in the case of complete reducibility of the first kind.* Let, by §3, no. 3, in the case of complete reducibility of the first kind,

$$\mathfrak{R}[Q] = Qe_1 + \cdots + Qe_n$$

be a representation as a direct sum of rings of rank one, and let Q be the smallest extension field in which such a representation is possible. The n components of $\mathfrak{R}$ are then given by $K_i e_i$, where each K_i is a subfield of Q and $\mathfrak{R}$ is homomorphic to K_i.

Taking $e_1, \dots, e_n$ as module basis, every element

$$\gamma = c_1 e_1 + \cdots + c_n e_n$$

is assigned the diagonal matrix with diagonal entries $c_1, \dots, c_n$. Consequently

$$S(\gamma) = c_1 + \cdots + c_n, \qquad N(\gamma) = c_1 \cdots c_n,$$

where $S(\gamma)$ is the trace and $N(\gamma)$ the norm with respect to the principal class. If $\alpha_1, \dots, \alpha_n$ is a module basis of $\mathfrak{R}$ consisting of elements of $\mathfrak{R}$, and

$$\alpha_j = \alpha_j^{(1)} e_1 + \cdots + \alpha_j^{(n)} e_n,$$

then

$$|S(\alpha_i \alpha_j)| = \left|\alpha_i^{(k)}\right|^2.$$

That is, the discriminant is represented as the square of the determinant of the conjugate basis elements. If in particular $\mathfrak{R}$ itself is a field, hence an extension field of the first kind of the subfield P, then the homomorphisms from $\mathfrak{R}$ to the K_i become isomorphisms, and the K_i are the conjugate fields with respect to P in the Galois extension field Q. The representation of trace, norm, and discriminant then becomes the familiar representation by conjugate elements. It should again be emphasized that $\mathfrak{R}$ is assumed only as a field equivalent to the K_i, not as a field already lying in the Galois extension field Q.

§7. The discriminant theorem for the orders of an algebraic number field or function field

1. The number fields and function fields under consideration all fit the following field type, with a fixed notion of integral elements. Let $\mathfrak{o}$ be a principal ideal domain without zero divisors and with identity (the ring of rational integers, the polynomial domain in one indeterminate with coefficients in a field, or the functional domain), and let Ω be its quotient field. Let K be a finite extension of the first kind of Ω, and let $\mathfrak{O}$ be the ring of all elements of K integral over $\mathfrak{o}$. Every subring of $\mathfrak{O}$ which contains $\mathfrak{o}$ is called an order; $\mathfrak{O}$ itself is called the principal order.

Every $\mathfrak{o}$-module in $\mathfrak{O}$, in particular every ideal of an order and the order itself, has a linearly independent module basis over $\mathfrak{o}$. If K has degree n over Ω, it is enough to restrict attention to orders of rank n over $\mathfrak{o}$. Every order $\mathfrak{T}$ is therefore a ring satisfying the assumptions of §4 and §5, and indeed a ring without zero divisors; hence the discriminant $D_{\mathfrak{T}}$ of the order is uniquely determined up to the square of a unit of $\mathfrak{o}$ as a factor. This discriminant is also, up to a unit of Ω, identical with the discriminant of the field K viewed as an Ω-module, and is therefore non-zero, since K was assumed to be an extension of the first kind (§6, no. 3).

At the same time the discriminant has the representation by the square of the determinant of conjugate basis elements given in §6, no. 4.

The ideal theory in $\mathfrak{T}$ is given by the theorem: in $\mathfrak{T}$, every non-zero ideal can be uniquely represented as a product of finitely many pairwise coprime primary ideals whose corresponding prime ideals have no proper divisor different from the unit ideal.

2. *Passage from the order to rings of finite rank over a field.* Let p be a prime element of $\mathfrak{o}$, so that the principal ideal derived from p in $\mathfrak{o}$ is a prime ideal $\mathfrak{o}p$ of $\mathfrak{o}$ different from the zero and unit ideals; let $\mathfrak{T}p$ denote correspondingly the principal ideal derived from p in $\mathfrak{T}$. The residue-class ring

$$\mathfrak{R} = \mathfrak{T}/\mathfrak{T}p$$

becomes a ring of rank n over a subfield P isomorphic to the field $\mathfrak{o}/\mathfrak{o}p$; $\mathfrak{T}$ is homomorphic to $\mathfrak{T}/\mathfrak{T}p$.

That $\mathfrak{R}$ contains a subfield isomorphic to $\mathfrak{o}/\mathfrak{o}p$ follows from the second isomorphism theorem: the subring $(\mathfrak{o}, \mathfrak{T}p)/\mathfrak{T}p$ is isomorphic to $\mathfrak{o}/[\mathfrak{o}, \mathfrak{T}p] = \mathfrak{o}/\mathfrak{o}p$. Let $\alpha_1, \ldots, \alpha_n$ be a linearly independent module basis of $\mathfrak{T}$ over $\mathfrak{o}$, and let (α_i) be the residue classes determined by these elements modulo $\mathfrak{T}p$. Any linear dependence between the (α_i) over P would arise from a congruence

$$c_1\alpha_1 + \cdots + c_n\alpha_n \equiv 0 \pmod{\mathfrak{T}p},$$

hence from a relation $c_1\alpha_1 + \cdots + c_n\alpha_n = py$ with $y \in \mathfrak{T}$. From the representation of y in the basis α_i and the uniqueness of this representation, it follows that every c_i is divisible by p; consequently the element (c_i) of P vanishes. Thus the (α_i) form a basis of $\mathfrak{R}$ over P.

Under the homomorphism the discriminant $D_{\mathfrak{T}}$ of $\mathfrak{T}$ passes to the discriminant of $\mathfrak{R}$, and the ideal decomposition of $\mathfrak{T}p$ passes to the decomposition of the zero ideal of $\mathfrak{R}$. If

$$\mathfrak{T}p = \mathfrak{q}_1 \cdots \mathfrak{q}_s = [\mathfrak{q}_1, \ldots, \mathfrak{q}_s]$$

is the representation of $\mathfrak{T}p$ in $\mathfrak{T}$, then it becomes, under the homomorphism,

$$(0) = [(\mathfrak{q}_1), \ldots, (\mathfrak{q}_s)]$$

in $\mathfrak{R}$. By the first isomorphism theorem, $\mathfrak{R}/(\mathfrak{q}_i)$ is isomorphic to $\mathfrak{T}/\mathfrak{q}_i$; hence $\mathfrak{q}_i$ and $(\mathfrak{q}_i)$ are simultaneously proper primary ideals or prime ideals, and the $(\mathfrak{q}_i)$ are pairwise coprime by homomorphism.

3. **Discriminant theorem.** *A prime element p of $\mathfrak{o}$ divides the discriminant $D_{\mathfrak{T}}$ of an order $\mathfrak{T}$ if and only if, in the decomposition of the principal ideal $\mathfrak{T}p$ into pairwise coprime primary components in $\mathfrak{T}$, at least one proper primary component or at least one prime ideal of the second kind occurs.*

If the residue field $\mathfrak{o}/\mathfrak{o}p$ is a perfect field, then every prime element p which divides $D_{\mathfrak{T}}$ – and only such a p – has at least one proper primary component. If $\mathfrak{o}/\mathfrak{o}p$ is perfect and $\mathfrak{T}$ is chosen to be the principal order $\mathfrak{D}$, then p divides the discriminant if and only if it is divisible by at least the square of a prime ideal of $\mathfrak{D}$.

For, by the correspondence between the ideal decomposition of $\mathfrak{T}p$ in $\mathfrak{T}$ and that of the zero ideal in $\mathfrak{R} = \mathfrak{T}/\mathfrak{T}p$, the occurrence of a proper primary component or of a prime ideal of the second kind in $\mathfrak{T}p$ means that $\mathfrak{R}$ is not completely reducible of the first kind. By §6, no. 3, this is the case if and only if the discriminant of $\mathfrak{R}$ vanishes, which by the homomorphism in no. 2 says precisely that $D_{\mathfrak{T}}$ is divisible by p.[61]

§8. The discriminant theorem for the orders of relative fields

If instead of the principal ideal domain $\mathfrak{o}$ one already takes as the starting ring the principal order of a number field or function field – more generally, a multiplication ring, i.e. a ring in which the ideal theory of the principal order holds – then in place of the discriminant $D_{\mathfrak{T}}$ there appears the discriminant ideal of $\mathfrak{T}$ with respect to $\mathfrak{o}$; the discriminant theorem transfers completely by means of the following considerations.

1. Let $\mathfrak{o}$ be a multiplication ring, that is, a ring without zero divisors and with identity, in which every ideal different from the zero and unit ideals can be represented uniquely as a power product of prime ideals, and where these prime ideals have no proper divisor different from the unit ideal. Let $\Omega, K, \mathfrak{D}, \mathfrak{T}$ be defined as in §7, no. 1; then in $\mathfrak{T}$ the ideal theory stated there holds.

[61] Example of the occurrence of prime ideals of the second kind: let $\mathfrak{o}$ be the derived functional domain, Ω its quotient field, and let K be the extension field obtained by adjoining $\sqrt{x}$. Consider the order $\mathfrak{T}$ with module basis $1, \sqrt{x}$ over $\mathfrak{o}$. Its discriminant is $\begin{vmatrix} 1 & \sqrt{x} \\ 1 & -\sqrt{x} \end{vmatrix}^2$. The principal ideal $\mathfrak{T}2$ remains a prime ideal in $\mathfrak{T}$, but one of the second kind, since $\mathfrak{T}/\mathfrak{T}2$ is isomorphic to $P[t]/(t^2 - x)$, where P is the corresponding residue field. In contrast, $\mathfrak{T}\sqrt{x}$ is proper primary. If one replaces one indeterminate by two, one obtains examples that are generated over P only after adjoining two elements.

In particular, if a multiplication ring $\mathfrak{o}$ has only one prime ideal different from the zero and unit ideals, then $\mathfrak{o}$ is a principal ideal domain. For if one chooses an element p whose principal ideal is divisible by exactly the first power of this prime ideal, then p generates the prime ideal itself; the power-product representation then implies that all other ideals too are principal.

2. *Trace and discriminant ideal.* The trace of every element γ of $\mathfrak{T}$ is defined as the trace of γ in K, where K is regarded as an Ω-module. Every system of n elements of K linearly independent over Ω may be used as a basis for producing the trace. If $\alpha_1, \ldots, \alpha_n$ is such a system, then, since K is assumed to be an extension of the first kind,

$$|S(\alpha_i \alpha_j)|$$

is a non-zero element of Ω, indeed the square of the determinant of the conjugate basis elements. From the integral closedness of $\mathfrak{o}$ in Ω it follows that this determinant is an element of $\mathfrak{o}$ as soon as all α_i belong to $\mathfrak{O}$.

Let $\tau_1, \ldots, \tau_n$ run through all systems of n elements of $\mathfrak{T}$, and form, for every such system, the determinant $|S(\tau_i \tau_j)|$. The ideal in $\mathfrak{o}$ derived from the totality of these determinants is called the discriminant ideal of the order $\mathfrak{T}$ with respect to $\mathfrak{o}$. If in particular the order has a linearly independent module basis $\alpha_1, \ldots, \alpha_n$, then the determinant $|S(\alpha_i \alpha_j)|$ may be taken as a basis of the discriminant ideal; the definition agrees with that of §7, no. 1.

3. *Passage to the quotient ring.* If $\mathfrak{a}$ is any ideal of $\mathfrak{o}$, the quotient ring $\mathfrak{T}_{\mathfrak{a}}$ is defined as the totality of elements of the quotient field of $\mathfrak{T}$ whose denominator is an element of $\mathfrak{T}$ prime to $\mathfrak{a}$. Apart from the zero and unit ideals, the ring $\mathfrak{T}_{\mathfrak{a}}$ has no prime ideals other than the extension ideals of the prime ideals belonging to $\mathfrak{a}$. There is a one-to-one correspondence, with isomorphism of residue-class rings, between the non-trivial ideals of $\mathfrak{T}_{\mathfrak{a}}$ and those ideals of $\mathfrak{T}$ whose corresponding prime ideals are among those belonging to $\mathfrak{a}$; moreover the zero and unit ideals correspond one-to-one.

The same considerations apply to the multiplication ring $\mathfrak{o}$. If in particular $\mathfrak{p}$ is a prime ideal, then $\mathfrak{o}_{\mathfrak{p}}$ has only one prime ideal different from the zero and unit ideals; since the residue-class rings are isomorphic, $\mathfrak{o}_{\mathfrak{p}}$ is, with $\mathfrak{o}$, a multiplication ring, and by no. 1 it is therefore a principal ideal domain. The basis of the prime ideal derived from $\mathfrak{p}$ becomes a prime element p of $\mathfrak{o}_{\mathfrak{p}}$. The extension ideal of any ideal $\mathfrak{c}$ of $\mathfrak{o}$ is generated by the primary component of $\mathfrak{c}$ belonging to $\mathfrak{p}$, and is therefore equal to $(p)^{\alpha}$ or to the unit ideal according as $\mathfrak{p}$ occurs in $\mathfrak{c}$ – to the α-th power – or does not occur in $\mathfrak{c}$. If the extensions of two ideals in all quotient rings $\mathfrak{o}_{\mathfrak{p}}$ agree, then the two ideals are identical.

4. *The discriminant ideal under passage to the quotient ring.* Let $\mathfrak{p}$ be a prime ideal of $\mathfrak{o}$ and put $\mathfrak{P} = \mathfrak{T}\mathfrak{p}$ for the extension ideal in $\mathfrak{T}$. The quotient ring $\mathfrak{T}_{\mathfrak{P}}$ possesses a module basis consisting of linearly independent elements over the principal ideal domain $\mathfrak{o}_{\mathfrak{p}}$; at the same time $\mathfrak{T}_{\mathfrak{P}}$ is a finite $\mathfrak{p}$-order in the extension field K of the quotient field of $\mathfrak{o}_{\mathfrak{p}}$. Thus in $\mathfrak{T}_{\mathfrak{P}}$, with respect to $\mathfrak{o}_{\mathfrak{p}}$, the discriminant theorem (§7, no. 3) holds.

The discriminant $D_{\mathfrak{T}_{\mathfrak{P}}}$ is then a basis of the extension ideal, taken in $\mathfrak{o}_{\mathfrak{p}}$, of the discriminant ideal of $\mathfrak{T}$ with respect to $\mathfrak{o}$. Namely, choose elements $\alpha_1, \ldots, \alpha_n$ of $\mathfrak{T}$ as an $\mathfrak{o}_{\mathfrak{p}}$-module basis of $\mathfrak{T}_{\mathfrak{P}}$; then $|S(\alpha_i \alpha_j)|$ belongs to the discriminant ideal, and every determinant $|S(\tau_i \tau_j)|$ in the quotient ring is divisible by $|S(\alpha_i \alpha_j)|$.

5. **General discriminant theorem.** *A prime ideal $\mathfrak{p}$ of a multiplication ring $\mathfrak{o}$ occurs in the discriminant ideal of an order with respect to $\mathfrak{o}$ if and only if, in the ideal decomposition of $\mathfrak{T}\mathfrak{p}$ into pairwise coprime primary components in $\mathfrak{T}$, at least one proper primary component or at least one prime ideal of the second kind occurs.*

From 3 and 4 it follows that a prime ideal $\mathfrak{p}$ of $\mathfrak{o}$ occurs in the discriminant ideal of $\mathfrak{T}$ if and only if the prime element p of $\mathfrak{o}_{\mathfrak{p}}$ divides the discriminant $D_{\mathfrak{T}_{\mathfrak{P}}}$. But this is identical with saying that the residue-class ring $\mathfrak{T}_{\mathfrak{P}}/\mathfrak{T}_{\mathfrak{P}}p$ is not completely reducible of the first kind; and since this residue-class ring is isomorphic to $\mathfrak{T}/\mathfrak{T}\mathfrak{p}$, the general discriminant theorem is proved.[62]

As a specialization to relative discriminants of number fields one obtains: a prime ideal $\mathfrak{p}$ of the principal order of a number field occurs in the relative discriminant of an extension field if and only if $\mathfrak{p}$ is divisible in the principal order of the extension field by the square of a prime ideal.

Göttingen, March 30, 1926.

32. Joint with R. Brauer: On minimal splitting fields of irreducible representations

Sitz. Ber. d. Preuß. Akad. d. Wiss. 1927, pp. 221–228.

[62]When $\mathfrak{o}$ is the principal order of an algebraic number field, this gives agreement with the relative discriminant defined by Hilbert, which is known to agree with Hecke's definition.

(Presented by Mr. Schur.)

The question of the number fields of smallest degree over a ground field P in which a representation of a finite group that is irreducible over P decomposes into absolutely irreducible constituents was first treated by I. Schur.[1] Suppose, for example, that P contains the character of such a constituent. Schur then showed that the degree of these fields relative to P – the index – agrees with the number of those absolutely irreducible constituents which all belong to the same class; that consequently the same field is already determined by the requirement of splitting off one absolutely irreducible constituent; and further that the degree of every field, relative to P, in which such a splitting occurs is a multiple of the index. All these fields are to be called splitting fields, also in the case where P does not contain the character. By an example Schur showed that the splitting fields of smallest degree need not be isomorphic. If P contains no corresponding simple character, then the splitting fields must comprise the field of such a character and its conjugates with respect to P; the absolutely irreducible representations belonging to different conjugate characters are indeed conjugate, but plainly belong to different classes. To split off a system of absolutely irreducible representations of one class, here too it suffices to take the field of the corresponding character and one of the corresponding splitting fields.

In what follows the question of splitting fields is pursued further. The splitting fields of smallest degree can be characterized by assigned non-commutative division rings; the general splitting fields by two-sided simple rings invariantly connected with these non-commutative division rings. Further, by means of the example of the quaternion division ring, it is shown that the degrees of minimal splitting fields are not bounded; here a splitting field is called minimal if none of its proper subfields is a splitting field. This unboundedness follows essentially from a number-theoretic existence theorem whose proof is supplied by H. Hasse in the immediately following note. The example, however, is also treated elementarily by computation – more in the way of verification; and thus, without using the general theory and the number-theoretic existence theorem, the unboundedness of the degrees is shown by proving elementarily a less far-reaching, but sufficient, existence theorem.[2] It should also be noted that in this example we are again dealing with splitting fields of a finite group, namely the quaternion group, which also underlies Schur's example. The representations given there are simultaneously (isomorphic) representations of the quaternion division ring.

1. Characterization of the splitting fields

First let us briefly assemble the basic concepts. Let $\mathfrak{S}$ denote a hypercomplex system with respect to a field P_0. By a representation Γ of $\mathfrak{S}$ – in P – one means a system of matrices with elements from P, in such a way that Γ becomes a homomorphic image of $\mathfrak{S}$, and that diagonal matrices p_0E are assigned to the elements p_0 from P_0; hence P must contain P_0. A representation Γ together with all its transforms $P^{-1}\Gamma P$ forms a representation class. The representations of finite groups G are included in the representations so defined; the assigned hypercomplex system $\mathfrak{S}$, the "group ring", consists of all "group numbers", i.e. of all linear combinations of the group elements, with coefficients from P_0, the original group elements being regarded as linearly independent and multiplication being defined by group multiplication and the laws of arithmetic. The notions "reducible", "irreducible", "completely reducible", and "absolutely irreducible" representations are defined as usual; the terminology is to be extended to representation classes. A system $\mathfrak{S}$ is called "completely reducible in P" if all its representation classes in P are completely reducible.

Now assume $\mathfrak{S}$ to be completely reducible in P_0; assume P_0 to be a perfect field, so that the same is true for every algebraic extension P of P_0. The representation classes remain completely reducible under every extension P of P_0.[3] The center of $\mathfrak{S}$ becomes a direct sum of finitely many fields; every extension P of P_0 isomorphic to such a field K is the field of a (simple) character. If P_0 is extended to a P isomorphic to K, then there splits off from $\mathfrak{S}$ a two-sided simple ring to which the absolutely irreducible representation

[1] Arithmetische Untersuchungen über endliche Gruppen, Sitzungsber. d. Berl. Ak. d. Wiss. 1906, p. 164. Beiträge zur Theorie der Gruppen linearer Substitutionen, Transact. of the Am. Math. Soc. Ser. 2, Vol. XV, 1909, p. 159. In the second paper the results are transferred to completely reducible hypercomplex systems. A representation is collected into one class together with all its transforms (similar representations).

[2] This example is due to E. Noether; it rests on a modification of one due to R. Brauer, in which the existence of a minimal splitting field whose degree exceeds the index was shown in general. The elementary treatment of the example is due to R. Brauer. The characterization of the splitting fields goes back to investigations of the authors, which otherwise pursue separate goals. For E. Noether the matter is a construction of representation theory on the basis of module and ideal theory, under general finiteness hypotheses; for R. Brauer it is the construction of non-commutative division rings of finite degree over a given commutative perfect ground field, with the help of factor systems. The authors arrived independently at the characterization of splitting fields of smallest degree, R. Brauer for perfect ground fields, E. Noether without this restriction. R. Brauer found the characterization of the general splitting fields in connection with a question of E. Noether concerning the possibility of non-commutative embedding; E. Noether was able here also to remove the restriction to a perfect ground domain.

[3] If P_0 is not perfect, complete reducibility is preserved if and only if the components K of the center become "extensions of the first kind" of P_0. Under this hypothesis all further consequences in the text remain valid.

class of the character is assigned. By Wedderburn's theorem this ring is the direct product of a matrix ring over P with a non-commutative division ring $\mathfrak{A}$, uniquely determined up to isomorphism, whose center is P and which has degree m^2 relative to P, where m denotes the Schur index belonging to the character. Every splitting field of $\mathfrak{G}$ – with respect to the representation belonging to the character – is at the same time one of $\mathfrak{A}$ with respect to the representations in P, and conversely. The conjugate representations of $\mathfrak{G}$ are obtained by starting from the corresponding two-sided simple ring over P_0; its assigned division ring $\mathfrak{A}_0$ becomes isomorphic to $\mathfrak{A}$ and has K as center. Thus the problem of splitting fields is completely reduced to that of the assigned non-commutative division ring $\mathfrak{A}$ and its splittings.[4] In particular one has the theorem:

All maximal commutative subfields of $\mathfrak{A}$ have the same degree m relative to P, where m denotes the Schur index. All these maximal commutative subfields are splitting fields of smallest degree, and every splitting field of smallest degree is contained among them. The degree of every splitting field is a multiple of m. The ring $\mathfrak{A}$ has only one absolutely irreducible representation class and likewise only one irreducible representation class with respect to P.

For the characterization of the general splitting fields, let $\mathfrak{A}_r$ denote the direct product of $\mathfrak{A}$ with the matrix ring of degree r^2 over P (the system of all r-rowed matrices with elements from P). Then $\mathfrak{A}_r$ is two-sided simple, with $\mathfrak{A}$ as assigned non-commutative division ring; and every two-sided simple ring, hypercomplex over P, to which $\mathfrak{A}$ is assigned, is obtained in this way. $\mathfrak{A}_r$ is also characterized as the system of all r-rowed matrices with elements from $\mathfrak{A}$.

Every splitting field of $\mathfrak{A}$ of degree mr – if such a field exists – is isomorphic to a maximal commutative subfield of $\mathfrak{A}_r$. Conversely, all maximal commutative subfields of $\mathfrak{A}_r$ are splitting fields, each of degree ms, where s is a divisor of r. In the regular case all maximal commutative subfields of $\mathfrak{A}_r$ have degree mr. Here by the regular case is meant: the ground field P has the property that for every commutative field Σ over P which is finite over P, there are still commutative extensions over Σ of arbitrary degree.[5]

Whether among these splitting fields, for $r > 1$, minimal ones also occur is not yet decided by the embedding in $\mathfrak{A}_r$. That this actually can be the case for infinitely many r is shown by the following example.

2. The unboundedness of the degrees of minimal splitting fields

For this unboundedness the quaternion division ring is now to be given as an example, regarded as a hypercomplex system over the field P of rational numbers, with the basis elements i, j, k besides the unit 1 and with the familiar relations:

$$i^2 = j^2 = k^2 = -1; \qquad ij = k; \quad jk = i; \quad ki = j; \quad ji = -k; \quad kj = -i; \quad ik = -j.$$

It will be shown that all and only those algebraic number fields are splitting fields for which, after adjunction, there exists an "idempotent" element r, so that $r^2 = r$, with r different from both the zero element and the unit element. These number fields can also be characterized as those in which (-1) is representable as a sum of three squares, and consequently as a sum of two squares.[6] Indeed, set

$$r = \alpha + \frac{1}{2}\beta i + \frac{1}{2}\gamma j + \frac{1}{2}\delta k;$$

then

$$r^2 = \left(\alpha^2 - \frac{1}{4}\beta^2 - \frac{1}{4}\gamma^2 - \frac{1}{4}\delta^2\right) + \alpha\beta i + \alpha\gamma j + \alpha\delta k,$$

so that $r^2 = r$ is equivalent to

$$4\alpha^2 - 4\alpha - \beta^2 - \gamma^2 - \delta^2 = 0; \quad 2\alpha\beta - \beta = 0; \quad 2\alpha\gamma - \gamma = 0; \quad 2\alpha\delta - \delta = 0,$$

and hence, since β, γ, δ are not to vanish simultaneously, equivalent to

$$(-1) = \beta^2 + \gamma^2 + \delta^2.$$

[4] This corresponds to Schur's reduction to the system of matrices over P which commute with the irreducible representation of $\mathfrak{G}$ in P. The transposes of these matrices give a representation of $\mathfrak{A}$ in P.

[5] Thus, for example, no regular case occurs when P is the field of all real numbers.

[6] That representability of (-1) as a sum of three squares always implies such representability as a sum of two follows from the identity

$$(c^2 + d^2)(a^2 + b^2 + c^2 + d^2) = (ac + bd)^2 + (ad - bc)^2 + (c^2 + d^2)^2,$$

which follows, for example, from the product theorem for norms of quaternions. Instead of requiring the existence of an idempotent element, it would also have sufficed for the splitting to require an element of vanishing norm.

That every such idempotent element produces a splitting, and that the existence of such an idempotent element is necessary for the splitting, follows from these facts: every irreducible representation of a completely reducible system is generated by a one-sided simple ideal, more precisely by the corresponding ideal class, and every such ideal class generates the representation. These one-sided simple ideals all occur in a one-sided direct-sum decomposition, and this decomposition determines the "primitive" idempotent elements that become the components of the unit. Conversely, every idempotent element gives a direct-sum decomposition $\mathfrak{S} = r\mathfrak{S} + (e-r)\mathfrak{S}$, where e denotes the unit; the "primitive" ones split off simple ideals.[7] A non-commutative division ring, regarded as a hypercomplex system with respect to its center, becomes – when extended to a hypercomplex system with respect to a splitting field – a direct sum of m absolutely simple one-sided ideals, where m is the Schur index.

In the case of the quaternion division ring, m is two; hence every idempotent element $\neq 0$ and $\neq 1$ already produces the decomposition into absolutely simple ideals to which the splitting into absolutely irreducible representation classes corresponds. This characterizes the indicated fields as splitting fields.

It follows further: all cyclic fields Ω_n of degree 2^n over the field of rational numbers in which (-1) can be represented as a sum of two squares are minimal splitting fields of the quaternion division ring. For a splitting field cannot be real, because of the representation of (-1) as a sum of squares; on the other hand, the subfield of Ω_n of degree 2^{n-1} – and therefore every proper subfield of Ω_n – is necessarily real, being the intersection of Ω_n with the field of all real algebraic numbers. Namely, Ω_n is a Galois field of degree two over this intersection, which therefore must coincide with the subfield of degree 2^{n-1}.

According to Hasse's existence theorem[8] there are such fields Ω_n for every n; the degrees of minimal splitting fields are not bounded. In fact, every power of the index occurs here as the degree of a minimal splitting field.[9]

3. Elementary treatment of the splitting fields of the quaternion division ring

Let $\mathfrak{D}$ denote the representation of the quaternion group formed from the following eight matrices:[10]

$$\pm E = \pm \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \pm I = \pm \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad \pm J = \pm \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \pm K = \pm \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

The group ring generated by $\mathfrak{D}$ is precisely a representation of the quaternion division ring regarded as a hypercomplex system; we shall show elementarily that $\mathfrak{D}$ is representable in a field K if and only if -1 is representable in K as a sum of two squares.

a) Let $\mathfrak{D}^* = T^{-1}\mathfrak{D}T$ be a representation rational over a field K, and put $I^* = T^{-1}IT$, $J^* = T^{-1}JT$. Since J and J^* are two similar matrices rational over K, there is also a transformation R rational over K which carries J^* into J,

$$R^{-1}J^*R = J.$$

If $\mathfrak{D}^*$ is replaced from the start by the likewise K-rational representation $R^{-1}\mathfrak{D}^*R$, one sees that, without restriction, one may assume $J^* = J$. Let

$$I^* = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \qquad (a, b, c, d \text{ numbers from } K).$$

[7] The connection between primitive idempotent elements and irreducible representations is already found in the dissertation of M. Herzberger: Über Systeme hyperkomplexer Größen, Chapter 4, Berlin 1923. That simple ideals generate irreducible representations is shown in the commutative case, for example, by E. Noether, Der Diskriminantensatz für Ordnungen, Crelle 157 (1926); in the non-commutative case by W. Krull, Theorie und Anwendung der verallgemeinerten Abelschen Gruppen, Heidelberger Berichte 1926. Cf. also the closing remarks in Speiser's group theory. If $a_1, \ldots, a_m$ denotes a basis of the ideal with respect to the splitting field, and c an arbitrary ring element, so that $a_i c = \gamma_{i1}a_1 + \cdots + \gamma_{im}a_m$, then (γ_{ik}) is the matrix assigned to c; the fact that all irreducible representations are generated by ideals lies deeper.

[8] The existence of Ω_n can already be read from the parameter representation of cyclic fields of degree 4, as given for example in Weber (Kleines Lehrbuch der Algebra, § 93).

[9] Subsequently R. Brauer was able to show further that there are even minimal splitting fields of the quaternion division ring of every even degree, hence of all degrees possible at all for splitting fields. Such a field $P(\Theta)$ may be defined for every $n \geq 2$ by the equation

$$f(\Theta) = \Theta^{2n} + a^2p^2\Theta^2 + b^2p^2\beta^2 = 0; \quad \text{therefore} \quad -1 = \left(\frac{ap}{\Theta^{n-1}}\right)^2 + \left(\frac{bp\beta}{\Theta^n}\right)^2,$$

where, by suitable choice of a, b, p, β, one can achieve that 1. $f(x)$ is irreducible for every n; 2. $P(\Theta)$ has only the one subfield $P(\Theta^2)$ different from P; 3. $P(\Theta^2)$ also has real fields among its conjugates, and thus cannot be a splitting field. One possible choice of a, b, p, β is, for example,

$$f(x) = x^{2n} + (2^5 \cdot 9)4x^2 + 9 \cdot 64.$$

The proof rests on a modification of a method of Furtwängler for constructing primitive equations (Math. Ann. 85, p. 34, § 3), but is rather laborious in the detailed execution.

[10] Cf. Sylvester, Math. Papers Vol. III p. 647, Vol. IV p. 122.

From $IJ = -JI$ it follows that $I^*J^* = -J^*I^*$, and because $J^* = J$ this gives

$$a = -d, \qquad b = c.$$

From $I^2 = -E$ it follows that $I^{*2} = -E$, hence

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}^2 = \begin{pmatrix} a^2+b^2 & 0 \\ 0 & a^2+b^2 \end{pmatrix},$$

so $a^2 + b^2 = -1$; consequently -1 is indeed representable in K as a sum of two squares.

b) Conversely, suppose that -1 is representable in a field K as a sum of two squares. If, say, $-1 = a^2 + b^2$, set

$$I^* = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}, \qquad J^* = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad K^* = I^*J^*.$$

A straightforward calculation shows that $\pm E$, $\pm I^*$, $\pm J^*$, $\pm K^*$ form a group rational over K and similar to $\mathfrak{D}$.[11]

It remains to show that for infinitely many n there are cyclic fields of degree 2^n which are minimal splitting fields of the quaternion group.

Let p be a rational prime such that, for some suitable $r > 0$,

$$(*) \qquad 2^r + 1 \equiv 0 \pmod p$$

and let 2^n be the highest power of 2 dividing $p - 1$.

First we show that primes p occur for infinitely many values of n. Let k be any positive rational integer, and let p be a prime divisor of $2^{2^k} + 1$. Then the congruence required for p is fulfilled with $r = 2^k$. Further, 2 (mod p) has exponent 2^{k+1}, and therefore, for the highest power 2^n dividing $p - 1$, the number n is greater than k. Thus in any case there are arbitrarily large n to which primes p correspond.

Let ε be a primitive p-th root of unity. We now show that in $P(\varepsilon)$ the number -1 can be represented as a sum of two squares,

$$-1 = a^2 + b^2 = (a + ib)(a - ib) \qquad (a, b \text{ numbers from } P(\varepsilon)).$$

This is equivalent to saying that in the field of $(4p)$-th roots of unity, $P(\varepsilon, i) = P(\varepsilon \cdot i)$, there are numbers whose relative norm with respect to $P(\varepsilon)$ is exactly -1.

The number $1 + i\varepsilon^\nu$ ($\nu = 1, 2, \ldots, p-1$) has relative norm $(1 + i\varepsilon^\nu)(1 - i\varepsilon^\nu) = 1 + \varepsilon^{2\nu}$; consequently all numbers $1 + \varepsilon^\nu$ ($\nu = 1, 2, \ldots, p-1$) are relative norms of numbers from $P(\varepsilon i)$ relative to the ground field $P(\varepsilon)$. Therefore the same holds for

$$\Pi = \prod_{\nu=0}^{r-1} (1 + \varepsilon^{2^\nu}) = (1+\varepsilon)(1+\varepsilon^2)(1+\varepsilon^4)\cdots(1+\varepsilon^{2^{r-1}}) = \sum_{\lambda=0}^{2^r-1} \varepsilon^\lambda.$$

If one adjoins to Π the term $\varepsilon^{2^r} = \varepsilon^{-1} = \varepsilon^{p-1}$ (by $(*)$), then $\Pi + \varepsilon^{p-1}$ becomes a sum of pieces $1 + \varepsilon + \varepsilon^2 + \cdots + \varepsilon^{p-1} = 0$, and consequently

$$\Pi = -\varepsilon^{-1}.$$

Now ε too is the relative norm of a number from $P(\varepsilon i)$, namely of $\varepsilon^{(p+1)/2}$. Hence the analogous statement holds for

$$\Pi\varepsilon = -1.$$

Therefore -1 is representable in $P(\varepsilon)$ as a sum of two squares;[12] $P(\varepsilon)$ is a splitting field for the quaternion group.

Let K be the cyclic field of degree 2^n contained in $P(\varepsilon)$. We assert that K too is a splitting field.[13] Let m be the index of the quaternion division ring with respect to K as ground field. Then $m = 1$ or $m = 2$. But since there is a splitting field $P(\varepsilon)$ of odd relative degree $(p-1)/2^n$ with respect to K, the second case is impossible; hence $m = 1$. This means precisely that K is a splitting field.

Thus there are, for infinitely many n, cyclic fields of degree 2^n which are minimal splitting fields.

[11]The characterization of the splitting fields can also be obtained easily with the help of factor systems.

[12]For primes of the form $8n \pm 3$ this argument is already found in E. Landau, Über die Darstellung definiter Funktionen durch Quadrate, Math. Ann. vol. 62, pp. 272–285.

[13]The choice of the cyclic splitting field of degree 2^n as a subfield of a cyclotomic field is modeled on Hasse's example.

33. Hypercomplex quantities and representation theory in arithmetic conception

Atti Congresso Bologna 2 (1928), pp. 71–73.

Introduction

The most important general theorems on hypercomplex systems go back to Molien (Math. Annalen vol. 41 and Dorpat Reports 1897). Shortly afterward, essentially independently of this, Frobenius developed the theory of hypercomplex systems and of their representations – in particular the representation theory of finite groups – in a unified way. The foundation is the concept, going back to Dedekind, of the group determinant, and more generally of the determinant of an arbitrary hypercomplex system. Frobenius shows that the different irreducible factors of this determinant (the coefficient domain is always the field of all complex numbers) correspond to the different irreducible representation classes, and that in this way all irreducible representation classes are exhausted. Such a representation class is completely characterized by its "character system"; in the case of finite groups this character system arises by decomposition of the determinant of the commutative hypercomplex system derived from the classes of conjugate elements of the group. This determinant decomposes into linear factors with the characters as coefficients – a direct generalization of the result first obtained by Dedekind, that the group determinant of a finite abelian group decomposes into linear factors whose coefficients are the various characters of the abelian group (correspondence with Frobenius).[14]

The totality of possible representations – perhaps only of the irreducible ones – of a given ring in a given, not necessarily commutative, division ring is under consideration.

The first problem is by far the simpler one; this is already shown by the fact that no special hypotheses of any kind are needed either on the ring to be represented or on the – in general non-commutative – representation division ring. The fact that the representation is by matrices of fixed degree supplies all the necessary finiteness hypotheses. The problem can be treated completely by means of the theory of groups with operators, and I shall therefore briefly discuss it.

A group $\mathfrak{M}$ is called a group with a domain of operators if, at the same time, there is given a system of symbols $\Theta, H, \ldots$ – the operators – such that the combination $\Theta(a), H(a), \ldots$ is uniquely defined for every a in $\mathfrak{M}$ and yields an element of $\mathfrak{M}$, and such that the distributive law is satisfied: $\Theta(ab) = \Theta(a)\Theta(b)$. Two groups with the same domain of operators are called operator-isomorphic if they are isomorphic in the usual sense and if, moreover, from the correspondence of a and $\bar{a}$ there follows also that of $\Theta(a)$ and $\Theta(\bar{a})$, of $H(a)$ and $H(\bar{a})$, and so on. If as subgroups one then admits only those which themselves admit the domain of operators (so that a group is called simple if it has no admissible proper normal divisor except the identity), then the Jordan-Hölder theorem on composition series remains valid in the form: if a group with operator domain has a composition series at all, then the system of factor groups is uniquely determined up to ordering in the sense of operator-isomorphism. Under the same hypothesis, there also holds the theorem on the decomposition, unique in the sense of operator-isomorphism, into directly indecomposable factors.

The connection with the representation problem is given by the fact that, among the groups with operators, one finds in particular ideals and modules, these being regarded as abelian groups with respect to addition, while the operators are given by multiplication with the ring elements. Every representation, however, is generated by a representation module, i.e. by a module with respect to the ring and the representation division ring. More precisely, every module class, i.e. every class of representation modules operator-isomorphic to one another, generates the same representation. Thus the question of uniqueness of the reduction of a representation is answered by the composition-series theorem and by the theorem on the direct product. Applied to the representation module, the composition series gives, for each matrix, a reduction of the form

$$\begin{pmatrix} B_1 & 0 & 0 \\ 0 & B_2 & 0 \\ A_{ik} & \cdots & B_r \end{pmatrix},$$

where the diagonal matrices corresponding to the factor groups generate an "irreducible" representation, i.e. one that itself no longer admits such a reduction.

Since the classes of these factor groups are uniquely determined, the same holds for the diagonal representations (uniquely in the sense of equivalence of representations). Analogously, the theorem on the direct

[14] What follows is a free elaboration, prepared by B. L. van der Waerden, of my lecture from the winter semester 1927/28. We undertook the preparation for print jointly. I am also indebted to B. L. van der Waerden for a number of critical remarks.

product gives the uniqueness of the "decomposition" into "indecomposable" constituents:

$$\begin{pmatrix} C_1 & & & \\ & C_2 & & \\ & & \ddots & \\ & & & C_s \end{pmatrix},$$

where each C_i can then still be reduced uniquely according to the composition-series theorem.

I would like to sketch the second problem for the case of representations of hypercomplex systems, more generally of rings, which satisfy the "double-chain theorem". The result is that every irreducible (simple) module class is an ideal class; hence the composition series from one-sided ideals already supply all irreducible representations through their factor groups. More precisely, the composition series in the residue-class ring modulo the radical suffice, the latter becoming a "completely reducible" ring. The problem is thereby reduced to the structural investigation of completely reducible rings; and thus it proves appropriate to take as starting point the representations in the automorphism division rings. This is to be understood as follows. All simple (right) ideals of a two-sided simple component of such a ring belong to the same class, and therefore have the same automorphism ring, which, because of the simplicity of the ideals, becomes a division ring; at the same time it is the automorphism ring of the simple left ideals. The simple ring itself becomes isomorphic to the system of all matrices over the automorphism division ring; and from this representation spring all representations with respect to a subfield, in particular with respect to the coefficient domain of the hypercomplex system, by replacing the elements of the automorphism division ring themselves by assigned matrices. Thus the Wedderburn structure theorem known in the case of hypercomplex systems is reinterpreted here through the concept of the automorphism division ring, a concept originating in general group theory, and is at the same time recognized as the source of the representation theory of hypercomplex systems. The new questions of the Galois theory of non-commutative division rings arise; they are closely connected with the question of "splitting fields". At the same time, an outlook is obtained toward structure theorems for general rings that are not completely reducible, by means of the automorphism rings of the indecomposable ideals. Group theory, in its extension to the most general groups with operators, here again proves itself to be the ordering principle.

34. Hypercomplex Quantities and Representation Theory

Math. Zs. 30 (1929), pp. 641–692.

Introduction

The most important general theorems on hypercomplex systems go back to Molien (Math. Annalen vol. 41 and the Dorpat reports of 1897). Essentially independently of this, Frobenius soon afterward developed the theory of hypercomplex systems and of their representations – in particular the representation theory of finite groups – as a unified theory. The basis is the concept, due to Dedekind, of the group determinant, more generally the determinant of an arbitrary hypercomplex system. Frobenius shows that to the various irreducible factors of this determinant (where the coefficient domain is always the field of all complex numbers) there correspond the various irreducible representation classes, and that in this way all irreducible representation classes are exhausted. Such a representation class is completely characterized by its "character system"; in the case of finite groups this character system arises by decomposing the determinant of the commutative hypercomplex system derived from the classes of conjugate elements of the group. This determinant decomposes into linear factors, with the characters as coefficients – a direct generalization of the result first obtained by Dedekind, namely that the group determinant of a finite Abelian group decomposes into linear factors whose coefficients are the different characters of the Abelian group (correspondence with Frobenius).[1]

These conceptually simple and transparent results, however, are obtained by Frobenius through laborious calculation. The later development aimed at a simplified derivation of these results, and at the same time at their extension when an arbitrary field is taken as coefficient domain. This later development proceeded in completely separate ways for hypercomplex systems and for representation theory. Hypercomplex systems received an arithmetical treatment through Wedderburn: every hypercomplex system is uniquely a direct sum of two-sided directly indecomposable rings; for systems without radical these indecomposable rings become isomorphic to the system of all n-rowed matrices with elements from an assigned, not necessarily commutative, division ring – determined essentially uniquely.[2] Representation theory received an elementary foundation through Burnside and I. Schur, who proceeded directly from a given representation, independently of the hypercomplex system, and operated with matrix theorems. In particular Schur treated the question of the number fields of smallest degree in which a representation irreducible over a given field decomposes absolutely. These absolutely irreducible constituents belong to finitely many conjugate representation classes. The degree of the number fields sought is equal to the product of the number of these classes by the index (the number of constituents in each of the conjugate classes). The chief auxiliary tool of Schur's theory is the system of all matrices commuting with an irreducible representation.[3]

In the following purely arithmetical foundation, hypercomplex quantities and representation theory again appear as one unified whole – as a special case of a general theory of non-commutative rings satisfying only certain finiteness conditions. More precisely, this is a theory of module and ideal classes with respect to these rings, with the main result that the irreducible module classes are already exhausted by the corresponding ideal classes; in particular, for rings without radical all module classes decompose into irreducible ones (become completely reducible). This is the arithmetical equivalent of Frobenius's result that the irreducible factors of the regular group (respectively system) determinant already exhaust the totality of irreducible representation classes, and that complete reducibility of representations holds for systems without radical. Namely, the consideration of the system determinant and its decomposition can be interpreted as a transition to the norm, whereas here the direct-sum decomposition of the ideals and their composition series are treated directly. Representation theory, by means of the concept of the representation module, becomes a theory of module classes.

The introduction of module and ideal classes has a purely group-theoretic basis. Modules and ideals are regarded as Abelian groups with respect to addition, restricted by the condition that they admit certain multiplications by ring elements: they form "groups with operators" (§1). For such groups, in place of the ordinary isomorphism there appears the "operator isomorphism" (§2); groups that are operator-isomorphic to one another (within a fixed domain) are collected into a class – a class concept that, for example in the case of the ideals of a number field, coincides with the usual one. The theory of groups with operators goes back to W. Krull and O. Schmidt (cf. note 6); it is developed systematically in Chapter I. The theorems that

[1]What follows is a free elaboration, prepared by B. L. van der Waerden, of my lectures of the winter semester 1927/28. We prepared the text for print jointly. I am also indebted to B. L. van der Waerden for a number of critical remarks.

[2]Cf. the references in Dickson, *Algebras and their Arithmetics* (German edition: *Algebren und ihre Zahlentheorie*, Zürich 1927).

[3]Cf. the references in Chapter 10 of Speiser's group theory.

remain valid here as well – on composition series, direct product (respectively sum), and completely reducible groups; uniqueness theorems in the sense of the class division just mentioned – form the basis of everything that follows.

They first yield the general uniqueness theorems for the irreducible diagonal constituents of arbitrary representations (Chapter III, §16), on the basis of the one-to-one correspondence of representation classes with the classes of representation modules, hence with classes of groups with operators. The further question of the totality of representations requires a more precise study of the structure of the rings to be represented, thus, in the special case, of hypercomplex systems. In Chapter II the results of Wedderburn are obtained anew and carried further, on the basis of the group-theoretic conception according to which one-sided ideals stand in the foreground. It turns out that the "multiple-chain theorem" for right ideals, or the identical "minimal condition" (in every set of right ideals there is at least one minimal member within the set), suffices as a finiteness condition.[4] One obtains the identity of the rings without radical satisfying the minimal condition with the (right) completely reducible rings with identity element. In such rings all simple right ideals of a two-sided indecomposable ring belong to the same class and therefore possess – as groups with operators – the same automorphism ring, which, because of the simplicity of the ideals, becomes a division ring. This automorphism division ring of the class is what mediates the matrix representation found by Wedderburn.

This matrix representation in the automorphism division ring also solves the representation problem in the completely reducible case, as soon as representation with respect to the automorphism division ring or to its subfields is required – in particular with respect to the coefficient domain of a hypercomplex system. There are – this is a direct consequence of the theorem reducing module classes to ideal classes – no further irreducible representations beyond the Wedderburn representation and those that arise from it when the elements of the automorphism division ring are replaced in the natural way by matrices over a subfield.

Thus in the completely reducible case there are as many different irreducible representation classes as ideal classes; their number agrees with the number of different two-sided indecomposable, hence simple, rings. This number is finally again identical with the number of indecomposable components of the center, which become commutative fields. In the case of hypercomplex systems with algebraically closed coefficient domain, these latter fields are completely determined by the different ring homomorphisms (irreducible representations) of the center; up to a numerical factor these are precisely the characters.

The results obtained also give, at the same time, a survey of the irreducible representations of systems with radical, insofar as these can be reduced to the representations of the residue class ring modulo the radical, which becomes a system without radical.

As will be shown later, for hypercomplex systems without radical the theory of "splitting fields" also follows from the automorphism division ring: that is, the theory of commutative extensions of the coefficient domain in which the representations decompose into absolutely irreducible representations – equivalently, in which a direct-sum decomposition into absolutely simple right ideals occurs. The splitting fields are identical with those of the automorphism division ring, and all splitting fields of smallest degree are isomorphic to the maximal commutative subfields of the automorphism division ring. The connection with Schur's investigations mentioned above is established by the fact that the transposes of the matrices commuting with the representation supply precisely a representation of the automorphism division ring.[5] These theorems are to be developed systematically in the framework of a Galois theory of non-commutative division rings.

The underlying concepts – operator homomorphism and automorphism ring – also yield the structure of general rings with radical that satisfy only the minimal condition; this will be carried out from another side.

Chapter I. Group-Theoretic Foundations

§ 1. Groups with operators

Let $\mathfrak{G}$ be a group (finite or infinite), with elements $a, b, \ldots$. By an operator domain Ω for the group $\mathfrak{G}$ is meant a set of new symbols $H, \Theta, \ldots$, such that to every a in $\mathfrak{G}$ and every Θ in Ω there is assigned a uniquely defined Θa in $\mathfrak{G}$, and such that the distributive law holds:

$$\Theta(ab) = \Theta a \cdot \Theta b.$$

[4]Wedderburn's methods of proof can be transferred if the "double chain theorem" is assumed. Cf. E. Artin, Zur Theorie der hyperkomplexen Zahlen, Hamb. Abh. 1927. Cf. also A. Suschkewitsch, Über die endlichen Gruppen ohne das Gesetz der eindeutigen Umkehrbarkeit, Math. Annalen 99 (1928), pp. 30–50, where parallel structure theorems are developed for (finite) domains with only one associative, non-invertible operation. The splitting of the "kernel" in Suschkewitsch into right and left groups corresponds to the representation of a two-sided simple ring with identity element as a sum of right and left ideals.

[5]Cf. R. Brauer–E. Noether, Über minimale Zerfällungskörper irreduzibler Darstellungen, Sitz.-Ber. Preuß. Akad. Wiss. 1927, p. 221.

Accordingly, every operator defines a homomorphism of the group into itself, where the group is mapped either onto itself or onto a subgroup. The identity element goes into the identity element, and inverses go into inverses.

An operator domain is called *absolute* if different operators also define different homomorphisms. From an arbitrary operator domain one obtains an absolute one by equating all those elements which generate the same homomorphism. An absolute operator domain is a one-to-one image of a subset of the set of all homomorphisms of the group into itself.

An *admissible subgroup* $\mathfrak{H}$ of $\mathfrak{G}$ – with respect to a fixed operator domain Ω – is a subgroup which admits the operator domain, that is, for which Θa lies in $\mathfrak{H}$ for every a in $\mathfrak{H}$ and Θ in Ω.

Examples. 1. Let the operators be the inner automorphisms: $\Theta a = c^{-1}ac$. The admissible subgroups are the normal divisors.

2. Let the operators be all automorphisms. The admissible subgroups are the "characteristic subgroups", which go into themselves under all automorphisms.[6]

3. Let $\mathfrak{G}$ be a ring, i.e. an Abelian group with respect to addition, in which a multiplication is also defined, with the properties

$$r(a+b) = ra + rb, \qquad (a+b)r = ar + br, \qquad ab \cdot c = a \cdot bc.$$

Every element r simultaneously defines two operators: the operators rx and xr. The admitted subgroups are the "ideals" $\mathfrak{a}$, namely: left-sided ideals, which admit the operations rx: $r\mathfrak{a} \subseteq \mathfrak{a}$; right-sided ideals, which admit the operations xr: $\mathfrak{a}r \subseteq \mathfrak{a}$; and two-sided ideals, which admit both operations. All ideals become trivial ($= \{0\}$ or $= \mathfrak{G}$) when the ring $\mathfrak{G}$ is a division ring, i.e. when $\mathfrak{G} - \{0\}$ is a group with respect to multiplication.[7]

4. *Modules with respect to a ring* $\mathfrak{o}$. Let $\mathfrak{o}$ be a ring, and $\mathfrak{M}$ an Abelian group written additively. Suppose a multiplication $r \cdot a$ of elements of $\mathfrak{o}$ with elements of $\mathfrak{M}$ is given; the product is to lie again in $\mathfrak{M}$. One requires:

$$\left.\begin{aligned} r(a+b) &= ra + rb,\\ rs \cdot a &= r \cdot sa,\\ (r+s)a &= ra + sa, \end{aligned}\right\} \qquad r, s \in \mathfrak{o}, \quad a, b \in \mathfrak{M}.$$

Then $\mathfrak{M}$ is called an $\mathfrak{o}$-module; more precisely, since the multipliers from $\mathfrak{o}$ are written on the left, a left $\mathfrak{o}$-module. The ring $\mathfrak{o}$ is at the same time an operator domain. If it is absolute, i.e. if different ring elements also yield different operations, one speaks of an *absolute multiplier domain* for the module $\mathfrak{M}$.

As above, one can pass from an arbitrary multiplier domain to the absolute one by equating the elements which generate the same operation. The absolute multiplier domain is again a ring. The admissible subgroups of a module $\mathfrak{M}$ are called *submodules*. If, in particular, $\mathfrak{o} = \mathfrak{M}$, then one obtains the left ideals again.[8]

5. *Bimodules.* If $\mathfrak{M}$ is simultaneously a left $\mathfrak{o}$-module and a right $\mathfrak{o}'$-module, and if moreover for a in $\mathfrak{o}$, α in $\mathfrak{M}$, and a' in $\mathfrak{o}'$ one always has

$$a \cdot \alpha a' = a\alpha \cdot a',$$

then $\mathfrak{M}$ is called a bimodule.

6. *The automorphism ring of an Abelian group.*[9] For the endomorphisms of an Abelian group (written additively) one can define an addition and a multiplication by the formulas

$$(H + \Theta)a = Ha + \Theta a, \qquad (H\Theta)a = H(\Theta a).$$

The operations so defined plainly again represent homomorphisms, and therefore again belong to the system. The system forms a ring, and the Abelian group can, according to 4, be regarded as a module with respect to this ring.

In what follows, by "subgroups" we shall always mean admissible subgroups. The intersection $\mathfrak{A} \cap \mathfrak{B}$ of two admissible subgroups is again an admissible subgroup. So is the product $\mathfrak{A}\mathfrak{B}$, provided at least one of the two subgroups $\mathfrak{A}, \mathfrak{B}$ is a normal divisor.

[6]The concepts explained here come from W. Krull, Über verallgemeinerte endliche Abelsche Gruppen, Math. Zeitschr. 23 (1925), pp. 161–196, and O. Schmidt, Über unendliche Gruppen mit endlicher Kette, Math. Zeitschr. 29 (1928), pp. 34–41.

[7]Thus neither for rings nor for division rings is the commutative law of multiplication assumed.

[8]In the same way one requires for right modules:

$$(a+b)r = ar + br, \qquad a \cdot rs = ar \cdot s, \qquad a(r+s) = ar + as.$$

[9]Cf. A. Châtelet, *Les Groupes Abéliens finis*, Paris, Gauthier-Villars, 1925, p. 99.

§ 2. The isomorphism theorems

A mapping of a group $\mathfrak{G}$ onto a group $\overline{\mathfrak{G}}$ – where $\mathfrak{G}$ and $\overline{\mathfrak{G}}$ are to possess the same operator domain – is called an operator homomorphism if, first, it is a homomorphism in the ordinary sense ($ab \mapsto \bar{a}\bar{b}$), and if, second, whenever a maps to $\bar{a}$, the element Θa maps to $\Theta\bar{a}$.[10] Notation: $\mathfrak{G} \sim \overline{\mathfrak{G}}$. If the correspondence is one-to-one, it is called an operator isomorphism. Notation: $\mathfrak{G} \simeq \overline{\mathfrak{G}}$. As for ordinary groups, one proves the homomorphism theorem:

If $\overline{\mathfrak{G}}$ is an operator-homomorphic image of $\mathfrak{G}$, then $\overline{\mathfrak{G}}$ is operator-isomorphic to a factor group $\mathfrak{G}/\mathfrak{N}$, where $\mathfrak{N}$ is an admissible normal divisor, consisting of all elements of $\mathfrak{G}$ that correspond to the identity in $\overline{\mathfrak{G}}$. Conversely, every admissible normal divisor $\mathfrak{N}$ defines a factor group $\mathfrak{G}/\mathfrak{N}$ which admits the operator domain of $\mathfrak{G}$ and is an operator-homomorphic image of $\mathfrak{G}$.

A group is called *simple* if it has no normal divisors other than itself and the identity group. Every homomorphism of a simple group is either an isomorphism, or else assigns the identity element to every element.

First isomorphism theorem. Let $\overline{\mathfrak{G}}$ be a homomorphic image of $\mathfrak{G}$, let $\overline{\mathfrak{A}}$ be a normal divisor of $\overline{\mathfrak{G}}$, and let $\mathfrak{A}$ be the totality of those elements of $\mathfrak{G}$ to which elements of $\overline{\mathfrak{A}}$ are assigned. Then $\mathfrak{A}$ is again a normal divisor, and

$$\overline{\mathfrak{G}}/\overline{\mathfrak{A}} \simeq \mathfrak{G}/\mathfrak{A}. \tag{1}$$

Proof. $\mathfrak{G} \sim \overline{\mathfrak{G}}$, $\overline{\mathfrak{G}} \sim \overline{\mathfrak{G}}/\overline{\mathfrak{A}}$, hence $\mathfrak{G} \sim \overline{\mathfrak{G}}/\overline{\mathfrak{A}}$. Therefore $\overline{\mathfrak{G}}/\overline{\mathfrak{A}}$ is isomorphic to a factor group in $\mathfrak{G}$; the corresponding normal divisor is the totality of elements that correspond to the identity in $\overline{\mathfrak{G}}/\overline{\mathfrak{A}}$; this is precisely $\mathfrak{A}$.

Addendum. If, according to the homomorphism theorem, one sets $\overline{\mathfrak{G}} = \mathfrak{G}/\mathfrak{N}$, then $\mathfrak{A}$ certainly contains $\mathfrak{N}$. From $\mathfrak{A}$ one can recover $\overline{\mathfrak{A}}$: $\overline{\mathfrak{A}} = \mathfrak{A}/\mathfrak{N}$. The assignment $\mathfrak{A} \leftrightarrow \overline{\mathfrak{A}}$ is one-to-one. Formula (1) can also be written

$$(\mathfrak{G}/\mathfrak{N})/(\mathfrak{A}/\mathfrak{N}) \simeq \mathfrak{G}/\mathfrak{A}.$$

Second isomorphism theorem. Let $\mathfrak{A}$ be a subgroup and $\mathfrak{B}$ a normal divisor of $\mathfrak{G}$. Then $\mathfrak{A} \cap \mathfrak{B}$ is a normal divisor in $\mathfrak{A}$, and

$$\mathfrak{A}\mathfrak{B}/\mathfrak{B} \simeq \mathfrak{A}/(\mathfrak{A} \cap \mathfrak{B}).$$

Proof. Under the homomorphism $\mathfrak{G} \sim \mathfrak{G}/\mathfrak{B}$, in particular the elements a of $\mathfrak{A}$ are assigned certain residue classes $a\mathfrak{B}$, which together make up the group $\mathfrak{A}\mathfrak{B}/\mathfrak{B}$. Thus $\mathfrak{A} \sim \mathfrak{A}\mathfrak{B}/\mathfrak{B}$, from which the assertion follows by the homomorphism theorem.

An operator-homomorphic mapping of a group $\mathfrak{G}$ onto itself or onto a subgroup is called an endomorphism of $\mathfrak{G}$. If $\mathfrak{G}$ is in particular an Abelian group (with operators), written additively, and if sum and product of homomorphisms are defined as above (§1, Example 6), then the operator homomorphisms of the group into itself form a ring, the automorphism ring.

The automorphism ring of a simple Abelian group (with operators) is a division ring.

Proof. Every homomorphism maps the group either onto itself or onto the zero group (since there are no other admissible subgroups). The homomorphisms that do not map everything to zero are, by what was remarked above about simple groups, isomorphisms, and the isomorphic mappings of a group onto itself form a group. Thus after omitting the zero operator the ring becomes a group; hence the ring itself is a division ring.

If, in particular, $\mathfrak{G}$ is a left $\mathfrak{o}$-module, and if the operator homomorphisms are written as right operators, then $\mathfrak{G}$ becomes a *bimodule with respect to $\mathfrak{o}$ on the left and the automorphism ring on the right.* For the fact that the new operators Γ were chosen as homomorphisms is expressed by the formulas

$$(a+b)\Gamma = a\Gamma + b\Gamma, \qquad ra \cdot \Gamma = r \cdot a\Gamma,$$

which (together with the remaining formulas already interpreted earlier) characterize the bimodule.

Conversely, if a bimodule is given, then these same formulas express that every element of the right multiplier domain induces an operator homomorphism with respect to the left operators, and conversely.

[10] One may extend the concept of operator homomorphism by allowing the groups $\mathfrak{G}, \overline{\mathfrak{G}}$ to have separate operator domains, the homomorphism then mapping not only $\mathfrak{G}$ onto $\overline{\mathfrak{G}}$ but also the operator domains onto one another, in such a way that, if a maps to $\bar{a}$ and Θ maps to $\overline{\Theta}$, then $\overline{\Theta}\bar{a}$ is the image of Θa. In this general form the concept of operator homomorphism also includes that of ring homomorphism; cf. §8.

§ 3. Composition series

If in $\mathfrak{G}$ there is a finite sequence of admissible subgroups

$$\mathfrak{G} > \mathfrak{A}_1 > \mathfrak{A}_2 > \cdots > \mathfrak{A}_r = \mathfrak{E} \tag{2}$$

($\mathfrak{E}$ is the group consisting of the identity element e alone), such that every $\mathfrak{A}_i$ is a normal divisor in the preceding member, and it is impossible to insert between two successive members of the sequence another subgroup with the same property, then the sequence is called a composition series.

The factor groups $\mathfrak{G}/\mathfrak{A}_1,\ \mathfrak{A}_1/\mathfrak{A}_2, \ldots,\ \mathfrak{A}_{r-1}/\mathfrak{E}$ are called composition factors. They are simple groups, i.e. they possess no admissible normal divisors other than themselves and the identity group.

If among the operators one includes, in particular, all inner automorphisms, then the sequence (2) consists entirely of normal divisors, and one calls it a principal series. If one includes all automorphisms, it is called a characteristic series. In the later examples of §1 this would mean composition series of ideals in $\mathfrak{o}$, respectively composition series of modules or bimodules.

Jordan–Hölder theorem. If for a group $\mathfrak{G}$ two different composition series exist,

$$\mathfrak{G} > \mathfrak{A}_1 > \mathfrak{A}_2 > \cdots > \mathfrak{A}_r = \mathfrak{E}, \qquad \mathfrak{G} > \mathfrak{B}_1 > \mathfrak{B}_2 > \cdots > \mathfrak{B}_s = \mathfrak{E},$$

then they have the same length, $r = s$, and the composition factors

$$\mathfrak{G}/\mathfrak{A}_1,\ \mathfrak{A}_1/\mathfrak{A}_2, \ldots, \mathfrak{A}_{r-1}/\mathfrak{E}$$

are, in some order, isomorphic to

$$\mathfrak{G}/\mathfrak{B}_1,\ \mathfrak{B}_1/\mathfrak{B}_2, \ldots, \mathfrak{B}_{s-1}/\mathfrak{E}.$$

For the proof, see for example E. Noether, Abstrakter Aufbau usw., Math. Annalen 96 (1926), §10, p. 57. At the same time it is proved there that:

If in $\mathfrak{G}$ there is a composition series, then a composition series can be drawn through every normal divisor $\mathfrak{H}$ of $\mathfrak{G}$.

The existence of a composition series is clear for finite groups, more generally for those groups in which a maximal condition and a minimal condition hold:

The *maximal condition* says that in every set of subgroups there is a maximal subgroup, i.e. one that is no longer contained in another subgroup of the set. Equivalently, every ascending chain of subgroups $\mathfrak{A}_1 \subset \mathfrak{A}_2 \subset \mathfrak{A}_3 \cdots$ breaks off after finitely many members.

The *minimal condition* says that in every set of subgroups there is a minimal subgroup, one that contains no other subgroup of the set; equivalently, every descending chain $\mathfrak{A}_1 \supset \mathfrak{A}_2 \supset \mathfrak{A}_3 \cdots$ breaks off after finitely many members.

If only normal divisors are admitted as subgroups (for example for Abelian groups), *then conversely the maximal and minimal conditions follow from the existence of the composition series.*

Proof. Every normal divisor has a composition series, whose length is called the length of the normal divisor. If $\mathfrak{A} \subset \mathfrak{B}$, then the length of $\mathfrak{A}$ is smaller than that of $\mathfrak{B}$, since a composition series for $\mathfrak{B}$ can be drawn through $\mathfrak{A}$. Thus every subgroup of shortest length is at the same time minimal, and every subgroup of greatest length is at the same time maximal.

§ 4. Direct products and intersections

A group $\mathfrak{G}$ is called the direct product of two factors, $\mathfrak{G} = \mathfrak{A} \times \mathfrak{B}$, if

1. $\mathfrak{A}, \mathfrak{B}$ are normal divisors in $\mathfrak{G}$,
2. $\mathfrak{A}\mathfrak{B} = \mathfrak{G}$,
3. $\mathfrak{A} \cap \mathfrak{B} = \mathfrak{E}$.

The definition is plainly equivalent to the following: Every element g of $\mathfrak{G}$ has a unique representation as $g = ab$, with a in $\mathfrak{A}$ and b in $\mathfrak{B}$, and the elements of $\mathfrak{A}$ commute with those of $\mathfrak{B}$: $ab = ba$.

A group $\mathfrak{G}$ is called the direct product of n factors, $\mathfrak{G} = \mathfrak{A}_1 \times \cdots \times \mathfrak{A}_n$, if, putting $\mathfrak{B}_i = \mathfrak{A}_1 \cdots \mathfrak{A}_{i-1}\mathfrak{A}_{i+1} \cdots \mathfrak{A}_n$, one has $\mathfrak{G} = \mathfrak{A}_i \times \mathfrak{B}_i$ directly for every i.

The definition is again equivalent to the following: Every g of $\mathfrak{G}$ is uniquely representable as

$$g = a_1 a_2 \cdots a_n, \qquad a_i \in \mathfrak{A}_i,$$

and the elements of $\mathfrak{A}_i$ commute with those of $\mathfrak{A}_k$.

The following facts are often used:

1. If $\mathfrak{A}_1 \times \cdots \times \mathfrak{A}_n = \mathfrak{R}$ and $\mathfrak{R} \times \mathfrak{H} = \mathfrak{G}$, then $\mathfrak{A}_1 \times \cdots \times \mathfrak{A}_n \times \mathfrak{H} = \mathfrak{G}$.

2. If $\mathfrak{G} = \mathfrak{A}_1 \times \cdots \times \mathfrak{A}_n = \mathfrak{C}_1 \times \cdots \times \mathfrak{C}_n$ and $\mathfrak{C}_i \subseteq \mathfrak{A}_i$, then $\mathfrak{C}_i = \mathfrak{A}_i$. (This follows from the representation of the elements of $\mathfrak{A}_i$ by means of the $\mathfrak{C}_j$.)

3. If $\mathfrak{G} = \mathfrak{A} \times \mathfrak{B}$ and $\mathfrak{R} \supseteq \mathfrak{A}$, then $\mathfrak{R} = \mathfrak{A} \times (\mathfrak{R} \cap \mathfrak{B})$. (This follows from the representation of the elements of $\mathfrak{R}$ in the form ab, where the second factor belongs both to $\mathfrak{R}$ and to $\mathfrak{B}$.)

4. If $\mathfrak{G} = \mathfrak{A} \times \mathfrak{B}$, then $\mathfrak{A} \sim \mathfrak{G}/\mathfrak{B}$ and $\mathfrak{B} \sim \mathfrak{G}/\mathfrak{A}$ (second isomorphism theorem). It follows further that:

5. If $\mathfrak{G} = \mathfrak{A} \times \mathfrak{B}$, and if $\mathfrak{A}$ and $\mathfrak{B}$ possess composition series of lengths m and n, then $\mathfrak{G}$ possesses a composition series of length $m + n$.

6. If $\mathfrak{A} \sim \overline{\mathfrak{A}}$ and $\mathfrak{B} \sim \overline{\mathfrak{B}}$, then $\mathfrak{A} \times \mathfrak{B} \sim \overline{\mathfrak{A}} \times \overline{\mathfrak{B}}$.

A group is called directly indecomposable if it cannot be represented as a direct product of factors $\neq \mathfrak{E}$.

It is clear that *every group satisfying the minimal condition is the direct product of finitely many directly indecomposable groups.*[11]

If the groups in question are written additively, then the notions product and direct product pass into sum and direct sum. Notation: for a sum of groups, $(\mathfrak{A}_1, \mathfrak{A}_2, \ldots)$; for a direct sum, $\mathfrak{A}_1 + \mathfrak{A}_2 + \cdots$.

The concept of "direct intersection" will not be used later, but the following theorems concerning the connection between direct intersection and direct product show how the ideal theory of the next chapters can also be formulated with intersections instead of sums, thereby restoring the connection with the familiar commutative theories.

A group $\mathfrak{D}$ is called the direct intersection of two groups $\mathfrak{R}$ and $\mathfrak{S}$ with respect to $\mathfrak{G}$ if

1. $\mathfrak{R}, \mathfrak{S}$ are normal divisors in $\mathfrak{G}$,
2. $\mathfrak{R} \cap \mathfrak{S} = \mathfrak{D}$,
3. $\mathfrak{R}\mathfrak{S} = \mathfrak{G}$.

Likewise $\mathfrak{D}$ is called the direct intersection of n groups $\mathfrak{S}_1, \ldots, \mathfrak{S}_n$ if, putting $\mathfrak{R}_i = \mathfrak{S}_1 \cap \cdots \cap \mathfrak{S}_{i-1} \cap \mathfrak{S}_{i+1} \cap \cdots \cap \mathfrak{S}_n$, the intersection $\mathfrak{D} = \mathfrak{R}_i \cap \mathfrak{S}_i$ is direct for every i.

From the definition it follows that $\mathfrak{D}$ must be a normal divisor in $\mathfrak{G}$. If $\mathfrak{D} = \mathfrak{S}_1 \cap \cdots \cap \mathfrak{S}_n$ is direct and, under the homomorphism $\mathfrak{G} \sim \mathfrak{G}/\mathfrak{D}$, the groups $\mathfrak{G}, \mathfrak{S}, \mathfrak{D}$ pass to $\overline{\mathfrak{G}}, \overline{\mathfrak{S}}, \mathfrak{E}$, then $\mathfrak{E} = \overline{\mathfrak{S}}_1 \cap \cdots \cap \overline{\mathfrak{S}}_n$ becomes direct. Conversely, from every such representation $\mathfrak{E} = \overline{\mathfrak{S}}_1 \cap \cdots \cap \overline{\mathfrak{S}}_n$ one returns to a direct representation $\mathfrak{D} = \mathfrak{S}_1 \cap \cdots \cap \mathfrak{S}_n$. Through this assignment theorems on direct intersections for arbitrary $\mathfrak{D}$ are reduced to the special case $\mathfrak{D} = \mathfrak{E}$.

In the case $\mathfrak{D} = \mathfrak{E}$ and two factors, the conditions for direct product and direct intersection coincide. For n factors there is the following one-to-one relation between direct product and intersection:

I. If $\mathfrak{G} = \mathfrak{A}_1 \times \cdots \times \mathfrak{A}_n = \mathfrak{A}_i \times \mathfrak{B}_i$, then $\mathfrak{E} = \mathfrak{B}_1 \cap \cdots \cap \mathfrak{B}_n = \mathfrak{B}_i \cap \mathfrak{C}_i$ directly, and $\mathfrak{C}_i = \mathfrak{A}_i$.

II. If $\mathfrak{E} = \mathfrak{S}_1 \cap \cdots \cap \mathfrak{S}_n = \mathfrak{R}_i \cap \mathfrak{S}_i$ directly, then $\mathfrak{G} = \mathfrak{R}_1 \times \cdots \times \mathfrak{R}_n = \mathfrak{R}_i \times \mathfrak{T}_i$, and $\mathfrak{T}_i = \mathfrak{S}_i$.

Proof of I. Let $\mathfrak{B} = \mathfrak{B}_1 \cap \cdots \cap \mathfrak{B}_n = \mathfrak{B}_i \cap \mathfrak{C}_i$; then $\mathfrak{C}_i \supseteq \mathfrak{A}_i$. We show $\mathfrak{C}_i \subseteq \mathfrak{A}_i$. If, for instance, c lies in $\mathfrak{C}_1$, then c lies in $\mathfrak{B}_2, \ldots, \mathfrak{B}_n$, so that the component representation with respect to the $\mathfrak{A}$ gives

$$c = a_1 a_2 \cdots a_n = a_1 e a_3 \cdots a_n = a_1 a_2 e \cdots a_n = \cdots = a_1 \cdots a_{n-1} e.$$

Since the product of all $\mathfrak{A}_i$ is direct, the representations agree; in every position except the first there is once an e, hence $c = a_1$, or $\mathfrak{C}_1 \subseteq \mathfrak{A}_1$, i.e. $\mathfrak{C}_1 = \mathfrak{A}_1$, and correspondingly $\mathfrak{C}_i = \mathfrak{A}_i$. It follows that $\mathfrak{B} = \mathfrak{B}_i \cap \mathfrak{C}_i = \mathfrak{B}_i \cap \mathfrak{A}_i = \mathfrak{E}$; moreover $\mathfrak{B}_i\mathfrak{C}_i = \mathfrak{B}_i\mathfrak{A}_i = \mathfrak{G}$, hence the intersection is direct.

Proof of II. Let $\mathfrak{H} = \mathfrak{R}_1 \cdots \mathfrak{R}_n = \mathfrak{R}_i\mathfrak{T}_i$; then $\mathfrak{T}_i \subseteq \mathfrak{S}_i$. We show $\mathfrak{S}_i \subseteq \mathfrak{T}_i$. Let, for instance, s lie in $\mathfrak{S}_1$. By means of $\mathfrak{G} = \mathfrak{S}_i \times \mathfrak{R}_i$ one obtains

$$s = s_1 e = s_2 r_2 = \cdots = s_n r_n.$$

Forming $t_1 = e r_2 \cdots r_n = t_i r_i$, and taking account of the elementwise commutativity of $\mathfrak{R}_i$ and $\mathfrak{S}_i$, one gets

$$s t_1^{-1} = s_i t_i^{-1},$$

which is therefore an element of all the $\mathfrak{S}_i$, and hence is equal to e. Thus $\mathfrak{S}_1 \subseteq \mathfrak{T}_1$, or $\mathfrak{T}_1 = \mathfrak{S}_1$, and correspondingly $\mathfrak{T}_i = \mathfrak{S}_i$. It follows that $\mathfrak{H} = \mathfrak{R}_i\mathfrak{T}_i = \mathfrak{R}_i\mathfrak{S}_i = \mathfrak{G}$ and $\mathfrak{R}_i \cap \mathfrak{T}_i = \mathfrak{R}_i \cap \mathfrak{S}_i = \mathfrak{E}$; hence $\mathfrak{G}$ is the direct product of the $\mathfrak{R}_i$.

[11] As W. Krull in the commutative case and O. Schmidt in general have shown (cf. note 6), under the hypothesis of the maximal and minimal conditions the representation is unique up to isomorphism. Since one can add all inner automorphisms to the operator domain without changing the concept of direct indecomposability, it is enough to require the finiteness conditions for normal divisors, or the existence of a principal series.

34. Hypercomplex Quantities and Representation Theory

Continuation: Chapter I, §§5–7, and Chapter II, §§8–14.

§ 5. Completely reducible groups

A group is called *completely reducible* if it is the direct product of finitely many simple groups:

$$\mathfrak{G} = \mathfrak{A}_1 \times \cdots \times \mathfrak{A}_n.$$

In this case the groups

$$\mathfrak{A}_1 \times \cdots \times \mathfrak{A}_n \supset \mathfrak{A}_1 \times \cdots \times \mathfrak{A}_{n-1} \supset \cdots \supset \mathfrak{A}_1 \supset \mathfrak{E}$$

form a composition series. Its length is n, and its composition factors are

$$\sim \mathfrak{A}_n,\ \mathfrak{A}_{n-1}, \ldots, \mathfrak{A}_1.$$

If $\mathfrak{G}$ is completely reducible, then every normal divisor is a direct factor, and the other factor can be chosen as a product of such simple groups as occur in a prescribed product decomposition of $\mathfrak{G}$. The normal divisor $\mathfrak{H}$ is itself completely reducible.

Proof. We have

$$\mathfrak{G} = \mathfrak{H}\mathfrak{A}_1\mathfrak{A}_2 \cdots \mathfrak{A}_n. \tag{3}$$

Put $\mathfrak{H}_i = \mathfrak{H}\mathfrak{A}_1 \cdots \mathfrak{A}_i$. Then $\mathfrak{H}_{i+1} = \mathfrak{H}_i\mathfrak{A}_{i+1}$. Either $\mathfrak{A}_{i+1} \leq \mathfrak{H}_i$, whence $\mathfrak{H}_{i+1} = \mathfrak{H}_i$; in that case we omit the factor $\mathfrak{A}_{i+1}$ from (3). Or $\mathfrak{H}_i \cap \mathfrak{A}_{i+1}$ is a proper normal divisor of $\mathfrak{A}_{i+1}$, hence is $\mathfrak{E}$, and then $\mathfrak{H}_i \times \mathfrak{A}_{i+1}$ is direct. After the superfluous factors have been omitted, (3) becomes

$$\mathfrak{G} = \mathfrak{H} \times \mathfrak{A}_{i_1} \times \cdots \times \mathfrak{A}_{i_r},$$

so that $\mathfrak{H}$ is a direct factor. Further,

$$\mathfrak{H} \sim \mathfrak{G}/(\mathfrak{A}_{i_1} \times \cdots \times \mathfrak{A}_{i_r}) \sim \mathfrak{A}_{j_1} \times \cdots \times \mathfrak{A}_{j_\mu},$$

where $\mathfrak{A}_{j_1}, \ldots, \mathfrak{A}_{j_\mu}$ are the remaining $\mathfrak{A}_i$. Thus $\mathfrak{H}$ is completely reducible.

Corollary. Every decomposition of a completely reducible group into directly indecomposable factors is a decomposition into simple factors, and therefore gives rise to a composition series; from this follows the unique determination of the factors up to isomorphism.

§ 6. Modules with respect to a field. Hypercomplex systems

An almost trivial example of completely reducible groups with operators is given by modules with finite basis with respect to a not necessarily commutative field.

Let $\mathfrak{G}$ be a right K-module, and let the identity element e of K be at the same time the identity operator: $ae = a$ for a in $\mathfrak{G}$. Every submodule aK derived from an a is simple, namely equal to the module derived from any one of its non-zero elements. Thus if $\mathfrak{H}$ is the submodule derived from some elements, and a is an element not belonging to $\mathfrak{H}$, then $\mathfrak{H} \cap aK = \mathfrak{E}$, so $\mathfrak{H} + aK$ is direct. Starting with any basis element a_1, one can therefore keep adjoining new basis elements $a_2, a_3, \ldots$, and obtains $\mathfrak{G}$ as the direct sum

$$\mathfrak{G} = a_1K + a_2K + \cdots + a_nK.$$

Thus $\mathfrak{G}$ is completely reducible.

The number n, the length of the composition series, is called the *rank* of $\mathfrak{G}$ with respect to K. Because of the uniqueness of the representation of the elements of $\mathfrak{G}$ (and because e has been assumed to be the identity operator), the basis $a_1, \ldots, a_n$ is linearly independent.

Every submodule $\mathfrak{H}$ is a direct summand:

$$\mathfrak{G} = \mathfrak{H} + \mathfrak{R} = (h_1K + \cdots + h_sK) + (r_1K + \cdots + r_{n-s}K),$$

or: *every linearly independent basis of $\mathfrak{H}$ can be completed to a linearly independent basis of $\mathfrak{G}$.* The r_i may even be chosen from the original basis elements a_i.

Let $(a_1, \ldots, a_n)$ and $(c_1, \ldots, c_n)$ be linearly independent bases. Then

$$c_k = \sum_i a_i \pi_{ik},$$

or, written in matrices,

$$(c_1, \ldots, c_n) = (a_1, \ldots, a_n)P, \qquad P = \begin{pmatrix} \pi_{11} & \cdots & \pi_{1n} \\ \vdots & & \vdots \\ \pi_{n1} & \cdots & \pi_{nn} \end{pmatrix}.$$

Conversely,

$$(a_1, \ldots, a_n) = (c_1, \ldots, c_n)Q.$$

It follows that

$$(a_1, \ldots, a_n) = (a_1, \ldots, a_n)PQ, \qquad (c_1, \ldots, c_n) = (c_1, \ldots, c_n)QP,$$

and, since the a_i and c_i are linearly independent,

$$PQ = QP = E.$$

Special K-modules that are also rings are the "hypercomplex systems."

A ring $\mathfrak{o}$ which is at the same time a right module with respect to a commutative field K is called a *hypercomplex system with respect to* K (also called an "algebra over K" in the literature) if:

1. its rank is finite (let a linearly independent basis be $u_1, \ldots, u_n$);
2. $ab \cdot x = a \cdot bx = ax \cdot b$. This is expressed by saying that K is joined commutatively with $\mathfrak{o}$;
3. the identity element e of K is also the identity operator: $ae = a$ for a in $\mathfrak{o}$.

Since

$$\Bigl(\sum_i u_i \alpha_i\Bigr)\Bigl(\sum_k u_k \beta_k\Bigr) = \sum_{i,k} (u_i u_k)(\alpha_i \beta_k),$$

the system is uniquely determined once, in addition to K, the *multiplication table* is given, i.e. once it is known how each product $u_i u_k$ is expressed through the u_j:

$$u_i u_k = \sum_j u_j \gamma_{ik}^j.$$

The γ_{ik}^j must satisfy the familiar relations following from the associative law.

If $\mathfrak{o}$, as we shall often assume, has an identity element e ($ea = ae = a$ for all a), then the elements x of K may be identified with ex, and K may be regarded as a subfield of $\mathfrak{o}$. The hypercomplex system may then also be described as a *ring of finite rank with respect to a field lying in the center.*

An example of a hypercomplex system is the *group ring* of a finite group: one takes the group elements as the basis elements u_i, and group multiplication as multiplication. The field K is arbitrary.

§ 7. Matrices

The square matrices (π_{ik}), with π_{ik} from a ring $\mathfrak{o}$, form a ring under the ordinary matrix multiplication $\sum_j \pi_{ij} \rho_{jk} = \sigma_{ik}$ and addition $\pi_{ik} + \rho_{ik} = \sigma_{ik}$.

A matrix with elements from a field K for which there is both a right inverse and a left inverse is called *regular.*

We saw in §6: *The transition matrix between two linearly independent bases of a K-module is regular.*

For regularity a right inverse suffices. Indeed, form a right K-module with linearly independent basis $(a_1, \ldots, a_n)$ (the module of linear forms in the indeterminates $a_1, \ldots, a_n$), and put

$$(c_1, \ldots, c_n) = (a_1, \ldots, a_n)P.$$

[0]Condition 2 says that multiplication by an element of K is a homomorphism for $\mathfrak{o}$ when $\mathfrak{o}$ is regarded both as an $\mathfrak{o}$-left-module and as an $\mathfrak{o}$-right-module. By virtue of 3. these homomorphisms are even isomorphisms.

Then

$$(c_1, \dots, c_n)P^{-1} = (a_1, \dots, a_n)PP^{-1} = (a_1, \dots, a_n),$$

so $(c_1, \dots, c_n)$ is again a basis, hence again linearly independent, and the matrix P is regular. Moreover the right inverse is equal to the left inverse.

Similarly, a left inverse suffices.

If P has a left inverse, then P is not a left zero divisor. Proof. From $PA = 0$ follows $A = P^{-1}PA = 0$.

Thus P also has only one right inverse, which by §6 coincides with the left inverse.

If, as usual, P_x denotes the matrix $(\pi_{ik}x)$, and C_{ik} denotes the matrix which has the identity element in the ik place and zeros everywhere else, then

$$(\pi_{ik}) = \sum_{i,k} C_{ik}\pi_{ik};$$

so the matrices over K form a K-module of rank n^2. The C_{ik} satisfy the relations

$$C_{ij}C_{jk} = C_{ik}, \qquad C_{ij}C_{\bar{j}k} = 0, \quad j \neq \bar{j}.$$

Thus the matrices over K form a finite K-module and, if K is commutative, a hypercomplex system.

Chapter II. Non-commutative ideal theory

§ 8. Homomorphism theorem for rings

If $\mathfrak{a}$ is a two-sided ideal in $\mathfrak{o}$, then the residue-class domain $\mathfrak{o}/\mathfrak{a}$ is not only an $\mathfrak{o}$-module but also a ring, as is easily seen. Every homomorphic image of $\mathfrak{o}$ is again a ring, and is isomorphic to a residue-class ring $\mathfrak{o}/\mathfrak{a}$, where $\mathfrak{a}$ is a two-sided ideal. The proof is as for groups.

In particular, if $\mathfrak{o}$ is a field, then there are no ideals other than (0) and $\mathfrak{o}$; hence here the homomorphism is either an isomorphism, or assigns zero to every element.

In particular, suppose the ring $\mathfrak{o}$ is a right K-module, where K is to be a ring elementwise commuting with $\mathfrak{o}$:

$$ab \cdot x = a \cdot bx = ax \cdot b$$

(for example, in the case of a hypercomplex system with respect to a commutative field K). Then as admissible right, left, or two-sided ideals one considers only those which are at the same time K-modules, and in addition to ring homomorphism one also requires operator homomorphism with respect to multiplication by K. The admissible ideals become bimodules with respect to $\mathfrak{o}$ and K. (For hypercomplex systems, by “ideals” without further qualification one always means only those admissible with respect to the underlying coefficient field K.) In this enlarged sense too, the homomorphism theorem above remains valid.

An example of ring homomorphism is the transition from a multiplier domain $\mathfrak{o}$ to the absolute multiplier domain (§1, Example 4). Thus the absolute multiplier domain is isomorphic to the residue-class ring $\mathfrak{o}/\mathfrak{d}$, where $\mathfrak{d}$ is the two-sided ideal of elements of $\mathfrak{o}$ which annihilate $\mathfrak{M}$.

From the validity of the homomorphism theorem there follow, as in §2, the first and second isomorphism theorems for ring and operator isomorphism.

All products $\mathfrak{A}\mathfrak{B}$ of subsets of $\mathfrak{o}$ are to be understood as module products: $\mathfrak{A}\mathfrak{B}$ is the totality of all sums $\sum ab$, with a in $\mathfrak{A}$, b in $\mathfrak{B}$, and $a\mathfrak{B}$ is the totality of all sums $\sum ab$, b in $\mathfrak{B}$. If $\mathfrak{B}$ is a right module with respect to some ring as operator domain, then the product $\mathfrak{A}\mathfrak{B}$, respectively $a\mathfrak{B}$, is also a right module. In particular: a right ideal, an admissible ideal with respect to a coefficient domain K, etc. If $(\mathfrak{A}, \mathfrak{B})$ denotes the sum of the modules $\mathfrak{A}$ and $\mathfrak{B}$, then the rules of calculation are

$$\mathfrak{A}\mathfrak{B} \cdot \mathfrak{C} = \mathfrak{A} \cdot \mathfrak{B}\mathfrak{C},$$

$$\mathfrak{A}(\mathfrak{B}, \mathfrak{C}) = (\mathfrak{A}\mathfrak{B}, \mathfrak{A}\mathfrak{C}), \qquad (\mathfrak{A}, \mathfrak{B})\mathfrak{C} = (\mathfrak{A}\mathfrak{C}, \mathfrak{B}\mathfrak{C}).$$

The direct sum is denoted by $\mathfrak{A} + \mathfrak{B}$ (§4).

In what follows we shall often consider rings satisfying a maximal or minimal condition (§3) for right ideals. Hypercomplex systems in particular belong to these, since by the convention just made only K-modules are admitted as ideals, and their rank therefore remains bounded (§6).

§ 9. Idempotent elements. Direct sum decomposition into right ideals

Let $\mathfrak{o}$ be a ring with identity element.

If $\mathfrak{r}$ is a right ideal, then $\mathfrak{r}\mathfrak{o} \subseteq \mathfrak{r}$, and because of the identity element even $\mathfrak{r}\mathfrak{o} = \mathfrak{r}$. Correspondingly for left ideals: $\mathfrak{o}\mathfrak{l} = \mathfrak{l}$.

An element c is called idempotent if $c^2 = c$.

If $\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_n$ is a decomposition into right ideals, and if

$$e = e_1 + \cdots + e_n, \qquad e_i \in \mathfrak{r}_i,$$

then

$$e_i^2 = e_i, \qquad e_i e_k = 0 \quad (i \neq k) \quad (\text{“orthogonality relations”}),$$

$$\mathfrak{r}_i = e_i \mathfrak{o}.$$

Proof. Let r be an element of $\mathfrak{r}_1$. Then

$$r = er = e_1 r + \cdots + e_n r, \qquad e_i r \in \mathfrak{r}_i,$$

but also

$$r = r + 0 + \cdots + 0,$$

so

$$e_1 r = r, \qquad e_i r = 0 \quad (i \neq 1).$$

From the first formula it follows that $\mathfrak{r}_1 \subseteq e_1\mathfrak{o}$; on the other hand $e_1\mathfrak{o} \subseteq \mathfrak{r}_1$, so $e_1\mathfrak{o} = \mathfrak{r}_1$. Specializing $r = e_1$ gives

$$e_1^2 = e_1, \qquad e_i e_1 = 0 \quad (i \neq 1).$$

The same holds when the index 1 is replaced by k.

Conversely, if $e = \sum e_i$, $e_i e_k = 0$ $(i \neq k)$, $e_i^2 = e_i$, and one puts $\mathfrak{r}_i = e_i\mathfrak{o}$, then $\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_n$.

Proof. Every element r of $\mathfrak{o}$ is

$$r = er = e_1 r + \cdots + e_n r,$$

hence

$$\mathfrak{o} = (e_1\mathfrak{o}, \ldots, e_n\mathfrak{o}).$$

But the representation is unique; for from

$$0 = e_1 a_1 + \cdots + e_n a_n$$

it follows, upon multiplication by e_i, that

$$e_i^2 a_i = e_i a_i = 0.$$

From a right representation

$$\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_n = e_1\mathfrak{o} + \cdots + e_n\mathfrak{o}$$

there accordingly follows a left representation

$$\mathfrak{o} = \mathfrak{l}_1 + \cdots + \mathfrak{l}_n = \mathfrak{o}e_1 + \cdots + \mathfrak{o}e_n.$$

If the $\mathfrak{r}_i$ are directly indecomposable, then so are the $\mathfrak{l}_i$. For a decomposition, say of $\mathfrak{l}_1$, would mean

$$\mathfrak{l}_1 = \mathfrak{o}e_1 = \mathfrak{o}e_1' + \mathfrak{o}e_1'', \qquad \mathfrak{o} = \mathfrak{o}e_1' + \mathfrak{o}e_1'' + \mathfrak{o}e_2 + \cdots + \mathfrak{o}e_n.$$

From this would result the right decomposition

$$\mathfrak{o} = e_1'\mathfrak{o} + e_1''\mathfrak{o} + e_2\mathfrak{o} + \cdots + e_n\mathfrak{o} = \mathfrak{r}_1' + \mathfrak{r}_1'' + \mathfrak{r}_2 + \cdots + \mathfrak{r}_n,$$

$$\mathfrak{r}_1 \cong \mathfrak{o}/(\mathfrak{r}_2 + \cdots + \mathfrak{r}_n) = \mathfrak{r}_1' + \mathfrak{r}_1'',$$

so $\mathfrak{r}_1$ would be decomposable.

[0]There is another converse: if $e_i e_k = 0$, $e_i^2 = e_i$, $c = \sum e_i$, then c is a left identity for the ring $e_1\mathfrak{o} + \cdots + e_n\mathfrak{o}$. The fact that the sum is direct is seen in exactly the same way as above.

§ 10. Decomposition into two-sided ideals

Let $\mathfrak{o}$ again be a ring with identity element, and let

$$\mathfrak{o} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_n \tag{1}$$

be a decomposition of $\mathfrak{o}$ into two-sided ideals. Then the orthogonality relations also hold for the ideals:

$$\mathfrak{a}_i\mathfrak{a}_k = 0 \quad (i \neq k), \qquad \mathfrak{a}_i^2 = \mathfrak{a}_i. \tag{2}$$

Proof. Let $\mathfrak{b}_1 = \mathfrak{a}_2 + \cdots + \mathfrak{a}_n$, $\mathfrak{o} = \mathfrak{a}_1 + \mathfrak{b}_1$. Then $\mathfrak{a}_1\mathfrak{b}_1 \leq \mathfrak{a}_1 \cap \mathfrak{b}_1 = 0$, so $\mathfrak{a}_1\mathfrak{b}_1 = 0$, and a fortiori

$$\mathfrak{a}_1\mathfrak{a}_i = 0 \quad (i \neq 1);$$

$$\mathfrak{a}_1 = \mathfrak{a}_1\mathfrak{o} = \mathfrak{a}_1(\mathfrak{a}_1 + \mathfrak{b}_1) = (\mathfrak{a}_1^2, \mathfrak{a}_1\mathfrak{b}_1) = \mathfrak{a}_1^2.$$

Conversely: *from (1) and (2) it follows that the modules $\mathfrak{a}_i$ are two-sided ideals. Proof.*

$$\mathfrak{o}\mathfrak{a}_i = (\mathfrak{a}_1 + \cdots + \mathfrak{a}_n)\mathfrak{a}_i = \mathfrak{a}_i^2 = \mathfrak{a}_i, \qquad \mathfrak{a}_i\mathfrak{o} = \mathfrak{a}_i.$$

Right ideals in the ring $\mathfrak{a}_i$ are right ideals in $\mathfrak{o}$. Proof.

$$\mathfrak{r}\mathfrak{o} = \mathfrak{r}(\mathfrak{a}_i + \mathfrak{b}_i) = (\mathfrak{r}\mathfrak{a}_i, \mathfrak{r}\mathfrak{b}_i) = (\mathfrak{r}, 0) = \mathfrak{r}.$$

If $\mathfrak{r}$ is a right ideal in $\mathfrak{o}$, then $\mathfrak{r}$ is a direct sum of right ideals $\mathfrak{r}\mathfrak{a}_i \leq \mathfrak{a}_i$. Proof. $\mathfrak{r} = \mathfrak{r}\mathfrak{o} = (\mathfrak{r}\mathfrak{a}_1, \ldots, \mathfrak{r}\mathfrak{a}_n)$, and $\mathfrak{r}\mathfrak{a}_i\mathfrak{o} = \mathfrak{r}\mathfrak{a}_i$, so the $\mathfrak{r}\mathfrak{a}_i$ are right ideals contained in $\mathfrak{r}$. Since $\mathfrak{r}\mathfrak{a}_i \leq \mathfrak{a}_i$, the sum is direct:

$$\mathfrak{r} = \mathfrak{r}\mathfrak{a}_1 + \cdots + \mathfrak{r}\mathfrak{a}_n.$$

In particular, if $\mathfrak{r}$ is directly indecomposable, then $\mathfrak{r}$ can have only one component; hence $\mathfrak{r}$ must lie in one of the $\mathfrak{a}_i$.

By means of these theorems, ideal theory in $\mathfrak{o}$ is controlled once the ideal theory of the individual $\mathfrak{a}_i$ is known. We shall therefore mostly restrict ourselves to two-sided indecomposable rings.

Two right ideals contained in different $\mathfrak{a}_i$ can never be operator-isomorphic. Proof. An ideal lying in $\mathfrak{a}_1$ is annihilated by all the other $\mathfrak{a}_k$, but not by $\mathfrak{a}_1$. By contrast, an ideal lying in $\mathfrak{a}_2$ is annihilated by $\mathfrak{a}_1$.

Thus if it is possible to decompose the ring $\mathfrak{o}$ into two-sided indecomposable ideals $\mathfrak{a}_1 + \cdots + \mathfrak{a}_n$, and each $\mathfrak{a}_i$ again into one-sided indecomposable right ideals $\mathfrak{r}_i' + \mathfrak{r}_i'' + \cdots$, then the operator-isomorphic $\mathfrak{r}$ are to be sought among those belonging to the same $\mathfrak{a}_i$. But it may happen that there are still different classes of $\mathfrak{r}$ – that is, classes not operator-isomorphic – in the same $\mathfrak{a}_i$, as the following example shows.

Let K be the field of rational numbers,

$$\mathfrak{o} = e_1K + e_2K + uK$$

a hypercomplex system, with the e_i and u commuting with the numbers of K, and with multiplication table

	e_1	e_2	u
e_1	e_1	0	u
e_2	0	e_2	0
u	0	u	0

The identity element is $e = e_1 + e_2$. The e_i satisfy the orthogonality relations. Thus the right decomposition is

$$\mathfrak{o} = e_1\mathfrak{o} + e_2\mathfrak{o} = (e_1, u) + (e_2),$$

and the left decomposition is

$$\mathfrak{o} = \mathfrak{o}e_1 + \mathfrak{o}e_2 = (e_1) + (e_2, u).$$

Since $e_2\mathfrak{o}$ and $\mathfrak{o}e_1$ are visibly indecomposable, $\mathfrak{o}e_2$ and $e_1\mathfrak{o}$ must be indecomposable as well.

A two-sided decomposition is impossible. For then one component would have to contain $e_1\mathfrak{o}$ and the other $e_2\mathfrak{o}$; the first would then contain u, but the second would contain $ue_2 = u$ as well.

But $e_1\mathfrak{o}$ and $e_2\mathfrak{o}$ are not operator-isomorphic, since they have different rank with respect to K.

Let $\mathfrak{o} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_n$, with the $\mathfrak{a}_i$ two-sided indecomposable. Then they are uniquely determined.

Proof. Suppose also $\mathfrak{o} = \mathfrak{c}_1 + \cdots + \mathfrak{c}_n$. Then

$$\mathfrak{a}_i = \mathfrak{a}_i\mathfrak{o} = (\mathfrak{a}_i\mathfrak{c}_1, \ldots, \mathfrak{a}_i\mathfrak{c}_n).$$

This latter sum is direct because $\mathfrak{a}_i\mathfrak{c}_k \leq \mathfrak{c}_k$ and $\leq \mathfrak{a}_i$; since, however, the $\mathfrak{a}_i$ are indecomposable, all $\mathfrak{a}_i\mathfrak{c}_k$ must be zero except one, $\mathfrak{a}_i\mathfrak{c}_{j_i}$. Thus

$$\mathfrak{a}_i = \mathfrak{a}_i\mathfrak{c}_{j_i}.$$

Similarly,

$$\mathfrak{c}_{j_i} = \mathfrak{o}\mathfrak{c}_{j_i} = \mathfrak{a}_1\mathfrak{c}_{j_i} + \cdots + \mathfrak{a}_n\mathfrak{c}_{j_i},$$

and since $\mathfrak{a}_i\mathfrak{c}_{j_i} \neq 0$, all the remaining terms are 0. Therefore

$$\mathfrak{c}_{j_i} = \mathfrak{a}_i\mathfrak{c}_{j_i} = \mathfrak{a}_i,$$

q.e.d.

§ 11. The center

The center $\mathfrak{Z}$ of a ring $\mathfrak{o}$ is the totality of all z commuting with all ring elements ($za = az$ for all a). $\mathfrak{Z}$ is a commutative ring.

Every one-sided ideal $\mathfrak{a}$ in $\mathfrak{o}$ has a "contraction ideal" $\mathfrak{a} \cap \mathfrak{Z}$ in $\mathfrak{Z}$. Since both $\mathfrak{a}$ and $\mathfrak{Z}$ are $\mathfrak{Z}$-modules, so is $\mathfrak{a} \cap \mathfrak{Z}$.

Every ideal $\mathfrak{A}$ in $\mathfrak{Z}$ has an extension ideal: the ideal $\mathfrak{a} = (\mathfrak{A}, \mathfrak{o}\mathfrak{A})$ generated by $\mathfrak{A}$ in $\mathfrak{o}$. This is two-sided, because

$$\mathfrak{a}\mathfrak{o} = (\mathfrak{o}\mathfrak{A}, \mathfrak{A})\mathfrak{o} = (\mathfrak{o}\mathfrak{A}\mathfrak{o}, \mathfrak{A}\mathfrak{o}) = (\mathfrak{o}^2\mathfrak{A}, \mathfrak{o}\mathfrak{A}) \subseteq \mathfrak{o}\mathfrak{A} \subseteq \mathfrak{a}.$$

If $\mathfrak{o}$ is assumed to have an identity element, one may write $\mathfrak{a} = \mathfrak{o}\mathfrak{A}$. In this case the following theorem holds:

Every two-sided decomposition of $\mathfrak{o}$ corresponds one-to-one to a decomposition of the center: from

$$\mathfrak{o} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_n, \qquad \mathfrak{A}_i = \mathfrak{a}_i \cap \mathfrak{Z} \tag{1}$$

there follows

$$\mathfrak{Z} = \mathfrak{A}_1 + \cdots + \mathfrak{A}_n, \qquad \mathfrak{o}\mathfrak{A}_i = \mathfrak{a}_i, \tag{2}$$

and conversely.

Proof. Let z be a central element, and let

$$z = z_1 + \cdots + z_n \tag{3}$$

be its decomposition according to (1). Then the z_i are central elements; for

$$za = z_1a + \cdots + z_na = az = az_1 + \cdots + az_n.$$

By directness of the sum, and since $\mathfrak{a}_i$ is two-sided, it follows that

$$z_ia = az_i.$$

Thus z_i lies in $\mathfrak{a}_i \cap \mathfrak{Z} = \mathfrak{A}_i$, so (3) gives the asserted sum decomposition. In particular, $e = \sum e_i$, hence the e_i are central elements. Finally

$$z = ze_1 + \cdots + ze_n,$$

and so

$$\mathfrak{A}_i = \mathfrak{Z}e_i, \qquad \mathfrak{o}\mathfrak{A}_i = \mathfrak{o}\mathfrak{Z}e_i = \mathfrak{o}e_i = \mathfrak{a}_i.$$

Conversely, if $\mathfrak{Z} = \mathfrak{A}_1 + \cdots + \mathfrak{A}_n$ is a decomposition of the center, then by §9

$$\mathfrak{A}_i = \mathfrak{Z}e_i, \qquad e_i^2 = e_i, \qquad e_ie_k = 0 \quad (i \neq k).$$

Hence, by the orthogonality relations §9,

$$\begin{aligned} \mathfrak{o} = \mathfrak{o}e &= \mathfrak{o}e_1 + \cdots + \mathfrak{o}e_n = \mathfrak{o}\mathfrak{Z}e_1 + \cdots + \mathfrak{o}\mathfrak{Z}e_n \\ &= \mathfrak{o}\mathfrak{A}_1 + \cdots + \mathfrak{o}\mathfrak{A}_n = \mathfrak{a}_1 + \cdots + \mathfrak{a}_n. \end{aligned}$$

If now $\mathfrak{a}_i \cap \mathfrak{Z} = \mathfrak{A}'_i$, then from the first part of the assertion one knows that

$$\mathfrak{Z} = \mathfrak{A}'_1 + \cdots + \mathfrak{A}'_n \quad \text{directly}.$$

But $\mathfrak{A}'_i \supseteq \mathfrak{A}_i$, hence $\mathfrak{A}'_i = \mathfrak{A}_i$ by §4, 2.

Corollary. $\mathfrak{a}_i$ and $\mathfrak{A}_i$ are simultaneously two-sided directly decomposable or indecomposable.

§ 12. Nilpotent ideals

An ideal $\mathfrak{c}$ is called *nilpotent* if $\mathfrak{c}^\rho = 0$.

Example. The ideal (u) in §10.

If there is a nilpotent right ideal $\mathfrak{r}$ in $\mathfrak{o}$, then there is also a nilpotent left ideal, indeed a two-sided ideal.

Proof. Let $\mathfrak{r}^\rho = 0$. Then $\mathfrak{o}\mathfrak{r}$, or, if $\mathfrak{o}$ contains no identity element and $\mathfrak{o}\mathfrak{r}$ could be zero, $(\mathfrak{o}\mathfrak{r}, \mathfrak{r})$, is a left ideal, and

$$(\mathfrak{o}\mathfrak{r})^\rho = \mathfrak{o}\mathfrak{r}\mathfrak{o}\mathfrak{r}\cdots\mathfrak{o}\mathfrak{r} \subseteq \mathfrak{o}\mathfrak{r}\mathfrak{r}\cdots\mathfrak{r} = \mathfrak{o}\mathfrak{r}^\rho = 0.$$

The sum of two nilpotent right ideals is again a nilpotent right ideal.

Proof. Let $\mathfrak{c}^\rho = 0$, $\mathfrak{d}^\sigma = 0$. In

$$(\mathfrak{c}, \mathfrak{d})^{\rho+\sigma-1} = (\ldots, \mathfrak{d}\mathfrak{c}\cdots\mathfrak{d}\cdots\mathfrak{c}, \ldots)$$

every term contains either at least ρ factors $\mathfrak{c}$ or at least σ factors $\mathfrak{d}$. In the first case one has

$$\mathfrak{d}\mathfrak{c}\cdots\mathfrak{d}\cdots\mathfrak{c}\cdots \subseteq \mathfrak{d}\mathfrak{c}\mathfrak{c}\cdots\mathfrak{c} = \mathfrak{d}\mathfrak{c}^\rho = 0;$$

in the latter case the same argument applies to the factors $\mathfrak{d}$. Thus

$$(\mathfrak{c}, \mathfrak{d})^{\rho+\sigma-1} = 0.$$

If now the maximal condition for right ideals in $\mathfrak{o}$ is satisfied, then there is a maximal nilpotent right ideal. It contains all other nilpotent right ideals; otherwise one could form its sum with such an ideal. Call it $\mathfrak{c}$. Since $\mathfrak{o}\mathfrak{c}$ is also a nilpotent right ideal, $\mathfrak{o}\mathfrak{c} \subseteq \mathfrak{c}$, so $\mathfrak{c}$ is a left ideal. Every nilpotent left ideal $\mathfrak{d}$ is likewise contained in $\mathfrak{c}$, since $(\mathfrak{d}, \mathfrak{d}\mathfrak{o})$ is a nilpotent right ideal. Thus $\mathfrak{d} \subseteq \mathfrak{c}$. *There is therefore a maximal two-sided nilpotent ideal $\mathfrak{c}$, or "radical," containing all the others, right- and left-sided.*

In the example of §10, (u) is the maximal nilpotent ideal.

If the radical is the zero ideal, one speaks of a "ring without radical" ("semisimple ring," "Dedekind system"). The residue-class ring modulo the radical is always a ring without radical.

If $\mathfrak{Z}$ is the center of $\mathfrak{o}$, and $\mathfrak{c}$ is the radical of $\mathfrak{o}$, then $\mathfrak{C} = \mathfrak{c} \cap \mathfrak{Z}$ is the radical of $\mathfrak{Z}$.

Proof. It is clear that $\mathfrak{C}$ is nilpotent. If there were a larger nilpotent ideal $\overline{\mathfrak{C}}$ in $\mathfrak{Z}$, then it could be extended to a nilpotent ideal $\bar{\mathfrak{c}}$ not wholly contained in $\mathfrak{c}$, since $((\overline{\mathfrak{C}}\mathfrak{o})^\rho = \overline{\mathfrak{C}}^\rho\mathfrak{o}^\rho = 0)$.

Consequence. If $\mathfrak{o}$ is a ring without radical, so is $\mathfrak{Z}$. The converse is not true, however, as the example in §10 shows. The center of $\mathfrak{o}/\mathfrak{c}$ contains a ring isomorphic to $\mathfrak{Z}/\mathfrak{C}$, but this can be a proper subring (example in §10).

§ 13. Completely reducible rings

By §5 a ring is called right completely reducible if it is a direct sum of finitely many simple right ideals.

A right completely reducible ring with identity element has no nilpotent ideal $\neq (0)$, hence no radical.

Proof. Every right ideal $\mathfrak{r}$ is a direct summand (§5):

$$\mathfrak{o} = \mathfrak{t} + \mathfrak{r} = e_1\mathfrak{o} + e_2\mathfrak{o}.$$

Since $e_2^2 = e_2$, also $e_2^\rho = e_2$. If $\mathfrak{r}$ were nilpotent, then $e_2^\rho = 0$, hence $e_2 = 0$, and so $\mathfrak{r} = 0$, q.e.d.

We now show the two converses. First: *A ring without radical satisfying the minimal condition for right ideals is completely reducible with respect to right ideals.*

Proof. Let $\mathfrak{t}$ be a minimal right ideal $\neq 0$. We shall show that $\mathfrak{t}$ contains an idempotent element.

Since $\mathfrak{t}^2 \subseteq \mathfrak{t}$ and $\mathfrak{t}^2 \neq 0$, one has $\mathfrak{t}^2 = \mathfrak{t}$, for $\mathfrak{t}^2$ is a right ideal. Hence there is an a in $\mathfrak{t}$ such that $a\mathfrak{t} \neq 0$. Then necessarily $a\mathfrak{t} = \mathfrak{t}$, because $a\mathfrak{t}$ is a right ideal.

The totality of all b in $\mathfrak{t}$ which are annihilated by a $(ab = 0)$ is a right ideal, and is not equal to $\mathfrak{t}$, hence is 0. Thus $ab = 0$ implies $b = 0$.

Because $\mathfrak{t} = a\mathfrak{t}$, a must have a representation $a = ac$, with $c \neq 0$. It follows that

$$ac = ac^2, \qquad a(c - c^2) = 0, \qquad c - c^2 = 0, \qquad c^2 = c.$$

We next show that $\mathfrak{t}$ is a direct summand. The set $c\mathfrak{o}$ is a non-zero right ideal contained in $\mathfrak{t}$, since $c^2 = c$ lies in $c\mathfrak{o}$; hence it is $\mathfrak{t}$. Every element r of $\mathfrak{o}$ has a representation

$$r = cr + (r - cr) \qquad \text{(one-sided Peirce decomposition).}$$

The elements cr form the ideal $\mathfrak{t}$; the elements $r - cr$ form another right ideal $\mathfrak{r}$, annihilated by c: $c\mathfrak{r} = 0$. Hence

$$\mathfrak{o} = (\mathfrak{t}, \mathfrak{r}).$$

Since the elements of $\mathfrak{t}$ are not annihilated by c, the sum is direct:

$$\mathfrak{o} = \mathfrak{t} + \mathfrak{r}. \tag{1}$$

Representing $\mathfrak{o}$, by the minimal condition, as a direct sum of directly indecomposable ideals,

$$\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_n,$$

the $\mathfrak{r}_i$ must be simple, i.e. minimal. For if, say, $\mathfrak{r}_1$ were not simple and $\mathfrak{t}$ were a minimal ideal in $\mathfrak{r}_1$, then (1) would give

$$\mathfrak{r}_1 = \mathfrak{t} + \mathfrak{r} \cap \mathfrak{r}_1 \qquad \S 4, 3,$$

so $\mathfrak{r}_1$ would be decomposable, a contradiction. Thus $\mathfrak{o}$ is completely reducible.

Second: *A ring without radical satisfying the minimal condition has an identity element.*

To show first the existence of a left identity, it is enough to prove the following: *if a right ideal* $\mathfrak{a} = c_1\mathfrak{o} \neq \mathfrak{o}$ *has a left identity* c_1, *then there is a right ideal* $c\mathfrak{o}$ *of greater length which likewise has a left identity* c. For, since complete reducibility, and therefore boundedness of all lengths, has already been shown, one reaches in finitely many steps a left identity for $\mathfrak{o}$ itself.

Let therefore $\mathfrak{a} = c_1\mathfrak{o}$ and $c_1^2 = c_1$. By the formula

$$r = c_1 r + (r - c_1 r)$$

we have, as above, a Peirce decomposition $\mathfrak{o} = c_1\mathfrak{o} + \mathfrak{r}$. Let c_2 be an idempotent element of $\mathfrak{r}$, as above; thus $c_2^2 = c_2$, and $c_1c_2 = 0$ since c_1 annihilates all elements of $\mathfrak{r}$. Put $e_1 = c_1$, $e_2 = c_2 - c_2c_1$. Then

$$e_1^2 = e_1, \quad e_1e_2 = 0, \quad e_2e_1 = 0, \quad e_2c_2 = e_2, \quad e_2 \neq 0, \quad e_2^2 = e_2.$$

By the remark made in the note above, $c = e_1 + e_2$ is a left identity for the ring

$$c\mathfrak{o} = e_1\mathfrak{o} + e_2\mathfrak{o},$$

which, since $e_2^2 = e_2 \neq 0$, has greater length than $\mathfrak{a}$ as a right ideal.

To show that the constructed left identity e is also a right identity, we make the Peirce decomposition into left ideals:

$$\mathfrak{o} = \mathfrak{o}e + \mathfrak{l}.$$

We have $\mathfrak{l}e = 0$, $\mathfrak{l} = e\mathfrak{l}$, hence $\mathfrak{l}^2 = \mathfrak{l}e\mathfrak{l} = 0$. Since by hypothesis no nilpotent left ideal can exist, $\mathfrak{l} = 0$. Therefore e is also a right identity, and hence an identity altogether.

Theorem. From right-sided complete reducibility and the existence of the identity follows two-sided complete reducibility:

$$\mathfrak{o} = \mathfrak{d}_1 + \cdots + \mathfrak{d}_s, \qquad \mathfrak{d}_i \text{ two-sided simple.}$$

As in the preceding proof, it is enough to show that every minimal two-sided ideal $\mathfrak{a}$ is a direct summand. As a right ideal, $\mathfrak{a}$ is a direct summand:

$$\mathfrak{o} = \mathfrak{a} + \mathfrak{r} = e_1\mathfrak{o} + e_2\mathfrak{o}.$$

The corresponding left decomposition is

$$\mathfrak{o} = \mathfrak{c} + \mathfrak{l} = \mathfrak{o}e_1 + \mathfrak{o}e_2.$$

It follows that

$$\mathfrak{r}\mathfrak{c} \subseteq \mathfrak{r}\mathfrak{a} \leq \mathfrak{r} \cap \mathfrak{a} = 0, \qquad \mathfrak{l}\mathfrak{a} = \mathfrak{o}e_2e_1\mathfrak{o} = 0,$$
$$(\mathfrak{a}\mathfrak{l})^2 = \mathfrak{a}\mathfrak{l}\mathfrak{a}\mathfrak{l} = 0, \qquad \text{hence } \mathfrak{a}\mathfrak{l} = 0;$$
$$\mathfrak{r} = \mathfrak{r}\mathfrak{o} = (\mathfrak{r}\mathfrak{c}, \mathfrak{r}\mathfrak{l}) = \mathfrak{r}\mathfrak{l}, \qquad \mathfrak{l} = \mathfrak{o}\mathfrak{l} = (\mathfrak{a}\mathfrak{l}, \mathfrak{r}\mathfrak{l}) = \mathfrak{r}\mathfrak{l}.$$

Thus $\mathfrak{r} = \mathfrak{l}$, so $\mathfrak{r}$ is two-sided; hence $\mathfrak{a}$ is a two-sided direct summand. From here on the proof is the same as for one-sided complete reducibility.

The $\mathfrak{d}_i$ are also two-sided simple as rings, since all ideals in $\mathfrak{d}_i$ are ideals in $\mathfrak{o}$. We now investigate their structure.

§ 14. Two-sided simple completely reducible rings with identity element

Let $\mathfrak{o}$ be such a ring:

$$\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_n = e_1\mathfrak{o} + \cdots + e_n\mathfrak{o}, \qquad \mathfrak{r}_i \text{ simple.}$$

Then

$$\mathfrak{o} = \mathfrak{l}_1 + \cdots + \mathfrak{l}_n = \mathfrak{o}e_1 + \cdots + \mathfrak{o}e_n, \qquad \mathfrak{l}_i \text{ directly indecomposable §9.}$$

Moreover $\mathfrak{l}_i\mathfrak{r}_i$ is a non-zero two-sided ideal (since e_i^2 lies in $\mathfrak{l}_i\mathfrak{r}_i$), hence is $\mathfrak{o}$.

Put $\mathfrak{A}_{ik} = \mathfrak{r}_i\mathfrak{l}_k$. Then

$$\mathfrak{o} = (\mathfrak{r}_1, \ldots, \mathfrak{r}_n) \cdot (\mathfrak{l}_1, \ldots, \mathfrak{l}_n) = (\ldots, \mathfrak{A}_{ij}, \ldots)$$

directly. For if $0 = \sum_{i,j} a_{ij}$, then, since a_{ij} lies in $\mathfrak{r}_i$ and $\sum \mathfrak{r}_i$ is direct,

$$0 = \sum_j a_{ij},$$

and then, because a_{ij} lies in $\mathfrak{l}_j$,

$$0 = a_{ij}.$$

Further, $\mathfrak{A}_{ij}\mathfrak{r}_j = \mathfrak{r}_i\mathfrak{l}_j\mathfrak{r}_j = \mathfrak{r}_i\mathfrak{o} = \mathfrak{r}_i$, so $\mathfrak{A}_{ij} \neq 0$. Let $a_{ij} \neq 0$ lie in $\mathfrak{A}_{ij}$. Then

$$a_{ij}\mathfrak{r}_j \subseteq \mathfrak{r}_i, \qquad a_{ij}\mathfrak{r}_j \neq 0 \text{ because } a_{ij}e_j = a_{ij},$$

and therefore

$$a_{ij}\mathfrak{r}_j = \mathfrak{r}_i.$$

If for every r_j in $\mathfrak{r}_j$ one puts

$$a_{ij}r_j = r_i,$$

then the map $r_j \mapsto r_i$ is an operator homomorphism from $\mathfrak{r}_j$ to $\mathfrak{r}_i$. The elements mapped to zero form an ideal contained in $\mathfrak{r}_j$, hence the zero ideal; therefore the map is an *isomorphism*. Thus:

Any two simple right ideals $\mathfrak{r}_i$ and $\mathfrak{r}_j$ occurring in a representation of $\mathfrak{o}$ are operator-isomorphic; the isomorphisms are mediated by the elements of $\mathfrak{A}_{ij}$. Since any two different simple right ideals $\mathfrak{r}, \mathfrak{r}'$ occur together in at least one representation of $\mathfrak{o}$, every two such ideals are isomorphic.

All homomorphisms from $\mathfrak{r}_j$ to $\mathfrak{r}_i$ are mediated by elements of $\mathfrak{A}_{ij}$.

Proof. If $e_j \mapsto a_{ij}$, then $e_j^2 \mapsto a_{ij}e_j$, so $a_{ij} = a_{ij}e_j$ lies in $\mathfrak{r}_i\mathfrak{l}_j = \mathfrak{A}_{ij}$. Further,

$$\mathfrak{r}_j = e_j\mathfrak{r}_j \mapsto a_{ij}\mathfrak{r}_j = \mathfrak{r}_i.$$

In particular, the homomorphisms of $\mathfrak{r}_i$ into itself are mediated by $\mathfrak{A}_{ii}$.

The product of two elements of $\mathfrak{A}_{ii}$ corresponds to the product of the homomorphisms, and the sum to the sum (definition: see §1, 6). Distinct a_{ii} give distinct automorphisms, for e_i goes to $a_{ii}e_i = a_{ii}$. Thus the ring $\mathfrak{A}_{ii}$ is isomorphic to the automorphism ring of $\mathfrak{r}_i$.

But the automorphism ring of a simple ideal is a field (§2), hence $\mathfrak{A}_{ii}$ is a field. Since all $\mathfrak{r}_i$ are operator-isomorphic, their automorphism rings are ring-isomorphic: $\mathfrak{A}_{ii}$ is uniquely determined by $\mathfrak{o}$ up to ring isomorphism. The possible different $\mathfrak{A}_{ii}$ are only different concrete realizations of the abstractly defined automorphism ring of the simple right ideals.

Main theorem. Every completely reducible two-sided simple ring $\mathfrak{o}$ with identity element is isomorphic to the ring of matrices of degree n over a field K. (The field K is isomorphic to the automorphism field of the right ideals of $\mathfrak{o}$.)

Proof. Construction of the matrix units c_{ik}. Let Γ_{11} be the identical automorphism of $\mathfrak{r}_1$, let Γ_{i1} be an arbitrary isomorphism $\Gamma_{i1}\mathfrak{r}_1 = \mathfrak{r}_i$, and finally set

$$\Gamma_{ik} = \Gamma_{i1}\Gamma_{k1}^{-1}.$$

Then in general

$$\Gamma_{ik}\Gamma_{kl} = \Gamma_{il}.$$

If Γ_{ik} is mediated by the element c_{ik} of $\mathfrak{A}_{ik}$, then further

$$c_{ik} = e_ic_{ik} = c_{ik}e_k, \qquad c_{ik}c_{kl}e_l = c_{il}e_l,$$

hence

$$c_{ik}c_{kl} = c_{il}, \qquad c_{ij}c_{kl} = c_{ij}e_je_kc_{kl} = 0 \quad (j \neq k).$$

The c_{ik} are therefore matrix units (§7).

Construction of K. The assignment

$$a_{ij} = c_{i1}a_{11}c_{1i}$$

assigns to every a_{11} in $\mathfrak{A}_{11}$ a "conjugate element" a_{ii} of $\mathfrak{A}_{ii}$.

This assignment is a ring isomorphism, since the isomorphisms Δ, Γ corresponding to a and c are connected by

$$\Delta_{ii} = \Gamma_{i1}\Delta_{11}\Gamma_{i1}^{-1},$$

a relation well known to represent a ring isomorphism. Now form the elements

$$\alpha = a_{11} + \cdots + a_{nn} = a_{11} + c_{21}a_{11}c_{12} + \cdots + c_{n1}a_{11}c_{1n}.$$

Each a_{11} determines exactly one α (uniquely because the sum is direct). The totality of the α is a field $K \sim \mathfrak{A}_{11}$, since from

$$\alpha = a_{11} + \cdots + a_{nn}, \qquad \beta = b_{11} + \cdots + b_{nn},$$

one obtains

$$\alpha + \beta = (a_{11} + b_{11}) + \cdots + (a_{nn} + b_{nn}), \qquad \alpha\beta = a_{11}b_{11} + \cdots + a_{nn}b_{nn};$$

so the assignment $a_{11} \mapsto \alpha$ is an isomorphism.

Further,

$$c_{ik}\alpha = c_{ik}a_{kk} = c_{ik}c_{k1}a_{11}c_{1k} = c_{i1}a_{11}c_{1k},$$

$$\alpha c_{ik} = a_{ii}c_{ik} = c_{i1}a_{11}c_{1i}c_{ik} = c_{i1}a_{11}c_{1k},$$

so every α commutes with all c_{ik}.

Finally, the same formulas give

$$c_{ik}K = c_{ik}\mathfrak{A}_{kk} = c_{ik}\mathfrak{r}_k\mathfrak{l}_k = \mathfrak{r}_i\mathfrak{l}_k = \mathfrak{A}_{ik},$$

and therefore

$$\mathfrak{o} = \sum c_{ik}K.$$

The assignment. If every element of $\mathfrak{o}$ is written in the form $\sum c_{ik}\alpha_{ik}$, and the matrix (α_{ik}) is assigned to it, the assignment is an isomorphism, as follows immediately from the definition of matrix multiplication. This proves the main theorem.

Converse. The ring of matrices of degree n over a field A is two-sided simple and completely reducible; A is isomorphic to the automorphism ring of the right ideals, and Ae is the K constructed in the preceding theorem.

Proof. Let the matrix units (§7) be c_{ik}. Put

$$\mathfrak{r}_i = \sum_k c_{ik}A.$$

Then plainly $\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_n$. The $\mathfrak{r}_i$ are simple right ideals, for every non-zero element $\sum c_{ik}\alpha_k$ generates the full $\mathfrak{r}_i$. Indeed, if $\alpha_j \neq 0$, then

$$\left(\sum c_{ik}\alpha_k\right) \cdot (c_{jl}\alpha_j^{-1}) = c_{il},$$

and $\sum c_{il}\beta_l$ runs through all elements of $\mathfrak{r}_i$.

The $\mathfrak{r}_i$ are operator-isomorphic: $\mathfrak{r}_i = c_{ik}\mathfrak{r}_k$.

It follows that $\mathfrak{o}$ is two-sided indecomposable; hence, by the theorem of §13 and the already proved complete reducibility, and since an identity element exists, it is two-sided simple. This is also seen from the fact that any $\sum\sum c_{ik}\alpha_{ik} \neq 0$ generates the entire two-sided ideal $\mathfrak{o}$.

Further,

$$\mathfrak{r}_i = c_{ii}\mathfrak{o}, \qquad \mathfrak{l}_i = \mathfrak{o}c_{ii},$$

$$e = \sum c_{ii}, \qquad c_{ii} = e_i,$$

[0]Recently B. L. van der Waerden found a simpler proof, operating directly with the abstract automorphism field, which is to appear in his book on algebra (Grundlehren d. Math. Wiss., Berlin: Julius Springer). [11 June 1929.]

$$\mathfrak{A}_{ii} = \mathfrak{r}_i \cap \mathfrak{l}_i = c_{ii}A \sim A.$$

Finally, putting $a_{11} = c_{11}\lambda$, one obtains

$$\alpha = a_{11} + c_{21}a_{11}c_{12} + \cdots = c_{11}\lambda + c_{22}\lambda + \cdots = e\lambda,$$

which proves everything asserted.

The center of $\mathfrak{o}$ is the center of K, hence a field.

Proof. $\mathfrak{Z}(K)$ consists of all ζ that commute with all x. But these also commute with all c_{ik}, and hence with all sums $\sum c_{ik}a_{ik}$. Thus

$$\mathfrak{Z}(K) \subseteq \mathfrak{Z}(\mathfrak{o}).$$

Conversely, let z lie in $\mathfrak{Z}(\mathfrak{o})$:

$$z = \sum_\mu \sum_\nu c_{\mu\nu}\gamma_{\mu\nu}.$$

Then

$$zc_{ik} = c_{ik}z, \qquad \sum_\mu c_{\mu k}\gamma_{\mu i} = \sum_\nu c_{i\nu}\gamma_{k\nu},$$

and so

$$\gamma_{\mu i} = 0 \ (\mu \neq i), \qquad c_{ik}\gamma_{ii} = c_{ik}\gamma_{kk}, \qquad \gamma_{ii} = \gamma_{kk} = \gamma;$$
$$z = \sum c_{ii}\gamma = e\gamma = \gamma.$$

Thus z lies in K and commutes with all x, hence z lies in $\mathfrak{Z}(K)$.

Since a matrix ring is of course also left completely reducible, and since every right completely reducible ring with identity element is a direct sum of matrix rings, it follows that:

Every right completely reducible ring with identity element is also left completely reducible, and conversely.

And: *The center of a completely reducible ring with identity element is a direct sum of commutative fields corresponding to the two-sided simple matrix rings.*

There remains the following theorem:

Any two different decompositions $\mathfrak{o} = \sum c_{ik}K$, $\mathfrak{o} = \sum c'_{ik}K'$ are transformed into one another by inner automorphisms $\alpha' \mapsto x^{-1}\alpha' x$, $c'_{ik} \mapsto x^{-1}c_{ik}x$.

Proof. Let

$$\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_n = \mathfrak{r}'_1 + \cdots + \mathfrak{r}'_n,$$

and let a, b be elements mediating two reciprocal isomorphisms of the ideals $\mathfrak{r}_1, \mathfrak{r}'_1$:

$$\mathfrak{r}_1 = a\mathfrak{r}'_1, \qquad \mathfrak{r}'_1 = b\mathfrak{r}_1, \qquad ab = c_{11}, \qquad ba = c'_{11}.$$

Put

$$x = \sum_i c_{i1}ac'_{1i}, \qquad y = \sum_i c'_{i1}bc_{1i}.$$

Then

$$xy = e, \qquad y = x^{-1}, \qquad x^{-1}c_{ik}x = c'_{ik}, \qquad x^{-1}\alpha x = \alpha'.$$

34. Hypercomplex Quantities and Representation Theory

Continuation and completion: Chapter III, §§15–19, and Chapter IV, §§20–26.

Chapter III. Module and representation theory

§ 15. Representations and representation modules

Let $\mathfrak{o}$ be a ring and let K be a ring with identity element. In later applications K will always be a field.

A representation of degree n of $\mathfrak{o}$ in K is a homomorphism

$$\mathfrak{o} \sim \mathfrak{D},$$

where $\mathfrak{D}$ is a ring of matrices of degree n over K.

By a *representation module* of $\mathfrak{o}$ with respect to K one understands a double module $\mathfrak{M}$ which is a left $\mathfrak{o}$-module and a right K-module,

$$\mathfrak{o}\mathfrak{M} \subseteq \mathfrak{M}, \qquad \mathfrak{M}K \subseteq \mathfrak{M},$$

which is also a direct sum of finitely many one-membered K-modules,

$$\mathfrak{M} = x_1K + \cdots + x_nK,$$

and in which the identity element of K is the identity operator.

Every representation module leads to a representation. Let $c \in \mathfrak{o}$ and

$$cx_k = \sum_i x_i\gamma_{ik},$$

or, in matrix form,

$$c(x_1, \ldots, x_n) = (x_1, \ldots, x_n)C, \qquad C = (\gamma_{ik}).$$

Then the matrices C form a representation of c, for

$$(b + c)x_k = \sum_i x_i(\beta_{ik} + \gamma_{ik}),$$

and

$$\begin{aligned} bcx_k &= b\sum_j x_j\gamma_{jk} = \sum_j bx_j\gamma_{jk} = \sum_j \sum_i x_i\beta_{ij}\gamma_{jk} \\ &= \sum_i x_i\Big(\sum_j \beta_{ij}\gamma_{jk}\Big), \end{aligned}$$

or

$$bc(x_1, \ldots, x_n) = (x_1, \ldots, x_n)BC.$$

Conversely, every representation $\mathfrak{D}$ of $\mathfrak{o}$ belongs to a representation module, indeed to a definite basis of that module. Namely, let $\mathfrak{M}$ be the totality of the formal linear forms

$$y = \sum_i x_i\alpha_i, \qquad \alpha_i \in K.$$

Then $\mathfrak{M}$ is a right K-module. Define further, if the element c is assigned the matrix $C = (\gamma_{ik})$,

$$cx_k = \sum_i x_i\gamma_{ik}, \qquad c\Big(\sum_k x_k\alpha_k\Big) = \sum_i x_i \sum_k \gamma_{ik}\alpha_k.$$

From the homomorphism relations

$$c + d \sim C + D, \qquad cd \sim CD$$

it follows that

$$(c + d)x_i = cx_i + dx_i, \qquad cdx_i = c(dx_i),$$

and the same is true for sums $\sum_i x_i\alpha_i$. The remaining double-module laws

$$c(y + z) = cy + cz, \qquad c(y\alpha) = (cy)\alpha$$

are trivial. Thus $\mathfrak{M}$ is a representation module and, by construction, belongs exactly to the representation $\mathfrak{o} \sim \mathfrak{D}$.

Let $(y_1, \ldots, y_n)$ be another basis for the same representation module:

$$(y_1, \ldots, y_n) = (x_1, \ldots, x_n)P, \qquad (x_1, \ldots, x_n) = (y_1, \ldots, y_n)P^{-1}.$$

If $b \sim B$ with respect to the x-basis, then with respect to the y-basis

$$b(y_1, \ldots, y_n) = b(x_1, \ldots, x_n)P = (x_1, \ldots, x_n)BP = (y_1, \ldots, y_n)P^{-1}BP.$$

One calls the representations $b \sim B$ and $b \sim P^{-1}BP$ *equivalent* and counts them in the same representation class. Since every regular matrix P transforms one basis of $\mathfrak{M}$ into another, we have proved:

Every representation module leads uniquely to one definite representation class.[15a)]

15) Representation modules with respect to commutative fields occur first in W. Krull, *Theorie und Anwendung der verallgemeinerten Abelschen Gruppen*, Sitzungsberichte der Heidelberger Akademie, 1926, 1st treatise.

15a) Added in proof (14 April 1929). As B. L. van der Waerden informed me, one can obtain an invariant connection independent of the special choice of basis by separating the notions "linear transformation" and "matrix." A linear transformation is a homomorphism of two modules of linear forms; a matrix is the expression, or representation, of this homomorphism for a particular choice of basis. I. Every representation module assigns to the multiplier domain $\mathfrak{o}$ a unique homomorphic system of linear transformations of the module into itself. Indeed, by the end of §2 the left multipliers of a double module $\mathfrak{M}$ generate operator homomorphisms of $\mathfrak{M}$ into itself with respect to the right operators (here multiplication by K), and the assignment is homomorphic. II. Every system of linear transformations of a linear-form module into itself that is homomorphic to $\mathfrak{o}$ leads to a representation module.

It is clear that two operator-isomorphic representation modules correspond to the same representation class. Conversely, if two representation modules $(x_1, \ldots, x_n)$ and $(\bar{x}_1, \ldots, \bar{x}_n)$ generate the same representation, then the assignment

$$x_i \mapsto \bar{x}_i, \qquad \sum_i x_i\alpha_i \mapsto \sum_i \bar{x}_i\alpha_i$$

gives an isomorphism. For the multiplication rule is completely determined by the matrices C.

Special case: if two bases of $\mathfrak{M}$ generate the same representation, they are carried into one another by an operator isomorphism of $\mathfrak{M}$ onto itself.

Equivalently, if $C = PCP^{-1}$ for all C and one fixed P, then the assignment

$$y_i \mapsto x_i, \qquad y_i = \sum_k x_k\pi_{ik}$$

is an isomorphism of $\mathfrak{M}$ onto itself. Conversely, every isomorphism $y_i \mapsto x_i$ of the double module $\mathfrak{M}$ onto itself is mediated by a matrix P which commutes with all C.

One may also extend the notion of representation to rings which are still commutatively linked with a commutative multiplier domain P (§8). In that case one takes only such representation rings K as contain P in their center, and one demands of the representation not merely that it be a ring homomorphism $\mathfrak{o} \sim \mathfrak{D}$, but that it be an operator homomorphism: from $a \sim A$ there should follow

$$a\rho \sim A\rho \qquad (\rho \in P).$$

For the representation modules this requirement means the additional rule

$$am \cdot \rho = a(m\rho) = a\rho \cdot m.$$

The module and the ring are then said to be *commutatively connected with* P.

§ 16. Reducible representations

If $\mathfrak{U}$ is a submodule of the representation module $\mathfrak{M}$, and if it is possible to choose a basis for $\mathfrak{M}$ consisting of a basis $z_1, \ldots, z_t$ of $\mathfrak{U}$, supplemented by $y_1, \ldots, y_r$, thus

$$\mathfrak{M} = y_1K + \cdots + y_rK + z_1K + \cdots + z_tK, \tag{2a}$$

then the representations have the form

$$C = \begin{pmatrix} R & 0 \\ S & T \end{pmatrix},$$

where the matrices T by themselves form a representation of degree t mediated by $\mathfrak{U}$, and the matrices R form a representation of degree r mediated by $\mathfrak{M}/\mathfrak{U}$.

Indeed,

$$(cy_1, \ldots, cy_r, cz_1, \ldots, cz_t) = (y_1, \ldots, y_r, z_1, \ldots, z_t)C.$$

Since the cz_i, as elements of $\mathfrak{U}$, are expressible through the z_i alone, the upper-right block of C is zero. If the other parts of C are named R, S, T in the order shown above, then in particular

$$(cz_1, \ldots, cz_t) = (z_1, \ldots, z_t)T;$$

hence the T form a representation mediated by $\mathfrak{U}$. Further,

$$(cy_1, \ldots, cy_r) \equiv (y_1, \ldots, y_r)R \pmod{\mathfrak{U}},$$

while the y_i form a basis linearly independent modulo $\mathfrak{U}$. Thus the R form a representation generated by $\mathfrak{M}/\mathfrak{U}$.

Conversely, if a "reducible" representation

$$C = \begin{pmatrix} R & 0 \\ S & T \end{pmatrix}$$

is given, with R and T square matrices, then in the associated representation module the products of each c with the last t basis elements $z_1, \dots, z_t$ are expressed through those elements alone; hence $\mathfrak{U} = (z_1, \dots, z_t)$ is a submodule.

Corollary. Let

$$\mathfrak{M} \supset \mathfrak{U}_1 \supset \mathfrak{U}_2 \supset \cdots \supset \mathfrak{U}_e = 0$$

be a composition series for $\mathfrak{M}$, and suppose that each $\mathfrak{U}_i$ is a direct summand of the preceding one as a K-module. Then one can choose a basis

$$z_{11}, \dots, z_{1r_1}, \quad z_{21}, \dots, z_{2r_2}, \quad \dots \quad z_{e1}, \dots, z_{er_e}$$

such that the z_{ik} with $i > \nu$ form a basis of $\mathfrak{U}_\nu$. The representations mediated by $\mathfrak{M}$ then have the form

$$C = \begin{pmatrix} R_{11} & 0 & \cdots & 0 \\ R_{12} & R_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ R_{1e} & R_{2e} & \cdots & R_{ee} \end{pmatrix}, \tag{2}$$

and the $R_{\nu\nu}$ form representations mediated by the composition factors $\mathfrak{U}_{\nu-1}/\mathfrak{U}_\nu$.

The individual representations $R_{\nu\nu}$ are irreducible, since the composition factors $\mathfrak{U}_{\nu-1}/\mathfrak{U}_\nu$ are simple.

If K is assumed to be a field, so that every submodule is a direct summand, then conversely every representation reduced as far as possible in the form (2) leads to a composition series. By the Jordan-Hölder theorem the composition factors are uniquely determined up to operator isomorphism. Hence the $R_{\nu\nu}$ are uniquely determined up to equivalent representations and up to their order.[16)]

16) In other words: if $\mathfrak{U}$ is a direct summand as a K-module. If K is a field, this hypothesis is always fulfilled.

If $\mathfrak{M} = \mathfrak{A} + \mathfrak{B}$ is the direct sum of two representation modules $\mathfrak{A} = (y_1, \dots, y_r)$ and $\mathfrak{B} = (z_1, \dots, z_s)$, then the representation mediated by the basis $(y_1, \dots, y_r, z_1, \dots, z_s)$ has the form

$$C = \begin{pmatrix} R & 0 \\ 0 & S \end{pmatrix},$$

where R denotes the representation of the element c mediated by $\mathfrak{A}$, and S the one mediated by $\mathfrak{B}$; conversely as well. If one first writes $\mathfrak{M}$ as a direct sum of directly indecomposable components and draws composition series in these as above, then for a suitable basis the representation has a block form whose large boxes correspond to the directly indecomposable components and whose diagonal subblocks correspond to the composition factors.

If again K is a field, every representation reduced as far as possible in this way conversely leads to a representation of the module by directly indecomposable summands and composition factors within them. By the theorem of Krull and Otto Schmidt, the classes of directly indecomposable representation components are uniquely determined up to order.

If the module $\mathfrak{M}$ is completely reducible, all boxes in the block form consist of a single irreducible representation, and one calls the representation completely reducible.

§ 17. Direct sum decompositions of representation modules for rings with identity element

Let $\mathfrak{M}$ be an $\mathfrak{o}$-module, with $\mathfrak{o}$ a ring with identity. Then

$$\mathfrak{M} = \mathfrak{M}_e + \mathfrak{M}_0,$$

where the identity is the identity operator on $\mathfrak{M}_e$ and the zero operator on $\mathfrak{M}_0$. If $\mathfrak{M}$ is also a right K-module, then $\mathfrak{M}_e$ and $\mathfrak{M}_0$ are also right K-modules.

Proof. Every $m \in \mathfrak{M}$ can be written as

$$m = em + (m - em).$$

Let the set of the em be $\mathfrak{M}_e$, and the set of the $m - em$ be $\mathfrak{M}_0$. These are plainly additive groups, and right K-modules if $\mathfrak{M}$ is one. Moreover

$$r(em) = erm,$$

so $\mathfrak{M}_e$ is also an $\mathfrak{o}$-module; and

$$r(m - em) = rm - rem = rm - rm,$$

so $\mathfrak{M}_0$ is an $\mathfrak{o}$-module. For the em, e is identity operator; for the $m - em$, it is zero operator. Thus only zero belongs to both $\mathfrak{M}_e$ and $\mathfrak{M}_0$, and the sum $\mathfrak{M}_e + \mathfrak{M}_0$ is direct.

For representation modules, $\mathfrak{M}_0$ gives the trivial representation, in which every element is assigned zero. Splitting off the summand $\mathfrak{M}_0$, there remains the representation mediated by $\mathfrak{M}_e$, in which the identity element is assigned the identity matrix. We can and will therefore restrict ourselves henceforth to such representations.

Let

$$\mathfrak{o} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_s$$

be a direct sum of two-sided ideals,

$$e = e_1 + \cdots + e_s$$

the corresponding decomposition of e, and let $\mathfrak{M}$ be an $\mathfrak{o}$-module on which e is identity operator. Then

$$\mathfrak{M} = \mathfrak{a}_1\mathfrak{M} + \cdots + \mathfrak{a}_s\mathfrak{M}$$

directly.

Indeed, plainly $\mathfrak{M} = \mathfrak{o}\mathfrak{M} = (\mathfrak{a}_1\mathfrak{M}, \ldots, \mathfrak{a}_s\mathfrak{M})$. Put $\mathfrak{b}_i = \mathfrak{a}_1 + \cdots + \widehat{\mathfrak{a}_i} + \cdots + \mathfrak{a}_s$. Then correspondingly $\mathfrak{b}_i$ is a two-sided ideal and

$$\mathfrak{M} = (\mathfrak{a}_i\mathfrak{M}, \mathfrak{b}_i\mathfrak{M}),$$

and this sum is direct, since e_i is identity operator on $\mathfrak{a}_i\mathfrak{M}$ and zero operator on $\mathfrak{b}_i\mathfrak{M}$.

On the basis of this theorem we shall usually restrict ourselves to representations of two-sided indecomposable rings.

§ 18. Module and representation theory of completely reducible rings

Let $\mathfrak{o}$ be completely reducible and two-sided simple, and let $\mathfrak{M}$ be a finite $\mathfrak{o}$-module for which the identity element of $\mathfrak{o}$ is identity operator. Then $\mathfrak{M}$ is completely reducible, and its simple constituents are operator-isomorphic to the simple left ideals $\mathfrak{l}_i$.

Proof. Let

$$\mathfrak{o} = \mathfrak{l}_1 + \cdots + \mathfrak{l}_n, \qquad \mathfrak{M} = (\mathfrak{o}m_1, \ldots, \mathfrak{o}m_k),$$

and therefore

$$\mathfrak{M} = (\ldots, \mathfrak{l}_i m_k, \ldots). \tag{4}$$

The non-zero $\mathfrak{l}_i m_k$ are isomorphic to $\mathfrak{l}_i$, since the assignment $a \mapsto am_k$ is an operator isomorphism. Hence the $\mathfrak{l}_i m_k$ are simple. Omitting from (4) those $\mathfrak{l}_i m_k$ already contained in the sum of the preceding ones, the sum becomes direct.

Consequently, if in addition $\mathfrak{M}$ is directly indecomposable, then $\mathfrak{M}$ is simple and isomorphic to some $\mathfrak{l}_i$.

Let Δ be the automorphism field of this simple $\mathfrak{M}$, and let Λ be that of $\mathfrak{l}_i$. Then, by the operator isomorphism of $\mathfrak{M}$ and $\mathfrak{l}_i$, we have the ring isomorphism $\Delta \simeq \Lambda$. By §1 one can regard $\mathfrak{M}$ and $\mathfrak{l}_i$ as double modules with Δ and Λ, respectively, as right domains; they are then also representation modules. The representations mediated by them therefore pass into one another under the ring isomorphism $\Delta \simeq \Lambda$. Thus for the study of this representation class we may restrict ourselves to the representation mediated by $\mathfrak{l}_i$ in its automorphism field.

Specifically, put a basis of matrix units for $\mathfrak{l}_1$ in the form

$$\mathfrak{l}_1 = (c_{11}, \ldots, c_{n1})$$

and realize the automorphism field Λ of the earlier §14 by the opposite field of $\mathfrak{o}$; in the earlier considerations right ideals are to be replaced by left ideals and automorphisms written on the right. If

$$c = \sum_{i,k} c_{ik}\alpha_{ik}$$

is an element of $\mathfrak{o}$, then

$$(cc_{11}, \dots, cc_{n1}) = (c_{11}, \dots, c_{n1})(\alpha_{ik}).$$

Thus the representation mediated by $\mathfrak{l}_1$ is given by the matrix (α_{ik}). For $\mathfrak{l}_\nu = (c_{1\nu}, \dots, c_{n\nu})$ one obtains exactly the same representation.

If $\mathfrak{o}$ is a direct sum of two-sided simple components, then these components are represented in turn isomorphically, while the remaining components act by zero.

§ 19. The simple composition factors in modules and representation modules

Let $\mathfrak{o}$ be a ring with identity, and let $\mathfrak{c}$ be its radical. In every simple module and every simple representation module over $\mathfrak{o}$, the elements of $\mathfrak{c}$ have annihilator effect; hence one obtains a module or representation module for $\mathfrak{o}/\mathfrak{c}$.

Indeed, $\mathfrak{c}\mathfrak{M}$ is again a module (and, in the representation case, also a right K-module). Since $\mathfrak{M}$ is simple, either $\mathfrak{c}\mathfrak{M} = \mathfrak{M}$ or $\mathfrak{c}\mathfrak{M} = 0$. In the former case $\mathfrak{c}^\rho\mathfrak{M} = \mathfrak{M}$ for every exponent ρ, contradicting nilpotence of $\mathfrak{c}$. Thus $\mathfrak{c}\mathfrak{M} = 0$.

It follows that the simple composition factors of arbitrary modules and representation modules are also generated by simple left ideals of $\mathfrak{o}/\mathfrak{c}$.

For representation modules commutatively connected with P, this means in particular that, on reducing an arbitrary representation by means of a composition series as in §16, only the diagonal representations mediated by simple left ideals of $\mathfrak{o}/\mathfrak{c}$ occur, apart from zeros.

No finiteness condition on $\mathfrak{o}$ beyond the hypotheses already placed on the representation over P is needed for these facts. Every representation is an isomorphic representation of its absolute multiplier domain. This absolute multiplier domain, as inverse image of a finite matrix ring, is itself a finite-rank P-module. Since the permitted ideals are by definition P-modules, the maximal and minimal conditions hold for it.

If P is a commutative field, this absolute multiplier domain is a hypercomplex system over P. This assumption is automatically fulfilled whenever non-zero representations occur on the diagonal: then P is isomorphic to a subfield of the automorphism field of a simple left ideal, and, in realization by the subfield K of $\mathfrak{o}/\mathfrak{c}$, the commutative connection forces it into the center of K.

Chapter IV. Representations of groups and hypercomplex systems

§ 20. Inclusion of hypercomplex systems

We now consider hypercomplex systems $\mathfrak{o}$ with respect to a commutative field P. By convention, only P-modules are counted as ideals in $\mathfrak{o}$. Thus the maximal and minimal conditions for left and right ideals are fulfilled, and the entire ring theory of Chapter II applies.

In what follows, by a representation of the hypercomplex system $\mathfrak{o}$ we always mean one in the field P, and more precisely one satisfying: from $a \sim A$ follows

$$a\rho \sim A\rho \qquad (\rho \in P),$$

i.e. module homomorphy with respect to P. For representation modules this reads

$$a\rho \cdot m = a(m\rho).$$

The results of §19 therefore apply. All irreducible representations are mediated by simple left ideals of

$$\mathfrak{o}_0 = \mathfrak{o}/\mathfrak{c},$$

where $\mathfrak{c}$ is the radical. Thus there are as many inequivalent irreducible representations as there are two-sided simple summands $\mathfrak{a}$ in the decomposition

$$\mathfrak{o}_0 = \mathfrak{a}_1 + \dots + \mathfrak{a}_s.$$

To determine explicitly the representation defined by such a left ideal, one first passes from the elements of $\mathfrak{o}$ to their residue classes in $\mathfrak{o}_0$. Multiplication by an element of P is an isomorphism for all ideals of $\mathfrak{o}_0$; hence P is isomorphic to a subfield $e^{(\nu)}P$ of the automorphism field K_ν of every simple left ideal. The representation mediated by $\mathfrak{l}_\nu$ over P is found by first considering the representation over this subfield $e^{(\nu)}P$

and then transferring it to P by the isomorphism $e^{(\nu)}P \simeq P$. The field K_ν has finite rank over P, so this is a representation in a finite-degree subfield of the automorphism field of $\mathfrak{l}_\nu$.

Let

$$c = c_1 + \cdots + c_s$$

be the decomposition of an element of $\mathfrak{o}_0$ into its two-sided components. For the representation mediated by $\mathfrak{l}_\nu$, only the matrix of c_ν is considered, since the other c_i annihilate $\mathfrak{l}_\nu$ and are represented by zero matrices. If one writes c_ν in matrix units as

$$c_\nu = \sum c_{ik}^{(\nu)} \alpha_{ik}^{(\nu)},$$

then every $\alpha_{ik}^{(\nu)}$ is to be replaced by the matrix that represents it in the representation of K_ν over P.

Since K_ν has finite rank over P, every element x of K_ν satisfies an equation $f(x) = 0$ with coefficients in $e^{(\nu)}P$, because its powers are linearly dependent. If P is algebraically closed, the equation splits into linear factors; since K_ν is a field, x is already a root of one of the linear factors, and hence lies in $e^{(\nu)}P$. Thus $K_\nu = e^{(\nu)}P$. In this case the matrices $(\alpha_{ik}^{(\nu)})$ themselves are the irreducible representations.

Together with the end of §19 this gives Burnside's theorem:

Let P be algebraically closed and commutatively connected with a ring $\mathfrak{o}$. Then every irreducible representation of degree n over P contains exactly n^2 linearly independent matrices.

From this follows at once the customary formulation: a system of matrices of degree n over P, closed under multiplication, is irreducible if and only if no non-trivial homogeneous linear relation

$$\sum \alpha_{ik} x_{ik} = 0$$

with coefficients in P exists that is satisfied by the entries x_{ik} of all the matrices. Indeed, adjoining all linear combinations, one obtains a ring and a P-module and can regard this ring as its own representation.

For the absolute multiplier domain is isomorphic to a two-sided simple completely reducible ring with identity, hence to a full matrix ring over the automorphism field of its left ideals. The algebraic closedness of P makes that automorphism field equal to P. The irreducible representation is generated by a simple left ideal. If this ideal has rank m, and the representation has degree n, then $m = n$, and in the matrix ring there are exactly n^2 linearly independent elements.

Similarly, the generalized Burnside theorem follows. Under the same hypotheses, if $\mathfrak{D}$ is a completely reducible representation decomposed into inequivalent components of degrees

$$n_1, n_2, \ldots, n_s,$$

then $\mathfrak{D}$ has rank

$$n_1^2 + n_2^2 + \cdots + n_s^2$$

with respect to P.

If $\mathfrak{o}$ is a hypercomplex system without radical over an arbitrary field, then $\mathfrak{o}$ is completely reducible and has an identity element. From §§17 and 18 it follows that every representation of $\mathfrak{o}$ is completely reducible.

Conversely, every completely reducible representation of a ring $\mathfrak{o}$ commutatively connected with P has as absolute multiplier domain a hypercomplex system without radical. For the absolute multiplier domain is isomorphic to a ring and P-module of matrices over P, hence is a hypercomplex system. The elements of its radical are represented by zero in every irreducible constituent, hence in the whole representation; therefore the radical is zero.

The *regular representation* of a hypercomplex system $\mathfrak{o}$ is the representation mediated by the unit ideal $\mathfrak{o}$ itself. For a group ring, one chooses specifically the group elements as basis. Since all left ideals of $\mathfrak{o}/\mathfrak{c}$ occur as composition factors in $\mathfrak{o}$, all irreducible representations already occur on the diagonal of the regular representation after the regular representation is reduced by means of a composition series, as in §16.

A representation is known when the representing matrices for the basis elements $u_1, \ldots, u_h$ are known. If the system is a group ring, so that the u_i are group elements, then every homomorphic matrix representation of the group is also a representation of the group ring. Thus the problem of representing a group is a special case of representing hypercomplex systems.

§ 21. Extension of the ground field. Representations of the center

Let

$$\mathfrak{o} = a_1 P + \cdots + a_h P$$

be a hypercomplex system, and let Ω be an extension field of P. One can form

$$\mathfrak{o}\Omega = a_1\Omega + \cdots + a_h\Omega$$

with the old multiplication rules for the basis elements.[18a] Then $\mathfrak{o}\Omega$ is again hypercomplex with respect to Ω.

Every representation of $\mathfrak{o}$ leads to a representation of $\mathfrak{o}\Omega$, since the representation of all elements is known as soon as the representations of the basis elements a_i are known.[19]

18a) More precisely: one introduces new symbols a_i with the old multiplication rules; $a_1\Omega + \cdots + a_h\Omega$ then contains a subring isomorphic to $\mathfrak{o}$, so that the new symbols may afterwards be identified with the old a_i.

19) Of course, here again only representations which are P-module homomorphisms are considered: from $a \sim A$ should follow $a\rho \sim A\rho$ for $\rho \in P$, and correspondingly for Ω.

An ideal or representation module, as well as the corresponding representation, is called *absolutely irreducible* if it remains irreducible after passage to the algebraically closed field.

If $\mathfrak{o}\Omega$ is without radical, then so is $\mathfrak{o}$; for a nilpotent ideal $\mathfrak{c}$ in $\mathfrak{o}$ would lead to an extended ideal $\mathfrak{c}\Omega$ in $\mathfrak{o}\Omega$. The converse is not true, as will be seen below.

The center of $\mathfrak{o}\Omega$ is equal to the extended center $\mathfrak{Z}\Omega$. If $\mathfrak{o}$ is completely reducible, then $\mathfrak{Z}$ is a direct sum of fields:

$$\mathfrak{Z} = \mathfrak{z}_1 + \cdots + \mathfrak{z}_s.$$

If $\mathfrak{o}\Omega$ remains completely reducible upon passage to Ω (that is, if no nilpotent ideal is added), then the new center decomposes as

$$\mathfrak{Z}\Omega = \mathfrak{z}_1\Omega + \cdots + \mathfrak{z}_s\Omega$$

again into a direct sum of fields. If Ω is algebraically closed, these fields have rank 1 over Ω and are isomorphic to Ω.

For the representations of the center $\mathfrak{Z}$ the following theorems hold.

Every irreducible representation of a commutative hypercomplex system $\mathfrak{Z}$ in the algebraically closed field Ω is of degree 1; equivalently, the irreducible representations are identical with the homomorphisms $\mathfrak{Z} \to \Omega$.

Proof. Every irreducible representation of $\mathfrak{Z}$ yields one of $\mathfrak{Z}\Omega$, hence also one of $\mathfrak{Z}\Omega/\mathfrak{c}$, where $\mathfrak{c}$ is the radical of $\mathfrak{Z}\Omega$. The quotient $\mathfrak{Z}\Omega/\mathfrak{c}$ is a commutative system without radical, hence by the end of §14 a direct sum of fields. These are of degree 1 because Ω is algebraically closed, and they generate all irreducible representations (§20).

At the same time, since equivalent representations of degree 1 necessarily become equal, the number of distinct homomorphisms $\mathfrak{Z} \to \Omega$ is equal to the rank of $\mathfrak{Z}\Omega/\mathfrak{c}$.

In the special case that $\mathfrak{Z}$ is a field over P, this gives the theorem: $\mathfrak{Z}$ is completely reducible, i.e. without radical, if and only if $\mathfrak{Z}$ is an extension of the first kind of P. For a field the homomorphisms are isomorphisms. Their number is the rank of $\mathfrak{Z}\Omega/\mathfrak{c}$, and it is equal to the field degree precisely when $\mathfrak{c} = 0$.

If

$$\mathfrak{Z} = \mathfrak{z}_1 + \cdots + \mathfrak{z}_s$$

is completely reducible, then in each representation the separate fields $\mathfrak{z}_i$ are also represented homomorphically; in an irreducible representation one of these fields is mapped isomorphically and the others by zero.

For systems without radical, the application of these facts to hypercomplex systems gives first:

If $\mathfrak{o}\Omega$, and therefore also $\mathfrak{o}$, is a system without radical, then in every absolutely irreducible representation of $\mathfrak{o}$ the central elements z are represented by diagonal matrices $E\xi$, and $z \mapsto \xi$ is a representation of degree 1 of $\mathfrak{Z}$ in Ω. The resulting correspondence between the absolutely irreducible representation classes of $\mathfrak{o}$ and those of $\mathfrak{Z}$ is one-to-one. Thus the number of these representation classes is equal to the rank of the center.[20]

20) Corresponding theorems hold not only for representations in the algebraically closed field Ω, but also for representations in the individual automorphism fields; this will be carried out elsewhere.

Indeed, the decomposition into two-sided indecomposable ideals

$$\mathfrak{o}\Omega = \mathfrak{a}_1 + \cdots + \mathfrak{a}_s$$

corresponds one-to-one to a decomposition of the center into fields

$$\mathfrak{Z}\Omega = \mathfrak{z}_1 + \cdots + \mathfrak{z}_s,$$

where $\mathfrak{z}_i$ is the center of $\mathfrak{a}_i$. In an irreducible representation of $\mathfrak{o}$, one $\mathfrak{a}_i$ is mapped isomorphically and the others by zero. The representation class is uniquely determined by $\mathfrak{a}_i$. The component $\mathfrak{a}_i$ is represented by a full matrix ring, and the center of such a full matrix ring consists of diagonal matrices $E\xi$. The assignment $z \mapsto \xi$ is a homomorphism of $\mathfrak{Z}$ in which one $\mathfrak{z}_i$ is mapped isomorphically and the others by zero. This proves the assertion.

For the question when a radical-free $\mathfrak{o}$ gives rise to a radical-free $\mathfrak{o}\Omega$, it follows that if $\mathfrak{Z}$ has a component $\mathfrak{Z}_i$ which is an extension of the second kind of P, then $\mathfrak{o}\Omega$ certainly has a radical.[21)] For $\mathfrak{Z}_i\Omega$ already has one in that case.

21) If $\mathfrak{Z}$ has only components of the first kind, then $\mathfrak{o}\Omega$ is without radical, as will likewise be proved later. For characteristic zero I. Schur already proved the equivalent theorem that irreducible representations over P remain completely reducible under every extension of the coefficient domain; see I. Schur, Beiträge zur Theorie der Gruppen linearer Substitutionen, *Trans. Amer. Math. Soc.* 15 (1909), p. 159.

§ 22. Application to abelian groups

Let

$$G = G_1 \times G_2 \times \cdots \times G_r$$

be a finite abelian group, decomposed into cyclic groups G_i of orders h_i. Its elements are

$$a = a_1^{\alpha_1} \cdots a_r^{\alpha_r}.$$

Let P be a field whose characteristic does not divide the group order

$$h = h_1 h_2 \cdots h_r.$$

The group ring $\mathfrak{Z}$ consists of all sums

$$\sum_{\alpha_1,\dots,\alpha_r} A_{\alpha_1\dots\alpha_r} a_1^{\alpha_1} \cdots a_r^{\alpha_r} \qquad (A \in P)$$

and is a homomorphic image of the polynomial ring $P[x_1,\dots,x_r]$ through $x_i \mapsto a_i$. Thus

$$\mathfrak{Z} \simeq P[x_1,\dots,x_r]/(x_1^{h_1}-1,\dots,x_r^{h_r}-1).$$

In the extension field Ω the polynomials $x_i^{h_i}-1$ split into distinct factors $x_i - \xi_i$, so the defining ideal is the product, or intersection, of distinct maximal ideals

$$(x_1 - \xi_1^{(\alpha_1)},\dots,x_r - \xi_r^{(\alpha_r)}),$$

one for each zero $(\xi_1^{(\alpha_1)},\dots,\xi_r^{(\alpha_r)})$. To this intersection corresponds a direct-sum decomposition

$$\mathfrak{Z}\Omega = \mathfrak{Z}_1 + \cdots + \mathfrak{Z}_h,$$

where each component is isomorphic to Ω and gives the representation

$$a_i \mapsto \xi_i^{(\alpha_i)}, \qquad \text{or more generally} \qquad a \mapsto \chi^{(\alpha)}(a).$$

These representations are the characters. Their number is $h_1 \cdots h_r = h$, as the general theory requires.

§ 23. Determinant of a hypercomplex system

Let

$$\mathfrak{o} = a_1 P + \cdots + a_h P$$

be a hypercomplex system. Adjoin to P the h indeterminates $x_1,\dots,x_h$ and form

$$\mathfrak{o}^* = a_1 P(x) + \cdots + a_h P(x),$$

where the old multiplication rules are used for the basis elements a_i. The x_i are to commute with the a_i, thereby fixing the rules of calculation in $\mathfrak{o}^*$.

In $\mathfrak{o}^*$ lies the "general element" of $\mathfrak{o}$,

$$w = a_1 x_1 + \cdots + a_h x_h.$$

If in a representation $a_i \mapsto A_i$, then w is assigned

$$W = A_1 x_1 + \cdots + A_h x_h.$$

The matrix W is called the system matrix belonging to the representation; if the a_i form a group and $\mathfrak{o}$ is the group ring, it is the group matrix. In the regular representation one has the regular system matrix.

The entries of W are linear forms in the x_i. The system determinant $|W|$ is therefore of degree n for a representation of degree n. In particular, the regular system determinant has degree h.

The system determinant is unchanged under passage to equivalent representations, since

$$|PWP^{-1}| = |P|\,|W|\,|P^{-1}| = |W|.$$

When one passes from the basis $(a_1, \ldots, a_h)$ to a new basis $(b_1, \ldots, b_h)$ and from $w = \sum a_i x_i$ to $w = \sum b_j y_j$, the new elements of W, and hence also the new determinant, are obtained from the old ones by a regular substitution of variables

$$x_i = \sum_j \rho_{ij} y_j.$$

If a composition series of the representation module is given, then for a suitable choice of basis the matrix W has lower block-triangular form,

$$W = \begin{pmatrix} W_1 & 0 & \cdots & 0 \\ * & W_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ * & * & \cdots & W_s \end{pmatrix}.$$

Among the determinants $|W_i|$ of an arbitrary representation, no other factors occur than those occurring in the regular representation of $\mathfrak{o}$, or even of $\mathfrak{o}/\mathfrak{c}$, where $\mathfrak{c}$ is the radical.

If P is algebraically closed, then the determinant $|W_i|$ belonging to an irreducible representation is a prime function of the x_i, and inequivalent representations give different prime factors.

Proof. Since all irreducible representations of $\mathfrak{o}$ are also representations of $\mathfrak{o}/\mathfrak{c}$, we may restrict ourselves to the ring without radical $\mathfrak{o}/\mathfrak{c}$. In it take as basis the matrix units $c_{ik}^{(\nu)}$; the general element is then

$$w = \sum_{\nu,i,k} c_{ik}^{(\nu)} x_{ik}^{(\nu)}.$$

The matrices of the irreducible representations are

$$W_\nu = (x_{ik}^{(\nu)}).$$

The functions $|W_\nu| = |x_{ik}^{(\nu)}|$ are known to be irreducible and are plainly distinct.

To compute the $|W_i|$, one can factor the regular system matrix of $\mathfrak{o}/\mathfrak{c}$. Each prime factor occurs as often as the degree of the corresponding irreducible representation, since the corresponding ideal occurs that many times in the composition series.

One can also start from the regular system matrix of $\mathfrak{o}$, but then each irreducible factor is obtained more often, namely as often as the corresponding simple ideal occurs as a composition factor among the left ideals. In this regular representation $\mathfrak{o}$ has been regarded as a left ideal. If $\mathfrak{o}$ is regarded as a right ideal, a second regular system matrix appears, the “antistrophic matrix” of Frobenius; it contains the same irreducible factors, namely the system determinants of all irreducible representations, though possibly with different exponents.

The system determinant of a commutative system splits into linear factors, because all irreducible representations have degree 1. These linear factors are the irreducible representations themselves and, for abelian groups, give the characters. This fact was Dedekind’s point of departure in his study of the group determinant of non-abelian groups.

§ 24. Traces and characters

If, in a representation $\mathfrak{o} \sim \mathfrak{D}$ of a hypercomplex system $\mathfrak{o}$, the element a is assigned the matrix A, put

$$\mathrm{Sp}_D(a) = \mathrm{Sp}\, A.$$

Traces are linear functions:

$$\mathrm{Sp}(c+d) = \mathrm{Sp}(c) + \mathrm{Sp}(d), \qquad \mathrm{Sp}(c\alpha) = \alpha\, \mathrm{Sp}(c).$$

Equivalent representations have the same traces. In a reducible representation the trace is the sum of the traces in the representations mediated by the composition factors. Indeed,

$$\mathrm{Sp}\begin{pmatrix} A_{11} & 0 & \cdots & 0 \\ * & A_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ * & * & \cdots & A_{ss} \end{pmatrix} = \mathrm{Sp}(A_{11}) + \mathrm{Sp}(A_{22}) + \cdots + \mathrm{Sp}(A_{ss}).$$

The *principal trace* is the trace in the regular representation. The *reduced trace* is the sum of the traces in the distinct irreducible representations.

If c is an element of the maximal nilpotent ideal $\mathfrak{c}$, then $\mathrm{Sp}(c) = 0$ for every representation.

Proof. It suffices to consider representations through the simple left ideals of $\mathfrak{o}/\mathfrak{c}$, since in any other representation only these composition factors occur, and the trace is additively composed. But c annihilates every element of $\mathfrak{o}/\mathfrak{c}$; hence the zero matrix is assigned to every $c \in \mathfrak{c}$, so $\mathrm{Sp}(c) = 0$.

If $\mathfrak{o}$ is two-sided simple and P algebraically closed, hence a matrix ring

$$\mathfrak{o} = \sum c_{ik} P,$$

then in the irreducible representation of $\mathfrak{o}$ the element

$$a = \sum c_{ik} \alpha_{ik}$$

is assigned the matrix (α_{ik}), so that

$$\mathrm{Sp}_{\mathrm{red}}(a) = \sum_i \alpha_{ii}, \qquad \mathrm{Sp}_{\mathrm{pr}}(a) = n\ \mathrm{Sp}_{\mathrm{red}}(a) = n \sum_i \alpha_{ii}.$$

In particular,

$$\mathrm{Sp}_{\mathrm{red}}(c_{ik}) = 0 \quad (i \neq k), \qquad \mathrm{Sp}_{\mathrm{red}}(c_{ii}) = e,$$

whereas

$$\mathrm{Sp}_{\mathrm{pr}}(c_{ik}) = 0 \quad (i \neq k), \qquad \mathrm{Sp}_{\mathrm{pr}}(c_{ii}) = ne.$$

Passing from P to the algebraically closed field Ω does not change the principal trace, since it can be computed from the same basis. This is not true of the reduced trace.

The traces of elements a of the system without radical $\mathfrak{o}$ in the absolutely irreducible representations are called characters and are denoted by $\chi(a)$, or by $\chi^{(\nu)}(a)$ when the representation is to be specified.

In an irreducible representation of degree n_ν, the central elements are represented, by §21, as diagonal matrices $E\theta^{(\nu)}$, where $z \mapsto \theta^{(\nu)}(z)$ is a representation of degree 1 of the center, or a homomorphism of the center into Ω. Therefore the trace has the value $n_\nu \theta^{(\nu)}(z)$. Thus the homomorphisms θ of the center are linked with the characters χ by

$$\chi^{(\nu)}(z) = n_\nu \theta^{(\nu)}(z). \tag{1}$$

In the commutative case $n_\nu = 1$, and the characters themselves give the homomorphisms.

If Ω has characteristic zero, as will always be assumed in what follows, one can divide (1) by n_ν:

$$\theta^{(\nu)}(z) = \frac{\chi^{(\nu)}(z)}{n_\nu}.$$

The homomorphism property of the $\theta^{(\nu)}(z)$ is expressed by

$$\frac{\chi^{(\nu)}(z+z')}{n_\nu} = \frac{\chi^{(\nu)}(z)}{n_\nu} + \frac{\chi^{(\nu)}(z')}{n_\nu},$$

and

$$\frac{\chi^{(\nu)}(zz')}{n_\nu} = \frac{\chi^{(\nu)}(z)}{n_\nu}\,\frac{\chi^{(\nu)}(z')}{n_\nu}.$$

A representation class is already uniquely determined by the traces of the matrices alone. To know the traces of all matrices it is of course enough to know the traces of the basis elements in the given representation.

Proof. The representation module R is known when one knows how often each irreducible representation module $\mathfrak{M}_\nu$ occurs in it as a direct summand. If this number is μ_ν, then the trace of $e^{(\nu)}$ in the representation is $\mu_\nu n_\nu$, where n_ν is the degree of the irreducible representation. Hence

$$\mu_\nu = \frac{\mathrm{Sp}(e^{(\nu)})}{n_\nu}.$$

§ 25. Discriminants

Let

$$\mathfrak{o} = a_1 P + \cdots + a_h P$$

be a hypercomplex system. The *discriminant matrix* is the matrix whose entries are the principal traces $\mathrm{Sp}(a_i a_j)$. The *reduced discriminant matrix* is the corresponding matrix for reduced traces, formed in the algebraically closed field. The determinant of the matrix is the discriminant, respectively the reduced discriminant.

The discriminant remains the same under extension of the ground field. When passing to another basis it is multiplied by the square of the transformation determinant.

Proof. Let

$$(b_1, \ldots, b_h) = (a_1, \ldots, a_h)P.$$

Then

$$(\mathrm{Sp}(a_i a_j)) = (\mathrm{Sp}(a_i b_j))P. \tag{1}$$

Likewise, if P^t denotes the transposed matrix,

$$(\mathrm{Sp}(a_i b_j)) = P^t(\mathrm{Sp}(b_i b_j)), \tag{1a}$$

and from (1) and (1a)

$$(\mathrm{Sp}(a_i a_j)) = P^t(\mathrm{Sp}(b_i b_j))P.$$

Passing to determinants gives the assertion.

The determinant is therefore determined only up to a square factor from P, but its vanishing or non-vanishing is an invariant property. It also follows from (1) that, up to a non-zero factor, the discriminant may be computed from two different bases, namely as $|\,\mathrm{Sp}(a_i b_j)|$.

The discriminant vanishes if $\mathfrak{o}$ possesses a nilpotent ideal $\mathfrak{c}$.

Proof. Choose basis elements for $\mathfrak{o}$ in the form

$$c_1, \ldots, c_r, d_1, \ldots, d_s,$$

where $c_1, \ldots, c_r$ form a basis of $\mathfrak{c}$. Then the discriminant matrix has the form

$$\begin{pmatrix} \mathrm{Sp}(c_i c_j) & \mathrm{Sp}(c_i d_j) \\ \mathrm{Sp}(d_i c_j) & \mathrm{Sp}(d_i d_j) \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & \mathrm{Sp}(d_i d_j) \end{pmatrix}$$

in the top block sense needed for the determinant; hence the determinant is zero. The same holds for the reduced discriminant.

Consequently, the discriminant also vanishes if a nilpotent ideal appears after passage to the algebraically closed field.

Let

$$\mathfrak{o} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_s$$

be a direct two-sided sum, and let $M, M_1, \ldots, M_s$ be the discriminant matrices of the rings $\mathfrak{o}, \mathfrak{a}_1, \ldots, \mathfrak{a}_s$. Then M is block diagonal,

$$M = \begin{pmatrix} M_1 & 0 & \cdots & 0 \\ 0 & M_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & M_s \end{pmatrix},$$

so

$$|M| = |M_1|\,|M_2| \cdots |M_s|.$$

Indeed, choose a basis of $\mathfrak{o}$ formed from bases of $\mathfrak{a}_1, \ldots, \mathfrak{a}_s$. If $a \in \mathfrak{a}_i$, then

$$\mathrm{Sp}_{\mathfrak{o}}(a) = \mathrm{Sp}_{\mathfrak{a}_i}(a),$$

where both times principal traces are meant, but in the rings $\mathfrak{o}$ and $\mathfrak{a}_i$ respectively. Further, $a_i a a_j = 0$ if the factors lie in different two-sided components; hence the mixed trace blocks are zero. The same argument applies to the reduced discriminant.

To form the discriminant of a matrix ring $\sum c_{ik}P$, choose two different bases: the c_{ik} once ordered by first indices and once by second indices. The product table then shows that the discriminant is

$$D_{\text{red}} = e \quad \text{for reduced traces,} \qquad D = n^{n^2}e \quad \text{for principal traces.}$$

Therefore, if over the algebraically closed extension field $\mathfrak{o}$ decomposes into matrix rings of degrees $n_1, \dots, n_s$, its discriminant is

$$D = n_1^{n_1^2} n_2^{n_2^2} \cdots n_s^{n_s^2} e,$$

and its reduced discriminant is

$$D_{\text{red}} = e.$$

Thus the reduced discriminant is always non-zero for systems without radical; the principal discriminant is non-zero exactly when no n_i is divisible by the characteristic of the field. Hence:

Non-vanishing of the reduced discriminant is necessary and sufficient for systems without radical after algebraic closure of the field P.

In characteristic zero, non-vanishing of the discriminant is necessary and sufficient for the non-appearance of a radical after algebraic closure of the field P.[22)]

22) For commutative systems compare E. Noether, *Diskriminantensatz für Ordnungen ...*, J. reine angew. Math. 157 (1927), pp. 82–104, §§4–6. The method of proof there is the same, but more complicated in detail; the transition from one basis to another (§4, 4 there) is to be replaced by the one given here at the beginning of this paragraph, for the determinant occurring there vanishes.

§ 26. Placement of the group ring

Let $a_1, \dots, a_h$ be the elements of a finite group, and form the group ring over a field whose characteristic does not divide h. Then the discriminant $D \neq 0$, and hence the group ring is a ring without radical.

Proof. First we show, with Sp henceforth meaning principal trace,

$$\mathrm{Sp}(e) = he, \qquad \mathrm{Sp}(a_i) = 0 \quad (a_i \neq e).$$

In the regular representation one has

$$e \mapsto \begin{pmatrix} e & 0 & \cdots & 0 \\ 0 & e & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e \end{pmatrix}, \qquad \mathrm{Sp}(e) = he.$$

For $a_i \neq e$ the products $a_i a_1, \dots, a_i a_h$ are just a permutation of $a_1, \dots, a_h$ with no fixed point; therefore the matrix by which these products are expressed in terms of $a_1, \dots, a_h$ has only zeros in the main diagonal, and so $\mathrm{Sp}(a_i) = 0$.

Take now the bases $a_1, \dots, a_h$ and $a_1^{-1}, \dots, a_h^{-1}$. The matrix $\left(\mathrm{Sp}(a_i a_j^{-1})\right)$ is diagonal with diagonal entries he; hence

$$D = h^h e \neq 0.$$

By the class of a group element a one means the sum formed in the group ring

$$K_a = \sum_s s^{-1} a s, \tag{1}$$

where the summation runs only over the distinct conjugates $s^{-1}as$. The classes K_a are central elements, since they commute with all group elements. They generate the center; for if an element $\sum_i \alpha_i a_i$ of the group ring commutes with all s, then

$$\sum_i \alpha_i a_i = s^{-1}\Big(\sum_i \alpha_i a_i\Big)s = \sum_i \alpha_i (s^{-1} a_i s),$$

so the coefficients are constant on conjugacy classes.

Thus the rank of the center is equal to the number of classes of conjugate group elements; hence the number of absolutely irreducible representations is also equal to this number of classes.

The matrices A and $S^{-1}AS$ have the same trace; therefore the group elements a and $s^{-1}as$ have the same trace. Taking traces on both sides of (1) yields

$$\chi(K_a) = h_a \chi(a),$$

where h_a is the number of elements in the class K_a. In words: the character of a group element is equal to the character of the class divided by the number of elements in the class.

(Eingegangen am 12. August 1928.)

35. On Maximal Domains of Integral Functions

Rec. Soc. Math. Moscou 36 (1929), pp. 65–72

The following investigations are closely related to Hilbert's problem on relatively integral functions. They arose from an occasional question of E. Hecke, who had been able to settle a special case directly by computation because he needed it as a lemma.

The question is this. By a maximal domain of integral functions in $x_1, \ldots, x_r$, namely polynomials, power series, or quotients of such, inside a given domain $\mathfrak{B}$, I mean one which can no longer be enlarged by adjoining integral denominators: whenever $ag(x)$ belongs to it, where $g(x) \in \mathfrak{B}$ and a is a nonzero rational integer, then $g(x)$ itself also belongs to it. Every integral domain $\mathfrak{A}$ in $\mathfrak{B}$ determines uniquely a maximal domain $M(\mathfrak{A})$; it consists of all functions $g(x) \in \mathfrak{B}$ for which there is at least one $a \neq 0$ such that $ag(x) \in \mathfrak{A}$.

The question is what can be said about the denominators a of this maximal domain $M(\mathfrak{A})$ when the original integral domain $\mathfrak{A}$ is finite, that is, when it consists of all integral polynomials in finitely many functions $f_1(x), \ldots, f_s(x)$ from $\mathfrak{B}$. Here $\mathfrak{B}$ is to be taken as all integral polynomials or power series in $x_1, \ldots, x_r$, or as quotients whose only denominators are those occurring in the $f_i(x)$ and their powers.

I show that, in the case of power series or their quotients subject to the additional hypothesis that only algebraic dependencies occur among the $f_i(x)$, the a may always be chosen so that only finitely many different prime numbers divide them, although in general to arbitrary powers. The latter point is immediately clear: if $ag(x)$ belongs to $\mathfrak{A}$, but $a^q g(x)^q$ does not for any exponent q, then all powers a^q occur, for example in $\mathfrak{A} = [2x]$.

The question of boundedness can only be formulated as follows: can one give a generally infinite system of generators for $M(\mathfrak{A})$ such that, in the associated denominators, the exponents of the prime numbers remain bounded? Since only finitely many primes occur, this question is, by known arguments, identical with the question of the finiteness of $M(\mathfrak{A})$, hence with Hilbert's problem on relatively integral functions in its integral form. I cannot decide that question by the methods used here.

The proof that only finitely many different primes p occur in the denominators a rests on the fact that to every such p there corresponds at least one relation modulo p among the functions $f_i(x)$ generating $\mathfrak{A}$, a relation with integral coefficients which is not identically satisfied in x. Equivalently, let $\mathfrak{o}$ denote the integral polynomial domain in indeterminates $u_1, \ldots, u_s$. Then $\mathfrak{A}$ is a homomorphic image of $\mathfrak{o}$ under $u_i \mapsto f_i(x)$. This homomorphism is mediated by a prime ideal $\mathfrak{a}$ of $\mathfrak{o}$, so that $\mathfrak{A} \simeq \mathfrak{o}/\mathfrak{a}$. Similarly $\mathfrak{B}_p$ is a homomorphic image of $\mathfrak{o}_p$, the subscript denoting reduction modulo p. This homomorphism is mediated by a prime ideal $\mathfrak{c}_p$ of $\mathfrak{o}_p$, containing $\mathfrak{a}_p$. The prime p occurs in a denominator of $M(\mathfrak{A})$ if and only if $\mathfrak{c}_p$ is a proper overideal of $\mathfrak{a}_p$. This can happen only if either the transcendence degree of $\mathfrak{A}_p$ drops relative to that of $\mathfrak{A}$, or $\mathfrak{a}_p$ loses its property of being prime. Only finitely many primes can do this, since even modulo p an algebraic dependence implies the vanishing of the functional determinant, and since $\mathfrak{a}$ is an absolute prime ideal and so loses primality modulo only finitely many p.

All the arguments remain valid, with slight modifications, if rational integers are replaced by the integers of a finite algebraic number field and prime numbers by its prime ideals.

§1. Functional Determinants over Coefficient Domains of Characteristic p

Let $f_1(x_1, \ldots, x_r), \ldots, f_t(x_1, \ldots, x_r)$ be rational functions, or more generally formal power series or quotients of such, with coefficients in a field P of characteristic p. The derivatives $\partial f_i / \partial x_j$ are also to be formally defined, with the usual rules of calculation preserved. Let Ω be an algebraically closed extension field of P. Then, just as in characteristic zero:

If there is an algebraic dependence among the $f_i(x)$ with coefficients in Ω, then the rank of the functional matrix

$$\left(\frac{\partial f_i}{\partial x_j}\right) \qquad (1 \leq i \leq t,\ 1 \leq j \leq r)$$

is less than t.

In the polynomial domain $\Omega[u_1, \ldots, u_t]$, let $\mathfrak{m}$ denote the prime ideal of all polynomials $G(u)$ for which $G(f) = 0$ identically in x. By hypothesis $\mathfrak{m} \neq 0$. Let $G(u) \neq 0$ be a polynomial of least degree in $\mathfrak{m}$; in particular G is irreducible. As usual one obtains

$$\sum_{i=1}^{t} \frac{\partial G}{\partial u_i}(f) \frac{\partial f_i}{\partial x_j} = 0, \qquad j = 1, \ldots, r.$$

Hence either the rank of the functional matrix is less than t, or all $(\partial G/\partial u_i)(f)$ vanish. Since G was chosen of least degree, it follows that all $\partial G/\partial u_i$ vanish. Thus

$$G(u) = H(u_1^p, \ldots, u_t^p).$$

Because Ω is algebraically closed, hence perfect, one further obtains $G(u) = (T(u))^p$, contradicting the choice of G. The proof also applies in characteristic zero, with the last step omitted.

Corollary. Let $F_1(x), \ldots, F_t(x)$ be integral functions, i.e. polynomials, power series, or quotients of such, whose functional matrix has rank exactly t. Choose the prime p so that it divides neither the denominators of the $F_i(x)$ nor all t-rowed determinants of the functional matrix. Then $F_1(x), \ldots, F_t(x)$ remain algebraically independent modulo p. Indeed, the functional matrix of the reduced functions is obtained by reducing the coefficients modulo p, and by the choice of p it still has rank t.

§2. Formulation of the Theorem

As in the introduction, let $\mathfrak{A}$ be an integral domain of integral functions in $x_1, \ldots, x_r$, formed from polynomials or power series with rational integral coefficients, or quotients of such. In the case of power series, they may be formal or convergent in some region; only the fact that the functions form an integral domain is used.

Let Q be the quotient field of $\mathfrak{A}$, and let $\mathfrak{B}$ be an integral domain between $\mathfrak{A}$ and Q. As above, $\mathfrak{M}$ is called maximal in $\mathfrak{B}$ if from $ag(x) \in \mathfrak{M}$, with $a \neq 0$ integral and $g(x) \in \mathfrak{B}$, it always follows that $g(x) \in \mathfrak{M}$. The maximal domain $M(\mathfrak{A})$ generated by $\mathfrak{A}$ in $\mathfrak{B}$ is the intersection of all maximal domains in $\mathfrak{B}$ containing $\mathfrak{A}$, and it consists of all $g(x) \in \mathfrak{B}$ for which some $a \neq 0$ satisfies $ag(x) \in \mathfrak{A}$.

Assume now that $\mathfrak{A}$ is a finite integral domain with identity, with $f_1(x), \ldots, f_s(x)$ as an integral basis; thus $\mathfrak{A}$ consists of all integral polynomials in the $f_i(x)$. Let $\mathfrak{B}$ consist of all integral polynomials or power series in x, or of quotients whose only denominators are those occurring among the finitely many $f_i(x)$. Assume also that the variables x actually occur in at least one of the $f_i(x)$; otherwise there is nothing to prove.

For polynomials or rational functions the quotient field Q has transcendence degree t, $1 \leq t \leq r$, over the field K of rational numbers. If, for instance, $f_1(x), \ldots, f_t(x)$ form an irreducible system, then Q is a finite algebraic extension of $K(f_1(x), \ldots, f_t(x))$. Also, the functional matrices of $f_1, \ldots, f_t$ and $f_1, \ldots, f_s$ both have rank t.

For power series or their quotients we impose the additional hypothesis that this structure of Q remains valid: if t is the rank of the functional matrix of $f_1, \ldots, f_s$ and also that of $f_1, \ldots, f_t$, then Q is again to be a finite algebraic extension of $K(f_1(x), \ldots, f_t(x))$.

Now let $\mathfrak{o}$ be the integral polynomial domain in $u_1, \ldots, u_s$. Then $\mathfrak{A}$ is a ring-homomorphic image of $\mathfrak{o}$ under $u_i \mapsto f_i(x)$. Therefore $\mathfrak{A}$ is ring-isomorphic to $\mathfrak{o}/\mathfrak{a}$, where $\mathfrak{a}$ is a prime ideal. The theorem is:

Theorem. Under the added hypothesis just stated, let $\mathfrak{o}_p, \mathfrak{a}_p, \mathfrak{A}_p, \mathfrak{B}_p$ denote the domains obtained from $\mathfrak{o}, \mathfrak{a}, \mathfrak{A}, \mathfrak{B}$ by passing to residue classes modulo p. Then, except for at most finitely many prime numbers p, the homomorphism from $\mathfrak{o}_p$ to $\mathfrak{A}_p$ is mediated by $\mathfrak{a}_p$; hence

$$\mathfrak{A}_p \simeq \mathfrak{o}_p/\mathfrak{a}_p \simeq \mathfrak{o}/(\mathfrak{a}, p\mathfrak{o}).$$

Corollary. If p is chosen so that $\mathfrak{A}_p \simeq \mathfrak{o}_p/\mathfrak{a}_p$, then $pg(x) \in \mathfrak{A}$ and $g(x) \in \mathfrak{B}$ imply $g(x) \in \mathfrak{A}$. Equivalently, $\mathfrak{A}$ is maximal in $\mathfrak{B}$ with respect to all primes except the finite exceptional set. Iterating gives: if a is divisible by none of the exceptional primes, then $ag(x) \in \mathfrak{A}$ and $g(x) \in \mathfrak{B}$ imply $g(x) \in \mathfrak{A}$.

§3. Proof of the Theorem

Lemma. The prime ideal $\mathfrak{a}$ mediating the homomorphism between $\mathfrak{o}$ and $\mathfrak{A}$ is an absolute prime ideal.

Let $\mathfrak{o}^*$ be the polynomial domain in $u_1, \ldots, u_s$ with arbitrary algebraic numbers as coefficients, and let $\mathfrak{a}^*$ be the ideal in $\mathfrak{o}^*$ generated by $\mathfrak{a}$. We must prove that $\mathfrak{a}^*$ is prime. It is enough to show that $\mathfrak{a}^*$ mediates the homomorphism from $\mathfrak{o}^*$ into the integral domain $\mathfrak{A}^*$ consisting of all polynomials in the $f_i(x)$ with algebraic coefficients.

Let $H(u) \in \mathfrak{o}^*$ and $H(f_1(x), \ldots, f_s(x)) = 0$. Let $\omega_1, \ldots, \omega_m$ be a linearly independent basis of the finite number field generated by the coefficients of H. After multiplying by a nonzero rational integer a, we may write

$$aH(u) = \omega_1 H_1(u) + \cdots + \omega_m H_m(u),$$

with $H_i(u) \in \mathfrak{o}$. From $H(f) = 0$ follows

$$\omega_1 H_1(f) + \cdots + \omega_m H_m(f) = 0,$$

and hence $H_i(f) = 0$ for all i. Thus all $H_i(u)$ lie in $\mathfrak{a}$, and $H(u) \in \mathfrak{a}^*$.

The proof of the theorem can now be phrased as follows. Choose p so that

I. $\mathfrak{A}_p$ is defined, II. $\mathfrak{a}_p$ remains prime, III. $\operatorname{trdeg}(\mathfrak{A}_p) = \operatorname{trdeg}(\mathfrak{A})$.

Then the homomorphism from $\mathfrak{o}_p$ to $\mathfrak{A}_p$ is mediated by $\mathfrak{a}_p$. Apart from finitely many exceptions these conditions hold. I is clear; II follows from the lemma and the fact that an absolute prime ideal loses primality modulo only finitely many primes; III follows from the corollary on functional determinants.

Let the homomorphism from $\mathfrak{o}_p$ to $\mathfrak{A}_p$ be mediated by a prime ideal $\mathfrak{c}_p$. Since $\mathfrak{a}_p \subseteq \mathfrak{c}_p$, the two ideals are equal precisely when their transcendence degrees agree. By the added hypothesis $\mathfrak{a}$ has transcendence degree t. Number the u_i so that the classes of $u_1, \ldots, u_t$ modulo $\mathfrak{a}$ form an irreducible system and the classes of $u_{t+1}, \ldots, u_s$ are algebraic over them. Then $\mathfrak{a}$ contains primitive integral polynomials

$$G_{t+1}(u_1, \ldots, u_t, u_{t+1}), \ldots, G_s(u_1, \ldots, u_t, u_s),$$

which reduce modulo p to nonzero polynomials. In each such polynomial the last variable actually occurs. Otherwise there would be an algebraic dependence among the classes of $u_1, \ldots, u_t$ modulo $\mathfrak{a}_p$. Therefore $\mathfrak{a}_p$ has transcendence degree t, is identical with $\mathfrak{c}_p$, and the theorem follows.

§4. Extension to Integral Algebraic Coefficients

The arguments remain valid, with slight changes, if rational integers are replaced by the integers of a finite algebraic number field K, and prime numbers by its prime ideals.

In the proof one must add to the exceptional prime ideals those which occur in the coefficients of the polynomials $G_{t+1}, \ldots, G_s$, since these can no longer be assumed primitive. This ensures that their reductions do not vanish. The condition of absolute irreducibility is unchanged.

The final corollary is then: if the integer a of K is divisible by none of the finitely many exceptional prime ideals, then $ag(x) \in \mathfrak{A}$ and $g(x) \in \mathfrak{B}$ imply $g(x) \in \mathfrak{A}$. If $\lambda_1, \ldots, \lambda_m$ are integers of K each divisible by exactly one exceptional prime ideal, but not by its square and not by the others, then it is enough to use denominators which are products of powers of these λ_i.

Let

$$(a) = \mathfrak{p}_1^{\alpha_1} \cdots \mathfrak{p}_m^{\alpha_m}$$

be the prime ideal decomposition of a, with the $\mathfrak{p}_i$ non-exceptional. Passing to the quotient ring R defined by (a), the ideals $\mathfrak{p}_i$ remain prime but become principal, and the factorization of (a) is unchanged. The proof of the corollary therefore remains as before: if $ag(x) \in \mathfrak{A}$, then after clearing a denominator prime to a, one obtains $g(x) \in \mathfrak{A}$.

For the statement about denominators, let R^* be the quotient ring in K defined by the finitely many exceptional prime ideals. In R^* every a is, up to a unit, a product of powers of the basis elements λ_i. Returning to integers gives

$$\gamma a = \beta\, \lambda_1^{\alpha_1} \cdots \lambda_m^{\alpha_m},$$

where γ, β are divisible by none of the exceptional prime ideals. Hence from $ag(x) \in \mathfrak{A}$ it follows that

$$\lambda_1^{\alpha_1} \cdots \lambda_m^{\alpha_m} g(x) \in \mathfrak{A}.$$

Notes. The finiteness results used here are the standard Hilbert–Noether finiteness theorems. For integral projective invariants of a system of ground forms, the result says that they can be represented by a full invariant system in the usual sense, with only finitely many different primes in the denominators. The transcendence degree of a system is the invariant number of algebraically independent functions on which the system algebraically depends; the transcendence degree of a prime ideal is that of its residue field. The absolute primality assertion is reduced to the known theorems on absolute irreducibility and elimination theory. For functional determinants in characteristic zero one may also use van der Waerden's purely algebraic proof. The additional hypothesis for power series is satisfied in the analytic special case when the chosen functions are analytically independent and the remaining ones algebraically dependent on them, but it was deliberately stated formally.

(Rec. Math. XXXVI:1; 1929)

36. Ideal Differentiation and the Different

J. Ber. d. DMV 39 (1930), p. 17

E. Noether, Göttingen: Ideal differentiation and the different.

It is shown how the different of an algebraic number field can be regarded as the differential quotient of a defining ideal. That this definition agrees with the usual one follows from structural theorems which are interesting in themselves and which, in an analogous sense, constitute a generalization of the Lagrange interpolation formula.

A detailed account is to appear in the *Mathematische Annalen.*

37. Normal Basis in Fields without Higher Ramification

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By a normal basis of a Galois number field K/k one means a basis of the principal order $\mathfrak{O}$ of K, relative to the principal order $\mathfrak{o}$ of k, consisting of conjugate elements. A necessary condition for the existence of such a basis is, as is known, that no higher ramification fields occur in K/k. This remains true even if one restricts to a place, i.e. passes to the p-adic extension k_p of k and the corresponding K_p of K. I prove below the converse:

For every place $\mathfrak{p}$ not dividing the degree n of K/k, a normal basis exists. In particular: if K/k has no higher ramification, then at every ramified place a normal basis exists; the discriminant there is the square of a group determinant.

To this last assertion one must add that, contrary to a conjecture of E. Artin, the passage to his conductors still requires further considerations: decomposing the group determinant according to irreducible characters yields generalized root numbers; a suitable grouping yields a factorization in the ground field enlarged by the characters. In the simplest cases I can identify these new factors with Artin's conductors by means of the ideal-theoretic decomposition of the root numbers. In the general case, where no normal basis exists, a splitting into generalized root numbers is still always possible by means of a composition series by Galois modules; from this one again obtains a corresponding decomposition in the enlarged ground field.

The proof of the existence of the normal basis rests on the fact that $\mathfrak{O}/\mathfrak{o}$, viewed as a Galois module, becomes operator-isomorphic to a one-sided ideal of the integral group ring extended by $\mathfrak{o}$. At all ordinary ramification places this group ring is a maximal order; ideals in maximal orders of p-adic semisimple systems are principal ideals. Returning from this to the Galois module gives precisely the existence of the normal basis. At the higher ramification places, the integral group ring passes to a nonmaximal order, and the ideal corresponding to $\mathfrak{O}/\mathfrak{o}$ becomes a nonprincipal ideal.

All these considerations remain valid for function fields of one variable, provided that the characteristic of the constant field does not divide the degree of K/k.

§1. The p-adically Extended Integral Group Ring

Let $\mathfrak{O}$ and $\mathfrak{o}_p$ denote the principal orders of an algebraic number field k and of its p-adic extension k_p, where p is a prime ideal of k. Let (G) denote the rational integral group ring and $[G]$ the integral group ring of a group G with n elements. By $(G)_k$, respectively $(G)_{k_p}$, we mean the group ring with coefficients in k, respectively k_p; analogously $[G]_\mathfrak{o}$ and $[G]_{\mathfrak{o}_p}$ are the corresponding extensions of the integral group ring to coefficients in $\mathfrak{o}$, respectively $\mathfrak{o}_p$.

Theorem 1. If p does not divide the number n of elements of G, then $[G]_{\mathfrak{o}_p}$ is a maximal order of $(G)_{k_p}$. If p divides n, then $[G]_{\mathfrak{o}_p}$ is not maximal.

The discriminant of the integral group ring, and hence of $[G]_\mathfrak{o}$ and $[G]_{\mathfrak{o}_p}$, is a power of n. Thus if p does not divide n, the discriminant of $[G]_{\mathfrak{o}_p}$ is a unit. This means that, while $\mathfrak{o}_p$ is fixed, $[G]_{\mathfrak{o}_p}$ cannot be enlarged to a larger order; such an enlargement would split off a nonunit square factor from the discriminant. This proves the first part.

If, on the other hand, p divides n, the direct sum corresponding to the identity representation,

$$E^0[G]_{\mathfrak{o}_p} + (1 - E^0)[G]_{\mathfrak{o}_p}, \qquad E^0 = \frac{1}{n}\sum_{S \in G} S,$$

is a proper extension of $[G]_{\mathfrak{o}_p}$.

Theorem 2. If p does not divide n, every integral or fractional ideal with respect to $[G]_{\mathfrak{o}_p}$ is principal. Indeed, by Theorem 1, $[G]_{\mathfrak{o}_p}$ is a maximal order in a p-adic semisimple system, and the ideals of such orders are principal.

§2. Galois Modules, Operator Isomorphism, Normal Basis

Let K/k be Galois with group G. For $z \in K$, let z^S denote the element obtained from z by the substitution $S \in G$. The substitutions in G induce operator automorphisms of K/k, regarded as a k-module. Thus K/k can be made into a module with respect to $(G)_k$ by setting

$$z\Big(\sum_i c_i S_i\Big) = \sum_i c_i z^{S_i}, \qquad z \in K,\ c_i \in k,\ S_i \in G.$$

Definition. Every $(G)_k$-module contained in K/k is called a rational Galois module. Similarly, the $[G]_{\mathfrak{o}}$-modules contained in K/k are called integral Galois modules; in particular $\mathfrak{O}/\mathfrak{o}$ is an integral Galois module.

Theorem 3. As a rational Galois module, K/k is operator-isomorphic to $(G)_k$. This follows from the fact that K/k always has a field normal basis, i.e. a k-basis consisting of conjugates z^S. The correspondence

$$\sum_i c_i S_i \longmapsto \sum_i c_i z^{S_i}$$

is an operator homomorphism, and since the ranks are equal it is an isomorphism.

Theorem 4. As an integral Galois module, $\mathfrak{O}/\mathfrak{o}$ is operator-isomorphic to a $[G]_{\mathfrak{o}}$-module in $(G)_k$ of maximum rank n. Likewise, $\mathfrak{O}_p/\mathfrak{o}_p$ is operator-isomorphic to a $[G]_{\mathfrak{o}_p}$-module of rank n in $(G)_{k_p}$.

The isomorphism of Theorem 3 assigns to the $[G]_{\mathfrak{o}}$-module $\mathfrak{O}/\mathfrak{o}$ a $[G]_{\mathfrak{o}}$-module of rank n in $(G)_k$. Locally one extends coefficients from k to k_p; the resulting system is semisimple, in general a sum of isomorphic fields.

Theorem 5. At every place p which does not divide the degree n of K/k, the module $\mathfrak{O}_p/\mathfrak{o}_p$ has a normal basis.

By Theorem 1, $[G]_{\mathfrak{o}_p}$ is a maximal order there; the $[G]_{\mathfrak{o}_p}$-modules of rank n in $(G)_{k_p}$ are integral or fractional ideals, hence principal by Theorem 2. If the ideal corresponding to $\mathfrak{O}_p/\mathfrak{o}_p$ is $W[G]_{\mathfrak{o}_p}$, then it has the $\mathfrak{o}_p$-basis WS_i. The operator isomorphism says that w^{S_i} is an $\mathfrak{o}_p$-basis of $\mathfrak{O}_p/\mathfrak{o}_p$, where w is the element corresponding to W. Thus the conjugate elements w^{S_i} form a normal basis.

If p divides n, the corresponding module cannot be principal, since in this case no normal basis can exist.

Additional remark to Theorem 3. From the operator isomorphism between $(G)_k$ and K/k follow Speiser's results on Klein's problem of forms: every representation of the Galois group G of K/k which is irreducible over k is generated by an irreducible rational Galois module from K/k; every absolutely irreducible representation is generated by a Galois module from K_Z/Z, where Z is a splitting field of the corresponding component of the group ring. If $v_1, \ldots, v_m$ is a basis of such a Galois module and $S \mapsto (s_{ij})$ is the corresponding representation, then

$$v_i^S = \sum_j s_{ij} v_j.$$

This is precisely Klein and Speiser's formulation.

§3. The Discriminant as Group Determinant in Fields without Higher Ramification

For decomposing the discriminant it is enough to consider its decomposition at the individual ramified places. In fields without higher ramification, only places with normal basis are involved.

Theorem 6. For Galois fields K/k without higher ramification, the discriminant at every ramified place is the square of the group determinant of the Galois group after replacing the indeterminates by a normal basis. Corresponding to the irreducible representations, the group determinant decomposes into generalized root numbers; the discriminant decomposes into ideals of the ground field enlarged by the characters, with conjugate complex characters corresponding to the same ideals.

With w^S as a normal basis, the discriminant Δ is the square of

$$D = \det(w^{ST})_{S,T\in G}.$$

This is the group determinant with the w^S in place of the indeterminates x_S. If

$$D = D_1 \cdots D_h$$

is the factorization according to the absolutely irreducible representations, then for a representation $S \mapsto \rho_i(S)$

$$D_i = \det\Big(\sum_{S\in G} w^S \rho_i(S)\Big).$$

Thus the D_i are generalized root numbers. For cyclic groups they are simply the Lagrange root numbers

$$\sum_{S\in G} w^S \chi_i(S).$$

To pass from the factorization of the group determinant to that of the discriminant, one groups each representation with its adjoint. If D_i and D_i' are the corresponding factors, put

$$\Delta_i = D_i D_i'.$$

Then $\Delta_i^T = \Delta_i$ for all $T \in G$; hence Δ_i lies in the ground field enlarged by the real subfield of the i-th character. The discriminant factorization

$$\Delta = \Delta_1 \cdots \Delta_h$$

is therefore a factorization in the field obtained by adjoining the real subfields of the characters, with conjugate complex characters corresponding to the same factors.

If one can show that the ideals generated by the Δ_i actually already lie in the base field k, and are equal for conjugate characters, then they agree with Artin's conductors, provided that to composite characters one assigns the corresponding products of powers of the Δ_i.

Theorem 7. Let k be the field of rational numbers, and let K/k be cyclic of prime degree l, prime to the discriminant, hence without higher ramification. Then the ideals generated by the Δ_λ for the nontrivial representations are all equal and agree with the conductor of K/k.

This follows from the known decomposition of root numbers. At a ramified place p of K/k, for k_ε, the field of l-th roots of unity, and K_ε, one has

$$(p) = \mathfrak{p}_1 \cdots \mathfrak{p}_{l-1}, \qquad \mathfrak{p}_i = \mathfrak{P}_i^l,$$

with $\mathfrak{p}_i$ in k_ε and $\mathfrak{P}_i$ in K_ε. Further, for the root numbers,

$$(\Omega_\lambda) = \mathfrak{P}_1^{r_1} \cdots \mathfrak{P}_{l-1}^{r_{l-1}}, \qquad (\Omega_{\bar\lambda}) = \mathfrak{P}_1^{l-r_1} \cdots \mathfrak{P}_{l-1}^{l-r_{l-1}},$$

where $r_1, \ldots, r_{l-1}$ is a permutation of $1, \ldots, l-1$. The root numbers Ω_λ are here identical with the factors D_λ of the group determinant; hence

$$(\Delta_\lambda) = (D_\lambda D_{\bar\lambda}) = (\Omega_\lambda \Omega_{\bar\lambda}) = \mathfrak{P}_1^l \cdots \mathfrak{P}_{l-1}^l = \mathfrak{p}_1 \cdots \mathfrak{p}_{l-1} = (p),$$

and therefore there is agreement with the conductor of K/k.

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38. Jointly with R. Brauer and H. Hasse: Proof of a Main Theorem in the Theory of Algebras

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At last our joint efforts have succeeded in proving the following theorem, which is of fundamental importance for the structure theory of algebras over algebraic number fields and beyond.

Main Theorem. Every normal division algebra over an algebraic number field is cyclic, or, equivalently, of Dickson type.

It is a special pleasure to present this result, essentially an achievement of the p-adic method, to Kurt Hensel, the founder of that method, on his seventieth birthday. Our proof consists of three reductions, each contributed by one of us.

H. Hasse gave the first reduction on the basis of his recently developed theory of cyclic algebras over algebraic number fields.

Reduction 1. The main theorem is proved once the following is shown:

I. Every everywhere split algebra over Ω is $\sim \Omega$.

Here and below, an "algebra A over Ω" means a normal simple algebra over the algebraic number field Ω, i.e. a simple hypercomplex system with center Ω. The notation $A \sim \Omega$ means that A is a full matrix algebra over Ω, equivalently that it belongs to the special similarity class determined by Ω itself as division algebra. An algebra is called everywhere split if for every prime place $\mathfrak{p}$ of Ω its $\mathfrak{p}$-adic extension $A_\mathfrak{p}$ is split:

$$A_\mathfrak{p} \sim \Omega_\mathfrak{p}.$$

Let D be a division algebra over Ω. Choose a cyclic field Z/Ω such that, for every prime place $\mathfrak{p}$, the $\mathfrak{p}$-degree of Z is a multiple of the $\mathfrak{p}$-index of D. Then for every prime divisor $\mathfrak{P}$ of $\mathfrak{p}$ in Z, the local field $Z_\mathfrak{P}$ splits $D_\mathfrak{p}$. Hence the algebra D_Z obtained by extending scalars from Ω to Z is everywhere split. If I is known, $D_Z \sim Z$. Thus D has a cyclic splitting field Z, hence is cyclically representable; moreover it has such a cyclic splitting field whose degree agrees with the degree of D. Therefore D is cyclic.

The second reduction was prompted decisively by R. Brauer, who informed H. Hasse by letter that the related question of the exact value of the exponent of a division algebra can be reduced, using Sylow's theorem, to the case of a solvable splitting field. The same idea gives the next reduction of the cyclicity question.

Reduction 2. Statement I is proved once the following is shown:

II. Every everywhere split algebra over Ω with a solvable splitting field is $\sim \Omega$.

Let A be everywhere split over Ω. Let K be a Galois splitting field for A, let p be a prime number, and let Σ be the Sylow field of K/Ω corresponding to p, i.e. the fixed field of a p-Sylow subgroup of the Galois group. Then A_Σ is an everywhere split algebra over Σ, and has the solvable splitting field K. By II, $A_\Sigma \sim \Sigma$. Thus Σ is a splitting field of A, and its degree over Ω is prime to p. Hence the index of A, being a divisor of that degree, is prime to p. Since this holds for every prime p, the index is 1, and $A \sim \Omega$.

The third reduction, which together with the first two leads to the final proof, was extracted by E. Noether from a communication of Hasse; independently, Brauer had also considered it.

Reduction 3. Statement II is proved once the following is shown:

III. Every everywhere split algebra over Ω with a cyclic splitting field of prime degree is $\sim \Omega$.

Let A be everywhere split and have a solvable splitting field K/Ω. Choose a tower

$$K = \Lambda_0 > \Lambda_1 > \cdots > \Lambda_r = \Omega$$

such that each Λ_i/Λ_{i+1} is cyclic of prime degree. Then A_{Λ_1} is everywhere split over Λ_1, and has the cyclic splitting field $K = \Lambda_0$ of prime degree. By III, $A_{\Lambda_1} \sim \Lambda_1$, so Λ_1 already splits A. Repeating the same argument down the tower gives $A \sim \Omega$. Thus II follows from III.

But III is known from Hasse's results. Hence the main theorem follows by reversing the reductions 3, 2, and 1. Noether further observes that III, in the more general form for cyclic splitting fields of arbitrary degree, is equivalent to Hasse's norm theorem. For Reduction 3 one needs only the prime-degree case, the original Hilbert–Furtwängler norm theorem. Thus Reduction 3 gives a new simple proof of Hasse's norm theorem. In Hasse's notation: if Z/Ω is cyclic and $\alpha \in \Omega$ satisfies

$$\left(\frac{\alpha, Z}{\mathfrak{p}}\right) = 1$$

for every prime place $\mathfrak{p}$, then every cyclic algebra

$$A = (\alpha, Z)$$

is everywhere split. Reduction 3 gives $A \sim \Omega$, and hence α is the norm of an element of Z.

Consequences

Theorem 1. The exponent of a normal simple algebra over an algebraic number field is equal to its index.

Theorem 2. The fundamental ideal of a proper normal division algebra over Ω is divisible by at least one prime place of Ω, and in fact by at least two.

The fundamental ideal may be defined as the reduced norm of the reduced different. An infinite prime place is included exactly when the algebra reduces there to the quaternion algebra, rather than to the real or complex number field. By Hasse's results, a prime place occurs in the fundamental ideal exactly when the $\mathfrak{p}$-index is not 1. If all $\mathfrak{p}$-indices were 1, the algebra would be everywhere split; then by I it would not be a proper division algebra. Nor can a single $\mathfrak{p}$-index alone be different from 1, because the norm residue symbols, whose orders are the local indices, satisfy the product theorem.

Theorem 3. For an algebra A over Ω, an algebraic number field K/Ω is a splitting field if and only if, for every prime divisor $\mathfrak{P}_i$ in K of every prime place $\mathfrak{p}$ of Ω, the $\mathfrak{P}_i$-degree $n_{\mathfrak{P}_i}$ of K is a multiple of the $\mathfrak{p}$-index $m_{\mathfrak{p}}$ of A.

If

$$\mathfrak{p} = \mathfrak{P}_1^{e_{\mathfrak{P}_1}} \cdots \mathfrak{P}_g^{e_{\mathfrak{P}_g}}, \qquad \mathrm{N}(\mathfrak{P}_i) = \mathfrak{p}^{\,f_{\mathfrak{P}_i}},$$

then

$$n_{\mathfrak{P}_i} = e_{\mathfrak{P}_i} f_{\mathfrak{P}_i}.$$

The $\mathfrak{p}$-index of A is the index $m_{\mathfrak{p}}$ of $A_{\mathfrak{p}}$.

The assertion follows because K splits A precisely when $A_{K_{\mathfrak{P}_i}} \sim K_{\mathfrak{P}_i}$ for all $\mathfrak{P}_i$. At a finite prime place let

$$A_{\mathfrak{p}} \sim (\alpha, W_{\mathfrak{p}})$$

be the distinguished cyclic local representation, where $W_{\mathfrak{p}}$ is the unramified field of degree $m_{\mathfrak{p}}$ over $\Omega_{\mathfrak{p}}$. Let B_i be the intersection and C_i the composite of $W_{\mathfrak{p}}$ and $K_{\mathfrak{P}_i}$. With

$$d_i = (m_{\mathfrak{p}}, f_{\mathfrak{P}_i}),$$

the extension $B_i/\Omega_{\mathfrak{p}}$ has degree d_i, and $K_{\mathfrak{P}_i}/B_i$ is unramified of degree $m_{\mathfrak{p}}/d_i$. The local algebra splits exactly when α is a norm from $K_{\mathfrak{P}_i}$ over B_i, which is equivalent to the valuation of α in $\mathfrak{P}_i$ being a multiple of $m_{\mathfrak{p}}/d_i$. Since

$$\left(\frac{m_{\mathfrak{p}}}{d_i}, \frac{f_{\mathfrak{P}_i}}{d_i}\right) = 1,$$

this is equivalent to $n_{\mathfrak{P}_i}$ being a multiple of $m_{\mathfrak{p}}$. For infinite prime places, unless $m_{\mathfrak{p}} = 1$, one has $m_{\mathfrak{p}} = 2$, and the condition says precisely that the corresponding local field is complex.

Theorem 4. *Decomposition theorem.* Let K/Ω be Galois, and let $\mathfrak{p}$ be a prime ideal of Ω not dividing the relative discriminant of K. The relative degree f of the prime divisors of $\mathfrak{p}$ in K is the least exponent for which

$$A^f \sim \Omega$$

for every algebra A in the subgroup of Brauer's group assigned to K.

Indeed, f is the local degree of K at $\mathfrak{p}$, while the $\mathfrak{p}$-index of A is the index, equivalently the exponent, of $A_{\mathfrak{p}}$; Theorem 3 supplies the local criterion. The existence of algebras with exact $\mathfrak{p}$-index f follows from Frobenius' density theorem and the construction of a cyclic algebra (α, Z) whose norm residue symbols at two suitably chosen prime places have reciprocal values of order f and are trivial elsewhere.

Theorem 5. *Uniqueness and ordering theorem.* $K \subseteq K'$ if and only if $\mathfrak{R}_K \subseteq \mathfrak{R}_{K'}$. In particular, the correspondence between Galois fields K and their algebra groups is one-to-one and order-preserving.

The implication $K \subseteq K' \Rightarrow \mathfrak{R}_K \subseteq \mathfrak{R}_{K'}$ is immediate. Conversely, the decomposition theorem gives the required divisibilities of relative degrees for all primes outside the relative discriminant; the standard analytic argument then yields $K \subseteq K'$.

Theorem 6. All absolutely irreducible representations of a finite group G can be realized over cyclotomic fields; in any case they can be realized over the field of n^h-th roots of unity for $n = |G|$ and sufficiently large h.

Passing to the rational integral group ring of G, the absolutely irreducible representations of G are the absolutely irreducible representations of the simple components G_i of that semisimple algebra. Their centers are the fields of the corresponding characters, hence cyclotomic fields. By Theorem 3 it suffices, at the finitely many prime divisors of the fundamental ideal, to make the local degrees divisible by the corresponding indices. The field of n^h-th roots of unity does this for sufficiently large h.

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39. Hypercomplex Systems in Their Relations to Commutative Algebra and Number Theory

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1. The theory of hypercomplex systems, or algebras, has advanced strongly in recent years; but only very recently has the significance of this theory for commutative questions become clear. I want to report on this significance of the noncommutative for the commutative, and I shall do so by following two classical questions going back to Gauss: the principal genus theorem and the closely related norm theorem. Over time these questions have repeatedly changed their formulation. In Gauss they appear as the conclusion of the theory of quadratic forms; then they play an essential role in the characterization of relatively cyclic and abelian number fields by class field theory; finally they can be stated as theorems on automorphisms and on the splitting of algebras. This last formulation at the same time transfers the theorems to arbitrary relatively Galois number fields.

The principle of applying the noncommutative to the commutative is this: by means of the theory of algebras one seeks invariant and simple formulations for known facts about quadratic forms or cyclic fields, formulations depending only on structural properties of algebras. Once such invariant formulations are proved, the transfer to arbitrary Galois fields is obtained automatically.

2. First, an overview of the methods and further results. The main difficulty lies in finding the right formulation for general Galois fields; without the hypercomplex method there was no starting point for this. In the examples mentioned above, the underlying passage to the noncommutative is obtained by considering field and group simultaneously through the *crossed product* and its multiplication constants, the *factor systems*. Such crossed products were first considered by Dickson, whereas the theory of factor systems was developed by Speiser, Schur, and R. Brauer from a quite different starting point, namely the question of absolutely irreducible representations. Only by combining the two theories did one obtain a sufficiently simple and far-reaching structure.

At the same time this gives new and transparent proofs of known facts: Hasse's hypercomplex proof of the reciprocity law for cyclic fields by means of an invariant formulation of his norm residue symbol; and Chevalley's hypercomplex foundation of local class field theory on the same basis, for which new algebraic theorems on factor systems were needed. One must add the limitation that the crossed-product method by itself apparently does not yield the full theory of Galois number fields: new results of Artin, proceeding from Hasse's proof in the sense of this principle, give only numerical equalities in place of full isomorphism theorems.

Methods giving a full isomorphism, namely operator isomorphism, already exist in algebra. They continue ideas of A. Speiser: the Galois field is viewed as a *Galois module*, i.e. as a module over the base field which admits the substitutions of the Galois group as operators. There is an operator isomorphism between the field and the group ring in the sense that elements correspond one-to-one, linear forms over the base field correspond, and the substitutions of the Galois group in the field correspond to multiplication in the group ring. A theorem formulated by me was proved by M. Deuring, who based on it a proof of Galois theory. Further structural theorems of Deuring run parallel to formal facts of Artin L-series and give a structural approach to Artin conductors. These Artin L-series and conductors, formed with general group characters, together with Speiser's work, form the first connection between number theory and representation theory.

3. I now examine the norm theorem and the principal genus theorem in detail. First, the definition of the crossed product. Let K/k be a Galois field of degree n, with group G. The crossed product means a simultaneous embedding of K and G in an algebra A in such a way that the automorphisms of K become inner. Let u_S denote symbols corresponding to the n group elements. First set up A as a module of linear forms of rank n over K:

$$A = u_{S_1}K + \cdots + u_{S_n}K,$$

i.e. A consists of all linear forms $u_{S_1}a_1 + \cdots + u_{S_n}a_n$, $a_i \in K$. The inner automorphism condition gives

$$u_S^{-1} z u_S = z^S, \qquad \text{or} \qquad z u_S = u_S z^S \quad (z \in K),$$

and the multiplication is determined by

$$u_S u_T = u_{ST} a_{S,T}, \qquad a_{S,T} \in K^*.$$

Associativity gives the factor-system condition

$$a_{S,T} a_{ST,R} = a_{S,TR}\, a_{T,R}^S.$$

The resulting A is called the crossed product of K with G with factor system $a_{S,T}$. One proves that A is a normal simple algebra over k, that K is a maximal commutative subfield, hence a splitting field, and conversely that for a given division algebra D there are always matrix rings D_r that can be obtained in this way as crossed products.

Replacing u_S by $v_S = u_S c_S$, $c_S \in K^*$, gives associated factor systems

$$a'_{S,T} = a_{S,T} \frac{c_S^T c_T}{c_{ST}}.$$

Associated factor systems are grouped into a class (a); similarly, all algebras similar to A are grouped into a class $\mathfrak{A}$. The classes $\mathfrak{A}$ and (a) correspond bijectively. The classes with fixed splitting field K form, under direct product, an abelian group isomorphic to the componentwise product of classes of factor systems. The identity element is the split algebra class, respectively the system of all transformation quantities

$$\frac{c_S^T c_T}{c_{ST}}.$$

4. For cyclic splitting fields this specializes to the norm concept. Let Z/k be cyclic, with S a generator of its group. One may let the powers of u correspond to the powers of S:

$$A = Z + uZ + \cdots + u^{n-1}Z,$$

$$zu = uz^S, \qquad u^n = a, \qquad a \in k^*,$$

and, if $v = uc$,

$$a' = a\,N(c).$$

Thus each factor system is represented by a single element $a \in k^*$, and one writes

$$A = (a, Z).$$

The identity class of factor systems is given by the norms from Z^*, so the group of algebra classes is isomorphic to

$$k^*/N(Z^*).$$

A cyclic algebra (a, Z) therefore splits if and only if a is the norm of an element of Z.

This connection between norm and splitting gives the formulation of the norm theorem as a theorem on split algebras: if an algebra splits at every place, then it splits absolutely. Here a place is defined, as in number theory, by replacing the base field k with its p-adic extension, and at infinite places by extension to the real numbers. For cyclic algebras this theorem says: if a is a local norm at every finite and infinite place, then a is the norm of a number of Z; equivalently, without passing to p-adic fields, if a is a norm residue modulo every prime ideal and satisfies the known sign conditions, then a is a global norm. This is precisely the norm theorem of class field theory. The general theorem on split algebras follows from this cyclic special case by purely algebraic-arithmetic arguments. One first important consequence is Hasse's theorem: every simple normal algebra over an algebraic number field is cyclic.

5. A second consequence is the principal genus theorem. Its invariant formulation rests on the fact that the defining relations of the crossed product are purely multiplicative, and therefore still make sense when K^* is replaced by an abelian group $\mathfrak{G}$ whose automorphism group contains a subgroup isomorphic to G. The module of linear forms is then replaced by the extension of G by $\mathfrak{G}$ in the sense of group theory. If $\mathfrak{G}$ is taken to be the group of all ideals of K, the factor systems become systems of ideals. A division into classes in $\mathfrak{G}$ induces a division into classes for the factor systems.

The identity class of factor systems is defined to consist of those elements $a_{S,T}$ which generate split algebras at all finite and infinite ramified places of K. Let the corresponding extension of G be denoted by G^*. Then the invariant formulation of the principal genus theorem is:

If, under this class division, the substitutions

$$v_S = u_S(c_S)$$

produce an automorphism of G^*, that is, if the transformation quantities

$$\frac{(c_S)(c_T)^S}{(c_{ST})}$$

belong to the identity ideal class of factor systems, then there is an ideal class $(\mathfrak{b})$ such that

$$(c_S) = \frac{(\mathfrak{b})}{(\mathfrak{b})^S}$$

for all $S \in G$. The hypothesis expresses the automorphism property; the fact that this automorphism is inner is expressed by

$$v_S = (\mathfrak{b})^{-1} u_S(\mathfrak{b}) = u_S(\mathfrak{b})^{1-S} = u_S(c_S).$$

In the cyclic special case one obtains: if $N([c])$ lies in the identity ideal class of factor systems, then (c) is a symbolic $(1-S)$-st power,

$$(c) = (\mathfrak{b})^{1-S}.$$

6. The theorem has been stated for full ideal classes; its content is unchanged if, as usual, one restricts to ideals prime to the ramified places. With this restriction, the principal genus defined here for quadratic fields becomes Gauss's principal genus: ideal classes correspond to quadratic forms, and the norms of classes correspond to the numbers represented by the forms. The identity class means that these represented numbers are quadratic residues at the ramified places; the associated forms therefore have the total character of the principal form and constitute Gauss's principal genus. It is known that the symbolic $(1-S)$-st power becomes duplication.

For cyclic fields the statement becomes: the principal genus consists of all ideal classes whose norms are norm residues at the finite and infinite ramified places. This is equivalent to being a norm residue modulo the conductor, and the usual theorem is obtained as a specialization.

Even for arbitrary Galois fields one can introduce a conductor composed only of the ramified places, in such a way that after normalizing the factor systems, at each such place the ray contains only elements of the identity class. This raises the question of the connection with Artin conductors, and hence with the theory of Galois modules, the second hypercomplex method. The future must show how far these two methods reach.

40. Noncommutative Algebras

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The main theorems of commutative algebra are, as is known, contained in Galois theory, preceded by the theory of adjunction fields and splitting fields: fields sufficient for a given polynomial to split off one linear factor, or to split completely into linear factors.

Here I develop the corresponding parts of algebra in the noncommutative case, especially in the hypercomplex case. In principle I work with noncommutative methods, namely representation in noncommutative fields. At the end I show how the above-mentioned theorems of commutative algebra can be proved in a completely parallel manner by means of representation in commutative fields.

The underlying representation theory, preceded by a short automorphism theory (§1), is a further development of the theory based on representation modules. In particular, alongside direct representation I consider reciprocal representation. Each can be reduced to the other, but the reciprocal representation is now generated by the reciprocal representation module. The advantage is that the reciprocal representation module, hence a bimodule, can also be regarded as a one-sided module over an extension ring (§2). Thus representation in noncommutative fields is reduced to ideal theory in the extension ring; the irreducible representation classes correspond to the irreducible ideal classes of the extension ring, in exact analogy with the facts which hold for a hypercomplex system itself when represented over its commutative coefficient field (§§3, 4).

From this point one obtains the structural theorems for matrix rings over noncommutative fields by observing (§5) that every subring gives a representation by this field, or equivalently a reciprocal representation by the reciprocally isomorphic field. This reciprocally isomorphic field is a first analogue of a minimal adjunction field and splitting field in the commutative case: it mediates all reciprocal representations of first degree, and, when the rank over the center is finite, also a full decomposition into direct summands of rank 1. This gives the Galois theory for fields (§6) and also expresses the fact (§7) that reciprocally isomorphic division algebras, i.e. fields of finite rank over their centers, generate inverse classes in R. Brauer's group of algebra classes.

A second proof of the Galois theory, valid more generally for simple systems, is almost an immediate consequence of a theorem on commuting subrings, which in turn is connected almost directly with the preceding automorphism considerations. Up to this point the arguments are purely noncommutative. But the question of commutative splitting fields is also treated noncommutatively by representing the splitting field through the division algebra, using the observation prepared in §5 (§7). The final part gives the transfer to the commutative case (§8) and the theory of splitting fields for arbitrary systems (§9).

R. Brauer based the theory of splitting fields on this representation theory in the commutative case, and on irrational factor systems. Because of this commutative foundation he, and later Albert, had to assume the center to be a perfect field, a restriction which is unnecessary in the noncommutative foundation. With the same restriction, again using representation theory in the commutative case, R. Brauer and K. Shoda further developed the theory.

Finally I mention that, for fields, a short account is found in van der Waerden, following my lecture course of summer 1928. Van der Waerden's account introduced several simplifications relative to that course; some of these I adopted in a second course, and some only here. The theorem on commuting subrings (§5) and the second proof of the noncommutative Galois theory based on it (§6, 1 and 2) were added only later. Certain arguments of the first course, the method of forming intersections of difference ideals, have the advantage, though they are more complicated, of transferring to systems of infinite rank and to integral systems. For commutative integral systems this leads to the connection between ideal differentiation and the different.

The principal application of the theory of splitting fields lies in the theory of crossed products and their factor systems, which in turn gives the basis for applications in number theory; this will not be pursued here.

§1. On Automorphisms, Modules, and Bimodules

Representation theory based on representation modules rests on the theory of the automorphism ring of abelian groups, with or without operators. The implicit arguments involved, namely the relations between mappings and calculation laws, are formulated here as a few simple propositions in order to avoid repetitions.

1. Multiplicative mapping. Associative law. Let $\mathfrak{G}$ first be a group without operators, and let $\mathfrak{A}$ be its absolute automorphism domain, i.e. the system of all homomorphisms of $\mathfrak{G}$ into itself. $\mathfrak{A}$ is closed under multiplication, since the product $\sigma\tau$ is defined by

$$g(\sigma\tau) = (g\sigma)\tau, \qquad g \in \mathfrak{G}, \quad \sigma, \tau \in \mathfrak{A}, \tag{1}$$

and plainly satisfies the associative law.

The group $\mathfrak{G}$ becomes a group with operators when a set $\mathfrak{B}$ of symbols $O, H, \ldots$ is given so that the products $gO, gH, \ldots$ are well-defined elements of $\mathfrak{G}$ and generate homomorphisms of $\mathfrak{G}$ into itself:

$$(gh)O = gO \cdot hO.$$

If the operator domain $\mathfrak{B}$ is multiplicatively closed, then the induced map into the automorphism domain is multiplicatively homomorphic if and only if the associative relation

$$g(OH) = (gO)H, \qquad g \in \mathfrak{G},\ O, H \in \mathfrak{B}, \tag{1a}$$

is satisfied. Correspondingly, a reciprocal homomorphism is obtained for left operators.

2. Operator-homomorphic mapping. If $\mathfrak{G}$ is a group with operators, operator-homomorphy is defined by

$$(gO)\sigma = (g\sigma)O, \tag{2}$$

and, for left operators, by

$$(Og)\sigma = O(g\sigma). \tag{2*}$$

For two operator domains $\mathfrak{B}$ and $\mathfrak{C}$, the elements of $\mathfrak{C}$ generate $\mathfrak{B}$-automorphisms, and at the same time the elements of $\mathfrak{B}$ generate $\mathfrak{C}$-automorphisms, if and only if the domains are commutatively connected with $\mathfrak{G}$, respectively if the running associative law is fulfilled:

$$(gO)H = (gH)O, \tag{2a}$$

respectively

$$(OH)g = O(Hg). \tag{2a*}$$

3. Modules and bimodules over rings. A right module $\mathfrak{M}$ over a ring $\mathfrak{R}$ is an additive abelian group with the elements of $\mathfrak{R}$ as right operators, satisfying the associativity relation and the distributive relations

$$(g+h)\rho = g\rho + h\rho, \qquad g(\rho+\sigma) = g\rho + g\sigma. \tag{3a}$$

The corresponding definitions hold for left modules.

There are two kinds of bimodules. Right modules over two rings $\mathfrak{R}$ and $\mathfrak{S}$ are called bimodules when, in addition to the individual module laws, one has

$$(m\rho)\sigma = (m\sigma)\rho.$$

An $\mathfrak{R}$-left, $\mathfrak{S}$-right module is a bimodule when the running associative law

$$\rho(m\sigma) = (\rho m)\sigma$$

is added.

If the operator domain $\mathfrak{R}$ of an additively written abelian group $\mathfrak{M}$ is a ring, then the map from $\mathfrak{R}$ into the absolute automorphism ring is ring-homomorphic precisely when $\mathfrak{M}$ is a right $\mathfrak{R}$-module; for left modules it is reciprocally ring-homomorphic. If $\mathfrak{M}$ is a right module over $\mathfrak{R}, \mathfrak{S}$, then $\mathfrak{M}$ is a bimodule precisely when $\mathfrak{R}$ maps homomorphically into a subring of the $\mathfrak{S}$-automorphism ring and $\mathfrak{S}$ maps homomorphically into a subring of the $\mathfrak{R}$-automorphism ring. For $\mathfrak{R}$-left, $\mathfrak{S}$-right bimodules, the $\mathfrak{R}$-map is reciprocal and the $\mathfrak{S}$-map direct.

In particular:

1) If $\mathfrak{R}$ is a ring with identity, then $\mathfrak{R}$ is directly isomorphic to its automorphism ring as a left $\mathfrak{R}$-module, and reciprocally isomorphic to its automorphism ring as a right $\mathfrak{R}$-module.

2) If $\mathfrak{R}$ is a full matrix ring over a generally noncommutative field A,

$$\mathfrak{R} = \sum c_{ik}A = \sum Ac_{ik},$$

then A is reciprocally isomorphic to the automorphism field of the simple right ideals, and directly isomorphic to the automorphism field of the simple left ideals.

4. Passage from a right bimodule to a one-sided module over a product ring. The significance of the right bimodule rests essentially on this passage.

Theorem. Let $\mathfrak{M}$ be a right module over a ring $\mathfrak{T}$ which contains two elementwise commuting subrings $\mathfrak{R}$ and $\mathfrak{S}$. Then $\mathfrak{M}$ may also be viewed as a bimodule over $\mathfrak{R}$ and $\mathfrak{S}$. Conversely, if a bimodule over $\mathfrak{R}$ and $\mathfrak{S}$ is given, and if there exists a product ring $\mathfrak{T}$ in which $\mathfrak{R}$ and $\mathfrak{S}$ commute elementwise, then $\mathfrak{M}$ can also be regarded as a right module over $\mathfrak{T}$, with the $\mathfrak{T}$-operation extending the given $\mathfrak{R}$- and $\mathfrak{S}$-operations.

The product ring in the absolute automorphism ring consists of all elements of the form

$$\sum_i \rho_i\sigma_i + \rho + \sigma$$

(and if $\mathfrak{R}, \mathfrak{S}$ have identities, the extra terms ρ, σ disappear). The homomorphisms

$$\mathfrak{R} \to \overline{\mathfrak{R}}, \qquad \mathfrak{S} \to \overline{\mathfrak{S}}$$

extend by

$$\sum_i \rho_i\sigma_i + \rho + \sigma \longmapsto \sum_i \overline{\rho}_i\,\overline{\sigma}_i + \overline{\rho} + \overline{\sigma}$$

to a homomorphism $\mathfrak{T} \to \overline{\mathfrak{T}}$. Hence

$$m\Big(\sum_i \rho_i\sigma_i + \rho + \sigma\Big) = \sum_i (m\rho_i)\sigma_i + m\rho + m\sigma$$

is well-defined and defines a $\mathfrak{T}$-module containing the given $\mathfrak{R}, \mathfrak{S}$-module structure.

§2. Reciprocal and Direct Representation

1. Modules of linear forms. An $\mathfrak{S}$-right module $\mathfrak{M}$, where $\mathfrak{S}$ is a ring with identity, is called a module of linear forms in $\mathfrak{S}$ if $\mathfrak{M}$ is a direct sum of n one-term $\mathfrak{S}$-modules,

$$\mathfrak{M} = m_1\mathfrak{S} + \cdots + m_n\mathfrak{S},$$

such that $m_i\mathfrak{S}$ is operator-isomorphic to $\mathfrak{S}$; thus $m_i s = 0$ always implies $s = 0$.

Theorem 1. If $\mathfrak{M}$ is a module of linear forms in $\mathfrak{S}$, then the $\mathfrak{S}$-automorphism ring $\mathfrak{U}$ of $\mathfrak{M}$ is reciprocally isomorphic, not merely homomorphic, to the ring $\overline{\mathfrak{U}}$ of all n-rowed matrices over $\mathfrak{S}$.

An arbitrary $\mathfrak{S}$-automorphism α of $\mathfrak{M}$ is completely described by

$$m_i \longmapsto \overline{m}_i = m_i\alpha;$$

hence

$$\sum_i m_i s_i \longmapsto \sum_i \overline{m}_i s_i = \sum_i (m_i\alpha)s_i = \sum_i (m_i s_i)\alpha.$$

Writing

$$(m_1\alpha, \ldots, m_n\alpha) = (m_1, \ldots, m_n)A,$$

the correspondence $\alpha \mapsto A$ is one-to-one, and

$$\alpha + \beta \mapsto A + B, \qquad \alpha\beta \mapsto BA.$$

Thus the isomorphism is reciprocal. Consequently:

Theorem 1'. The automorphism ring of a module of linear forms of degree n in $\mathfrak{S}$ admits an isomorphic, faithful reciprocal representation by the full matrix ring of n-rowed matrices over $\mathfrak{S}$.

2. Representation and representation module. A module of linear forms $\mathfrak{M}$ in $\mathfrak{S}$ is called a reciprocal representation module of $\mathfrak{R}$ in $\mathfrak{S}$ if $\mathfrak{M}$ is a bimodule over $\mathfrak{R}$ and $\mathfrak{S}$ and at the same time a right module over both $\mathfrak{R}$ and $\mathfrak{S}$. It is called a direct representation module if $\mathfrak{M}$ is a bimodule and at the same time a left $\mathfrak{R}$-module and a right $\mathfrak{S}$-module.

Theorem 2. Every reciprocal, respectively direct, representation module generates a class of equivalent reciprocal, respectively direct, representations of $\mathfrak{R}$ in $\mathfrak{S}$, and all such representations are obtained in this way.

For a reciprocal representation module, $\mathfrak{R}$ maps directly homomorphically into a subring of the $\mathfrak{S}$-automorphism ring of $\mathfrak{M}$; Theorem 1' then gives a reciprocal representation. Conversely, a reciprocal representation, composed with the reciprocal isomorphism into the $\mathfrak{S}$-automorphism ring, gives a direct homomorphism from $\mathfrak{R}$, hence a representation module. The direct case is obtained by reversing the homomorphy. Direct representations of $\mathfrak{R}$ are reciprocal representations of the ring reciprocal to $\mathfrak{R}$, and conversely.

3. Passage from a reciprocal representation module to a module over an extension ring. For representation modules the transition theorem reads as follows.

If $\mathfrak{M}$ is a right module over a ring $\mathfrak{T}$ which contains two elementwise commuting subrings $\mathfrak{R}$ and $\mathfrak{S}$, and if $\mathfrak{M}$ is a module of linear forms over $\mathfrak{S}$, then $\mathfrak{M}$ may also be regarded as a reciprocal representation module of $\mathfrak{R}$ in $\mathfrak{S}$. Conversely, a reciprocal representation module may be regarded as a $\mathfrak{T}$-module whenever the product ring $\mathfrak{T}$ exists.

If $\mathfrak{M}$ is a reciprocal representation module of $\mathfrak{R}$ in $\mathfrak{S}$, and also a $\mathfrak{T}$-module with $\mathfrak{T}$ as product ring, then $\mathfrak{R},\mathfrak{S}$-isomorphism of $\mathfrak{M}$ to a module $\mathfrak{N}$ is equivalent to $\mathfrak{T}$-isomorphism. Thus each class of isomorphic $\mathfrak{T}$-modules corresponds to a reciprocal representation class of $\mathfrak{R}$ in $\mathfrak{S}$, and conversely.

Theorem on commuting matrices. Let $\mathfrak{R} \mapsto \mathfrak{R}^*$ be a reciprocal representation of degree n of $\mathfrak{R}$ in $\mathfrak{S}$, and let $\mathfrak{B}^*$ be the ring of all n-rowed matrices over $\mathfrak{S}$ which commute elementwise with $\mathfrak{R}^*$. Then $\mathfrak{B}^*$ gives a reciprocally isomorphic representation of the $\mathfrak{R},\mathfrak{S}$-automorphisms of the representation module generated by it, i.e. the $\mathfrak{S}$-automorphisms which are at the same time $\mathfrak{R}$-automorphisms; if the product ring $\mathfrak{T}$ exists, these are the $\mathfrak{T}$-automorphisms.

The $\mathfrak{R},\mathfrak{S}$-automorphisms amount to a correspondence $m_i \mapsto m_i'$ which produces the same representation. The elements m_i' need not form an $\mathfrak{S}$-basis of $\mathfrak{M}$; correspondingly, the matrices in $\mathfrak{B}^*$ need not be invertible. In this extended sense

$$(m_1', m_2', \dots, m_n')a = (m_1', m_2', \dots, m_n')A$$

remains correct, although A is no longer uniquely determined by a.

The diagonal matrices $E \cdot s$ give a directly isomorphic representation of $\mathfrak{S}$, the identity representation of $\mathfrak{S}$. The correspondence from $\mathfrak{T}$ to matrices in $\mathfrak{S}$ defined by $\mathfrak{R}$ and $\mathfrak{S}$ is therefore not a representation of the ring $\mathfrak{T}$, but an extension of the reciprocal representation of $\mathfrak{R}$ to one which is operator-homomorphic over $\mathfrak{S}$:

$$\sum_i r_i s_i \longmapsto \sum_i R_i s_i.$$

In particular, the reciprocal representation of $\mathfrak{R}$ is operator-homomorphic with respect to the intersection $[\mathfrak{R},\mathfrak{S}]$ of $\mathfrak{R}$ and $\mathfrak{S}$, which lies in the center of $\mathfrak{R}$.

§3. Modules with Respect to a Field

In what follows only representations in generally noncommutative fields will be involved. We therefore collect a few simple results on modules of linear forms over a field.

1. Normal basis of a submodule with respect to a given basis of the full module. Let A be a field and

$$\mathfrak{N} = x_1 A + \cdots + x_n A$$

a module of linear forms of rank n. Let

$$\mathfrak{L} = z_1 A + \cdots + z_\ell A$$

be a submodule of rank $\ell \leq n$. The z_i are called a normal basis with respect to the x_i if they are of the form

$$z_i = x_i - (x_{\ell+1}\alpha_{i,\ell+1} + \cdots + x_n \alpha_{i,n}), \qquad i = 1, \dots, \ell.$$

Every module $\mathfrak{L}$ has, after a suitable numbering of the x_i, such a normal basis. Indeed,

$$\mathfrak{N} = \mathfrak{L} + x_{\ell+1}A + \cdots + x_nA,$$

and so

$$x_i \equiv x_{\ell+1}\alpha_{i,\ell+1} + \cdots + x_n\alpha_{i,n} \pmod{\mathfrak{L}}, \qquad i = 1, \ldots, \ell.$$

Thus the corresponding z_i lie in $\mathfrak{L}$ and exhaust $\mathfrak{L}$.

2. Extension module. Let A again be a field, and let P be a subfield. An A-module N of rank n is called an extension module of a P-module M, written $N = M_A$, if M is a submodule of N of the same rank. If

$$M = y_1P + \cdots + y_nP,$$

then

$$M_A = y_1A + \cdots + y_nA.$$

Conversely, for every P-module M, the extension module exists uniquely up to module isomorphism.

Corollary a). If $z_1, \ldots, z_s$ are linearly independent over P in M, then they are also linearly independent over A in M_A. Thus every P-module

$$T = z_1P + \cdots + z_sP$$

inside M generates an extension module

$$T_A = z_1A + \cdots + z_sA \subset M_A.$$

Corollary b). Every P-module T inside M is a contraction module, i.e. the intersection of its extension module with M:

$$T = T_A \cap M.$$

3. Theorem on invariant modules. Let $N = M_A$ be the extension module of a P-module M, and let P be the full invariant field for a group $\mathfrak{G}$ of ring automorphisms of A. The group $\mathfrak{G}$ is defined as an operator domain on N by

$$Gm = m, \qquad m \in M,$$

and

$$G\Big(\sum_i m_i\alpha_i\Big) = \sum_i m_i\, G(\alpha_i).$$

Lemma: M consists precisely of those elements of N fixed by $\mathfrak{G}$.

Theorem. The extension modules L_A of submodules L of M, and only these, are admissible submodules with respect to $\mathfrak{G}$ as operator domain.

One direction is clear. Conversely, let L be admissible for $\mathfrak{G}$, and let $z_1, \ldots, z_\ell$ be a normal basis of L with respect to a P-basis $x_1, \ldots, x_n$ of M. For every $G \in \mathfrak{G}$,

$$G(z_i) = x_i - (x_{\ell+1}G(\alpha_{i,\ell+1}) + \cdots + x_nG(\alpha_{i,n}))$$

again lies in L. Comparing coefficients in $x_1, \ldots, x_\ell$ gives $G(z_i) = z_i$. Hence, by the lemma, the z_i lie in M, and L is the extension of the P-module

$$T = z_1P + \cdots + z_\ell P$$

in M, with $T = L \cap M$.

§4. Hypercomplex Systems and Their Representation Classes

1. Extension theorem. We now combine the representation theorems of §2 with the module theorems of §3. Instead of arbitrary rings $\mathfrak{R}$, we consider hypercomplex systems with identity $S, T, \ldots$ over a commutative field P:

$$S = x_1 P + \cdots + x_n P.$$

Instead of arbitrary representation rings $\mathfrak{S}$, we consider only fields $A, B, \ldots$, unless otherwise stated with center P. In this case there always exists the product ring in which S and A commute elementwise, with intersection P; this is the direct product $S \times A$, which is at the same time the extension module S_A of S:

$$S_A = x_1 A + \cdots + x_n A.$$

It will therefore also be called the extension ring.

Under this specialization, the theorem on invariant modules becomes:

Extension theorem. Every two-sided ideal of S_A is the extension ideal of an ideal of S, namely the extension of its intersection with S.

Indeed, let the chosen group $\mathfrak{G}$ be the group of inner automorphisms of A. Then P, as center of A, is the full invariant field, while the elementwise commutativity of A with S says that $\mathfrak{G}$ is extended to S_A so that S is the full invariant domain. Every two-sided ideal $\mathfrak{a}$ of S_A is an admissible subgroup, since $\alpha^{-1}\mathfrak{a}\alpha \subseteq \mathfrak{a}$, and so it is the extension of its intersection module $\mathfrak{a} \cap S$, which is an ideal in S.

Addendum: If S is commutative, then S is the center of S_A. More generally, the center of S is also the center of S_A. By $A_r, B_n, \ldots$ we shall always mean matrix rings of the indicated degree over $A, B, \ldots$:

$$A_r = \sum A c_{ik} = \sum c_{ik} A.$$

We shall use essentially the extension theorem for simple systems:

If S is simple, then S_A is also simple:

$$S_A = \sum B c_{ik} = \sum c_{ik} B = B_t,$$

with associated field B, whose center contains P. Here A, and hence B, may have infinite rank over P; only S is assumed hypercomplex. If S and A have common center P, then P is also the center of S_A. More generally, if S is a simple hypercomplex system, then the direct product $S \times A_r$ is simple.

2. Representation classes. By a reciprocal or direct representation of a hypercomplex system we mean, as usual, a representation not equal to the zero representation and operator-homomorphic with respect to the coefficient field P.

Theorem on representation classes. Let S be hypercomplex with identity and coefficient field P, and let A be a field with center P. Then the number of distinct irreducible reciprocal representation classes of S in A is equal to the number of classes of simple right ideals in the quotient ring of S_A by its radical. If S is a simple system, then it has exactly one irreducible reciprocal and one irreducible direct representation class in A.

By the corollary to the transition theorem, the simple reciprocal representation classes and the simple module classes over S_A correspond one-to-one. Since S_A has finite rank over the field A, it satisfies the maximum and minimum conditions for one-sided ideals. If S is simple, then S_A is simple; hence there is only one simple one-sided ideal class, and so only one irreducible reciprocal representation class. The reciprocal system gives the corresponding assertion for direct representations.

3. Rank relation. If S is simple and

$$S_A = \sum B c_{ik}$$

has finite rank both over A and over B, then

$$n = rt, \qquad n = (S : P), \qquad t^2 = (S_A : B), \tag{1}$$

where r is the degree of the irreducible reciprocal representation.

4. Sharpening of the rank relation when A has finite rank over P. In this case the three relations are

$$(S : P) = n = rt, \tag{1}$$

$$(B : P)\, t = (A : P)\, r, \tag{2}$$

$$(S : P)(B : P) = (A : P)\, r^2. \tag{3}$$

Here (2) denotes the rank over P of a simple right ideal of S_A, expressed by the ranks over B and A; (3) follows from (2) by multiplying by r, using (1).

§5. Application of the Representation Theorems to Matrix Rings

The reciprocal representation of S in A considered in §4 can also be regarded as a direct homomorphism of S into a subring of A_r, where $\overline{A}$ denotes the field reciprocally isomorphic to A. The principle of the following paragraphs is the converse one: to reduce the investigation of matrix rings A_r to representation theory in the reciprocally isomorphic field A.

1. Reducible and irreducible embedding in A_r. Let A and $\overline{A}$ be reciprocally isomorphic fields, of finite or infinite rank over their center P. The simple system S over P is called irreducibly, respectively reducibly, embeddable in A_r if S has an irreducible, respectively reducible, reciprocal representation of degree r in A. If S is irreducibly embeddable in A_r, then it is reducibly embeddable in all matrix rings A_{rs}, $s > 1$. The isomorphism always extends the identity of P; by §4, 3, r divides the rank n of S over P.

In this way one obtains all simple subrings containing P, of finite rank over P, in the various A_r. For every such subring is reciprocally isomorphic to a subring of $\overline{A}_r$ and therefore has a reducible or irreducible reciprocal representation in A.

2. Theorem on inner automorphisms. Let $S^{(1)}$ and $S^{(2)}$ be two simple subrings of A_f containing P, of finite rank over P. If $S^{(1)}$ and $S^{(2)}$ are isomorphic over P, then this isomorphism is induced by an inner automorphism of A_f.

Indeed, $S^{(1)}$ and $S^{(2)}$ give generally reducible embeddings of a simple system S in A_f, hence direct representations of degree f in A. They belong to the same reducible direct representation class; the transforming matrix induces the automorphism. If A itself has finite rank over P, so that it is hypercomplex, this becomes the familiar theorem: two simple subsystems of A_f containing the center P, and isomorphic over P, are carried into one another by an inner automorphism of A_f. In particular, every automorphism of A_f is inner.

3. Theorem on elementwise commuting subrings. Let S be a simple system contained in A_f and containing P, and let R be the totality of all elements of A_f which commute elementwise with S. Then R is also a matrix ring:

$$R = B_s.$$

The associated fields B of S_A and B of R are reciprocally isomorphic. The intersection of R and S is the center of S. The ring R is a field if and only if S is irreducibly embedded in A_f.

By §2, 3, the ring R is directly isomorphic to the automorphism ring of the reciprocal representation module $\mathfrak{M}$ of S in A. This is the automorphism ring of $\mathfrak{M}$, regarded as an S_A-module. If $\mathfrak{M}$ decomposes into s simple S_A-modules and one writes, as in §4, 1,

$$S_A = \sum Bc_{ik},$$

then R is isomorphic to $\sum Bc_{ik}$, where the two fields are reciprocally isomorphic. Thus $R = B$ is a field precisely when $s = 1$, i.e. when S is irreducibly embedded. By definition, $R \cap S$ is the center of S.

4. Sharpened commutation theorem for hypercomplex matrix rings. In this case the rank relations of §4, 4 give the sharpening:

The simple subsystems of A_f which contain P, where A has finite rank over its center P, split into pairs $S, \overline{S}$, such that $\overline{S}$ consists exactly of all elements which commute elementwise with S, and conversely. The associated fields of S_A and $\overline{S}$ are reciprocally isomorphic, as are those of S and $\overline{S}_A$. The intersection of S

and $\overline{S}$ is the common center. One subring is a field if and only if the other is irreducibly embedded. The product of the ranks over P of S and $\overline{S}$ is the rank of A_f.

If $f = rs = r's'$, where S is irreducibly embeddable in A_r and $\overline{S}$ in $A_{r'}$, then the associated fields of S and $\overline{S}$ become isomorphic to a pair of commuting subrings of A_g, with $g = ss'$.

The product assertion follows directly from the rank relation. If S is irreducibly embedded, then $\overline{S}$ is a field and is reciprocally isomorphic to B, where $S_A = B_t$; the rank relation (3) of §4, 4 gives the assertion. If S is reducibly embedded, so $\overline{S} = B_s$, then $f = rs$, and multiplying (3) by s^2 gives the assertion. This also shows that $\overline{S}$ is precisely the totality of the elements commuting elementwise with S, and the remaining assertions follow from the theorem on commuting subrings.

§6. Galois Theory of Simple Systems

The sharpened commutation theorem of §5, 4 expresses the Galois theory of simple systems with respect to the center as ground domain. We consider first division rings, and then simple systems in general.

1. Galois theory of division rings of finite rank over the center. The Galois group $\mathfrak{G}$ of A is defined as the group of all automorphisms extending the identity of the center P, hence, by §5, 2, as the group of all inner automorphisms. Thus

$$\mathfrak{G} \simeq A^*/P^*,$$

where A^* and P^* denote the multiplicative groups of nonzero elements of A and P. Subgroups $\mathfrak{H}$ of $\mathfrak{G}$ therefore correspond one-to-one to subgroups H^* of A^* containing P^*, by

$$\mathfrak{H} \sim H^*/P^*.$$

A subgroup $\mathfrak{H}$ of $\mathfrak{G}$ is called closed if, after adjoining zero, H^* becomes a division ring H. The fact that C admits the group $\mathfrak{H}$ is equivalent to saying that C commutes elementwise with H. Hence the commutation theorem of §5, 4 becomes the following.

Main theorem of the Galois theory of noncommutative division rings. The division rings C between P and A and the closed subgroups $\mathfrak{H}$ of $\mathfrak{G}$ correspond bijectively in such a way that C is the full invariant division ring of $\mathfrak{H}$, and $\mathfrak{H}$ is the full invariant group of C.

2. Galois theory of simple systems. If A_f is a simple system with center P, the Galois group $\mathfrak{G}$ is again the group of inner automorphisms, hence isomorphic to

$$A_f^*/P^*,$$

where now A_f^* denotes the multiplicative group of the regular elements of A_f. A subgroup $\mathfrak{H} \sim H^*/P^*$ of $\mathfrak{G}$ is called simple-closed if the subring H generated by H^* is a simple system and if H^* is the set of all regular elements of H. The passage from the commutation theorem to Galois theory is supplied by the following lemma.

Lemma. Every simple system in A_f containing P is generated by the group H^* of its regular elements. This is clear if P has infinitely many elements: the "general element" formed with indeterminates is regular, and by suitable specialization one obtains regular basis elements over P. The lemma holds in general by Shoda.[1]

By the lemma, commutation of S with a simple system $\mathfrak{S}$ is again equivalent to saying that S admits the automorphisms induced by the regular elements of $\mathfrak{S}$. Thus the commutation theorem of §5, 4 becomes also here the following.

Main theorem of the Galois theory of simple systems. The simple systems S in A_f containing P and the simple-closed subgroups $\mathfrak{H}$ of $\mathfrak{G}$ correspond bijectively in such a way that S is the full invariant domain of $\mathfrak{H}$, and $\mathfrak{H}$ is the full invariant group of S.

3. Proof by means of the extension principle. For the case of a division ring I give a second proof of the main theorem. It rests on the principle of extending isomorphisms, i.e. representations of first degree, and since it uses no count of ranks it makes fewer finiteness assumptions. In particular it shows in what direction a transfer to division rings S of infinite rank will be possible. The proof also does not use the fact that there is only one irreducible reciprocal representation class; it therefore applies exactly in the same way in the commutative case, §8, 2. I first state some lemmas.

[1] Shoda, work cited in note 3); there the introduction of indeterminates is avoided.

Hypotheses. The simple system S over P is to admit a reciprocal representation of first degree in A, and is therefore a division ring. Here A may have finite or infinite rank over its center P. A decomposition of S_A into n simple operator-isomorphic right ideals, as in §4, 3, is given by

$$S_A = r_1 + \cdots + r_n = e_1 S_A + \cdots + e_n S_A = e_1 A + \cdots + e_n A;$$

the reciprocal representation generated by e_i is therefore defined by

$$e_i \sigma = e_i \sigma_i.$$

Lemma 1. The n reciprocal representations generated by $e_1, \ldots, e_n$ are distinct, hence are distinct representations of the same representation class. Thus S has at least as many distinct representations as its rank.

To prove this, we extend the representations from S, as at the end of §2, to correspondences of S_A which are operator-homomorphic over A. Here these correspondences are defined by

$$e_i w = e_i \omega$$

with $\omega \in A$, for each $w \in S_A$. If e_i and e_j, $i \neq j$, generated the same representation, one would have $e_i w = e_j w$ for every $w \in S_A$. This is impossible because $e_i e_i = e_i$ and $e_j e_i = 0$.

Lemma 2. Let T be a division ring between P and S, and let s be the rank of S over T. Then every reciprocal isomorphism of T into A admits at least s different extensions.

Let the reciprocal isomorphism, i.e. the reciprocal representation of T in A, be mediated by a decomposition

$$T_A = E_1 T_A + \cdots + E_h T_A = E_1 A + \cdots + E_h A, \qquad E_i t = E_i \tau,$$

where the E_i are idempotents. This is no restriction, since a representation generated by a basis element $E_i \alpha$ is also generated by $\alpha^{-1} E_i \alpha$, hence by an idempotent. Right multiplication by S gives

$$S_A = E_1 S_A + \cdots + E_h S_A.$$

The $E_i S_A$ all have rank s over A. Indeed the rank of $E_i S_A$ is at most s, as follows by substituting a T-basis into S, using the commutativity of S and A:

$$S = T w_1 + \cdots + T w_s,$$

$$E_i S_A = E_i T_A w_1 + \cdots + E_i T_A w_s = E_i w_1 A + \cdots + E_i w_s A.$$

Since the sum S_A has rank $n = hs$, every $E_i S_A$ has exactly rank s. Thus

$$E_i S_A = r_{i1} + \cdots + r_{is} = e_{i1} A + \cdots + e_{is} A,$$

where the s representations generated by $e_{i1}, \ldots, e_{is}$ are all distinct by Lemma 1. They are all extensions of the representation of T generated by E_i, since $E_i t = E_i \tau$ implies

$$(e_{i1} + \cdots + e_{is}) t = (e_{i1} + \cdots + e_{is}) \tau$$

and hence $e_{ij} t = e_{ij} \tau$, by the direct decomposition.

Definition. The partition $\mathfrak{I}_T$ of the reciprocal isomorphisms $\mathfrak{I}$ from S into A, induced by T, is defined as follows: precisely those isomorphisms in $\mathfrak{I}$ are regarded as equivalent which induce the same isomorphism on T.

Main theorem. If $\mathfrak{I}_T$ is the partition induced by T, then T is a maximal subfield relative to this partition.

For if Z is an intermediate division ring between T and S, of degree l over T, then by Lemma 2 every class of $\mathfrak{I}_T$ splits into at least l classes of $\mathfrak{I}_Z$. The transition from this form of the theorem to the usual one arises by multiplying the elements of the set $\mathfrak{I}$ by the inverse of any fixed one of them; the set then becomes the automorphism group, and a class in $\mathfrak{I}_T$ becomes the invariant group of T. The statement that closed subgroups are full invariant groups is included in this: for $\mathfrak{H} \sim H^*/P^*$, one simply puts the division ring H in place of the intermediate division ring T.

§7. The Class of Similar Algebras. Splitting Fields

From now on only division rings $A, \overline{A}$ and matrix rings A_f of finite rank over their center P will occur. Thus they are simple normal algebras over P, and will be called algebras over P, or simply algebras; $A, \overline{A}$ are then division algebras. All A_f with the same associated A are collected into one class of similar algebras A, A_f. Isomorphic A's are identified.

1. The group of algebra classes. If A, B are algebras over P, then the same is true of their product, the direct product over P; for by the end of §3,

$$A_f \times_P B_g = C_h,$$

where C is a uniquely determined, up to isomorphism, division algebra with center P. Multiplication of A_f, B_g by matrix rings over P shows that this multiplication is uniquely determined for the classes, which therefore form a multiplicatively closed, commutative system. Moreover R. Brauer's theorem holds: the classes of similar algebras form an abelian group under direct product. The system has an identity class, consisting of matrix rings over P, that is, of the algebras similar to P. Also each class (A) has the inverse $(\overline{A}) = (A)^{-1}$, consisting of the algebras similar to $\overline{A}$, where $\overline{A}$ is reciprocally isomorphic to A. This follows from the commutation theorem §5, 3 by the specialization $S = A$. For A can be embedded irreducibly in itself; one obtains $B = P$ and $A_A = P_f$.[2]

2. Splitting fields of an algebra class. The theory of splitting fields again rests entirely on the principle stated at the beginning of §5: the commutative extension fields are represented in A or in $\overline{A}$. We first need the following lemma.

Lemma. If A is a division ring with center P, and Z is a finite commutative extension field of P, then A_Z is simple with center Z. For by §4, 1 this holds for the system $Z_{\overline{A}}$, reciprocally isomorphic to A_Z. Thus for every class (A) of algebras over P similar to A, the field Z gives an extension class $(A)_Z$ of simple algebras over Z similar to A_Z.

The associated division algebra D of an extension class follows immediately from the commutation theorem of §5, 3.

Theorem on the associated division algebra of an extension class. Let $(A)_Z$ be an extension class, and let Z denote an irreducible embedding, i.e. representation, of Z in A_f. Then

$$(A)_Z = (D),$$

where D is the set of all elements of A_f which commute elementwise with Z. One only has to replace the simple system S in §5, 3 by Z, observing that Z_A and A_Z are reciprocally isomorphic. If Z is embedded reducibly, the totality of elements commuting elementwise with Z is a matrix ring over D.

A commutative extension field Z of P is called a splitting field of the class (A) if the extension class $(A)_Z$ splits completely, that is, is equal to the identity class over Z, the class of all matrix rings over Z.

Theorem on the characterization of splitting fields. A finite commutative extension field Z of P is a splitting field of the class (A) if and only if its irreducible embedding in A_f gives a maximal commutative subfield of A_f.

Indeed, the property that Z be a splitting field of (A) is equivalent to $(D) = (Z)$. Hence, by the theorem on the associated division algebra, the irreducible embedding Z must be a maximal commutative subfield. If this condition is fulfilled, then necessarily $D = Z$, since adjoining any $a \in D$ to Z gives a commutative extension field.

Remarks. 1. It follows at the same time that a maximal commutative subfield Z, when irreducibly embedded, generates a maximal commutative subring. For the elements commuting elementwise with Z form a division ring.

2. The theorem also gives the existence of splitting fields, since the associated division algebra A certainly has maximal commutative subfields.

[2] The finer theorems on the group of algebra classes use the theory of factor systems; that will not be taken up here. Notice that up to this point no considerations concerning commutative fields have occurred. For example, the fact that the rank $(A : P)$ is a square has neither been used nor proved.

3. Rank relations, index, infinite commutative extensions. The rank statement from §5, 4 first gives the known fact that the rank of a simple algebra over its center is a square. For if Z is maximal commutative in A, then, since every embedding in A is irreducible,

$$(Z:P)(Z:P)=(A:P),$$

hence

$$(A:P)=m^2, \qquad (Z:P)=m,$$

where m is Schur's index; it is also the degree of all splitting fields embeddable in the division algebra A itself. For the degree n of a general splitting field one obtains

$$n=mr,$$

for instance from $(A_r:P)=m^2r^2$, or also from the rank relation §4, 3, since the index m is also the number of simple components of A_Z, which becomes a matrix ring over Z of rank m^2. More generally, if $A_Z \simeq D_l$, if L is the irreducible embedding of A in A_r, and if d^2 is the rank $(D:L)$ of the division algebra D over its center Z, then

$$l\,d=mr.$$

Indeed, by §5, 4 and the theorem on the associated division algebra from 2.,

$$(A_r:P)=(L:P)(D:P),$$

so

$$m^2r^2=l^2d^2.$$

From the existence of splitting fields follow the facts for infinite commutative, algebraic or transcendental, extension fields Ω of P: the extension class $(A)_\Omega$ is a class of simple algebras with center Ω. If in particular Ω is algebraically closed, then A_Ω is a matrix ring of degree m. Thus the index is equal to the "absolute number of components." For if Z is a splitting field of A and Ω' is the compositum of Ω and Z, then $A_{\Omega'}$ is a matrix ring over Ω', hence has no radical and has center Ω'. Therefore A_Ω also has no radical and its center is Ω; any element in the center of A_Ω not belonging to Ω would also lie in the center of $A_{\Omega'}$. Hence A_Ω is a simple algebra over Ω, and certainly a matrix ring over Ω if Ω is algebraically closed.[3]

4. Existence of separable splitting fields. Every class (A) has separable splitting fields, indeed such that can be embedded in A itself.[4]

Let the ground field P have characteristic p, and let the index be

$$m=sp^f$$

with s prime to p. If Z is a splitting field of A of degree m, and Z_1 is the extension of first kind contained in it, so that

$$(Z:Z_1)=p^f,$$

then the index of $A_{Z_1}\sim D$ is p^f. We prove the existence of a separable extension field Z_2 of Z_1 such that the index of D_{Z_2} is a proper divisor of p^f. Now D_Ω, with Ω algebraically closed, has no radical by 3.; the reduced discriminant of D therefore does not vanish. Hence there is at least one element $d\in D$ with nonzero reduced trace. This d cannot already lie in the ground field Z_1, because the trace of every element $\alpha\in Z_1$ is $p^f\alpha$, hence vanishes. Thus $Z_2=Z_1(d)$ is a proper, indeed separable, extension field; the index of D_{Z_2} is a proper divisor of p^f. Repeating this finitely many times gives a separable splitting field of (A) whose degree m agrees with the index of A.

§8. Splitting Fields, Decomposition Fields, and Galois Theory in the Commutative Case

In order to pass from the splitting fields of systems simple over their center to those of arbitrary simple systems, one still has to develop the splitting theory of the center; a Galois theory parallel to §6, 3 is to be added to it. All fields and systems occurring in this paragraph are commutative.

[3] In general there are also "minimal" splitting fields, i.e. such that no proper subfield is a splitting field, of all possible degrees. Compare Brauer–Noether, the work cited in note 2).

[4] Compare G. Köthe, Über Schiefkörper mit Unterkörpern zweiter Art über dem Zentrum, Journ. f. Math. 166 (1932), pp. 182–184. The present, much simpler proof goes back to a remark of M. Zorn.

1. Splitting and decomposition fields of commutative simple systems. Let the commutative system Z be simple over P, hence a field, and let Ω be an algebraically closed extension field of P. It is known that Z_Ω remains a system without radical if and only if Z is separable, i.e. an extension of first kind, over P. For only then does it have as many isomorphic maps of first degree into Ω as its rank over P, hence as many distinct representations, which in this case coincide with representation classes.

The possibility that a radical occurs in Z_Ω, so that a direct decomposition into absolutely simple components need not take place, leads to the following definitions. An extension field Λ of P is called a splitting field if Z_Λ splits into composition factors of first degree; in other words, if all absolutely irreducible representations already lie in Λ. An extension field T of P is called a decomposition field if Z_T splits off at least one composition factor of first degree; in other words, if at least one absolutely irreducible representation of Z exists in T.

Splitting and decomposition fields are characterized by the following theorem.

Theorem. A field T is a decomposition field if and only if it contains a subfield Z' isomorphic to Z. A field Λ is a splitting field if and only if it contains a subfield Γ isomorphic to the Galois field belonging to Z.

This follows directly from the definitions of splitting and decomposition fields in terms of absolutely irreducible representations. In particular, in analogy with the noncommutative case, one obtains the following corollary: a field Z' is a minimal decomposition field, i.e. no proper subfield is a decomposition field, if and only if it is isomorphic to Z. A minimal decomposition field is at the same time a splitting field if and only if Z is normal, hence a Galois field over P.

This is the reason for calling a simple algebra normal over its center. In a fixed algebraically closed Ω over P, there is only one minimal splitting field, namely the Galois field belonging to Z, in contrast to the generally infinitely many nonisomorphic fields in the noncommutative case. This has its source in the uniqueness of the direct decomposition in the commutative case, in contrast to the noncommutative case where uniqueness is only up to operator isomorphism; equivalently, in the finitely many different absolutely irreducible representations in the commutative case, as opposed to the infinitely many different ones in the noncommutative case, although they belong to the same class.

2. Hypercomplex proof of Galois theory in the case of separable fields Z/P. In exact analogy with §6, 3, the theory of isomorphisms holds for arbitrary Z separable over P; if Z is Galois, one can then pass to the usual automorphism group.

Hypotheses. The simple system Z over P is a finite separable extension field, Ω denotes a finite or infinite splitting field over P, $Z' = Z^{(1)}$ a subfield isomorphic to Z, and Γ the corresponding Galois field. By 1. there is a direct decomposition

$$Z_\Omega = r^{(1)} + \cdots + r^{(n)} = e^{(1)} Z_\Omega + \cdots + e^{(n)} Z_\Omega = e^{(1)}\Omega + \cdots + e^{(n)}\Omega.$$

The representation $Z \to Z^{(i)}$ generated by $e^{(i)}$, with $Z^{(i)} \subseteq \Gamma$, is defined by

$$e^{(i)} z = e^{(i)} z^{(i)}.$$

Lemma 1. The n representations generated by $e^{(1)}, \ldots, e^{(n)}$ are all distinct. The proof is as in §6, 3, or also follows from 1.; they belong to distinct representation classes.

Lemma 2. If T is an intermediate field between P and Z, and s is the rank of Z over T, then every isomorphism of T into Ω admits at least, and hence exactly, s extensions.

The proof is as in §6, 3. The strengthening of "at least" to "exactly" follows from the uniqueness of the decomposition of Z_Ω, according to which Z has not merely at least but exactly n isomorphisms in Ω.

As in §6, 3, one defines the partition $\mathfrak{I}_T$ induced by T of the finite set $\mathfrak{I}$ of isomorphisms of Z: precisely those isomorphisms in $\mathfrak{I}$ are regarded as equivalent in $\mathfrak{I}_T$ which induce the same isomorphism on T. One proves the following.

Main theorem. If $\mathfrak{I}_T$ is the partition induced by T, then T is a maximal subfield with respect to this partition.

From here, in the case of a Galois Z, one passes to the automorphism group and to the usual formulation exactly as in §6, 3 by composing with the reciprocal of an isomorphism. The corresponding assertion for subgroups follows from the consideration of idempotents, which is of interest in itself.

3. The idempotents of Z_Ω. Let S run through the n substitutions which carry Z into the individual $Z^{(i)}$, that is, the compositions of the isomorphisms $Z \to Z_1$ and $Z \to Z^{(i)}$, where the first is the reciprocal of $Z \to Z_1$. The field Z need not be Galois. For elements $a \in Z_\Omega$, the substitutions S are defined as operators by stipulating that they induce the identity on Z. Thus for each a the conjugate a^S lying in $Z_{(i)}$ is defined. With these conventions one has the following theorem.

Theorem. The n idempotents $e^{(i)}$ of Z_Ω are conjugate:

$$e^{(i)} = e^S, \qquad e = e^{(1)};$$

they lie in the n conjugate extension systems $Z\Omega$.

Indeed, applying S carries the defining relations

$$ez = ez^{(1)}, \qquad e^2 = e$$

to

$$e^S z = e^S z^{(i)}, \qquad (e^S)^2 = e^S.$$

By uniqueness of the decomposition the idempotents are themselves uniquely determined; hence $e^S = e^{(i)}$. Since Z is also a decomposition field, so that the representation $ez = ez^{(1)}$ is already mediated, e already lies in $Z\Omega$, and consequently e^S lies in $Z^S\Omega$.

If Z is Galois, the part of the main theorem of Galois theory concerning subgroups follows immediately. If $\mathfrak{H}$ is a subgroup of the Galois group $\mathfrak{G}$, then, by the uniqueness of the components of a direct sum, the element

$$\sum_{H \in \mathfrak{H}} e^H$$

admits all substitutions from $\mathfrak{H}$ and no others. Since the group $\mathfrak{G}$ was defined for Z, this is the Galois theory of Z; by isomorphism it also settles the theory for Z'.

4. Connection of the idempotents with complementary bases. Again Z is not assumed to be Galois.

Theorem. Let $a_1, \dots, a_n$ be a basis of Z over P, and write the idempotent as

$$e = a_1\beta_1 + \cdots + a_n\beta_n,$$

so that

$$e^S = a_1\beta_1^S + \cdots + a_n\beta_n^S.$$

Then

$$\beta_1^S, \dots, \beta_n^S$$

are the images, under the map mediated by e^S, of the complementary basis $b_1, \dots, b_n$ to $a_1, \dots, a_n$.

For the map mediated by e^S, $e^S x = e^S x^S$, in particular $e^S a_i = e^S a_i^S$, the relations

$$e^S e^S = e^S, \qquad e^S e^R = 0 \quad (S \neq R)$$

give the assignments $e^S \mapsto 1$, $e^R \mapsto 0$. Hence

$$1 = a_1^S\beta_1^S + \cdots + a_n^S\beta_n^S, \qquad 0 = a_1^R\beta_1^S + \cdots + a_n^R\beta_n^S \quad (S \neq R).$$

Written in matrix form, this is the defining equation for the complementary bases:

$$\begin{pmatrix} a_1^1 & \cdots & a_n^1 \\ \vdots & \ddots & \vdots \\ a_1^n & \cdots & a_n^n \end{pmatrix} \begin{pmatrix} \beta_1^S \\ \vdots \\ \beta_n^S \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}.$$

The passage to the trace definition follows because, for

$$a_i = \sum_S a_i^S e^S, \qquad b_i = \sum_S b_i^S e^S,$$

the matrix relation, after interchanging rows, becomes

$$\delta_{ij} = \mathrm{Sp}(a_i b_j).$$

Here the a_i and b_i lie in Z, since they admit all substitutions S.[5,6]

The contragredience of the complementary bases follows here from the fact that the idempotents e^S are invariants of Z, independent of the basis chosen. Thus all

$$a_1\beta_1^S + \cdots + a_n\beta_n^S$$

are transformed into themselves when one passes to another basis

$$a_1', \ldots, a_n'$$

of Z/P.

§9. Splitting Fields and Decomposition Fields of Arbitrary Systems

1. Definitions and reduction to the simple case. The definitions given in §8 correspond in the general case to the following. Let S be hypercomplex over P. A commutative extension field Λ of P is called a splitting field of S if, in a composition series by one-sided ideals, S_Λ splits into absolutely simple composition factors; in other words, if all irreducible representations of S in Λ are already absolutely irreducible. An extension field T of P is called a decomposition field if S_T splits off at least one absolutely simple composition factor; equivalently, if at least one representation irreducible in T is already absolutely irreducible.

These definitions plainly contain the ones given in §8 for the commutative case and in §7 for simple algebras. The latter is true because A_Λ, for every commutative extension of the center, is completely reducible, so that composition factors and components of the direct sum decomposition coincide. Full matrix rings over the center, and only these, give absolutely simple components, hence also absolutely irreducible representations. Moreover A_Λ is two-sided simple, hence decomposes into operator-isomorphic components. Therefore the splitting off of one absolutely simple component already gives complete splitting: decomposition fields and splitting fields coincide.

From the definitions the following facts follow immediately: the splitting fields of S are the composita obtained by taking one splitting field for each representation irreducible in P; a decomposition field of S is any decomposition field of a representation irreducible in P. Because of the one-to-one correspondence between the simple systems, namely the components of the residue-class ring modulo the radical, and the representations irreducible in P, it is enough to consider simple systems. The results will then follow by combining §7 and §8.

2. Splitting fields and decomposition fields of simple systems. We first consider the case of a simple system A with separable center Z over P. From §8, 1 and §8, 3 one obtains the following. The two-sided decomposition of A_Ω corresponding to the direct decomposition of Z_Ω, where Ω denotes a splitting field of Z, gives n conjugate components

$$e^S A_\Omega$$

of A_Ω, each isomorphic to A, which become simple algebras over their centers

$$e^S Z_\Omega = e^S Z.$$

The substitutions S are defined as operators for A_Ω by stipulating that they induce the identity on A.

Indeed, by §8, 3 the idempotents are conjugate; the same is therefore true of the components $e^S A_\Omega$, by the definition of the substitutions for A. Since A is two-sided simple, $e^S A_\Omega$ is ring-isomorphic to A.

[5] Compare Representation Theory §25.

[6] Connections between complementary bases and idempotents occur first in Dedekind, Zur Theorie der aus n Haupteinheiten gebildeten komplexen Größen; compare vol. II of the collected works, pp. 4–5.

Moreover $e^S A_\Omega = e^S A_{Z^S}$, because $e^S Z_\Omega = e^S Z$; hence the center coincides with the coefficient domain, and the components are normal algebras.

The results of §7, 2 now further imply: if A is a simple system over P with separable center Z over P, then A_Ω too is without radical, hence completely reducible; here Ω may be any finite or infinite extension field. For by §7, 2, $e^S A_\Omega$ remains simple after every extension of coefficients, and therefore has no radical. Hence the same holds for the direct sum; this is A_Ω, or, if Ω is not a splitting field of Z, is obtained by coefficient extension from A_Ω.

From §7, 2 follows also the following characterization.

Characterization of decomposition fields and splitting fields. If A is a division algebra with separable center Z, then the irreducible embeddings of the decomposition fields are given by all and only the maximal commutative subfields of the A_f containing Z. The splitting fields are all and only the composita of a decomposition field with a splitting field of the center, in particular with the Galois field belonging to the center. A minimal decomposition field may be a splitting field even when the center is not Galois.

A decomposition field must contain a field isomorphic to Z; the assertion about embeddings of decomposition fields now follows from the preceding facts and from §7, 2. A splitting field must moreover contain a splitting field Ω of Z. But every decomposition field over Ω is at the same time a splitting field, since the components $e^S A_\Omega$ are simple and have a center isomorphic to Ω. If Z is Galois, minimal decomposition fields and splitting fields therefore coincide. But even in the non-Galois case, unlike the commutative case, there may be minimal decomposition fields which are at the same time splitting fields, namely whenever such a minimal decomposition field contains the Galois field belonging to Z. Example: a quaternion division algebra with center isomorphic to $P(\sqrt{2})$, P the field of rational numbers; the Galois field belonging to $P(\sqrt{2})$ is a minimal decomposition and splitting field.

Finally, if the center is inseparable, then Z_Ω, and with it A_Ω, is a system with radical. Let $\mathfrak{C}$ be the radical of Z_Ω, and let $\mathfrak{C}A_\Omega$ be the extension ideal in A_Ω. Then $Z_\Omega/\mathfrak{C}$ has a direct decomposition into conjugate components, in exact analogy with §8, 2. This gives, just as above, the direct decomposition of $A_\Omega/\mathfrak{C}A_\Omega$, in such a way that the components $e^S A_\Omega$ are isomorphic to A, with center $e^S Z^\Omega$. Thus the theorems on decomposition and splitting fields remain valid. The count of ranks further shows that the composition factors, corresponding to a composition series of $\mathfrak{C}A_\Omega$, are operator-isomorphic to $A_\Omega/\mathfrak{C}A_\Omega$, not merely homomorphic.

Expressed for irreducible representations, this gives the summary: if a representation of a hypercomplex system irreducible over P is given, then the representation is absolutely completely reducible if and only if the associated center is separable over P. In every case, the splitting and decomposition fields of the representation can be characterized as above by embedding into the associated simple system. In particular, the lowest degree of a decomposition field is the product of the degree of the center and the index of the associated algebra class over the center; the degree of every decomposition field is a multiple of this. The representation decomposes, in the sense of composition series, into as many distinct but conjugate absolutely irreducible representation classes as the degree of the largest separable field contained in the center. Each class occurs as many times as is given by the product of the index and the degree of the center over this separable extension.

For a perfect ground field, where only separable extension fields occur, this returns the known results of I. Schur, with the addition of the embedding characterization which already occurs in R. Brauer.

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41. The Principal Genus Theorem for Relative Galois Number Fields

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In recent times it has become clear that noncommutative methods, in particular the theory of algebras, make it possible to formulate and prove familiar theorems on relative cyclic and abelian number fields for arbitrary relative Galois fields. I recall above all the norm theorem, which in general can be expressed as a theorem on split algebras. In what follows I show that, correspondingly, the principal genus theorem too can be formulated hypercomplexly, as a theorem on inner automorphisms or on crossed representations. The proof follows from the above-mentioned theorem on split algebras by purely algebraic-arithmetical considerations, whereas in the former theorem the norm theorem in the cyclic case, and already for prime degree, remains the transcendental core.

For better understanding of the formulation I first put forward the principal genus theorem "in the minimal sense", not used below: it is an elementary algebraic theorem which, in the cyclic special case, passes into Hilbert's familiar Theorem 90, according to which $N(a)$ is equal to one if and only if $a = b^{1-S}$. I formulate even this theorem as one about inner automorphisms or crossed representations, and prove it by means of the general theorems on inner automorphisms and crossed product representations of algebras.

In the actual principal genus theorem, the crossed product of the ideal-class group and the Galois group replaces the latter construction, but with a finer induced division of the factor systems into ideal classes. This captures an analogue of the ray-class division of class-field theory. Since general theorems on automorphisms and representations are still absent here, the principal genus theorem may be regarded as a first step in this direction. I therefore give a reduction, as mentioned above, by means of the theorem on split algebras. In this way the theorem is reduced to the corresponding theorem for ideals rather than ideal classes; the Brandt theorems on the interrelation of the maximal orders of an algebra may be regarded here as the analogue of the automorphism theorems. Together with the study of maximal orders of crossed products, they give a proof essentially parallel to the proof in the minimal case, though more complicated because one must pass to the individual places. I therefore prefer the almost trivial proof suggested by E. Artin, which is an even simpler analogue here of Speiser's proof.

§1. Principal Genus Theorem in the Minimal Case

In this paragraph k denotes an arbitrary commutative field, not necessarily a number field; K/k is a separable Galois extension of degree n, and $\mathfrak{G}$ is its Galois group.

I first recall the familiar facts about crossed products. The crossed product means a simultaneous embedding of K and $\mathfrak{G}$ in an algebra A such that the automorphisms of K become inner. If $u_{S_1}, \dots, u_{S_n}$ are symbols corresponding to the n group elements, first set A, as a module of linear forms of rank n over K, equal to

$$A = u_{S_1}K + \cdots + u_{S_n}K. \tag{1}$$

By demanding the inner automorphism generated by u_S, more generally by $u_S K^*$, A becomes a ring, hence an algebra of rank n^2 over k. The demand is expressed by

$$u_S^{-1} z u_S = z^S, \qquad \text{or} \qquad z u_S = u_S z^S \tag{2}$$

for every $z \in K$, and by

$$u_S u_T = u_{ST} a_{S,T}, \qquad a_{S,T} \in K^*. \tag{3}$$

The associative law is equivalent to

$$a_{S,T}^R a_{ST,R} = a_{S,TR} a_{T,R}. \tag{4}$$

Defining the product of two arbitrary elements by

$$\Big(\sum_S u_S b_S\Big)\Big(\sum_T u_T c_T\Big) = \sum_{S,T} u_{ST}\, a_{S,T}\, b_S^T c_T,$$

one obtains the crossed product of K with $\mathfrak{G}$ for the factor system $a_{S,T}$, written

$$A = (a_{S,T}, K, \mathfrak{G}).$$

One proves that A is a simple normal algebra over k, hence a matrix ring D_r of degree r over the associated division algebra D, and that K is a maximal commutative subfield, hence a splitting field. Conversely, for every such normal division algebra D there are matrix rings D_r which arise in this way as crossed products.

If one passes from u_S to $v_S = u_S c_S$, with $c_S \in K^*$, the associated factor systems are

$$\bar{a}_{S,T} = a_{S,T} \frac{c_S^T c_T}{c_{ST}}. \tag{5}$$

In particular the factor systems

$$\frac{c_S^T c_T}{c_{ST}}$$

are associated with one; these are called transformation quantities. Associated factor systems are collected into a class (a). The classes (a) correspond one-to-one to direct product formations of an abelian group; under direct product they form an abelian group, isomorphic to the componentwise product of classes of factor systems. The identity element is the split algebra class, i.e. the system of all transformation quantities. This is the R. Brauer group of algebra classes.

For cyclic algebras one obtains the following special case. If Z/k is cyclic and S is a generating substitution, the powers of u correspond to the powers of S, and

$$A = Z + uZ + \cdots + u^{n-1}Z, \tag{1'}$$

$$zu = uz^S, \qquad u^n = \alpha, \tag{2'–3'}$$

with

$$\alpha \in k^*, \qquad \bar{\alpha} = \alpha\, N(c) \quad \text{if } v = uc. \tag{4'–5'}$$

Thus each factor system here consists of a single element of the ground field, written $A = (\alpha, Z, S)$; the identity class is given by norms from Z^*, and the group of algebra classes is isomorphic to $k^*/N(Z^*)$.

The formulation of the principal genus theorem in the minimal case uses the fact that relations (2)–(5) are purely multiplicative and hence also define the extension group $\mathfrak{G}^*$ of $\mathfrak{G}$, consisting of the elements in the n complexes $u_S K^*$:

$$\mathfrak{G}^* = \{\, u_{S_1}K^*, \ldots, u_{S_n}K^* \,\}. \tag{6}$$

Here K^* is an abelian normal divisor of $\mathfrak{G}^*$, and $\mathfrak{G}^*/K^* \simeq \mathfrak{G}$. The group $\mathfrak{G}^*$ consists of exactly the regular elements g^* which transform K into itself, i.e. $g^{*-1}Kg^* = K$. Thus the ring automorphisms of K are produced by all and only the elements of $\mathfrak{G}^*$. Indeed, if $v = uw$ induces such an inner automorphism of K, then w commutes with every element of K, and so lies in K^*, since K is maximal commutative.

Principal genus theorem in the minimal case. First form: every group automorphism of $\mathfrak{G}^*$ extending the identity automorphism of K^* is inner and is produced by an element of K^*. Equivalently, if the transformation quantities

$$\frac{c_S^T c_T}{c_{ST}}$$

all have the value one, then the c_S are symbolic $(1-S)$-th powers:

$$c_S = b^{1-S} = \frac{b}{b^S} \qquad (S \in \mathfrak{G}).$$

Third form: the group $\mathfrak{G}$ has only one crossed representation class of degree one in K^* belonging to factor system one.

A representation $u_S \mapsto C_S$ is called crossed with factor system $a_{S,T}$ if

$$C_S^T C_T = C_{ST} a_{S,T}$$

holds. Two representations belong to the same class if

$$C_S = B^{-S} D_S B.$$

I prove the first form and then show its equivalence to the second and third forms. The proof rests on the fact that every group automorphism of the indicated kind extends to a ring automorphism of A, and that such an automorphism is known to be inner. Let v_S be the images of the u_S under this automorphism. Then $\sum v_S K$ again forms the crossed product of $\mathfrak{G}$ with K for the same factor system, because by assumption all relations (2)–(4) remain valid. The map

$$\sum u_S b_S \longmapsto \sum v_S b_S$$

is a ring automorphism fixing all elements of K. Hence it is produced by an element of K^*.

The passage to the second form is based on the fact that, by (2), the v_S must have the form $v_S = u_S c_S$. Since the factor systems are also preserved, (5) says that the corresponding transformation quantities must all be one. Conversely, (2)–(5) show that the substitution $v_S = u_S c_S$ produces an automorphism of the required kind. The first form therefore yields

$$v_S = b^{-1} u_S b = u_S \frac{b}{b^S}, \qquad c_S = b^{1-S}.$$

In the cyclic case the hypothesis says in particular $N(c) = 1$, and $c = b^{1-S}$ is the familiar theorem. The third form is immediately equivalent to the second: the hypothesis says that $u_S \mapsto c_S$ is a crossed representation of degree one with factor system one, and $c_S = b^{1-S}$ says that this representation is equivalent to the identity representation.

§2. The Principal Genus Theorem

Preliminary remarks on class divisions. In the principal genus theorem of class-field theory for cyclic relative fields, two different divisions into classes occur: absolute ideal classes in the upper field and ray classes in the lower field. In the general case this corresponds to the fact that a division of the ideals of the upper field into classes induces a finer division of the factor systems into ideal classes.

Let $\mathfrak{I}$ first denote the group of all ideals of K. Since $\mathfrak{G}$ induces automorphisms of $\mathfrak{I}$, the extension by $\mathfrak{G}$ is defined by relations (2)–(5), and consists of the n complexes $\{\cdots u_S \mathfrak{I} \cdots\}$. Passing from $\mathfrak{I}$ to the group of absolute ideal classes by imposing equivalence, the n complexes $u_S \bar{\mathfrak{I}}$ are still defined, because $\mathfrak{G}$ also induces automorphisms of the ideal-class group, the principal class being transformed into itself. Thus the equivalence relation from $\mathfrak{I}$ to $\bar{\mathfrak{I}}$ transfers from $\{\cdots u_S \mathfrak{I} \cdots\}$ to $\{\cdots u_S \bar{\mathfrak{I}} \cdots\}$. In particular u_S is equivalent to all products $u_S(c_S)$, where (c_S) runs through principal ideals.

If now one requires the product relations (3) to be unambiguous in this equivalence sense, this says that all transformation quantities arising from principal ideals become uniquely equivalent. In other words:

In the system of the n complexes $\{\cdots u_S \bar{\mathfrak{I}} \cdots\}$, where $\bar{\mathfrak{I}}$ is the group of absolute ideal classes of K and H denotes the principal class, the product relations (3) for the $u_S H$ are unambiguously defined if and only if, in the induced ideal-class division of the factor systems, the principal class contains all transformation quantities arising from principal ideals.

Definition. The induced ideal-class division of the factor systems is fixed by requiring the principal class to consist of all principal ideals $(a_{S,T})$ for which, for at least one system of basis elements $a_{S,T}$, the algebra

$$(a_{S,T}, K, \mathfrak{G})$$

splits at all ramified places of K.

These principal ideals form a group under the product of factor systems; the group contains the transformation quantities

$$\frac{(c_S)^T (c_T)}{(c_{ST})},$$

because the factor systems $c_S^T c_T / c_{ST}$ produce algebras which split everywhere.

Formulation of the principal genus theorem. I state the theorem, corresponding to the minimal theorem, in three equivalent forms.

First form: with the induced ideal-class division of the factor systems taken as basis, suppose that the substitution

$$v_S = u_S c_S$$

produces an automorphism of the n complexes $\{\cdots u_S \bar{\mathfrak{I}} \cdots\}$, i.e. that the same relations (3) hold for the v_S in the sense of this class division. Then this automorphism is inner and is produced by an ideal class b.

Second form: if the transformation quantities formed from the ideal classes c_S,

$$\frac{c_S^T c_T}{c_{ST}},$$

all belong to the principal class of the induced class division of the factor systems, then the vectors $\{\cdots c_S \cdots\}$ of ideal classes form the principal genus, and the classes c_S are symbolic $(1-S)$-th powers: there is an ideal class b such that

$$c_S = b^{1-S} = \frac{b}{b^S} \qquad (S \in \mathfrak{G}).$$

Third form: with the induced ideal-class division for the factor systems, the group $\mathfrak{G}$ has only one crossed representation class of degree one in the ideal-class group which belongs to the identity class of factor systems.

The equivalence of the three forms follows as in §1. The passage from the first to the second form is even simpler here, because the automorphism is already assumed to be produced by $v_S = u_S c_S$. For the third form one only has to note that representations in the ideal-class group are defined purely multiplicatively; this imposes no restriction for representations of degree one.

Proof of the principal genus theorem. The proof of the second form rests on two lemmas.

Lemma 1. If the transformation quantities formed from the ideals c_S,

$$\frac{c_S^T c_T}{c_{ST}},$$

give the unit ideal, then the c_S are symbolic $(1-S)$-th powers:

$$c_S = b^{1-S} \qquad (S \in \mathfrak{G}).$$

This is equivalent to the first form when the automorphism there is assumed to be produced by $v_S = u_S c_S$. Proof: $b = \sum c_S$ has the required property; the sum is the module sum, i.e. the greatest common divisor. Indeed,

$$b = \sum_S c_S, \qquad b^T = \sum_S c_S^T, \qquad b^T c_T = \sum_S c_S^T c_T = \sum_S c_{ST} = b,$$

and therefore $c_T = b^{1-T}$.

Lemma 2. Let the algebra

$$A = (a_{S,T}, K, \mathfrak{G})$$

split at all finite and infinite ramified places of K, and suppose that the principal ideals $(a_{S,T})$ are transformation quantities of ideals:

$$(a_{S,T}) = \frac{c_S^T c_T}{c_{ST}}.$$

Then A splits everywhere, absolutely. Thus the $(a_{S,T})$ are transformation quantities of principal ideals:

$$(a_{S,T}) = \frac{(d_S)^T (d_T)}{(d_{ST})}.$$

Conversely, every algebra which splits everywhere satisfies this condition.

The proof rests on the known behavior of A under p-adic extension. Let k_p be the p-adic extension of k, and $K_\mathfrak{P}$ the $\mathfrak{P}$-adic extension of K, where $\mathfrak{P}$ is a prime ideal above p. Let A_p be the extension of A obtained by passing from k to k_p. Let $\mathfrak{Z}$ be the decomposition group of $\mathfrak{P}$, and $a_\mathfrak{Z}$ the corresponding part of the factor system. Then

$$A_p \sim (a_\mathfrak{Z}, K_\mathfrak{P}/k_p, \mathfrak{Z}).$$

If p is unramified, so that $\mathfrak{Z}$ is cyclic, this is the distinguished representation of A_p by means of the unramified inertia field $K_\mathfrak{P}/k_p$.

It remains to show that, under the hypothesis of Lemma 2, A_p also splits in this unramified case. For infinite places $K_\mathfrak{P} = k_p$, and A_p therefore splits. For finite p it follows first that the factor system $a_\mathfrak{Z}$

is associated with a factor system consisting of units. In passing to the normalized cyclic representation $A_p = (\alpha, K_{\mathfrak{P}}/k_p, R)$, with R a generator of $\mathfrak{Z}$, the factor system is expressed by the relations

$$u_{\mathfrak{P}_i} u_{\mathfrak{P}_j} = u_{\mathfrak{P}_{i+j}} \varepsilon_{\mathfrak{P}_i, \mathfrak{P}_j}.$$

Putting $u_R = u$, one obtains

$$u^i = u_R^i \, \varepsilon_R \varepsilon_{R^2} \cdots \varepsilon_{R^{i-1}},$$

and in particular $u_E = \varepsilon_E$, where E is the order of $\mathfrak{Z}$. Since in the unramified extension $K_{\mathfrak{P}}/k_p$ all units are norms, A_p splits. Hence A splits everywhere, and thus, by the theorem on split algebras, splits absolutely.

Equivalently, the hypothesis implies that $a_{S,T}$ are transformation quantities of elements,

$$a_{S,T} = \frac{d_S^T d_T}{d_{ST}}, \qquad d_S \in K^*.$$

In weakened form this says that the $(a_{S,T})$ are transformation quantities of principal ideals. Conversely, if A splits everywhere, then the principal ideals $(a_{S,T})$ are transformation quantities of ideals at every place; this is just another formulation of the hypothesis in the theorem on split algebras.

The two lemmas now give the proof of the principal genus theorem. Let c_S be an ideal in the class c_S. The hypothesis says, by the definition of the induced class division, that the transformation quantities of the c_S become principal ideals $(a_{S,T})$ satisfying the hypotheses of Lemma 2. Hence there are principal ideals (d_S) such that the transformed quantities

$$\frac{c_S^T c_T}{c_{ST}}$$

become the unit ideal. The ideals $c_S/(d_S)$, which lie in the same classes c_S, therefore satisfy the hypothesis of Lemma 1. Thus an ideal class b exists with

$$c_S = b^{1-S}.$$

This means that the classes c_S are symbolic $(1-S)$-th powers of the class b.

In the cyclic case the principal genus theorem passes into the familiar theorem as soon as, after normalization, the principal class of the induced class division becomes the group of principal ideals (α) for which α is a norm residue at all ramified places.

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42. Split Crossed Products and Their Maximal Orders

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In the simplest case of split crossed products, namely those which yield split algebras and hence have factor systems associated with one, the relations can be followed rather explicitly. This remains true even for non-Galois splitting fields, i.e. maximal commutative subfields. The simple algebraic facts belonging here are developed in §1. In §2, with an algebraic number field as ground field, I give explicit representations of all maximal orders and of their ideals by means of modules and complementary modules of the underlying splitting field k. I further give a division into “domains”, in which all maximal orders are put into the same domain when they have the same intersection with k. Thus the domains correspond one-to-one to the orders in k; the principal order is assigned the principal domain. The maximal orders of a domain are transformed into one another by ideals formed from the modules of the corresponding order; in the principal domain these are the extensions of ideals of k. In §3 I pass to arbitrary simple algebras. By refining the division into domains, requiring not only the same intersection but also agreement of the p-adic components of the maximal orders at ramified places, one can transfer the theorems obtained in the split case. In particular, the maximal orders of a principal domain are then also transformed into one another by extensions of ideals of k.

The theorems on the principal domain are found in Chevalley and Hasse; Chevalley also shows that without the refined division concerning the ramified places the theorems no longer hold in general.

§I. Matrix Units of Split Crossed Products and Complementary Bases of the Splitting Fields

In this paragraph Ω denotes an arbitrary ground field. Let k/Ω be a Galois separable extension of degree n, and let $\mathfrak{G}$ be the Galois group with elements $S, T, \ldots$; let $u_S, u_T, \ldots$ be the corresponding operators. We consider crossed products with factor system one, hence split algebras

$$K = \mathfrak{G} \times k = u_{S_1}k + \cdots + u_{S_n}k, \qquad u_S u_T = u_{ST},$$

with

$$z u_S = u_S z^S \qquad (z \in k).$$

Put

$$E_{\mathfrak{G}} = \sum_{S \in \mathfrak{G}} u_S.$$

Then factor system one gives

$$E_{\mathfrak{G}} u_S = u_S E_{\mathfrak{G}} = E_{\mathfrak{G}}, \tag{1}$$

$$z E_{\mathfrak{G}} = \sum_S u_S z^S, \tag{2a}$$

and

$$E_{\mathfrak{G}} z E_{\mathfrak{G}} = E_{\mathfrak{G}} \sum_S u_S z^S = E_{\mathfrak{G}} \ \mathrm{Sp}(z), \tag{2b}$$

where $\mathrm{Sp}(z)$ is the trace of z as an element of k/Ω.

Theorem. If $a_1, \ldots, a_n$ is a basis of k/Ω, and $\bar{a}_1, \ldots, \bar{a}_n$ is the complementary basis, then

$$c_{ik} = \bar{a}_i E_{\mathfrak{G}} a_k$$

is a system of matrix units of K. Thus K can be represented as the module product

$$K = k E_{\mathfrak{G}} k = k\Big(\sum_S u_S\Big)k = \sum_{i,k} (\bar{a}_i E_{\mathfrak{G}} a_k)\, \Omega.$$

Indeed, by (2b),

$$c_{ij} c_{hk} = \bar{a}_i E_{\mathfrak{G}} a_j \bar{a}_h E_{\mathfrak{G}} a_k = \bar{a}_i E_{\mathfrak{G}} a_k \ \mathrm{Sp}(a_j \bar{a}_h) = c_{ik} \ \mathrm{Sp}(a_j \bar{a}_h),$$

which is precisely the matrix-unit property, since the complementary bases are defined by

$$\mathrm{Sp}(a_j \bar{a}_h) = \delta_{jh}.$$

Remark 1. If Z is any commutative extension field of Ω, everything remains valid when a_i is a basis of k_Z/Z and K_Z becomes a direct sum of fields. The substitutions of the group are then defined by their inducing the identity on Z.

Remark 2. That $K = kE_{\mathfrak{G}}k$ also follows directly from the fact that the right-hand side is a nonzero ideal of the simple system K.

Now assume that the ground field Ω has characteristic zero. The representation $K = kE_{\mathfrak{G}}k$ of the split algebra remains valid even when k is a non-Galois splitting field of K/Ω. One passes to the Galois splitting field $\bar{k}$. Let $\bar{k}$ be the Galois field belonging to k, $\bar{\mathfrak{G}}$ its group, $\bar{u}_S$ the corresponding operators, and $\bar{K} = \bar{k}E_{\bar{\mathfrak{G}}}\bar{k}$ a split crossed product. Let the subfield k correspond to the subgroup $\mathfrak{H}$ of order h. Set

$$\bar{\mathfrak{G}} = \mathfrak{H}S_1 + \cdots + \mathfrak{H}S_n = S_1^{-1}\mathfrak{H} + \cdots + S_n^{-1}\mathfrak{H}, \qquad E_{\mathfrak{H}} = \frac{1}{h}\sum_{H \in \mathfrak{H}} u_H,$$

and

$$E_{\mathfrak{G}} = \frac{1}{h} E_{\bar{\mathfrak{G}}} = \sum_i u_{S_i}, \qquad u_{S_i} = E_{\mathfrak{H}} \bar{u}_{S_i}.$$

Then $E_{\mathfrak{H}}$ generates the identity representation of $\mathfrak{H}$, while $E_{\mathfrak{G}}$, like $E_{\bar{\mathfrak{G}}}$, generates the identity representation of $\bar{\mathfrak{G}}$. For these $E_{\mathfrak{G}}$ and u_{S_i}, relation (1) holds on one side and (2) holds completely. First,

$$E_{\mathfrak{G}} = E_{\mathfrak{G}} E_{\mathfrak{H}} = E_{\mathfrak{H}} E_{\mathfrak{G}} = E_{\mathfrak{H}} E_{\mathfrak{G}} E_{\mathfrak{H}}, \tag{3}$$

$$z E_{\mathfrak{H}} = E_{\mathfrak{H}} z, \tag{4}$$

and hence

$$z u_{S_i} = u_{S_i} z^{S_i}. \tag{5}$$

Equation (2a) follows from (5) by summation, and then (2b) follows, while (1) holds on one side:

$$E_{\mathfrak{G}} u_{S_i} = E_{\mathfrak{G}}.$$

The proof of $K = k E_{\mathfrak{G}} k$ is now exactly the same as before. As a full matrix ring over Ω, K contains every field of degree n as a subfield, in particular k.

For §3 I also note that the passage from K to $\bar{K}$ is possible even when K does not split. Besides k, the field $\bar{k}$ is also a splitting field, and K is therefore similar to $\bar{K}$, the crossed product of $\bar{\mathfrak{G}}$ with $\bar{k}$. Since k is a splitting field, the subgroup $\mathfrak{H}$ corresponding to k has factor system one, so the idempotent $E_{\mathfrak{H}}$ is as above. It follows conversely that K is isomorphic to $E_{\mathfrak{H}} \bar{K} E_{\mathfrak{H}}$. This algebra is isomorphic to the automorphism ring of the left ideal $\bar{K} E_{\mathfrak{H}}$, and hence is similar to $\bar{K}$; the rank agrees with that of K. In the split case this gives again

$$E_{\mathfrak{H}} \bar{K} E_{\mathfrak{H}} = E_{\mathfrak{H}} \bar{k} E_{\bar{\mathfrak{G}}} \bar{k} E_{\mathfrak{H}} = (E_{\mathfrak{H}} \bar{k} E_{\mathfrak{H}}) E_{\bar{\mathfrak{G}}} (E_{\mathfrak{H}} \bar{k} E_{\mathfrak{H}}) = k E_{\mathfrak{G}} k.$$

§II. The Maximal Orders of Split Crossed Products and Their Division into Domains

From now on Ω is a finite algebraic number field, K a split algebra of degree n over Ω, and k a Galois or non-Galois maximal commutative subfield, so that $K = k E_{\mathfrak{G}} k$. Let $a, \bar{a}$ be a pair of complementary finite $\mathfrak{o}$-modules in k, where $\mathfrak{o}$ is the principal order of Ω and o the principal order of k. The structure of the maximal orders of K relative to k is described by the following theorems.

Theorem 1. The module product

$$\mathcal{O}_a = \bar{a} E_{\mathfrak{G}} a$$

is a maximal order of K; in particular

$$\mathcal{O} = \mathcal{O}_o = o E_{\mathfrak{G}} o$$

is a maximal order.

Theorem 2. The transformation of $\mathcal{O}$ into $\mathcal{O}_a$ is effected by the left ideal

$$\mathcal{L} = o E_{\mathfrak{G}} a$$

and by its reciprocal right ideal

$$\mathcal{R} = \bar{a} E_{\mathfrak{G}} o$$

with respect to $\mathcal{O}$.

Remark. The occurrence of the two complementary modules $a, \bar{a}$ in the product $\mathcal{L}\mathcal{R}$ leads to the reciprocal relation precisely because of the trace formation caused by $E_{\mathfrak{G}}$.

Theorem 3. All left and right ideals with respect to $\mathcal{O}$ are exhausted by those described in Theorem 2. Consequently the orders $\mathcal{O}_a$ exhaust all maximal orders in K. Any two maximal orders $\mathcal{O}_a$ and $\mathcal{O}_b$ are transformed into one another by

$$\mathcal{L}_{ab} = \bar{a} E_{\mathfrak{G}} b, \qquad \mathcal{R}_{ba} = \bar{b} E_{\mathfrak{G}} a,$$

and these exhaust the left and right ideals with respect to $\mathcal{O}_a$.

Theorem 4. As $\mathcal{O}_a$ runs through all maximal orders of K, the intersection

$$[\mathcal{O}_a, k]$$

runs through all orders of k; this intersection is in each case the order of a. If all maximal orders belonging to the same order in k are grouped into one domain, then the transformations within a domain are given by left ideals $\bar{a}E_{\mathfrak{G}}b$ with respect to $\mathcal{O}_a$ and by reciprocal right ideals, where a and b are modules of the corresponding order in k.

Theorem 4a. In particular, the transformation of the maximal orders of the principal domain, i.e. the domain of all maximal orders containing the principal order o of k, is effected by ideals

$$aE_{\mathfrak{G}}b,$$

where a and b are ideals of k. Equivalently: if $\mathcal{L}$ is a left ideal of a maximal order of the principal domain and o is contained in the right order of $\mathcal{L}$, then $\mathcal{L}$ is the extension of an o-ideal.

The proofs rest on passage to the individual places. It is enough to pass to the quotient ring at p, where p runs through all prime ideals of Ω. If $\mathcal{C}$ is any $\mathfrak{o}$-module in K, let its component $\mathcal{C}_p$ denote the extended module $\mathcal{C}\mathfrak{o}_p$. Then:

Lemma. $\mathcal{C}$ is the intersection of all extension modules $\mathcal{C}_p$. If $w \in \mathcal{C}_p$, there is a $\sigma \in \mathfrak{o}$, not divisible by p, such that $\sigma w \in \mathcal{C}$. Conversely, if w lies in the intersection of all $\mathcal{C}_p$ and g is the ideal of all $\sigma \in \mathfrak{o}$ with $\sigma w \in \mathcal{C}$, then g is divisible by no p, so $g = \mathfrak{o}$. No hypothesis on the rank of $\mathcal{C}$ is required.

Consequences. Every module is determined by its components. If $\mathcal{D}$ is a proper overmodule of $\mathcal{C}$, at least one $\mathcal{D}_p$ is a proper overmodule of $\mathcal{C}_p$. In particular, if $\mathcal{M}$ is an order in K, and all its components $\mathcal{M}_p$ are maximal orders over $\mathfrak{o}_p$, then $\mathcal{M}$ itself is maximal.

Proof of Theorem 1. $\mathcal{O}_a$ is an order: it is a finite $\mathfrak{o}$-module of maximum rank. Moreover,

$$\mathcal{O}_a^2 = \bar{a}E_{\mathfrak{G}}a\,\bar{a}E_{\mathfrak{G}}a = \bar{a}E_{\mathfrak{G}}a \;\mathrm{Sp}(a\bar{a}) \subseteq \bar{a}E_{\mathfrak{G}}a = \mathcal{O}_a.$$

For maximality it suffices, by the consequence above, to prove maximality of every component. Since $\mathfrak{o}_p$ is a principal ideal ring, a_p and $\bar{a}_p$ have independent complementary bases $a_1, \ldots, a_n$ and $\bar{a}_1, \ldots, \bar{a}_n$. The products $\bar{a}_iE_{\mathfrak{G}}a_k$ form matrix units by §1; hence

$$\mathcal{O}_{a,p} = \sum_{i,k} (\bar{a}_iE_{\mathfrak{G}}a_k)\mathfrak{o}_p$$

is maximal. Since $\mathcal{O}_a$ is maximal, $\mathrm{Sp}(a\bar{a}) = \mathfrak{o}$ follows.

Proof of Theorem 2. From the trace rule one obtains

$$\begin{aligned}
\mathcal{OL} &= oE_{\mathfrak{G}}o\,oE_{\mathfrak{G}}a = oE_{\mathfrak{G}}a = \mathcal{L},\\
\mathcal{RO} &= \bar{a}E_{\mathfrak{G}}o\,oE_{\mathfrak{G}}o = \bar{a}E_{\mathfrak{G}}o = \mathcal{R},\\
\mathcal{LR} &= oE_{\mathfrak{G}}a\,\bar{a}E_{\mathfrak{G}}o = oE_{\mathfrak{G}}o = \mathcal{O},\\
\mathcal{RL} &= \bar{a}E_{\mathfrak{G}}o\,oE_{\mathfrak{G}}a = \bar{a}E_{\mathfrak{G}}a = \mathcal{O}_a.
\end{aligned}$$

Proof of Theorem 3. Let $\mathcal{L}$ be a left ideal with respect to $\mathcal{O}$. Then

$$\mathcal{L} = oE_{\mathfrak{G}}\mathcal{L} = (oE_{\mathfrak{G}}l_1, \ldots, oE_{\mathfrak{G}}l_n)$$

for an $\mathfrak{o}$-basis $l_1, \ldots, l_n$ of $\mathcal{L}$. Writing $l_\mu = \sum_i u_{S_i}z_{i\mu}$ gives $E_{\mathfrak{G}}l_\mu = E_{\mathfrak{G}}a_\mu$, and hence $\mathcal{L} = oE_{\mathfrak{G}}a$. If $\mathcal{L}$ is regular, then a has maximum rank n, and $\bar{a}E_{\mathfrak{G}}o$ is the reciprocal right ideal. Composition gives the general assertion.

Proof of Theorem 4. The intersection $t = [k, \mathcal{O}_a]$ is an order. As a subset of $\mathcal{O}_a$, it is a right multiplier domain and is the largest order in o with this property; it therefore contains the order o_a of a. To prove $t = o_a$, it suffices to prove equality componentwise. If $\tau \in t_p$, then τc_{ik}, for every $c_{ik} = \bar{a}_iE_{\mathfrak{G}}a_k$, is a linear combination of the c_{ij} with coefficients in $\mathfrak{o}_p$. On the other hand $a_k\tau$ is a linear combination of the a_j with coefficients in $\mathfrak{o}_p$. By the uniqueness of the representation through the c_{ij}, these coefficients must lie in $\mathfrak{o}_p$. Thus τ lies in the order of a_p. This proves that t is the order of a.

Proof of Theorem 4a. This is the specialization of Theorem 4 to the principal domain. Starting with $\mathcal{O} = oE_{\mathfrak{G}}o$, if o is contained in the right order of $\mathcal{L} = oE_{\mathfrak{G}}a$, then $\bar{a}E_{\mathfrak{G}}a$ belongs to the principal domain. Therefore a is an ideal of k, and $\mathcal{L} = \mathcal{O}a$. Starting from $\mathcal{O}_a$, correspondingly $\mathcal{L} = \bar{a}E_{\mathfrak{G}}b$, and $a^{-1}b$ is an ideal of k; hence $\mathcal{L} = \mathcal{O}_a a^{-1}b$.

§III. Maximal Orders of Arbitrary Crossed Products

The theorems derived for split crossed products remain valid in the general case once the division into domains is refined. The facts about the principal domain then hold exactly; the remaining facts must be formulated place by place. Here a place is to be understood as a p-adic extension.

Definition. Let K be a simple normal algebra over Ω, and let k be a maximal commutative subfield. A domain relative to k consists of all maximal orders of K which have the same intersection with k and which, in addition, have the same p-adic components at the ramified places of K. If the intersection with k is the principal order, one obtains a principal domain.

For principal domains:

Theorem 1. The maximal orders of a principal domain are transformed into one another by transformation with extension ideals of ideals of k. Equivalently: the left ideals of the maximal orders of a principal domain which are prime to the different and contain o in their right order are extensions of ideals of k.

Theorem 2. If K has prime degree, there is only one principal domain; it is transformed into itself by extension ideals of the k-ideals. The corresponding ideals of a fixed maximal order $\mathcal{O}$ are the ideals of k extended by the two-sided ideals of $\mathcal{O}$.

Proof. Two maximal orders of a principal domain differ only at finitely many places; by definition these are places where K splits. There K has the form considered in §§1–2. This is clear when k is Galois; if the factor system is associated with one, it may be replaced by the system one. By §1 this also holds in the non-Galois case. Thus at every such place, and hence globally, the ideals are extension ideals.

If K has prime degree, it is either split or a division algebra. In the latter case it remains a division algebra at all ramified places and therefore has only one principal domain, so transformation by extension ideals again applies. The most general transformation is obtained by composing the extension ideals with those which transform the maximal order into itself, namely with two-sided ideals; these, as is known, differ from central ideals only at the ramified places, hence differ only by extension ideals. This proves the remaining assertions.

Also:

Theorem 3. The maximal orders of an arbitrary domain are transformed into one another by ideals prime to the different which, at every place, are composed from modules of the corresponding order as described in §2, Theorem 3. For a fixed ideal the components of these modules differ from the order at only finitely many places.

This follows directly from §2; one only has to replace the operators u_S at the individual places by corresponding equivalent operators.

Whether, for every order of k, there are maximal orders having that order as intersection remains to be investigated more closely at the ramified places. A principal domain always exists, as follows from the crossed, or possibly generalized, product representation: it gives an order containing o once the factor systems are taken integral.

A further question is to what extent the arithmetic division into domains can characterize the algebraic splitting fields. In other words: do different splitting fields always produce different domain decompositions, or already different principal domains? And do the principal domains of all splitting fields perhaps exhaust all maximal orders?

That such a characterization is certainly impossible by a single maximal order is shown by the simplest examples. For instance, the maximal order

$$\left[1, i, j, \frac{1+i+j+k}{2}\right]$$

of the quaternion field contains, besides the principal order $[1, i]$, also the principal order

$$\left[1, \frac{1+i+j+k}{2}\right]$$

of the field of third roots of unity.

Göttingen, August 1932.

43. Ideal Differentiation and the Different

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Introduction

The principal theorem of ramification theory for algebraic number fields says, as is well known, that the different, the ramification ideal, is divisible at least by the $(e-1)$-st power of a prime ideal which occurs to the e-th power in the prime number p; precisely by the $(e-1)$-st power when e is not divisible by p, and by a higher power when e is divisible by p.

This is analogous to the fact that the derivative $f'(x)$ of a polynomial $f(x)$ is divisible at least by the $(e-1)$-st power of a linear factor which occurs in $f(x)$ to the e-th power; precisely by that power when e is not divisible by the characteristic of the coefficient domain, and by a higher power when e is divisible by the characteristic.

I shall show that this is more than a formal analogy. The different can be regarded as the differential quotient of a defining ideal of the number field K in an associated integral polynomial domain in $x_1, \ldots, x_n$, the differential quotient being taken at the point $x = \omega$, that is, at $x_1 = \omega_1, \ldots, x_n = \omega_n$, where $\omega_1, \ldots, \omega_n$ is a module basis of the system of algebraic integers of K, the principal order $\mathfrak{o}$. The defining ideal $\mathfrak{M}$ is the totality of the relations among the ω_i, namely all integral polynomials $f(x)$ such that $f(\omega) = 0$. The differential quotient $\mathfrak{M}'[x \to \omega]$ is defined as a difference quotient at $x = \omega$.

Indeed, because $f(\omega) = 0$, regard $\mathfrak{M}$ as consisting of all differences $f(x) - f(\omega)$. Form the difference ideal

$$\mathfrak{B} = (x_1 - \omega_1, \ldots, x_n - \omega_n)$$

and the difference quotient

$$\mathfrak{A} = \mathfrak{M} : \mathfrak{B},$$

where the quotient is the usual ideal quotient, taken in the polynomial domain in the x_i with coefficients in the integers of K, hence in $\mathfrak{o}$. The different $\mathfrak{D}$ is then defined by

$$\mathfrak{D} = \mathfrak{A}[x \to \omega].$$

The extension of the coefficient domain is necessary in order for the difference ideal and the difference quotient to be defined. Thus this is the direct generalization of the differential quotient of a polynomial in one variable,

$$f'(\xi) = \left.\frac{f(x) - f(\xi)}{x - \xi}\right|_{x=\xi},$$

where here as well the polynomial domain must be enlarged by adjoining ξ so that the differences are defined. In fact the ideal differential quotient becomes the ordinary polynomial differential quotient as soon as the basis of the ω_i consists of the powers of a single element.

The definition just given therefore attaches directly to familiar facts. It introduces, however, non-invariant intermediate objects, since the ideal $\mathfrak{M}$ depends on the chosen basis. An equivalent, completely invariant definition is obtained by replacing the integral polynomial domain by a ring $\mathfrak{O}$ isomorphic to the principal order $\mathfrak{o}$; the coefficient domain of $\mathfrak{O}$ is then extended by $\mathfrak{o}$ to form $\mathfrak{O}_{\mathfrak{o}}$. The difference ideal is generated by all differences $x - \xi$, where x and ξ correspond under the isomorphism from $\mathfrak{O}$ to $\mathfrak{o}$; the difference quotient is the quotient of the zero ideal by this ideal; and the different is obtained by replacing each x in the difference quotient by the corresponding ξ.

This invariant definition is placed first below. It defines at once the different of an arbitrary commutative ring relative to a subring, subject only to hypotheses ensuring that coefficient extension is possible. These hypotheses are automatically satisfied, for example, when a module basis exists, possibly after passage to suitable quotient rings. Thus one obtains, in particular, the different of an arbitrary order in a number field, the different of a residue class ring of an order modulo a prime-power, relative differents, and differents for algebraic functions of several indeterminates.

The agreement of this differential definition with the usual definition follows from a structural analysis of the Galois extension ring attached to the number field by direct product formation. This analysis is based on a direct-sum decomposition. In the special case of a residue class ring modulo a polynomial in one indeterminate, it becomes an analysis of the Lagrange interpolation formula.

More precisely, introduce a ring $\mathfrak{K}$ isomorphic to the number field K and containing $\mathfrak{o}$, and extend its coefficient field, the rational field, by the Galois closure F of K. The extended system $\mathfrak{K}_F$ is a direct sum of n simple components corresponding to the n isomorphisms of K. These components are extensions of the difference quotient and of its conjugates; at the same time the zero ideal of $\mathfrak{K}_F$ is the intersection of the n ideals obtained from the difference ideal and its conjugates. The simple components are of the form $Fe^{(i)}$, where the $e^{(i)}$ are the components of the identity. Since $e^{(i)}$ goes over to the identity under $x = \xi$, the difference quotient is $\mathfrak{d}e^{(i)}$, where $\mathfrak{d}$ denotes the different of $\mathfrak{o}$. Thus $\mathfrak{d}$ consists of precisely those elements of the field which, when multiplied by $e^{(i)}$, lie in the direct product of the order with itself.

To recover the definition by the complementary module, observe that $e^{(i)}$ is a linear form in the basis $x_1, \ldots, x_n$ corresponding to the ω_i. The coefficients of the x_i in the $e^{(i)}$ form a basis of the complementary module of $\mathfrak{o}$ and of its conjugates. Since the difference quotient has coefficients in $\mathfrak{o}$, the different is the quotient of $\mathfrak{o}$ by its complementary module. This is Dedekind's definition. The construction is not restricted to the principal order; it characterizes the different of every order as the quotient of the order by its complementary module.

In the principal-order case, the agreement with the definition by a fundamental equation follows from the fact that the ideal differential quotient becomes the polynomial differential quotient when the basis consists of powers of one element. The conceptual relation between this definition and Dedekind's is thereby obtained.

With a fundamental equation one obtains directly the principal ramification theorem mentioned at the start. For the exact determination of exponents one considers the different $\mathfrak{d}[p^t]$ of the residue class ring of the principal order modulo p^t, with t sufficiently large; $t = n$ is enough, and for quadratic fields is also necessary. The crucial point is that the different $\mathfrak{d}$, reduced modulo p^t, passes exactly into $\mathfrak{d}[p^t]$. One may then restrict to the residue class ring modulo $\mathfrak{p}^t$. This ring is isomorphic to a residue class ring modulo a polynomial in one indeterminate with coefficients modulo p^t; differentiating this polynomial gives, as O. Ore showed by another route, the exact survey of possible exponents by means of the supplement numbers.

Finally, the structural analysis also gives the known theorem that the different of an overfield of K is the product of the relative different and the different of K. This is a consequence of the corresponding multiplication property for the identity components of the direct-sum decomposition, and therefore for complementary modules. All the preceding facts hold in the same manner for relative differents after passage to quotient rings, and for algebraic function fields with coefficient field of characteristic zero. In characteristic p, and for integral algebraic functions, the structural properties remain, but the ramification theory changes because inseparable extensions can occur.

§1. Extension of the Coefficient Domain of a Ring, or Direct Product Formation. Defining Ideals

In this section general rings are considered; multiplication is not assumed commutative. Unless the contrary is explicitly stated, however, the existence of an identity element is assumed.

1. Definition of the direct product. A ring $\mathfrak{O} \times_{\mathfrak{h}} \mathfrak{o}$, or $\mathfrak{O}_{\mathfrak{o}}$, is called the direct product of $\mathfrak{O}$ and $\mathfrak{o}$ over $\mathfrak{h}$, or the ring obtained from $\mathfrak{O}$ by extension of the coefficient domain $\mathfrak{h}$ to $\mathfrak{o}$, if the following conditions hold.

1. It contains $\mathfrak{O}$ and $\mathfrak{o}$ as subrings and is generated as their product.
2. The intersection of $\mathfrak{O}$ and $\mathfrak{o}$ is $\mathfrak{h}$; the identity lies in $\mathfrak{h}$.
3. The elements of $\mathfrak{O}$ commute elementwise with the elements of $\mathfrak{o}$. Hence every element is a finite bilinear combination
$$x_1 y_1 + \cdots + x_m y_m, \qquad x_i \in \mathfrak{O}, \quad y_i \in \mathfrak{o}.$$
4. Equality is only that forced formally by the relations in the two factors. Thus a relation
$$\sum_i x_i y_i = 0$$
must be reducible, by adding formally vanishing expressions $(yh)\delta - y(h\delta)$, to relations inside $\mathfrak{O}$ and inside $\mathfrak{o}$ separately.

These requirements determine equality, addition, and multiplication uniquely from the corresponding laws in the factors. Isomorphic data give isomorphic direct products. Conversely, any ring which is a product of $\mathfrak{O}$ and $\mathfrak{o}$ with these two factors commuting elementwise is a homomorphic image of the direct product.

2. Criterion for existence. The direct product need not exist. For example, take $\mathfrak{h} = \mathbb{Z}$, $\mathfrak{O} = \mathfrak{h}[x]$ with relation $1 - 2x = 0$, and $\mathfrak{o} = \mathfrak{h}[y]$ with relation $1 - 2y = 0$. No common overring can have intersection exactly $\mathfrak{h}$, since in any such ring

$$0 = (1 - 2x)y - x(1 - 2y) = y - x.$$

The necessary and sufficient condition is that there be at least one ring R containing $\mathfrak{O}$ and $\mathfrak{o}$, in which their intersection is $\mathfrak{h}$ and in which they commute elementwise. If the direct product exists, it itself is such a ring. Conversely, one constructs the direct product from formal bilinear combinations with the equality described above; the embedding ring R supplies the required intersection property.

3. Defining ideals. A system S of elements z of $\mathfrak{O}$ is a generating system of $\mathfrak{O}$ over $\mathfrak{h}$ if $\mathfrak{O} = \mathfrak{h}[S]$. If $\mathfrak{h}$ lies in the center of $\mathfrak{O}$, then $\mathfrak{O}$ is a homomorphic image of the noncommutative polynomial domain $\mathfrak{h}\{Z\}$, where the indeterminates Z correspond to the elements of S. Thus

$$\mathfrak{O} \simeq \mathfrak{h}\{Z\}/\mathfrak{M},$$

where $\mathfrak{M}$ is the two-sided ideal of all polynomials $F(Z)$ for which $F(z) = 0$ in $\mathfrak{O}$. This is the defining ideal. If there is a single generator z and $\mathfrak{M}$ is principal, a generator $G(Z)$ with $G(z) = 0$ is a defining equation.

4. The defining ideal of a direct product. Let $\mathfrak{O} \simeq \mathfrak{h}\{Z\}/\mathfrak{M}$. In $\mathfrak{o}\{Z\}$ let $\mathfrak{M}_{\mathfrak{o}}$ be the ideal generated by $\mathfrak{M}$ after extension of coefficients. If the direct product $\mathfrak{O}_{\mathfrak{o}}$ exists, then

$$\mathfrak{O}_{\mathfrak{o}} \simeq \mathfrak{o}\{Z\}/\mathfrak{M}_{\mathfrak{o}}.$$

Thus the defining relations in the direct product are exactly those obtained by extending the coefficients in the defining ideal.

§2. Rings with Independent Module Bases. Construction of the Direct Product

1. Construction. A system T of elements t of $\mathfrak{O}$ is an $\mathfrak{h}$-module basis if every element of $\mathfrak{O}$ is a finite linear combination

$$h_1 t_1 + \cdots + h_m t_m, \qquad h_i \in \mathfrak{h}.$$

It is independent if $\sum h_i t_i = 0$ implies all $h_i = 0$. If $\mathfrak{O}$ has an independent $\mathfrak{h}$-module basis containing the identity, then the direct product $\mathfrak{O}_{\mathfrak{o}}$ exists for every extension ring $\mathfrak{o}$ of $\mathfrak{h}$ whose intersection with $\mathfrak{O}$ is $\mathfrak{h}$ and whose elements commute with those of $\mathfrak{O}$. It is the ring of all expressions

$$y_1 t_1 + \cdots + y_m t_m, \qquad y_i \in \mathfrak{o},$$

with equality by coefficients in the basis T.

2. Extension and contraction of modules. For an $\mathfrak{h}$-module $\mathfrak{B} \subset \mathfrak{O}$, let $\mathfrak{B}_{\mathfrak{o}}$ be the $\mathfrak{o}$-module generated by it in $\mathfrak{O}_{\mathfrak{o}}$. For an $\mathfrak{o}$-module $\mathfrak{C} \subset \mathfrak{O}_{\mathfrak{o}}$, its contraction is $[\mathfrak{C}, \mathfrak{O}] = \mathfrak{C} \cap \mathfrak{O}$. If $\mathfrak{B}$ has an independent $\mathfrak{h}$-module basis extendable to such a basis of $\mathfrak{O}$, then

$$\mathfrak{B} = [\mathfrak{B}_{\mathfrak{o}}, \mathfrak{O}].$$

The proof is simply comparison of coefficients in the extended basis.

3. A sufficient criterion. If there exists an extension ring $\mathfrak{f}$ of $\mathfrak{h}$ such that the direct products $\mathfrak{O}_{\mathfrak{f}}$ and $\mathfrak{o}_{\mathfrak{f}}$ exist over $\mathfrak{h}$, and such that $\mathfrak{O}_{\mathfrak{f}} \times_{\mathfrak{f}} \mathfrak{o}_{\mathfrak{f}}$ exists over $\mathfrak{f}$, then $\mathfrak{O} \times_{\mathfrak{h}} \mathfrak{o}$ exists. Indeed the latter product gives an embedding ring and

$$[\mathfrak{O}, \mathfrak{o}] \subseteq [\mathfrak{O}_{\mathfrak{f}}, \mathfrak{o}_{\mathfrak{f}}] = \mathfrak{f}, \qquad [\mathfrak{O}, \mathfrak{f}] = \mathfrak{h}.$$

§3. The Different of a Ring over a Subring. Differential Quotient of a Defining Ideal

From now on all rings are commutative. The rings $\mathfrak{O}$ and $\mathfrak{o}$ are assumed to be isomorphic, equivalent extensions of $\mathfrak{h}$, and the direct product $\mathfrak{O}_{\mathfrak{o}}$ is assumed to exist.

1. Definition of the different. Let x run through the elements of $\mathfrak{O}$, and let ξ be the corresponding element of $\mathfrak{o}$. The ideal

$$\mathfrak{B} = \{\dots, x - \xi, \dots\}$$

generated by all differences is the difference ideal. The difference quotient is

$$\mathfrak{A} = (0) : \mathfrak{B}.$$

The different of $\mathfrak{o}$ over $\mathfrak{h}$ is

$$\mathfrak{d} = \mathfrak{A}[x \to \xi],$$

and the different of $\mathfrak{O}$ is $\mathfrak{D} = \mathfrak{A}[\xi \to x]$.

2. Differential quotient of a defining ideal. Let $\mathfrak{M}$ be the defining ideal of $\mathfrak{O}$ over $\mathfrak{h}$ with respect to generators z. In $\mathfrak{o}[Z]$ put

$$\mathfrak{B}_Z = \{\dots, Z - \xi, \dots\}, \qquad \mathfrak{C} = \mathfrak{M}_{\mathfrak{o}} : \mathfrak{B}_Z.$$

Equivalently,

$$\mathfrak{C} = \{\dots, F(Z) - F(\xi), \dots\} : \{\dots, Z - \xi, \dots\}.$$

The differential quotient $\mathfrak{M}'$ is $\mathfrak{C}[\xi \to Z]$. It contains $\mathfrak{M}$, and

$$\mathfrak{M}'[Z \to z] = \mathfrak{D}, \qquad \mathfrak{M}'[Z \to \xi] = \mathfrak{d}.$$

3. A defining equation. Suppose $\mathfrak{O}$ has a defining equation

$$G(z) = 0, \qquad G(Z) = Z^n + h_1 Z^{n-1} + \cdots + h_n, \quad h_i \in \mathfrak{h}.$$

Then the different is defined and is principal, with generator

$$G'(\xi) \quad \text{respectively} \quad G'(z),$$

Moreover

$$\mathfrak{M}' = (G'(Z), G(Z)).$$

The reason is that $1, z, \dots, z^{n-1}$ is an independent module basis and that

$$C(Z, \xi) = \frac{G(Z) - G(\xi)}{Z - \xi}$$

generates the difference quotient. Substitution $\xi \to Z$ gives $G'(Z)$.

§4. The Different of a Direct Sum

Assume

$$\mathfrak{O} = R_1 + \cdots + R_r, \qquad e = e_1 + \cdots + e_r, \qquad R_i R_j = 0 \ (i \neq j),$$

a direct sum of ideals, where the e_i are the components of the identity. Assume also that each R_i has an independent $\mathfrak{h}_i$-module basis containing e_i, with $\mathfrak{h}_i = \mathfrak{h} e_i$, and that

$$[\mathfrak{h}, \mathfrak{O}_i] = 0$$

holds for the corresponding complementary summands. Then $\mathfrak{O}$ itself has an independent $\mathfrak{h}$-module basis assembled from those of the R_i, and the different of $\mathfrak{O}$ over $\mathfrak{h}$ is the direct sum of the differents of the R_i over the $\mathfrak{h}_i$.

If the corresponding decomposition of $\mathfrak{o}$ is

$$\mathfrak{o} = r_1 + \cdots + r_r, \qquad 1 = \varepsilon_1 + \cdots + \varepsilon_r,$$

then

$$\mathfrak{O}_{\mathfrak{o}} = \sum_{i,j} R_i \varepsilon_j.$$

The component $R_i\varepsilon_i$ carries the direct product $R_i \times r_i$, and the component of the global difference quotient on it is

$$\mathfrak{A}e_i\varepsilon_i = (0) : \mathfrak{B}e_i\varepsilon_i.$$

Therefore, writing $\mathfrak{D}_i$ and $\mathfrak{d}_i$ for the component differents,

$$\mathfrak{D} = \mathfrak{D}_1 + \cdots + \mathfrak{D}_r, \qquad \mathfrak{d} = \mathfrak{d}_1 + \cdots + \mathfrak{d}_r.$$

The proof consists in passing between the direct-sum components by coefficient comparison in the independent bases, verifying the sharpened intersection relations

$$[\mathfrak{o}, \mathfrak{D}_i] = 0,$$

the induced isomorphisms

$$\mathfrak{o} \sim \mathfrak{o}e_i, \qquad R_i \sim R_i\varepsilon_i, \qquad \mathfrak{h}_i \sim \mathfrak{h}_i e_i.$$

§5. The Galois Extension Ring of a Field. Structure Theorems

1. The rings involved. Let K be a separable field extension of degree n over a ground field P. Let F be a Galois field containing the n conjugate fields

$$K^{(1)}, \ldots, K^{(n)}, \qquad K = K^{(1)},$$

and generated by their product. Let $\mathfrak{K}$ be an extension of P isomorphic to K, equivalent over P, and such that $[\mathfrak{K}, F] = P$. Since $\mathfrak{K}$ has a finite independent P-basis, the direct products

$$\mathfrak{K}_K, \qquad \mathfrak{K}_F, \qquad K_F$$

exist. The ring $\mathfrak{K}_F$ is called the Galois extension ring of $\mathfrak{K}$.

Every ideal of $\mathfrak{K}$ is the contraction of its extension in $\mathfrak{K}_K$ or $\mathfrak{K}_F$, and every ideal of $\mathfrak{K}_K$ is the contraction of its extension in $\mathfrak{K}_F$. This is the module-basis theorem of §2 over fields.

Write

$$\mathfrak{B}^{(i)} = \{\ldots, x - \xi^{(i)}, \ldots\}$$

for the conjugate difference ideals, and

$$\mathfrak{A}^{(i)} = (0) : \mathfrak{B}^{(i)}$$

for their difference quotients. Their extensions to $\mathfrak{K}_F$ are denoted by $\mathfrak{B}_F^{(i)}$ and $\mathfrak{A}_F^{(i)}$.

2. Direct-sum decomposition. The structure theorem is:

$$(0) = \bigcap_{i=1}^{n} \mathfrak{B}_F^{(i)}, \qquad \mathfrak{K}_F = \mathfrak{A}_F^{(1)} + \cdots + \mathfrak{A}_F^{(n)}$$

as a direct sum. In particular,

$$\mathfrak{K}_K = \mathfrak{A} + \mathfrak{B}$$

as a direct sum, and its zero ideal is the intersection of the corresponding difference ideal with the difference quotient.

The residue class ring modulo each $\mathfrak{B}_F^{(i)}$ has rank one over F, and by separability the ideals $\mathfrak{B}_F^{(i)}$ are distinct and pairwise coprime. Hence the identity decomposes uniquely as

$$e = e^{(1)} + \cdots + e^{(n)},$$

where

$$e^{(i)} \in \bigcap_{j \neq i} \mathfrak{B}_F^{(j)}, \qquad e^{(i)} \equiv 1 \pmod{\mathfrak{B}_F^{(i)}}.$$

The simple components are $Fe^{(i)}$.

3. Components as conjugates. If

$$x = \xi^{(1)}e^{(1)} + \cdots + \xi^{(n)}e^{(n)}$$

is the component representation of an element $x \in \mathfrak{K}$, then the components are conjugate. Thus $\xi^{(1)}, \ldots, \xi^{(n)}$ are the n conjugate values of x under the isomorphisms of $\mathfrak{K}$ to the conjugate fields $K^{(i)}$. If B is an absolute module in $\mathfrak{K}$ and $B^{(i)}$ its components in $\mathfrak{K}_F$, then

$$B = [\mathfrak{K}, B^{(1)} + \cdots + B^{(n)}].$$

4. Complementary bases. Each P-basis $t_1, \ldots, t_n$ of $\mathfrak{K}$ has a complementary basis $T_1, \ldots, T_n$. It is obtained from the expansion of the identity components

$$e^{(i)} = A_1^{(i)}t_1 + \cdots + A_n^{(i)}t_n.$$

If $\alpha_1^{(i)}, \ldots, \alpha_n^{(i)}$ are the conjugate bases corresponding to the t_j, then the matrices

$$(\alpha_j^{(i)})_{i,j} \quad \text{and} \quad (A_j^{(i)})_{i,j}$$

are reciprocal:

$$(\alpha_j^{(i)})(A_j^{(i)})^t = E_n.$$

The complement of the complementary basis is the original basis. Under a change $(s) = (t)P$, the complementary bases transform contragrediently:

$$(S) = P^{-1}(T).$$

If

$$c = C_1t_1 + \cdots + C_nt_n = c_1T_1 + \cdots + c_nT_n,$$

then, with Sp denoting the sum of conjugates,

$$C_i = \mathrm{Sp}(cT_i), \qquad c_i = \mathrm{Sp}(ct_i).$$

5. A defining equation. The planned discussion of the special case of a defining equation and of Lagrange's interpolation formula was left blank in the manuscript. The preceding structure theorem gives the invariant form of the result.

§6. The Different of an Order

The structure theorems for fields are now sharpened in the integral sense. Let $\mathfrak{h}$ be an integrally closed subring of P whose quotient field is P. Let $\mathfrak{O}$ be an $\mathfrak{h}$-order in $\mathfrak{K}$, meaning a subring containing $\mathfrak{h}$, generating $\mathfrak{K}$ as quotient field, and possessing a finite, not necessarily independent, $\mathfrak{h}$-module basis. Let $\mathfrak{o}$ be the corresponding order in K, and $\mathfrak{o}^{(i)}$ its conjugates. Let $\mathfrak{g}$ be the ring generated by these conjugate orders in F.

The direct products $\mathfrak{O}_\mathfrak{o}$ and $\mathfrak{O}_\mathfrak{g}$ exist. Thus the different of $\mathfrak{O}$ and of $\mathfrak{o}$ over $\mathfrak{h}$ is defined. Write $\mathfrak{B}$ for the difference ideal, $\mathfrak{A} = (0) : \mathfrak{B}$ for the difference quotient, and

$$\mathfrak{D} = \mathfrak{A}[\xi \to x], \qquad \mathfrak{d} = \mathfrak{A}[x \to \xi]$$

for the differents.

The structure theorem for the Galois extension ring of orders says that the difference ideal and quotient of the order extend to the corresponding ideals in the field case, and that the quotient for the order is the contraction of the extended quotient. Further,

$$\mathfrak{A} = \mathfrak{d}e^{(1)}, \qquad \mathfrak{A}^{(i)} = \mathfrak{d}^{(i)}e^{(i)}.$$

Hence

$$\mathfrak{d} = [\mathfrak{o}, \mathfrak{A}^{(1)} + \cdots + \mathfrak{A}^{(n)}].$$

The different $\mathfrak{d}$ consists exactly of those elements $x \in K$ for which $xe^{(1)}$ lies in $\mathfrak{O}_\mathfrak{o}$. It is nonzero.

The complementary module. If $\mathfrak{O}$ has an independent $\mathfrak{h}$-basis $t_1, \ldots, t_n$, then the complementary elements $T_1, \ldots, T_n$ form the basis of an $\mathfrak{h}$-module $\mathfrak{C}$, the complementary module of $\mathfrak{O}$. Then

$$\mathfrak{d} = \mathfrak{o} : \mathfrak{C}.$$

For, if

$$e^{(1)} = A_1 t_1 + \cdots + A_n t_n,$$

then $\mathfrak{d}$ is precisely the set of all elements d such that $dA_i \in \mathfrak{o}$ for all i. In the special case of a regular order with basis

$$1, z, \ldots, z^{n-1},$$

one obtains

$$\mathfrak{d} = (f'(z)).$$

Equivalently, the complementary module $\mathfrak{C}$ consists of all elements c of $\mathfrak{K}$ for which the traces

$$\mathrm{Sp}(co), \qquad o \in \mathfrak{O},$$

are integral, that is, lie in $\mathfrak{h}$.

Supplement. When the order has an independent $\mathfrak{h}$-module basis, the difference quotient $\mathfrak{A}$ is the intersection of all conjugate difference ideals except the one used:

$$\mathfrak{A} = [\mathfrak{O}_{\mathfrak{o}}, \mathfrak{B}^{(2)} \cap \cdots \cap \mathfrak{B}^{(n)}].$$

This follows because the difference ideal itself is then the contraction of its extension.

Fundamental equation. Suppose a fundamental equation exists, so that after adjoining indeterminates and passing to a quotient ring the order has a basis $1, u, \ldots, u^{n-1}$. If

$$G(u) = 0$$

is a defining equation, then the different is obtained from the polynomial derivative $G'(u)$. Under integral closure the required multiplication of contents of coefficient ideals holds, and ramification theory can be treated by considering the fundamental equation modulo p^t.

§7. Ramification Theory of the Principal Order modulo p^t

Let a number field be given, with $\mathfrak{h}$ its ring of integers; or let one have a relative field over the quotient ring of the principal order by a prime ideal, with p the basis element of that prime ideal. After adjoining indeterminates $u_1, \ldots, u_n$, both the principal order and the residue class ring modulo p^t possess the $\mathfrak{h}$-module basis

$$1, u, \ldots, u^{n-1}.$$

Thus in both cases the different is given by $G'(u)$. The different of the principal order passes modulo p^t into the different of the residue class ring modulo p^t.

If

$$p = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r},$$

then the residue class ring modulo p^t is the direct sum of the components corresponding to the residue class rings modulo $\mathfrak{p}_i^{te_i}$:

$$R = Re_1 + \cdots + Re_r.$$

By §4, the different of this direct sum is the direct sum of the component differents. Thus it is enough to treat a residue class ring modulo a single power $\mathfrak{p}^N$.

Choose ξ so that

$$\varphi(\xi) \equiv 0 \pmod{p}, \qquad \varphi(\xi) \not\equiv 0 \pmod{p^2},$$

where $\varphi(x)$ is a prime function modulo p of degree f. For $t \geq 2$ replace ξ by $\xi^* = \xi e_1$, where e_1 is the first idempotent occurring in the decomposition of the residue class ring modulo p^t. Then

$$\mathfrak{p} = (p, \varphi(\xi^*)).$$

A basis modulo p^t is

$$\xi^\mu \varphi(\xi)^\nu, \qquad \mu = 0, 1, \ldots, f-1, \quad \nu = 0, 1, \ldots, e-1.$$

Indeed, from $(\varphi(\xi), p) = \mathfrak{p}$ and $(p) = \mathfrak{p}^e$ follows

$$\varphi(\xi) = pM(\xi), \qquad M(\xi) \not\equiv 0 \pmod{p},$$

so higher powers can be successively expressed through lower ones. The defining equation for this basis has the form

$$F(x) = \varphi(x)^e + pM(x),$$

where $M(x)$ is not divisible by $\varphi(x)$ modulo p. Consequently the different of the residue class ring is given by the polynomial derivative $F'(\xi)$. This reduces the determination of exponents to Ore's results, and in particular to the supplement numbers. By passing to the quotient ring the same applies to relative differents; one adds only the statement that the different of an overfield is the product of the different of the base field and the relative different.

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